

# AMC 8

## §1 AMC 8 1985

1. (AMC 8 1985)  $\frac{3 \times 5}{9 \times 11} \times \frac{7 \times 9 \times 11}{3 \times 5 \times 7} =$  (A) 1 (B) 0 (C) 49 (D)  $\frac{1}{49}$  (E) 50

**Answer:** A

### Solution 1

By the commutative property:

$$\frac{3 \times 5}{9 \times 11} \times \frac{7 \times 9 \times 11}{3 \times 5 \times 7} = \frac{3 \times 5 \times 7 \times 9 \times 11}{9 \times 11 \times 3 \times 5 \times 7} = \frac{3 \times 5 \times 7 \times 9 \times 11}{3 \times 5 \times 7 \times 9 \times 11} = \boxed{\text{(A)} 1}$$

### Solution 2

Notice that the  $9 \times 11$  in the denominator of the first fraction cancels with the same term in the second fraction, the 7s in the numerator and denominator of the second fraction cancel, and the  $3 \times 5$  in the numerator of the first fraction cancels with the same term in the denominator second fraction. Then everything in the expression cancels, leaving us with  $\boxed{\text{(A)} 1}$ .

### Solution 3 (Brute Force)

(Note: This method is highly time consuming and should only be used as a last resort in math, also please do not do this.)  $3 \times 5 \times 7 \times 9 \times 11 = 10395$   $9 \times 11 \times 3 \times 5 \times 7 = 10395$

Thus, the answer is 1, or  $\boxed{\text{(A)} 1}$

### Solution 4 (Easy, similar to solution 1)

Notice the two fractions are multiplied, so we can consider them as 1 fraction. All terms cancel out, leaving us with  $\boxed{\text{(A)} 1}$ . [shunyipanda](#)

### Solution 5 (Capital Pi)

This fraction can be simplified to

$$\frac{\prod_{k=1}^5 (2k+1)}{\prod_{k=1}^5 (2k+1)}$$

The numerator and denominator are the same, so the answer is  $\boxed{\text{(A)} 1}$ . [shunyipanda](#) Note: please can somebody make this look better

2. (AMC 8 1985)  $90+91+92+93+94+95+96+97+98+99 =$  (A) 845 (B) 945 (C) 1005 (D) 10

**Answer:** B

### Solution 1

Rearrange the terms to form 5 pairs, each of which sums to 189:

$$\begin{aligned} 90 + 91 + \cdots + 98 + 99 &= (90 + 99) + (91 + 98) + (92 + 97) + (93 + 96) + (94 + 95) \\ &= 189 + 189 + 189 + 189 + 189 \\ &= 189 \cdot 5 \\ &= \boxed{\text{(B)} 945} \end{aligned}$$

### Solution 2

Rewrite this sum as:

$$\begin{aligned} 90 + 91 + 92 + \cdots + 98 + 99 &= 90 + (90 + 1) + (90 + 2) + \cdots + (90 + 8) + (90 + 9) \\ &= 90 \cdot 10 + (1 + 2 + \cdots + 8 + 9) \\ &= 900 + \frac{9 \cdot 10}{2} \\ &= \boxed{\text{(B)} 945} \end{aligned}$$

shunyipanda (Minor edits)

### Solution 3

Rewrite this sum as:

$$\begin{aligned} 90 + 91 + \cdots + 98 + 99 &= (100 - 10) + (100 - 9) + \cdots + (100 - 2) + (100 - 1) \\ &= 100 \cdot 10 - (10 + 9 + \cdots + 2 + 1) \\ &= 1000 - \frac{10 \cdot 11}{2} \\ &= \boxed{\text{(B)} 945} \end{aligned}$$

### Solution 4 (Arithmetic Series Formula)

Recall that the sum of an arithmetic series is  $\frac{n}{2}(a_1 + a_n)$ , where  $n$  is the number of terms,  $a_1$  is the first term and  $a_n$  is the last term.

Therefore, the sum of this series is  $\frac{10}{2}(90 + 99) = \boxed{\text{(B)} 945}$ .

### Solution 5 (Difference of Triangular Numbers)

Notice that this expression is the sum of the first 99 positive integers minus the sum of the first 89 positive integers, that is,  $\frac{99 \cdot 100}{2} - \frac{89 \cdot 90}{2} = 4950 - 4005 = \boxed{\text{(B)} 945}$ .

### Solution 6 (Answer Choices, Estimation)

Since all the terms are between 90 and 100, the sum must be between  $10 \cdot 90 = 900$  and  $10 \cdot 100 = 1000$ . Only  $\boxed{\text{(B)} 945}$  is between 900 and 1000.

### Solution 7 (Brute Force)

Adding all the terms yields  $\boxed{\text{(B)} 945}$ .

3. (AMC 8 1985)  $\frac{10^7}{5 \times 10^4} =$  (A) .002      (B) .2      (C) 20      (D) 200      (E) 2000

Answer: D

### Solution 1

Cancelling powers of 10 yields:

$$\frac{10^7}{5 \times 10^4} = \frac{10^3}{5} = \frac{1000}{5} = \boxed{\text{(D)} 200}$$

(Reverted a Mistake)

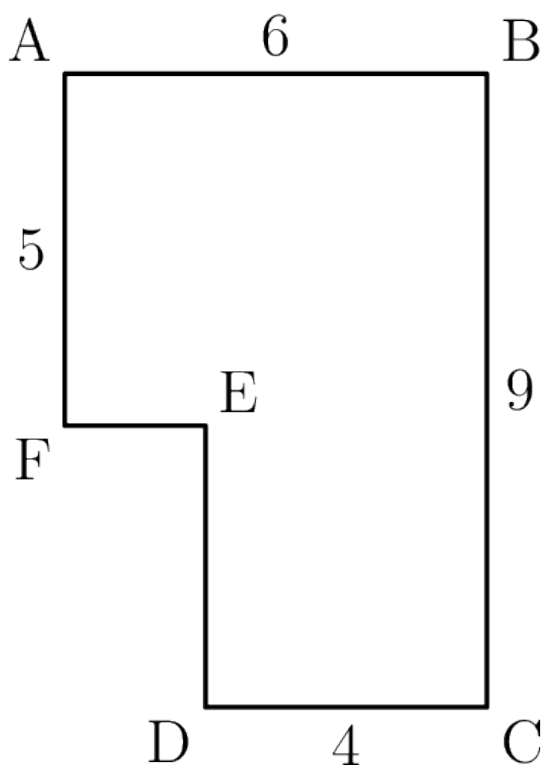
**Solution 2 (Brute Force)**

$$\frac{10^7}{5 \times 10^4} = \frac{10000000}{50000} = \boxed{\text{(D)} 200}$$

**Solution 3 (Reasoning)**

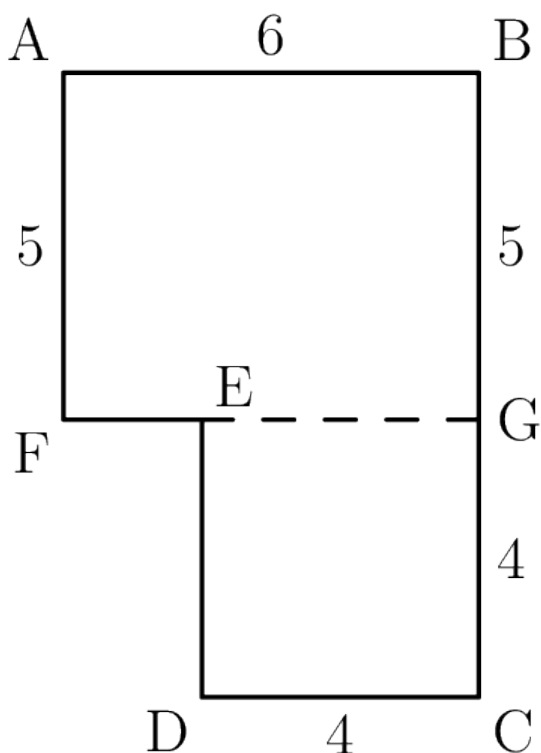
As we can see, the numerator is way larger than the denominator so answer choices (A) .002, (B) .2, and (C) 20 are eliminated.  $\frac{10^7}{10^4}$  is 1000, but in this question the expression,  $\frac{10^7}{5 \times 10^4}$ , is smaller than 1000 since the denominator is larger than  $10^4$ . Therefore the answer is  $\boxed{\text{(D)} 200}$ . [shunyipanda](#)

4. (AMC 8 1985) The area of polygon  $ABCDEF$ , in square units, is  
 (A) 24      (B) 30      (C) 46      (D) 66      (E) 74



**Answer:** C

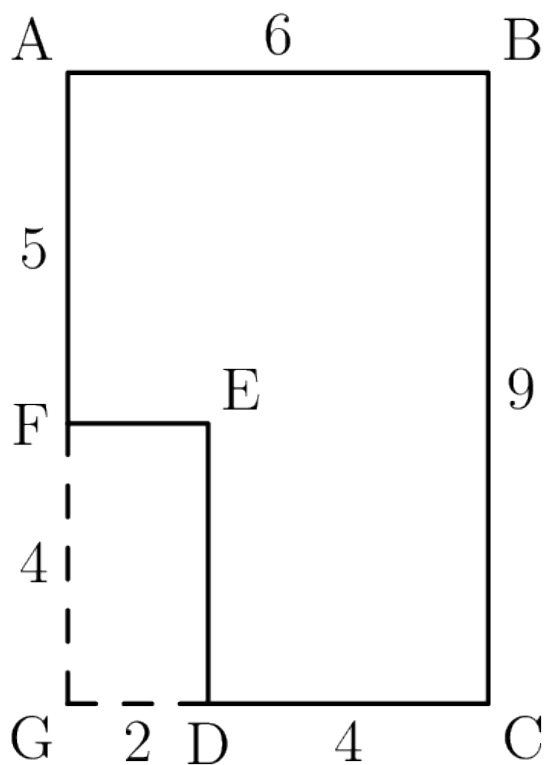
**Solution 1 (Adding Rectangular Areas)**



Extend  $\overline{FE}$  to meet  $\overline{BC}$  at  $G$ , dividing  $ABCDEF$  into rectangles  $ABGF$  and  $EGCD$ . Then  $BG = AF = 5$  and  $GC = 9 - 5 = 4$ .

So  $ABGF$  has area  $6 \cdot 5 = 30$ , while  $EGCD$  has area  $4 \cdot 4 = 16$ . Therefore, the total area of  $ABCDEF$  is  $30 + 16 = \boxed{\text{(C) } 46}$ .

**Solution 2 (Subtracting Rectangular Areas)**

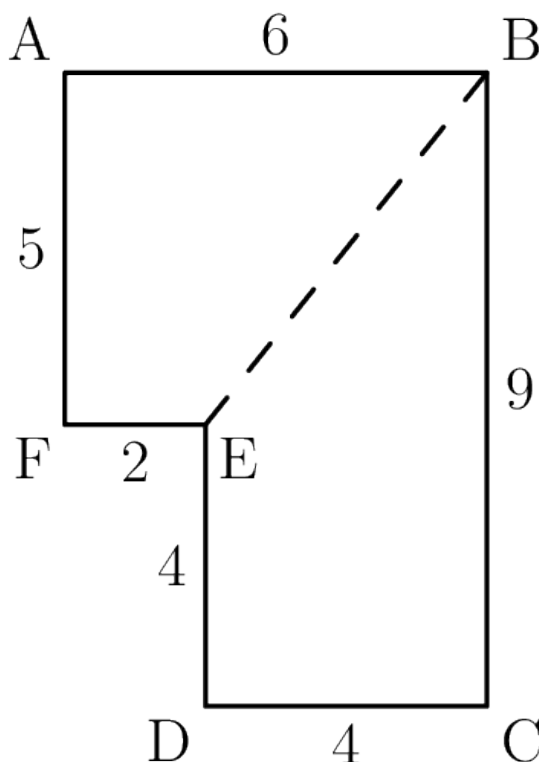


Extend  $\overline{CD}$  and  $\overline{AF}$  to meet at  $G$ . Then  $FG = 9 - 5 = 4$  and  $GD = 6 - 4 = 2$ .

The area of  $ABCDEF$  is the difference between the area of rectangle  $ABCG$  and the area of rectangle  $FEDG$ .

Since the area of  $ABCG$  is  $6 \cdot 9 = 54$ , while the area of  $FEDG$  is  $4 \cdot 2 = 8$ , the area of  $ABCDEF$  is  $54 - 8 = \boxed{\text{(C) } 46}$ .

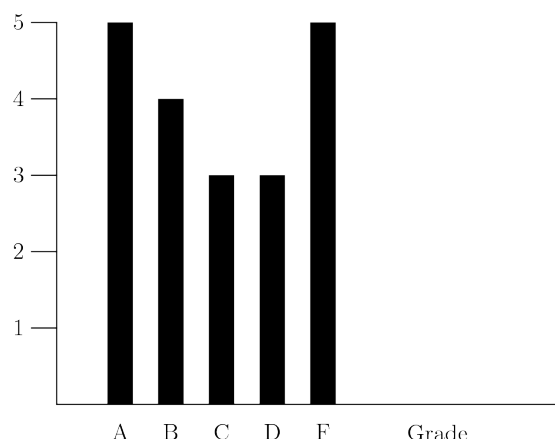
**Solution 3 (Trapezoids)**



First, notice  $FE = 6 - 4 = 2$  and  $ED = 9 - 5 = 4$ .

Draw  $\overline{BE}$  to divide  $ABCDEF$  into trapezoids  $ABEF$  and  $BCDE$ . Then the area of  $ABEF = \frac{5(6+2)}{2} = 20$  and the area of  $BCDE$  is  $\frac{4(4+9)}{2} = 26$ , so the area of  $ABCDEF$  is  $20 + 26 = \boxed{\text{(C) } 46}$ .

5. (AMC 8 1985) The bar graph shows the grades in a mathematics class for the last grading period. If  $A$ ,  $B$ ,  $C$ , and  $D$  are satisfactory grades, what fraction of the grades shown in the graph are satisfactory?
- (A)  $\frac{1}{2}$     (B)  $\frac{2}{3}$     (C)  $\frac{3}{4}$     (D)  $\frac{4}{5}$     (E)  $\frac{9}{10}$



**Answer:** C

**Solution 1**

Of  $5 + 4 + 3 + 3 + 5 = 20$  students in the class,  $5 + 4 + 3 + 3 = 15$  received satisfactory grades. So  $\frac{15}{20} = \boxed{\text{(C)} \frac{3}{4}}$  of the class received satisfactory grades.

**Solution 2**

Of  $5 + 4 + 3 + 3 + 5 = 20$  students in the class, 5 did not receive satisfactory grades. That is,  $\frac{5}{20} = \frac{1}{4}$  of the students did not receive satisfactory grades, and thus  $1 - \frac{1}{4} = \boxed{\text{(C)} \frac{3}{4}}$  did.

6. (AMC 8 1985) A ream of paper containing 500 sheets is 5 cm thick. Approximately how many sheets of this type of paper would there be in a stack 7.5 cm high?

(A) 250      (B) 550      (C) 667      (D) 750      (E) 1250

**Answer:** D

**Solution 1**

Each sheet is  $\frac{5}{500} = 0.01$  cm thick, so a 7.5 cm stack contains  $\frac{7.5}{0.01} = \boxed{\text{(D)} 750}$  sheets.

**Solution 2**

Since each sheet has the same thickness, the number of sheets in a stack is directly proportional to the thickness of a stack.

Let  $x$  be the number of sheets in a 7.5 cm stack. Then  $\frac{5}{500} = \frac{7.5}{x}$ , so  $5x = 3750$  and thus  $x = \boxed{\text{(D)} 750}$ .

**Solution 3**

Using dimensional analysis with the conversion factor  $\frac{500 \text{ sheets}}{5 \text{ cm}}$  yields  $(7.5 \text{ cm}) \left( \frac{500 \text{ sheets}}{5 \text{ cm}} \right) = \boxed{\text{(D)} 750}$  sheets.

**Solution 4**

Each centimeter of a stack contains  $\frac{500}{5} = 100$  sheets of paper, so a 7.5 cm stack must contain  $7.5 \cdot 100 = \boxed{\text{(D)} 750}$  sheets of paper. [shunyipanda](#)

**Solution 5**

Since a 7.5 cm stack is 1.5 times as thick as a 5 cm stack, a 7.5 cm stack contains  $1.5 \cdot 500 = \boxed{\text{(D)} 750}$  sheets of paper. **Shriya2**

7. (AMC 8 1985) A “stair-step” figure is made of alternating black and white squares in each row. Rows 1 through 4 are shown. All rows begin and end with a white square. The number of black squares in the 37th row is

(A) 34      (B) 35      (C) 36      (D) 37      (E) 38

**Answer:** C

**Solution 1**

The first row has 0 black squares, while each row after the first has 1 more black square than the previous row. Therefore, the second row has 1 black square, the third row has 2 black squares, and so on, until the 37<sup>th</sup> row has  $\boxed{\text{(C)} 36}$  black squares.

**Solution 2**

Let  $f(n)$  be the number of black squares in the  $n^{\text{th}}$  row. Since each row has 1 more black square than the previous row,  $f(n)$  is linear with slope 1, that is,  $f(n) = n + b$  for some constant  $b$ . Since  $f(1) = 0$ ,  $0 = 1 + b$  and  $b = -1$ . So  $f(n) = n - 1$  and the 37<sup>th</sup> row has  $f(37) = 37 - 1 = \boxed{\text{(C)} 36}$  black squares.

8. (AMC 8 1985) If  $a = -2$ , the largest number in the set  $\{-3a, 4a, \frac{24}{a}, a^2, 1\}$  is

(A)  $-3a$       (B)  $4a$       (C)  $\frac{24}{a}$       (D)  $a^2$       (E) 1

**Answer:** A

**Solution 1**

Substituting  $a = -2$  yields  $\{-3a, 4a, \frac{24}{a}, a^2, 1\} = \{(-3)(-2), 4(-2), \frac{24}{-2}, (-2)^2, 1\} = \{6, -8, -12, 4, 1\}$ .

The largest element of this set is therefore 6, that is,  $\boxed{\text{(A)} -3a}$ .

**Solution 2**

Since  $a$  is negative,  $-3a$ ,  $a^2$ , and 1 must be positive while  $4a$  and  $\frac{24}{a}$  are negative. Evaluating the positive expressions shows that  $\boxed{\text{(A)} -3a} = 6$  is the largest.

9. (AMC 8 1985) The product of the 9 factors  $(1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{4}) \cdots (1 - \frac{1}{10}) =$

(A)  $\frac{1}{10}$       (B)  $\frac{1}{9}$       (C)  $\frac{1}{2}$       (D)  $\frac{10}{11}$       (E)  $\frac{11}{2}$

**Answer:** A

**Solution 1**

Simplifying each factor and cancelling yields  $(1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{4}) \cdots (1 - \frac{1}{10}) = (\frac{1}{2})(\frac{2}{3})(\frac{3}{4}) \cdots (\frac{9}{10}) = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10} = \boxed{\text{(A)} \frac{1}{10}}$ .

**Solution 2 (Induction, Generalization)**

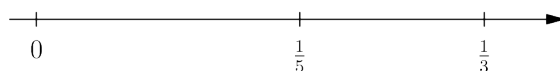
It can be proven by induction that  $(1 - \frac{1}{2}) \cdots (1 - \frac{1}{n}) = \frac{1}{n}$  for all integers  $n \geq 2$ .

For the base case, if  $n = 2$ , then this product is simply  $1 - \frac{1}{2} = \frac{1}{2}$ .

For the inductive case, assume that  $(1 - \frac{1}{2}) \cdots (1 - \frac{1}{k}) = \frac{1}{k}$ . Then  $(1 - \frac{1}{2}) \cdots (1 - \frac{1}{k+1}) = (1 - \frac{1}{2}) \cdots (1 - \frac{1}{k})(1 - \frac{1}{k+1}) = (\frac{1}{k})(\frac{k}{k+1}) = \frac{k}{k(k+1)} = \frac{1}{k+1}$ . This completes the induction.

Applying this lemma for  $n = 10$  yields (A)  $\frac{1}{10}$ .

10. (AMC 8 1985) The fraction halfway between  $\frac{1}{5}$  and  $\frac{1}{3}$  (on the number line) is



- (A)  $\frac{1}{4}$     (B)  $\frac{2}{15}$     (C)  $\frac{4}{15}$     (D)  $\frac{53}{200}$     (E)  $\frac{8}{15}$

**Answer:** C

**Solution 1**

The fraction halfway between  $\frac{1}{5}$  and  $\frac{1}{3}$  is their average, which is  $\frac{1}{2} \left( \frac{1}{5} + \frac{1}{3} \right) = \frac{1}{2} \left( \frac{8}{15} \right) =$  (C)  $\frac{4}{15}$

**Solution 2**

- For any two fractions that have the property  $\frac{a}{b}$  and  $\frac{a}{c}$  where  $a$  is equal for both fractions;

Set aside the  $a$ . Take the average of  $b$  and  $c$ . For this example, the average of 3 and 5 is 4. Then multiply those numbers.  $3 \cdot 5 = 15$ . The number that is exactly in between  $\frac{a}{b}$  and  $\frac{a}{c}$  will be  $\frac{a \cdot ((b+c)/2)}{b \cdot c}$ .

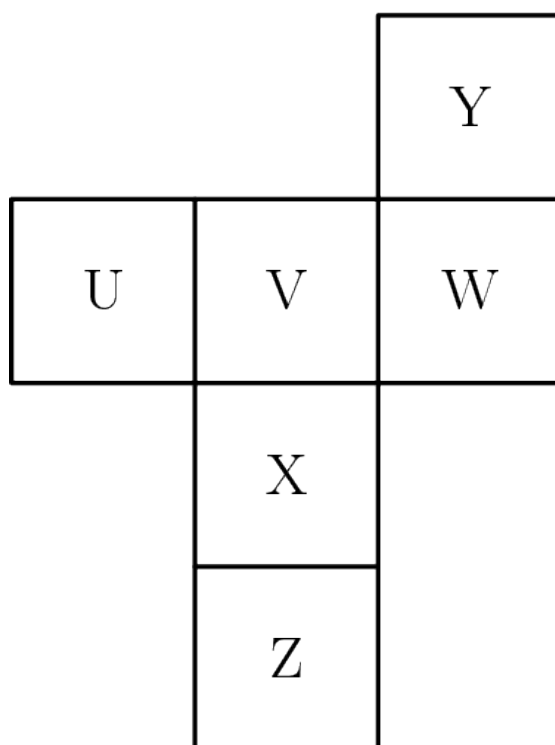
In this example, it would be  $\frac{1 \cdot ((3+5)/2)}{3 \cdot 5}$ , which simplifies to  $\frac{4}{15}$ .

Therefore, the answer is C

Edit by mathmagical

11. (AMC 8 1985) A piece of paper containing six joined squares labeled as shown in the diagram is folded along the edges of the squares to form a cube. The label of the face opposite the face labeled X is





- (A) Z    (B) U    (C) V    (D) W    (E) Y

**Answer:** E

**Solution 1**

Notice that two faces of a cube are opposite each other if and only if they share no vertices. The diagram shows that X shares at least one vertex with each of U, V, W, and Z, so by process of elimination X must be opposite **(E) Y**.

**Solution 2**

To find the face opposite X, we can find the faces sharing an edge with X, so the only face remaining will be the opposite face.

Clearly, V and Z share an edge with X. Also, the faces V, X, and W share a common vertex, therefore X shares an edge with W. Similarly, the faces U, V, and X share a common vertex, so X shares an edge with U.

The only face X doesn't share an edge with is Y, which is choice **(E)**.

12. (AMC 8 1985) A square and a triangle have equal perimeters. The lengths of the three sides of the triangle are 6.2 cm, 8.3 cm and 9.5 cm. The area of the square is  
 (A)  $24 \text{ cm}^2$     (B)  $36 \text{ cm}^2$     (C)  $48 \text{ cm}^2$     (D)  $64 \text{ cm}^2$     (E)  $144 \text{ cm}^2$

**Answer:** B

**Solution**

The perimeter of the triangle is  $6.2 + 8.3 + 9.5 = 24$  cm. Therefore, the perimeter of the square is also 24 cm, so its side length is  $\frac{24}{4} = 6$  cm and its area is  $6^2 =$   
**(B)  $36 \text{ cm}^2$** .

13. (AMC 8 1985) If you walk for 45 minutes at a rate of 4 mph and then run for 30 minutes at a rate of 10 mph, how many miles will you have gone at the end of one hour and 15 minutes?

- (A) 3.5 miles    (B) 8 miles    (C) 9 miles    (D)  $25\frac{1}{3}$  miles    (E) 480 miles

**Answer:** B

**Solution**

Since you walked for 45 minutes, or  $\frac{3}{4}$  hour, you walked  $\frac{3}{4} \cdot 4 = 3$  miles. Since you ran for 30 minutes, or  $\frac{1}{2}$  hour, you ran  $\frac{1}{2} \cdot 10 = 5$  miles. Therefore, you traveled a total of  $3 + 5 = \boxed{\text{(B) 8 miles}}$ .

14. (AMC 8 1985) The difference between a 6.5% sales tax and a 6% sales tax on an item priced at \$20 before tax is

- (A) \$.01 (B) \$.10 (C) \$.50 (D) \$1 (E) \$10

**Answer:** B

**Solution 1**

With a 6.5% sales tax, an item costs 1.065 times its listed price, while with a 6% sales tax, it costs 1.06 times its listed price. The difference between these prices is therefore  $1.065 - 1.06 = .005$  times its listed price. For this item, this difference is  $.005 \cdot \$20 = \boxed{\text{(B) $.10}}$ .

**Solution 2**

With a 6.5% sales tax, the item costs  $1.065 \cdot \$20 = \$21.30$ . With a 6% sales tax, the item costs  $1.06 \cdot \$20 = \$21.20$ . So the difference is  $\$21.30 - \$21.20 = \boxed{\text{(B) $.10}}$ .

15. (AMC 8 1985) How many whole numbers between 100 and 400 contain the digit 2?

- (A) 100    (B) 120    (C) 138    (D) 140    (E) 148

**Answer:** C

**Solution 1 (Complementary Counting)**

Since 400 does not contain the digit 2, only the set of integers from 100 to 399 needs to be considered. Notice that this is the set of all three-digit integers with 1, 2, or 3 as their first digit.

Since there are 3 choices for the first digit and 10 choices for each of the second and third digits, this set contains  $3 \cdot 100 = 300$  numbers.

For a three-digit integer between 100 and 399 that does not contain the digit 2, there are  $3 - 1 = 2$  choices for the first digit and  $10 - 1 = 9$  choices for each of the second and third digit, for a total of  $2 \cdot 9 \cdot 9 = 162$  numbers.

Therefore,  $300 - 162 = \boxed{\text{(C) 138}}$  of these numbers contain the digit 2.

**Solution 2**

All 100 integers from 200 to 299 start with 2.

Of the integers from 100 to 400 not starting with 2, 20 have a second digit of 2:  $\{120, 121, 122, \dots, 129, 320, 321, 322, \dots, 329\}$ , while 20 have a third digit of 2:  $\{102, 112, 122, \dots, 192, 302, 312, 322, \dots, 392\}$ . Of these, 122 and 322 have both a second and third digit of 2, so a total of  $20 + 20 - 2 = 38$  integers between 100 and 400 not starting with 2 contain the digit 2.

Therefore, a total of  $100 + 38 = \boxed{\text{(C) 138}}$  integers between 100 and 400 contain the digit 2.

16. (*AMC 8 1985*) The ratio of boys to girls in Mr. Brown's math class is 2 : 3. If there are 30 students in the class, how many more girls than boys are in the class?
- (A) 1      (B) 3      (C) 5      (D) 6      (E) 10

**Answer:** D

**Solution 1**

Let  $x$  be half the number of boys. Then there are  $2x$  boys and  $3x$  girls. So  $2x + 3x = 5x = 30$  and thus  $x = 6$ . Therefore, between the number of girls and the number of boys is  $3x - 2x = x = \boxed{\text{(D) } 6}$ .

**Solution 2**

Since the ratio of boys to girls is 2 : 3,  $\frac{2}{2+3} = \frac{2}{5}$  of the students are boys and  $\frac{3}{2+3} = \frac{3}{5}$  are girls. Thus the difference between the number of boys and the number of girls is  $\frac{3}{5} - \frac{2}{5} = \frac{1}{5}$  of the number of students in the class, that is  $\frac{1}{5} \cdot 30 = \boxed{\text{(D) } 6}$ .

17. (*AMC 8 1985*) If your average score on your first six mathematics tests was 84 and your average score on your first seven mathematics tests was 85, then your score on the seventh test was

(A) 86      (B) 88      (C) 90      (D) 91      (E) 92

**Answer:** D

**Solution 1**

Since your average score on your first 6 tests is 84, your total score on your first 6 tests is  $6 \cdot 84 = 504$ .

Since your average score on your first 7 tests is 85, your total score on your first 7 tests is  $7 \cdot 85 = 595$ .

Therefore, your score on your 7<sup>th</sup> test was  $595 - 504 = \boxed{\text{(D) } 91}$ .

**Solution 2**

Let's remove the condition that the average of the first seven tests is 85, and say the 7th test score was a 0. Then, the average of the first seven tests would be

$$\frac{6 \times 84}{7} = 72$$

If we increase the seventh test score by 7, the average will increase by  $\frac{7}{7} = 1$ . We need the average to increase by  $85 - 72 = 13$ , so the seventh test score is  $7 \times 13 = 91$  more than 0, which is clearly  $\boxed{\text{(D) } 91}$ .

**Solution 3 (Answer Choices)**

Since 84 is divisible by 7, your total score on your first 6 tests, that is,  $6 \cdot 84$ , is divisible by 7. Also, your total score on your first 7 tests, that is,  $85 \cdot 7$ , must be divisible by 7. Therefore, your score on your 7<sup>th</sup> test is also divisible by 7. Of the answer choices, only  $\boxed{\text{(D) } 91}$  is divisible by 7.

18. (*AMC 8 1985*) Nine copies of a certain pamphlet cost less than \$10.00 while ten copies of the same pamphlet (at the same price) cost more than \$11.00. How much does one copy of this pamphlet cost?

(A) \$1.07 (B) \$1.08 (C) \$1.09 (D) \$1.10 (E) \$1.11

**Answer:** E

**Solution**

Let  $p$  be the cost of one pamphlet in dollars. The first restriction implies  $9p < 10$ , that is,  $p < 1.1111\dots$ , while the second restriction implies  $10p > 11$ , that is,  $p > 1.10$ . Thus  $1.10 < p < 1.1111\dots$ , and thus  $p = \boxed{\text{(E)} \$1.11}$ , since the price must be a whole number of cents.

Note: the second restriction alone eliminates all the incorrect answer choices.

19. (AMC 8 1985) If the length and width of a rectangle are each increased by 10%, then the perimeter of the rectangle is increased by  
 (A) 1%      (B) 10%      (C) 20%      (D) 21%      (E) 40%

**Answer:** B

**Solution 1**

Let  $w$ ,  $l$ , and  $p$  be the original width, length, and perimeter of the rectangle, respectively. Then  $p = 2(w + l)$ .

The width and length are increased to  $1.1w$  and  $1.1l$ , respectively. Therefore, the new perimeter is  $2(1.1w + 1.1l) = 2 \cdot 1.1(w + l) = 1.1 \cdot 2(w + l) = 1.1p$ , or  $\boxed{\text{(B)} 10\%}$  more than the original perimeter.

**Solution 2**

Since each side is scaled by a factor of 1.1, the new rectangle is similar to the original, and therefore its perimeter is also scaled by a factor of 1.1, that is, a  $\boxed{\text{(B)} 10\%}$  increase.

**Solution 3**

Suppose the original rectangle is a square of side length 10, so its perimeter is  $4 \cdot 10 = 40$ . After scaling, its sides will have length  $1.1 \cdot 10 = 11$ , so its perimeter will be  $11 \cdot 4 = 44$ . Therefore, the new perimeter is  $\frac{44}{40} = 1.1$  times the original perimeter, a  $\boxed{\text{(B)} 10\%}$  increase.

20. (AMC 8 1985) In a certain year, January had exactly four Tuesdays and four Saturdays. On what day did January 1 fall that year?  
 (A) Monday      (B) Tuesday      (C) Wednesday      (D) Friday      (E) Saturday

**Answer:** C

**Solution 1**

January has 31 days, that is, 4 weeks followed by 3 extra days. Since there are only 4 Tuesdays and 4 Saturdays, the 3 extra days must include neither Tuesday nor Saturday. This is only possible if the extra days are Wednesday through Friday. Therefore, January 29 (and thus January 1) is a  $\boxed{\text{(C)} \text{Wednesday}}$ .

**Solution 2 (Answer Choices)**

If January 1 was on Monday or Tuesday, then there would be 5 Tuesdays. If January 1 was on Friday or Saturday, there would be 5 Saturdays. This eliminates all choices except  $\boxed{\text{(C)} \text{Wednesday}}$ .

21. (AMC 8 1985) Mr. Green receives a 10% raise every year. His salary after four such raises has gone up by what percent?  
 (A) less than 40%      (B) 40%      (C) 44%      (D) 45%      (E) more than 45%

**Answer:** E

**Solution 1**

Each year, Mr. Green's salary is multiplied by 1.1. Therefore, after 4 years, his salary will be  $1.1^4 = 1.4641$  times his current salary, for a total increase of 46.41%, which is (E) more than 45%.

**Solution 2**

Suppose Mr. Green's salary is originally \$100.00. After 1 year, his salary will be \$110.00. After 2 years, his salary will be \$121.00. After 3 years, his salary will be \$133.10. After 4 years, his salary will be \$146.41, which is 1.4641 times his original salary. This is a 46.41% increase, so the answer is (E) more than 45%.

22. (AMC 8 1985) Assume every 7-digit whole number is a possible telephone number except those that begin with 0 or 1. What fraction of telephone numbers begin with 9 and end with 0?

(A)  $\frac{1}{63}$       (B)  $\frac{1}{80}$       (C)  $\frac{1}{81}$       (D)  $\frac{1}{90}$       (E)  $\frac{1}{100}$

**Answer:** B

**Solution 1**

For a valid phone number, there are  $10 - 2 = 8$  possibilities for the first digit and 10 for each of the other 6 digits. Therefore, there are  $8 \cdot 10^6 = 8000000$  valid phone numbers.

For a phone number that starts with 9 and ends with 0, there are 10 possibilities for each of the middle 5 digits. Therefore, there are  $10^5 = 100000$  phone numbers that start with 9 and end with 0.

Therefore,  $\frac{100000}{8000000} =$  (B)  $\frac{1}{80}$  of valid phone numbers start with 9 and end with 0.

**Solution 2**

Notice that selecting a random valid phone number is identical to randomly and independently selecting each digit of the phone number.

There are  $10 - 2 = 8$  possibilities for the first digit and 10 possibilities for the last digit, so the probability of selecting a phone number with first digit 9 and last digit

0 is  $\frac{1}{8} \cdot \frac{1}{10} =$  (B)  $\frac{1}{80}$ .

23. (AMC 8 1985) King Middle School has 1200 students. Each student takes 5 classes a day. Each teacher teaches 4 classes. Each class has 30 students and 1 teacher. How many teachers are there at King Middle School?

(A) 30      (B) 32      (C) 40      (D) 45      (E) 50

**Answer:** E

**Solution 1**

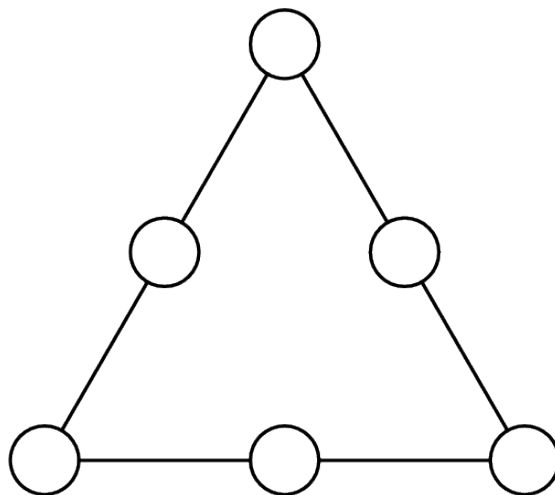
Each of the 1200 students attends 5 class sessions, so they collectively attend a total of  $1200 \cdot 5 = 6000$  class sessions. Since 30 students attend each class session, there must be  $\frac{6000}{30} = 200$  class sessions held. Since each class has 1 teacher, the teachers collectively host a total of  $200 \cdot 1 = 200$  class sessions. Since each teacher hosts 4 class sessions, there must be  $\frac{200}{4} =$  (E) 50 teachers.

**Solution 2 (Similar to above solution)**

Each class is 30 students so, there are 40 classes in total, that need to take 200 periods every day.

As each teacher is only taking 4 periods, there will be needed  $\frac{200}{4} = 50$  teachers.

24. (AMC 8 1985) In a magic triangle, each of the six whole numbers 10–15 is placed in one of the circles so that the sum,  $S$ , of the three numbers on each side of the triangle is the same. The largest possible value for  $S$  is



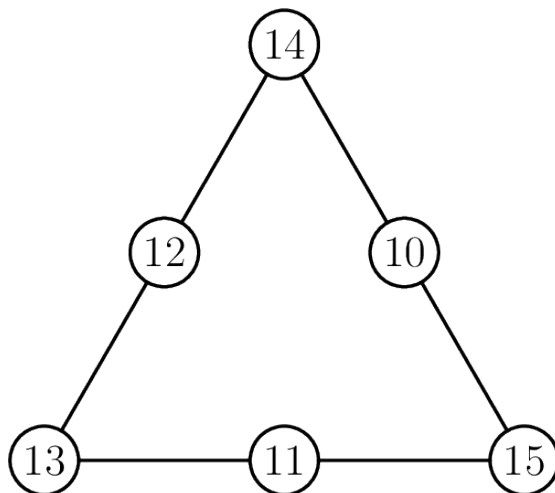
- (A) 36      (B) 37      (C) 38      (D) 39      (E) 40

**Answer:** D

**Solution 1**

Let  $x$  be the sum of the vertices and  $y$  be the sum of the midpoints. Then  $x + y = 10 + 11 + 12 + 13 + 14 + 15 = 75$ . Also, since each vertex is included in 2 sides while each midpoint is included in 1 side,  $3S = 2x + y = x + (x + y) = x + 75$ , so  $S = \frac{x}{3} + 25$ . Therefore, maximizing  $x$  also maximizes  $S$ .

Since  $x \leq 13 + 14 + 15 = 42$ ,  $S \leq \frac{42}{3} + 25 = 39$ . Achieving  $S = \boxed{\text{(D) } 39}$  is possible using the following configuration:

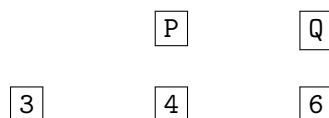


**Solution 2 (Answer Choices)**

As in Solution 1, notice that if  $x$  is the sum of the vertices, then  $S = \frac{x}{3} + 25$ . Trying  $S = 40$ , we find that  $x = 45$  which is too large. Trying  $S = \boxed{\text{(D)} 39}$ , we find that  $x = 42$  which works as shown in the configuration in Solution 1.

-sethecool, edited by j314andrews

25. (AMC 8 1985) Five cards are lying on a table as shown.



Each card has a letter on one side and a whole number on the other side. Jane said, "If a vowel is on one side of any card, then an even number is on the other side." Mary showed Jane was wrong by turning over one card. Which card did Mary turn over?

- (A) 3      (B) 4      (C) 6      (D) P      (E) Q

**Answer:** A

**Solution 1**

Recall that a counterexample to the statement "If  $X$ , then  $Y$ " is a situation in which  $X$  is true but  $Y$  is false. Therefore, a counterexample to Jane's statement would be a card with a vowel on one side and an odd number on the other side. Of the cards shown, only  $\boxed{\text{(A)} 3}$  could fit this description.

**Solution 2 (Exploitation of Multiple Choice)**

The only attributes of letters and numbers mentioned in Jane's statement are vowel/consonant and odd/even. Only one answer is correct, so since 4 and 6 are both even numbers, and P and Q are both consonants, none of those answers could be correct. Thus  $\boxed{\text{(A)} 3}$  must be correct.

RCFX - Minor Changes

## §2 AMC 8 1986

1. (AMC 8 1986) In July 1861, 366 inches of rain fell in Cherrapunji, India. What was the average rainfall in inches per hour during that month?

- (A)  $\frac{366}{31 \times 24}$       (B)  $\frac{366 \times 31}{24}$       (C)  $\frac{366 \times 24}{31}$       (D)  $\frac{31 \times 24}{366}$       (E)  $366 \times 31 \times 24$

**Answer:** A

**Solution 1**

July has 31 days, or  $31 \times 24$  hours. Therefore, the average rainfall was  $\boxed{\text{(A)} \frac{366}{31 \times 24}}$  inches per hour.

**Solution 2 (Dimensional Analysis)**

We can express the number of inches of rain in a **ratio**,  $\frac{366 \text{ inches}}{1 \text{ month}}$ . We need to find to find the number of hours, but first we need to find the number of days. We know there are 31 days in July, so we get

$$\frac{366 \text{ inches}}{1 \text{ month}} \times \frac{1 \text{ month}}{31 \text{ days}}$$

We know there are 24 hours in a day, so we get

$$\frac{366 \text{ inches}}{1 \text{ month}} \times \frac{1 \text{ month}}{31 \text{ days}} \times \frac{1 \text{ day}}{24 \text{ hours}}$$

And simplifying gets us  $\frac{366 \text{ inches}}{31 \times 24 \text{ hours}}$ , which is **(A)**.

Note that in the simplifying step, we noted that the month and day units cancel, leaving us with only inches per hour.

2. (AMC 8 1986) Which of the following numbers has the largest reciprocal?

(A)  $\frac{1}{3}$     (B)  $\frac{2}{5}$     (C) 1    (D) 5    (E) 1986

**Answer:** A

**Solution 1**

Smaller positive real numbers have larger reciprocals. Of these numbers, **(A)**  $\frac{1}{3}$  is the smallest.

**Solution 2**

The reciprocals of the answer choices are 3,  $2\frac{1}{2}$ , 1,  $\frac{1}{5}$ , and  $\frac{1}{1986}$ . Since 3 is the largest,

**(A)**  $\frac{1}{3}$  is correct.

3. (AMC 8 1986) The smallest sum one could get by adding three different numbers from the set  $\{7, 25, -1, 12, -3\}$  is

(A)  $-3$     (B)  $-1$     (C) 3    (D) 5    (E) 21

**Answer:** C

**Solution**

The smallest possible sum is found by adding the 3 smallest elements of the set. This sum is  $(-3) + (-1) + 7 = \mathbf{(C) 3}$ .

4. (AMC 8 1986) The product  $(1.8)(40.3 + .07)$  is closest to

(A) 7    (B) 42    (C) 74    (D) 84    (E) 737

**Answer:** C

**Solution 1**

$(1.8)(40.3 + .07) = (1.8)(40.37)$ , which is slightly more than  $(1.8)(40) = 72$ . So this expression is closest to **(C) 74**.

-abcde26, edited by j314andrews

**Solution 2 (Brute Force)**

$(1.8)(40.3 + .07) = (1.8)(40.37) = 72.666$ , which is closest to **(C) 74**.

5. (AMC 8 1986) A contest began at noon one day and ended 1000 minutes later. At what time did the contest end?

(A) 10:00 p.m.    (B) midnight    (C) 2:30 a.m.    (D) 4:40 a.m.    (E) 6:40 a.m.

**Answer:** D

**Solution**



The contest lasts 1000 minutes, which is  $\frac{1000}{60} = 16\frac{2}{3}$  hours or 16 hours and 40 minutes. Since the contest starts at 12:00 p.m., it ends at **(D)** 4:40 a.m.

**Solution 2**

There are 60 minutes in an hour. So, we can easily eliminate some of the choices by noting that noon is exactly 720 minutes away from midnight. Since  $720 < 1000$ , we know that it cannot be A or B. Because midnight is 720 minutes away, we know that the contest ended  $1000 - 720 = 280$  minutes after midnight. The highest multiple of 60 that will fit into 280 is 240, which is  $4 \times 60$ , and the remainder is 40 minutes, meaning that the contest ended at 4 : 40 a.m. 4 : 40 is **(D)**

6. (AMC 8 1986)  $\frac{2}{1-\frac{2}{3}} =$  (A)  $-3$  (B)  $-\frac{4}{3}$  (C)  $\frac{2}{3}$  (D) 2 (E) 6

**Answer:** E

**Solution**

$$\frac{2}{1-\frac{2}{3}} = 2 \div \left(1 - \frac{2}{3}\right) = 2 \div \frac{1}{3} = 2 \cdot 3 = \textbf{(E) } 6$$

7. (AMC 8 1986) How many whole numbers are between  $\sqrt{8}$  and  $\sqrt{80}$ ?  
(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

**Answer:** B

**Solution 1**

Note that  $2 < \sqrt{8} < 3$  and  $8 < \sqrt{80} < 9$ . Therefore, only the integers from 3 to 8 inclusive are between  $\sqrt{8}$  and  $\sqrt{80}$ , for a total of **(B)** 6 integers.

**Linear Approximation using Calculus** While this may not be the best, most efficient way to solve this problem, it can prove to be quite useful in similar problems. Using Calculus, we can use tangent-line approximation to approximate these square roots. First, we understand that these roots are of the form  $f(x) = \sqrt{x}$ . Given the values to approximate, 8, and 80, we find the closest square number to them, namely, 9, and 81.

Now, we take the derivative of  $f(x)$  to get  $f'(x) = \frac{1}{2\sqrt{x}}$ . We can approximate  $f(x)$ , to get  $f(x) \cong f(x) + f'(x) \cdot \Delta x$ , where  $x$  is the perfect square closest to the the input value (in this case 8), and  $\Delta x$  is the difference between  $x$  and the input value.

The two cases are as follows:

| input | $x$ value | $\Delta x$     |
|-------|-----------|----------------|
| 8     | 9         | $8 - 9 = -1$   |
| 80    | 81        | $80 - 81 = -1$ |

Now, we can plug in our values into the equation, getting:

$$\begin{aligned} f(8) &\cong f(9) + f'(9) \cdot -1 \implies \sqrt{9} + \frac{1}{2\sqrt{9}} \cdot -1 \implies 3 - \frac{1}{6} \\ f(80) &\cong f(81) + f'(81) \cdot -1 \implies \sqrt{81} + \frac{1}{2\sqrt{81}} \cdot -1 \implies 9 - \frac{1}{18} \end{aligned}$$

Thus, we see that our answer lies in the amount of whole numbers between  $3 - \frac{1}{6}$ , and  $9 - \frac{1}{18}$ , and as we are counting whole numbers, we can round each expression to narrow our interval to  $[3, 8]$ . Therefore, we see that our answer is  $\boxed{6 (B)}$ .

Note: this solution is definitely not the fastest solution, and in a competition, you should probably use intuition to see the closest numbers in the interval of those square roots, and solve the problem from there. Cheetahboy93

8. (AMC 8 1986) In the product shown, B is a digit. The value of B is

$$\begin{array}{r} B2 \\ \times 7B \\ \hline 6396 \end{array}$$

- (A) 3      (B) 5      (C) 6      (D) 7      (E) 8

**Answer:** E

**Solution 1**

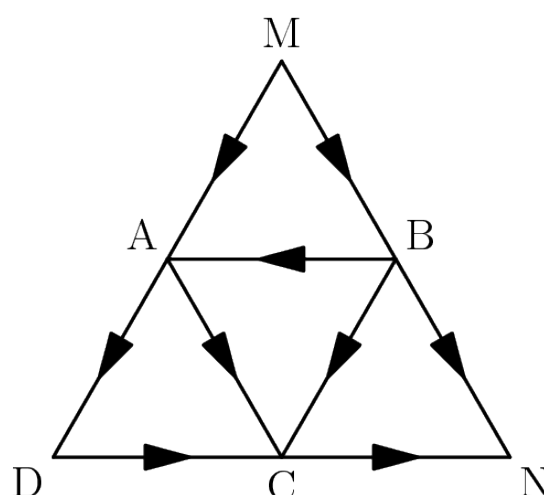
Recall that the units digit of the product of any two positive integers can be found by taking the units digit of each integer, multiplying these two digits, and taking the units digit of this product.

In this case,  $2 \times B$  has a units digit of 6, so  $B$  is either 3 or 8. If  $B = 3$ , then the product is  $32 \times 73 < 40 \times 80 = 3200$ , which is too small. Thus  $B = \boxed{(E) 8}$ , which works.

**Solution 2 (Quadratic Equation)**

This equation is equivalent to  $(10B + 2)(70 + B) = 6396$ . Collecting all terms and factoring yields  $10B^2 + 702B - 6256 = (B - 8)(10B + 782) = 0$ .  $B = \boxed{(E) 8}$  is the only positive solution to this equation.

9. (AMC 8 1986) Using only the paths and the directions shown, how many different routes are there from M to N?



- (A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** E

**Solution 1**

After one step, the possible routes are  $M \rightarrow A$  and  $M \rightarrow B$ .

After two steps, the possible routes are  $M \rightarrow A \rightarrow D$ ,  $M \rightarrow A \rightarrow C$ ,  $M \rightarrow B \rightarrow A$ ,  $M \rightarrow B \rightarrow C$ , and  $M \rightarrow B \rightarrow N$  (complete).

After three steps, the possible routes are  $M \rightarrow A \rightarrow D \rightarrow C$ ,  $M \rightarrow A \rightarrow C \rightarrow N$  (complete),  $M \rightarrow B \rightarrow A \rightarrow D$ ,  $M \rightarrow B \rightarrow A \rightarrow C$ , and  $M \rightarrow B \rightarrow C \rightarrow N$  (complete).

After four steps, the possible routes are  $M \rightarrow A \rightarrow D \rightarrow C \rightarrow N$  (complete),  $M \rightarrow B \rightarrow A \rightarrow D \rightarrow C$ , and  $M \rightarrow B \rightarrow A \rightarrow C \rightarrow N$  (complete).

After five steps, the only possible route is  $M \rightarrow B \rightarrow A \rightarrow D \rightarrow C \rightarrow N$  (complete).

Therefore, there (E) 6 paths from  $M$  to  $N$ .

**Solution 2**

There is 1 path from  $C$  to  $N$ , since the only edge leaving  $C$  leads to  $N$ .

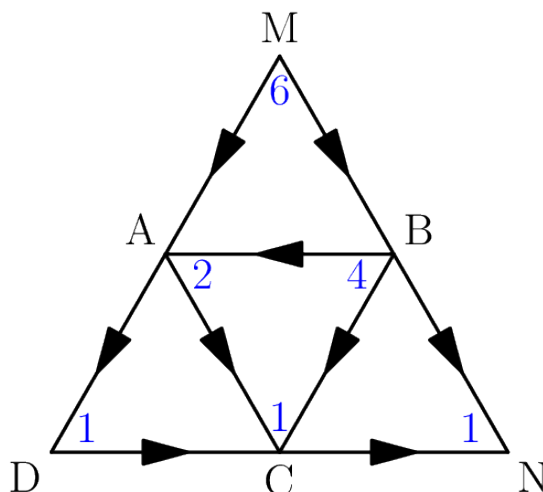
Since the only edge leaving  $D$  leads to  $C$ , there is only 1 path from  $D$  to  $N$ .

There are 2 edges which leave  $A$ ; one leads to  $C$  and the other leads to  $D$ . Therefore, there are  $1 + 1 = 2$  paths from  $A$  to  $N$ .

There are 3 edges which leave  $B$ ; one leads to  $A$ , one leads to  $C$ , and one leads to  $N$ . Therefore, there are  $2 + 1 + 1 = 4$  paths from  $B$  to  $N$ .

There are 2 edges which leave  $M$ ; one leads to  $A$  and the other leads to  $B$ . Therefore, there are  $4 + 2 =$ (E) 6 paths from  $M$  to  $N$ .

The following diagram shows the number of paths to  $N$  from each starting point.



**Note** This is an example of a directed acyclic graph (DAG); it has no cycles (paths that start and end at the same point). A topological sort of this graph is  $M, B, A, D, C, N$ .

10. (AMC 8 1986) A picture 3 feet across is hung in the center of a wall that is 19 feet wide. How many feet from the end of the wall is the nearest edge of the picture?  
 (A)  $1\frac{1}{2}$       (B) 8      (C)  $9\frac{1}{2}$       (D) 16      (E) 22

**Answer:** B

**Solution 1**

The wall is 19 feet wide and the picture is 3 feet wide, the uncovered parts of the wall total 16 feet in width. Since there is the same amount of empty space on each side of the picture, there is  $16 \div 2 = \boxed{\text{(B)} 8}$  feet of space on each side of the picture.

**Solution 2**

Let  $x$  be the distance in feet between an edge of the wall and the nearest edge of the picture. Since there are  $x$  feet on both sides of picture, the total width of the wall is  $2x + 3$  feet. So  $2x + 3 = 19$  and therefore  $x = \boxed{\text{(B)} 8}$ .

11. (AMC 8 1986) If  $A * B$  means  $\frac{A+B}{2}$ , then  $(3 * 5) * 8$  is  
 (A) 6      (B) 8      (C) 12      (D) 16      (E) 30

**Answer:** A

**Solution**

Evaluating yields  $(3 * 5) * 8 = \left(\frac{3+5}{2}\right) * 8 = 4 * 8 = \frac{4+8}{2} = \boxed{\text{(A)} 6}$ .

12. (AMC 8 1986) The table below displays the grade distribution of the 30 students in a mathematics class on the last two tests. For example, exactly one student received a 'D' on Test 1 and a 'C' on Test 2 (see circled entry). What percent of the students received the same grade on both tests?

|        |   | Test 2 |   |   |   |   |
|--------|---|--------|---|---|---|---|
|        |   | A      | B | C | D | F |
| Test 1 | A | 2      | 2 | 1 | 0 | 0 |
|        | B | 1      | 4 | 3 | 0 | 0 |
|        | C | 1      | 3 | 5 | 2 | 0 |
|        | D | 0      | 0 | 1 | 1 | 1 |
|        | F | 0      | 0 | 2 | 1 | 0 |

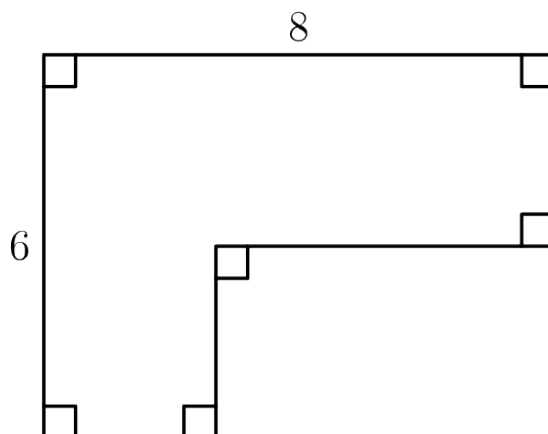
- (A) 12%      (B) 25%      (C)  $33\frac{1}{3}\%$       (D) 40%      (E) 50%

**Answer:** D

**Solution**

Of the 30 students,  $2 + 4 + 5 + 1 + 0 = 12$  got the same grade on both tests. So  $\frac{12}{30} = 0.4 = \boxed{\text{(D)} 40\%}$  of the class got the same grade on both tests.

13. (AMC 8 1986) The perimeter of the polygon shown is



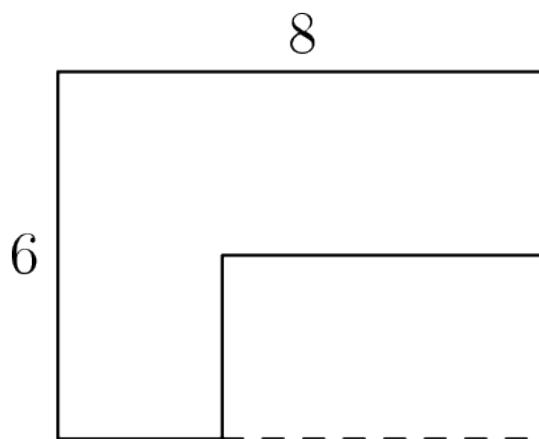
- (A) 14    (B) 20    (C) 28    (D) 48    (E) cannot be determined from the information given

**Answer:** C

**Solution 1**

The unlabeled vertical sides must have total length 6 while the unlabeled horizontal sides must have total length 8. Thus the unlabeled sides have total length 14, so the total perimeter is  $14 + 6 + 8 = \boxed{\text{(C) } 28}$ .

**Solution 2**



Extend two unlabeled sides of the original figure form a  $6 \times 8$  rectangle, as shown. Notice that the extensions added to these two sides are the same lengths as the other two unlabeled sides of the original figure, so the  $6 \times 8$  rectangle has the same perimeter as the original figure. Therefore, the perimeter of the original figure is  $2(6 + 8) = \boxed{\text{(C) } 28}$ .

14. (AMC 8 1986) If  $200 \leq a \leq 400$  and  $600 \leq b \leq 1200$ , then the largest value of the quotient  $\frac{b}{a}$  is  
 (A)  $\frac{3}{2}$     (B) 3    (C) 6    (D) 300    (E) 600

**Answer:** C

**Solution**

Since  $a$  and  $b$  are both positive,  $\frac{b}{a}$  is maximized if  $b$  is maximized and  $a$  is minimized. The maximum possible value of  $b$  is 1200 and the minimum possible value of  $a$  is 200. In this case,  $\frac{b}{a} = \frac{1200}{200} = \boxed{\text{(C) } 6}$ .

15. (*AMC 8 1986*) Sale prices at the Ajax Outlet Store are 50% below original prices. On Saturdays an additional discount of 20% off the sale price is given. What is the Saturday price of a coat whose original price is

180

?

- (A) \$54      (B) \$72      (C) \$90      (D) \$108      (E) \$110

**Answer:** B

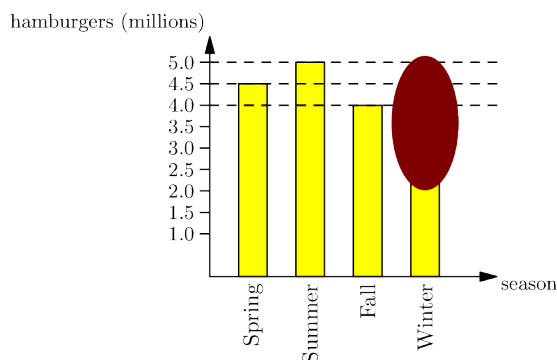
**Solution 1**

The sale price of the coat is  $(1 - 0.5)(\$180) = (0.5)(\$180) = \$90$ . The Saturday price is  $(1 - 0.2)(\$90) = (0.8)(\$90) = \boxed{\text{(B) } \$72}$ .

**Solution 2**

After applying both discounts to an item, its price would be  $(1 - 0.5)(1 - 0.2) = 0.5 \cdot 0.8 = 0.4$  times its original price. Therefore, the Saturday price of the coat is  $0.4 \cdot \$180 = \boxed{\text{(B) } \$72}$ .

16. (*AMC 8 1986*) A bar graph shows the number of hamburgers sold by a fast food chain each season. However, the bar indicating the number sold during the winter is covered by a smudge. If exactly 25% of the chain's hamburgers are sold in the fall, how many million hamburgers are sold in the winter?



- (A) 2.5      (B) 3      (C) 3.5      (D) 4      (E) 4.5

**Answer:** A

**Solution 1**

During the fall, 4 million hamburgers were sold. Since this is 25%, or  $\frac{1}{4}$  of the company's hamburgers sold, a total of  $4 \div \frac{1}{4} = 16$  million hamburgers were sold in total. Since  $4.5 + 5 + 4 = 13.5$  million hamburgers were sold in the other three seasons,  $16 - 13.5 = \boxed{\text{(A) } 2.5}$  million hamburgers were sold in the winter.

**Solution 2**

Let  $x$  be the number of hamburgers sold in the winter, in millions. Then a total of  $4.5 + 5 + 4 + x = 13.5 + x$  million hamburgers were sold. Since 25% of the hamburgers were sold in the fall,  $0.25(13.5 + x) = 4$ . Solving this equation yields  $x = \boxed{\text{(A) } 2.5}$ .

17. (AMC 8 1986) Let  $o$  be an odd whole number and let  $n$  be any whole number. Which of the following statements about the whole number  $(o^2 + no)$  is always true?

(A) it is always odd      (B) it is always even (C) it is even only if  $n$  is even      (D) it is odd only if  $n$  is even  
(E) it is odd only if  $n$  is even

**Answer:** E

**Solution 1**

Since  $o$  is odd,  $o^2$  must be odd.

If  $n$  is odd, then  $no$  is odd, so  $o^2 + no$  is even.

If  $n$  is even, then  $no$  is even, so  $o^2 + no$  is odd.

Therefore, of the answer choices, only **(E) it is odd only if  $n$  is even** accurately describes  $o^2 + no$ .

**Solution 2**

We are given that  $o \equiv 1 \pmod{2}$ , so in mod 2 we have

$$1^2 + 1(n) = n + 1$$

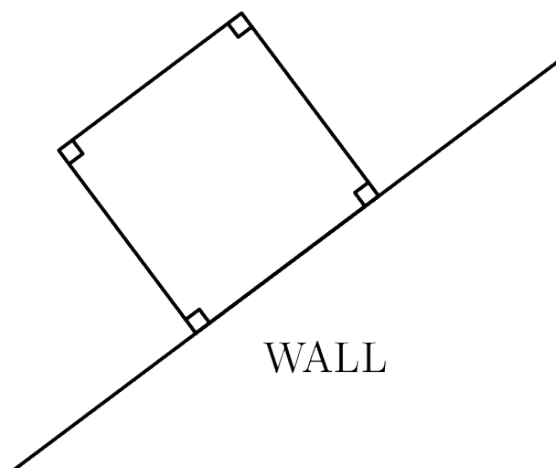
which is odd only if  $n$  is even  $\rightarrow$  **E**

**Solution 3 (Answer Choices)**

If  $o = 1$  and  $n = 2$ , then  $o^2 + no = 3$ , which eliminates (B) and (D). If  $o = 1$  and  $n = 1$ , then  $o^2 + no = 2$ , which eliminates (A) and (C). This leaves

**(E) it is odd only if  $n$  is even** as the correct answer.

18. (AMC 8 1986) A rectangular grazing area is to be fenced off on three sides using part of a 100 meter rock wall as the fourth side. Fence posts are to be placed every 12 meters along the fence including the two posts where the fence meets the rock wall. What is the fewest number of posts required to fence an area 36 m by 60 m?



(A) 11      (B) 12      (C) 13      (D) 14      (E) 16

**Answer:** B

**Solution 1**

To minimize the total length of the fence, one of the longer (60 m) sides of the rectangle must touch the wall. In this case, the total length of the fence is  $36 + 60 + 36 = 132$  m. Since pairs of consecutive posts are 12 m apart, there are  $132 \div 12 = 11$  intervals between consecutive posts, and therefore  $11 + 1 = \boxed{\text{(B)} 12}$  posts are needed.

**Solution 2**

As in Solution 1, one of the 60 m sides must touch the wall. The 60 m side not touching the wall has  $60 \div 12 + 1 = 6$  posts. Each of the 36 m sides has  $36 \div 12 + 1 = 4$  posts. Since 2 posts are at intersections of 36 m and 60 m sides, there are a total of  $6 + 2 \cdot 4 - 2 = \boxed{\text{(B)} 12}$  posts.

19. (AMC 8 1986) At the beginning of a trip, the mileage odometer read 56,200 miles. The driver filled the gas tank with 6 gallons of gasoline. During the trip, the driver filled his tank again with 12 gallons of gasoline when the odometer read 56,560. At the end of the trip, the driver filled his tank again with 20 gallons of gasoline. The odometer read 57,060. To the nearest tenth, what was the car's average miles-per-gallon for the entire trip?
- (A) 22.5      (B) 22.6      (C) 24.0      (D) 26.9      (E) 27.5

**Answer:** D

**Solution**

The car used 12 gallons of fuel during the first part of the trip and 20 gallons during the second part of the trip, for a total of 32 gallons. The total distance traveled during the trip was the difference between the final and initial odometer readings, that is,  $57060 - 56200 = 860$  miles. Therefore, the car's fuel efficiency was  $860 \div 32 \approx \boxed{\text{(D)} 26.9}$  miles per gallon.

20. (AMC 8 1986) The value of the expression  $\frac{(304)^5}{(29.7)(399)^4}$  is closest to
- (A) .003      (B) .03      (C) .3      (D) 3      (E) 30

**Answer:** D

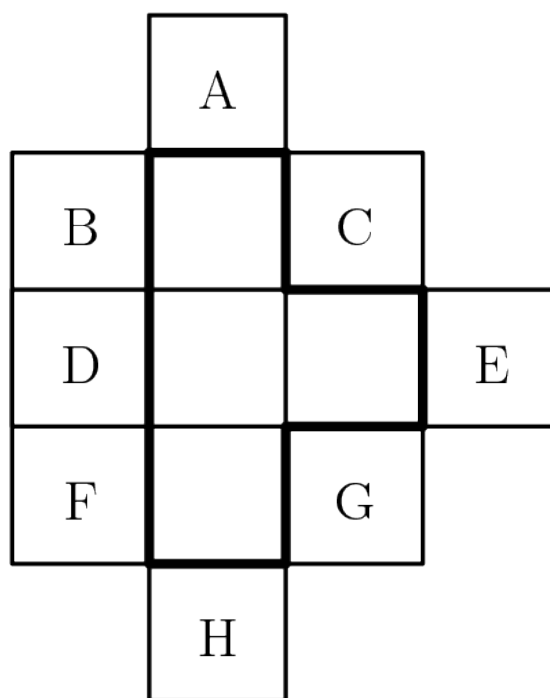
**Solution**

By rounding,  $\frac{(304)^5}{(29.7)(399)^4} \approx \frac{300^5}{30 \cdot 400^4} = \frac{3^5 \cdot 10^{10}}{3 \cdot 4^4 \cdot 10^9} = \frac{3^4 \cdot 10}{4^4} = \frac{810}{256} \approx \boxed{\text{(D)} 3}$ .

Note: The actual value of this expression, rounded to 8 decimal places, is 3.44921198.

21. (AMC 8 1986) Suppose one of the eight lettered identical squares is included with the four squares in the T-shaped figure outlined. How many of the resulting figures can be folded into a topless cubical box?





- (A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** E

**Solution 1**

Suppose the existing net is folded so that the square immediately to the right of  $D$  is on the bottom. Then only the top and left sides of the folded cube would be missing.  $B$ ,  $D$  and  $F$  could be added as the left side of the cube, while  $A$ ,  $E$ , and  $H$  could be added as the right side of the cube.  $C$  and  $G$  could not be added, as at most three faces can share a vertex. Therefore, **(E) 6** of the lettered faces could be added to the existing net.

**Solution 2**

Fold the four squares into the four sides of a cube. Then, there are six edges "open" (for lack of better term). For each open edge, we can add a square/side, so the answer is  $6 \rightarrow$  **(E)**.

22. (*AMC 8 1986*) Alan, Beth, Carlos, and Diana were discussing their possible grades in mathematics class this grading period. Alan said, "If I get an A, then Beth will get an A." Beth said, "If I get an A, then Carlos will get an A." Carlos said, "If I get an A, then Diana will get an A." All of these statements were true, but only two of the students received an A. Which two received A's?

- (A) Alan, Beth      (B) Beth, Carlos      (C) Carlos, Diana      (D) Alan, Diana      (E) Beth, Diana

**Answer:** C

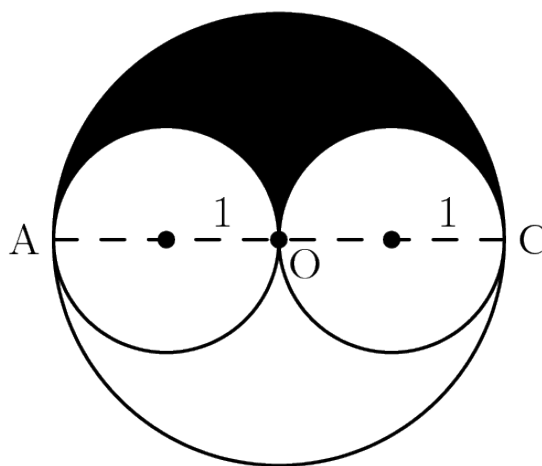
**Solution**

If Alan got an A, then Beth, and therefore Carlos and Diana would also get A's. But only two students actually got A's, so Alan did not get an A.

If Beth got an A, then Carlos, and therefore Diana would get also get A's. Once again, this is impossible since only two students got A's. Therefore, Beth did not get an A.

By process of elimination, **(C)** Carlos, Diana must have gotten A's.

23. (AMC 8 1986) The large circle has diameter  $AC$ . The two small circles have their centers on  $AC$  and just touch at  $O$ , the center of the large circle. If each small circle has radius 1, what is the value of the ratio of the area of the shaded region to the area of one of the small circles?



- (A) between  $\frac{1}{2}$  and 1      (B) 1      (C) between 1 and  $\frac{3}{2}$       (D) between  $\frac{3}{2}$  and 2      (E) cannot be determined

**Answer:** B

**Solution 1**

The small circle has radius 1, thus its area is  $\pi$ .

The large circle has radius 2, thus its area is  $4\pi$ .

The area of the semicircle above  $AC$  is then  $2\pi$ .

The part that is not shaded are two small semicircles. Together, these form one small circle, hence their total area is  $\pi$ . This means that the area of the shaded part is  $2\pi - \pi = \pi$ . This is equal to the area of a small circle, hence the correct answer is **(B) 1**.

**Solution 2**

Let  $A$  be the area of one of the small circles. Since all circles are similar, the large circle's area is  $2^2 \cdot A = 4A$ . So, the area inside the large circle but outside the small circles is  $4A - 2A = 2A$ . Therefore, the shaded region must then be  $\frac{2A}{2} = A$ , so the desired ratio is  $\frac{A}{A} = \mathbf{(B) 1}$ .

24. (AMC 8 1986) The 600 students at King Middle School are divided into three groups of equal size for lunch. Each group has lunch at a different time. A computer randomly assigns each student to one of three lunch groups. The probability that three friends, Al, Bob, and Carol, will be assigned to the same lunch group is approximately
- (A)  $\frac{1}{27}$       (B)  $\frac{1}{9}$       (C)  $\frac{1}{8}$       (D)  $\frac{1}{6}$       (E)  $\frac{1}{3}$

**Answer:** B

**Solution 1**

Suppose Al, Bob and Carol are the first three students to be randomly assigned groups, in that order. After Al is assigned to a group, his group has 199 open seats,

out of a total of 599 open seats across all the groups. Therefore, the probability Bob is also assigned to this group is  $\frac{199}{599}$ , which is slightly less than  $\frac{1}{3}$ . Similarly, if Al and Bob are assigned to the same group, the probability Carol is also assigned to this group is  $\frac{198}{598}$ , which is also slightly less than  $\frac{1}{3}$ . So, the probability all three of them end up in the same group is  $\frac{199}{599} \cdot \frac{198}{598}$ , which is slightly less than  $\frac{1}{3} \cdot \frac{1}{3} = \boxed{\text{(B)} \frac{1}{9}}$ .

25. (AMC 8 1986) Which of the following sets of whole numbers has the largest average?

(A) multiples of 2 between 1 and 101      (B) multiples of 3 between 1 and 101 (C) multiples of 4 between 1 and 101  
(D) multiples of 5 between 1 and 101      (E) multiples of 6 between 1 and 101

**Answer:** D

**Solution 1**

Let  $f(n)$  be the average of the multiples of  $n$  between 1 and 101. There must be  $\left\lfloor \frac{101}{n} \right\rfloor$  (the greatest integer less than or equal to  $\frac{101}{n}$ ) multiples of  $n$  between 1 and 101. Let  $g(n) = \left\lfloor \frac{101}{n} \right\rfloor$ . Then the multiples of  $n$  between 1 and 101 must be  $\{n, 2n, \dots, g(n) \cdot n\}$ , so their average is  $f(n) = \frac{n+2n+\dots+g(n) \cdot n}{g(n)} = \frac{n}{g(n)}(1 + 2 + \dots + g(n)) = \frac{n}{g(n)} \cdot \frac{g(n)(g(n)+1)}{2} = \frac{n(g(n)+1)}{2}$ . So  $f(2) = 51$ ,  $f(3) = 51$ ,  $f(4) = 52$ ,  $f(5) = 52.5$ , and  $f(6) = 51$ , thus the set of (D) multiples of 5 between 1 and 101 has the greatest average.

**Solution 2**

The elements of each answer choice form arithmetic sequence. Since the sum of any  $n$ -term arithmetic sequence  $\{a_1, \dots, a_n\}$  is  $\frac{n}{2}(a_1 + a_n)$ , its average is  $\frac{1}{2}(a_1 + a_n)$ , that is, the average of its first and last terms. Therefore, the averages of (A), (B), (C), (D), and (E) are  $\frac{1}{2}(2 + 100) = 51$ ,  $\frac{1}{2}(3 + 99) = 51$ ,  $\frac{1}{2}(4 + 100) = 52$ ,  $\frac{1}{2}(5 + 100) = 52.5$ , and  $\frac{1}{2}(6 + 96) = 51$ , respectively. So of these, the set of (D) multiples of 5 between 1 and 101 has the largest average.

### §3 AMC 8 1987

1. (AMC 8 1987)  $.4 + .02 + .006 =$  (A) .012      (B) .066      (C) .12      (D) .24      (E) .426

**Answer:** E

**Solution 1**

$$.4 + .02 + .006 = .400 + .020 + .006 = \boxed{\text{(E)} .426}$$

**Solution 2**

$$.4 + .02 + .006 = \frac{4}{10} + \frac{2}{100} + \frac{6}{1000} = \frac{400}{1000} + \frac{20}{1000} + \frac{6}{1000} = \frac{426}{1000} = \boxed{\text{(E)} .426}.$$

2. (AMC 8 1987)  $\frac{2}{25} =$  (A) 0.008      (B) 0.08      (C) 0.8      (D) 1.25      (E) 12.5

**Answer:** B

**Solution**

$$\frac{2}{25} = \frac{2 \cdot 4}{25 \cdot 4} = \frac{8}{100} = \boxed{\text{(B)} .08}$$

3. (AMC 8 1987)  $2(81+83+85+87+89+91+93+95+97+99) =$  (A) 1600      (B) 1650      (C) 1700      (D) 1750      (E) 1800

**Answer:** E

**Solution 1**

Rearranging terms yields  $2(81 + 83 + 85 + 87 + 89 + 91 + 93 + 95 + 97 + 99)$   
 $= 2((81+99)+(83+97)+(85+95)+(87+93)+(89+91)) = 2(180+180+180+180+180)$   
 $= 2 \cdot 5 \cdot 180 = \boxed{\text{(E)} 1800}.$

**Solution 2**

Recall that the sum of an arithmetic series  $a_1 + \dots + a_n = \frac{n}{2}(a_1 + a_n)$ , so this sum is  $2\left(\frac{10}{2}(81 + 99)\right) = \boxed{\text{(E)} 1800}.$

4. (*AMC 8 1987*) Martians measure angles in clerts. There are 500 clerts in a full circle. How many clerts are there in a right angle?

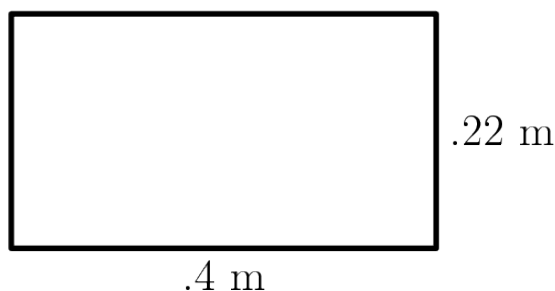
(A) 90      (B) 100      (C) 125      (D) 180      (E) 250

**Answer:** C

**Solution**

A right angle is  $\frac{1}{4}$  of a circle, so it has  $\frac{1}{4} \cdot 500 = \boxed{\text{(C)} 125}$  clerts.

5. (*AMC 8 1987*) The area of the rectangular region is



(A) .088 m<sup>2</sup>      (B) .62 m<sup>2</sup>      (C) .88 m<sup>2</sup>      (D) 1.24 m<sup>2</sup>      (E) 4.22 m<sup>2</sup>

**Answer:** A

**Solution**

The area is  $.4 \times .22 = \boxed{\text{(A)} 0.088}$  m<sup>2</sup>.

6. (*AMC 8 1987*) The smallest product one could obtain by multiplying two numbers in the set  $\{-7, -5, -1, 1, 3\}$  is

(A)  $-35$       (B)  $-21$       (C)  $-15$       (D)  $-1$       (E)  $3$

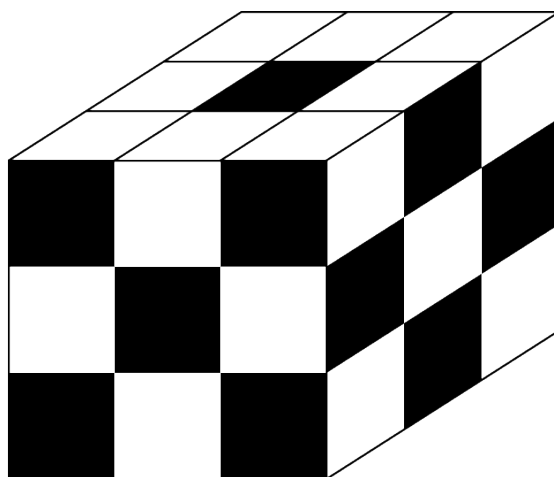
**Answer:** B

**Solution**

Choosing the positive and negative numbers with the highest absolute values minimizes this product, which in this case is  $-7 \cdot 3 = \boxed{\text{(B)} -21}.$

7. (*AMC 8 1987*) The large cube shown is made up of 27 identical sized smaller cubes. For each face of the large cube, the opposite face is shaded the same way. The total number of smaller cubes that must have at least one face shaded is

(A) 10      (B) 16      (C) 20      (D) 22      (E) 24



**Answer:** C

**Solution 1**

Notice that no two shaded tiles are adjacent. Therefore, each small cube has at most one face shaded, and the number of small cubes with at least one face shaded is the same as the number of shaded tiles on the entire cube. Since there are 10 visible shaded tiles, and the hidden faces of the entire cube are identical to the shown faces, there must be  $2 \cdot 10 = \boxed{\text{(C) } 20}$  small cubes with at least one face shaded.

**Solution 2**

The top and bottom layers each have 2 cubes with no shaded sides, which are located in the middle of the front and back rows. The middle layer has 3 cubes with no shaded sides, which are all in the middle row. Therefore,  $2 \cdot 2 + 3 = 7$  cubes have no shaded sides, and  $27 - 7 = \boxed{\text{(C) } 20}$  cubes have at least one shaded side.

8. (AMC 8 1987) If A and B are nonzero digits, then the number of digits (not necessarily different) in the sum of the three whole numbers is

$$\begin{array}{r} 9 \quad 8 \quad 7 \quad 6 \\ A \quad 3 \quad 2 \\ B \quad 1 \\ \hline \end{array}$$

- (A) 4      (B) 5      (C) 6      (D) 9      (E) depends on the values of A and B

**Answer:** B

**Solution**

The smallest possible sum occurs if  $A = B = 1$ , so it is  $9876 + 132 + 11 = 10019$ . The largest possible sum occurs if  $A = B = 9$ , so it is  $9876 + 932 + 91 = 10899$ . Therefore, the sum is between 10019 and 10899, so it has  $\boxed{\text{(B) } 5}$  digits.

9. (AMC 8 1987) When finding the sum  $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$ , the least common denominator used is
- (A) 120      (B) 210      (C) 420      (D) 840      (E) 5040

**Answer:** C

**Solution**

The least common denominator is  $\text{lcm}(2, 3, 4, 5, 6, 7) = \text{lcm}(2, 3, 2^2, 5, 2 \cdot 3, 7) = 2^2 \cdot 3 \cdot 5 \cdot 7 = \boxed{\text{(C)} 420}$ .

10. (AMC 8 1987)  $4(299) + 3(299) + 2(299) + 298 =$  (A) 2889 (B) 2989 (C) 2991 (D) 2999 (E)

**Answer:** B

**Solution**

Using the distributive property,  $4(299) + 3(299) + 2(299) + 298 = 4(299) + 3(299) + 2(299) + 1(299) - 1 = (4 + 3 + 2 + 1)(299) - 1 = 10(299) - 1 = \boxed{\text{(B)} 2989}$ .

11. (AMC 8 1987) The sum  $2\frac{1}{7} + 3\frac{1}{2} + 5\frac{1}{19}$  is between  
(A) 10 and  $10\frac{1}{2}$  (B)  $10\frac{1}{2}$  and 11 (C) 11 and  $11\frac{1}{2}$  (D)  $11\frac{1}{2}$  and 12 (E) 12 and  $12\frac{1}{2}$

**Answer:** B

**Solution 1**

Since  $\frac{1}{7} < \frac{1}{4}$  and  $\frac{1}{19} < \frac{1}{4}$ , it follows that  $2\frac{1}{7} + 3\frac{1}{2} + 5\frac{1}{19} < 2\frac{1}{4} + 3\frac{1}{2} + 5\frac{1}{4} = 11$ . Also,  $2\frac{1}{7} + 3\frac{1}{2} + 5\frac{1}{19} > 2 + 3\frac{1}{2} + 5 = 10\frac{1}{2}$ . Therefore, the sum is between  $\boxed{\text{(B)} 10\frac{1}{2} \text{ and } 11}$ .

**Solution 2**

This sum is  $2 + \frac{1}{2} + 3 + \frac{1}{7} + 5 + \frac{1}{19} = 2 + 3 + 5 + \frac{1}{2} + \frac{1}{7} + \frac{1}{19} = 10\frac{1}{2} + \frac{1}{7} + \frac{1}{19} = 10\frac{1}{2} + \frac{26}{133}$ .

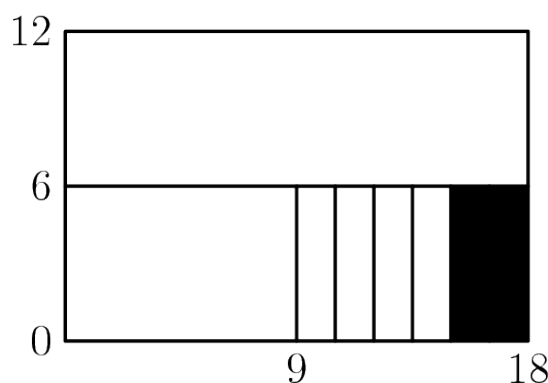
Since  $\frac{26}{133} < \frac{1}{2}$ , this sum must be between  $\boxed{\text{(B)} 10\frac{1}{2} \text{ and } 11}$ .

Anabel.disher (talk) 12:49, 18 June 2025 (EDT)

**Solution 3**

The total sum is  $10 + \frac{1}{2} + \frac{1}{7} + \frac{1}{19} = 10 + \frac{9}{14} + \frac{1}{19} = 10 + \frac{185}{266} = 10\frac{185}{266}$ , which is between  $\boxed{\text{(B)} 10\frac{1}{2} \text{ and } 11}$ .

12. (AMC 8 1987) What fraction of the large 12 by 18 rectangular region is shaded?



- (A)  $\frac{1}{108}$  (B)  $\frac{1}{18}$  (C)  $\frac{1}{12}$  (D)  $\frac{2}{9}$  (E)  $\frac{1}{3}$

**Answer:** C

**Solution 1**

The rectangle is divided into upper and lower halves, and then the lower half is divided into left and right halves. The lower right section is therefore  $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$  of the area of the entire rectangle. Since this section is divided into 6 equal rectangles, 2

of which are shaded,  $\frac{1}{3}$  of the lower-right section is shaded, that is,  $\frac{1}{3} \cdot \frac{1}{4} = \boxed{\text{(C)} \frac{1}{12}}$  of the large rectangle.

**Solution 2**

The total area of the rectangle is  $12 \cdot 18 = 216$ . The  $x$ -coordinate of the left edge of the shaded area is  $\frac{4}{6} = \frac{2}{3}$  of the way from 9 to 18, which is  $9 + \frac{2}{3} \cdot 9 = 15$ . Therefore, the shaded area is  $6 \cdot 3 = 18$ . Therefore,  $\frac{18}{216} = \boxed{\text{(C)} \frac{1}{12}}$  of the entire rectangle is shaded.

13. (AMC 8 1987) Which of the following fractions has the largest value?

(A)  $\frac{3}{7}$     (B)  $\frac{4}{9}$     (C)  $\frac{17}{35}$     (D)  $\frac{100}{201}$     (E)  $\frac{151}{301}$

**Answer:** E

**Solution**

Notice that  $\frac{3}{7} < \frac{3}{6} = \frac{1}{2}$ ,  $\frac{4}{9} < \frac{4}{8} = \frac{1}{2}$ ,  $\frac{17}{35} < \frac{17}{34} = \frac{1}{2}$ , and  $\frac{100}{201} < \frac{100}{200} = \frac{1}{2}$ , but  $\frac{151}{301} > \frac{151}{302} = \frac{1}{2}$ . Since  $\boxed{\text{(E)} \frac{151}{301}}$  is the only answer choice greater than  $\frac{1}{2}$ , it must be the largest.

14. (AMC 8 1987) A computer can do 10,000 additions per second. How many additions can it do in one hour?

(A) 6 million    (B) 36 million    (C) 60 million    (D) 216 million    (E) 360 million

**Answer:** B

**Solution**

Using dimensional analysis,  $\frac{10000 \text{ additions}}{1 \text{ second}} \cdot \frac{3600 \text{ seconds}}{1 \text{ hour}} = \boxed{\text{(B)} 36 \text{ million}}$  additions/hour.

15. (AMC 8 1987) The sale ad read: "Buy three tires at the regular price and get the fourth tire for \$3." Sam paid \$240 for a set of four tires at the sale. What was the regular price of one tire?

(A) \$59.25    (B) \$60    (C) \$70    (D) \$79    (E) \$80

**Answer:** D

**Solution 1**

Let  $x$  be the regular price of a tire. Then  $3x + \$3 = \$240$ , so  $x = \boxed{\text{(D)} \$79}$ .

**Solution 2**

Since Sam paid \$240 for all 4 tires, which included \$3 for the last one, he paid  $\$240 - \$3 = \$237$  for the first 3 tires at regular price. Therefore, the regular price of a tire is  $\$237 \div 3 = \boxed{\text{(D)} \$79}$ .

16. (AMC 8 1987) Joyce made 12 of her first 30 shots in the first three games of this basketball season, so her seasonal shooting average was 40%. In her next game, she took 10 shots and raised her seasonal shooting average to 50%. How many of these 10 shots did she make?

(A) 2    (B) 3    (C) 5    (D) 6    (E) 8

**Answer:** E

**Solution**

After the fourth game, Joyce had taken 40 total shots, 50% of which she made, so she made  $40 \times .5 = 20$  shots. Since she made 12 shots in the first three games, she made  $20 - 12 = \boxed{\text{(E)} 8}$  shots in the last game.

17. (AMC 8 1987) Abby, Bret, Carl, and Dana are seated in a row of four seats numbered #1 to #4. Joe looks at them and says:

"Bret is next to Carl." "Abby is between Bret and Carl." However each one of Joe's statements is false. Bret is actually sitting in seat #3. Who is sitting in seat #2?

- (A) Abby      (B) Bret      (C) Carl      (D) Dana      (E) There is not enough information to be sure

**Answer:** D

**Solution**

Bret sits in seat #3.

—      —      **Bret**      —

Since Carl is not next to Bret, he sits in seat #1.

**Carl**      —      Bret      —

Since Abby is not between Bret and Carl, she sits in seat #4.

Carl      —      Bret      **Abby**

This leaves  $\boxed{\text{(D)} \text{ Dana}}$  to sit in seat #2.

Carl      **Dana**      Bret      Abby

18. (AMC 8 1987) Half the people in a room left. One third of those remaining started to dance. There were then 12 people who were not dancing. The original number of people in the room was what?

- (A) 24      (B) 30      (C) 36      (D) 42      (E) 72

**Answer:** C

**Solution**

Let the original number of people in the room be  $x$ . Half of them left, so  $\frac{x}{2}$  of them are left in the room.

After that, one third of this group is dancing, so  $\frac{x}{2} - \frac{1}{3} \left( \frac{x}{2} \right) = \frac{x}{3}$  people are not dancing.

This is given to be 12, so

$$\frac{x}{3} = 12 \Rightarrow x = 36$$

$\boxed{\text{C}}$

**Solution 2**

First note that of the  $\frac{1}{2}$  people remaining in the room,  $\frac{2}{3}$  are not dancing. Therefore  $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$  of the original amount of people in the room is 12. The answer is  $\boxed{\text{C}}$ .



19. (AMC 8 1987) A calculator has a squaring key  $x^2$  which replaces the current number displayed with its square. For example, if the display is 000003 and the  $x^2$  key is depressed, then the display becomes 000009. If the display reads 000002, how many times must you depress the  $x^2$  key to produce a displayed number greater than 500?

(A) 4      (B) 5      (C) 8      (D) 9      (E) 250

**Answer:** A

**Solution**

We just brute force this:

$$\begin{aligned} 2 &\rightarrow 4 \\ &\rightarrow 16 \\ &\rightarrow 256 \\ &\rightarrow 65536 > 500 \end{aligned}$$

Clearly we need to press the button 4 times, so **A**

20. (AMC 8 1987) "If a whole number  $n$  is not prime, then the whole number  $n - 2$  is not prime." A value of  $n$  which shows this statement to be false is

(A) 9      (B) 12      (C) 13      (D) 16      (E) 23

**Answer:** A

**Solution**

To show that the conditional statement "If  $P$ , then  $Q$ " is false, we must find a counterexample where the starting condition ( $P$ ) is true, but the conclusion ( $Q$ ) is false.

In this problem:

- Condition  $P$ :  $n$  is not prime.
- Conclusion  $Q$ :  $n - 2$  is not prime.

Therefore, a counterexample requires a number  $n$  that is **not prime** but results in an  $n - 2$  that is **prime** (making the conclusion false).

Let's test the choices:

- **(A)** 9: 9 is not prime, and  $9 - 2 = 7$ , which is prime. This breaks the rule, so 9 is a valid counterexample.
- **(B)** 12: 12 is not prime, and  $12 - 2 = 10$ , which is not prime. This follows the rule.
- **(C)** 13: 13 is prime, so it fails the initial condition.
- **(D)** 16: 16 is not prime, and  $16 - 2 = 14$ , which is not prime. This follows the rule.
- **(E)** 23: 23 is prime, so it fails the initial condition.

Only **(A)** 9 is a counterexample.

21. (AMC 8 1987) Suppose  $n^*$  means  $\frac{1}{n}$ , the reciprocal of  $n$ . For example,  $5^* = \frac{1}{5}$ . How many of the following statements are true?

i)  $3^* + 6^* = 9^*$  ii)  $6^* - 4^* = 2^*$  iii)  $2^* \cdot 6^* = 12^*$  iv)  $10^* \div 2^* = 5^*$   
 (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

**Answer:** C

**Solution**

Statement (i) is false, since  $\frac{1}{3} + \frac{1}{6} = \frac{1}{2} \neq \frac{1}{9}$ .

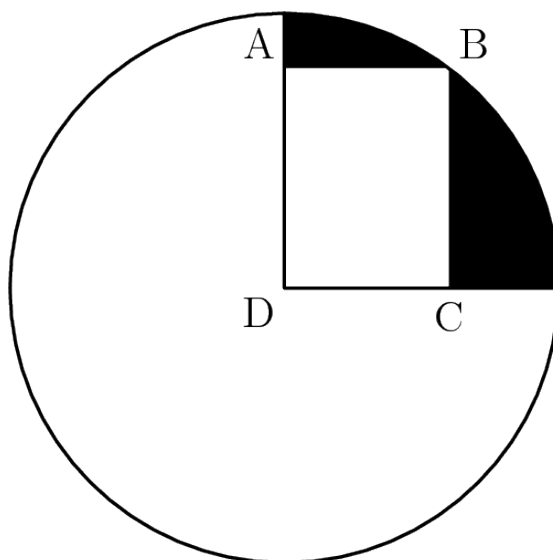
Statement (ii) is false, since  $\frac{1}{6} - \frac{1}{4} = -\frac{1}{12} \neq \frac{1}{2}$ .

Statement (iii) is true, since  $\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$ .

Statement (iv) is true, since  $\frac{1}{10} \div \frac{1}{2} = \frac{1}{5}$ .

So **(C) 2** of these statements are true.

22. (AMC 8 1987) ABCD is a rectangle, D is the center of the circle, and B is on the circle. If AD = 4 and CD = 3, then the area of the shaded region is between



- (A) 4 and 5 (B) 5 and 6 (C) 6 and 7 (D) 7 and 8 (E) 8 and 9

**Answer:** D

**Solution**

Since  $ABCD$  is a rectangle,  $BC = AD = 4$ . So by the Pythagorean theorem,  $BD = \sqrt{BC^2 + CD^2} = \sqrt{3^2 + 4^2} = 5$ . So, the circle has radius 5 and total area  $25\pi$ . Consequently, the quarter circle bounded by  $\angle D$  has area  $\frac{25\pi}{4}$ .

Since  $ABCD$  has area  $3 \cdot 4 = 12$ , the shaded area is  $\frac{25\pi}{4} - 12 \approx \frac{25 \cdot 3.14}{4} - 12 = 7.625$ , so the area is between **(D) 7 and 8**.

23. (AMC 8 1987) Assume the adjoining chart shows the 1980 U.S. population, in millions, for each region by ethnic group. To the nearest percent, what percent of the U.S. Black population lived in the South?

|       | NE | MW | South | West |
|-------|----|----|-------|------|
| White | 42 | 52 | 57    | 35   |
| Black | 5  | 5  | 15    | 2    |
| Asian | 1  | 1  | 1     | 3    |
| Other | 1  | 1  | 2     | 4    |

- (A) 20%    (B) 25%    (C) 40%    (D) 56%    (E) 80%

**Answer:** D

**Solution**

The total U.S. Black population was  $5 + 5 + 15 + 2 = 27$  million. So  $\frac{15}{27} = \frac{5}{9} \approx$   
(D) 56% of the U.S. Black population lived in the South.

24. (*AMC 8 1987*) A multiple choice examination consists of 20 questions. The scoring is +5 for each correct answer,  $-2$  for each incorrect answer, and 0 for each unanswered question. John's score on the examination is 48. What is the maximum number of questions he could have answered correctly?

- (A) 9    (B) 10    (C) 11    (D) 12    (E) 16

**Answer:** D

**Solution 1**

Let  $x$  be the number of questions John answered correctly. Since he answered at most  $20 - x$  questions incorrectly, he scored at least  $5x - 2(20 - x) = 7x - 40$  points. So  $7x - 40 \leq 48$ , and thus  $x \leq \frac{88}{7} = 12\frac{4}{7}$ . Therefore, he answered at most (D) 12 questions correctly, which would result in a score of 48 if he answered 6 questions incorrectly.

**Solution 2**

Since John scored 48 points, and each correct answer scores 5 points, he answered at least 10 questions correctly.

If he answered 10 questions correctly, he gained 50 points from correct answers, so he lost 2 points from incorrect answers. Therefore, he could have scored 48 points by answering 10 questions correctly and 1 question incorrectly.

If he answered 11 questions correctly, he gained 55 points from correct answers, so he lost 7 points from incorrect answers. This is impossible since 7 is not divisible by 2.

If he answered 12 questions correctly, he gained 60 points from correct answers, so he lost 12 points from incorrect answers. Therefore, he could have scored 48 points by answering 12 questions correctly and 6 questions incorrectly.

If he answered 13 or more questions correctly, he gained at least 65 points from correct answers, so he lost at least 17 points from incorrect answers. So, he answered at least 9 questions incorrectly, which would be impossible since the test only has 20 questions.

Therefore, he answered at most (D) 12 questions correctly.

**Solution 3 (Answer Choices)**

If John answered 16 questions correctly, he gained 80 points from correct answers, so he lost 32 points from incorrect answers. So, he answered 16 questions incorrectly, which is impossible since the test only had 20 questions.

If he answered 12 questions correctly, he gained 60 points from correct answers, so he lost 12 points from incorrect answers. So, he answered 6 questions incorrectly, which is possible.

Since all the other answer choices are lower than 12, he answered at most (D) 12 questions correctly.

25. (*AMC 8 1987*) Ten balls numbered 1 to 10 are in a jar. Jack reaches into the jar and randomly removes one of the balls. Then Jill reaches into the jar and randomly removes a different ball. The probability that the sum of the two numbers on the balls removed is even is

(A)  $\frac{4}{9}$       (B)  $\frac{9}{19}$       (C)  $\frac{1}{2}$       (D)  $\frac{10}{19}$       (E)  $\frac{5}{9}$

**Answer:** A

**Solution 1**

At first, there are 5 even balls and 5 odd balls. Therefore, after Jack chooses a ball, the jar will have 4 balls with the same parity (odd/even) as Jack's ball and 5 balls with the opposite parity. For the sum to be even, Jill must choose a ball with the same parity as Jack's ball. The probability that this happens is (A)  $\frac{4}{9}$ .

**Solution 2**

Jack has 10 choices for his ball, and after he makes his selection Jill will have 9 choices for hers. Therefore, there are  $10 \cdot 9 = 90$  combinations of selections, and each combination is equally likely.

For the sum of their numbers to be even, Jack and Jill must either both select odd numbers or both select even numbers.

If they both select even numbers, Jack has 5 choices for his ball, and Jill will have 4 choices for hers, for a total of  $4 \cdot 5 = 20$  combinations of selections.

If they both select odd numbers, Jack has 5 choices for his ball, and Jill will have 4 choices for hers, for a total of  $4 \cdot 5 = 20$  combinations of selections.

Therefore, of the 90 combinations of selections,  $20 + 20 = 40$  yield an even sum.

Therefore, the probability that the sum is even is (A)  $\frac{4}{9}$ .

**Solution 3 - Complementary Counting**

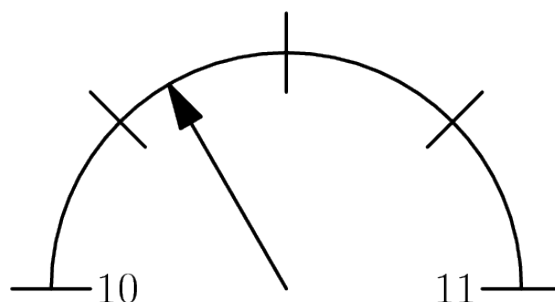
An alternative approach would be to calculate the probability the sum of the two numbers is odd, and subtract that from the total. We see that there is only one way to get an odd sum, where the two numbers must be of opposite parity.

Here, we have two cases  $\rightarrow$  the two numbers are, respectively, (even, odd), and (odd, even). The probability to get an even number as the first number is  $\frac{5}{10} = \frac{1}{2}$ , and an odd number as the second number gives us a probability of  $\frac{5}{9}$ . Multiplying the two, we get  $\frac{5}{18}$ , and because both cases have equal probability, we have the probability for the sum to be odd is  $\frac{10}{18} = \frac{5}{9}$ .

Finally, subtracting this from 1 gives us  $\frac{4}{9}$  (A). Cheetahboy93

§4 AMC 8 1988

1. (AMC 8 1988) The diagram shows part of a scale of a measuring device. The arrow indicates an approximate reading of  $m$ . What is the value of  $m$ ?



- (A) 10.05    (B) 10.15    (C) 10.25    (D) 10.3    (E) 10.6

**Answer:** D

**Solution**

Since the scale is divided into 4 equal sections, the marks to the left and right the arrow must be 10.25 and 10.5, respectively. The only answer choice strictly between 10.25 and 10.5 is **(D) 10.3**.

2. (AMC 8 1988) The product  $8 \times .25 \times 2 \times .125 =$  (A)  $\frac{1}{8}$     (B)  $\frac{1}{4}$     (C)  $\frac{1}{2}$     (D) 1    (E) 2

**Answer:** C

**Solution**

Converting decimals to fractions yields  $8 \times .25 \times 2 \times .125 = \frac{8}{1} \times \frac{1}{4} \times \frac{2}{1} \times \frac{1}{8} = \frac{16}{32} =$   
**(C)  $\frac{1}{2}$** .

3. (AMC 8 1988) Evaluate the following expression:

$$\frac{1}{10} + \frac{2}{20} + \frac{3}{30} = \text{(A) } .1 \quad \text{(B) } .123 \quad \text{(C) } .2 \quad \text{(D) } .3 \quad \text{(E) } .6$$

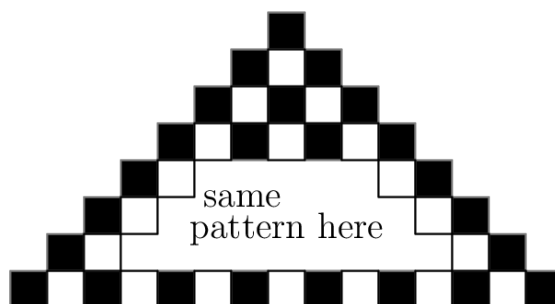
**Answer:** D

**Solution**

Each fraction simplifies to  $\frac{1}{10}$ , so this sum is  $\frac{1}{10} + \frac{1}{10} + \frac{1}{10} = \frac{3}{10} =$  **(D) .3**.

4. (AMC 8 1988) The figure consists of alternating light and dark squares. The number of dark squares exceeds the number of light squares by

- (A) 7    (B) 8    (C) 9    (D) 10    (E) 11



**Answer:** B

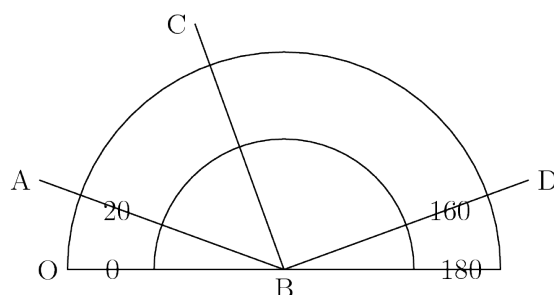
**Solution 1**

Notice that in each row, the number of black squares is 1 more than the number of white squares. Since this figure has 8 rows, it has **(B) 8** more black squares than white squares.

**Solution 2**

Notice that after the first row, each row has 1 more white square and 1 more black square than the previous row. So, this figure has  $1+2+\cdots+7+8$  black squares and  $1+2+\cdots+7$  white squares. Therefore, there are  $(1+2+\cdots+7+8)-(1+2+\cdots+7) =$  **(B) 8** more black squares than white squares.

5. (AMC 8 1988) If  $\angle CBD$  is a right angle, then this protractor indicates that the measure of  $\angle ABC$  is approximately



- (A)  $20^\circ$     (B)  $40^\circ$     (C)  $50^\circ$     (D)  $70^\circ$     (E)  $120^\circ$

**Answer:** C

**Solution**

$\angle OBA = 20^\circ$ ,  $\angle CBD = 90^\circ$ , and  $\angle OBD = 160^\circ$ . So  $\angle ABC = 160^\circ - 20^\circ - 90^\circ =$  **(C)  $50^\circ$** .

6. (AMC 8 1988)  $\frac{(.2)^3}{(.02)^2} =$  (A) .2    (B) 2    (C) 10    (D) 15    (E) 20

**Answer:** E

**Solution 1**

Converting the decimals to fractions yields  $\frac{(.2)^3}{(.02)^2} = \left(\frac{1}{5}\right)^3 \div \left(\frac{1}{50}\right)^2 = \frac{1}{125} \div \frac{1}{2500} = \frac{1}{125} \cdot \frac{2500}{1} = \frac{2500}{125} =$  **(E) 20**.

**Solution 2**

Expanding yields  $\frac{(.2)^3}{(.02)^2} = \frac{.2 \times .2 \times .2}{.02 \times .02} = \frac{.2}{.02} \times \frac{.2}{.02} \times .2 = 10 \times 10 \times .2 =$  **(E) 20**.

**Solution 3**

Using scientific notation,  $\frac{(0.2)^3}{(0.02)^2} = \frac{(2 \times 10^{-1})^3}{(2 \times 10^{-2})^2} = \frac{8 \times 10^{-3}}{4 \times 10^{-4}} = 2 \times 10 =$  **(E) 20**.

7. (AMC 8 1988)  $2.46 \times 8.163 \times (5.17 + 4.829)$  is closest to  
(A) 100    (B) 200    (C) 300    (D) 400    (E) 500

**Answer:** B

**Solution 1**

$$2.46 \times 8.163 \times (5.17 + 4.829) = 2.46 \times 8.163 \times 9.999 \approx 2.5 \times 8 \times 10 = \boxed{\text{(B)} 200}.$$

- GaloisTorrent, edited by j314andrews,

8. (AMC 8 1988) Betty used a calculator to find the product  $0.075 \times 2.56$ . She forgot to enter the decimal points. The calculator showed 19200. If Betty had entered the decimal points correctly, the answer would have been

(A) .0192      (B) .192      (C) 1.92      (D) 19.2      (E) 192

**Answer:** B

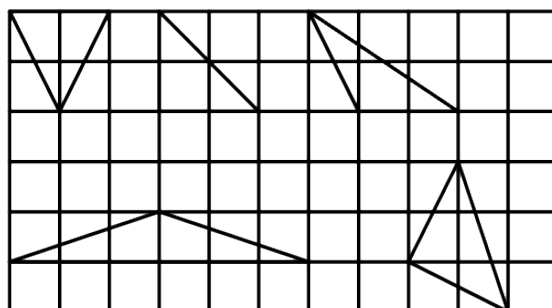
**Solution 1**

Betty calculated  $75 \times 256 = 19200$ , when she should have calculated  $\frac{75}{1000} \times \frac{256}{100} = \frac{19200}{100000} = \frac{192}{1000} = \boxed{\text{(B)} .192}$ .

**Solution 2**

Since 0.075 has 3 decimal places and 2.56 has 2 decimal places, the decimal point in the product should be moved  $3 + 2 = 5$  places to the left. Therefore, the correct product is  $.19200 = \boxed{\text{(B)} .192}$ .

9. (AMC 8 1988) An isosceles triangle is a triangle with two sides of equal length. How many of the five triangles on the square grid below are isosceles?



(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** D

**Solution**

Going from top left to bottom right, the first triangle has two sides of length  $\sqrt{5}$  by the Pythagorean theorem, so it is isosceles.

The second triangle has two sides of length 2, so it is isosceles.

The third triangle has sides of length 2,  $\sqrt{5}$ , and  $\sqrt{13}$ , so it is not isosceles.

The fourth triangle has two sides of length  $\sqrt{10}$  so it is isosceles.

The fifth triangle has two sides of length  $\sqrt{5}$ , so it is isosceles.

So,  $\boxed{\text{(D)} 4}$  of these triangles are isosceles.

10. (AMC 8 1988) Chris's birthday is on a Thursday this year. What day of the week will it be 60 days after her birthday?

(A) Monday      (B) Wednesday      (C) Thursday      (D) Friday      (E) Saturday

**Answer:** A

**Solution 1**

Since 60 days is 8 weeks and 4 days, the day 60 days after Chris' birthday will be 4 days after a Thursday, which is **(A) Monday**.

**Solution 2 (Modular Arithmetic)**

Since  $60 \equiv 4 \pmod{7}$ , the day 60 days after Chris' birthday will be 4 days after a Thursday, which is **(A) Monday**.

11. (AMC 8 1988)  $\sqrt{164}$  is  
 (A) 42 (B) less than 10 (C) between 10 and 11 (D) between 11 and 12 (E) between 12 and 13

**Answer:** E

**Solution**

Since  $12^2 = 144 < 164 < 169 = 13^2$ , it follows that  $12 < \sqrt{164} < 13$ . That is,  $\sqrt{164}$  is **(E) between 12 and 13**.

12. (AMC 8 1988) Suppose the estimated 20 billion dollar cost to send a person to the planet Mars is shared equally by the 250 million people in the U.S. Then each person's share is  
 (A) 20 dollars (B) 50 dollars (C) 80 dollars (D) 100 dollars (E) 125 dollars

**Answer:** C

**Solution 1**

The cost per person is  $\frac{\$20 \text{ billion}}{250 \text{ million}} = \frac{\$20000 \text{ million}}{250 \text{ million}} = \textbf{(C) \$80}$ .

**Solution 2**

The cost per person is  $\frac{\$20 \text{ billion}}{250 \text{ million}} = \frac{\$20 \times 10^9}{250 \times 10^6} = \frac{\$20 \times 10^3}{250} = \frac{\$20000}{250} = \textbf{(C) \$80}$ .

13. (AMC 8 1988) If rose bushes are spaced about 1 foot apart, approximately how many bushes are needed to surround a circular patio whose radius is 12 feet?  
 (A) 12 (B) 38 (C) 48 (D) 75 (E) 450

**Answer:** D

**Solution**

The circumference of the patio is  $2\pi \cdot 12 = 24\pi \approx 24 \cdot 3.14 = 75.36$ . Since the bushes are spaced about 1 foot apart, about **(D) 75** are needed.

14. (AMC 8 1988)  $\diamond$  and  $\Delta$  are whole numbers and  $\diamond \times \Delta = 36$ . The largest possible value of  $\diamond + \Delta$  is  
 (A) 12 (B) 13 (C) 15 (D) 20 (E) 37

**Answer:** E

**Solution 1**

Since  $\diamond$  and  $\Delta$  can be swapped to get the same sum, only cases where  $\diamond \geq \Delta$  need to be considered. These cases are shown in the table:

| $\diamond$ | $\Delta$ | $\diamond + \Delta$ |
|------------|----------|---------------------|
| 36         | 1        | 37                  |
| 18         | 2        | 20                  |
| 12         | 3        | 15                  |
| 9          | 4        | 13                  |
| 6          | 6        | 12                  |



So **(E) 37** is the largest possible sum.

**Solution 2**

We notice that the function  $f(x) = x + \frac{36}{x}$  is convex with a minimum at  $x = 6$ , so we seek to pick a value of  $x$  far from 6. To ensure that the answers are positive integers, we let  $x = 36$ , from which we find that  $f(x) = 37$ , the maximum.

-GaloisTorrent

**Solution 3 (Answer Choices)**

Notice that  $1 \times 36 = 36$  and  $1 + 36 = 37$ . Since **(E) 37** is the highest answer choice it must be correct.

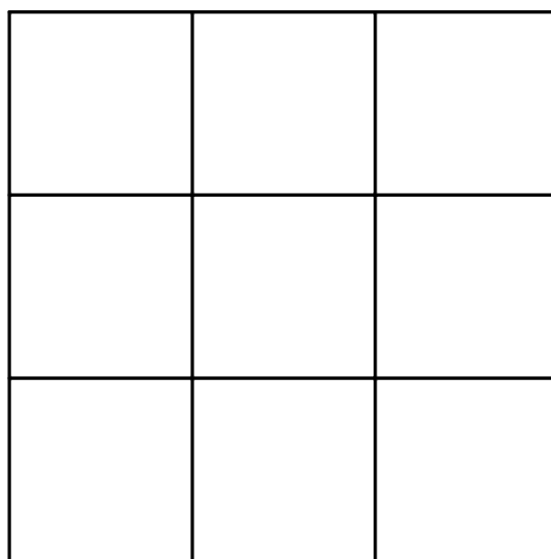
15. (AMC 8 1988) The reciprocal of  $\left(\frac{1}{2} + \frac{1}{3}\right)$  is  
 (A)  $\frac{1}{6}$     (B)  $\frac{2}{5}$     (C)  $\frac{6}{5}$     (D)  $\frac{5}{2}$     (E) 5

**Answer:** C

**Solution**

$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ , and the reciprocal of  $\frac{5}{6}$  is **(C)  $\frac{6}{5}$** .

16. (AMC 8 1988) Placing no more than one X in each small square, what is the greatest number of X's that can be put on the grid shown without getting three X's in a row vertically, horizontally, or diagonally?  
 (A) 2    (B) 3    (C) 4    (D) 5    (E) 6



**Answer:** E

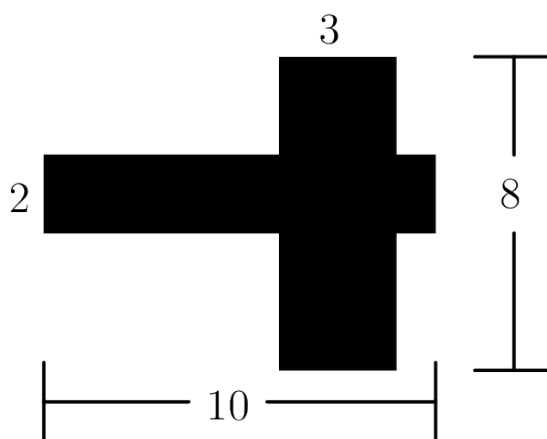
**Solution**

It is impossible to place 7 or more X's, as there would be 2 or fewer empty squares and thus not every row could have an empty square.

It is possible to place **(E) 6** X's as follows:

|   |   |   |
|---|---|---|
|   | X | X |
| X |   | X |
| X | X |   |

17. (AMC 8 1988) The shaded region formed by the two intersecting perpendicular rectangles, in square units, is



- (A) 23    (B) 38    (C) 44    (D) 46    (E) unable to be determined from the information given

**Answer:** B

**Solution**

The area of the  $2 \times 10$  rectangle is 20, while the area of the  $8 \times 3$  rectangle is 24. If these two areas were added together, however, the area of the intersection, which is a  $2 \times 3$  rectangle, would be counted twice. Since the intersection has area 6, the area of this figure must be  $20 + 24 - 6 = \boxed{\text{(B) } 38}$ .

18. (AMC 8 1988) The average weight of 6 boys is 150 pounds and the average weight of 4 girls is 120 pounds. The average weight of the 10 children is

- (A) 135 pounds    (B) 137 pounds    (C) 138 pounds    (D) 140 pounds    (E) 141 pounds

**Answer:** C

**Solution**

The total weight of the boys is  $6 \times 150 = 900$  pounds, and the total weight of the girls is  $4 \times 120 = 480$  pounds. Therefore, the total weight of all 10 children

is  $900 + 480 = 1380$  pounds, and their average weight is  $1380 \div 10 = \boxed{\text{(C)} 138}$  pounds.

19. (AMC 8 1988) What is the 100th number in the arithmetic sequence: 1, 5, 9, 13, 17, 21, 25, ...?

(A) 397      (B) 399      (C) 401      (D) 403      (E) 405

**Answer:** A

**Solution 1**

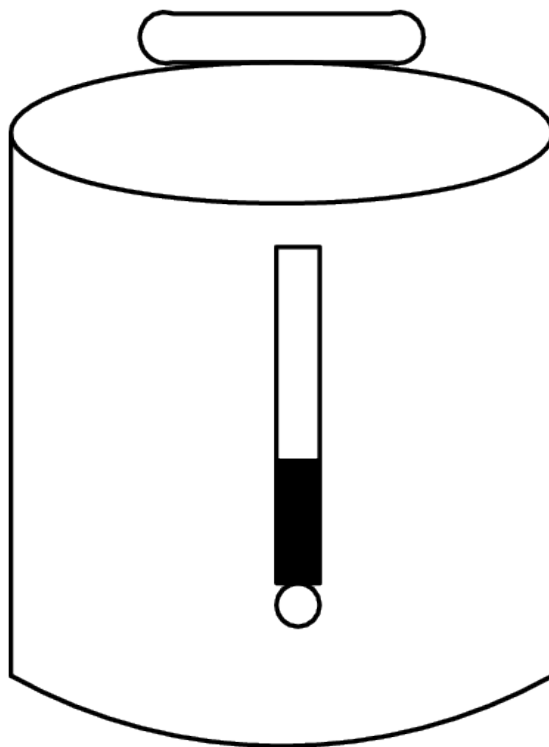
The common difference of this arithmetic sequence is  $5 - 1 = 4$ . From the first term, it takes 99 steps to reach the 100th term, so the 100th term is  $1 + 4 \cdot 99 = \boxed{\text{(A)} 397}$ .

**Solution 2**

As in solution 1, the common difference is 4. Recall that the  $n$ th term of an arithmetic sequence with first term  $a$  and common difference  $d$  is  $a + d(n - 1) = 1 + 4(100 - 1) = \boxed{\text{(A)} 397}$ .

20. (AMC 8 1988) The glass gauge on a cylindrical coffee maker shows that there are 45 cups left when the coffee maker is 36% full. How many cups of coffee does it hold when it is full?

(A) 80      (B) 100      (C) 125      (D) 130      (E) 262



**Answer:** C

**Solution**

Let  $x$  be the number of cups of coffee that the coffee maker can hold when it is full. Then  $.36x = 45$ , so  $x = \boxed{\text{(C)} 125}$ .

21. (AMC 8 1988) A fifth number,  $n$ , is added to the set  $\{3, 6, 9, 10\}$  to make the mean of the set of five numbers equal to its median. The number of possible values of  $n$  is

- (A) 1      (B) 2      (C) 3      (D) 4      (E) more than 4

**Answer:** C

**Solution**

The mean of the set  $\{3, 6, 9, 10, n\}$  is  $\frac{3+6+9+10+n}{5} = \frac{28+n}{5}$ .

If  $n < 6$ , the median is 6. Solving  $\frac{28+n}{5} = 6$  yields  $n = 2$ , which is valid since  $2 < 6$ .

If  $6 \leq n < 9$ , the median is  $n$ . Solving  $\frac{28+n}{5} = n$  yields  $n = 7$ , which is valid since  $6 \leq 7 < 9$ .

If  $n \geq 9$ , the median is 9. Solving  $\frac{28+n}{5} = 9$  yields  $n = 17$ , which is valid since  $17 \geq 9$ .

Therefore, there are (C) 3 possible values for  $n$ .

22. (AMC 8 1988) Tom's Hat Shoppe increased all original prices by 25%. Now the shoppe is having a sale where all prices are 20% off these increased prices. Which statement best describes the sale price of an item?

- (A) The sale price is 5% higher than the original price. (B) The sale price is higher than the original price.  
 (C) The sale price is higher than the original price, but by more than 5%. (D) The sale price is lower.  
 (E) The sale price is the same as the original price.

**Answer:** E

**Solution 1**

Let  $x$  be the original price of the item. The shoppe originally increased this to  $1.25x$ .

The sale brings it down to  $(0.8)(1.25x) = x$ , so (E) The sale price is the same as the original price.

**Solution 2**

Suppose the original price of an item was \$100. The shoppe originally increased the price to  $1.25 \cdot \$100 = \$125$ , while the sale reduced it to  $0.8 \cdot \$125 = \$100$ , so

(E) The sale price is the same as the original price.

23. (AMC 8 1988) Maria buys computer disks at a price of 4 for  $< \text{dollar} > 5$  and sells them at a price of 3 for  $< \text{dollar} > 5$ . How many computer disks must she sell in order to make a profit of  $< \text{dollar} > 100$ ?

- (A) 100      (B) 120      (C) 200      (D) 240      (E) 1200

**Answer:** D

**Solution 1**

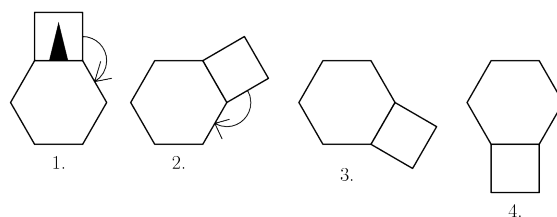
Maria buys at  $\frac{5}{4}$  dollars per disk and sells at  $\frac{5}{3}$  dollars per disk, so her profit is  $\frac{5}{3} - \frac{5}{4} = \frac{5}{12}$  dollars per disk. Thus to make a profit of \$100 she must sell  $100 \div \frac{5}{12} = \text{span style="border: 1px solid black; padding: 0 2px;">(C) 240 disks. This is valid since 240 is divisible by both 3 and 4, and therefore Maria will not have any disks left over.$

**Solution 2**

If Maria buys 12 disks, she buys 3 sets of 4 so she pays \$15. If Maria sells 12 disks, she sells 4 sets of 3 so she gets \$20. Therefore, for every 12 disks Maria sells, she profits \$5.

To profit \$100, she must sell  $\frac{100}{5} = 20$  batches of 12 disks, for a total of (C) 240 disks.

24. (AMC 8 1988)



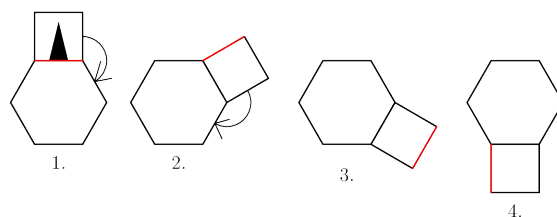
The square in the first diagram "rolls" clockwise around the fixed regular hexagon until it reaches the bottom. In which position will the solid triangle be in diagram 4?



**Answer:** A

**Solution 1**

In this diagram, the side containing the base of the triangle is shown in red. After 4 steps, it will be on the left side of the square, so **(A)** is correct.



**Solution 2**

In each step, the square rotates by an angle of  $150^\circ$  clockwise. Therefore, in 3 steps, the square rotates  $450^\circ$  clockwise, which is  $90^\circ$  more than a full rotation. Since the triangle originally pointed upward, after rotating  $90^\circ$  clockwise it will point rightward. Therefore, **(A)** is correct.

25. (AMC 8 1988) A **palindrome** is a whole number that reads the same forwards and backwards. If one neglects the colon, certain times displayed on a digital watch are palindromes. Three examples are: **1 : 01**, **4 : 44**, and **12 : 21**. How many times during a 12-hour period will be palindromes?

(A) 57    (B) 60    (C) 63    (D) 90    (E) 93

**Answer:** A

**Solution**

Consider two cases:

Case 1: the hour is 9 or less. In this case, the time has 3 digits, so it is in the form  $A : BA$ . Since  $A$  can be any digit from 1 to 9 while  $B$  can be any digit from 0 to 5, there are  $9 \cdot 6 = 54$  three-digit palindromes.

Case 2: the hour is 10 or more. In this case, the time has 4 digits, so it is in the form  $AB : BA$ . Since  $AB$  can be 10, 11, or 12, there are 3 four-digit palindromes.

Therefore, there are  $54 + 3 = \mathbf{(A) 57}$  total palindromes.

# §5 AMC 8 1989

1. (*AMC 8 1989*)  $(1+11+21+31+41)+(9+19+29+39+49) =$  (A) 150 (B) 199 (C) 200 (D) 249

**Answer:** E

**Solution 1**

Rearranging terms yields  $(1 + 49) + (11 + 39) + (21 + 29) + (31 + 19) + (41 + 9) = 50 + 50 + 50 + 50 + 50 = 50 \cdot 5 = \boxed{\text{(E)} 250}$ .

**Solution 2**

Alternatively, the terms can be arranged to get  $(1 + 9) + (11 + 19) + (21 + 29) + (31 + 39) + (41 + 49) = 10 + 30 + 50 + 70 + 90 = \boxed{\text{(E)} 250}$

2. (*AMC 8 1989*)  $\frac{2}{10} + \frac{4}{100} + \frac{6}{1000} =$  (A) .012 (B) .0246 (C) .12 (D) .246 (E) 246

**Answer:** D

**Solution**

$$\frac{2}{10} + \frac{4}{100} + \frac{6}{1000} = \frac{200}{1000} + \frac{40}{1000} + \frac{6}{1000} = \frac{246}{1000} = \boxed{\text{(D)} .246}$$

3. (*AMC 8 1989*) Which of the following numbers is the largest?

(A) .99 (B) .9099 (C) .9 (D) .909 (E) .9009

**Answer:** A

**Solution 1**

When written with 4 decimal places, the answer choices are .9900, .9099, .9000, .9090, and .9009. So  $\boxed{\text{(A)} .99}$  must be greater than all the other answer choices, whose first two decimal places are .90.

**Solution 2**

The answer choices sorted from largest to smallest are  $.99 > .9099 > .909 > .9009 > .9$ , so choice  $\boxed{\text{(A)} .99}$  is the largest.

**Solution 3**

If .9 is subtracted from each answer choice, the answer choices become .09, .0099, 0, .009, and .0009. Of these, all except .09 are less than .01, so  $\boxed{\text{(A)} .99}$  is the largest.

4. (*AMC 8 1989*) Estimate to determine which of the following numbers is closest to  $\frac{401}{.205}$ .

(A) .2 (B) 2 (C) 20 (D) 200 (E) 2000

**Answer:** E

**Solution 1**

$$\text{By rounding, } \frac{401}{.205} \approx \frac{400}{.2} = \boxed{\text{(E)} 2000}$$

**Solution 2**

Using scientific notation estimated to the first significant digit, we see the numerator rounds to  $\frac{4 \cdot 10^2}{2 \cdot 10^{-1}}$ . Subtracting the exponents gives  $2 - (-1) = 3$ . The answer choice that has 3 zeros to the right of the significant digit is  $\boxed{\text{E}}$

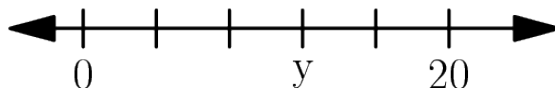
5. (*AMC 8 1989*)  $-15 + 9 \times (6 \div 3) =$  (A)  $-48$  (B)  $-12$  (C)  $-3$  (D)  $3$  (E)  $12$

**Answer:** D

**Solution**

By order of operations,  $-15 + 9 \times (6 \div 3) = -15 + 9 \times 2 = -15 + 18 = \boxed{\text{(D)} 3}$ .

6. (*AMC 8 1989*) If the markings on the number line are equally spaced, what is the number  $y$ ?



- (A) 3 (B) 10 (C) 12 (D) 15 (E) 16

**Answer:** C

**Solution 1**

Since there are 5 gaps between 0 and 20, each gap has length  $\frac{20}{5} = 4$ .

Since there are 3 gaps between 0 and  $y$ ,  $y = 0 + 3 \times 4 = \boxed{\text{(C)} 12}$ .

**Solution 2**

There are 5 gaps between 0 and 20, while there are 3 gaps between 0 and  $y$ . So  $y$  is  $\frac{3}{5}$  of the way from 0 to 20, that is,  $y = \frac{3}{5} \cdot 20 = \boxed{\text{(C)} 12}$ .

-stjwyl, edited by j314andrews

7. (*AMC 8 1989*) If the value of 20 quarters and 10 dimes equals the value of 10 quarters and  $n$  dimes, then  $n =$  (A) 10 (B) 20 (C) 30 (D) 35 (E) 45

**Answer:** D

**Solution 1**

The total value of 20 quarters and 10 dimes is  $20 \cdot \$0.25 + 10 \cdot \$0.10 = \$6.00$ .

The value of 10 quarters is  $10 \cdot \$0.25 = \$2.50$ .

Therefore, the value of  $n$  dimes is  $\$6.00 - \$2.50 = \$3.50$ , so  $n = \$3.50 \div \$0.10 = \boxed{\text{(D)} 35}$

**Solution 2**

$$20(25) + 10(10) = 10(25) + n(10)$$

$$600 = 250 + 10n$$

$$35 = n \implies \boxed{\text{D}}$$

**Solution 3**

The  $n$  dimes' values need to sum to 10 quarters and 10 dimes.

$$10n = 10 \cdot 25 + 10 \cdot 10$$

we can divide both sides by 10.

$$n = 25 + 10 = 35$$

So, our answer is  $\boxed{\text{D}}$ —stjwyl

8. (*AMC 8 1989*)  $(2 \times 3 \times 4) \left( \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) =$  (A) 1 (B) 3 (C) 9 (D) 24 (E) 26

**Answer:** E

**Solution 1**

By the distributive property, this expression becomes  $3 \times 4 + 2 \times 4 + 2 \times 3 =$  **(E) 26**.

**Solution 2**

Simplifying the left factor yields  $24 \left( \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) = 12 + 8 + 6 =$  **(E) 26**.

-fn106068, edited by j314andrews

**Solution 3 (Estimation, Answer Choices)**

Since  $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} > \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$ , it follows that  $(2 \times 3 \times 4) \left( \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) > 2 \times 3 \times 4 \times 1 = 24$ .

The only answer choice greater than 24 is **(E) 26**.

9. (*AMC 8 1989*) There are 2 boys for every 3 girls in Ms. Johnson's math class. If there are 30 students in her class, what percent of them are boys?

(A) 12% (B) 20% (C) 40% (D) 60% (E)  $66\frac{2}{3}\%$

**Answer:** C

**Solution 1**

Since the ratio of boys to girls is  $2 : 3$ ,  $\frac{2}{2+3} = \frac{2}{5} =$  **(C) 40%** of the students are boys. The number of students in the class is irrelevant.

**Solution 2**

$$\frac{30}{2+3} = \frac{30}{5} = 6,$$

and that is the base unit. Since there are 2 boys for every 5 people,  $6 \cdot 2 = 12$ . Applying this into a fraction gives  $\frac{12}{30} = \frac{2}{5} = 40\% =$  **(C)**.

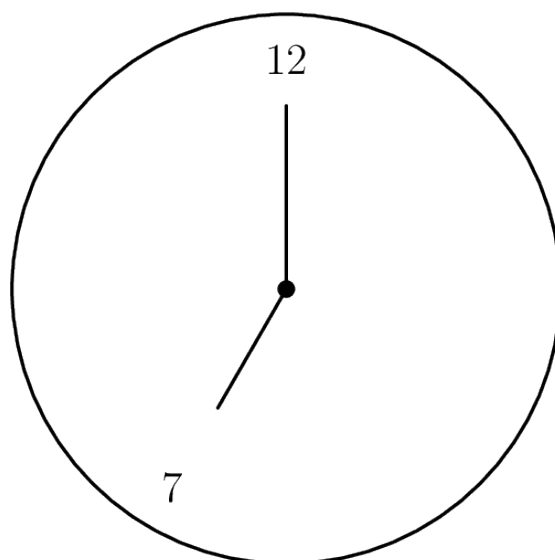
10. (*AMC 8 1989*) What is the number of degrees in the smaller angle between the hour hand and the minute hand on a clock that reads seven o'clock?

(A)  $50^\circ$  (B)  $120^\circ$  (C)  $135^\circ$  (D)  $150^\circ$  (E)  $165^\circ$

**Answer:** D

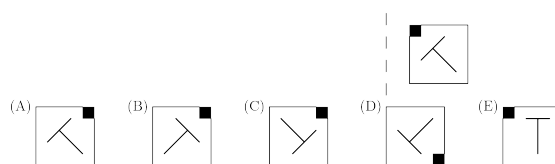
**Solution**





The minute hand points at 12 while the hour hand points at 7. Therefore, the angle between the hands is  $\frac{5}{12}$  of a full circle, or  $\frac{5}{12} \cdot 360^\circ = \boxed{\text{(D)} 150^\circ}$

11. (AMC 8 1989) Which of the five "T-like shapes" would be symmetric to the one shown with respect to the dashed line?



**Answer:** B

**Solution**

The reflection (mirror image) of the figure across the line is  $\boxed{\text{(B)}}$ .

12. (AMC 8 1989)  $\frac{1-\frac{1}{3}}{1-\frac{1}{2}} =$  (A)  $\frac{1}{3}$  (B)  $\frac{2}{3}$  (C)  $\frac{3}{4}$  (D)  $\frac{3}{2}$  (E)  $\frac{4}{3}$

**Answer:** E

**Solution**

$$\left(1 - \frac{1}{3}\right) \div \left(1 - \frac{1}{2}\right) = \frac{2}{3} \div \frac{1}{2} = \frac{2}{3} \cdot \frac{2}{1} = \boxed{\text{(E)} \frac{4}{3}}.$$

13. (AMC 8 1989)  $\frac{9}{7 \times 53} =$  (A)  $\frac{9}{7 \times 53}$  (B)  $\frac{9}{7 \times .53}$  (C)  $\frac{9}{.7 \times 5.3}$  (D)  $\frac{9}{7 \times .53}$  (E)  $\frac{.09}{.07 \times .53}$

**Answer:** A

**Solution**

Notice that  $\frac{9}{7 \times 53} = \frac{.1 \times 9}{.1 \times 7 \times 53} = \boxed{\text{(A)} \frac{.9}{.7 \times 53}}$ . Choices (B) and (E) are 100 times the original expression while choices (C) and (D) are 10 times the original expression.

14. (AMC 8 1989) When placing each of the digits 2, 4, 5, 6, 9 in exactly one of the boxes of this subtraction problem, what is the smallest difference that is possible?  
 (A) 58 (B) 123 (C) 149 (D) 171 (E) 176

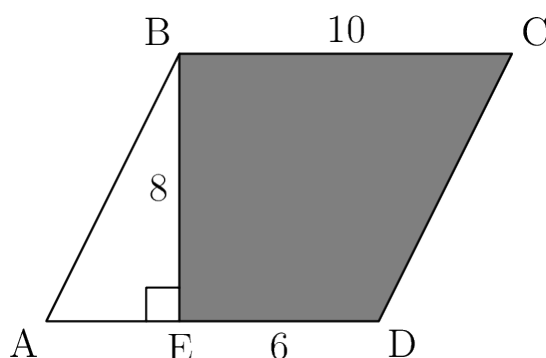
$$\begin{array}{r} \square \square \square \\ - \quad \square \square \\ \hline \end{array}$$

**Answer:** C

**Solution**

The smallest possible difference is found by using the smallest possible three-digit number and the largest possible two-digit number. In this case, this difference is  $245 - 96 = \boxed{\text{(C)} 149}$ .

15. (AMC 8 1989) The area of the shaded region  $BEDC$  in parallelogram  $ABCD$  is



- (A) 24      (B) 48      (C) 60      (D) 64      (E) 80

**Answer:** D

**Solution 1**

The area of  $BEDC$  can be found by subtracting the area of  $\triangle AEB$  from that of  $ABCD$ .

$ABCD$  is a parallelogram, so its area is  $10 \cdot 8 = 80$ .

Also,  $AD = BC = 10$ , so  $AE$  is 4. So  $\triangle AEB$  has area  $\frac{1}{2} \cdot 4 \cdot 8 = 16$ .

Therefore,  $BEDC$  has area  $80 - 16 = \boxed{\text{(D)} 64}$ .

**Solution 2**

Notice that  $BEDC$  is a trapezoid. Therefore, its area is  $\frac{1}{2}(8)(6 + 10) = \boxed{\text{(D)} 64}$ .

16. (AMC 8 1989) In how many ways can 47 be written as the sum of two primes?  
(A) 0      (B) 1      (C) 2      (D) 3      (E) more than 3

**Answer:** A

**Solution**

Since 47 is odd, if two integers sum to 47, then one is odd while the other is even. The even number must be 2 as there are no other even prime numbers. But this would require the odd number to be  $47 - 2 = 45$ , which is not prime. Therefore, there  $\boxed{\text{(A)} 0}$  ways to write 47 as the sum of two primes.

17. (AMC 8 1989) The number  $N$  is between 9 and 17. The average of 6, 10, and  $N$  could be  
(A) 8      (B) 10      (C) 12      (D) 14      (E) 16

**Answer:** B

**Solution**

The smallest possible average is  $\frac{6+10+9}{3} = \frac{25}{3} = 8\frac{1}{3}$ , while the highest possible average is  $\frac{6+10+17}{3} = \frac{33}{3} = 11$ . The only answer choice between  $8\frac{1}{3}$  and 11 is **(B) 10**, which would be the average if  $N = 14$ .

18. (AMC 8 1989) Many calculators have a reciprocal key  $\left[\frac{1}{x}\right]$  that replaces the current number displayed with its reciprocal. For example, if the display is  $\boxed{00004}$  and the  $\left[\frac{1}{x}\right]$  key is pressed, then the display becomes  $\boxed{000.25}$ . If  $\boxed{00032}$  is currently displayed, what is the fewest positive number of times you must depress the  $\left[\frac{1}{x}\right]$  key so the display again reads  $\boxed{00032}$ ?
- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** B

**Solution 1**

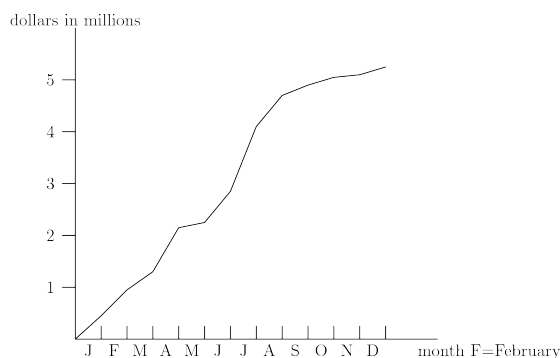
After pressing the reciprocal key once, the calculator will display the decimal representation of  $\frac{1}{32}$  (which is 0.03125). After pressing the reciprocal key again, the calculator will display  $1 \div \frac{1}{32} = 32$ .

So, after **(B) 2** presses the calculator will display 32 again. (Since  $\frac{1}{32} = 0.03125$  is a terminating decimal, there will not be any rounding.)

**Solution 2**

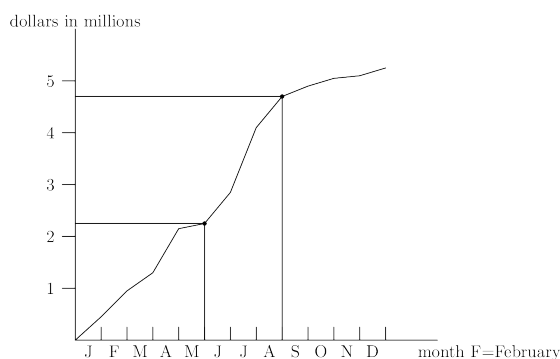
Let  $f(x) = \frac{1}{x}$ . Then  $f(f(x)) = 1 \div \frac{1}{x} = x$ . Therefore, if  $x \neq \frac{1}{x}$  (that is,  $x \neq \pm 1$ ), a calculator displaying  $x$  will first display  $x$  again after **(B) 2** presses.

19. (AMC 8 1989) The graph below shows the total accumulated dollars (in millions) spent by the Surf City government during 1988. For example, about .5 million had been spent by the beginning of February and approximately 2 million by the end of April. Approximately how many millions of dollars were spent during the summer months of June, July, and August?
- (A) 1.5      (B) 2.5      (C) 3.5      (D) 4.5      (E) 5.5



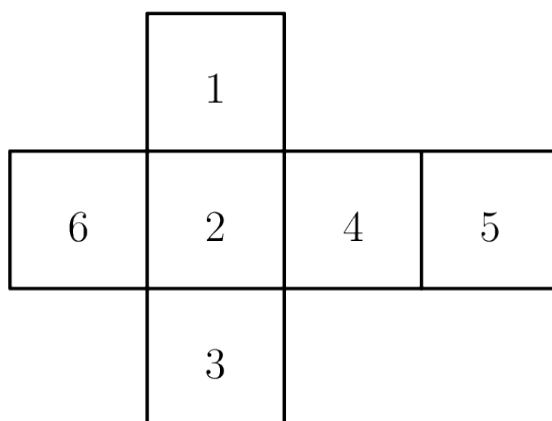
**Answer:** B

**Solution**



From the graph, about \$2.2 million was spent by the start of June and about \$4.7 million was spent by the end of August. So about  $4.7 - 2.2 = \boxed{\text{(B)} 2.5}$  million dollars were spent between these two months.

20. (*AMC 8 1989*) The figure may be folded along the lines shown to form a number cube. Three number faces come together at each corner of the cube. What is the largest sum of three numbers whose faces come together at a corner?



- (A) 11      (B) 12      (C) 13      (D) 14      (E) 15

**Answer:** D

**Solution**

The highest sum of three different numbers on this cube is  $4 + 5 + 6 = 15$ . But 4 and 6 are on opposite faces, so the faces labeled 4, 5, and 6 will not meet at a corner.

The next highest sum of three different numbers on this cube is  $3 + 5 + 6 = 14$ . These faces meet at a corner, so the answer is  $\boxed{\text{(D)} 14}$ .

21. (*AMC 8 1989*) Jack had a bag of 128 apples. He sold 25% of them to Jill. Next he sold 25% of those remaining to June. Of those apples still in his bag, he gave the shiniest one to his teacher. How many apples did Jack have then?
- (A) 7      (B) 63      (C) 65      (D) 71      (E) 111

**Answer:** D

**Solution**

After selling apples to Jill, Jack has  $0.75 \cdot 128 = 96$  apples left. After selling apples to June, he has  $0.75 \cdot 96 = 72$  apples left. After giving an apple to his teacher, he has  $72 - 1 = \boxed{\text{(D)} 71}$  apples left.

22. (AMC 8 1989) The letters A, J, H, S, M, E and the digits 1, 9, 8, 9 are "cycled" separately as follows and put together in a numbered list:

AJHSME 1989

1. JHSMEA 9891
2. HSMEAJ 8919
3. SMEAJH 9198

.....

What is the number of the line on which AJHSME 1989 will appear for the first time?

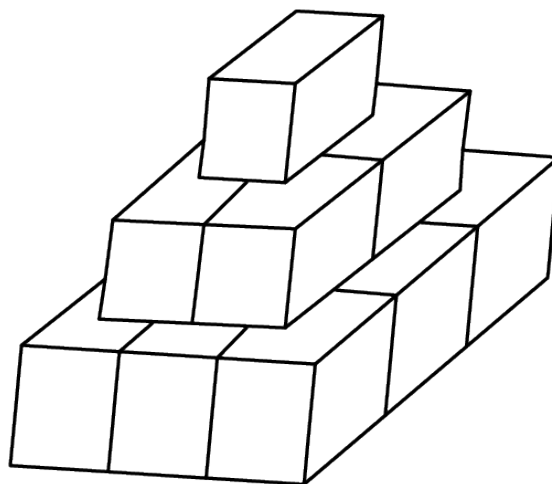
- (A) 6      (B) 10      (C) 12      (D) 18      (E) 24

**Answer:** C

**Solution**

Every 4th line includes 1989 while every 6th line includes AJHSME. Therefore, a line has AJSHME 1989 if and only if its number is divisible by both 4 and 6. Therefore, the number of the first such line is  $\text{lcm}(4, 6) = \boxed{\text{(C)} 12}$ .

23. (AMC 8 1989) An artist has 14 cubes, each with an edge of 1 meter. She stands them on the ground to form a sculpture as shown. She then paints the exposed surface of the sculpture. How many square meters does she paint?
- (A) 21      (B) 24      (C) 33      (D) 37      (E) 42



**Answer:** C

**Solution**

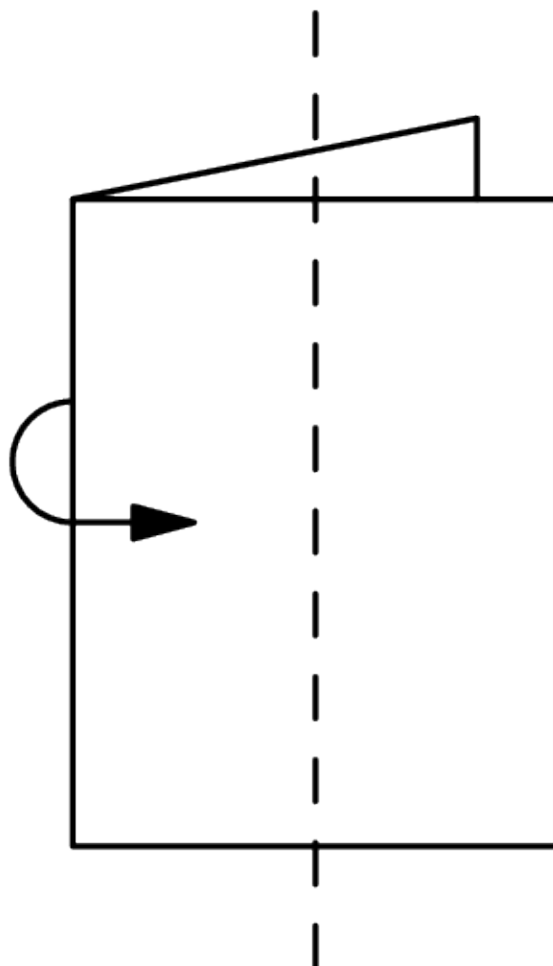
The top view of this figure forms a  $3 \text{ m} \times 3 \text{ m}$  square, so it has area  $9 \text{ m}^2$ .

The left, right, front, and rear views each consist of six  $1 \text{ m} \times 1 \text{ m}$  squares stacked in a pyramid shape, so each of these views has an area of  $6 \text{ m}^2$ . So, these views have a total area of  $4 \cdot 6 = 24 \text{ m}^2$ .

Therefore, the total painted area is  $9 + 24 = \boxed{\text{(C) } 33}$   $\text{m}^2$ . Note that the bottom is not painted.

24. (AMC 8 1989) Suppose a square piece of paper is folded in half vertically. The folded paper is then cut in half along the dashed line. Three rectangles are formed—a large one and two small ones. What is the ratio of the perimeter of one of the small rectangles to the perimeter of the large rectangle?

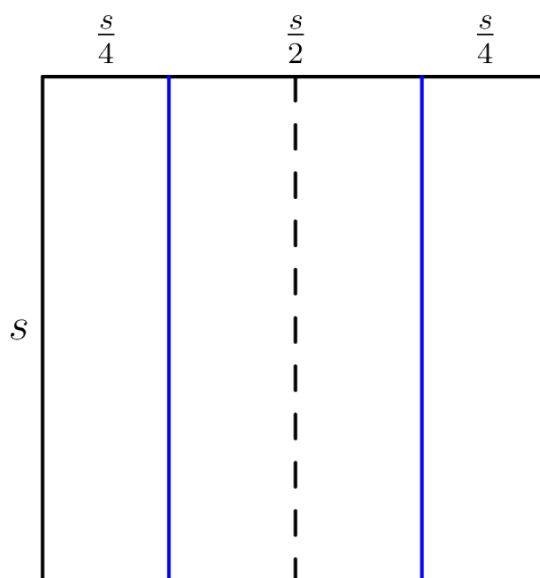
(A)  $\frac{1}{2}$     (B)  $\frac{2}{3}$     (C)  $\frac{3}{4}$     (D)  $\frac{4}{5}$     (E)  $\frac{5}{6}$



**Answer:** E

**Solution 1**

In this diagram, the dashed segment is the fold while the blue segments are cuts.



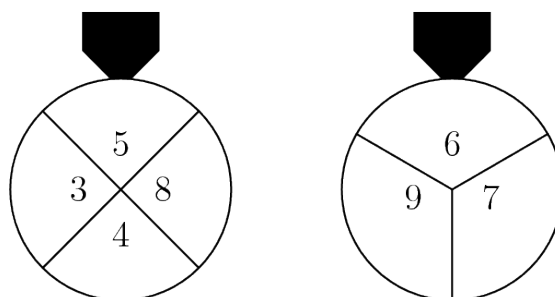
Let  $s$  be the side length of the square. After being cut, the paper will be separated into two  $s \times \frac{s}{4}$  rectangles and a  $s \times \frac{s}{2}$  rectangle. The larger rectangle has perimeter  $2(s + \frac{s}{2}) = 3s$  while the smaller rectangle has perimeter  $2(s + \frac{s}{4}) = \frac{5s}{2}$ . Therefore, the ratio of the perimeters is  $(\frac{5s}{2}) \div (3s) = \boxed{\text{(E)} \frac{5}{6}}$ .

### Solution 2

Suppose the square has side length 4. Then the paper would be separated into two  $4 \times 1$  rectangles and a  $4 \times 2$  rectangle. The larger rectangle has perimeter  $2(4 + 2) = 12$  and the smaller rectangle has perimeter  $2(4 + 1) = 10$ , so the ratio of the perimeters is  $\frac{10}{12} = \boxed{\text{(E)} \frac{5}{6}}$ .

25. (AMC 8 1989) Every time these two wheels are spun, two numbers are selected by the pointers. What is the probability that the sum of the two selected numbers is even?

(A)  $\frac{1}{6}$     (B)  $\frac{3}{7}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{3}$     (E)  $\frac{5}{7}$



**Answer:** C

### Solution 1

For the sum to be even, both numbers must have the same parity (i.e. both are odd or both are even).

The first spinner has 2 odd and 2 even numbers, so regardless of the second spinner's outcome, the probability the first spinner's number has the same parity as the second spinner's number, and thus the probability of the sum being even, is  $\boxed{(C) \frac{1}{2}}$ .

### Solution 2

For the sum to be even, either both numbers are odd or both numbers are even.

The probability that both numbers are even is  $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ , while the probability that both numbers are odd is  $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$ . So the probability that the sum is even is

$$\frac{1}{6} + \frac{1}{3} = \boxed{(C) \frac{1}{2}}.$$

### Solution 3 (Complementary Counting)

We can complement count this, so we are looking for odd results. This happens when they have different **parity**. We can find the probability of an odd sum by using casework.

Case 1: The left spinner is odd. That has  $\frac{1}{2}$ , and the right has probability of  $\frac{1}{3}$  for an even number, of  $\frac{1}{6}$ .

Case 2: The left spinner is even. That has  $\frac{1}{2}$ , and the right has probability of  $\frac{2}{3}$  for an odd number, of  $\frac{1}{3}$ .

$$\text{The sum is } \frac{1}{6} + \frac{1}{3} = \frac{1}{2} \implies \boxed{(C)}$$

## §6 AMC 8 1990

1. (*AMC 8 1990*) What is the smallest sum of two 3-digit numbers that can be obtained by placing each of the six digits 4, 5, 6, 7, 8, 9 in one of the six boxes in this addition problem?

$$\begin{array}{r} \square \quad \square \quad \square \\ + \quad \square \quad \square \quad \square \\ \hline \end{array}$$

- (A) 947      (B) 1037      (C) 1047      (D) 1056      (E) 1245

**Answer:** C

### Solution 1

The minimum sum is found by choosing smaller digits to have higher place values. In this case, 4 and 5 should be hundreds digits, 6 and 7 should be tens digits, and 8 and 9 should be units digits. The minimum sum is therefore  $468 + 579 = \boxed{(C) 1047}$ .

### Solution 2

Let the two 3-digit numbers be  $\overline{abc}$  and  $\overline{def}$ . Their sum is  $100(a + d) + 10(b + e) + (c + f)$ .



To minimize this sum,  $a + d$  should be minimized, which happens if  $\{a, d\} = \{4, 5\}$ . By similar logic using the remaining digits, the sum is minimized if  $\{b, e\} = \{6, 7\}$  while  $\{c, f\} = \{8, 9\}$ . The minimum sum is therefore  $100(9) + 10(13) + (17) =$   
**(C) 1047**

2. (AMC 8 1990) Which digit of .12345, when changed to 9, gives the largest number?

(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** A

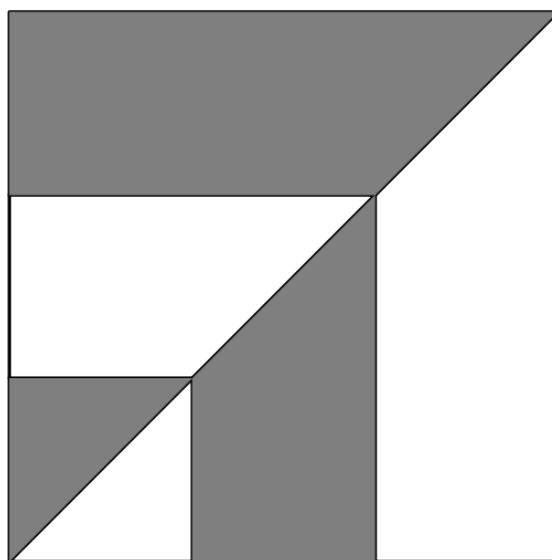
**Solution 1**

Since **(A) 1** has the highest place value, changing it to 9 will result in the largest number.

**Solution 2**

Changing 1 to 9 results in a number greater than .9 while changing any other digit to 9 results in a number less than .2. Therefore, changing **(A) 1** results in the largest number.

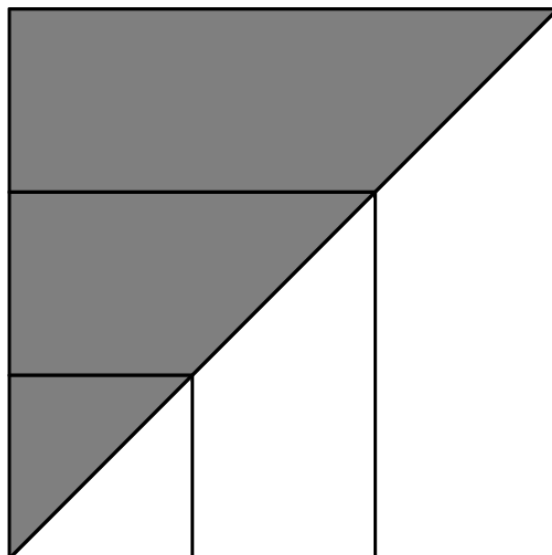
3. (AMC 8 1990) What fraction of the square is shaded?



(A)  $\frac{1}{3}$       (B)  $\frac{2}{5}$       (C)  $\frac{5}{12}$       (D)  $\frac{3}{7}$       (E)  $\frac{1}{2}$

**Answer:** E

**Solution 1**



Swap the two trapezoids in the middle layer, as shown. The shaded region and unshaded region now form congruent triangles, **(E)**  $\frac{1}{2}$  of the square is shaded.

**Solution 2**

Notice that the figure is made of 3 pairs of congruent shapes, and each of these pairs has 1 shaded and 1 unshaded shape. Therefore, **(E)**  $\frac{1}{2}$  of the square is shaded.

4. (*AMC 8 1990*) Which of the following could **not** be the unit's digit [one's digit] of the square of a whole number?  
 (A) 1      (B) 4      (C) 5      (D) 6      (E) 8

**Answer:** E

**Solution 1**

Let  $n$  be a whole number.

If  $n$  ends in 0, then  $n^2$  ends in 0.

If  $n$  ends in 1 or 9, then  $n^2$  ends in 1.

If  $n$  ends in 2 or 8, then  $n^2$  ends in 4.

If  $n$  ends in 3 or 7, then  $n^2$  ends in 9.

If  $n$  ends in 4 or 6, then  $n^2$  ends in 6.

If  $n$  ends in 5, then  $n^2$  ends in 5.

Therefore,  $n^2$  cannot end in **(E)** 8.

**Solution 2**

Notice that  $1^2 = 1$ ,  $2^2 = 4$ ,  $5^2 = 25$ , and  $4^2 = 16$ , to eliminate (A), (B), (C), and (D) respectively. So **(E)** 8 must be correct.

5. (*AMC 8 1990*) Which of the following is closest to the product  $(.48017)(.48017)(.48017)$ ?  
 (A) 0.011      (B) 0.110      (C) 1.10      (D) 11.0      (E) 110

**Answer:** B

**Solution 1**

Since .48017 is between .4 and .5,  $(.48017)^3$  must be between  $(.4)^3 = .064$  and  $(.5)^3 = .125$ . So .48017 must be closest to **(B) 0.110**.

**Solution 2**

Since .48017 is a bit less than .5,  $(.48017)^3$  should be a bit less than  $(.5)^3 = .125$ . Of the answer choices, **(B) 0.110** is a bit less than .125 so it is the closest.

6. (AMC 8 1990) Which of these five numbers is the largest?

(A)  $13579 + \frac{1}{2468}$       (B)  $13579 - \frac{1}{2468}$       (C)  $13579 \times \frac{1}{2468}$       (D)  $13579 \div \frac{1}{2468}$       (E) 13579.2468

**Answer:** D

**Solution**

Choices (A) and (E) are slightly more than 13579.

Choice (B) is slightly less than 13579.

Choice (C), which is equal to  $\frac{13579}{2468}$ , is much less than 13579.

Choice (D), which is equal to  $13579 \times 2468$ , is much larger than 13579.

So **(D)**  $13579 \div \frac{1}{2468}$  is by far the largest of the answer choices.

7. (AMC 8 1990) When three different numbers from the set  $\{-3, -2, -1, 4, 5\}$  are multiplied, the largest possible product is

(A) 10      (B) 20      (C) 30      (D) 40      (E) 60

**Answer:** C

**Solution**

For the product of 3 real numbers to be positive, either all 3 numbers are positive, or 1 number is positive and 2 numbers are negative.

Since there are only 2 positive numbers, the only way to get a positive product is to select 2 negative numbers and 1 positive number. The product is maximized by selecting the 2 negative numbers with the largest absolute values and the positive number with the largest absolute value. In this case the product would be  $(-3)(-2)(5) = \mathbf{(C) 30}$ .

8. (AMC 8 1990) A dress originally priced at 80 dollars was put on sale for 25% off. If 10% tax was added to the sale price, then the total selling price (in dollars) of the dress was

(A) 45 dollars      (B) 52 dollars      (C) 54 dollars      (D) 66 dollars      (E) 68 dollars

**Answer:** D

**Solution**

The sale price is  $0.75 \cdot \$80 = \$60$ , so the total price is  $1.1 \cdot \$60 = \mathbf{(D) \$66}$ .

9. (AMC 8 1990) The grading scale shown is used at Jones Junior High. The fifteen scores in Mr. Freeman's class were:

89, 72, 54, 97, 77, 92, 85, 74, 75,  
63, 84, 78, 71, 80, 90.

In Mr. Freeman's class, what percent of the students received a grade of C?

|    |          |
|----|----------|
| A: | 93 - 100 |
| B: | 85 - 92  |
| C: | 75 - 84  |
| D: | 70 - 74  |
| F: | 0 - 69   |

- (A) 20%    (B) 25%    (C) 30%    (D)  $33\frac{1}{3}\%$     (E) 40%

**Answer:** D

**Solution**

Of the 15 students in Mr. Freeman's class, 5 students (those with scores of 77, 75, 84, 78, and 80) received a grade of C. So  $\frac{5}{15} = \boxed{\text{(D)} 33\frac{1}{3}\%}$  of his students received a grade of C.

10. (*AMC 8 1990*) On this monthly calendar, the date behind one of the letters is added to the date behind C. If this sum equals the sum of the dates behind A and B, then the letter is

|  | Tues. | Wed. | Thurs. | Fri. | Sat. |
|--|-------|------|--------|------|------|
|  |       |      | C      | A    |      |
|  |       |      | Q      |      |      |
|  | S     | B    | P      | T    | R    |

- (A) P    (B) Q    (C) R    (D) S    (E) T

**Answer:** A

**Solution 1**

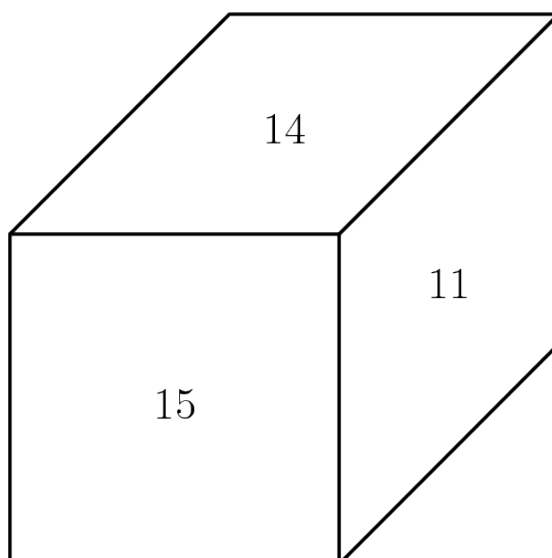
Since the date behind C is 1 less than the date behind A, the date added to the date behind C should be 1 more than the date behind B, which would be the date behind  $\boxed{\text{(A)} P}$ .

**Solution 2**

Let the date behind C be  $x$ , and  $y$  be the date added to C. Then the date behind A is  $x + 1$ , while the date behind B is  $x + 13$ . So  $(x + 1) + (x + 13) = x + y$ , and therefore  $y = x + 14$ , which is the date behind  $\boxed{\text{(A)} P}$ .

-motorfinn, edited by j314andrews

11. (*AMC 8 1990*) The numbers on the faces of this cube are consecutive whole numbers. The sums of the two numbers on each of the three pairs of opposite faces are equal. The sum of the six numbers on this cube is



- (A) 75      (B) 76      (C) 78      (D) 80      (E) 81

**Answer:** E

**Solution**

Since 11 and 15 are among the 6 consecutive whole numbers, the numbers must be either  $\{11, 12, 13, 14, 15, 16\}$  or  $\{10, 11, 12, 13, 14, 15\}$ .

If the numbers were  $\{10, 11, 12, 13, 14, 15\}$ , each pair of opposite faces would sum to  $\frac{10+11+12+13+14+15}{3} = 25$ . But 11 would be opposite 14, which is not the case on this cube.

Therefore, the numbers are  $\{11, 12, 13, 14, 15, 16\}$ , which works if 16, 13, and 12 are placed opposite 11, 14, and 15 respectively. In this case, the sum is  $11 + 12 + 13 + 14 + 15 + 16 = \boxed{\text{(E)} 81}$ .

12. (AMC 8 1990) There are twenty-four 4-digit numbers that use each of the four digits 2, 4, 5, and 7 exactly once. Listed in numerical order from smallest to largest, the number in the 17th position in the list is
- (A) 4527      (B) 5724      (C) 5742      (D) 7245      (E) 7524

**Answer:** B

**Solution**

Each digit appears as the thousands digit in  $3! = 6$  numbers. Therefore, the 13th through 18th numbers start with 5, and thus the 17th number on the list is the 5th number starting with 5. The numbers starting with 5 from smallest to largest, are 5247, 5274, 5427, 5472, 5724, 5742, so  $\boxed{\text{(B)} 5724}$  is 17th on the list.

13. (AMC 8 1990) One proposal for new postage rates for a letter was 30 cents for the first ounce and 22 cents for each additional ounce (or fraction of an ounce). The postage for a letter weighing 4.5 ounces was
- (A) 96 cents      (B) 1.07 dollars      (C) 1.18 dollars      (D) 1.20 dollars      (E) 1.40 dollars

**Answer:** C

**Solution**

There are  $4.5 - 1 = 3.5$  additional ounces, which would round up to paying for 4 additional ounces. Therefore, the total cost is  $\$0.30 + 4 \cdot \$0.22 = \boxed{\text{(B)} \$1.18}$ .

14. (AMC 8 1990) A bag contains only blue balls and green balls. There are 6 blue balls. If the probability of drawing a blue ball at random from this bag is  $\frac{1}{4}$ , then the number of green balls in the bag is

(A) 12      (B) 18      (C) 24      (D) 30      (E) 36

**Answer:** B

**Solution 1**

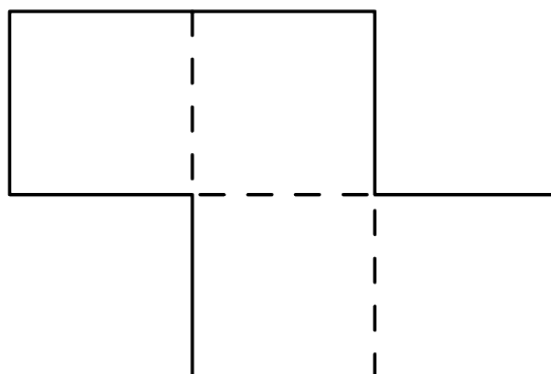
Since  $\frac{1}{4}$  of the balls are blue, there are  $4 \times 6 = 24$  total balls. Thus  $24 - 6 = \boxed{\text{(B)} 18}$  balls are green.

**Solution 2**

Let  $x$  be the number of green balls in the bag. Then the probability of selecting a blue ball is  $\frac{6}{6+x} = \frac{1}{4}$ . Solving this equation yields  $x = \boxed{\text{(B)} 18}$ .

- goldenn, edited by j314andrews

15. (AMC 8 1990) The area of this figure is  $100 \text{ cm}^2$ . Its perimeter is



[figure consists of four identical squares]

(A) 20 cm      (B) 25 cm      (C) 30 cm      (D) 40 cm      (E) 50 cm

**Answer:** E

**Solution**

Since the area of the whole figure is  $100 \text{ cm}^2$ , each square has an area of  $\frac{100}{4} = 25 \text{ cm}^2$  and a side length of  $\sqrt{25} = 5 \text{ cm}$ .

The boundary of this figure is made of 10 segments, each of length 5 cm, so its perimeter is  $10 \cdot 5 = \boxed{\text{(E)} 50 \text{ cm}}$ .

16. (AMC 8 1990)  $1990 - 1980 + 1970 - 1960 + \cdots - 20 + 10 =$  (A)  $-990$       (B)  $-10$       (C) 990      (D) 1000      (E) 1990

**Answer:** D

**Solution 1**

This sum has  $\frac{1990}{10} = 199$  terms. Grouping the sum as  $(1990 - 1980) + (1970 - 1960) + \cdots + (30 - 20) + 10$  creates  $\frac{198}{2} + 1 = 100$  groups which each sum to 10, so the total sum is  $100 \cdot 10 = \boxed{\text{(D)} 1000}$ .

**Solution 2**

As in Solution 1, this sum has 199 terms. Grouping the sum as  $1990 + (-1980 + 1970) + (-1960 + 1950) + \dots + (-20 + 10)$  creates  $\frac{198}{2} = 99$  groups which each sum to  $-10$ , so the total sum is  $1990 + (99)(-10) = \boxed{\text{(D)} 1000}$ .

**Solution 3**

Grouping the terms as  $(1990 + 10) - (1980 + 20) + \dots + (1030 + 970) - (1020 + 980) + (1010 + 990) - 1000$  shows that the sum is  $(2000 - 2000) + \dots + (2000 - 2000) + 2000 - 1000 = 0 + \dots + 0 + 1000 = \boxed{\text{(D)} 1000}$ .

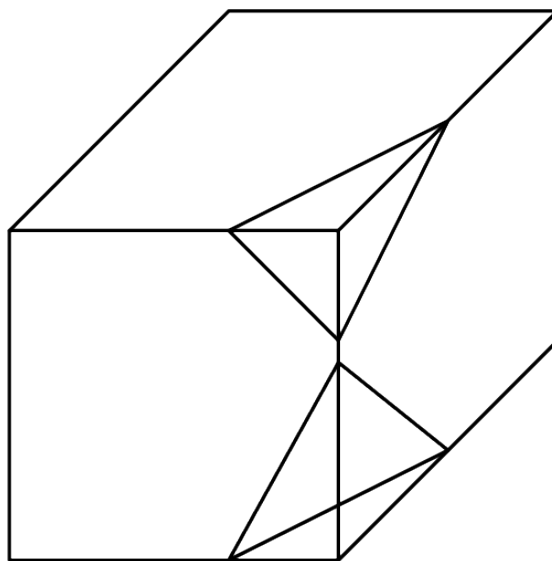
17. (AMC 8 1990) A straight concrete sidewalk is to be 3 feet wide, 60 feet long, and 3 inches thick. How many cubic yards of concrete must a contractor order for the sidewalk if concrete must be ordered in a whole number of cubic yards?
- (A) 2      (B) 5      (C) 12      (D) 20      (E) more than 20

**Answer:** A

**Solution**

The sidewalk is 1 yard wide, 20 yards long, and  $\frac{1}{12}$  yard thick, so its volume is  $1 \cdot 20 \cdot \frac{1}{12} = 1\frac{2}{3}$  cubic yards. So  $\boxed{\text{(A)} 2}$  cubic yards must be ordered.

18. (AMC 8 1990) Each corner of a rectangular prism is cut off. Two (of the eight) cuts are shown. How many edges does the new figure have?



- (A) 24      (B) 30      (C) 36      (D) 42      (E) 48

**Answer:** C

**Solution**

The new figure retains the middle part of each of the 12 edges of the cube and gains 3 new edges from each cut. Therefore, the new figure has  $12 + 3 \cdot 8 = \boxed{\text{(C)} 36}$  edges.

19. (AMC 8 1990) There are 120 seats in a row. What is the fewest number of seats that must be occupied so the next person to be seated must sit next to someone?

- (A) 30      (B) 40      (C) 41      (D) 60      (E) 119

**Answer:** B**Solution 1**

A qualifying arrangement with 40 occupied seats can be created by dividing the 120-seat row into 40 groups of 3 adjacent seats and placing a person in the middle seat of each group.

It is impossible to find a qualifying arrangement with 39 or fewer occupied seats. Suppose the row of seats is initially empty, and each person who sits down must choose an empty seat that is not next to an occupied seat. Then each person who sits down will eliminate at most 3 available seats from people who sit down later. Therefore, after 39 people sit down, there must be at least  $120 - 3 \cdot 39 = 3$  available seats, and thus a qualifying arrangement with 39 or fewer occupied seats does not exist.

Therefore, a qualifying arrangement has at least (B) 40 occupied seats.

**Solution 2**

Consider a row of chairs with  $n$  occupied seats such that each empty seat is next to an occupied seat. Then there are at most 2 empty seats between any of the  $n - 1$  pairs of consecutive occupied seats, at most 1 empty seat before the first occupied seat and at most 1 empty seat after the last occupied seat. Therefore, the row must have at most  $n + 2(n - 1) + 2 = 3n$  chairs in total.

Therefore, a row with 39 or fewer occupied seats may have at most  $39 \cdot 3 = 117$  seats, which is too few. A row with (B) 40 occupied seats and 120 total seats can be created by placing 1 empty seat on each end and 2 empty seats between each pair of consecutive occupied seats.

20. (AMC 8 1990) The annual incomes of 1,000 families range from 8200 dollars to 98,000 dollars. In error, the largest income was entered on the computer as 980,000 dollars. The difference between the mean of the incorrect data and the mean of the actual data is

- (A) 882 dollars      (B) 980 dollars      (C) 1078 dollars      (D) 482,000 dollars      (E) 882,000 dollars

**Answer:** A**Solution 1**

The sum of the incorrectly entered incomes is  $\$980000 - \$98000 = \$882000$  higher than the sum of the actual incomes. Therefore, the mean of the incorrect incomes is  $\frac{\$882000}{1000} = \text{\textbf{(A) \$882}}$  higher than the mean of the actual incomes.

**Solution 2**

Let  $S$  be the sum of the correctly entered incomes. The mean of the actual incomes is  $\frac{S+98000}{1000}$ , and for the incorrect data the mean is  $\frac{S+980000}{1000}$ . The difference between these means is  $\frac{S+980000}{1000} - \frac{S+98000}{1000} = \text{\textbf{(A) \$882}}$ .

21. (AMC 8 1990) A list of 8 numbers is formed by beginning with two given numbers. Each new number in the list is the product of the two previous numbers. Find the first number if the last three are shown:

\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, 16, 64, 1024



- (A)  $\frac{1}{64}$     (B)  $\frac{1}{4}$     (C) 1    (D) 2    (E) 4

**Answer:** B

**Solution**

Each missing term must be the quotient of the term two entries after it and the term immediately after it. Therefore, the fifth term is 4, the fourth term is 4, the third is 1, the second is 4, and the first is  $\frac{1}{4}$ .

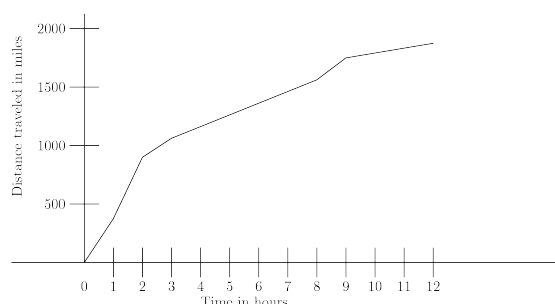
22. (AMC 8 1990) Several students are seated at a large circular table. They pass around a bag containing 100 pieces of candy. Each person receives the bag, takes one piece of candy and then passes the bag to the next person. If Chris takes the first and last piece of candy, then the number of students at the table could be
- (A) 10    (B) 11    (C) 19    (D) 20    (E) 25

**Answer:** B

**Solution**

Since the bag had 100 candies at first, it must have been passed  $100 - 1 = 99$  times. Since the bag started and ended at the same student, the number of students at the table must be a factor of 99. Of the answer choices, only **(B) 11** is a factor of 99.

23. (AMC 8 1990) The graph relates the distance traveled [in miles] to the time elapsed [in hours] on a trip taken by an experimental airplane. During which hour was the average speed of this airplane the largest?



- (A) first (0-1)    (B) second (1-2)    (C) third (2-3)    (D) ninth (8-9)    (E) last (11-12)

**Answer:** B

**Solution**

The time when the average speed is greatest is when the slope of the graph is steepest. This is in the second hour  $\rightarrow$  **(B)**.

24. (AMC 8 1990) Three  $\Delta$ 's and a  $\diamond$  will balance nine  $\bullet$ 's. One  $\Delta$  will balance a  $\diamond$  and a  $\bullet$ .



How many  $\bullet$ 's will balance the two  $\diamond$ 's in this balance?



- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** C

**Solution 1**

Consider the scale in which 3  $\triangle$ 's and 1  $\diamond$  balance 9  $\bullet$ 's.

After replacing each  $\triangle$  on this scale with 1  $\bullet$  and 1  $\diamond$ , the scale will remain balanced. This scale now has 4  $\diamond$ 's and 3  $\bullet$ 's balancing 9  $\bullet$ 's.

After removing 3  $\bullet$ 's from each side of the scale will remain balanced. This scale now has 4  $\diamond$ 's balancing 6  $\bullet$ 's.

Finally, if each side of the scale is divided by 2, the scale will remain balanced. Therefore, 2  $\diamond$ 's balances **(C) 3**  $\bullet$ 's.

**Solution 2**

For simplicity, suppose  $\triangle = a$ ,  $\diamond = b$  and  $\bullet = c$ . Then,

$$3a + b = 9c$$

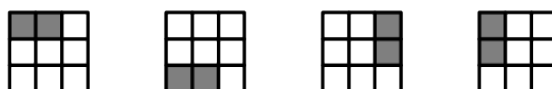
$$a = b + c$$

and we want to know what  $2b$  is in terms of  $c$ . Substituting the second equation into the first, we have

$$4b = 6c \Rightarrow 2b = 3c$$

Thus, we need 3  $\bullet$ 's  $\rightarrow$  **(C)**.

25. (AMC 8 1990) How many different patterns can be made by shading exactly two of the nine squares? Patterns that can be matched by flips and/or turns are not considered different. For example, the patterns shown below are **not** considered different.



- (A) 3      (B) 6      (C) 8      (D) 12      (E) 18

**Answer:** C

**Solution 1**

We break this into cases.

*Case 1: At least one square is a vertex:* **WLOG**, suppose one of them is in the upper-left corner. Then, consider the diagonal through that square. The two squares on that diagonal could be the second square, or the second square is on one side of the diagonal.

The square is reflectionally symmetric about this diagonal, so we only consider the squares on one side, giving another three possibilities.

In this case, there are  $2 + 3 = 5$  distinct squares.

*Case 2: At least one square is on an edge, but no square is on a vertex:* There are clearly two edge-edge combinations and one edge-center combination, so this case has 3 squares.

In total, there are  $3 + 5 = 8$  distinct squares  $\rightarrow$  **C**

### Solution 2 (Burnside's Lemma)

Consider each of the 8 symmetries of the square.

The identity transformation preserves all  $\binom{9}{2} = 36$  arrangements.

The  $90^\circ$  rotations do not preserve any arrangements.

The  $180^\circ$  rotation preserves 4 arrangements: those with either opposite corners or the centers of opposite edges shaded.

The horizontal and vertical reflections each preserve 6 arrangements: the 3 in which the shaded squares are mirror images of each other and the 3 in which both shaded squares are on the line of reflection.

The diagonal reflections each preserve 6 arrangements: the 3 in which the shaded squares are mirror images of each other and the 3 in which both shaded squares are on the line of reflection.

Therefore, by Burnside's lemma, there are  $\frac{1}{8}(36 + 4 + 2 \cdot 6 + 2 \cdot 6) = \boxed{\text{(C)} 8}$  different configurations if reflections and rotations are considered the same.

-j314andrews

## §7 AMC 8 1991

1. (*AMC 8 1991*)  $1,000,000,000,000 - 777,777,777,777 =$  (A) 222,222,222,222 (B) 222,222,222,223  
(D) 322,222,222,223 (E) 333,333,333,333

**Answer:** B

2. (*AMC 8 1991*)  $\frac{16+8}{4-2} =$  (A) 4 (B) 8 (C) 12 (D) 16 (E) 20

**Answer:** C

**Solution**

$$\frac{16+8}{4-2} = \frac{24}{2} = \boxed{\text{(C)} 12}.$$

3. (*AMC 8 1991*) Two hundred thousand times two hundred thousand equals  
(A) four hundred thousand (B) four million (C) forty thousand  
(D) four hundred million (E) forty billion

**Answer:** E

**Solution**

Writing the numbers in scientific notation, we have

$$\begin{aligned} (2 \times 10^5) \times (2 \times 10^5) &= 4 \times 10^{10} \\ &= 40 \times 10^9 \\ &= \text{forty billion} \rightarrow \boxed{\text{E}}. \end{aligned}$$

4. (AMC 8 1991) If  $991+993+995+997+999 = 5000 - N$ , then  $N =$  (A) 5 (B) 10 (C) 15 (D) 20

**Answer:** E

**Solution**

$$991 + 993 + 995 + 997 + 999 = 5000 - N$$

$$\Rightarrow (1000 - 9) + (1000 - 7) + (1000 - 5) + (1000 - 3) + (1000 - 1) = 5000 - N$$

$$\Rightarrow 5 \times 1000 - (1 + 3 + 5 + 7 + 9) = 5000 - N$$

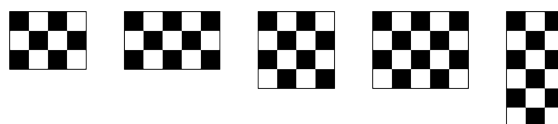
$$\Rightarrow 5000 - 25 = 5000 - N$$

$$\Rightarrow N = 25 \rightarrow \boxed{\text{E}}.$$

5. (AMC 8 1991) A "domino" is made up of two small squares:



Which of the "checkerboards" illustrated below CANNOT be covered exactly and completely by a whole number of non-overlapping dominoes?



- (A)  $3 \times 4$  (B)  $3 \times 5$  (C)  $4 \times 4$  (D)  $4 \times 5$  (E)  $6 \times 3$

**Answer:** B

**Solution**

For a bunch of dominoes to completely tile a board, the board must have an even number of squares. The  $3 \times 5$  board clearly does not, so  $\boxed{\text{B}}$  cannot be tiled completely.

6. (AMC 8 1991) Which number in the array below is both the largest in its column and the smallest in its row? (Columns go up and down, rows go right and left.)

|    |   |    |    |   |
|----|---|----|----|---|
| 10 | 6 | 4  | 3  | 2 |
| 11 | 7 | 14 | 10 | 8 |
| 8  | 3 | 4  | 5  | 9 |
| 13 | 4 | 15 | 12 | 1 |
| 8  | 2 | 5  | 9  | 3 |

- (A) 1 (B) 6 (C) 7 (D) 12 (E) 15

**Answer:** C

**Solution**

The largest numbers in the first, second, third, fourth and fifth columns are 13, 7, 15, 12, 9 respectively. Of these, only 7 is the smallest in its row  $\rightarrow$  **C**.

7. (AMC 8 1991) The value of  $\frac{(487,000)(12,027,300)+(9,621,001)(487,000)}{(19,367)(.05)}$  is closest to  
 (A) 10,000,000 (B) 100,000,000 (C) 1,000,000,000 (D) 10,000,000,000 (E) 100,000,000,000

**Answer:** D

**Solution**

We can make the approximations

$$\begin{aligned} 487,000 &\approx 500,000 \\ 12,027,300 &\approx 12,000,000 \\ 9,621,001 &\approx 10,000,000 \\ 19,367 &\approx 20,000. \end{aligned}$$

Using these instead of the original numbers for an estimate, we have

$$\begin{aligned} \frac{(500,000)(12,000,000) + (10,000,000)(500,000)}{(20000)(.05)} &= \frac{500,000 \times 22,000,000}{1000} \\ &= 500,000 \times 22,000 \\ &= 1.1 \times 10^{10} \\ &\approx 10,000,000,000 \rightarrow \mathbf{D}. \end{aligned}$$

"Google says the answer is 10,887,305,816.1" xxsc

8. (AMC 8 1991) What is the largest quotient that can be formed using two numbers chosen from the set  $\{-24, -3, -2, 1, 2, 8\}$ ?  
 (A)  $-24$  (B)  $-3$  (C)  $8$  (D)  $12$  (E)  $24$

**Answer:** D

**Solution**

Let the two chosen numbers be  $a$  and  $b$ . To maximize the quotient, we first have either  $a, b > 0$  or  $a, b < 0$ , and from there we maximize  $|a|$  and minimize  $|b|$ .

For the case  $a, b < 0$ , we have  $a = -24$  and  $b = -2$ , which gives us  $(-24)/(-2) = 12$ . For the case  $a, b > 0$ , we have  $a = 8$  and  $b = 1$ , which gives us  $8/1 = 8$ .

Since  $12 > 8$ , our answer is **D**.

9. (AMC 8 1991) How many whole numbers from 1 through 46 are divisible by either 3 or 5 or both?  
 (A) 18 (B) 21 (C) 24 (D) 25 (E) 27

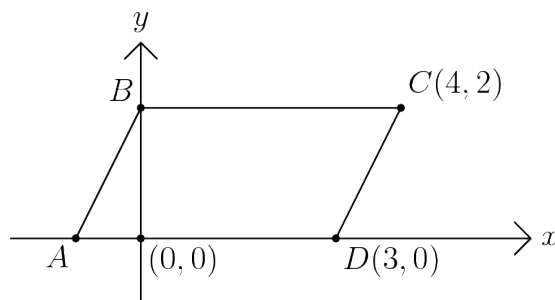
**Answer:** B

**Solution**

There are  $\left\lfloor \frac{46}{3} \right\rfloor = 15$  numbers divisible by 3,  $\left\lfloor \frac{46}{5} \right\rfloor = 9$  numbers divisible by 5, so at first we have  $15 + 9 = 24$  numbers that are divisible by 3 or 5, except we counted the multiples of  $\text{LCM}(3, 5) = 15$  twice, once for 3 and once for 5.

There are  $\left\lfloor \frac{46}{15} \right\rfloor = 3$  numbers divisible by 15, so there are  $24 - 3 = 21$  numbers divisible by 3 or 5.  $\rightarrow$  **B**

10. (AMC 8 1991) The area in square units of the region enclosed by parallelogram  $ABCD$  is



- (A) 6      (B) 8      (C) 12      (D) 15      (E) 18

**Answer:** B

**Solution**

The base is  $\overline{BC} = 4$ . The height has a length of the difference of the y-coordinates of A and B, which is 2. Therefore the area is  $4 \cdot 2 = 8 \Rightarrow \boxed{\text{B}}$ .

11. (AMC 8 1991) There are several sets of three different numbers whose sum is 15 which can be chosen from  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ . How many of these sets contain a 5?

- (A) 3      (B) 4      (C) 5      (D) 6      (E) 7

**Answer:** B

**Solution**

Let the three-element set be  $\{a, b, c\}$  and suppose that  $a = 5$ .

We need  $b + c = 10$  and  $b \neq c$ . This gives us four solutions, so there are 4 sets with a 5 also with the desired properties  $\rightarrow \boxed{\text{B}}$ .

12. (AMC 8 1991) If  $\frac{2+3+4}{3} = \frac{1990+1991+1992}{N}$ , then  $N =$  (A) 3      (B) 6      (C) 1990      (D) 1991      (E) 1

**Answer:** D

**Solution 1**

Note that for all integers  $n \neq 0$ ,

$$\frac{(n-1) + n + (n+1)}{n} = 3.$$

Thus, we must have  $N = 1991 \rightarrow \boxed{\text{D}}$ .

**Solution 2**

As we know that  $\frac{1990+1991+1992}{n}$  has to be some multiple of  $\frac{2+3+4}{3}$ , then we know that the first equation is  $995(1990/2)$  times bigger than the second one(in my solution), so the bottom must be  $3 \cdot 995 = \boxed{\text{(D)1991}}$ -fn106068

13. (AMC 8 1991) How many zeros are at the end of the product

$$25 \times 25 \times 25 \times 25 \times 25 \times 25 \times 25 \times 8 \times 8 \times 8?$$

- (A) 3      (B) 6      (C) 9      (D) 10      (E) 12

**Answer:** C

**Solution**

$$\begin{aligned}(5^2)^7 \times (2^3)^3 &= 5^{14} \times 2^9 \\ &= 5^5 \times 10^9\end{aligned}$$

Since  $5^5$  doesn't end in a 0, we can conclude that the product ends in 9 zeroes  
→ **C**.

14. (*AMC 8 1991*) Several students are competing in a series of three races. A student earns 5 points for winning a race, 3 points for finishing second and 1 point for finishing third. There are no ties. What is the smallest number of points that a student must earn in the three races to be guaranteed of earning more points than any other student?
- (A) 9      (B) 10      (C) 11      (D) 13      (E) 15

**Answer:** D

**Solution**

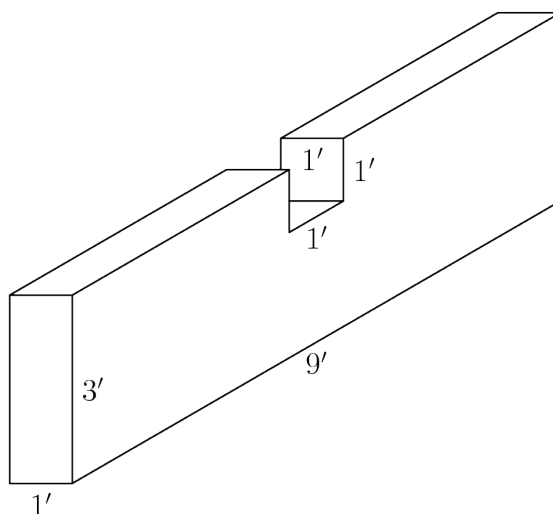
It is possible for two students to get 11 points: one student could win the first two races and finish third in the third race (scoring  $5 + 5 + 1 = 11$ ), while another student could finish second in the first two races and win the third race (scoring  $3 + 3 + 5 = 11$ ).

To score 13 or more points, a student must win at least 2 races. So any other student could win at most 1 race and thus score at most  $5 + 3 + 3 = 11$  points. Therefore, **(D) 13** is the minimum score a student needs to guarantee outscoring all other students.

Note: it is impossible for a student to score exactly 12 points.

-stjwyl, edited by j314andrews

15. (*AMC 8 1991*) All six sides of a rectangular solid were rectangles. A one-foot cube was cut out of the rectangular solid as shown. The total number of square feet in the surface of the new solid is how many more or less than that of the original solid?



- (A) 2 less      (B) 1 less      (C) the same      (D) 1 more      (E) 2 more

**Answer:** C

**Solution**

Initially, that one-foot cube contributed 3 faces to the surface area of the whole solid. When it was removed, these 3 faces are removed, but there are 3 new faces where the cube was carved out, so the net change is  $0 \rightarrow \boxed{\text{C}}$ .

16. (*AMC 8 1991*) The 16 squares on a piece of paper are numbered as shown in the diagram. While lying on a table, the paper is folded in half four times in the following sequence:

- (1) fold the top half over the bottom half
- (2) fold the bottom half over the top half
- (3) fold the right half over the left half
- (4) fold the left half over the right half.

Which numbered square is on top after step 4?

|    |    |    |    |
|----|----|----|----|
| 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

- (A) 1      (B) 9      (C) 10      (D) 14      (E) 16

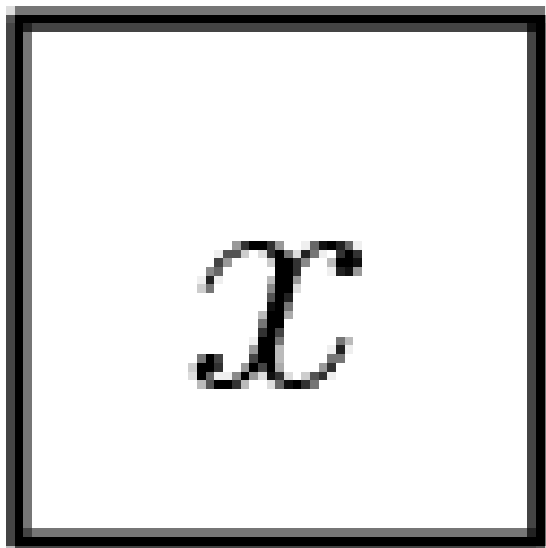
**Answer:** B

**Solution**

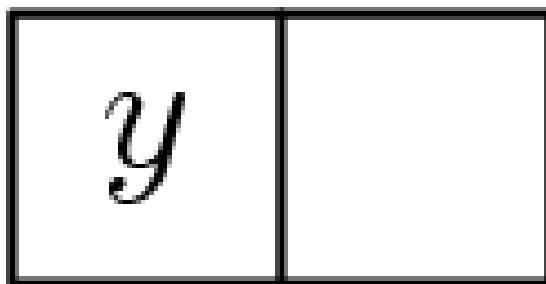
Suppose we undo each of the four folds, considering just the top square until we completely unfold the paper.  $x$  will be marked in the square if the face that shows after all the folds is face up,  $y$  if that face is facing down.

Step 0:





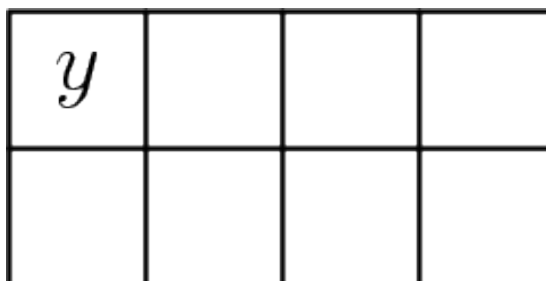
Step 1:



Step 2:



Step 3:



Step 4:

|     |  |  |  |
|-----|--|--|--|
|     |  |  |  |
|     |  |  |  |
| $y$ |  |  |  |
|     |  |  |  |

The marked square is in the same spot as the number 9  $\rightarrow$  B.

17. (AMC 8 1991) An auditorium with 20 rows of seats has 10 seats in the first row. Each successive row has one more seat than the previous row. If students taking an exam are permitted to sit in any row, but not next to another student in that row, then the maximum number of students that can be seated for an exam is
- (A) 150      (B) 180      (C) 200      (D) 400      (E) 460

**Answer:** C

**Solution**

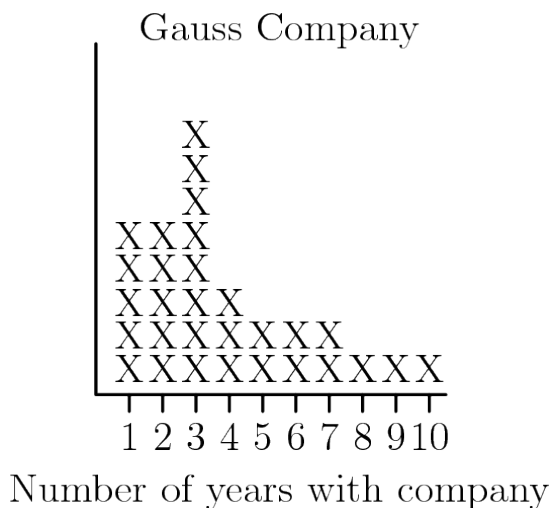
We first note that if a row has  $n$  seats, then the maximum number of students that can be seated in that row is  $\lceil \frac{n}{2} \rceil$ , where  $\lceil x \rceil$  is the smallest integer greater than or equal to  $x$ . If a row has  $2k$  seats, clearly we can only fit  $k$  students in that row. If a row has  $2k + 1$  seats, we can fit  $k + 1$  students by putting students at the ends and then alternating between skipping a seat and putting a student in.

For each row with  $10 + k$  seats, there is a corresponding row with  $29 - k$  seats. The sum of the maximum number of students for these rows is

$$\left\lceil \frac{10 + k}{2} \right\rceil + \left\lceil \frac{29 - k}{2} \right\rceil = 20.$$

There are  $20/2 = 10$  pairs of rows, so the maximum number of students for the exam is  $20 \times 10 = 200 \rightarrow$  C.

18. (AMC 8 1991) The vertical axis indicates the number of employees, but the scale was accidentally omitted from this graph. What percent of the employees at the Gauss company have worked there for 5 years or more?



- (A) 9%      (B)  $23\frac{1}{3}\%$       (C) 30%      (D)  $42\frac{6}{7}\%$       (E) 50%

**Answer:** C

**Solution**

Let  $x$  be the number of employees represented by each X. The graph has a total of 30 X's, while the columns corresponding to 5 or more years at the company have a combined 9 X's. So of the  $30x$  employees,  $9x$  have worked at the company for at least 5 years, which is  $\frac{9x}{30x} = \frac{3}{10} = \boxed{\text{(C) } 30\%}$ .

19. (AMC 8 1991) The average (arithmetic mean) of 10 different positive whole numbers is 10. The largest possible value of any of these numbers is
- (A) 10      (B) 50      (C) 55      (D) 90      (E) 91

**Answer:** C

**Solution**

If the average of the numbers is 10, then their sum is  $10 \times 10 = 100$ .

To maximize the largest number of the ten, we minimize the other nine. Since they must be distinct, positive whole numbers, we let them be 1, 2, 3, 4, 5, 6, 7, 8, 9. Their sum is 45.

The sum of nine of the numbers is 45, and the sum of all ten is 100 so the last number must be  $100 - 45 = 55 \rightarrow \boxed{\text{C}}$ .

20. (AMC 8 1991) In the addition problem, each digit has been replaced by a letter. If different letters represent different digits then  $C =$

$$\begin{array}{r}
 A \quad B \quad C \\
 \phantom{A} \quad A \quad B \\
 + \phantom{A} \phantom{B} \quad A \\
 \hline
 3 \quad 0 \quad 0
 \end{array}$$

- (A) 1      (B) 3      (C) 5      (D) 7      (E) 9

**Answer:** A

**Solution 1**

If  $A \geq 3$  this sum would be larger than 300, which is too large. If  $A \leq 1$  this sum would be at most  $199 + 19 + 1 = 219$ , which is too small. So  $A = 2$ .

If  $B \geq 8$  this sum would be at least  $280 + 28 + 2 = 310$ , which is too large. If  $B \leq 6$  this sum would be at most  $269 + 26 + 2 = 297$ , which is too small. So  $B = 7$ .

This forces  $C = \boxed{(A) 1}$ , for a sum of  $271 + 27 + 2 = 300$ .

**Solution 2**

From this we have

$$111A + 11B + C = 300.$$

Clearly,  $A < 3$ . Since  $B, C \leq 9$ ,

$$111A > 201 \Rightarrow A \geq 2.$$

Thus,  $A = 2$  and  $11B + C = 78$ . From here it becomes clear that  $B = 7$  and  $C = 1 \rightarrow \boxed{A}$ .

**Solution 3**

Using logic,  $a + b + c = 10$ , therefore  $b + a + 1$ (from the carry over) = 10. So  $b + a = 9A + 1 = 3$

Thus,  $A = 2$  and  $11B + C = 78$ . From here it becomes clear that  $B = 7$  and  $C = 1 \rightarrow \boxed{A}$ .

21. (AMC 8 1991) For every  $3^\circ$  rise in temperature, the volume of a certain gas expands by 4 cubic centimeters. If the volume of the gas is 24 cubic centimeters when the temperature is  $32^\circ$ , what was the volume of the gas in cubic centimeters when the temperature was  $20^\circ$ ?

- (A) 8      (B) 12      (C) 15      (D) 16      (E) 40

**Answer:** A

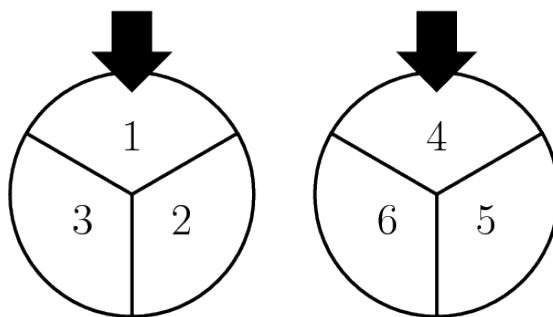
**Solution 1**

The temperature increased by  $12^\circ$ , so it increased by  $3^\circ$  four times. Since each  $3^\circ$  increase caused the volume of the gas to expand by 4 cubic centimeters, the gas expanded by a total of  $4 \cdot 4 = 16$  cubic centimeters. Therefore, the original volume of the gas was  $24 - 16 = \boxed{\text{(A)} 8}$  cubic centimeters.

### Solution 2

Let  $x$  be the temperature in degrees and  $y$  be the volume of the gas. Since a  $3^\circ$  increase in the temperature causes a 4 cubic centimeter expansion, the relationship between  $x$  and  $y$  is linear and has slope  $\frac{4}{3}$ . Using point-slope form with the point  $(32, 24)$ , the equation of this relationship is  $y = \frac{4}{3}(x - 32) + 24$ . Substituting  $x = 20$  gives  $y = \boxed{\text{(A)} 8}$ .

22. (AMC 8 1991) Each spinner is divided into 3 equal parts. The results obtained from spinning the two spinners are multiplied. What is the probability that this product is an even number?



- (A)  $\frac{1}{3}$     (B)  $\frac{1}{2}$     (C)  $\frac{2}{3}$     (D)  $\frac{7}{9}$     (E) 1

**Answer:** D

### Solution 1 (Complementary Counting)

The probability that the product is even can be found by subtracting the probability that the product is odd from 1. This approach is known as complementary counting.

The product of two numbers is odd if and only if both numbers are odd. The probability the left spinner lands on an odd number is  $\frac{2}{3}$ , while the probability the right spinner lands on an odd number is  $\frac{1}{3}$ . So, the probability that the product is

odd is  $\frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$ , while the probability that the product is even is  $1 - \frac{2}{9} = \boxed{\text{(D)} \frac{7}{9}}$ .

### Solution 2 (Inclusion-Exclusion)

The product of two numbers is even if and only if at least one number is even. The probability the left spinner lands on an even number is  $\frac{1}{3}$ . The probability that the right spinner lands on an even number is  $\frac{2}{3}$ . The probability that both spinners land on even numbers is  $\frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$ . By the principle of inclusion-exclusion, the

probability that the product is even is  $\frac{1}{3} + \frac{2}{3} - \frac{2}{9} = \boxed{\text{(D)} \frac{7}{9}}$ .

23. (AMC 8 1991) The Pythagoras High School band has 100 female and 80 male members. The Pythagoras High School orchestra has 80 female and 100 male members. There are 60 females who are members in both band and orchestra. Altogether, there are 230 students who are in either band or orchestra or both. The number of males who are in the band but NOT in the orchestra is

- (A) 10      (B) 20      (C) 30      (D) 50      (E) 70

**Answer:** A

**Solution 1**

There are 230 students in either band or orchestra, and  $80 + 100 = 180$  who are in orchestra. So,  $230 - 180 = 50$  students are in band but not orchestra.

Of the 100 female band members, 60 are in orchestra, so  $100 - 60 = 40$  are not in orchestra. Therefore, of the 50 students who are in band but not orchestra, 40 are female, so  $50 - 40 = \boxed{\text{(A)} 10}$  are male.

**Solution 2**

By the principle of inclusion-exclusion,  $100 + 80 - 60 = 120$  female students are in either band or orchestra. So  $230 - 120 = 110$  male students are in either band or orchestra. Of these 110 male students, 100 are in orchestra, so  $\boxed{\text{(A)} 10}$  are in band but not orchestra.

24. (AMC 8 1991) A cube of edge 3 cm is cut into  $N$  smaller cubes, not all the same size. If the edge of each of the smaller cubes is a whole number of centimeters, then  $N =$  (A) 4      (B) 8      (C) 12      (D) 16      (E) 20

**Answer:** E

**Solution**

The smaller cubes have side lengths less than 3 cm, so they must have side lengths of either 1 cm or 2 cm. Since not all smaller cubes are the same size, at least 1 of them has side length 2 cm. However, there cannot be more than 1 cube of side length 2 cm, since the smaller cubes cannot overlap.

Therefore, there is 1 cube with side length 2 cm. Since the original cube had volume of  $27 \text{ cm}^3$ , while the 2 cm cube has volume  $8 \text{ cm}^3$ , the 1 cm cubes have total volume of  $19 \text{ cm}^3$ . So there are 19 cubes with side length 1 cm, and therefore  $1 + 19 = \boxed{\text{(E)} 20}$  smaller cubes in total.

25. (AMC 8 1991) An equilateral triangle is originally painted black. Each time the triangle is changed, the middle fourth of each black triangle turns white. After five changes, what fractional part of the original area of the black triangle remains black?



- (A)  $\frac{1}{1024}$       (B)  $\frac{15}{64}$       (C)  $\frac{243}{1024}$       (D)  $\frac{1}{4}$       (E)  $\frac{81}{256}$

**Answer:** C

**Solution**

Each change reduces the black area to  $\frac{3}{4}$  of its former value. Therefore, after 5 changes,  $\left(\frac{3}{4}\right)^5 = \boxed{\text{(C)} \frac{243}{1024}}$  of the triangle's area remains black.

-edited by stjwyl, j314andrews

# §8 AMC 8 1992

1. (AMC 8 1992)  $\frac{10 - 9 + 8 - 7 + 6 - 5 + 4 - 3 + 2 - 1}{1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9} =$  (A)  $-1$  (B)  $1$  (C)  $5$  (D)  $9$  (E)  $1$

**Answer:** B

**Solution**

$$\begin{aligned}\frac{10 - 9 + 8 - 7 + 6 - 5 + 4 - 3 + 2 - 1}{1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9} &= \frac{(10 - 9) + (8 - 7) + (6 - 5) + (4 - 3) + (2 - 1)}{1 + (-2 + 3) + (-4 + 5) + (-6 + 7) + (-8 + 9)} \\ &= \frac{1 + 1 + 1 + 1 + 1}{1 + 1 + 1 + 1 + 1} \\ &= 1 \rightarrow \boxed{\text{B}}.\end{aligned}$$

2. (AMC 8 1992) Which of the following is not equal to  $\frac{5}{4}$ ?

(A)  $\frac{10}{8}$  (B)  $1\frac{1}{4}$  (C)  $1\frac{3}{12}$  (D)  $1\frac{1}{5}$  (E)  $1\frac{10}{40}$

**Answer:** D

**Solution 1**

The fraction in question is equal to 1.25. Expressing all of the choices we have,

(A) 1.25 (B) 1.25 (C) 1.25 (D) 1.2 (E) 1.25.

It's clear then that choice  $\boxed{\text{D}}$  is the answer.

**Solution 2 (low on time)**

A quick glance shows that  $\boxed{\text{D}}$  is the only answer with a denominator that does not have 4 as a factor, so it is unlikely that it will simplify into  $\frac{5}{4}$ .

3. (AMC 8 1992) What is the largest difference that can be formed by subtracting two numbers chosen from the set  $\{-16, -4, 0, 2, 4, 12\}$ ?

(A) 10 (B) 12 (C) 16 (D) 28 (E) 48

**Answer:** D

**Solution**

The largest difference can be found by subtracting the smallest number from the largest number. Therefore, the largest difference is  $12 - (-16) = \boxed{\text{(D)} 28}$ .

4. (AMC 8 1992) During the softball season, Judy had 35 hits. Among her hits were 1 home run, 1 triple and 5 doubles. The rest of her hits were singles. What percent of her hits were singles?

(A) 28% (B) 35% (C) 70% (D) 75% (E) 80%

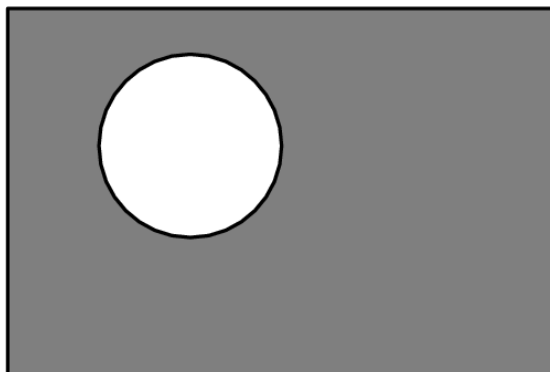
**Answer:** E

**Solution**

From the information given,  $35 - 5 - 1 - 1 = 28$  hits were single. Thus, the percentage was

$$\frac{28}{35} = 80\% \rightarrow \boxed{\text{E}}.$$

5. (*AMC 8 1992*) A circle of diameter 1 is removed from a  $2 \times 3$  rectangle, as shown. Which whole number is closest to the area of the shaded region?



- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** E

**Solution**

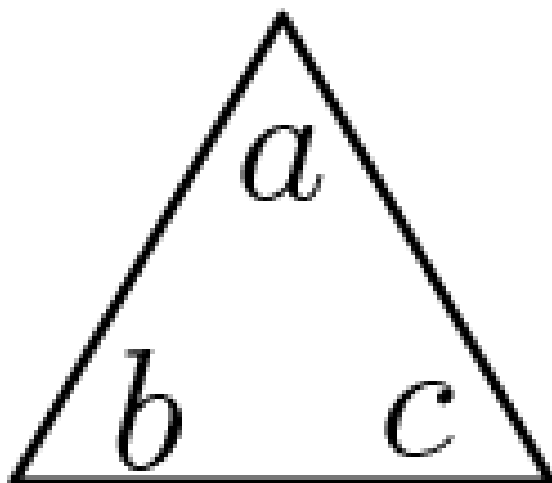
The area of the shaded region is the area of the circle subtracted from the area of the rectangle.

The diameter of the circle is 1, so the radius is  $1/2$  and the area is

$$(1/2)^2\pi = \pi/4.$$

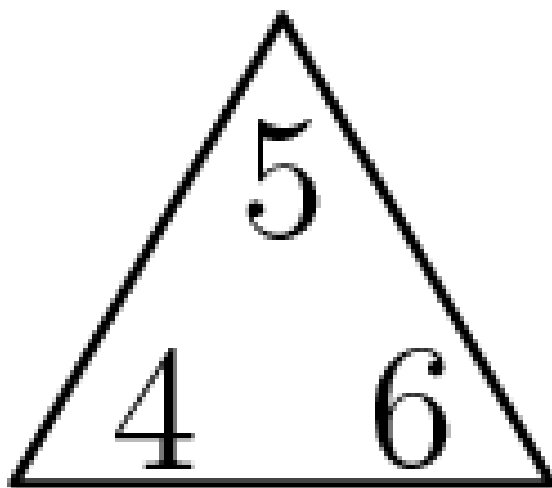
The rectangle obviously has area  $2 \times 3 = 6$ , so the area of the shaded region is  $6 - \pi/4$ . This is closest to 5  $\rightarrow$  **E**.

6. (*AMC 8 1992*) Suppose that

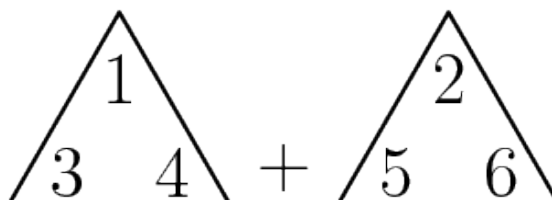


means  $a + b - c$ . For example,





is  $5 + 4 - 6 = 3$ . Then the sum



is

- (A)  $-2$     (B)  $-1$     (C)  $0$     (D)  $1$     (E)  $2$

**Answer:** D

**Solution**

The first triangle represents  $1 + 3 - 4 = 0$ , and the 2nd triangle represents  $2 + 5 - 6 = 1$ . Since we have to add the 2 triangles together, the final answer is 1, which is D.

7. (*AMC 8 1992*) The digit-sum of 998 is  $9 + 9 + 8 = 26$ . How many 3-digit whole numbers, whose digit-sum is 26, are even?  
 (A) 1    (B) 2    (C) 3    (D) 4    (E) 5

**Answer:** A

**Solution**

The highest digit sum for three-digit numbers is  $9 + 9 + 9 = 27$ . Therefore, the only possible digit combination is 9, 9, 8. Of course, of the three possible numbers, only 998 works. Thus, the answer is (A) 1.

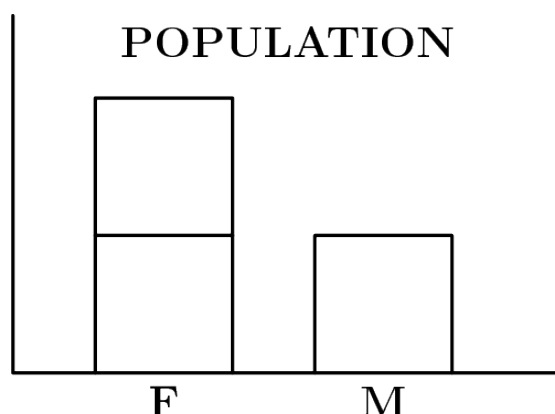
8. (*AMC 8 1992*) A store owner bought 1500 pencils at \$0.10 each. If he sells them for \$0.25 each, how many of them must he sell to make a profit of exactly \$100.00?  
 (A) 400    (B) 667    (C) 1000    (D) 1500    (E) 1900

**Answer:** C

**Solution**

$1500 \times 0.1 = 150$ , so the store owner is \$150 below profit. Therefore he needs to sell  $150 + 100 = 250$  dollars worth of pencils. Selling them at \$0.25 each gives  $250/0.25 =$  (C) 1000.

9. (*AMC 8 1992*) The population of a small town is 480. The graph indicates the number of females and males in the town, but the vertical scale-values are omitted. How many males live in the town?



- (A) 120    (B) 160    (C) 200    (D) 240    (E) 360

**Answer:** B

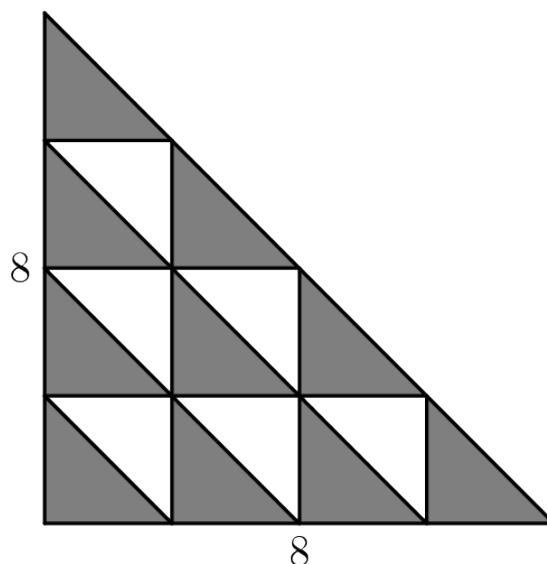
**Solution**

The graph shows that the ratio of men to total population is  $\frac{1}{3}$ , so the total number of men is  $\frac{1}{3} \times 480 = \boxed{\text{(B) } 160}$ .

**Solution 2**

The graph shows 3 equal squares, each with value  $x$ . So  $3x = 480$ , so  $x = \boxed{160}$ .

10. (*AMC 8 1992*) An isosceles right triangle with legs of length 8 is partitioned into 16 congruent triangles as shown. The shaded area is



- (A) 10    (B) 20    (C) 32    (D) 40    (E) 64

**Answer:** B

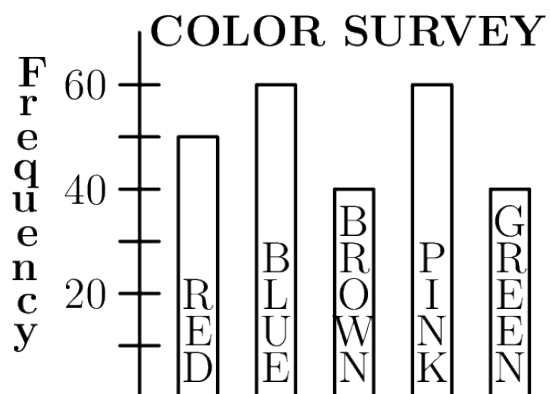
**Solution 1**

Because the smaller triangles are congruent, the shaded area take  $\frac{10}{16}$  of the largest triangles area, which is  $\frac{8 \times 8}{2} = 32$ , so the shaded area is  $\frac{10}{16} \times 32 = \boxed{\text{(B) } 20}$ .

**Solution 2**

Each of the triangle has side length of  $\frac{1}{4} \times 8 = 2$ , so the area is  $\frac{1}{2} \times 2 \times 2 = 2$ . Because there are 10 triangles is the shaded area, its area is  $2 \times 10 = \boxed{\text{(B) } 20}$ .

11. (AMC 8 1992) The bar graph shows the results of a survey on color preferences. What percent preferred blue?



- (A) 20%      (B) 24%      (C) 30%      (D) 36%      (E) 42%

**Answer: B**

**Solution**

The total frequency is  $50 + 60 + 40 + 60 + 40 = 250$ , with the blue frequency of 60. Therefore, the percentage that preferred blue is  $\frac{60}{250} = \boxed{\text{(B) } 24\%}$ .

12. (AMC 8 1992) The five tires of a car (four road tires and a full-sized spare) were rotated so that each tire was used the same number of miles during the first 30,000 miles the car traveled. For how many miles was each tire used?

- (A) 6000      (B) 7500      (C) 24,000      (D) 30,000      (E) 37,500

**Answer: C**

**Solution**

In the 30,000 miles, four tires were always used at one time, so the amount of miles the five tires were used in total is  $30,000 \times 4 = 120,000$ . Five tires were used and each was used equally, so each tire was used for  $\frac{120,000}{5} = \boxed{\text{(C) } 24,000}$ .

13. (AMC 8 1992) Five test scores have a mean (average score) of 90, a median (middle score) of 91 and a mode (most frequent score) of 94. The sum of the two lowest test scores is

- (A) 170      (B) 171      (C) 176      (D) 177      (E) not determined by the information given

**Answer: B**

**Solution**

Because there was an odd number of scores, 91 must be the middle score. Since there are two scores above 91 and 94 appears the most frequent (so at least twice) and  $94 > 91$ , 94 appears twice. Also, the sum of the five numbers is  $90 \times 5 = 450$ . Thus, the sum of the lowest two scores is  $450 - 91 - 94 - 94 = \boxed{\text{(B) } 171}$ .

14. (AMC 8 1992) When four gallons are added to a tank that is one-third full, the tank is then one-half full. The capacity of the tank in gallons is  
 (A) 8      (B) 12      (C) 20      (D) 24      (E) 48

**Answer:** D

**Solution**

Four gallons is  $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$  of the tank. Thus, capacity of the tank in gallons is  $6 \cdot 4 = \boxed{\text{(D) } 24}$ .

15. (AMC 8 1992) What is the 1992<sup>nd</sup> letter in this sequence?

ABCDEDCBAABCDEDCBAABCDEDCBAABCDEDC...

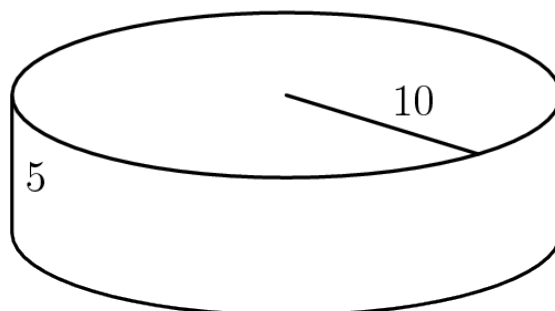
- (A) A      (B) B      (C) C      (D) D      (E) E

**Answer:** C

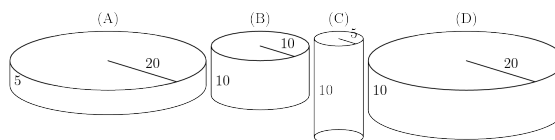
**Solution**

The sequence consists of repeated strings of 9 letters, ABCDEDCBA. When 1992 is divided by 9, the remainder is 3, and so the 1992<sup>nd</sup> letter in the sequence is the 3<sup>rd</sup> letter of the smaller string,  $\boxed{\text{(C) } C}$ .

16. (AMC 8 1992)



Which cylinder has twice the volume of the cylinder shown above?



- (E) None of the above

**Answer:** B

**Solution 1**

Recall that a cylinder with radius  $r$  and height  $h$  has volume  $V = \pi r^2 h$ .

The original cylinder thus has volume  $500\pi$ , while the cylinders in choices (A), (B), (C), and (D) have volumes  $2000\pi$ ,  $1000\pi$ ,  $250\pi$ , and  $4000\pi$  respectively. So only the cylinder in choice  $\boxed{\text{(B)}}$  has twice the volume of the original cylinder.

**Solution 2**

The volume of a cylinder is directly proportional to both its height and the square of its radius.

The cylinder in choice (A) has the same height as and twice the radius of the original, so its volume is  $2^2 = 4$  times that of the original.

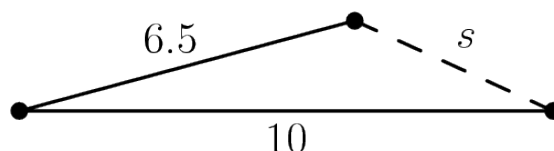
The cylinder in choice (B) has twice the height of and the same radius as the original, so its volume is 2 times that of the original.

The cylinder in choice (C) has twice the height of and half the radius of the original, so its volume is  $2 \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{2}$  of that of the original.

The cylinder in choice (D) has twice the height of and twice the radius of the original, so its volume is  $2 \cdot 2^2 = 8$  times that of the original.

Therefore, only the cylinder in choice **(B)** has twice the volume of the original cylinder.

17. (AMC 8 1992) The sides of a triangle have lengths 6.5, 10, and  $s$ , where  $s$  is a whole number. What is the smallest possible value of  $s$ ?



- (A) 3    (B) 4    (C) 5    (D) 6    (E) 7

**Answer:** B

**Solution**

By the triangle inequality,  $6.5 + s > 10$  and therefore  $s > 3.5$ . The smallest whole number  $s$  that satisfies this inequality is **(B) 4**.

18. (AMC 8 1992) On a trip, a car traveled 80 miles in an hour and a half, then was stopped in traffic for 30 minutes, then traveled 100 miles during the next 2 hours. What was the car's average speed in miles per hour for the 4-hour trip?
- (A) 45    (B) 50    (C) 60    (D) 75    (E) 90

**Answer:** A

**Solution**

The car traveled a total of  $80 + 100 = 180$  miles on the trip, so its average speed was  $180 \div 4 = \mathbf{(A) 45}$  miles per hour.

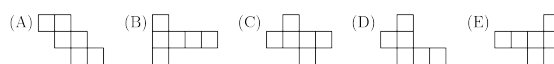
19. (AMC 8 1992) The distance between the 5<sup>th</sup> and 26<sup>th</sup> exits on an interstate highway is 118 miles. If any two exits are at least 5 miles apart, then what is the largest number of miles there can be between two consecutive exits that are between the 5<sup>th</sup> and 26<sup>th</sup> exits?
- (A) 8    (B) 13    (C) 18    (D) 47    (E) 98

**Answer:** C

**Solution**

There are 21 pairs of consecutive exits. To find the maximum number of miles of one of these, the other 20 must be equal to the minimum number yielding a total of  $(5)(20) = 100$  miles. The longest distance must be  $118 - 100 = \boxed{\text{(C) } 18}$ .

20. (AMC 8 1992) Which pattern of identical squares could NOT be folded along the lines shown to form a cube?

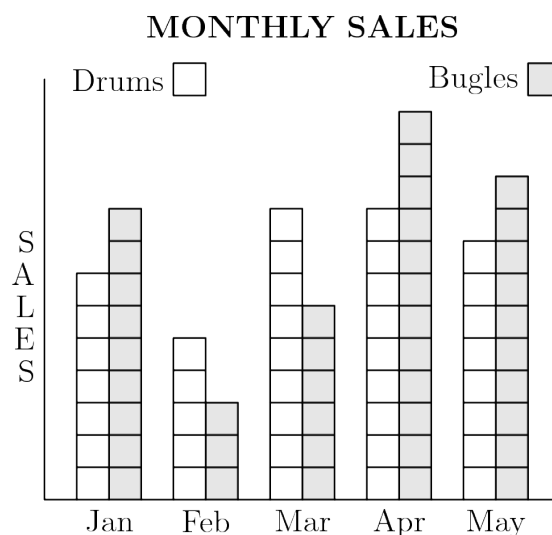


**Answer:** D

**Solution**

If pattern  $\boxed{\text{(D)}}$  were to be folded into a cube, the topmost square would overlap the rightmost square.

21. (AMC 8 1992) Northside's Drum and Bugle Corps raised money for a trip. The drummers and bugle players kept separate sales records. According to the double bar graph, in what month did one group's sales exceed the other's by the greatest percent?



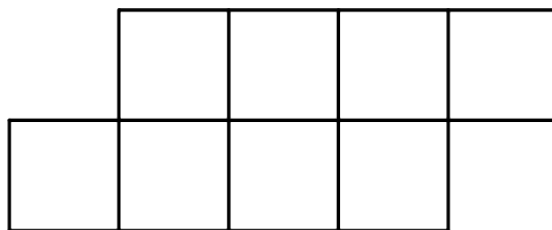
- (A) Jan      (B) Feb      (C) Mar      (D) Apr      (E) May

**Answer:** B

**Solution**

The ratio of the sales of the group with more sales and the sales of the group with less sales is largest in  $\boxed{\text{(B) Feb}}$  which is  $5/3 \approx 1.67$ .

22. (AMC 8 1992) Eight  $1 \times 1$  square tiles are arranged as shown so their outside edges form a polygon with a perimeter of 14 units. Two additional tiles of the same size are added to the figure so that at least one side of each tile is shared with a side of one of the squares in the original figure. Which of the following could be the perimeter of the new figure?

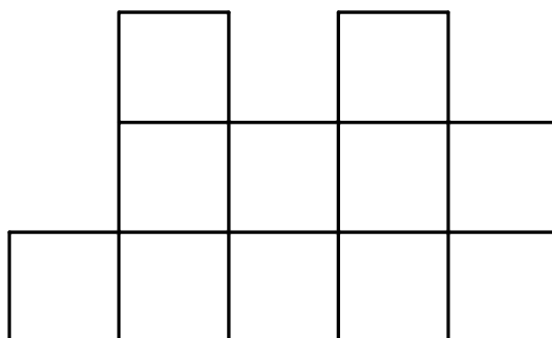


- (A) 15    (B) 17    (C) 18    (D) 19    (E) 20

**Answer:** C

**Solution 1**

One such figure would be



The perimeter of this figure is (C) 18.

**Solution 2**

If we attach a square to one side, one side will be covered, and 3 more sides will be added, resulting in a +2 perimeter. If we attach a square to two sides, 2 sides are covered and 2 sides are attached, so the perimeter don't change. There is no possibility of attaching a square to 3 sides, so we can get 14, 16, 18 as the perimeter so the only answer is (C) 18.

23. (AMC 8 1992) If two dice are tossed, the probability that the product of the numbers showing on the tops of the dice is greater than 10 is

- (A)  $\frac{3}{7}$     (B)  $\frac{17}{36}$     (C)  $\frac{1}{2}$     (D)  $\frac{5}{8}$     (E)  $\frac{11}{12}$

**Answer:** B

**Solution**

When the first dice is 1, there are no possibilities for the second dice.

When the first dice is 2, the second dice can be 6, giving one possibility.

When the first dice is 3, the second dice can be 4, 5, 6, giving three possibilities.

When the first dice is 4, the second dice can be 3, 4, 5, 6, giving four possibilities.

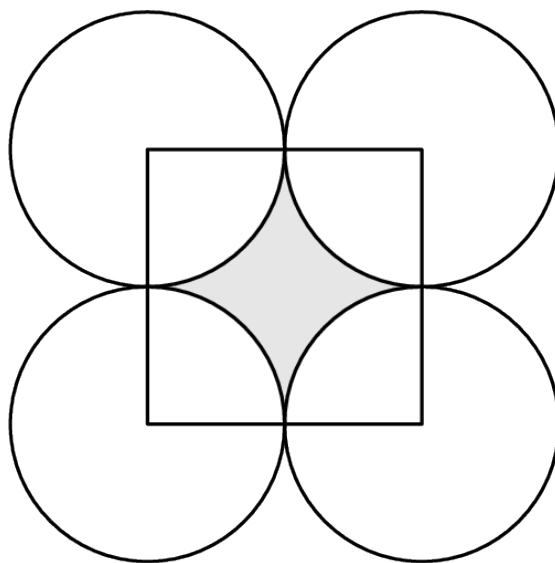
When the first dice is 5, the second dice can be 3, 4, 5, 6, giving four possibilities.

When the first dice is 6, the second dice can be 2, 3, 4, 5, 6, giving five possibilities.

The total number of ways for the product to greater than 10 is  $1 + 3 + 4 + 4 + 5 = 17$  and the total number of possibilities is  $(6)(6) = 36$ , yielding a probability of

- (B)  $\frac{17}{36}$ .

24. (AMC 8 1992) Four circles of radius 3 are arranged as shown. Their centers are the vertices of a square. The area of the shaded region is closest to



- (A) 7.7      (B) 12.1      (C) 17.2      (D) 18      (E) 27

**Answer:** A

**Solution**

The square has side length 6, so its area is 36.

The unshaded region inside the square consists of 4 quarter-circles of radius 3, so it has the same area as a circle of radius 3, or  $9\pi$ .

Therefore, the shaded area is  $36 - 9\pi \approx 36 - 9(3.14) = 7.74$ , which is closest to **(A) 7.7**.

25. (AMC 8 1992) One half of the water is poured out of a full container. Then one third of the remainder is poured out. Continue the process: one fourth of the remainder for the third pouring, one fifth of the remainder for the fourth pouring, etc. After how many pourings does exactly one tenth of the original water remain?
- (A) 6      (B) 7      (C) 8      (D) 9      (E) 10

**Answer:** D

**Solution**

1) Model the amount left in the container as follows:

After the first pour  $\frac{1}{2}$  remains, after the second  $\frac{1}{2} \times \frac{2}{3}$  remains, etc.

This becomes the product  $\frac{1}{2} \times \frac{1}{3} \times \frac{3}{4} \times \cdots \times \frac{9}{10}$ .

Note that the terms cancel out leaving  $\frac{1}{10}$ .

Now all that remains is to count the number of terms or pourings, as the numerators form an arithmetic sequence with a common difference of 1 and endpoints (1,9), the number of pourings is **(D) 9**.



# §9 AMC 8 1993

1. (*AMC 8 1993*) Which pair of numbers does NOT have a product equal to 36?

(A)  $\{-4, -9\}$       (B)  $\{-3, -12\}$       (C)  $\left\{\frac{1}{2}, -72\right\}$       (D)  $\{1, 36\}$       (E)  $\left\{\frac{3}{2}, 24\right\}$

**Answer:** C

## **Solution 1**

Let's calculate each of the answer choices and see which one DOES NOT equal 36.

$A$  comes out to be  $-4 \times -9 = 36$ ,

$B$  equals  $-3 \times -12 = 36$ ,

$C$  is  $\frac{1}{2} \times -72 = -36$ ,

$D$  simplifies to  $1 \times 36 = 36$ ,

and  $E$  equals  $\frac{3}{2} \times 24 = 3 \times 12 = 36$ .

Thus, our answer is  $\boxed{C}$ .

## **Solution 2**

In order for a product to be positive (36), the numbers should either be both positive or both negative. Looking at our answer choices, the only option that does not fit this description is  $\boxed{C}$ .

2. (*AMC 8 1993*) When the fraction  $\frac{49}{84}$  is expressed in simplest form, then the sum of the numerator and the denominator will be

(A) 11      (B) 17      (C) 19      (D) 33      (E) 133

**Answer:** C

## **Solution**

$$\begin{aligned}\frac{49}{84} &= \frac{7^2}{2^2 \cdot 3 \cdot 7} \\ &= \frac{7}{2^2 \cdot 3} \\ &= \frac{7}{12}.\end{aligned}$$

The sum of the numerator and denominator is  $7 + 12 = 19 \rightarrow \boxed{C}$ .

3. (*AMC 8 1993*) Which of the following numbers has the largest prime factor?

(A) 39      (B) 51      (C) 77      (D) 91      (E) 121

**Answer:** B

## **Solution**

A. The prime factors of 39 are 3 and 13. Therefore, the largest prime factor is 13.

B. The prime factors of 51 are 3 and 17. Therefore, the largest prime factor is 17.

C. The prime factors of 77 are 7 and 11. Therefore, the largest prime factor is 11.

D. The prime factors of 91 are 7 and 13. Therefore, the largest prime factor is 13.

E. The prime factors of 121 are 11. Therefore, the largest prime factor is 11.

So, the answer is  $B$

4. (*AMC 8 1993*)  $1000 \times 1993 \times 0.1993 \times 10 =$  (A)  $1.993 \times 10^3$  (B) 1993.1993 (C)  $(199.3)^2$  (D) 1,993

**Answer:** E

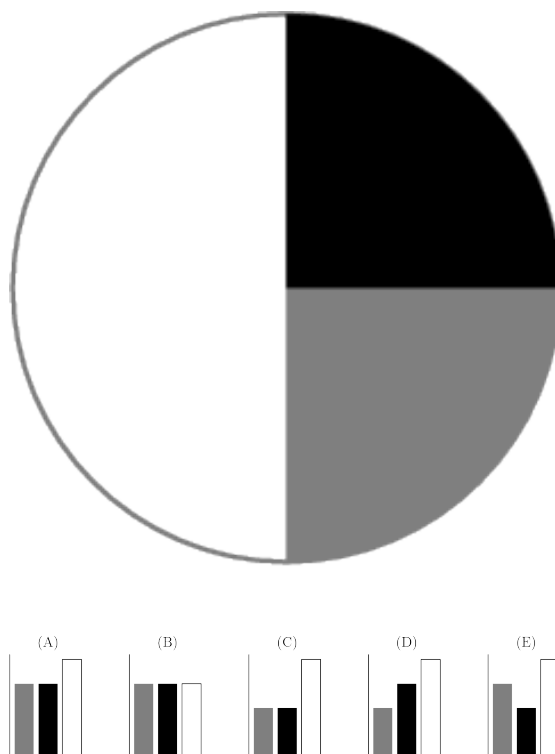
**Solution**

$$1000 \times 10 = 10^4$$

$$0.1993 = 1993 \times 10^{-4}$$

$$1993 \times 1993 \times 10^{-4} \times 10^4 = \boxed{\text{(E)} (1993)^2}$$

5. (*AMC 8 1993*) Which one of the following bar graphs could represent the data from the circle graph?



**Answer:** C

**Solution**

Since the white portion of the graph is twice as much as the black and gray, its bar in the bar graph will need to be twice as much as the black and gray. Note that the black and gray portions are equal. This cancels choices *A*, *B*, *D*, *E*. The final answer is now C.

6. (*AMC 8 1993*) A can of soup can feed 3 adults or 5 children. If there are 5 cans of soup and 15 children are fed, then how many adults would the remaining soup feed?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 25

**Answer:** B

**Solution**

A can of soup will feed 5 children so 15 children are feed by 3 cans of soup. Therefore, there are  $5 - 3 = 2$  cans for adults, so  $3 \times 2 = \boxed{\text{(B)} 6}$  adults are fed.

7. (AMC 8 1993)  $3^3 + 3^3 + 3^3 =$  (A)  $3^4$  (B)  $9^3$  (C)  $3^9$  (D)  $27^3$  (E)  $3^{27}$

**Answer:** A

**Solution**

$$3^3 + 3^3 + 3^3 = 3 \times 3^3 = \boxed{\text{(A)} 3^4}$$

8. (AMC 8 1993) To control her blood pressure, Jill's grandmother takes one half of a pill every other day. If one supply of medicine contains 60 pills, then the supply of medicine would last approximately

(A) 1 month (B) 4 months (C) 6 months (D) 8 months (E) 1 year

**Answer:** D

**Solution**

If Jill's grandmother takes one half of a pill every other day, she takes a pill every 4 days. Since she has 60 pills, the supply will last  $60 \times 4 = 240$  days which is about

$\boxed{\text{(D)} 8 \text{ months}}$ .

9. (AMC 8 1993) Consider the operation  $*$  defined by the following table:

| $*$ | 1 | 2 | 3 | 4 |
|-----|---|---|---|---|
| 1   | 1 | 2 | 3 | 4 |
| 2   | 2 | 4 | 1 | 3 |
| 3   | 3 | 1 | 4 | 2 |
| 4   | 4 | 3 | 2 | 1 |

For example,  $3*2 = 1$ . Then  $(2*4)*(1*3) =$  (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

**Answer:** D

**Solution 1**

Using the chart,  $(2*4) = 3$  and  $(1*3) = 3$ . Therefore,  $(2*4)*(1*3) = 3*3 = \boxed{\text{(D)} 4}$ .

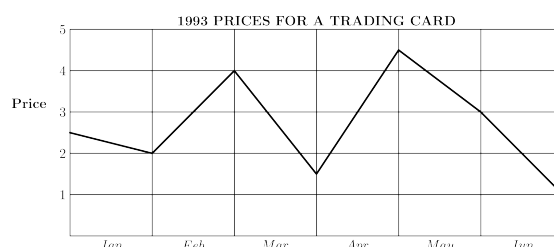
**Solution 2**

By the chart, we can see that the " $*$ " operation is actually multiplication modulo 5.

Thus, we can do  $(2*4)*(1*3) \rightarrow (2 \cdot 4) \cdot (1 \cdot 3) = 8 \cdot 3 = 24 \rightarrow \boxed{\text{(D)} 4}$

-JeffersonJ

10. (AMC 8 1993) This line graph represents the price of a trading card during the first 6 months of 1993.



The greatest monthly drop in price occurred during

(A) January (B) March (C) April (D) May (E) June

**Answer:** B

**Solution**

The graph shows the following price changes:

Jan:  $-0.50$

Feb:  $+2.00$

Mar:  $-2.50$

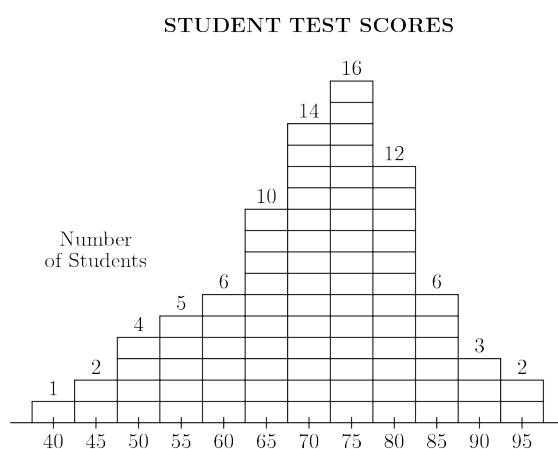
Apr:  $+3.00$

May:  $-0.50$

Jun:  $-2.00$

Therefore, (B) March has the largest price drop.

11. (AMC 8 1993) Consider this histogram of the scores for 81 students taking a test:



The median is in the interval labeled

- (A) 60    (B) 65    (C) 70    (D) 75    (E) 80

**Answer: C**

**Solution**

Since 81 students took the test, the median is the score of the 41<sup>st</sup> student. The five rightmost intervals include  $2 + 3 + 6 + 12 + 16 = 39$  students, so the 41<sup>st</sup> one must lie in the next interval, which is (C) 70.

12. (AMC 8 1993) If each of the three operation signs,  $+$ ,  $-$ ,  $\times$ , is used exactly ONCE in one of the blanks in the expression

$$5 \_ 4 \_ 6 \_ 3$$

then the value of the result could equal

- (A) 9    (B) 10    (C) 15    (D) 16    (E) 19

**Answer: E**

**Solution**

There are a reasonable number of ways to place the operation signs, so guess and check to find that  $5 - 4 + 6 \times 3 =$  (E) 19.

13. (*AMC 8 1993*) The word "HELP" in block letters is painted in black with strokes 1 unit wide on a 5 by 15 rectangular white sign with dimensions as shown. The area of the white portion of the sign, in square units, is



- (A) 30      (B) 32      (C) 34      (D) 36      (E) 38

**Answer:** D

**Solution**

Count the number of black squares in each letter. H has 11, E has 11, L has 7, and P has 10, giving the number of black squares to be  $11 + 11 + 7 + 10 = 39$ . The total number of squares is  $(15)(5) = 75$  and the number of white squares is  $75 - 39 = \boxed{\text{(D) } 36}$ .

14. (*AMC 8 1993*) The nine squares in the table shown are to be filled so that every row and every column contains each of the numbers 1, 2, 3. Then  $A + B =$

|   |   |   |
|---|---|---|
| 1 |   |   |
|   | 2 | A |
|   |   | B |

- (A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** C

**Solution**

The square connected both to 1 and 2 cannot be the same as either of them, so must be 3.

|   |   |   |
|---|---|---|
| 1 | 3 |   |
|   | 2 | A |
|   |   | B |

The last square in the top row cannot be either 1 or 3, so it must be 2.

|   |   |   |
|---|---|---|
| 1 | 3 | 2 |
|   | 2 | A |
|   |   | B |

The other two squares in the rightmost column with A and B cannot be two, so they must be 1 and 3 and therefore have a sum of  $1 + 3 = \boxed{\text{(C) } 4}$ .

15. (*AMC 8 1993*) The arithmetic mean (average) of four numbers is 85. If the largest of these numbers is 97, then the mean of the remaining three numbers is

- (A) 81.0      (B) 82.7      (C) 83.0      (D) 84.0      (E) 84.3

**Answer:** A

**Solution**

Say that the four numbers are  $a, b, c, 97$ . Then  $\frac{a+b+c+97}{4} = 85$ . What we are trying to find is  $\frac{a+b+c}{3}$ . Solving,

$$\frac{a+b+c+97}{4} = 85$$

$$a+b+c+97 = 340$$

$$a+b+c = 243$$

$$\frac{a+b+c}{3} = \boxed{\text{(A) } 81}$$

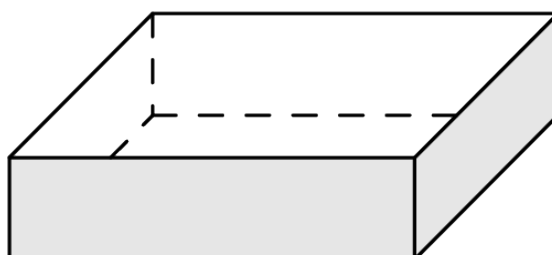
16. (AMC 8 1993)  $\frac{1}{1 + \frac{1}{2 + \frac{1}{3}}} =$  (A)  $\frac{1}{6}$  (B)  $\frac{3}{10}$  (C)  $\frac{7}{10}$  (D)  $\frac{5}{6}$  (E)  $\frac{10}{3}$

**Answer:** C

**Solution**

$$\frac{1}{1 + \frac{1}{2 + \frac{1}{3}}} = \frac{1}{1 + \frac{1}{\frac{7}{3}}} = \frac{1}{1 + \frac{3}{7}} = \frac{1}{\frac{10}{7}} = \boxed{\text{(C) } \frac{7}{10}}$$

17. (AMC 8 1993) Square corners, 5 units on a side, are removed from a 20 unit by 30 unit rectangular sheet of cardboard. The sides are then folded to form an open box. The surface area, in square units, of the interior of the box is



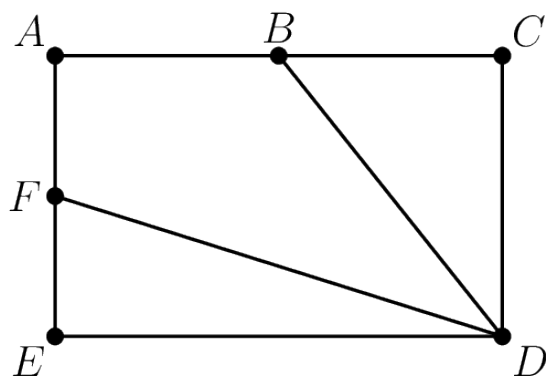
- (A) 300 (B) 500 (C) 550 (D) 600 (E) 1000

**Answer:** B

**Solution**

If the sides of the open box are folded down so that a flat sheet with four corners cut out remains, then the revealed surface would have the same area as the interior of the box. This is equal to the area of the four corners subtracted from the area of the original sheet, which is  $((20)(30) - 4(5)(5)) = 600 - 100 = \boxed{\text{(B) } 500}$ .

18. (AMC 8 1993) The rectangle shown has length  $AC = 32$ , width  $AE = 20$ , and  $B$  and  $F$  are midpoints of  $\overline{AC}$  and  $\overline{AE}$ , respectively. The area of quadrilateral  $ABDF$  is

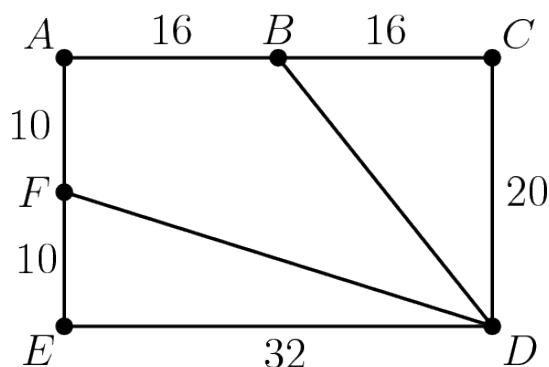


- (A) 320      (B) 325      (C) 330      (D) 335      (E) 340

**Answer:** A

**Solution**

The area of the quadrilateral  $ABDF$  is equal to the areas of the two right triangles  $\triangle BCD$  and  $\triangle EFD$  subtracted from the area of the rectangle  $ABCD$ . Because  $B$  and  $F$  are midpoints, we know the dimensions of the two right triangles.



$$(20)(32) - \frac{(16)(20)}{2} - \frac{(10)(32)}{2} = 640 - 160 - 160 = \boxed{\text{(A) } 320}$$

19. (*AMC 8 1993*)  $(1901 + 1902 + 1903 + \cdots + 1993) - (101 + 102 + 103 + \cdots + 193) =$   
 (A) 167,400      (B) 172,050      (C) 181,071      (D) 199,300      (E) 362,142

**Answer:** A

**Solution**

We see that  $1901 = 1800 + 101$ ,  $1902 = 1800 + 102$ , etc. Each term in the first set of numbers is 1800 more than the corresponding term in the second set; Because there are 93 terms in the first set, the expression can be paired up as follows and simplified:

$$(1901 - 101) + (1902 - 102) + (1903 - 103) + \cdots + (1993 - 193) = 1800 + 1800 + \cdots + 1800 = (1800)(93) = \boxed{\text{(A) } 167,400}$$

20. (*AMC 8 1993*) When  $10^{93} - 93$  is expressed as a single whole number, the sum of the digits is

- (A) 10      (B) 93      (C) 819      (D) 826      (E) 833

**Answer:** D

**Solution**

$$10^2 - 93 = 7$$

$$10^3 - 93 = 907$$

$$10^4 - 93 = 9907$$

This can be generalized into  $10^n - 93$  is equal to  $n - 2$  nines followed by the digits 07. Then  $10^{93} - 93$  is equal to 91 nines followed by 07. The sum of the digits is equal to  $9(91) + 7 = 819 + 7 = \boxed{\text{(D) 826}}$ .

21. (AMC 8 1993) If the length of a rectangle is increased by 20% and its width is increased by 50%, then the area is increased by
- (A) 10%      (B) 30%      (C) 70%      (D) 80%      (E) 100%

**Answer:** D

**Solution**

If a rectangle had dimensions  $10 \times 10$  and area 100, then its new dimensions would be  $12 \times 15$  and area 180. The area is increased by  $180 - 100 = 80$  or  $80/100 = \boxed{\text{(D) 80\%}}$ .

**Alternate Solution** Let the dimensions of the rectangle be  $x \times y$ . This rectangle has area  $xy$ . The new dimensions would be  $1.2x \times 1.5y$ , so the area is  $= 1.8xy$ , which is  $\boxed{80\%}$  more than the original area.

22. (AMC 8 1993) Pat Peano has plenty of 0's, 1's, 3's, 4's, 5's, 6's, 7's, 8's and 9's, but he has only twenty-two 2's. How far can he number the pages of his scrapbook with these digits?
- (A) 22      (B) 99      (C) 112      (D) 119      (E) 199

**Answer:** D

**Solution**

There is 1 two in the one-digit numbers.

The number of two-digit numbers with a two in the tens place is 10 and the number with a two in the ones place is 9. Thus the digit two is used  $10 + 9 = 19$  times for the two digit numbers.

Now, Pat Peano only has  $22 - 1 - 19 = 2$  remaining twos. You must subtract 1 because 22 is counted twice. The last numbers with a two that he can write are 102 and 112. He can continue numbering the last couple pages without a two until 120, with the last number he writes being  $\boxed{\text{(D) 119}}$ .

23. (AMC 8 1993) Five runners,  $P$ ,  $Q$ ,  $R$ ,  $S$ ,  $T$ , have a race, and  $P$  beats  $Q$ ,  $P$  beats  $R$ ,  $Q$  beats  $S$ , and  $T$  finishes after  $P$  and before  $Q$ . Who could NOT have finished third in the race?
- (A)  $P$  and  $Q$       (B)  $P$  and  $R$       (C)  $P$  and  $S$       (D)  $P$  and  $T$       (E)  $P, S$  and  $T$



**Answer:** C

**Solution**

First, note that  $P$  beats  $T$ , who beats  $Q$ , who beats  $S$ . Also,  $P$  beats  $R$ .

So  $P$  finishes first, and  $S$  finishes no earlier than fourth. Thus neither  $P$  nor  $S$  finishes third.

$Q$  finishes third if the runners finish in the order  $P, T, Q, S, R$ .

$R$  finishes third if the runners finish in the order  $P, T, R, Q, S$ .

$T$  finishes third if the runners finish in the order  $P, R, T, Q, S$ .

Thus (C)  $P$  and  $S$  are the only runners who certainly do not finish third.

24. (AMC 8 1993) What number is directly above 142 in this array of numbers?

|    |    |    |     |   |   |
|----|----|----|-----|---|---|
|    |    |    | 1   |   |   |
|    |    | 2  | 3   | 4 |   |
|    | 5  | 6  | 7   | 8 | 9 |
| 10 | 11 | 12 | ... |   |   |

- (A) 99      (B) 119      (C) 120      (D) 121      (E) 122

**Answer:** C

**Solution 1**

Notice that a number in row  $k$  is  $2k$  less than the number directly below it. For example, 5, which is in row 3, is  $(2)(3) = 6$  less than the number below it, 11.

From row 1 to row  $k$ , there are  $k \left( \frac{1+(-1+2k)}{2} \right) = k^2$  numbers in those  $k$  rows. Because there are  $12^2 = 144$  numbers up to the 12th row, 142 is in the  $k^{\text{th}}$  row. The number directly above is in the 11th row, and is 22 less than 142. Thus the number directly above 142 is  $142 - 22 = \boxed{\text{(C) } 120}$ .

**Solution 2**

Writing a couple more rows, the last number in each row ends in a perfect square. Thus 142 is two left from the last number in its row, 144. One left and one up from 144 is the last number of its row, also a perfect square, and is 121. This is one right and one up from 142, so the number directly above 142 is one less than 121, or (C) 120.

Remark: The last number in each row ends in a perfect square because there is an odd amount of numbers in each row and the sum of the first  $n$  odd numbers is always a perfect square.

**Solution 3 (eyeball it)**

We can notice that even numbers are above even numbers so we can narrow our answers down to 2 solutions, 120 and 122. Notice each number on the right end of each row is a perfect square. The perfect square closest to these is 121 so 122 is the first number on the next row. Notice 142 is the third to last number on its row so it is too far away from 122, so our answer is (C) 120.

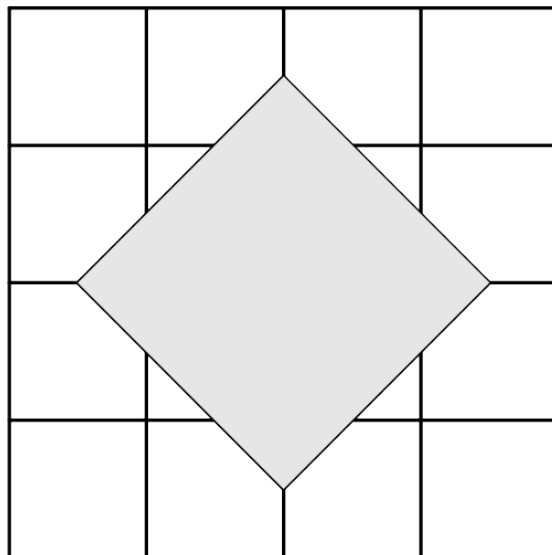
25. (AMC 8 1993) A checkerboard consists of one-inch squares. A square card, 1.5 inches on a side, is placed on the board so that it covers part or all of the area of each of  $n$  squares. The maximum possible value of  $n$  is

- (A) 4 or 5      (B) 6 or 7      (C) 8 or 9      (D) 10 or 11      (E) 12 or more

**Answer:** E

**Solution**

Using the **Pythagorean Theorem**, the diagonal of the square  $\sqrt{(1.5)^2 + (1.5)^2} = \sqrt{4.5} > 2$ . Because this is longer than 2 (length of the sides of two adjacent squares), the card can be placed like so, covering 12 squares.  $\rightarrow$  **(E) 12 or more**.



## §10 AMC 8 1994

1. (*AMC 8 1994*) Which of the following is the largest?

- (A)  $\frac{1}{3}$       (B)  $\frac{1}{4}$       (C)  $\frac{3}{8}$       (D)  $\frac{5}{12}$       (E)  $\frac{7}{24}$

**Answer:** D

**Solution 1**

The least common denominator is  $\text{lcm}(3, 4, 8, 12, 24) = 24$ . The answer choices (A) to (E) are thus  $\frac{8}{24}$ ,  $\frac{6}{24}$ ,  $\frac{9}{24}$ ,  $\frac{10}{24}$ , and  $\frac{7}{24}$ , so **(D)  $\frac{5}{12}$**  is the largest.

**Solution 2**

C and D are the only answer choices where the numerator is close to half of the denominator.  $\frac{3}{8} = .375$  and  $\frac{5}{12} \approx .41$ . Thus the answer is **(D)  $\frac{5}{12}$** .

2. (*AMC 8 1994*)  $\frac{1}{10} + \frac{2}{10} + \frac{3}{10} + \frac{4}{10} + \frac{5}{10} + \frac{6}{10} + \frac{7}{10} + \frac{8}{10} + \frac{9}{10} + \frac{55}{10} =$  (A)  $4\frac{1}{2}$       (B) 6.4      (C) 9      (D) 10

**Answer:** D

**Solution**

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = \frac{(9)(10)}{2} = 45 \quad \frac{45+55}{10} = \frac{100}{10} = \text{(D) } 10$$

3. (*AMC 8 1994*) Each day Maria must work 8 hours. This does not include the 45 minutes she takes for lunch. If she begins working at 7:25 A.M. and takes her lunch break at noon, then her working day will end at

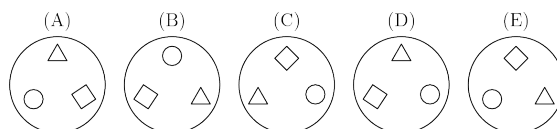
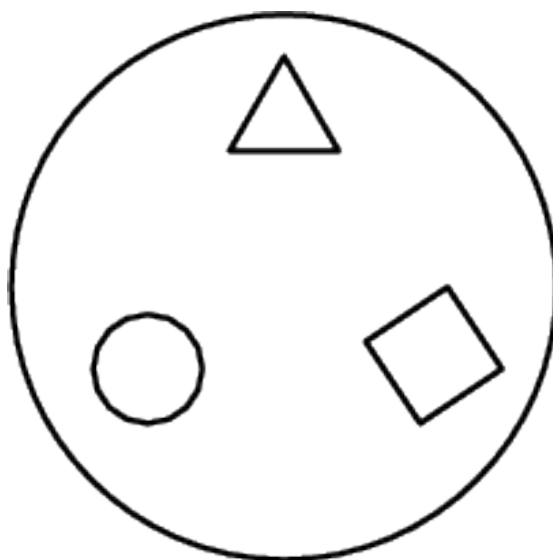
- (A) 3:40 P.M.    (B) 3:55 P.M.    (C) 4:10 P.M.    (D) 4:25 P.M.    (E) 4:40 P.M.

**Answer:** C

**Solution**

8 hours from 7:25 AM is 15:25 or 3:25 PM. 45 minutes from 25 minutes is 10 minutes after the hour, so her working day ends at **(C) 4:10 P.M.**

4. (*AMC 8 1994*) Which of the following represents the result when the figure shown below is rotated clockwise  $120^\circ$  around its center?



**Answer:** B

**Solution**

Rotating the circle  $120^\circ$  is  $1/3$  of a full turn. Going  $1/3$  clockwise of a full loop around the circle, the triangle would replace the square, the square would replace, the circle, and the circle would replace the triangle, **(B)**.

5. (*AMC 8 1994*) Given that 1 mile = 8 furlongs and 1 furlong = 40 rods, the number of rods in one mile is  
 (A) 5    (B) 320    (C) 660    (D) 1760    (E) 5280

**Answer:** B

**Solution**

$$(1 \text{ mile}) \left( \frac{8 \text{ furlongs}}{1 \text{ mile}} \right) \left( \frac{40 \text{ rods}}{1 \text{ furlong}} \right) = \boxed{\text{(B) } 320} \text{ rods}$$

6. (*AMC 8 1994*) The unit's digit (one's digit) of the product of any six consecutive positive whole numbers is

- (A) 0    (B) 2    (C) 4    (D) 6    (E) 8

**Answer:** A**Solution 1**

Any set of 6 consecutive positive integers contains at least one multiple of 5 and at least one multiple of 2. Thus the product of any 6 consecutive positive integers is divisible by  $5 \cdot 2 = 10$ , so its units digit is (A) 0.

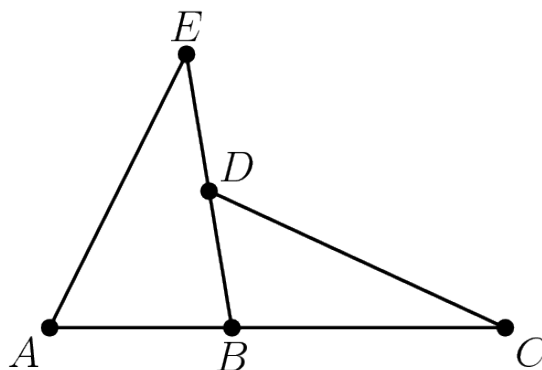
**Solution 2**

Since this question has a single correct answer choice, notice that  $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$ , which has units digit (A) 0.

**Solution 3**

Since this question has a single correct answer choice, notice that  $5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$  is divisible by 10, so its units digit is (A) 0.

7. (AMC 8 1994) If  $\angle A = 60^\circ$ ,  $\angle E = 40^\circ$  and  $\angle C = 30^\circ$ , then  $\angle BDC =$



- (A)
- $40^\circ$
- (B)
- $50^\circ$
- (C)
- $60^\circ$
- (D)
- $70^\circ$
- (E)
- $80^\circ$

**Answer:** B**Solution 1**

The sum of the angles in a triangle is  $180^\circ$ . We can find  $\angle ABE = 80^\circ$ , so  $\angle CBD = 180 - 80 = 100^\circ$ .

$$\angle BDC = 180 - 100 - 30 = \boxed{\text{(B)} 50^\circ}$$

**Solution 2**

The sum of angles of a quadrilateral is  $360^\circ$ . Reflex  $\angle EDC = 360 - 60 - 40 - 30 = 230^\circ$ .  $\angle BDC$  is a straight angle less than Reflex  $\angle EDC$ .

$$\angle BDC = 230 - 180 = \boxed{\text{(B)} 50^\circ}$$

8. (AMC 8 1994) For how many three-digit whole numbers does the sum of the digits equal 25?

- (A) 2    (B) 4    (C) 6    (D) 8    (E) 10

**Answer:** C**Solution**

Because  $8 + 8 + 8 = 24$ , it follows that one of the digits must be a 9. The other two digits then have a sum of  $25 - 9 = 16$ . The groups of digits that produce a sum of 25 are 799, 889 and can be arranged as follows

$$799, 979, 997, 889, 898, 988$$

The number of configurations is  $\boxed{\text{(C) } 6}$ .

9. (*AMC 8 1994*) A shopper buys a 100 dollar coat on sale for 20% off. An additional 5 dollars are taken off the sale price by using a discount coupon. A sales tax of 8% is paid on the final selling price. The total amount the shopper pays for the coat is  
 (A) 81.00 dollars      (B) 81.40 dollars      (C) 82.00 dollars      (D) 82.08 dollars      (E) 82.40 dollars

**Answer:** A

**Solution**

After the 20% sale, the coat costs  $100(0.8) = 80$  dollars. Then 5 dollars are taken off for a cost of  $80 - 5 = 75$ . Adding on the sales tax, the final amount is  $(75)(1.08) = \boxed{\text{(A) } 81.00 \text{ dollars}}$ .

10. (*AMC 8 1994*) For how many positive integer values of  $N$  is the expression  $\frac{36}{N+2}$  an integer?  
 (A) 7      (B) 8      (C) 9      (D) 10      (E) 12

**Answer:** A

**Solution**

We should list all the positive divisors of 36 and count them. By trial and error, the divisors of 36 are found to be 1, 2, 3, 4, 6, 9, 12, 18, 36, for a total of 9. However, 1 and 2 can't be equal to  $N + 2$  for a POSITIVE integer  $N$ , so the number of possibilities is  $\boxed{\text{(A) } 7}$ .

**Solution 2**

To find the number of positive divisors of 36, first prime factorize 36 to get  $2^2 * 3^2$ . Then add 1 to the power of both 2 and 3 to get 3. Multiply  $3 * 3$  to get 9. Since the problem is asking only for positive integer values of  $N$ , subtract 2 from 9 (since  $2 - 2$  and  $1 - 2$  result in integers that are not positive) to get  $\boxed{\text{(A) } 7}$ .

11. (*AMC 8 1994*) Last summer 100 students attended basketball camp. Of those attending, 52 were boys and 48 were girls. Also, 40 students were from Jonas Middle School and 60 were from Clay Middle School. Twenty of the girls were from Jonas Middle School. How many of the boys were from Clay Middle School?  
 (A) 20      (B) 32      (C) 40      (D) 48      (E) 52

**Answer:** B

**Solution**

Make a table with the information given.

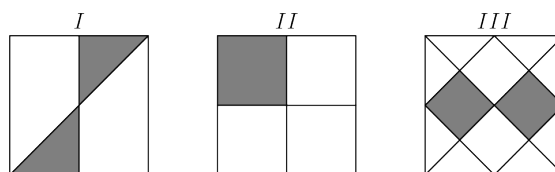
|       | Jonas | Clay | total |
|-------|-------|------|-------|
| boys  |       |      | 52    |
| girls | 20    |      | 48    |
| total | 40    | 60   | 100   |

Because the first two columns must add up to the third column, and the same with the rows, the rest of the empty boxes can be filled in.

|       | Jonas | Clay | total |
|-------|-------|------|-------|
| boys  | 20    | 32   | 52    |
| girls | 20    | 28   | 48    |
| total | 40    | 60   | 100   |

The number of boys from Clay is (B) 32.

12. (*AMC 8 1994*) Each of the three large squares shown below is the same size. Segments that intersect the sides of the squares intersect at the midpoints of the sides. How do the shaded areas of these squares compare?



- (A) The shaded areas in all three are equal. (B) Only the shaded areas of *I* and *II* are equal.  
 (C) Only the shaded areas of *I* and *III* are equal. (D) Only the shaded areas of *II* and *III* are equal.  
 (E) The shaded areas of *I*, *II* and *III* are all different.

**Answer:** A

**Solution**

Square II clearly has  $\frac{1}{4}$  shaded. Partitioning square I into eight right triangles also shows  $\frac{1}{4}$  of it is shaded. Lastly, square III can be partitioned into sixteen triangles, and because four are shaded,  $\frac{1}{4}$  of the total square is shaded. (A).

13. (*AMC 8 1994*) The number halfway between  $\frac{1}{6}$  and  $\frac{1}{4}$  is  
 (A)  $\frac{1}{10}$     (B)  $\frac{1}{5}$     (C)  $\frac{5}{24}$     (D)  $\frac{7}{24}$     (E)  $\frac{5}{12}$

**Answer:** C

**Solution**

The number halfway between is the average.

$$\frac{\frac{1}{6} + \frac{1}{4}}{2} = \frac{\frac{2}{12} + \frac{3}{12}}{2} = \text{(C) } \frac{5}{24}$$

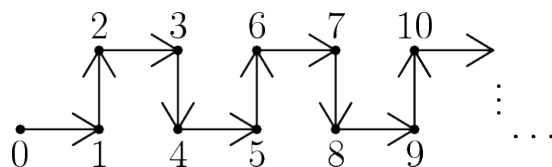
14. (*AMC 8 1994*) Two children at a time can play pairball. For 90 minutes, with only two children playing at a time, five children take turns so that each one plays the same amount of time. The number of minutes each child plays is  
 (A) 9    (B) 10    (C) 18    (D) 20    (E) 36

**Answer:** E

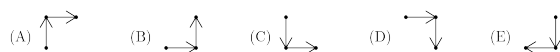
**Solution**

There are  $2 \times 90 = 180$  minutes of total playing time. Divided equally among the five children, each child gets  $180/5 = \text{(E) } 36$  minutes.

15. (AMC 8 1994) If this path is to continue in the same pattern:



then which sequence of arrows goes from point 425 to point 427?



**Answer:** A

**Solution**

Notice the pattern from 0 to 4 repeats for every four arrows. Any number that has a remainder of 0 when divided by 4 corresponds to 0.

The remainder when 425 is divided by 4 is 1. The arrows from point 425 to point 427 correspond to points 1 and 3, which have the same pattern as (A).

16. (AMC 8 1994) The perimeter of one square is 3 times the perimeter of another square. The area of the larger square is how many times the area of the smaller square?

(A) 2      (B) 3      (C) 4      (D) 6      (E) 9

**Answer:** E

**Solution**

Let  $a$  be the sidelength of one square, and  $b$  be the sidelength of the other, where  $a > b$ . If the perimeter of one is 3 times the other's, then  $a = 3b$ . The area of the larger square over the area of the smaller square is

$$\frac{a^2}{b^2} = \frac{(3b)^2}{b^2} = \frac{9b^2}{b^2} = \boxed{\text{(E) } 9}$$

17. (AMC 8 1994) Pauline Bunyan can shovel snow at the rate of 20 cubic yards for the first hour, 19 cubic yards for the second, 18 for the third, etc., always shoveling one cubic yard less per hour than the previous hour. If her driveway is 4 yards wide, 10 yards long, and covered with snow 3 yards deep, then the number of hours it will take her to shovel it clean is closest to

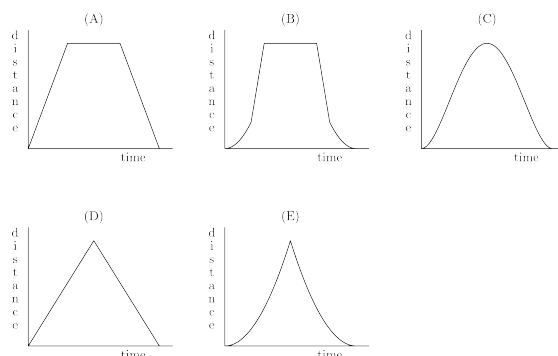
(A) 4      (B) 5      (C) 6      (D) 7      (E) 12

**Answer:** D

**Solution**

Her driveway has  $(4)(10)(3) = 120$  cubic yards of snow. After the first hour she would have  $120 - 20 = 100$  cubic yards, then  $100 - 19 = 81$ ,  $81 - 18 = 63$ ,  $63 - 17 = 46$ ,  $46 - 16 = 30$ ,  $30 - 15 = 15$ , and  $15 - 14 = 1$  cubic yard after the seventh hour. It will take her a little more than seven hours to shovel it clean, which is closest to (D) 7.

18. (*AMC 8 1994*) Mike leaves home and drives slowly east through city traffic. When he reaches the highway he drives east more rapidly until he reaches the shopping mall where he stops. He shops at the mall for an hour. Mike returns home by the same route as he came, driving west rapidly along the highway and then slowly through city traffic. Each graph shows the distance from home on the vertical axis versus the time elapsed since leaving home on the horizontal axis. Which graph is the best representation of Mike's trip?

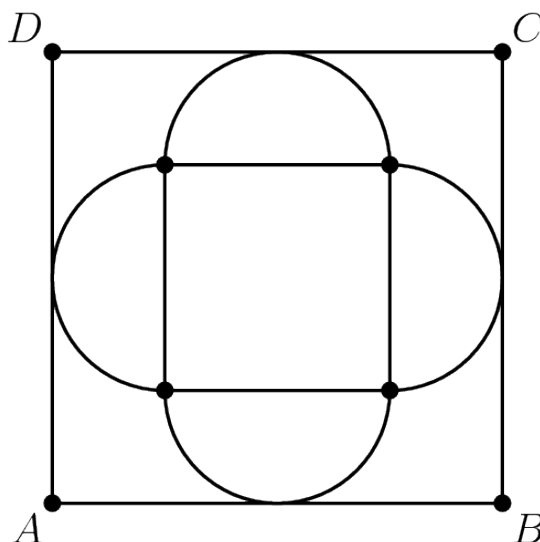


**Answer:** B

**Solution**

When Mike is shopping at the mall, his distance doesn't change, so there should be a flat plateau shape on the graph. This rules out *C, D, E*. The portion of graph *A* in which the distance is changing is linear, inconsistent with how he changes speed from the city and the highway. The best representation of his travels is graph **(B)**.

19. (*AMC 8 1994*) Around the outside of a 4 by 4 square, construct four semicircles (as shown in the figure) with the four sides of the square as their diameters. Another square,  $ABCD$ , has its sides parallel to the corresponding sides of the original square, and each side of  $ABCD$  is tangent to one of the semicircles. The area of the square  $ABCD$  is



- (A) 16      (B) 32      (C) 36      (D) 48      (E) 64

**Answer:** E



**Solution**

The radius of each semicircle is 2, half the sidelength of the square. The line straight down the middle of square  $ABCD$  is the sum of two radii and the length of the smaller square, which is equivalent to its side length. The area of  $ABCD$  is  $(4 + 2 + 2)^2 = \boxed{\text{(E) } 64}$ .

20. (*AMC 8 1994*) Let  $W, X, Y$  and  $Z$  be four different digits selected from the set  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ .

If the sum  $\frac{W}{X} + \frac{Y}{Z}$  is to be as small as possible, then  $\frac{W}{X} + \frac{Y}{Z}$  must equal

- (A)  $\frac{2}{17}$       (B)  $\frac{3}{17}$       (C)  $\frac{17}{72}$       (D)  $\frac{25}{72}$       (E)  $\frac{13}{36}$

**Answer:** D

**Solution**

$$\frac{W}{X} + \frac{Y}{Z} = \frac{WZ + XY}{XZ}$$

Small fractions have small numerators and large denominators. To maximize the denominator, let  $X = 8$  and  $Z = 9$ .

$$\frac{9W + 8Y}{72}$$

To minimize the numerator, let  $W = 1$  and  $Y = 2$ .

$$\frac{9 + 16}{72} = \boxed{\text{(D)} \rightarrow \frac{25}{72}}$$

**Solution 2**

To make the smallest fraction, you need the lowest numerator and the highest denominator. So, take the first 2 and last 2 digits of the set, which are 1, 2 and 8, 9. Balance the equations to be "even". Since 1 is smaller than 2, put it over 8. You get  $\frac{1}{8} + \frac{2}{9} = \frac{25}{72}$ , or  $\boxed{\text{D}}$ .

-goldenn

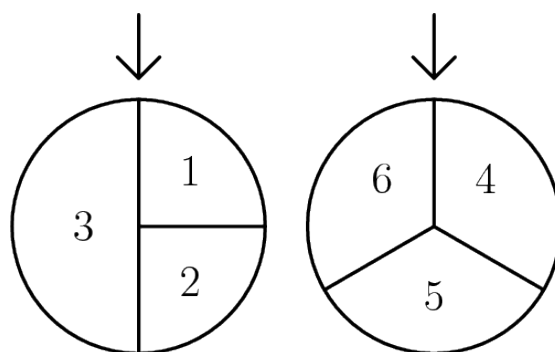
21. (*AMC 8 1994*) A gumball machine contains 9 red, 7 white, and 8 blue gumballs. The least number of gumballs a person must buy to be sure of getting four gumballs of the same color is
- (A) 8      (B) 9      (C) 10      (D) 12      (E) 18

**Answer:** C

**Solution**

If a person gets three gumballs of each of the three colors, that is, 9 gumballs, then the 10<sup>th</sup> gumball must be the fourth one for one of the colors. Therefore, the person must buy  $\boxed{\text{(C) } 10}$  gumballs.

22. (*AMC 8 1994*) The two wheels shown below are spun and the two resulting numbers are added. The probability that the sum is even is



- (A)  $\frac{1}{6}$     (B)  $\frac{1}{4}$     (C)  $\frac{1}{3}$     (D)  $\frac{5}{12}$     (E)  $\frac{4}{9}$

**Answer:** D

**Solution**

An even sum occurs when an even is added to an even or an odd is added to an odd. Looking at the areas of the regions, the chance of getting an even in the first wheel is  $\frac{1}{4}$  and the chance of getting an odd is  $\frac{3}{4}$ . On the second wheel, the chance of getting an even is  $\frac{2}{3}$  and an odd is  $\frac{1}{3}$ , resulting in an answer of D.

$$\frac{1}{4} \cdot \frac{2}{3} + \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{6} + \frac{1}{4} = \boxed{\text{(D)} \frac{5}{12}}$$

23. (AMC 8 1994) If  $X$ ,  $Y$  and  $Z$  are different digits, then the largest possible 3-digit sum for

$$\begin{array}{r} X \quad X \quad X \\ \quad Y \quad X \\ + \quad \quad X \\ \hline \end{array}$$

has the form

- (A)  $XXY$     (B)  $XYZ$     (C)  $YYX$     (D)  $YYZ$     (E)  $ZZY$

**Answer:** D

**Solution**

The sum can be rewritten as  $113X + 10Y$ . To get the largest possible sum, we maximize the hundreds digit,  $X$ . If  $X = 9$ , the sum is a 4-digit number, so we let  $X = 8$  and  $113(8) + 10Y = 904 + 10Y$ . To continue maximizing this sum, we can let  $Y = 9$ , a different digit from  $X$ , and  $904 + 10(9) = 994$ , which has the form

$$\boxed{\text{(D)} YYZ}.$$

24. (AMC 8 1994) A 2 by 2 square is divided into four 1 by 1 squares. Each of the small squares is to be painted either green or red. In how many different ways can the painting be accomplished so that no green square shares its top or right side with any red square? There may be as few as zero or as many as four small green squares.

- (A) 4    (B) 6    (C) 7    (D) 8    (E) 16

**Answer:** B

**Solution 1**

If a green square cannot share its top or right side with a red square, then a red square can not share its bottom or left side with a green square. Let us split this up into several cases.

Case 1: There are no green squares. This can be done in 1 way.

Case 2: There is one green square and three red squares. This can only be done when the green square's top and right edges are against the edge, so there is 1 way.

Case 3: There are two green squares and two red squares. This happens when the two green squares are in the two top squares or two right squares, so there are 2 ways.

Case 4: There are three green squares and one red square. Similar to case 2, this happens when the red square's left and bottom edges are against the edge, so there is 1 way.

Case 5: There are four green squares and zero red squares. 1 way.

$$1 + 1 + 2 + 1 + 1 = \boxed{\text{(B)} 6}$$

### Solution 2 (Generalization)

Suppose the grid has  $m$  rows and  $n$  columns.

Since a green square's top edge may not border a red square, each column, from top to bottom, must consist of  $k$  green squares followed by  $m - k$  red squares, where  $k$  is an integer such that  $0 \leq k \leq m$ . Also, since a green square's right edge may not border a red square, each column except the leftmost must contain at least as many green squares as the column directly to its left. Any coloring satisfying these two constraints is valid.

Notice that each valid coloring can be found by tracing a path from the top left corner to the bottom right corner using edges of the grid, coloring all squares above and to the right of the path green, and coloring all the squares to the right of the path red.

There are  $\binom{m+n}{n}$  paths through the grid, and thus there are also  $\binom{m+n}{n}$  valid to color it. So, there are  $\binom{4}{2} = \boxed{\text{(B)} 6}$  valid ways to color a  $2 \times 2$  grid.

-j314andrews

25. (AMC 8 1994) Find the sum of the digits in the answer to

$$\underbrace{9999 \dots 99}_{94 \text{ nines}} \times \underbrace{4444 \dots 44}_{94 \text{ fours}}$$

where a string of 94 nines is multiplied by a string of 94 fours.

- (A) 846      (B) 855      (C) 945      (D) 954      (E) 1072

**Answer:** A

### Solution 1

Notice that:

$$9 \cdot 4 = 36 \text{ and } 3 + 6 = 9 = 9 \cdot 1 \quad 99 \cdot 44 = 4356 \text{ and } 4 + 5 + 3 + 6 = 18 = 9 \cdot 2$$

So the sum of the digits of  $x$  9s times  $x$  4s is simply  $x \cdot 9$  (Try to find the proof that it works for all values of  $x$  MATHWIZARD10).

Therefore the answer is  $94 \cdot 9 = \boxed{\text{(A)} 846}$ .

## Solution 2

$$\underbrace{9999 \cdots 99}_{94 \text{ nines}} \cdot \underbrace{4444 \cdots 44}_{94 \text{ fours}} = (10^{94} - 1) \cdot \underbrace{4444 \cdots 44}_{94 \text{ fours}} = \underbrace{444 \cdots 4}_{94 \text{ fours}} \underbrace{000 \cdots 0}_{94 \text{ zeros}} - \underbrace{4444 \cdots 44}_{94 \text{ fours}} = \underbrace{444 \cdots 4}_{93 \text{ fours}} \underbrace{3}_{93 \text{ fours}} \underbrace{555 \cdots 5}_{93 \text{ fives}} \underbrace{6}_{93 \text{ fives}}$$

$$4 \cdot 93 + 3 + 5 \cdot 93 + 6 = 9 \cdot 94 = \boxed{\text{(A)} 846}$$

## §11 AMC 8 1995

1. (*AMC 8 1995*) Walter has exactly one penny, one nickel, one dime and one quarter in his pocket. What percent of one dollar is in his pocket?

(A) 4%      (B) 25%      (C) 40%      (D) 41%      (E) 59%

**Answer:** D

**Solution**

Walter has  $1 + 5 + 10 + 25 = 41$  cents in his pocket. There are 100 cents in a dollar. Therefore, Walter has  $\frac{41}{100} \cdot 100\% = 41\%$  of a dollar in his pocket, and the right answer is  $\boxed{D}$ .

2. (*AMC 8 1995*) Jose is 4 years younger than Zack. Zack is 3 years older than Inez. Inez is 15 years old. How old is Jose?

(A) 8      (B) 11      (C) 14      (D) 16      (E) 22

**Answer:** C

**Solution**

Zack is 3 years older than Inez, who is 15. Therefore, Zack is  $15 + 3 = 18$  years old.

Jose is 4 years younger than Zack, who is 18. Therefore, Jose is  $18 - 4 = 14$  years old, and the answer is  $\boxed{C}$ .

3. (*AMC 8 1995*) Which of the following operations has the same effect on a number as multiplying by  $\frac{3}{4}$  and then dividing by  $\frac{3}{5}$ ?

(A) dividing by  $\frac{4}{3}$       (B) dividing by  $\frac{9}{20}$       (C) multiplying by  $\frac{9}{20}$       (D) dividing by  $\frac{5}{4}$       (E) multiplying by  $\frac{5}{4}$

**Answer:** E

**Solution**

Dividing by  $\frac{3}{5}$  is the same as multiplying by its reciprocal,  $\frac{5}{3}$ .

Multiplying a number by two numbers is the same as multiplying the number by the product of those two numbers. Thus, multiplying by  $\frac{3}{4}$  and then  $\frac{5}{3}$  is the same as multiplying by  $\frac{3 \cdot 5}{4 \cdot 3} = \frac{5}{4}$ , and the answer is  $\boxed{E}$ .

4. (*AMC 8 1995*) A teacher tells the class,

*"Think of a number, add 1 to it, and double the result. Give the answer to your partner. Partner, subtract 1 from the number you are given and double the result to get your answer."*

Ben thinks of 6, and gives his answer to Sue. What should Sue's answer be?

(A) 18      (B) 24      (C) 26      (D) 27      (E) 30

**Answer:** C

**Solution**

Ben adds 1 to 6, getting  $1 + 6 = 7$ . Then Ben doubles it, to get  $7 \cdot 2 = 14$ . He gives this number to Sue. Then Sue subtracts 1 from the number, to get  $14 - 1 = 13$ . Finally, Sue doubles the result to get  $13 \cdot 2 = 26$ , and the answer is  $\boxed{C}$ .

5. (AMC 8 1995) Find the smallest whole number that is larger than the sum

$$2\frac{1}{2} + 3\frac{1}{3} + 4\frac{1}{4} + 5\frac{1}{5}.$$

- (A) 14      (B) 15      (C) 16      (D) 17      (E) 18

**Answer:** C

**Solution 1**

Adding the whole numbers gives  $2 + 3 + 4 + 5 = 14$ .

Adding the fractions gives  $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{30}{60} + \frac{20}{60} + \frac{15}{60} + \frac{12}{60} = \frac{77}{60}$ . This will create one more whole, and a fraction that is less than 1. Thus, the smallest whole number that is less than 15 plus some fractional part is 16, and the answer is  $\boxed{C}$ .

**Solution 2**

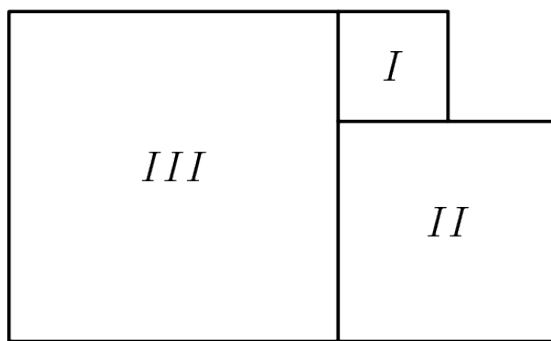
Convert the fractional parts to decimals, and approximate the answer.  $2.5 + 3.33 + 4.25 + 5.2 = 15.28$ , and the answer is 16, which is  $\boxed{C}$ .

**Solution 3 (Speedy)**

Clearly, break the mixed numbers into  $2 + 3 + 4 + 5 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$ .

This is the same as  $14 + (\frac{1}{2} + \frac{1}{3}) + (\frac{1}{4} + \frac{1}{5})$ . Notice that clearly,  $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ , and  $1 > \frac{1}{4} + \frac{1}{5} > \frac{1}{6}$  (clearly,  $\frac{1}{4} > \frac{1}{6}$ ). This means that the fractional parts is between 1 and 2, so when that is added to 14, the original expression is between 15, and 16. So the least integer that is bigger than this sum is clearly 16. So select  $\boxed{C}$ .

6. (AMC 8 1995) Figures *I*, *II*, and *III* are squares. The perimeter of *I* is 12 and the perimeter of *II* is 24. The perimeter of *III* is



- (A) 9      (B) 18      (C) 36      (D) 72      (E) 81

**Answer:** C

**Solution 1**

Since the perimeter of *I* is 12, each side is  $\frac{12}{4} = 3$ .

Since the perimeter of *II* is 24, each side is  $\frac{24}{4} = 6$ .

The side of  $III$  is equal to the sum of the sides of  $I$  and  $II$ . Therefore, the side of  $III$  is  $3 + 6 = 9$ .

Since  $III$  is also a square, it has an perimeter of  $9 \cdot 4 = 36$ , and the answer is  $\boxed{C}$ .

**Solution 2**

Let a side of  $I$  equal  $x$ , and let a side of  $II$  equal  $y$ . The perimeter of  $I$  is  $4x$ , and the perimeter of  $II$  is  $4y$ . One side of  $III$  has length  $x + y$ , so the perimeter is  $4x + 4y$ , which just so happens to be the sum of the perimeters of  $I$  and  $II$ , giving us  $12 + 24 = 36$ , or answer  $\boxed{C}$ .

7. (AMC 8 1995) At Clover View Junior High, one half of the students go home on the school bus. One fourth go home by automobile. One tenth go home on their bicycles. The rest walk home. What fractional part of the students walk home?

(A)  $\frac{1}{16}$       (B)  $\frac{3}{20}$       (C)  $\frac{1}{3}$       (D)  $\frac{17}{20}$       (E)  $\frac{9}{10}$

**Answer:** B

**Solution**

The students who walk home are

$1 - \frac{1}{2} - \frac{1}{4} - \frac{1}{10} = \frac{1}{2} - \frac{1}{4} - \frac{1}{10} = \frac{1}{4} - \frac{1}{10} = \frac{5}{20} - \frac{2}{20} = \frac{3}{20}$  of the entire student population. Therefore, the answer is  $\boxed{B}$ .

8. (AMC 8 1995) An American traveling in Italy wishes to exchange American money (dollars) for Italian money (lire). If  $3000 \text{ lire} = 1.60$ , *how much lire will the traveler receive in exchange for*

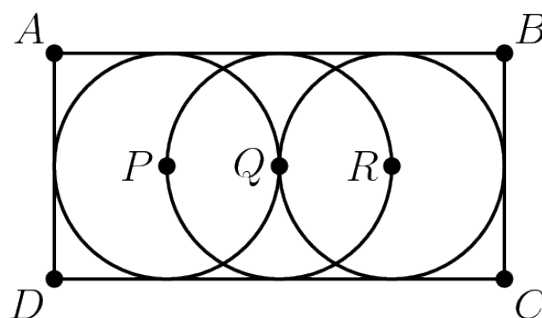
(A) 180      (B) 480      (C) 1800      (D) 1875      (E) 4875

**Answer:** D

**Solution**

$\frac{3000 \text{ lire}}{1.6 \text{ dollars}} = \frac{x \text{ lire}}{1 \text{ dollar}}$   $x = 1875$ , and the answer is  $\boxed{D}$

9. (AMC 8 1995) Three congruent circles with centers  $P$ ,  $Q$ , and  $R$  are tangent to the sides of rectangle  $ABCD$  as shown. The circle centered at  $Q$  has diameter 4 and passes through points  $P$  and  $R$ . The area of the rectangle is



(A) 16      (B) 24      (C) 32      (D) 64      (E) 128

**Answer:** C

**Solution 1**

If circle  $Q$  has diameter 4, then so do congruent circles  $P$  and  $R$ . Draw a diameter through  $P$  parallel to  $AD$ . The diameter will be congruent to  $AD$ , and thus  $AD = 4$ , which is the height of the rectangle.

Draw a horizontal line  $PQR$  that extends to the sides of the rectangle. This line is 2 diameters long, so it has length  $4 \cdot 2 = 8$ . It is parallel and congruent to  $AB$ , so the width of the rectangle is 8.

Thus, the area of the rectangle is  $lw = 4 \cdot 8 = 32$ , and the answer is  $\boxed{C}$ .

### Solution 2

The area of circle  $P$  and circle  $R$  almost fill the rectangle. Circle  $P$  has radius  $\frac{4}{2} = 2$ , and so does circle  $R$ . Thus, the sum of their areas is  $\pi \cdot 2^2 + \pi \cdot 2^2 = 8\pi$ . Since  $\pi \approx 3.14$ , the area of the circles is just over 24. The area of the rectangle is greater than the area of the circles. 64 would be too high, as the two circles appear to take up much more than half the area of the rectangle. But 24 is too low, as it is less than the area of the two circles. Thus, the only reasonable answer is  $\boxed{C}$ , which is 32.

10. (AMC 8 1995) A jacket and a shirt originally sold for 80 dollars and 40 dollars, respectively. During a sale Chris bought the 80 dollar jacket at a 40% discount and the 40 dollar shirt at a 55% discount. The total amount saved was what percent of the total of the original prices?

(A) 45%      (B)  $47\frac{1}{2}\%$       (C) 50%      (D)  $79\frac{1}{6}\%$       (E) 95%.

**Answer:** A

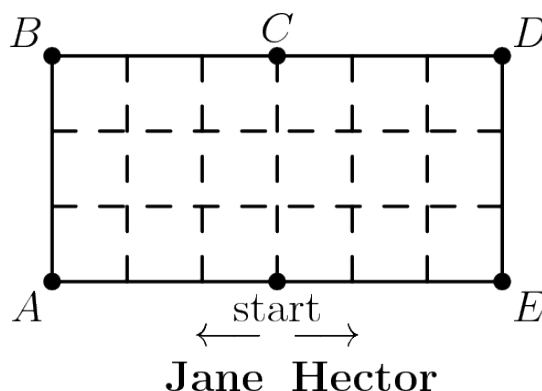
### Solution

Chris would have spent  $40 + 80 = 120$  dollars if there were no sale.

The sale means that he saved  $80(0.40) + 40(0.55) = 32 + 22 = 54$  dollars.

He saved  $\frac{54}{120} \cdot 100\% = 0.45 \cdot 100\% = 45\%$  off the total of the original prices, and the answer is  $\boxed{A}$ .

11. (AMC 8 1995) Jane can walk any distance in half the time it takes Hector to walk the same distance. They set off in opposite directions around the outside of the 18-block area as shown. When they meet for the first time, they will be closest to



(A) A      (B) B      (C) C      (D) D      (E) E

**Answer:** D

### Solution 1

Counting around, when Jane walks 12 steps, she will be at  $D$ . When Hector walks 6 steps, he will also be at  $D$ . Since Jane has walked twice as many steps as Hector, they will reach this spot at the same time. Thus, the answer is  $\boxed{D}$ .

**Solution 2**

We count to find that the perimeter of the rectangle is 18. We divide by 3 to see how far Hector has walked when he meets Jane, this distance is 6. We then move 6 steps counterclockwise to find the point they met or  $\boxed{D}$ .

12. (AMC 8 1995) A *lucky* year is one in which at least one date, when written in the form month/day/year, has the following property: *The product of the month times the day equals the last two digits of the year.* For example, 1956 is a lucky year because it has the date 7/8/56 and  $7 \times 8 = 56$ . Which of the following is NOT a lucky year?

(A) 1990      (B) 1991      (C) 1992      (D) 1993      (E) 1994

**Answer:** E

**Solution**

We examine only the factors of 90, 91, 92, 93, and 94 that are less than 13, because for a year to be lucky, it must have at least one factor between 1 and 12 to represent the month.

90 has factors of 1, 2, 3, 5, 6, 9, and 10. Dividing 90 by the last number 10, gives 9. Thus, 10/9/90 is a valid date, and 1990 is a lucky year.

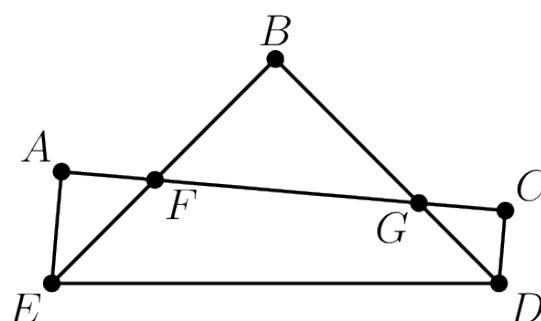
91 has factors of 1 and 7. Dividing 91 by 7 gives 13, and 7/13/91 is a valid date. Thus, 1991 is a lucky year.

92 has factors of 1, 2 and 4. Dividing 92 by 4 gives 23, and 4/23/92 is a valid date. Thus, 1992 is a lucky year.

93 has factors of 1 and 3. Dividing 93 by 3 gives 31, and 3/31/93 is a valid date. Thus, 1993 is a lucky year.

94 has factors of 1 and 2. Dividing 94 by 2 gives 47, and no month has 47 days. Thus, 1994 is not a lucky year, and the answer is  $\boxed{E}$ .

13. (AMC 8 1995) In the figure,  $\angle A$ ,  $\angle B$ , and  $\angle C$  are right angles. If  $\angle AEB = 40^\circ$  and  $\angle BED = \angle BDE$ , then  $\angle CDE =$



(A)  $75^\circ$       (B)  $80^\circ$       (C)  $85^\circ$       (D)  $90^\circ$       (E)  $95^\circ$

**Answer:** E

**Solution**

Because  $\angle BED = \angle BDE$ ,  $\angle B = 90^\circ$ , and  $\triangle BED$  is a triangle, we get:

$$\angle B + \angle BED + \angle BDE = 180$$



$$90 + \angle BED + \angle BED = 180$$

$$2\angle BED = 90$$

$$\angle BED = \angle BDE = 45^\circ$$

So  $\angle AED = \angle AEB + \angle BED = 40 + 45 = 85^\circ$ . Since ACDE is a quadrilateral, the sum of its angles is 360. Therefore:

$$\angle A + \angle C + \angle CDE + \angle AED = 360$$

$$90 + 90 + \angle CDE + 85 = 360$$

$$\angle CDE = 95^\circ \text{ (E)}$$

14. (AMC 8 1995) A team won 40 of its first 50 games. How many of the remaining 40 games must this team win so it will have won exactly 70% of its games for the season?

(A) 20      (B) 23      (C) 28      (D) 30      (E) 35

**Answer:** B

**Solution 1**

The team will play a total of  $50 + 40 = 90$  games, so to finish with a 70% win percentage it must win  $0.7 \cdot 90 = 63$  games. Therefore, it must win  $63 - 40 = \boxed{\text{(B) } 23}$  of the remaining games.

**Solution 2**

Let  $x$  be the number of wins in the team's remaining 40 games. Then the team wins  $40 + x$  of its 90 games.

Since  $70\% = \frac{7}{10}$ , it follows that  $\frac{40+x}{90} = \frac{7}{10}$ . Multiplying both sides of this equation by 90 yields  $40 + x = 63$ , so  $x = \boxed{\text{(B) } 23}$ .

15. (AMC 8 1995) What is the 100<sup>th</sup> digit to the right of the decimal point in the decimal form of  $4/37$ ?

(A) 0      (B) 1      (C) 2      (D) 7      (E) 8

**Answer:** B

**Solution**

$$\frac{4}{37} = \frac{12}{111} = \frac{108}{999} = 0.108108108\dots$$

Since this repeats every three digits, digit number  $x = \text{digit number } (x \bmod 3)$ , and the 100th digit =  $(100 \bmod 3)\text{th digit} = 1\text{st digit} = \boxed{\text{(B) } 1}$

16. (AMC 8 1995) Students from three middle schools worked on a summer project.

- Seven students from Allen school worked for 3 days.
- Four students from Balboa school worked for 5 days.
- Five students from Carver school worked for 9 days.

The total amount paid for the students' work was 774. Assuming each student received the same amount for a day's work, how much did the students from Balboa school earn altogether?

- (A) 9.00 dollars      (B) 48.38 dollars      (C) 180.00 dollars      (D) 193.50 dollars      (E) 258.00 dollars

**Answer:** C

**Solution**

The students combined for  $7 \cdot 3 + 4 \cdot 5 + 5 \cdot 9 = 86$  student workdays. Therefore, each student earned  $\$774 \div 86 = \$9$  per workday. Since the Balboa students completed  $5 \cdot 4 = 20$  workdays, they earned a total of  $20 \cdot \$9 = \boxed{\text{(C) } \$180.00}$ .

17. (AMC 8 1995) The table below gives the percent of students in each grade at Annville and Cleona elementary schools:

|                  | <u>K</u> | <u>1</u> | <u>2</u> | <u>3</u> | <u>4</u> | <u>5</u> | <u>6</u> |
|------------------|----------|----------|----------|----------|----------|----------|----------|
| <b>Annville:</b> | 16%      | 15%      | 15%      | 14%      | 13%      | 16%      | 11%      |
| <b>Cleona:</b>   | 12%      | 15%      | 14%      | 13%      | 15%      | 14%      | 17%      |

Annville has 100 students and Cleona has 200 students. In the two schools combined, what percent of the students are in grade 6?

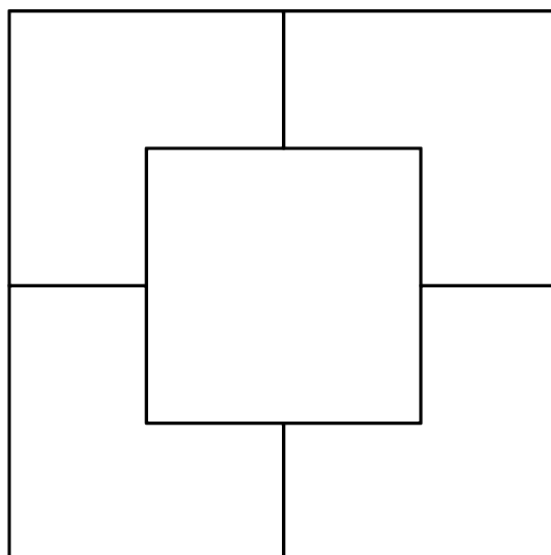
- (A) 12%      (B) 13%      (C) 14%      (D) 15%      (E) 28%

**Answer:** D

**Solution**

By the tables, Annville has 11 6th graders and Cleona has 34. Together they have 45 6th graders and 300 total students, so the percent is  $\frac{45}{300} = \frac{15}{100} = \boxed{\text{(D) } 15\%}$

18. (AMC 8 1995) The area of each of the four congruent L-shaped regions of this 100-inch by 100-inch square is  $\frac{3}{16}$  of the total area. How many inches long is the side of the center square?



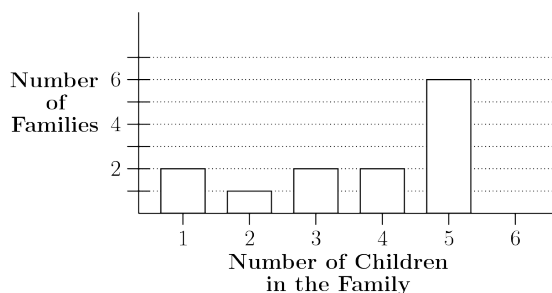
- (A) 25      (B) 44      (C) 50      (D) 62      (E) 75

**Answer:** C

**Solution**

The area taken up by the L's is  $4 * \frac{3}{16} = \frac{3}{4}$  of the area of the whole square. What remains has  $\frac{1}{4}$  of the area of the larger square.  $\frac{100*100}{4} = \frac{100}{2} * \frac{100}{2} = 50 * 50$  is the area of the smaller square, so its side length is 50. (C)

19. (AMC 8 1995) The graph shows the distribution of the number of children in the families of the students in Ms. Jordan's English class. The median number of children in the family for this distribution is



- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer: D**

**Solution**

Counting, there are thirteen total families. The middle number is 7th in either direction, and it is easy to see from the right side that this number is 4 (D)

20. (AMC 8 1995) Diana and Apollo each roll a standard die obtaining a number at random from 1 to 6. What is the probability that Diana's number is larger than Apollo's number?

- (A)  $\frac{1}{3}$       (B)  $\frac{5}{12}$       (C)  $\frac{4}{9}$       (D)  $\frac{17}{36}$       (E)  $\frac{1}{2}$

**Answer: B**

**Solution 1**

Note that the probability of Diana rolling a number larger than Apollo's is the same as the probability of Apollo's being more than Diana's. If we denote this common probability  $D$ , then  $2D + P(\text{Apollo}=\text{Diana}) = 1$ . Now all we need to do is find  $P(\text{Apollo}=\text{Diana})$ . There are  $6(6) = 36$  possibilities total, and 6 of those have  $\text{Apollo}=\text{Diana}$ , so  $P(\text{Apollo}=\text{Diana}) = \frac{6}{36} = \frac{1}{6}$ . Going back to our first equation and solving for  $D$ , we get

$$2D + \frac{1}{6} = 1$$

$$2D = \frac{5}{6}$$

$$D = \frac{5}{12} \Rightarrow \text{(B)}$$

**Solution 2**

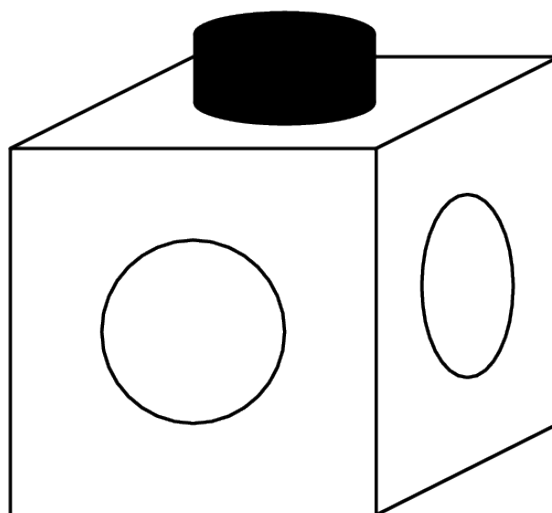
We can use simple casework to solve this problem too. There are six cases based on Apollo's Roll. Apollo Rolls a 1: Diana could roll a 2, 3, 4, 5, or 6. Apollo Rolls a 2: Diana could roll a 3, 4, 5, or 6. Apollo Rolls a 3: Diana could roll a 4, 5, or 6. Apollo Rolls a 4: Diana could roll a 5 or 6. Apollo Rolls a 5: Diana could roll a 6. Apollo

Rolls a 6: There are no successful outcomes. The total amount of successful cases is  $5 + 4 + 3 + 2 + 1 = 15$ . The total amount of possible cases is  $6(6) = 36$ . Therefore, the probability of Diana rolling a bigger number is  $\frac{15}{36} = \frac{5}{12} \Rightarrow$  (B)

**Solution 3 (2 seconds, quick)**

Define a "win" for Player 1 as rolling a greater number than Player 2. Because potential ties make up  $\frac{6}{36} = \frac{1}{6}$  of all 36 combinations, Apollo and Diana "split" the wins in the remaining  $1 - \frac{1}{6} = \frac{5}{6}$ . By symmetry, each person wins half of these, so the probability Diana wins is  $\frac{\frac{5}{6}}{2} = \frac{5}{12} \Rightarrow$  (B)

21. (AMC 8 1995) A plastic snap-together cube has a protruding snap on one side and receptacle holes on the other five sides as shown. What is the smallest number of these cubes that can be snapped together so that only receptacle holes are showing?



- (A) 3      (B) 4      (C) 5      (D) 6      (E) 8

**Answer:** B

**Solution**

It is rather easy to see that the only way that you can possibly have three cubes without protruding snaps is to have them in a triangle formation. However, in this case, the exterior angles are  $120^\circ$ , whereas the outside angle of the cubes are  $90^\circ$  each. A square like formation will work in the case of 4 cubes, so the answer is (B).

22. (AMC 8 1995) The number 6545 can be written as a product of a pair of positive two-digit numbers. What is the sum of this pair of numbers?  
(A) 162      (B) 172      (C) 173      (D) 174      (E) 222

**Answer:** A

**Solution 1**

The prime factorization of 6545 is  $5 \cdot 7 \cdot 11 \cdot 17 = 385 \cdot 17$ , which contains a three digit number, but we want 6545 to be expressed as  $ab \times cd$ . Now we do trial and error:

$$5 \cdot 7 = 35, 11 \cdot 17 = 187 \text{ X}$$

$$5 \cdot 11 = 55, 7 \cdot 17 = 119 \text{ X}$$

$$5 \cdot 17 = 85, 7 \cdot 11 = 77 \checkmark$$

$$85 + 77 = \boxed{\text{(A) } 162}$$

**Solution 2**

Notice that  $6545 = 6561 - 16 = 81^2 - 4^2 = (81 - 4)(81 + 4)$ . Thus, the sum is  $(81 - 4) + (81 + 4) = 2 \times 81 = \boxed{\text{(A) } 162}$ .

This solution is motivated by noticing that a large perfect square, 6561, is close to 6545. In fact, this number is  $3^8$ , which makes it easily recognized.

- GaloisTorrent

**Solution 3**

Let  $x$  and  $y$  be the numbers, with  $x \leq y$ . Since  $xy = 6545$  is odd, both  $x$  and  $y$  are odd.

Since  $y < 100$ ,  $x > \frac{6545}{100} = 65.45$ . Thus  $66 \leq x, y \leq 99$ .

Since 6545 is divisible by 5, one of  $x, y$  is divisible by 5. Therefore, either  $x$  or  $y$  is in the set  $\{75, 85, 95\}$ . The only divisor of 6545 in this set is 85, and  $6545 \div 85 = 77$ .

Therefore,  $x = 77$  and  $y = 85$ , so  $x + y = \boxed{\text{(A) } 162}$ .

-j314andrews

23. (AMC 8 1995) How many four-digit whole numbers are there such that the leftmost digit is odd, the second digit is even, and all four digits are different?

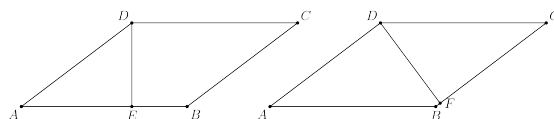
(A) 1120      (B) 1400      (C) 1800      (D) 2025      (E) 2500

**Answer:** B

**Solution**

Count from left to right: There are 5 choices for the first digit, 5 choices for the second, 8 remaining choices for the third, and 7 remaining for the fourth, so there are  $5 * 5 * 8 * 7 = \boxed{\text{(B) } 1400}$  numbers.

24. (AMC 8 1995) In parallelogram  $ABCD$ ,  $\overline{DE}$  is the altitude to the base  $\overline{AB}$  and  $\overline{DF}$  is the altitude to the base  $\overline{BC}$ . [Note: Both pictures represent the same parallelogram.] If  $DC = 12$ ,  $EB = 4$ , and  $DE = 6$ , then  $DF =$



(A) 6.4      (B) 7      (C) 7.2      (D) 8      (E) 10

**Answer:** C

**Solution 1**

The area of  $ABCD$  is  $DC \cdot DE = 12 \cdot 6 = 72$ . Since  $DE = 6$  and  $AE = AB - BE = 12 - 4 = 8$ , by the Pythagorean Theorem,  $AD = \sqrt{6^2 + 8^2} = 10$ .

Since the area of  $ABCD$  can also be written as  $AD \cdot DF$ , it follows that  $10 \cdot DF = 72$ , so  $DF = \boxed{\text{(C) } 7.2}$ .

**Solution 2**

Since  $AE = 12 - 4 = 8$ , by the Pythagorean theorem,  $AD = \sqrt{6^2 + 8^2} = 10$ .

Since  $\angle A = \angle C$  and  $\angle DEA = \angle DFC = 90^\circ$ ,  $\triangle DEA \sim \triangle DFC$ .

Thus  $\frac{DE}{DA} = \frac{DF}{DC}$ , so  $\frac{6}{10} = \frac{DF}{12}$  and thus  $DF = \boxed{\text{(C)} 7.2}$

-j314andrews

25. (AMC 8 1995) Buses from Dallas to Houston leave every hour on the hour. Buses from Houston to Dallas leave every hour on the half hour. The trip from one city to the other takes 5 hours. Assuming the buses travel on the same highway, how many Dallas-bound buses does a Houston-bound bus pass in the highway (not in the station)?

(A) 5      (B) 6      (C) 9      (D) 10      (E) 11

**Answer:** D

### Solution 1

Let  $B$  be any Houston-bound bus. Let  $X$  be the first bus to arrive in Dallas after  $B$  leaves Dallas. At the time  $B$  leaves,  $X$  is 0.5 hours from Dallas and thus left Houston 4.5 hours before  $B$  left Dallas. Let  $Y$  be the last bus to leave Houston before  $B$  arrives in Houston. Then  $Y$  must leave Houston 4.5 hours after  $B$  left Dallas. Therefore,  $Y$  leaves Houston 9 hours after  $X$ . Since  $B$  must pass every bus between  $X$  and  $Y$  inclusive,  $B$  passes a total of  $9 + 1 = \boxed{\text{(D)} 10}$  Dallas-bound buses.

### Solution 2

Say you are on the Houston-bound bus that left at 12:30 in the afternoon, looking out the window to see how many buses you pass. At 12:45 pm, the Dallas bus that left at 8:00 am is 4:45 away (Note -  $a : b - a$  is for hrs. and  $b$  is for min.) from Dallas, and therefore 15 minutes from Houston. Your bus is also 15 minutes from Houston, so you are delighted at the first bus you have passed.

At 1:15 pm, the 9:00 am Dallas bus meets you again, being 4:15 away from Dallas and therefore 45 minutes from Houston. After a while you might notice that the buses meet you in 30 minute intervals after 12:45 pm; indeed, the bus that left an hour later than the bus most recently has traveled 30 minutes closer, and so have you, for a total of an hour closer, and when you passed the most recent bus, the bus after it had indeed left an hour later. You decide to predict how many buses you pass by finding all valid times that are 30 minute intervals from 12:45. They are 12:45, 1:15, 1:45, 2:15, 2:45, 3:15, 3:45, 4:15, 4:45, 5:15, and 5:30, your arrival time. Hence, you will pass  $\boxed{\text{(D)} 10}$  Dallas-bound buses.

## §12 AMC 8 1996

1. (AMC 8 1996) How many positive factors of 36 are also multiples of 4?

(A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** B

### Solution

The factors of 36 are 1, 2, 3, 4, 6, 9, 12, 18, and 36.

The multiples of 4 up to 36 are 4, 8, 12, 16, 20, 24, 28, 32 and 36.

Only 4, 12 and 36 appear on both lists, so the answer is 3, which is option  $\boxed{\text{B}}$ .

### Solution 2

$36 = 4^1 \cdot 3^2$ . All possible factors of 36 will be here, except for ones divisible by 2 and not by 4.  $(1 + 1) \cdot (2 + 1) = 6$ . Subtract factors not divisible by 4, which are 1,  $3^1$ , and  $3^2$ .  $6 - 3 = 3$ , which is  $\boxed{B}$ .

**Solution 3**

Divide 36 by 4, and the remaining factors, when multiplied by 4, will be factors of 36.

$36 \div 4 = 9$ , which has 3 factors, giving us option  $\boxed{B}$ .

2. (*AMC 8 1996*) Jose, Thuy, and Kareem each start with the number 10. Jose subtracts 1 from the number 10, doubles his answer, and then adds 2. Thuy doubles the number 10, subtracts 1 from her answer, and then adds 2. Kareem subtracts 1 from the number 10, adds 2 to his number, and then doubles the result. Who gets the largest final answer?
- (A) Jose      (B) Thuy      (C) Kareem      (D) Jose and Thuy      (E) Thuy and Kareem

**Answer:** C

**Solution**

Jose gets  $10 - 1 = 9$ , then  $9 \cdot 2 = 18$ , then  $18 + 2 = 20$ .

Thuy gets  $10 \cdot 2 = 20$ , then  $20 - 1 = 19$ , and then  $19 + 2 = 21$ .

Kareem gets  $10 - 1 = 9$ , then  $9 + 2 = 11$ , and then  $11 \cdot 2 = 22$ .

Thus, Kareem gets the highest number, and the answer is  $\boxed{C}$ .

3. (*AMC 8 1996*) The 64 whole numbers from 1 through 64 are written, one per square, on a checkerboard (an 8 by 8 array of 64 squares). The first 8 numbers are written in order across the first row, the next 8 across the second row, and so on. After all 64 numbers are written, the sum of the numbers in the four corners will be
- (A) 130      (B) 131      (C) 132      (D) 133      (E) 134

**Answer:** A

**Solution**

Obviously 1 is in the top left corner, 8 is in the top right corner, and 64 is in the bottom right corner. To find the bottom left corner, subtract 7 from 64 which is 57. Adding the results gives  $1 + 8 + 57 + 64 = 130$  which is answer  $\boxed{A}$ .

4. (*AMC 8 1996*)  $\frac{2 + 4 + 6 + \cdots + 34}{3 + 6 + 9 + \cdots + 51} =$  (A)  $\frac{1}{3}$       (B)  $\frac{2}{3}$       (C)  $\frac{3}{2}$       (D)  $\frac{17}{3}$       (E)  $\frac{34}{3}$

**Answer:** B

**Solution 1**

First, notice that each number in the numerator is a multiple of 2, and each number in the denominator is a multiple of 3. This suggests that each expression can be factored. Factoring gives:

$$\frac{2(1 + 2 + 3 + \cdots + 17)}{3(1 + 2 + 3 + \cdots + 17)}$$

Since all the material in the parentheses is the same, the common factor in the numerator and the denominator may be cancelled, leaving  $\frac{2}{3}$ , which is option  $\boxed{B}$ .

**Solution 2**

There are 17 terms in the numerator  $2 + 4 + 6 + \dots + 30 + 32 + 34$ . Consider adding those terms, but in a different order. Start with the last two terms,  $2 + 34$ , and then add the next two terms on the outside,  $4 + 32$ , and continue. You will get 8 pairs of numbers that add to 36, while the  $9^{\text{th}}$  number in the middle will be alone. That number is 18. Adding all the numbers gives  $8 \cdot 36 + 18 = 306$ .

Similarly, the denominator has 17 terms of  $3 + 6 + 9 + \dots + 45 + 48 + 51$ . There are 8 pairs of numbers that add up to 54, with the  $9^{\text{th}}$  number in the center being 27. The total of all the numbers is  $8 \cdot 54 + 27 = 459$ .

The answer is  $\frac{306}{459}$ . Eyeballing the options, the fraction is clearly under 1, but more than  $\frac{1}{2}$ . Thus, the answer must be  $\frac{2}{3}$ , or  $\boxed{B}$ . Alternately, you can do the work by factoring out a 9 in the numerator to give  $\frac{34}{51}$ . Factoring out 17 will give the desired answer.

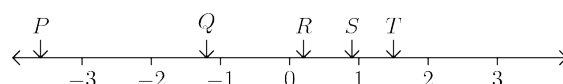
### Solution 3

Start by finding a pattern:

$$\frac{2}{3} = \frac{2}{3} \cdot \frac{2+4}{3+6} = \frac{6}{9} = \frac{2}{3} \cdot \frac{2+4+6}{3+6+9} = \frac{12}{18} = \frac{2}{3}$$

Each step doesn't seem to change the value of the fraction, so  $\boxed{B}$  is the right answer.

5. (AMC 8 1996) The letters  $P$ ,  $Q$ ,  $R$ ,  $S$ , and  $T$  represent numbers located on the number line as shown.



Which of the following expressions represents a negative number?

- (A)  $P - Q$     (B)  $P \cdot Q$     (C)  $\frac{S}{Q} \cdot P$     (D)  $\frac{R}{P \cdot Q}$     (E)  $\frac{S + T}{R}$

**Answer:** A

### Solution 1

First, note that  $P < Q < 0 < R < S < T$ . Thus,  $P$  and  $Q$  are negative, while  $R$ ,  $S$ , and  $T$  are positive.

Option  $A$  is the difference of two numbers. If the first number is lower than the second number, the answer will be negative. In this case,  $P < Q$ , so  $P - Q < 0$ , and  $P - Q$  is thus indeed a negative number. So option  $\boxed{A}$  is correct.

Option  $B$  is the product of two negatives, which is always positive.

Option  $C$  starts with the quotient of a positive and a negative. This is negative. But this negative quotient is then multiplied by a negative number, which gives a positive number.

Option  $D$  has the product of two negatives in the denominator. Thus, the denominator is positive. Additionally, the numerator is (barely) positive. A positive over a positive will be positive.

Option  $E$  has the sum of two positives in the numerator. That will be positive. The denominator is also positive. Again, the quotient of two positives is positive.

### Solution 2

Estimate the numbers, and calculate.



$$P \approx -3.5 \quad Q \approx -1.1 \quad R \approx 0.1 \quad S \approx 0.9 \quad T \approx 1.5$$

For option *A*, the value of  $P - Q = -3.5 - (-1.1) = -3.5 + 1.1 = -2.4$ . Thus A is the right answer.

$$\text{For option } B, PQ = -3.5 \cdot -1.1 = +3.85$$

$$\text{For option } C, \frac{S}{Q} \cdot P = \frac{0.9}{-1.1} \cdot -3.5 \approx -0.818 \cdot -3.5 \approx +2.836$$

$$\text{For option } D, \frac{R}{PQ} = \frac{0.1}{3.85} \approx +0.026$$

$$\text{For option } E, \frac{S+T}{R} = \frac{0.9+1.5}{0.1} = \frac{2.4}{0.1} = +24$$

6. (AMC 8 1996) What is the smallest result that can be obtained from the following process?

- Choose three different numbers from the set  $\{3, 5, 7, 11, 13, 17\}$ .
- Add two of these numbers.
- Multiply their sum by the third number.

(A) 15      (B) 30      (C) 36      (D) 50      (E) 56

**Answer:** C

**Solution**

Since we want the smallest possible result, and we are only adding and multiplying positive numbers over 1, we can "prune" the set to the three smallest numbers  $\{3, 5, 7\}$ . Using bigger numbers will create bigger sums and bigger products.

From there, compute the 3 ways you can do the two operations:

$$(3 + 5)7 = 8 \cdot 7 = 56 \quad (3 + 7)5 = 10 \cdot 5 = 50 \quad (7 + 5)3 = 12 \cdot 3 = 36$$

The smallest number is 36, giving an answer of C

7. (AMC 8 1996) Brent has goldfish that quadruple (become four times as many) every month, and Gretel has goldfish that double every month. If Brent has 4 goldfish at the same time that Gretel has 128 goldfish, then in how many months from that time will they have the same number of goldfish?

(A) 4      (B) 5      (C) 6      (D) 7      (E) 8

**Answer:** B

**Solution 1**

Call this month "Month 0". Make a table of the fish that Brent and Gretel have each month.

|       |   |       |   |        |   |   |   |   |     |   |   |    |   |     |   |   |    |   |     |   |   |     |   |      |   |   |      |   |      |  |
|-------|---|-------|---|--------|---|---|---|---|-----|---|---|----|---|-----|---|---|----|---|-----|---|---|-----|---|------|---|---|------|---|------|--|
| Month | / | Brent | / | Gretel | 0 | / | 4 | / | 128 | 1 | / | 16 | / | 256 | 2 | / | 64 | / | 512 | 3 | / | 256 | / | 1024 | 4 | / | 1024 | / | 2048 |  |
| 5     | / | 4096  | / | 4096   |   |   |   |   |     |   |   |    |   |     |   |   |    |   |     |   |   |     |   |      |   |   |      |   |      |  |

You could create a similar table without doing all of the calculations by converting all the goldfish into powers of 2. In this table, you could increase Brent's goldfish by two powers of 2, while increasing Gretel's fish by one power of 2.

Either way, in 5 months they will have the same number of fish, giving an answer of B

**Solution 2**

Create two equations.

Brent starts with 4 goldfish and they quadruple each month, giving us  $y = 4 \cdot 4^x$  or  $y = 2^{2x+2}$ .

Gretel starts with 128 goldfish and they double each month, giving us  $y = 128 \cdot 2^x$  or  $y = 2^{x+7}$ .

Using substitution,  $2^{2x+2} = 2^{x+7}$ . Thus  $2x + 2 = x + 7$  giving us  $x = 5$ , which is B.

8. (AMC 8 1996) Points  $A$  and  $B$  are 10 units apart. Points  $B$  and  $C$  are 4 units apart. Points  $C$  and  $D$  are 3 units apart. If  $A$  and  $D$  are as close as possible, then the number of units between them is

(A) 0      (B) 3      (C) 9      (D) 11      (E) 17

**Answer:** B

**Solution**

If  $AB = 10$  and  $BC = 4$ , then  $(10 - 4) \leq AC \leq (10 + 4)$  by the **triangle inequality**. In the triangle inequality, the equality is only reached when the "triangle"  $ABC$  is really a degenerate triangle, and  $ABC$  are collinear.

Simplifying, this means the smallest value  $AC$  can be is 6.

Applying the triangle inequality on  $ACD$  with  $AC = 6$  and  $CD = 3$ , we know that  $6 - 3 \leq AD \leq 6 + 3$  when  $AC$  is minimized. If  $AC$  were larger, then  $AD$  could be larger, but we want the smallest  $AD$  possible, and not the largest. Thus,  $AD$  must be at least 3, but cannot be smaller than 3. Therefore, B is the answer.

This answer comes when  $ABCD$  are all on a line, with  $A = 0, B = 10, C = 6$ , and  $D = 3$ .

9. (AMC 8 1996) If 5 times a number is 2, then 100 times the reciprocal of the number is

(A) 2.5      (B) 40      (C) 50      (D) 250      (E) 500

**Answer:** D

**Solution**

If 5 times a number is 2, then  $5x = 2 \rightarrow x = \frac{2}{5}$ , and the number is  $\frac{2}{5}$ .

If the number is  $\frac{2}{5}$ , then 100 times its reciprocal is 100 times  $\frac{5}{2}$ , which is 250, giving an answer of D.

10. (AMC 8 1996) When Walter drove up to the gasoline pump, he noticed that his gasoline tank was  $\frac{1}{8}$  full. He purchased 7.5 gallons of gasoline for \$10. With this additional gasoline, his gasoline tank was then  $\frac{5}{8}$  full. The number of gallons of gasoline his tank holds when it is full is

(A) 8.75      (B) 10      (C) 11.5      (D) 15      (E) 22.5

**Answer:** D

**Solution**

The tank started at  $\frac{1}{8}$  full, and ended at  $\frac{5}{8}$  full. Therefore, Walter filled  $\frac{5}{8} - \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$  of the tank.

If Walter fills half the tank with 7.5 gallons, then Walter can fill two halves of the tank (or a whole tank) with  $7.5 \times 2 = 15$  gallons, giving an answer of D

11. (AMC 8 1996) Let  $x$  be the number

$$0.\underbrace{0000\dots0000}_{1996 \text{ zeros}}1,$$

where there are 1996 zeros after the decimal point. Which of the following expressions represents the largest number?

- (A)  $3 + x$       (B)  $3 - x$       (C)  $3 \cdot x$       (D)  $3/x$       (E)  $x/3$

**Answer:** D

**Solution**

Estimate each of the options.

$A$  will give a number that is just over 3.

$B$  will give a number that is just under 3. This eliminates  $B$ , because  $A$  is bigger.

$C$  will give a number that is barely over 0, since it is three times a tiny number. This eliminates  $C$ , because  $A$  is bigger.

$D$  will give a huge number.  $\frac{1}{x}$  will get very, very large in magnitude when  $x$  gets close to zero. You can see this by examining the sequence  $x = 0.1$ , which gives 10 as the reciprocal,  $x = 0.01$ , which gives 100 as the reciprocal, and  $x = 0.001$ , which gives 1000 as the reciprocal. Thus,  $D$  will be huge, and this eliminates  $A$ .

$E$  will give a small number, since you're dividing a tiny number into thirds. This eliminates  $E$ , and gives  $\boxed{D}$  as the answer.

12. (AMC 8 1996) What number should be removed from the list

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11$$

so that the average of the remaining numbers is 6.1?

- (A) 4      (B) 5      (C) 6      (D) 7      (E) 8

**Answer:** B

**Solution 1**

Adding all of the numbers gives us  $\frac{11 \cdot 12}{2} = 66$  as the current total. Since there are 11 numbers, the current average is  $\frac{66}{11} = 6$ . We need to take away a number from the total and then divide the result by 10 because there will only be 10 numbers left to give an average of 6.1. Setting up the equation:

$$\frac{66-x}{10} = 6.1 \quad 66 - x = 61 \quad x = 5$$

Thus, the answer is  $\boxed{B}$

**Solution 2**

Similar to the first solution, the current total is 66. Since there are 11 numbers on the list, taking 1 number away will leave 10 numbers. If those 10 numbers have an average of 6.1, then those 10 numbers must have a sum of  $10 \times 6.1 = 61$ . Thus, the number that was removed must be  $66 - 61 = 5$ , and the answer is  $\boxed{B}$ .

13. (AMC 8 1996) In the fall of 1996, a total of 800 students participated in an annual school clean-up day. The organizers of the event expect that in each of the years 1997, 1998, and 1999, participation will increase by 50% over the previous year. The number of participants the organizers will expect in the fall of 1999 is

- (A) 1200    (B) 1500    (C) 2000    (D) 2400    (E) 2700

**Answer:** E

**Solution 1**

If the participation increases by 50%, then it is the same as saying participation is multiplied by a factor of  $100\% + 50\% = 1 + 0.5 = 1.5$ .

In 1997, participation will be  $800 \cdot 1.5 = 1200$ .

In 1998, participation will be  $1200 \cdot 1.5 = 1800$

In 1999, participation will be  $1800 \cdot 1.5 = 2700$ , giving an answer of E.

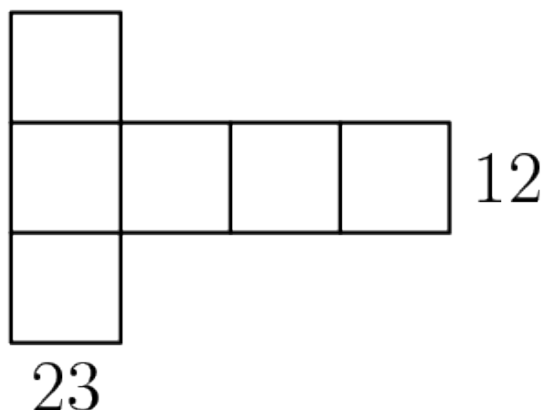
**Solution 2**

Since the percentage increase is the same each year, this is an example of exponential growth with a base of 1.5. In three years, there will be  $1.5^3 = \frac{27}{8}$  times as many participants. Multiplying this by the 800 current participants, there are 2700 participants, and the answer is E.

14. (AMC 8 1996) Six different digits from the set

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

are placed in the squares in the figure shown so that the sum of the entries in the vertical column is 23 and the sum of the entries in the horizontal row is 12. The sum of the six digits used is



- (A) 27    (B) 29    (C) 31    (D) 33    (E) 35

**Answer:** B

**Solution**

Looking at the vertical column, the three numbers sum to 23. If we make the numbers on either end 9 and 8 in some order, the middle number will be 6. This is the minimum for the intersection.

Looking at the horizontal row, the four numbers sum to 12. If we minimize the three numbers on the right to 123, the first number has a maximum value of 6. This is the maximum for the intersection

Thus, the minimum of the intersection is 6, and the maximum of the intersection is 6. This means the intersection must be 6, and the other numbers must be 9 and 8 in the column, and 123 in the row. The sum of all the numbers is  $12 + 23 - 6 =$  (B) 29.

15. (*AMC 8 1996*) The remainder when the product  $1492 \cdot 1776 \cdot 1812 \cdot 1996$  is divided by 5 is

(A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** E

**Solution 1**

To determine a remainder when a number is divided by 5, you only need to look at the last digit. If the last digit is 0 or 5, the remainder is 0. If the last digit is 1 or 6, the remainder is 1, and so on.

To determine the last digit of  $1492 \cdot 1776 \cdot 1812 \cdot 1996$ , you only need to look at the last digit of each number in the product. Thus, we compute  $2 \cdot 6 \cdot 2 \cdot 6 = 12^2 = 144$ . The last digit of the number  $1492 \cdot 1776 \cdot 1812 \cdot 1996$  is also 4, and thus the remainder when the number is divided by 5 is also 4, which gives an answer of  $\boxed{E}$ .

**Solution 2**

$1492 \cdot 1776 \cdot 1812 \cdot 1996 \pmod{5} \equiv 2 \cdot 1 \cdot 2 \cdot 1 \pmod{5} = 4$ , which is option  $\boxed{E}$ .

16. (*AMC 8 1996*)  $1 - 2 - 3 + 4 + 5 - 6 - 7 + 8 + 9 - 10 - 11 + \cdots + 1992 + 1993 - 1994 - 1995 + 1996 =$  (A)  $-998$       (B)  $-1$       (C)  $0$       (D)  $1$       (E)  $998$

**Answer:** C

**Solution**

Put the numbers in groups of 4:

$$(1 - 2 - 3 + 4) + (5 - 6 - 7 + 8) + (9 - 10 - 11 + 12) + \cdots + (1993 - 1994 - 1995 + 1996)$$

The first group has a sum of 0.

The second group increases the two positive numbers on the end by 1, and decreases the two negative numbers in the middle by 1. Thus, the second group also has a sum of 0.

Continuing the pattern, every group has a sum of 0, and thus the entire sum is 0, giving an answer of  $\boxed{C}$ .

**Solution 2**

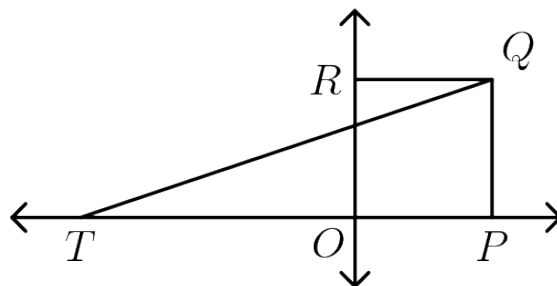
Let any term of the series be  $t_n$ . Realize that at every  $n \equiv 0 \pmod{4}$ , the sum of the series is 0. For  $t_{1996}$  we know  $1996 \equiv 0 \pmod{4}$  so the solution is  $\boxed{C}$ .

**Solution 3**

This solution is slightly more detailed than Solution 2, even though they are essentially the same.

Compute the sum of the first and last term, the second and second last term, the third and third last term. We get  $(1 + 1996) + (-2 - 1995) + (-3 - 1994) + (4 + 1993)$  and so on. We see that each pair's sum alternates between 1997 and  $-1997$ . We see that every two consecutive pair cancels out. For example  $(1 + 1996) + (-2 - 1995)$  gives a result of 0. There are a total of 1996 terms, there are 998 pairs, and there are 499 pairs which result in a sum of 1997 each and 499 pairs which result in a sum of  $-1997$  each. These all cancel out to a sum of 0, and thus the entire sum is 0, giving an answer of  $\boxed{C}$ .

17. (*AMC 8 1996*) Figure  $OPQR$  is a square. Point  $O$  is the origin, and point  $Q$  has coordinates  $(2,2)$ . What are the coordinates for  $T$  so that the area of triangle  $PQT$  equals the area of square  $OPQR$ ?



NOT TO SCALE

- (A)  $(-6, 0)$     (B)  $(-4, 0)$     (C)  $(-2, 0)$     (D)  $(2, 0)$     (E)  $(4, 0)$

**Answer:** C

**Solution**

The area of  $\square OPQR$  is  $2^2 = 4$ .

The area of  $\triangle PQT$  is  $\frac{1}{2}bh = \frac{1}{2} \cdot PT \cdot PQ$ .

If we set the areas equal, the area of  $\triangle PQT$  is 4. Also, note that  $PQ = 2$ . Plugging those in, we get:

$$4 = \frac{1}{2} \cdot PT \cdot 2 \cdot PT = 4$$

If  $PT = 4$ , and  $PO = 2$ , then  $OT = 2$ , and  $T$  must be 2 units to the left of the origin. This would be  $(-2, 0)$ , giving answer **C**.

18. (AMC 8 1996) Ana's monthly salary was 2000 in May. In June she received a 20% raise.

- (A) 1920 dollars    (B) 1980 dollars    (C) 2000 dollars    (D) 2020 dollars    (E) 2040 dollars

**Answer:** A

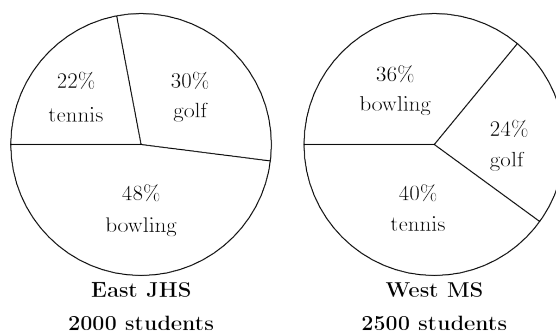
**Solution**

Notice that a 20% raise translates to Ana's monthly salary multiplied by  $(100 + 20)\% = 120\%$ , and a 20% pay cut translates to her salary multiplied by  $100\% - 20\% = 80\%$ , so

$$2000 \cdot 120\% = 2400$$

$$2400 \cdot 80\% = 1920 \rightarrow \mathbf{A}$$

19. (AMC 8 1996) The pie charts below indicate the percent of students who prefer golf, bowling, or tennis at East Junior High School and West Middle School. The total number of students at East is 2000 and at West, 2500. In the two schools combined, the percent of students who prefer tennis is



- (A) 30%      (B) 31%      (C) 32%      (D) 33%      (E) 34%

**Answer:** C**Solution**

In the first school,  $2000 \cdot 22\% = 2000 \cdot 0.22 = 440$  students prefer tennis.

In the second school,  $2500 \cdot 40\% = 2500 \cdot 0.40 = 1000$  students prefer tennis.

In total,  $440 + 1000 = 1440$  students prefer tennis out of a total of  $2000 + 2500 = 4500$  students

This means  $\frac{1440}{4500} \cdot 100\% = \frac{32}{100} \cdot 100\% = 32\%$  of the students in both schools prefer tennis, giving answer (C) 32%.

20. (AMC 8 1996) Suppose there is a special key on a calculator that replaces the number  $x$  currently displayed with the number given by the formula  $1/(1-x)$ . For example, if the calculator is displaying 2 and the special key is pressed, then the calculator will display -1 since  $1/(1-2) = -1$ . Now suppose that the calculator is displaying 5. After the special key is pressed 100 times in a row, the calculator will display

- (A)
- $-0.25$
- (B) 0      (C) 0.8      (D) 1.25      (E) 5

**Answer:** A**Solution**

We look for a pattern, hoping this sequence either settles down to one number, or that it forms a cycle that repeats.

After 1 press, the calculator displays  $\frac{1}{1-5} = -\frac{1}{4}$

After 2 presses, the calculator displays  $\frac{1}{1-(-\frac{1}{4})} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$

After 3 presses, the calculator displays  $\frac{1}{1-\frac{4}{3}} = \frac{1}{-\frac{1}{3}} = -3$

Thus, every three presses, the display will be 5. On press  $3 \cdot 33 = 99$ , the display will be 5. One more press will give  $-\frac{1}{4}$ , which is answer A.

21. (AMC 8 1996) How many subsets containing three different numbers can be selected from the set

$$\{89, 95, 99, 132, 166, 173\}$$

so that the sum of the three numbers is even?

- (A) 6      (B) 8      (C) 10      (D) 12      (E) 24

**Answer:** D**Solution**

To have an even sum with three numbers, we must add either  $E + O + O$ , or  $E + E + E$ , where  $O$  represents an odd number, and  $E$  represents an even number.

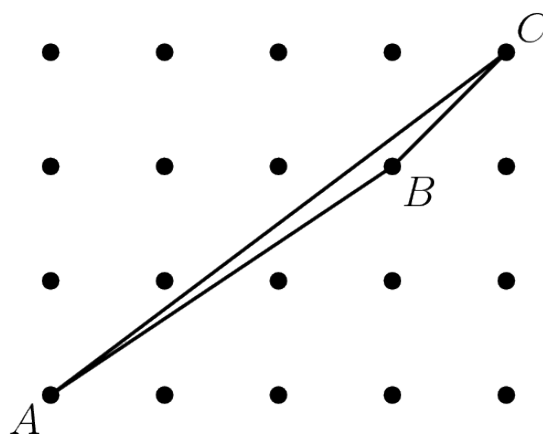
Since there are not three even numbers in the given set,  $E + E + E$  is impossible. Thus, we must choose two odd numbers, and one even number.

There are 2 choices for the even number.

There are 4 choices for the first odd number. There are 3 choices for the last odd number. But the order in which we pick these 2 numbers doesn't matter, so this overcounts the pairs of odd numbers by a factor of 2. Thus, we have  $\frac{4 \cdot 3}{2} = 6$  choices for a pair of odd numbers.

In total, there are 2 choices for an even number, and 6 choices for the odd numbers, giving a total of  $2 \cdot 6 = 12$  possible choices for a 3-element set that has an even sum. This is option **D**.

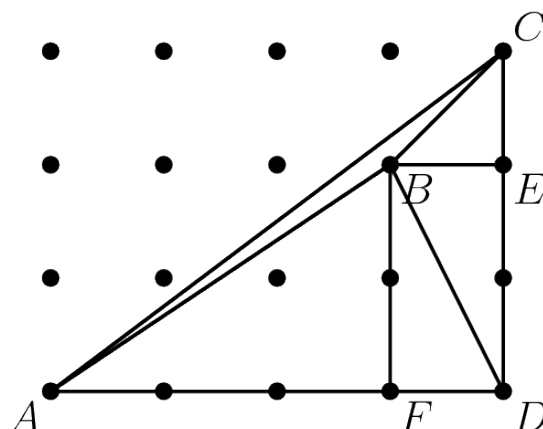
22. (AMC 8 1996) The horizontal and vertical distances between adjacent points equal 1 unit. The area of triangle  $ABC$  is



- (A)  $1/4$     (B)  $1/2$     (C)  $3/4$     (D) 1    (E)  $5/4$

**Answer:** B

**Solution 1**



$\triangle ADC$  takes up half of the  $4 \times 3$  grid, so it has area of 6.

$\triangle ABD$  has height of  $BF = 2$  and a base of  $AD = 4$ , for an area of  $\frac{1}{2} \cdot 2 \cdot 4 = 4$ .

$\triangle CBD$  has height of  $BE = 1$  and a base of  $CD = 3$ , for an area of  $\frac{1}{2} \cdot 1 \cdot 3 = \frac{3}{2}$

Note that  $\triangle ABC$  can be found by taking  $\triangle ADC$ , and subtracting off  $\triangle ABD$  and  $\triangle CBD$ .

Thus, the area of  $\triangle ABC = 6 - 4 - \frac{3}{2} = \frac{1}{2}$ , and the answer is **B**.

There are other equivalent ways of dissecting the figure; right triangles  $\triangle ABF$ ,  $\triangle BCE$  and rectangle  $\square BEDF$  can also be used. You can also use  $\triangle BEC$  and trapezoid  $ADBE$ .

**Solution 2**

Using the **Shoelace Theorem**, and labelling the points  $A(0,0)$ ,  $B(3,2)$ ,  $C(4,3)$ , we find the area is:



$(0, 0)$   $(3, 2)$   $(4, 3)$   $(0, 0)$

Area  $= \frac{1}{2} = |(3 \cdot 0 + 4 \cdot 2 + 0 \cdot 3) - (0 \cdot 2 + 3 \cdot 3 + 4 \cdot 0)| = \frac{1}{2}$ , which is option **B**.

### Solution 3

Using **Pick's Theorem**, we can simply use the number of coordinate points to get the formula. In application, we get the formula  $\frac{3}{2} + 0 - 1$ , which equals  $\frac{1}{2}$ , giving us our answer **B**.

### Solution 4

Using the Pythagorean Theorem, we know  $BC$  is  $\sqrt{2}$ . Using  $BC$  as a base of the triangle, the height is  $\frac{1}{2}\sqrt{2}$ . The area would be  $\frac{1}{2} \cdot \frac{1}{2}\sqrt{2} \cdot \sqrt{2}$ , which equals  $\frac{1}{2}$ , so the answer is **B**.

23. (AMC 8 1996) The manager of a company planned to distribute a \$50 bonus to each employee from the company fund, but the fund contained \$5 less than what was needed. Instead the manager gave each employee a \$45 bonus and kept the remaining \$95 in the company fund. The amount of money in the company fund before any bonuses were paid was

(A) \$945      (B) \$950      (C) \$955      (D) \$990      (E) \$995

**Answer:** E

### Solution 1

Let  $x$  be the number of employees at the company. Then the total amount of money in the company fund, in dollars, can be written as either  $50x - 5$  or  $45x + 95$ .

So  $50x - 5 = 45x + 95$ , and thus  $x = 20$ . Substituting  $x = 20$  into either  $50x - 5$  or  $45x + 95$  shows that the total amount of money in the company fund is **(E) \$995**.

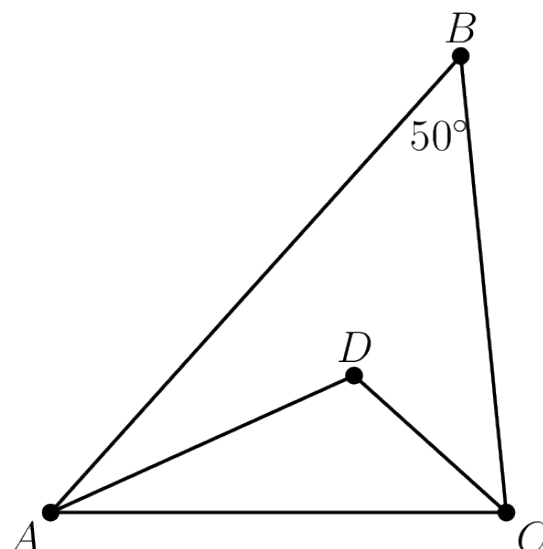
### Solution 2 (Answer Choices)

Since the number of employees is an integer, the amount of money in the fund, in dollars, must be 5 less than a multiple of 50. This eliminates (B), (C), and (D).

Also, the amount of money in the fund, in dollars, must also be 95 more than a multiple of 45. This eliminates (A).

Therefore, the correct answer is **(E) \$995**.

24. (AMC 8 1996) The measure of angle  $ABC$  is  $50^\circ$ ,  $\overline{AD}$  bisects angle  $BAC$ , and  $\overline{DC}$  bisects angle  $BCA$ . The measure of angle  $ADC$  is



- (A)  $90^\circ$       (B)  $100^\circ$       (C)  $115^\circ$       (D)  $122.5^\circ$       (E)  $125^\circ$

**Answer:** C

**Solution**

Let  $\angle CAD = \angle BAD = x$ , and let  $\angle ACD = \angle BCD = y$

From  $\triangle ABC$ , we know that  $50 + 2x + 2y = 180$ , leading to  $x + y = 65$ .

From  $\triangle ADC$ , we know that  $x + y + \angle D = 180$ . Plugging in  $x + y = 65$ , we get  $\angle D = 180 - 65 = 115$ , which is answer C.

**Solution 2**

Construct  $\overline{BE}$  through  $D$  and intersects  $\overline{AC}$  at point  $E$

By Exterior Angle Theorem,

$$\angle ADE = \angle ABD + \angle BAD$$

Similarly,

$$\angle EDC = \angle DBC + \angle BCD$$

Thus,

$$\angle ADC = \angle ABC + \angle BAD + \angle BCD$$

Let  $\angle ADC = x$

Because  $\overline{AD}$  and  $\overline{CD}$  are angle bisectors,

$$\begin{aligned} 180^\circ - x &= \angle BAD + \angle BCD \\ &= x - 50^\circ \end{aligned}$$

$$x = 115^\circ$$

Thus, the answer is C

25. (AMC 8 1996) A point is chosen at random from within a circular region. What is the probability that the point is closer to the center of the region than it is to the boundary of the region?

- (A)  $1/4$       (B)  $1/3$       (C)  $1/2$       (D)  $2/3$       (E)  $3/4$

**Answer:** A

**Solution 1**

Draw a circle with a radius of 2. Draw a concentric circle with radius 1. The circumference of this inner circle is the set of all points that are 1 from the center, and 1 from the outer circle meaning the set of points are equidistant from the center of the circles to the outside of the big circle.

The inside of smaller circle is the set of all points that are closer to the center of the region than to the boundary of the outer circle. The "washer" region that is outside the circle of radius 1, but inside the circle of radius 2, is the set of all points that are closer to the boundary than to the center of the circle.

If you select a random point in a region of area  $B$ , the probability that the point is in a smaller subregion  $A$  is the ratio  $\frac{A}{B}$ . In this case,  $B = \pi \cdot 2^2 = 4\pi$ , and  $A = \pi \cdot 1^2 = \pi$ , and the ratio of areas is  $\frac{\pi}{4\pi} = \frac{1}{4}$ , and the answer is A.

**Solution 1.1**

Alternatively, we could also let the radius of the entire circular region be  $R$ , the radius of the region where it is closer to the center be  $r$ .

The probability is  $\frac{1}{4}$ , or answer choice A

## §13 AMC 8 1997

1. (*AMC 8 1997*)  $\frac{1}{10} + \frac{9}{100} + \frac{9}{1000} + \frac{7}{10000} =$  (A) 0.0026      (B) 0.0197      (C) 0.1997      (D) 0.26      (E) 0.2697

**Answer:** C

**Solution**

Convert each fraction to a decimal.

$$\frac{1}{10} + \frac{9}{100} + \frac{9}{1000} + \frac{7}{10000} = 0.1 + 0.09 + 0.009 + 0.0007 = 0.1997 \quad \boxed{\text{(C)}}$$

2. (*AMC 8 1997*) Ahn chooses a two-digit integer, subtracts it from 200, and doubles the result. What is the largest number Ahn can get?
- (A) 200      (B) 202      (C) 220      (D) 380      (E) 398

**Answer:** D

**Solution**

The smallest two-digit integer he can subtract from 200 is 10. This will give the largest result for that first operation, and doubling it will keep it as the largest number possible.

$$200 - 10 = 190$$

$$190 \times 2 = 380$$

(D)

3. (*AMC 8 1997*) Which of the following numbers is the largest?
- (A) 0.97      (B) 0.979      (C) 0.9709      (D) 0.907      (E) 0.9089

**Answer:** B

**Solution**

Using the process of elimination:

Tenths digit: All tenths digits are equal, at 9.

Hundreths digit:  $A$ ,  $B$ , and  $C$  all have the same hundreths digit of 7, and it is greater than the hundredths of  $D$  or  $E$  (which is 0). Eliminate both  $D$  and  $E$ .

Thousandths digit:  $B$  has the largest thousandths digit of the remaining answers, and is the correct answer.  $A$  has an "invisible" thousandths digit of 0, while  $C$  also has a thousandths digit of 0.

**(B)**

4. (*AMC 8 1997*) Julie is preparing a speech for her class. Her speech must last between one-half hour and three-quarters of an hour. The ideal rate of speech is 150 words per minute. If Julie speaks at the ideal rate, which of the following number of words would be an appropriate length for her speech?
- (A) 2250      (B) 3000      (C) 4200      (D) 4350      (E) 5650

**Answer:** E

**Solution**

Since there are 60 minutes in an hour, we need a number of words between  $(\frac{1}{2} \times 60) \times 150 = 4500$  and  $(\frac{3}{4} \times 60) \times 150 = 6750$ . The only such choice is **(E)**.

5. (*AMC 8 1997*) There are many two-digit multiples of 7, but only two of the multiples have a digit sum of 10. The sum of these two multiples of 7 is
- (A) 119      (B) 126      (C) 140      (D) 175      (E) 189

**Answer:** A

**Solution 1 (brute force)**

Writing out all two digit numbers that have a digital sum of 10, you get 19, 28, 37, 46, 55, 64, 73, 82, and 91. The two numbers on that list that are divisible by 7 are 28 and 91. Their sum is  $28 + 91 = 119$ , choice **(A)**.

**Solution 2**

Writing out all the two digit multiples of 7, you get 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, and 91. Again you find 28 and 91 have a digital sum of 10, giving answer **(A)**.

You may notice that adding 7 either increases the digital sum by 7, or decreases it by 2, depending on whether there is carrying or not.

6. (*AMC 8 1997*) In the number 74982.1035 the value of the *place* occupied by the digit 9 is how many times as great as the value of the *place* occupied by the digit 3?
- (A) 1,000      (B) 10,000      (C) 100,000      (D) 1,000,000      (E) 10,000,000

**Answer:** C

**Solution 1**

The digit 9 is 5 places to the left of the digit 3. Thus, it has a place value that is  $10^5 = 100,000$  times greater. **(C)**

**Solution 2**

The digit 9 is in the 100s place. The digit 3 is in the  $\frac{1}{1000}$ ths place. The ratio of these two numbers is  $\frac{100}{\frac{1}{1000}} = 100 \times 1000 = 100,000$ , and the answer is **(C)**.

7. (*AMC 8 1997*) The area of the smallest square that will contain a circle of radius 4 is

(A) 8      (B) 16      (C) 32      (D) 64      (E) 128

**Answer:** D

**Solution**

Draw a square circumscribed around the circle. (Alternately, the circle is inscribed in the square.) If the circle has radius 4, it has diameter 8. Two of the diameters of the circle will run parallel to the sides of the square. Thus, the smallest square that contains it has side length 8, and area  $8 \times 8 = 64$ . (D)

8. (*AMC 8 1997*) Walter gets up at 6:30 a.m., catches the school bus at 7:30 a.m., has 6 classes that last 50 minutes each, has 30 minutes for lunch, and has 2 hours additional time at school. He takes the bus home and arrives at 4:00 p.m. How many minutes has he spent on the bus?

(A) 30      (B) 60      (C) 75      (D) 90      (E) 120

**Answer:** B

**Solution**

There are  $4\frac{1}{2}$  hours from 7:30 a.m. to noon, and 4 hours from noon to 4 p.m., meaning Walter was away from home for a total of  $8\frac{1}{2}$  hours, or  $8.5 \times 60 = 510$  minutes.

Tallying up the times he was not on the bus, he has  $6 \times 50 = 300$  minutes in classes, 30 minutes at lunch, and  $2 \times 60 = 120$  minutes of "additional time". That is a total of  $300 + 30 + 120 = 450$  minutes that Walter was away from home, but not on the bus.

Therefore, Walter spent  $510 - 450 = 60$  minutes on the bus, and the answer is B

9. (*AMC 8 1997*) Three students, with different names, line up in single file. What is the probability that they are in alphabetical order from front-to-back?

(A)  $\frac{1}{12}$       (B)  $\frac{1}{9}$       (C)  $\frac{1}{6}$       (D)  $\frac{1}{3}$       (E)  $\frac{2}{3}$

**Answer:** C

**Solution 1**

There are 3 ways to pick the first student, 2 ways to pick the second student, and 1 way to pick the last student, for a total of  $3! = 3 \times 2 \times 1 = 6$  ways to line the students up.

Only 1 of those ways is alphabetical. Thus, the probability is  $\frac{1}{6}$  or C

**Solution 2**

Give the students uncreative names like Abby Adams, Bob Bott, and Carol Crock:  $A$ ,  $B$ , and  $C$ , replacing the alphabetically first student's name with  $A$ , and the alphabetically last student's name with  $C$ . List all the ways they can line up:

$ABC \ ACB \ BAC \ BCA \ CAB \ CBA$

Only 1 of those 6 lists is in alphabetical order, giving an answer of  $\frac{1}{6}$  or C

**Solution 3**

We look at the first place in the line, there is only one correct person who belongs in this place, but without the restriction there can be three different people who

can go in this spot. So we have a  $\frac{1}{3}$  chance of getting the first place right. Now look at the second place, there is only one correct person who belongs in this place, but without the restriction there can be two different people who can go in this place, as one person is already in the first place. So we have a  $\frac{1}{2}$  chance of getting the second place right. Now look at the third place, there is only one correct person who belongs in this place, but since all the other spots are filled there is only one person who can be in this place at all. So we have a  $\frac{1}{1}$  chance of getting the third place right. Since the total probability of getting the order depends on the probability of each place having the right person we should multiply these three fractions.  $\frac{1}{3} \times \frac{1}{2} \times \frac{1}{1}$  is the probability of getting the order right. Simplifying this we get that the answer is  $\frac{1}{6}$  or  $\boxed{C}$ . blankbox

10. (AMC 8 1997) What fraction of this square region is shaded? Stripes are equal in width, and the figure is drawn to scale.



- (A)  $\frac{5}{12}$     (B)  $\frac{1}{2}$     (C)  $\frac{7}{12}$     (D)  $\frac{2}{3}$     (E)  $\frac{5}{6}$

**Answer:** C

### Solution 1

Fill in grid lines to make the shape look like a small 6x6 checkerboard.

There are  $6 \times 6 = 36$  squares total. The smallest shaded region has 3 squares. The medium shaded region has 7 squares. The largest shaded region has 11 squares. The total shaded area is  $3 + 7 + 11 = 21$  squares.

Thus,  $\frac{21}{36} = \frac{7}{12}$  of the square is shaded, and the answer is  $\boxed{C}$

### Solution 2

Similar to the first solution, fill in the gridlines to make 36 squares total.

Instead of counting shaded squares, count unshaded squares. There are  $1 + 5 + 9 = 15$  unshaded squares.

Thus  $\frac{15}{36} = \frac{5}{12}$  of the square is unshaded, and  $1 - \frac{5}{12} = \frac{7}{12}$  of the square is shaded.

11. (AMC 8 1997) Define  $\boxed{N}$  as the number of whole number divisors of  $N$ . For example,  $\boxed{3} = 2$  because 3 has two divisors, 1 and 3. Find the value of

$$\boxed{11} \times \boxed{20}.$$

- (A) 6      (B) 8      (C) 12      (D) 16      (E) 24

**Answer:** A

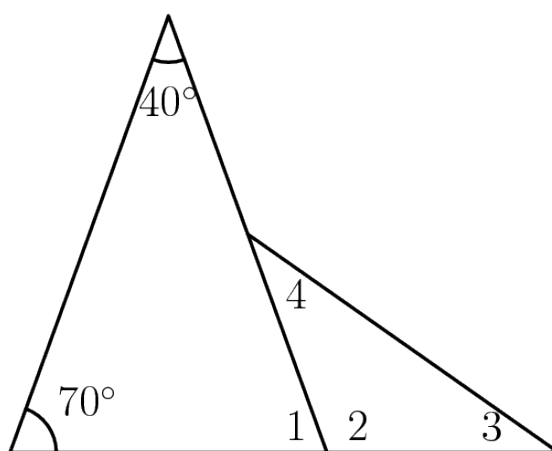
**Solution**

$\boxed{11} \times \boxed{20}$  11 has 2 factors, and 20 has  $\{1, 2, 4, 5, 10, 20\}$  as factors, for a total of 6 factors. Plugging  $\boxed{11} = 2$  and  $\boxed{20} = 6$ :

$\boxed{2 \times 6} \boxed{12}$  12 has factors of  $\{1, 2, 3, 4, 6, 12\}$ , so  $\boxed{12} = 6$ , and the answer is  $\boxed{A}$

12. (AMC 8 1997)  $\angle 1 + \angle 2 = 180^\circ$   $\angle 3 = \angle 4$

Find  $\angle 4$ .



- (A)  $20^\circ$       (B)  $25^\circ$       (C)  $30^\circ$       (D)  $35^\circ$       (E)  $40^\circ$

**Answer:** D

**Solution**

Using the left triangle, we have:

$$\angle 1 + 70 + 40 = 180 \quad \angle 1 = 180 - 110 \quad \angle 1 = 70$$

Using the given fact that  $\angle 1 + \angle 2 = 180$ , we have  $\angle 2 = 180 - 70 = 110$ .

Finally, using the right triangle, and the fact that  $\angle 3 = \angle 4$ , we have:

$$\angle 2 + \angle 3 + \angle 4 = 180 \quad 110 + \angle 3 + \angle 4 = 180 \quad 110 + \angle 4 + \angle 4 = 180 \quad 2\angle 4 = 70 \quad \angle 4 = 35$$

Thus, the answer is  $\boxed{D}$

**Side Note to Make Solution Faster** You can use Exterior Angle Theorem to directly say that  $\angle 1 = \angle 3 + \angle 4$  instead of needing to use up time by doing  $\angle 2 + \angle 3 + \angle 4 = 180$ , so on. Using exterior angle theorem goes straight to  $\angle 1 = \angle 3 + \angle 4$ , and therefore,  $2(\angle 4) = 70^\circ$ , as we know  $180^\circ - 40^\circ - 70^\circ = \angle 1 = 70^\circ$ , and  $\angle 3 = \angle 4$ . This means that  $\angle 4 = 35^\circ$ , so the answer is  $\boxed{D}$ .

13. (AMC 8 1997) Three bags of jelly beans contain 26, 28, and 30 beans. The ratios of yellow beans to all beans in each of these bags are 50%, 25%, and 20%, respectively. All three bags of candy are dumped into one bowl. Which of the following is closest to the ratio of yellow jelly beans to all beans in the bowl?

- (A) 31%    (B) 32%    (C) 33%    (D) 35%    (E) 95%

**Answer:** A

**Solution**

In bag  $A$ , there are 26 jellybeans, and 50% are yellow. That means there are  $26 \times 50\% = 26 \times 0.50 = 13$  yellow jelly beans in this bag.

In bag  $B$ , there are 28 jellybeans, and 25% are yellow. That means there are  $28 \times 25\% = 28 \times 0.25 = 7$  yellow jelly beans in this bag.

In bag  $C$ , there are 30 jellybeans, and 20% are yellow. That means there are  $30 \times 20\% = 30 \times 0.20 = 6$  yellow jelly beans in this bag.

In all three bags, there are  $13 + 7 + 6 = 26$  yellow jelly beans in total, and  $26 + 28 + 30 = 84$  jelly beans of all types in total.

Thus,  $\frac{26}{84} = \frac{26}{84} \cdot 100\% = \frac{13}{42} \cdot 100\% = 30.9\%$  of all jellybeans are yellow. Thus, the correct answer is A.

14. (*AMC 8 1997*) There is a set of five positive integers whose average (mean) is 5, whose median is 5, and whose only mode is 8. What is the difference between the largest and smallest integers in the set?

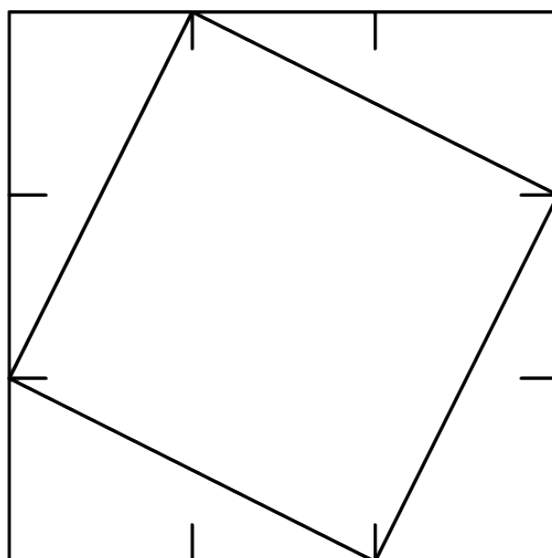
- (A) 3    (B) 5    (C) 6    (D) 7    (E) 8

**Answer:** D

**Solution**

Let  $a$  and  $b$  be the two smallest integers, with  $a \leq b$ . Since 5 is the median and 8 is the unique mode, the integers, from smallest to largest, are  $a, b, 5, 8, 8$ . Since the mean is 5, the sum of all five integers is  $5 \cdot 5 = 25$ , and thus  $a + b = 4$ . Since 8 is the unique mode,  $a \neq b$ . So  $a = 1$  and  $b = 3$ , and thus the range is  $8 - 1 =$ (D) 7.

15. (*AMC 8 1997*) Each side of the large square in the figure is trisected (divided into three equal parts). The corners of an inscribed square are at these trisection points, as shown. The ratio of the area of the inscribed square to the area of the large square is



- (A)  $\frac{\sqrt{3}}{3}$     (B)  $\frac{5}{9}$     (C)  $\frac{2}{3}$     (D)  $\frac{\sqrt{5}}{3}$     (E)  $\frac{7}{9}$



**Answer:** B

**Solution 1**

Define 1 unit as the distance from vertex of the large square to the closest trisection point. Then the large square has side length 3, so its area is 9.

Each of the four right triangles shown has area  $\frac{1}{2} \cdot 2 \cdot 1 = 1$ , so the area of the small square is  $9 - 4 \cdot 1 = 5$ . The ratio of the areas of the two squares is thus  $\boxed{\text{(B)} \frac{5}{9}}$ .

**Solution 2**

Let the side of the big square be  $3x$ . The area of this square is  $3x \cdot 3x = 9x^2$

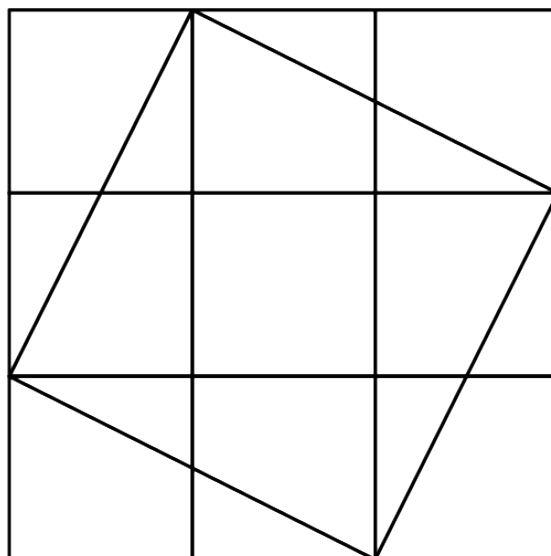
You can use the Pythagorean Theorem on the any of the right triangles to find the length of the side of the inscribed square. One leg is  $x$ , while the other leg is  $2x$ . Thus, the length of the hypotenuse  $c$ , which is also the side of the inscribed square, is:

$$a^2 + b^2 = c^2 \quad x^2 + (2x)^2 = c^2 \quad x^2 + 4x^2 = c^2 \quad 5x^2 = c^2$$

Once you get  $c^2$ , you can stop, because  $c^2$  is the area of the inscribed square. The ratio of the areas of the small square to the big square is  $\frac{5x^2}{9x^2} = \frac{5}{9}$ , giving option  $\boxed{B}$ .

**Solution 3**

Fill in the gridlines of the diagram.



Let each grid space have length 1. The inscribed square consists of four congruent triangles and one square.

The triangles each have area  $\frac{1}{2} \cdot 2 \cdot 1 = 1$ , and in total they cover  $4 \cdot 1 = 4$  square units of area. The central square's area is  $1 \cdot 1 = 1$ . Therefore, the area of the inscribed square is  $4 + 1 = 5$ .

The big square contains  $3 \cdot 3 = 9$  square units of area. Thus, the ratio is equal to  $\frac{5}{9}$ , or  $\boxed{B}$ .

16. (AMC 8 1997) Penni Precisely buys 100worthofstockineachofthreecompanies :  
AlabamaAlmonds, BostonBeans, andCaliforniaCauliflower. Afteroneyear, AAwasup20

- (A)  $A = B = C$     (B)  $A = B < C$     (C)  $C < B = A$     (D)  $A < B < C$     (E)  $B < A < C$

**Answer:** E

**Solution 1**

AA is  $100 \cdot 120\% = 100 \cdot 1.2 = 120$  after one year. After the second year, AA is  $120 \cdot 0.8 = 96$ .

BB is  $100 \cdot 75\% = 100 \cdot 0.75 = 75$  after one year. After the second year, BB is  $75 \cdot 1.25 = 93.75$

CC remains unchanged throughout, and stays at 100.

Thus,  $B < A < C$ , and the right answer is E

**Solution 2**

AA will be  $100 \cdot \frac{6}{5} \cdot \frac{4}{5} = 100 \cdot \frac{24}{25}$  at the end.

BB will be  $100 \cdot \frac{3}{4} \cdot \frac{5}{4} = 100 \cdot \frac{15}{16}$  at the end.

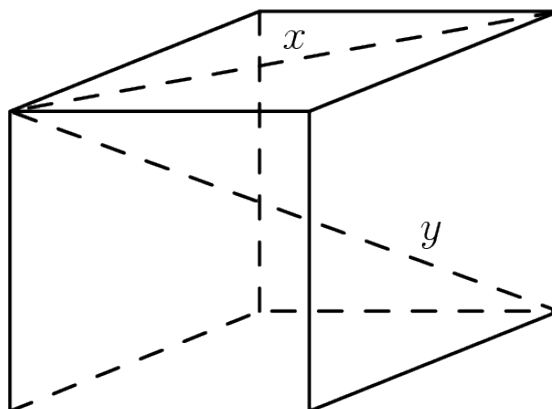
CC will be unchanged at 100.

Since all the fractions are under 1,  $C$  will be highest value.

Since  $\frac{24}{25}$  is only  $\frac{1}{25}$  away from 1, while  $\frac{15}{16}$  is  $\frac{1}{16}$  away from 1,  $\frac{24}{25}$  is closer to 1, and will be closer to the original value.

Thus,  $B < A < C$ , and the right answer is E

17. (AMC 8 1997) A cube has eight vertices (corners) and twelve edges. A segment, such as  $x$ , which joins two vertices not joined by an edge is called a diagonal. Segment  $y$  is also a diagonal. How many diagonals does a cube have?



- (A) 6    (B) 8    (C) 12    (D) 14    (E) 16

**Answer:** E

**Solution 1**

On each face, there are 2 diagonals like  $x$ . There are 6 faces on a cube. Thus, there are  $2 \times 6 = 12$  diagonals that are "x-like".

Every "y-like" diagonal must connect the bottom of the cube to the top of the cube. Thus, for each of the 4 bottom vertices of the cube, there is a different "y-like" diagonal. So there are 4 "y-like" diagonals.

This gives a total of  $12 + 4 = 16$  diagonals on the cube, which is answer E.

**Solution 2**

There are 8 vertices on a cube (A, B, C, D, E, F, G, H). If you pick one vertex, it will have 7 line segments associated with it, so you will have  $\frac{8 \cdot 7}{2} = 28$  segments within the cube. The division by 2 is necessary because you counted both the segment from A to B and the segment from B to A.

But not all 28 of these segments are diagonals. Some are edges. There are 4 edges on the top, 4 edges on the bottom, and 4 edges that connect the top to the bottom. So there are 12 edges total, meaning that there are  $28 - 12 = 16$  segments that are not edges. All of these segments are diagonals, and thus the answer is  $\boxed{E}$ .

**Solution 3**

Consider picking one point on the corner of the cube. That point has 3 "x-like" diagonals, and 1 "y-like" diagonal that ends on the opposite vertex. Thus, each vertex has  $3 + 1 = 4$  diagonals associated with it. There are 8 vertices on the cube, giving a total of  $8 \cdot 4 = 32$  diagonals.

However, each diagonal was counted as both a "starting point" and an "ending point". So there are really  $\frac{32}{2} = 16$  diagonals, giving an answer of  $\boxed{E}$ .

18. (AMC 8 1997) At the grocery store last week, small boxes of facial tissue were priced at 4 boxes for 5. *This week they are on sale at 5 boxes for 4.* The percent decrease in the price per box during the sale was closest to
- (A) 30%      (B) 35%      (C) 40%      (D) 45%      (E) 65%

**Answer:** B

**Solution**

Last week, each box was  $\frac{5}{4} = 1.25$

This week, each box is  $\frac{4}{5} = 0.80$

Percent decrease is given by  $\frac{X_{old} - X_{new}}{X_{old}} \cdot 100\%$

This, the percent decrease is  $\frac{1.25 - 0.8}{1.25} \cdot 100\% = \frac{45}{125} \cdot 100\% = 36\%$ , which is closest to  $\boxed{B}$

19. (AMC 8 1997) If the product  $\frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} \cdot \dots \cdot \frac{a}{b} = 9$ , what is the sum of  $a$  and  $b$ ?
- (A) 11      (B) 13      (C) 17      (D) 35      (E) 37

**Answer:** D

**Solution 1**

Notice that the numerator of the first fraction cancels out the denominator of the second fraction, and the numerator of the second fraction cancels out the denominator of the third fraction, and so on.

The only numbers left will be  $a$  in the numerator from the last fraction and 2 in the denominator from the first fraction. (The  $b$  will cancel with the numerator of the preceeding number.) Thus,  $\frac{a}{2} = 9$ , and  $a = 18$ .

Since the numerator is always one more than the denominator,  $b = a - 1 = 17$ , and  $a + b = 18 + 17 = 35$ , giving an answer of  $\boxed{D}$

**Solution 2**

Find a pattern. If  $a = 4$ , then the expression is just the first two terms, which is 2.

If  $a = 5$ , then the expression is the first three terms, giving 2.5.

If  $a = 6$ , the expression is the first four terms, giving 3.

If  $a = 7$ , the expression will be the first five terms, giving 3.5.

Conjecture that the expression is always going to equal  $\frac{a}{2}$ , and thus when  $a = 18$ , the expression will be 9, as desired.

As above, when  $a = 18$ ,  $b = 17$ , and the sum is 35, or  $\boxed{D}$

20. (AMC 8 1997) A pair of 8-sided dice have sides numbered 1 through 8. Each side has the same probability (chance) of landing face up. The probability that the product of the two numbers that land face-up exceeds 36 is

(A)  $\frac{5}{32}$       (B)  $\frac{11}{64}$       (C)  $\frac{3}{16}$       (D)  $\frac{1}{4}$       (E)  $\frac{1}{2}$

**Answer:** A

**Solution**

There are  $8 \times 8 = 64$  combinations to examine.

If one die is 1, then even with an 8 on the other die, no combinations will work.

If one die is 2, then even with an 8 on the other die, no combinations will work.

If one die is 3, then even with an 8 on the other die, no combinations will work.

If one die is 4, then even with an 8 on the other die, no combinations will work.

If one die is 5, then the other die must be an 8 to have a product over 36. Thus, (5, 8) works.

If one die is 6, then the other die must be either 7 or 8 to have a product over 36. Thus, (6, 7) and (6, 8) both work.

If one die is 7, then the other die can be 6, 7, or 8 to have a product over 36. Thus, (7, 6), (7, 7), and (7, 8) all work.

If one die is 8, then the other die can be 5, 6, 7, or 8 to have a product over 36. Thus, (8, 5), (8, 6), (8, 7), and (8, 8) work.

There are a total of  $1 + 2 + 3 + 4 = 10$  combinations that work out of a total of 64 possibilities.

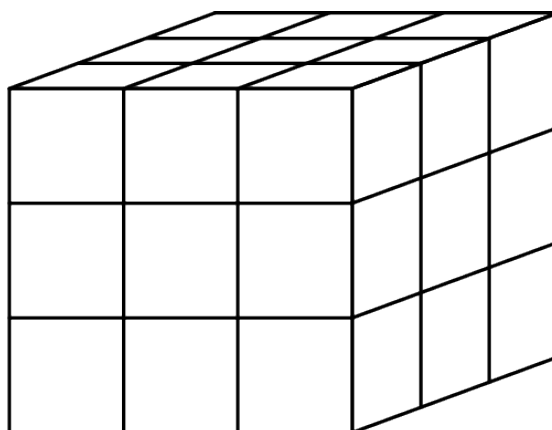
Thus, the answer is  $\frac{10}{64} = \frac{5}{32}$ , and the answer is  $\boxed{A}$

But this may be slightly complicated, so you can also do solution 2 (scroll to bottom).

**Solution 2**

Make a chart with all products. 10 work out of 64. Simplify for  $5/32$  or  $\boxed{A}$ .

21. (AMC 8 1997) Each corner cube is removed from this  $3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$  cube. The surface area of the remaining figure is



- (A) 19 sq.cm      (B) 24 sq.cm      (C) 30 sq.cm      (D) 54 sq.cm      (E) 72 sq.cm

**Answer:** D

**Solution**

The original cube has 6 square surfaces that each have an area of  $3^2 = 9$ , for a total surface area of  $6 \cdot 9 = 54$ .

Since no two corner cubes touch, we can examine the effect of removing each corner cube individually.

Each corner cube contributes 3 faces each of surface area 1 to the big cube, so the surface area is decreased by 3 when the cube is removed.

However, when the cube is removed, 3 faces on the  $3 \times 3 \times 3$  cube will be revealed, increasing the surface area by 3.

Thus, the surface area does not change with the removal of a corner cube, and it remains 54, which is answer D.

22. (AMC 8 1997) A two-inch cube ( $2 \times 2 \times 2$ ) of silver weighs 3 pounds and is worth 200. How much is a three-inch cube of silver worth?

- (A) 300 dollars      (B) 375 dollars      (C) 450 dollars      (D) 560 dollars      (E) 675 dollars

**Answer:** E

**Solution**

The two-inch cube has a volume of 8 cubic inches, and the three-inch cube has a volume of 27 cubic inches. Thus, the three-inch cube has a weight that is  $\frac{27}{8}$  times that of the two-inch cube. Then its value is  $\frac{27}{8} \cdot 200 = \boxed{\text{(E) 675 dollars}}$ .

23. (AMC 8 1997) There are positive integers that have these properties:

- the sum of the squares of their digits is 50, and
- each digit is larger than the one to its left.

The product of the digits of the largest integer with both properties is

- (A) 7      (B) 25      (C) 36      (D) 48      (E) 60

**Answer:** C

**Solution**

Five-digit numbers will have a minimum of  $1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$  as the sum of their squares if the five digits are distinct and non-zero. If there is a zero, it will be forced to the left by rule #2.

No digit will be greater than 7, as  $8^2 = 64$ .

Trying four digit numbers  $WXYZ$ , we have  $w^2 + x^2 + y^2 + z^2 = 50$  with  $0 < w < x < y < z < 8$   $z = 7$  will not work, since the other digits must be at least  $1^2 + 2^2 + 3^2 = 14$ , and the sum of the squares would be over 50.

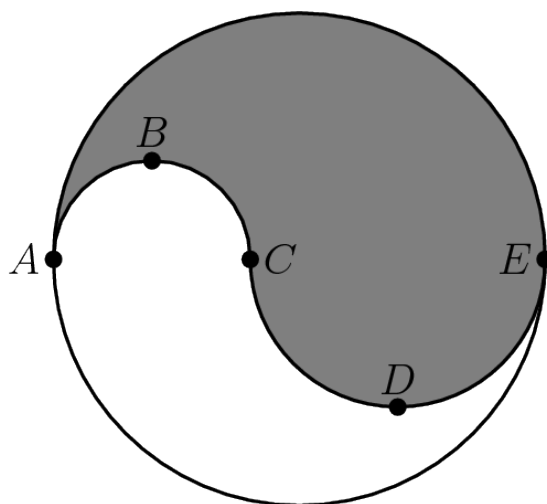
$z = 6$  will give  $w^2 + x^2 + y^2 = 14$ .  $(w, x, y) = (1, 2, 3)$  will work, giving the number 1236. No other number with  $z = 6$  will work, as  $w, x$ , and  $y$  would have to be greater.

$z = 5$  will give  $w^2 + x^2 + y^2 = 25$ .  $y = 4$  forces  $x = 3$  and  $w = 0$ , which has a leading zero, and then we have 345 which is a 3-digit number.

$z = 4$  can only give the number 1234, which does not satisfy the condition of the problem.

Thus, the number in question is 1236, and the product of the digits is 36, giving  $\boxed{C}$  as the answer.

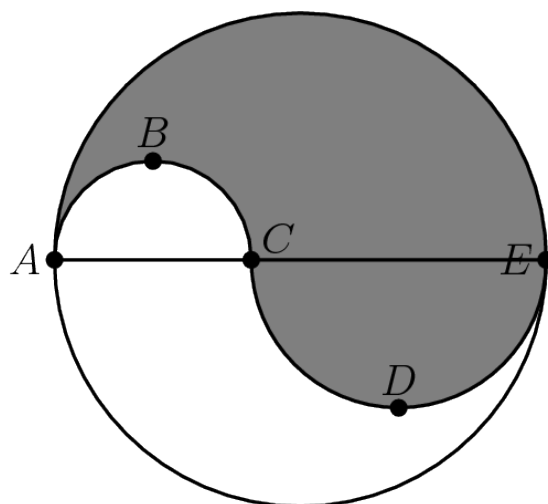
24. (AMC 8 1997) Diameter  $ACE$  is divided at  $C$  in the ratio 2 : 3. The two semicircles,  $ABC$  and  $CDE$ , divide the circular region into an upper (shaded) region and a lower region. The ratio of the area of the upper region to that of the lower region is



- (A) 2 : 3      (B) 1 : 1      (C) 3 : 2      (D) 9 : 4      (E) 5 : 2

**Answer:** C

**Solution**



Draw  $\overline{AE}$  to divide the big circle in half. Assign  $AC = 4$  and  $CE = 6$  so that the radii work out to be integers. ( $4x$  and  $6x$  can be used instead, but the  $x$  will cancel in the ratio.)

The shaded region is equal to the area of semicircle  $\widehat{AE}$  on top, plus the area of the semicircle  $CDE$  on the bottom, minus the area of semicircle  $\widehat{ABC}$  on top.

The radii of those three semicircles are 5, 3, and 2, respectively. Thus, the area of the shaded region is  $\frac{1}{2}\pi \cdot 5^2 + \frac{1}{2}\pi \cdot 3^2 - \frac{1}{2}\pi \cdot 2^2 = \frac{\pi}{2}(5^2 + 3^2 - 2^2) = 15\pi$

The total area of the circle is  $\pi \cdot 5^2 = 25\pi$ . Thus, the unshaded area is  $25\pi - 15\pi = 10\pi$ . Therefore the ratio of shaded:unshaded is  $15\pi : 10\pi = \boxed{(C) 3 : 2}$ .

25. (AMC 8 1997) All of the even numbers from 2 to 98 inclusive, excluding those ending in 0, are multiplied together. What is the rightmost digit (the units digit) of the product?  
 (A) 0      (B) 2      (C) 4      (D) 6      (E) 8

**Answer:** D

**Solution 1**

All the tens digits of the product will be irrelevant to finding the units digit. Thus, we are searching for the units digit of  $(2 \cdot 4 \cdot 6 \cdot 8) \cdot (2 \cdot 4 \cdot 6 \cdot 8) \cdot (2 \cdot 4 \cdot 6 \cdot 8) \cdot \dots$

There will be 10 groups of 4 numbers. The number now can be rewritten as  $(2 \cdot 4 \cdot 6 \cdot 8)^{10}$ . Multiplying, we get  $384^{10}$ .

Again, we can disregard the tens and hundreds digit of 384, since we only want the units digit of the number, leaving  $4^{10}$ .

Now, we try to find a pattern to the units digit of  $4^n$ . To compute this quickly, we once again discard all tens digits and higher.

$$4^1 = 4.$$

$$4^2 = 4 \cdot 4 = 1\textbf{6}, \text{ discard the 1.}$$

$$4^3 = 1 \cdot 4 = \textbf{4} \quad 4^4 = 4 \cdot 4 = 1\textbf{6}, \text{ discard the 1.}$$

$$4^5 = 1 \cdot 4 = \textbf{4}$$

Those equalities are, in reality, congruences  $\pmod{10}$ .

Thus, the pattern of the units digits is  $\{4, 6, 4, 6, 4, 6, 4, 6, 4, 6\}$ . The cycle repeats so that term  $n$  is the same as term  $n + 2$ . The tenth number in the cycle is 6, giving an answer of  $\boxed{D}$ .

### Solution 2

Again, the value we seek is equal to  $(2 \cdot 4 \cdot 6 \cdot 8)^{10} \pmod{10}$ . We can use equivalence to simplify.

$$\begin{aligned} (2 \cdot 4 \cdot 6 \cdot 8)^{10} \pmod{10} &\equiv (2 \cdot 4 \cdot (-4) \cdot (-2))^{10} \pmod{10} \equiv 64^{10} \pmod{10} \equiv 4^{10} \pmod{10} \\ &\equiv 2^{20} \pmod{10} \equiv 8^6 \cdot 2^2 \pmod{10} \equiv (-2)^6 \cdot 2^2 \pmod{10} \equiv 2^8 \pmod{10} \equiv 256 \pmod{10} \equiv 6 \pmod{10} \end{aligned}$$

Thus our answer is  $\boxed{D}$ .

## §14 AMC 8 1998

1. (AMC 8 1998) For  $x = 7$ , which of the following is the smallest?

(A)  $\frac{6}{x}$     (B)  $\frac{6}{x+1}$     (C)  $\frac{6}{x-1}$     (D)  $\frac{x}{6}$     (E)  $\frac{x+1}{6}$

**Answer:** B

### Solution 1

Plugging  $x$  in for every answer choice would give

(A)  $\frac{6}{7}$     (B)  $\frac{6}{8}$     (C)  $\frac{6}{6}$     (D)  $\frac{7}{6}$     (E)  $\frac{8}{6}$

From here, we can see that the smallest is answer choice  $\boxed{B}$ .

### Solution 2

Note that  $\frac{6}{x+1} < \frac{6}{x} < \frac{6}{x-1}$  (for  $x > 1$ ) and  $\frac{x}{6} < \frac{x+1}{6}$ . Therefore, we just need to compare  $\frac{6}{x+1}$  and  $\frac{x}{6}$ . Plugging in  $x = 7$ , we get  $\frac{3}{4}$  and  $\frac{7}{6}$ , respectively, with  $\frac{3}{4} < \frac{7}{6}$ . Thus, the answer is  $\boxed{(B) \frac{6}{x+1}}$ .

2. (AMC 8 1998) If  $\frac{a}{c} \Big| \frac{b}{d} = a \cdot d - b \cdot c$ , what is the value of  $\frac{3}{1} \Big| \frac{4}{2}$ ?

(A)  $-2$     (B)  $-1$     (C)  $0$     (D)  $1$     (E)  $2$

**Answer:** E

### Solution

Plugging in values for  $a$ ,  $b$ ,  $c$ , and  $d$ , we get

$$a = 3, b = 4, c = 1, d = 2,$$

$$a \times d = 3 \times 2 = 6 \quad b \times c = 4 \times 1 = 4 \quad 6 - 4 = 2 \quad \boxed{E}$$

3. (AMC 8 1998)  $\frac{\frac{3}{8} + \frac{7}{8}}{\frac{4}{5}} =$  (A) 1    (B)  $\frac{25}{16}$     (C) 2    (D)  $\frac{43}{20}$     (E)  $\frac{47}{16}$

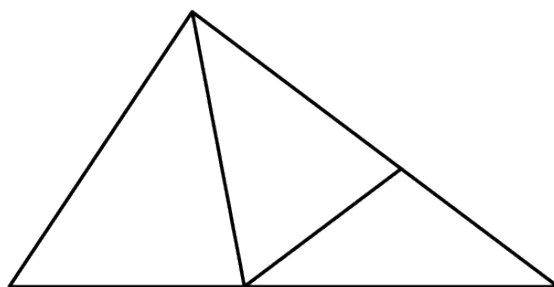
**Answer:** B

### Solution

$$\frac{\frac{3}{8} + \frac{7}{8}}{\frac{4}{5}} = \frac{\frac{10}{8}}{\frac{4}{5}} = \frac{\frac{5}{4}}{\frac{4}{5}} = \frac{5}{4} \cdot \frac{5}{4} = \frac{25}{16} = \boxed{B}$$



4. (*AMC 8 1998*) How many triangles are in this figure? (Some triangles may overlap other triangles.)



- (A) 9      (B) 8      (C) 7      (D) 6      (E) 5

**Answer:** E

**Solution**

By inspection, we have that there 5 triangles: Each of the 3 small triangles, 1 medium triangle made of the rightmost two small triangles, and the 1 large triangle.

E

5. (*AMC 8 1998*) Which of the following numbers is largest?

- (A) 9.12344      (B) 9.123 $\overline{4}$       (C) 9.12 $\overline{34}$       (D) 9. $\overline{1234}$       (E) 9. $\overline{1234}$

**Answer:** B

**Solution 1**

The first six decimal places of the answer choices are as follows:

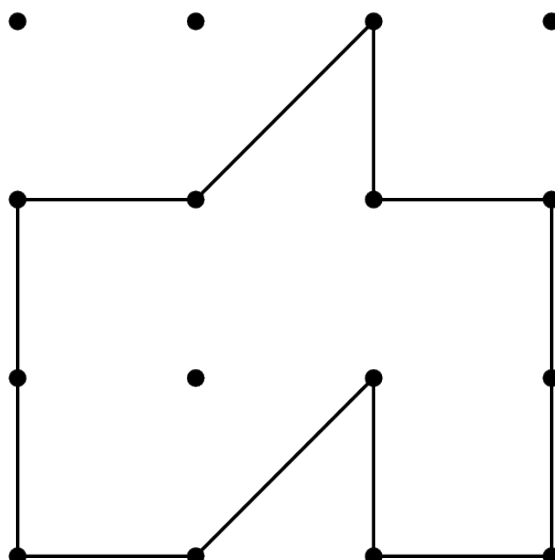
- (A): 9.123440
- (B): 9.123444
- (C): 9.123434
- (D): 9.123423
- (E): 9.123412

So (B) 9.123 $\overline{4}$  is the largest of these numbers.

**Solution 2**

Looking at the fifth digit after the decimal of each, we find that choices *C*, *D*, and *E* are eliminated. Since *B* has a non-zero sixth digit after the decimal, unlike *A*, we conclude that B is the largest number.

6. (*AMC 8 1998*) Dots are spaced one unit apart, horizontally and vertically. The number of square units enclosed by the polygon is



- (A) 5      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** B

**Solution 1**

We could count the area contributed by each square on the  $3 \times 3$  grid:

Top-left: 0

Top: Triangle with area  $\frac{1}{2}$

Top-right: 0

Left: Square with area 1

Center: Square with area 1

Right: Square with area 1

Bottom-left: Square with area 1

Bottom: Triangle with area  $\frac{1}{2}$

Bottom-right: Square with area 1

Adding all of these together, we get 6 which is the same as B

**Solution 2**

By Pick's Theorem, we get the formula,  $A = I + \frac{b}{2} - 1$  where  $I$  is the number of lattice points in the interior and  $b$  being the number of lattice points on the boundary. In this problem, we can see that  $I = 1$  and  $B = 12$ . Substituting gives us  $A = 1 + \frac{12}{2} - 1 = 6$  Thus, the answer is (B) 6

**Solution 3**

Notice that the extra triangle on the top with area  $1/2$  can be placed (like a jigsaw puzzle) at the bottom of the grid where there is a triangular hole, also with area  $1/2$ . This creates a  $2 \times 3$  rectangle, with a area of 6. The answer is (B) 6 sakshamsethi

7. (AMC 8 1998)  $100 \times 19.98 \times 1.998 \times 1000 =$  (A)  $(1.998)^2$       (B)  $(19.98)^2$       (C)  $(199.8)^2$       (D)  $(1998)^2$

**Answer:** D

**Solution**

$$19.98 \times 100 = 1998 \quad 1.998 \times 1000 = 1998 \quad 1998 \times 1998 = (1998)^2 = \boxed{D}$$

8. (AMC 8 1998) A child's wading pool contains 200 gallons of water. If water evaporates at the rate of 0.5 gallons per day and no other water is added or removed, how many gallons of water will be in the pool after 30 days?

(A) 140      (B) 170      (C) 185      (D) 198.5      (E) 199.85

**Answer:** C

**Solution**

Over the 30 days, a total of  $30 \cdot 0.5 = 15$  gallons of water evaporate. So  $200 - 15 =$  (C) 185 gallons remain.

9. (AMC 8 1998) For a sale, a store owner reduces the price of a \$10 scarf by 20%. Later the price is lowered again by half the reduced price. The price is now

(A) 2.00 dollars      (B) 3.75 dollars      (C) 4.00 dollars      (D) 4.90 dollars      (E) 6.40 dollars

**Answer:** C

**Solution 1**

After the first discount, the price of the scarf is  $0.8 \cdot \$10 = \$8$ .

After the second discount, the price is  $\frac{1}{2} \cdot \$8 =$  (C) \$4.00.

**Solution 2**

The first discount multiplies the price by  $1 - 0.2 = 0.8$ , while the second discount multiplies the price by  $\frac{1}{2}$ . So the two discounts multiply the price by  $0.8 \cdot \frac{1}{2} = 0.4$ , and thus the final price of the scarf is  $0.4 \cdot \$10 =$  (C) \$4.00.

10. (AMC 8 1998) Each of the letters W, X, Y, and Z represents a different integer in the set  $\{1, 2, 3, 4\}$ , but not necessarily in that order. If  $\frac{W}{X} - \frac{Y}{Z} = 1$ , then the sum of W and Y is
- (A) 3      (B) 4      (C) 5      (D) 6      (E) 7

**Answer:** E

**Solution**

There are different ways to approach this problem, and I'll start with the different factor of the numbers of the set  $\{1, 2, 3, 4\}$ .

1 has factor 1.

2 has factors 1 and 2    3 has factors 1 and 3    4 has factors 1, 2, and 4.

From here, we note that even though all numbers have the factor 1, only 4 has another factor other than 1 in the set (ie. 2)

We could therefore have one fraction be  $\frac{4}{2}$  and another  $\frac{3}{1}$ .

The sum of the numerators is  $4 + 3 = 7 =$  E

11. (AMC 8 1998) Harry has 3 sisters and 5 brothers. His sister Harriet has S sisters and B brothers. What is the product of S and B?
- (A) 8      (B) 10      (C) 12      (D) 15      (E) 18

**Answer:** C

**Solution**

Since Harry has 3 sisters and 5 brothers, his parents have a total of 3 daughters and 6 sons. Therefore, his sister Harriet has 2 sisters and 6 brothers, and thus  $S \cdot B = 2 \cdot 6 =$  (C) 12.

12. (AMC 8 1998)  $2\left(1 - \frac{1}{2}\right) + 3\left(1 - \frac{1}{3}\right) + 4\left(1 - \frac{1}{4}\right) + \cdots + 10\left(1 - \frac{1}{10}\right) =$  (A) 45 (B) 49 (C) 50 (D)

**Answer:** A

**Solution**

Taking the first product, we have

$$\left(1 - \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{1}{2} \times 2 = 1$$

Looking at the second, we get

$$\left(1 - \frac{1}{3}\right) = \frac{2}{3} \cdot \frac{2}{3} \times 3 = 2$$

We seem to be going up by 1.

Just to check,

$$1 - \frac{1}{n} = \frac{n-1}{n} \cdot \frac{n-1}{n} \times n = n - 1$$

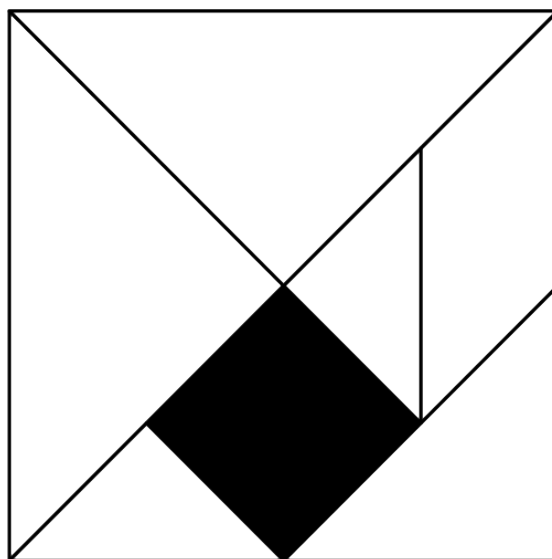
Now that we have discovered the pattern, we have to find the last term.

$$1 - \frac{1}{10} = \frac{9}{10} \cdot \frac{9}{10} \times 10 = 9$$

The sum of all numbers from 1 to 9 is

$$\frac{9 \cdot 10}{2} = 45 = \boxed{A}$$

13. (AMC 8 1998) What is the ratio of the area of the shaded square to the area of the large square? (The figure is drawn to scale)



- (A)  $\frac{1}{6}$  (B)  $\frac{1}{7}$  (C)  $\frac{1}{8}$  (D)  $\frac{1}{12}$  (E)  $\frac{1}{16}$

**Answer:** C

**Solution 1**

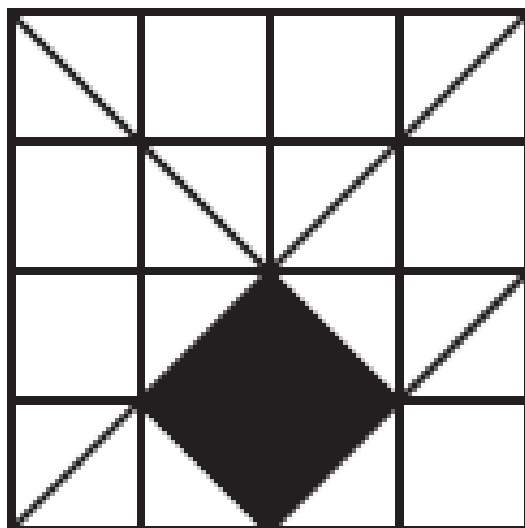
We can divide the large square into quarters by diagonals.

Then, in  $\frac{1}{4}$  the area of the big square, the little square would have  $\frac{1}{2}$  the area.

$$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8} = \boxed{C}$$

**Solution 2**

Answer: C



Divide the square into 16 smaller squares as shown. The shaded square is formed from 4 half-squares, so its area is 2. The ratio 2 to 16 is  $1/8$ .

14. (AMC 8 1998) At Annville Junior High School, 30% of the students in the Math Club are in the Science Club, and 80% of the students in the Science Club are in the Math Club. There are 15 students in the Science Club. How many students are in the Math Club?

(A) 12      (B) 15      (C) 30      (D) 36      (E) 40

**Answer:** E

**Solution**

If 80% of the people in the science club of 15 people are in the Math Club,  $\frac{4}{5} \times 15 = 12$  people are in the both the Math Club and the Science Club.

These 12 people make up 30% of the Math Club.

Setting up a proportion:

$$12 \cdot 1.00 = 0.30 \cdot x \quad \frac{12}{0.3} = 40 = x$$

There are  $40 = \boxed{E}$  people in the Math Club.

15. (AMC 8 1998) **Don't Crowd the Isles** Problems 15, 16, and 17 all refer to the following:

In the very center of the Irenic Sea lie the beautiful Nisos Isles. In 1998 the number of people on these islands is only 200, but the population triples every 25 years. Queen Irene has decreed that there must be at least 1.5 square miles for every person living in the Isles. The total area of the Nisos Isles is 24,900 square miles.

Estimate the population of Nisos in the year 2050.

(A) 600      (B) 800      (C) 1000      (D) 2000      (E) 3000

**Answer:** D

**Solution**

The population triples every 25 years from 200, and there are 50 years between 2000 and 2050, so we will have  $200 \times 3 = 600 \times 3 = 1800$ .

This is an underestimate, since there are actually 2 more years for the island's population to grow.

Therefore, 1800 can be rounded to (D) 2000

16. (AMC 8 1998) **Don't Crowd the Isles** Problems 15, 16, and 17 all refer to the following:

In the very center of the Irenic Sea lie the beautiful Nisos Isles. In 1998 the number of people on these islands is only 200, but the population triples every 25 years. Queen Irene has decreed that there must be at least 1.5 square miles for every person living in the Isles. The total area of the Nisos Isles is 24,900 square miles.

Estimate the year in which the population of Nisos will be approximately 6,000.

- (A) 2050      (B) 2075      (C) 2100      (D) 2125      (E) 2150

**Answer:** B

**Solution 1**

We could triple the population every 25 years and make a chart:

Year: 2000 Population: 200

Year: 2025 Population: 600

Year: 2050 Population: 1800

Year: 2075 Population: 5400

Year: 2100 Population: 16200

The closest year is 2075, or  $\boxed{B}$

**Solution 2**

We could find out how many periods of 25 years we need to triple by dividing our total from our present number.

$$\frac{6000}{200} = 30$$

The power of 3 that 30 is closest to is  $27 = 3^3$

Therefore, after 3 periods, we will be closest to 6000.

$\boxed{B}$

17. (AMC 8 1998) **Don't Crowd the Isles** Problems 15, 16, and 17 all refer to the following:

In the very center of the Irenic Sea lie the beautiful Nisos Isles. In 1998 the number of people on these islands is only 200, but the population triples every 25 years. Queen Irene has decreed that there must be at least 1.5 square miles for every person living in the Isles. The total area of the Nisos Isles is 24,900 square miles.

In how many years, approximately, from 1998 will the population of Nisos be as much as Queen Irene has proclaimed that the islands can support?

- (A) 50 yrs.      (B) 75 yrs.      (C) 100 yrs.      (D) 125 yrs.      (E) 150 yrs.

**Answer:** C

**Solution 1**

We can divide the total area by how much will be occupied per person:

$$\frac{24900 \text{ acres}}{1.5 \text{ acres per person}} = 16600 \text{ people can stay on the island at its maximum capacity.}$$

We can divide 16600 by the current population in 1998 which is 200 to see by what factor the population increases:

$\frac{16600}{200} = 83$ -fold increase in population.

Thus, the population increases by a factor 83. This is very close to  $3 \times 3 \times 3 \times 3 = 81$ , and so there are about 4 triplings of the island's population.

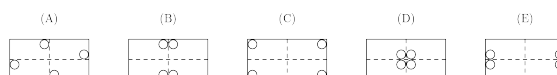
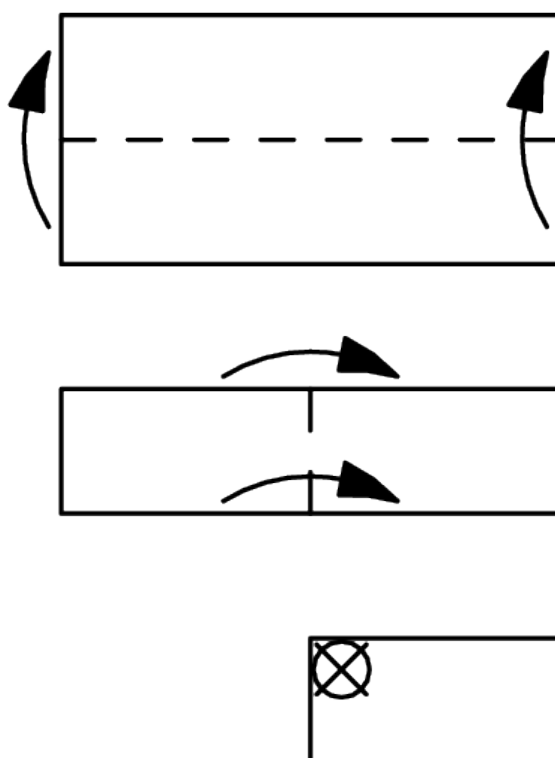
It takes  $4 \times 25 = 100 = \boxed{C}$  years to triple the island's population four times in succession.

### Solution 2

We can continue the pattern, and because the pattern increases numbers rapidly, it won't be hard.  $\frac{24900 \text{ acres}}{1.5 \text{ acres per person}} = 16600$  people can live on the island at its maximum capacity.

1998 : 200 people   2023 : 600 people   2048 : 1800 people   2073 : 5400 people   2098 : 16200 people  
After the year 2098, it will not be possible for the next increase to occur, because when 16200 is tripled, it is way more than the maximum capacity. Thus, the answer is  $2098 - 1998 = 100$ , or  $\boxed{C}$

18. (AMC 8 1998) As indicated by the diagram below, a rectangular piece of paper is folded bottom to top, then left to right, and finally, a hole is punched at X. What does the paper look like when unfolded?



**Answer:** B

### Solution 1

By reversing the folds, we find that the holes will be

- 1) In the middle (in relation from left to right)
- 2) At separating poles (up and down)

This is pictured in  $\boxed{B}$

**Solution 2**

If you unfold the paper, the hole you punched will not move. The only answer that keeps the hole in the upper-right quadrant in the same place is  $\boxed{B}$ .

**Solution 3**

You can check each answer choice by folding the paper the same way it was folded before the hole was punched. The only answer choice that has the hole in the right spot is  $\boxed{B}$ .

19. (AMC 8 1998) Tamika selects two different numbers at random from the set  $\{8, 9, 10\}$  and adds them. Carlos takes two different numbers at random from the set  $\{3, 5, 6\}$  and multiplies them. What is the probability that Tamika's result is greater than Carlos' result?

(A)  $\frac{4}{9}$       (B)  $\frac{5}{9}$       (C)  $\frac{1}{2}$       (D)  $\frac{1}{3}$       (E)  $\frac{2}{3}$

**Answer:** A

**Solution**

The different possible (and equally likely) values Tamika gets are:

$$8 + 9 = 17 \quad 8 + 10 = 18 \quad 9 + 10 = 19$$

The different possible (and equally likely) values Carlos gets are:

$$3 \times 5 = 15 \quad 3 \times 6 = 18 \quad 5 \times 6 = 30$$

The probability that if Tamika had the sum 17 her sum would be greater than Carlos's set is  $\frac{1}{3}$ , because 17 is only greater than 15

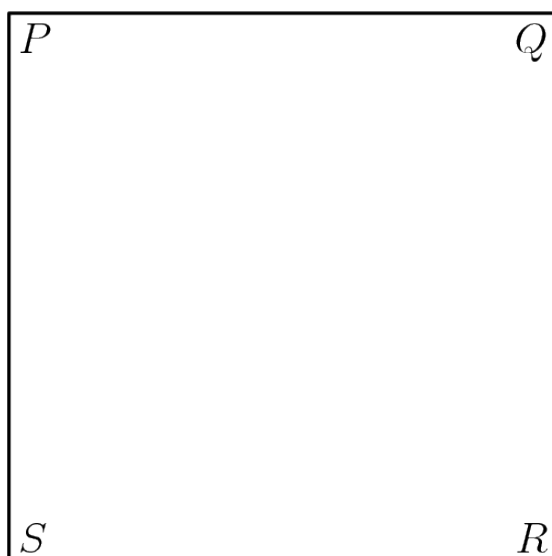
The probability that if Tamika had the sum 18 her sum would be greater than Carlos's set is  $\frac{1}{3}$ , because 18 is only greater than 15

The probability that if Tamika had the sum 19 her sum would be greater than Carlos's set is  $\frac{2}{3}$ , because 19 is greater than both 15 and 18

Each sum has a  $\frac{1}{3}$  possibility of being chosen, so we have

$$\frac{1}{3} \left( \frac{1}{3} + \frac{1}{3} + \frac{2}{3} \right) = \frac{1}{3} \left( \frac{4}{3} \right) = \frac{4}{9} = \boxed{A}$$

20. (AMC 8 1998) Let  $PQRS$  be a square piece of paper.  $P$  is folded onto  $R$  and then  $Q$  is folded onto  $S$ . The area of the resulting figure is 9 square inches. Find the perimeter of square  $PQRS$ .





- (A) 9      (B) 16      (C) 18      (D) 24      (E) 36

**Answer:** D

**Solution**

After both folds are completed, the square would become a triangle that has an area of  $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$  of the original square.

Since the area is 9 square inches for  $\frac{1}{4}$  of the square,  $9 \times 4 = 36$  square inches is the area of square  $PQRS$ .

The length of the side of a square that has an area of 36 square inches is  $\sqrt{36} = 6$  inches.

Each side is 6 inches, so the total perimeter is  $6 \times 4 = 24 = \boxed{D}$ .

21. (AMC 8 1998) A  $4 \times 4 \times 4$  cubical box contains 64 identical small cubes that exactly fill the box. How many of these small cubes touch a side or the bottom of the box?

- (A) 48      (B) 52      (C) 60      (D) 64      (E) 80

**Answer:** B

**Solution 1**

The box is filled with four  $4 \times 4$  layers of cubes.

All 16 cubes in the bottom layer touch the bottom of the box.

In each of the remaining 3 layers, all the cubes except for those in the center  $2 \times 2$  square touch the side of the box. Thus 12 cubes in each of these layers touch the box.

Therefore, a total of  $16 + 3 \cdot 12 = \boxed{(B) 52}$  cubes touch the box.

**Solution 2**

The box has a total of  $4^3 = 64$  cubes, arranged in four  $4 \times 4$  layers.

The only cubes which do not touch the box are those in the middle  $2 \times 2$  squares of each of the top 3 layers. So  $2 \cdot 2 \cdot 3 = 12$  cubes do not touch the box, and thus  $64 - 12 = \boxed{(B) 52}$  cubes touch the box.

**Solution 3**

Each small cube would have dimensions  $1 \times 1 \times 1$  making each cube a unit cube.

If there are 16 cubes per face and there are 5 faces we are counting, we have  $16 \times 5 = 80$  cubes.

Some cubes are on account of overlap between different faces.

We could reduce this number by subtracting the overlap areas, which could mean subtracting 4 cubes from each side and 12 from the bottom.

$$80 - (4 \times 4 + 12) = 80 - 28 = 52 = \boxed{B}$$

**Solution 4**

We can use casework to solve this.

Case 1: The cubes on the sides

There are four sides of a cube which means 64 cubes, but there will be an overlap of 16 cubes ( $4 \text{ sides} \times 4 \text{ cubes on each side}$ ). So there are  $16 \times 4 - 16 = 64 - 16 = 48$  cubes on the sides.

Case 2: The cubes on the bottom

There are 16 cubes on the bottom, but by counting the cubes on the sides, 12 of the cubes are already accounted for.  $16 - 12 = 4$  cubes on the bottom that aren't already counted.

$48 + 4 = \boxed{\text{(B)} 52}$  cubes touching at least one side. Since there is no top, we don't need to count the top face cubes.

22. (AMC 8 1998) Terri produces a sequence of positive integers by following three rules. She starts with a positive integer, then applies the appropriate rule to the result, and continues in this fashion.

Rule 1: If the integer is less than 10, multiply it by 9.

Rule 2: If the integer is even and greater than 9, divide it by 2.

Rule 3: If the integer is odd and greater than 9, subtract 5 from it.

A sample sequence: 23, 18, 9, 81, 76, . . .

Find the 98<sup>th</sup> term of the sequence that begins 98, 49, . . . (A) 6 (B) 11 (C) 22 (D) 27 (E) 54

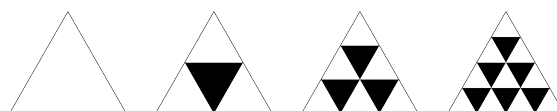
**Answer:** D

**Solution**

Since each term after the first depends solely on the previous term, the sequence will repeat the same subsequence endlessly once a term appears twice.

The first terms of the sequence are 98, 49, 44, 22, 11, 6, 54, 27, 22, and thus the 5-term subsequence 22, 11, 6, 54, 27 repeats endlessly starting at the 4<sup>th</sup> term. Since the 4<sup>th</sup> term is 22, the 94<sup>th</sup> term is also 22, and thus the 98<sup>th</sup> term is  $\boxed{\text{(D)} 27}$ .

23. (AMC 8 1998) If the pattern in the diagram continues, what fraction of the interior would be shaded in the eighth triangle?



- (A)  $\frac{3}{8}$  (B)  $\frac{5}{27}$  (C)  $\frac{7}{16}$  (D)  $\frac{9}{16}$  (E)  $\frac{11}{45}$

**Answer:** C

**Solution**

Notice that the  $n^{\text{th}}$  triangle contains  $n$  rows of congruent smaller triangles, and for each integer  $k$  from 1 to  $n$  inclusive, the  $k^{\text{th}}$  row contains  $k$  unshaded and  $k - 1$  shaded small triangles.

Therefore, the 8<sup>th</sup> triangle contains 8 rows, which contain a total of  $1 + 2 + 3 + \dots + 8 = 36$  unshaded small triangles and  $0 + 1 + 2 + \dots + 7 = 28$  shaded small triangles. Therefore,

$\frac{28}{28+36} = \frac{28}{64} = \boxed{\text{(C)} \frac{7}{16}}$  of the 8<sup>th</sup> triangle is shaded.

24. (AMC 8 1998) A rectangular board of 8 columns has squares numbered beginning in the upper left corner and moving left to right so row one is numbered 1 through 8, row two is 9 through 16, and so on. A student shades square 1, then skips one square and shades square 3, skips two squares and shades square 6, skips 3 squares and shades square 10, and continues in this way until there is at least one shaded square in each column. What is the number of the shaded square that first achieves this result?

- Answer: E**

The numbers that are shaded are the triangular numbers, which are numbers in the form  $\frac{n(n+1)}{2}$  for positive integers. They can also be generated by starting with 1, and adding 1, 2, 3, 4... as in the description of the problem.

So, instead of adding 1, 2, 3, 4, 5, 6, 7, 8, 9, 10..., we can effectively either add 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3... or subtract 7, 6, 5, 4, 3, 2, 1, 0, 7, 6, 5... if we are only concerned about remainders when divided by 8. We will pick the number that keeps the terms on the list between 1 and 8. We get:

Finally, a term with 0 is found, and checking, all numbers 1 through 7 are also on the right side of the list. This means the last term in our sequence is the first time that column 8 is shaded. There are 15 terms in the sequence, leading to an answer of  $\frac{15 \cdot 16}{2} = 120$ , which is choice **E**.

Note that the triangular numbers up to 120 are 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, 120. When you divide each of those numbers by 8, all remainders must be present. We first search for number(s) that are evenly divisible by 8; if two such numbers exist, we search for numbers that leave a remainder of 1, etc.

Quickly scanning the list, only 6, 10, 28, 36, 66, 78 and 120 are even. That smaller list doesn't have any multiples of 8 until it hits 120. So **(E)** 120 must be the answer.

### Solution 3

The student shades cells containing triangular numbers in increasing order starting from 1. Thus the first cells the student shades contain 1, 3, 6, 10, 15, 21, and 28, so they are in the first, third, sixth, second, seventh, fifth, and fourth columns respectively.

So after shading 7 cells, only the eighth column, whose cells contain multiples of 8, lacks a shaded cell. Therefore, the last cell the student shades contains the smallest triangular number divisible by 8.

Recall that the integer  $n$  is triangular if and only if  $n = \frac{k(k+1)}{2}$  for some positive integer  $k$ . So if  $n$  is divisible by 8,  $2n = k(k+1)$  is divisible by 16. Since only one of  $k$  and  $k+1$  can be even, either  $k$  or  $k+1$  must be divisible by 16. The smallest such integer  $k$  such this holds is 15, and thus the smallest triangular number which is divisible by 8 is **(E) 120**.

### Solution 4 (Answer Choices)

The squares the student shades contain the triangular numbers, so the sixth shaded square is 21, which is in the fifth column, while the seventh shaded square is 28, which is in the fourth column. So after the student has shaded 7 squares, only the eighth column, which contains all positive integers which are divisible by 8, has no unshaded squares. Therefore, the last square the student shades must contain a number divisible by 8.

Of the answer choices, only 64 and 120 are divisible by 8. Since 64 is not a triangular number, the correct answer is **(E) 120**.

25. (AMC 8 1998) Three generous friends, each with some cash, redistribute their money as follows: Amy gives enough money to Jan and Toy to double the amount that each has. Jan then gives enough to Amy and Toy to double their amounts. Finally, Toy gives Amy and Jan enough to double their amounts. If Toy has 36 when they begin and 36 when they end, what is the total amount that all three friends have?
- (A) 108 dollars      (B) 180 dollars      (C) 216 dollars      (D) 252 dollars      (E) 288 dollars

**Answer: D**

### Solution

After Ami gives money to Toy and Jan, Toy has  $2 \cdot \$36 = \$72$ .

After Jan gives money to Ami and Toy, Toy has  $2 \cdot \$72 = \$144$ .

Therefore, Toy gives  $\$144 - \$36 = \$108$  to Jan and Ami. Since Jan and Ami each doubled their money, they must have had a total of  $\$108$  before Toy gave them money and thus now have a total of  $2 \cdot \$108 = \$216$ .

Therefore, the three friends have a total of  $\$216 + \$36 = \mathbf{(D) \$252}$ .

## §15 AMC 8 1999

1. (AMC 8 1999)  $(6?3) + 4 - (2 - 1) = 5$  To make this statement true, the question mark between the 6 and the 3 should be replaced by
- (A)  $\div$       (B)  $\times$       (C)  $+$       (D)  $-$       (E) None of these

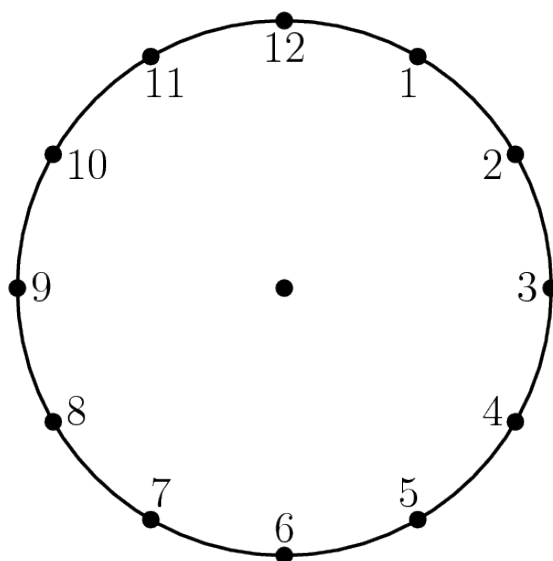
**Answer: A**

### Solution

Simplifying the given expression, we get:  $(6?3) = 2$ .

At this point, it becomes clear that it should be **(A)  $\div$** .

2. (AMC 8 1999) What is the degree measure of the smaller angle formed by the hands of a clock at 10 o'clock?



- (A) 30    (B) 45    (C) 60    (D) 75    (E) 90

**Answer:** C

**Solution 1**

At 10 : 00, the hour hand will be on the 10 while the minute hand on the 12.

This makes them  $\frac{1}{6}$ th of a circle apart, and  $\frac{1}{6} \cdot 360^\circ = \boxed{\text{(C) } 60}$ .

**Solution 2**

We know that the full clock is a circle, and therefore has 360 degrees. Considering that there are 12 numbers, the distance between one number will be  $360 \div 12 = 30$ . If the time is 10 : 00, then the hour hand will be on 10, and the minute hand will be on, 12, making them 2 numbers apart, so they will be 60 degrees apart, or  $\boxed{\text{(C) } 60}$

3. (AMC 8 1999) Which triplet of numbers has a sum NOT equal to 1?

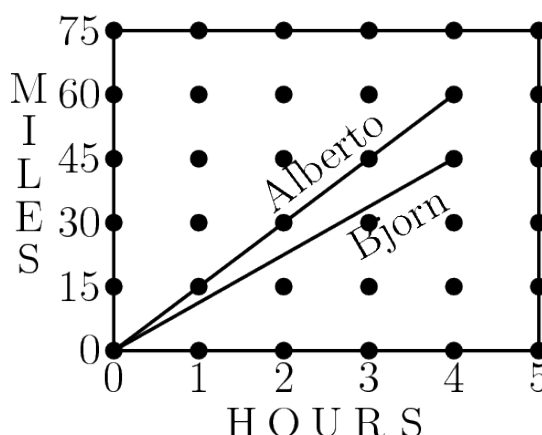
- (A)  $(1/2, 1/3, 1/6)$     (B)  $(2, -2, 1)$     (C)  $(0.1, 0.3, 0.6)$     (D)  $(1.1, -2.1, 1.0)$     (E)  $(-3/2, -5/2,$

**Answer:** D

**Solution**

By adding each triplet, we can see that  $\boxed{\text{(D)}}$  gives us 0, not 1, as our sum.

4. (AMC 8 1999) The diagram shows the miles traveled by bikers Alberto and Bjorn. After four hours, about how many more miles has Alberto biked than Bjorn?



- (A) 15      (B) 20      (C) 25      (D) 30      (E) 35

**Answer:** A

**Solution**

After 4 hours, we see that Bjorn biked 45 miles, and Alberto biked 60. Thus the answer is  $60 - 45 = 15$  A.

**Solution 2**

We see that each dot is 15 units away from the nearest one above it. So the answer is A.

5. (*AMC 8 1999*) A rectangular garden 60 feet long and 20 feet wide is enclosed by a fence. To make the garden larger, while using the same fence, its shape is changed to a square. By how many square feet does this enlarge the garden?
- (A) 100      (B) 200      (C) 300      (D) 400      (E) 500

**Answer:** D

**Solution 1**

We need the same perimeter as a 60 by 20 rectangle, but the greatest area we can get. right now the perimeter is 160. To get the greatest area while keeping a perimeter of 160, the sides should all be 40. that means an area of 1600. Right now, the area is  $20 \times 60$  which is 1200.  $1600 - 1200 = 400$  which is D.

**Solution 2**

We need the same perimeter as a 60 by 20 rectangle, but in a square. Right now the perimeter is 160, since a square has equal sides and 160 divided by 4 is 40, the sides should all be 40, that means an area of 1600. Before, the area is  $20 \times 60$  which is 1200.  $1600 - 1200 = 400$  which is D.

6. (*AMC 8 1999*) Bo, Coe, Flo, Joe, and Moe have different amounts of money. Neither Joe nor Bo has as much money as Flo. Both Bo and Coe have more than Moe. Joe has more than Moe, but less than Bo. Who has the least amount of money?
- (A) Bo      (B) Coe      (C) Flo      (D) Joe      (E) Moe

**Answer:** E

**Solution**

Use logic to solve this problem. You don't actually need to use any equations.

Neither Jo nor Bo has as much money as Flo. So Flo clearly does not have the least amount of money. Rule out Flo.

Both Bo and Coe have more than Moe. Rule out Bo and Coe; they clearly do not have the least amount of money.

Jo has more than Moe. Rule out Jo.

The only person who has not been ruled out is Moe. So **(E)** is the answer.

7. (AMC 8 1999) The third exit on a highway is located at milepost 40 and the tenth exit is at milepost 160. There is a service center on the highway located three-fourths of the way from the third exit to the tenth exit. At what milepost would you expect to find this service center?

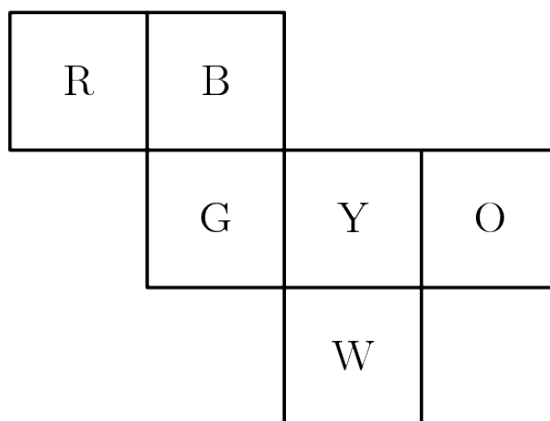
(A) 90      (B) 100      (C) 110      (D) 120      (E) 130

**Answer:** E

**Solution**

There are  $160 - 40 = 120$  miles between the third and tenth exits, so the service center is at milepost  $40 + (3/4)(120) = 40 + 90 = \mathbf{(E) 130}$ .

8. (AMC 8 1999) Six squares are colored, front and back, (R = red, B = blue, O = orange, Y = yellow, G = green, and W = white). They are hinged together as shown, then folded to form a cube. The face opposite the white face is



(A) B      (B) G      (C) O      (D) R      (E) Y

**Answer:** A

**Solution 1**

When G is arranged to be the base, B is the back face and W is the front face. Thus, **(A) B** is opposite W.

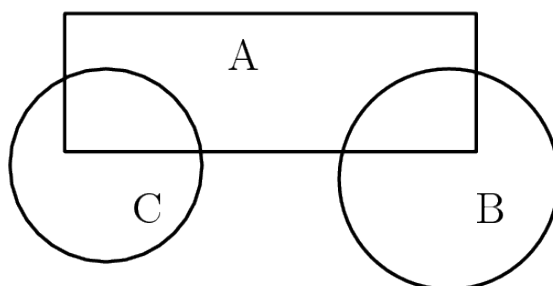
**Solution 2**

Let Y be the top and fold G, O, and W down. Then **(A) B** will fold to become the back face and be opposite W.

**Solution 3 (Last Resort)**

If you're out of time or checking your work, I guess you can just rip a net from some paper (no scissors though). Like I said, this is a last resort and not very convenient. **(A) B** would be the answer.

9. (*AMC 8 1999*) Three flower beds overlap as shown. Bed A has 500 plants, bed B has 450 plants, and bed C has 350 plants. Beds A and B share 50 plants, while beds A and C share 100. The total number of plants is



- (A) 850      (B) 1000      (C) 1150      (D) 1300      (E) 1450

**Answer:** C

**Solution 1**

Plants shared by two beds have been counted twice, so the total is  $500 + 450 + 350 - 50 - 100 = \boxed{\text{(C) } 1150}$ .

**Solution 2**

Bed A has 350 plants it doesn't share with B or C. Bed B has 400 plants it doesn't share with A or C. And C has 250 it doesn't share with A or B. The total is  $350 + 400 + 250 + 50 + 100 = \boxed{\text{(C) } 1150}$  plants.

10. (*AMC 8 1999*) A complete cycle of a traffic light takes 60 seconds. During each cycle the light is green for 25 seconds, yellow for 5 seconds, and red for 30 seconds. At a randomly chosen time, what is the probability that the light will NOT be green?

- (A)  $\frac{1}{4}$       (B)  $\frac{1}{3}$       (C)  $\frac{5}{12}$       (D)  $\frac{1}{2}$       (E)  $\frac{7}{12}$

**Answer:** E

**Solution 1**

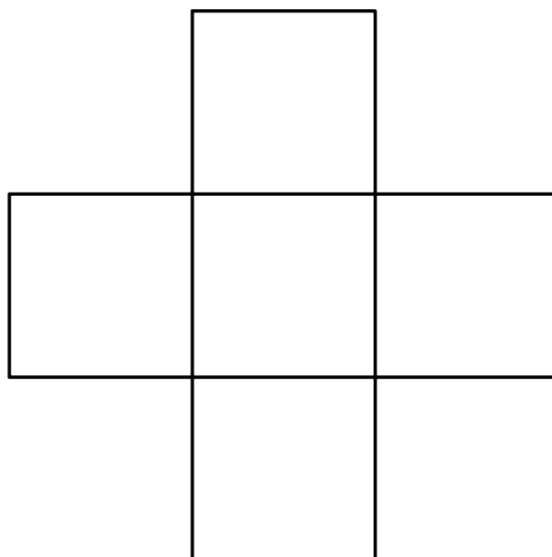
$$\frac{\text{time not green}}{\text{total time}} = \frac{R + Y}{R + Y + G} = \frac{35}{60} = \boxed{\text{(E) } \frac{7}{12}}$$

**Solution 2**

The probability of green is  $\frac{25}{60} = \frac{5}{12}$ , so the probability of not green is  $1 - \frac{5}{12} = \boxed{\text{(E) } \frac{7}{12}}$ .

11. (*AMC 8 1999*) Each of the five numbers 1, 4, 7, 10, and 13 is placed in one of the five squares so that the sum of the three numbers in the horizontal row equals the sum of the three numbers in the vertical column. The largest possible value for the horizontal or vertical sum is





- (A) 20      (B) 21      (C) 22      (D) 24      (E) 30

**Answer:** D

**Solution 1**

The largest sum occurs when 13 is placed in the center. This sum is  $13 + 10 + 1 = 13 + 7 + 4 = \boxed{\text{(D) } 24}$ . Note: Two other common sums, 18 and 21, are also possible.

**Solution 2**

Since the horizontal sum equals the vertical sum, twice this sum will be the sum of the five numbers plus the number in the center. When the center number is 13, the sum is the largest,

$$[10 + 4 + 1 + 7 + 2(13)] = 2S \quad 48 = 2S \quad S = \boxed{\text{(D) } 24}$$

The other four numbers are divided into two pairs with equal sums.

12. (*AMC 8 1999*) The ratio of the number of games won to the number of games lost (no ties) by the Middle School Middies is  $11/4$ . To the nearest whole percent, what percent of its games did the team lose?
- (A) 24      (B) 27      (C) 36      (D) 45      (E) 73

**Answer:** B

**Solution**

The ratio means that for every 11 games won, 4 are lost, so the team has won  $11x$  games, lost  $4x$  games, and played  $15x$  games for some positive integer  $x$ . The percentage of games lost is just  $\frac{4x}{15x} \times 100 = \frac{4}{15} \times 100 = 26.\bar{6}\% \approx \boxed{\text{(B) } 27\%}$

**Alternate Solution** The Won/Lost ratio is  $11/4$  so, for some number  $N$ , the team won  $11N$  games and lost  $4N$  games. Thus, the team played  $15N$  games and the fraction of games lost is  $\frac{4N}{15N} = \frac{4}{15} \approx 0.27 = \boxed{27\%}$ . -Clara Garza

13. (*AMC 8 1999*) The average age of the 40 members of a computer science camp is 17 years. There are 20 girls, 15 boys, and 5 adults. If the average age of the girls is 15 and the average age of the boys is 16, what is the average age of the adults?

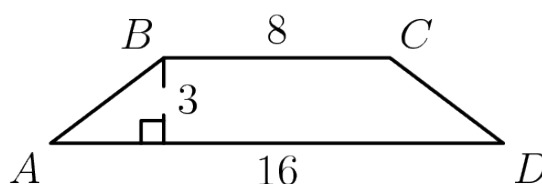
- (A) 26      (B) 27      (C) 28      (D) 29      (E) 30

**Answer:** C

**Solution**

First, find the total amount of the girl's ages and add it to the total amount of the boy's ages. It equals  $(20)(15) + (15)(16) = 540$ . The total amount of everyone's ages can be found from the average age,  $17 \cdot 40 = 680$ . Then you do  $680 - 540 = 140$  to find the sum of the adult's ages. The average age of an adult is divided among the five of them,  $140 \div 5 = \boxed{(C) 28}$ .

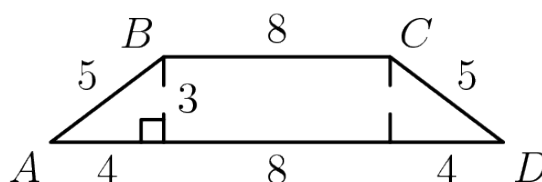
14. (AMC 8 1999) In trapezoid  $ABCD$ , the sides  $AB$  and  $CD$  are equal. The perimeter of  $ABCD$  is



- (A) 27      (B) 30      (C) 32      (D) 34      (E) 48

**Answer:** D

**Solution**



There is a rectangle present, with both horizontal bases being 8 units in length. The excess units on the bottom base must then be  $16 - 8 = 8$ . The fact that  $AB$  and  $CD$  are equal in length indicate, by the Pythagorean Theorem, that these excess lengths are equal. There are two with a total length of 8 units, so each is 4 units. The triangle has a hypotenuse of 5, because the triangles are 3-4-5 right triangles. So, the sides of the trapezoid are 8, 5, 16, and 5. Adding those up gives us the perimeter,  $8 + 5 + 16 + 5 = 13 + 21 = \boxed{(D) 34}$  units.

15. (AMC 8 1999) Bicycle license plates in Flatville each contain three letters. The first is chosen from the set C,H,L,P,R, the second from A,I,O, and the third from D,M,N,T.

When Flatville needed more license plates, they added two new letters. The new letters may both be added to one set or one letter may be added to one set and one to another set. What is the largest possible number of ADDITIONAL license plates that can be made by adding two letters?

- (A) 24      (B) 30      (C) 36      (D) 40      (E) 60

**Answer:** D

**Solution 1 (Complementary Counting)**

There are currently 5 choices for the first letter, 3 choices for the second letter, and 4 choices for the third letter, for a total of  $5 \cdot 3 \cdot 4 = 60$  license plates.

Adding 2 letters to the start gives  $7 \cdot 3 \cdot 4 = 84$  plates.

Adding 2 letters to the middle gives  $5 \cdot 5 \cdot 4 = 100$  plates.

Adding 2 letters to the end gives  $5 \cdot 3 \cdot 6 = 90$  plates.

Adding a letter to the start and middle gives  $6 \cdot 4 \cdot 4 = 96$  plates.

Adding a letter to the start and end gives  $6 \cdot 3 \cdot 5 = 90$  plates.

Adding a letter to the middle and end gives  $5 \cdot 4 \cdot 5 = 100$  plates.

You can get at most 100 license plates total, giving an additional  $100 - 60 = 40$  plates, making the answer  $\boxed{D}$

### Solution 2

Using the same logic as above, the number of combinations of plates is simply the product of the size of each set of letters.

In general, when three numbers have the same fixed sum, their product will be maximal when they are as close together as possible. This is a 3D analogue of the fact that a rectangle with fixed perimeter maximizes its area when the sides are equal (ie when it becomes a square). In this case, no matter where you add the letters, there will be  $5 + 3 + 4 + 2 = 14$  letters in total. If you divide them as evenly as possible among the three groups, you get 5, 5, 4, which is a possible situation.

As before, the answer is  $5 \cdot 5 \cdot 4 - 5 \cdot 3 \cdot 4 = 40$ , and the correct choice is  $\boxed{D}$

### Solution 3

Before new letters were added, five different letters could have been chosen for the first position, three for the second, and four for the third. This means that  $5 \cdot 3 \cdot 4 = 60$  plates could have been made.

If two letters are added to the second set, then  $5 \cdot 5 \cdot 4 = 100$  plates can be made. If one letter is added to each of the second and third sets, then  $5 \cdot 4 \cdot 5 = 100$  plates can be made. None of the other four ways to place the two letters will create as many plates. So,  $100 - 60 = \boxed{40}$  ADDITIONAL plates can be made. So the correct choice is  $\boxed{D}$

16. (AMC 8 1999) Tori's mathematics test had 75 problems: 10 arithmetic, 30 algebra, and 35 geometry problems. Although she answered 70% of the arithmetic, 40% of the algebra, and 60% of the geometry problems correctly, she did not pass the test because she got less than 60% of the problems right. How many more problems would she have needed to answer correctly to earn a 60% passing grade?
- (A) 1      (B) 5      (C) 7      (D) 9      (E) 11

**Answer:** B

### Solution

First, calculate how many of each type of problem she got right:

Arithmetic:  $70\% \cdot 10 = 0.70 \cdot 10 = 7$

Algebra:  $40\% \cdot 30 = 0.40 \cdot 30 = 12$

Geometry:  $60\% \cdot 35 = 0.60 \cdot 35 = 21$

Altogether, Tori answered  $7 + 12 + 21 = 40$  questions correct. To get a 60% on her test overall, she needed to get  $60\% \cdot 75 = 0.60 \cdot 75 = 45$  questions right.

Therefore, she needed to answer  $45 - 40 = 5$  more questions to pass, so the correct answer is  $\boxed{(B)5}$

17. (*AMC 8 1999*) Problems 17, 18, and 19 refer to the following:

At Central Middle School the 108 students who take the 1999 AMC 8 meet in the evening to talk about problems and eat an average of two cookies apiece. Walter and Gretel are baking Bonnie's Best Bar Cookies this year. Their recipe, which makes a pan of 15 cookies, lists these items:  $1\frac{1}{2}$  cups flour, 2 eggs, 3 tablespoons butter,  $\frac{3}{4}$  cups sugar, and 1 package of chocolate drops. They will make only full recipes, not partial recipes.

Walter can buy eggs by the half-dozen. How many half-dozens should he buy to make enough cookies? (Some eggs and some cookies may be left over.)

- (A) 1      (B) 2      (C) 5      (D) 7      (E) 15

**Answer:** C

**Solution**

If 108 students eat 2 cookies on average, there will need to be  $108 \cdot 2 = 216$  cookies. There are 15 cookies per pan, meaning there needs to be  $\frac{216}{15} = 14.4$  pans. However, since half-recipes are forbidden, we need to round up and make  $\lceil \frac{216}{15} \rceil = 15$  pans.

1 pan requires 2 eggs, so 15 pans require  $2 \cdot 15 = 30$  eggs. Since there are 6 eggs in a half dozen, we need  $\frac{30}{6} = 5$  half-dozens of eggs, and the answer is  $\boxed{C}$ .

18. (*AMC 8 1999*) Problems 17, 18, and 19 refer to the following:

At Central Middle School the 108 students who take the AMC 8 meet in the evening to talk about problems and eat an average of two cookies apiece. Walter and Gretel are baking Bonnie's Best Bar Cookies this year. Their recipe, which makes a pan of 15 cookies, lists these items:  $1\frac{1}{2}$  cups flour, 2 eggs, 3 tablespoons butter,  $\frac{3}{4}$  cups sugar, and 1 package of chocolate drops. They will make only full recipes, not partial recipes.

They learn that a big concert is scheduled for the same night and attendance will be down 25%. How many recipes of cookies should they make for their smaller party?

- (A) 6      (B) 8      (C) 9      (D) 10      (E) 11

**Answer:** E

**Solution 1**

If 108 students eat 2 cookies on average, there will need to be  $108 \cdot 2 = 216$  cookies. But with the smaller attendance, you will only need  $100\% - 25\% = 75\%$  of these cookies, or  $75\% \cdot 216 = 0.75 \cdot 216 = 162$  cookies.

162 cookies requires  $\frac{162}{15} = 10.8$  batches. However, since half-batches are forbidden, we must round up to get  $\lceil \frac{162}{15} \rceil = 11$  batches, and the correct answer is  $\boxed{E}$ .

**Solution 2**

If there were 108 students before, with the 25% of students missing, there will be 75% of 108 students left. This is  $75\% \cdot 108 = 0.75 \cdot 108 = 81$  students. These students eat  $81 \cdot 2 = 162$  cookies. Follow the logic of the second paragraph above to find that there needs to be 11 batches, and the correct answer is  $\boxed{E}$ .

19. (*AMC 8 1999*) Problems 17, 18, and 19 refer to the following:

At Central Middle School the 108 students who take the AMC 8 meet in the evening to talk about problems and eat an average of two cookies apiece. Walter and Gretel

are baking Bonnie's Best Bar Cookies this year. Their recipe, which makes a pan of 15 cookies, lists these items:  $1\frac{1}{2}$  cups flour, 2 eggs, 3 tablespoons butter,  $\frac{3}{4}$  cups sugar, and 1 package of chocolate drops. They will make only full recipes, not partial recipes.

The drummer gets sick. The concert is cancelled. Walter and Gretel must make enough pans of cookies to supply 216 cookies. There are 8 tablespoons in a stick of butter. How many sticks of butter will be needed? (Some butter may be left over, of course.)

- (A) 5      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** B

**Solution**

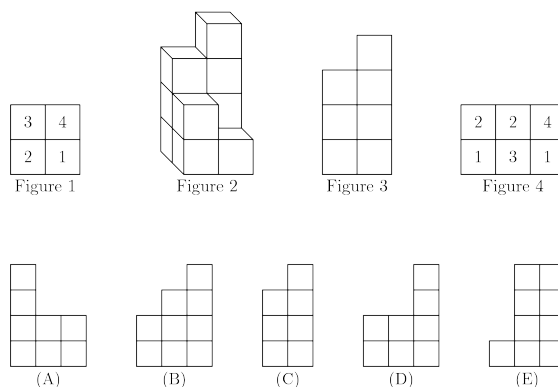
For 216 cookies, you need to make  $\frac{216}{15} = 14.4$  pans. Since fractional pans are forbidden, round up to make  $\lceil \frac{216}{15} \rceil = 15$  pans.

There are 3 tablespoons of butter per pan, meaning  $3 \cdot 15 = 45$  tablespoons of butter are required for 15 pans.

Each stick of butter has 8 tablespoons, so we need  $\frac{45}{8} = 5.625$  sticks of butter. However, we must round up again because partial sticks of butter are forbidden! Thus, we need  $\lceil \frac{45}{8} \rceil = 6$  sticks of butter, and the answer is **B**.

20. (AMC 8 1999) Figure 1 is called a "stack map." The numbers tell how many cubes are stacked in each position. Fig. 2 shows these cubes, and Fig. 3 shows the view of the stacked cubes as seen from the front.

Which of the following is the front view for the stack map in Fig. 4?



**Answer:** B

**Solution**

The third view is a direct, head-on view of the cubes. Thus, you will only see the highest (or, in these cases, higher) tower in each up-down column. For figure 4:

The highest tower in the first up-down column is 2 in the upper-left box.

The highest tower in the second up-down column is the 3 in the lower-middle box.

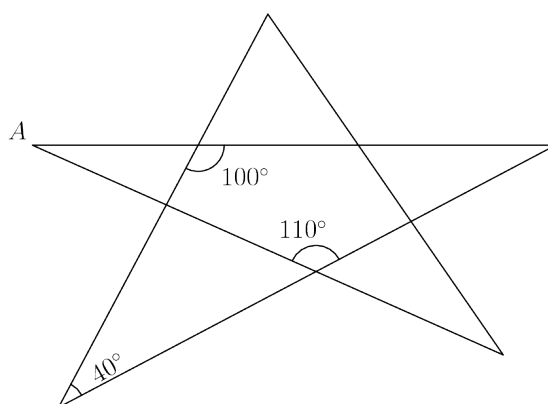
The highest tower in the third up-down column is the 4 in the upper-right box.

Thus, the head-on view of this tower should have 2 boxes on the left, 3 in the middle, and 4 on the right.

Diagram **B** shows this description.

As trivia, option C shows the stack from the point of view of an observer on the right, facing towards the left.

21. (AMC 8 1999) The degree measure of angle  $A$  is



- (A) 20      (B) 30      (C) 35      (D) 40      (E) 45

**Answer:** B

**Solution 1**

Angle-chasing using the small triangles:

Use the line below and to the left of the  $110^\circ$  angle to find that the rightmost angle in the small lower-left triangle is  $180 - 110 = 70^\circ$ .

Then use the small lower-left triangle to find that the remaining angle in that triangle is  $180 - 70 - 40 = 70^\circ$ .

Use congruent vertical angles to find that the lower angle in the smallest triangle containing  $A$  is also  $70^\circ$ .

Next, use line segment  $AB$  to find that the other angle in the smallest triangle containing  $A$  is  $180 - 100 = 80^\circ$ .

The small triangle containing  $A$  has a  $70^\circ$  angle and an  $80^\circ$  angle. The remaining angle must be  $180 - 70 - 80 = 30^\circ$ , **B**

**Solution 2 (Quick and Easy, using larger triangles)**

The third angle of the triangle containing the  $100^\circ$  angle and the  $40^\circ$  angle is  $180^\circ - 100^\circ - 40^\circ = 40^\circ$ . It follows that  $A$  is the third angle of the triangle consisting of the found  $40^\circ$  angle and the given  $110^\circ$  angle. Thus,  $A$  is a  $180^\circ - 110^\circ - 40^\circ = 30^\circ$  angle, and so the answer is **30°, B**.

22. (AMC 8 1999) In a far-off land three fish can be traded for two loaves of bread and a loaf of bread can be traded for four bags of rice. How many bags of rice is one fish worth?

- (A)  $\frac{3}{8}$       (B)  $\frac{1}{2}$       (C)  $\frac{3}{4}$       (D)  $2\frac{2}{3}$       (E)  $3\frac{1}{3}$

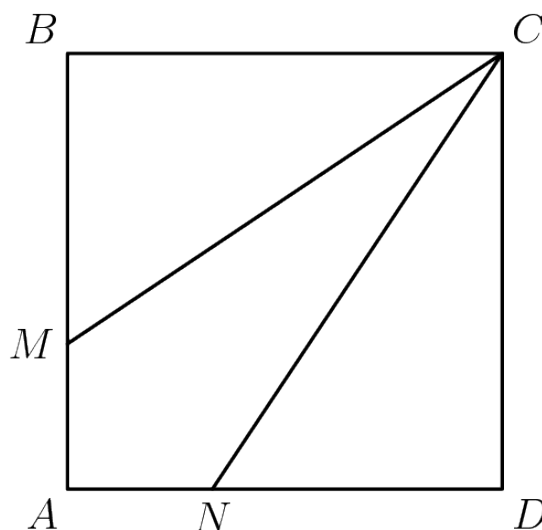
**Answer:** D

**Solution**

Let  $f$  represent one fish,  $l$  a loaf of bread, and  $r$  a bag of rice. Then:  $3f = 2l$ ,  $l = 4r$

Substituting  $l$  from the second equation back into the first gives us  $3f = 8r$ . So each fish is worth  $\frac{8}{3}$  bags of rice, or  $2\frac{2}{3} \Rightarrow$  **D**.

23. (AMC 8 1999) Square  $ABCD$  has sides of length 3. Segments  $CM$  and  $CN$  divide the square's area into three equal parts. How long is segment  $CM$ ?



- (A)  $\sqrt{10}$     (B)  $\sqrt{12}$     (C)  $\sqrt{13}$     (D)  $\sqrt{14}$     (E)  $\sqrt{15}$

**Answer:** C

**Solution**

Since the square has side length 3, the area of the entire square is 9.

The segments divide the square into 3 equal parts, so the area of each part is  $9 \div 3 = 3$ .

Since  $\triangle CBM$  has area 3 and base  $CB = 3$ , using the area formula for a triangle:

$$A_{tri} = \frac{1}{2}bh \quad 3 = \frac{1}{2}3h \quad h = 2$$

Thus, height  $BM = 2$ .

Since  $\triangle CBM$  is a right triangle,  $CM = \sqrt{BM^2 + BC^2} = \sqrt{2^2 + 3^2} = \boxed{\text{(C) } \sqrt{13}}$ .

**Solution 2**

Connect  $AC$ ,  $S_{\triangle AMC} = S_{\triangle ANC}$ . To satisfied the three area is equal, we have  $2S_{\triangle AMC} = S_{\triangle BMC}$ ,  $2S_{\triangle ANC} = S_{\triangle DNC}$ . Thus,  $AM = AN = \frac{1}{2}BM = \frac{1}{2}ND = 1$ .  $BM = 2, BC = 3, MC = \boxed{\text{(C) } \sqrt{13}}$ .

24. (AMC 8 1999) When  $1999^{2000}$  is divided by 5, the remainder is

- (A) 0    (B) 1    (C) 2    (D) 3    (E) 4

**Answer:** B

**Solution 1**

Note that the units digits of the powers of 9 have a pattern:  $9^1 = 9$ ,  $9^2 = 81$ ,  $9^3 = 729$ ,  $9^4 = 6561$ , and so on. Since all natural numbers with the same last digit have the same remainder when divided by 5, the entire number doesn't matter, just the last digit. For even powers of 9, the number ends in a 1. Since the exponent is even, the final digit is 1. Note that all natural numbers that end in 1 have a remainder of 1 when divided by 5. So, our answer is  $\boxed{\text{(B) } 1}$ .

**Solution 2**

Write 1999 as  $2000 - 1$ . We are taking  $(2000 - 1)^{2000} \pmod{10}$ . Using the binomial theorem, we see that ALL terms in this expansion are divisible by 2000, which is in turn divisible by 5, except for the very last term, which is just  $(-1)^{2000}$ . This is clear because the binomial expansion is just choosing how many 2000s and how many  $-1$ s there are for each term. Using this, we can take the entire polynomial mod 10, which leaves just  $(-1)^{2000} = \boxed{(B) 1}$ .

**Solution 3**

As  $1999 \equiv -1 \pmod{5}$ , we have  $1999^{2000} \equiv (-1)^{2000} \equiv 1 \pmod{5}$ . Thus, the answer is  $\boxed{(B) 1}$ .

**Solution 4**

A sum/product of any two natural numbers has the same remainder, when divided by 5, as the sum/product of their remainders. Thus, we can use the basic definition of an exponent and view the problem as  $1999 \cdot 1999 \cdots 1999$ .

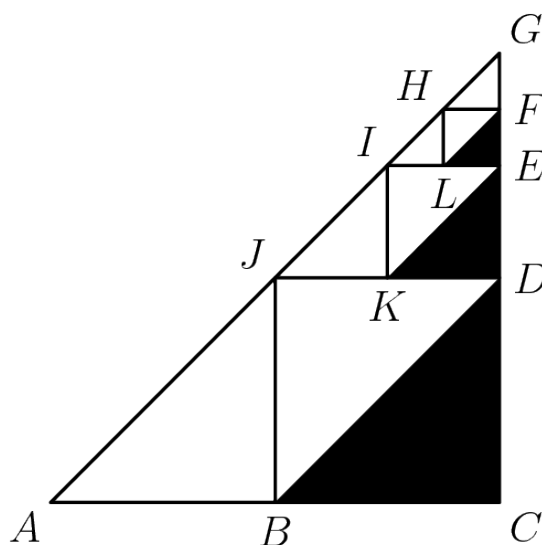
Using the fact stated in the first sentence, we see that the remainder of 1999, when divided by 5, is 4. The problem can now be viewed as finding the remainder of  $4^{2000}$  when it is divided by 5, which is already much simpler.

Now, toying with the simplified problem a little, see notice that powers of 4 alternate from ending in 4 and 6. We notice that even powers of 4 always end in 6, and also that 2000 is even. Thus  $4^{2000}$  must end in 6, which, when divided by 5 gives a remainder of  $\boxed{(B) 1}$ .

**Solution 5**

By Fermat's Little Theorem,  $1999^{2000} \equiv 1999^{2000 \pmod{5-1}} \equiv 1999^0 \equiv 1 \pmod{5}$ . Thus the answer is  $\boxed{(B) 1}$ .

25. (AMC 8 1999) Points  $B$ ,  $D$ , and  $J$  are midpoints of the sides of right triangle  $ACG$ . Points  $K$ ,  $E$ ,  $I$  are midpoints of the sides of triangle  $JDG$ , etc. If the dividing and shading process is done 100 times (the first three are shown) and  $AC = CG = 6$ , then the total area of the shaded triangles is nearest



- (A) 6      (B) 7      (C) 8      (D) 9      (E) 10

**Answer: A**



### Solution 1

Since  $\triangle FGH$  is fairly small relative to the rest of the diagram, we can make an underestimate by using the current diagram. All triangles are right-isosceles triangles.

$$CD = \frac{CG}{2} = 3, DE = \frac{CD}{2} = \frac{3}{2}, EF = \frac{DE}{2} = \frac{3}{4}$$

$$CB = CD = 3, DK = DE = \frac{3}{2}, EL = EF = \frac{3}{4}$$

$$[CBD] = \frac{1}{2}3^2 = \frac{9}{2}$$

$$[DKE] = \frac{1}{2}\left(\frac{3}{2}\right)^2 = \frac{9}{8}$$

$$[ELF] = \frac{1}{2}\left(\frac{3}{4}\right)^2 = \frac{9}{32}$$

The sum of the shaded regions is  $\frac{9}{2} + \frac{9}{8} + \frac{9}{32} = \frac{189}{32} \approx 5.9$ . 5.9 is an underestimate, as some portion (but not all) of  $\triangle FGH$  will be shaded in future iterations.

If you shade all of  $\triangle FGH$ , this will add an additional  $\frac{9}{32}$  to the area, giving  $\frac{198}{32} \approx 6.2$ , which is an overestimate.

Thus,  $\boxed{(A) 6}$  is the only answer that is both over the underestimate and under the overestimate.

### Solution 2

In iteration 1, congruent triangles  $\triangle ABJ$ ,  $\triangle BDJ$ , and  $\triangle BDC$  are created, with one of them being shaded.

In iteration 2, three more congruent triangles are created, with one of them being shaded.

As the process continues indefinitely, in each row,  $\frac{1}{3}$  of each triplet of new congruent triangles will be shaded. The "fourth triangle" at the top ( $\triangle FGH$  in the diagram) will gradually shrink,

leaving about  $\frac{1}{3}$  of the area shaded. This means  $\frac{1}{3} \left( \frac{1}{2}6 \cdot 6 \right) = 6$  square units will be shaded when the process goes on indefinitely, giving  $\boxed{A}$ .

### Solution 3

Using Solution 1 as a template, note that the sum of the areas forms a **geometric series**:

$$\frac{9}{2} + \frac{9}{8} + \frac{9}{32} + \frac{9}{128} + \dots$$

This is the sum of a geometric series with first term  $a_1 = \frac{9}{2}$  and common ratio  $r = \frac{1}{4}$ . This is the easiest way to do this problem.

The sum of an infinite geometric series with  $|r| < 1$  is shown by the formula.  $S_\infty = \frac{a_1}{1-r}$ . Insert the values to get  $\frac{\frac{9}{2}}{1-\frac{1}{4}} = \frac{9}{2} \cdot \frac{4}{3} = 6$ , giving an answer of  $\boxed{A}$ .

### Solution 4

Find the area of the bottom triangle, which is 4.5. Notice that the area of the triangles is divided by 4 every time.  $4.5 \cdot 5/4 \approx 5.7$ , and  $5.7 + (1/4) \approx 5.9$ . We can clearly see that the sum is approaching

### Solution 5

The area of  $DBC$  is clearly  $\frac{1}{3}$  of  $AJDC$ , the area of  $DEK$  is  $\frac{1}{3}$  of  $JIED$ , if the process is going to infinity, the shaded triangles will be  $\frac{1}{3}$  of the triangle  $ACG$ . However, 100 times is much enough. The answer is  $\frac{1}{3} \times 6 \times 6 \times \frac{1}{2} = \boxed{(A)6}$ .

### Solution 6

Because all of the triangles are isosceles, this gives us the area of one triangle with side length  $s$ ,  $A = \frac{1}{2}s^2$ . Logically,  $s$  keeps on getting  $\frac{1}{2}$  as small as the previous one ( $3, \frac{3}{2}, \frac{3}{4}$ , etc) Meaning for triangle number  $n$ ,  $s = 3(\frac{1}{2})^n$ . Putting this into our original equation,  $A = \frac{1}{2}(3(\frac{1}{2})^n)^2 = \frac{9}{2}(\frac{1}{2})^{2n} = 9(\frac{1}{2})^{2n+1}$ . This means the total area is  $9 \sum_{n=1}^{\infty} \frac{1}{2}^{2n+1}$ , using exponent laws, we can simplify this to  $\frac{9}{2} \sum_{n=1}^{\infty} \frac{1}{4}^n$ .

The formula for a geometric series like this is  $\sum_{n=1}^{\infty} a * r^n = \frac{a}{1-r}$ . Using this, we get  $\frac{\frac{9}{2}}{\frac{4}{4}-\frac{1}{4}} = \frac{9}{2} \times \frac{4}{3} = 3 \times 2 = \boxed{(A)6}$ .

### Solution 7

Notice how you can fit  $\triangle KDE$  into the top right corner of square  $BCDJ$ ? Repeat this process by putting smaller triangles into the corner of the square and eventually, you should see that this converges to just  $\frac{1}{4}$  of the square.  $\triangle BCD$  is  $\frac{1}{2}$  of the square and rest of the triangles make  $\frac{1}{4}$  which totals to  $\frac{3}{4}$  of the square. Square  $BCDJ$  has side length 3 so it's area is 9, and  $\frac{3}{4}$  of 9 is  $\boxed{(A)6}$ .

### Solution 8

As in Solution 3, the area form the geometric series  $\frac{9}{2} + \frac{9}{8} + \frac{9}{32} + \frac{9}{128} + \dots$ . If this is  $x$ , then  $4x$  is  $18 + \frac{9}{2} + \frac{9}{8} + \frac{9}{32} + \frac{9}{128} + \dots$ . Then we take away  $x$  to get  $3x = 18$ . When we solve for  $x$  we get  $x = 6$ , so the correct answer is  $\boxed{A}$ .

## §16 AMC 8 2000

1. (*AMC 8 2000*) Aunt Anna is 42 years old. Caitlin is 5 years younger than Brianna, and Brianna is half as old as Aunt Anna. How old is Caitlin?  
(A) 15      (B) 16      (C) 17      (D) 21      (E) 37

**Answer:** B

### Solution

Brianna is  $\frac{1}{2} \cdot 42 = 21$  years old, so Caitlin is  $21 - 5 = \boxed{(B) 16}$  years old.

2. (*AMC 8 2000*) Which of these numbers is less than its reciprocal?  
(A)  $-2$       (B)  $-1$       (C)  $0$       (D)  $1$       (E)  $2$

**Answer:** A

### Solution 1

The reciprocal of  $-2$  is  $-\frac{1}{2}$ . Since  $-2 < -\frac{1}{2}$ ,  $\boxed{(A) -2}$  is correct.

The reciprocal of  $-1$  is  $-1$ . Since  $-1 = -1$ , (B) is incorrect.

The reciprocal of  $0$  is undefined, so (C) is incorrect.

The reciprocal of  $1$  is  $1$ . Since  $1 = 1$ , (D) is incorrect.

The reciprocal of  $2$  is  $\frac{1}{2}$ . Since  $2 > \frac{1}{2}$ , (E) is incorrect.

### Solution 2

Let  $x$  be a real number that is less than its reciprocal, that is,  $x < \frac{1}{x}$ .

Since  $\frac{1}{x}$  must be defined,  $x \neq 0$ .

If  $x$  is positive, then  $x^2 < 1$ , so  $0 < x < 1$ .

If  $x$  is negative, then  $x^2 > 1$ , so  $x < -1$ .

Therefore,  $x < \frac{1}{x}$  if and only if either  $x < -1$  or  $0 < x < 1$ . Of the answer choices, only **(A)  $-2$**  is a possible value of  $x$ .

### Solution 3

Starting again with  $x < \frac{1}{x}$ , we avoid multiplication by  $x$ . Instead, move everything to the left, and find a common denominator:

$$x < \frac{1}{x} \quad x - \frac{1}{x} < 0 \quad \frac{x^2-1}{x} < 0 \quad \frac{(x+1)(x-1)}{x} < 0$$

Divide this expression at  $x = -1$ ,  $x = 0$ , and  $x = 1$ , as those are the three points where the expression on the left will "change sign".

If  $x < -1$ , all three of those terms will be negative, and the inequality is true. Therefore,  $(-\infty, -1)$  is part of our solution set.

If  $-1 < x < 0$ , the  $(x + 1)$  term will become positive, but the other two terms remain negative. Thus, there are no solutions in this region.

If  $0 < x < 1$ , then both  $(x + 1)$  and  $x$  are positive, while  $(x - 1)$  remains negative. Thus, the entire region  $(0, 1)$  is part of the solution set.

If  $1 < x$ , then all three terms are positive, and there are no solutions.

At all three "boundary points", the function is either 0 or undefined. Therefore, the entire solution set is  $(-\infty, -1) \cup (0, 1)$ , and the only option in that region is  $x = -2$ , leading to **(A)**.

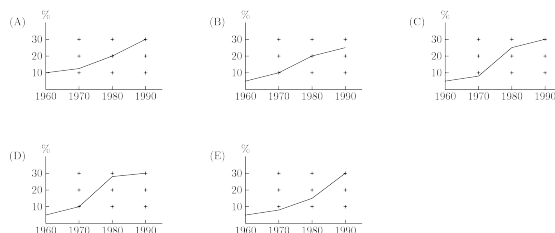
3. (AMC 8 2000) How many whole numbers lie in the interval between  $\frac{5}{3}$  and  $2\pi$ ?  
 (A) 2      (B) 3      (C) 4      (D) 5      (E) infinitely many

**Answer: D**

### Solution

Note that  $1 < \frac{5}{3} < 2$  while  $6 < 2\pi < 7$ . Therefore, the interval from  $\frac{5}{3}$  to  $2\pi$  contains **(D) 5** whole numbers: 2, 3, 4, 5, and 6.

4. (AMC 8 2000) In 1960 only 5% of the working adults in Carlin City worked at home. By 1970 the "at-home" work force increased to 8%. In 1980 there were approximately 15% working at home, and in 1990 there were 30%. The graph that best illustrates this is



**Answer: E**

### Solution 1

The data are 1960(5%), 1970(8%), 1980(15%), and 1990(30%). Only one of these graphs has the answer and that is choice **(E)**.

**Solution 2**

We can look at the graphs and note that the only one that has all the correct points is  $\boxed{E}$

5. (AMC 8 2000) Each principal of Lincoln High School serves exactly one 3-year term. What is the maximum number of principals this school could have during an 8-year period?
- (A) 2      (B) 3      (C) 4      (D) 5      (E) 8

**Answer:** C

**Solution 1**

Suppose the current principal's term will end in 1 year. Of the next 8 years, the current principal will serve the first year; the next principal will serve the second, third, and fourth years; a third principal will serve the fifth, sixth, and seventh years; and a fourth principal will serve the eighth year. Therefore, 4 different principals will serve in the next 8 years.

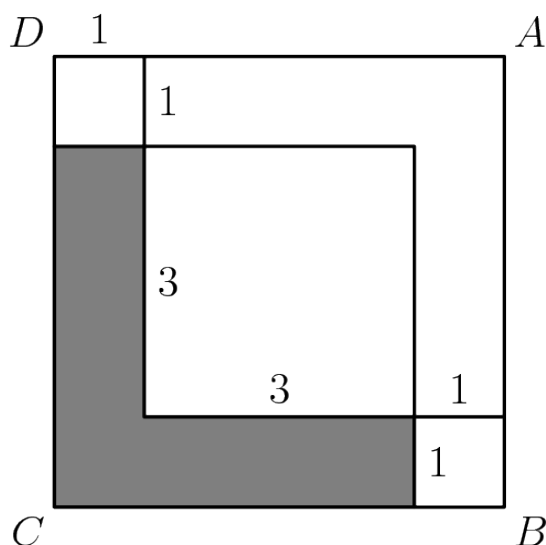
Consider a period that includes at least 5 different principals. During this period, all principals except the first and last must serve complete terms. So at least 3 principals serve complete terms, and thus this period is at least 9 years long. Therefore, an 8-year period may not have 5 or more different principals.

Thus there can be at most  $\boxed{(C) 4}$  principals in an 8-year period.

**Solution 2**

In an 8-year period, at most  $\left\lfloor \frac{8}{3} \right\rfloor = 2$  principals could serve complete terms. Furthermore, at most 2 principals could serve incomplete terms during this period: those who are in office at the start and end of this period. Therefore, at most  $2 + 2 = \boxed{(C) 4}$  principals could be active during this period (which would happen if the acting principal at the start of the period has 1 year left of their term).

6. (AMC 8 2000) Figure  $ABCD$  is a square. Inside this square three smaller squares are drawn with the side lengths as labeled. The area of the shaded  $L$ -shaped region is



- (A) 7      (B) 10      (C) 12.5      (D) 14      (E) 15

**Answer:** A

**Solution 1**

The side of the large square is  $1 + 3 + 1 = 5$ , so the area of the large square is  $5^2 = 25$ .

The area of the middle square is  $3^2$ , and the sum of the areas of the two smaller squares is  $2 * 1^2 = 2$ .

Thus, the big square minus the three smaller squares is  $25 - 9 - 2 = 14$ . This is the area of the two congruent L-shaped regions.

So the area of one L-shaped region is  $\frac{14}{2} = 7$ , and the answer is A.

**Solution 2**

The shaded area can be divided into two regions: one rectangle that is 1 by 3, and one rectangle that is 4 by 1. (Or the reverse, depending on which rectangle the 1 by 1 square is "joined" to.) Either way, the total area of these two regions is  $3 + 4 = 7$ , and the answer is A.

**Solution 3**

Chop the entire 5 by 5 region into 25 squares like a piece of graph paper. When you draw all the lines, you can count that only 7 of the small 1 by 1 squares will be shaded, giving A as the answer.

**Solution 4**

In the bottom left corner of the 5 by 5 square there is a 4 by 4 square which has an area of  $4 \cdot 4 = 16$ . In the top right of that 4 by 4 square is a 3 by 3 square with an area of  $3 \cdot 3 = 9$ . When we remove the 3 by 3 square from the 4 by 4 square we get the L-shaped figure so our answer is  $16 - 9 =$ (A) 7

7. (AMC 8 2000) What is the minimum possible product of three different numbers of the set  $\{-8, -6, -4, 0, 3, 5, 7\}$ ?
- (A)  $-336$       (B)  $-280$       (C)  $-210$       (D)  $-192$       (E) 0

**Answer:** B

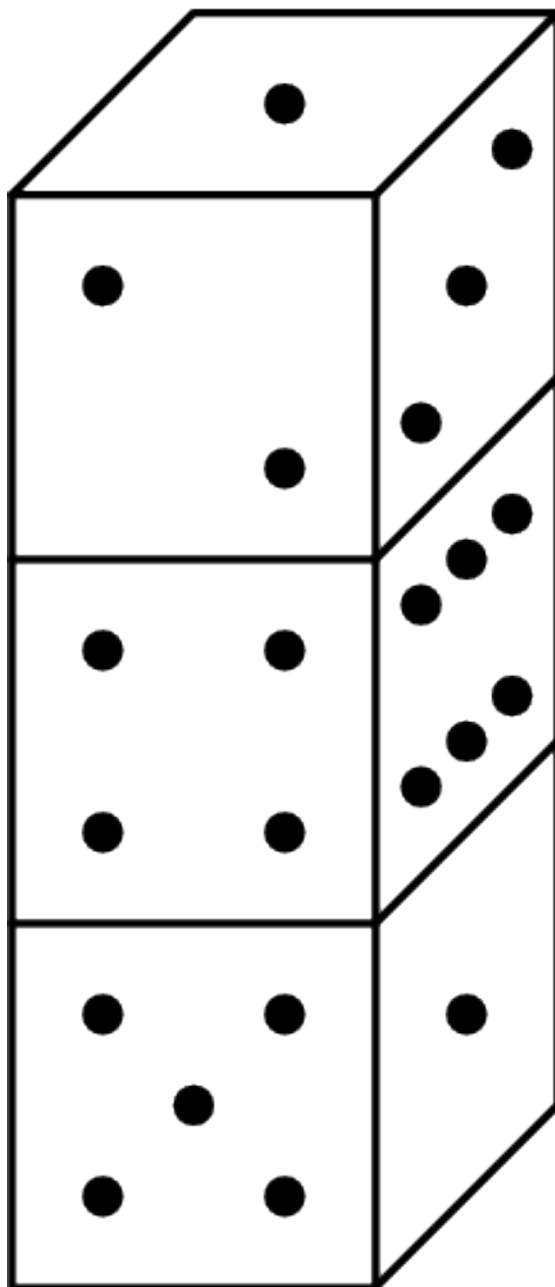
**Solution**

If the product of 3 selected numbers is negative, either all 3 numbers are negative or 1 number is negative and 2 numbers are positive.

The minimum product of 3 negative numbers from this set is  $(-8)(-6)(-4) = -192$ , while the minimum product of 1 negative number and 2 positive numbers from this set is  $(-8)(5)(7) = -280$ .

Therefore, the minimum product of 3 numbers from this set is (B)  $-280$ .

8. (AMC 8 2000) Three dice with faces numbered 1 through 6 are stacked as shown. Seven of the eighteen faces are visible, leaving eleven faces hidden (back, bottom, between). The total number of dots NOT visible in this view is



- (A) 21      (B) 22      (C) 31      (D) 41      (E) 53

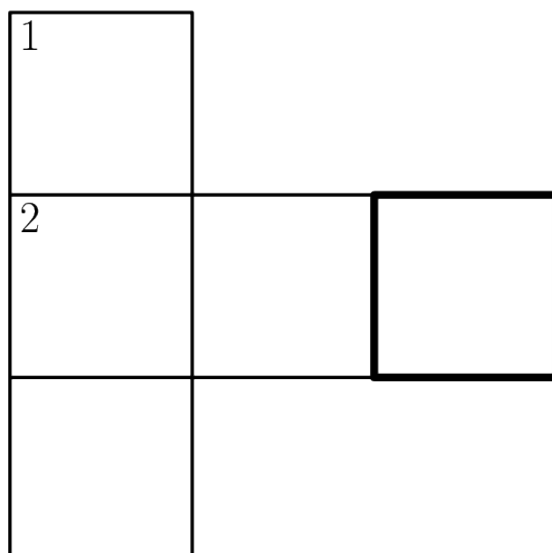
**Answer:** D

**Solution**

The numbers on one die total  $1 + 2 + 3 + 4 + 5 + 6 = 21$ , so the numbers on the three dice total 63. Numbers 1, 1, 2, 3, 4, 5, 6 are visible, and these total 22. This leaves  $63 - 22 = \boxed{\text{(D) } 41}$  not seen.

9. (*AMC 8 2000*) Three-digit powers of 2 and 5 are used in this "cross-number" puzzle. What is the only possible digit for the outlined square?

|               |             |
|---------------|-------------|
| <b>ACROSS</b> | <b>DOWN</b> |
| 2. $2^m$      | 1. $5^n$    |



- (A) 0      (B) 2      (C) 4      (D) 6      (E) 8

**Answer:** D

**Solution**

The 3-digit powers of 5 are 125 and 625, so space 2 is filled with a 2. The only 3-digit power of 2 beginning with 2 is 256, so the outlined block is filled with a **(D) 6**.

10. (*AMC 8 2000*) Ara and Shea were once the same height. Since then Shea has grown 20% while Ara has grown half as many inches as Shea. Shea is now 60 inches tall. How tall, in inches, is Ara now?

- (A) 48      (B) 51      (C) 52      (D) 54      (E) 55

**Answer:** E

**Solution**

Shea has grown 20%, if  $x$  was her original height, then  $1.2x = 60$ , so she was originally  $\frac{60}{1.2} = 50$  inches tall which is a  $60 - 50 = 10$  inch increase. Ara also started off at 50 inches. Since Ara grew half as much as Shea, Ara grew  $\frac{10}{2} = 5$  inches. Therefore, Ara is now  $50 + 5 = 55$  inches tall which is choice **(E)**.

11. (*AMC 8 2000*) The number 64 has the property that it is divisible by its units digit. How many whole numbers between 10 and 50 have this property?

- (A) 15      (B) 16      (C) 17      (D) 18      (E) 20

**Answer:** C

**Solution**

Casework by the units digit  $u$  will help organize the answer.

$u = 0$  gives no solutions, since no real numbers are divisible by 0.  $u = 1$  has 4 solutions, since all numbers are divisible by 1.

$u = 2$  has 4 solutions, since every number ending in 2 is even (ie divisible by 2).

$u = 3$  has 1 solution: 33.  $\pm 10$  or  $\pm 20$  will retain the units digit, but will stop the number from being divisible by 3.  $\pm 30$  is the smallest multiple of 10 that will keep the number divisible by 3, but those numbers are 3 and 63, which are out of the range of the problem.

$u = 4$  has 2 solutions: 24 and 44. Adding or subtracting 10 will kill divisibility by 4, since 10 is not divisible by 4.

$u = 5$  has 4 solutions: every number ending in 5 is divisible by 5.

$u = 6$  has 1 solution: 36.  $\pm 10$  or  $\pm 20$  will kill divisibility by 3, and thus kill divisibility by 6.

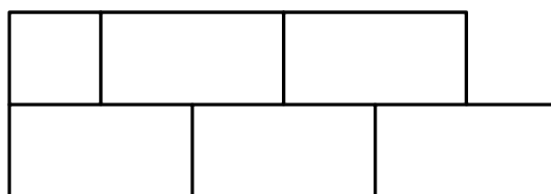
$u = 7$  has no solutions. The first multiples of 7 that end in 7 are 7 and 77, but both are outside of the range of this problem.

$u = 8$  has 1 solution: 48.  $\pm 10, \pm 20, \pm 30$  will all kill divisibility by 8 since 10, 20, and 30 are not divisible by 8.

$u = 9$  has no solutions. 9 and 99 are the smallest multiples of 9 that end in 9.

Totalling the solutions, we have  $0 + 4 + 4 + 1 + 2 + 4 + 1 + 0 + 1 + 0 = 17$  solutions, giving the answer  $\boxed{C}$ , which is 17.

12. (AMC 8 2000) A block wall 100 feet long and 7 feet high will be constructed using blocks that are 1 foot high and either 2 feet long or 1 foot long (no blocks may be cut). The vertical joins in the blocks must be staggered as shown, and the wall must be even on the ends. What is the smallest number of blocks needed to build this wall?



- (A) 344      (B) 347      (C) 350      (D) 353      (E) 356

**Answer:** D

**Solution**

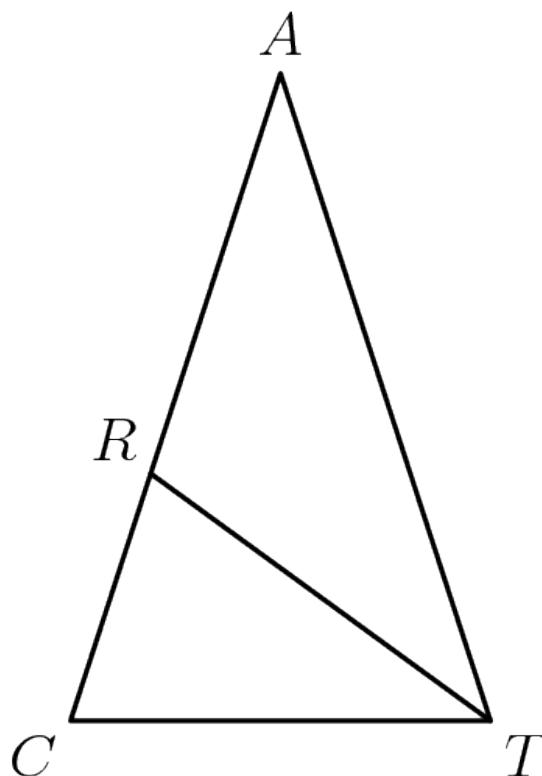
Since the bricks are 1 foot high, there will be 7 rows. To minimize the number of blocks used, rows 1, 3, 5, and 7 will look like the bottom row of the picture, which takes  $\frac{100}{2} = 50$  bricks to construct. Rows 2, 4, and 6 will look like the upper row pictured, which has 49 2-foot bricks in the middle, and 2 1-foot bricks on each end for a total of 51 bricks.

Four rows of 50 bricks and three rows of 51 bricks totals  $4 \cdot 50 + 3 \cdot 51 = 200 + 153 = 353$  bricks, giving the answer  $\boxed{D}$ .

If you wanted to be extra sure, you can realize that  $700/2 = 350$  blocks. However, this is impossible, as you need them to be staggered.

13. (AMC 8 2000) In triangle  $CAT$ , we have  $\angle ACT = \angle ATC$  and  $\angle CAT = 36^\circ$ . If  $\overline{TR}$  bisects  $\angle ATC$ , then  $\angle CRT =$





- (A)  $36^\circ$     (B)  $54^\circ$     (C)  $72^\circ$     (D)  $90^\circ$     (E)  $108^\circ$

**Answer:** C

**Solution**

In  $\triangle ACT$ , the three angles sum to  $180^\circ$ , and  $\angle C = \angle T$ .  $\angle CAT + \angle ATC + \angle ACT = 180$ .  $36 + \angle ATC + \angle ATC = 180$ .  $2\angle ATC = 144$ .  $\angle ATC = 72$ .

Since  $\angle ATC$  is bisected by  $\overline{TR}$ ,  $\angle RTC = \frac{72}{2} = 36$ .

Now focusing on the smaller  $\triangle RTC$ , the sum of the angles in that triangle is  $180^\circ$ , so:

$$\begin{aligned} \angle RTC + \angle TCR + \angle CRT &= 180 \\ 36 + \angle ACT + \angle CRT &= 180 \\ 36 + \angle ATC + \angle CRT &= 180 \\ 36 + 72 + \angle CRT &= 180 \\ \angle CRT &= 72^\circ, \text{ giving the answer } \boxed{C} \end{aligned}$$

14. (AMC 8 2000) What is the units digit of  $19^{19} + 99^{99}$ ?

- (A) 0    (B) 1    (C) 2    (D) 8    (E) 9

**Answer:** D

**Solution**

Finding a pattern for each half of the sum, even powers of 19 have a units digit of 1, and odd powers of 19 have a units digit of 9. So,  $19^{19}$  has a units digit of 9.

Powers of 99 have the exact same property, so  $99^{99}$  also has a units digit of 9.  $9 + 9 = 18$  which has a units digit of 8, so the answer is  $\boxed{D}$ .

**Solution 2**

Using modular arithmetic:

$$99 \equiv 9 \equiv -1 \pmod{10}$$

Similarly,

$$19 \equiv 9 \equiv -1 \pmod{10}$$

We have

$$(-1)^{19} + (-1)^{99} = -1 + -1 \equiv \boxed{\text{(D)} 8} \pmod{10}$$

-ryjs

### Solution 3

Experimentation gives

any number ending with  $9^{\text{something even}}$  = has units digit 1

any number ending with  $9^{\text{something odd}}$  = has units digit 9

Using this we have

$$\begin{aligned} 19^{19} + 99^{99} \\ 9^{19} + 9^{99} \end{aligned}$$

Both 19 and 99 are odd, so we are left with

$$9 + 9 = 18,$$

which has units digit  $\boxed{\text{(D)} 8}$ .-ryjs

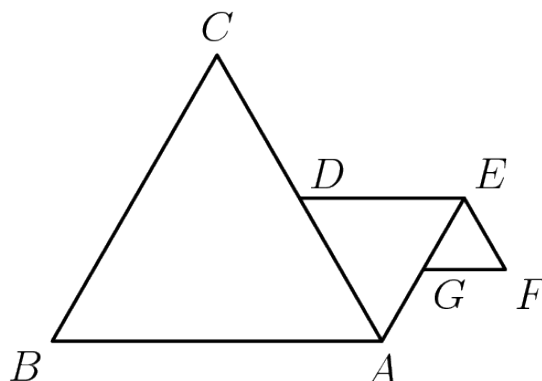
### Solution 4

$$19^{19} + 99^{99} = 3697296376497267726571879056288054405956687642817411024302599724235525704552775$$

which ends in 8. :) If you are really stupid and have time (like about 3 hours), this is a real brainless bash.

-hefei417

15. (AMC 8 2000) Triangles  $ABC$ ,  $ADE$ , and  $EFG$  are all equilateral. Points  $D$  and  $G$  are midpoints of  $\overline{AC}$  and  $\overline{AE}$ , respectively. If  $AB = 4$ , what is the perimeter of figure  $ABCDEFG$ ?



- (A) 12    (B) 13    (C) 15    (D) 18    (E) 21

**Answer:** C

### Solution 1

The large triangle  $ABC$  has sides of length 4. The medium triangle has sides of length 2. The small triangle has sides of length 1. There are 3 segment sizes, and all segments depicted are one of these lengths.

Starting at  $A$  and going clockwise, the perimeter is:

$AB + BC + CD + DE + EF + FG + GA$   $4 + 4 + 2 + 2 + 1 + 1 + 1$  15, thus the answer is  $\boxed{C}$

### Solution 2

The perimeter of  $ABCDEFG$  is the perimeter of the three triangles, minus segments  $AD$  and  $EG$ , which are on the interior of the figure. Because each of these segments is on two triangles, each segment must be subtracted two times.

As in solution 1, the sides of the triangles are 4, 2, and 1, and the perimeters of the triangles are thus 12, 6, and 3.

The perimeter of the three triangles is  $12 + 6 + 3 = 21$ . Subtracting the two segments  $AD$  and  $EG$  two times, the perimeter of  $ABCDEFG$  is  $21 - 2 - 1 - 2 - 1 = 15$ , and the answer is  $\boxed{C}$ .

### Solution 3

First, remove segments  $DA$ , and  $EG$ . Notice that segments  $DE$ , and  $AG$  can fit into  $DA$ , and  $EG$ , respectively. This means we only need to account for the perimeters of triangles  $ABC$ , and  $EFG$ , whose perimeters are 12, and 3 respectively. Adding these perimeters gives us  $12 + 3 = 15$ , or answer choice  $\boxed{C}$ . Electric\_Cheetah\_88

16. (AMC 8 2000) In order for Mateen to walk a kilometer (1000m) in his rectangular backyard, he must walk the length 25 times or walk its perimeter 10 times. What is the area of Mateen's backyard in square meters?
- (A) 40      (B) 200      (C) 400      (D) 500      (E) 1000

**Answer:** C

### Solution

The length  $L$  of the rectangle is  $\frac{1000}{25} = 40$  meters. The perimeter  $P$  is  $\frac{1000}{10} = 100$  meters. Since  $P_{rect} = 2L + 2W$ , we plug values in to get:

$$100 = 2 \cdot 40 + 2W \quad 100 = 80 + 2W \quad 2W = 20 \quad W = 10 \text{ meters}$$

Since  $A_{rect} = LW$ , the area is  $40 \cdot 10 = 400$  square meters or  $\boxed{C}$ .

17. (AMC 8 2000) The operation  $\otimes$  is defined for all nonzero numbers by  $a \otimes b = \frac{a^2}{b}$ . Determine  $[(1 \otimes 2) \otimes 3] - [1 \otimes (2 \otimes 3)]$ .
- (A)  $-\frac{2}{3}$       (B)  $-\frac{1}{4}$       (C) 0      (D)  $\frac{1}{4}$       (E)  $\frac{2}{3}$

**Answer:** A

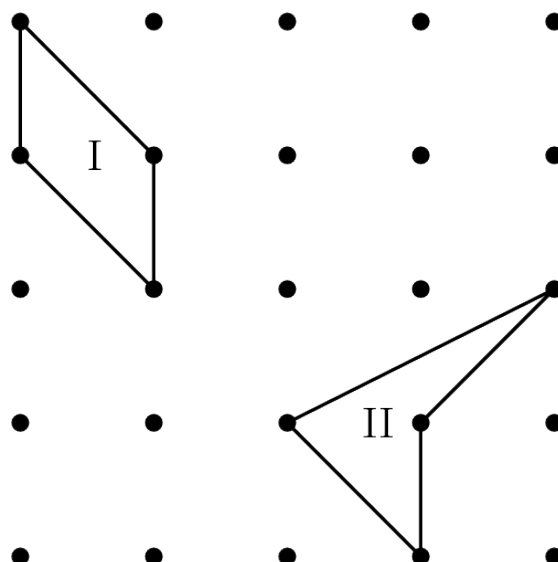
### Solution

Follow PE(MD)(AS), doing the innermost parentheses first.

$$[(1 \otimes 2) \otimes 3] - [1 \otimes (2 \otimes 3)] \quad \left[\frac{1^2}{2} \otimes 3\right] - \left[1 \otimes \frac{2^2}{3}\right] \quad \left[\frac{1}{2} \otimes 3\right] - \left[1 \otimes \frac{4}{3}\right] \quad \left[\frac{(\frac{1}{2})^2}{3}\right] - \left[\frac{1^2}{(\frac{4}{3})}\right] \quad \left[\frac{1}{4} \cdot \frac{1}{3}\right] - \left[\frac{3}{4}\right]$$

$$\frac{1}{12} - \frac{3}{4} \quad \frac{1}{12} - \frac{9}{12} \quad \frac{-8}{12} \quad -\frac{2}{3}, \text{ which is answer } \boxed{A}$$

18. (AMC 8 2000) Consider these two geoboard quadrilaterals. Which of the following statements is true?



- (A) The area of quadrilateral I is more than the area of quadrilateral II. (B) The area of quadrilateral I is less than the area of quadrilateral II. (C) The quadrilaterals have the same area and the same perimeter. (D) The quadrilaterals have the same area, but the perimeter of I is more than the perimeter of II. (E) The quadrilaterals have the same area, but the perimeter of I is less than the perimeter of II.

**Answer:** E

### Solution 1

First consider the area of the two figures. Assume that the pegs are 1 unit apart. Divide region I into two triangles by drawing a horizontal line on the second row from the top. Shifting the bottom triangle up one unit will create a square, and this square has the same area as region I. Thus, region I has an area of 1 square unit.

Draw a horizontal line on figure II on the fourth row from the top to divide the figure into two triangles. Temporarily move the top of the higher triangle one peg to the left. This will preserve the area of the triangle, as you are keeping both the height and the base of the triangle equal. Now you have two congruent triangles that are half of the area of a unit square. Thus, region II also has an area of 1 square unit, and the areas are equal.

To compare the perimeters, note that region I has two different side lengths: two sides are 1 unit apart, and the other two sides are  $\sqrt{2}$  apart, as they are the diagonal of a unit square. The total perimeter is  $2 + 2\sqrt{2}$ .

Note that region II has three different side lengths. One side is a unit length, while two sides are  $\sqrt{2}$ . For the perimeters to be equal, the last side must be of unit length. But the last side in region II is clearly longer than 1 unit, so we can say that the perimeter of region II is greater than the perimeter of region I without calculating it.

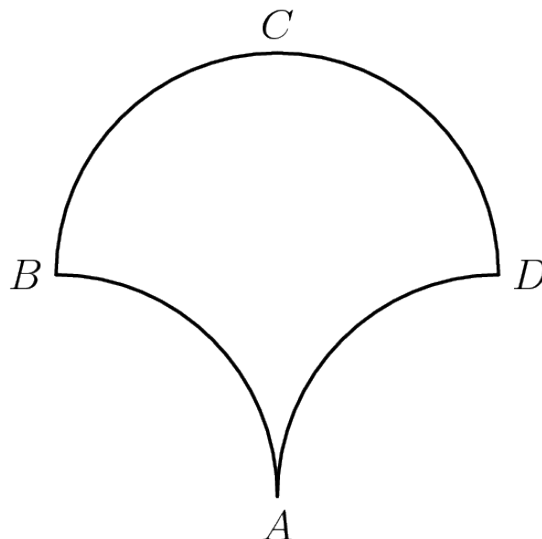
Thus, the correct answer is E.

(The perimeter of region II is  $1 + 2\sqrt{2} + \sqrt{5}$ , since the last side is the diagonal of a 2 by 1 rectangle, and can be found with the pythagorean theorem as  $\sqrt{2^2 + 1^2} = \sqrt{5}$ . This does not need to be found for this problem, as you can do a one-to-one correspondence with three of the four sides of the figure as outlined, and just compare the last remaining side of each figure.)

### Solution 2

Using the Pick's Theorem, The area of both figures is 1. This rules out answer choices A and B. When using the Pythagorean Theorem, we find that the perimeter of I is  $2 + 2(\sqrt{2})$  and the perimeter of II is  $1 + 2(\sqrt{2}) + \sqrt{5}$ . Since  $\sqrt{5} > 1$ , the perimeter of II is greater, therefore the correct answer is E. Yaay!

19. (AMC 8 2000) Three circular arcs of radius 5 units bound the region shown. Arcs  $AB$  and  $AD$  are quarter-circles, and arc  $BCD$  is a semicircle. What is the area, in square units, of the region?



- (A) 25      (B)  $10 + 5\pi$       (C) 50      (D)  $50 + 5\pi$       (E)  $25\pi$

**Answer:** C

**Solution 1**

Draw two squares: one that has opposing corners at  $A$  and  $B$ , and one that has opposing corners at  $A$  and  $D$ . These squares share side  $\overline{AO}$ , where  $O$  is the center of the large semicircle.

These two squares have a total area of  $2 \cdot 5^2$ , but have two quarter circle "bites" of radius 5 that must be removed. Thus, the bottom part of the figure has area

$$2 \cdot 25 - 2 \cdot \frac{1}{4}\pi \cdot 5^2 = 50 - \frac{25\pi}{2}$$

This is the area of the part of the figure underneath  $\overline{BD}$ . The part of the figure over  $\overline{BD}$  is just a semicircle with radius 5, which has area of  $\frac{1}{2}\pi \cdot 5^2 = \frac{25\pi}{2}$

Adding the two areas gives a total area of (c)50

**Solution 2**

Draw line  $\overline{BD}$ . Then draw  $\overline{CO}$ , where  $O$  is the center of the semicircle. You have two quarter circles on top, and two quarter circle-sized "bites" on the bottom. Move the pieces from the top to fit in the bottom like a jigsaw puzzle. You now have a rectangle with length  $\overline{BD}$  and height  $\overline{AO}$ , which are equal to 10 and 5, respectively. Thus, the total area is 50, and the answer is C.

20. (AMC 8 2000) You have nine coins: a collection of pennies, nickels, dimes, and quarters having a total value of

1.02

, with at least one coin of each type. How many dimes must you have?

- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** A

**Solution 1**

Since you have one coin of each type,  $1 + 5 + 10 + 25 = 41$  cents are already determined, leaving you with a total of  $102 - 41 = 61$  cents remaining for 5 coins.

You must have 1 more penny. If you had more than 1 penny, you must have at least 6 pennies to leave a multiple of 5 for the nickels, dimes, and quarters. But you only have 5 more coins to assign.

Now you have  $61 - 1 = 60$  cents remaining for 4 coins, which may be nickels, quarters, or dimes. If you have only one more dime, that leaves 50 cents in 3 nickels or quarters, which is impossible. If you have two dimes, that leaves 40 cents for 2 nickels or quarters, which is again impossible. If you have three dimes, that leaves 30 cents for 1 nickel or quarter, which is still impossible. And all four remaining coins being dimes will not be enough.

Therefore, you must have no more dimes to assign, and the 60 cents in 4 coins must be divided between the quarters and nickels. We quickly see that 2 nickels and 2 quarters work. Thus, the total count is 2 quarters, 2 nickels, 1 penny, plus one more coin of each type that we originally subtracted. Double-checking, that gives a total  $2 + 2 + 1 + 4 = 9$  coins, and a total of  $2 \cdot 25 + 2 \cdot 5 + 1 + (1 + 5 + 10 + 25) = 102$  cents.

There is only 1 dime in that combo, so the answer is A.

**Solution 2 (Faster)**

We see that there must be 102 cents, so therefore there's at least 2 pennies. That leaves 7 coins. We assume that there are 3 quarters, leaving 25 cents with 4 coins left. If all 4 are nickels, that would only be 20 cents, missing 5. Therefore, one nickel must be changed into 1 dime, so the answer is A

Solution by [ILoveMath31415926535](#)

**Solution 3**

It is clear that there should only be 2 pennies; any more would take up too many coins, and the limit is 9. Now we have

$$1$$

left, and 7 coins to use. Looking at the quarters, we can make methodical guesses. If there is 1 quarter, then we would have to make

$$0.75$$

with 6 coins. We take a few educated guesses for the nickel and dime combos, and see that 1 quarter will not work. Trying values for 2 quarters, we see that this will not work either. When we reach 3 quarters, the remaining is

$$0.25$$

made from 4 coins. We try with 2 dimes, which does not work (it only takes 3 coins) and we try with 1 dime. After trying 2 pennies, 3 quarters, 1 dime, and 3 nickels, it is evident that this combo works. Therefore, the answer is A

- It might seem that this solution takes time, but in reality, it can be done very quickly, given ones desire to solve a problem quickly

21. (AMC 8 2000) Keiko tosses one penny and Ephraim tosses two pennies. The probability that Ephraim gets the same number of heads that Keiko gets is:

- (A)  $\frac{1}{4}$     (B)  $\frac{3}{8}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{3}$     (E)  $\frac{3}{4}$

**Answer:** B

**Solution**

Divide it into 2 cases:

1) Keiko and Ephraim both get 0 heads: This means that they both roll all tails, so there is only 1 way for this to happen.

2) Keiko and Ephraim both get 1 head: For Keiko, there is only 1 way for this to happen because he is only flipping 1 penny, but for Ephraim, there are 2 ways since there are 2 choices for when he can flip the head. So, in total there are  $2 \cdot 1 = 2$  ways for this case.

Thus, in total there are 3 ways that work. Since there are 2 choices for each coin flip (Heads or Tails), there are  $2^3 = 8$  total ways of flipping 3 coins.

Thus, since all possible coin flips of 3 coins are equally likely, the probability is

$$\boxed{(B) \frac{3}{8}}.$$

**Solution 2**

Let  $K(n)$  be the probability that Keiko gets  $n$  heads, and let  $E(n)$  be the probability that Ephraim gets  $n$  heads.

$$K(0) = \frac{1}{2} \quad K(1) = \frac{1}{2} \quad K(2) = 0 \quad (\text{Keiko only has one penny!})$$

$$E(0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \quad E(1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = 2 \cdot \frac{1}{4} = \frac{1}{2} \quad (\text{because Ephraim can get HT or TH})$$

$$E(2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

The probability that Keiko gets 0 heads and Ephraim gets 0 heads is  $K(0) \cdot E(0)$ . Similarly for 1 head and 2 heads. Thus, we have:

$$P = K(0) \cdot E(0) + K(1) \cdot E(1) + K(2) \cdot E(2) \quad P = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} + 0 \cdot \frac{1}{4} \quad P = \frac{3}{8}$$

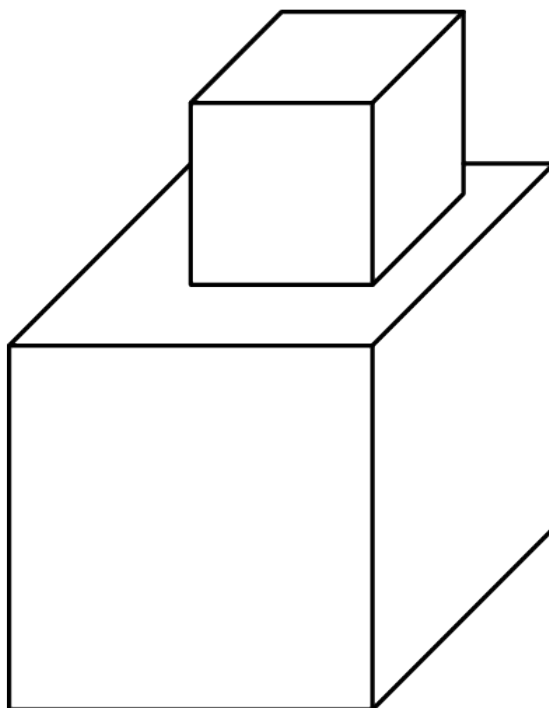
Thus the answer is  $\boxed{B}$ .

**Solution 3**

Keiko can only get 1 head or one tail. That means Ephraim has to also get 1 head or one tail. There are 3 cases  $HHT, HTH, TTT$  and each have a  $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$

chance. Adding all 3 ways makes a total of  $\boxed{(B) \frac{3}{8}}$

22. (AMC 8 2000) A cube has edge length 2. Suppose that we glue a cube of edge length 1 on top of the big cube so that one of its faces rests entirely on the top face of the larger cube. The percent increase in the surface area (sides, top, and bottom) from the original cube to the new solid formed is closest to



- (A) 10    (B) 15    (C) 17    (D) 21    (E) 25

**Answer:** C

**Solution 1**

The original cube has 6 faces, each with an area of  $2 \cdot 2 = 4$  square units. Thus the original figure had a total surface area of 24 square units.

The new figure has the original surface, with 6 new faces that each have an area of 1 square unit, for a total surface area of 6 additional square units added to it. But 1 square unit of the top of the bigger cube, and 1 square unit on the bottom of smaller cube, is not on the surface, and does not count towards the surface area.

The total surface area is therefore  $24 + 6 - 1 - 1 = 28$  square units.

The percent increase in surface area is  $\frac{SA_{new} - SA_{old}}{SA_{old}} \cdot 100\% = \frac{28 - 24}{24} \cdot 100\% \approx 16.67\%$ , giving the closest answer as C.

**Solution 2(Similar to 1)**

The original cube has 6 faces, each with an area of  $2 \cdot 2 = 4$  square units. Thus the original figure had a total surface area of 4 times 6 square units. Notice that the little cube covers a bit of the larger cube but this is replaced by the top of the little cube. Therefore, the actual increase comes from the four sides of the cube giving  $1 \text{ squared} = 1$ ,  $1 (4) = 4$ . So 4 over the original 4 times 6.  $4/4(6) = 1/6$  which is about 16.67% giving C..

**Solution 3**

To be honest, we could just bash this one out to get about 20%, which is C. Yaaaay! BRO, who ever made this is tweaking, if you bashed it to 20 you would choose D so this is not bashable. Tsk Tsk Plus they never said how to bash  $\neg$  D is closer to 20 sir lock in





determine  $\angle B$  or  $\angle D$  alone. If you see  $\triangle BFD$ , then you can see that the problem is solved quickly after determining  $\angle BFD$ .

But start with  $\triangle AGF$ , since that's where most of our information is. Looking at  $\triangle AGF$ , since  $\angle AFG = \angle AGF$ , and  $\angle A = 20$ , we can write:

$$\angle A + \angle AFG + \angle AGF = 180 \quad 20 + 2\angle AFG = 180 \quad \angle AFG = 80$$

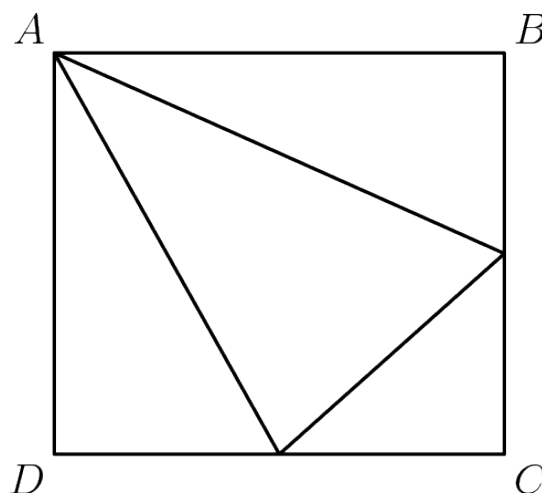
By noting that  $\angle AFG$  and  $\angle GFD$  make a straight line, we know

$$\angle AFG + \angle GFD = 180 \quad 80 + \angle GFD = 180 \quad \angle GFD = 100$$

Ignoring all other parts of the figure and looking only at  $\triangle BFD$ , you see that  $\angle B + \angle D + \angle BFD = 180$ . But  $\angle BFD$  is the same as  $\angle GFD$ . Therefore:

$$\angle B + \angle D + \angle GFD = 180 \quad \angle B + \angle D + 100 = 180 \quad \angle B + \angle D = 80^\circ, \text{ and the answer is thus } \boxed{D}$$

25. (AMC 8 2000) The area of rectangle  $ABCD$  is 72. If point  $A$  and the midpoints of  $\overline{BC}$  and  $\overline{CD}$  are joined to form a triangle, the area of that triangle is



- (A) 21      (B) 27      (C) 30      (D) 36      (E) 40

**Answer:** B

### Solution 1

To quickly solve this multiple choice problem, make the (not necessarily valid, but very convenient) assumption that  $ABCD$  can have any dimension. Give the rectangle dimensions of  $AB = CD = 12$  and  $BC = AD = 6$ , which is the easiest way to avoid fractions. Labeling the right midpoint as  $M$ , and the bottom midpoint as  $N$ , we know that  $DN = NC = 6$ , and  $BM = MC = 3$ .

$$\begin{aligned} [\triangle ADN] &= \frac{1}{2} \cdot 6 \cdot 6 = 18 & [\triangle MNC] &= \frac{1}{2} \cdot 3 \cdot 6 = 9 & [\triangle ABM] &= \frac{1}{2} \cdot 12 \cdot 3 = 18 \\ [\triangle AMN] &= [\square ABCD] - [\triangle ADN] - [\triangle MNC] - [\triangle ABM] & [\triangle AMN] &= 72 - 18 - 9 - 18 & [\triangle AMN] &= 27, \text{ and the answer is } \boxed{B} \end{aligned}$$

### Solution 2

The above answer is fast, but satisfying, and assumes that the area of  $\triangle AMN$  is independent of the dimensions of the rectangle. All in all, it's a very good answer though. However this is an alternative if you don't get the above answer. Label  $AB = CD = l$  and  $BC = DA = h$

Labelling  $m$  and  $n$  as the right and lower midpoints respectively, and redoing all the work above, we get:

$$[\triangle ADN] = \frac{1}{2} \cdot h \cdot \frac{l}{2} = \frac{lh}{4} \quad [\triangle MNC] = \frac{1}{2} \cdot \frac{l}{2} \cdot \frac{h}{2} = \frac{lh}{8} \quad [\triangle ABM] = \frac{1}{2} \cdot l \cdot \frac{h}{2} = \frac{lh}{4} \quad [\triangle AMN] = [\square ABCD] - [\triangle ADN] - [\triangle MNC] - [\triangle ABM] \quad [\triangle AMN] = lh - \frac{lh}{4} - \frac{lh}{8} - \frac{lh}{4} \\ [\triangle AMN] = \frac{3}{8}lh = \frac{3}{8} \cdot 72 = 27, \text{ and the answer is } \boxed{B}$$

### Solution 3

Let's assume WLOG that the sides of the rectangle are 9 and 8. The area of the 3 triangles would then be  $8 \cdot \frac{9}{2} \cdot \frac{1}{2} = 18$ ,  $4 \cdot \frac{9}{2} \cdot \frac{1}{2} = 9$ ,  $4 \cdot 9 \cdot \frac{1}{2} = 18$ . Adding these up, we get 45, and subtracting that from 72, we get 27, so the answer is  $\boxed{B}$

## §17 AMC 8 2001

1. (AMC 8 2001) John's shop class is making a golf trophy. He has to paint 300 dimples on a golf ball. If it takes him 2 seconds to paint one dimple, how many minutes will he need to do his job?

(A) 4      (B) 6      (C) 8      (D) 10      (E) 12

**Answer:** D

### Solution

It will take him  $300 \cdot 2 = 600$  seconds to paint all the dimples.

This is equivalent to  $\frac{600}{60} = 10$  minutes  $\Rightarrow \boxed{D}$ .

2. (AMC 8 2001) I'm thinking of two whole numbers. Their product is 24 and their sum is 11. What is the larger number?

(A) 3      (B) 4      (C) 6      (D) 8      (E) 12

**Answer:** D

### Solution 1

Let the numbers be  $x$  and  $y$ . Then we have  $x + y = 11$  and  $xy = 24$ . Solving for  $x$  in the first equation yields  $x = 11 - y$ , and substituting this into the second equation gives  $(11 - y)(y) = 24$ . Simplifying this gives  $-y^2 + 11y = 24$ , or  $y^2 - 11y + 24 = 0$ . This factors as  $(y - 3)(y - 8) = 0$ , so  $y = 3$  or  $y = 8$ , and the corresponding  $x$  values are  $x = 8$  and  $x = 3$ . These are essentially the same answer: one number is 3 and one number is 8, so the largest number is 8,  $\boxed{D}$ .

### Solution 2

Use the answers to attempt to "reverse engineer" an appropriate pair of numbers. Looking at option A, guess that one of the numbers is 3. If the sum of two numbers is 11 and one is 3, then other must be  $11 - 3 = 8$ . The product of those numbers is  $3 \cdot 8 = 24$ , which is the second condition of the problem, so our numbers are 3 and 8.

However, 3 is the smaller of the two numbers, so the answer is 8 or  $\boxed{D}$ .

### Solution 3

We go through the divisor pairs of 24 to see which two numbers sum to 11. The numbers clearly cannot be negative. If one was negative, then the other must also be negative in order to multiply to a positive product, but it would be impossible for the numbers to add up to a positive sum. So, we look at the positive divisor

pairs of 24, namely 1 and 24, 2 and 12, 3 and 8, and 4 and 6. The only pair of numbers that sums to 11 is 3 and 8. The larger number is 8, so the answer is **D**.

3. (AMC 8 2001) Granny Smith has 63. Elberta has 2 more than Anjou and Anjou has one-third as much as Granny Smith. How many dollars does Elberta have?

(A) 17      (B) 18      (C) 19      (D) 21      (E) 23

**Answer:** E

**Solution**

Since Anjou has  $\frac{1}{3}$  the amount of money as Granny Smith and Granny Smith has

63

, Anjou has  $\frac{1}{3} \times 63 = 21$  dollars. Elberta has

2

more than this, so she has

23

, or **E**.

4. (AMC 8 2001) The digits 1, 2, 3, 4 and 9 are each used once to form the smallest possible **even** five-digit number. The digit in the tens place is

(A) 1      (B) 2      (C) 3      (D) 4      (E) 9

**Answer:** E

**Solution 1**

Since the number is even, the last digit must be 2 or 4. To make the smallest possible number, the ten-thousands digit must be as small as possible, so the ten-thousands digit is 1. Similarly, the thousands digit has second priority, so it must also be as small as possible once the ten-thousands digit is decided, so the thousands digit is 2. Similarly, the hundreds digit needs to be the next smallest number, so it is 3. However, for the tens digit, we can't use 4, since we already used 2 and the number must be even, so the units digit must be 4 and the tens digit is 9, **E** (The number is 12394.)

**Solution 2 (Faster)**

We know that the smallest possible number is 12349. However, 12349 is an odd number. Thus, by shifting the 4 (the largest even number) into the unit's place and rearranging the rest of the digits from least to greatest, we arrive at 12394. Thus, the digit in the tens place is **E** 9.

5. (AMC 8 2001) On a dark and stormy night Snoopy suddenly saw a flash of lightning. Ten seconds later he heard the sound of thunder. The speed of sound is 1088 feet per second and one mile is 5280 feet. Estimate, to the nearest half-mile, how far Snoopy was from the flash of lightning.

(A) 1      (B)  $1\frac{1}{2}$       (C) 2      (D)  $2\frac{1}{2}$       (E) 3

**Answer:** C

**Solution**

During the 10 seconds, the sound traveled  $1088 \times 10 = 10880$  feet from the lightning to Snoopy. This is equivalent to  $\frac{10880}{5280} \approx 2$  miles, **C**.

6. (AMC 8 2001) Six trees are equally spaced along one side of a straight road. The distance from the first tree to the fourth is 60 feet. What is the distance in feet between the first and last trees?

(A) 90      (B) 100      (C) 105      (D) 120      (E) 140

**Answer:** B

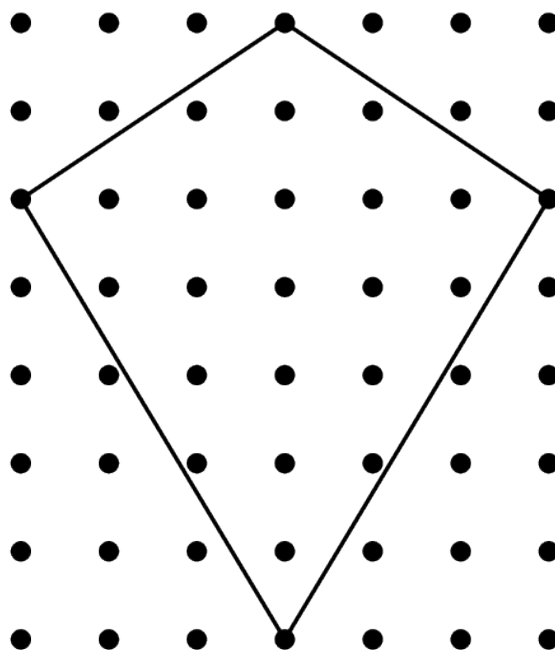
**Solution**

There are 3 spaces between the 1st and 4th trees, so each of these spaces has  $\frac{60}{3} = 20$  feet. Between the first and last trees there are 5 spaces, so the distance between them is  $20 \times 5 = 100$  feet, **[B]**.

7. (AMC 8 2001)

*Problems 7, 8 and 9 are about these kites.*

To promote her school's annual Kite Olympics, Genevieve makes a small kite and a large kite for a bulletin board display. The kites look like the one in the diagram below. For her small kite Genevieve draws the kite on a one-inch grid. For the large kite she triples both the height and width of the entire grid.



What is the number of square inches in the area of the small kite?

(A) 21      (B) 22      (C) 23      (D) 24      (E) 25

**Answer:** A

**Solution 1**

The area of a kite is half the product of its diagonals. The diagonals have lengths of 6 and 7, so the area is  $\frac{(6)(7)}{2} = 21$ , **[A]**.

**Solution 2**

Drawing in the diagonals of the kite will form four right triangles on the "inside" part of the grid. Drawing in the border of the 7 by 6 grid will form four right triangles on the "outside" part of the grid. Since each right triangle on the inside

can be paired with a congruent right triangle that is on the outside, the area of the kite is half the total area of the grid, or  $\frac{(6)(7)}{2} = 21$ , **(A)**.

**Solution 3**

Pick's Theorem states:

$$\frac{\text{number of boundary points}}{2} + \text{number of interior points} - 1$$

as the area of a figure on a grid. Counting, we see there are 4 boundary points and 20 interior points. Therefore, we have

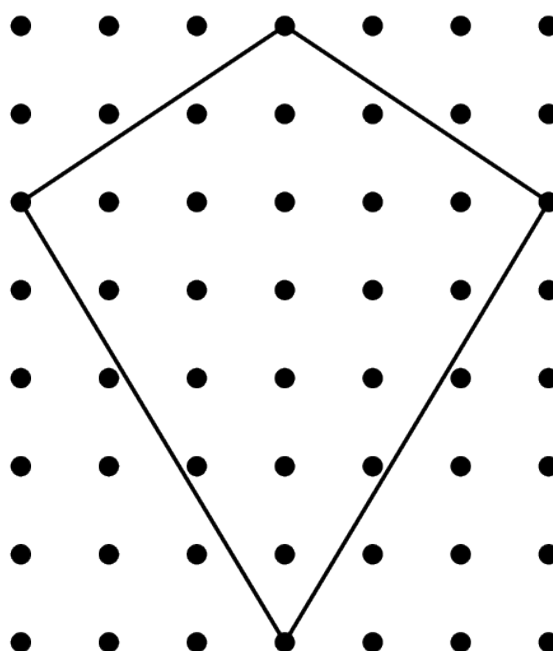
$$\frac{4}{2} + 20 - 1 \implies 20 + 1 \implies 21.$$

Hence, the answer is **(A)**

8. (AMC 8 2001)

*Problems 7, 8 and 9 are about these kites.*

To promote her school's annual Kite Olympics, Genevieve makes a small kite and a large kite for a bulletin board display. The kites look like the one in the diagram below. For her small kite Genevieve draws the kite on a one-inch grid. For the large kite she triples both the height and width of the entire grid.



Genevieve puts bracing on her large kite in the form of cross-connecting opposite corners of the kite. How many inches of bracing material does she need?

- (A) 30    (B) 32    (C) 35    (D) 38    (E) 39

**Answer:** E

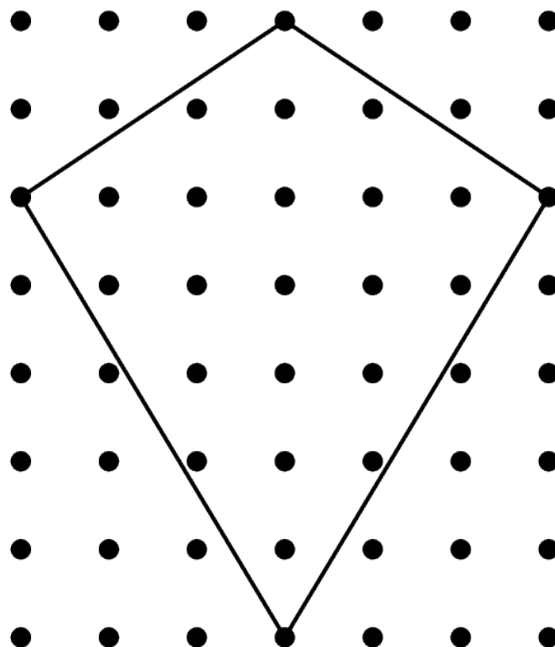
**Solution**

Each diagonal of the large kite is 3 times the length of the corresponding diagonal of the short kite since it was made with a grid 3 times as long in both height and width. The diagonals of the small kite are 6 and 7, so the diagonals of the large kite are  $3 \cdot 6 = 18$  and  $3 \cdot 7 = 21$ , and the amount of bracing Genevieve needs is the sum of these lengths, which is 39, **(E)**

9. (AMC 8 2001)

Problems 7, 8 and 9 are about these kites.

To promote her school's annual Kite Olympics, Genevieve makes a small kite and a large kite for a bulletin board display. The kites look like the one in the diagram below. For her small kite Genevieve draws the kite on a one-inch grid. For the large kite she triples both the height and width of the entire grid.



The large kite is covered with gold foil. The foil is cut from a rectangular piece that just covers the entire grid. How many square inches of waste material are cut off from the four corners?

- (A) 63    (B) 72    (C) 180    (D) 189    (E) 264

**Answer:** D

**Solution**

The large grid has dimensions three times that of the small grid, so its dimensions are  $3(6) \times 3(7)$ , or  $18 \times 21$ , so the area is  $(18)(21) = 378$ . The area of the kite is half of the area of the rectangle as you can see, so the area of the waste material is also half the area of the rectangle. Thus, the area of the waste material is  $378/2 = \boxed{189 \text{ (D)}}$ .

10. (AMC 8 2001) A collector offers to buy quarters for 2000% of their value. At that rate how much will Bryden get for his four quarters?

- (A) 20 dollars    (B) 50 dollars    (C) 200 dollars    (D) 500 dollars    (E) 2000 dollars

**Answer:** A

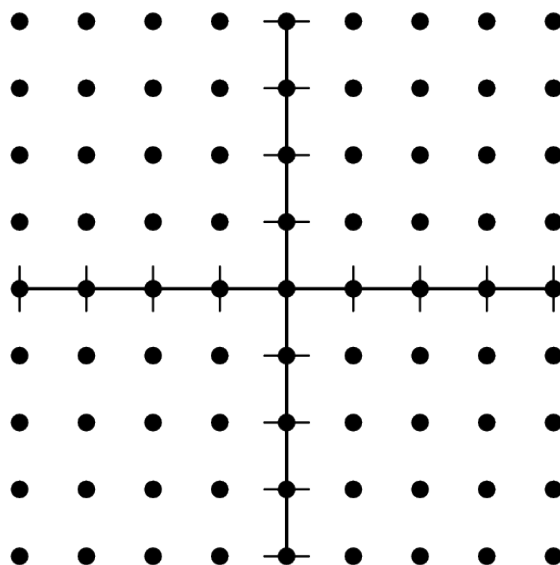
**Solution**

2000% is equivalent to  $20 \times 100\%$ . Therefore, 2000% of a number is the same as 20 times that number. 4 quarters is 1 dollar, so Bryden will get  $20 \times 1 = 20$  dollars,  $\boxed{\text{A}}$ .

**Solution 2**

Since 2000% is just  $\frac{2000}{100}$ , we can multiply that by 100, because four quarters is a 100 cents. After the multiplication, we get 2000. Since our answer is in cents right now, we need to convert it to dollars, which would be (A) 20 dollars.

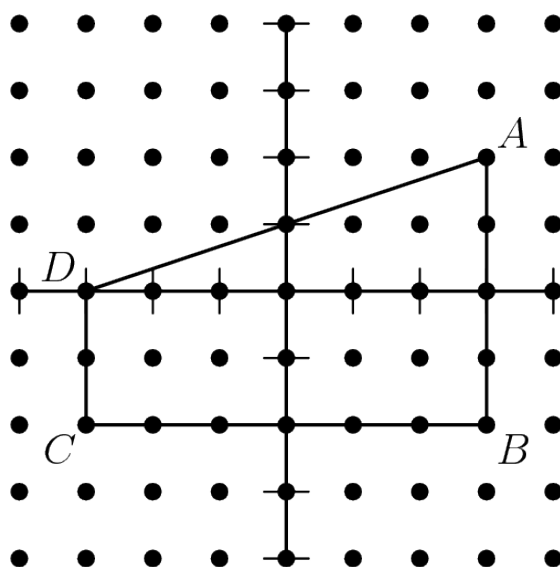
11. (AMC 8 2001) Points  $A$ ,  $B$ ,  $C$  and  $D$  have these coordinates:  $A(3, 2)$ ,  $B(3, -2)$ ,  $C(-3, -2)$  and  $D(-3, 0)$ . The area of quadrilateral  $ABCD$  is



- (A) 12      (B) 15      (C) 18      (D) 21      (E) 24

**Answer:** C

**Solution 1**



This quadrilateral is a trapezoid, because  $AB \parallel CD$  but  $BC$  is not parallel to  $AD$ . The area of a trapezoid is the product of its height and its median, where the median is the average of the side lengths of the bases. The two bases are  $AB$  and  $CD$ , which have lengths 2 and 4, respectively, so the length of the median is  $\frac{2+4}{2} = 3$ .  $CB$  is perpendicular to the bases, so it is the height, and has length 6. Therefore, the area of the trapezoid is  $(3)(6) = 18$ , C



**Solution 2**

Using the diagram above, the figure can be divided along the x-axis into two familiar regions that do not overlap: a right triangle and a rectangle. Since the areas do not overlap, the area of the entire trapezoid is the sum of the area of the triangle and the area of the rectangle.

$$A_{trap} = A_{tri} + A_{rect} \quad A_{trap} = \frac{1}{2}bh + lw \quad A_{trap} = \frac{1}{2} \cdot 6 \cdot 2 + 6 \cdot 2 \quad A_{trap} = 6 + 12 = 18 \rightarrow \boxed{C}$$

**Solution 3**

Using the diagram in solution 1 we can use the Pick's Theorem to solve this problem. There are twelve lattice points inside the quadrilateral and fourteen lattice points on the quadrilateral. Applying the Pick's Theorem,  $12 + \frac{14}{2} - 1 = 18 \rightarrow \boxed{C}$

**Solution 4 (Can someone make an asymptote for this)**

We can add a 2 by 6 triangle to make a 4\*6 rectangle which has area 24. Then subtracting the area of the triangle, we get  $24 - (2 \cdot 6 / 2) = 24 - 6 = 18$ , answer C

12. (AMC 8 2001) If  $a \otimes b = \frac{a+b}{a-b}$ , then  $(6 \otimes 4) \otimes 3 =$  (A) 4 (B) 13 (C) 15 (D) 30 (E) 72

**Answer:** A

**Solution 1**

$$6 \otimes 4 = \frac{6+4}{6-4} = \frac{10}{2} = 5. \quad 5 \otimes 3 = \frac{5+3}{5-3} = \frac{8}{2} = \boxed{(A) 4}$$

**Solution 2 (Overkill)**

When you expand the general form of  $(a \otimes b) \otimes c$ , you get

$$\begin{aligned} (a \otimes b) \otimes c &= \frac{a \otimes b + c}{a \otimes b - c} \\ (a \otimes b) \otimes c &= \frac{\frac{a+b}{a-b} + c}{\frac{a+b}{a-b} - c} \\ (a \otimes b) \otimes c &= \frac{\frac{a+b+ac-bc}{a-b}}{\frac{a+b-ac+bc}{a-b}} \\ (a \otimes b) \otimes c &= \frac{a+b+ac-bc}{a+b-ac+bc} \end{aligned}$$

Now, substituting  $a = 6$ ,  $b = 4$ , and  $c = 3$ :

$$\begin{aligned} (6 \otimes 4) \otimes 3 &= \frac{6+4+18-12}{6+4-18+12} \\ (6 \otimes 4) \otimes 3 &= \frac{16}{4} \\ (6 \otimes 4) \otimes 3 &= 4 \end{aligned}$$

(A)

13. (*AMC 8 2001*) Of the 36 students in Richelle's class, 12 prefer chocolate pie, 8 prefer apple, and 6 prefer blueberry. Half of the remaining students prefer cherry pie and half prefer lemon. For Richelle's pie graph showing this data, how many degrees should she use for cherry pie?

(A) 10      (B) 20      (C) 30      (D) 50      (E) 72

**Answer:** D

**Solution**

There are 36 students in the class: 12 prefer chocolate pie, 8 prefer apple pie, and 6 prefer blueberry pie. Therefore,  $36 - 12 - 8 - 6 = 10$  students prefer cherry pie or lemon pie. Half of these prefer each, so 5 students prefer cherry pie. This means that  $\frac{5}{36}$  of the students prefer cherry pie, so  $\frac{5}{36}$  of the full  $360^\circ$  should be used for cherry pie. This is  $(\frac{5}{36})(360^\circ) = 50^\circ$ , **(D)**

14. (*AMC 8 2001*) Tyler has entered a buffet line in which he chooses one kind of meat, two different vegetables and one dessert. If the order of food items is not important, how many different meals might he choose?

- Meat: beef, chicken, pork
- Vegetables: baked beans, corn, potatoes, tomatoes
- Dessert: brownies, chocolate cake, chocolate pudding, ice cream

(A) 4      (B) 24      (C) 72      (D) 80      (E) 144

**Answer:** C

**Solution 1**

There are 3 possibilities for the meat and 4 possibilities for the dessert, for a total of  $4 \times 3 = 12$  possibilities for the meat and the dessert. There are 4 possibilities for the first vegetable and 3 possibilities for the second, but order doesn't matter, so we overcounted by a factor of 2. For example, we counted 'baked beans and corn' and 'corn and baked beans' as 2 different possibilities, so the total possibilities for the two vegetables is  $\frac{4 \times 3}{2} = 6$ , and the total number of possibilities is  $12 \times 6 = \mathbf{(C) 72}$ .

**Solution 2 (combinatorics)**

To find the total ways, we multiply all the combinations of all 3 items:

For meat, it's asking how many ways can you pick 1 item out of 3 or  $\binom{3}{1} = \frac{3!}{(3-1)!1!} = \frac{3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 1} = 3$  (plain logic.)

For vegetables, it's the amount of ways you can pick 2 items out of 4 or  $\binom{4}{2} = \frac{4!}{(4-2)!2!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 2 \cdot 1} = \frac{12}{2} = 6$

For dessert, it's the amount of ways you can pick 1 item out of 4 or  $\binom{4}{1} = \frac{4!}{(4-1)!1!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 1} = 4$  (also logic).

Therefore, it's  $3 \times 6 \times 4 = 12 \times 6 = \mathbf{(C) 72}$ .

15. (*AMC 8 2001*) Homer began peeling a pile of 44 potatoes at the rate of 3 potatoes per minute. Four minutes later Christen joined him and peeled at the rate of 5 potatoes per minute. When they finished, how many potatoes had Christen peeled?

(A) 20      (B) 24      (C) 32      (D) 33      (E) 40

**Answer:** A

**Solution**

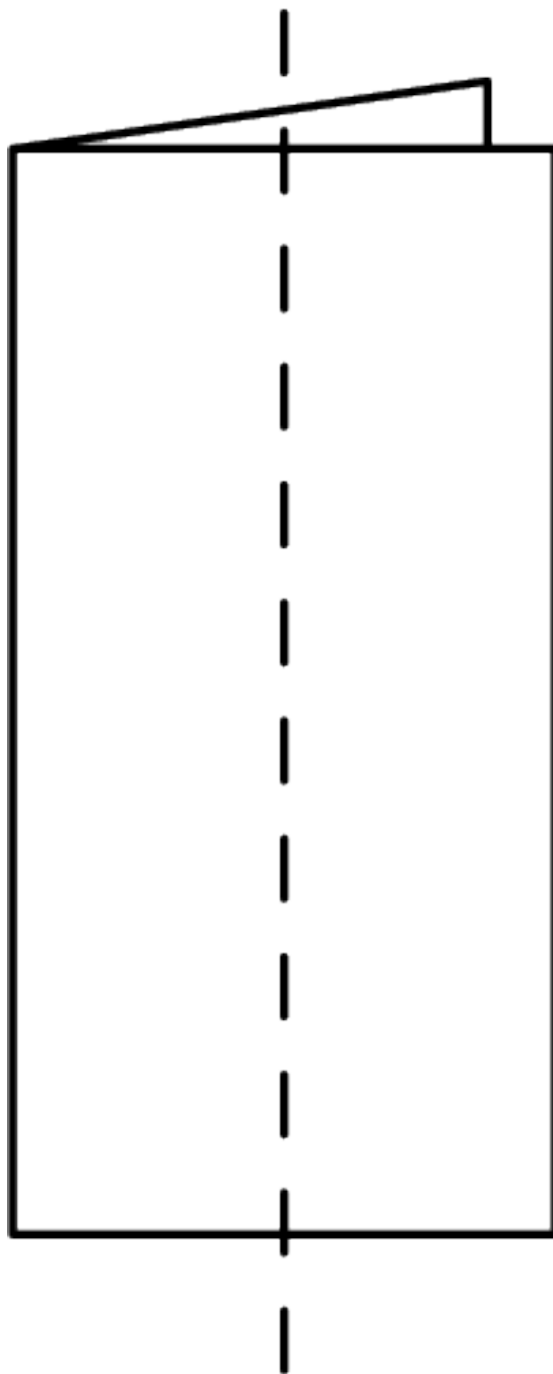
After the 4 minutes of Homer peeling alone, he had peeled  $4 \times 3 = 12$  potatoes. This means that there are  $44 - 12 = 32$  potatoes left. Once Christen joins him, the two are peeling potatoes at a rate of  $3 + 5 = 8$  potatoes per minute. So, they finish peeling after another  $\frac{32}{8} = 4$  minutes. In these 4 minutes, Christen peeled  $4 \times 5 = \boxed{\text{A}}$  20 potatoes.

**Solution 2**

After the 4 minutes of Homer peeling alone, he had peeled  $4 \times 3 = 12$  potatoes. This means that there are  $44 - 12 = 32$  potatoes left. Once Christen joins him, the two are peeling potatoes at a rate of  $3 + 5 = 8$  potatoes per minute. Because Christen's rate is 5, she peels  $\frac{5}{8}$  of 32, or 20 potatoes.

-N.

16. (*AMC 8 2001*) A square piece of paper, 4 inches on a side, is folded in half vertically. Both layers are then cut in half parallel to the fold. Three new rectangles are formed, a large one and two small ones. What is the ratio of the perimeter of one of the small rectangles to the perimeter of the large rectangle?



- (A)  $\frac{1}{3}$     (B)  $\frac{1}{2}$     (C)  $\frac{3}{4}$     (D)  $\frac{4}{5}$     (E)  $\frac{5}{6}$

**Answer:** E

**Solution 1**

The smaller rectangles each have the same height as the original square, but have  $\frac{1}{4}$  the length, since the paper is folded in half and then cut in half the same way. The larger rectangle has the same height as the original square but has  $\frac{1}{2}$  the length, since the paper is cut in half after the fold but the fold retains both sides of the larger rectangle. Therefore, the smaller rectangles have dimensions  $4 \times 1$  and the larger rectangle has dimensions  $4 \times 2$ . The ratio of their perimeters is  $\frac{2(4+1)}{2(4+2)} = \frac{5}{6}$ , E

**Solution 2**

A simpler way to solve the problem is by brute-forcing it. We know that the side length of the original paper is 4 inches by 4 inches. The smaller paper would have a side length of 4 inches by  $4 \cdot \frac{1}{4} = 1$  inches, having a perimeter of  $2(4 + 1) = 2(5) = 10$  inches. The larger rectangle would have a side length of 4 inches by  $4 \cdot \frac{1}{2} = 2$  inches, having a perimeter of  $2(4 + 2) = 2(6) = 12$  inches. Thus, the answer is  $\frac{10}{12} = \frac{5}{6}$ , i.e. **E**

17. (AMC 8 2001) For the game show *Who Wants To Be A Millionaire?*, the dollar values of each question are shown in the following table (where K = 1000).

|          |     |     |     |     |    |    |    |    |     |     |     |      |      |      |
|----------|-----|-----|-----|-----|----|----|----|----|-----|-----|-----|------|------|------|
| Question | 1   | 2   | 3   | 4   | 5  | 6  | 7  | 8  | 9   | 10  | 11  | 12   | 13   | 14   |
| Value    | 100 | 200 | 300 | 500 | 1K | 2K | 4K | 8K | 16K | 32K | 64K | 125K | 250K | 500K |

Between which two questions is the percent increase of the value the smallest?

- (A) From 1 to 2      (B) From 2 to 3      (C) From 3 to 4      (D) From 11 to 12      (E) From 14 to 15

**Answer: B**

**Solution**

Notice that in two of the increases, the dollar amount doubles. The increases in which this is not true is 2 to 3, 3 to 4, and 11 to 12. We can disregard 11 to 12 since that increase is almost 2 times. The increase from 2 to 3 is  $\frac{300-200}{200} = \frac{1}{2}$ , so it's only multiplied by a factor of 1.5. The increase from 3 to 4 is  $\frac{500-300}{300} = \frac{2}{3}$ , so it is multiplied by a factor of 1.6. Therefore, the smallest percent increase is From 2 to 3, **B**

18. (AMC 8 2001) Two dice are thrown. What is the probability that the product of the two numbers is a multiple of 5?

- (A)  $\frac{1}{36}$       (B)  $\frac{1}{18}$       (C)  $\frac{1}{6}$       (D)  $\frac{11}{36}$       (E)  $\frac{1}{3}$

**Answer: D**

**Solution 1**

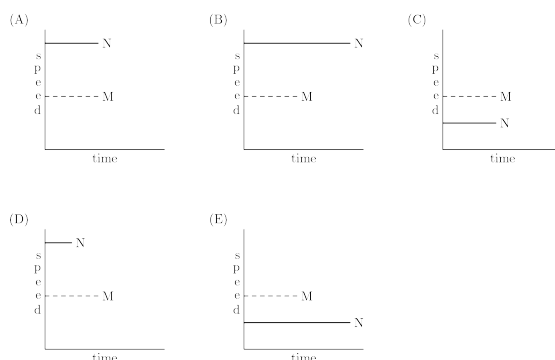
This is equivalent to asking for the probability that at least one of the numbers is a multiple of 5, since if one of the numbers is a multiple of 5, then the product with it and another integer is also a multiple of 5, and if a number is a multiple of 5, then since 5 is prime, one of the factors must also have a factor of 5, and 5 is the only multiple of 5 on a die, so one of the numbers rolled must be a 5. To find the probability of rolling at least one 5, we can find the probability of not rolling a 5 and subtract that from 1, since you either roll a 5 or not roll a 5. The probability of not rolling a 5 on either dice is  $\left(\frac{5}{6}\right)\left(\frac{5}{6}\right) = \frac{25}{36}$ . Therefore, the probability of rolling at least one five, and thus rolling two numbers whose product is a multiple of 5, is  $1 - \frac{25}{36} = \frac{11}{36}$ , **D**

**Solution 2 (quick easy)**

The only way to get a multiple of 5 is to have at least one 5. If the first dice rolls a 5, there are 6 ways to get a multiple of 5. If the second dice rolls a 5, there are also 6 ways. However, one case is repeated: both dice roll a 5. Therefore, there are  $6 + 6 - 1 = 11$ , and there is a total of  $6 \times 6$  ways, so the probability is (D)  $\frac{11}{36}$

Solution by [IHateMath314265169](#)

19. (AMC 8 2001) Car M traveled at a constant speed for a given time. This is shown by the dashed line. Car N traveled at twice the speed for the same distance. If Car N's speed and time are shown as solid line, which graph illustrates this?



**Answer: D**

**Solution 1**

Since car N has twice the speed, it must be twice as high on the speed axis. Also, since cars M and N travel at the same distance but car N has twice the speed, car N must take half the time. Therefore, line N must be half the size of line M. Since the speeds are constant, both lines are horizontal. Reviewing the graphs, we see that the only one satisfying these conditions is graph **D**.

**Solution 2**

Since Car N goes at a faster speed than Car M, it will take less time to get to the same destination. Reviewing the graphs, we see that only one graph fits this criteria, graph **D**.

20. (AMC 8 2001) Kaleana shows her test score to Quay, Marty and Shana, but the others keep theirs hidden. Quay thinks, "At least two of us have the same score." Marty thinks, "I didn't get the lowest score." Shana thinks, "I didn't get the highest score." List the scores from lowest to highest for Marty (M), Quay (Q) and Shana (S).

(A) S,Q,M      (B) Q,M,S      (C) Q,S,M      (D) M,S,Q      (E) S,M,Q

**Answer: A**

**Solution**

Since the only other score Quay knows is Kaleana's, and he knows that two of them have the same score, Quay and Kaleana must have the same score, and  $K = Q$ . Marty knows that he didn't get the lowest score, and the only other score he knows is Kaleana's, so Marty must know that Kaleana must have a lower score than him, and  $M > K$ . Finally, Shana knows that she didn't get the highest score, and the only other score she knows is Kaleana's, so Shana must know that Kaleana must have a higher score than her, and  $S < K$ . Putting these together and substituting  $Q$  for  $K$ , we have  $S < Q < M$ , and from least to greatest this is S, Q, M, therefore the answer is **A**.

21. (AMC 8 2001) The mean of a set of five different positive integers is 15. The median is 18. The maximum possible value of the largest of these five integers is
- (A) 19      (B) 24      (C) 32      (D) 35      (E) 40

**Answer:** D

**Solution**

Since there is an odd number of terms, the median is the number in the middle, specifically, the third largest number is 18, and there are 2 numbers less than 18 and 2 numbers greater than 18. The sum of these integers is  $5(15) = 75$ , since the mean is 15. To make the largest possible number with a given sum, the other numbers must be as small as possible. The two numbers less than 18 must be positive and distinct, so the smallest possible numbers for these are 1 and 2. The number right after 18 also needs to be as small as possible, so it must be 19. This means that the remaining number, the maximum possible value for a number in the set, is  $75 - 1 - 2 - 18 - 19 = 35$ , (D) 35.

22. (*AMC 8 2001*) On a twenty-question test, each correct answer is worth 5 points, each unanswered question is worth 1 point and each incorrect answer is worth 0 points. Which of the following scores is **NOT** possible?

(A) 90      (B) 91      (C) 92      (D) 95      (E) 97

**Answer:** E

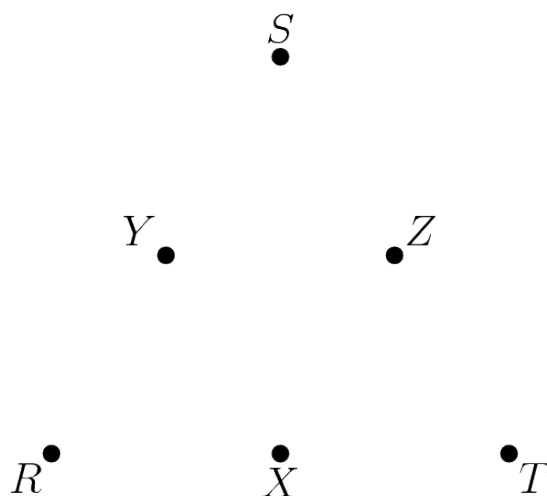
**Solution**

The highest possible score is if you get every answer right, to get  $5(20) = 100$ . The second highest possible score is if you get 19 questions right and leave the remaining one blank, to get a  $5(19) + 1(1) = 96$ . Therefore, no score between 96 and 100, exclusive, is possible, so 97 is not possible, E.

**Solution 2**

We can equivalently construct the following rules: You have 100 point at first, but if you give the wrong answer, you will lose 5 points, if you don't answer a question you will lose 4 points. Obviously, you can lose 10 points, 9 points, 8 points, 5 points or 4 points, but you cannot lose 3 points. The answer is E.

23. (*AMC 8 2001*) Points  $R$ ,  $S$  and  $T$  are vertices of an equilateral triangle, and points  $X$ ,  $Y$  and  $Z$  are midpoints of its sides. How many noncongruent triangles can be drawn using any three of these six points as vertices?



(A) 1      (B) 2      (C) 3      (D) 4      (E) 20

**Answer: D**

**Solution 1 (Complementary Counting)**

There are 6 points in the figure, and 3 of them are needed to form a triangle, so there are  $\binom{6}{3} = 20$  possible triplets of the 6 points.

**Case 1: Triangles congruent to  $\triangle RST$**  There is obviously only 1  $\triangle RST$  itself.

**Case 2: Triangles congruent to  $\triangle SYZ$**  There are 4  $\triangle SYZ$  of the form  $\triangle SYZ$ ,  $\triangle RXY$ ,  $\triangle TXZ$ , and  $\triangle XYZ$ .

**Case 3: Triangles congruent to  $\triangle RSX$**  There are 6  $\triangle RSX$  of the form  $\triangle RSX$ ,  $\triangle TSX$ ,  $\triangle STY$ ,  $\triangle RTY$ ,  $\triangle RSZ$ , and  $\triangle RTZ$ .

**Case 4: Triangles congruent to  $\triangle SYX$**  There are again 6  $\triangle SYX$  of the form  $\triangle SYX$ ,  $\triangle SZX$ ,  $\triangle TYZ$ ,  $\triangle TYX$ ,  $\triangle RXZ$ , and  $\triangle RYZ$ .

However, if we add these up, we accounted for only  $1 + 4 + 6 + 6 = 17$  of the 20 possible triplets. We see that the remaining triplets don't even form triangles; they are  $\triangle SYR$ ,  $\triangle RXT$ , and  $\triangle TZS$ . Adding these 3 into the total yields for all of the possible triplets, so we see that there are only 4 possible non-congruent, non-degenerate triangles,  $\boxed{D}$ .

**Solution 2 (Brute Force)**

We can do casework in this problem like the solution above, but that would take too much time. Instead, we see that we can split this equilateral dot triangle to two halves by dropping an altitude from the top vertex of the big triangle. Using the smaller triangle we won't have to worry about extra unneeded cases. We can see that there are three distinct triangles in the half, and combining this with the larger equilateral triangle our answer is  $3 + 1 = \boxed{4}$ , which is  $\boxed{D}$ .

–FIREDRAGONMATH16

**Solution 3 (Elimination)**

Notice that 20 is obviously too high (There are only 20 ways to choose 3 of the points to form a triangle or a line in total!) and you can count 4 distinct triangles quickly:

$\triangle RYX$ ,  $\triangle RYT$ ,  $\triangle RYZ$ ,  $\triangle RST$

**Solution 4 (Some symmetry)**

Note that there is only one triangle  $\triangle XYZ$  that does not use any vertex from the largest triangle. By symmetry, we assume that  $\triangle RST$  is congruent to  $\triangle SYZ$ ,  $\triangle SRX$ ,  $\triangle SRT$ ,  $\triangle SXZ$ ,  $\triangle RYX$ ,  $\triangle RYT$ ,  $\triangle RYZ$ .

–Sean Y

**Solution 5 (FREESTYLE!)**

Simply count all possible such triangles, while keeping in mind that the maximum length of any such triangle is 2 units:

(2,1,1) lengths

(2,1,2) lengths

(2,2,2) lengths

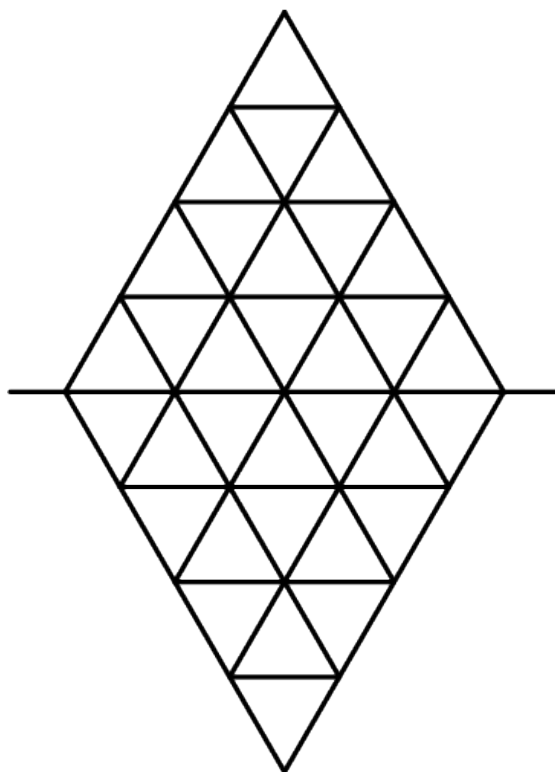
(1,1,1) lengths



To avoid stuff being congruent, we know that there are 4 different such triangles. So, our answer is  $\boxed{4}$ .

- cheltstudent B)

24. (AMC 8 2001) Each half of this figure is composed of 3 red triangles, 5 blue triangles and 8 white triangles. When the upper half is folded down over the centerline, 2 pairs of red triangles coincide, as do 3 pairs of blue triangles. There are 2 red-white pairs. How many white pairs coincide?



- (A) 4      (B) 5      (C) 6      (D) 7      (E) 9

**Answer:** B

**Solution**

Each half has 3 red triangles, 5 blue triangles, and 8 white triangles. There are also 2 pairs of red triangles, so 2 red triangles on each side are used, leaving 1 red triangle, 5 blue triangles, and 8 white triangles remaining on each half. Also, there are 3 pairs of blue triangles, using 3 blue triangles on each side, so there is 1 red triangle, 2 blue triangles, and 8 white triangles remaining on each half. Also, we have 2 red-white pairs. This obviously can't use 2 red triangles on one side, since there is only 1 on each side, so we must use 1 red triangle and 1 white triangle per side, leaving 2 blue triangles and 7 white triangles apiece. The remaining blue triangles cannot be matched with other blue triangles since that would mean there were more than 3 blue pairs, so the remaining blue triangles must be paired with white triangles, yielding 4 blue-white pairs, one for each of the remaining blue triangles. This uses 2 blue triangles and 2 white triangles on each side, leaving 5 white triangles apiece, which must be paired with each other, so there are 5 white-white pairs,  $\boxed{B}$ .

25. (AMC 8 2001) There are 24 four-digit whole numbers that use each of the four digits 2,

4, 5 and 7 exactly once. Only one of these four-digit numbers is a multiple of another one. Which of the following is it?

- (A) 5724      (B) 7245      (C) 7254      (D) 7425      (E) 7542

**Answer:** D

**Solution 1**

The smallest number formed from these four digits is 2457 while the largest is 7542. Since  $7542 < 2457 \cdot 4$ , the correct answer must be either 2 or 3 times another number containing these four digits.

By inspection, **(D) 7425** is correct since  $7425 \div 3 = 2475$ .

(A) is incorrect since  $5724 \div 2 = 2862$  and  $5724 \div 3 = 1908$ .

(B) is incorrect since  $7245 \div 3 = 2415$ .

(C) is incorrect since  $7254 \div 2 = 3627$  and  $7254 \div 3 = 2418$ .

(E) is incorrect since  $7542 \div 2 = 3771$  and  $7542 \div 3 = 2514$ .

**Solution 2**

Let  $ABCD$  and  $WXYZ$  be four-digit integers such that  $\{A, B, C, D\} = \{W, X, Y, Z\} = \{2, 4, 5, 7\}$ , and  $WXYZ$  is a multiple of  $ABCD$ . As in solution 1, either  $ABCD \times 2 = WXYZ$  or  $ABCD \times 3 = WXYZ$ .

Suppose  $ABCD \times 2 = WXYZ$ .

Then  $A = 2$ , since otherwise  $ABCD \times 2$  is too large. If  $D = 4$  or  $D = 5$ , then  $Z = 8$  or  $Z = 0$  which is invalid. So  $D = 7$  and  $Z = 4$ . Since  $W$  is at least 4 but less than 6,  $W = 5$ . If  $B = 4$ ,  $2 \times ABCD$  is too small, so  $B = 5$ . Thus  $C = 7$  and  $ABCD = 2547$ . But in this case  $2 \times ABCD = 5094$ , which is invalid.

Thus  $ABCD \times 3 = WXYZ$ .

Then  $A = 2$ , since otherwise  $ABCD \times 3$  is too large. If  $B = 7$ , then  $ABCD > 2700$  and  $WXYZ > 3 \times 2700 = 8100$  which is too large. If  $D = 7$ , then  $Z = 1$  which is invalid. So  $C = 7$ . If  $B = 4$  and  $D = 5$ , then  $WXYZ = 3 \times 2574 = 7722$  which is invalid. Therefore,  $B = 5$  and  $D = 4$ , so  $WXYZ = 3 \times 2475 = \mathbf{(D) 7425}$ .

-j314andrews

**Solution 3**

Let  $ABCD$  and  $WXYZ$  be four-digit integers such that  $\{A, B, C, D\} = \{W, X, Y, Z\} = \{2, 4, 5, 7\}$ , and  $WXYZ$  is a multiple of  $ABCD$ . As in solution 1, either  $ABCD \times 2 = WXYZ$  or  $ABCD \times 3 = WXYZ$ .

Suppose  $ABCD \times 2 = WXYZ$ .

Since  $ABCD \geq 2457$ ,  $WXYZ \geq 2457 \times 2 = 4914$ . Note that 5274 is the smallest possible even value of  $WXYZ$  which is at least 4914, so  $WXYZ \geq 5274$ .

Since  $WXYZ \leq 7542$ ,  $ABCD \leq 7542 \div 2 = 3761$ . Note that 2754 is the largest possible value of  $ABCD$  which is at most 3761, so  $ABCD \leq 2754$ .

Since  $ABCD \leq 2754$ ,  $ABCD \leq 2754 \times 2 = 5508$ . Note that 5472 is the largest possible even value of  $WXYZ$  which is at most 5508, so  $WXYZ \leq 5472$ .

Since  $WXYZ \leq 5472$ ,  $ABCD \leq 5472 \div 2 = 2736$ . Note that 2574 is the largest possible value of  $ABCD$  which is at most 2736, so  $ABCD \leq 2574$ .

Since  $ABCD \leq 2574$ ,  $WXYZ \leq 2574 \times 2 = 5148$ . But it is impossible for both  $WXYZ \geq 5274$  and  $WXYZ \leq 5148$ , and thus  $ABCD \times 2 \neq WXYZ$ .

Therefore,  $ABCD \times 3 = WXYZ$ .

Since  $ABCD \geq 2457$ ,  $WXYZ \geq 2457 \times 3 = 7371$ . Note that 7425 is the smallest possible value of  $WXYZ$  which is at least 7371, so  $WXYZ \geq 7425$ .

Since  $WXYZ \leq 7542$ ,  $ABCD \leq 7542 \div 3 = 2514$ . Note that 2475 is the largest possible value of  $ABCD$  which is at most 2514, so  $ABCD \leq 2475$ .

Since  $ABCD \leq 2475$ ,  $WXYZ \leq 2475 \times 3 = 7425$ . Since both  $WXYZ \geq 7425$  and  $WXYZ \leq 7425$ ,  $WXYZ = \boxed{\text{(D)} 7425}$  (and thus  $ABCD = 7425 \div 3 = 2475$ ).

-j314andrews

## §18 AMC 8 2002

1. (AMC 8 2002) A circle and two distinct lines are drawn on a sheet of paper. What is the largest possible number of points of intersection of these figures?

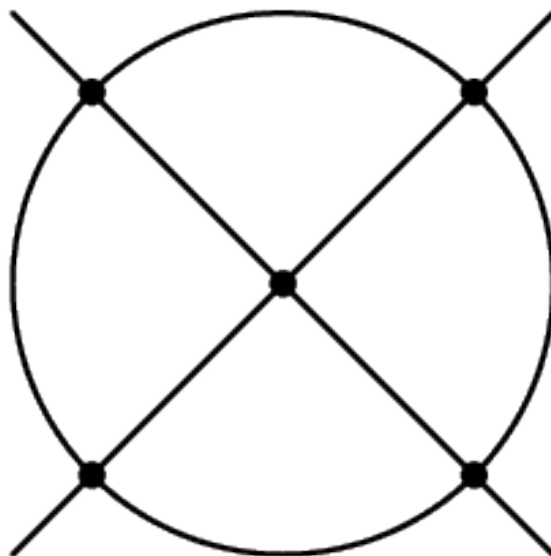
(A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** D

**Solution**

Each line may intersect the circle at most twice, while the two lines may intersect each other at most once. This yields a total of  $2 \cdot 2 + 1 = \boxed{\text{(D)} 5}$  intersections.

A possible configuration is shown below:



2. (AMC 8 2002) How many different combinations of 5 bills and 2 bills can be used to make a total of 17? Order does not matter in this problem.

(A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** A

**Solution**

Since four \$5 bills total to \$20, at most three \$5 bills may be used.

If three \$5 bills are used, the \$2 bills must total to \$2, so one \$2 bill may be used.

If two \$5 bills are used, the \$2 bills must total to \$7, which is impossible since 7 is odd.

If one \$5 bill is used, the \$2 bills must total to \$12, so six \$2 bills may be used.

If no \$5 bills are used, the \$2 bills must total to \$17, which is impossible since 17 is odd.

Thus there are **(A) 2** such combinations of bills that total to \$17.

3. (AMC 8 2002) What is the smallest possible average of four distinct positive even integers?

(A) 3      (B) 4      (C) 5      (D) 6      (E) 7

**Answer: C**

**Solution**

The smallest average is achieved by using the four smallest positive even integers, which are 2, 4, 6, and 8. Their average is  $\frac{2+4+6+8}{4} = \frac{20}{4} = \mathbf{(C) 5}$ .

4. (AMC 8 2002) The year 2002 is a palindrome (a number that reads the same from left to right as it does from right to left). What is the product of the digits of the next year after 2002 that is a palindrome?

(A) 0      (B) 4      (C) 9      (D) 16      (E) 25

**Answer: B**

**Solution**

The next palindrome is 2112, so the answer is  $2 \cdot 1 \cdot 1 \cdot 2 = \mathbf{(B) 4}$ .

5. (AMC 8 2002) Carlos Montado was born on Saturday, November 9, 2002. On what day of the week will Carlos be 706 days old?

(A) Monday      (B) Wednesday      (C) Friday      (D) Saturday      (E) Sunday

**Answer: C**

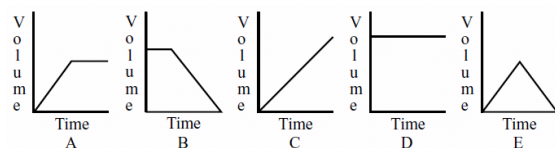
**Solution 1**

After 700 days, or 100 weeks, it will be Saturday. Thus after 706 days, it will be 6 days after Saturday, which is **(C) Friday**.

**Solution 2**

After 707 days, or 101 weeks, it will be Saturday. Thus after 706 days, it will be 1 day before Saturday, which is **(C) Friday**.

6. (AMC 8 2002) A birdbath is designed to overflow so that it will be self-cleaning. Water flows in at the rate of 20 milliliters per minute and drains at the rate of 18 milliliters per minute. One of these graphs shows the volume of water in the birdbath during the filling time and continuing into the overflow time. Which one is it?



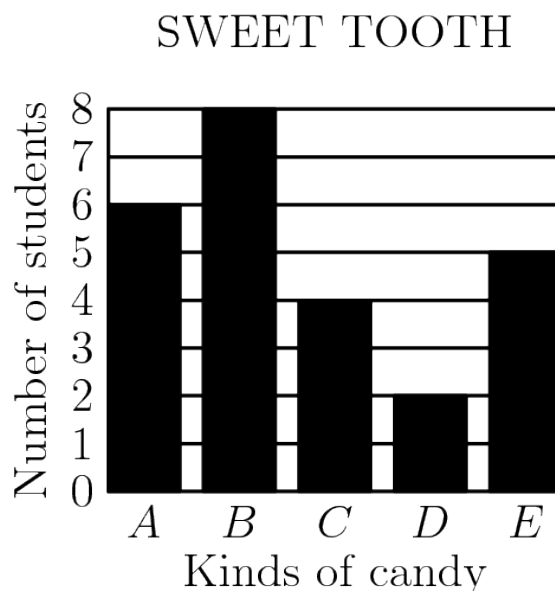
(A) A      (B) B      (C) C      (D) D      (E) E

**Answer: A**

**Solution**

At first, the volume of water in the birdbath increases by  $20 - 18 = 2$  millimeters per minute. Once the birdbath is full, the volume of water inside remains constant. This is best represented by graph (A) A.

7. (AMC 8 2002) The students in Mrs. Sawyer's class were asked to do a taste test of five kinds of candy. Each student chose one kind of candy. A bar graph of their preferences is shown. What percent of her class chose candy E?



- (A) 5      (B) 12      (C) 15      (D) 16      (E) 20

**Answer:** E

**Solution**

According to the graph, 5 students out of  $6 + 8 + 4 + 2 + 5 = 25$  total chose candy E. Thus  $\frac{5}{25} = 0.2 = \text{span style="border: 1px solid black; padding: 0 2px;">(E) 20\%$  chose candy E.

8. (AMC 8 2002) **Juan's Old Stamping Grounds** Problems 8,9 and 10 use the data found in the accompanying paragraph and table:

Juan organizes the stamps in his collection by country and by the decade in which they were issued. The prices he paid for them at a stamp shop were: Brazil and France, 6 cents each, Peru 4 cents each, and Spain 5 cents each. (Brazil and Peru are South American countries and France and Spain are in Europe.)

Number of Stamps by Decade

| Country | 50s | 60s | 70s | 80s |
|---------|-----|-----|-----|-----|
| Brazil  | 4   | 7   | 12  | 8   |
| France  | 8   | 4   | 12  | 15  |
| Peru    | 6   | 4   | 6   | 10  |
| Spain   | 3   | 9   | 13  | 9   |

Juan's Stamp Collection

How many of his European stamps were issued in the '80s?

- (A) 9      (B) 15      (C) 18      (D) 24      (E) 42

**Answer:** D

**Solution**

Juan has 15 French and 9 Spanish stamps from the '80s, for a total of  $15 + 9 = \boxed{\text{(D)} 24}$  European stamps from the '80s.

9. (*AMC 8 2002*) **Juan's Old Stamping Grounds** Problems 8,9 and 10 use the data found in the accompanying paragraph and table:

Juan organizes the stamps in his collection by country and by the decade in which they were issued. The prices he paid for them at a stamp shop were: Brazil and France, 6 cents each, Peru 4 cents each, and Spain 5 cents each. (Brazil and Peru are South American countries and France and Spain are in Europe.)

Number of Stamps by Decade

| Country | 50s | 60s | 70s | 80s |
|---------|-----|-----|-----|-----|
| Brazil  | 4   | 7   | 12  | 8   |
| France  | 8   | 4   | 12  | 15  |
| Peru    | 6   | 4   | 6   | 10  |
| Spain   | 3   | 9   | 13  | 9   |

Juan's Stamp Collection

His South American stamps issued before the '70s cost him

- (A) \$0.40      (B) \$1.06      (C) \$1.80      (D) \$2.38      (E) \$2.64

**Answer:** B

**Solution**

Juan has a total of  $4 + 7 = 11$  Brazilian and  $6 + 4 = 10$  Peruvian stamps from the '50s and '60s. These stamps cost a total of  $11 \cdot \$0.06 + 10 \cdot \$0.04 = \boxed{\text{(B)} \$1.06}$

10. (*AMC 8 2002*) **Juan's Old Stamping Grounds** Problems 8,9 and 10 use the data found in the accompanying paragraph and table:

Juan organizes the stamps in his collection by country and by the decade in which they were issued. The prices he paid for them at a stamp shop were: Brazil and France, 6 cents each, Peru 4 cents each, and Spain 5 cents each. (Brazil and Peru are South American countries and France and Spain are in Europe.)

Number of Stamps by Decade

| Country | 50s | 60s | 70s | 80s |
|---------|-----|-----|-----|-----|
| Brazil  | 4   | 7   | 12  | 8   |
| France  | 8   | 4   | 12  | 15  |
| Peru    | 6   | 4   | 6   | 10  |
| Spain   | 3   | 9   | 13  | 9   |

Juan's Stamp Collection

The average price of his '70s stamps is closest to

- (A) 3.5 cents      (B) 4 cents      (C) 4.5 cents      (D) 5 cents      (E) 5.5 cents

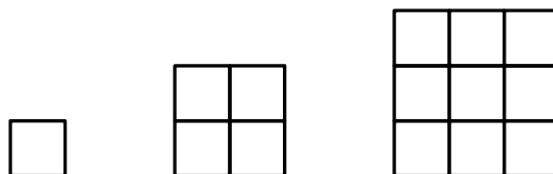
**Answer:** E

**Solution**

Juan has a total of  $12 + 12 + 6 + 13 = 43$  stamps from the '70s, and these stamps cost a total of  $12 \cdot \$0.06 + 12 \cdot \$0.06 + 6 \cdot \$0.04 = 13 \cdot \$0.05 = \$2.33$ .

The average price of these stamps is therefore  $\$2.33 \div 43 \approx \$0.0542$ , which is closest to **(E) 5.5cent**.

11. (AMC 8 2002) A sequence of squares is made of identical square tiles. The edge of each square is one tile length longer than the edge of the previous square. The first three squares are shown. How many more tiles does the seventh square require than the sixth?



- (A) 11      (B) 12      (C) 13      (D) 14      (E) 15

**Answer:** C

**Solution 1**

For all positive integers  $n$ , the  $n^{\text{th}}$  square has side length  $n$  and thus consists of  $n^2$  tiles. The seventh square thus has  $7^2 = 49$  tiles while the sixth square has  $6^2 = 36$  tiles. Therefore, the seventh square has  $49 - 36 = \mathbf{(C) 13}$  more tiles than the sixth square.

**Solution 2**

As in solution 1, the sixth square has side length 6 and the seventh square has side length 7. A square of side length 6 may be transformed into a square of side length 7 by adding a row of 6 tiles below, a column of 6 tiles to the right, and 1 tile directly below the new column and to the right of the new row, for a total of  $6 + 6 + 1 = \mathbf{(C) 13}$  added tiles.

**Solution 3**

As in solution 1, the seventh square has  $7^2$  tiles while the sixth square has  $6^2$  tiles. Thus the seventh square has  $7^2 - 6^2 = (7 + 6)(7 - 6) = 13 \cdot 1 = \mathbf{(C) 13}$  more tiles than the sixth square.

12. (AMC 8 2002) A board game spinner is divided into three regions labeled  $A$ ,  $B$  and  $C$ . The probability of the arrow stopping on region  $A$  is  $\frac{1}{3}$  and on region  $B$  is  $\frac{1}{2}$ . The probability of the arrow stopping on region  $C$  is

- (A)  $\frac{1}{12}$       (B)  $\frac{1}{6}$       (C)  $\frac{1}{5}$       (D)  $\frac{1}{3}$       (E)  $\frac{2}{5}$

**Answer:** B

**Solution**

The total of the probabilities of the arrow stopping on  $A$ ,  $B$ , and  $C$  is 1, so the probability of the arrow stopping on  $C$  is  $1 - \frac{1}{2} - \frac{1}{3} = \mathbf{(B) \frac{1}{6}}$ .

13. (AMC 8 2002) For his birthday, Bert gets a box that holds 125 jellybeans when filled to capacity. A few weeks later, Carrie gets a larger box full of jellybeans. Her box is twice as high, twice as wide and twice as long as Bert's. Approximately, how many jellybeans did Carrie get?

- (A) 250      (B) 500      (C) 625      (D) 750      (E) 1000

**Answer:** E

**Solution**

Since each dimension of Carrie's box is twice the corresponding dimension of Bert's box, Carrie's box has  $2 \cdot 2 \cdot 2 = 8$  times the volume of Bert's Box. Therefore, Carrie's box can hold  $8 \cdot 125 = \boxed{\text{(E)} 1000}$  jellybeans.

14. (AMC 8 2002) A merchant offers a large group of items at 30% off. Later, the merchant takes 20% off these sale prices. The total discount is  
 (A) 35%      (B) 44%      (C) 50%      (D) 56%      (E) 60%

**Answer:** B

**Solution 1**

Let's assume that each item is 100 dollars. First we take off 30% off of 100 dollars.  $100 \cdot 0.7 = 70$

Next, we take off the extra 20% as asked by the problem.  $70 \cdot 0.80 = 56$

So the final price of an item is \$56. We have to do  $100 - 56$  because 56 was the final price and we wanted the discount. So the final answer is 44%, which is answer choice  $\boxed{\text{(B)} 44\%}$ .

**Solution 2**

Let  $x$  be the price of an item on sale. When the item is 30% off, its new price is

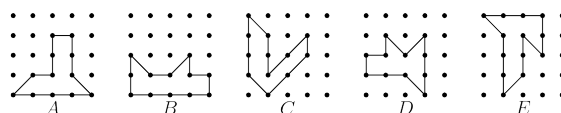
$$(1 - 0.3)x = 0.7x.$$

When 20% is taken off of that price, the item's final sales price is

$$(1 - 0.2) \cdot 0.7x = 0.8 \cdot 0.7x = 0.56x = (1 - 0.44)x.$$

Therefore, the item was 44% off, so the answer is  $\boxed{\text{(B)}}$ .

15. (AMC 8 2002) Which of the following polygons has the largest area?



- (A) A      (B) B      (C) C      (D) D      (E) E

**Answer:** E

**Solution 1**

Each polygon can be partitioned into unit squares and right triangles with sidelength 1. Count the number of boxes enclosed by each polygon, with the unit square being 1, and the triangle being .5. A has 5, B has 5, C has 5, D has 4.5, and E has 5.5. Therefore, the polygon with the largest area is  $\boxed{\text{(E)} E}$ .

**Solution 2 (Under 15 seconds)**

Pick's Theorem:

$$A = I + \frac{1}{2}B - 1$$

where  $I$  is the number of lattice points in the interior and  $B$  being the number of lattice points on the boundary.

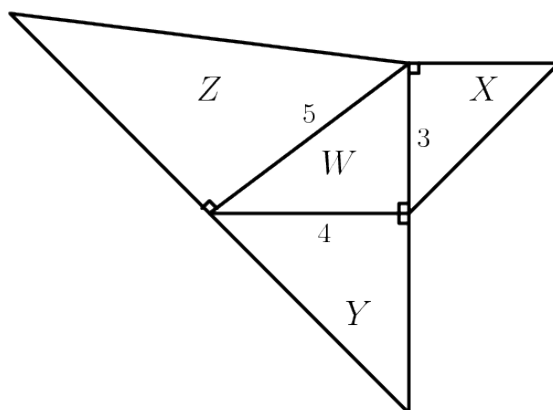
Ok, so we want to find the polygon with the largest area, first notice that not a single polygon has a lattice point in the interior. Now look at the formula:



$$A = \frac{1}{2}B - 1$$

So now just look at the polygon with the most lattice points on the boundary, and that is figure E, with 13 lattice points on the boundary. So the polygon with the largest area is **(E) E**.

16. (AMC 8 2002) Right isosceles triangles are constructed on the sides of a 3-4-5 right triangle, as shown. A capital letter represents the area of each triangle. Which one of the following is true?



- (A)  $X + Z = W + Y$  (B)  $W + X = Z$  (C)  $3X + 4Y = 5Z$  (D)  $X + W = \frac{1}{2}(Y + Z)$  (E)  $X + Y = Z$

**Answer:** E

### Solution 1

The area of a right triangle can be found by using the legs of triangle as the base and height. In the three isosceles triangles, the length of their second leg is the same as the side that is connected to the 3-4-5 triangle.

$$\begin{aligned} W &= (3)(4)/2 = 6 \\ X &= (3)(3)/2 = 4.5 \\ Y &= (4)(4)/2 = 8 \\ Z &= (5)(5)/2 = 12.5 \end{aligned}$$

Plugging into the answer choices, the only that works is **(E)  $X + Y = Z$** .

### Solution 2

Looking at the diagram, we notice that three right isosceles triangles on one right triangle reminds us of the Pythagorean theorem, since each right isosceles triangle is actually half of a square. Each square's area represents a side length squared, so the squares on the legs of the right triangle adds to the square on the hypotenuse. This gives  $2X + 2Y = 2Z$ . Then, dividing by 2 we get  $X + Y = Z$ , which is one of the answer choices. Since there can only be one correct answer, and there is already one, we see that the answer must be **(E)  $X + Y = Z$** .

17. (AMC 8 2002) In a mathematics contest with ten problems, a student gains 5 points for a correct answer and loses 2 points for an incorrect answer. If Olivia answered every problem and her score was 29, how many correct answers did she have?

(A) 5      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** C**Solution 1**

We can try to guess and check to find the answer. If she got five right, her score would be  $(5 * 5) - (5 * 2) = 15$ . If she got six right her score would be  $(6 * 5) - (2 * 4) = 22$ . That's close, but it's still not right! If she got 7 right, her score would be  $(7 * 5) - (2 * 3) = 29$ . Thus, our answer is (C) 7. avamarora

**Solution 2**

We can start with the full score, 50, and subtract not only 2 points for each incorrect answer but also the 5 points we gave her credit for. This expression is equivalent to her score, 29. Let  $x$  be the number of questions she answers correctly. Then, we will represent the number incorrect by  $10 - x$ .

$$50 - 7(10 - x) = 29$$

$$50 - 70 + 7x = 29$$

$$7x = 49$$

$$x = \text{(C) 7}$$

**Solution 3**

Suppose she got  $x$  questions right. Then she got  $10 - x$  questions wrong. Since she gains 5 points for a correct answer and loses 2 for an incorrect one, we can solve  $5x - 2(10 - x) = 29$  to get that  $x = \text{(C) 7}$ .

**Solution 4**

We see that Olivia's score is odd. Since subtracting multiples of 2 (even numbers) does not change a number's parity (odd or even), Olivia's score from only her correct answers must be odd. Then, we test odd multiples of 5 greater than 29 to see which one works. The smallest odd multiple of 5 greater than 29 is 35, meaning 7 answers were correct and the remaining 3 answers were wrong. We see that the final score is

$$7(5) - 3(2) = 35 - 6 = 29,$$

which is the score Olivia got. Thus, the answer is (C)7.

**Solution 5**

Suppose  $x$  represents the number of correctly answered questions and  $y$  represents the number of incorrectly answered questions.

We can model the situation with the following system of equations:

$$\begin{cases} x + y = 10 \\ 5x - 2y = 29 \end{cases}$$

Multiply the first equation by 2 to get:

$$2x + 2y = 20$$

Now add this equation to the second equation:

$$\begin{aligned}(2x + 2y) + (5x - 2y) &= 20 + 29 \\ 7x &= 49\end{aligned}$$

Solve for  $x$ :

$$x = 7$$

-N.

18. (AMC 8 2002) Gage skated 1 hr 15 min each day for 5 days and 1 hr 30 min each day for 3 days. How long would he have to skate the ninth day in order to average 85 minutes of skating each day for the entire time?
- (A) 1 hr      (B) 1 hr 10 min      (C) 1 hr 20 min      (D) 1 hr 40 min      (E) 2 hr

**Answer:** E

**Solution 1**

Converting into minutes and adding, we get that she skated  $75 * 5 + 90 * 3 + x = 375 + 270 + x = 645 + x$  minutes total, where  $x$  is the amount she skated on day 9. Dividing by 9 to get the average, we get  $\frac{645+x}{9} = 85$ . Solving for  $x$ ,

$$645 + x = 765$$

$$x = 120$$

Now we convert back into hours and minutes to get (E) 2 hr.

**Solution 2**

For the first five days, each day you are 10 minutes short of 85 minutes. And for the next three days, you are 5 minutes above 85 minutes. So in total you are missing  $3 * 5 - 5 * 10$ , which equals to negative 35. So on the ninth day, to have an average of 85 minutes, Gage need to skate for  $85 + 35$  minutes, which is 120 minutes, or (E) 2 hr.

19. (AMC 8 2002) How many whole numbers between 99 and 999 contain exactly one 0?
- (A) 72      (B) 90      (C) 144      (D) 162      (E) 180

**Answer:** D

**Solution 1**

Numbers with exactly one zero have the form  $\overline{a0b}$  or  $\overline{ab0}$ , where the  $a, b \neq 0$ . There are  $(9 \cdot 1 \cdot 9) + (9 \cdot 9 \cdot 1) = 81 + 81 = 162$  such numbers, hence our answer is (D).

**Solution 2 (Complementary Counting)**

(Whole numbers between 99 and 999)-(Whole numbers which do not contain exactly one 0)= (How many whole numbers between 99 and 999 contain exactly one 0).

1) How many whole numbers are between 99 and 999,  $999 - 99 = 900$ , so there are 900 numbers between 99 and 999.

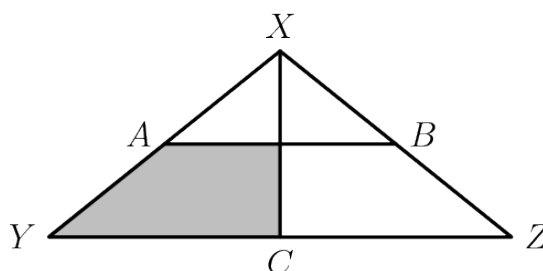
2a) How many whole numbers in this range do not contain the digit 0, there are  $10 - 1 = 9$  possible digits for each digit in this three digit number,  $9^3 = 729$ . So there are 729 numbers in this range which do not contain the digit 0.

2b) How many whole numbers in this range contain the digit 0 2 times, there are  $10 - 1 = 9$  possible digits for the first digit and the other two digits have to be the digit 0. So there are 9 of these numbers.

So there are  $900 - (729 + 9)$  whole numbers between 99 and 999 that contain exactly one 0.

Simplifying the expression we get that there are 162 whole numbers between 99 and 999 that contain exactly one 0. So the answer is D.

20. (AMC 8 2002) The area of triangle  $XYZ$  is 8 square inches. Points  $A$  and  $B$  are midpoints of congruent segments  $\overline{XY}$  and  $\overline{XZ}$ . Altitude  $\overline{XC}$  bisects  $\overline{YZ}$ . The area (in square inches) of the shaded region is



- (A)  $1\frac{1}{2}$     (B) 2    (C)  $2\frac{1}{2}$     (D) 3    (E)  $3\frac{1}{2}$

**Answer:** D

**Solution 1**

We know the area of triangle  $XYZ$  is 8 square inches. The area of a triangle can also be represented as  $\frac{bh}{2}$  or in this problem  $\frac{XC \cdot YZ}{2}$ . By solving, we have

$$\frac{XC \cdot YZ}{2} = 8,$$

$$XC \cdot YZ = 16.$$

With SAS congruence, triangles  $XCY$  and  $XCZ$  are congruent. Hence, triangle  $XCY = \frac{8}{2} = 4$ . (Let's say point  $D$  is the intersection between line segments  $XC$  and  $AB$ .) We can find the area of the trapezoid  $ADCY$  by subtracting the area of triangle  $XAD$  from 4.

We find the area of triangle  $XAD$  by the  $\frac{bh}{2}$  formula-  $\frac{XD \cdot AD}{2} = \frac{\frac{XC}{2} \cdot AD}{2}$ .  $AD$  is  $\frac{1}{4}$  of  $YZ$  from solution 1. The area of  $XAD$  is

$$\frac{\frac{XC}{2} \cdot \frac{YZ}{4}}{2} = \frac{16}{16} = 1$$

.

Therefore, the area of the shaded area- trapezoid  $ADCY$  has area  $4 - 1 = \boxed{(D) 3}$ .

- sarah07

21. (AMC 8 2002) Harold tosses a nickel four times. The probability that he gets at least as many heads as tails is
- (A)  $\frac{5}{16}$     (B)  $\frac{3}{8}$     (C)  $\frac{1}{2}$     (D)  $\frac{5}{8}$     (E)  $\frac{11}{16}$

**Answer:** E

**Solution 1**

Case 1: There are two heads and two tails. There are  $\binom{4}{2} = 6$  ways to choose which two tosses are heads, and the other two must be tails.

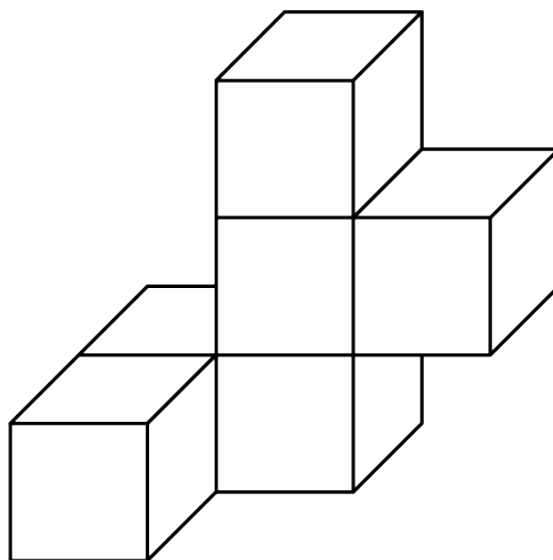
Case 2: There are three heads, one tail. There are  $\binom{4}{1} = 4$  ways to choose which of the four tosses is a tail.

Case 3: There are four heads and no tails. This can only happen 1 way.

There are a total of  $2^4 = 16$  possible configurations, giving a probability of  $\frac{6+4+1}{16} =$

(E)  $\frac{11}{16}$ .

22. (AMC 8 2002) Six cubes, each an inch on an edge, are fastened together, as shown. Find the total surface area in square inches. Include the top, bottom and sides.



- (A) 18      (B) 24      (C) 26      (D) 30      (E) 36

**Answer:** C

**Solution**

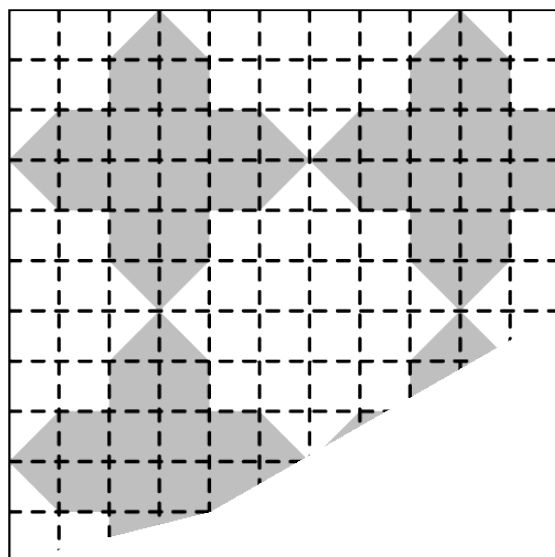
Count the number of sides that are not exposed, where a cube is connected to another cube and subtract it from the total number of faces. There are 5 places with two adjacent cubes, covering 10 sides, and  $(6)(6) = 36$  faces. The exposed surface area is  $36 - 10 =$  (C) 26.

**Solution 2**

We can count the number of showing faces from each side. One thing that we notice is that the front face has the same number of squares as the back face, the side faces have the same surface area, etc. Therefore, we are looking for  $2(\text{front surface area} + \text{side surface area} + \text{top surface area})$ . We find that this is  $2(5 + 4 + 4) = 2 * 13 =$  (C) 26.

Solution by [ILoveMath31415926535](#)

23. (AMC 8 2002) A corner of a tiled floor is shown. If the entire floor is tiled in this way and each of the four corners looks like this one, then what fraction of the tiled floor is made of darker tiles?



- (A)  $\frac{1}{3}$     (B)  $\frac{4}{9}$     (C)  $\frac{1}{2}$     (D)  $\frac{5}{9}$     (E)  $\frac{5}{8}$

**Answer:** B

**Solution**

The same pattern is repeated for every  $6 \times 6$  tile. Looking closer, there is also symmetry of the top  $3 \times 3$  square, so the fraction of the entire floor in dark tiles is the same as the fraction in the square. Counting the tiles, there are 4 dark tiles, and 9 total tiles, giving a fraction of  $\boxed{\text{(B)} \frac{4}{9}}$ .

24. (AMC 8 2002) Miki has a dozen oranges of the same size and a dozen pears of the same size. Miki uses her juicer to extract 8 ounces of pear juice from 3 pears and 8 ounces of orange juice from 2 oranges. She makes a pear-orange juice blend from an equal number of pears and oranges. What percent of the blend is pear juice?
- (A) 30    (B) 40    (C) 50    (D) 60    (E) 70

**Answer:** B

**Solution 1**

A pear gives  $\frac{8}{3}$  ounces of juice per pear. An orange gives  $\frac{8}{2} = 4$  ounces of juice per orange. If the pear-orange juice blend used one pear and one orange each, the percentage of pear juice would be

$$\frac{\frac{8}{3}}{\frac{8}{3} + 4} \times 100 = \frac{8}{8 + 12} \times 100 = \boxed{\text{(B)} 40}$$

**Solution 2**

Since it doesn't matter how many pears and oranges there are, you can make the number of them whatever you like. In this case, we could use 6, because it's the LCM of 2 and 3. Then for the 6 pears, there are  $\frac{6}{3} \cdot 8 = 16$  ounces of pear juice. For the 6 oranges, there are  $\frac{6}{2} \cdot 8 = 24$  ounces of orange juice. Since we are looking for the percent of pear juice, we need to do  $16/(16 + 24) = 16/40$ . Simplifying, we get  $\frac{2}{5}$ . Hence the answer is  $\boxed{\text{(B)} 40}$ .

25. (*AMC 8 2002*) Loki, Moe, Nick and Ott are good friends. Ott had no money, but the others did. Moe gave Ott one-fifth of his money, Loki gave Ott one-fourth of his money and Nick gave Ott one-third of his money. Each gave Ott the same amount of money. What fractional part of the group's money does Ott now have?

(A)  $\frac{1}{10}$     (B)  $\frac{1}{4}$     (C)  $\frac{1}{3}$     (D)  $\frac{2}{5}$     (E)  $\frac{1}{2}$

**Answer:** B

**Solution 1**

Since Ott gets equal amounts of money from each friend, we can say that he gets  $x$  dollars from each friend. This means that Moe has  $5x$  dollars, Loki has  $4x$  dollars, and Nick has  $3x$  dollars. The total amount is  $12x$  dollars, and since Ott gets  $3x$  dollars total,

$$\frac{3x}{12x} = \frac{3}{12} = \boxed{\text{(B)} \frac{1}{4}}.$$

**Solution 2**

We can assign any natural number to the price that Ott's friends gave him. For this example, we will use 5.

Moe gave Ott a fifth of his money, and also five dollars. So in total Moe has 25 dollars. Loki gave Ott a fourth of his money, and also five dollars. So in total Loki has 20 dollars. Nick gave Ott a third of his money, and also five dollars. So in total Nick has 15 dollars.  $5 \cdot 3 = 15$  which is the total money the group gave Ott.

$25 + 20 + 15 = 60$  is the group's balance. Therefore, the fraction of the group's money that Ott now has is  $\frac{15}{60} = \boxed{\text{(B)} \frac{1}{4}}.$

## §19 AMC 8 2003

1. (*AMC 8 2003*) Jamie counted the number of edges of a cube, Jimmy counted the numbers of corners, and Judy counted the number of faces. They then added the three numbers. What was the resulting sum?
- (A) 12    (B) 16    (C) 20    (D) 22    (E) 26

**Answer:** E

**Solution**

On a cube, there are 12 edges, 8 corners, and 6 faces. Adding them up gets  $12 + 8 + 6 = \boxed{\text{(E)} 26}.$

2. (*AMC 8 2003*) Which of the following numbers has the smallest prime factor?
- (A) 55    (B) 57    (C) 58    (D) 59    (E) 61

**Answer:** C

**Solution**

The smallest prime factor is 2, and since 58 is the only multiple of 2, the answer is  $\boxed{\text{(C)} 58}.$

3. (*AMC 8 2003*) A burger at Ricky C's weighs 120 grams, of which 30 grams are filler. What percent of the burger is not filler?
- (A) 60%    (B) 65%    (C) 70%    (D) 75%    (E) 90%

**Answer:** D

**Solution**

There are 30 grams of filler, so there are  $120 - 30 = 90$  grams that aren't filler.

$$\frac{90}{120} = \frac{3}{4} = \boxed{\text{(D) } 75\%}.$$

4. (*AMC 8 2003*) A group of children riding on bicycles and tricycles rode past Billy Bob's house. Billy Bob counted 7 children and 19 wheels. How many tricycles were there?

(A) 2      (B) 4      (C) 5      (D) 6      (E) 7

**Answer:** C

**Solution 1**

If all the children were riding bicycles, there would be  $2 \times 7 = 14$  wheels. Each tricycle adds an extra wheel and  $19 - 14 = 5$  extra wheels are needed, so there are

$\boxed{\text{(C) } 5}$  tricycles.

**Solution 2**

Setting up an equation, we have  $a + b = 7$  children and  $3a + 2b = 19$ . Solving for the variables, we get,  $a = \boxed{\text{(C) } 5}$  tricycles.

5. (*AMC 8 2003*) If 20% of a number is 12, what is 30% of the same number?

(A) 15      (B) 18      (C) 20      (D) 24      (E) 30

**Answer:** B

**Solution**

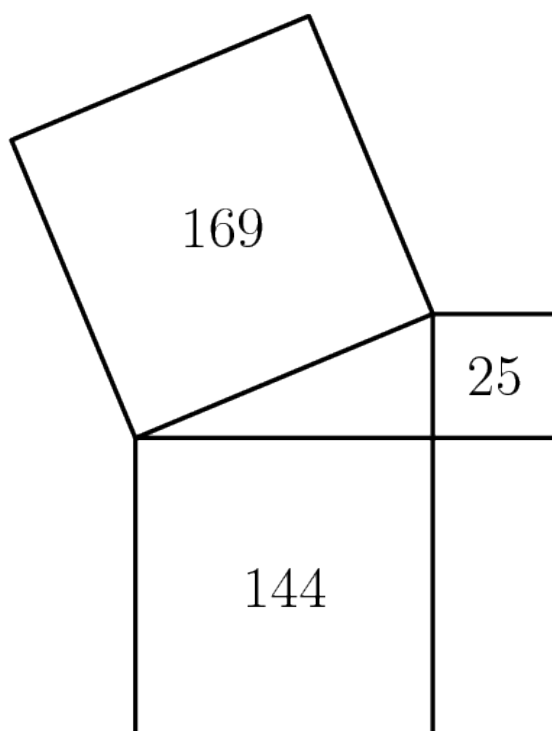
20% of a number is equal to  $\frac{1}{5}$  of that number. Let  $n$ =the number

$$\frac{1}{5}n = 12 \text{ Multiply both sides by 5}$$

$$n = 60 \text{ 30\% of } n \text{ is equal to } \frac{3}{10}n = \frac{3}{10} \cdot 60 = 3 \cdot 6 = \boxed{\text{(B) } 18}$$

6. (*AMC 8 2003*) Given the areas of the three squares in the figure, what is the area of the interior triangle?





- (A) 13      (B) 30      (C) 60      (D) 300      (E) 1800

**Answer:** B

**Solution**

The sides of the squares are 5, 12 and 13 for the square with area 25, 144 and 169, respectively. The legs of the interior triangle are 5 and 12, so the area is  $\frac{5 \times 12}{2} = \boxed{\text{(B) } 30}$

(note: the  $ab/2$  area method only works because the converse Pythagorean theorem holds for the triple  $[5, 12, 13]$ . Therefore, we can find the solution this way because we know the triangle is right megaboy6679)

7. (*AMC 8 2003*) Blake and Jenny each took four 100-point tests. Blake averaged 78 on the four tests. Jenny scored 10 points higher than Blake on the first test, 10 points lower than him on the second test, and 20 points higher on both the third and fourth tests. What is the difference between Jenny's average and Blake's average on these four tests?

- (A) 10      (B) 15      (C) 20      (D) 25      (E) 40

**Answer:** A

**Solution 1**

Blake scored a total of  $4 \times 78 = 312$  points. Jenny scored  $10 - 10 + 20 + 20 = 40$  points higher than Blake, so her average is  $\frac{312+40}{4} = 88$ . the difference is  $88 - 78 = \boxed{\text{(A) } 10}$ .

**Solution 2**

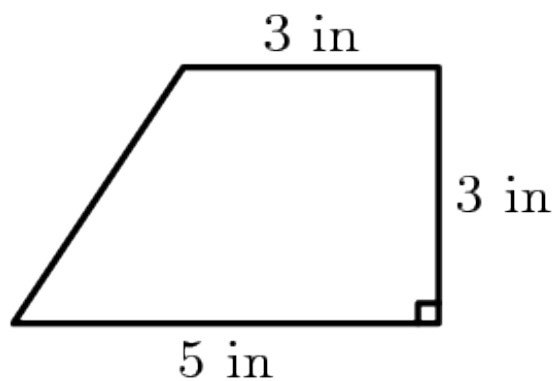
The total point difference between Blake's and Jenny's scores is  $10 - 10 + 20 + 20 = 40$ . The average of it is  $\frac{40}{4} = \boxed{\text{(A) } 10}$ .

8. (*AMC 8 2003*) **Bake Sale**

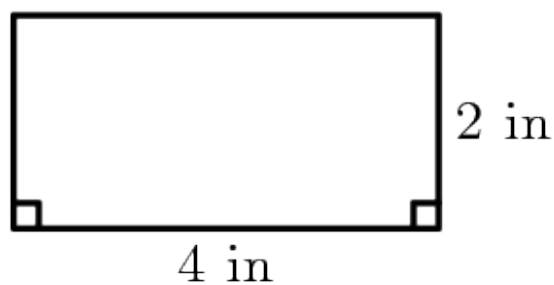
(Problems 8, 9, and 10 use the data found in the accompanying paragraph and figures)

Four friends, Art, Roger, Paul and Trisha, bake cookies, and all cookies have the same thickness. The shapes of the cookies differ, as shown.

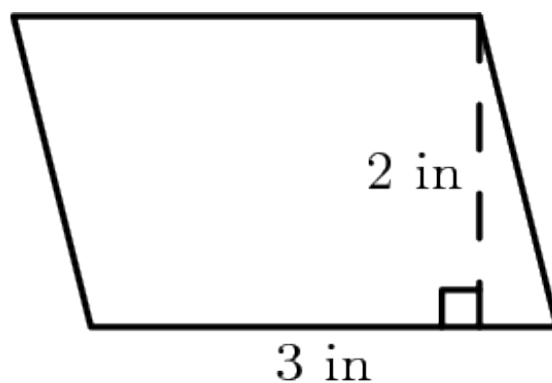
- o Art's cookies are trapezoids.



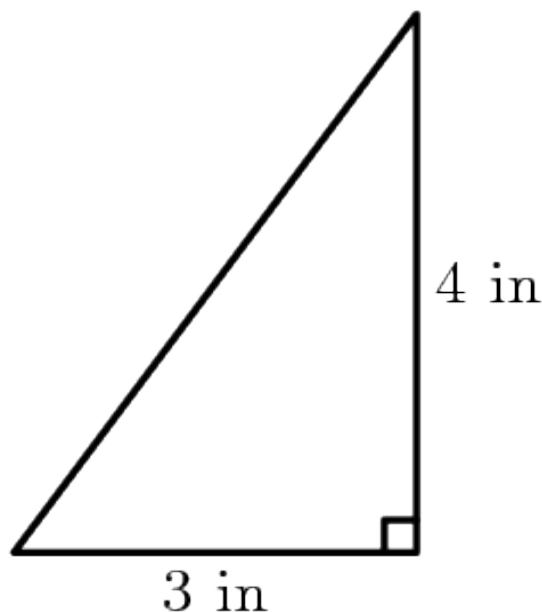
- o Roger's cookies are rectangles.



- o Paul's cookies are parallelograms.



- o Trisha's cookies are triangles.



Each friend uses the same amount of dough, and Art makes exactly 12 cookies. Who gets the fewest cookies from one batch of cookie dough?

- (A) Art    (B) Roger    (C) Paul    (D) Trisha    (E) There is a tie for fewest.

**Answer:** A

**Solution**

Out of all the cookies, Art's has an area of  $12 \text{ in}^2$ , which was the greatest area out of all the cookies' areas. Roger's cookie had an area of  $8 \text{ in}^2$ , and both Paul and Trisha's cookies had an area of  $6 \text{ in}^2$ . This means **(A) Art** makes the fewest cookies, since his cookie area is the greatest. The answer is not that there is a tie between Paul and Trisha because they can make the most cookies with a given amount of cookie dough, not the least.

9. (*AMC 8 2003*) Each friend uses the same amount of dough, and Art makes exactly 12 cookies. Art's cookies sell for 60 cents each. To earn the same amount from a single batch, how much should one of Roger's cookies cost in cents?

- (A) 18    (B) 25    (C) 40    (D) 75    (E) 90

**Answer:** C

**Solution**

The area of one of Art's cookies is  $3 \cdot 3 + \frac{2 \cdot 3}{2} = 9 + 3 = 12$ . As he has 12 cookies in a batch, the amount of dough each person used is  $12 \cdot 12 = 144$ . Roger's cookies have an area of  $\frac{144}{2.4} = \frac{144}{8} = 18$  cookies in a batch. In total, the amount of money Art will earn is  $12 \cdot 60 = 720$ . Thus, the amount Roger would need to charge per cookie is  $\frac{720}{18} = \textbf{(C) } 40$ .

10. (*AMC 8 2003*) How many cookies will be in one batch of Trisha's cookies?

- (A) 10    (B) 12    (C) 16    (D) 18    (E) 24

**Answer:** E

**Solution 1**

Art's cookies have areas of  $3 \cdot 3 + \frac{2 \cdot 3}{2} = 9 + 3 = 12$ . There are 12 cookies in one of Art's batches so everyone used  $12 \cdot 12 = 144$  in<sup>2</sup> of dough. Trisha's cookies have an area of  $\frac{3 \cdot 4}{2} = 6$  so she has  $\frac{144}{6} = \boxed{\text{(E)} 24}$  cookies per batch.

**Solution 2**

Using the same method as the previous solution you can find that Art's cookies have a area of 12 and Trisha's cookies have an area of 6. Because they both used the same amount of dough and Trisha's cookies use half the dough compared to Art's you can just multiply the total number of cookies Art makes by 2 to balance out the dough used. This gives  $12 \cdot 2$  which is  $\boxed{\text{(E)} 24}$ .

11. (*AMC 8 2003*) Business is a little slow at Lou's Fine Shoes, so Lou decides to have a sale. On Friday, Lou increases all of Thursday's prices by 10%. Over the weekend, Lou advertises the sale: "Ten percent off the listed price. Sale starts Monday." How much does a pair of shoes cost on Monday that cost 40 dollars on Thursday?

(A) 36      (B) 39.60      (C) 40      (D) 40.40      (E) 44

**Answer:** B

**Solution**

On Friday, the shoes would cost  $40 \cdot 1.1 = 44$  dollars. Then on Monday, the shoes would cost  $44 - \frac{44}{10} = 44 - 4.4 = \boxed{\text{(B)} 39.60}$ .

12. (*AMC 8 2003*) When a fair six-sided die is tossed on a table top, the bottom face cannot be seen. What is the probability that the product of the numbers on the five faces that can be seen is divisible by 6?

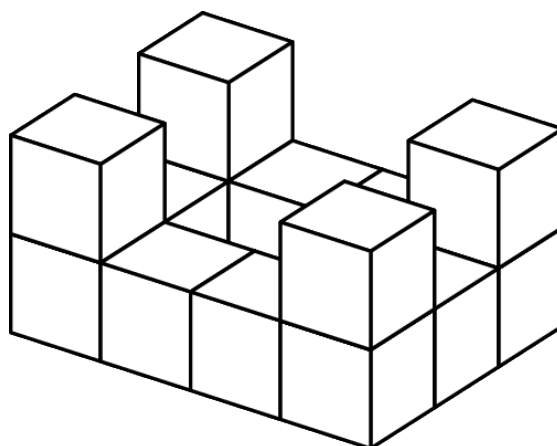
(A)  $\frac{1}{3}$       (B)  $\frac{1}{2}$       (C)  $\frac{2}{3}$       (D)  $\frac{5}{6}$       (E) 1

**Answer:** E

**Solution**

We have six cases: each different case, every one where a different number cannot be seen. The rolls that omit numbers one through five are all something times six: an example would be where the number you cannot see is one, so the product should be  $2 \times 3 \times 4 \times 5 \times 6$ , and so product should be divisible by six. The roll that omits six on the other hand is  $1 \times 2 \times 3 \times 4 \times 5$ , which has  $2 \times 3$ , also equal to six. We can see that all of them have a factor of 6 and therefore are divisible by six, so the solution should be **(E)**

13. (*AMC 8 2003*) Fourteen white cubes are put together to form the figure below. The complete surface of the figure, including the bottom, is painted red. The figure is then separated into individual cubes. How many of the individual cubes have exactly four red faces?



- (A) 4    (B) 6    (C) 8    (D) 10    (E) 12

**Answer:** B

**Solution**

This is the number cubes that are adjacent to another cube on exactly two sides. The bottom corner cubes are connected on three sides, and the top corner cubes are connected on one. The number we are looking for is the number of middle cubes, which is **(B) 6**.

14. (AMC 8 2003) In this addition problem, each letter stands for a different digit.

$$\begin{array}{r} T \quad W \quad O \\ + \quad T \quad W \quad O \\ \hline F \quad O \quad U \quad R \end{array}$$

If  $T = 7$  and the letter  $O$  represents an even number, what is the only possible value for  $W$ ?

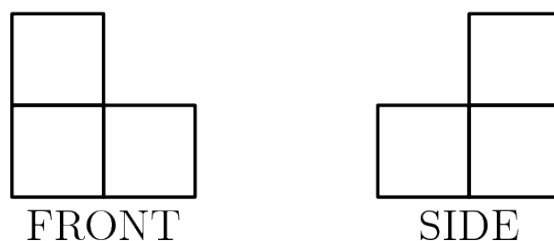
- (A) 0    (B) 1    (C) 2    (D) 3    (E) 4

**Answer:** D

**Solution**

Since both  $T$ 's are 7, then  $O$  has to equal 4, because  $7 + 7 = 14$ . Then,  $F$  has to equal 1. To get  $R$ , we do  $4 + 4$  (since  $O = 4$ ) to get  $R = 8$ . The value for  $W$  then has to be a number less than 5, otherwise it will change the value of  $O$ , and can't be a number that has already been used, like 4 or 1. The only other possibilities are 2 and 3. 2 doesn't work because it makes  $U = 4$ , which is what  $O$  already equals. So, the only possible value of  $W$  is **(D) 3**.

15. (AMC 8 2003) A figure is constructed from unit cubes. Each cube shares at least one face with another cube. What is the minimum number of cubes needed to build a figure with the front and side views shown?

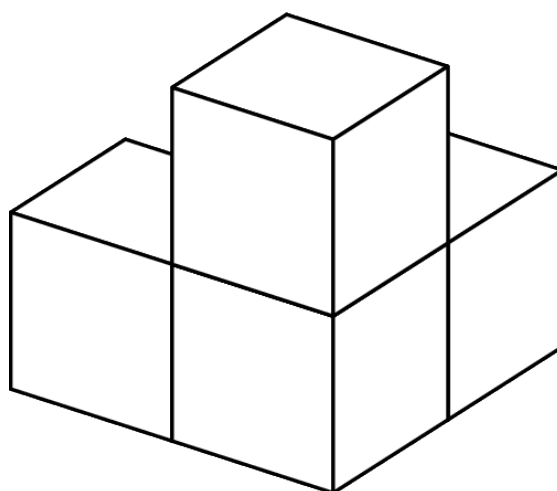


- (A) 3      (B) 4      (C) 5      (D) 6      (E) 7

**Answer:** B

**Solution**

In order to minimize the amount of cubes needed, we must match up as many squares of our given figures with each other to make different sides of the same cube. One example of the solution with **(B) 4** cubes. Notice the corner cube cannot be removed for a figure of 3 cubes because at least one face of a cube must be touching another face.



16. (*AMC 8 2003*) Ali, Bonnie, Carlo, and Dianna are going to drive together to a nearby theme park. The car they are using has 4 seats: 1 driver's seat, 1 front passenger seat, and 2 back passenger seats. Bonnie and Carlo are the only ones who know how to drive the car. How many possible seating arrangements are there?

- (A) 2      (B) 4      (C) 6      (D) 12      (E) 24

**Answer:** D

**Solution**

There are only 2 people who can go in the driver's seat—Bonnie and Carlo. Any of the 3 remaining people can go in the front passenger seat. There are 2 people who can go in the first back passenger seat, and the remaining person must go in the last seat. Thus, there are  $2 \cdot 3 \cdot 2$  or 12 ways. The answer is then **(D) 12**.

**Solution 2 (Quick)**

If there weren't any extra requirements, there would be 24 combinations. However, there are only 2, which is half of 4, ways to put the people for the first seat by restriction. Therefore, half of 24 is **(D) 12**.

Solution by [ILoveMath31415926535](#)

17. (*AMC 8 2003*) The six children listed below are from two families of three siblings each. Each child has blue or brown eyes and black or blond hair. Children from the same family have at least one of these characteristics in common. Which two children are Jim's siblings?

| Child    | Eye Color | Hair Color |
|----------|-----------|------------|
| Benjamin | Blue      | Black      |
| Jim      | Brown     | Blonde     |
| Nadeen   | Brown     | Black      |
| Austin   | Blue      | Blonde     |
| Tevyn    | Blue      | Black      |
| Sue      | Blue      | Blonde     |

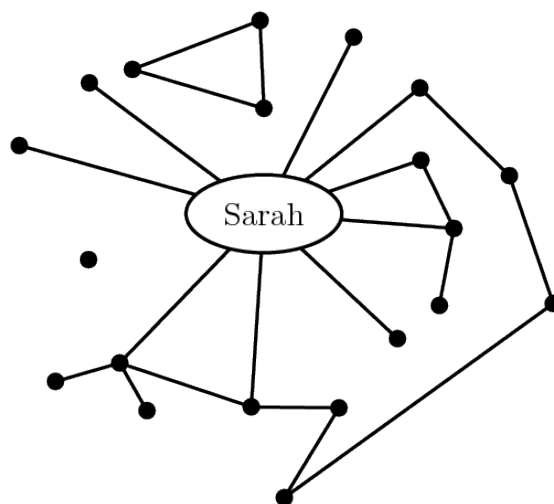
- (A) Nadeen and Austin      (B) Benjamin and Sue      (C) Benjamin and Austin      (D) Nadeen and Tevyn  
 (E) Austin and Sue

**Answer:** E

**Solution**

Jim has brown eyes and blonde hair. If you look for anybody who has brown eyes **or** blonde hair, you find that Nadeen, Austin, and Sue are Jim's possible siblings. However, the children have at least one common characteristics. Since Austin and Sue both have blonde hair, Nadeen is ruled out and therefore **(E) Austin and Sue** are his siblings. You can also see that in the hair color column, there are three black haired people and three blond haired people. Since Jim has blond hair, all his siblings must be the other two with blond hair.

18. (AMC 8 2003) Each of the twenty dots on the graph below represents one of Sarah's classmates. Classmates who are friends are connected with a line segment. For her birthday party, Sarah is inviting only the following: all of her friends and all of those classmates who are friends with at least one of her friends. How many classmates will not be invited to Sarah's party?



- (A) 1      (B) 4      (C) 5      (D) 6      (E) 7

**Answer:** D

**Solution**

There are 3 people who are friends with only each other who won't be invited, plus 1 person who has no friends, and 2 people who are friends of friends of friends who won't be invited. So the answer is **(D) 6**.

19. (AMC 8 2003) How many integers between 1000 and 2000 have all three of the numbers 15, 20, and 25 as factors?

(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** C

**Solution 1**

Find the least common multiple of 15, 20, 25 by turning the numbers into their prime factorization.

$$15 = 3 \cdot 5, 20 = 2^2 \cdot 5, 25 = 5^2$$

Gather all necessary multiples  $3, 2^2, 5^2$  when multiplied gets 300. The multiples of 300 – 300, 600, 900, 1200, 1500, 1800, 2100. The number of multiples between 1000 and 2000 is **(C) 3**.

20. (AMC 8 2003) What is the measure of the acute angle formed by the hands of the clock at 4:20 PM?

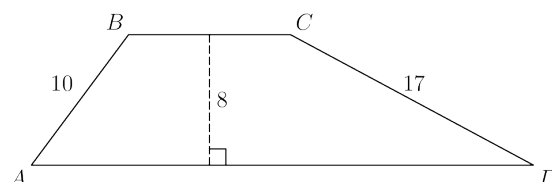
(A) 0      (B) 5      (C) 8      (D) 10      (E) 12

**Answer:** D

**Solution 1**

Imagine the clock as a circle. The minute hand will be at the 4 at 20 minutes past the hour. The central angle formed between 4 and 5 is 30 degrees (since it is  $\frac{1}{12}$  of a full circle, 360). By 4 : 20, the hour hand would have moved  $\frac{1}{3}$  way from 4 to 5 since  $\frac{20}{60}$  is reducible to  $\frac{1}{3}$ . One third of the way from 4 to 5 is one third of 30 degrees, which is 10 degrees past the 4. Recall that the minute hand is at the 4, so the angle between them is **(D) 10**.

21. (AMC 8 2003) The area of trapezoid  $ABCD$  is  $164 \text{ cm}^2$ . The altitude is 8 cm,  $AB$  is 10 cm, and  $CD$  is 17 cm. What is  $BC$ , in centimeters?



(A) 9      (B) 10      (C) 12      (D) 15      (E) 20

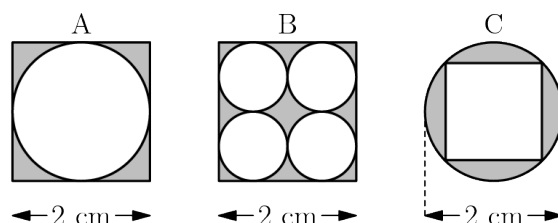
**Answer:** B

**Solution**

Using the formula for the area of a trapezoid, we have  $164 = 8(\frac{BC+AD}{2})$ . Thus  $BC+AD = 41$ . Drop perpendiculars from  $B$  to  $AD$  and from  $C$  to  $AD$  and let them hit  $AD$  at  $E$  and  $F$  respectively. Note that each of these perpendiculars has length 8. From the Pythagorean Theorem,  $AE = 6$  and  $DF = 15$  thus  $AD = BC + 21$ . Substituting back into our original equation we have  $BC + BC + 21 = 41$  thus  $BC =$  **(B) 10**

22. (AMC 8 2003) The following figures are composed of squares and circles. Which figure has a shaded region with largest area?





- (A) A only      (B) B only      (C) C only      (D) both A and B      (E) all are equal

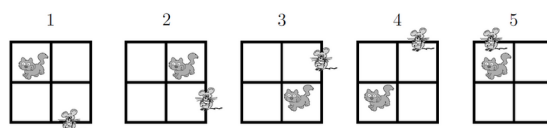
**Answer:** C

**Solution**

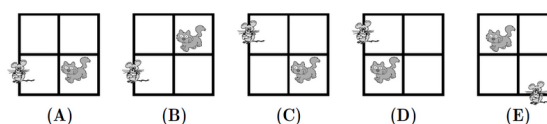
First we have to find the area of the shaded region in each of the figures. In figure **A** the area of the shaded region is the area of the circle subtracted from the area of the square. That is  $2^2 - 1^2\pi = 4 - \pi$ . In figure **B** the area of the shaded region is the sum of the areas of the 4 circles subtracted from the area of the square. That is  $2^2 - 4\left(\left(\frac{1}{2}\right)^2\pi\right) = 4 - 4\left(\frac{\pi}{4}\right) = 4 - \pi$ . In figure **C** the area of the shaded region is the area of the square subtracted from the area of the circle. The diameter of the circle and the diagonal of the square are equal to 2. We can easily find the area of the square using the area formula  $\frac{d_1d_2}{2}$ . So the area of the shaded region is  $1^2\pi - \frac{2 \cdot 2}{2} = \pi - 2$ . Clearly the largest area that we found among the three shaded regions is  $\pi - 2$ . Thus the answer is C.

- Note - If you don't know the diagonal formula, you can also find the length of the square using the Pythagorean Theorem (set up an equation and solve for  $x$ ), which is  $\sqrt{2}$ , and get that the area is 2, and continue as shown in the solution.

23. (AMC 8 2003) In the pattern below, the cat moves clockwise through the four squares and the mouse moves counterclockwise through the eight exterior segments of the four squares.



If the pattern is continued, where would the cat and mouse be after the 247th move?



**Answer:** A

**Solution**

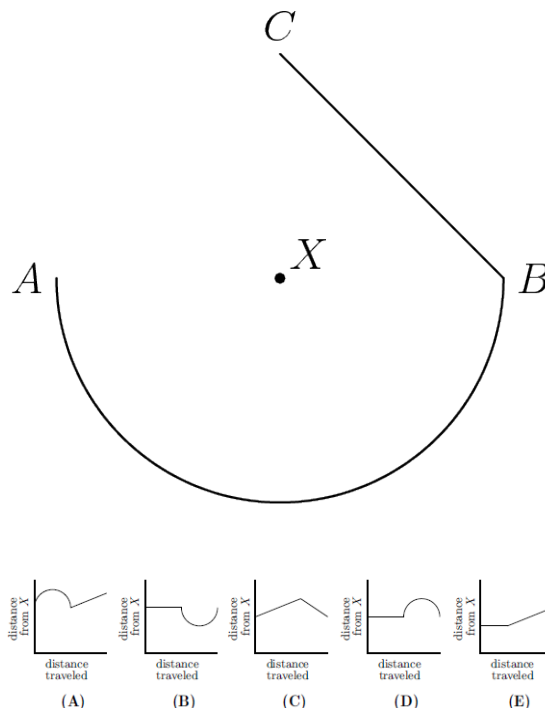
Break this problem into two parts: where the cat will be after the 247<sup>th</sup> move, and where the mouse will be.

The cat has four possible positions in 1 cycle which are repeated every four moves. 247 has a remainder of 3 when divided by 4. This corresponds to the position the cat has after the 3rd move, which is the bottom right corner.

Similarly, the mouse has eight possible positions in 1 cycle that repeat every eight moves. 247 has a remainder of 7 when divided by 8. This corresponds to the position the mouse has after the 7th move which is bottom left corner.

The only arrangement with the mouse in that position and the cat in the bottom right square is **(A)**.

24. (AMC 8 2003) A ship travels from point  $A$  to point  $B$  along a semicircular path, centered at Island  $X$ . Then it travels along a straight path from  $B$  to  $C$ . Which of these graphs best shows the ship's distance from Island  $X$  as it moves along its course?

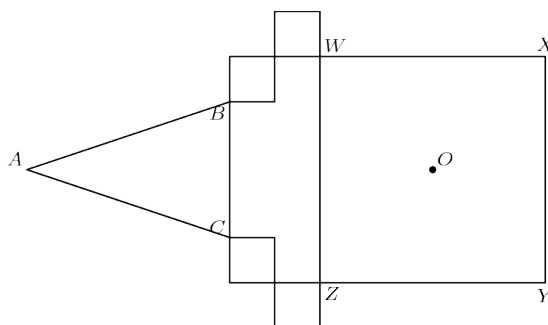


**Answer:** B

### Solution

The distance from Island  $X$  to any point on the semicircle will always be constant. On the graph, this will represent a straight line. The distance between Island  $X$  and line  $BC$  will not be constant though. We can easily prove that the distance between  $X$  and line  $BC$  will represent a curve. As the ship travels from  $B$  to  $C$ , the distance between the ship and Island  $X$  will first decrease until it reaches the point  $Y$  so that  $\overline{XY}$  is perpendicular to  $\overline{BC}$ , and then increase afterwards. Hence the answer choice that fits them all is **(B)**.

25. (AMC 8 2003) In the figure, the area of square  $WXYZ$  is  $25 \text{ cm}^2$ . The four smaller squares have sides 1 cm long, either parallel to or coinciding with the sides of the large square. In  $\triangle ABC$ ,  $AB = AC$ , and when  $\triangle ABC$  is folded over side  $\overline{BC}$ , point  $A$  coincides with  $O$ , the center of square  $WXYZ$ . What is the area of  $\triangle ABC$ , in square centimeters?



- (A)  $\frac{15}{4}$     (B)  $\frac{21}{4}$     (C)  $\frac{27}{4}$     (D)  $\frac{21}{2}$     (E)  $\frac{27}{2}$

**Answer:** C

**Solution**

We see that  $XY = 5$ , the vertical distance between  $B$  and  $X$  is 1, and the vertical distance between  $C$  and  $Y$  is 1. Therefore,  $BC = 5 - 1 - 1 = 3$ . We are given that the length of the altitude of  $\triangle ABC$  is equal to the distance between  $\overline{BC}$  and  $O$ , which is  $1 + 1 + \frac{5}{2} = \frac{9}{2}$ . So the area of  $\triangle ABC$  is  $\frac{1}{2} \left( 3 \cdot \frac{9}{2} \right)$ , which is (C)  $\frac{27}{4}$ .

## §20 AMC 8 2004

1. (*AMC 8 2004*) On a map, a 12-centimeter length represents 72 kilometers. How many kilometers does a 17-centimeter length represent?

- (A) 6    (B) 102    (C) 204    (D) 864    (E) 1224

**Answer:** B

**Solution**

We set up the proportion  $\frac{12\text{cm}}{72\text{km}} = \frac{17\text{cm}}{x\text{km}}$ . Thus  $x = 102 \Rightarrow$  (B) 102

**Solution 2**

Since we know 12 centimeters on the map is equal to 72 kilometers, 1 centimeter on the map is simply  $72/12$ , which is 6. The problem is asking how many kilometers a 17-centimeter length represents, so we multiply 6 by 17 which is simply 102, your answer.

2. (*AMC 8 2004*) How many different four-digit numbers can be formed by rearranging the four digits in 2004?

- (A) 4    (B) 6    (C) 16    (D) 24    (E) 81

**Answer:** B

**Solution 1**

We can solve this problem easily, just by calculating how many choices there are for each of the four digits. First off, we know there are only 2 choices for the first digit, because 0 isn't a valid choice, or the number would be a 3-digit number, which is not what we want. We have 3 choices for the second digit, since we already used up one of the digits, and 2 choices for the third, and finally just 1 choice for the fourth and final one.  $2 \cdot 3 \cdot 2 \cdot 1$  is 12, but there are 2 zeros that have been counted as different numbers, so divide by 2 to get (B) 6.

**Solution 2**

Note that the four-digit number must start with either a 2 or a 4. The four-digit numbers that start with 2 are 2400, 2040, and 2004. The four-digit numbers that start with 4 are 4200, 4020, and 4002 which gives us a total of **(B) 6**.

### Solution 3

In order for the resulting numbers to have four digits, they cannot start with 0. Therefore, both zeroes must be in the last three places. There are  $\binom{3}{2} = 3$  ways to choose which two of the last three places are zeroes. Then there are  $2 \cdot 1 = 2$  ways to arrange the 2 and the 4 in the remaining two places, giving us a total of  $3 \cdot 2 = \mathbf{(B) 6}$ .

3. (*AMC 8 2004*) Twelve friends met for dinner at Oscar's Overstuffed Oyster House, and each ordered one meal. The portions were so large, there was enough food for 18 people. If they shared, how many meals should they have ordered to have just enough food for the 12 of them?

(A) 8    (B) 9    (C) 10    (D) 15    (E) 18

**Answer:** A

### Solution

Set up the proportion  $\frac{12 \text{ meals}}{18 \text{ people}} = \frac{x \text{ meals}}{12 \text{ people}}$ . Solving for  $x$  gives us  $x = \mathbf{(A) 8}$ .

4. (*AMC 8 2004*) Ms. Hamilton's eighth-grade class wants to participate in the annual three-person-team basketball tournament. Lance, Sally, Joy, and Fred are chosen for the team. In how many ways can the three starters be chosen?

(A) 2    (B) 4    (C) 6    (D) 8    (E) 10

**Answer:** B

### Solution

There are  $\binom{4}{3}$  ways to choose three starters. Thus the answer is **(B) 4**.

### Solution 2

We can choose 3 people by eliminating one from a set of 4 one at a time and the other three get selected. There are 4 ways to remove a person from a group of four (without considering order), so there are **(B) 4** ways to choose three people, where order doesn't matter.

5. (*AMC 8 2004*) Ms. Hamilton's eighth-grade class wants to participate in the annual three-person-team basketball tournament. The losing team of each game is eliminated from the tournament. If sixteen teams compete, how many games will be played to determine the winner?

(A) 4    (B) 7    (C) 8    (D) 15    (E) 16

**Answer:** D

### Solution 1

Note that the winning team will be the only team that wins all of the games. Therefore, to find the total number of games to determine the winner has a 1:1 correspondence to the number of ways to determine the losers. (Think of it this way: If you want to select two balls from a bag of 6 balls, it is analogous to selecting the 4 balls that you don't want to select, both are 6 choose 2.). There are 15 losing teams, and since each round is unique, there are 15 total rounds.

**Solution 2**

There will be 8 games the first round, 4 games the second round, 2 games the third round, and 1 game in the final round, giving us a total of  $8 + 4 + 2 + 1 = 15$  games.

**(D) 15.**

6. (*AMC 8 2004*) After Sally takes 20 shots, she has made 55% of her shots. After she takes 5 more shots, she raises her percentage to 56%. How many of the last 5 shots did she make?

(A) 1    (B) 2    (C) 3    (D) 4    (E) 5

**Answer:** C

**Solution**

Sally made  $0.55 * 20 = 11$  shots originally. Letting  $x$  be the number of shots she made, we have  $\frac{11+x}{25} = 0.56$ . Solving for  $x$  gives us  $x =$  **(C) 3**

7. (*AMC 8 2004*) An athlete's target heart rate, in beats per minute, is 80% of the theoretical maximum heart rate. The maximum heart rate is found by subtracting the athlete's age, in years, from 220. To the nearest whole number, what is the target heart rate of an athlete who is 26 years old?

(A) 134    (B) 155    (C) 176    (D) 194    (E) 243

**Answer:** B

**Solution**

The maximum heart rate is  $220 - 26 = 194$  beats per minute. The target heart rate is then  $0.8 * 194 \approx$  **(B) 155** beats per minute.

8. (*AMC 8 2004*) Find the number of two-digit positive integers whose digits total 7.

(A) 6    (B) 7    (C) 8    (D) 9    (E) 10

**Answer:** B

**Solution**

The numbers are 16, 25, 34, 43, 52, 61, 70 which gives us a total of **(B) 7**.

9. (*AMC 8 2004*) The average of the five numbers in a list is 54. The average of the first two numbers is 48. What is the average of the last three numbers?

(A) 55    (B) 56    (C) 57    (D) 58    (E) 59

**Answer:** D

**Solution**

Let the 5 numbers be  $a, b, c, d$ , and  $e$ . Thus  $\frac{a+b+c+d+e}{5} = 54$  and  $a+b+c+d+e = 270$ . Since  $\frac{a+b}{2} = 48$ ,  $a+b = 96$ . Substituting back into our original equation, we have  $96 + c + d + e = 270$  and  $c + d + e = 174$ . Dividing by 3 gives the average of **(D) 58**.

10. (*AMC 8 2004*) Handy Aaron helped a neighbor  $1\frac{1}{4}$  hours on Monday, 50 minutes on Tuesday, from 8:20 to 10:45 on Wednesday morning, and a half-hour on Friday. He is paid \$3 per hour. How much did he earn for the week?

(A) \$8    (B) \$9    (C) \$10    (D) \$12    (E) \$15

**Answer:** E

**Solution**

Let's convert everything to minutes and add them together. On Monday he worked for  $\frac{5}{4} \cdot 60 = 75$  minutes. On Tuesday he worked 50 minutes. On Wednesday he worked for 2 hours 25 minutes, or  $2(60) + 25 = 145$  minutes. On Friday he worked  $\frac{60}{2} = 30$  minutes. This adds up to  $75 + 50 + 145 + 30 = 300$  minutes, or  $300/60 = 5$  hours and  $5 \cdot 3 = \boxed{\text{(E)} \$15}$ .

11. (AMC 8 2004) The numbers  $-2, 4, 6, 9$  and  $12$  are rearranged according to these rules:

<pre>The largest isn't first, but it is in one of the first three places. The smallest isn't last, but it is  
The median isn't first or last. </pre>What is the average of the first and last numbers?

- (A) 3.5      (B) 5      (C) 6.5      (D) 7.5      (E) 8

**Answer:** C

**Solution**

From rule 1, the largest number,  $12$ , can be second or third. From rule 2, because there are five places, the smallest number  $-2$  can either be third or fourth. The median,  $6$  can be second, third, or fourth. Because we know the middle three numbers, the first and last numbers are  $4$  and  $9$ , disregarding their order. Their average is  $(4 + 9)/2 = \boxed{\text{(C)} 6.5}$ .

12. (AMC 8 2004) Niki usually leaves her cell phone on. If her cell phone is on but she is not actually using it, the battery will last for 24 hours. If she is using it constantly, the battery will last for only 3 hours. Since the last recharge, her phone has been on 9 hours, and during that time she has used it for 60 minutes. If she doesn't talk any more but leaves the phone on, how many more hours will the battery last?

- (A) 7      (B) 8      (C) 11      (D) 14      (E) 15

**Answer:** B

**Solution**

When not being used, the cell phone uses up  $\frac{1}{24}$  of its battery per hour. When being used, the cell phone uses up  $\frac{1}{3}$  of its battery per hour. Since Niki's phone has been on for 9 hours, of those 8 simply on and 1 with Niki using her phone,  $8(\frac{1}{24}) + 1(\frac{1}{3}) = \frac{2}{3}$  of its battery has been used up. To drain the remaining  $\frac{1}{3}$  the phone can last for  $\frac{\frac{1}{3}}{\frac{1}{24}} = \boxed{\text{(B)} 8}$  more hours without being used.

13. (AMC 8 2004) Amy, Bill and Celine are friends with different ages. Exactly one of the following statements is true.

<pre>I. Bill is the oldest. II. Amy is not the oldest. III. Celine is not the youngest. </pre>Rank the friends from the oldest to the youngest.

- (A) Bill, Amy, Celine      (B) Amy, Bill, Celine      (C) Celine, Amy, Bill  
(D) Celine, Bill, Amy      (E) Amy, Celine, Bill

**Answer:** E

**Solution**

If Bill is the oldest, then Amy is not the oldest, and both statements I and II are true, so statement I is not the true one.

If Amy is not the oldest, and we know Bill cannot be the oldest, then Celine is the oldest. This would mean she is not the youngest, and both statements II and III are true, so statement II is not the true one.

Therefore, statement III is the true statement, and both I and II are false. From this, Amy is the oldest, Celine is in the middle, and lastly Bill is the youngest. This order is (E) Amy, Celine, Bill.

### Solution 2

Bashing through/ eliminating the options

The first statement tells us that Bill is the oldest. In the answer choices, you can see that the first answer is the only option where Bill is the oldest. Bill being the oldest is false (as Amy not being the oldest in the answer choice is also a true statement, but we cannot have two true statements per our conditions), so we can eliminate that answer. Now we know that Bill isn't the oldest and could be the youngest or the second oldest. Don't forget, we have TWO false statements and only one true statement. The second statement where Amy is not the oldest is another false statement. So now we know Amy is the oldest, followed by Celine whom is the second oldest, and Bill, whom is the youngest. Thus, the answer is

(E) Amy, Celine, Bill.

–Angela–

(amyxin0316)

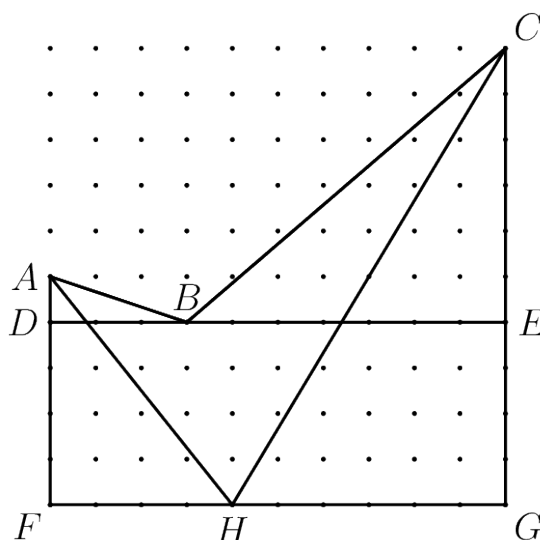
14. (AMC 8 2004) What is the area enclosed by the geoboard quadrilateral below?



- (A) 15      (B)  $18\frac{1}{2}$       (C)  $22\frac{1}{2}$       (D) 27      (E) 41

**Answer:** C

**Solution 1**



Divide the shape up as above.

$$Area = [DEGF] + [ABD] + [BCE] - [AFH] - [CGH] = 4 \cdot 10 + \frac{1}{2} \cdot 1 \cdot 3 + \frac{1}{2} \cdot 7 \cdot 6 - \frac{1}{2} \cdot 5 \cdot 4 - \frac{1}{2} \cdot 6 \cdot 10 = 40 + \frac{3}{2} + 21$$

### Solution 2

Let the bottom left corner be  $(0, 0)$ . The points would then be  $(4, 0)$ ,  $(0, 5)$ ,  $(3, 4)$ , and  $(10, 10)$ . Applying the [Shoelace Theorem](#),

$$Area = \frac{1}{2} \begin{vmatrix} 4 & 0 \\ 0 & 5 \\ 3 & 4 \\ 10 & 10 \\ 4 & 0 \end{vmatrix} = \frac{1}{2} |(20 + 30) - (15 + 40 + 40)| = \frac{1}{2} |50 - 95| = \boxed{\text{(C)} 22\frac{1}{2}}$$

### Solution 3

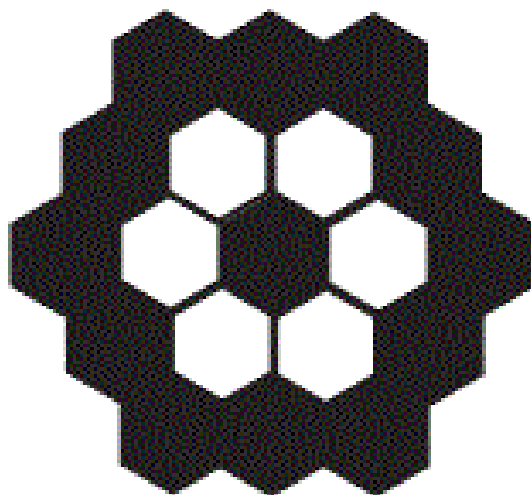
The figure contains 21 interior points and 5 boundary points. Using [Pick's Theorem](#), the area is

$$21 + \frac{5}{2} - 1 = \boxed{\text{(C)} 22\frac{1}{2}}$$

### Ishan Also See

15. (*AMC 8 2004*) Thirteen black and six white hexagonal tiles were used to create the figure below. If a new figure is created by attaching a border of white tiles with the same size and shape as the others, what will be the difference between the total number of white tiles and the total number of black tiles in the new figure?





- (A) 5    (B) 7    (C) 11    (D) 12    (E) 18

**Answer:** C

**Solution**

The first ring around the middle tile has 6 tiles, and the second has 12. From this pattern, the third ring has 18 tiles. Of these,  $6 + 18 = 24$  are white and  $1 + 12 = 13$  are black, with a difference of  $24 - 13 = \boxed{\text{(C) } 11}$ .

16. (*AMC 8 2004*) Two 600 mL pitchers contain orange juice. One pitcher is  $\frac{1}{3}$  full and the other pitcher is  $\frac{2}{5}$  full. Water is added to fill each pitcher completely, then both pitchers are poured into one large container. What fraction of the mixture in the large container is orange juice?
- (A)  $\frac{1}{8}$     (B)  $\frac{3}{16}$     (C)  $\frac{11}{30}$     (D)  $\frac{11}{19}$     (E)  $\frac{11}{15}$

**Answer:** C

**Solution 1**

The first pitcher contains  $600 \cdot \frac{1}{3} = 200$  mL of orange juice. The second pitcher contains  $600 \cdot \frac{2}{5} = 240$  mL of orange juice. In the large pitcher, there is a total of  $200 + 240 = 440$  mL of orange juice and  $600 + 600 = 1200$  mL of fluids, giving a fraction of  $\frac{440}{1200} = \boxed{\text{(C) } \frac{11}{30}}$ .

**Solution 2**

We have the first pitcher having  $\frac{1}{3}$  orange juice, and the other having  $\frac{2}{5}$ , so we can simply add together the fractions and get  $\frac{11}{15}$ , but since we poured 2 containers together, so we divide by 2 and get  $\boxed{\text{(C) } \frac{11}{30}}$ .

17. (*AMC 8 2004*) Three friends have a total of 6 identical pencils, and each one has at least one pencil. In how many ways can this happen?
- (A) 1    (B) 3    (C) 6    (D) 10    (E) 12

**Answer:** D

**Solution 1**

For each person to have at least one pencil, assign one pencil to each of the three friends so that you have 3 left. In partitioning the remaining 3 pencils into 3 distinct groups, use **Ball-and-urn** to find the number of possibilities is  $\binom{3+3-1}{3-1} = \binom{5}{2} = \boxed{(D) 10}$ .

Solution by phoenixfire Minor Edit by Yuvag

### Solution 2

Like in solution 1, for each person to have at least one pencil, assign one of the pencils to each of the three friends so that you have 3 left. In partitioning the remaining 3 pencils into 3 distinct groups, use number of non-negative integral solutions. Let the three friends be  $a, b, c$  respectively.

$$a + b + c = 3 \text{ The total being 3 and 2 plus signs, which implies } \binom{3+2}{2} = \binom{5}{2} = \boxed{(D) 10}.$$

Solution by phoenixfire Minor Edit by Yuvag

### Solution 3

For each of the 3 People to have at least one pencil when distributing 6 pencil amongst them, we can use another formula from the **Ball-and-urn** counting technique, shown below:

for  $n$  = number of items, and  $s$  = slots:

$$\binom{n-1}{s-1}$$

Now we can plug in our values,

number of items = 6, and slots = 3:

$$\binom{6-1}{3-1} = \binom{5}{2} = \boxed{(D) 10}.$$

Solution by Yuvag

### Solution 4

Like in solution 1 and solution 2, assign one pencil to each of the three friends so that you have 3 left. In partitioning the remaining 3 pencils into 3 distinct groups use casework. Let the three friends be  $a, b, c$  respectively.

$$a + b + c = 3,$$

Case 1 :  $a = 0$ ,

$$b + c = 3,$$

$$b = 0, 1, 2, 3,$$

$$c = 3, 2, 1, 0,$$

$\boxed{4}$  solutions.

Case 2 :  $a = 1$ ,

$$1 + b + c = 3,$$

$$b + c = 2,$$

$$b = 0, 1, 2,$$

$$c = 2, 1, 0,$$

$\boxed{3}$  solutions.

Case 3 :  $a = 2$ ,

$$2 + b + c = 3,$$

$$b + c = 1,$$

$$b = 0, 1,$$

$$c = 1, 0,$$

2 solutions.

Case 4 :  $a = 3$ ,

$$3 + b + c = 3,$$

$$b + c = 0,$$

$$b = 0,$$

$$c = 0,$$

1 solution.

Therefore there will be a total of  $4 + 3 + 2 + 1 = \boxed{\text{(D)} 10}$  solutions. Solution by phoenixfire

18. (*AMC 8 2004*) Five friends compete in a dart-throwing contest. Each one has two darts to throw at the same circular target, and each individual's score is the sum of the scores in the target regions that are hit. The scores for the target regions are the whole numbers 1 through 10. Each throw hits the target in a region with a different value. The scores are: Alice 16 points, Ben 4 points, Cindy 7 points, Dave 11 points, and Ellen 17 points. Who hits the region worth 6 points?

(A) Alice      (B) Ben      (C) Cindy      (D) Dave      (E) Ellen

**Answer:** A

**Solution 1**

The only way to get Ben's score is with  $1 + 3 = 4$ . Cindy's score can be made of  $3 + 4$  or  $2 + 5$ , but since Ben already hit the 3, Cindy hit  $2 + 5 = 7$ . Similarly, Dave's darts were in the region  $4 + 7 = 11$ . Lastly, because there is no 7 left, (A) Alice must have hit the regions  $6 + 10 = 16$  and Ellen  $8 + 9 = 17$ .

**Solution 2**

Taking the 2 largest scores to narrow out the points we get

Ellen with either 10, 7 or 9, 8 and Alice with either 10, 6 or 9, 7, the 2 pairs that work are Alice=10, 6 Ellen=9, 8, therefore Alice scored the 6.

This way, we don't need to check anyway if anyone else got 6.

19. (*AMC 8 2004*) A whole number larger than 2 leaves a remainder of 2 when divided by each of the numbers 3, 4, 5, and 6. The smallest such number lies between which two numbers?

(A) 40 and 49      (B) 60 and 79      (C) 100 and 129      (D) 210 and 249      (E) 320 and 369

**Answer:** B

**Solution**

The smallest number divisible by 3, 4, 5, and 6, or their least common multiple, can be found to be 60. When 2 is added to a multiple of number, its remainder when divided by that number is 2. The number we are looking for is therefore 62, and between (B) 60 and 79.

20. (*AMC 8 2004*) Two-thirds of the people in a room are seated in three-fourths of the chairs. The rest of the people are standing. If there are 6 empty chairs, how many people are in the room?

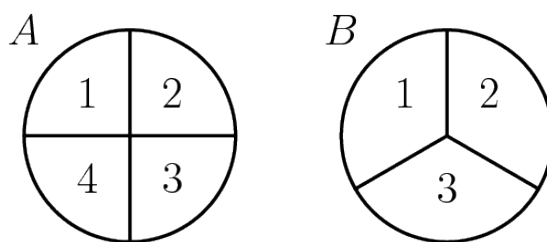
(A) 12      (B) 18      (C) 24      (D) 27      (E) 36

**Answer:** D

**Solution 1**

Working backwards, if  $\frac{3}{4}$  of the chairs are taken and 6 are empty, then there are three times as many taken chairs as empty chairs, or  $3 \cdot 6 = 18$ . If  $x$  is the number of people in the room and  $\frac{2}{3}$  are seated, then  $\frac{2}{3}x = 18$  and  $x = \boxed{\text{(D)}27}$ .

21. (*AMC 8 2004*) Spinners  $A$  and  $B$  are spun. On each spinner, the arrow is equally likely to land on each number. What is the probability that the product of the two spinners' numbers is even?



(A)  $\frac{1}{4}$       (B)  $\frac{1}{3}$       (C)  $\frac{1}{2}$       (D)  $\frac{2}{3}$       (E)  $\frac{3}{4}$

**Answer:** D

**Solution**

An even number comes from multiplying an even and even, even and odd, or odd and even. Since an odd number only comes from multiplying an odd and odd, there are less cases and it would be easier to find the probability of spinning two odd numbers from 1. Multiply the independent probabilities of each spinner getting an odd number together and subtract it from 1.

$$1 - \frac{2}{4} \cdot \frac{2}{3} = 1 - \frac{1}{3} = \boxed{\text{(D)} \frac{2}{3}}$$

**Solution 2**

We can make a chart and then we see that the 12 possibilities: 1, 2, 3, 2, 4, 6, 3, 6, 9, 4, 8, and 12. Out of these only 8 work; thus the probability is

$$\boxed{\text{(D)} \frac{2}{3}}$$

**Solution 3**

We do a little bit of casework. In order to get a product that's even, we need at least one even number. First, we consider the probability of getting an even number on the first spinner, and then multiply by 1 because the second spinner can be anything.

$$\frac{1}{2} \cdot 1 = \frac{1}{2}$$

Next, we look at the chance that we don't get an even on the first spinner, but get an even on the second spinner (we don't do the probability of even on the second spinner multiplied by one because we would be double counting both spinners are even).

$$\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$$

Add these two together to get the total probability, and we get

$$\frac{1}{2} + \frac{1}{6} = \frac{4}{6} = \boxed{\text{(D)} \frac{2}{3}}$$

22. (AMC 8 2004) At a party there are only single women and married men with their wives. The probability that a randomly selected woman is single is  $\frac{2}{5}$ . What fraction of the people in the room are married men?

(A)  $\frac{1}{3}$     (B)  $\frac{3}{8}$     (C)  $\frac{2}{5}$     (D)  $\frac{5}{12}$     (E)  $\frac{3}{5}$

**Answer:** B

**Solution 1**

Assume arbitrarily (and WLOG) there are 5 women in the room, of which  $5 \cdot \frac{2}{5} = 2$  are single and  $5 - 2 = 3$  are married. Each married woman came with her husband, so there are 3 married men in the room as well for a total of  $5 + 3 = 8$  people. The

fraction of the people that are married men is  $\boxed{\text{(B)} \frac{3}{8}}$ .

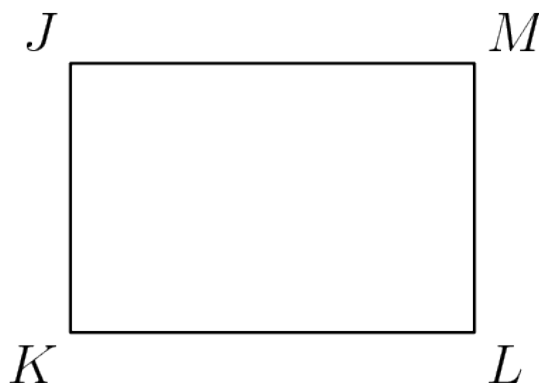
**Solution 2 (Algebraic)**

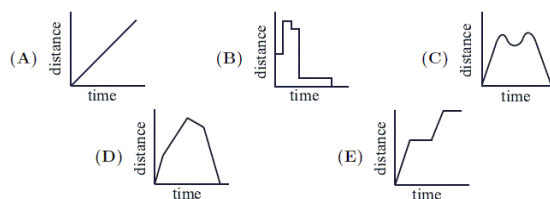
Let single women be  $S$ , married wives to be  $W$ , and men to be  $M$ . As such, the number of men is equal to married woman since each husband has his own wife or  $W = M$ . We are asked to find the probability of men in the total group or  $\frac{M}{S+W+M}$ .

We are also given that out of all the women,  $\frac{2}{5}$  are single or  $\frac{S}{S+W} = \frac{2}{5}$ . Rewrite this as  $S = \frac{2}{5}(S+W)$ ,  $S = \frac{2S}{5} + \frac{2W}{5}$ , isolating  $S$  gives us  $\frac{3S}{5} = \frac{2W}{5}$  or  $S = \frac{2M}{3}$  (because  $W = M$ ).

Now, substituting gives us  $\frac{M}{\frac{2M}{3}+M+M} = \frac{M}{\frac{2M}{3}+\frac{6M}{3}} = \frac{M}{\frac{8M}{3}} = \boxed{\text{(B)} \frac{3}{8}}$  RandomMathGuy500

23. (AMC 8 2004) Tess runs counterclockwise around rectangular block  $JKLM$ . She lives at corner  $J$ . Which graph could represent her straight-line distance from home?





**Answer:** D

### Solution 1

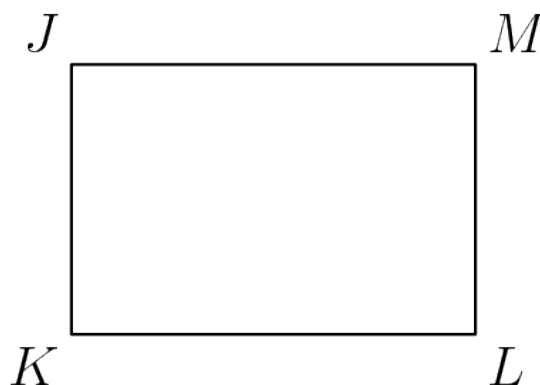
For her distance to be represented as a constant horizontal line, Tess would have to be running in a circular shape with her home as the center. Since she is running around a rectangle, this is not possible, ruling out *B* and *E* with straight lines. Because *JL* is the diagonal of the rectangle, and *L* is at the middle distance around the perimeter, her maximum distance should be in the middle of her journey. The maximum in *A* is at the end, and *C* has two maximums, ruling both out. Thus the answer is **(D)**.

### Solution 2

Even though Solution 1 may find an answer, it is never proven that *D* actually describes the distance to Tess' home.

It may be easier to compare the distance traveled by Tess rather than the time spent, if we suppose that Tess moves at constant speed then the graphs distance-time and distance-position are equivalent.

Let  $x$  be the linear distance walked by Tess, and  $d$  her distance from home  $J$ . We will divide this problem into four parts (sections  $JK$ ,  $KL$ ,  $LM$  and  $MJ$ ).



Section  $JK$ : At any point of the segment  $JK$ , the distance of Tess from home is described by the function  $d_{JK}(x) = x_J$  where  $x_J$  is the linear distance traveled by Tess from point  $J$ .

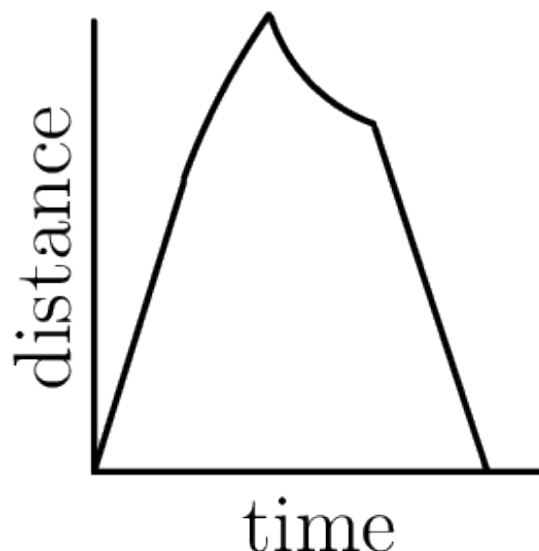
Section  $KL$ : From  $K$  to  $L$ , the distance from home is described by a right triangle, where  $d$  is the hypotenuse. In this situation the distance follows the function  $d_{KL}(x) = \sqrt{JK^2 + x_K^2}$ , which is a curve (this was obtained by applying the Pythagorean Theorem).

Section  $LM$ : Following an analogue reasoning to Section  $KL$ , the distance function in this case results  $d_{LM}(x) = \sqrt{MJ^2 + (LM - x_L)^2}$  which is also a curve.

Section  $MJ$ : Analogue to Section  $JK$ , we obtain a linear function  $d_{MJ}(x) = MJ - x_M$

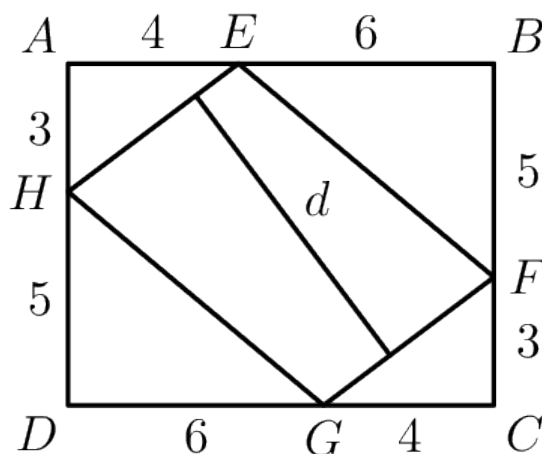
None of the shown answers in the multiple choice describe the real distance between Tess and home along the path.

The real graph should have a shape of the following kind:



Which is slightly different from the expected solution  $D$ .

24. (AMC 8 2004) In the figure,  $ABCD$  is a rectangle and  $EFGH$  is a parallelogram. Using the measurements given in the figure, what is the length  $d$  of the segment that is perpendicular to  $\overline{HE}$  and  $\overline{FG}$ ?



- (A) 6.8    (B) 7.1    (C) 7.6    (D) 7.8    (E) 8.1

**Answer:** C

**Solution**

The area of the parallelogram can be found in two ways. The first is by taking rectangle  $ABCD$  and subtracting the areas of the triangles cut out to create parallelogram  $EFGH$ . This is

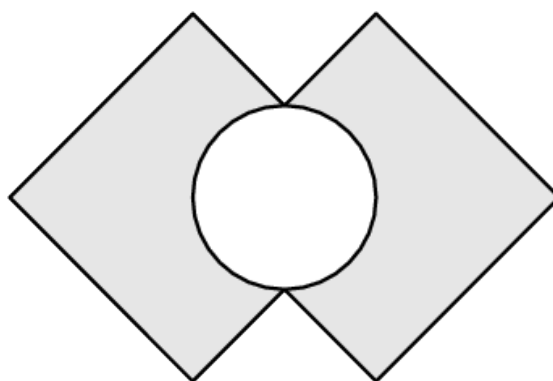
$$(4 + 6)(5 + 3) - 2 \cdot \frac{1}{2} \cdot 6 \cdot 5 - 2 \cdot \frac{1}{2} \cdot 3 \cdot 4 = 80 - 30 - 12 = 38$$

The second way is by multiplying the base of the parallelogram such as  $\overline{FG}$  with its altitude  $d$ , which is perpendicular to both bases.  $\triangle FGC$  is a 3-4-5 triangle so  $\overline{FG} = 5$ . Set these two representations of the area together.

$$5d = 38 \rightarrow d = \boxed{\text{(C) } 7.6}$$

:)

25. (AMC 8 2004) Two  $4 \times 4$  squares intersect at right angles, bisecting their intersecting sides, as shown. The circle's diameter is the segment between the two points of intersection. What is the area of the shaded region created by removing the circle from the squares?



- (A)  $16 - 4\pi$       (B)  $16 - 2\pi$       (C)  $28 - 4\pi$       (D)  $28 - 2\pi$       (E)  $32 - 2\pi$

**Answer:** D

### Solution 1

If the circle was shaded in, the intersection of the two squares would be a smaller square with half the sidelength, 2. The area of this region would be the two larger squares minus the area of the intersection, the smaller square. This is  $4^2 + 4^2 - 2^2 = 28$ .

The diagonal of this smaller square created by connecting the two points of intersection of the squares is the diameter of the circle. This value can be found with Pythagorean or a  $45^\circ - 45^\circ - 90^\circ$  triangle to be  $2\sqrt{2}$ . The radius is half the diameter,  $\sqrt{2}$ . The area of the circle is  $\pi r^2 = \pi(\sqrt{2})^2 = 2\pi$ .

The area of the shaded region is the area of the two squares minus the area of the circle which is  $\boxed{\text{(D) } 28 - 2\pi}$ .

## §21 AMC 8 2005

1. (AMC 8 2005) Connie multiplies a number by 2 and gets 60 as her answer. However, she should have divided the number by 2 to get the correct answer. What is the correct answer?
- (A) 7.5      (B) 15      (C) 30      (D) 120      (E) 240

**Answer:** B

### Solution



If  $x$  is the number, then  $2x = 60$  and  $x = 30$ . Dividing the number by 2 yields  $\frac{30}{2} = \boxed{\text{(B) } 15}$ .

A second way to do it is to divide the number by 4, as you multiplied by 2 when you were supposed to divide by 2. So,  $\frac{60}{4} = \boxed{\text{(B) } 15}$ .

2. (*AMC 8 2005*) Karl bought five folders from Pay-A-Lot at a cost of \$2.50 each. Pay-A-Lot had a 20%-off sale the following day. How much could Karl have saved on the purchase by waiting a day?
- (A) \$1.00      (B) \$2.00      (C) \$2.50      (D) \$2.75      (E) \$5.00

**Answer:** C

**Solution**

Karl paid  $5 \cdot 2.50 = \$12.50$ . 20% of this cost that he saved is  $12.50 \cdot .2 = \boxed{\text{(C) } \$2.50}$ .

**Solution 2**

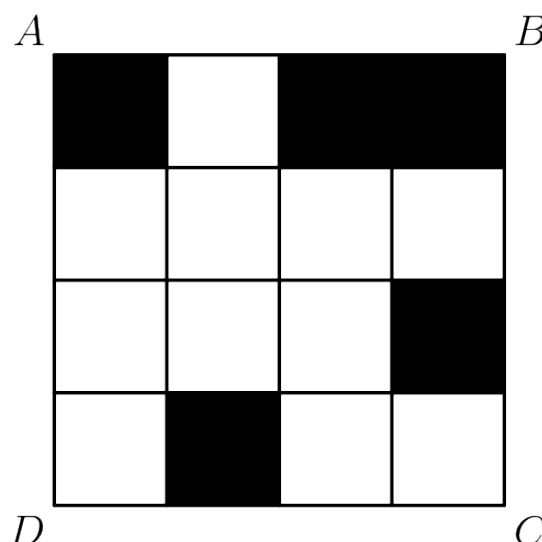
Each folder can also be  $5/2$  dollars, and 20% can be shown as  $(1/5)$ . We can multiply  $(5/2) \cdot (1/5) = (1/2)$ .  $(1/2)$  is also 50 cents or the amount of money that is saved after the 20% discount. So each folder is  $2.50 - 0.50 = \$2$ . Since Karl bought 5 folders all of the folders after the discount is  $(5)(2) = 10$ , and the money bought before the discount is  $(5)(2.50) = \$12.50$ . To find the money Karl saves all we have to do is subtract  $12.50 - 10 = 2.50$ . Thus the answer is  $\boxed{\text{(C) } \$2.50}$ .

**Solution 3**

Having a 20% off sale is equivalent to a "Buy 4 get 1 free" sale. Thus, he would have saved the price of 1 folder, which is  $\boxed{\text{(C) } \$2.50}$ .

-JeffersonJ

3. (*AMC 8 2005*) What is the minimum number of small squares that must be colored black so that a line of symmetry lies on the diagonal  $\overline{BD}$  of square  $ABCD$ ?



- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** D

**Solution**

Rotating square  $ABCD$  counterclockwise  $45^\circ$  so that the line of symmetry  $BD$  is a vertical line makes it easier to see that (D) 4 squares need to be colored to match its corresponding square.

4. (*AMC 8 2005*) A square and a triangle have equal perimeters. The lengths of the three sides of the triangle are 6.1 cm, 8.2 cm and 9.7 cm. What is the area of the square in square centimeters?

(A) 24      (B) 25      (C) 36      (D) 48      (E) 64

**Answer:** C

**Solution**

The perimeter of the triangle is  $6.1 + 8.2 + 9.7 = 24$  cm. A square's perimeter is four times its sidelength, since all its sidelengths are equal. If the square's perimeter is 24, the sidelength is  $24/4 = 6$ , and the area is  $6^2 = \text{span style="border: 1px solid black; padding: 0 2px;">(C) 36.$

5. (*AMC 8 2005*) Soda is sold in packs of 6, 12 and 24 cans. What is the minimum number of packs needed to buy exactly 90 cans of soda?

(A) 4      (B) 5      (C) 6      (D) 8      (E) 15

**Answer:** B

**Solution**

Start by buying the largest packs first. After three 24-packs,  $90 - 3(24) = 18$  cans are left. After one 12-pack,  $18 - 12 = 6$  cans are left. Then buy one more 6-pack. The total number of packs is  $3 + 1 + 1 = \text{span style="border: 1px solid black; padding: 0 2px;">(B) 5.$

6. (*AMC 8 2005*) Suppose  $d$  is a digit. For how many values of  $d$  is  $2.00d5 > 2.005$ ?

(A) 0      (B) 4      (C) 5      (D) 6      (E) 10

**Answer:** C

**Solution**

We see that 2.0055 works but 2.0045 does not. The digit  $d$  can be numbers 5 through 9, which is (C) 5 values.

7. (*AMC 8 2005*) Bill walks  $\frac{1}{2}$  mile south, then  $\frac{3}{4}$  mile east, and finally  $\frac{1}{2}$  mile south. How many miles is he, in a direct line, from his starting point?

(A) 1      (B)  $1\frac{1}{4}$       (C)  $1\frac{1}{2}$       (D)  $1\frac{3}{4}$       (E) 2

**Answer:** B

**Solution**

We have a right-angle triangle with sides  $3/4$  and 1. We can divide the simple pythagorean theorem identity 3,4,5 by 4 to get  $3/4$ , 1,  $5/4$ . This shows us the

answer is B  $1\frac{1}{4}$ .

8. (*AMC 8 2005*) Suppose  $m$  and  $n$  are positive odd integers. Which of the following must also be an odd integer?

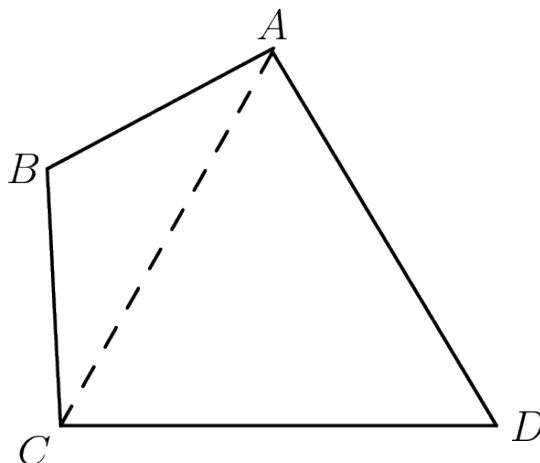
(A)  $m + 3n$       (B)  $3m - n$       (C)  $3m^2 + 3n^2$       (D)  $(nm + 3)^2$       (E)  $3mn$

**Answer:** E

**Solution**

Assume **WLOG** that  $m$  and  $n$  are both 1. Plugging into each of the choices, we get 4, 2, 6, 16, and 3. The only odd integer is **(E)  $3mn$** .

9. (AMC 8 2005) In quadrilateral  $ABCD$ , sides  $\overline{AB}$  and  $\overline{BC}$  both have length 10, sides  $\overline{CD}$  and  $\overline{DA}$  both have length 17, and the measure of angle  $ADC$  is  $60^\circ$ . What is the length of diagonal  $\overline{AC}$ ?



- (A) 13.5    (B) 14    (C) 15.5    (D) 17    (E) 18.5

**Answer:** D

**Solution 1**

Because  $\overline{AD} = \overline{CD}$ ,  $\triangle ADC$  is an isosceles triangle with  $\angle DAC = \angle DCA$ . Angles in a triangle add up to  $180^\circ$ , and since  $\angle ADC = 60^\circ$ , the other two angles are also  $60^\circ$ , and  $\triangle ADC$  is an equilateral triangle. Therefore  $\overline{AC} = \overline{DA} = \mathbf{(D) 17}$ .

**Solution 2**

We can divide  $\overline{CD}$  in half and connect this point to A, dividing  $\triangle ADC$  in half. This means the base will be  $\frac{17}{2}$  and the hypotenuse will be 17. By using the Pythagorean's Theorem, we see that if the base and height are shared, the hypotenuse should be the same. This tells us that the length of  $\overline{AC} = \mathbf{(D) 17}$ .

**Solution 3 (Easiest)**

Take an equilateral triangle with side length 17.  $\triangle ADC$  is congruent to this by  $SAS$ , hence it is equilateral. The answer is **(D) 17**.

10. (AMC 8 2005) Joe had walked half way from home to school when he realized he was late. He ran the rest of the way to school. He ran 3 times as fast as he walked. Joe took 6 minutes to walk half way to school. How many minutes did it take Joe to get from home to school?

- (A) 7    (B) 7.3    (C) 7.7    (D) 8    (E) 8.3

**Answer:** D

**Solution 1**

We can use the equation  $d = rt$  where  $d$  is the distance,  $r$  is the rate, and  $t$  is the time. The distances he ran and walked are equal, so  $r_r t_r = r_w t_w$ , where  $r_r$  is the rate at which he ran,  $t_r$  is the time for which he ran,  $r_w$  is the rate at which he

walked, and  $t_w$  is the time for which he walked. Because he runs three times faster than he walks,  $r_r = 3r_w$ . We want to find the time he ran,  $t_r = \frac{r_w t_w}{t_r} = \frac{(r_w)(6)}{3r_w} = 2$  minutes. He traveled for a total of  $6 + 2 = \boxed{\text{(D)} 8}$  minutes.

### Solution 2

We know that it took Joe 6 minutes to walk halfway. If he ran 3 times as fast as that, he went  $\frac{6}{3} = 2$  minutes for the other half (as  $t = \frac{d}{s}$  and if we multiply  $s$  by 3 we are essentially dividing  $t$  by 3). Therefore, we have  $6 + 2 = 8$  minutes.

### Solution 3

According to  $d = rt$ , and if we assume the distance from home to school is 1 mile, then  $\frac{1}{2} = 6r$ , meaning  $r = \frac{1}{12}$  mph. Then, given that he runs 3 times the speed he walks, he runs at  $\frac{3}{12}$  mph or  $\frac{1}{4}$  mph. Also, note that he only has  $\frac{1}{2}$  mile left to reach school. Therefore, once again using  $d = rt$ ,  $\frac{1}{2} = \frac{1}{4}t$ , and  $t = 2$ . Combining the given 6 minutes and the deduced 2 minutes, we reach  $\boxed{8 \text{ minutes}}$ .

-N.

11. (AMC 8 2005) The sales tax rate in Bergville is 6%. During a sale at the Bergville Coat Closet, the price of a coat is discounted 20% from its 90.00 price. Two clerks, Jack and Jill, calculate and adds 6% sales tax, then subtracts 20% from this total. Jill brings up 90.00, subtracts 20

(A) − \$1.06      (B) − \$0.53      (C) \$0      (D) \$0.53      (E) \$1.06

Answer: C

### Solution

The price Jack rings up is  $\$(90.00)(1.06)(0.80)$ . The price Jill rings up is  $\$(90.00)(0.80)(1.06)$ . By the commutative property of multiplication, these quantities are the same, and the difference is  $\boxed{\text{(C)} \$0}$ .

Reminder: Commutative states  $abc = bac = acb = \dots$  and is the same in any order. This only works for addition and multiplication.

12. (AMC 8 2005) Big Al, the ape, ate 100 bananas from May 1 through May 5. Each day he ate six more bananas than on the previous day. How many delicious bananas did Big Al eat on May 5?

(A) 20      (B) 22      (C) 30      (D) 32      (E) 34

Answer: D

### Solution

There are 5 days from May 1 to May 5. The number of bananas he eats each day is an arithmetic sequence. He eats  $n$  bananas on May 5, and  $n - 4(6) = n - 24$  bananas on May 1. The sum of this arithmetic sequence is equal to 100.

$$\frac{n + n - 24}{2} \cdot 5 = 100$$

$$n - 12 = 20$$

$$n = \boxed{\text{(D)} 32}$$

### Solution 2

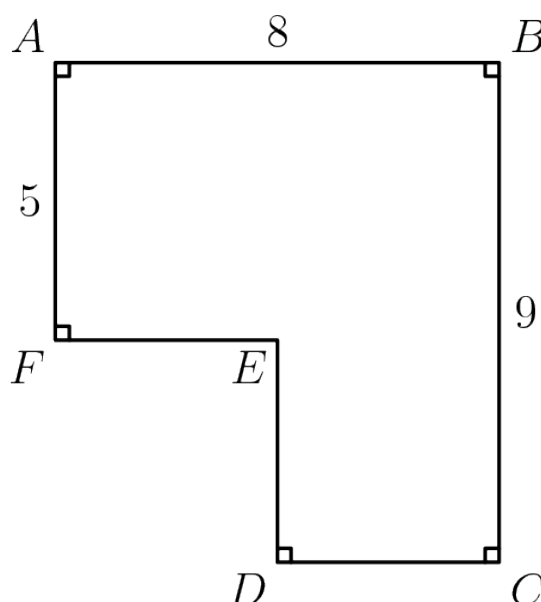
There are 5 days from May 1 to May 5. If we set the first day as  $n$ , the second day can be expressed as  $n + 6$ , the third as  $n + 12$ , and so on, for five days. The sum  $n + (n + 6) + (n + 12) + (n + 18) + (n + 24)$  is equal to 100, as stated in the problem. We can write a very simple equation, that is:  $5n + 60 = 100$ . Now all we do is just solve.  $5n = 40$ , so Big Al eats 8 bananas on the first day. The fifth day,  $n + 24$ , is then 32, which is your answer.

### Solution 3 (Lightning Quick)

Simply realize that the middle term of the arithmetic sequence is the arithmetic mean of all terms, which is simply  $\frac{100}{5} = 20$ . This means that the number of bananas the ape ate on May 5th is just  $20 + 6 \cdot 2 = 32$ . Select **D**.

You can do this quicker than you read it if you truly master arithmetic sequences.

13. (AMC 8 2005) The area of polygon  $ABCDEF$  is 52 with  $AB = 8$ ,  $BC = 9$  and  $FA = 5$ . What is  $DE + EF$ ?



- (A) 7    (B) 8    (C) 9    (D) 10    (E) 11

**Answer:** C

### Solution

Notice that  $AF + DE = BC$ , so  $DE = 4$ . Let  $O$  be the intersection of the extensions of  $AF$  and  $DC$ , which makes rectangle  $ABCO$ . The area of the polygon is the area of  $FEDO$  subtracted from the area of  $ABCO$ .

$$\text{Area} = 52 = 8 \cdot 9 - EF \cdot 4$$

Solving for the unknown,  $EF = 5$ , therefore  $DE + EF = 4 + 5 = \boxed{\text{(C) } 9}$ .

### Solution 2

Similar to solution 1,  $AF + DE = BC$ , so  $DE = 4$ . Split the polygon into two rectangles by extending the  $DE$  so it intersects  $AB$ . Let's say the length of  $FE$  is equal to  $x$ . We can form the equation:

$$5x + 9(8 - x) = 52$$

. We see that  $x = 5$ , so we can add  $5 + 4 = \boxed{\text{(C)} 9}$ .

- ArjunBhumula

14. (AMC 8 2005) The Little Twelve Basketball League has two divisions, with six teams in each division. Each team plays each of the other teams in its own division twice and every team in the other division once. How many games are scheduled?

(A) 80      (B) 96      (C) 100      (D) 108      (E) 192

**Answer:** B

**Solution**

Within each division, there are  $\binom{6}{2} = 15$  pairings, and each of these games happens twice. The same goes for the other division so that there are  $4(15) = 60$  games within their own divisions. The number of games between the two divisions is  $(6)(6) = 36$ . Together there are  $60 + 36 = \boxed{\text{(B)} 96}$  conference games.

**Solution 2**

Each team plays 10 games in its own division and 6 games against teams in the other division. So each of the 12 teams plays 16 conference games. Because each game involves two teams, there are  $\frac{12 \times 16}{2} = \boxed{96}$  games scheduled. aopsav (Credit to AoPS Alcumus)

**Solution 3**

Suppose division 1 has teams  $A, B, C, D, E, F$ . Suppose division 2 has teams  $G, H, I, J, K, L$ . Then, for interdivisional games in Division 1,  $A$  has 10 games scheduled. Because  $A$  has already played  $B$  twice in those 10 games,  $B$  will not have to account for  $A$ , making its scheduled games 8. This continues until  $F$  has 0 games scheduled. The same will happen for division 2 in interdivisional games. So,  $2(10 + 8 + 6 + 4 + 2) = 60$ . Now, for cross-divisional games, team  $A$  will play  $G, H, I, J, K$ , and  $L$  once. Each team from division 1 plays 6 cross-divisional games, for a total of  $6 \cdot 6 = 36$  games. Division 2 will NOT be counted like done for division 1 since we have already considered the games. This is the same concept applied for interdivisional games. (This is also the concept applied in the Handshake Formula.) Now,  $30 + 30 + 36 = \boxed{96}$  games scheduled.

15. (AMC 8 2005) How many different isosceles triangles have integer side lengths and perimeter 23?

(A) 2      (B) 4      (C) 6      (D) 9      (E) 11

**Answer:** C

**Solution**

Let  $b$  be the base of the isosceles triangles, and let  $a$  be the lengths of the other legs. From this,  $2a + b = 23$  and  $b = 23 - 2a$ . From triangle inequality,  $2a > b$ , then plug in the value from the previous equation to get  $2a > 23 - 2a$  or  $a > 5.75$ . The maximum value of  $a$  occurs when  $b = 1$ , in which from the first equation  $a = 11$ . Thus,  $a$  can have integer side lengths from 6 to 11, and there are  $\boxed{\text{(C)} 6}$  triangles.

16. (AMC 8 2005) A five-legged Martian has a drawer full of socks, each of which is red, white or blue, and there are at least five socks of each color. The Martian pulls out one sock at a time without looking. How many socks must the Martian remove from the drawer to be certain there will be 5 socks of the same color?

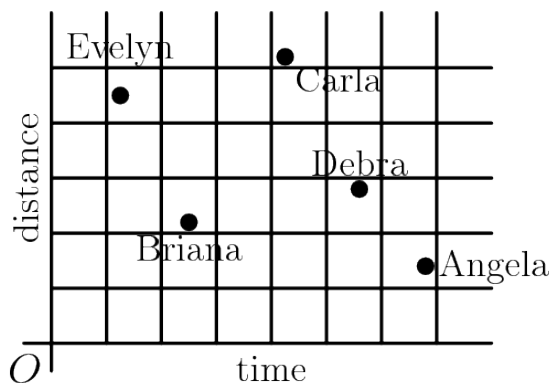
(A) 6      (B) 9      (C) 12      (D) 13      (E) 15

**Answer:** D

**Solution**

The Martian can pull out 12 socks, 4 of each color, without having 5 of the same kind yet. However, the next one he pulls out must be the fifth of one of the colors so he must remove **(D) 13** socks.

17. (AMC 8 2005) The results of a cross-country team's training run are graphed below. Which student has the greatest average speed?



- (A) Angela    (B) Briana    (C) Carla    (D) Debra    (E) Evelyn

**Answer:** E

**Solution**

Average speed is distance over time, or the slope of the line through the point and the origin. **(E) Evelyn** has the steepest line, and runs the greatest distance for the shortest amount of time.

18. (AMC 8 2005) How many three-digit numbers are divisible by 13?

- (A) 7    (B) 67    (C) 69    (D) 76    (E) 77

**Answer:** C

**Solution 1**

Let  $k$  be any positive integer so that  $13k$  is a multiple of 13. For the smallest three-digit number,  $13k > 100$  and  $k > \frac{100}{13} \approx 7.7$ . For the greatest three-digit number,  $13k < 999$  and  $k < \frac{999}{13} \approx 76.8$ . The number  $k$  can range from 8 to 76 so there are **(C) 69** three-digit numbers.

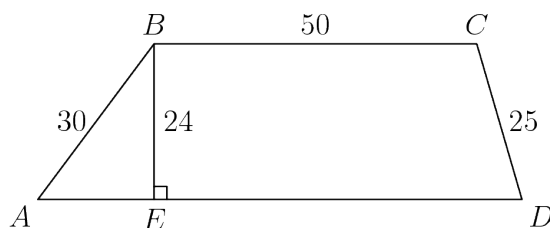
**Solution 2**

3 digits number start from 100 and ends at 999.1 to 100 total multiple of 13 =  $\frac{100}{13}$

1 to 999 total multiple of 13 =  $\frac{999}{13}$

Hence  $\frac{999}{13} - \frac{100}{13} = 76 - 7 = 69$

19. (AMC 8 2005) What is the perimeter of trapezoid  $ABCD$ ?



(A) 180      (B) 188      (C) 196      (D) 200      (E) 204

**Answer:** A

**Solution**

Draw altitudes from  $B$  and  $C$  to base  $AD$  to create a rectangle and two right triangles. The side opposite  $BC$  is equal to 50. The bases of the right triangles can be found using Pythagorean or special triangles to be 18 and 7. Add it together to get  $AD = 18 + 50 + 7 = 75$ . The perimeter is  $75 + 30 + 50 + 25 = \boxed{\text{(A) } 180}$ .

20. (AMC 8 2005) Alice and Bob play a game involving a circle whose circumference is divided by 12 equally-spaced points. The points are numbered clockwise, from 1 to 12. Both start on point 12. Alice moves clockwise and Bob, counterclockwise. In a turn of the game, Alice moves 5 points clockwise and Bob moves 9 points counterclockwise. The game ends when they stop on the same point. How many turns will this take?

(A) 6      (B) 8      (C) 12      (D) 14      (E) 24

**Answer:** A

**Solution**

Alice moves  $5k$  steps and Bob moves  $9k$  steps, where  $k$  is the turn they are on. Alice and Bob coincide when the number of steps they move collectively,  $14k$ , is a multiple of 12. Since this number must be a multiple of 12, as stated in the previous sentence, 14 has a factor 2,  $k$  must have a factor of 6. The smallest number of turns that is a multiple of 6 is  $\boxed{\text{(A) } 6}$ .

Another solution (Modular Arithmetic) Step 1: Write their positions after  $k$  turns.

Alice: moves 5 steps clockwise each turn. Position after  $k$  turns:

$$A(k) = 12 + 5k \pmod{12}$$

Bob: moves 9 steps counterclockwise each turn. Moving 9 steps counterclockwise is the same as moving  $+3$  clockwise (since  $12 - 9 = 3$ ). Position after  $k$  turns:

$$B(k) = 12 + 3k \pmod{12}$$

Step 2: Find when they meet.

$$A(k) \equiv B(k) \pmod{12}$$

So:

$$12 + 5k \equiv 12 + 3k \pmod{12}$$

Cancel the 12's:

$$5k \equiv 3k \pmod{12} \Rightarrow 2k \equiv 0 \pmod{12}$$

That means  $2k$  is divisible by 12:

$$2k \equiv 0 \pmod{12} \Rightarrow k \equiv 0 \pmod{6}$$

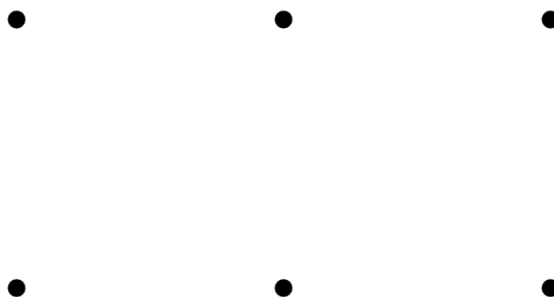
So the smallest positive  $k$  is:

$$k = 6$$

(True) They meet after 6 turns  $\rightarrow$  Answer: (A) 6 - timi821

21. (AMC 8 2005) How many distinct triangles can be drawn using three of the dots below as vertices?





- (A) 9      (B) 12      (C) 18      (D) 20      (E) 24

**Answer:** C

**Solution 1**

The number of ways to choose three points to make a triangle is  $\binom{6}{3} = 20$ . However, two\* of these are a straight line so we subtract 2 to get (C) 18.

- Note: We are assuming that there are no degenerate triangles in this problem, and that is why we subtract two.

**Solution 2**

**Case 1: One vertex is on the top 3 points**

Here, there is  $\binom{3}{1} = 3$  ways to choose the vertex on the top and  $\binom{3}{2} = 3$  ways the choose the 2 on the bottom, so there is  $3 \cdot 3 = 9$  triangles.

**Case 2: One vertex is on the bottom 3 points**

By symmetry, there is 9 triangles.

Otherwise, the triangle is degenerate.

Our final answer is  $9 + 9 =$  (C) 18 Bobbert

22. (AMC 8 2005) A company sells detergent in three different sized boxes: small (S), medium (M) and large (L). The medium size costs 50% more than the small size and contains 20% less detergent than the large size. The large size contains twice as much detergent as the small size and costs 30% more than the medium size. Rank the three sizes from best to worst buy.

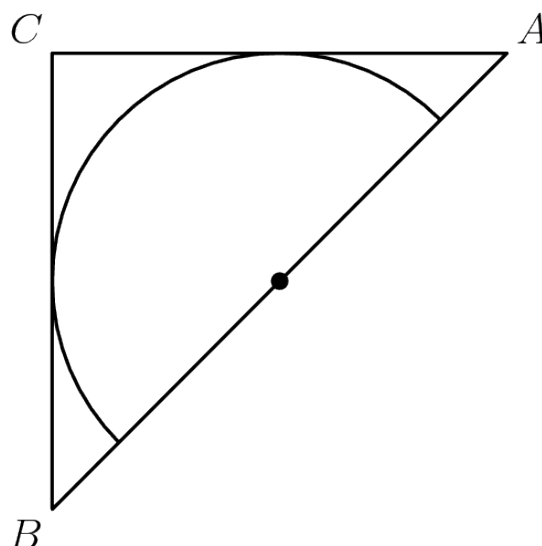
- (A) SML      (B) LMS      (C) MSL      (D) LSM      (E) MLS

**Answer:** E

**Solution**

WLOG (Without Loss of Generality), suppose the small size costs \$1 and has 5 oz. The medium size then costs \$1.50 and has 8 oz. The large size costs \$1.95 and has 10 oz. The small, medium, and large sizes cost, respectively, 0.200, 0.188, and \$0.195 per oz. The sizes from best to worst buy are (E) MLS.

23. (AMC 8 2005) Isosceles right triangle  $ABC$  encloses a semicircle of area  $2\pi$ . The circle has its center  $O$  on hypotenuse  $\overline{AB}$  and is tangent to sides  $\overline{AC}$  and  $\overline{BC}$ . What is the area of triangle  $ABC$ ?



- (A) 6      (B) 8      (C)  $3\pi$       (D) 10      (E)  $4\pi$

**Answer:** B

**Solution**

First, we notice half a square so first let's create a square. Once we have a square, we will have a full circle. This circle has a diameter of 4 which will be the side of the square. The area would be  $4 \cdot 4 = 16$ . Divide 16 by 2 to get the original shape and you get 8

**Solution 2 (Super Fast)**

Well, the question states that it is an Isosceles triangle, so 2 of the side lengths (not the hypotenuse) must be the same. Let us call them  $x$ , and the formula for finding the area of a triangle is  $\frac{1}{2}(a)(b)$ , if we implement this we get  $\frac{1}{2}(x)(x) = \frac{x^2}{2}$ . The only answer which satisfies this condition is B therefore we have solved the question. We can double check as  $8 \times 2 = 16$  and the root of  $16 = 4$ .

(B) 8.

24. (AMC 8 2005) A certain calculator has only two keys  $[+1]$  and  $[x2]$ . When you press one of the keys, the calculator automatically displays the result. For instance, if the calculator originally displayed "9" and you pressed  $[+1]$ , it would display "10." If you then pressed  $[x2]$ , it would display "20." Starting with the display "1," what is the fewest number of keystrokes you would need to reach "200"?

- (A) 8      (B) 9      (C) 10      (D) 11      (E) 12

**Answer:** B

**Solution 1 (Now rigorous)**

We can start at 200 and work our way down to 1. We want to press the button that multiplies by 2 the most, but since we are going down instead of up, we divide by 2 instead. If we come across an odd number, then we will subtract that number by 1. Notice

$$200 \div 2 = 100, \quad 100 \div 2 = 50, \quad 50 \div 2 = 25, \quad 25 - 1 = 24, \quad (\text{Cannot divide})$$

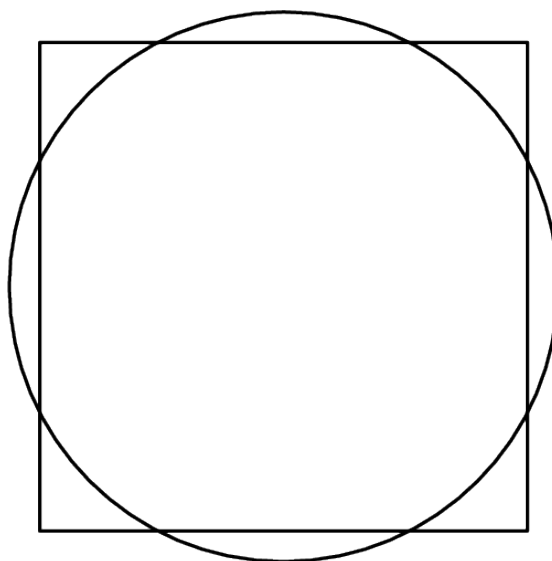
Since we've reached 1, it's clear that the answer should be (B) 9 - Javapost. Because we only subtracted 1 when we had to, this is optimal. Roy2020 minor edits by IncredibleCheetah53

**Solution 2 (duodecimal)**

We make two key observations. First, pressing  $[x2]$  appends a 0 to the end of a number's binary representation. Second, pressing  $[x2]$  and then pressing  $[+1]$  appends to the end of a number's binary representation.

The base-ten number 200 is represented as  $11001000_2$  in binary. Therefore, the five 0s contribute 5 button presses. Similarly, each of the three 1s each contribute 2 button presses, although since we initially started with a 1 we can ignore one of the 1s. Thus, the answer is  $5 + 2 \cdot 2 = \boxed{\text{(B)} 9}$ .

25. (AMC 8 2005) A square with side length 2.0 and a circle share the same center. The total area of the regions that are inside the circle and outside the square is equal to the total area of the regions that are outside the circle and inside the square. What is the radius of the circle?



- (A)  $\frac{2}{\sqrt{\pi}}$     (B)  $\frac{1+\sqrt{2}}{2}$     (C)  $\frac{3}{2}$     (D)  $\sqrt{3}$     (E)  $\sqrt{\pi}$

**Answer:** A

**Solution 1**

Let the region within the circle and square be  $a$ . In other words, it is the area inside the circle **and** the square. Let  $r$  be the radius. We know that the area of the circle minus  $a$  is equal to the area of the square, minus  $a$ .

We get:

$$\pi r^2 - a = 4 - a \quad r^2 = \frac{4}{\pi} \quad r = \frac{2}{\sqrt{\pi}}$$

So the answer is  $\boxed{\text{(A)} \frac{2}{\sqrt{\pi}}}$ .

**Solution 2**

We realize that since the areas of the regions outside of the circle and the square are equal to each other, the area of the circle must be equal to the area of the square.

$$\pi r^2 = 4 \quad r^2 = \frac{4}{\pi} \quad r = \frac{2}{\sqrt{\pi}}$$

So the answer is  $\boxed{\text{(A)} \frac{2}{\sqrt{\pi}}}$ .

**§22 AMC 8 2006**

1. (*AMC 8 2006*) Mindy made three purchases for \$1.98, \$5.04, and \$9.89. What was her total, to the nearest dollar?

(A) 10      (B) 15      (C) 16      (D) 17      (E) 18

**Answer:** D

**Solution**

The three prices round to \$2, \$5, and \$10, which have a sum of **(D) 17**.

We know that there will not be a rounding error, as the total amount rounded is clearly less than \$0.50.

2. (*AMC 8 2006*) On the AMC 8 contest Billy answers 13 questions correctly, answers 7 questions incorrectly and doesn't answer the last 5. What is his score?

(A) 1      (B) 6      (C) 13      (D) 19      (E) 26

**Answer:** C

**Solution**

As the AMC 8 only rewards 1 point for each correct answer, everything is irrelevant except the number Billy answered correctly, **(C) 13**.

3. (*AMC 8 2006*) Elisa swims laps in the pool. When she first started, she completed 10 laps in 25 minutes. Now she can finish 12 laps in 24 minutes. By how many minutes has she improved her lap time?

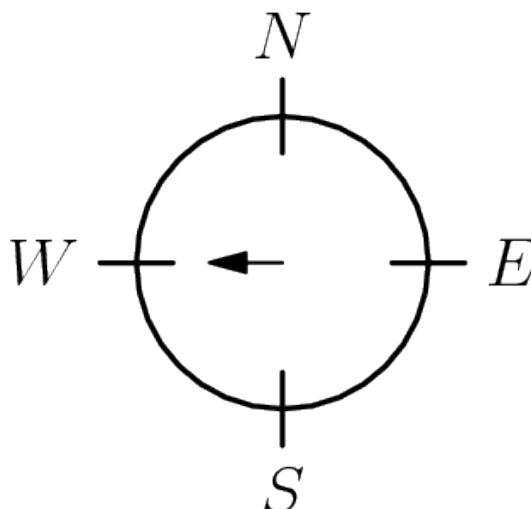
(A)  $\frac{1}{2}$       (B)  $\frac{3}{4}$       (C) 1      (D) 2      (E) 3

**Answer:** A

**Solution**

When Elisa started, she finished a lap in  $\frac{25}{10} = 2.5$  minutes. Now, she finishes a lap in  $\frac{24}{12} = 2$  minutes. The difference is  $2.5 - 2 =$  **(A)  $\frac{1}{2}$** .

4. (*AMC 8 2006*) Initially, a spinner points west. Chenille moves it clockwise  $2\frac{1}{4}$  revolutions and then counterclockwise  $3\frac{3}{4}$  revolutions. In what direction does the spinner point after the two moves?



- (A) north    (B) east    (C) south    (D) west    (E) northwest

**Answer:** B

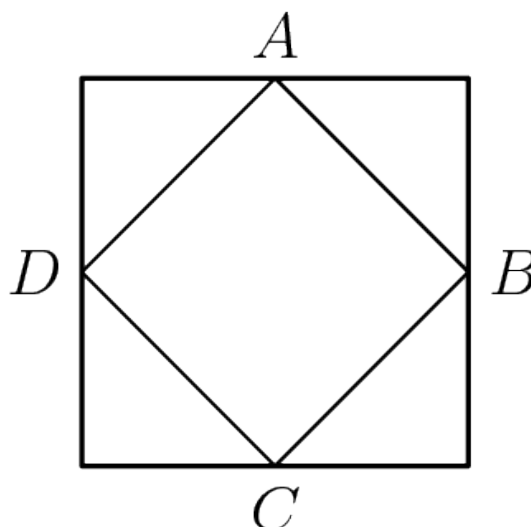
**Solution 1**

If the spinner goes clockwise  $2\frac{1}{4}$  revolutions and then counterclockwise  $3\frac{3}{4}$  revolutions, it ultimately goes counterclockwise  $1\frac{1}{2}$  which brings the spinner pointing **(B) east**.

**Solution 2 (Minor improvement)**

Note that full revolutions do not matter, so this is equivalent to going clockwise  $\frac{1}{4}$  revolutions and then counterclockwise  $\frac{3}{4}$  revolutions, making it ultimately go counterclockwise  $\frac{1}{2}$ , having the spinner point **(B) east**.

5. (AMC 8 2006) Points  $A, B, C$  and  $D$  are midpoints of the sides of the larger square. If the larger square has area 60, what is the area of the smaller square?



- (A) 15    (B) 20    (C) 24    (D) 30    (E) 40

**Answer:** D

**Solution 1**

Drawing segments  $AC$  and  $BD$ , the number of triangles outside square  $ABCD$  is the same as the number of triangles inside the square. Thus areas must be equal so the area of  $ABCD$  is half the area of the larger square which is  $\frac{60}{2} = \boxed{\text{(D)} 30}$ .

**Solution 2**

If the side length of the larger square is  $x$ , the side length of the smaller square is  $\frac{\sqrt{2} \cdot x}{2}$ . Therefore the area of the smaller square is  $\frac{x^2}{2}$ , half of the larger square's area,  $x^2$ .

Thus, the area of the smaller square in the picture is  $\frac{60}{2} = \boxed{\text{(D)} 30}$ .

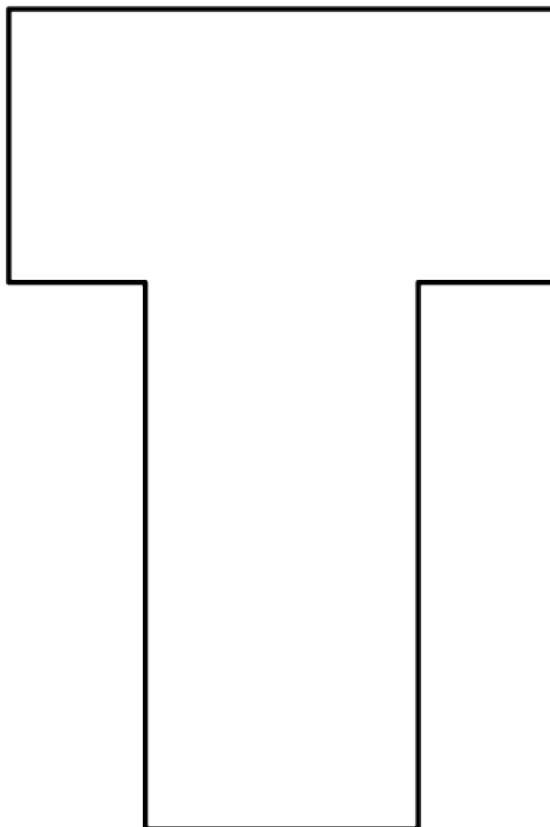
**Solution 3**

The smaller square forms four congruent isosceles right triangles whose legs measure half of the larger square's side length. Since the larger square has an area of 60, its sides measure  $\sqrt{60} = 2\sqrt{15}$ . Therefore, the legs of the right triangles measure  $\sqrt{15}$ . Now we can use the Pythagorean Theorem to find the length of the hypotenuses, or the smaller square's sides. We have

$$\sqrt{\sqrt{15}^2 + \sqrt{15}^2} = \sqrt{15 + 15} = \sqrt{30}.$$

Squaring this, the area of the smaller square is  $\boxed{\text{(D)} 30}$ .

6. (AMC 8 2006) The letter T is formed by placing two  $2 \times 4$  inch rectangles next to each other, as shown. What is the perimeter of the T, in inches?



- (A) 12      (B) 16      (C) 20      (D) 22      (E) 24

**Answer:** C

**Solution**

If the two rectangles were separate, the perimeter would be  $2(2(2 + 4)) = 24$ . It is easy to see that their connection erases 2 from each of the rectangles, so the final perimeter is  $24 - 2 \times 2 = \boxed{\text{(C) } 20}$ .

7. (AMC 8 2006) Circle  $X$  has a radius of  $\pi$ . Circle  $Y$  has a circumference of  $8\pi$ . Circle  $Z$  has an area of  $9\pi$ . List the circles in order from smallest to largest radius.  
(A)  $X, Y, Z$       (B)  $Z, X, Y$       (C)  $Y, X, Z$       (D)  $Z, Y, X$       (E)  $X, Z, Y$

**Answer:** B

**Solution**

Not using the formulas of circles,  $C = 2\pi r$  and  $A = \pi r^2$ , we find that circle  $Y$  has a radius of 4 and circle  $Z$  has a radius of 3. Also, circle  $X$  has a radius of  $\pi$ . Thus, the order from smallest to largest radius is  $\boxed{\text{(B) } Z, X, Y}$ .

8. (AMC 8 2006) The table shows some of the results of a survey by radiostation KACL. What percentage of the males surveyed listen to the station?

|         | Listen | Don't Listen | Total |
|---------|--------|--------------|-------|
| Males   | ?      | 26           | ?     |
| Females | 58     | ?            | 96    |
| Total   | 136    | 64           | 200   |

- (A) 39      (B) 48      (C) 52      (D) 55      (E) 75

**Answer:** E

**Solution**

Filling out the chart, it becomes

|         | Listen | Don't Listen | Total |
|---------|--------|--------------|-------|
| Males   | 78     | 26           | 104   |
| Females | 58     | 38           | 96    |
| Total   | 136    | 64           | 200   |

Thus, the percentage of males surveyed that listen to the station is  $100 \cdot \frac{78}{104} \% = \boxed{\text{(E) } 75\%}$ .

**Solution 2**

Instead of filling out the entire chart, you can find the number of male listeners which is  $200 - 96 = 104$ . Since we know the fraction of males that don't listen to the station is  $\frac{26}{104}$ , we know that the percent of males that DO listen is much larger than 50%, so the only answer is  $\boxed{\text{(E) } 75\%}$ .

9. (AMC 8 2006) What is the product of  $\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \cdots \times \frac{2006}{2005}$  ?  
(A) 1      (B) 1002      (C) 1003      (D) 2005      (E) 2006

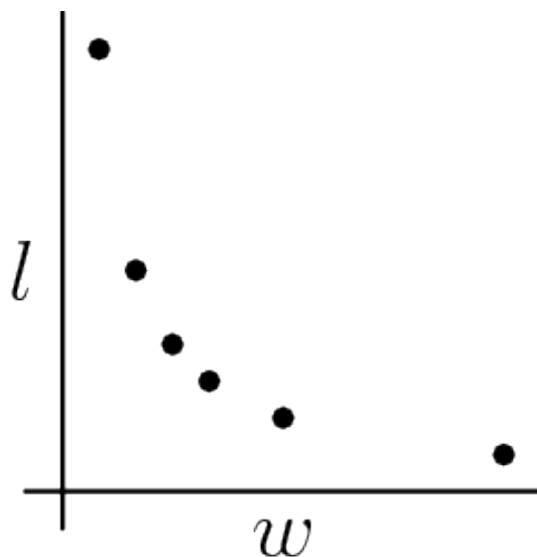
**Answer:** C

**Solution**

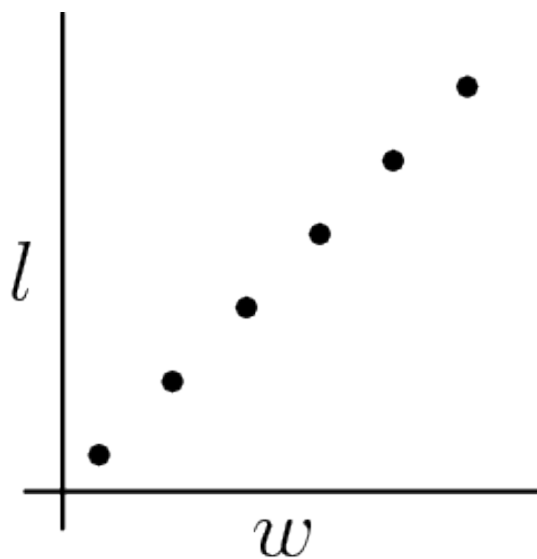
The numerator in each fraction cancels out with the denominator of the next fraction. There are only two numbers that didn't cancel:  $\frac{2006}{2} = \boxed{\text{(C) } 1003}$ .

10. (AMC 8 2006) Jorge's teacher asks him to plot all the ordered pairs  $(w, l)$  of positive integers for which  $w$  is the width and  $l$  is the length of a rectangle with area 12. What should his graph look like?

(A)

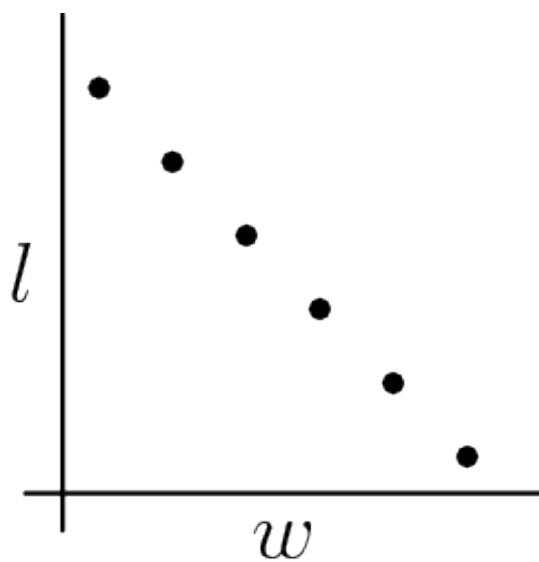


(B)

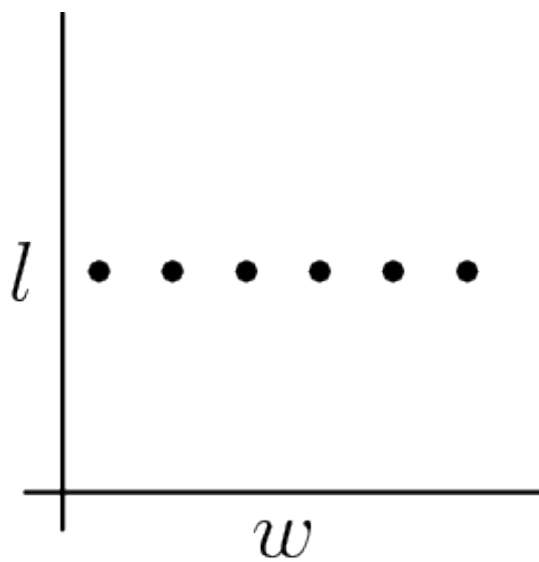


(C)

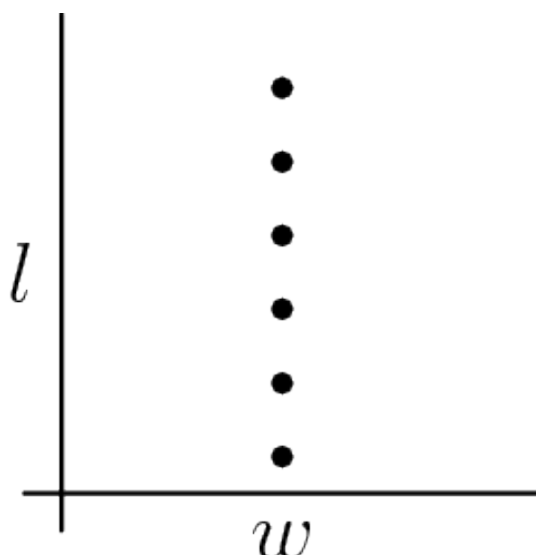




(D)



(E)



**Answer:** A

**Solution**

The length of the rectangle will relate invertly to the width, specifically using the theorem  $l = \frac{12}{w}$ . The only graph that could represent a inverted relationship is **(A)**. (The rest are linear graphs that don't represent indirect relationships, therefore they are incorrect.)

11. (*AMC 8 2006*) How many two-digit numbers have digits whose sum is a perfect square?  
**(A)** 13      **(B)** 16      **(C)** 17      **(D)** 18      **(E)** 19

**Answer:** C

**Solution**

There is 1 integer whose digits sum to 1: 10.

There are 4 integers whose digits sum to 4: 13, 22, 31, and 40.

There are 9 integers whose digits sum to 9: 18, 27, 36, 45, 54, 63, 72, 81, and 90.

There are 3 integers whose digits sum to 16: 79, 88, and 97.

Two digits cannot sum to 25 or any greater square since the greatest sum of digits of a two-digit number is  $9 + 9 = 18$ .

Thus, the answer is  $1 + 4 + 9 + 3 = \mathbf{(C)17}$ .

12. (*AMC 8 2006*) Antonette gets 70% on a 10-problem test, 80% on a 20-problem test and 90% on a 30-problem test. If the three tests are combined into one 60-problem test, which percent is closest to her overall score?  
**(A)** 40      **(B)** 77      **(C)** 80      **(D)** 83      **(E)** 87

**Answer:** D

**Solution**

$$70\% \cdot 10 = 7 \quad 80\% \cdot 20 = 16 \quad 90\% \cdot 30 = 27$$

Adding them up gets  $7 + 16 + 27 = 50$ . The overall percentage correct would be  $\frac{50}{60} = \frac{5}{6} = 5 \cdot 16.\bar{6} = 83.\bar{3} \approx \mathbf{(D) 83}$ .

13. (*AMC 8 2006*) Cassie leaves Escanaba at 8:30 AM heading for Marquette on her bike. She bikes at a uniform rate of 12 miles per hour. Brian leaves Marquette at 9:00 AM heading for Escanaba on his bike. He bikes at a uniform rate of 16 miles per hour. They both bike on the same 62-mile route between Escanaba and Marquette. At what time in the morning do they meet?

(A) 10 : 00      (B) 10 : 15      (C) 10 : 30      (D) 11 : 00      (E) 11 : 30

**Answer:** D

**Solution 1**

If Cassie leaves  $\frac{1}{2}$  an hour earlier than Brian, when Brian starts, the distance between them will be  $62 - \frac{12}{2} = 56$ . Every hour, they will get  $12 + 16 = 28$  miles closer.  $\frac{56}{28} = 2$ , so 2 hours from 9 : 00 AM is when they meet, which is (D) 11 : 00.

**Solution 2 (Basic Algebra)**

Let  $x$  be the # of hours after Cassie leaves, when both of them meet. In that  $x$  hours, Cassie will travel  $12x$  miles. But, Brian will travel  $16(x - 1/2)$  miles as he starts  $1/2$  hours after Cassie. These two distances sum to the total distance of 62 miles as they meet here, yielding the equation,  $12x + 16(x - 1/2) = 62$ . After distribution 16 we get  $12x + 16x - 8 = 62$ . After combining like terms and adding 8 to both sides we get  $28x = 70$ , and  $x = 5/2$ . So they meet  $5/2$  hours after 8:30 a.m. which is (D) 11 : 00.

- LearnForEver

14. (*AMC 8 2006*) Problems 14, 15 and 16 involve Mrs. Reed's English assignment.

A Novel Assignment

The students in Mrs. Reed's English class are reading the same 760-page novel. Three friends, Alice, Bob and Chandra, are in the class. Alice reads a page in 20 seconds, Bob reads a page in 45 seconds and Chandra reads a page in 30 seconds.

If Bob and Chandra both read the whole book, Bob will spend how many more seconds reading than Chandra?

(A) 7,600      (B) 11,400      (C) 12,500      (D) 15,200      (E) 22,800

**Answer:** B

**Solution**

The information is the same for Problems 14,15, and 16. Therefore, we shall only use the information we need. All we need for this problem is that there's 760 pages, Bob reads a page in 45 seconds and Chandra reads a page in 30 seconds. A lot of people will find how long it takes Bob to read the book, how long it takes Chandra to read the book, and then find the seconds. However, if we just set up the expression, we can find an easier way.

$$760 \cdot 45 - 760 \cdot 30 = 760(45 - 30) = 760(15) = \span style="border: 1px solid black; padding: 2px;">(\mathbf{B})11,400$$

15. (*AMC 8 2006*) Problems 14, 15 and 16 involve Mrs. Reed's English assignment.

A Novel Assignment

The students in Mrs. Reed's English class are reading the same 760-page novel. Three friends, Alice, Bob and Chandra, are in the class. Alice reads a page in 20 seconds, Bob reads a page in 45 seconds and Chandra reads a page in 30 seconds.

Chandra and Bob, who each have a copy of the book, decide that they can save time by "team reading" the novel. In this scheme, Chandra will read from page 1 to a certain page and Bob will read from the next page through page 760, finishing the book. When they are through they will tell each other about the part they read. What is the last page that Chandra should read so that he and Bob spend the same amount of time reading the novel?

- (A) 425    (B) 444    (C) 456    (D) 484    (E) 506

**Answer:** C

**Solution 1**

Same as the previous problem, we only use the information we need. Note that it's not just Chandra reads half of it and Bob reads the rest since they have different reading rates. In this case, we set up an equation and solve.

Let  $x$  be the number of pages that Chandra reads.

$30x = 45(760 - x)$  Distribute the 45  $30x = 45(760) - 45x$  Add  $45x$  to both sides

$75x = 45(760)$  Divide both sides by 15 to make it easier to solve

$5x = 3(760)$  Divide both sides by 5  $x = 3(152) = \boxed{\text{(C)}456}$

**Solution 2 (a bit faster)**

Chandra and Bob read at a rate of 30 : 45 seconds per page, respectively. Simplifying that gets us Bob reads 2 pages for every 3 pages that Chandra reads. Therefore Chandra should read  $\frac{3}{2+3} = \frac{3}{5}$  of the book.  $\frac{3}{5} \cdot 760 = \boxed{\text{(C)}456}$

16. (AMC 8 2006) Problems 14, 15 and 16 involve Mrs. Reed's English assignment.

A Novel Assignment

The students in Mrs. Reed's English class are reading the same 760-page novel. Three friends, Alice, Bob and Chandra, are in the class. Alice reads a page in 20 seconds, Bob reads a page in 45 seconds and Chandra reads a page in 30 seconds.

Before Chandra and Bob start reading, Alice says she would like to team read with them. If they divide the book into three sections so that each reads for the same length of time, how many seconds will each have to read?

- (A) 6400    (B) 6600    (C) 6800    (D) 7000    (E) 7200

**Answer:** E

**Solution**

The amount of pages Bob, Chandra, and Alice will read is in the ratio 4:6:9. Therefore, Bob, Chandra, and Alice read 160, 240, and 360 pages respectively. They would also be reading for the same amount of time because the ratio of the pages read was based on the time it takes each of them to read a page. Therefore, the amount of seconds each person reads is simply  $160 \cdot 45 = \boxed{\text{(E)} 7200}$ .

**Solution 2**

We can solve this using equations. Let Bob read  $x$  pages and Chandra read  $y$  pages. Then Alice would read  $760 - x - y$  pages. Then we can set up the equations using the time it takes them to read a page:  $45x = 30y = 20(760 - x - y)$ . Since we only have two variables, we only need two equations. Let's use  $45x = 30y$  and  $45x = 20(760 - x - y)$ . We can simplify  $45x = 30y$  down to  $3x = 2y$  and

$45x = 20(760 - x - y)$  down to  $9x = 4(760 - x - y)$ . We can then solve the equations accordingly:

$$9x = 3040 - 4x - 4y$$

$$13x = 3040 - 4y$$

From the equation  $3x = 2y$ , we know that  $4y = 6x$ .

$$13x = 3040 - 6x$$

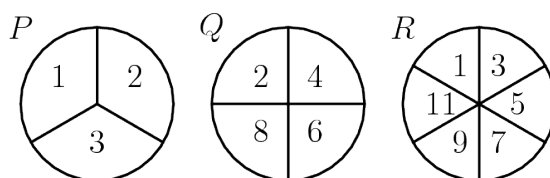
$$19x = 3040$$

$$x = 160$$

Now that we know  $x$ , we can multiply it by 45 (Bob's speed) to get the answer:

$$160 \times 45 = \boxed{\text{(E)} 7200}$$

17. (AMC 8 2006) Jeff rotates spinners  $P$ ,  $Q$  and  $R$  and adds the resulting numbers. What is the probability that his sum is an odd number?



- (A)  $\frac{1}{4}$     (B)  $\frac{1}{3}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{3}$     (E)  $\frac{3}{4}$

**Answer:** B

**Solution**

In order for Jeff to have an odd number sum, the numbers must either be Odd + Odd + Odd or Even + Even + Odd. We easily notice that we cannot obtain Odd + Odd + Odd because spinner  $Q$  contains only even numbers. Therefore we must work with Even + Even + Odd and spinner  $Q$  will give us one of our even numbers. We also see that spinner  $R$  only contains odd, so spinner  $R$  must give us our odd number. We still need one even number from spinner  $P$ . There is only 1 even number: 2. Since spinning the required numbers are automatic on the other spinners, we only have to find the probability of spinning a 2 in spinner  $P$ , which

is clearly  $\boxed{\text{(B)} \frac{1}{3}}$

18. (AMC 8 2006) A cube with 3-inch edges is made using 27 cubes with 1-inch edges. Nineteen of the smaller cubes are white and eight are black. If the eight black cubes are placed at the corners of the larger cube, what fraction of the surface area of the larger cube is white?

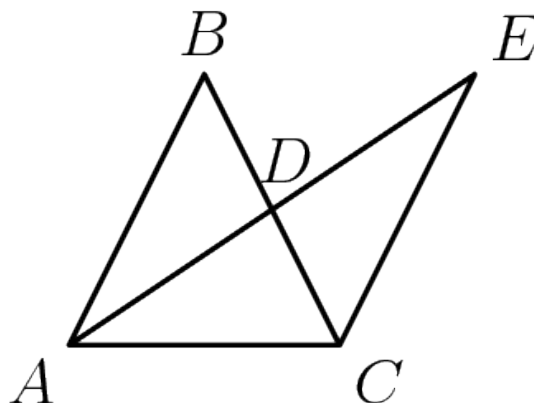
- (A)  $\frac{1}{9}$     (B)  $\frac{1}{4}$     (C)  $\frac{4}{9}$     (D)  $\frac{5}{9}$     (E)  $\frac{19}{27}$

**Answer:** D

**Solution**

The surface area of the cube is  $6(3)(3) = 54$ . Each of the eight black cubes has 3 faces on the outside, making  $3(8) = 24$  black faces. Therefore there are  $54 - 24 = 30$  white faces. To find the ratio, we evaluate  $\frac{30}{54} = \boxed{\text{(D)} \frac{5}{9}}$ .

19. (AMC 8 2006) Triangle  $ABC$  is an isosceles triangle with  $\overline{AB} = \overline{BC}$ . Point  $D$  is the midpoint of both  $\overline{BC}$  and  $\overline{AE}$ , and  $\overline{CE}$  is 11 units long. Triangle  $ABD$  is congruent to triangle  $ECD$ . What is the length of  $\overline{BD}$ ?



- (A) 4      (B) 4.5      (C) 5      (D) 5.5      (E) 6

**Answer:** D

**Solution**

Since triangle  $ABD$  is congruent to triangle  $ECD$  and  $\overline{CE} = 11$ ,  $\overline{AB} = 11$ . Since  $\overline{AB} = \overline{BC}$ ,  $\overline{BC} = 11$ . Because point  $D$  is the midpoint of  $\overline{BC}$ ,  $\overline{BD} = \frac{\overline{BC}}{2} = \frac{11}{2} = \boxed{\text{(D)} 5.5}$ . We can say this because of CPCTC; corresponding parts of congruent triangles are congruent.

20. (AMC 8 2006) A singles tournament had six players. Each player played every other player only once, with no ties. If Helen won 4 games, Ines won 3 games, Janet won 2 games, Kendra won 2 games and Lara won 2 games, how many games did Monica (the sixth player) win?

- (A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** C

**Solution**

Since there are 6 players, a total of  $\frac{6(6-1)}{2} = 15$  games are played. So far,  $4 + 3 + 2 + 2 + 2 = 13$  games finished (one person won from each game), so Monica needs to win  $15 - 13 = \boxed{\text{(C)} 2}$ .

**SOLUTION NUMBER 2 (Similar to the first one)** There are six players, so to find the total number of games played, we must do  $5+4+3+2+1=15$ . Next we add  $4+3+2+2+2=13$ . Since, every win represents one game and there are 15 games, Monica has to win  $15-13=2$  games. So the answer is (c).

-This is by someone

**SOLUTION NUMBER 2 (Similar to the first one)** There are six players, so to find the total number of games played, we must do  $5+4+3+2+1=15$ . Next

we add  $4+3+2+2+2=13$ . Since, every win represents one game and there are 15 games, Monica has to win  $15-13=2$  games. So the answer is (c).

-This is by someone.

21. (AMC 8 2006) An aquarium has a rectangular base that measures 100 cm by 40 cm and has a height of 50 cm. The aquarium is filled with water to a depth of 37 cm. A rock with volume  $1000 \text{ cm}^3$  is then placed in the aquarium and completely submerged. By how many centimeters does the water level rise?

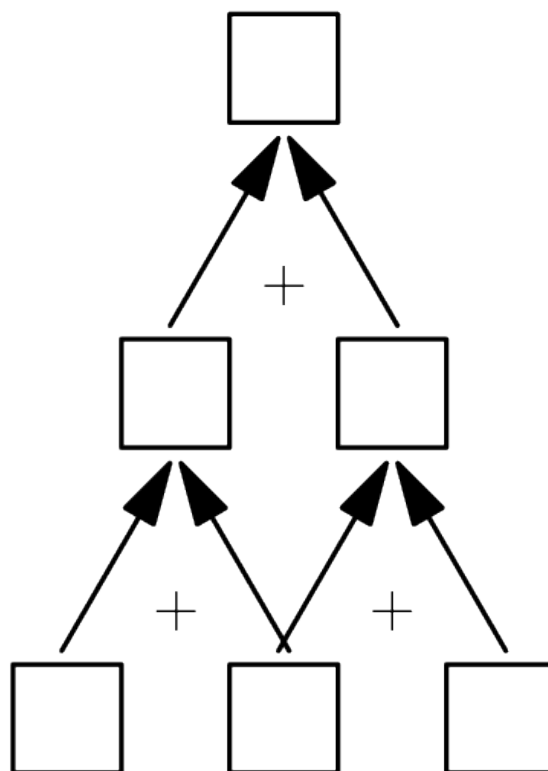
(A) 0.25    (B) 0.5    (C) 1    (D) 1.25    (E) 2.5

**Answer:** A

**Solution**

The water level will rise 1cm for every  $100 \cdot 40 = 4000 \text{ cm}^2$ . Since 1000 is  $\frac{1}{4}$  of 4000, the water will rise  $\frac{1}{4} \cdot 1 = \boxed{\text{(A) } 0.25}$

22. (AMC 8 2006) Three different one-digit positive integers are placed in the bottom row of cells. Numbers in adjacent cells are added and the sum is placed in the cell above them. In the second row, continue the same process to obtain a number in the top cell. What is the difference between the largest and smallest numbers possible in the top cell?



(A) 16    (B) 24    (C) 25    (D) 26    (E) 35

**Answer:** D

**Solution**

If the lower cells contain  $A, B$  and  $C$ , then the second row will contain  $A + B$  and  $B + C$ , and the top cell will contain  $A + 2B + C$ . To obtain the smallest sum, place 1 in the center cell and 2 and 3 in the outer ones. The top number will be 7. For

the largest sum, place 9 in the center cell and 7 and 8 in the outer ones. This top number will be 33. The difference is  $33 - 7 = \boxed{\text{(D)} 26}$ .

23. (AMC 8 2006) A box contains gold coins. If the coins are equally divided among six people, four coins are left over. If the coins are equally divided among five people, three coins are left over. If the box holds the smallest number of coins that meets these two conditions, how many coins are left when equally divided among seven people?

(A) 0      (B) 1      (C) 2      (D) 3      (E) 5

**Answer:** A

**Solution 1(not dumb and bad)**

This is a **modular arithmetic** problem.

Let the number of coins be  $N$ . Then:

$$N \equiv 4 \pmod{6}$$

$$N \equiv 3 \pmod{5}$$

Solving this system, the smallest such  $N$  is:

$$N = 28$$

Now, dividing 28 by 7:

$$28 \div 7 = 4 \text{ with a remainder of } \boxed{0}$$

This is the number of coins left when the box is divided equally among seven people.

**Solution 2**

The counting numbers that leave a remainder of 4 when divided by 6 are 4, 10, 16, 22, 28, 34,  $\dots$

The counting numbers that leave a remainder of 3 when divided by 5 are 3, 8, 13, 18, 23, 28, 33,  $\dots$

So 28 is the smallest possible number of coins that meets both conditions. Because 28 is divisible by 7, there are  $\boxed{\text{(A)} 0}$  coins left when they are divided among seven people.

**Solution 3**

If there were two more coins in the box, the number of coins would be divisible by both 6 and 5. The smallest number that is divisible by 6 and 5 is 30, so the smallest possible number of coins in the box is 28 and the remainder when divided by 7 is  $\boxed{\text{(A)} 0}$ .

24. (AMC 8 2006) In the multiplication problem below  $A$ ,  $B$ ,  $C$ , and  $D$  are different digits. What is  $A + B$ ?

$$\begin{array}{r} A \ B \ A \\ \times \quad C \ D \\ \hline C \ D \ C \ D \end{array}$$

(A) 1      (B) 2      (C) 3      (D) 4      (E) 9

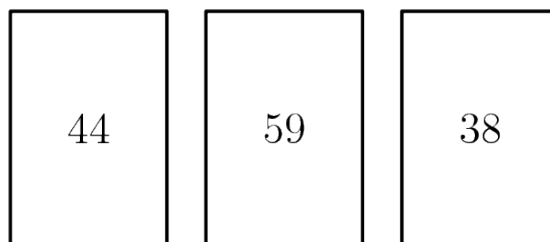


**Answer:** A

**Solution 1**

$CDCD = CD \odot 101$ , so  $ABA = 101$ . Therefore,  $A = 1$  and  $B = 0$ , so  $A + B = 1 + 0 = \boxed{\text{(A)} 1}$ .

25. (AMC 8 2006) Barry wrote 6 different numbers, one on each side of 3 cards, and laid the cards on a table, as shown. The sums of the two numbers on each of the three cards are equal. The three numbers on the hidden sides are prime numbers. What is the average of the hidden prime numbers?



- (A) 13      (B) 14      (C) 15      (D) 16      (E) 17

**Answer:** B

**Solution 1**

Let  $s$  be the sum of the two numbers on any card. Then the hidden prime numbers are  $s - 44$ ,  $s - 59$ , and  $s - 38$ .

If  $s$  were even, then both  $s - 44$  and  $s - 38$  would be even, which is impossible since 2 is the only even prime number.

Therefore,  $s$  is odd. Since  $s - 59$  is even and prime,  $s - 59 = 2$  and thus  $s = 61$ .

Thus the hidden prime numbers are 17, 2, and 23, so their average is  $\frac{17+2+23}{3} =$

$\boxed{\text{(B)} 14}$ .

**Solution 2 (Answer Choices)**

Let  $x$  be the number on the other side of the card showing 38. Then the sum of the two numbers on each card is  $x + 38$ , so the number on the other side of the card showing 59 is  $x - 21$  while the number on the other side of the card showing 44 is  $x - 6$ .

Let  $a$  be the average of the hidden numbers. Then  $a = \frac{1}{3}(x + (x - 21) + (x - 6)) = x - 9$ , so  $x = a + 9$  and thus  $a + 9$  is prime.

Of the answer choices, only  $a = \boxed{\text{(B)} 14}$  yields a prime value of  $a + 9$ .

**Solution 3**

Say that the 3 hidden prime numbers are  $x$ ,  $y$  and  $z$  for cards 1, 2, and 3 respectively. Therefore, since  $x + 44 = z + 38 = y + 59$ ,  $y = x - 15$ ,  $z = x + 6$ . Now we note that if  $y$  is odd, then  $x$  must be even and  $z$  must be even too, which won't work since 2 is the only distinct even prime. Therefore,  $y$  must be even so  $y$  must be 2 so  $x = 17$ ,  $z = 23$  and the average is  $\frac{42}{3} = \boxed{\text{(B)} 14}$ .

**§23 AMC 8 2007**

1. (*AMC 8 2007*) Theresa's parents have agreed to buy her tickets to see her favorite band if she spends an average of 10 hours per week helping around the house for 6 weeks. For the first 5 weeks she helps around the house for 8, 11, 7, 12 and 10 hours. How many hours must she work for the final week to earn the tickets?

(A) 9      (B) 10      (C) 11      (D) 12      (E) 13

**Answer:** D

**Solution 1**

Let  $x$  be the number of hours she must work for the final week. We are looking for the average, so

$$\frac{8 + 11 + 7 + 12 + 10 + x}{6} = 10$$

Solving gives:

$$\frac{48 + x}{6} = 10$$

$$48 + x = 60$$

$$x = 12$$

So, the answer is (D) 12

**Solution 2**

Use average deviation:

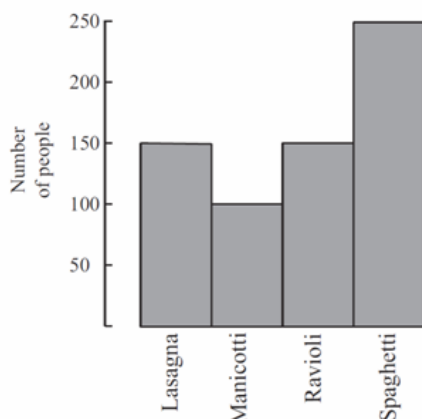
The average is 10 hour per day. If work 8 hours then it is 2 hours short; if work 11 hours then there is 1 hour surplus, the last day need to cancel out the collective deviation from the previous 5 days.

So we got

$$-2 + 1 - 3 + 2 + 0 = -2$$

The last day need to have +2 deviation to cancel out the -2 collective deviation to get 10 as average value, so (D) 12.

2. (*AMC 8 2007*) 650 students were surveyed about their pasta preferences. The choices were lasagna, manicotti, ravioli and spaghetti. The results of the survey are displayed in the bar graph. What is the ratio of the number of students who preferred spaghetti to the number of students who preferred manicotti?



- (A)  $\frac{2}{5}$     (B)  $\frac{1}{2}$     (C)  $\frac{5}{4}$     (D)  $\frac{5}{3}$     (E)  $\frac{5}{2}$

**Answer:** E

**Solution**

The answer is  $\frac{\text{number of students who preferred spaghetti}}{\text{number of students who preferred manicotti}}$

So,

$$\frac{250}{100}$$

Simplify,

$$\frac{5}{2}$$

The answer is (E)  $\frac{5}{2}$

3. (AMC 8 2007) What is the sum of the two smallest prime factors of 250?

- (A) 2    (B) 5    (C) 7    (D) 10    (E) 12

**Answer:** C

**Solution**

The prime factorization of 250 is  $2 \cdot 5^3$ . The smallest two are 2 and 5.  $2 + 5 = \boxed{\text{(C) } 7}$ .

4. (AMC 8 2007) A haunted house has six windows. In how many ways can Georgie the Ghost enter the house by one window and leave by a different window?

- (A) 12    (B) 15    (C) 18    (D) 30    (E) 36

**Answer:** D

**Solution**

Georgie can enter the haunted house through any of the six windows. Then, he can leave through any of the remaining five windows.

So, Georgie has a total of  $6 \cdot 5$  ways he can enter the house by one window and leave by a different window.

Therefore, we have (D) 30 ways.

5. (AMC 8 2007) Chandler wants to buy a \$500 mountain bike. For his birthday, his grandparents send him \$50, his aunt sends him \$35 and his cousin gives him \$15. He earns \$16 per week for his paper route. He will use all of his birthday money

and all of the money he earns from his paper route. In how many weeks will he be able to buy the mountain bike?

- (A) 24      (B) 25      (C) 26      (D) 27      (E) 28

**Answer:** B

**Solution**

Let  $x$  be the number of weeks.

Thus, we have the equation  $50 + 35 + 15 + 16x = 500$ .

Simplify,

$$100 + 16x = 500 \quad 16x = 400 \quad x = 25$$

The answer is (B) 25

6. (*AMC 8 2007*) The average cost of a long-distance call in the USA in 1985 was 41 cents per minute, and the average cost of a long-distance call in the USA in 2005 was 7 cents per minute. Find the approximate percent decrease in the cost per minute of a long-distance call.

- (A) 7      (B) 17      (C) 34      (D) 41      (E) 80

**Answer:** E

**Solution**

The percent decrease is (the amount of decrease)/(original amount)

the amount of decrease is  $41 - 7 = 34$

so the percent decrease is  $\frac{34}{41}$  which is about (E) 80%

7. (*AMC 8 2007*) The average age of 5 people in a room is 30 years. An 18-year-old person leaves the room. What is the average age of the four remaining people?

- (A) 25      (B) 26      (C) 29      (D) 33      (E) 36

**Answer:** D

**Solution 1**

Let  $x$  be the average of the remaining 4 people.

The equation we get is  $\frac{4x+18}{5} = 30$

Simplify,

$$4x + 18 = 150 \quad 4x = 132 \quad x = 33$$

Therefore, the answer is (D) 33

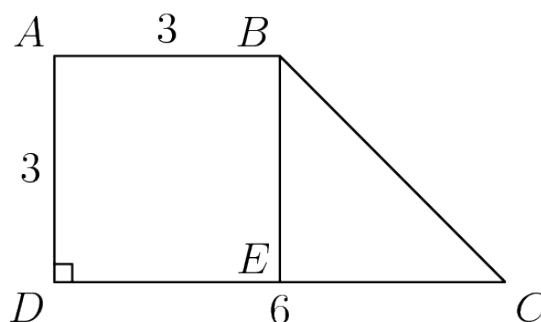
**Solution 2**

Since an 18 year old left from a group of people averaging 30, The remaining people must total  $30 - 18 = 12$  years older than 30. Therefore, the average is  $\frac{12}{4} = 3$  years over 30. Giving us (D) 33

**Solution 3**

The total ages would be  $30 * 5 = 150$ . Then, if one 18 year old leaves, we subtract 18 from 150 and get 132. Then, we divide 132 by 4 to get the new average, (D) 33

8. (AMC 8 2007) In trapezoid  $ABCD$ ,  $\overline{AD}$  is perpendicular to  $\overline{DC}$ ,  $AD = AB = 3$ , and  $DC = 6$ . In addition,  $E$  is on  $\overline{DC}$ , and  $\overline{BE}$  is parallel to  $\overline{AD}$ . Find the area of  $\triangle BEC$ .



- (A) 3    (B) 4.5    (C) 6    (D) 9    (E) 18

**Answer:** B

**Solution 1 (Area Formula for Triangles)**

Clearly,  $ABED$  is a square with side-length 3. By segment subtraction, we have  $EC = DC - DE = 6 - 3 = 3$ .

The area of  $\triangle BEC$  is

$$\frac{1}{2} \cdot EC \cdot BE = \frac{1}{2} \cdot 3 \cdot 3 = \boxed{\text{(B) } 4.5}.$$

Aplus95 (Solution)

**Solution 2 (Area Subtraction)**

Clearly,  $ABED$  is a square with side-length 3.

Let the brackets denote areas. We apply area subtraction to find the area of  $\triangle BEC$ :

$$\begin{aligned} [BEC] &= [ABCD] - [ABED] \\ &= \frac{AB + CD}{2} \cdot AD - AB^2 \\ &= \frac{3 + 6}{2} \cdot 3 - 3^2 \\ &= \boxed{\text{(B) } 4.5}. \end{aligned}$$

MRENTHUSIASM

**Solution 3 (Cheese, Don't use in competition unless stuck)**

4.5 is the only one that isn't an integer, and is the odd one out.

$$\boxed{\text{(B) } 4.5}.$$

9. (AMC 8 2007) To complete the grid below, each of the digits 1 through 4 must occur once in each row and once in each column. What number will occupy the lower right-hand square?

|   |   |   |   |
|---|---|---|---|
| 1 |   | 2 |   |
| 2 | 3 |   |   |
|   |   |   | 4 |
|   |   |   |   |

- (A) 1    (B) 2    (C) 3    (D) 4    (E) cannot be determined

**Answer:** B

**Solution**

The number in the first row, last column must be a 3 due to the fact if a 3 was in the first row, second column, there would be two threes in that column. By the same reasoning, the number in the second row, last column has to be a 1. Therefore the number in the lower right-hand square is **(B) 2**.

**Solution 2**

Note how the first and second row already contain a 2. Since the third row, last column already has a 4, the only possible place a 2 could be in is the bottom right square. Thus our answer is **(B) 2**.

10. (AMC 8 2007) For any positive integer  $n$ , define  $\boxed{n}$  to be the sum of the positive factors of  $n$ . For example,  $\boxed{6} = 1 + 2 + 3 + 6 = 12$ . Find  $\boxed{\boxed{11}}$ .
- (A) 13    (B) 20    (C) 24    (D) 28    (E) 30

**Answer:** D

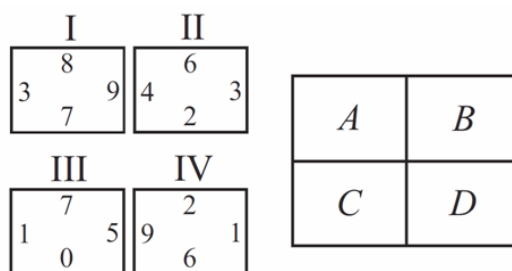
**Solution**

We have

$$\begin{aligned}
 \boxed{\boxed{11}} &= \boxed{1 + 11} \\
 &= \boxed{12} \\
 &= 1 + 2 + 3 + 4 + 6 + 12 \\
 &= 28,
 \end{aligned}$$

from which the answer is **(D) 28**.

11. (AMC 8 2007) Tiles *I*, *II*, *III* and *IV* are translated so one tile coincides with each of the rectangles *A*, *B*, *C* and *D*. In the final arrangement, the two numbers on any side common to two adjacent tiles must be the same. Which of the tiles is translated to Rectangle *C*?



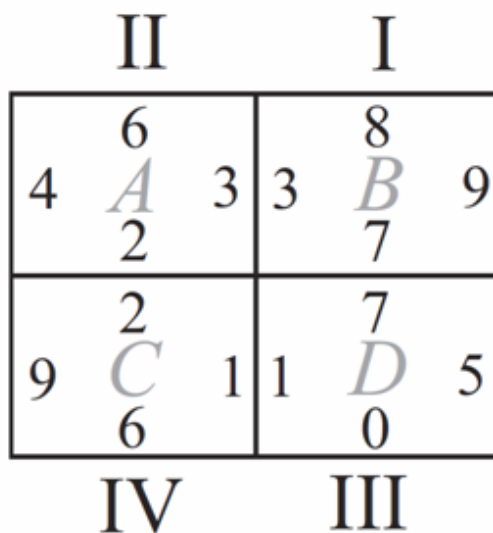
- (A) *I*    (B) *II*    (C) *III*    (D) *IV*    (E) cannot be determined

**Answer:** D

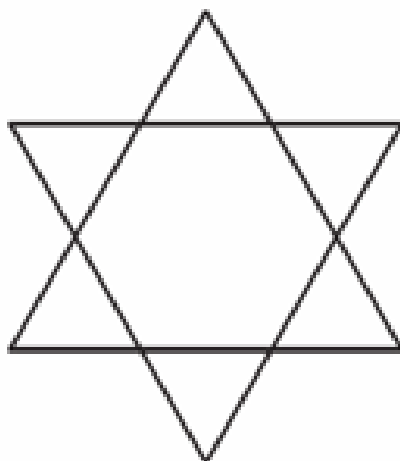
**Solution**

We first notice that tile III has a 0 on the bottom and a 5 on the right side. Since no other tile has a 0 or a 5, Tile III must be in rectangle *D*. Tile III also has a 1 on the left, so Tile IV must be in Rectangle *C*.

The answer is (D)



12. (*AMC 8 2007*) A unit hexagram is composed of a regular hexagon of side length 1 and its 6 equilateral triangular extensions, as shown in the diagram. What is the ratio of the area of the extensions to the area of the original hexagon?



- (A) 1 : 1    (B) 6 : 5    (C) 3 : 2    (D) 2 : 1    (E) 3 : 1

**Answer:** A

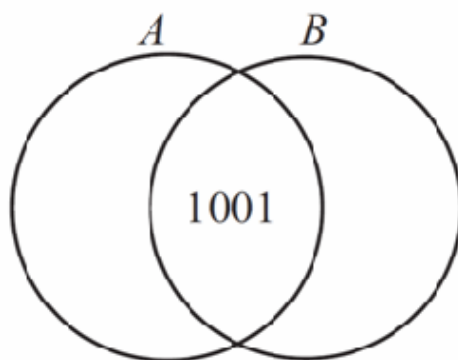
**Solution**

The six equilateral triangular extensions fit perfectly into the hexagon meaning the answer is **(A) 1 : 1**.

**Solution 2**

Split the hexagon into six small equilateral triangles. You will see that the six outer triangles can be folded to the hexagon, so the answer is **(A) 1 : 1**.

13. (AMC 8 2007) Sets  $A$  and  $B$ , shown in the Venn diagram, have the same number of elements. Their union has 2007 elements and their intersection has 1001 elements. Find the number of elements in  $A$ .



- (A) 503    (B) 1006    (C) 1504    (D) 1507    (E) 1510

**Answer:** C

**Solution**



Let  $x$  be the number of elements in  $A$  and  $B$  which is equal.

Then we could form equation  $2x - 1001 = 2007$   $2x = 3008$   $x = 1504$ .

The answer is (C) 1504

### Solution 2

Let  $x$  be the number of elements in  $A$  not including the intersection.  $2007 - 1001 = 1006$  total elements excluding the intersection. Since we know that  $A = B$ , we can find that  $x = \frac{1006}{2} = 503$ . Now we need to add the intersection.  $503 + 1001 =$  (C)1504.

14. (AMC 8 2007) The base of isosceles  $\triangle ABC$  is 24 and its area is 60. What is the length of one of the congruent sides?

(A) 5      (B) 8      (C) 13      (D) 14      (E) 18

Answer: C

### Solution 1

The area of a triangle is shown by  $\frac{1}{2}bh$ . We set the base equal to 24, and the area equal to 60, and we get the triangle's height, or altitude, to be 5. In this isosceles triangle, the height bisects the base, so by using the Pythagorean Theorem,  $a^2 + b^2 = c^2$ , we can solve for one of the legs of the triangle (it will be the hypotenuse,  $c$ ).  $a = 12$ ,  $b = 5$ ,  $c = 13$ . The answer is (C) 13.

### Solution 2 (Heron's Formula)

According to **Heron's Formula**, setting side  $a$  as 24, we have

$$\sqrt{s(s-24)(s-b)(s-c)} = 60$$

where  $s$  is the triangle's semiperimeter (i.e.  $\frac{a+b+c}{2}$ ). Since the triangle is isosceles,  $b = c$ , so we can rewrite  $s$  as  $\frac{24+2b}{2} = 12 + b$ . Substituting and solving the equation and taking the positive solution for  $b$ ,

$$\sqrt{(12+b)(-12+b)(12)(12)} = 60$$

$$\sqrt{144(144-b^2)} = 60$$

$$144(144-b^2) = 3600$$

$$-b^2 = -169$$

$$b = \span style="border: 1px solid black; padding: 2px;">(C) 13$$

15. (AMC 8 2007) Let  $a, b$  and  $c$  be numbers with  $0 < a < b < c$ . Which of the following is impossible?

(A)  $a + c < b$       (B)  $a \cdot b < c$       (C)  $a + b < c$       (D)  $a \cdot c < b$       (E)  $\frac{b}{c} = a$

Answer: A

### Solution

According to the given rules, every number needs to be positive. Since  $c$  is always greater than  $b$ , adding a positive number ( $a$ ) to  $c$  will always make it greater than  $b$ .

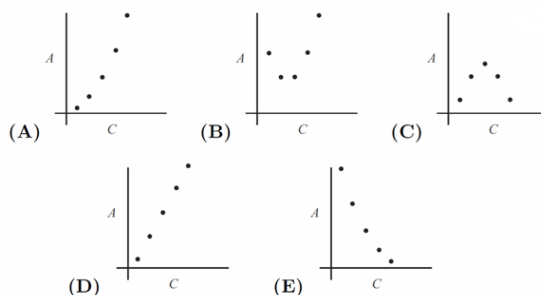
Therefore, the answer is (A)  $a + c < b$

### Solution 2

We can test numbers into the inequality we're given. The simplest is  $0 < 1 < 2 < 3$ . We can see that  $3 + 1 > 2$ , so (A)  $a + c < b$  is correct.

—jason.ca

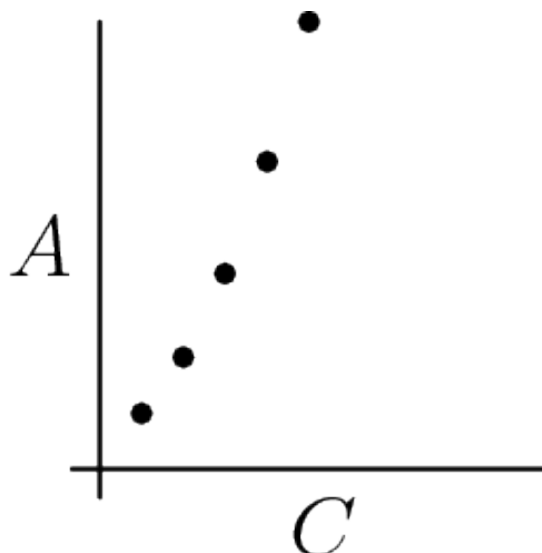
16. (AMC 8 2007) Amanda draws five circles with radii 1, 2, 3, 4 and 5. Then for each circle she plots the point  $(C, A)$ , where  $C$  is its circumference and  $A$  is its area. Which of the following could be her graph?



**Answer:** A

### Solution 1

The circumference of a circle is obtained by simply multiplying the radius by  $2\pi$ . So, the  $C$ -coordinate (in this case, it is the  $x$ -coordinate) will increase at a steady rate. The area, however, is obtained by squaring the radius and multiplying it by  $\pi$ . Since squares do not increase in an evenly spaced arithmetic sequence, the increase in the  $A$ -coordinates (aka the  $y$ -coordinates) will be much more significant. The answer is (A),



.-RBANDA

**Solution 2 (parametric plot)**

Since we know that the formula for circumference of a circle is  $2\pi r$ , and for the area of a circle it is  $\pi r^2$ , any coordinate is simply  $(2\pi r, \pi r^2)$ , notice that, in more similar terms, it is  $(2\pi t, \pi t^2)$ , which is a parametric plot. So let:

$$X(t) = 2\pi t$$

$$Y(t) = \pi t^2$$

Solving for  $t$  gives  $\frac{X(t)}{2\pi} = t$ , thus letting  $Y(t) = y$  and  $X(t) = x$ ,

$$y = \pi \left( \frac{x^2}{4\pi^2} \right)$$

$$y = \frac{x^2}{4\pi}$$

Since the domain of the function is  $x \geq 0$ , note that the function is always increasing quadratically. The only answer choice that fits this function is (A)

17. (AMC 8 2007) A mixture of 30 liters of paint is 25% red tint, 30% yellow tint and 45% water. Five liters of yellow tint are added to the original mixture. What is the percent of yellow tint in the new mixture?

(A) 25      (B) 35      (C) 40      (D) 45      (E) 50

**Answer:** C

**Solution**

Since 30% of the original 30 liters of paint was yellow, and 5 liters of yellow paint were added to make the new mixture, there are  $9 + 5 = 14$  liters of yellow tint in the new mixture. Since only 5 liters of paint were added to the original 30, there are a total of 35 liters of paint in the new mixture. This gives 40% of yellow tint in the new mixture, which is (C) 40.

18. (AMC 8 2007) The product of the two 99-digit numbers

303,030,303,...,030,303 and 505,050,505,...,050,505

has thousands digit  $A$  and units digit  $B$ . What is the sum of  $A$  and  $B$ ?

(A) 3      (B) 5      (C) 6      (D) 8      (E) 10

**Answer:** D

**Solution**

We can first make a small example to find out  $A$  and  $B$ . So,

$$303 \times 505 = 153015$$

The ones digit plus thousands digit is  $5 + 3 = 8$ .

Note that the ones and thousands digits are, added together, 8. (and so on...) So the answer is (D) 8 This is a direct multiplication way.

**Solution 2 (FASTER!)**

Just solve it by hand to get 153321489657825994162330498666835003171339507675844012180348516684

Then, the ones digit plus thousands digit is  $5 + 3 = 8$ . So the answer is (D) 8

frfr best way this is what i did during the amc 8im that old call me unccccc67

19. (AMC 8 2007) Pick two consecutive positive integers whose sum is less than 100. Square both of those integers and then find the difference of the squares. Which of the following could be the difference?

(A) 2    (B) 64    (C) 79    (D) 96    (E) 131

**Answer:** C

**Solution 1**

Let the smaller of the two numbers be  $x$ . Then, the problem states that  $(x+1)+x < 100$ .  $(x+1)^2 - x^2 = x^2 + 2x + 1 - x^2 = 2x + 1$ .  $2x + 1$  is obviously odd, so only answer choices C and E need to be considered.

$2x + 1 = 131$  contradicts the fact that  $2x + 1 < 100$ , so the answer is (C)79.

**Solution 2**

Since for two consecutive numbers  $a$  and  $b$ , the difference between their squares is  $a^2 - b^2 = (a+b)(a-b)$ , which equals to  $a+b$ , because  $a$  and  $b$  are consecutive. Because they are consecutive, one number must be even, and the other odd. Since the sum of an even and an odd number is always odd, and the sum of  $a$  and  $b$  is less than 100, you can eliminate all answers except for (C)79.

**Solution 3**

From the question we can make  $a^2 - b^2$  and we can let  $a$  be  $b-1$  where  $a+b < 100$ . First we factor  $a^2 - b^2$  to  $(a+b)(a-b)$  and then plug in  $a = b-1$  to have:  $(a+a+1)(a-a+1) = (2a+1)(1) = 2a+1 = a+b$ . since  $a+b$  will always be an odd number, we can narrow the answers down to (C) and (E). Since 131 would be too big for our range of  $a+b < 100$ , we would choose (C)79. note that this solution would be better if the question was asking for the values of  $a$  and  $b$

20. (AMC 8 2007) Before district play, the Unicorns had won 45% of their basketball games. During district play, they won six more games and lost two, to finish the season having won half their games. How many games did the Unicorns play in all?
- (A) 48    (B) 50    (C) 52    (D) 54    (E) 60

**Answer:** A

**Solution 1**

At the beginning of the problem, the Unicorns had played  $y$  games and they had won  $x$  of these games. From the information given in the problem, we can say that  $\frac{x}{y} = 0.45$ . Next, the Unicorns win 6 more games and lose 2 more, for a total of  $6+2=8$  games played during district play. We are told that they end the season having won half of their games, or 0.5 of their games. We can write another equation:  $\frac{x+6}{y+8} = 0.5$ . This gives us a system of equations:  $\frac{x}{y} = 0.45$  and  $\frac{x+6}{y+8} = 0.5$ . We first multiply both sides of the first equation by  $y$  to get  $x = 0.45y$ . Then, we multiply both sides of the second equation by  $(y+8)$  to get  $x+6 = 0.5(y+8)$ . Applying the Distributive Property gives yields  $x+6 = 0.5y+4$ . Now we substitute  $0.45y$  for  $x$  to get  $0.45y+6 = 0.5y+4$ . Solving gives us  $y = 40$ . Since the problem asks for the total number of games, we add on the last 8 games to get the solution (A) 48.

**Solution 2 (Answer Choices)**

We can check each answer choice from left to right to see which one is correct. Suppose the Unicorns played 48 games in total. Then, after district play, they would have won 24 games. Now, consider the situation before district play. The Unicorns

would have won 18 games out of 40. Converting to a percentage,  $18/40 = 45\%$ . Thus, the answer is **(A)48**.

Note: If A didn't work, we would have similarly tested the other choices until we found one that did.

### Solution 3 (Using Logic)

We know that  $45\% = \frac{9}{20}$ . Therefore, the number of games before district play must be a multiple of 20 in order for the number of games won to be an integer. The Unicorns played  $6 + 2 = 8$  more games during district play. The only answer choice that is 8 more than a multiple of 20 is **(A)48**.

### Solution 4

At the start of the problem we figure out the Unicorns have won 45% of their games, which can be represented by  $\frac{9}{20}$ . Since this is a simplified fraction and not necessarily the amount of games they have played, we can add a variable (in this case I am using  $x$ ) to the numerator and denominator to get  $\frac{9x}{20x}$ . We then know that during the district play they have won 6 more games and lost 2 of them for a total of  $6+2=8$  games played. We then can add it to our fraction we have built to make  $\frac{9x+6}{20x+8}$ . We also know that the fraction is equal to  $\frac{1}{2}$ . Putting these equations together we get  $\frac{9x+6}{20x+8} = \frac{1}{2}$ . Now we can cross multiply to come out with  $18x + 12 = 20x + 8$ . We now solve for  $x$ :  $18x + 12 = 20x + 8 \rightarrow 4 = 2x \rightarrow 2 = x$ . After we get  $x$  which comes out to be 2 we then plug it back into the denominator to get  $20(2) + 8$  which is equal to 48, so the answer must be **(A)48**.

21. (AMC 8 2007) Two cards are dealt from a deck of four red cards labeled  $A, B, C, D$  and four green cards labeled  $A, B, C, D$ . A winning pair is two of the same color or two of the same letter. What is the probability of drawing a winning pair?
- (A)  $\frac{2}{7}$       (B)  $\frac{3}{8}$       (C)  $\frac{1}{2}$       (D)  $\frac{4}{7}$       (E)  $\frac{5}{8}$

**Answer:** D

### Solution 1

Notice that, no matter which first card you choose, there are exactly 4 cards that either have the same color or letter as it. Since there are 7 cards left to choose from, the probability is **(D)** $\frac{4}{7}$ . -theepiccarrot7

### Solution 2

We can use casework to solve this.

Case 1: Same letter

After choosing any letter, there are seven cards left, and only one of them will produce a winning pair. Therefore, the probability is  $\frac{1}{7}$ .

Case 2: Same color

After choosing any letter, there are seven cards left. Three of them will make a winning pair, so the probability is  $\frac{3}{7}$ .

Now that we have the probability for both cases, we can add them:  $\frac{1}{7} + \frac{3}{7} = \textbf{(D)}\frac{4}{7}$ .

22. (AMC 8 2007) A lemming sits at a corner of a square with side length 10 meters. The lemming runs 6.2 meters along a diagonal toward the opposite corner. It stops,

makes a 90 degree right turn and runs 2 more meters. A scientist measures the shortest distance between the lemming and each side of the square. What is the average of these four distances in meters?

- (A) 2    (B) 4.5    (C) 5    (D) 6.2    (E) 7

**Answer:** C

**Solution 1**

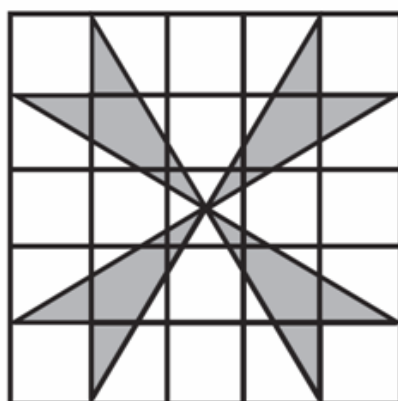
The shortest segments would be perpendicular to the square. The lemming went  $x$  meters horizontally and  $y$  meters vertically. No matter how much it went, the lemming would have been  $x$  and  $y$  meters from the sides and  $10 - x$  and  $10 - y$  meters from the remaining two. To find the average, add the lengths of the four segments and divide by four:  $\frac{x+10-x+y+10-y}{4} = \boxed{\text{(C) } 5}$ .

- Note that from any point in the square, the average distance from one side to the other is half of the square's side length.

**Solution 2**

For any point in the square, the sum of its distance to the left edge and right edge is equal to 10, and the sum of its distance to the up edge and down edge is also equal to 10. Thus, the answer is  $\boxed{\text{(C) } 5}$ , and the moving progress is misguide at all.

23. (AMC 8 2007) What is the area of the shaded part shown in the 5 x 5 grid?



- (A) 4    (B) 6    (C) 8    (D) 10    (E) 12

**Answer:** B

**Solution 1**

The area of the square around the pinwheel is 25. The area of the pinwheel is equal to the square — the white space. Each of the four triangles have a base of 3 units and a height of 2.5 units, and so their combined area is 15 units squared. Then the unshaded space consists of the four triangles with total area of 15, and there are four white corner squares. Therefore the area of the pinwheel is  $25 - (15 + 4)$  which is  $\boxed{\text{(B) } 6}$

**Solution 2**

The pinwheel is composed of 8 congruent obtuse triangles whose base measures length 1 and height measures length 1.5. Using the area formula for triangles, the pinwheel has an area of

$$8\left(\frac{1}{2} \cdot 1 \cdot 1.5\right) = 8(0.75) = \boxed{\text{(B) } 6}.$$

24. (AMC 8 2007) A bag contains four pieces of paper, each labeled with one of the digits "1, 2, 3" or "4", with no repeats. Three of these pieces are drawn, one at a time without replacement, to construct a three-digit number. What is the probability that the three-digit number is a multiple of 3?

(A)  $\frac{1}{4}$     (B)  $\frac{1}{3}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{3}$     (E)  $\frac{3}{4}$

**Answer:** C

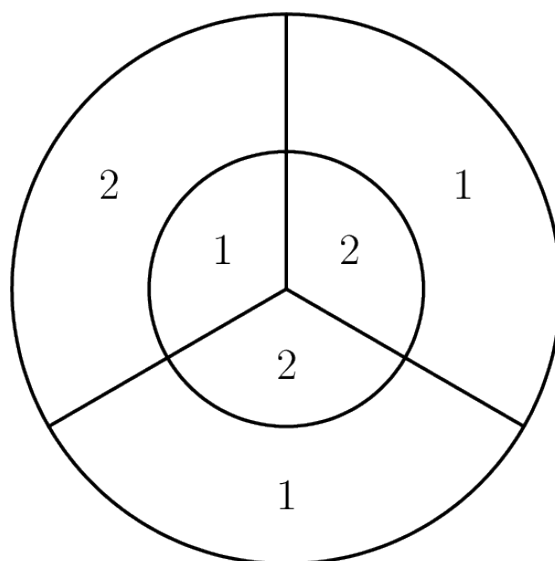
**Solution 1**

The number of ways to form a 3-digit number is  $4 \cdot 3 \cdot 2 = 24$ . The combination of digits that give us multiples of 3 are (1,2,3) and (2,3,4), as the integers in the subsets have a sum which is divisible by 3. The number of 3-digit numbers that contain these numbers is  $3! + 3! = 12$ . Therefore, the probability is  $\frac{12}{24} = \boxed{\frac{1}{2}}$ .

**Solution 2**

Only (1,2,3) and (2,3,4) are possible combinations that sum to 3. The total ways to choose three numbers from the four numbers is 4 choose 3 which equals 4. The answer is  $\frac{2}{4} = \boxed{\frac{1}{2}}$ .

25. (AMC 8 2007) On the dart board shown in the figure, the outer circle has radius 6 and the inner circle has a radius of 3. Three radii divide each circle into three congruent regions, with point values shown. The probability that a dart will hit a given region is proportional to the area of the region. When two darts hit this board, the score is the sum of the point values in the regions. What is the probability that the score is odd?



(A)  $\frac{17}{36}$     (B)  $\frac{35}{72}$     (C)  $\frac{1}{2}$     (D)  $\frac{37}{72}$     (E)  $\frac{19}{36}$

**Answer:** B

**Solution**

To get an odd sum, we must add an even number and an odd number (a 1 and a 2). So we have a little casework to do. Before we do that, we also have to figure out some relative areas. You could either find the absolute areas (as is done below for completeness), or apply some proportional reasoning and symmetry to realize the relative areas (which is really all that matters).

The area of this entire smaller circle is  $9\pi$ . The area of each smaller section then must be  $\frac{9\pi}{3}$  or  $3\pi$ . The larger sections trisect a "ring" which is the difference of two circles, one with radius 3, the other radius 6. So, the area of the ring (*annulus*) is  $36\pi - 9\pi$  or  $27\pi$ . The area of each larger section must be  $\frac{27\pi}{3}$  or  $9\pi$ . Note that the area of the whole circle is  $36\pi$ .

One smaller section and two larger sections contain an odd number (that is, 1). So the probability of throwing an odd number is  $3\pi + (2 \cdot 9\pi) = 21\pi$ . Since the area of the whole circle is  $36\pi$ , the probability of getting an odd is  $\frac{21\pi}{36\pi} = \frac{21}{36} = \frac{7}{12}$ .

Since the remaining sections contain even numbers (that is, 2), the probability of throwing an even is the complement, or  $1 - \frac{7}{12} = \frac{5}{12}$ .

Now, the two cases: You could either get an odd then an even, or an even then an odd.

Case 1: Odd, then even. Multiply the probabilities to get  $\frac{7}{12} \cdot \frac{5}{12} = \frac{35}{144}$ .

Case 2: Even, then odd. Multiply the probabilities to get  $\frac{5}{12} \cdot \frac{7}{12} = \frac{35}{144}$ . Notice that this is the same.

Thus, the total probability of an odd sum is  $\frac{35}{144} \cdot \frac{2}{1} = \boxed{\text{(B)} = \frac{35}{72}}$ .

## §24 AMC 8 2008

1. (*AMC 8 2008*) Susan had 50 dollars to spend at the carnival. She spent 12 dollars on food and twice as much on rides. How many dollars did she have left to spend?  
(A) 12      (B) 14      (C) 26      (D) 38      (E) 50

**Answer:** B

**Solution**

If Susan spent 12 dollars, then twice that much on rides, then she spent  $12 + 12 \times 2 = 36$  dollars in total. We subtract 36 from 50 to get  $\boxed{\text{(B)} 14}$

2. (*AMC 8 2008*) The ten-letter code BEST OF LUCK represents the ten digits 0 – 9, in order. What 4-digit number is represented by the code word CLUE?  
(A) 8671      (B) 8672      (C) 9781      (D) 9782      (E) 9872

**Answer:** A

**Solution**

We can derive that  $C = 8$ ,  $L = 6$ ,  $U = 7$ , and  $E = 1$ . Therefore, the answer is  $\boxed{\text{(A)} 8671}$  edited by OwenTang

**Solution 2**



We can easily see that  $C = 8$ , which makes only (A) 8761 and (B) 8762 possible. Since the only difference between answer choices A and B is the last digit and that the last digit in the code word "CLUE" is E, we can just find that  $E = 1$  and that the answer is **(A) 8671**. solution by SuperVince1

3. (AMC 8 2008) If February is a month that contains Friday the 13<sup>th</sup>, what day of the week is February 1?

(A) Sunday (B) Monday (C) Wednesday (D) Thursday (E) Saturday

**Answer:** A

**Solution**

We can go backwards by days, but we can also backwards by weeks. If we go backwards by weeks, we see that February 6 is a Friday. If we now go backwards by days, February 1 is a **(A) Sunday**.

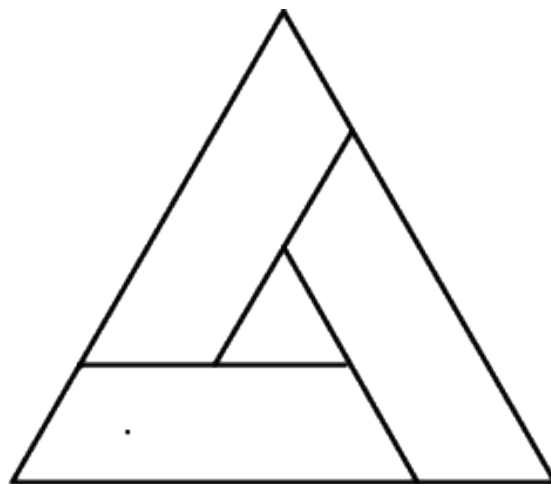
**Solution 2**

Since the days of the week repeat every 7 days and  $1 - 13 \equiv 2 \pmod{7}$ , the day of the week of the day 2 days after February 13 is the same as the day of the week of February 1. Since we know that February 13 is a Friday, the day of the week of the day 2 days after February 13 is a Sunday so we get that February 1 is a **(A) Sunday**.

**Solution 3**

We can go back 2 weeks and say that Friday is the -1st of February. Going forward 2 days, we get that February 1st is Sunday. So our answer is **(A) Sunday**.

4. (AMC 8 2008) In the figure, the outer equilateral triangle has area 16, the inner equilateral triangle has area 1, and the three trapezoids are congruent. What is the area of one of the trapezoids?



(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

**Answer:** C

**Solution**

The area outside the small triangle but inside the large triangle is  $16 - 1 = 15$ . This is equally distributed between the three trapezoids. Each trapezoid has an area of  $15/3 = \mathbf{(C) 5}$ .

**Solution 2**

You can roughly see that 1 trapezoid = 5 triangles = 5 Area since 1 triangle is 1 area so **(C) 5**

5. (AMC 8 2008) Barney Schwinn notices that the odometer on his bicycle reads 1441, a palindrome, because it reads the same forward and backward. After riding 4 more hours that day and 6 the next, he notices that the odometer shows another palindrome, 1661. What was his average speed in miles per hour?

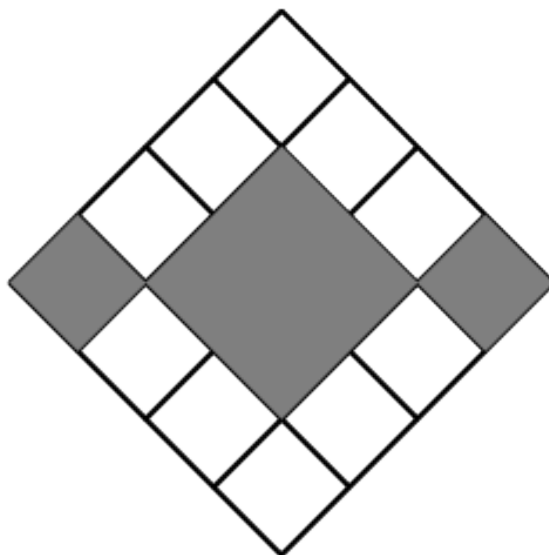
(A) 15      (B) 16      (C) 18      (D) 20      (E) 22

**Answer:** E

**Solution**

Barney travels  $1661 - 1441 = 220$  miles in  $4 + 6 = 10$  hours for an average of  $220/10 = \mathbf{(E) 22}$  miles per hour.

6. (AMC 8 2008) In the figure, what is the ratio of the area of the gray squares to the area of the white squares?



(A) 3 : 10      (B) 3 : 8      (C) 3 : 7      (D) 3 : 5      (E) 1 : 1

**Answer:** D

**Solution**

Dividing the gray square into four smaller squares, there are 6 gray tiles and 10 white tiles, giving a ratio of **(D) 3 : 5**.

7. (AMC 8 2008) If  $\frac{3}{5} = \frac{M}{45} = \frac{60}{N}$ , what is  $M + N$ ?

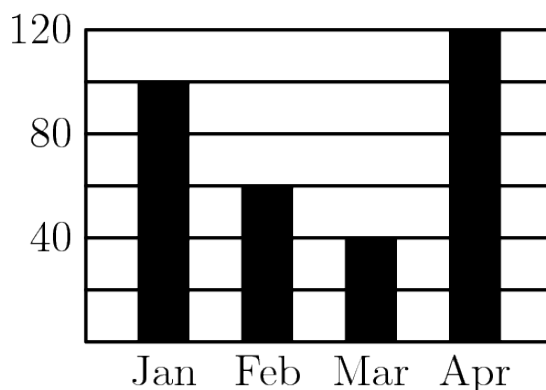
(A) 27      (B) 29      (C) 45      (D) 105      (E) 127

**Answer:** E

**Solution**

Separate into two equations  $\frac{3}{5} = \frac{M}{45}$  and  $\frac{3}{5} = \frac{60}{N}$  and solve for the unknowns.  $M = 27$  and  $N = 100$ , therefore  $M + N = \mathbf{(E) 127}$ .

8. (AMC 8 2008) Candy sales from the Boosters Club from January through April are shown. What were the average sales per month in dollars?



- (A) 60      (B) 70      (C) 75      (D) 80      (E) 85

**Answer:** D

**Solution**

There are a total of  $100 + 60 + 40 + 120 = 320$  dollars of sales spread through 4 months, for an average of  $320/4 = \boxed{\text{(D) } 80}$ .

9. (AMC 8 2008) In 2005 Tycoon Tammy invested 100 dollars for two years. During the first year her investment suffered a 15% loss, but during the second year the remaining investment showed a 20% gain. Over the two-year period, what was the change in Tammy's investment?

- (A) 5% loss      (B) 2% loss      (C) 1% gain      (D) 2% gain      (E) 5% gain

**Answer:** D

**Solution**

After the 15% loss, Tammy has  $100 \cdot 0.85 = 85$  dollars. After the 20% gain, she has  $85 \cdot 1.2 = 102$  dollars. This is an increase in 2 dollars from her original 100 dollars, a  $\boxed{\text{(D) } 2\% \text{ gain}}$ .

10. (AMC 8 2008) The average age of the 6 people in Room A is 40. The average age of the 4 people in Room B is 25. If the two groups are combined, what is the average age of all the people?

- (A) 32.5      (B) 33      (C) 33.5      (D) 34      (E) 35

**Answer:** D

**Solutions**

The total of all their ages over the number of people is

$$\frac{6 \cdot 40 + 4 \cdot 25}{6 + 4} = \frac{340}{10} = \boxed{\text{(D) } 34}.$$

11. (AMC 8 2008) Each of the 39 students in the eighth grade at Lincoln Middle School has one dog or one cat or both a dog and a cat. Twenty students have a dog and 26 students have a cat. How many students have both a dog and a cat?

- (A) 7      (B) 13      (C) 19      (D) 39      (E) 46

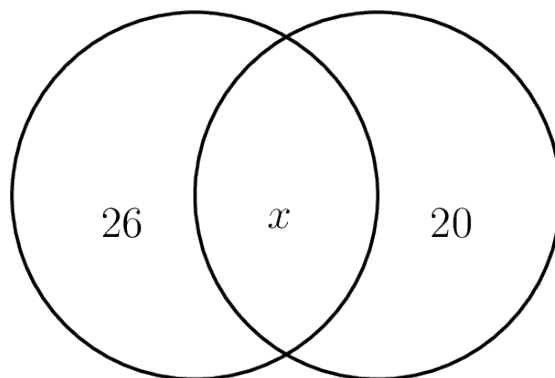
**Answer:** A

**Solution 1**

The union of two sets is equal to the sum of each set minus their intersection. The number of students that have both a dog and a cat is  $20 + 26 - 39 = \boxed{\text{(A)} 7}$ .

**Solution 2 (Venn Diagram)**

We create a diagram:



Let  $x$  be the number of students with both a dog and a cat.

Therefore, we have

$$26 + 20 - x = 39$$

$$46 - x = 39$$

$$x = \boxed{\text{(A)} 7}$$

12. (AMC 8 2008) A ball is dropped from a height of 3 meters. On its first bounce it rises to a height of 2 meters. It keeps falling and bouncing to  $\frac{2}{3}$  of the height it reached in the previous bounce. On which bounce will it rise to a height less than 0.5 meters?

(A) 3    (B) 4    (C) 5    (D) 6    (E) 7

**Answer:** C

**Solution**

Each bounce is  $\frac{2}{3}$  times the height of the previous bounce. The first bounce reaches 2 meters, the second  $\frac{4}{3}$ , the third  $\frac{8}{9}$ , the fourth  $\frac{16}{27}$ , and the fifth  $\frac{32}{81}$ . Half of 81 is 40.5, so the ball does not reach the required height on bounce  $\boxed{\text{(C)} 5}$ .

13. (AMC 8 2008) Mr. Harman needs to know the combined weight in pounds of three boxes he wants to mail. However, the only available scale is not accurate for weights less than 100 pounds or more than 150 pounds. So the boxes are weighed in pairs in every possible way. The results are 122, 125 and 127 pounds. What is the combined weight in pounds of the three boxes?

(A) 160    (B) 170    (C) 187    (D) 195    (E) 354

**Answer:** C

**Solution 1**

Each box is weighed twice during this, so the combined weight of the three boxes is half the weight of these separate measures:

$$\frac{122 + 125 + 127}{2} = \frac{374}{2} = \boxed{\text{(C) } 187}.$$

**Solution 2**

Using variables  $a$ ,  $b$ , and  $c$  to denote the boxes, with  $a \leq b \leq c$ . It is obvious that  $a + b = 122$  and  $a + c = 125$ , since these are the two smallest pairs. Subtracting the former equation from the latter results in  $c - b = 3$ . Additionally,  $c + b = 127$ . Solving for  $c$  and  $b$  gives  $c = 65$  and  $b = 62$ , so we can thus find  $a = 60$ . Solving  $60 + 62 + 65 = \boxed{\text{(C) } 187}$

**Solution 3**

We use variables  $x$ ,  $y$ , and  $z$  to denote the three boxes. We know that  $x + y = 122$ ,  $x + z = 125$ , and  $y + z = 127$ . Note that the order of the variables or equations doesn't matter as long as they are consistent. By adding all three equations, we obtain  $2x + 2y + 2z = 122 + 125 + 127 = 374$ . Dividing by 2, we obtain the sum  $x + y + z$  is  $\boxed{\text{(C) } 187}$ .

14. (AMC 8 2008) Three A's, three B's, and three C's are placed in the nine spaces so that each row and column contain one of each letter. If A is placed in the upper left corner, how many arrangements are possible?

|   |  |  |
|---|--|--|
| A |  |  |
|   |  |  |
|   |  |  |

- (A) 2    (B) 3    (C) 4    (D) 5    (E) 6

**Answer:** C

**Solution**

There are 2 ways to place the remaining As, 2 ways to place the remaining Bs, and 1 way to place the remaining Cs for a total of  $(2)(2)(1) = \boxed{\text{(C) } 4}$ .

15. (AMC 8 2008) In Theresa's first 8 basketball games, she scored 7, 4, 3, 6, 8, 3, 1 and 5 points. In her ninth game, she scored fewer than 10 points and her points-per-game

average for the nine games was an integer. Similarly in her tenth game, she scored fewer than 10 points and her points-per-game average for the 10 games was also an integer. What is the product of the number of points she scored in the ninth and tenth games?

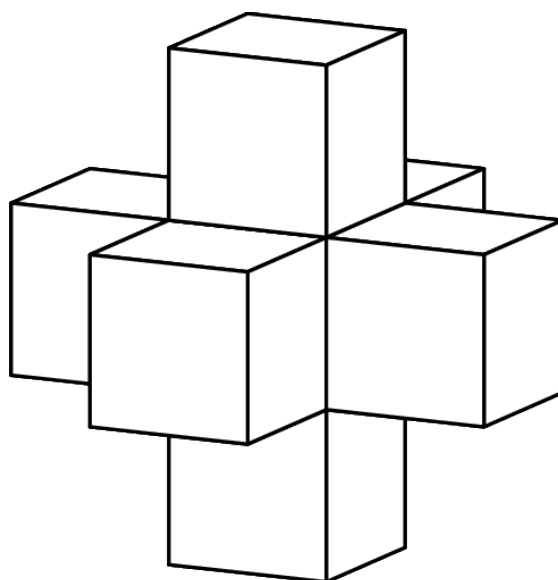
- (A) 35    (B) 40    (C) 48    (D) 56    (E) 72

**Answer:** B

**Solution**

The total number of points from the first 8 games is  $7 + 4 + 3 + 6 + 8 + 3 + 1 + 5 = 37$ . We have to make this a multiple of 9 by scoring less than 10 points. The smallest multiple of 9 that is greater than 37 is 45.  $45 - 37 = 8$ . Now we have to add a number to get a multiple of 10. The next multiple of 10 is 50, which means that we add 5. Multiplying these together, you get  $8 \cdot 5 = \boxed{\text{(B) } 40}$ .

16. (AMC 8 2008) A shape is created by joining seven unit cubes, as shown. What is the ratio of the volume in cubic units to the surface area in square units?



- (A) 1 : 6    (B) 7 : 36    (C) 1 : 5    (D) 7 : 30    (E) 6 : 25

**Answer:** D

**Solution**

The volume is of seven unit cubes which is 7. The surface area is the same for each of the protruding cubes which is  $5 \cdot 6 = 30$ . The ratio of the volume to the surface area is  $\boxed{\text{(D) } 7 : 30}$ .

17. (AMC 8 2008) Ms. Osborne asks each student in her class to draw a rectangle with integer side lengths and a perimeter of 50 units. All of her students calculate the area of the rectangle they draw. What is the difference between the largest and smallest possible areas of the rectangles?

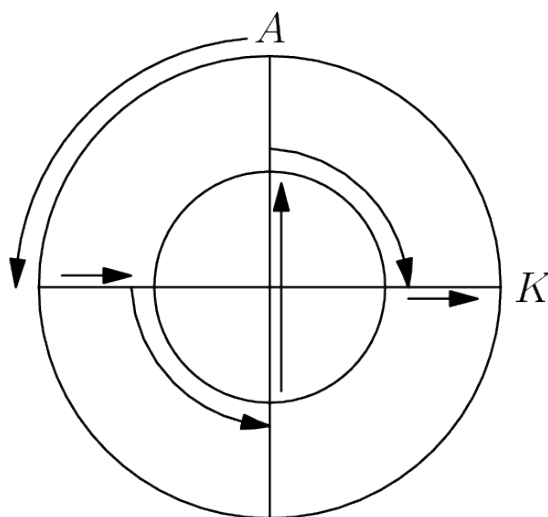
- (A) 76    (B) 120    (C) 128    (D) 132    (E) 136

**Answer:** D

**Solution**

$$156 - 24 = \boxed{\text{(D)} 132}.$$

18. (*AMC 8 2008*) Two circles that share the same center have radii 10 meters and 20 meters. An aardvark runs along the path shown, starting at  $A$  and ending at  $K$ . How many meters does the aardvark run?



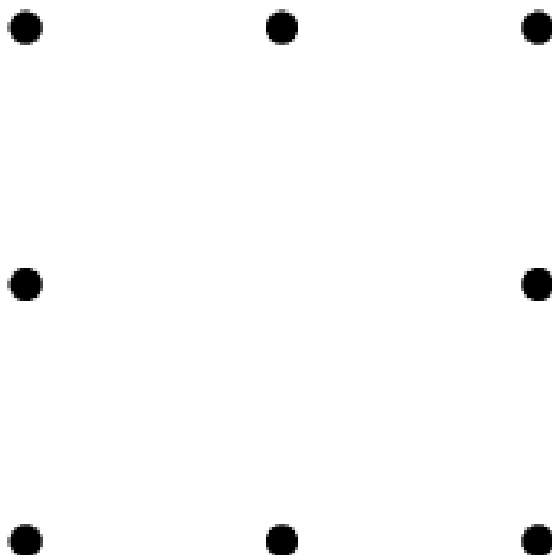
- (A)  $10\pi + 20$       (B)  $10\pi + 30$       (C)  $10\pi + 40$       (D)  $20\pi + 20$   
(E)  $20\pi + 40$

**Answer:** E

### Solution

We will deal with this part by part:  
 Part 1:  $\frac{1}{4}$  circumference of big circle =  $\frac{2\pi r}{4} = \frac{\pi r}{2} = \frac{20\pi}{2} = 10\pi$   
 Part 2: Big radius minus small radius =  $20 - 10 = 10$   
 Part 3:  $\frac{1}{4}$  circumference of small circle =  $\frac{\pi r}{2} = \frac{10\pi}{2} = 5\pi$   
 Part 4: Diameter of small circle:  $2 * 10 = 20$   
 Part 5: Same as part 3:  $5\pi$   
 Part 6: Same as part 2:  $10$   
 Total:  $10\pi + 10 + 5\pi + 20 + 5\pi + 10 = \boxed{E = 20\pi + 40}$

19. (*AMC 8 2008*) Eight points are spaced around at intervals of one unit around a  $2 \times 2$  square, as shown. Two of the 8 points are chosen at random. What is the probability that the two points are one unit apart?



- (A)  $\frac{1}{4}$     (B)  $\frac{2}{7}$     (C)  $\frac{4}{11}$     (D)  $\frac{1}{2}$     (E)  $\frac{4}{7}$

**Answer:** B

**Solution 1**

The two points are one unit apart at 8 places around the edge of the square. There are  $\binom{8}{2} = 28$  ways to choose two points. The probability is

$$\frac{8}{28} = \boxed{\text{(B)} \frac{2}{7}}$$

**Solution 2**

Arbitrarily pick a point in the grid. Clearly, we see two options for the other point to be placed, so the answer is  $\boxed{\text{(B)} \frac{2}{7}}$ .

20. (AMC 8 2008) The students in Mr. Neatkin's class took a penmanship test. Two-thirds of the boys and  $\frac{3}{4}$  of the girls passed the test, and an equal number of boys and girls passed the test. What is the minimum possible number of students in the class?

- (A) 12    (B) 17    (C) 24    (D) 27    (E) 36

**Answer:** B

**Solution**

Let  $b$  be the number of boys and  $g$  be the number of girls.

$$\frac{2}{3}b = \frac{3}{4}g \Rightarrow b = \frac{9}{8}g$$

For  $g$  and  $b$  to be integers,  $g$  must cancel out with the denominator, and the smallest possible value is 8. This yields 9 boys. The minimum number of students is  $8 + 9 = \boxed{\text{(B)} 17}$ .

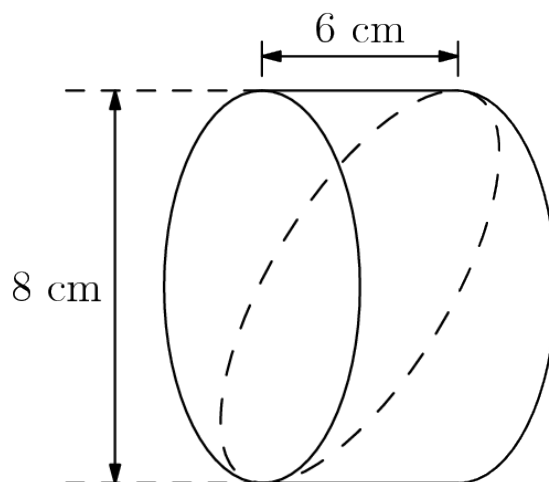
**Solution 2**

We know that  $\frac{2}{3}B = \frac{3}{4}G$  or  $\frac{6}{9}B = \frac{6}{8}G$ . So, the ratio of the number of boys to girls is  $8 : 9$  (Not  $9 : 8$  because the numbers 8 and 9 are in the denominators of the



fractions). The smallest total number of students is  $9 + 8 = \boxed{\text{(B)} 17}$ . DY, edited by SuperVince1

21. (AMC 8 2008) Jerry cuts a wedge from a 6-cm cylinder of bologna as shown by the dashed curve. Which answer choice is closest to the volume of his wedge in cubic centimeters?



- (A) 48    (B) 75    (C) 151    (D) 192    (E) 603

**Answer:** C

**Solution 1**

The slice is cutting the cylinder into two equal wedges with equal volume. The cylinder's volume is  $\pi r^2 h = \pi(4^2)(6) = 96\pi$ . The volume of the wedge is half this which is  $48\pi \approx \boxed{\text{(C)} 151}$ .

**Solution 2**

The volume of the whole cylinder is  $96\pi$ . We approximate  $\pi$  to be 3, meaning our answer will be slightly too small.  $96 * 3 = 288$ , and we know the wedge cuts the cylinder perfectly in half.  $288/2 = 144$ , so the closest answer choice is  $\boxed{\text{(C)} 151}$ .

22. (AMC 8 2008) For how many positive integer values of  $n$  are both  $\frac{n}{3}$  and  $3n$  three-digit whole numbers?

- (A) 12    (B) 21    (C) 27    (D) 33    (E) 34

**Answer:** A

**Solution 1**

Instead of finding  $n$ , we find  $x = \frac{n}{3}$ . We want  $x$  and  $9x$  to be three-digit whole numbers. The smallest three-digit whole number is 100, so that is our minimum value for  $x$ , since if  $x \in \mathbb{Z}^+$ , then  $9x \in \mathbb{Z}^+$ . The largest three-digit whole number divisible by 9 is 999, so our maximum value for  $x$  is  $\frac{999}{9} = 111$ . There are 12 whole numbers in the closed set  $[100, 111]$ , so the answer is  $\boxed{\text{(A)} 12}$ .

- ColtsFan10

**Solution 2**

We can set the following inequalities up to satisfy the conditions given by the question,  $100 \leq \frac{n}{3} \leq 999$ , and  $100 \leq 3n \leq 999$ . Once we simplify these and combine

the restrictions, we get the inequality,  $300 \leq n \leq 333$ . Now we have to find all multiples of 3 in this range for  $\frac{n}{3}$  to be an integer. We can compute this by setting  $\frac{n}{3} = x$ , where  $x \in \mathbb{Z}^+$ . Substituting  $x$  for  $n$  in the previous inequality, we get,  $100 \leq x \leq 111$ , and there are  $111 - 100 + 1$  integers in this range giving us the answer, **(A) 12**.

- kn07

### Solution 3

We can create a list of the positive integers  $n$  that fulfill the requirement of  $\frac{n}{3}$  and  $3n$  are three-digit whole numbers. The first number of this list must be 300 since  $\frac{300}{3} = 100$  is the smallest positive integer that satisfies this requirement. The last number of this list must be 333 since  $3 \cdot 333 = 999$  is the largest positive integer that satisfies this requirement. Since the problem requires  $\frac{n}{3}$  and  $3n$  must be whole numbers, the other numbers must be multiples of 3 (just like 300 and 333), so the list would look like this:

$$300, 303, 306, \dots, 333$$

To put this list in to a countable form we must put it in a form similar to  $1, 2, 3, \dots, n$ . So, we manipulate it as follows:

$$300 - 300, 303 - 300, 306 - 300, \dots, 333 - 300 \Rightarrow 0, 3, 6, \dots, 33$$

$$\frac{0}{3}, \frac{3}{3}, \frac{6}{3}, \dots, \frac{33}{3} \Rightarrow 0, 1, 2, \dots, 11$$

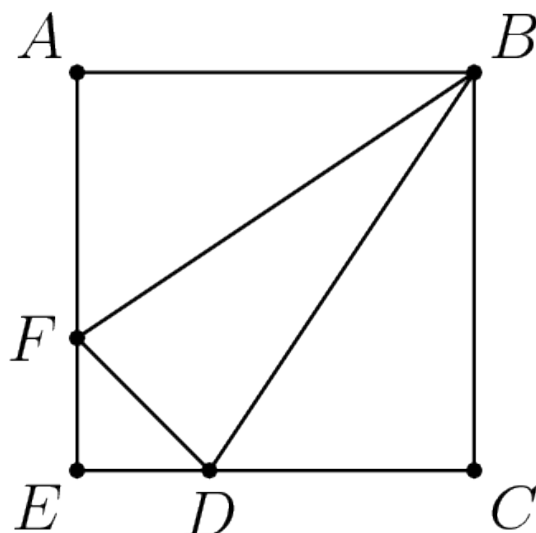
$$0 + 1, 1 + 1, 1 + 2, \dots, 11 + 1 \Rightarrow 1, 2, 3, \dots, 12$$

Now we can tell that there are 12 positive integers which satisfies the two requirements, so the answer is **(A) 12**.

### Solution 4

We have to find the range for all numbers first. We can first find the smallest digit in which  $\frac{n}{3}$  is a 3 digit number that  $3n$  is a 3 digit number which is  $\frac{300}{3} = 100$ , then we find the biggest number in which  $\frac{n}{3}$  is a 3 digit number and  $3n$  is a 3 digit number which is  $\frac{333}{3} = 111$ . Now we know that all integers from 100 to 111 are possible values and the amount of integer values in that range is **(A) 12** (note I divided the numbers by 3 for convenience)

23. (AMC 8 2008) In square  $ABCE$ ,  $AF = 2FE$  and  $CD = 2DE$ . What is the ratio of the area of  $\triangle BFD$  to the area of square  $ABCE$ ?

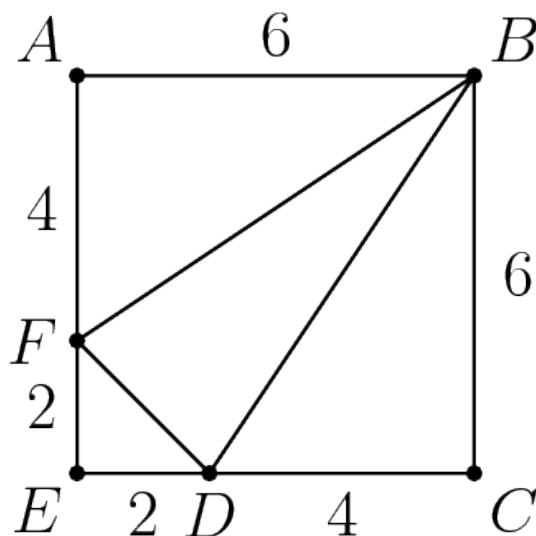


- (A)  $\frac{1}{6}$     (B)  $\frac{2}{9}$     (C)  $\frac{5}{18}$     (D)  $\frac{1}{3}$     (E)  $\frac{7}{20}$

**Answer:** C

**Solution 1**

The area of  $\triangle BFD$  is the area of square  $ABCE$  subtracted by the area of the three triangles around it. Arbitrarily assign the side length of the square to be 6.



The ratio of the area of  $\triangle BFD$  to the area of  $ABCE$  is

$$\frac{36 - 12 - 12 - 2}{36} = \frac{10}{36} = \boxed{\text{(C)} \frac{5}{18}}$$

**Solution 2**

Say that  $\overline{FE}$  has length  $x$ , and that from there we can infer that  $\overline{AF} = 2x$ . We also know that  $\overline{ED} = x$ , and that  $\overline{DC} = 2x$ . The area of triangle  $BFD$  is the square's area subtracted from the area of the excess triangles, which is simply these equations:

$$9x^2 - (3x^2 + \frac{x^2}{2} + 3x^2)$$

$$\frac{9x^2 - 6.5x^2}{2.5x^2}$$

Thus, the area of the triangle is  $2.5x^2$ . We can now put the ratio of triangle  $BFD$ 's area to the area of the square  $ABCE$  as a fraction. We have:

$$\frac{2.5x^2}{9x^2} = \frac{2.5x^2}{9x^2} = \frac{2.5}{9} = \frac{5}{18}$$

Thus, our answer is (C)  $\frac{5}{18}$

24. (*AMC 8 2008*) Ten tiles numbered 1 through 10 are turned face down. One tile is turned up at random, and a die is rolled. What is the probability that the product of the numbers on the tile and the die will be a square?

(A)  $\frac{1}{10}$       (B)  $\frac{1}{6}$       (C)  $\frac{11}{60}$       (D)  $\frac{1}{5}$       (E)  $\frac{7}{30}$

**Answer:** C

**Solution**

The numbers can at most multiply to be 60. The squares less than 60 are 1, 4, 9, 16, 25, 36, and 49. The possible pairs are (1, 1), (1, 4), (2, 2), (4, 1), (3, 3), (9, 1), (4, 4), (8, 2), (5, 5), and (9, 4). There are 11 choices and 60 possibilities giving a probability of (C)  $\frac{11}{60}$ .

25. (*AMC 8 2008*) Margie's winning art design is shown. The smallest circle has radius 2 inches, with each successive circle's radius increasing by 2 inches. Which of the following is closest to the percent of the design that is black?



(A) 42      (B) 44      (C) 46      (D) 47      (E) 49

**Answer:** A

**Solution**

Let the smallest circle be 1, the second smallest circle be 2, the third smallest circle be 3, etc.

| circle # | radius | area     |
|----------|--------|----------|
| 1        | 2      | $4\pi$   |
| 2        | 4      | $16\pi$  |
| 3        | 6      | $36\pi$  |
| 4        | 8      | $64\pi$  |
| 5        | 10     | $100\pi$ |
| 6        | 12     | $144\pi$ |

The entire circle's area is  $144\pi$ . The area of the black regions is  $(100 - 64)\pi + (36 - 16)\pi + 4\pi = 60\pi$ . The percentage of the design that is black is  $\frac{60\pi}{144\pi} = \frac{5}{12} \approx \boxed{\text{(A)} 42}$ .

## §25 AMC 8 2009

1. (*AMC 8 2009*) Bridget bought a bag of apples at the grocery store. She gave half of the apples to Ann. Then she gave Cassie 3 apples, keeping 4 apples for herself. How many apples did Bridget buy?

(A) 3    (B) 4    (C) 7    (D) 11    (E) 14

**Answer:** E

**Solution**

If we set up an equation, we find out  $x = (3 + 4) \cdot 2$  because 3 apples were left after giving half, then four away. We can simplify the equations to  $x = 7 \cdot 2 = 14$ . The answer is (E) 14.

**Solution 2**

You can work backwards and add 3 apples that she gave to Cassie to the 4 she currently has, which results in 7, and then multiply by 2 since she gave half the apples to Ann, resulting in (E) 14.

Or you can use algebra: let  $x$  be the number of apples, we have  $3 + 4 + x/2 = x$  solving for  $x$ , we get 14 resulting in (E) 14. Anabel.disher

2. (*AMC 8 2009*) On average, for every 4 sports cars sold at the local dealership, 7 sedans are sold. The dealership predicts that it will sell 28 sports cars next month. How many sedans does it expect to sell?

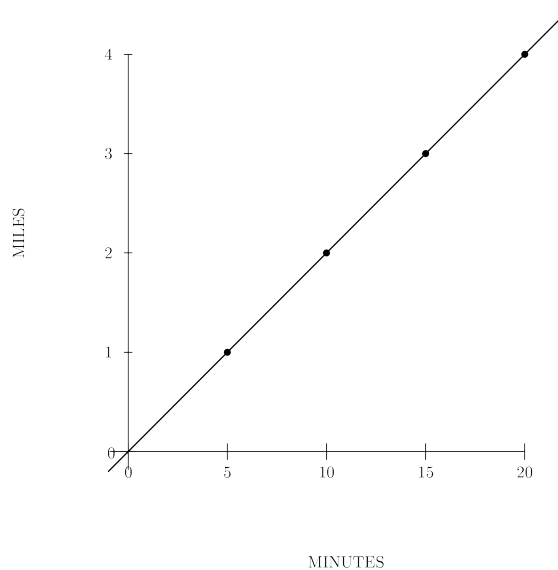
(A) 7    (B) 32    (C) 35    (D) 49    (E) 112

**Answer:** D

**Solution**

This means the ratio is 4 : 7. If the ratio now is 28 :  $x$ , then that means  $x = \boxed{\text{(D)} 49}$ .

3. (*AMC 8 2009*) The graph shows the constant rate at which Suzanna rides her bike. If she rides a total of a half an hour at the same speed, how many miles would she have ridden?



- (A) 5      (B) 5.5      (C) 6      (D) 6.5      (E) 7

**Answer:** C

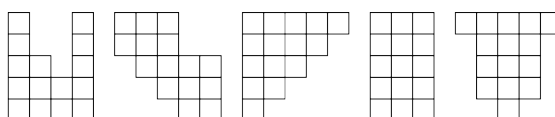
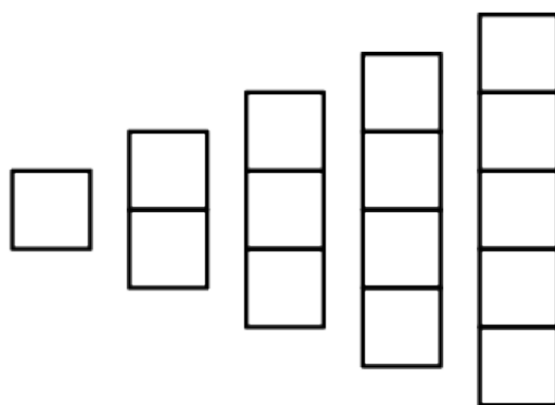
**Solution**

Suzanna's speed is  $\frac{1}{5}$ . This means she runs  $\frac{1}{5} \cdot 30 = \boxed{\text{(C)} 6}$

**Solution 2**

From the graph, we can see that every 5 minutes Suzanna goes, her distance increases by 1. Since half an hour is 10 minutes away, she would go 2 more miles.  $4 + 2$  is 6, so the answer is  $\boxed{\text{(C)} 6}$

4. (AMC 8 2009) The five pieces shown below can be arranged to form four of the five figures shown in the choices. Which figure **cannot** be formed?



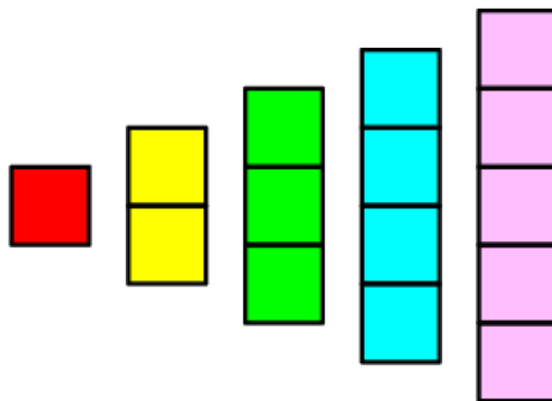
- (A)                      (B)                      (C)                      (D)                      (E)

**Answer:** B

**Solution**

The answer is **(B)** because the longest piece cannot fit into the figure.

Note that the five pieces can be arranged to form the figures in **(A)**, **(C)**, **(D)**, and **(E)**, as shown below:



**Solution 2**

We can quickly see on A that  $4+1, 3, 2, 5$  works but on B you can see that there are 3 rows of 3 and we only have 1 3's and 1  $2+1$ 's so 1 row of 3 is left blank so **(B)**, you can use this as a check for solution 1 or as a quick strategy.

5. (*AMC 8 2009*) A sequence of numbers starts with 1, 2, and 3. The fourth number of the sequence is the sum of the previous three numbers in the sequence:  $1 + 2 + 3 = 6$ . In the same way, every number after the fourth is the sum of the previous three numbers. What is the eighth number in the sequence?

**(A)** 11      **(B)** 20      **(C)** 37      **(D)** 68      **(E)** 99

**Answer:** D

**Solution**

List them out, adding the three previous numbers to get the next number,

1, 2, 3, 6, 11, 20, 37, **(D)** 68

6. (*AMC 8 2009*) Steve's empty swimming pool will hold 24,000 gallons of water when full. It will be filled by 4 hoses, each of which supplies 2.5 gallons of water per minute. How many hours will it take to fill Steve's pool?

**(A)** 40      **(B)** 42      **(C)** 44      **(D)** 46      **(E)** 48

**Answer:** A

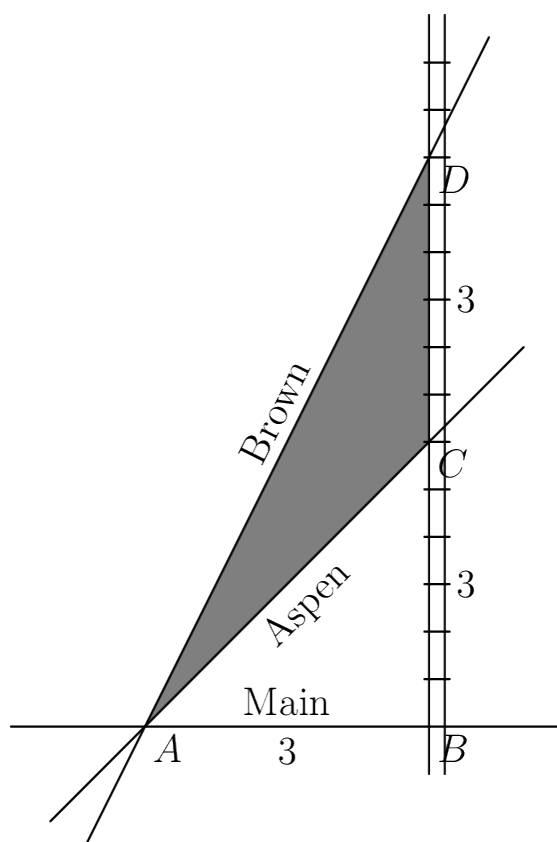
**Solution**

Each of the four hoses hose fills  $24,000/4 = 6,000$  gallons of water. At the rate it goes at it will take  $6,000/2.5 = 2400$  minutes, or **(A)** 40 hours.

**Solution 2**

If all four hoses fill 2.5 gallons a minute, every minute 10 gallons would be added. Since every hour has 60 minutes, 600 gallons of water would be added every hour.  $24000/600 =$  **(A)** 40 hours.

7. (AMC 8 2009) The triangular plot of ACD lies between Aspen Road, Brown Road and a railroad. Main Street runs east and west, and the railroad runs north and south. The numbers in the diagram indicate distances in miles. The width of the railroad track can be ignored. How many square miles are in the plot of land ACD?



- (A) 2    (B) 3    (C) 4.5    (D) 6    (E) 9

**Answer:** C

**Solution**

The area of a triangle is  $\frac{1}{2}bh$ . If we let  $CD$  be the base of the triangle, then the height is  $AB$ , and the area is  $\frac{1}{2} \cdot 3 \cdot 3 = \boxed{\text{(C) } 4.5}$ .

**Solution 2**

We can see that there is a big triangle encasing  $ACD$ . The area of that triangle is  $\frac{1}{2} \cdot 3 \cdot 6 = 9$ . We can easily see that triangle  $ABC$  is  $\frac{1}{2} \cdot 3 \cdot 3$ , which is 4.5.  $9 - 4.5$  is 4.5, so the answer is  $\boxed{\text{(C) } 4.5}$ .

8. (AMC 8 2009) The length of a rectangle is increased by 10% and the width is decreased by 10%. What percent of the old area is the new area?

- (A) 90    (B) 99    (C) 100    (D) 101    (E) 110

**Answer:** B

**Solution**

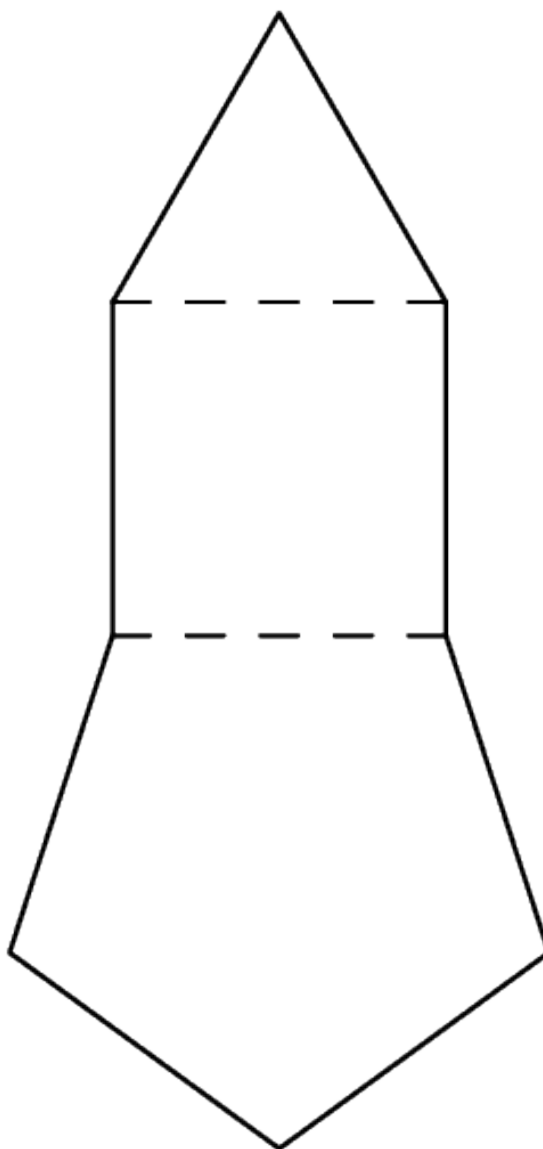
In a rectangle with dimensions  $10 \times 10$ , the new rectangle would have dimensions  $11 \times 9$ . The ratio of the new area to the old area is  $99/100 = \boxed{\text{(B) } 99}$ .



**Solution 2**

If you take the length as  $x$  and the width as  $y$  then  $A(\text{OLD}) = xy$   $A(\text{NEW}) = 1.1x \times .9y = .99xy$   $.99/1 = 99\%$

9. (*AMC 8 2009*) Construct a square on one side of an equilateral triangle. On one non-adjacent side of the square, construct a regular pentagon, as shown. On a non-adjacent side of the pentagon, construct a hexagon. Continue to construct regular polygons in the same way, until you construct an octagon. How many sides does the resulting polygon have?



- (A) 21      (B) 23      (C) 25      (D) 27      (E) 29

**Answer:** B

**Solution**

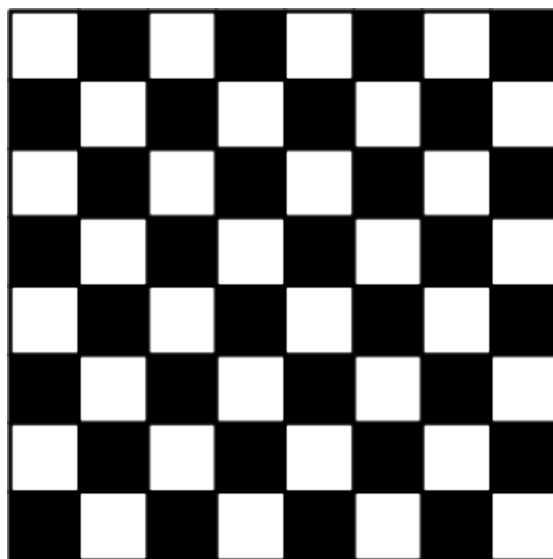
Of the six shapes used to create the polygon, the triangle and octagon are adjacent to the others on one side, and the others are adjacent on two sides. In the triangle and octagon  $3 + 8 - 2(1) = 9$  sides are on the outside of the final polygon. In the

other shapes  $4 + 5 + 6 + 7 - 4(2) = 14$  sides are on the outside. The resulting polygon has  $9 + 14 = \boxed{\text{(B)} 23}$  sides.

**Solution 2**

We can quickly see a pattern if we draw out the other shapes. Every shape will have two of its sides taken out except the triangle and octagon. We can then make the expression  $2 + 2 + 3 + 4 + 5 + 7$  which is  $\boxed{\text{(B)} 23}$

10. (AMC 8 2009) On a checkerboard composed of 64 unit squares, what is the probability that a randomly chosen unit square does not touch the outer edge of the board?



- (A)  $\frac{1}{16}$     (B)  $\frac{7}{16}$     (C)  $\frac{1}{2}$     (D)  $\frac{9}{16}$     (E)  $\frac{49}{64}$

**Answer:** D

**Solution**

There are  $8^2 = 64$  total squares. There are  $(8 - 1)(4) = 28$  unit squares on the perimeter and therefore  $64 - 28 = 36$  NOT on the perimeter. The probability of choosing one of those squares is  $\frac{36}{64} = \boxed{\text{(D)} \frac{9}{16}}$ .

**Solution 2**

The squares that don't touch the border are in a 6 by 6 square. Therefore, the number of squares that don't touch the border is  $6 * 6 = 36$ . There are 64 choices total. Therefore, the answer is  $36/64$ , which simplifies to  $\boxed{\text{(D)} \frac{9}{16}}$ .

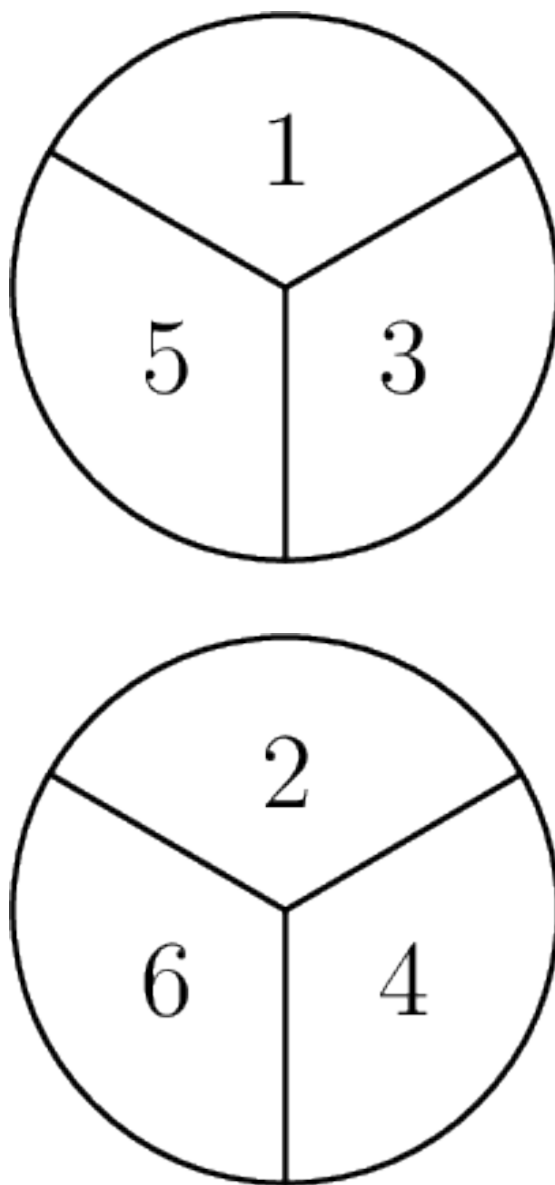
11. (AMC 8 2009) The Amaco Middle School bookstore sells pencils costing a whole number of cents. Some seventh graders each bought a pencil, paying a total of \$1.43. Some of the 30 sixth graders each bought a pencil, and they paid a total of \$1.95. How many more sixth graders than seventh graders bought a pencil?
- (A) 1    (B) 2    (C) 3    (D) 4    (E) 5

**Answer:** D

**Solution**

Because the pencil costs a whole number of cents, the cost must be a factor of both 143 and 195. They can be factored into  $11 \cdot 13$  and  $3 \cdot 5 \cdot 13$ . The common factor cannot be 1 or there would have to be more than 30 sixth graders, so the pencil costs 13 cents. The difference in costs that the sixth and seventh graders paid is  $195 - 143 = 52$  cents, which is equal to  $52/13 = \boxed{\text{(D)} 4}$  sixth graders.

12. (AMC 8 2009) The two spinners shown are spun once and each lands on one of the numbered sectors. What is the probability that the sum of the numbers in the two sectors is prime?



- (A)  $\frac{1}{2}$     (B)  $\frac{2}{3}$     (C)  $\frac{3}{4}$     (D)  $\frac{7}{9}$     (E)  $\frac{5}{6}$

**Answer:** D

**Solution**

The possible sums are

|   | 1 | 3 | 5  |
|---|---|---|----|
| 2 | 3 | 5 | 7  |
| 4 | 5 | 7 | 9  |
| 6 | 7 | 9 | 11 |

Only 9 is not prime, so there are 7 prime numbers and 9 total numbers for a probability of **(D)**  $\frac{7}{9}$ .

13. (AMC 8 2009) A three-digit integer contains one of each of the digits 1, 3, and 5. What is the probability that the integer is divisible by 5?

(A)  $\frac{1}{6}$     (B)  $\frac{1}{3}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{3}$     (E)  $\frac{5}{6}$

**Answer:** B

**Solution 1**

The three digit numbers are 135, 153, 351, 315, 513, 531. The numbers that end in 5 are divisible are 5, and the probability of choosing those numbers is **(B)**  $\frac{1}{3}$ .

**Solution 2**

The number is divisible by 5 if and only if the number ends in 5 (also 0, but that case can be ignored, as none of the digits are 0) If we randomly arrange the three digits, the probability of the last digit being 5 is **(B)**  $\frac{1}{3}$ .

Note: The last sentence is true because there are 3 randomly-arrangeable numbers)

14. (AMC 8 2009) Austin and Temple are 50 miles apart along Interstate 35. Bonnie drove from Austin to her daughter's house in Temple, averaging 60 miles per hour. Leaving the car with her daughter, Bonnie rode a bus back to Austin along the same route and averaged 40 miles per hour on the return trip. What was the average speed for the round trip, in miles per hour?

(A) 46    (B) 48    (C) 50    (D) 52    (E) 54

**Answer:** B

**Solution**

The way to Temple took  $\frac{50}{60} = \frac{5}{6}$  hours, and the way back took  $\frac{50}{40} = \frac{5}{4}$  for a total of  $\frac{5}{6} + \frac{5}{4} = \frac{25}{12}$  hours. The trip is  $50 \cdot 2 = 100$  miles. The average speed is  $\frac{100}{25/12} = \mathbf{(B) 48}$  miles per hour.

**Solution 2**

We calculate the harmonic mean of Austin and Temple. Plugging in, we have:  $\frac{2ab}{a+b} = \frac{2 \cdot 60 \cdot 40}{60+40} = \frac{4800}{100} = \mathbf{(B) 48}$  miles per hour.

**Solution 3**

The 50 miles part of this question is redundant. We find that the average speed is skewed towards the lower speed, since you spend more time in the lower speed. So, we take the LCM of 40 and 60, getting 120. So, we do  $(40 + 40 + 40 + 60 + 60)/5$ , calculation the ratio of the parts and taking the mean. Solving, we get  $240/5 = \mathbf{(B) 48}$  -themathgood

15. (*AMC 8 2009*) A recipe that makes 5 servings of hot chocolate requires 2 squares of chocolate,  $\frac{1}{4}$  cup sugar, 1 cup water and 4 cups milk. Jordan has 5 squares of chocolate, 2 cups of sugar, lots of water and 7 cups of milk. If she maintains the same ratio of ingredients, what is the greatest number of servings of hot chocolate she can make?

(A)  $5\frac{1}{8}$       (B)  $6\frac{1}{4}$       (C)  $7\frac{1}{2}$       (D)  $8\frac{3}{4}$       (E)  $9\frac{7}{8}$

**Answer:** D

**Solution**

Assuming excesses of the other ingredients, the chocolate can make  $\frac{5}{2} \cdot 5 = 12.5$  servings, the sugar can make  $\frac{2}{\frac{1}{4}} \cdot 5 = 40$  servings, the water can make unlimited servings, and the milk can make  $\frac{7}{4} \cdot 5 = 8.75$  servings. Limited by the amount of milk, Jordan can make at most  $\boxed{\text{(D)} 8\frac{3}{4}}$  servings.

16. (*AMC 8 2009*) How many 3-digit positive integers have digits whose product equals 24?

(A) 12      (B) 15      (C) 18      (D) 21      (E) 24

**Answer:** D

**Solution**

With the digits listed from least to greatest, the 3-digit integers are 138, 146, 226, 234. 226 can be arranged in  $\frac{3!}{2!} = 3$  ways, and the other three can be arranged in  $3! = 6$  ways. There are  $3 + 6(3) = \boxed{\text{(D)} 21}$  3-digit positive integers.

17. (*AMC 8 2009*) The positive integers  $x$  and  $y$  are the two smallest positive integers for which the product of 360 and  $x$  is a square and the product of 360 and  $y$  is a cube. What is the sum of  $x$  and  $y$ ?

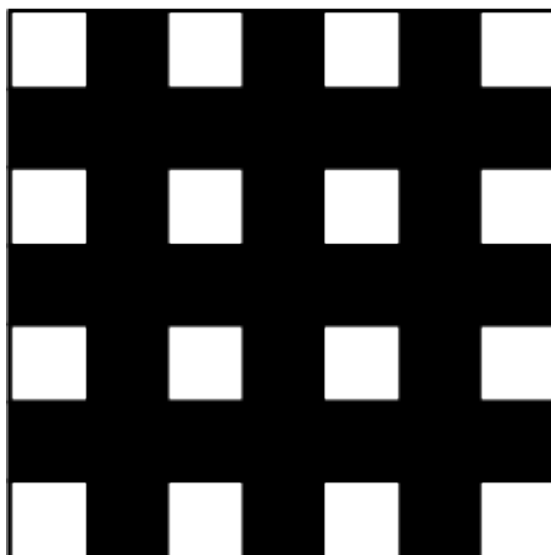
(A) 80      (B) 85      (C) 115      (D) 165      (E) 610

**Answer:** B

**Solution**

Take the prime factorization of 360.  $360 = 2^3 \cdot 3^2 \cdot 5$ . We want  $x$  to be as small as possible. And you want  $x \cdot 360$  to be a square. So  $x = 2 \cdot 5 = 10$ .  $y$  is similar.  $y = 3 \cdot 5^2 = 3 \cdot 25 = 75$ . So,  $x + y = 75 + 10 = 85$ , or  $\boxed{\text{(B)} 85}$ .

18. (*AMC 8 2009*) The diagram represents a 7-foot-by-7-foot floor that is tiled with 1-square-foot black tiles and white tiles. Notice that the corners have white tiles. If a 15-foot-by-15-foot floor is to be tiled in the same manner, how many white tiles will be needed?



- (A) 49      (B) 57      (C) 64      (D) 96      (E) 126

**Answer:** C

**Solution 1**

In a 1-foot-by-1-foot floor, there is 1 white tile. In a 3-by-3, there are 4. Continuing on, you can deduce the  $n^{\text{th}}$  positive odd integer floor has  $n^2$  white tiles. 15 is the 8<sup>th</sup> odd integer, so there are (C) 64 white tiles.

**Solution 2**

After testing a couple of cases, we find that the number of white squares is  $(s/2)^2$ , where  $s$  is the side length of the square and  $s/2$  is rounded up to the next whole number. Therefore,  $15/2$  rounded up is 8, so  $8^2 = 64$ , or (C) 64.

19. (AMC 8 2009) Two angles of an isosceles triangle measure  $70^\circ$  and  $x^\circ$ . What is the sum of the three possible values of  $x$ ?

- (A) 95      (B) 125      (C) 140      (D) 165      (E) 180

**Answer:** D

**Solution**

There are 3 cases: where  $x^\circ$  is a base angle with the  $70^\circ$  as the other angle, where  $x^\circ$  is a base angle with  $70^\circ$  as the vertex angle, and where  $x^\circ$  is the vertex angle with  $70^\circ$  as a base angle.

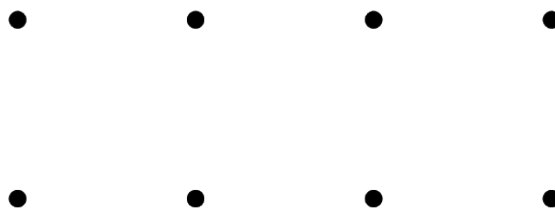
Case 1:  $x^\circ$  is a base angle with  $70^\circ$  as the other angle: Here,  $x = 70$ , since base angles are congruent.

Case 2:  $x^\circ$  is a base angle with  $70^\circ$  as the vertex angle: Here, the 2 base angles are both  $x^\circ$ , so we can use the equation  $2x + 70 = 180$ , which simplifies to  $x = 55$ .

Case 3:  $x^\circ$  is the vertex angle with  $70^\circ$  as a base angle: Here, both base angles are  $70^\circ$ , since base angles are congruent. Thus, we can use the equation  $x + 140 = 180$ , which simplifies to  $x = 40$ .

Adding up all the cases, we get  $70 + 55 + 40 = 165$ , so the answer is (D) 165.

20. (AMC 8 2009) How many non-congruent triangles have vertices at three of the eight points in the array shown below?



- (A) 5    (B) 6    (C) 7    (D) 8    (E) 9

**Answer:** D

**Solution**

Assume the base of the triangle is on the bottom four points because a congruent triangle can be made by reflecting the base on the top four points. For a triangle with a base of length 1, there are 3 triangles. For a triangle with a base of length 2, there are 3 triangles. For length 3, there are 2. In total, the number of non-congruent triangles is  $3 + 3 + 2 = \boxed{\text{(D) } 8}$ .

21. (AMC 8 2009) Andy and Bethany have a rectangular array of numbers greater than 0 with 40 rows and 75 columns. Andy adds the numbers in each row. The average of his 40 sums is  $A$ . Bethany adds the numbers in each column. The average of her 75 sums is  $B$ . What is the value of  $\frac{A}{B}$ ?

- (A)  $\frac{64}{225}$     (B)  $\frac{8}{15}$     (C) 1    (D)  $\frac{15}{8}$     (E)  $\frac{225}{64}$

**Answer:** D

**Solution**

Note that  $40A = 75B = \text{sum of the numbers in the array}$ . This means that  $\frac{A}{B} = \boxed{\text{(D) } \frac{15}{8}}$  using basic algebraic manipulation.

22. (AMC 8 2009) How many whole numbers between 1 and 1000 do not contain the digit 1?

- (A) 512    (B) 648    (C) 720    (D) 728    (E) 800

**Answer:** D

**Solution (Super Fast!)**

Note that this is the same as finding how many numbers with up to three digits do not contain 1.

Since there are 10 total possible digits, and only one of them is not allowed (1), each place value has its choice of 9 digits, for a total of  $9 * 9 * 9 = 729$  such numbers. However, we overcounted by one; 0 is not between 1 and 1000, so there are  $\boxed{\text{(D) } 728}$  numbers.

Note: As stated above, 020 is equivalent to 20, but 000 is not between 1 and 1000.

**Solution 2 (Easy Casework)**

We consider the 3 cases, where the number is 1, 2 or 3 digits.

Case 1 : The number has a 1 digit. Well, 1-9 all work except for 1, so  $9 - 1 = 8$  numbers, for numbers that are 1 digit.

Case 2 : The number is has 2 digits. Consider the two digit number. Since the tens digit can't be 0 or 1, we have 8 choices for the tens digit. For the ones digit, we

have 9, since the ones digit can't be a 1. This gives us a total of  $8 * 9 = 72$  two-digit numbers.

Case 3 : The number has 3 digits. Using similar logic to Case 2, we have  $8 * 9 * 9$  choices for the third number.

We add the cases up getting  $648 + 72 + 8$  which gives us (D) 728 numbers total.

23. (AMC 8 2009) On the last day of school, Mrs. Awesome gave jelly beans to her class. She gave each boy as many jelly beans as there were boys in the class. She gave each girl as many jelly beans as there were girls in the class. She brought 400 jelly beans, and when she finished, she had six jelly beans left. There were two more boys than girls in her class. How many students were in her class?

(A) 26      (B) 28      (C) 30      (D) 32      (E) 34

**Answer:** B

**Solution 1(Algebraic)**

If there are  $x$  girls, then there are  $x + 2$  boys. She gave each girl  $x$  jellybeans and each boy  $x + 2$  jellybeans, for a total of  $x^2 + (x + 2)^2$  jellybeans. She gave away  $400 - 6 = 394$  jellybeans.

$$\begin{aligned}x^2 + (x + 2)^2 &= 394 \\x^2 + x^2 + 4x + 4 &= 394 \\2x^2 + 4x &= 390 \\x^2 + 2x &= 195\end{aligned}$$

From here, we can see that  $x = 13$  as  $13^2 + 26 = 195$ , so there are 13 girls,  $13 + 2 = 15$  boys, and  $13 + 15 =$  (B) 28 students.

Note by FrankensteindDied: The user who made this solution did not mention their name.

**Solution 2(Answer Choices)**

We can use the answer choices to our advantage; we can take the number of girls as  $x$ , and the number of boys as  $x + 2$ . If we test out (A),  $x$  is 12, and  $x + 2$  is 14.  $400 - 6 = 394$  jellybeans were given out, but  $12^2 + 14^2$  is not equal to 394. Thus, (A) is not the answer. Next, we try (B).  $x$  is 13,  $x + 2$  is 15. We see that  $13^2 + 15^2 = 394$ , so there are  $13 + 15 =$  (B) 28 students.

24. (AMC 8 2009) The letters  $A$ ,  $B$ ,  $C$  and  $D$  represent digits. If 
$$\begin{array}{r} A \quad B \\ + \quad C \quad A \\ \hline D \quad A \end{array}$$
 and

$$\begin{array}{r} A \quad B \\ - \quad C \quad A \\ \hline A \end{array}$$
, what digit does  $D$  represent?

(A) 5      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** E

**Solution**

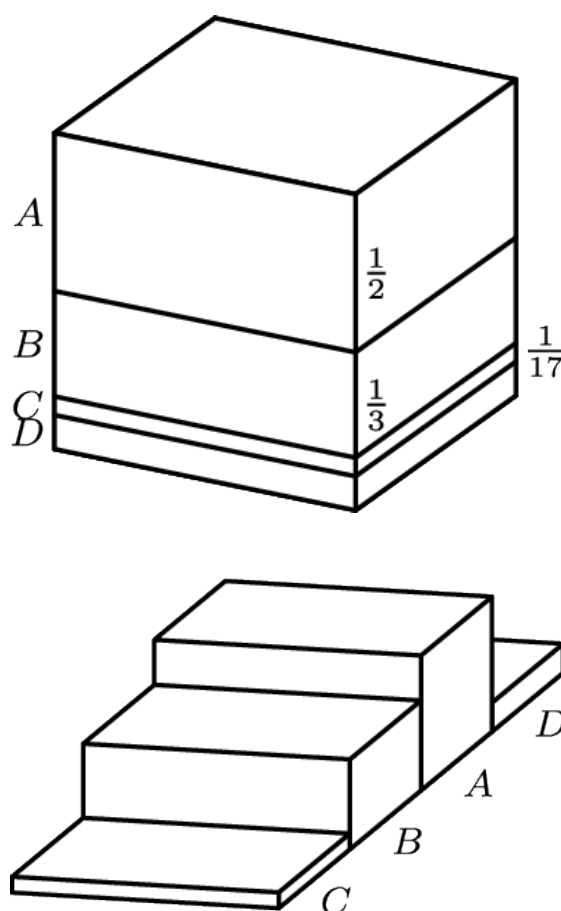


Because  $B + A = A$ ,  $B$  must be 0. Next, because  $B - A = A \implies 0 - A = A$ , we get  $A = 5$  as the "0" mentioned above is actually 10 in this case.

Now we can rewrite  $\frac{A}{D} + \frac{C}{A}$  as  $\frac{5}{5} + \frac{0}{5}$ . Therefore,  $D = 5 + C$ .

Finally,  $A - 1 - C = 0 \implies A = C + 1 \implies C = 4$ , So we have  $D = 5 + C \implies D = 5 + 4 = \boxed{\text{(E)} 9}$ .

25. (AMC 8 2009) A one-cubic-foot cube is cut into four pieces by three cuts parallel to the top face of the cube. The first cut is  $\frac{1}{2}$  foot from the top face. The second cut is  $\frac{1}{3}$  foot below the first cut, and the third cut is  $\frac{1}{17}$  foot below the second cut. From the top to the bottom the pieces are labeled A, B, C, and D. The pieces are then glued together end to end as shown in the second diagram. What is the total surface area of this solid in square feet?



- (A) 6    (B) 7    (C)  $\frac{419}{51}$     (D)  $\frac{158}{17}$     (E) 11

**Answer:** E

**Solution 1**

The areas of the tops of A, B, C, and D in the figure formed has sum  $1 + 1 + 1 + 1 = 4$  as do the bottoms. Thus, the total so far is 8. Now, one of the sides has an area of one, since it combines all of the heights of A, B, C, and D, which is 1. The other side is also the same. Thus the total area now is 10. From the front, the surface area is half, because if you looked at it straight from the front it would look exactly

like a side of  $A$ , with a surface area of half. From the back, it is the same thing. Thus, the total surface area is  $10 + \frac{1}{2} + \frac{1}{2} = 11$ , or **(E) 11**.

**Solution 2 (Faster method that relies on the MCQ choices)**

The top parts and the bottom parts sum to 8. The sides add on another 2. Therefore, the only logical answer would be **(E) 11** since it is the only answer greater than 10.

**Solution 3 (Fast and logical)**

We know that the surface area of the original cube is 6, because each side is length 1. If we were to then spread out the pieces separately, we would add 2 for each of the faces we pull apart because we are creating 2 new faces. This happens 3 times, so  $6 + 2 + 2 + 2 = 12$ . Now, we must subtract the overlap - they are mashed together. We need to subtract 2 times the overlap for each of B, C, and D. This simplifies to  $2(B + C + D)$ . Since we know that A is  $1/2$ ,  $B + C + D$  is also  $1/2$ .  $1/2 * 2 = 1$ , and  $12 - 1 = \mathbf{(E) 11}$ .

## §26 AMC 8 2010

1. (*AMC 8 2010*) At Euclid Middle School, the mathematics teachers are Miss Germain, Mr. Newton, and Mrs. Young. There are 11 students in Mrs. Germain's class, 8 students in Mr. Newton's class, and 9 students in Mrs. Young's class taking the AMC 8 this year. How many mathematics students at Euclid Middle School are taking the contest?

(A) 26      (B) 27      (C) 28      (D) 29      (E) 30

**Answer:** C

**Solution**

Given that these are the only math teachers at Euclid Middle School and we are told how many from each class are taking the AMC 8, we simply add the three given numbers to find the total.  $11 + 8 + 9 = \mathbf{(C) 28}$

2. (*AMC 8 2010*) If  $a@b = \frac{a \times b}{a+b}$  for  $a, b$  positive integers, then what is  $5@10$ ?

(A)  $\frac{3}{10}$       (B) 1      (C) 2      (D)  $\frac{10}{3}$       (E) 50

**Answer:** D

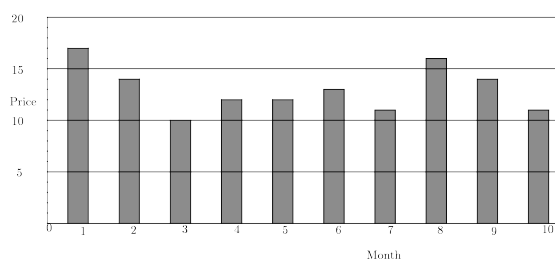
**Solution**

Substitute  $a = 5$  and  $b = 10$  into the expression for  $a@b$  to get:

$$5@10 = \frac{5 \times 10}{5+10} = \frac{50}{15} = \frac{10}{3}$$

Thus, answer choice **(D)  $\frac{10}{3}$**  is correct.

3. (*AMC 8 2010*) The graph shows the price of five gallons of gasoline during the first ten months of the year. By what percent is the highest price more than the lowest price?



- (A) 50      (B) 62      (C) 70      (D) 89      (E) 100

**Answer:** C

**Solution**

The highest price was in Month 1, which was \$17. The lowest price was in Month 3, which was \$10. 17 is  $\frac{17}{10} \cdot 100 = 170\%$  of 10, and is  $170 - 100 = 70\%$  more than 10. Therefore, the answer is **(C) 70**

4. (AMC 8 2010) What is the sum of the mean, median, and mode of the numbers 2, 3, 0, 3, 1, 4, 0, 3?

- (A) 6.5      (B) 7      (C) 7.5      (D) 8.5      (E) 9

**Answer:** C

**Solution**

Putting the numbers in numerical order we get the list 0, 0, 1, 2, 3, 3, 3, 4. The mode is 3. The median is  $\frac{2+3}{2} = 2.5$ . The average is  $\frac{0+0+1+2+3+3+3+4}{8} = \frac{16}{8} = 2$ . The sum of all three is  $3 + 2.5 + 2 =$  **(C) 7.5**

5. (AMC 8 2010) Alice needs to replace a light bulb located 10 centimeters below the ceiling in her kitchen. The ceiling is 2.4 meters above the floor. Alice is 1.5 meters tall and can reach 46 centimeters above the top of her head. Standing on a stool, she can just reach the light bulb. What is the height of the stool, in centimeters?

- (A) 32      (B) 34      (C) 36      (D) 38      (E) 40

**Answer:** B

**Solution**

Convert everything to the same unit. Since the answer is in centimeters, change meters to centimeters by moving the decimal place two places to the right.

The ceiling is 240 centimeters above the floor. The combined height of Alice and the light bulb when she reaches for it is  $10 + 150 + 46 = 206$  centimeters. That means the stool's height needs to be  $240 - 206 =$  **(B) 34**

**Solution 2**

Convert everything to the same unit. Since the answer is in centimeters, change meters to centimeters by multiplying by 100.

So, the ceiling is 240 centimeters above the floor. 10 cm below means the light is at 230 cm. Because Alice is 150 cm, and she can reach 46 cm **above her head**, her reach height is  $46 + 150 = 196$ . The height of the stool is the difference of the heights;  $230 - 196 = 34$  cm.

-N.

6. (*AMC 8 2010*) Which of the following figures has the greatest number of lines of symmetry?

(A) equilateral triangle (B) non-square rhombus (C) non-square rectangle (D) isosceles trapezoid  
(E) square

**Answer:** E

**Solution**

An equilateral triangle has 3 lines of symmetry. A non-square rhombus has 2 lines of symmetry. A non-square rectangle has 2 lines of symmetry. An isosceles trapezoid has 1 line of symmetry. A square has 4 lines of symmetry.

Therefore, the answer is (E) square.

7. (*AMC 8 2010*) Using only pennies, nickels, dimes, and quarters, what is the smallest number of coins Freddie would need so he could pay any amount of money less than a dollar?

(A) 6 (B) 10 (C) 15 (D) 25 (E) 99

**Answer:** B

**Solution**

To make any units digit, you can use 4 pennies and 1 nickel. Using 3 quarters, you can make 75, so you will need 2 more dimes. Therefore, you will need

$$4 + 1 + 2 + 3 = B(10) \text{ coins.}$$

8. (*AMC 8 2010*) As Emily is riding her bicycle on a long straight road, she spots Emerson skating in the same direction  $\frac{1}{2}$  mile in front of her. After she passes him, she can see him in her rear mirror until he is  $\frac{1}{2}$  mile behind her. Emily rides at a constant rate of 12 miles per hour, and Emerson skates at a constant rate of 8 miles per hour. For how many minutes can Emily see Emerson?

(A) 2 (B) 6 (C) 12 (D) 15 (E) 16

**Answer:** D

**Solution**

Because they are both moving in the same direction, Emily is riding relative to Emerson  $12 - 8 = 4$  mph. Now we can look at it as if Emerson is not moving at all [on his skateboard] and Emily is riding at 4 mph. It takes her

$$\frac{1}{2} \text{ mile} \cdot \frac{1 \text{ hour}}{4 \text{ miles}} = \frac{1}{8} \text{ hour}$$

to ride the  $\frac{1}{2}$  mile to reach him, and then the same amount of time to be  $\frac{1}{2}$  mile ahead of him. This totals to

$$2 \cdot \frac{1}{8} \text{ hour} \cdot \frac{60 \text{ minutes}}{1 \text{ hour}} = \text{(D) 15 minutes}$$

9. (*AMC 8 2010*) Ryan got 80% of the problems correct on a 25-problem test, 90% on a 40-problem test, and 70% on a 10-problem test. What percent of all the problems did Ryan answer correctly?

(A) 64 (B) 75 (C) 80 (D) 84 (E) 86

**Answer:** D

**Solution**

Ryan answered  $(0.8)(25) = 20$  problems correct on the first test,  $(0.9)(40) = 36$  on the second, and  $(0.7)(10) = 7$  on the third. This amounts to a total of  $20 + 36 + 7 = 63$  problems correct. The total number of problems is  $25 + 40 + 10 = 75$ . Therefore, the percentage is  $\frac{63}{75} = 84\% \rightarrow \boxed{\text{(D)} 84}$

10. (AMC 8 2010) Six pepperoni circles will exactly fit across the diameter of a 12-inch pizza when placed. If a total of 24 circles of pepperoni are placed on this pizza without overlap, what fraction of the pizza is covered by pepperoni?

(A)  $\frac{1}{2}$     (B)  $\frac{2}{3}$     (C)  $\frac{3}{4}$     (D)  $\frac{5}{6}$     (E)  $\frac{7}{8}$

**Answer:** B

**Solution**

The pepperoni circles' diameter is 2, since  $\frac{12}{6} = 2$ . From that we see that the area of the 24 circles of pepperoni is  $\left(\frac{2}{2}\right)^2 (24\pi) = 24\pi$ . The large pizza's area is  $6^2\pi$ .

Therefore, the ratio is  $\frac{24\pi}{36\pi} = \boxed{\text{(B)} \frac{2}{3}}$

11. (AMC 8 2010) The top of one tree is 16 feet higher than the top of another tree. The heights of the two trees are in the ratio 3 : 4. In feet, how tall is the taller tree?

(A) 48    (B) 64    (C) 80    (D) 96    (E) 112

**Answer:** B

**Solution 1(algebra solution)**

Let the height of the taller tree be  $h$  and let the height of the smaller tree be  $h - 16$ . Since the ratio of the smaller tree to the larger tree is  $\frac{3}{4}$ , we have  $\frac{h-16}{h} = \frac{3}{4}$ . Solving for  $h$  gives us  $h = 64 \Rightarrow \boxed{\text{(B)} 64}$

**Solution 2**

To answer this problem, you have to make it so that we have the same proportion as 3:4, but the difference between them is 16. Since the two numbers in the ratio are consecutive (difference of 1), if we multiply both of them by 16, we would get a difference of 16 between them. So, it would be 48:64 and since we need to find the height of the taller tree, we get  $h = 64 \Rightarrow \boxed{\text{(B)} 64}$

**Solution 3 (another algebra solution)**

$$s + 16 = t \quad 3t = 4s$$

Solving by substitution,  $\frac{3}{4}t + 16 = t$   $16 = \frac{1}{4}t$   $t = \boxed{\text{(B)} 64}$

**Solution 4 (ANOTHER ALGEBRA ONE)**

Assume the ratio is represented through  $x$ . Then,  $4x = 3x + 16$ , meaning

12. (AMC 8 2010) Of the 500 balls in a large bag, 80% are red and the rest are blue. How many of the red balls must be removed from the bag so that 75% of the remaining balls are red?

(A) 25    (B) 50    (C) 75    (D) 100    (E) 150

**Answer:** D

**Solution 1(logical solution)**

Since 80 percent of the 500 balls are red, there are 400 red balls. Therefore, there must be 100 blue balls. For the 100 blue balls to be 25% or  $\frac{1}{4}$  of the bag, there must be 400 balls in the bag so 100 red balls must be removed. The answer is **(D) 100**.

**Solution 2(algebra solution)**

We could also set up a proportion. Since we know there are 400 red balls, we let the amount of red balls removed be  $x$ , so  $\frac{400-x}{500-x} = \frac{3}{4}$ . Cross-multiplying gives us  $1600 - 4x = 1500 - 3x \implies x = 100$ , so our answer is **(D) 100**.

**Solution 3**

Before any balls are taken out of the bag, there are 400 red balls. Let the number of red balls we take out of the bag be  $x$ . Since we want the number of red ball to be 75% of the remaining balls, we can set up the equation:  $0.75(500-x) = 400-x$ . In words, 75% of the total remaining balls are equal to the remaining red balls. Solving we get  $x = 100$ , so our answer is **(D) 100**. J.L.L

13. (AMC 8 2010) The lengths of the sides of a triangle in inches are three consecutive integers. The length of the shortest side is 30% of the perimeter. What is the length of the longest side?
- (A) 7      (B) 8      (C) 9      (D) 10      (E) 11

**Answer:** E

**Solution 1(algebra solution)**

Let  $n$ ,  $n+1$ , and  $n+2$  be the lengths of the sides of the triangle. Then the perimeter of the triangle is  $n + (n+1) + (n+2) = 3n+3$ . Using the fact that the length of the smallest side is 30% of the perimeter, it follows that:

$n = 0.3(3n+3) \implies n = 0.9n + 0.9 \implies 0.1n = 0.9 \implies n = 9$ . The longest side is then  $n+2 = 11$ . Thus, answer choice **(E) 11** is correct.

**Solution 2**

Since the length of the shortest side is a whole number and is equal to  $\frac{3}{10}$  of the perimeter, it follows that the perimeter must be a multiple of 10. Adding the two previous integers to each answer choice, we see that  $11+10+9=30$ . Thus, answer choice **(E) 11** is correct.

14. (AMC 8 2010) What is the sum of the prime factors of 2010?
- (A) 67      (B) 75      (C) 77      (D) 201      (E) 210

**Answer:** C

**Solution**

First, we must find the prime factorization of 2010.  $2010 = 2 \cdot 3 \cdot 5 \cdot 67$ . We add the factors up to get **(C) 77**.

15. (AMC 8 2010) A jar contains five different colors of gumdrops: 30% are blue, 20% are brown, 15% red, 10% yellow, and the other 30 gumdrops are green. If half of the blue gumdrops are replaced with brown gumdrops, how many gumdrops will be brown?
- (A) 35      (B) 36      (C) 42      (D) 48      (E) 64

**Answer:** C

**Solution 1**

We do  $100 - 30 - 20 - 15 - 10$  to find the percent of gumdrops that are green. We find that 25% of the gumdrops are green. That means there are 120 gumdrops. If we replace half of the blue gumdrops with brown gumdrops, then 35% of the jar's gumdrops are brown.  $\frac{35}{100} \cdot 120 = 42 \Rightarrow \boxed{\text{(C) } 42}$

**Solution 2**

We first calculate the percentage of gumdrops that are green. Subtracting all the other percentages from 100%, we have:  $100 - 30 - 20 - 15 - 10 = 25\%$ .

Thus, 25% of the gumdrops are green.

Next, we set up a proportion:  $\frac{30}{25\%} = \frac{x}{30\%}$  to find the number of blue gumdrops. Cross-multiplying, we solve for  $x$ : 36

So there are 36 blue gumdrops.

We divide the number of blue gumdrops by 2 to get 18

Now, we calculate the number of brown marbles using the proportion:

$$\frac{30}{25\%} = \frac{y}{20\%},$$

where  $y$  is the number of brown marbles. Solving, we find  $y = 24$ .

Adding these together gives:

$$24 + 18 = 42.$$

Thus, the answer is  $\boxed{\text{(C) } 42}$

16. (AMC 8 2010) A square and a circle have the same area. What is the ratio of the side length of the square to the radius of the circle?

(A)  $\frac{\sqrt{\pi}}{2}$     (B)  $\sqrt{\pi}$     (C)  $\pi$     (D)  $2\pi$     (E)  $\pi^2$

**Answer:** B

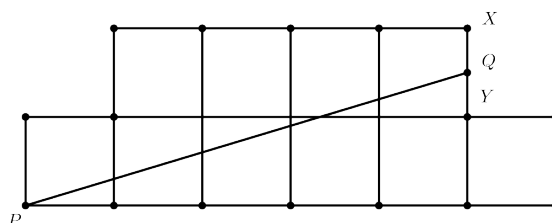
**Solution**

Let the side length of the square be  $s$ , and let the radius of the circle be  $r$ . Thus we have  $s^2 = r^2\pi$ . Dividing each side by  $r^2$ , we get  $\frac{s^2}{r^2} = \pi$ . Since  $\left(\frac{s}{r}\right)^2 = \frac{s^2}{r^2}$ , we have  $\frac{s}{r} = \sqrt{\pi} \Rightarrow \boxed{\text{(B) } \sqrt{\pi}}$

**Solution 2**

Let the area of both the circle and square be  $\pi$ . We know that the radius of a circle with area  $\pi$  is 1 and the side length of the square is  $\sqrt{\pi}$ . Now we can calculate the ratio  $\frac{\sqrt{\pi}}{1} \Rightarrow \boxed{\text{(B) } \sqrt{\pi}}$

17. (AMC 8 2010) The diagram shows an octagon consisting of 10 unit squares. The portion below  $\overline{PQ}$  is a unit square and a triangle with base 5. If  $\overline{PQ}$  bisects the area of the octagon, what is the ratio  $\frac{XQ}{QY}$ ?



- (A)  $\frac{2}{5}$       (B)  $\frac{1}{2}$       (C)  $\frac{3}{5}$       (D)  $\frac{2}{3}$       (E)  $\frac{3}{4}$

**Answer:** D

**Solution 1**

We see that half the area of the octagon is 5. We see that the triangle area is  $5 - 1 = 4$ . That means that  $\frac{5h}{2} = 4 \rightarrow h = \frac{8}{5}$ .

$$QY = \frac{8}{5} - 1 = \frac{3}{5}$$

Meaning,  $\frac{\frac{2}{5}}{\frac{3}{5}} = \boxed{\text{(D)} \frac{2}{3}}$

**Solution 2**

Like stated in solution 1, we know that half the area of the octagon is 5.

After moving the square from the bottom right to the top left, the area of the resulting trapezoid is  $5 + 1 = 6$ .

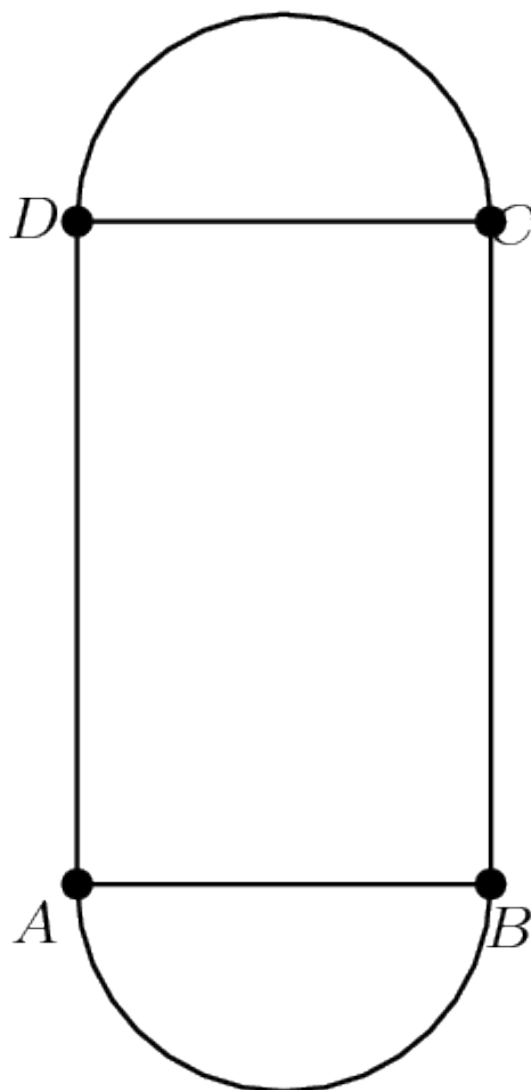
$5(XQ + 2)/2 = 6$ . Solving for  $XQ$ , we get  $XQ = 2/5$ .

Subtracting  $2/5$  from 1, we get  $QY = 3/5$ .

Therefore, the answer comes out to  $\boxed{\text{(D)} \frac{2}{3}}$

18. (AMC 8 2010) A decorative window is made up of a rectangle with semicircles on either end. The ratio of  $AD$  to  $AB$  is  $3 : 2$ , and  $AB$  is 30 inches. What is the ratio of the area of the rectangle to the combined areas of the semicircles?





- (A)  $2 : 3$     (B)  $3 : 2$     (C)  $6 : \pi$     (D)  $9 : \pi$     (E)  $30 : \pi$

**Answer:** C

**Solution 1**

We can set a proportion:

$$\frac{AD}{AB} = \frac{3}{2}$$

We substitute  $AB$  with 30 and solve for  $AD$ .

$$\frac{AD}{30} = \frac{3}{2}$$

$$AD = 45$$

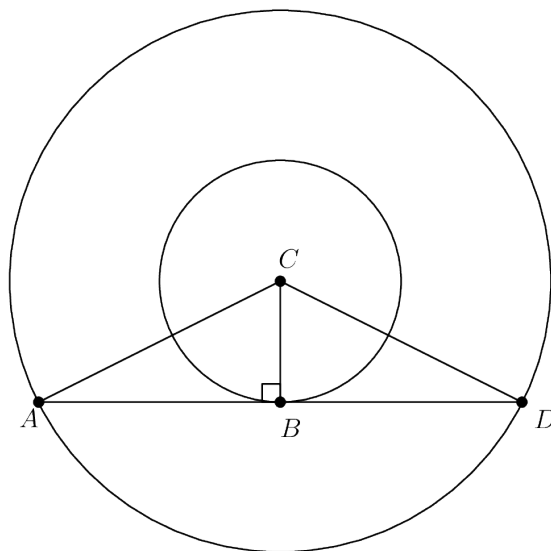
We calculate the combined area of semicircle by putting together semicircle  $AB$  and  $CD$  to get a circle with radius 15. Thus, the area is  $225\pi$ . The area of the rectangle is  $30 \cdot 45 = 1350$ . We calculate the ratio:

$$\frac{1350}{225\pi} = \frac{6}{\pi} \Rightarrow \boxed{\text{(C)} 6 : \pi}$$

**Solution 2 (more efficient)**

We can solve this without the measurement of 30 inches. Let the constant of proportionality between the sides of the rectangle be  $k$ .  $\frac{A_{rect}}{A_{circle}} = \frac{3k \cdot 2k}{\pi(\frac{2k}{2})^2} = \frac{6k^2}{\pi k^2} = \frac{6}{\pi} \Rightarrow \boxed{\text{(C)} 6 : \pi}$

19. (AMC 8 2010) The two circles pictured have the same center  $C$ . Chord  $\overline{AD}$  is tangent to the inner circle at  $B$ ,  $AC$  is 10, and chord  $\overline{AD}$  has length 16. What is the area between the two circles?



- (A)  $36\pi$     (B)  $49\pi$     (C)  $64\pi$     (D)  $81\pi$     (E)  $100\pi$

**Answer:** C

**Solution**

Since  $\triangle ACD$  is isosceles,  $CB$  bisects  $AD$ . Thus  $AB = BD = 8$ . From the Pythagorean Theorem,  $CB = 6$ . Thus the area between the two circles is  $100\pi - 36\pi = 64\pi$   $\boxed{\text{(C)} 64\pi}$

Note: The length  $AC$  is necessary information, as this tells us the radius of the larger circle. The area of the annulus is  $\pi(AC^2 - BC^2) = \pi AB^2 = 64\pi$ .

20. (AMC 8 2010) In a room,  $\frac{2}{5}$  of the people are wearing gloves, and  $\frac{3}{4}$  of the people are wearing hats. What is the minimum number of people in the room wearing both a hat and a glove?

- (A) 3    (B) 5    (C) 8    (D) 15    (E) 20

**Answer:** A

**Solution**

Let  $x$  be the number of people wearing both a hat and a glove. Since the number of people wearing a hat or a glove must be whole numbers, the number of people in the room must be a multiple of 4 and 5. Since we are trying to find the minimum  $x$ , we must use the *least* common multiple.  $\text{lcm}(4, 5) = 20$ . Thus, we can say

that there are 20 people in the room, all of which are wearing at least a hat or a glove. (Any people wearing neither item would unnecessarily increase the number of people in the room.)

It follows that there are  $\frac{2}{5} \cdot 20 = 8$  people wearing gloves and  $\frac{3}{4} \cdot 20 = 15$  people wearing hats. Then by applying the Principle of Inclusion and Exclusion (PIE), the total number of people in the room wearing either a hat or a glove or both is  $8 + 15 - x = 23 - x$ , where  $x$  is the number wearing both. Since everyone in the room is wearing at least one item (see above),  $23 - x = 20$ , and so  $x = \boxed{\text{(A)} 3}$ .

### Solution 2

We know that  $\frac{2}{5} = 40\%$ . (Let us call this Set A)

We also know that  $\frac{3}{4} = 75\%$  (Let us call this Set B)

We know that the total  $\#(A \text{ union } B)$  is equal to 100%. Using the Principle of Inclusion and Exclusion for two sets we know that  $\#(A \text{ union } B) = \#(A) + \#(B) - \#(A \text{ intersection } B)$ . So,  $100\% = 40\% + 75\% - \#(A \text{ intersection } B)$ . So  $\#(A \text{ intersection } B) = 15\%$ . We now find the smallest integer that is 15% of another integer. So the smallest integer that satisfies this is :A(3)

- In case you did not know what ' $\#$ ' is it is the cardinality or size of the set.

21. (AMC 8 2010) Hui is an avid reader. She bought a copy of the best seller *Math is Beautiful*. On the first day, Hui read  $\frac{1}{5}$  of the pages plus 12 more, and on the second day she read  $\frac{1}{4}$  of the remaining pages plus 15 pages. On the third day she read  $\frac{1}{3}$  of the remaining pages plus 18 pages. She then realized that there were only 62 pages left to read, which she read the next day. How many pages are in this book?

(A) 120      (B) 180      (C) 240      (D) 300      (E) 360

**Answer:** C

### Solution 1 (algebra solution)

Let  $x$  be the number of pages in the book. After the first day, Hui had  $\frac{4x}{5} - 12$  pages left to read. After the second, she had  $\left(\frac{3}{4}\right)\left(\frac{4x}{5} - 12\right) - 15 = \frac{3x}{5} - 24$  left. After the third, she had  $\left(\frac{2}{3}\right)\left(\frac{3x}{5} - 24\right) - 18 = \frac{2x}{5} - 34$  left. This is equivalent to 62.

$$\begin{aligned}\frac{2x}{5} - 34 &= 62 \\ 2x - 170 &= 310 \\ 2x &= 480 \\ x &= \boxed{\text{(C)} 240}\end{aligned}$$

### Solution 2 (working backwards)

On the last day, Hui read 62 pages. Working backwards, she read 18 pages more (80 total). 80 is  $\frac{2}{3}$  of the remaining pages, so the third morning Hui had 120 pages to read. Add back the 15 and you find 135 must be  $\frac{3}{4}$  of the remaining pages, so there were 180 pages on the second morning. Add 12 more, then 192 is  $\frac{4}{5}$  of the total book, which must be  $\boxed{\text{(C)} 240}$  pages long.

22. (AMC 8 2010) The hundreds digit of a three-digit number is 2 more than the units digit. The digits of the three-digit number are reversed, and the result is subtracted from the original three-digit number. What is the units digit of the result?

(A) 0      (B) 2      (C) 4      (D) 6      (E) 8

**Answer:** E

**Solution 1**

Let the hundreds, tens, and units digits of the original three-digit number be  $a$ ,  $b$ , and  $c$ , respectively. We are given that  $a = c + 2$ . The original three-digit number is equal to  $100a + 10b + c = 100(c + 2) + 10b + c = 101c + 10b + 200$ . The hundreds, tens, and units digits of the reversed three-digit number are  $c$ ,  $b$ , and  $a$ , respectively. This number is equal to  $100c + 10b + a = 100c + 10b + (c + 2) = 101c + 10b + 2$ . Subtracting this expression from the expression for the original number, we get  $(101c + 10b + 200) - (101c + 10b + 2) = 198$ . Thus, the units digit in the final result is **(E) 8**

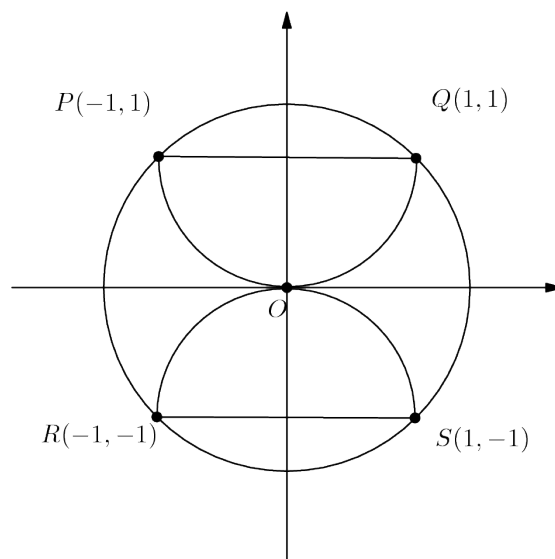
**Solution 2, Faster and Simpler**

The result must hold for any three-digit number with hundreds digit being 2 more than the units digit. 301 is such a number. Evaluating, we get  $301 - 103 = 198$ . Thus, the units digit in the final result is **(E) 8**. This is much faster than the first solution, and note that this method works for any number that has a hundreds digit 2 more than the units digit. AtomicBoy

**Solution 3**

Set the units digit of the original number as  $x$ . Thus, its hundreds digit is  $x + 2$ . After the digits are reversed, the hundreds digit of the original number will be the units digit of the new number. Since  $x - (x + 2) = -2$ , we can do regrouping and "borrow" 1 from the tens digit and bring it to the units digit as a 10. Therefore, the units digit will end up as  $-2 + 10 =$  **(E) 8**

23. (AMC 8 2010) Semicircles  $POQ$  and  $ROS$  pass through the center of circle  $O$ . What is the ratio of the combined areas of the two semicircles to the area of circle  $O$ ?



(A)  $\frac{\sqrt{2}}{4}$       (B)  $\frac{1}{2}$       (C)  $\frac{2}{\pi}$       (D)  $\frac{2}{3}$       (E)  $\frac{\sqrt{2}}{2}$

**Answer:** B

**Solution**

By the Pythagorean Theorem, the radius of the larger circle turns out to be  $\sqrt{1^2 + 1^2} = \sqrt{2}$ . Therefore, the area of the larger circle is  $(\sqrt{2})^2\pi = 2\pi$ . Using the coordinate plane given, we find that the radius of each of the two semicircles to be 1. So, the area of the two semicircles is  $1^2\pi = \pi$ . Finally, the ratio of the combined areas of the two semicircles to the area of circle  $O$  is  $\boxed{\text{(B)} \frac{1}{2}}$ .

24. (AMC 8 2010) What is the correct ordering of the three numbers,  $10^8$ ,  $5^{12}$ , and  $2^{24}$ ?  
 (A)  $2^{24} < 10^8 < 5^{12}$  (B)  $2^{24} < 5^{12} < 10^8$  (C)  $5^{12} < 2^{24} < 10^8$  (D)  $10^8 < 5^{12} < 2^{24}$   
 (E)  $10^8 < 2^{24} < 5^{12}$

**Answer:** A

**Solution 1**

Since all of the exponents are multiples of four, we can simplify the problem by taking the fourth root of each number. Evaluating we get  $10^2 = 100$ ,  $5^3 = 125$ , and  $2^6 = 64$ . Since  $64 < 100 < 125$ , it follows that  $\boxed{\text{(A)} 2^{24} < 10^8 < 5^{12}}$  is the correct answer.

**Solution 2**

First, let us make all exponents equal to 8. Then, it will be easy to order the numbers without doing any computations.  $10^8$  is as fine as it is. We can rewrite  $2^{24}$  as  $(2^3)^8 = 8^8$ . Then we can rewrite  $5^{12}$  as  $(5^{\frac{3}{2}})^8 = (\sqrt{125})^8$ . We take the eighth root of all of these to get 10, 8,  $\sqrt{125}$ . Obviously,  $8 < 10 < \sqrt{125}$ , so the answer is  $\boxed{\text{(A)} 2^{24} < 10^8 < 5^{12}}$ . Solution by Math

**Solution 3**

We know that  $10^8 = (5 \cdot 2)^8 = 5^8 \cdot 2^8$ . We also know that  $5^{12} = 5^{(8+4)} = 5^8 \cdot 5^4$ . If we remove the common factor of  $5^8$  from both expressions, we are left with  $2^8$ , which equals 256, and  $5^4$ , which equals 625. So we know that  $5^{12}$  is bigger than  $10^8$ . Now we need to figure out which is bigger,  $10^8$  or  $2^{24}$ . To do this, we rewrite  $2^{24}$  as  $(2^3)^8 = (8)^8$ , which is clearly less than  $10^8$ . Therefore,  $\boxed{\text{(A)} 2^{24} < 10^8 < 5^{12}}$  is the correct answer.

By naman14

25. (AMC 8 2010) Everyday at school, Jo climbs a flight of 6 stairs. Jo can take the stairs 1, 2, or 3 at a time. For example, Jo could climb 3, then 1, then 2. In how many ways can Jo climb the stairs?  
 (A) 13 (B) 18 (C) 20 (D) 22 (E) 24

**Answer:** E

**Solution 1**

A dynamics programming approach is quick and easy. The number of ways to climb one stair is 1. There are 2 ways to climb two stairs: 1,1 or 2. For 3 stairs, there are 4 ways: (1,1,1)(1,2)(2,1)(3)

For four stairs, consider what step they came from to land on the fourth stair. They could have hopped straight from the 1st, done a double from #2, or used a single step from #3. The ways to get to each of these steps are 1 + 2 + 4 = 7 ways to get to step 4. The pattern can then be extended: 4 steps: 1 + 2 + 4 = 7 ways. 5 steps: 2 + 4 + 7 = 13 ways. 6 steps: 4 + 7 + 13 = 24 ways.

Thus, there are  $\boxed{\text{(E)} 24}$  ways to get to step 6.

**Solution 2 (Generating a Function)**

We first start by doing some simple casework to try and find the number of ways we can get to the 1st, 2nd, and 3rd Stairs. The number of ways to climb one stair is 1. There are 2 ways to climb two stairs: 1,1 or 2. For 3 stairs, there are 4 ways: (1,1,1)(1,2)(2,1)(3)

Next we must try to generalize a recursive function to find the number of ways to climb the stairs. We start out with  $f(n)$ , which denotes the number of ways to climb the stairs. Then we must realize, To reach the  $n$ th stair, Jeff must take a 1 step from the  $f(n-1)$ th stair, a double step from the  $f(n-2)$ th stair, or a triple step from the  $f(n-3)$ th stair. Therefore, the number of ways of reaching the  $n$ th step is the number of ways of reaching the  $f(n-1)$ th plus the number of ways of reaching the  $f(n-2)$ th plus the number of ways of reaching the  $f(n-3)$ th stair. So we have our generalized recursive function:

$$f(n) = f(n-1) + f(n-2) + f(n-3)$$

Now that we have the function and a few of the values we can try to find  $f(6)$

$$f(1) = 1$$

$$f(2) = 2$$

$$f(3) = f(3-1) + f(3-2) + f(3-3) = f(2) + f(1) + f(0) = 2 + 1 + 1 = 4$$

$$f(4) = f(4-1) + f(4-2) + f(4-3) = f(3) + f(2) + f(1) = 4 + 2 + 1 = 7$$

$$f(5) = f(5-1) + f(5-2) + f(5-3) = f(4) + f(3) + f(2) = 7 + 4 + 2 = 13$$

$$f(6) = f(6-1) + f(6-2) + f(6-3) = f(5) + f(4) + f(3) = 13 + 7 + 4 = 24$$

Thus there are 24 ways to get to the 6th stair so the answer is (E) 24

**Solution 3 (Combinatorics)**

There are 7 ways to break down the 6 stairs. They are:

(1) 1, 1, 1, 1, 1, 1

(2) 2, 1, 1, 1, 1

(3) 2, 2, 1, 1

(4) 3, 1, 1, 1

(5) 2, 2, 2

(6) 3, 2, 1

(7) 3, 3

Let's count how many different orders in each way:

(1) 1

(2)  $5C1 = 5$

(3)  $4C2 = 6$

(4)  $4C1 = 4$

(5) 1

(6)  $3P3 = 6$

(7) 1

Thus, the sum of them is  $1 + 5 + 6 + 4 + 1 + 6 + 1 =$  (E)24.

**§27 AMC 8 2011**

1. (*AMC 8 2011*) Margie bought 3 apples at a cost of 50 cents per apple. She paid with a 5-dollar bill. How much change did Margie receive?

(A) \$1.50    (B) \$2.00    (C) \$2.50    (D) \$3.00    (E) \$3.50

**Answer:** E

**Solution**

50 cents is equivalent to \$0.50. Then the three apples cost  $3 \times \$0.50 = \$1.50$ . The change Margie receives is  $\$5.00 - \$1.50 =$  **(E) \$3.50**

-Mathloveralldxxx5223

2. (*AMC 8 2011*) Karl's rectangular vegetable garden is 20 feet by 45 feet, and Makenna's is 25 feet by 40 feet. Which of the following statements are true?

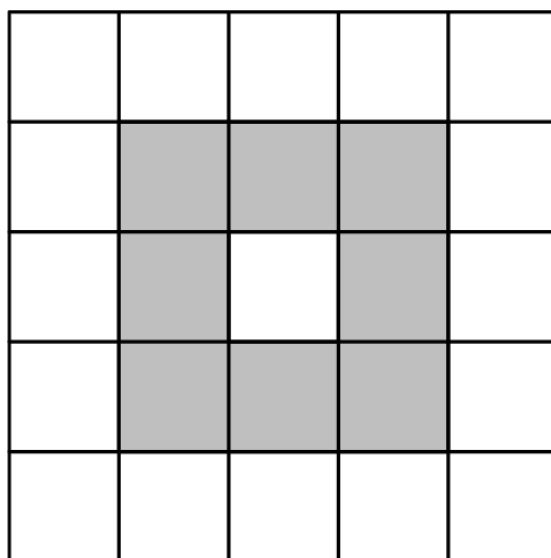
(A) Karl's garden is larger by 100 square feet. (B) Karl's garden is larger by 25 square feet.  
(C) The gardens are the same size. (D) Makenna's garden is larger by 25 square feet.  
(E) Makenna's garden is larger by 100 square feet.

**Answer:** E

**Solution**

The area of a rectangle is given by the formula length times width. Karl's garden is  $20 \times 45 = 900$  square feet and Makenna's garden is  $25 \times 40 = 1000$  square feet. Since  $1000 > 900$ , Makenna's garden is larger by  $1000 - 900 = 100$  square feet.  
 $\Rightarrow$  **(E) Makenna's garden is larger by 100 square feet.**

3. (*AMC 8 2011*) Extend the square pattern of 8 black and 17 white square tiles by attaching a border of black tiles around the square. What is the ratio of black tiles to white tiles in the extended pattern?

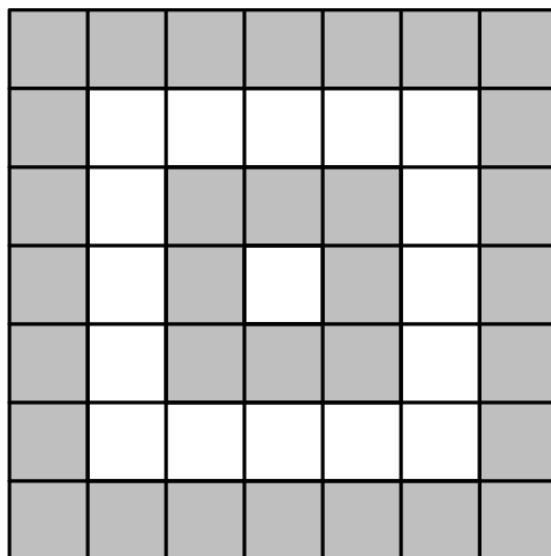


(A) 8 : 17    (B) 25 : 49    (C) 36 : 25    (D) 32 : 17    (E) 36 : 17

**Answer:** D

**Solution 1**

One way of approaching this is drawing the next circle of boxes around the current square.



We can now count the number of black and white tiles; 32 black tiles and 17 white tiles. This means the answer is **(D)** 32 : 17.

### Solution 2

If we did not want to draw a diagram though that is probably simpler in this case, we can imagine the last border of black tiles. We see that each side length (if the side length of a square is 1) increases by 2 each time we go out one layer. Therefore, we know that the next border will have a side length of 7. That border has  $7^2 - 5^2 = 24$  black squares in it. The other black border has  $3^2 - 1^2 = 8$  squares in it, so there are 32 black squares in all. The other ones must all be white, so there are  $49 - 32 = 17$  white squares. Thus, we get our answer of **(D)** 32 : 17.

### Solution 3 (somewhat fake)

Notice that in total, once we add the border there will be  $7^2$  squares total. So far as there is only one option that sums to 49 and we know that there are an integer number of boxes, it has to be **(D)** 32 : 17. math\_genius\_11

4. (AMC 8 2011) Here is a list of the numbers of fish that Tyler caught in nine outings last summer:

2, 0, 1, 3, 0, 3, 3, 1, 2.

Which statement about the mean, median, and mode is true?

(A) median < mean < mode      (B) mean < mode < median

(C) mean < median < mode      (D) median < mode < mean

(E) mode < median < mean

**Answer:** C

### Solution

First, put the numbers in increasing order.



0, 0, 1, 1, 2, 2, 3, 3, 3

The mean is  $\frac{0+0+1+1+2+2+3+3+3}{9} = \frac{15}{9}$ , the median is 2, and the mode is 3. Because,  $\frac{15}{9} < 2 < 3$ , the answer is **(C)** mean < median < mode

5. (AMC 8 2011) What time was it 2011 minutes after midnight on January 1, 2011?  
 (A) January 1 at 9:31 PM (B) January 1 at 11:51 PM (C) January 2 at 3:11 AM  
 (D) January 2 at 9:31 AM (E) January 2 at 6:01 PM

**Answer:** D

**Solution 1**

There are 60 minutes in an hour.  $2011/60 = 33\text{r}31$ , or 33 hours and 31 minutes. There are 24 hours in a day, so the time is 9 hours and 31 minutes after midnight on January 2, 2011.  $\Rightarrow$  **(D)** January 2 at 9:31AM

**Solution 2**

Like the previous solution, 2011 minutes is 33 hours and 31 minutes. That means it has to be January 1 at 9:31PM or January 2 at 9:31AM because they are the only ones that end with 31. But, the answer can't be in January 1 because there are 24 hours in a day and 33 is greater than 24. So, the answer is  $\Rightarrow$  **(D)** January 2 at 9:31AM.

-RealCXY

**Solution 3 (Estimate)**

The answer choices are relatively far apart, so we can use estimation to solve this problem.  $2100/60$  is around 35 hours, or about 1.5 days. The only answer choice around 1.5 days from the midnight of January 1, 2011 is **(D)** January 2 at 9:31AM.

MegaBoy6679 :D 23:31, 26 December 2022 (EST)

6. (AMC 8 2011) In a town of 351 adults, every adult owns a car, motorcycle, or both. If 331 adults own cars and 45 adults own motorcycles, how many of the car owners do not own a motorcycle?  
 (A) 20 (B) 25 (C) 45 (D) 306 (E) 351

**Answer:** D

**Solution 1**

By **PIE**, the number of adults who own both cars and motorcycles is  $331 + 45 - 351 = 25$ . Out of the 331 car owners, 25 of them own motorcycles and  $331 - 25 =$  **(D)** 306 of them don't.

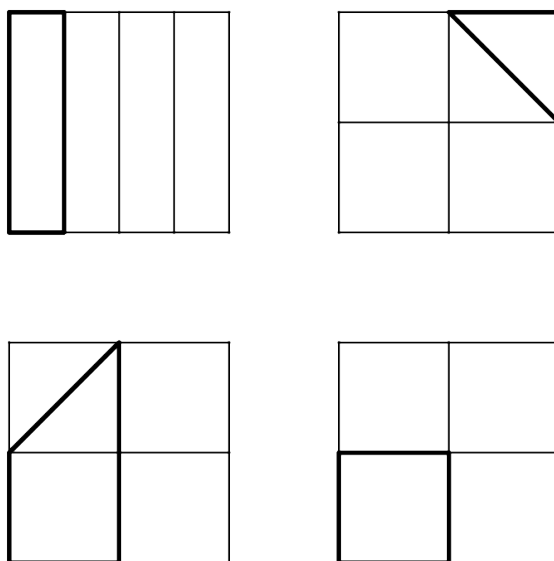
**Solution 2**

There are 351 total adults, and 45 own a motorcycle. The number of adults that don't own a motorcycle is  $351 - 45 = 306$ . Since everyone owns a car or motorcycle and one who doesn't own a motorcycle owns a car, the answer is **(D)** 306.

**Solution 3 (answer choices)**

Clearly, we can eliminate answer choice E, for at least one adult owns motorcycles. It is also fairly obvious that the answer must be in the 300 range, giving us **(D)** 306

7. (AMC 8 2011) Each of the following four large congruent squares is subdivided into combinations of congruent triangles or rectangles and is partially bolded. What percent of the total area is partially bolded?



- (A)  $12\frac{1}{2}$     (B) 20    (C) 25    (D)  $33\frac{1}{3}$     (E)  $37\frac{1}{2}$

**Answer:** C

**Solution**

Assume that the area of each square is 1. Then, the area of the bolded region in the top left square is  $\frac{1}{4}$ . The area of the top right bolded region is  $\frac{1}{8}$ . The area of the bottom left bolded region is  $\frac{3}{8}$ . And the area of the bottom right bolded region is  $\frac{1}{4}$ . Add the four fractions:  $\frac{1}{4} + \frac{1}{8} + \frac{3}{8} + \frac{1}{4} = 1$ . The four squares together have an area of 4, so the percentage bolded is  $\frac{1}{4} \cdot 100 = \boxed{\text{(C) } 25}$ .

8. (AMC 8 2011) Bag A has three chips labeled 1, 3, and 5. Bag B has three chips labeled 2, 4, and 6. If one chip is drawn from each bag, how many different values are possible for the sum of the two numbers on the chips?

- (A) 4    (B) 5    (C) 6    (D) 7    (E) 9

**Answer:** B

**Solution 1**

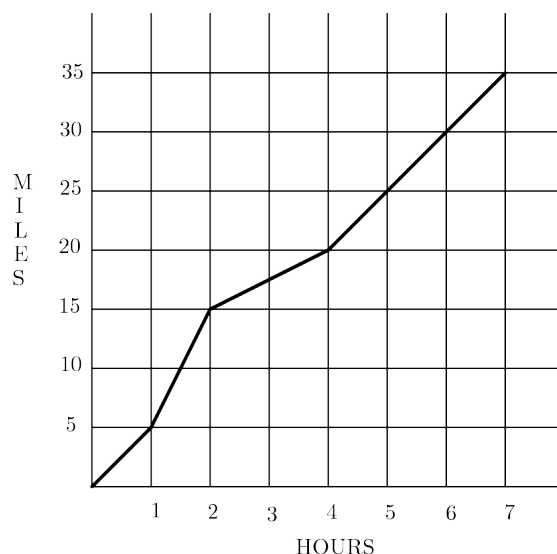
By adding a number from Bag A and a number from Bag B together, the values we can get are 3, 5, 7, 5, 7, 9, 7, 9, 11. Therefore the number of different values is  $\boxed{\text{(B) } 5}$ .

**Solution 2**

The sum of an even number added to an odd number is always odd. The smallest possible sum is 3, and the largest possible sum is 11. The odd numbers in between can be achieved by replacing chips with  $\pm 2$  within the same bag. Therefore, we can conclude that there are  $(11 - 3)/2(*) + 1 = \boxed{\text{(B) } 5}$  possible sums.

- Note that we are only accounting for odd numbers, thus the  $/2$ .

9. (AMC 8 2011) Carmen takes a long bike ride on a hilly highway. The graph indicates the miles traveled during the time of her ride. What is Carmen's average speed for her entire ride in miles per hour?



- (A) 2      (B) 2.5      (C) 4      (D) 4.5      (E) 5

**Answer:** E

**Solution**

We observe the graph and see that the shape of the graph does not matter. We only want the total time it took Carmen and the total distance she traveled. Based on the graph, Carmen traveled 35 miles for 7 hours. Therefore, her average speed is **(E) 5**

10. (AMC 8 2011) The taxi fare in Gotham City is 2.40 for the first  $1\frac{1}{2}$  mile and additional mileage charged at the rate 0.20 for each additional 0.1 mile. You plan to give the driver a 2 tip. How many miles can you ride for 10?

- (A) 3.0      (B) 3.25      (C) 3.3      (D) 3.5      (E) 3.75

**Answer:** C

**Solution 1 (Algebra)**

Let  $x$  be the number of miles you ride. The number of miles you ride after the first half mile is  $x - 0.5$ . We can write this equation:

$$10 = 2.4 + 0.2 \times \frac{x - 0.5}{0.1} + 2$$

$$5.6 = 2(x - 0.5)$$

$$2.8 = x - 0.5$$

$$x = \textbf{(C) } 3.3$$

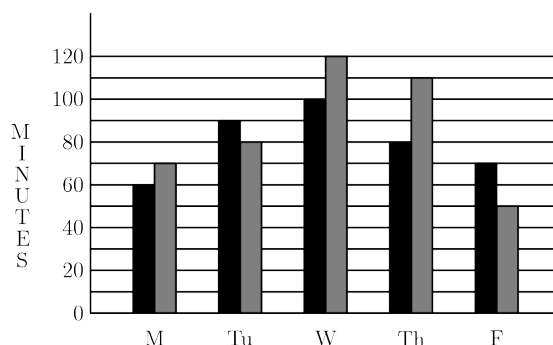
**Solution 2 (Table)**

| <i>Miles</i> | <i>Money</i> | <i>Remark</i>         |
|--------------|--------------|-----------------------|
| 0            | 8            | 2 for tip             |
| 0.5          | 5.6          | -2.4 for first 0.5 mi |
| 3.3          | 0            | -5.6 for 2.8 mi       |

Miles travelled = **(C) 3.3**

Solution by Anshulb, formatted by Cheetahboy93

11. (*AMC 8 2011*) The graph shows the number of minutes studied by both Asha (black bar) and Sasha (grey bar) in one week. On the average, how many more minutes per day did Sasha study than Asha?



- (A) 6    (B) 8    (C) 9    (D) 10    (E) 12

**Answer:** A

**Solution**

Average the differences between each day. We get 10, -10, 20, 30, -20. We find the average of this list to get **(A) 6**. ( In case you were wondering, the way to calculate the average is  $\frac{(10+(-10)+20+30+(-20))}{5} = \frac{30}{5} = 6$ . So the answer is indeed **(A) 6** )

**Solution 2**

This solution may take longer to do than the first solution. In total, Asha studied for 400 minutes a week (80 minutes per day) and Sasha studied for 430 minutes a week (86 minutes per day).  $86 - 80 = 6$ . Therefore, the answer is **(A) 6**.

-MiracleMaths

12. (*AMC 8 2011*) Angie, Bridget, Carlos, and Diego are seated at random around a square table, one person to a side. What is the probability that Angie and Carlos are seated opposite each other?

- (A)  $\frac{1}{4}$     (B)  $\frac{1}{3}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{3}$     (E)  $\frac{3}{4}$

**Answer:** B

**Solution 1**

If we designate a person to be on a certain side, then all placements of the other people can be considered unique. WLOG, assign Angie to be on the side. There are then  $3! = 6$  total seating arrangements. If Carlos is across from Angie, there are only  $2! = 2$  ways to fill the remaining two seats. Then the probability Angie and Carlos are seated opposite each other is  $\frac{2}{6} = \mathbf{(B) \frac{1}{3}}$ .

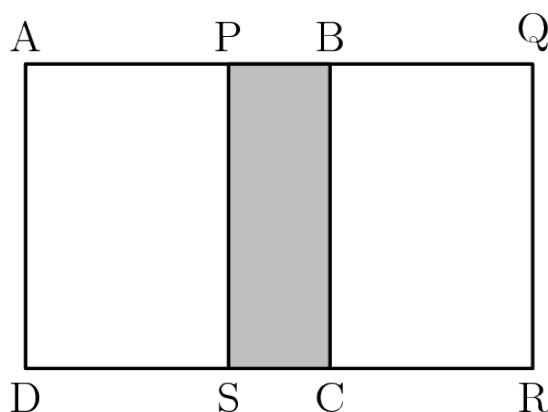
**Solution 2**

If we seat Angie first, there would be only one out of three ways Carlos can sit across from Angie. So the final answer is (B)  $\frac{1}{3}$

**Solution 3 (cheese)**

Three people can be seated opposite of Angie (Bridget, Carlos, or Diego). Only one of these is Carlos. Thus, the answer is (B)  $\frac{1}{3}$ .- @scthecool

13. (*AMC 8 2011*) Two congruent squares,  $ABCD$  and  $PQRS$ , have side length 15. They overlap to form the 15 by 25 rectangle  $AQRD$  shown. What percent of the area of rectangle  $AQRD$  is shaded?



- (A) 15      (B) 18      (C) 20      (D) 24      (E) 25

**Answer:** C

**Solution 2**

The length of BP is 5. the ratio of the areas is  $\frac{5 \cdot 15}{25 \cdot 15} = \frac{5}{25} = 20\%$ -Megacleverstarfish15

**Solution 3(similar to Solution 2)**

To find the overlap length, we do the total length of the squares and subtract 25(side length of figure).  $(15 + 15) - 25 = 5$ , so the overlap length is 5. To find what percentage of  $AQRD$  is shaded, we divide the shaded part by the area of the  $AQRD$ , so the percentage is  $\frac{15 \cdot 5}{15 \cdot 25} = \frac{5}{25} = \frac{1}{5} = \frac{20}{100} = 20\%$ , so the answer is

(C) 20.

14. (*AMC 8 2011*) There are 270 students at Colfax Middle School, where the ratio of boys to girls is 5 : 4. There are 180 students at Winthrop Middle School, where the ratio of boys to girls is 4 : 5. The two schools hold a dance and all students from both schools attend. What fraction of the students at the dance are girls?

- (A)  $\frac{7}{18}$       (B)  $\frac{7}{15}$       (C)  $\frac{22}{45}$       (D)  $\frac{1}{2}$       (E)  $\frac{23}{45}$

**Answer:** C

**Solution**

At Colfax Middle School, there are  $\frac{4}{9} \times 270 = 120$  girls. At Winthrop Middle School, there are  $\frac{5}{9} \times 180 = 100$  girls. The ratio of girls to the total number of students is

$$\frac{120+100}{270+180} = \frac{220}{450} = \boxed{\text{(C)} \frac{22}{45}}$$

### Solution 2 (Guess and Check)

Since we know there are 270 kids at Colfax Middle School and we know the ratio of boys to girls is  $5 : 4$ , we can use guess and check to find the number of girls at the dance (and vice versa).  $5 : 4 = 150 : 120$  and  $4 : 5 = 80 : 100$ . Now that the ratios sum to 270 and 180, we know that in Colfax there are 120 girls and in Winthrop, there are 100 girls.  $120 + 100 = 220$  total girls and  $270 + 180 = 450$  total people.

$$\frac{220}{450} = \boxed{\text{(C)} \frac{22}{45}}$$

### Solution 3 (Educated Guess)

We know that since 270 is greater than 180 and how the ratios are opposites, we know that there will be more boys than girls, so the fraction of students at the dance that are girls should be less than  $\frac{1}{2}$ . We also know that the amount of girls and the amount of boys are close, so the fraction of students at the dance that are girls should be really close to  $\frac{1}{2}$ . The only answer choice that/best satisfies these

conditions is  $\boxed{\text{(C)} \frac{22}{45}}$ .

15. (AMC 8 2011) How many digits are in the product  $4^5 \cdot 5^{10}$ ?

(A) 8      (B) 9      (C) 10      (D) 11      (E) 12

**Answer:** D

**Solution**

$$4^5 \cdot 5^{10} = 2^{10} \cdot 5^{10} = 10^{10}.$$

That is one 1 followed by ten 0's, which is  $\boxed{\text{(D)} 11}$  digits.

### Solution 2

$4^5$  has 4 digits while  $5^{10}$  has 7 digits. This means that  $4^5 \cdot 5^{10}$  has a total of  $7 + 4 = \boxed{\text{(D)} 11}$  digits.

16. (AMC 8 2011) Let  $A$  be the area of the triangle with sides of length 25, 25, and 30. Let  $B$  be the area of the triangle with sides of length 25, 25, and 40. What is the relationship between  $A$  and  $B$ ?

(A)  $A = \frac{9}{16}B$       (B)  $A = \frac{3}{4}B$       (C)  $A = B$       (D)  $A = \frac{4}{3}B$       (E)  $A = \frac{16}{9}B$

**Answer:** C

### Solution 1 (Using the Pythagorean Theorem)

#### 25-25-30

We can draw the altitude for the side with length 30. By HL Congruence, the two triangles formed are congruent. Thus the altitude splits the side with length 30 into two segments with length 15. By the **Pythagorean Theorem**, we have

$$15^2 + x^2 = 25^2$$

$$\begin{aligned}
 x^2 &= 25^2 - 15^2 \\
 x^2 &= (25 + 15)(25 - 15) \\
 x^2 &= 40 \cdot 10 \\
 x^2 &= 400 \\
 x &= \sqrt{400} \\
 x &= 20
 \end{aligned}$$

Thus we have two 15-20-25 right triangles.

### 25-25-40

We can draw the altitude for the side with length 40. By HL Congruence, the two triangles formed are congruent. Thus the altitude splits the side with length 40 into two segments with length 20. From the 25-25-30 case, we know that the other side length is 15, so we have two 15-20-25 right triangles. Let the area of a 15-20-25 right triangle be  $x$ .

$$a = 2x$$

$$b = 2x$$

$$\boxed{(C) \ A = B}$$

### Solution 2 (Using Heron's Formula)

Using Heron's formula, we can calculate the area of the two triangles. The formula states that

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

where  $s$  is the semiperimeter of a triangle with side lengths  $a$ ,  $b$ , and  $c$ .

For the 25-25-30 triangle,

$$s = \frac{25 + 25 + 30}{2} = 40$$

Therefore,

$$A = \sqrt{40 \cdot 15 \cdot 15 \cdot 10} = 300$$

For the 25-25-40 triangle,

$$s = \frac{25 + 25 + 40}{2} = 45$$

Therefore,

$$B = \sqrt{45 \cdot 20 \cdot 20 \cdot 5} = 300$$

Hence,

$$A = B \implies \boxed{(C)}$$

### Solution 3 (Simple)

**Assume** the triangles are right. Then, the 25-25-30 has hypotenuse 30, whereas 25-25-40 has hypotenuse 40. Because the formula for a triangles' area is  $(bh)/2$ , the hypotenuse does not matter. So, since the base and height are the same, the area will also be the same meaning

$$A = B \implies \boxed{(C)}$$

17. (AMC 8 2011) Let  $w$ ,  $x$ ,  $y$ , and  $z$  be whole numbers. If  $2^w \cdot 3^x \cdot 5^y \cdot 7^z = 588$ , then what does  $2w + 3x + 5y + 7z$  equal?

(A) 21      (B) 25      (C) 27      (D) 35      (E) 56

**Answer:** A

**Solution**

The **prime factorization** of 588 is  $2^2 \cdot 3 \cdot 7^2$ . We can see  $w = 2$ ,  $x = 1$ , and  $z = 2$ . Because  $5^0 = 1$ ,  $y = 0$ .

$$2w + 3x + 5y + 7z = 4 + 3 + 0 + 14 = \boxed{\text{(A)} 21}$$

18. (AMC 8 2011) A fair 6-sided die is rolled twice. What is the probability that the first number that comes up is greater than or equal to the second number?

(A)  $\frac{1}{6}$       (B)  $\frac{5}{12}$       (C)  $\frac{1}{2}$       (D)  $\frac{7}{12}$       (E)  $\frac{5}{6}$

**Answer:** D

**Solution**

There are  $6 \cdot 6 = 36$  ways to roll the two dice, and 6 of them result in two of the same number. Out of the remaining  $36 - 6 = 30$  ways, the number of rolls where the first dice is greater than the second should be the same as the number of rolls where the second dice is greater than the first. In other words, there are  $30/2 = 15$  ways the first roll can be greater than the second. The probability the first number is greater than or equal to the second number is

$$\frac{15 + 6}{36} = \frac{21}{36} = \boxed{\text{(D)} \frac{7}{12}}$$

**Solution 2**

If we list out some ways the first dice is greater than the second dice, we might see a pattern.

If dice 2 is 1, then dice 1 can be 1, 2, 3, 4, 5, or 6.

If dice 2 is 2, then dice 1 can be 2, 3, 4, 5, or 6.

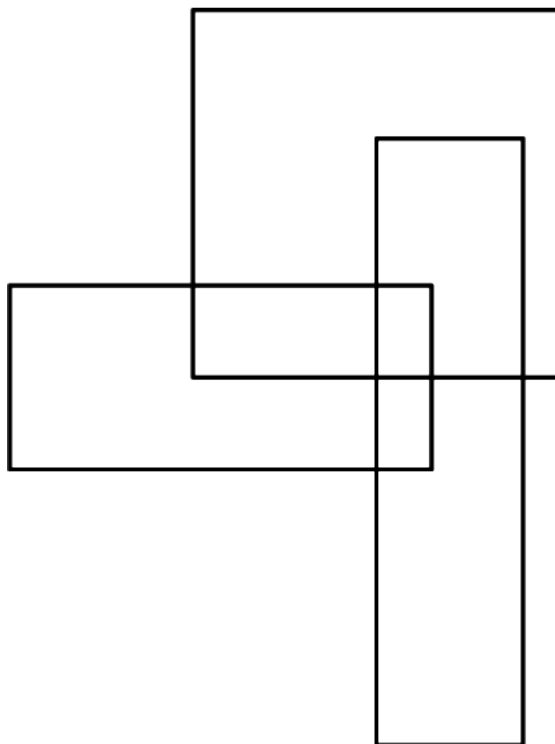
If dice 2 is 3, then dice 1 can be 3, 4, 5, or 6.

In the first case, dice 1 had six values it could land on. In the second case, dice 1 had five values it could land on. In the third case, dice 1 had four values it could land on. If we keep going, we would eventually get the pattern  $6 + 5 + 4 + 3 + 2 + 1 = 21$  as the number of total values dice 1 could land on if it's greater or equal to dice 2.

Since there are 36 ways two dice could land, our answer is  $\frac{21}{36}$ , or  $\boxed{\text{(D)} \frac{7}{12}}$ .

19. (AMC 8 2011) How many rectangles are in this figure?





- (A) 8    (B) 9    (C) 10    (D) 11    (E) 12

**Answer:** D

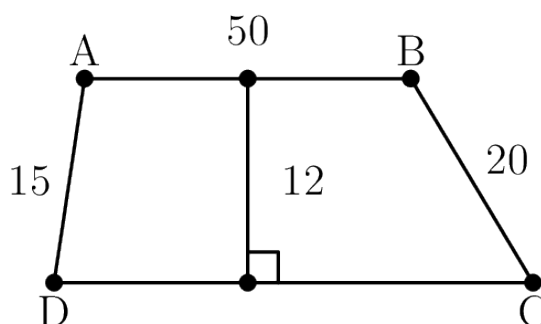
**Solution 1 (Splitting It Up)**

The figure can be divided into 7 sections. The number of rectangles with just one section is 3. The number of rectangles with two sections is 5. There are none with only three sections. The number of rectangles with four sections is 3.  $3 + 5 + 3 = \boxed{\text{(D)} 11}$

**Solution 2 (Count by Reduction)**

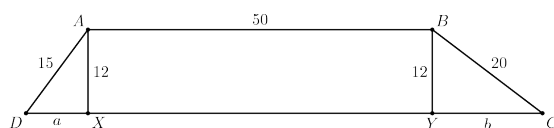
We can remove the 3 big blocks of rectangles one by one.  $7$  (left)  $+ 3$  (bottom)  $+ 1 = 11$  are removed in total.

20. (AMC 8 2011) Quadrilateral  $ABCD$  is a trapezoid,  $AD = 15$ ,  $AB = 50$ ,  $BC = 20$ , and the altitude is 12. What is the area of the trapezoid?



- (A) 600    (B) 650    (C) 700    (D) 750    (E) 800

**Answer:** D

**Solution**

If you draw altitudes from  $A$  and  $B$  to  $CD$ , the trapezoid will be divided into two right triangles and a rectangle. You can find the values of  $a$  and  $b$  with the **Pythagorean theorem**.

$$a = \sqrt{15^2 - 12^2} = \sqrt{81} = 9$$

$$b = \sqrt{20^2 - 12^2} = \sqrt{256} = 16$$

$ABYX$  is a rectangle so  $XY = AB = 50$ .

$$CD = a + XY + b = 9 + 50 + 16 = 75$$

The area of the trapezoid is

$$12 \cdot \frac{(50 + 75)}{2} = 6(125) = \boxed{\text{(D) } 750}$$

21. (AMC 8 2011) Students guess that Norb's age is 24, 28, 30, 32, 36, 38, 41, 44, 47, and 49. Norb says, "At least half of you guessed too low, two of you are off by one, and my age is a prime number." How old is Norb?

(A) 29      (B) 31      (C) 37      (D) 43      (E) 48

**Answer: C**

**Solution 1**

At least half the guesses are too low, so Norb's age must be greater than 36.

If two of the guesses are off by one, then his age is in between two guesses whose difference is 2. It could be 31, 37, or 48, but because his age is greater than 36 it can only be 37 or 48.

Lastly, Norb's age is a prime number so the answer must be **(C) 37**.

**Solution 2 (Alternative approach)**

Since two guesses are off by one, we know that both  $x + 1$  and  $x - 1$  are in the list where  $x$  is the age of Norb. Now, we know that  $x + 1$  and  $x - 1$  are 28 and 30, 30 and 32, 36 and 38 and 47 and 49. From these values, we know that  $x$  must be 29, 31, and 37. Since half of the guesses are too low, 24, 28, 30, 32, and 36 are all too low so we can eliminate all numbers in our list lesser than or equal to 36. Therefore, our list has only 37 left so the answer is **(C) 37**.

**Solution 3 (Using the Answer Choices)**

If his age is 29, then there won't be half of them who guessed less than his age. If his age is 31, then there still won't be half of them who guessed less than his age. If his age is 37, then there are half of them who guessed too low, half of them who guessed too high, and there are two people who guessed off by one. So, our answer

is probably (C) 37, but let's still check the other two choices. If his age is 43, then there won't be two people who were off by one. Finally, if his age is 48, then his age won't be a prime number. Therefore, our answer is **(C) 37**.

22. (AMC 8 2011) What is the **tens** digit of  $7^{2011}$ ?

(A) 0      (B) 1      (C) 3      (D) 4      (E) 7

**Answer:** D

**Solution 1**

Since we want the tens digit, we can find the last two digits of  $7^{2011}$ . We can do this by using modular arithmetic.

$$7^1 \equiv 07 \pmod{100}.$$

$$7^2 \equiv 49 \pmod{100}.$$

$$7^3 \equiv 43 \pmod{100}.$$

$$7^4 \equiv 01 \pmod{100}.$$

We can write  $7^{2011}$  as  $(7^4)^{502} \times 7^3$ . Using this, we can say:

$$7^{2011} \equiv (7^4)^{502} \times 7^3 \equiv 7^3 \equiv 343 \equiv 43 \pmod{100}.$$

From the above, we can conclude that the last two digits of  $7^{2011}$  are 43. Since they have asked us to find the tens digit, our answer is **(D) 4**.

-Ilovefruits

**Solution 2**

The last two digits of  $7^{2011}$  can be found by taking  $\pmod{100}$  of  $7^{2011}$ . By **Euler's Totient Theorem**, we have that  $7^{\phi(100)} \equiv 7^{40} \equiv 1 \pmod{100}$ , so we can conclude that  $7^{2011} \equiv 7^{40^{50} \cdot 11} \equiv 7^{11} \pmod{100}$ . We can find that the modular inverse of 7  $\pmod{100}$  is 43, and since  $7^{12} \equiv 01 \pmod{100}$ , we can conclude that  $7^{11} \equiv 7^{12} \cdot 7^{-1} \equiv 1 \cdot 43 \equiv 43 \pmod{100}$ . The problem asked for the tens digit, so our answer is **(D) 4**. Baihly2024

23. (AMC 8 2011) How many 4-digit positive integers have four different digits, where the leading digit is not zero, the integer is a multiple of 5, and 5 is the largest digit?

(A) 24      (B) 48      (C) 60      (D) 84      (E) 108

**Answer:** D

**Solution 1**

We can separate this into two cases. If an integer is a multiple of 5, the last digit must be either 0 or 5.

Case 1: The last digit is 5. The leading digit can be 1, 2, 3, or 4. Because the second digit can be 0 but not the leading digit, there are also 4 choices. The third digit cannot be the leading digit or the second digit, so there are 3 choices. The number of integers in this case is  $4 \cdot 4 \cdot 3 \cdot 1 = 48$ .

Case 2: The last digit is 0. Because 5 is the largest digit, one of the remaining three digits must be 5. There are 3 ways to choose which digit should be 5. The remaining digits can be 1, 2, 3, or 4, but since they have to be different there are  $4 \cdot 3$  ways to choose. The number of integers in this case is  $1 \cdot 3 \cdot 4 \cdot 3 = 36$ .

Therefore, the answer is  $48 + 36 =$  **(D) 84**.

24. (AMC 8 2011) In how many ways can 10001 be written as the sum of two primes?  
 (A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** A

**Solution**

For the sum of two numbers to be odd, one must be odd and the other must be even, because all odd numbers are of the form  $2n + 1$  where  $n$  is an integer, and all even numbers are of the form  $2m$  where  $m$  is an integer.

$$2n + 1 + 2m = 2m + 2n + 1 = 2(m + n) + 1$$

and  $m + n$  is an integer because  $m$  and  $n$  are both integers. The only even prime number is 2, so our only combination could be 2 and 9999. But, 9999 is clearly divisible by 3, so the number of ways 10001 can be written as the sum of two primes is (A) 0.

**Solution 2(Simple)**

First, we noticed that 10001 is equal to 5000+5001, if you subtract  $n$  to 5000 and add  $n$  to 5001, you always get an even number, even number is never a prime number except 2. We also see that whenever an addend is an odd number, the other addend will be even, so having an odd number as an addend is not possible, other than 9999 and 2, because 2 is a prime. We try 2 and 9999 but we can see 9999 is divisible by 3 and 9 clearly. So the answer is (A)0

**Solution 3 (assumed previous knowledge)**

It is helpful to know and understand the Goldbach Conjecture - that every even number can be written as the sum of 2 primes - and also, that the **odd numbers** that are the sum of two primes are exactly two more than a prime. This is because to make the sum of two numbers odd, you must have one even and one odd. There is only one even prime, which is two, so the sum will be of the form  $2 + p$ . Hence, the odd numbers that are the sum of two primes are exactly 2 more than a prime. Relating to the problem, 10001 is not 2 more than a prime, because  $10001 - 2 = 9999$  and we can easily see that 9999 is divisible by 3. Therefore, 10001 cannot be written as the sum of two primes, and so therefore the answer is (A) 0 mk

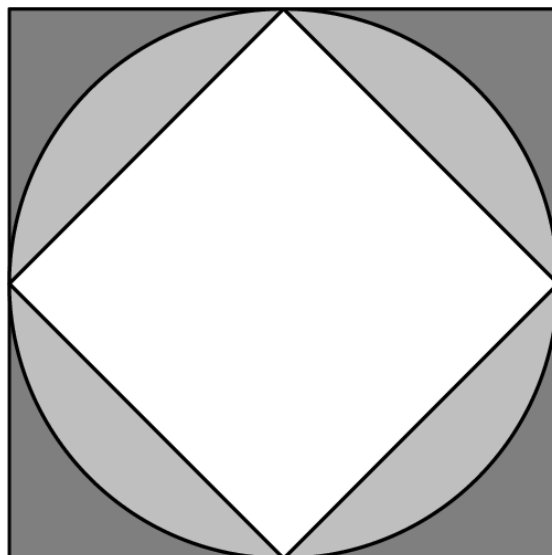
Note: the Goldbach Conjecture isn't solved, however you can assume it to be true for basically all numbers you see in competitions because many supercomputers tried to find a counterexample (went really high) but they didn't.

For proof based competitions where the graders actually like read your solutions don't use it they won't accept the conjecture as a solution

**Solution 4 (similar to solution 1)**

Our equation can be rewritten to  $Odd = Even + Odd$ . Since both addends have to be prime, one has to be even and one prime, which means it's 2. Plugging this in, the other number has to be 9999 which is not prime (divisible by 9) meaning there are (A) 0 answers.

25. (AMC 8 2011) A circle with radius 1 is inscribed in a square and circumscribed about another square as shown. Which fraction is closest to the ratio of the circle's shaded area to the area between the two squares?



- (A)  $\frac{1}{2}$     (B) 1    (C)  $\frac{3}{2}$     (D) 2    (E)  $\frac{5}{2}$

**Answer:** A

**Solution 1**

The area of the smaller square is one half of the product of its diagonals. Note that the distance from a corner of the smaller square to the center is equivalent to the circle's radius so the diagonal is equal to the diameter:  $2 \cdot 2 \cdot \frac{1}{2} = 2$ .

The circle's shaded area is the area of the smaller square subtracted from the area of the circle:  $\pi - 2$ .

If you draw the diagonals of the smaller square, you will see that the larger square is split 4 congruent half-shaded squares. The area between the squares is equal to the area of the smaller square: 2.

Approximating  $\pi$  to 3.14, the ratio of the circle's shaded area to the area between the two squares is about

$$\frac{\pi - 2}{2} \approx \frac{3.14 - 2}{2} = \frac{1.14}{2} \approx \boxed{\text{(A)} \frac{1}{2}}$$

**Solution 2**

For the ratio of the circle's shaded area to the area between the squares to be 1, they would have to be approximately the same size. For any ratio larger than that, the circle's shaded area must be greater. However, we can clearly see that the circle's shaded area is part of the area between the squares, and is approximately

$$\boxed{\text{(A)} \frac{1}{2}}$$

. The ratio must be less than 1, because the area inside the circle is less than the total shaded area, and the only answer choice that fits that is  $\frac{1}{2}$ . Note that this solution is not rigorous, because we still should show that the ratio is less than  $\frac{3}{4}$ .

**Solution 3**

Set the side length of the bigger square to be 8. Then the area of the big square is  $8^2 = 64$  and the area of the small square  $(4\sqrt{2})^2 = 32$ , making the difference 32.

The area of the circle is  $4^2$  times  $\pi$  which is  $16\pi$  or about 48. Knowing the area of the small square is 32.  $48 - 32$  is 16. The area of the big square is 64. So  $32/64$  is  $1/2$ , or **(A)**  $\frac{1}{2}$ .

**Solution 4 (Non-rigorous speed solution)**

Note that the shaded area of the circle is existent only between the gap between the two squares. This implies that the shaded area is less than the gap, and the only answer that shows that is **(A)**  $\frac{1}{2}$ .

- by nickelslordm

## §28 AMC 8 2012

1. (*AMC 8 2012*) Rachelle uses 3 pounds of meat to make 8 hamburgers for her family. How many pounds of meat does she need to make 24 hamburgers for a neighborhood picnic?

**(A)** 6      **(B)**  $6\frac{2}{3}$       **(C)**  $7\frac{1}{2}$       **(D)** 8      **(E)** 9

**Answer:** E

**Solution 1**

Since Rachelle uses 3 pounds of meat to make 8 hamburgers, she uses  $\frac{3}{8}$  pounds of meat to make one hamburger. She'll need 24 times that amount of meat for 24 hamburgers, or  $\frac{3}{8} \cdot 24 = \mathbf{(E) 9}$ .

**Solution 2**

We have the proportion  $\frac{3}{8} = \frac{x}{24}$ . To solve for  $x$ , multiply the left side of the equation by  $\frac{3}{3}$ . This gives us  $\frac{9}{24} = \frac{x}{24}$ . For  $\frac{9}{24}$  to equal  $\frac{x}{24}$ , we find that  $x = 9$ . The answer is then **(E)**. Andrew\_Lu with some small edits from: HAwaii\_lu

2. (*AMC 8 2012*) In the country of East Westmore, statisticians estimate there is a baby born every 8 hours and a death every day. To the nearest hundred, how many people are added to the population of East Westmore each year?

**(A)** 600      **(B)** 700      **(C)** 800      **(D)** 900      **(E)** 1000

**Answer:** B

**Solution**

There are  $24 \text{ hours} \div 8 \text{ hours} = 3$  births and one death everyday in East Westmore. Therefore, the population increases by  $3 - 1 = 2$  people everyday. Thus, there are  $2 \times 365 = 730$  people added to the population every year. Rounding, we find the answer is **(B) 700**.

3. (*AMC 8 2012*) On February 13 *The Oshkosh Northwestern* listed the length of daylight as 10 hours and 24 minutes, the sunrise was 6 : 57AM, and the sunset as 8 : 15PM. The length of daylight and sunrise were correct, but the sunset was wrong. When did the sun really set?

**(A)** 5 : 10PM      **(B)** 5 : 21PM      **(C)** 5 : 41PM      **(D)** 5 : 57PM      **(E)** 6 : 03PM

**Answer:** B

**Solution 1**

The problem wants us to find the time of sunset and gives us the length of daylight and time of sunrise. So all we have to do is add the length of daylight to the time of sunrise to obtain the answer. Convert 10 hours and 24 minutes into 10 : 24 in order to add easier.

Adding, we find that the time of sunset is  $6 : 57\text{AM} + 10 : 24 \implies 17 : 21 \implies$   
**(B) 5 : 21PM**.

### Solution 2 (Answer choices)

The length of daylight is 10 hours and 24 minutes, but for now, we can just look at the 24 minutes. Add 24 minutes to 57 part of sunrise, and you see that the sunset time must end in a 21. We see that the only option that ends with 21 is  $\implies$  **(B) 5 : 21PM**.

4. (AMC 8 2012) Peter's family ordered a 12-slice pizza for dinner. Peter ate one slice and shared another slice equally with his brother Paul. What fraction of the pizza did Peter eat?

(A)  $\frac{1}{24}$       (B)  $\frac{1}{12}$       (C)  $\frac{1}{8}$       (D)  $\frac{1}{6}$       (E)  $\frac{1}{4}$

**Answer:** C

### Solution 1

Peter ate  $1 + \frac{1}{2} = \frac{3}{2}$  slices. The pizza has 12 slices total. Taking the ratio of the amount of slices Peter ate to the amount of slices in the pizza, we find that Peter ate  $\frac{\frac{3}{2} \text{ slices}}{12 \text{ slices}} =$  **(C)  $\frac{1}{8}$**  of the pizza.

### Solution 2

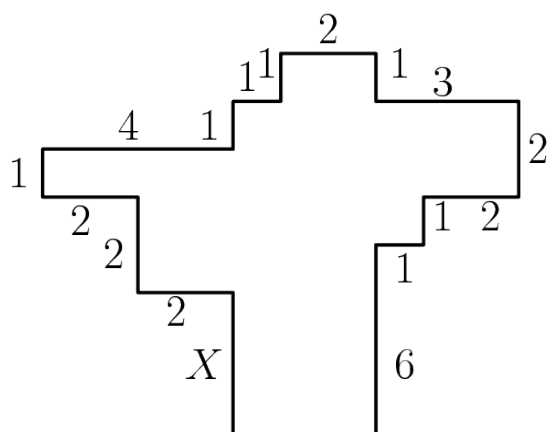
An alternative way of solving this problem is to add the slices separately. When Peter takes a full slice, he takes  $\frac{1}{12}$  of the pizza. When he shares the slice with Paul, he splits a slice of pizza into two, the equation is  $\frac{1}{2}$  of  $\frac{1}{12}$ , which is  $\frac{1}{24}$ . Adding gives:

$$\frac{1}{12} + \frac{1}{24} = \textbf{(C) } \frac{1}{8}$$

### Solution 3

Just like the above solutions, we know that Peter ate  $\frac{3}{2}$  slices, which is just 1.5 in decimals. There are 12 slices in total, so the fraction would be  $\frac{1.5}{12}$ . To remove the decimal, we can multiply it by 2 to get  $\frac{3}{24}$ , which is just **(C)  $\frac{1}{8}$**  of the pizza.

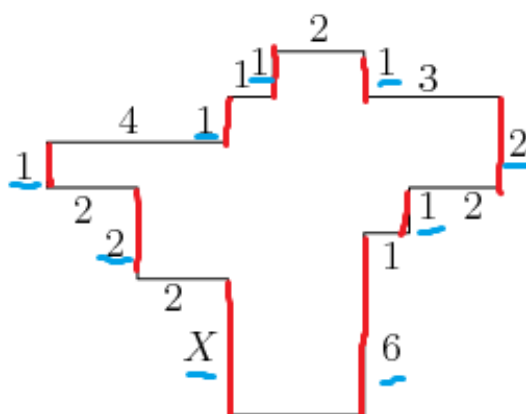
5. (AMC 8 2012) In the diagram, all angles are right angles and the lengths of the sides are given in centimeters. Note that the diagram is not drawn to scale. What is the length of  $X$ , in centimeters?



- (A) 1    (B) 2    (C) 3    (D) 4    (E) 5

**Answer:** E

**Solution 1**



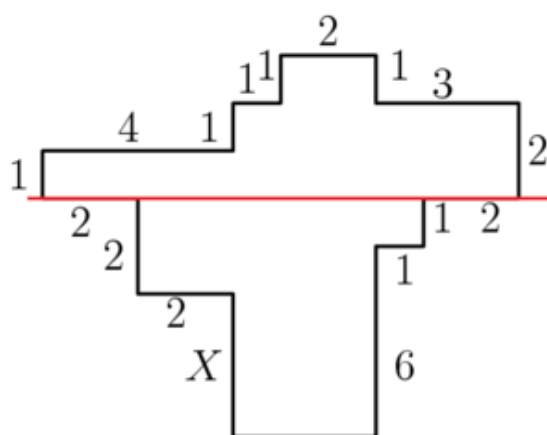
$$1 + 1 + 1 + 2 + X = 1 + 2 + 1 + 6$$

$$5 + X = 10$$

$$X = 5$$

Thus, the answer is **(E) 5**.

**Solution 2**





Note that we only need to consider the value below the marked red line, so we have the equation:

$$X + 2 = 6 + 1$$

$$X = 5$$

Hence, the answer is **(E) 5**.

6. (*AMC 8 2012*) A rectangular photograph is placed in a frame that forms a border two inches wide on all sides of the photograph. The photograph measures 8 inches high and 10 inches wide. What is the area of the border, in square inches?

(A) 36      (B) 40      (C) 64      (D) 72      (E) 88

**Answer:** E

**Solution**

In order to find the area of the frame, we need to subtract the area of the photograph from the area of the photograph and the frame together. The area of the photograph is  $8 \times 10 = 80$  square inches. The height of the whole frame (including the photograph) would be  $8 + 2 + 2 = 12$ , and the width of the whole frame,  $10 + 2 + 2 = 14$ . Therefore, the area of the whole figure would be  $12 \times 14 = 168$  square inches. Subtracting the area of the photograph from the area of both the frame and photograph, we find the answer to be  $168 - 80 = \mathbf{(E) 88}$ .

7. (*AMC 8 2012*) Isabella must take four 100-point tests in her math class. Her goal is to achieve an average grade of 95 on the tests. Her first two test scores were 97 and 91. After seeing her score on the third test, she realized she can still reach her goal. What is the lowest possible score she could have made on the third test?

(A) 90      (B) 92      (C) 95      (D) 96      (E) 97

**Answer:** B

**Solution 1**

Isabella wants an average grade of 95 on her 4 tests; this also means that she wants the sum of her test scores to be at least  $95 \times 4 = 380$  (if she goes over this number, she'll be over her goal!). She's already taken two tests, which sum to  $97 + 91 = 188$ , which means she needs 192 more points to achieve her desired average. In order to minimize the score on the third test, we assume that Isabella will receive all 100 points on the fourth test. Therefore, the lowest Isabella could have scored on the third test would be  $192 - 100 = \mathbf{(B) 92}$ .

**Solution 2**

Isabella needs to lose a max of 20. When she got 97 and 91, the max goes down to  $20 - 3 - 9 = 8$ . If we get 100 on her last test, then she will get **(B) 92** on the third test.

**Solution 3**

Let the sum of the first three test scores be  $x$ . If it's still possible to get an average of 95 after the third test, then the lowest score possible would be

$$\frac{x + 100}{4} = 95.$$

Then, solving the equation, we get

$$x = 280.$$

Given that the previous test scores are 97 and 91, we can subtract those score from it, which would give that the lowest score possible for the test is

$$280 - 97 - 91 = \boxed{\text{(B)} 92}$$

8. (AMC 8 2012) A shop advertises everything is "half price in today's sale." In addition, a coupon gives a 20% discount on sale prices. Using the coupon, the price today represents what percentage off the original price?

(A) 10      (B) 33      (C) 40      (D) 60      (E) 70

**Answer:** D

**Solution 1: With Algebra**

Let the original price of an item be  $x$ .

First, everything is half-off, so the price is now  $\frac{x}{2} = 0.5x$ .

Next, the extra coupon applies 20% off on the *sale price*, so the price after this discount will be  $100\% - 20\% = 80\%$  of what it was before. (Notice how this is not applied to the original price; if it were, the solution would be applying  $50\% + 20\% = 70\%$  off the original price.)

$$80\% \cdot 0.5x = \frac{4}{5} \cdot 0.5x = 0.4x$$

The price of the item after all discounts have been applied is  $0.4x = 40\% \cdot x$ . However, we need to find the percentage off the original price, not the current percentage of the original price. We then subtract  $40\%x$  from  $100\%x$  (the original price of the item), to find the answer,  $\boxed{\text{(D)} 60}$ .

**Solution 2: Fakesolving**

Since the problem implies that the percentage off the original price will be the same for every item in the store, **fakesolving** is applicable here. Say we are buying an item worth 10 dollars, a convenient number to work with. First, it is clear that we'll get 50% off, which makes the price then 5 dollars. Taking 20% off of 5 dollars gives us 4 dollars. Therefore, we have saved a total of  $\frac{10-4}{10} = \frac{6}{10} = \frac{60}{100} = \boxed{\text{(D)} 60}\%$ .

**Solution 3:**

The percent off would be 50% of 20% because it has an additional 20% off of half the price. The answer to that would be 10%. You could do  $10\% + 50\%$  to find the answer. This answer is  $\boxed{\text{(D)} 60}$ .

**Solution 4:**

The price is 50% off, which means the price would be  $100\% - 50\% = 50\%$  of what it was before. Then, you get 20% off. Sidenote: A common mistake is finding 20% of the price, but forgetting to subtract it. 20% off the price is  $100\% - 20\% = 80\%$ . Now, we can multiply the two discounts as decimals.  $0.5 \times 0.8 = 0.4 = 40\%$ . Our final answer is  $100\% - 40\% = \boxed{\text{(D)} 60\%}$

9. (AMC 8 2012) The Fort Worth Zoo has a number of two-legged birds and a number of four-legged mammals. On one visit to the zoo, Margie counted 200 heads and 522 legs. How many of the animals that Margie counted were two-legged birds?

(A) 61      (B) 122      (C) 139      (D) 150      (E) 161

**Answer:** C

**Solution 1: Algebra**

Let the number of two-legged birds be  $x$  and the number of four-legged mammals be  $y$ . We can now use systems of equations to solve this problem.

Write two equations:

$$2x + 4y = 522 \quad x + y = 200$$

Now multiply the latter equation by 2.

$$2x + 4y = 522 \quad 2x + 2y = 400$$

By subtracting the second equation from the first equation, we find that  $2y = 122 \implies y = 61$ . Since there were 200 heads, meaning that there were 200 animals, there were  $200 - 61 = \boxed{\text{(C) } 139}$  two-legged birds.

**Solution 2: Cheating the System**

First, we "assume" there are 200 two-legged birds only, and 0 four-legged mammals. Of course, this poses a problem, as then there would only be  $200 \cdot 2 = 400$  legs.

Now we have to do some swapping—for every two-legged bird we swap for a four-legged mammal, we gain 2 legs. For example, if we swapped one bird for one mammal, giving 199 birds and 1 mammal, there would be  $400 + 1(2) = 402$  legs. If we swapped two birds for two mammals, there would be  $400 + 2(2) = 404$  legs. If we swapped 50 birds for 50 mammals, there would be  $400 + 50(2) = 500$  legs.

**Solution 3: Add two legs for each bird**

Let's add 2 legs for each bird and assume each bird has four-legged as each mammal. So the total legs of these birds and mammals would be  $4 \cdot 200 = 800$ . Actually, there were only 522 legs. The difference between these two numbers exactly gives us the number of all the legs we added for all birds:  $800 - 522 = 278$ . Because each bird was added by 2 legs, so the total number of birds would be  $278/2 = \boxed{\text{(C) } 139}$  two-legged birds. —LarryFlora

10. (AMC 8 2012) How many 4-digit numbers greater than 1000 are there that use the four digits of 2012?

(A) 6      (B) 7      (C) 8      (D) 9      (E) 12

**Answer:** D

**Solution 1**

For this problem, all we need to do is find the amount of valid 4-digit numbers that can be made from the digits of 2012, since all of the valid 4-digit number will always be greater than 1000. The best way to solve this problem is by using casework.

There can be only two leading digits, namely 1 or 2.

When the leading digit is 1, you can make  $\frac{3!}{2!1!} \implies 3$  such numbers.

When the leading digit is 2, you can make  $3! \implies 6$  such numbers.

Summing the amounts of numbers, we find that there are  $\boxed{\text{(D) } 9}$  such numbers.

**Solution 2**

Notice that the first digit cannot be 0, as the number is greater than 1000. Therefore, there are three digits that can be in the thousands.

The rest three digits of the number have no restrictions, and therefore there are  $3! \implies 6$  for each leading digit.

Since the two 2's are indistinguishable, there are  $\frac{3 \cdot 6}{2}$  such numbers  $\implies \boxed{\text{(D)} 9}$ .

Note by algebramaster2: 1 and 2 are the only digits that can be in the thousands place as 0 in the thousands place will not make the number 4 digit anymore.

### Solution 3 (a simpler version of Solution 2)

We can list out the four digits in the number. The first digit of the number can't be 0 since the number is greater than 1000. This leaves us with three integers for the digit in the thousands place. We have no restrictions in the hundreds place except for the fact that it can't be the integer we chose for the thousands place. Continuing this pattern, there are 3 choices for the thousands place, 3 for the hundreds, 2 for the tens, and 1 for the ones. Adding this up we get 9 which is choice  $\boxed{\text{(D)}}$ .

—jason.ca

### Solution 4 (Complementary Counting)

There are  $\frac{4!}{2!}$  ways to arrange the numbers in 2012. The number of ways to arrange 2012 with 0 as the first digit is 3 because there are three places to put the 1 and the 2s are the same.  $12 - 3 = 9$  which is choice  $\boxed{\text{(D)}}$ .

11. (AMC 8 2012) The mean, median, and unique mode of the positive integers 3, 4, 5, 6, 6, 7, and  $x$  are all equal. What is the value of  $x$ ?

(A) 5      (B) 6      (C) 7      (D) 11      (E) 12

**Answer:** D

### Solution 1: Guess Check

We can eliminate answer choices (A) 5 and (C) 7, because of the above statement. Now we need to test the remaining answer choices.

Case 1:  $x = 6$

Mode: 6

Median: 6

Mean:  $\frac{37}{7}$

Since the mean does not equal the median or mode, (B) 6 can also be eliminated.

Case 2:  $x = 11$

Mode: 6

Median: 6

Mean: 6

We are done with this problem, because we have found when  $x = 11$ , the condition is satisfied. Therefore, the answer is  $\boxed{\text{(D)} 11}$ .

### Solution 2: Algebra

Notice that the mean of this set of numbers, in terms of  $x$ , is:

$$\frac{3+4+5+6+6+7+x}{7} = \frac{31+x}{7}$$

Because we know that the mode must be 6 (it can't be any of the numbers already listed, as shown above, and no matter what  $x$  is, either 6 or a new number, it will

not affect 6 being the mode), and we know that the mode must equal the mean, we can set the expression for the mean and 6 equal:

$$\begin{aligned}\frac{31+x}{7} &= 6 \\ 31+x &= 42 \\ x &= \boxed{\text{(D) } 11}\end{aligned}$$

### Solution 3: Balance scale

We know the unique mode must be 6, so the mean must be the same number 6. Let's imagine a scale. 6 exactly stands the mid-point of the scale. Numbers of 3, 4, 5 represent the left side "weights" of the scale. Numbers of 6, 7,  $x$  represent the right side "weights" of the scale. On the left side, the difference of the three "weights" between 6 are  $-3, -2, -1$ , respectively. It gives us the total difference is  $-6$ . In order to allow the scale to keep balance, on the right side, the total difference must be  $+6$ . Because we have already known the difference of the right side "weights" between 6 is  $0 + 1 = 1$ , partially, so the difference between 6 and unknown  $x$  must be  $+6 - 1 = +5$ . It exactly gives us the answer:  $x = 6 + 5 = \boxed{\text{(D) } 11}$ . —LarryFlora

### Solution 4: Must be an integer(Super fast)

Since all the answer choices are integers, we know that  $\frac{3+4+5+6+6+7+x}{7}$ , which simplifies to  $\frac{31+x}{7}$ , must be an integer. The only answer choice that satisfies this is  $\boxed{\text{(D) } 11}$ .

Edit: The only answer choice that keeps  $\frac{31+x}{7}$  equal to the mode, 6, is  $\boxed{\text{(D) } 11}$

12. (AMC 8 2012) What is the units digit (ones place digit) of  $13^{2012}$ ?

(A) 1    (B) 3    (C) 5    (D) 7    (E) 9

**Answer:** A

### Solution 1

The problem wants us to find the units digit of  $13^{2012}$ , therefore, we can eliminate the tens digit of 13, because the tens digit will not affect the final result. So our new expression is  $3^{2012}$ . Now we need to look for a pattern in the units digit.

$$3^1 \Rightarrow 3 \quad 3^2 \Rightarrow 9 \quad 3^3 \Rightarrow 7 \quad 3^4 \Rightarrow 1 \quad 3^5 \Rightarrow 3$$

We observe that there is a pattern for the units digit which recurs every four powers of three. Using this pattern, we can subtract 1 from 2012 and divide by 4. The remainder is the power of three that we are looking for, plus one. 2011 divided by 4 leaves a remainder of 3, so the answer is the units digit of  $3^{3+1}$ , or  $3^4$ . Thus, we find that the units digit of  $13^{2012}$  is  $\boxed{\text{(A) } 1}$ .

### Solution 2

Ignore the tens digit of 13, we find a pattern in the units digit that  $3^4 \Rightarrow 1$ . We also find 2012 can be divided by 4 evenly, which is  $2012/4 = 503$ . So  $3^{2012} = (3^4)^{503}$ . Because the units digit of  $3^4 \Rightarrow 1$  so the units digit  $1^{503} \Rightarrow 1$ . Thus, the units digit of  $13^{2012}$  is  $\boxed{\text{(A) } 1}$ . —LarryFlora

13. (AMC 8 2012) Jamar bought some pencils costing more than a penny each at the school bookstore and paid \$1.43. Sharona bought some of the same pencils and paid \$1.87. How many more pencils did Sharona buy than Jamar?

- (A) 2    (B) 3    (C) 4    (D) 5    (E) 6

**Answer:** C**Solution 1**

We assume that the price of the pencils remains constant. Convert \$1.43 and \$1.87 to cents. Since the price of the pencils is more than one penny, we can find the price of one pencil (in cents) by taking the greatest common divisor of 143 and 187, which is 11. Therefore, Jamar bought  $\frac{143}{11} \Rightarrow 13$  pencils and Sharona bought  $\frac{187}{11} \Rightarrow 17$  pencils. Thus, Sharona bought  $17 - 13 = \boxed{\text{(C) } 4}$  more pencils than Jamar.

**Solution 2**

We find the difference between 1.43 and 1.87 is  $1.87 - 1.43 = 0.44$ , which is the extra cost of Sharona's pencils than Jamar's pencils. Because the difference between the amounts of the pencils they bought must be divided evenly by 0.44, looking into the answers, 2 or 4 is possibly correct. It gives us the price of each pencil should be  $0.44/2 = 0.22$  or  $0.44/4 = 0.11$ , respectively. Then we find only 0.11 can be divided evenly by 1.43 and 1.87. So the answer is  $\boxed{\text{(C) } 4}$

**Solution 3 (quick solution)**

You can quickly tell, if you have memorized the multiplication table, that 187 and 143 are both multiples of 11 (Also GCF). Additionally, the difference is 44 cents. That means that there are  $\boxed{\text{(C) } 4}$  more pencils purchased by Sharona because  $11 \times 4 = 44$ .

14. (AMC 8 2012) In the BIG N, a middle school football conference, each team plays every other team exactly once. If a total of 21 conference games were played during the 2012 season, how many teams were members of the BIG N conference?
- (A) 6    (B) 7    (C) 8    (D) 9    (E) 10

**Answer:** B**Solution 1**

This problem is very similar to a handshake problem. We use the formula  $\frac{n(n-1)}{2}$  to usually find the number of games played (or handshakes). Now we have to use the formula in reverse.

So we have the equation  $\frac{n(n-1)}{2} = 21$ . Solving, we find that the number of teams in the BIG N conference is  $\boxed{\text{(B) } 7}$ .

**Solution 2**

(If someone understands what I'm trying to do here and can explain it better, please edit it) We know that every team has to play a game with every other team, so we just need to find out how many consecutive numbers, 1 to  $x$ , can fit into 21. We know that  $6 + 5 + 4 + 3 + 2 + 1 = 21$ , and since this doesn't count to 7th team that shook hands with the other 6, we know that there are  $\boxed{\text{(B) } 7}$  teams in the BIG N conference.

EDIT: So basically reiterating this, we are using a factorial and adding one because the team that they are talking about is not shaking the hands. Also as the other person said, we need to see how many consecutive numbers can add up to 21.

**Solution 3**

Lets say there are  $x$  teams in total and we number the teams with letters. Team A plays  $x - 1$  games against the other teams. Team B plays  $x - 1$  minus the game it played with team A, making it  $x - 2$  games. Team C plays  $x - 3$  games, and so on. Finding the sum of the consecutive numbers 1 to  $x - 1$  which equals 21 gives  $x = 7$ . So the correct answer choice is **(B) 7**.

A great way to visualize this problem is to draw circles and connecting lines between them. If you have 6 circles, you can draw 5 lines connecting to the first circle, then 4 more distinct lines connecting to the second circle, 3 more to the third, and so on until you have the second last circle drawing 1 line to the last circle.

15. (AMC 8 2012) The smallest number greater than 2 that leaves a remainder of 2 when divided by 3, 4, 5, or 6 lies between what numbers?

(A) 40 and 50      (B) 51 and 55      (C) 56 and 60      (D) 61 and 65      (E) 66 and 99

**Answer:** D

### Solution

To find the answer to this problem, we need to find the least common multiple of 3, 4, 5, 6 and add 2 to the result. To calculate the least common multiple, we need to find the prime factorization for each of them.  $3 = 3^1$ ,  $4 = 2^2$ ,  $5 = 5^1$ , and  $6 = 2^1 * 3^1$ . So the least common multiple of the four numbers is  $2^2 * 3^1 * 5^1 = 60$ , and by adding 2, we find that that such number is 62. Now we need to find the only given range that contains 62. The only such range is answer **(D)**, and so our final answer is **(D) 61 and 65**.

### Solution 2(Chinese Remainder Theorem)

We have to find the smallest number  $x$  for  $x \equiv 2 \pmod{3}$ ,  $x \equiv 2 \pmod{4}$ , and  $x \equiv 2 \pmod{5}$ . We are excluding six because it won't make our divisors relatively prime. For the first equation,  $x \equiv 2 \pmod{3}$  holds remainder( $r$ ) of 2. Now we have to find  $M$ . Our  $M$  which is all divisors multiplied by each other divided by the divisor we are using. So our  $M$  is  $\frac{3 \cdot 4 \cdot 5}{3} = 20$ . We now need to find the inverse of  $a$  times  $M$  and solve  $a$ . Doing this we can get  $20a \equiv 1 \pmod{3}$ , and the smallest value of  $a$  where this is true is 2. Now we multiply every value we have which gets us  $2 \cdot 20 \cdot 2$ . Then we repeat for mod 4 and mod 5. For mod 4 the numbers  $r, M, a$  are 2, 15, and 3 respectively. For mod 5 we get 2, 12, and 8. Now we have to multiply all our values and we get 80 (mod 3), 90 (mod 4), and 192 (mod 5). Now we do  $80 + 90 + 192 \pmod{5 \cdot 4 \cdot 3}$ . This is  $362 \pmod{60}$ . This means  $x \equiv 2 \pmod{60}$ , but since it needs to be greater than 2, we get 62, which divided by six also has a remainder of 2. So 62 is in the range of **(D) 61 and 65**.

If CRT didn't make sense, this video is very good:

16. (AMC 8 2012) Each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9 is used only once to make two five-digit numbers so that they have the largest possible sum. Which of the following could be one of the numbers?

(A) 76531      (B) 86724      (C) 87431      (D) 96240      (E) 97403

**Answer:** C

### Solution 1

In order to maximize the sum of the numbers, the numbers must have their digits ordered in decreasing value. There are only two numbers from the answer choices

with this property: 76531 and 87431. To determine the answer we will have to use estimation and the first two digits of the numbers.

For 76531 the number that would maximize the sum would start with 98. The first two digits of 76531 (when rounded) are 77. Adding 98 and 77, we find that the first three digits of the sum of the two numbers would be 175.

For 87431 the number that would maximize the sum would start with 96. The first two digits of 87431 (when rounded) are 87. Adding 96 and 87, we find that the first three digits of the sum of the two numbers would be 183.

From the estimations, we can say that the answer to this problem is **(C) 87431**.  
p.s. USE INTUITION, see answer choices before solving any question -little\_master

### Solution 2

In order to determine the largest number possible, we have to evenly distribute the digits when adding. The two numbers that show an example of this are 97531 and 86420. The digits can be interchangeable between numbers because we only care about the actual digits.

The first digit must be either 9 or 8. This immediately knocks out **(A)** 76531.

The second digit must be either 7 or 6. This doesn't cancel any choices.

The third digit must be either 5 or 4. This knocks out **(B)** 86724 and **(D)** 96240.

The fourth digit must be 3 or 2. This cancels out **(E)** 97403.

This leaves us with **(C) 87431**.

### Solution 3

If we use intuition, we know that the digits will be IN ORDER, to maximize the number. This eliminates **(B)**, **(D)**, and **(E)**. Additionally, the 2 two numbers must have 9 or 8 for the first digit to maximize the sum, eliminating **(A)**. This leaves **(C) 87431** -written by little\_master purely, not copied from anywhere

17. (AMC 8 2012) A square with integer side length is cut into 10 squares, all of which have integer side length and at least 8 of which have area 1. What is the smallest possible value of the length of the side of the original square?

**(A)** 3      **(B)** 4      **(C)** 5      **(D)** 6      **(E)** 7

**Answer:** B

### Solution

The first answer choice **(A)** 3, can be eliminated since there must be 10 squares with integer side lengths. We then test the next smallest side length, which is 4. The square with area 16 can be partitioned into 8 squares with area 1 and two squares with area 4, which satisfies all the conditions of the problem. Therefore, the smallest possible value of the length of the side of the original square is **(B) 4**.

18. (AMC 8 2012) What is the smallest positive integer that is neither prime nor square and that has no prime factor less than 50?

**(A)** 3127      **(B)** 3133      **(C)** 3137      **(D)** 3139      **(E)** 3149

**Answer:** A

### Solution



The problem states that the answer cannot be a perfect square or have prime factors less than 50. Therefore, the answer will be the product of at least two different primes greater than 50. The two smallest primes greater than 50 are 53 and 59. Multiplying these two primes, we obtain the number 3127, which is also the smallest number on the list of answer choices.

So we are done, and the answer is **(A) 3127**.

19. (AMC 8 2012) In a jar of red, green, and blue marbles, all but 6 are red marbles, all but 8 are green, and all but 4 are blue. How many marbles are in the jar?

(A) 6    (B) 8    (C) 9    (D) 10    (E) 12

**Answer:** C

**Solution 1 (Trial and Error)**

6 are blue and green -  $b + g = 6$  8 are red and blue -  $r + b = 8$  4 are red and green -  $r + g = 4$

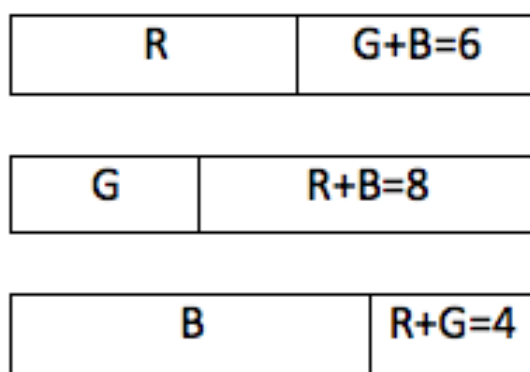
We can do trial and error. Let's make blue 5. That makes green 1 and red 3 because  $6 - 5 = 1$  and  $8 - 5 = 3$ . To check this, let's plug 1 and 3 into  $r + g = 4$ , which works. Now count the number of marbles -  $5 + 3 + 1 = 9$ . So the answer is **(C) 9**.

**Solution 2**

We already knew the facts: 6 are blue and green, meaning  $b + g = 6$ ; 8 are red and blue, meaning  $r + b = 8$ ; 4 are red and green, meaning  $r + g = 4$ . Then we need to add these three equations:  $b + g + r + b + r + g = 2(r + g + b) = 6 + 8 + 4 = 18$ . It gives us all of the marbles are  $r + g + b = 18/2 = 9$ . So the answer is **(C) 9**.  
LarryFlora

**Solution 3 (Venn Diagrams)**

We may draw three Venn diagrams to represent these three cases, respectively.



Let the amount of all the marbles be  $x$ , meaning  $R + G + B = x$ .

The Venn diagrams give us the equation:  $x = (x - 6) + (x - 8) + (x - 4)$ . So  $x = 3x - 18$ ,  $x = 18/2 = 9$ . Thus, the answer is **(C) 9**.  
LarryFlora

**Solution 4**

-similar to solution 3 but you can skip Venn Diagrams

Let  $x$  equal total marbles

Then:

$$\text{red} = x - 6$$

$$\text{Green} = x - 8$$

$$\text{Blue} = x - 4$$

$$\text{Then } x = r + g + b$$

$$\text{So } (x - 4) + (x - 8) + (x - 4) = x$$

Simplifying, you get  $x = 9$  so 9 marbles, answer choice C

#### Solution 4 (Answer Choices)

Since we know all but 8 marbles in the jar are green, the jar must have at least 9 marbles. Then we can just start from C and keep going. If there are 9 marbles total, there are 3 red marbles ( $9 - 6$ ), 1 green marble ( $9 - 8$ ), and 5 blue marbles ( $9 - 4$ ). Since we assumed there were 9 marbles and  $3 + 1 + 5 = 9$ , the answer is (C) 9.

#### Solution 5 (Algebra)

Let  $x$  be the number of total marbles. There are  $x - 6$  red marbles,  $x - 8$  green marbles, and  $x - 4$  blue marbles. We can create an equation:  $(x - 6) + (x - 8) + (x - 4) = x$ . Solving, we get  $x = 9$ , which means the total number of marbles is (C) 9. -J.L.L (Feel free to edit)

#### Solution 6

Let  $x$  be the number of total marbles,  $r$  be the number of red marbles,  $g$  be the number of green marbles, and  $b$  be the number of blue marbles. Then we have  $x - r = 6$ ,  $x - g = 8$ ,  $x - b = 4$ , and  $r + g + b = x$ . Adding the first three equations together, we get  $3x - r - g - b = 18$  or  $3x - (r + g + b) = 18$ . Substituting in the fourth equation, we have  $3x - x = 18 \implies$  (C) 9.

20. (AMC 8 2012) What is the correct ordering of the three numbers  $\frac{5}{19}$ ,  $\frac{7}{21}$ , and  $\frac{9}{23}$ , in increasing order?

(A)  $\frac{9}{23} < \frac{7}{21} < \frac{5}{19}$  (B)  $\frac{5}{19} < \frac{7}{21} < \frac{9}{23}$  (C)  $\frac{9}{23} < \frac{5}{19} < \frac{7}{21}$  (D)  $\frac{5}{19} < \frac{9}{23} < \frac{7}{21}$  (E)  $\frac{7}{21} < \frac{5}{19} < \frac{9}{23}$

Answer: B

#### Solution 1

The value of  $\frac{7}{21}$  is  $\frac{1}{3}$ . Now we give all the fractions a common denominator.

$$\frac{5}{19} \implies \frac{345}{1311} \quad \frac{1}{3} \implies \frac{437}{1311} \quad \frac{9}{23} \implies \frac{513}{1311}$$

Ordering the fractions from least to greatest, we find that they are in the order listed. Therefore, our final answer is (B)  $\frac{5}{19} < \frac{7}{21} < \frac{9}{23}$ .

#### Solution 2

Change  $\frac{7}{21}$  into  $\frac{1}{3}$ ;

$$\begin{aligned} \frac{1}{3} \cdot \frac{5}{5} &= \frac{5}{15} \\ \frac{5}{15} &> \frac{5}{19} \\ \frac{7}{21} &> \frac{5}{19} \end{aligned}$$

And

$$\begin{aligned}\frac{1}{3} \cdot \frac{9}{9} &= \frac{9}{27} \\ \frac{9}{27} &< \frac{9}{23} \\ \frac{7}{21} &< \frac{9}{23}\end{aligned}$$

Therefore, our answer is **(B)**  $\frac{5}{19} < \frac{7}{21} < \frac{9}{23}$ .

**Solution 3 (quick and easy)**

We know that  $\frac{5}{19}$  is  $\frac{14}{19}$  away from 1,  $\frac{7}{21}$  is  $\frac{14}{21}$  away from 1, and  $\frac{9}{23}$  is  $\frac{14}{23}$  away from 1. Since  $\frac{14}{19}$  is the largest, we know that it is the farthest away from 0, and  $\frac{14}{23}$  is the smallest, so it is the closest to 0. Therefore, our answer is **(B)**  $\frac{5}{19} < \frac{7}{21} < \frac{9}{23}$ .

**Solution 4**

Let  $\frac{a}{b} < 1$ , so  $b > a$ . We claim that  $\frac{a}{b} < \frac{a+2}{b+2}$ . Cross-multiplying gives  $a(b+2) < b(a+2) \Rightarrow ab + 2a < ab + 2b \Rightarrow 2a < 2b \Rightarrow a < b$ , which is true.

Since all fractions are less than 1, this applies here: and  $\frac{5}{19} < \frac{7}{21}$  and  $\frac{7}{21} < \frac{9}{23}$ .

Thus the answer is **(B)**  $\frac{5}{19} < \frac{7}{21} < \frac{9}{23}$ .

Note: The value 2 is not special, this works for any positive number. Just make sure that the fraction is less than 1 and you add it to both the numerator and denominator.

**Solution 5**

Just solve  $\frac{5}{19}$  and  $\frac{9}{23}$  to 2 or 3 decimal places, since  $\frac{7}{21} = \frac{1}{3}$ . You get 0.26 and 0.39 respectively, so,  $\frac{5}{19} < \frac{7}{21}$  and  $\frac{7}{21} < \frac{9}{23}$ .

hence the answer is **(B)**  $\frac{5}{19} < \frac{7}{21} < \frac{9}{23}$ .

-N.

21. (AMC 8 2012) Marla has a large white cube that has an edge of 10 feet. She also has enough green paint to cover 300 square feet. Marla uses all the paint to create a white square centered on each face, surrounded by a green border. What is the area of one of the white squares, in square feet?

(A)  $5\sqrt{2}$     (B) 10    (C)  $10\sqrt{2}$     (D) 50    (E)  $50\sqrt{2}$

**Answer:** D

**Solution**

If Marla evenly distributes her 300 square feet of paint between the 6 faces, each face will get  $300 \div 6 = 50$  square feet of paint. The surface area of one of the faces of the cube is  $10^2 = 100$  square feet. Therefore, there will be  $100 - 50 =$  **(D)** 50 square feet of white on each side.

22. (AMC 8 2012) Let  $R$  be a set of nine distinct integers. Six of the elements are 2, 3, 4, 6, 9, and 14. What is the number of possible values of the median of  $R$ ?
- (A) 4    (B) 5    (C) 6    (D) 7    (E) 8

**Answer:** D

**Solution 1**

Let the values of the missing integers be  $x, y, z$ . We will find the bound of the possible medians.

The smallest possible median will happen when we order the set as  $\{x, y, z, 2, 3, 4, 6, 9, 14\}$ . The median is 3.

The largest possible median will happen when we order the set as  $\{2, 3, 4, 6, 9, 14, x, y, z\}$ . The median is 9.

Therefore, all possible medians must be between 3 and 9 inclusive, yielding **(D) 7** possible medians.

23. (AMC 8 2012) An equilateral triangle and a regular hexagon have equal perimeters. If the area of the triangle is 4, what is the area of the hexagon?

(A) 4    (B) 5    (C) 6    (D)  $4\sqrt{3}$     (E)  $6\sqrt{3}$

**Answer:** C

**Solution 1 (recommended)**

Let the perimeter of the equilateral triangle be  $3s$ . The side length of the equilateral triangle would then be  $s$  and the sidelength of the hexagon would be  $\frac{s}{2}$ .

A hexagon contains six equilateral triangles. One of these triangles would be similar to the large equilateral triangle in the ratio  $1 : 4$ , since the sidelength of the small equilateral triangle is half the side length of the large one. Thus, the area of one of the small equilateral triangles is 1. The area of the hexagon is then  $1 \times 6 =$  **(C) 6**.

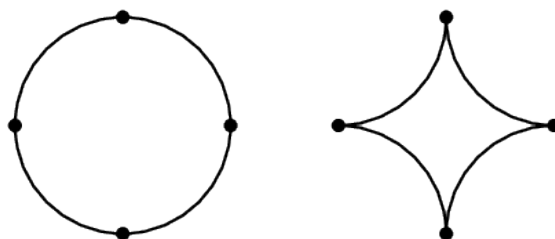
**Solution 2 (Hard)**

Let the side length of the small triangle be  $2x$ , so the side length of the hexagon is  $x$ . By the Pythagorean theorem, the height of the triangle is  $x\sqrt{3}$ , so the area of the triangle is  $\frac{x^2}{\sqrt{3}}$ . We are given that the area is 4, so  $x$  must be  $\frac{2}{\sqrt[3]{3}}$ . Therefore, by the Pythagorean theorem, the height of one of the triangles in the hexagon (made by subdividing the hexagon into 6 congruent triangles), is  $\frac{\sqrt{3}}{\sqrt[3]{3}}$ , so the area is  $\frac{\frac{2}{\sqrt[3]{3}} \times \frac{\sqrt{3}}{\sqrt[3]{3}}}{2}$ , which equals 1. Since there are 6 such triangles in the hexagon, the area of the hexagon is **(C) 6**.

**Solution 3 (Formula)**

The area of an equilateral triangle is  $\frac{\sqrt{3}}{4}a^2$ , where  $a$  is the side. Since a hexagon is six equilateral triangles combined, the area of a hexagon is  $\left(\frac{3\sqrt{3}}{2}\right)a^2$ . If we assume the side of the hexagon is 1, and the side of the triangle is 2 (perimeter is the same but sides are 6:3), then the area of the triangle will be  $4\sqrt{3}$ . We killed it is a  $\sqrt{3} : 4$  ratio of the assumed triangle area to the actual triangle area. This indicates the area of the hexagon will also be a ratio of  $\sqrt{3} : 4$ , so the area of the assumed hexagon is  $\frac{3\sqrt{3}}{2}$ . Therefore, the area of the actual hexagon will be  $\frac{\frac{3\sqrt{3}}{2} \cdot 4}{\sqrt{3}}$ . Simplifying the result will give us **(C) 6**.

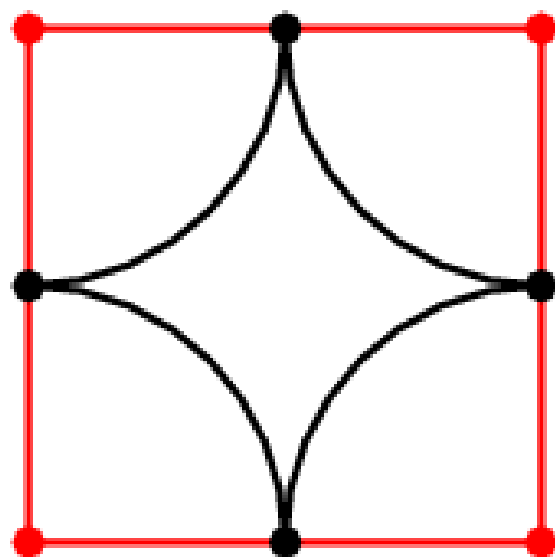
24. (AMC 8 2012) A circle of radius 2 is cut into four congruent arcs. The four arcs are joined to form the star figure shown. What is the ratio of the area of the star figure to the area of the original circle?



- (A)  $\frac{4-\pi}{\pi}$     (B)  $\frac{1}{\pi}$     (C)  $\frac{\sqrt{2}}{\pi}$     (D)  $\frac{\pi-1}{\pi}$     (E)  $\frac{3}{\pi}$

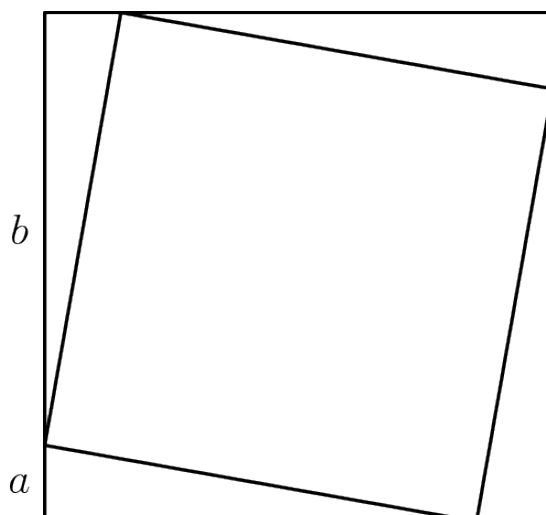
**Answer:** A

**Solution**



Draw a square around the star figure. The side length of this square is 4, because the side length is the diameter of the circle. The square forms 4-quarter circles around the star figure. This is the equivalent of one large circle with radius 2, meaning that the total area of the quarter circles is  $4\pi$ . The area of the square is 16. Thus, the area of the star figure is  $16 - 4\pi$ . The area of the circle is  $4\pi$ . Taking the ratio of the two areas, we find the answer is (A)  $\frac{4-\pi}{\pi}$ .

25. (AMC 8 2012) A square with area 4 is inscribed in a square with area 5, with one vertex of the smaller square on each side of the larger square. A vertex of the smaller square divides a side of the larger square into two segments, one of length  $a$ , and the other of length  $b$ . What is the value of  $ab$ ?



- (A)  $\frac{1}{5}$     (B)  $\frac{2}{5}$     (C)  $\frac{1}{2}$     (D) 1    (E) 4

**Answer: C**

**Solution 1 - Recommended for Contest**

The total area of the four congruent triangles formed by the squares is  $5 - 4 = 1$ . Therefore, the area of one of these triangles is  $\frac{1}{4}$ . The height of one of these triangles is  $a$  and the base is  $b$ . Using the formula for area of the triangle, we have  $\frac{ab}{2} = \frac{1}{4}$ .

Multiply by 2 on both sides to find that the value of  $ab$  is (C)  $\frac{1}{2}$ .

**Solution 2 (Algebra)**

We see that we want  $ab$ , so instead of solving for  $a$  and  $b$ , we find a way to get an expression with  $ab$ .

By the Triple Perpendicularity Model, all four triangles are congruent.

By the Pythagorean Theorem,

$$\sqrt{a^2 + b^2} = \sqrt{4}$$

$$\text{Thus, } \sqrt{a^2 + b^2} = 2$$

$$\text{As } a + b = \sqrt{5},$$

$$a^2 + 2ab + b^2 = 5$$

$$\text{So, } \sqrt{5 - 2ab} = 2$$

Simplifying,

$$5 - 2ab = 4 - 2ab = -1 \quad ab = \frac{1}{2} \quad \text{or} \quad \text{(C) } \frac{1}{2}$$

**Solution 3 (similar to solution 2)**

We know that each side of a square is equal, and each the area of a square can be expressed as the side squared. We can let the outside square with area 5's side be  $x$ . We get the equation  $x^2 = 5$ . Simplifying this we get  $x = \sqrt{5}$ .

We can then create the equation  $a + b = \sqrt{5}$ .

Using the same tactic we get that the side length of the inner square is 2. By the Pythagorean Theorem,

$$a^2 + b^2 = 2^2.$$

We can then express this expression as

$$(a + b)^2 - 2ab = 4.$$

We recall that  $a + b = \sqrt{5}$  and substitute it into our current equation:

$$(\sqrt{5})^2 - 2ab = 4.$$

We further simplify this to  $ab = 1/2$  which is  $\boxed{\text{(C)} \frac{1}{2}}$ .

#### Solution 4 (The Long Way)

Visually,  $a + b = \sqrt{5}$  (1) and we can tell that all the triangles are congruent. That means the long side of all the triangles is  $b$  and the short side will always be  $a$ . Notice that the hypotenuse is the side of the smaller square (2). Using the Pythagorean theorem,  $a^2 + b^2 = 2^2$  or  $a^2 + b^2 = 4$  (2).

Now we just have a system of equations:

Isolate  $b$  in (1)  $\rightarrow b = \sqrt{5} - a$ , and plug it in to (2).

$$a^2 + (5 - a)^2 = 4, \text{ Simplifying should get you to } 2a^2 - 2a\sqrt{5} + 1 = 0$$

Using the quadratic formula,  $a = \frac{2\sqrt{5} \pm \sqrt{20-8}}{4}$

Simplifying the squareroots,  $a = \frac{\sqrt{5} + \sqrt{3}}{2}$ .

This yields  $b$  to be  $\frac{\sqrt{5} - \sqrt{3}}{2}$  via substituting into (1)

$$ab = \frac{(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})}{2 \cdot 2} \text{ (difference of squares)} = \frac{5-3}{4} = \boxed{\text{(C)} \frac{1}{2}}$$

#### Solution 3 (Pythagorean Theorem Vieta's Formula)

Proceed as in solution 3 until you get the equation  $2a^2 - 2a\sqrt{5} + 1 = 0$ . The two solutions to this equation will be the long and short side of the triangle we are looking for, or  $a$  and  $b$  from the problem statement. Using Vieta's formula, their product is the constant term divided by the quadratic term, or  $\boxed{\text{(C)} \frac{1}{2}}$ .

## §29 AMC 8 2013

- (AMC 8 2013) Danica wants to arrange her model cars in rows with exactly 6 cars in each row. She now has 23 model cars. What is the smallest number of additional cars she must buy in order to be able to arrange all her cars this way?  
(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** A

#### Solution 1

Since each row will have 6 cars, Danica should find the smallest multiple of 6 that is greater than 23 and find the difference between 23 and that number.

We can see that  $6 \times 3 = 18 < 23$ , but  $6 \times 4 = 24 > 23$ . This is 1 more than 23, so the answer is  $\boxed{\text{(A)} 1}$ .

- (AMC 8 2013) A sign at the fish market says, "50% off, today only: half-pound packages for just 3 per package." What is the regular price for a full pound of fish, in dollars? (Assume that the

- (A) 6      (B) 9      (C) 10      (D) 12      (E) 15

**Answer:** D

**Solution 1**

50% off the price of half a pound of fish is \$3, so 100

**Solution 2**

Suppose a full pound at normal price costs  $x$  dollars. Then, with the 50% off deal, the full pound would cost  $x/2$  dollars. A half pound of this would cost  $x/4$  dollars (including the 50% off). 50% off half a pound is 3, so we form the equation  $x/4 = 3 \Rightarrow x = \boxed{\text{(D) } 12}$ .

3. (AMC 8 2013) What is the value of  $4 \cdot (-1 + 2 - 3 + 4 - 5 + 6 - 7 + \cdots + 1000)$ ?

- (A)  $-10$       (B)  $0$       (C)  $1$       (D)  $500$       (E)  $2000$

**Answer:** E

**Solution 1**

We group the addends inside the parentheses two at a time:

$$\begin{aligned} -1 + 2 - 3 + 4 - 5 + 6 - 7 + \cdots + 1000 &= (-1 + 2) + (-3 + 4) + (-5 + 6) + \cdots + (-999 + 1000) \\ &= \underbrace{1 + 1 + 1 + \cdots + 1}_{500 \text{ 1's}} \\ &= 500. \end{aligned}$$

Then the desired answer is  $4 \times 500 = \boxed{\text{(E) } 2000}$ .

**Solution 2(Solution 1 Worded Differently)**

Basically  $4 \times (-1+2-3+4\ldots-999+1000)$  is just  $4 \times (1+1+1+1+\ldots+1+1)$  [There are 500 1's because  $1000/2$ ] so  $4 \times 500 = 4 \times 500 = \boxed{\text{(E) } 2000}$ .

4. (AMC 8 2013) Eight friends ate at a restaurant and agreed to share the bill equally. Because Judi forgot her money, each of her seven friends paid an extra 2.50 to cover her portion of the total bill. What

- (A) \$120      (B) \$128      (C) \$140      (D) \$144      (E) \$160

**Answer:** C

**Solution 1**

Since Judi's 7 friends had to pay \$2.50 extra each to cover the total amount that Judi should have paid, we multiply  $2.50 \cdot 7 = 17.50$  is the bill Judi would have paid if she had money. Hence, to calculate the total amount, we multiply  $17.50 \cdot 8 = \boxed{\text{(C) } 140}$  to find the total the 8 friends paid.

**Solution 2**

Let  $m$  be the total bill the 8 friends split up; each person was supposed to pay  $\$ \frac{m}{8}$ . Since Judi forgot her money, we have to add  $\$ \frac{m}{8} + 2.50$  to get the amount of money each of the 7 friends have to pay. After multiplying by 7, we have  $7(\frac{m}{8} + 2.50) = \frac{7m}{8} + 17.5$ . This is equal to  $m$  because both sides represent the total bill. Solving for  $m$ :

$$\frac{7m}{8} + 17.5 = m$$

$$\frac{m}{8} = 17.5$$

$$m = \boxed{\text{(C) } 140}.$$



5. (*AMC 8 2013*) Hammie is in the 6<sup>th</sup> grade and weighs 106 pounds. Her quadruplet sisters are tiny babies and weigh 5, 5, 6, and 8 pounds. Which is greater, the average (mean) weight of these five children or the median weight, and by how many pounds?

(A) median, by 60      (B) median, by 20      (C) average, by 5      (D) average, by 15      (E) average, by 20

**Answer:** E

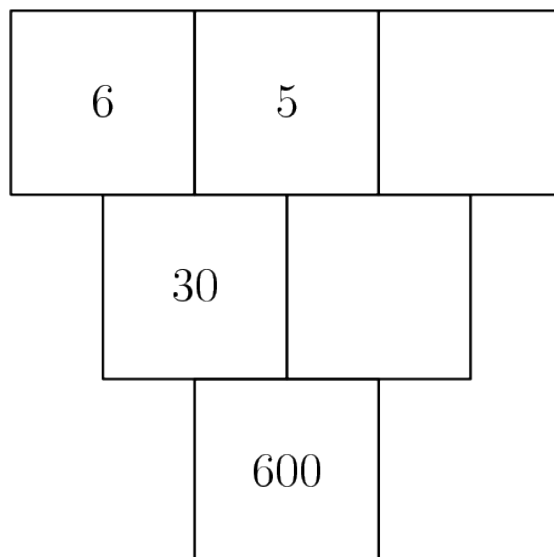
**Solution**

Listing the elements from least to greatest, we have (5, 5, 6, 8, 106), we see that the median weight is 6 pounds. The average weight of the five kids is  $\frac{5+5+6+8+106}{5} = \frac{130}{5} = 26$ .

Hence,

$$26 - 6 = \boxed{\text{(E) average, by 20}}.$$

6. (*AMC 8 2013*) The number in each box below is the product of the numbers in the two boxes that touch it in the row above. For example,  $30 = 6 \times 5$ . What is the missing number in the top row?

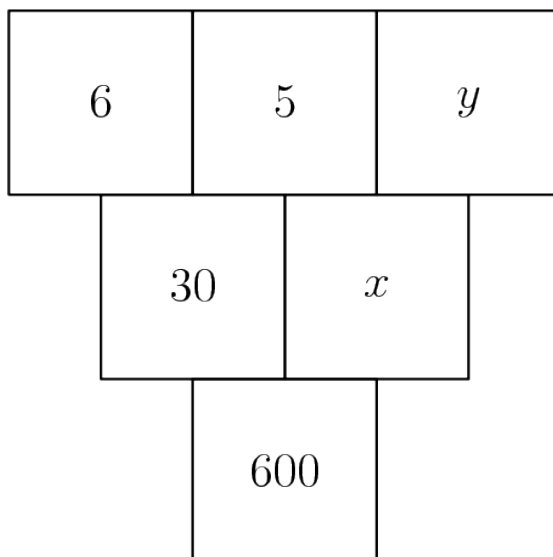


(A) 2      (B) 3      (C) 4      (D) 5      (E) 6

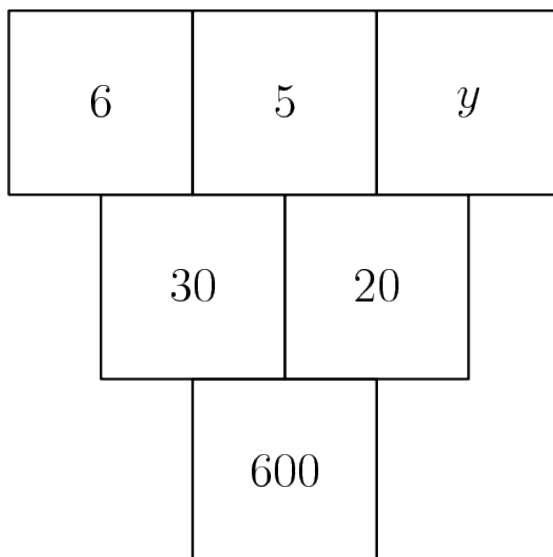
**Answer:** C

**Solution 1 (Working Backwards)**

Let the value in the empty box in the middle row be  $x$ , and the value in the empty box in the top row be  $y$ .  $y$  is the answer we're looking for.



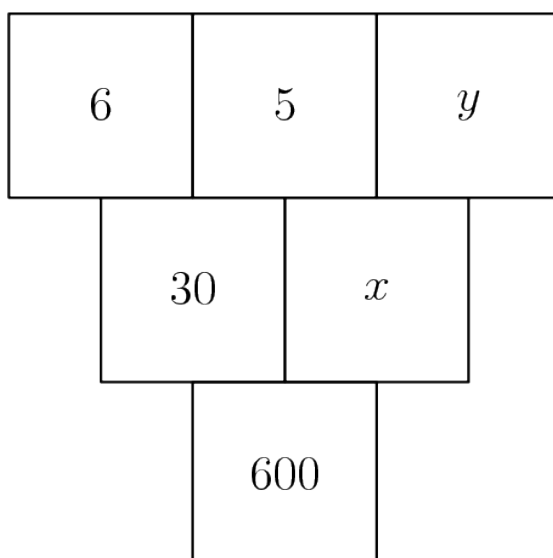
From the diagram,  $600 = 30x$ , making  $x = 20$ .



It follows that  $20 = 5y$ , so  $y = \boxed{\text{(C) } 4}$ .

**Solution 2 (Jumping to the Start)**

Another way to do this problem is to realize what makes up the bottommost number. This method doesn't work quite as well for this problem, but in a larger tree, it might be faster. Again, let the value in the empty box in the middle row be  $x$ , and the value in the empty box in the top row be  $y$ .  $y$  is the answer we're looking for.



We can write some equations:

$$600 = 30x$$

$$30 = 6 \cdot 5$$

$$x = 5y$$

Now we can substitute into the first equation using the two others:

$$600 = (6 \cdot 5)(5y)$$

$$600 = 6 \cdot 5 \cdot 5 \cdot y$$

$$600 = 6 \cdot 25 \cdot y$$

$$600 = 150y$$

$$\boxed{\text{(C) } 4} = y$$

### Solution 3(Brute Force)

Just try all the answer choices until you get 600

7. (AMC 8 2013) Trey and his mom stopped at a railroad crossing to let a train pass. As the train began to pass, Trey counted 6 cars in the first 10 seconds. It took the train 2 minutes and 45 seconds to clear the crossing at a constant speed. Which of the following was the most likely number of cars in the train?

(A) 60    (B) 80    (C) 100    (D) 120    (E) 140

**Answer:** C

### Solution 1

Clearly, for every 5 seconds, 3 cars pass. It's more convenient to have everything in seconds: 2 minutes and 45 seconds =  $2 \cdot 60 + 45 = 165$  seconds. We then set up a ratio:

$$\frac{3}{5} = \frac{x}{165}$$

$$3(165) = 5x$$

$$x = 3(33) = 99 \approx \boxed{\text{(C) } 100}.$$

### Solution 2

2 minutes and 45 seconds is equal to  $120 + 45 = 165$  seconds.

Since 6 cars pass at around 10 seconds, there are about  $\left\lfloor \frac{165}{10} \right\rfloor = 16$  groups of 6 cars.

There are about  $16 \cdot 6 = 96$  cars, so the closest answer choice is **(C) 100**.

P.S: If you got this wrong or wasted time on this question, don't worry. I did too.  
-PearlSeal34

8. (AMC 8 2013) A fair coin is tossed 3 times. What is the probability of at least two consecutive heads?

(A)  $\frac{1}{8}$     (B)  $\frac{1}{4}$     (C)  $\frac{3}{8}$     (D)  $\frac{1}{2}$     (E)  $\frac{3}{4}$

**Answer:** C

**Solution 1**

There are  $2^3 = 8$  ways to flip the coins, in order. There are two ways to get exactly two consecutive heads: HHT and THH. There is only one way to get three consecutive heads:

HHH. Therefore, the probability of flipping at least two consecutive heads is **(C)  $\frac{3}{8}$** .

**Solution 2**

Let's use **complementary counting**. To start with, the unfavorable outcomes (in this case, not getting 2 consecutive heads) are: TTT, HTH, and THT. The probability of these three outcomes is  $\frac{1}{8}$ ,  $\frac{1}{4}$ , and  $\frac{1}{4}$ , respectively. So the rest is exactly the probability of

flipping at least two consecutive heads:  $1 - \frac{1}{8} - \frac{1}{4} - \frac{1}{4} = \frac{3}{8}$ . It is the answer **(C)  $\frac{3}{8}$** .

**Solution 3**

We can list out all the ways to flip a coin three times: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT. Out of them, only HHH, HHT, THH, have at least two consecutive heads. Since there are three ways to flip at least two consecutive heads, and eight total choices, the answer is

**(C)  $\frac{3}{8}$** .

9. (AMC 8 2013) The Incredible Hulk can double the distance it jumps with each succeeding jump. If its first jump is 1 meter, the second jump is 2 meters, the third jump is 4 meters, and so on, then on which jump will it first be able to jump more than 1 kilometer?

(A) 9<sup>th</sup>    (B) 10<sup>th</sup>    (C) 11<sup>th</sup>    (D) 12<sup>th</sup>    (E) 13<sup>th</sup>

**Answer:** C

**Solution**

This is a geometric sequence in which the common ratio is 2. To find the jump that would be over a 1000 meters, we note that  $2^{10} = 1024$ .

However, because the first term is  $2^0 = 1$  and not  $2^1 = 2$ , the solution to the problem is  $10 - 0 + 1 =$  **(C) 11<sup>th</sup>**

10. (AMC 8 2013) What is the ratio of the least common multiple of 180 and 594 to the greatest common factor of 180 and 594?

(A) 110    (B) 165    (C) 330    (D) 625    (E) 660

**Answer:** C

**Solution 1**

To find either the LCM or the GCF of two numbers, always prime factorize first.

The prime factorization of  $180 = 3^2 \times 5 \times 2^2$ .

The prime factorization of  $594 = 3^3 \times 11 \times 2$ .

Then, to find the LCM, we have to find the greatest power of all the numbers there are; if one number is one but not the other, use it (this is  $3^3, 5, 11, 2^2$ ). Multiply all of these to get 5940.

For the GCF of 180 and 594, use the least power of all of the numbers that are in both factorizations and multiply.  $3^2 \times 2 = 18$ .

Thus the answer  $= \frac{5940}{18} = \boxed{(C) 330}$ .

### Solution 2

We start off with a similar approach as the original solution. From the prime factorizations, the GCF is 18.

It is a well known fact that  $\gcd(m, n) \times \text{lcm}(m, n) = |mn|$ . So we have,  $18 \times \text{lcm}(180, 594) = 594 \times 180$ .

Dividing by 18 yields  $\text{lcm}(180, 594) = 594 \times 10 = 5940$ .

Therefore,  $\frac{\text{lcm}(180, 594)}{\gcd(180, 594)} = \frac{5940}{18} = \boxed{(C) 330}$ .

### Solution 3

From Solution 1, the prime factorization of  $180 = 2^2 \cdot 3^2 \cdot 5$ . The prime factorization of  $594 = 2 \cdot 3^3 \cdot 11$ . Hence,  $\text{lcm}(180, 594) = 2^2 \cdot 3^3 \cdot 5 \cdot 11$ , and  $\gcd(180, 594) = 2 \cdot 3^2$ . Therefore,  $\frac{\text{lcm}(180, 594)}{\gcd(180, 594)} = \frac{2^2 \cdot 3^3 \cdot 5 \cdot 11}{2 \cdot 3^2} = 2 \cdot 3 \cdot 5 \cdot 11 = 330 \implies \boxed{(C) 330}$

11. (AMC 8 2013) Ted's grandfather used his treadmill on 3 days this week. He went 2 miles each day. On Monday he jogged at a speed of 5 miles per hour. He walked at the rate of 3 miles per hour on Wednesday and at 4 miles per hour on Friday. If Grandfather had always walked at 4 miles per hour, he would have spent less time on the treadmill. How many minutes less?

(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** D

### Solution

We use that fact that  $d = rt$ . Let  $d$  = distance,  $r$  = rate or speed, and  $t$  = time. In this case, let  $x$  represent the time.

On Monday, he was at a rate of 5 m.p.h. So,  $5x = 2$  miles  $\implies x = \frac{2}{5}$  hours.

For Wednesday, he walked at a rate of 3 m.p.h. Therefore,  $3x = 2$  miles  $\implies x = \frac{2}{3}$  hours.

On Friday, he walked at a rate of 4 m.p.h. So,  $4x = 2$  miles  $\implies x = \frac{2}{4} = \frac{1}{2}$  hours.

Adding up the hours yields  $\frac{2}{5}$  hours +  $\frac{2}{3}$  hours +  $\frac{1}{2}$  hours =  $\frac{47}{30}$  hours.

We now find the amount of time Grandfather would have taken if he walked at 4 m.p.h per day. Set up the equation,  $4x = 2$  miles  $\times 3$  days  $\implies x = \frac{3}{2}$  hours.

To find the amount of time saved, subtract the two amounts:  $\frac{47}{30}$  hours -  $\frac{3}{2}$  hours =  $\frac{1}{15}$  hours. To convert this to minutes, we multiply by 60.

Thus, the solution to this problem is  $\frac{1}{15} \times 60 = \boxed{(D) 4}$

**Short Solution In Two Short Steps!** 1. Calculate the minutes he used: Monday - 1 mile = 12 minutes. Thus, he used  $12 \times 2 = 24$  minutes. Wednesday - 1 mile = 20 minutes.

Thus, he used  $20 \times 2 = 40$  minutes. Friday - 1 mile = 15 minutes. Thus, he use  $15 \times 2 = 30$  minutes.  $24 + 40 + 30 = 94$

2. Minus the minutes he could've used: From Friday's experience, we know two miles takes 30 minutes.  $30 \times 3 = 90$  minutes.  $94 - 90 = 4$ .

Thus, the answer is **(D) 4**

12. (AMC 8 2013) At the 2013 Winnebago County Fair a vendor is offering a "fair special" on sandals. If you buy one pair of sandals at the regular price of 50, you get a second pair at a 40% discount, and a third pair at half the regular price. Javier took advantage of the "fair special" to buy three pairs of sandals. What percentage of the 150 dollar regular price did he save?

(A) 25%      (B) 30%      (C) 33%      (D) 40%      (E) 45%

**Answer:** B

**Solution**

First, find the amount of money one will pay for three sandals without the discount. We have  $\$50 \times 3$  sandals = \$150.

Then, find the amount of money using the discount:  $50 + 0.6 \times 50 + \frac{1}{2} \times 50 = \$105$ .

Finding the percentage yields  $\frac{105}{150} = 70\%$ .

To find the percent saved, we have  $100\% - 70\% = \mathbf{(B) 30\%}$

**Solution 2**

Step 1: Find the price of each sandal The first sandal is 50 dollars. The second sandal is  $(1 - 0.4)50 = 30$  The third sandal is  $\frac{1}{2} \times 50 = 25$  Step 2 : Sum them up  $50 + 30 + 25 = 105$  Step 2 : Divide by 150  $105/150 = 21/30 = 7/10 = 70\text{percent}$ . Step 3 : Subtract from 100 percent to find the amount the saved  $70\text{percent} = B) 30\text{percent}$

-JasonDaGoat

13. (AMC 8 2013) When Clara totaled her scores, she inadvertently reversed the units digit and the tens digit of one score. By which of the following might her incorrect sum have differed from the correct one?

(A) 45      (B) 46      (C) 47      (D) 48      (E) 49

**Answer:** A

**Solution(educated guess)**

Let the two digits be  $a$  and  $b$ .

The correct score was  $10a + b$ . Clara misinterpreted it as  $10b + a$ . The difference between the two is  $|9a - 9b|$  which factors into  $|9(a - b)|$ . Therefore, since the difference is a multiple of 9, the only answer choice that is a multiple of 9 is **(A) 45**.

**Solution 2(brute force elimination)**

We can simply test out numbers to see which one works. We can see that Clara's score can't be a multiple of ten because the reverse of the score is a one-digit number, which is too small for the answer choices. After testing multiples, the answer should be **(A) 45**.

14. (AMC 8 2013) Abe holds 1 green and 1 red jelly bean in his hand. Bob holds 1 green, 1 yellow, and 2 red jelly beans in his hand. Each randomly picks a jelly bean to show the other. What is the probability that the colors match?

- (A)  $\frac{1}{4}$     (B)  $\frac{1}{3}$     (C)  $\frac{3}{8}$     (D)  $\frac{1}{2}$     (E)  $\frac{2}{3}$

**Answer:** C

**Solution**

The favorable responses are either they both show a green bean or they both show a red bean. The probability that both show a green bean is  $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$ . The probability that both show a red bean is  $\frac{1}{2} \cdot \frac{2}{4} = \frac{1}{4}$ . Therefore the probability is  $\frac{1}{4} + \frac{1}{8} = \boxed{\text{(C)} \frac{3}{8}}$

**Solution 2**

We can list out all the combinations and we get this:  $GG, GY, GR_1, GR_2, RG, RY, RR_1, RR_2$ . There is a total of 8 combinations and 3 that are the same. Hence, we yield the answer  $\boxed{\text{C}}$ .

15. (AMC 8 2013) If  $3^p + 3^4 = 90$ ,  $2^r + 44 = 76$ , and  $5^3 + 6^s = 1421$ , what is the product of  $p$ ,  $r$ , and  $s$ ?

- (A) 27    (B) 40    (C) 50    (D) 70    (E) 90

**Answer:** B

**Solution 1: Solving**

First, we're going to solve for  $p$ . Start with  $3^p + 3^4 = 90$ . Then, change  $3^4$  to 81. Subtract 81 from both sides to get  $3^p = 9$  and see that  $p$  is 2. Now, solve for  $r$ . Since  $2^r + 44 = 76$ ,  $2^r$  must equal 32, so  $r = 5$ . Now, solve for  $s$ .  $5^3 + 6^s = 1421$  can be simplified to  $125 + 6^s = 1421$  which simplifies further to  $6^s = 1296$ . Therefore,  $s = 4$ .  $prs$  equals  $2 * 5 * 4$  which equals 40. So, the answer is  $\boxed{\text{(B)} 40}$ .

**Solution 2: Process of Elimination**

First, we solve for  $s$ . As Solution 1 perfectly states,  $5^3 + 6^s = 1421$  can be simplified to  $125 + 6^s = 1421$  which simplifies further to  $6^s = 1296$ . Therefore,  $s = 4$ . We know that you cannot take a root of any of the numbers raised to  $p$ ,  $r$ , or  $s$  and get a rational answer, and none of the answer choices are irrational, so that rules out the possibility that  $p$ ,  $r$ , or  $s$  is a fraction. The only answer choice that is divisible by 4 is  $\boxed{\text{(B)} 40}$ .

16. (AMC 8 2013) A number of students from Fibonacci Middle School are taking part in a community service project. The ratio of 8<sup>th</sup>-graders to 6<sup>th</sup>-graders is 5 : 3, and the ratio of 8<sup>th</sup>-graders to 7<sup>th</sup>-graders is 8 : 5. What is the smallest number of students that could be participating in the project?

- (A) 16    (B) 40    (C) 55    (D) 79    (E) 89

**Answer:** E

**Solution 1: Algebra**

We multiply the first ratio by 8 on both sides, and the second ratio by 5 to get the same number for 8th graders, in order that we can put the two ratios together:

$$5 : 3 = 5(8) : 3(8) = 40 : 24 \quad 8 : 5 = 8(5) : 5(5) = 40 : 25$$

Therefore, the ratio of 8th graders to 7th graders to 6th graders is 40 : 25 : 24. Since the ratio is in lowest terms, the smallest number of students participating in the project is  $40 + 25 + 24 = \boxed{\text{(E)} 89}$ .

**Solution 2: Fakesolving**

The number of 8th graders has to be a multiple of 8 and 5, so assume it is 40 (the smallest possibility). Then there are  $40 * \frac{3}{5} = 24$  6th graders and  $40 * \frac{5}{8} = 25$  7th graders. The numbers of students is  $40 + 24 + 25 = \boxed{\text{(E)} 89}$

17. (AMC 8 2013) The sum of six consecutive positive integers is 2013. What is the largest of these six integers?

(A) 335      (B) 338      (C) 340      (D) 345      (E) 350

**Answer:** B

**Solution 1**

The arithmetic mean of these numbers is  $\frac{2013}{6} = \frac{671}{2} = 335.5$ . Therefore the numbers are 333, 334, 335, 336, 337, 338, so the answer is  $\boxed{\text{(B)} 338}$ .

**Solution 2**

Let the 4<sup>th</sup> number be  $x$ . Then our desired number is  $x + 2$ .

Our integers are  $x - 3, x - 2, x - 1, x, x + 1, x + 2$ , so we have that  $6x - 3 = 2013 \implies x = \frac{2016}{6} = 336 \implies x + 2 = \boxed{\text{(B)} 338}$ .

**Solution 3**

Let the first term be  $x$ . Our integers are  $x, x + 1, x + 2, x + 3, x + 4, x + 5$ . We have,  $6x + 15 = 2013 \implies x = 333 \implies x + 5 = \boxed{\text{(B)} 338}$ .

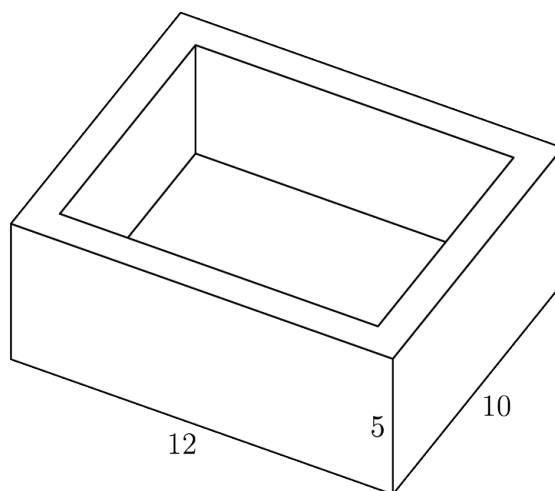
**Solution 4**

Since there are 6 numbers, we divide 2013 by 6 to find the mean of the numbers.  $\frac{2013}{6} = 335\frac{1}{2}$ . Then,  $335\frac{1}{2} + \frac{1}{2} = 336$  (the fourth number). Fifth: 337; Sixth:  $\boxed{338}$ .

**Solution 5**

Let the 6<sup>th</sup> number be  $x$ . Then our list is:  $x - 6 + x - 5 + x - 4 + x - 3 + x - 2 + x - 1 = 2013$ . Simplifying this gets you  $6x - 21 = 2013 \implies 6x = 2034$ , which means that  $x = \boxed{\text{(B)} 338}$ .

18. (AMC 8 2013) Isabella uses one-foot cubical blocks to build a rectangular fort that is 12 feet long, 10 feet wide, and 5 feet high. The floor and the four walls are all one foot thick. How many blocks does the fort contain?



(A) 204      (B) 280      (C) 320      (D) 340      (E) 600

**Answer:** B



**Solution 1**

There are  $10 \cdot 12 = 120$  cubes on the base of the box. Then, for each of the 4 layers above the bottom (as since each cube is 1 foot by 1 foot by 1 foot and the box is 5 feet tall, there are 4 feet left), there are  $9 + 11 + 9 + 11 = 40$  cubes. Hence, the answer is  $120 + 4 \cdot 40 = \boxed{\text{(B)} 280}$ .

**Solution 2 (Complementary Counting)**

We can just calculate the volume of the prism that was cut out of the original  $12 \times 10 \times 5$  box. Each interior side of the fort will be 2 feet shorter than each side of the outside. Since the floor is 1 foot, the height will be 4 feet. So the volume of the interior box is  $10 \times 8 \times 4 = 320 \text{ ft}^3$ .

The volume of the original box is  $12 \times 10 \times 5 = 600 \text{ ft}^3$ . Therefore, the number of blocks contained in the fort is  $600 - 320 = \boxed{\text{(B)} 280}$

19. (AMC 8 2013) Bridget, Cassie, and Hannah are discussing the results of their last math test. Hannah shows Bridget and Cassie her test, but Bridget and Cassie don't show theirs to anyone. Cassie says, 'I didn't get the lowest score in our class,' and Bridget adds, 'I didn't get the highest score.' What is the ranking of the three girls from the highest score to the lowest score?
- (A) Hannah, Cassie, Bridget      (B) Hannah, Bridget, Cassie  
 (C) Cassie, Bridget, Hannah      (D) Cassie, Hannah, Bridget  
 (E) Bridget, Cassie, Hannah

**Answer:** D

**Solution**

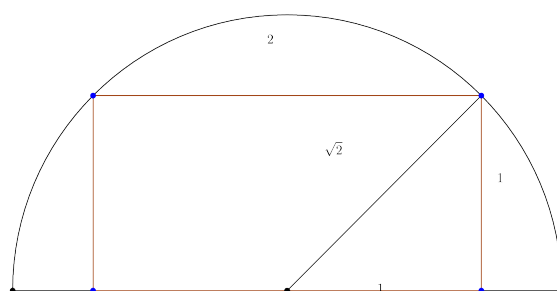
If Hannah did better than Cassie, there would be no way she could know for sure that she didn't get the lowest score in the class. Therefore, Hannah did worse than Cassie. Similarly, if Hannah did worse than Bridget, there is no way Bridget could have known that she didn't get the highest in the class. Therefore, Hannah did better than Bridget, so our order is  $\boxed{\text{(D) Cassie, Hannah, Bridget}}$

Note: It could be said that Cassie did better than Hannah, and Bridget knew this, and she had the same score as Hannah, so Bridget didn't get the highest score. This may be a reason that MAA didn't put this answer choice in.

20. (AMC 8 2013) A  $1 \times 2$  rectangle is inscribed in a semicircle with longer side on the diameter. What is the area of the semicircle?
- (A)  $\frac{\pi}{2}$       (B)  $\frac{2\pi}{3}$       (C)  $\pi$       (D)  $\frac{4\pi}{3}$       (E)  $\frac{5\pi}{3}$

**Answer:** C

**Solution**



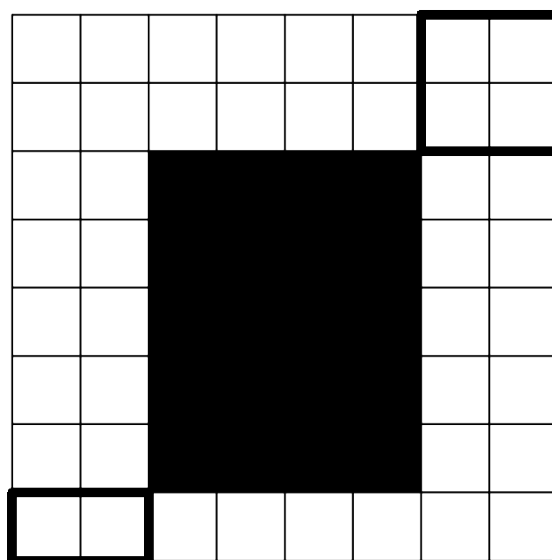
A semicircle has symmetry, so the center is exactly at the midpoint of the 2 side on the rectangle, making the radius, by the Pythagorean Theorem,  $\sqrt{1^2 + 1^2} = \sqrt{2}$ . The area is  $\frac{2\pi}{2} = \boxed{\text{(C)} \pi}$ .

**List of steps** 1. Draw a picture, realize the radius comes from the midpoint of rectangle. Using the pythagorean theorm, solve for the radius with is square root of 2. Solve for the area of the circle using the radius pi times radius squared, with will give you 2pi. Divide by 2, getting pi-Jason Da Goat

21. (AMC 8 2013) Samantha lives 2 blocks west and 1 block south of the southwest corner of City Park. Her school is 2 blocks east and 2 blocks north of the northeast corner of City Park. On school days she bikes on streets to the southwest corner of City Park, then takes a diagonal path through the park to the northeast corner, and then bikes on streets to school. If her route is as short as possible, how many different routes can she take?
- (A) 3      (B) 6      (C) 9      (D) 12      (E) 18

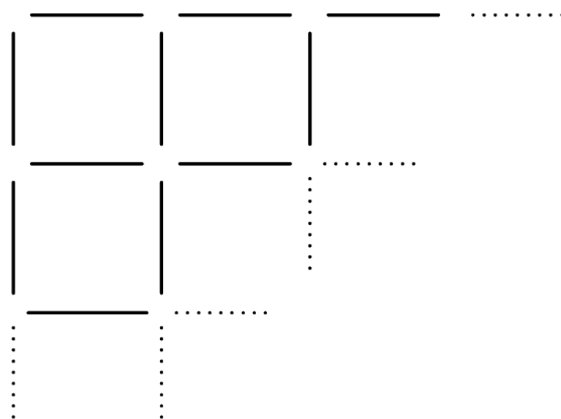
**Answer:** E

**Solution**



Using **combinations**, we get that the number of ways to get from Samantha's house to City Park is  $\binom{3}{1} = \frac{3!}{1!2!} = 3$ , and the number of ways to get from City Park to school is  $\binom{4}{2} = \frac{4!}{2!2!} = \frac{4 \cdot 3}{2} = 6$ . Since there's one way to go through City Park (just walking straight through), the number of different ways to go from Samantha's house to City Park to school  $3 \cdot 6 = \boxed{\text{(E)} 18}$ .

22. (AMC 8 2013) Toothpicks are used to make a grid that is 60 toothpicks long and 32 toothpicks wide. How many toothpicks are used altogether?



- (A) 1920      (B) 1952      (C) 1980      (D) 2013      (E) 3932

**Answer:** E

### Solution 1

There are 61 vertical columns with a length of 32 toothpicks, and there are 33 horizontal rows with a length of 60 toothpicks, because 32 and 60 are the number of intervals. You can verify this by trying a smaller case, i.e. a  $3 \times 4$  grid of toothpicks, with  $3 \times 3$  and  $2 \times 4$ .

Thus, our answer is  $61 \cdot 32 + 33 \cdot 60 = \boxed{\text{(E)} 3932}$ .

### Solution 2 - Common Sense

With a quick mental calculation,  $60 \cdot 30$  yields 1800, which is roughly where 4 of our 5 answer choices lie in. However, we can tell that each square would require at most 2 toothpicks that uniquely belong to itself, so the answer would be  $60 \cdot 30 \cdot 2$  which would be roughly  $\boxed{\text{(E)} 3932}$ .

-pearlseal34, Minor Edit by MagentaButterfly29

### Solution 3 - Deriving a Formula

If you wanted to find the number of toothpicks in a  $x$  by  $y$  rectangle you will notice a pattern. We can separate the toothpicks into 2 groups, vertical and horizontal toothpicks and add them together. There is  $x(y+1)$  vertical toothpick that are vertical and  $y(x+1)$  horizontal toothpicks. If my explanation doesn't make sense try drawing it out. You can add these two for the total amount of toothpicks. Now we can plug in 60 into  $x$  and 32 into  $y$  and solve.  $60(32+1)+32(60+1)= 60(32)+60+32(60)+32=3932$  which is  $\boxed{\text{(E)} 3932}$ .

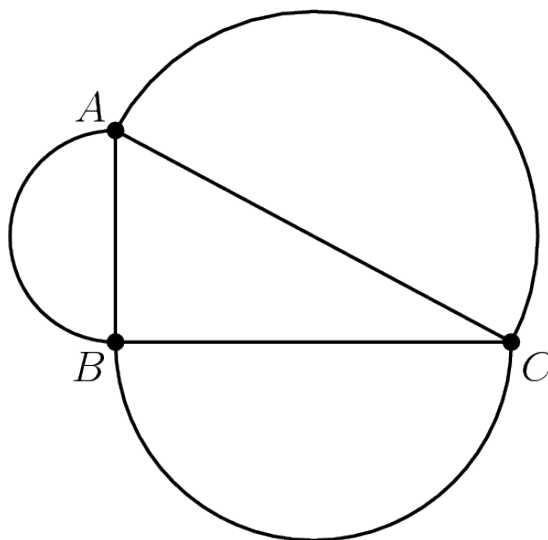
-ZenJacob

### Solution 4 - Border vs. Internal toothpicks

The total area of the  $60 \times 32$  grid is 1920 square units. If each square was separated, we would need  $1920 \cdot 4 = 7680$  toothpicks. However, once all the squares are joined to form a grid, each internal toothpick forms the edge of two adjacent squares. The perimeter/border of the grid contains  $60 + 60 + 32 + 32 = 184$  sticks. So,  $7680 - 184 = 7496$  toothpicks are internal toothpicks. Since each internal toothpick forms the edge of two adjacent squares, we need only  $7496 / 2 = 3748$  of them. So, the total toothpicks needed are 3748 (internal) + 184 (border) which is  $\boxed{\text{(E)} 3932}$ .

-AbbasK

23. (AMC 8 2013) Angle  $ABC$  of  $\triangle ABC$  is a right angle. The sides of  $\triangle ABC$  are the diameters of semicircles as shown. The area of the semicircle on  $\overline{AB}$  equals  $8\pi$ , and the arc of the semicircle on  $\overline{AC}$  has length  $8.5\pi$ . What is the radius of the semicircle on  $\overline{BC}$ ?



- (A) 7      (B) 7.5      (C) 8      (D) 8.5      (E) 9

**Answer:** B

**Solution 1**

If the semicircle on  $\overline{AB}$  were a full circle, the area would be  $16\pi$ .

$\pi r^2 = 16\pi \Rightarrow r^2 = 16 \Rightarrow r = +4$ , therefore the diameter of the first circle is 8.

The arc of the largest semicircle is  $8.5\pi$ , so if it were a full circle, the circumference would be  $17\pi$ . So the diameter = 17.

By the Pythagorean theorem, the other side has length 15, so the radius is (B) 7.5

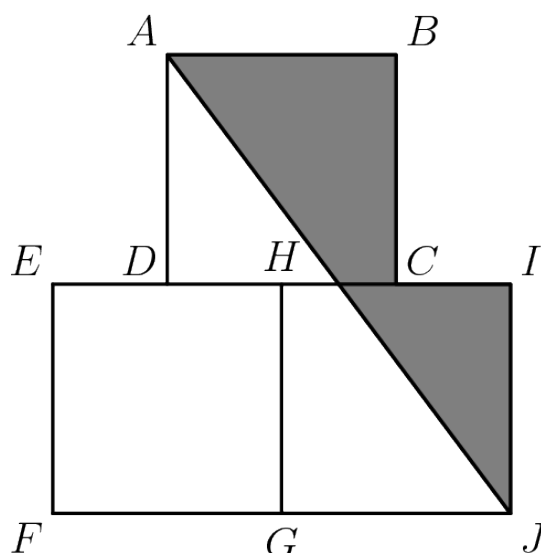
**Brief Explanation** SavannahSolver got a diameter of 17 because the given arc length of the semicircle was  $8.5\pi$ . The arc length of a semicircle can be calculated using the formula  $\pi r$ , where  $r$  is the radius. let's use the full circumference formula for a circle, which is  $2\pi r$ . Since the semicircle is half of a circle, its arc length is  $\pi r$ , which was given as  $8.5\pi$ . Solving for  $r$ , we get  $r = 8.5$ . Therefore, the diameter, which is  $2r$ , is  $2 \cdot 8.5 = 17$ . Then, the other steps to solve the problem will be the same as mentioned above by SavannahSolver the answer is (B) 7.5

. - TheNerdWhoIsNerdy.Minor edits by -Coin1

**Solution 2**

We go as in Solution 1, finding the diameter of the circle on  $\overline{AC}$  and  $\overline{AB}$ . Then, an extended version of the theorem says that the sum of the semicircles on the left is equal to the biggest one, so the area of the largest is  $\frac{289\pi}{8}$ , and the middle one is  $\frac{289\pi}{8} - \frac{64\pi}{8} = \frac{225\pi}{8}$ , so the radius is  $\frac{15}{2} = \span style="border: 1px solid black; padding: 0 2px;">(B) 7.5.$

24. (AMC 8 2013) Squares  $ABCD$ ,  $EFGH$ , and  $GHIJ$  are equal in area. Points  $C$  and  $D$  are the midpoints of sides  $IH$  and  $HE$ , respectively. What is the ratio of the area of the shaded pentagon  $AJICB$  to the sum of the areas of the three squares?



- (A)  $\frac{1}{4}$     (B)  $\frac{7}{24}$     (C)  $\frac{1}{3}$     (D)  $\frac{3}{8}$     (E)  $\frac{5}{12}$

**Answer:** C

**Solution 1**

Let point X be the point that intersects line EI. We can split our original triangle into trapezoid ABCX and triangle XIJ. WLOG (Without Loss Of Generality), AB equals 1 unit. Then since, X is the midpoint of line AJ, and point A is 1.5 units horizontally from J, the midpoint X will be 0.75 units away horizontally from A and thus 0.25 units horizontally from C. Therefore XC equals 0.25 units. Using the area formulas for trapezoids and triangles, we calculate the area of ABCX to be 0.625 and XIJ to be 0.375. The combined areas (which are equivalent to our original triangle) equal 1. Therefore, the ratio of the area of the hexagon to the three squares is 1:3 because the area of the three squares is 3. The answer is (C)  $\frac{1}{3}$  - TheNerdwhoIsNerdy.

Edits for Clarity and Accuracy by RamanujanIsBetter

**Solution 2 (extremely simple)**

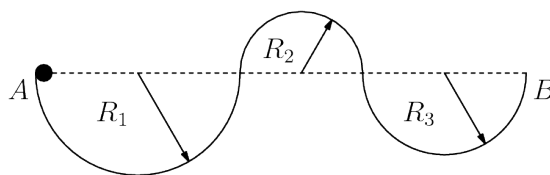
Extend line AD such that it intersects line FG. Call the intersection K. If we assign an arbitrary value to the side lengths of the squares, such as 2, the base of the now formed triangle AJK would have a length of 3, while the height would be 4. The area of triangle AJK would then be 6, and the rectangle EDKF would have an area of 2.  $6 + 2 = 8$ , and since all of the squares' areas add up to a total of 12 (if each has side length 2), we obtain  $(12 - 8)/12$  for the shaded part of the diagram which yields  $4/12 =$  (C)  $\frac{1}{3}$

**Solution 3 (Also very simple)**

Name the intersection of line segment AJ and CH as K. Notice that triangle ADK is congruent to triangle JIK by AAS (Angle-Angle-Side), so they have the same area. Therefore, the shaded area is just the area of the topmost square which is (C)  $\frac{1}{3}$  of the whole.

25. (AMC 8 2013) A ball with diameter 4 inches starts at point A to roll along the track shown. The track is comprised of 3 semicircular arcs whose radii are  $R_1 = 100$  inches,

$R_2 = 60$  inches, and  $R_3 = 80$  inches, respectively. The ball always remains in contact with the track and does not slip. What is the distance the center of the ball travels over the course from A to B?



- (A)  $238\pi$     (B)  $240\pi$     (C)  $260\pi$     (D)  $280\pi$     (E)  $500\pi$

**Answer:** A

**Solution 1**

The total length of all of the arcs is  $100\pi + 80\pi + 60\pi = 240\pi$ . Since we want the path from the center, the actual distance will be subtracted by  $2\pi$  because it's already half the circumference through semicircle A, which needs to go half the circumference extra through semicircle B, and it's already half the circumference through semicircle C, and the circumference is  $4\pi$ . Therefore, the answer is  $240\pi - 2\pi = \boxed{\text{(A)} 238\pi}$ .

### §30 AMC 8 2014

1. (*AMC 8 2014*) Harry and Terry are each told to calculate  $8 - (2 + 5)$ . Harry gets the correct answer. Terry ignores the parentheses and calculates  $8 - 2 + 5$ . If Harry's answer is  $H$  and Terry's answer is  $T$ , what is  $H - T$ ?

- (A)  $-10$     (B)  $-6$     (C)  $0$     (D)  $6$     (E)  $10$

**Answer:** A

**Solution**

We have  $H = 8 - 7 = 1$  and  $T = 8 - 2 + 5 = 11$ . Clearly  $1 - 11 = -10$ , so our answer is  $\boxed{\text{(A)} -10}$ .

**Solution 2**

We use PEMDAS,  $8 - (2 + 5)$  is the same as  $8 - 7$  which is 1,  $8 - 2 + 5$  is the same as  $6 + 5$  because  $8 - 2$  comes first, and the sum is 11.  $H - T$  is  $1 - 11$  which is  $\boxed{\text{(A)} -10}$ .

**Solution 3**

Harry, who calculates it correctly, gets  $8 - 7 = 1$ . Terry, who calculates it without parenthesis gets  $8 - 2 + 5$  which is  $6 + 5 = 11$ .

$$1 - 11 = -10$$

-JasonDaGoat

2. (*AMC 8 2014*) Paul owes Paula 35 cents and has a pocket full of 5-cent coins, 10-cent coins, and 25-cent coins that he can use to pay her. What is the difference between the largest and the smallest number of coins he can use to pay her?

- (A) 1    (B) 2    (C) 3    (D) 4    (E) 5

**Answer:** E

**Solution**

The fewest amount of coins that can be used is 2 (a quarter and a dime). The greatest amount is 7, if he only uses nickels. Therefore we have  $7 - 2 = \boxed{\text{(E)} 5}$ .

**Solution 2**

Quarter and Dime 2 coins All nickels  $35/5 = 7$  coins

$$7 \text{ coins} - 2 \text{ coins} = 5 \text{ coins}$$

-JasonDaGoat

3. (*AMC 8 2014*) Isabella had a week to read a book for a school assignment. She read an average of 36 pages per day for the first three days and an average of 44 pages per day for the next three days. She then finished the book by reading 10 pages on the last day. How many pages were in the book?

(A) 240    (B) 250    (C) 260    (D) 270    (E) 280

**Answer:** B

**Solution**

Isabella read  $3 \cdot 36 + 3 \cdot 44$  pages in the first 6 days. Although this can be calculated directly, it is simpler to calculate it as  $3 \cdot (36 + 44) = 3 \cdot 80$ , which gives that she read 240 pages. On her last day, she read 10 more pages for a total of  $240 + 10 = \boxed{\text{(B)} 250}$  pages.

4. (*AMC 8 2014*) The sum of two prime numbers is 85. What is the product of these two prime numbers?

(A) 85    (B) 91    (C) 115    (D) 133    (E) 166

**Answer:** E

**Solution**

Since the two prime numbers sum to an odd number, one of them must be even. The only even prime number is 2. The other prime number is  $85 - 2 = 83$ , and the product of these two numbers is  $83 \cdot 2 = \boxed{\text{(E)} 166}$ .

5. (*AMC 8 2014*) Margie's car can go 32 miles on a gallon of gas, and gas currently costs 4 per gallon. How many miles can Margie drive on 20?

(A) 64    (B) 128    (C) 160    (D) 320    (E) 640

**Answer:** C

**Solution**

Margie can afford  $20/4 = 5$  gallons of gas. She can go  $32 \cdot 5 = \boxed{\text{(C)} 160}$  miles on this amount of gas.

6. (*AMC 8 2014*) Six rectangles each with a common base width of 2 have lengths of 1, 4, 9, 16, 25, and 36. What is the sum of the areas of the six rectangles?

(A) 91    (B) 93    (C) 162    (D) 182    (E) 202

**Answer:** D

**Solution**

The sum of the areas is equal to  $2 \cdot 1 + 2 \cdot 4 + 2 \cdot 9 + 2 \cdot 16 + 2 \cdot 25 + 2 \cdot 36$ . This is equal to  $2(1 + 4 + 9 + 16 + 25 + 36)$ , which is equal to  $2 \cdot 91$ . This is equal to our final answer of  $\boxed{\text{(D)} 182}$ .

**Solution 2**

we can just multiply the common width 2 by each of the lengths 1 by 1, the sum would be 182. This is slow and grouping the lengths is easier to. The answer is still **(D) 182**.

**Solution 3**

The formula for a consecutive perfect squared sum is  $S = \frac{n(n+1)(2n+1)}{6}$ , where  $n$  is the number of terms from 1.

Multiplying by the constant length 2 for area gives  $S = \frac{n(n+1)(2n+1)}{3}$ .

Plugging in  $n = 6$  gives **(D) 182**.

7. (AMC 8 2014) There are four more girls than boys in Ms. Raub's class of 28 students. What is the ratio of number of girls to the number of boys in her class?

(A) 3 : 4      (B) 4 : 3      (C) 3 : 2      (D) 7 : 4      (E) 2 : 1

**Answer:** B

**Solution 1**

Let  $g$  being the number of girls in the class. The number of boys in the class is equal to  $g - 4$ . Since the total number of students is equal to 28, we get  $g + g - 4 = 28$ . Solving this equation, we get  $g = 16$ . There are  $16 - 4 = 12$  boys in our class, and our answer is  $16 : 12 = \mathbf{(B) 4 : 3}$ .

**Solution 2**

To make the amount of boys and girls equal,  $28 - 4 = 24$ .  $24/2 = 12$ . The girls would need to be  $12 +$  the 4 that we subtracted  $= 16$ . The boys would be 12. The ratio of girls to boys would be  $16 : 12$ , but simplified would be  $4 : 3$ . Thus, the answer is  $4 : 3$ .

**Solution 3**

We should be able to divide 28 by the sum of the ratios. If we do guess and check, the only two that complete these conditions is 3:4 and 4:3 since they add up to 7. But, the number of girls has to be more than the boys, so the answer is **(B) 4 : 3**.

8. (AMC 8 2014) Eleven members of the Middle School Math Club each paid the same integer amount for a guest speaker to talk about problem solving at their math club meeting. In all, they paid their guest speaker \$1A2. What is the missing digit A of this 3-digit number?

(A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** D

**Solution 1**

Since all the eleven members paid the same amount, that means that the total must be divisible by 11. We can do some trial-and-error to get  $A = 3$ , so our answer is **(D) 3** SparklyFlowers

**Solution 2**

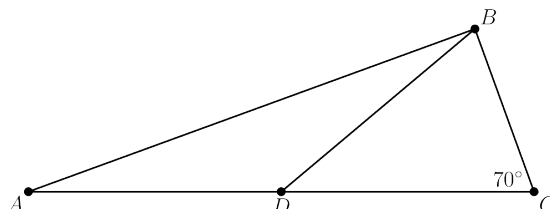
We know that a number is divisible by 11 if the odd digits added together minus the even digits added together (or vice versa) is a multiple of 11. Thus, we have  $1 + 2 - A =$  a multiple of 11. The only multiple that works here is 0, as  $11 \cdot 0 = 0$ . Thus,  $A = \mathbf{(D) 3}$  fn106068



**Solution 3**

If you know your multiplication tables up to  $12 \times 12$ , you know that the out of the choices provided, the only number with 1A2 that is divisible by 11 is 132 or  $12 \times 11$ . So the answer is **(D) 3**. Jupiter10

9. (AMC 8 2014) In  $\triangle ABC$ ,  $D$  is a point on side  $\overline{AC}$  such that  $BD = DC$  and  $\angle BCD$  measures  $70^\circ$ . What is the degree measure of  $\angle ADB$ ?



- (A) 100      (B) 120      (C) 135      (D) 140      (E) 150

**Answer:** D

**Solution**

Using angle chasing is a good way to solve this problem.  $BD = DC$ , so  $\angle DBC = \angle DCB = 70^\circ$ , because it is an isosceles triangle. Then  $\angle CDB = 180 - (70 + 70) = 40$ . Since  $\angle ADB$  and  $\angle BDC$  are supplementary,  $\angle ADB = 180 - 40 = \mathbf{(D) 140}$ .

10. (AMC 8 2014) The first AMC 8 was given in 1985 and it has been given annually since that time. Samantha turned 12 years old the year that she took the seventh AMC 8. In what year was Samantha born?
- (A) 1979      (B) 1980      (C) 1981      (D) 1982      (E) 1983

**Answer:** A

**Solution 1**

The seventh AMC 8 would have been given in 1991. If Samantha was 12 then, that means she was born 12 years ago, so she was born in  $1991 - 12 = 1979$ .

Our answer is **(A) 1979**

corrections made by DrDominic

**Solution 2**

Since she was 12 when she took the seventh AMC 8, she should be  $12 - 6 = 6$  years old when the first AMC 8 occurred. Therefore, she was born or was 'age 0' in  $1985 - 6 = \mathbf{(A) 1979}$  SweetMango77

corrections made by DrDominic

11. (AMC 8 2014) Jack wants to bike from his house to Jill's house, which is located three blocks east and two blocks north of Jack's house. After biking each block, Jack can continue either east or north, but he needs to avoid a dangerous intersection one block east and one block north of his house. In how many ways can he reach Jill's house by biking a total of five blocks?
- (A) 4      (B) 5      (C) 6      (D) 8      (E) 10

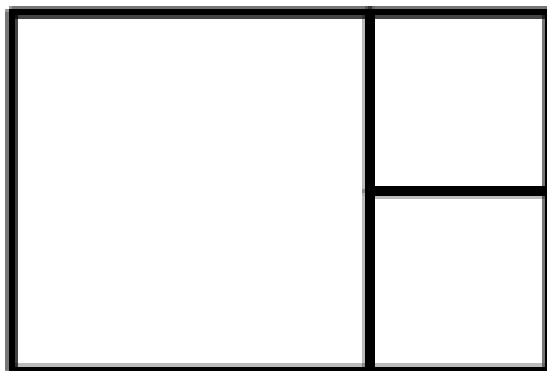
**Answer:** A

**Solution 1**

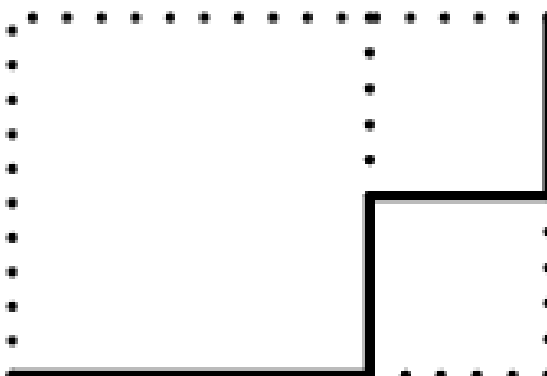
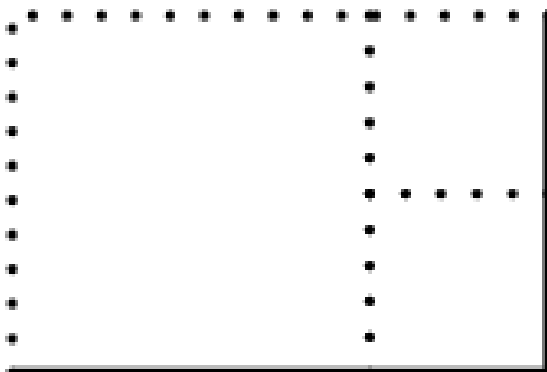
We can apply complementary counting and count the paths that DO go through the blocked intersection, which is  $\binom{2}{1}\binom{3}{1} = 6$ . There are a total of  $\binom{5}{2} = 10$  paths, so there are  $10 - 6 = 4$  paths possible. (A)4 is the correct answer.

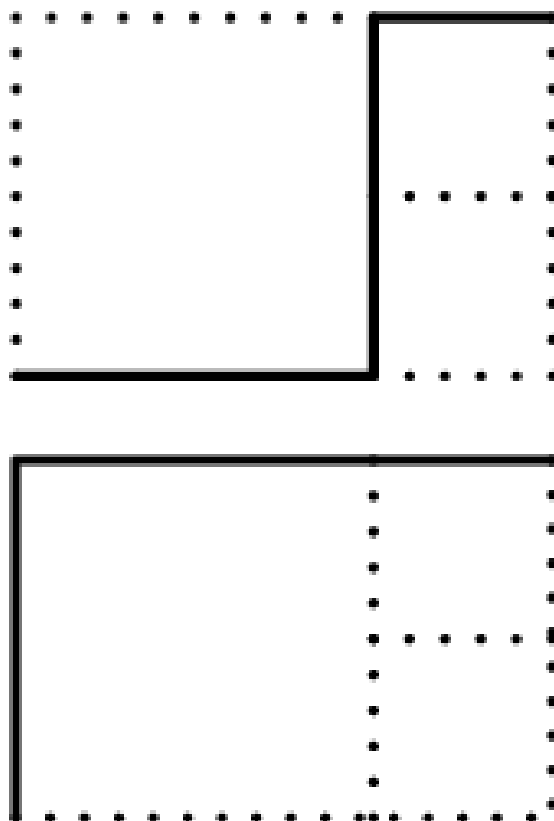
**Solution 2**

We can make a diagram of the roads available to Jack.



Then, we can simply list the possible routes.





There are 4 possible routes, so our answer is A.

Note: This is not recommended in more complicated problems (e.g. Jill's house is 1000 blocks east and 400 blocks north of Jack's house).

12. (*AMC 8 2014*) A magazine printed photos of three celebrities along with three photos of the celebrities as babies. The baby pictures did not identify the celebrities. Readers were asked to match each celebrity with the correct baby pictures. What is the probability that a reader guessing at random will match all three correctly as a fraction?

(A)  $\frac{1}{9}$     (B)  $\frac{1}{6}$     (C)  $\frac{1}{4}$     (D)  $\frac{1}{3}$     (E)  $\frac{1}{2}$

**Answer:** B

**Solution 1**

For the first celebrity, you have a  $\frac{1}{3}$  chance of picking the photo. Given you get the picture right, you now have a  $\frac{1}{2}$  chance of picking the next photo. If you get both of them right, you are guaranteed to get the third right. Thus the probability is  $\frac{1}{2} \cdot \frac{1}{3}$  which is  $\frac{1}{6}$ .-JasonDaGoat

**Solution 2**

Let's call the celebrities A, B, and C. There is a  $\frac{1}{3}$  chance that celebrity A's picture will be selected, and a  $\frac{1}{3}$  chance that his baby picture will be selected. That means there are two celebrities left. There is now a  $\frac{1}{2}$  chance that celebrity B's picture will be selected, and another  $\frac{1}{2}$  chance that his baby picture will be selected. This leaves a  $\frac{1}{1}$  chance for the last celebrity, so the total probability is  $\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{36}$ . However, the order of the celebrities doesn't matter, so the final probability will be

$$3! \cdot \frac{1}{36} = \boxed{\text{(B)} \frac{1}{6}}.$$

**Solution 3**

There is a  $\frac{1}{3}$  chance that the reader will choose the correct baby picture for the first person. Next, the second person gives a  $\frac{1}{2}$  chance, and the last person leaves only 1 choice. Thus, the probability is  $\frac{1}{3 \cdot 2} = \boxed{\text{(B)} \frac{1}{6}}$ .

**Solution 4**

There are  $3!$  ways assign the pictures to each of the celebrities. There is one favorable outcome where all of them are matched correctly, so the answer is  $\boxed{\text{(B)} \frac{1}{6}}$ .

13. (AMC 8 2014) If  $n$  and  $m$  are integers and  $n^2 + m^2$  is even, which of the following is impossible?

(A)  $n$  and  $m$  are even      (B)  $n$  and  $m$  are odd      (C)  $n + m$  is even      (D)  $n + m$  is odd      (E) none of these are impossible

**Answer:** D

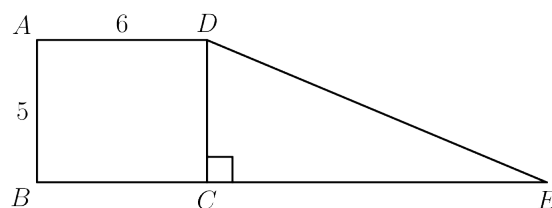
**Solution**

Since  $n^2 + m^2$  is even, either both  $n^2$  and  $m^2$  are even, or they are both odd. Therefore,  $n$  and  $m$  are either both even or both odd, since the square of an even number is even and the square of an odd number is odd. As a result,  $n + m$  must be even. The answer, then, is  $n + m$ ,  $\boxed{\text{(D)}}$ , is odd.

**Solution 2**

We can assign values of  $n$  and  $m$  as 1,3. Then  $1^2 + 3^2$  is 10. We can also assign values of  $n$  and  $m$  to be even, like 2 and 4. After calculating, we can determine that  $n^2 + m^2$  could be odd or even. Then, we can conclude that  $n + m$  is not odd. Hence our answer choice is  $\boxed{\text{(D)}}$  - TheNerdWhoIsNerdy.

14. (AMC 8 2014) Rectangle  $ABCD$  and right triangle  $DCE$  have the same area. They are joined to form a trapezoid, as shown. What is  $DE$ ?



(A) 12      (B) 13      (C) 14      (D) 15      (E) 16

**Answer:** B

**Solution 1**

The area of  $\triangle CDE$  is  $\frac{DC \cdot CE}{2}$ . The area of  $ABCD$  is  $AB \cdot AD = 5 \cdot 6 = 30$ , which also must be equal to the area of  $\triangle CDE$ , which, since  $DC = 5$ , must in turn equal  $\frac{5 \cdot CE}{2}$ . Through transitivity, then,  $\frac{5 \cdot CE}{2} = 30$ , and  $CE = 12$ . Then, using the Pythagorean Theorem, you should be able to figure out that  $\triangle CDE$  is a 5–12–13 triangle, so  $DE = \boxed{13}$ , or  $\boxed{\text{(B)}}$ .

**Solution 2**

The area of the rectangle is  $5 \times 6 = 30$ . Since the parallel line pairs are identical,  $DC = 5$ . Let  $CE$  be  $x$ .  $\frac{5x}{2} = 30$  is the area of the right triangle. Solving for  $x$ , we get  $x = 12$ . According to the Pythagorean Theorem, we have a 5 - 12 - 13 triangle. So, the hypotenuse  $DE$  has to be  $\boxed{(B)}$ .

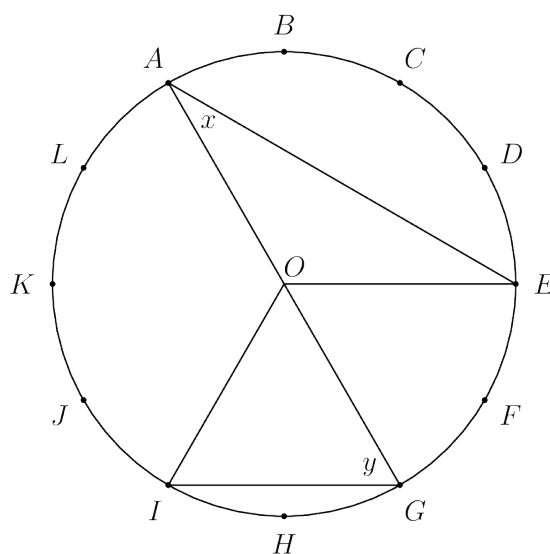
**Solution 3**

This problem can be solved with the Pythagorean Theorem ( $a^2 + b^2 = c^2$ ). We know  $AB = DC$ , so  $DC = 5$ .  $CE$  is twice the length of  $AD$ , so  $CE = 12$ .  $5^2 + 12^2 = c^2$ .  $5^2 = 25$ .  $12^2 = 144$ .  $25 + 144 = 169$ . 169 has a square root of 13, so the hypotenuse or  $DE$  is 13. The answer is  $\boxed{(B)}$ .

—MiracleMaths

Note: Another way to find out that  $CE$  is 12 is by using logic. Since  $DC = 5$ , we can reason that the triangle is a 5 - 12 - 13 triangle, because the only Pythagorean Triple with 5 as a leg(not hypotenuse, as that would be the case for a 3 - 4 - 5 right triangle), is 5 - 12 - 13. We know  $DE$  is the hypotenuse, so it can't be 12, which means  $CE$  has to be 12, which, in turn, means that  $DE$  has to be: 13  $\boxed{(B)}$ .

15. (AMC 8 2014) The circumference of the circle with center  $O$  is divided into 12 equal arcs, marked the letters  $A$  through  $L$  as seen below. What is the number of degrees in the sum of the angles  $x$  and  $y$ ?



- (A) 75      (B) 80      (C) 90      (D) 120      (E) 150

**Answer:** C

**Solution 1**

The measure of an inscribed angle is half the measure of its corresponding central angle. Since each unit arc is  $\frac{1}{12}$  of the circle's circumference, each unit central angle measures  $\frac{360^\circ}{12} = 30^\circ$ . From this,  $\angle EOG = 60^\circ$ , so  $x = 30^\circ$ . Also,  $\angle AOI = 120^\circ$ , so  $y = 60^\circ$ . The number of degrees in the sum of both angles is  $30 + 60 = \boxed{(C) 90}$ .

**Solution 2**

Since  $\triangle AOE$  is isosceles and  $\angle AOE = \frac{4}{12} \cdot 360^\circ = 120^\circ$ ,  $x = 30^\circ$ . Since  $\triangle GOI$  is isosceles and  $\angle GOI = \frac{2}{12} \cdot 360^\circ = 60^\circ$ ,  $y = 60^\circ$ . The number of degrees in the sum

of both angles is  $30 + 60 = \boxed{(C) 90}$ .

16. (AMC 8 2014) The "Middle School Eight" basketball conference has 8 teams. Every season, each team plays every other conference team twice (home and away), and each team also plays 4 games against non-conference opponents. What is the total number of games in a season involving the "Middle School Eight" teams?

(A) 60      (B) 88      (C) 96      (D) 144      (E) 160

**Answer:** B

**Solution**

Within the conference, there are 8 teams, so there are  $\binom{8}{2} = 28$  pairings of teams, and each pair must play two games, for a total of  $28 \cdot 2 = 56$  games within the conference.

Each team also plays 4 games outside the conference, and there are 8 teams, so there are a total of  $4 \cdot 8 = 32$  games outside the conference.

Therefore, the total number of games is  $56 + 32 = \boxed{(B) 88}$ .

**Solution 2**

We can also figure out the number of games the 8 teams play within the conference with the formula  $n(n - 1)/2$ , which is used for these kind of match questions, where Team A playing Team B is the same as Team B playing Team A. In this formula,  $n$  is the number of teams. So we will do  $(8 \times 7 / 2) \times 2$ , as each team plays the other team twice. The 2s cancel out, to yield  $8 \times 7$ , which is 56 games. Then, we can just proceed as above to find the number of games the 8 teams play against 4 teams outside the conference, which is  $8 \times 4 = 32$ . Summing these values up yields 88, which is answer choice B.

17. (AMC 8 2014) George walks 1 mile to school. He leaves home at the same time each day, walks at a steady speed of 3 miles per hour, and arrives just as school begins. Today he was distracted by the pleasant weather and walked the first  $\frac{1}{2}$  mile at a speed of only 2 miles per hour. At how many miles per hour must George run the last  $\frac{1}{2}$  mile in order to arrive just as school begins today?

(A) 4      (B) 6      (C) 8      (D) 10      (E) 12

**Answer:** B

**Solution 1**

Note that on a normal day, it takes him  $\frac{1}{3}$  hour to get to school. However, today it took  $\frac{\frac{1}{2} \text{ mile}}{2 \text{ mph}} = \frac{1}{4}$  hour to walk the first  $\frac{1}{2}$  mile. That means that he has  $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$  hours left to get to school, and  $\frac{1}{2}$  mile left to go. Therefore, his speed must be  $\frac{\frac{1}{2} \text{ mile}}{\frac{1}{12} \text{ hour}} = \boxed{6 \text{ mph}}$ , so  $\boxed{(B) 6}$  is the answer.

18. (AMC 8 2014) Four children were born at City Hospital yesterday. Assume each child is equally likely to be a boy or a girl. Which of the following outcomes is most likely?

(A) All 4 are boys  
 (B) All 4 are girls  
 (C) 2 are girls and 2 are boys  
 (D) 3 are of one gender and 1 is of the other gender

(E) All of these outcomes are equally likely

**Answer:** D

**Solution 1**

We'll just start by breaking cases down. The probability of A occurring is  $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$ .

The probability of B occurring is  $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$ .

The probability of C occurring is  $\binom{4}{2} \cdot \left(\frac{1}{2}\right)^4 = \frac{3}{8}$ , because we need to choose 2 of the 4 slots to be girls.

For D, there are two possible cases, 3 girls and 1 boy or 3 boys and 1 girl. The probability of the first case is  $\binom{4}{1} \cdot \left(\frac{1}{2}\right)^4 = \frac{1}{4}$  because we need to choose 1 of the 4 slots to be a boy. However, the second case has the same probability because we are choosing 1 of the 4 children to be a girl, so the total probability is  $\frac{1}{4} \cdot 2 = \frac{1}{2}$ .

So out of the four fractions, D is the largest. So our answer is (D) 3 are of one gender and 1 is of the other.

19. (AMC 8 2014) A cube with 3-inch edges is to be constructed from 27 smaller cubes with 1-inch edges. Twenty-one of the cubes are colored red and 6 are colored white. If the 3-inch cube is constructed to have the smallest possible white surface area showing, what fraction of the surface area is white?

(A)  $\frac{5}{54}$       (B)  $\frac{1}{9}$       (C)  $\frac{5}{27}$       (D)  $\frac{2}{9}$       (E)  $\frac{1}{3}$

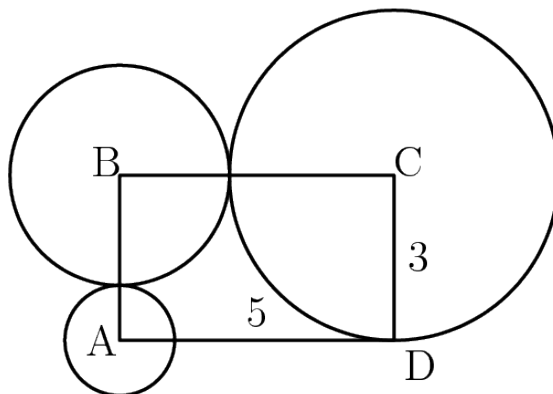
**Answer:** A

**Solution**

For the least possible surface area that is white, we should have 1 cube in the center, and the other 5 with only 1 face exposed. This gives 5 square inches of white, surface area. Since the cube has a surface area of 54 square inches, our

answer is (A)  $\frac{5}{54}$ .

20. (AMC 8 2014) Rectangle  $ABCD$  has sides  $CD = 3$  and  $DA = 5$ . A circle with a radius of 1 is centered at  $A$ , a circle with a radius of 2 is centered at  $B$ , and a circle with a radius of 3 is centered at  $C$ . Which of the following is closest to the area of the region inside the rectangle but outside all three circles?



(A) 3.5      (B) 4.0      (C) 4.5      (D) 5.0      (E) 5.5

**Answer:** B

**Solution 1**

The area in the rectangle but outside the circles is the area of the rectangle minus the area of all three of the quarter circles in the rectangle.

The area of the rectangle is  $3 \cdot 5 = 15$ . The area of all 3 quarter circles is  $\frac{\pi}{4} + \frac{\pi(2)^2}{4} + \frac{\pi(3)^2}{4} = \frac{14\pi}{4} = \frac{7\pi}{2}$ . Therefore the area in the rectangle but outside the circles is  $15 - \frac{7\pi}{2}$ .  $\pi$  is approximately  $\frac{22}{7}$ , and substituting that in will give  $15 - 11 = \boxed{\text{(B) } 4.0}$

**Solution 1.5 (People who don't know how to do the area of a sector)**

Instead of calculating the area of the rectangle minus the area of the 3 quarter-circles that are in it, we calculate the

area of the 3 circles, not just the quarter-circles.  $3^2\pi$  is equal to  $9\pi$ ,  $2^2\pi$  squared is equal to  $4\pi$ , and

$1^2\pi$  is equal to  $1$ . It is obvious that they are all quarter-circles, as mentioned in solution 1, so you divide each of

them by 4. So they become  $\pi$ ,  $\frac{9}{4}\pi$ , and  $\frac{1}{4}\pi$ . Adding them together, it's  $\frac{14}{4}\pi$ .

Now you can calculate  $15 - \frac{14}{4}\pi$ , and you'll get around 4.000000000000001, which is closest to  $\boxed{\text{(B) } 4.0}$

21. (AMC 8 2014) The 7-digit numbers 74A52B1 and 326AB4C are each multiples of 3. Which of the following could be the value of  $C$ ?

(A) 1      (B) 2      (C) 3      (D) 5      (E) 8

**Answer:** A

**Solution 1**

Since both numbers are divisible by 3, the sum of their digits has to be divisible by three.  $7 + 4 + 5 + 2 + 1 = 19$ . To be a multiple of 3,  $A + B$  has to be either 2 or 5 or 8... and so on. We add up the numerical digits in the second number;  $3 + 2 + 6 + 4 = 15$ . We then add two of the selected values, 5 to 15, to get 20. We then see that  $C = 1, 4$  or  $7, 10$ ... and so on, otherwise the number will not be divisible by three. We then add 8 to 15, to get 23, which shows us that  $C = 1$  or  $4$  or  $7$ ... and so on. To be a multiple of three, we select a few of the common numbers we got from both these equations, which could be 1, 4, and 7. However, in the answer choices, there is no 7 or 4 or anything greater than 7, but there is a 1, so  $\boxed{\text{(A) } 1}$  is our answer. Also, you could just look up the answer. It's much easier and totally isn't wrong in any way. :)

**Concept Solution(Solution 2)** for a number to be divisible by 3, the sum of its digits must be divisible by 3. In the number 74A52B1,  $7+4+A+5+2+B+1$  must be divisible by 3. We get  $19+A+B$ . 21 is the closest multiple of 3 meaning that  $A=2$ ,  $B=0$ , order doesn't matter. Now we plug in those values for 326AB4C. It will be  $3+2+6+2+0+4+C$  to get  $17+C$ . The closest multiple of 3 is 18. So that means  $17+C=18$ . Solving for C, our answer is  $\boxed{\text{(A) } 1}$  -TheNerdWhoIsNerdy.

**Solution 3 (Modulo Arithmetic)**

If  $n$  is divisible by 3, the sum of it's digits should also be divisible by 3. This means that  $19 + A + B \pmod{3} \equiv 0$  or  $A + B + 1 \pmod{3} \equiv 0$ . For equation



2,  $15 + A + B + C \pmod{3} \equiv 0$  or  $A + B + C \pmod{3} \equiv 0$ . Logically, you can see the correlation between our first and second equations so we can make the assumption,  $C = \boxed{\text{(A)} 1}$  RandomMathGuy500

22. (AMC 8 2014) A 2-digit number is such that the product of the digits plus the sum of the digits is equal to the number. What is the units digit of the number?

(A) 1    (B) 3    (C) 5    (D) 7    (E) 9

**Answer:** E

**Solution**

We can think of the number as  $10a + b$ , where  $a$  is the tens digit and  $b$  is the unit digit. Since the number is equal to the product of the digits ( $ab$ ) plus the sum of the digits ( $a + b$ ), we can say that  $10a + b = ab + a + b$ . We can simplify this to  $10a = ab + a$ , which factors to  $(10)a = (b + 1)a$ . Dividing by  $a$ , we have that  $b + 1 = 10$ . Therefore, the units digit,  $b$ , is  $\boxed{\text{(E)} 9}$

23. (AMC 8 2014) Three members of the Euclid Middle School girls' softball team had the following conversation.

Ashley: I just realized that our uniform numbers are all 2-digit primes.

Bethany: And the sum of your two uniform numbers is the date of my birthday earlier this month.

Caitlin: That's funny. The sum of your two uniform numbers is the date of my birthday later this month.

Ashley: And the sum of your two uniform numbers is today's date.

What number does Caitlin wear?

(A) 11    (B) 13    (C) 17    (D) 19    (E) 23

**Answer:** A

**Solution 1**

The maximum amount of days any given month can have is 31, and the smallest two-digit primes are 11, 13, and 17. There are a few different sums that can be deduced from the following numbers, which are 24, 30, and 28, all of which represent the three days. Therefore, since Bethany says that the other two people's uniform numbers are earlier, so that means Caitlin and Ashley's numbers must add up to 24. Similarly, Caitlin says that the other two people's uniform numbers are later, so the sum must add up to 30. This leaves 28 as today's date. From this, Caitlin was referring to the uniform wearers 13 and 17, telling us that her number is 11, giving our solution as  $\boxed{\text{(A)} 11}$ .

24. (AMC 8 2014) One day the Beverage Barn sold 252 cans of soda to 100 customers, and every customer bought at least one can of soda. What is the maximum possible median number of cans of soda bought per customer on that day?

(A) 2.5    (B) 3.0    (C) 3.5    (D) 4.0    (E) 4.5

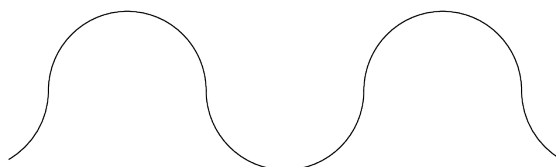
**Answer:** C

**Solution**

In order to maximize the median, we need to make the first half of the numbers as small as possible. Since there are 100 people, the median will be the average of the

50th and 51st largest amount of cans per person. To minimize the first 49, they would each have one can. Subtracting these 49 cans from the 252 cans gives us 203 cans left to divide among 51 people. Taking  $\frac{203}{51}$  gives us 3 and a remainder of 50. Seeing this, the largest number of cans the 50th person could have is 3, which leaves 4 to the rest of the people. The average of 3 and 4 is 3.5. Thus our answer is (C) 3.5.

25. (AMC 8 2014) A straight one-mile stretch of highway, 40 feet wide, is closed. Robert rides his bike on a path composed of semicircles as shown. If he rides at 5 miles per hour, how many hours will it take to cover the one-mile stretch?



Note: 1 mile = 5280 feet

- (A)  $\frac{\pi}{11}$     (B)  $\frac{\pi}{10}$     (C)  $\frac{\pi}{5}$     (D)  $\frac{2\pi}{5}$     (E)  $\frac{2\pi}{3}$

**Answer:** B

### Solution 1

There are two possible interpretations of the problem: that the road as a whole is 40 feet wide, or that each lane is 40 feet wide. Both interpretations will arrive at the same result. However, let us stick with the first interpretation for simplicity. Each lane must then be 20 feet wide, so Robert must be riding his bike in semicircles with radius 20 feet and diameter 40 feet. Since the road is 5280 feet long, over the whole mile, Robert rides  $\frac{5280}{40} = 132$  semicircles in total. Were the semicircles full circles, their circumference would be  $2\pi \cdot 20 = 40\pi$  feet; as it is, the circumference of each is half that, or  $20\pi$  feet. Therefore, over the stretch of highway, Robert rides a total of  $132 \cdot 20\pi = 2640\pi$  feet, which is equivalent to  $\frac{\pi}{2}$  miles. Robert rides at 5 miles per hour, so divide the  $\frac{\pi}{2}$  miles by 5 mph (because  $t = \frac{d}{r}$  and time = distance/rate) to arrive at (B)  $\frac{\pi}{10}$  hours.

Edit: If you are confused about the lanes, watch the video below :)

## §31 AMC 8 2015

1. (AMC 8 2015) How many square yards of carpet are required to cover a rectangular floor that is 12 feet long and 9 feet wide? (There are 3 feet in a yard.)  
(A) 12    (B) 36    (C) 108    (D) 324    (E) 972

**Answer:** A

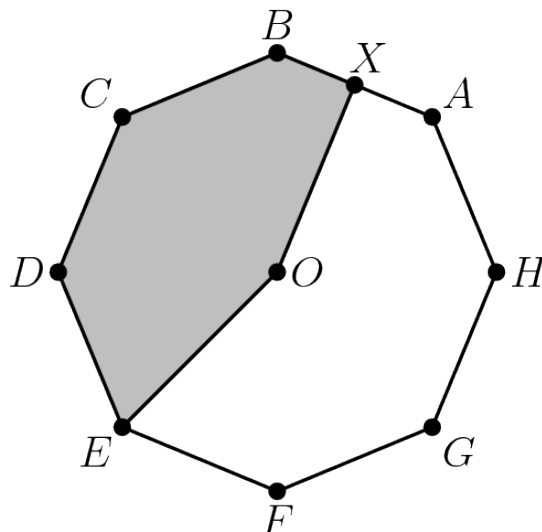
### Solution 1

We need  $12 \cdot 9$  square feet of carpet to cover the floor. Since there are 9 square feet in a square yard, we divide this by 9 to get (A) 12 square yards.

### Solution 2

Since there are 3 feet in a yard, we divide 9 by 3 to get 3 yards, and 12 by 3 to get 4 yards. To find the area of the carpet, we then multiply these two values together to get  $3 \cdot 4 = \boxed{\text{(A)} 12}$  square yards.

2. (AMC 8 2015) Point  $O$  is the center of the regular octagon  $ABCDEFGH$ , and  $X$  is the midpoint of the side  $\overline{AB}$ . What fraction of the area of the octagon is shaded?



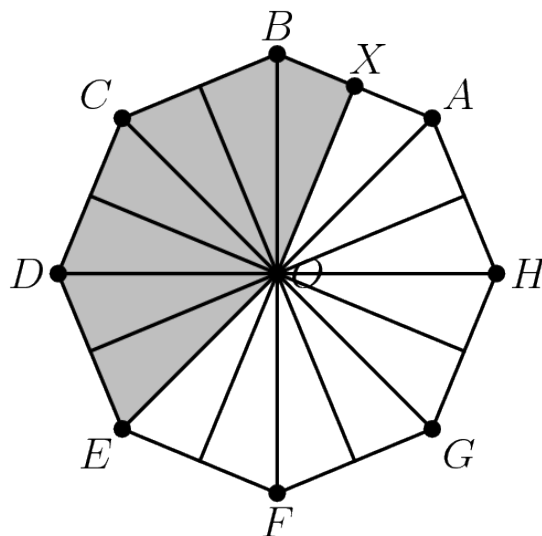
- (A)  $\frac{11}{32}$  (B)  $\frac{3}{8}$  (C)  $\frac{13}{32}$  (D)  $\frac{7}{16}$  (E)  $\frac{15}{32}$

**Answer:** D

**Solution 1**

Since octagon  $ABCDEFGH$  is a regular octagon, it is split into 8 equal parts, such as triangles  $\triangle ABO$ ,  $\triangle BCO$ ,  $\triangle CDO$ , etc. These parts, since they are all equal, are  $\frac{1}{8}$  of the octagon each. The shaded region consists of 3 of these equal parts plus half of another, so the fraction of the octagon that is shaded is  $\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} = \boxed{\text{(D)} \frac{7}{16}}$ .

**Solution 2**

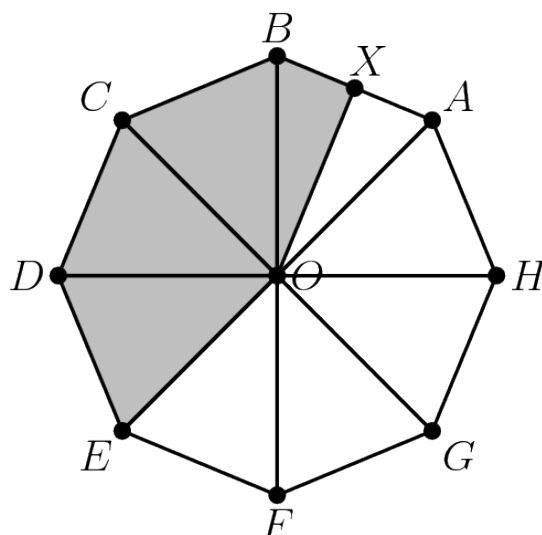


The octagon has been divided up into 16 identical triangles (and thus they each have equal area). Since the shaded region occupies 7 out of the 16 total triangles, the answer is **(D)**  $\frac{7}{16}$ .

-Flare

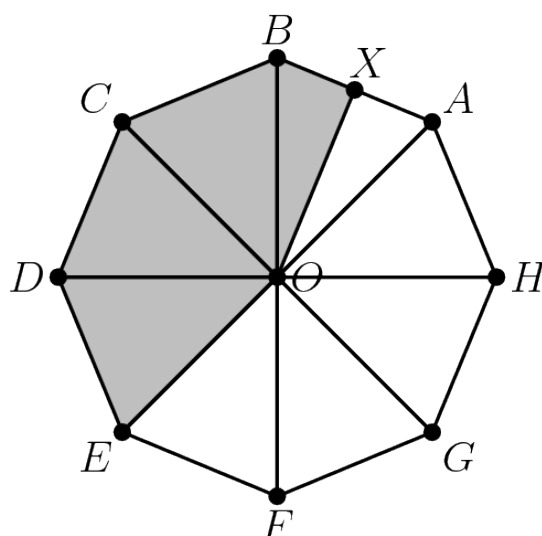
### Solution 3

For starters, what I find helpful is to divide the whole octagon up into triangles as shown here:



Now, it is just a matter of counting the larger triangles. Remember that  $\triangle BOX$  and  $\triangle XOA$  are not full triangles and are only half for these purposes. We count it up and we get a total of  $\frac{3.5}{8}$  of the shape shaded. We then simplify it to get our answer of **(D)**  $\frac{7}{16}$ .

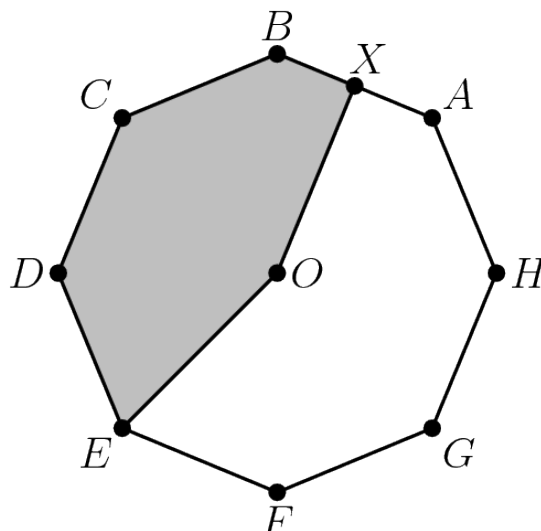
### Solution 4



We can divide the octagon into 8 parts and pretend that the area is 64. We know

that  $X$  is the midpoint of  $BA$  and that each space between two points is 8 because  $64/8=8$ . This means that  $BX=4$  because  $8/2=4$ . Then, we add that to  $3*8$  because there are 3 spaces between points that are each 8. After that, you turn it into a fraction,  $28/64$ , and simplify to get **(D)**  $\frac{7}{16}$ .

**Solution 5 (Quick and Efficient)**



We see that the midpoint of  $\overline{AB}$  is  $X$ . We can also notice that point  $X$  to point  $E$  \*ALMOST\* makes the octagon half shaded. Since we know an octagon has 8 sides,  $X$  is one-half of  $\overline{AB}$  which means instead of counting in eights, we need to count in sixteenths. Now back to the problem. Let's say that the midpoint of  $\overline{EF}$  is  $Y$ , then  $\overline{XY}$  cuts the octagon perfectly in half, but since there is no  $Y$ , we are now  $\frac{1}{16}$  away from one-half. We know that  $\frac{1}{2} = \frac{8}{16}$ , so we can just do  $\frac{8}{16} - \frac{1}{16}$  which gets us, for the final answer, **(D)**  $\frac{7}{16}$ .

3. (AMC 8 2015) Jack and Jill are going swimming at a pool that is one mile from their house. They leave home simultaneously. Jill rides her bicycle to the pool at a constant speed of 10 miles per hour. Jack walks to the pool at a constant speed of 4 miles per hour. How many minutes before Jack arrives does Jill arrive?
- (A) 5    (B) 6    (C) 8    (D) 9    (E) 10

**Answer:** D

**Solution**

Using  $d = rt$ , we can set up an equation for when Jill arrives at the swimming pool:  
 $1 = 10t$

Solving for  $t$ , we get that Jill gets to the pool in  $\frac{1}{10}$  of an hour, which is 6 minutes. Doing the same for Jack, we get that

Jack arrives at the pool in  $\frac{1}{4}$  of an hour, which in turn is 15 minutes. Thus, Jill has to wait  $15 - 6 =$  **(D)** 9

minutes for Jack to arrive at the pool.

**Solution 2**

T=D/sJill:  $(1/10) \times 60$  because in minutes, is equal to 6 min Jack:  $(1/4) \times 60$  is 15 minutes.  $15 - 6$  is 9, so our answer is  $15 - 6 = \boxed{\text{(D)} 9}$  - TheNerdWhoIsNerdy.

4. (AMC 8 2015) The Centerville Middle School chess team consists of two boys and three girls. A photographer wants to take a picture of the team to appear in the local newspaper. She decides to have them sit in a row with a boy at each end and the three girls in the middle. How many such arrangements are possible?
- (A) 2      (B) 4      (C) 5      (D) 6      (E) 12

**Answer:** E

**Solution**

There are  $2! = 2$  ways to order the boys on the ends, and there are  $3! = 6$  ways to order the girls in the middle. We get the answer to be  $2 \cdot 6 = \boxed{\text{(E)} 12}$ .

5. (AMC 8 2015) Billy's basketball team scored the following points over the course of the first 11 games of the season:

42, 47, 53, 53, 58, 58, 58, 61, 64, 65, 73

If his team scores 40 in the 12th game, which of the following statistics will show an increase?

- (A) range      (B) median      (C) mean      (D) mode      (E) mid-range

**Answer:** A

**Solution 1**

When they score a 40 on the next game, the range increases from  $73 - 42 = 31$  to  $73 - 40 = 33$ . This means the  $\boxed{\text{(A)} \text{ range}}$  increased.

Note: The range is defined to be the difference of the largest element and the smallest element of a set.

**Solution 2**

Because 40 is less than the score of every game they've played so far, the measures of center will never rise. Only measures of spread, such as the  $\boxed{\text{(A)} \text{ range}}$ , may increase.

6. (AMC 8 2015) In  $\triangle ABC$ ,  $AB = BC = 29$ , and  $AC = 42$ . What is the area of  $\triangle ABC$ ?

- (A) 100      (B) 420      (C) 500      (D) 609      (E) 701

**Answer:** B

**Solution 1**

We know the semi-perimeter of  $\triangle ABC$  is  $\frac{29+29+42}{2} = 50$ . Next, we use Heron's Formula to find that the area of the triangle is just  $\sqrt{50(50-29)^2(50-42)} = \sqrt{50 \cdot 21^2 \cdot 8} = \boxed{\text{(B)} 420}$ .

**Solution 2 (easier)**

Splitting the isosceles triangle in half, we get a right triangle with hypotenuse 29 and leg 21. Using the Pythagorean Theorem, we know the height is  $\sqrt{29^2 - 21^2} = 20$ .

Now that we know the height, the area is  $\frac{(20)(42)}{2} = \boxed{\text{(B)} 420}$ .

7. (AMC 8 2015) Each of two boxes contains three chips numbered 1, 2, 3. A chip is drawn randomly from each box and the numbers on the two chips are multiplied. What is the probability that their product is even?

(A)  $\frac{1}{9}$     (B)  $\frac{2}{9}$     (C)  $\frac{4}{9}$     (D)  $\frac{1}{2}$     (E)  $\frac{5}{9}$

**Answer:** E

**Solution 1**

(This solution is similar to Solution 2.) Let's make this a problem with boxes.

In total, there are 9 products derived from these numbers (because 3 numbers per box). This would be our denominator.

Every time a spinner lands on 2, we get an even product. These are (1, 2), (2, 1), (2, 2), (2, 3), and finally (3, 2).

Going back, we see there are  $3 \cdot 3 = 9$  possible combinations, and we have 5 evens, the final answer is

$$\boxed{\text{(E)} \frac{5}{9}}.$$

**Solution 2**

You can also make this problem into a spinner problem. You have the first spinner with 3 equally divided

sections: 1, 2, and 3. You make a second spinner that is identical to the first, with 3 equal sections of

1, 2, and 3. If the first spinner lands on 1, it must land on two for the result to be even. You write down the first

combination of numbers: (1, 2). Next, if the spinner lands on 2, it can land on any number on the second

spinner. We now have the combinations of (1, 2), (2, 1), (2, 2), and (2, 3). Finally, if the first spinner ends on 3, we

have (3, 2). Since there are  $3 \cdot 3 = 9$  possible combinations, and we have 5 evens, the final answer is

$$\boxed{\text{(E)} \frac{5}{9}}.$$

**Solution 3**

We can also list out the numbers. Box A has chips 1, 2, and 3, and Box B also has chips 1, 2, and 3. Chip 1 (from Box A)

could be with 3 partners from Box B. This is also the same for chips 2 and 3 from Box A.  $3 + 3 + 3 = 9$  total sums. Chip 1 could be multiplied with 2 other chips to make an even product, just like chip 3. Chip 2 can only multiply with 1 chip.

$2 + 2 + 1 = 5$ . The answer is  $\boxed{\text{(E)} \frac{5}{9}}.$

**Solution 4**

Here is another way:

Let's start by finding the denominator: Total choices. There are 3 chips we can choose from in the 1st box, and 3 chips we can choose from in the 2nd box. We do  $3 \cdot 3$ , and get 9. Now - to find the numerator: Desired choices. To get an even number, we need to pick 2 from at least one of the boxes. There are 2 choices as to

finding which box we will draw the 2 from. Then, we have 3 choices from the other box to pick any of the other chips, 1, 2, and 3.

$$\frac{3 \cdot 2}{9} = \frac{6}{9}$$

However, we are over counting the (2, 2) configuration twice, and so, we subtract that one configuration from our total.  $\frac{6}{9} - \frac{1}{9}$ . Thus, our answer is **(E)**  $\frac{5}{9}$ .

### Solution 5

This might take longer to solve, but you definitely will have the right answer. You first list out all the possible combinations regardless if the product is even or odd.  $1 \cdot 1 = 1$ .  $1 \cdot 2 = 2$ .  $1 \cdot 3 = 3$ .  $2 \cdot 1 = 2$ .  $2 \cdot 2 = 4$ .  $2 \cdot 3 = 6$ .  $3 \cdot 1 = 3$ .  $3 \cdot 2 = 6$ .  $3 \cdot 3 = 9$ . There are 9 possible combinations and 5 of the combinations' products are even. So, our answer is **(E)**  $\frac{5}{9}$ .

### Solution 6

We can use complementary counting to solve this problem. If we want to find the number of ways for the product to be odd, then the chip chosen from either box has to be 1 or 3. Since there are 2 ways out of 3 to pick an even chip from a box, the probability of picking an even chip from a box is  $\frac{2}{3}$ . Since there are 2 boxes, we can find that the answer is  $1 - \frac{2}{3} \cdot \frac{2}{3} =$  **(E)**  $\frac{5}{9}$ .

8. (AMC 8 2015) What is the smallest whole number larger than the perimeter of any triangle with a side of length 5 and a side of length 19?
- (A) 24      (B) 29      (C) 43      (D) 48      (E) 57

**Answer:** D

### Solution

We know from the **Triangle Inequality** that the last side,  $s$ , fulfills  $s < 5 + 19 = 24$ . Adding  $5 + 19$  to both sides of the inequality, we get  $s + 5 + 19 < 48$ , and because  $s + 5 + 19$  is the perimeter of our triangle, **(D)** 48 is our answer.

### Solution 2

$s$ , fulfills  $s < 5 + 19 = 24$ , meaning that  $s < 24$ . The greatest value of  $s$  can be 23.  $23 + 5 + 19$  is 47. Since 48 is our nearest answer choice, the answer is **(D)** 48. -TheNerdWhoIsNerdy.

minor edit by themathkidwoohooooo: the greatest value of  $s$  is  $24 - e$ , where  $e$  represents an infinitesimal (a constant smaller than any positive number but not zero or negative. it is between the positives and 0. another way to think of it is  $\frac{1}{\infty}$ ). NOT 23!!!!!!!!!!!!!!

9. (AMC 8 2015) On her first day of work, Janabel sold one widget. On day two, she sold three widgets. On day three, she sold five widgets, and on each succeeding day, she sold two more widgets than she had sold on the previous day. How many widgets in total had Janabel sold after working 20 days?
- (A) 39      (B) 40      (C) 210      (D) 400      (E) 401

**Answer:** D

### Solution 1



First, we recognize that the number of widgets Janabel sells each day forms an arithmetic sequence. On day 1, she sells 1 widget, on day 2, she sells 3 widgets, on day 3, she sells 5 widgets, and so on. This sequence can be described by the first term  $a = 1$  and the common difference  $d = 2$ .

Step 1: Find the number of widgets sold on day 20

The formula for the  $n$ -th term of an arithmetic sequence is:

$$a_n = a + (n - 1)d$$

For  $n = 20$ :

$$a_{20} = 1 + (20 - 1) \cdot 2 = 1 + 38 = 39$$

So, Janabel sells 39 widgets on day 20.

Step 2: Calculate the total number of widgets sold in 20 days

The sum  $S_n$  of the first  $n$  terms of an arithmetic sequence is given by:

$$S_n = \frac{n}{2} \cdot (a + a_n)$$

For  $n = 20$ :

$$S_{20} = \frac{20}{2} \cdot (1 + 39) = 10 \cdot 40 = 400$$

Therefore, the total number of widgets sold after 20 days is:  $(40 \cdot 10) = \boxed{\text{(D)} 400}$

### Solution 2

First, we have to find how many widgets she makes on Day 20. We can write the linear equation  $y = -1 + 2x$  to represent this situation. Then, we can plug in 20 for  $x$ :  $y = -1 + 2(20) - y = -1 + 40 - y = 39$ . The sum of 1, 3, 5, ... 39 is  $\frac{(1 + 39)(20)}{2} = \boxed{\text{(D)} 400}$ .

### Solution 3 ( !!FASTEST SOLUTION!! )

The sum is just the first 20 odd counting/natural numbers, which is  $20^2 = \boxed{\text{(D)} 400}$ .

Note: The sum of the first  $x$  odd numbers is  $x^2$ .

### Solution 4

We can easily find out she makes  $2 \cdot 20 - 1 = 39$  widgets on Day 20. Then, we make the sum of 1, 3, 5, ..., 35, 37, 39 by adding in this way:  $(1 + 39) + (3 + 37) + (5 + 35) + \dots + (19 + 21)$ , which include 10 pairs of 40. So, the sum of 1, 3, 5, ... 39 is  $(40 \cdot 10) = \boxed{\text{(D)} 400}$ .—LarryFlora

10. (AMC 8 2015) How many integers between 1000 and 9999 have four distinct digits?  
 (A) 3024      (B) 4536      (C) 5040      (D) 6480      (E) 6561

**Answer:** B

### Solution 1

There are 9 choices for the first number, since it cannot be 0, there are only 9 choices left for the second number since it must differ from the first, 8 choices for the third number, since it must differ from the first two, and 7 choices for the fourth number, since it must differ from all three. This means there are  $9 \times 9 \times 8 \times 7 = \boxed{\text{(B)} 4536}$  integers between 1000 and 9999 with four distinct digits.

If you're too lazy to do  $9 \times 9 \times 8 \times 7 = 4536$ , we can notice  $9 \times 9 = 81$  and  $8 \times 7 = 56$ . The unit digit must be 6. Therefore, the answer is the only answer choice with a unit digit of 6, which is B.

11. (*AMC 8 2015*) In the small country of Mathland, all automobile license plates have four symbols. The first symbol must be a vowel ( $A, E, I, O$ , or  $U$ ), the second and third symbols must be two different letters among the 21 non-vowels in the alphabet, and the fourth symbol must be a digit (0 through 9). If the symbols are chosen at random subject to these conditions, what is the probability (In fractions) that the plate will read "AMC8"?

(A)  $\frac{1}{22,050}$       (B)  $\frac{1}{21,000}$       (C)  $\frac{1}{10,500}$       (D)  $\frac{1}{2,100}$       (E)  $\frac{1}{1,050}$

**Answer:** B

**Solution 1**

There is one favorable case, which is the license plate says "AMC8." We must now find how many total cases there are. There are 5 choices for the first letter (since it must be a vowel), 21 choices for the second letter (since it must be of 21 consonants), 20 choices for the third letter (since it must differ from the second letter), and 10 choices for the digit. This leads to  $5 \cdot 21 \cdot 20 \cdot 10 = 21,000$  total possible license plates. Therefore, the probability of a license plate saying "AMC8"

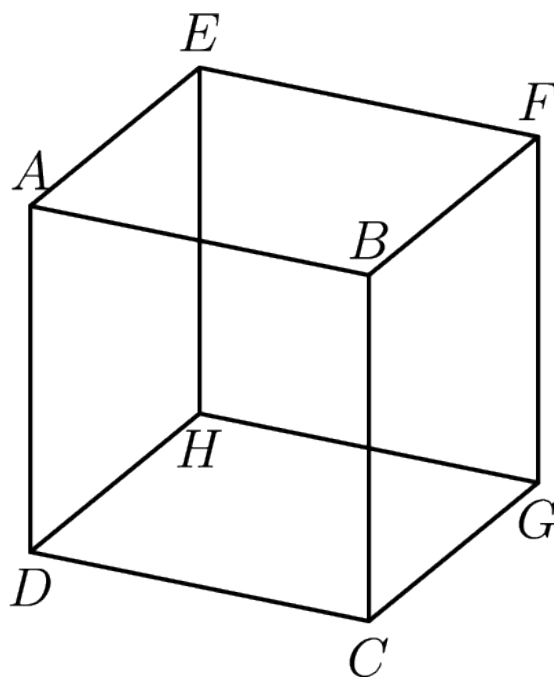
is (B)  $\frac{1}{21,000}$ .

**Solution 2**

The probability of choosing A as the first letter is  $\frac{1}{5}$ . The probability of choosing M next is  $\frac{1}{21}$ . The probability of choosing C as the third letter is  $\frac{1}{20}$  (since there are 20 other consonants to choose from other than M). The probability of having 8 as the last number is  $\frac{1}{10}$ . We multiply all these to obtain  $\frac{1}{5} \cdot \frac{1}{21} \cdot \frac{1}{20} \cdot \frac{1}{10} =$

$$\frac{1}{5 \times 21 \times 20 \times 10} = \frac{1}{21 \times 100 \times 10} = \text{(B) } \frac{1}{21,000}.$$

12. (*AMC 8 2015*) How many pairs of parallel edges, such as  $\overline{AB}$ , and  $\overline{GH}$ , or  $\overline{EH}$ , and  $\overline{FG}$ , does a cube have? Repeating parallel edges do not count toward the final answer.



- (A) 6    (B) 12    (C) 18    (D) 24    (E) 36

**Answer:** C

**Solution 1**

We first count the number of pairs of parallel lines that are in the same direction as  $\overline{AB}$ . The pairs of parallel lines are  $\overline{AB}$  and  $\overline{EF}$ ,  $\overline{CD}$  and  $\overline{GH}$ ,  $\overline{AB}$  and  $\overline{CD}$ ,  $\overline{EF}$  and  $\overline{GH}$ ,  $\overline{AB}$  and  $\overline{GH}$ , and  $\overline{CD}$  and  $\overline{EF}$ . These are 6 pairs total. We can do the same for the lines in the same direction as  $\overline{AE}$  and  $\overline{AD}$ . This means there are  $6 \cdot 3 = \boxed{\text{(C) } 18}$  total pairs of parallel lines.

**Solution 2**

Look at any edge, let's say  $\overline{AB}$ . There are three ways we can pair  $\overline{AB}$  with another edge.  $\overline{AB}$  and  $\overline{EF}$ ,  $\overline{AB}$  and  $\overline{HG}$ , and  $\overline{AB}$  and  $\overline{DC}$ . There are 12 edges on a cube. 3 times 12 is 36. We have to divide by 2 because every pair is counted twice, so  $\frac{36}{2}$  is  $\boxed{\text{(C) } 18}$  total pairs of parallel lines.

**Solution 3**

We can use the feature of 3-Dimension in a cube to solve the problem systematically. For example, in the 3-D of the cube,  $\overline{AB}$ ,  $\overline{BC}$ , and  $\overline{BF}$  have 4 different parallel edges respectively. So it gives us the total pairs of parallel lines are  $\binom{4}{2} \cdot 3 = \boxed{\text{(C) } 18}$ .

–LarryFlora

**Solution 4**

Our first case are lines on the same plane. There are two pairs of parallel lines for a square, and there are 6 squares, which give us  $6 \cdot 2 = 12$  cases.

The next cases we need to analyze are the parallel lines on opposite planes (lines such as  $\overline{EF}$  and  $\overline{DC}$ ). There are 2 pairs for every 2 planes, and there are 6 planes in total, giving us  $2 \cdot 3 = 6$  cases.

Adding up the cases, we get  $12 + 6 = \boxed{\text{(C) } 18}$

13. (*AMC 8 2015*) How many subsets of two elements can be removed from the set  $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$  so that the mean (average) of the remaining numbers is 6?

(A) 1    (B) 2    (C) 3    (D) 5    (E) 6

**Answer:** D

**Solution 1**

Since there will be 9 elements after removal, and their mean is 6, we know their sum is 54. We also know that the sum of the set pre-removal is 66. Thus, the sum of the 2 elements removed is  $66 - 54 = 12$ . There are only (D) 5 subsets of 2 elements that sum to 12:  $\{1, 11\}, \{2, 10\}, \{3, 9\}, \{4, 8\}, \{5, 7\}$ .

**Solution 2**

We can simply remove 5 subsets of 2 numbers while leaving only 6 behind. The average of this one-number set is still 6, so the answer is (D) 5.

-tryanotherangle

14. (*AMC 8 2015*) Which of the following integers cannot be written as the sum of four consecutive odd integers?

(A) 16    (B) 40    (C) 72    (D) 100    (E) 200

**Answer:** D

**Solution 1**

Let our 4 numbers be  $n, n + 2, n + 4, n + 6$ , where  $n$  is odd. Then, our sum is  $4n + 12$ . The only answer choice that cannot be written as  $4n + 12$ , where  $n$  is odd, is (D) 100.

**Solution 2**

If the four consecutive odd integers are  $2n - 3, 2n - 1, 2n + 1$  and  $2n + 3$ ; then, the sum is  $8n$ . All the integers are divisible by 8 except (D) 100.

**Solution 3**

If the four consecutive odd integers are  $a, a + 2, a + 4$  and  $a + 6$ , the sum is  $4a + 12$ , and  $4a + 12$  divided by 4 gives  $a + 3$ . This means that  $a + 3$  must be even. The only integer that does not give an even integer when divided by 4 is 100, so the answer is (D) 100.

**Solution 4**

From Solution 1, we have the sum of the 4 numbers to be equal to  $4n + 12$ . Taking mod 8 gives us  $4n + 4 \equiv b \pmod{8}$  for some residue  $b$  and for some odd integer  $n$ . Since  $n \equiv 1 \pmod{2}$ , we can express it as the equation  $n = 2a + 1$  for some integer  $a$ . Multiplying 4 to each side of the equation yields  $4n = 8a + 4$ , and taking mod 8 gets us  $4n \equiv 4 \pmod{8}$ , so  $b = 0$ . All the answer choices except choice D is a multiple of 8, and since 100 satisfies all the conditions the answer is (D) 100.

**Solution 5**

Since they want CONSECUTIVE odd numbers, it won't be hard to just list the sums out:

$$16 = 1 + 3 + 5 + 7$$

$$40 = 7 + 9 + 11 + 13$$

$$72 = 15 + 17 + 19 + 21$$

$$200 = 47 + 49 + 51 + 53$$

All of the answer choices can be a sum of consecutive odd numbers except 100, so the answer is (D)100.

15. (AMC 8 2015) At Euler Middle School, 198 students voted on two issues in a school referendum with the following results: 149 voted in favor of the first issue and 119 voted in favor of the second issue. If there were exactly 29 students who voted against both issues, how many students voted in favor of both issues?

(A) 49      (B) 70      (C) 79      (D) 99      (E) 149

**Answer:** D

**Solution 1**

Let:

- $A$  be the number of students who voted in favor of the first issue,
- $B$  be the number of students who voted in favor of the second issue,
- $A \cap B$  be the number of students who voted in favor of both issues.

We are given:

- $A = 149$
- $B = 119$
- 29 students voted against both issues, so the number of students who voted for at least one issue is:

$$198 - 29 = 169$$

By the principle of inclusion and exclusion:

$$A \cup B = A + B - A \cap B$$

Substitute known values:

$$169 = 149 + 119 - A \cap B$$

$$169 = 268 - A \cap B$$

$$A \cap B = 268 - 169 = \span style="border: 1px solid black; padding: 0 2px;">(D) 99$$

**Solution 2**

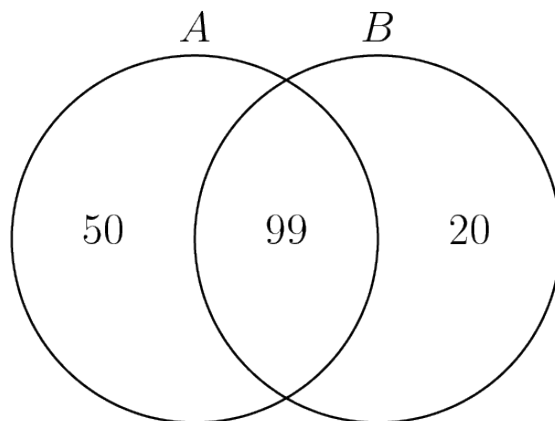
We can see that this is a Venn Diagram Problem.

First, we analyze the information given. There are 198 students. Let's use  $A$  as the first issue and  $B$  as the second issue.

149 students were for  $A$ , and 119 students were for  $B$ . There were also 29 students against both  $A$  and  $B$ .

Solving this without a Venn Diagram, we subtract 29 away from the total, 198. Out of the remaining 169, we have 149 people for  $A$  and 119 people for  $B$ . We add this up to get 268. Since that is more than what we need, we subtract 169 from 268 to get

(D) 99.



Note: One could use the Principle of Inclusion-Exclusion in a similar way to achieve the same result.

### Solution 3

There are  $198 - 149 = 49$  students who voted against the first issue. Out of these,  $49 - 29 = 20$  voted for the second issue. We also find that 50 students voted against the second issue but in favor of the first in a similar way. This means that  $149 - 50 = 99$  students voted in favor of both issues.

16. (AMC 8 2015) In a middle-school mentoring program, a number of the sixth graders are paired with a ninth-grade student as a buddy. No ninth grader is assigned more than one sixth-grade buddy. If  $\frac{1}{3}$  of all the ninth graders are paired with  $\frac{2}{5}$  of all the sixth graders, what fraction of the total number of sixth and ninth graders have a buddy?

(A)  $\frac{2}{15}$       (B)  $\frac{4}{11}$       (C)  $\frac{11}{30}$       (D)  $\frac{3}{8}$       (E)  $\frac{11}{15}$

Answer: B

### Solution 1 (Using equations)

Let the number of sixth graders be  $s$ , and the number of ninth graders be  $n$ . Thus,  $\frac{n}{3} = \frac{2s}{5}$ , which simplifies to  $n = \frac{6s}{5}$ . Since we are trying to find the value of  $\frac{\frac{n}{3} + \frac{2s}{5}}{n+s}$ , we can just substitute  $\frac{6s}{5}$  for  $n$  into the equation. We then get a value of  $\frac{\frac{6s}{5} + \frac{2s}{5}}{\frac{6s}{5} + s} = \frac{\frac{8s}{5}}{\frac{11s}{5}} = \frac{8s}{11s} = \frac{8}{11}$ .

### Solution 2 (Easy)

We see that the minimum number of ninth graders is 6, because if there are 3 then there is 1 ninth-grader with a buddy, which would mean there are 2.5 sixth graders, which is impossible (of course unless you really do have half of a person). With 6 ninth-graders, 2 of them are in the buddy program, so there  $\frac{2}{6} = \frac{1}{3}$  of the ninth-graders are paired with  $\frac{2}{5}$  of the sixth-graders.

total, two of whom have a buddy. Thus, the desired fraction is  $\frac{2}{6+6} = \frac{2}{12} = \frac{1}{6}$ .

17. (AMC 8 2015) Jeremy's father drives his son to school in rush hour traffic in 20 minutes. One day, there is no traffic, so his father can drive him 18 miles per hour faster and gets him to school in 12 minutes. How far in miles is it to school?

(A) 4      (B) 6      (C) 8      (D) 9      (E) 12

**Answer:** D**Solution 1**For starters, we identify  $d$  as distance and  $v$  as velocity (speed)Writing the equation gives us:  $\frac{d}{v} = \frac{1}{3}$  and  $\frac{d}{v+18} = \frac{1}{5}$ .This gives  $d = \frac{1}{5}v + 3.6 = \frac{1}{3}v$ , which gives  $v = 27$ , which then gives  $d = \boxed{\text{(D)} 9}$ .**Solution 2** $d = rt$ ,  $d = \frac{1}{3} \times r = \frac{1}{5} \times (r + 18)$   $\frac{r}{3} = \frac{r}{5} + \frac{18}{5}$   $10r = 270$  so  $r = 27$ , plug into the first one and it's  $\boxed{\text{(D)} 9}$  miles to school.**Solution 3**We set up an equation in terms of  $d$  the distance and  $x$  the speed In miles per hour.We have  $d = \left(\frac{1}{3}\right)(x) = \left(\frac{1}{5}\right)(x + 18)$ , giving

$$(5)(x) = (3)(x + 18)$$

$$5x = 3x + 54$$

$$2x = 54$$

$$x = 27$$

Hence,  $d = \frac{27}{3} = \boxed{\text{(D)} 9}$ .**Solution 4**Since it takes  $\frac{3}{5}$  of the original time for him to get to school when there is no traffic, the speed must be  $\frac{5}{3}$  of the speed in no traffic or  $\frac{2}{3}$  more. Letting  $x$  be the rate and we know that  $\frac{5}{3}x = x + 18$ , so we have  $\frac{2x}{3} = 18$  miles per hour. Solving for  $x$  gives us 27 miles per hour. Because 20 minutes is a third of an hour, the distance would then be 9 miles  $\boxed{\text{(D)} 9}$ .**Solution 5**When driving in rush hour traffic, he drives 20 minutes for one distance ( $1d$ ) to the school. It means he drives 60 minutes for 3 distances ( $3d$ ) to the school. When driving in no traffic hours, he drives 12 minutes for one distance ( $1d$ ) to the school. It means he drives 60 minutes for 5 distances ( $5d$ ) to the school. Subtracting these two situations, it gives us  $5d - 3d = 18 = 2d$ , then  $d = \frac{18}{2} = 9$ . So the distance to the school would be  $\boxed{\text{(D)} 9}$  miles. LarryFlora**Solution 6**(Ratios)**requires edits**In rush hour traffic, we can create this ratio for each distance we are going to try supposing the distance is  $d$ : $d : 20$  hours

In no traffic, we can do the same:

 $d : 12$  minutes

We want the ratio to be the distance to 60 minutes:

 $d : 20 \text{ minutes} = 3d : 60 \text{ minutes}$

$$d : 12 \text{ minutes} = 5d : 60 \text{ minutes}$$

This shows the speed in miles per hour. This means that the speed in the 2nd ratio has to be 18 faster than the first ratio. We get

$$3d + 18 = 5d \cdot d = 9, \text{ or } \boxed{\text{(D)} 9}.$$

**Solution 7 (Solution 3 but explained better)**

Distance equals speed multiplied by time. Let  $x/60$  represent the speed when Jeremy's father is driving through traffic. Let  $(x+18)/60$  represent the speed when there is no traffic. Because distance equals speed multiplied by time,  $20x/60$  would represent the distance to school. The "20" in this expression represents the 20 minutes it takes Jeremy's father to drive to school through the traffic, and "60" means 60 minutes. Anyway,  $20x/60$  simplifies to  $x/3$ . Using the same distance equation, the expression  $12(x+18)/60$  also represents the distance to school.  $(x+18)/60$  represents the speed when there is no traffic, and the "12" means 12 minutes it takes to get to school when there is no traffic.  $12(x+18)/60$  simplifies to  $(x+18)/5$ . Now, since both expressions equate to the distance to school, you get the equation  $x/3 = (x+18)/5$ . Solving for  $x$  gives you 27. This means that Jeremy's father drives 27 miles per hour in traffic. To find the distance to school, simply take 27 and plug it into the expressions  $(x+18)/5$  or  $x/3$ . Both give you 9, which is the distance to school. Evžen D.

18. (AMC 8 2015) An arithmetic sequence is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, 2, 5, 8, 11, 14 is an arithmetic sequence with five terms, in which the first term is 2 and the constant added is 3. Each row and each column in this  $5 \times 5$  array is an arithmetic sequence with five terms. The square in the center is labelled  $X$  as shown. What is the value of  $X$ ?

|    |  |     |  |    |
|----|--|-----|--|----|
| 1  |  |     |  | 25 |
|    |  |     |  |    |
|    |  | $X$ |  |    |
|    |  |     |  |    |
| 17 |  |     |  | 81 |

- (A) 21      (B) 31      (C) 36      (D) 40      (E) 42

**Answer:** B

**Solution 1**

We begin filling in the table. The top row has a first term 1 and a fifth term 25, so we have the common difference is  $\frac{25-1}{4} = 6$ . This means we can fill in the first row



of the table:

|    |   |     |    |    |
|----|---|-----|----|----|
| 1  | 7 | 13  | 19 | 25 |
|    |   |     |    |    |
|    |   | $X$ |    |    |
|    |   |     |    |    |
| 17 |   |     |    | 81 |

The fifth row has a first term of 17 and a fifth term of 81, so the common difference is  $\frac{81-17}{4} = 16$ . We can fill in the fifth row of the table as shown:

|    |    |     |    |    |
|----|----|-----|----|----|
| 1  | 7  | 13  | 19 | 25 |
|    |    |     |    |    |
|    |    | $X$ |    |    |
|    |    |     |    |    |
| 17 | 33 | 49  | 65 | 81 |

We must find the third term of the arithmetic sequence with a first term of 13 and a fifth term of 49. The common difference of this sequence is  $\frac{49-13}{4} = 9$ , so the third term is  $13 + 2 \cdot 9 = \boxed{\text{(B)} 31}$ .

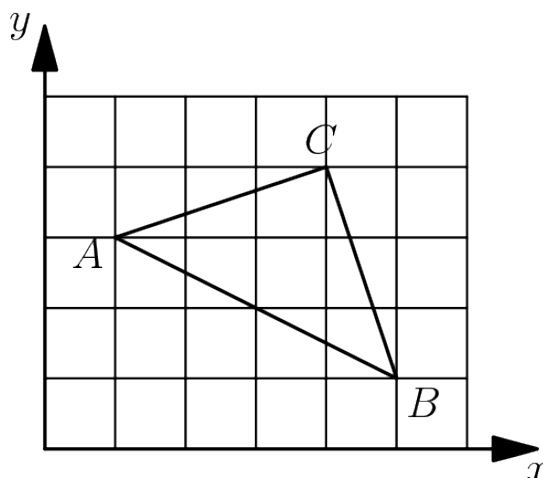
**Solution 2**

The middle term of the first row is  $\frac{25+1}{2} = 13$ , since the middle number is just the average in an arithmetic sequence. Similarly, the middle of the bottom row is  $\frac{17+81}{2} = 49$ . Applying this again for the middle column, the answer is  $\frac{49+13}{2} = \boxed{\text{(B)} 31}$ .

**Solution 3**

The value of  $X$  is simply the average of the average values of both diagonals that contain  $X$ . This is  $\frac{\frac{1+81}{2} + \frac{17+25}{2}}{2} = \frac{\frac{82}{2} + \frac{42}{2}}{2} = \frac{41+21}{2} = \boxed{\text{(B)} 31}$

19. (AMC 8 2015) A triangle with vertices as  $A = (1, 3)$ ,  $B = (5, 1)$ , and  $C = (4, 4)$  is plotted on a  $6 \times 5$  grid. What fraction of the grid is covered by the triangle?



- (A)  $\frac{1}{6}$     (B)  $\frac{1}{5}$     (C)  $\frac{1}{4}$     (D)  $\frac{1}{3}$     (E)  $\frac{1}{2}$

**Answer:** A

**Solution 1**

The area of  $\triangle ABC$  is equal to half the product of its base and height. By the Pythagorean Theorem, we find its height is  $\sqrt{1^2 + 2^2} = \sqrt{5}$ , and its base is  $\sqrt{2^2 + 4^2} = \sqrt{20}$ . We multiply these and divide by 2 to find the area of the triangle is  $\frac{\sqrt{5 \cdot 20}}{2} = \frac{\sqrt{100}}{2} = \frac{10}{2} = 5$ . Since the grid has an area of 30, the fraction of the grid covered by the triangle is  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$ .

**Solution 2**

Note angle  $\angle ACB$  is right; thus, the area is  $\sqrt{1^2 + 3^2} \times \sqrt{1^2 + 3^2} \times \frac{1}{2} = 10 \times \frac{1}{2} = 5$ ; thus, the fraction of the total is  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$ .

**Solution 3 (Shoelace Theorem)**

By the **Shoelace Theorem**, the area of  $\triangle ABC = \left| \frac{1}{2}(15 + 4 + 4 - 1 - 20 - 12) \right| = \left| \frac{1}{2}(-10) \right| = 5$ .

This means the fraction of the total area is  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$ .

**Solution 4 (easiest solution)**

The smallest rectangle that follows the grid lines and completely encloses  $\triangle ABC$  has an area of 12, where  $\triangle ABC$  splits the rectangle into four triangles. The area of  $\triangle ABC$  is therefore  $12 - \left( \frac{4 \cdot 2}{2} + \frac{3 \cdot 1}{2} + \frac{3 \cdot 1}{2} \right) = 12 - \left( 4 + \frac{3}{2} + \frac{3}{2} \right) = 12 - 7 = 5$ . That means that  $\triangle ABC$  takes up  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$  of the grid.

**Solution 5 (Pick's Theorem, very much recommended to learn this)**

Using **Pick's Theorem**, the area of the triangle is  $4 + \frac{4}{2} - 1 = 5$ . Therefore, the triangle takes up  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$  of the grid.

**Solution 6 (Heron's Formula, Not Recommended)**

We can find the lengths of the sides by using the **Pythagorean Theorem**. Then, we apply **Heron's Formula** to find the area.

$$\sqrt{\left(\frac{\sqrt{10} + \sqrt{10} + 2\sqrt{5}}{2}\right) \left(\frac{\sqrt{10} + \sqrt{10} + 2\sqrt{5}}{2} - \sqrt{10}\right) \left(\frac{\sqrt{10} + \sqrt{10} + 2\sqrt{5}}{2} - \sqrt{10}\right) \left(\frac{\sqrt{10} + \sqrt{10} + 2\sqrt{5}}{2} - \sqrt{10}\right)}$$

This simplifies to

$$\sqrt{(\sqrt{10} + \sqrt{5})(\sqrt{10} + \sqrt{5} - \sqrt{10})(\sqrt{10} + \sqrt{5} - \sqrt{10})(\sqrt{10} + \sqrt{5} - 2\sqrt{5})}.$$

Again, we simplify to get

$$\sqrt{(\sqrt{10} + \sqrt{5})(\sqrt{5})(\sqrt{5})(\sqrt{10} - \sqrt{5})}.$$

The middle two terms inside the square root multiply to 5, and the first and last terms inside the square root multiply to  $\sqrt{10}^2 - \sqrt{5}^2 = 10 - 5 = 5$ . This means that the area of the triangle is

$$\sqrt{5 \cdot 5} = 5.$$

The area of the grid is  $6 \cdot 5 = 30$ . Thus, the answer is  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$ .

**Solution 7 (Simple Deduction)**

First, count the number of 1x1 squares inside the main triangle, both fully enclosed and partially (you should count 10). Then, upon closer inspection, most of the squares that are not a single unit on the triangle can be created by connecting another shape. The only exception is one square that is a single unit and one that would need 2 shapes connected to it to make a single unit. On average, you need to connect 2 squares to make a unit. Knowing this, if there are 10 squares and you require 2 shapes to make a unit, 10 divided by 2 equals 5, which is the area.

5 is 1/6 of 30 (the total of the graph) and so the final answer is  $\frac{5}{30} = \boxed{\text{(A)} \frac{1}{6}}$ .

-Themathnerd3.14

20. (AMC 8 2015) Ralph went to the store and bought 12 pairs of socks for a total of \$24. Some of the socks he bought cost \$1 a pair, some of the socks he bought cost \$3 a pair, and some of the socks he bought cost \$4 a pair. If he bought at least one pair of each type, how many pairs of \$1 socks did Ralph buy?

(A) 4      (B) 5      (C) 6      (D) 7      (E) 8

**Answer: D**

**Solution 1**

So, let there be  $x$  pairs of \$1 socks,  $y$  pairs of \$3 socks, and  $z$  pairs of \$4 socks.

We have  $x + y + z = 12$ ,  $x + 3y + 4z = 24$ , and  $x, y, z \geq 1$ .

Now, we subtract to find  $2y + 3z = 12$ , and  $y, z \geq 1$ . It follows that  $2y$  is a multiple of 3 and  $3z$  is a multiple of 3. Since sum of 2 multiples of 3 = multiple of 3, so we must have  $2y = 6$ .

Therefore,  $y = 3$ , and it follows that  $z = 2$ . Now,  $x = 12 - y - z = 12 - 3 - 2 = \boxed{\text{(D)} 7}$ , as desired.

### Solution 2

Since the total cost of the socks was \$24 and Ralph bought 12 pairs, the average cost of each pair of socks is  $\frac{\$24}{12} = \$2$ .

There are two ways to make packages of socks that average to \$2. You can have:

- Two \$1 pairs and one \$4 pair (package adds up to \$6)
- One \$1 pair and one \$3 pair (package adds up to \$4)

Now, we need to solve

$$6a + 4b = 24,$$

where  $a$  is the number of \$6 packages and  $b$  is the number of \$4 packages. We see our only solution (that has at least one of each pair of sock) is  $a = 2, b = 3$ , which yields the answer of  $2 \times 2 + 3 \times 1 = \boxed{\text{(D)} 7}$ .

### Solution 3

Since there are 12 pairs of socks, and Ralph bought at least one pair of each, there are  $12 - 3 = 9$  pairs of socks left. Also, the sum of the three pairs of socks is  $1 + 3 + 4 = 8$ . This means that there are  $24 - 8 = 16$  dollars left. If there are only 1 dollar socks left, then we would have  $9 \cdot 1 = 9$  dollars wasted, which leaves 7 more dollars. If we replace one pair with a 3 dollar pair, then we would waste an additional 2 dollars. If we replace one pair with a 4 dollar pair, then we would waste an additional 3 dollars. The only way 7 can be represented as a sum of 2s and 3s is  $2 + 2 + 3$ . If we change 3 pairs, we would have 6 pairs left. Adding the one pair from previously, we have  $\boxed{\text{(D)} 7}$  pairs.

### Solution 4

Let the amount of 1 dollar socks be  $a$ , 3 dollar socks be  $b$ , and 4 dollar socks be  $c$ . We then know that  $a + b + c = 12$  and  $a + 3b + 4c = 24$ . We can make  $a + b + c = 12$  into  $a = 12 - b - c$  and then plug that into the other equation, producing  $12 - b - c + 3b + 4c = 24$  which simplifies to  $2b + 3c = 12$ . It's not hard to see  $b = 3$  and  $c = 2$ . Now that we know  $b$  and  $c$ , we know that  $a = 7$ , meaning the number of 1 dollar socks Ralph bought is  $\boxed{\text{(D)} 7}$ .

### Solution 5 (Guess and check)

If Ralph bought one sock of each kind, he already used \$8, so there are \$16 left and 9 socks. If we split the \$16 into four \$4 sections, (as it is the smallest possible number that 1, 3, 4, can make in different ways that in all use at least each of the numbers once,) if Ralph bought a \$3 pair, he would need to buy a \$1 pair in order for it to add up to a multiple of four. Similarly, if Ralph bought a \$1 pair, he would either need to buy three \$1 pairs or a \$3 pair. If Ralph bought a \$4 pair, it would already make a group. Now, the problem is just how we can split 9 into 4 groups of 1, 2, or 4. We clearly see that  $1 + 2 + 2 + 4 = 9$ , or a \$4 pair, two \$3 pairs, and six \$1 pairs. Because we subtracted the necessary one of each kind, there are two \$4 pairs, three \$3 pairs, and seven \$1 pairs. Therefore, the number of \$1 pairs Ralph bought is  $\boxed{\text{(D)} 7}$ . strongstephen

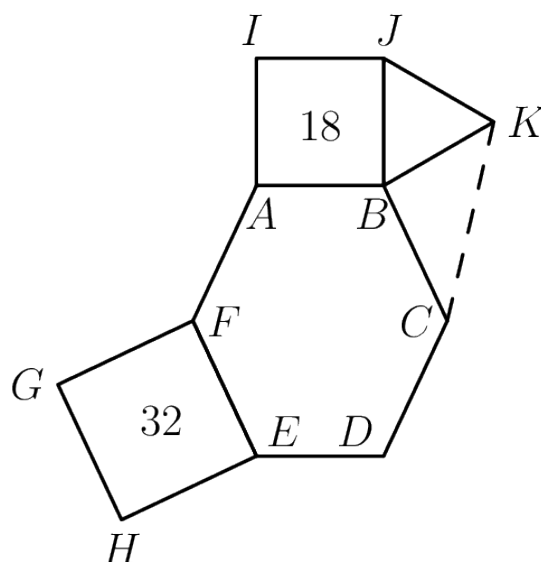
# Solution 6

To simplify the problem, we can reduce the price of each sock by \$1, resulting in the prices being \$0, \$2 and \$3. This means the total price Ralph spends becomes  $\$24 - 12 \times \$1 = \$12$ . Now the \$0 socks aren't worth anything, but we still need to keep in mind there are at least one of them, so the question becomes: at most 11 of \$2 and \$3 socks add up to \$12. It is obvious that  $3 \times \$2 + 2 \times \$3 = \$12$ , so the number of \$0 socks (actually \$1 socks) is  $12 - 3 - 2 = 7$ .

Note: this is very similar to Solution 1, but a non algebraic way of thinking about it.

-bbidev

21. (AMC 8 2015) In the given figure hexagon  $ABCDEF$  is equiangular,  $ABJI$  and  $FEHG$  are squares with areas 18 and 32 respectively,  $\triangle JBK$  is equilateral and  $FE = BC$ . What is the area of  $\triangle KBC$ ?



- (A)  $6\sqrt{2}$     (B) 9    (C) 12    (D)  $9\sqrt{2}$     (E) 32

**Answer:** C

# Solution 1

Clearly, since  $\overline{FE}$  is a side of a square with area 32,  $\overline{FE} = \sqrt{32} = 4\sqrt{2}$ . Now, since  $\overline{FE} = \overline{BC}$ , we have  $\overline{BC} = 4\sqrt{2}$ .

We know that  $\overline{JB}$  is a side of a square with area 18, so  $\overline{JB} = \sqrt{18} = 3\sqrt{2}$ . Since  $\triangle JBK$  is equilateral,  $\overline{BK} = \overline{JB} = 3\sqrt{2}$ .

Lastly,  $\triangle KBC$  is a right triangle. We see that  $\angle JBA + \angle ABC + \angle CBK + \angle KBJ = 360^\circ \rightarrow 90^\circ + 120^\circ + \angle CBK + 60^\circ = 360^\circ \rightarrow \angle CBK = 90^\circ$ , so  $\triangle KBC$  is a right triangle with legs  $3\sqrt{2}$  and  $4\sqrt{2}$ . Now, its area is  $\frac{3\sqrt{2} \cdot 4\sqrt{2}}{2} = \frac{24}{2} = \boxed{\text{(C) } 12}$ .

# Solution 2

Since  $\overline{FE} = \sqrt{32}$ , and  $\overline{FE} = \overline{BC}$ ,  $\overline{BC} = 4\sqrt{2}$ . Meanwhile,  $\overline{JB} = 3\sqrt{2}$ , and since  $\triangle JBK$  is equilateral,  $\overline{BK} = 3\sqrt{2}$ . If  $ABCDEF$  is equiangular,  $\angle ABC = \frac{180 \cdot (n-2)}{n} = 120^\circ$ , where  $n$  is the number of sides of the shape. Adding all the angles

around  $B$  gives  $270^\circ$ , so  $\angle KBC = 360 - 90 = 270^\circ$ . Because  $\triangle KBC$  is right, the area of  $\triangle KBC = \frac{4\sqrt{2} \cdot 3\sqrt{2}}{2} = 12$ . Therefore, the answer is **(C) 12**. strongstephen

Note: Another, more complicated way to determine if angle KBC is right is to extend line AB to a point L on triangle KBC. Now, since we know that angle IJB is right, and we know that line IJ now acts as a transversal to lines IJ and AL. That means angle IJB (90 degrees) must be equal to angle JBL by the alternate exterior angle rule. Since we know that triangle JKB is equilateral, angle JKB will be 60 degrees, hence angle KBL will be  $90 - 60 = 30$  degrees. Now, we move on to figure out ABCDEF. Through the solution above, we know that angle ABC is 120 degrees, so we can use the same line AL, which will result in the formation of a linear pair of angles: angles ABC and CBL. Since the angles in a linear pair always add up to 180, and angle ABC is 120, angle CBL =  $180 - 120 = 60$  degrees. Notice that angles CBL and KBL together make up angle KBC. Hence, by the value of 30 degrees derived for angle KBL, angle KBC = angle KBL + angle CBL = 30 degrees + 60 degrees = 90 degrees.

### Solution 3

Square ABJI's side length is  $\sqrt{18} = 3\sqrt{2}$ , and since Triangle JKB is equilateral, that means that the side length of Triangle JKB would be  $3\sqrt{2}$ , which means that KB, the height of Triangle KBC would be  $3\sqrt{2}$ . Square FEHG's side length would then be  $\sqrt{32} = 4\sqrt{2}$ , and because line FE = line BC, The base of Triangle KBC would be  $4\sqrt{2}$ . The formula to finding the area of a triangle is the  $\frac{\text{Height} \cdot \text{Base}}{2}$ , which means that Triangle KBC's area would just be  $\frac{3\sqrt{2} \cdot 4\sqrt{2}}{2}$ , and the answer is **12**.

22. (AMC 8 2015) On June 1, a group of students are standing in rows, with 15 students in each row. On June 2, the same group is standing with all of the students in one long row. On June 3, the same group is standing with just one student in each row. On June 4, the same group is standing with 6 students in each row. This process continues through June 12 with a different number of students per row each day. However, on June 13, they cannot find a new way of organizing the students. What is the smallest possible number of students in the group?
- (A) 21      (B) 30      (C) 60      (D) 90      (E) 1080

**Answer:** C

### Solution 1

Since we know the number must be a multiple of 15, we can eliminate A. We also know that after 12 days, the students can't find any more arrangements, meaning the number has 12 factors. Now, we just list the factors of every number, starting with 30:

$$30 = 1 \cdot 30, 2 \cdot 15, 3 \cdot 10, 5 \cdot 6$$

$$60 = 1 \cdot 60, 2 \cdot 30, 3 \cdot 20, 4 \cdot 15, 5 \cdot 12, 6 \cdot 10$$

60 has 12 factors, so the answer is **(C) 60**.

Note - A faster way to find the factors of a number is really helpful in competitions, as you don't want to waste time listing them all. What we can do is find the prime factorization of a number. We'll use 60 as an example here, so we take  $60 = 2 \cdot 2 \cdot 3 \cdot 5$  or  $60 = 2^2 \cdot 3^1 \cdot 5^1$ . Now, we just add one to the power of each factor. So since we have powers of 2, 1, and 1, we turn those into 3, 2, and 2. Finally, we just multiply them together to get  $3 \cdot 2 \cdot 2 = 12$  as the amount of factors in 60.

**Solution 2**

We know that the number of students in the group has to be a multiple of 15, so we can eliminate A. However, on June 13, they cannot find a new way of organizing the students meaning that the number has only 12 factors. The prime factorization of 60 is equal to  $2^2 \times 3^1 \times 5^1$ . Using the factor counting formula, we have to add one to each exponent and multiply them together to find the number of factors.  $(2 + 1) \times (1 + 1) \times (1 + 1) = 3 \times 2 \times 2 = 12$  factors making **(C) 60** the answer.

**Solution 3 (Combination of Solutions)**

On the 1st day, all the students are in an arrangement such that there are 15 students per row. On the 4th day, all the students are in an arrangement such that there are 6 students per row. This tells us that the # of students is a multiple of both 15 and 6. Since we can arrange the students for 12 days, this tells us that the # of students has 12 factors. We can also tell that the # of students is a multiple of 30, as 30 is the *LCM* of 6 and 15.

We now test answer choice B.

$$30 = 2 * 3 * 5 \Rightarrow 30 \text{ has } 2 * 2 * 2 = 8 \text{ factors.}$$

Let's test answer choice C.

$$60 = 2^2 * 3 * 5 \Rightarrow 60 \text{ has } 3 * 2 * 2 = 12 \text{ factors.}$$

We know that **(C) 60** is our answer.

23. (AMC 8 2015) Tom has twelve slips of paper which he wants to put into five cups labeled A, B, C, D, E. He wants the sum of the numbers on the slips in each cup to be an integer. Furthermore, he wants the five integers to be consecutive and increasing from A to E. The numbers on the papers are 2, 2, 2, 2.5, 2.5, 3, 3, 3, 3, 3.5, 4, and 4.5. If a slip with 2 goes into cup E and a slip with 3 goes into cup B, then the slip with 3.5 must go into what cup?

(A) A    (B) B    (C) C    (D) D    (E) E

**Answer:** D

**Solution**

The numbers have a sum of  $6 + 5 + 12 + 4 + 8 = 35$ , which averages to 7, which means A, B, C, D, and E will have the values 5, 6, 7, 8 and 9, respectively. Now, it's the process of elimination: Cup A will have a sum of 5, so putting a 3.5 slip in the cup will leave  $5 - 3.5 = 1.5$ ; however, all of our slips are bigger than 1.5, so this is impossible. Cup B has a sum of 6, but we are told that it already has a 3 slip, leaving  $6 - 3 = 3$ , which is too small for the 3.5 slip. Cup C is a little bit trickier but still manageable. It must have a value of 7, so adding the 3.5 slip leaves room for  $7 - 3.5 = 3.5$ . This looks good at first as we have slips smaller than that, but upon closer inspection, we see that no slip fits exactly. And the smallest sum of two slips is  $2 + 2 = 4$ , which is too big, so this case is also impossible. Cup E has a sum of 9, but we are told it already has a 2 slip. So, we are left with  $9 - 2 = 7$ , which is identical to Cup C and thus also impossible. With all other choices removed, we are left with the answer: Cup **(D) D**.

24. (AMC 8 2015) A baseball league consists of two four-team divisions. Each team plays every other team in its division  $N$  games. Each team plays every team in the

other division  $M$  games with  $N > 2M$  and  $M > 4$ . Each team plays a 76-game schedule. How many games does a team play within its own division?

- (A) 36      (B) 48      (C) 54      (D) 60      (E) 72

**Answer:** B

**Solution 1**

On one team they play  $3N$  games in their division and  $4M$  games in the other. This gives  $3N + 4M = 76$ .

Since  $M > 4$  we start by trying  $M=5$ . This doesn't work because 56 is not divisible by 3.

Next,  $M = 6$  does not work because 52 is not divisible by 3.

We try  $M = 7$  does work by giving  $N = 16$ ,  $M = 7$  and thus  $3 \times 16 = \boxed{\text{(B)} 48}$  games in their division.

$M = 10$  seems to work, until we realize this gives  $N = 12$ , but  $N > 2M$  so this will not work.

**Solution 2**

$76 = 3N + 4M > 10M$ , giving  $M < 7.6$ . Since  $M > 4$ , we have  $M = 5, 6, 7$ . Since  $4M$  is  $1 \pmod{3}$ , we must have  $M$  equal to  $1 \pmod{3}$ , so  $M = 7$ .

This gives  $3N = 48$ , as desired. The answer is  $\boxed{\text{(B)} 48}$ .

**Solution 3**

Notice that each team plays  $N$  games against each of the three other teams in its division. So that's  $3N$ .

Since each team plays  $M$  games against each of the four other teams in the other division, that's  $4M$ .

So  $3N + 4M = 76$ , with  $M > 4$ ,  $N > 2M$ .

Let's start by solving this Diophantine equation. In other words,  $N = \frac{76-4M}{3}$ .

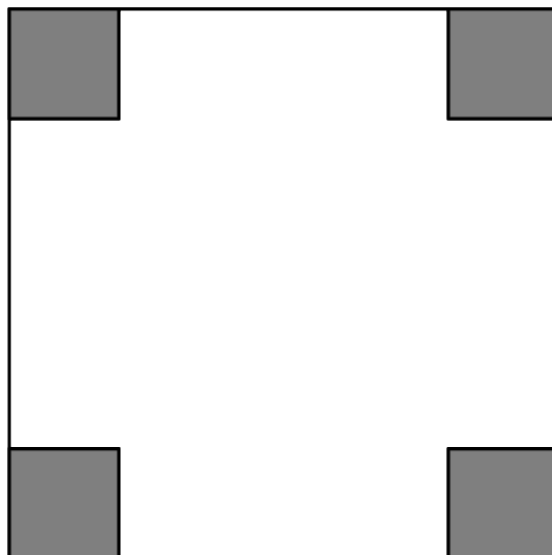
So  $76 - 4M \equiv 0 \pmod{3}$  (remember:  $M$  must be divisible by 3 for  $N$  to be an integer!). Therefore, after reducing 76 to 1 and  $-4M$  to  $2M$  (we are doing things in  $\pmod{3}$ ), we find that  $M \equiv 1 \pmod{3}$ .

Since  $M > 4$ , so the minimum possible value of  $M$  is 7. However, remember that  $N > 2M$ ! To find the greatest possible value of  $M$ , we assume that  $N = 2M$  and that is the upper limit of  $M$  (excluding that value because  $N > 2M$ ). Plugging  $N = 2M$  in,  $10M = 76$ . So  $M < 7.6$ . Since you can't have 7.6 games, we know that we can only check  $M = 7$  since we know that since  $M > 4$ ,  $M < 7.6$ ,  $M \equiv 1 \pmod{3}$ . After checking  $M = 7$ , we find that it will work.

So  $M = 7$ ,  $N = 16$ . So each team plays 16 games against each team in its division. Since they are asking for games in its division, which equals  $3n = 48$ . Select  $\boxed{\text{(B)} 48}$ . (P.S: This might be too complicated).

25. (AMC 8 2015) One-inch squares are cut from the corners of this 5 inch square. What is the area in square inches of the largest square that can fit into the remaining space?



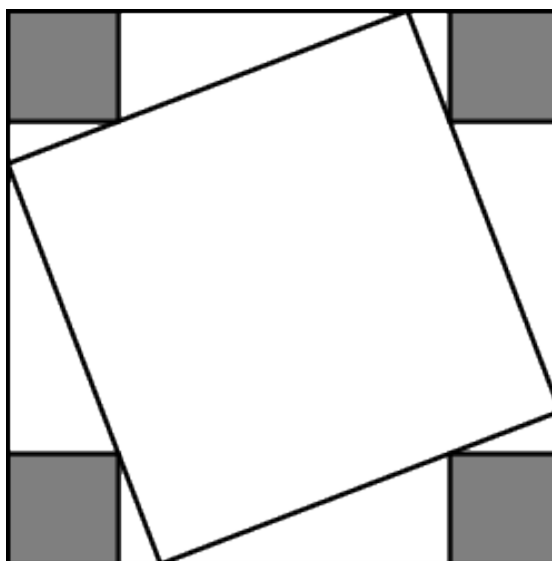


- (A) 9      (B)  $12\frac{1}{2}$       (C) 15      (D)  $15\frac{1}{2}$       (E) 17

**Answer:** C

**Solution 1**

We can draw a diagram as shown:



Let us focus on the 4 big triangles that, together with the inscribed square, fill in the large square. If we zoom in on one of the four triangles, we can see that it is composed of a small unit square in the corner of the large square and two triangles, one smaller than the other. We are going to focus specifically on the smaller of the two triangles. This triangle is similar to the big triangle itself by AA similarity (because the two sides of a square are parallel. To prove this fact, draw a diagonal of the square and find congruent triangles). Let the shorter leg of the big triangle be  $x$ ; then  $\frac{x}{x-1} = \frac{5-x}{1}$ .

$$x = -x^2 + 6x - 5$$

$$x^2 - 5x + 5 = 0$$

$$x = \frac{5 \pm \sqrt{(-5)^2 - (4)(1)(5)}}{2}$$

$$x = \frac{5 \pm \sqrt{5}}{2}$$

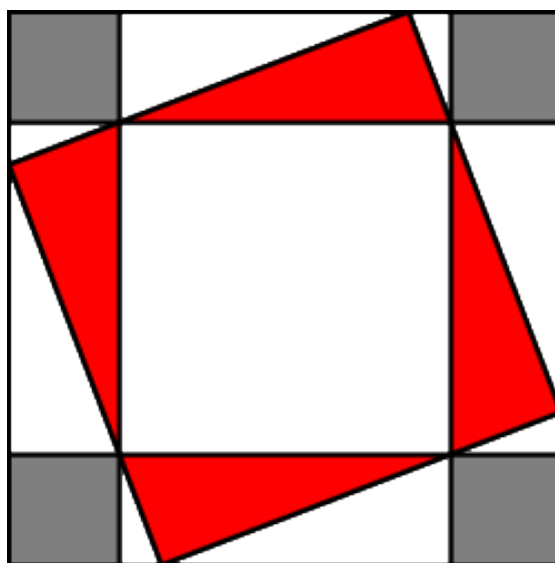
Thus,  $x = \frac{5 - \sqrt{5}}{2}$ , because by symmetry,  $x < \frac{5}{2}$ . Note that the other solution we got, namely,  $x = \frac{5 + \sqrt{5}}{2}$ , is the length of the segment  $5 - x$ .  $x$  and  $5 - x$  together sum to 5, the side of the length of the large square, and similarly, the sum of the solutions is 5. This solution is a result of the symmetry of the problem; if we had set the longer leg of the big triangle to be  $x$ , then we would solve the same quadratic to find the same roots, the only difference being that we take the other root.

This means the area of each triangle is  $\frac{5 - \sqrt{5}}{2} * (5 - \frac{5 - \sqrt{5}}{2}) * \frac{1}{2} = \frac{5}{2}$ . Thus, the area of the square is  $25 - (4 * \frac{5}{2}) = 15 \implies \boxed{\text{(C)}}$ .

Don't use this in a contest scenario, and only use it when practicing math skills. :)

**Solution 2 (Contest Solution, most recommended and fastest)**

We draw a square as shown:



We want to find the area of the biggest square. The area of this square is composed of the center white square and the four red triangles. Because the inner square has an area of  $(5 - 2)^2$ , 3, squared, 9, it also has a length of  $\sqrt{9} = 3$ . The heights of each of the red triangles are 1 (because the gray squares have lengths of one), and the area of one triangle is namely  $\frac{3 \cdot 1}{2}$ . Thus, the combined area of the four triangles is  $4 \cdot \frac{3}{2} = 6$ . Furthermore, the area of the smaller square is 9. We add these to see that the area of the large square is  $9 + 6 = 15 \implies \boxed{\text{(C)}}$ .

**Solution 3**

Let us find the area of the triangles and the unit squares: on each side, there are two triangles. They both have one leg of length 1, and let's label the other legs  $x$  for one of the triangles and  $y$  for the other. Note that  $x + y = 3$ . The area of each of the triangles is  $\frac{x}{2}$  and  $\frac{y}{2}$ , and there are 4 of each. So now, we need to find  $4(\frac{x}{2}) + 4(\frac{y}{2})$ .

Expanding out the fractions, we get  $4\left(\frac{x}{2}\right) + 4\left(\frac{y}{2}\right) = 2x + 2y = 2(x + y) = 2(3) = 6$ . So, the area of the square we need is  $25 - (4 + 6) = 15 \Rightarrow \boxed{\text{(C)}}$ .

**Solution 4**

Proceed as in solution 1 to get  $x^2 - 5x + 5 = 0$ .

Notice that the side length of the square is  $\sqrt{x^2 + (5 - x)^2}$  by the Pythagorean Theorem. Thus, the area of the square is

$$\begin{aligned} & \left( \sqrt{x^2 + (5 - x)^2} \right)^2 \\ &= x^2 + x^2 - 10x + 25 \\ &= (2x^2 - 10x + 10) + 15 \\ &= 2(x^2 - 5x + 5) + 15 \\ &= 2(0) + 15 \\ &= \boxed{15} \\ &\Rightarrow \boxed{\text{(C)}} \end{aligned}$$

**Solution 5 (Fast with no explanations)**

By inspection, we can see that the area of the largest square would be the area of the middle square formed by the vertices of the shaded squares plus half the combined area of the four smaller rectangles. Our desired answer is  $9 + \frac{3 \cdot 4}{2} = 15 \Rightarrow \boxed{\text{(C)}}$ .

## §32 AMC 8 2016

1. (*AMC 8 2016*) The longest professional tennis match lasted a total of 11 hours and 5 minutes. How many minutes is that?  
(A) 605      (B) 655      (C) 665      (D) 1005      (E) 1105

**Answer:** C

**Solution 1**

It is best to split 11 hours and 5 minutes into 2 parts, one of 11 hours and another of 5 minutes. We know that there is 60 minutes in a hour. Therefore, there are  $11 \cdot 60 = 660$  minutes in 11 hours. Adding the second part (the 5 minutes) we get  $660 + 5 = \boxed{\text{(C) 665}}$ .

**Solution 2**

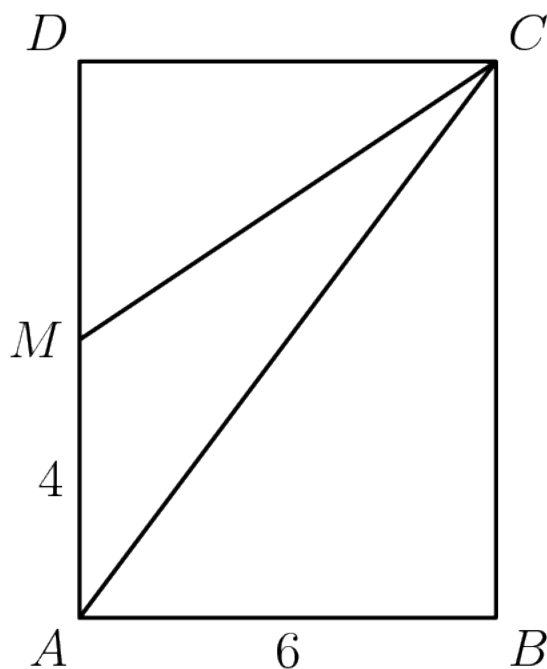
The best method comes when you remember your multiplication tables. Thus trivial, we get our answer  $\boxed{\text{(C) 665}}$ .

**Solution 3**

11 hours 5 min =  $(11 \cdot 60) + 5 \text{ min} = 665 \text{ min}$ , therefore  $\boxed{\text{(C) 665}}$ .

Carmen

2. (AMC 8 2016) In rectangle  $ABCD$ ,  $AB = 6$  and  $AD = 8$ . Point  $M$  is the midpoint of  $\overline{AD}$ . What is the area of  $\triangle AMC$ ?



- (A) 12      (B) 15      (C) 18      (D) 20      (E) 24

**Answer:** A

**Solution 1**

Using the triangle area formula for triangles:  $A = \frac{bh}{2}$ , where  $A$  is the area,  $b$  is the base, and  $h$  is the height. This equation gives us  $A = \frac{4 \cdot 6}{2} = \frac{24}{2} = \boxed{\text{(A)} 12}$ .

**Solution 2**

A triangle with the same height and base as a rectangle is half of the rectangle's area. This means that a triangle with half of the base of the rectangle and also the same height means its area is one quarter of the rectangle's area. Therefore, we get  $\frac{48}{4} = \boxed{\text{(A)} 12}$ .

**Solution 3(a check)**

We can find the area of the entire rectangle, DCBA to be  $8 \cdot 6 = 48$  and find DCM area to be  $\frac{6 \cdot 4}{2} = 12$  and BCA to be  $\frac{6 \cdot 8}{2} = 24$ ,  $48 - 12 - 24 = \boxed{\text{(A)} 12}$ .

**Solution 4**

A triangle is half of a rectangle. So since M is the midpoint of DA, we can see that triangle DAC is half of the whole rectangle. And it is also true that triangle MAC is half of triangle DAC, so triangle MAC would just be  $\frac{1}{4}$  of the whole rectangle. Since the rectangle's area is 48,  $\frac{1}{4}$  of 48 would be 12. Which gives us the answer  $\boxed{\text{(A)} 12}$ .

**Solution 5 (complicated way)**

We can subtract the total areas of triangles DCM and ABC from the rectangle ABCD. For triangle DCM, the base is 4 and the height is 6, so we multiply 4 and 6, then divide by 2 to get 12. For triangle ABC, the base is 6 and the height is 8,

so we multiply 4 and 8, then divide by 2 to get 24. We add 24 and 12 to get 36. Then, we calculate the area of rectangle ABCD, which is 48. We subtract 36 from 48, resulting in **(A) 12**.

-BananaBall00

3. (*AMC 8 2016*) Four students take an exam. Three of their scores are 70, 80, and 90. If the average of their four scores is 70, then what is the remaining score?  
**(A) 40**      **(B) 50**      **(C) 55**      **(D) 60**      **(E) 70**

**Answer:** A

**Solution 1**

Let  $r$  be the remaining student's score. We know that the average, 70, is equal to  $\frac{70+80+90+r}{4}$ . We can use basic algebra to solve for  $r$ :

$$\frac{70 + 80 + 90 + r}{4} = 70$$

$$\frac{240 + r}{4} = 70$$

$$240 + r = 280$$

$$r = 40$$

giving us the answer of **(A) 40**.

**Solution 2**

Since 90 is 20 more than 70, and 80 is 10 more than 70, for 70 to be the average, the other number must be 30 less than 70, or **(A) 40**.

4. (*AMC 8 2016*) When Cheenu was a boy, he could run 15 miles in 3 hours and 30 minutes. As an old man, he can now walk 10 miles in 4 hours. How many minutes longer does it take for him to travel a mile now compared to when he was a boy?  
**(A) 6**      **(B) 10**      **(C) 15**      **(D) 18**      **(E) 30**

**Answer:** B

**Solution 1**

When Cheenu was a boy, he could run 15 miles in 3 hours and 30 minutes =  $3 \times 60 + 30$  minutes = 210 minutes, thus running  $\frac{210}{15} = 14$  minutes per mile. Now that he is an old man, he can walk 10 miles in 4 hours =  $4 \times 60$  minutes = 240 minutes, thus walking  $\frac{240}{10} = 24$  minutes per mile. Therefore, it takes him **(B) 10** minutes longer to walk a mile now compared to when he was a boy.

**Solution 2**

From the question, the old man can travel five miles in two hours, so we can set both speeds to 15 miles. We can see that Cheenu as an old man takes 2 hours and 30 minutes for him to travel 15 miles, which is also 150 minutes. We can then divide this by fifteen, which gives us 10, thus the answer is **(B) 10**

5. (*AMC 8 2016*) The number  $N$  is a two-digit number.
- When  $N$  is divided by 9, the remainder is 1.
  - When  $N$  is divided by 10, the remainder is 3.

What is the remainder when  $N$  is divided by 11?

- (A) 0    (B) 2    (C) 4    (D) 5    (E) 7

**Answer:** E

**Solution 1**

From the second bullet point, we know that the second digit must be 3, for a number divisible by 10 ends in zero. Since there is a remainder of 1 when  $N$  is divided by 9, the multiple of 9 must end in a 2 for it to have the desired remainder (mod 10). We now look for this one:

$$9(1) = 9$$

$$9(2) = 18$$

$$9(3) = 27$$

$$9(4) = 36$$

$$9(5) = 45$$

$$9(6) = 54$$

$$9(7) = 63$$

$$9(8) = 72$$

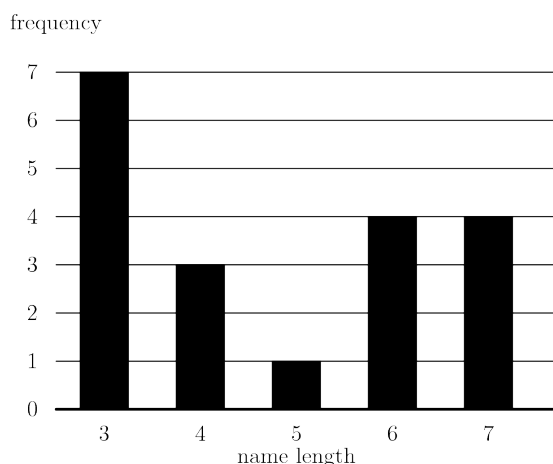
The number  $72 + 1 = 73$  satisfies both conditions. We subtract the biggest multiple of 11 less than 73 to get the remainder. Thus,  $73 - 11(6) = 73 - 66 = \boxed{\text{(E)} 7}$ .

**Solution 2**

We know that the number has to be one more than a multiple of 9, because of the remainder of one, and the number has to be 3 more than a multiple of 10, which means that it has to end in a 3. Now, if we just list the first few multiples of 9 adding one to the number we get: 10, 19, 28, 37, 46, 55, 64, 73, 82, 91. As we can see from these numbers, the only one that has a three in the units place is 73, thus we divide 73 by 11, getting 6 *R*7, hence,  $\boxed{\text{(E)} 7}$ .

We could also remember that, for a two-digit number to be divisible by 9, the sum of its digits has to be equal to 9. Since the number is one more than a multiple of 9, the multiple we are looking for has a ones digit of 2, and therefore a tens digit of  $9 - 2 = 7$ , and then we could proceed as above.

6. (AMC 8 2016) The following bar graph represents the length (in letters) of the names of 19 people. What is the median length of these names?



- (A) 3    (B) 4    (C) 5    (D) 6    (E) 7

**Answer:** B

**Solution 1**

We first notice that the median name will be the  $(19 + 1)/2 = 10^{\text{th}}$  name. The  $10^{\text{th}}$  name is **(B) 4**.

**Solution 2**

To find the median length of a name from a bar graph, we must add up the number of names. Doing so gives us  $7 + 3 + 1 + 4 + 4 = 19$ . Thus the index of the median length would be the 10th name. Since there are 7 names with length 3, and 3 names with length 4, the 10th name would have 4 letters. Thus our answer is **(B) 4**.

7. (AMC 8 2016) Which of the following numbers is not a perfect square?

(A)  $1^{2016}$     (B)  $2^{2017}$     (C)  $3^{2018}$     (D)  $4^{2019}$     (E)  $5^{2020}$

**Answer:** B

**Solution 1**

Our answer must have an odd exponent in order for it to not be a square. Because 4 is a perfect square,  $4^{2019}$  is also a perfect square, so our answer is **(B)  $2^{2017}$** .

**Solution 2 (more detailed explanation of Solution 1)**

Any even integer  $m$  can be divided into  $2n$  where  $n$  is an integer as well. Therefore, (A), (C), and (E) are immediately disqualified, as when  $m$  is even,  $a^m$  can always be written as  $(a^{\frac{m}{2}})^2$  as  $(a^m)^n = a^{mn}$  and vice versa. Only odd exponents are non-squares, therefore.  $4 = 2^2$ , which means  $4^{2019} = (2^2)^{2019} = (2^{2019})^2$  by the  $(a^m)^n = a^{mn}$  rule. The only option left is **(B)  $2^{2017}$** .

8. (AMC 8 2016) Find the value of the expression:

$$100 - 98 + 96 - 94 + 92 - 90 + \cdots + 8 - 6 + 4 - 2.$$

(A) 20    (B) 40    (C) 50    (D) 80    (E) 100

**Answer:** C

**Solution 1**

We can group each subtracting pair together:

$$(100 - 98) + (96 - 94) + (92 - 90) + \cdots + (8 - 6) + (4 - 2).$$

After subtracting, we have:

$$2 + 2 + 2 + \cdots + 2 + 2 = 2(1 + 1 + 1 + \cdots + 1 + 1).$$

There are 50 even numbers, therefore there are  $\frac{50}{2} = 25$  even pairs. Therefore the sum is  $2 \cdot 25 =$  **(C) 50**

**Solution 2**

Since our list does not end with one, we divide every number by 2 and we end up with

$$50 - 49 + 48 - 47 + \cdots + 4 - 3 + 2 - 1$$

We can group each subtracting pair together:

$$(50 - 49) + (48 - 47) + (46 - 45) + \dots + (4 - 3) + (2 - 1).$$

There are now 25 pairs of numbers, and the value of each pair is 1. This sum is 25. However, we divided by 2 originally so we will multiply  $2 \cdot 25$  to get the final answer of **(C) 50**

### Solution 3

Note: This solution is very similar to Solution 1, but simplified.

The value of the expression increases by 2 after every pair as shown:

$$\text{Pair 1: } [100-98= 2]$$

$$\text{Pair 1 + 2: } [2+96-94= 4]$$

$$\text{Pair 1 + 2 + 3: } [4+92-90= 6]$$

Since there is 25 pairs in the expression, the value will increase by two 25 times. Thus, the sum of the expression is  $2 \cdot 25 = \mathbf{(C) 50}$

9. (AMC 8 2016) What is the sum of the distinct prime integer divisors of 2016?

(A) 9      (B) 12      (C) 16      (D) 49      (E) 63

**Answer: B**

### Solution 1

The prime factorization is  $2016 = 2^5 \times 3^2 \times 7$ . Since the problem is only asking us for the distinct prime factors, we have 2, 3, 7. Their desired sum is then **(B) 12**.

10. (AMC 8 2016) Suppose that  $a * b$  means  $3a - b$ . What is the value of  $x$  if

$$2 * (5 * x) = 1$$

(A)  $\frac{1}{10}$       (B) 2      (C)  $\frac{10}{3}$       (D) 10      (E) 14.

**Answer: D**

### Solution 1

Let us plug in  $(5 * x) = 1$  into  $3a - b$ . Thus it would be  $3(5) - x$ . Now we have  $2 * (15 - x) = 1$ . Plugging  $2 * (15 - x)$  into  $3a - b$ , we have  $6 - 15 + x = 1$ . Solving for  $x$  we have

$$-9 + x = 1$$

$$x = \mathbf{(D) 10}$$

### Solution 2

Let us set a variable  $y$  equal to  $5 * x$ . Solving for  $y$  in the equation  $3(2) - y = 1$ , we see that  $y$  is equal to five. By substitution, we see that  $5 * x = 5$ . Solving for  $x$  in the equation  $5(3) - x = 5$  we get

$$x = \mathbf{(D) 10}$$



11. (AMC 8 2016) Determine how many two-digit numbers satisfy the following property: when the number is added to the number obtained by reversing its digits, the sum is 132. (A) 5 (B) 7 (C) 9 (D) 11 (E) 12

**Answer:** B

**Solution 1**

We can see that the original number can be written as  $10a + b$ , where  $a$  represents the tens digit and  $b$  represents the units digit. When this number is added to the number obtained by reversing its digits, which is  $10b + a$ , the sum would be  $11a + 11b$ . From this, we can construct the equation  $11a + 11b = 132$ , which simplifies to  $a + b = 12$ . Since there are 7 pairs of such digits  $a$  and  $b$ ,  $(3, 9), (4, 8), (5, 7), (6, 6), (7, 5), (8, 4), (9, 3)$ , the answer would be **(B) 7**.

**Solution 2**

We can set the number as  $ab$  where  $a$  is the tens digit and  $b$  is the ones digit. So now the equation will be  $ab + ba = 132$ .  $a + b$  has to have a remainder of 2 when divided by 10 so it will be  $a + b \equiv 2 \pmod{10}$ . We also know  $a < 10$  and  $b < 10$ . So  $a + b$  can either be 2 or 12.  $a + b$  cannot be 2 because then there will be only 3 numbers that work and that isn't in the answer choice. So  $a + b = 12$ . To check this we can do  $ab + ba = 132$  which equals to  $(a + b)0 + (a + b) = 132$  and since we said  $a + b = 12$  we get  $120 + 12 = 132$  which is true. So we have  $a + b = 12$  and  $a < 10$  and  $b < 10$ . If  $a$  is 9 then  $b = 3$  and if  $a = 3$  then  $b = 9$  calculating how many pairs are in between you get 7. So the answer is **(B) 7**.

**Solution 3**

Like similar solutions, we can say that  $10a + b$  is a two digit number, and therefore we have  $10a + b + 10b + a = 132$  which implies  $11a + 11b = 132$  which implies  $a + b = 12$ .

We then use stars and bars to find out that there are  $\binom{13}{1} = 13$  different possible sums.

However, we want to satisfy  $0 \leq a \leq 9$  and  $0 \leq b \leq 9$ , therefore we see that the sums  $0 + 12, 1 + 11, 2 + 10$ , and  $12 + 0, 11 + 1$ , and  $10 + 2$  do not work, and our answer is  $13 - 6 = \mathbf{(B) 7}$ .

**Solution 4**

The two digit number  $10A + B$  + the reversed number  $10B + A$  is equal to  $11A + 11B = 132$ . Dividing both sides by 11 gives  $A + B = 12$ . Since we can't have  $1 + 11, 2 + 10$  etc. meaning there are **(B) 7** solutions.

12. (AMC 8 2016) Jefferson Middle School has the same number of boys and girls.  $\frac{3}{4}$  of the girls and  $\frac{2}{3}$  of the boys went on a field trip. What fraction of the students on the field trip were girls?

(A)  $\frac{1}{2}$  (B)  $\frac{9}{17}$  (C)  $\frac{7}{13}$  (D)  $\frac{2}{3}$  (E)  $\frac{14}{15}$

**Answer:** B

**Solution 1**

Let there be  $b$  boys and  $g$  girls in the school. We see  $g = b$ , which means  $\frac{3}{4}b + \frac{2}{3}b = \frac{17}{12}b$  kids went on the trip and  $\frac{3}{4}b$  kids are girls. So, the answer is  $\frac{\frac{3}{4}b}{\frac{17}{12}b} = \frac{9}{17}$ , which is

**(B)  $\frac{9}{17}$** .

**Solution 2**

Using WLOG (Without loss of generativity), Let there be 12 boys and 12 girls in the school. Now we can do  $\frac{3}{4} \times 12 + \frac{2}{3} \times 12$  to get the total number of students going to the field trip to be 17. Since we already know the number of girls to be 9.

We have our answer to be  $\frac{9}{17}$ . So, the answer is **(B)**  $\frac{9}{17}$ .

13. (AMC 8 2016) Two different numbers are randomly selected from the set  $\{-2, -1, 0, 3, 4, 5\}$  and multiplied together. What is the probability that the product is 0?

(A)  $\frac{1}{6}$       (B)  $\frac{1}{5}$       (C)  $\frac{1}{4}$       (D)  $\frac{1}{3}$       (E)  $\frac{1}{2}$

**Answer:** D

**Solution 1**

1. Identify the total number of ways to select two different numbers from the set:

The set has 6 elements. The number of ways to choose 2 different numbers from 6 is given by the combination formula:  $\binom{6}{2} = \frac{6 \times 5}{2 \times 1} = 15$ .

2. Identify the favorable outcomes:

For the product to be zero, one of the chosen numbers must be zero. The set contains one zero (0). To have a product of zero, we need to choose 0 and any other number from the remaining five numbers  $-2, -1, 3, 4, 5$ .

The number of ways to choose 0 and one other number from the remaining five is 5.

3. Calculate the probability:

The probability is the number of favorable outcomes divided by the total number of outcomes: Probability =  $\frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{5}{15} = \frac{1}{3}$ .

Thus, the probability that the product is 0 is **(D)**  $\frac{1}{3}$ .

**Solution 2 (Complementary Counting)**

Because the only way the product of the two numbers is 0 is if one of the numbers we choose is 0, we calculate the probability of NOT choosing a 0. We get  $\frac{5}{6} \cdot \frac{4}{5} = \frac{2}{3}$ .

Therefore our answer is  $1 - \frac{2}{3} = \frac{1}{3}$  **(D)**  $\frac{1}{3}$ .

**Solution 3(One Sentence!!!)**

$$1/6 * \text{Any other number} + \text{Any other number} * 1/6 = 1/6 + 1/6 = \text{(D)} \frac{1}{3}$$

-JasonDaGoat

$$1/6 \text{ Chance of } 0 * \text{A number in the set} + 1/6 \text{ chance of A number in the set} * 0 = 1/6 + 1/6 = \text{(D)} \frac{1}{3}$$

- edited by TTDBJ

**Solution 4**

To get a product of 0, one of the numbers must be 0, which means that we have  $2P1 = 2$  possible combinations of a random number from the set, and 0. There is 6 total possible products, giving us  $\frac{1}{3}$  **(D)**  $\frac{1}{3}$ .-Electric\_Cheetah\_\_88

14. (AMC 8 2016) Karl's car uses a gallon of gas every 35 miles, and his gas tank holds 14 gallons when it is full. One day, Karl started with a full tank of gas, drove 350 miles, bought 8 gallons of gas, and continued driving to his destination. When he arrived, his gas tank was half full. How many miles did Karl drive that day?

(A) 525    (B) 560    (C) 595    (D) 665    (E) 735

**Answer:** A

**Solution**

Since he uses a gallon of gas every 35 miles, he had used  $\frac{350}{35} = 10$  gallons after 350 miles. Therefore, after the first leg of his trip he had  $14 - 10 = 4$  gallons of gas left. Then, he bought 8 gallons of gas, which brought him up to 12 gallons of gas in his gas tank. When he arrived, he had  $\frac{1}{2} \cdot 14 = 7$  gallons of gas. So he used 5 gallons of gas on the second leg of his trip. Therefore, the second part of his trip covered  $5 \cdot 35 = 175$  miles. Adding this to the 350 miles, we see that he drove  $350 + 175 = \boxed{\text{(A) } 525}$  miles.

15. (AMC 8 2016) What is the largest power of 2 that is a divisor of  $13^4 - 11^4$ ?

(A) 8    (B) 16    (C) 32    (D) 64    (E) 128

**Answer:** C

**Solution 1**

First, we use difference of squares on  $13^4 - 11^4 = (13^2)^2 - (11^2)^2$  to get  $13^4 - 11^4 = (13^2 + 11^2)(13^2 - 11^2)$ . Using difference of squares again and simplifying, we get  $(169 + 121)(13 + 11)(13 - 11) = 290 \cdot 24 \cdot 2 = (2 \cdot 8 \cdot 2) \cdot (3 \cdot 145)$ . Realizing that we don't need the right-hand side because it doesn't contain any factor of 2, we see that the greatest power of 2 that is a divisor  $13^4 - 11^4$  is  $\boxed{\text{(C) } 32}$ .

**Solution 2 (variant of Solution 1)**

Just like in the above solution, we use the difference-of-squares factorization, but only once to get  $13^4 - 11^4 = (13^2 - 11^2)(13^2 + 11^2)$ . We can then compute that this is equal to  $48 \cdot 290$ . Note that  $290 = 2 \cdot 145$  (we don't need to factorize any further as 145 is already odd) thus the largest power of 2 that divides 290 is only  $2^1 = 2$ , while  $48 = 2^4 \cdot 3$ , so the largest power of 2 that divides 48 is  $2^4 = 16$ . Hence, the largest power of 2 that is a divisor of  $13^4 - 11^4$  is  $2 \cdot 16 = \boxed{\text{(C) } 32}$ .

**Solution 3 (Lifting the exponent)**

Let  $n = 13^4 - 11^4$ . We wish to find the largest power of 2 that divides  $n$ .

Denote  $v_p(k)$  as the largest exponent of  $p$  in the prime factorization of  $n$ . In this problem, we have  $p = 2$ .

By the Lifting the Exponent Lemma on  $n$ ,

$$\begin{aligned} v_2(13^4 - 11^4) &= v_2(13 - 11) + v_2(4) + v_2(13 + 11) - 1 \\ &= v_2(2) + v_2(4) + v_2(24) - 1 \\ &= 1 + 2 + 3 - 1 = 5. \end{aligned}$$

Therefore, exponent of the largest power of 2 that divides  $13^4 - 11^4$  is 5, so the largest power of 2 that divides this number is  $2^5 = \boxed{\text{(C) } 32}$ .

**Solution 4 (Brute Force)**

We can simply take 13 to the 4th power, which is 28561. We subtract that by 11 to the 4th power, which is 14641 (You can use Pascal's Triangle to find this). Finally, subtract the numbers to get 13920.

To test the options, since we need the largest one, we can go from top down. Testing, we see that both D and E are decimals, and 32 works. So, our answer is **(C) 32**.

-themathgood

Note: This is not the fastest way, and is not recommended.

### Solution 5

$13^4 - 11^4$  is  $(13^2 - 11^2)(13^2 + 11^2)$  which is  $(169 - 121)(169 + 121)$  which is  $(48)(290)$  which has 5 factors of 2 making **(C) 32**.

16. (*AMC 8 2016*) Annie and Bonnie are running laps around a 400-meter oval track. They started at the same time, at the same place, but Annie has pulled ahead because she runs 25% faster than Bonnie. How many laps will Annie have run when she first passes Bonnie?

(A)  $1\frac{1}{4}$       (B)  $3\frac{1}{3}$       (C) 4      (D) 5      (E) 25

**Answer:** D

### Solution 1

Each lap Bonnie runs, Annie runs another quarter lap, so Bonnie will run four laps before she is overtaken because it takes 4 of  $\frac{1}{4}$  of the total lap. to total one lap. When Bonnie runs 4 laps, Annie has gained one full lap and would overtake Bonnie. That means Annie will have run **(D) 5** laps.

### Solution 2

Saying that Bonnie runs 40 meters per second, then Annie will run 50 meters per second. It will take Bonnie 10 seconds to run 1 lap and it would take 8 seconds for Annie to do 1 lap. The LCM of 8 and 10 is 40. In 40 seconds Bonnie will do 4 laps and Annie will do 5 laps. Since Annie did 1 more lap she passed Bonnie. So the answer is **(D) 5** laps.

17. (*AMC 8 2016*) An ATM password at Fred's Bank is composed of four digits from 0 to 9, with repeated digits allowable. If no password may begin with the sequence 9, 1, 1, then how many passwords are possible?

(A) 30      (B) 7290      (C) 9000      (D) 9990      (E) 9999

**Answer:** D

### Solution 1

For the first three digits, there are  $10^3 - 1 = 999$  combinations since 911 is not allowed. For the final digit, any of the 10 numbers are allowed.  $999 \cdot 10 = 9990 \rightarrow$  **(D) 9990**.

### Solution 2

Counting the prohibited cases, we find that there are 10 of them. This is because, when we start with 9, 1, and 1, we can have any of the 10 digits for the last digit. So, our answer is  $10^4 - 10 =$  **(D) 9990**.

18. (*AMC 8 2016*) In an All-Area track meet, 216 sprinters enter a 100-meter dash competition. The track has 6 lanes, so only 6 sprinters can compete at a time.

At the end of each race, the five non-winners are eliminated, and the winner will compete again in a later race. How many races are needed to determine the champion sprinter?

- (A) 36      (B) 42      (C) 43      (D) 60      (E) 72

**Answer:** C

**Solution 1**

From any  $n$ -th race, only  $\frac{1}{6}$  will continue on. Since we wish to find the total number of races, a column representing the races over time is ideal. Starting with the first race:

$$\begin{aligned}\frac{216}{6} &= 36 \\ \frac{36}{6} &= 6 \\ \frac{6}{6} &= 1\end{aligned}$$

Adding all of the numbers in the second column yields **(C) 43**

**Solution 2**

Every race eliminates 5 players. The winner is decided when there is only 1 runner left. You can construct the equation:  $216 - 5x = 1$ . Thus, 215 players have to be eliminated. Therefore, we need  $\frac{215}{5}$  games to decide the winner, or **(C) 43**

19. (AMC 8 2016) The sum of 25 consecutive even integers is 10,000. What is the largest of these 25 consecutive integers?

- (A) 360      (B) 388      (C) 412      (D) 416      (E) 424

**Answer:** E

**Solution 1**

Let  $n$  be the 13th consecutive even integer that's being added up. By doing this, we can see that the sum of all 25 even numbers will simplify to  $25n$  since  $(n - 2k) + \cdots + (n - 4) + (n - 2) + (n) + (n + 2) + (n + 4) + \cdots + (n + 2k) = 25n$ . Now,  $25n = 10000 \rightarrow n = 400$ . Remembering that this is the 13th integer, we wish to find the 25th, which is  $400 + 2(25 - 13) = \mathbf{(E) 424}$ .

**Solution 2**

Let  $x$  be the smallest number. The equation will become,  $x + (x + 2) + (x + 4) + \cdots + (x + 48) = 10,000$ . After you combine like terms, you get  $25x + (50 * 12) = 10,000$  which turns into  $10,000 - 600 = 25x$ .  $25x = 9400$ , so  $x = 376$ . Then, you add  $376 + 48 = \mathbf{(E) 424}$ .

Edit(dnathan12): For clarity, you add 48, because you want the largest number in the list. 376 is the smallest number AfterglowBlaziken

20. (AMC 8 2016) The least common multiple of  $a$  and  $b$  is 12, and the least common multiple of  $b$  and  $c$  is 15. What is the least possible value of the least common multiple of  $a$  and  $c$ ?

- (A) 20      (B) 30      (C) 60      (D) 120      (E) 180

**Answer:** A

**Solution**

We wish to find possible values of  $a$ ,  $b$ , and  $c$ . By finding the greatest common factor of 12 and 15, we can find that  $b$  is 3. Moving on to  $a$  and  $c$ , in order to minimize them, we wish to find the least such that the least common multiple of  $a$  and 3 is 12,  $\rightarrow 4$ . Similarly, with 3 and  $c$ , we obtain 5. The least common multiple of 4 and 5 is 20  $\rightarrow$  **(A)20**

### Solution 2

The factors of 12 are 12,6,4,3,2,1. The factors of 15 are 1,3,5,15. The 2 numbers that repeat are 1 and 3 so  $b$  either has to be 1 or 3. If  $b$  is 3 then  $a$  is 4 and  $c$  is 5 and the least common multiple of 4 and 5 are 20. We don't have to test 1 because 20 is the lowest answer, so if  $b$  equaling 1 resulted in the least common multiple being less than 20 then the correct answer won't be there. So the answer is **(A)20**.

### Solution 3 (Trial Error)

Let's say that  $a = 4, b = 3, c = 5$  (because  $LCM(3, 4) = 12$  and  $LCM(3, 5) = 15$ ), this works so clearly  $GCF(4, 5) =$  **(A)20**. JasonDaMoat, Minor edited by Quetzalcoatlus

21. (AMC 8 2016) A top hat contains 3 red chips and 2 green chips. Chips are drawn randomly, one at a time without replacement, until all 3 of the reds are drawn or until both green chips are drawn. What is the probability that the 3 reds are drawn?

(A)  $\frac{3}{10}$       (B)  $\frac{2}{5}$       (C)  $\frac{1}{2}$       (D)  $\frac{3}{5}$       (E)  $\frac{7}{10}$

**Answer:** B

### Solution 1

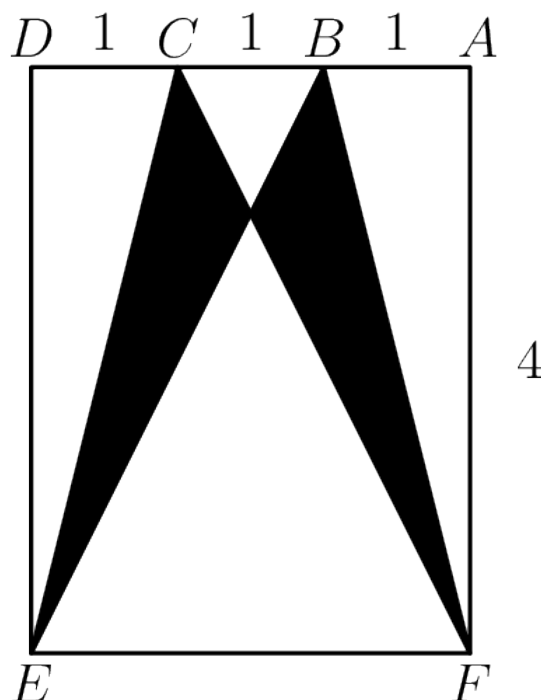
We put five chips randomly in order and then pick the chips from the left to the right. To find the number of ways to rearrange the three red chips and two green chips, we solve for  $\binom{5}{2} = 10$ . However, we notice that whenever the last chip we draw is red, we pick both green chips before we pick the last red chip. Similarly, when the last chip is green, we pick all three red chips before the last green chip. This means that the last chip must be green in all the situations that work. This means we are left with finding the number of ways to rearrange three red chips and one green chip, which is  $\binom{4}{3} = 4$ . Because a green chip will be last 4 out of the 10 situations, our answer is **(B)  $\frac{2}{5}$** .

### Solution 2

We can use basic probability to find the answer to this question. The probability of the first chip being red is  $\frac{3}{5}$  because there are 3 red chips and 5 chips in total. The probability of the second chip being red is  $\frac{2}{4}$  because there are now 2 red chips and 4 total chips. The probability of the third chip being red is  $\frac{1}{3}$  because there are now 1 red chips and 3 total chips. If we multiply these fractions, we get  $\frac{1}{10}$ . To get the answer of  $\frac{2}{5}$  we need to multiply this product by 4 because there are 4 different draws that these red chips can be chosen on. They can be chosen on the 1st 2nd, 3rd, or 4th draws because if a red chip was chosen on the 5th draw, 2 green chips would have already been chosen. So our answer is  $\frac{1}{10}$  multiplied by 4 which would leave us with the answer of **(B)  $\frac{2}{5}$** .

**Faster Solution (Similar to Solution 1)** Randomly arrange them in a row of 5. There is a  $\frac{3}{5}$  chance that the last one will be red. If the last one is red, 3 reds will not be drawn first, so there is a **(B)**  $\frac{2}{5}$  chance that 3 reds will come out first.

22. (AMC 8 2016) Rectangle  $DEFA$  below is a  $3 \times 4$  rectangle with  $DC = CB = BA = 1$ . What is the area of the "bat wings" (shaded region)?



- (A) 2    (B)  $2\frac{1}{2}$     (C) 3    (D)  $3\frac{1}{2}$     (E) 4

**Answer:** C

**Solution 1**

The area of trapezoid  $CBFE$  is  $\frac{1+3}{2} \cdot 4 = 8$ . Next, we find the height of each triangle to calculate their area. The two non-colored isosceles triangles are similar, and are in a  $3 : 1$  ratio by AA similarity (alternate interior and vertical angles) so the height of the larger is 3, while the height of the smaller one is 1. Thus, their areas are  $\frac{1}{2}$  and  $\frac{9}{2}$ . Subtracting these areas from the trapezoid, we get  $8 - \frac{1}{2} - \frac{9}{2} = \boxed{3}$ . Therefore, the answer to this problem is **(C) 3**.

**Solution 2**

Plot the point  $G$  where the two "wings" intersect. Now notice how  $\triangle CBG \sim \triangle EFG$ . Since the length of  $\overline{CB}$  is one third that of  $\overline{EF}$ , then that means  $\triangle EFG$ 's height is 3 times bigger than  $\triangle CBG$ . Since both of their heights ( $h$ ) add up to four, then we have the equation  $3h + h = 4 \implies h = 1$ . Now that we now the height and length of both triangles, we can use complementary counting, Area – Unshaded Region.

Total Area = 12  $[\triangle CDE] = 2 [\triangle ABF] = 2 [\triangle CBG] = \frac{1}{2} [\triangle EFG] = \frac{9}{2}$   
 Unshaded Region = 9  $\implies$  "Bat Wings" = **(C) 3**

23. (AMC 8 2016) Two congruent circles centered at points  $A$  and  $B$  each pass through the other circle's center. The line containing both  $A$  and  $B$  is extended to intersect

the circles at points  $C$  and  $D$ . The circles intersect at two points, one of which is  $E$ . What is the degree measure of  $\angle CED$ ?

- (A) 90      (B) 105      (C) 120      (D) 135      (E) 150

**Answer:** C

**Solution 1**

Observe that  $\triangle EAB$  is equilateral (all are radii of congruent circles). Therefore,  $m\angle AEB = m\angle EAB = m\angle EBA = 60^\circ$ . Since  $CD$  is a straight line, we conclude that  $m\angle EBD = 180^\circ - 60^\circ = 120^\circ$ . Since  $BE = BD$  (both are radii of the same circle),  $\triangle BED$  is isosceles, meaning that  $m\angle BED = m\angle BDE = 30^\circ$ . Similarly,  $m\angle AEC = m\angle ACE = 30^\circ$ .

Now,  $\angle CED = m\angle AEC + m\angle AEB + m\angle BED = 30^\circ + 60^\circ + 30^\circ = 120^\circ$ . Therefore, the answer is **(C) 120**.

**Graph for Solution 1** [https://drive.google.com/file/d/1sNkKxLi7\\_SpxihTp0QrOcD3Js14buvd8/view](https://drive.google.com/file/d/1sNkKxLi7_SpxihTp0QrOcD3Js14buvd8/view)  
Quetzalcoatlus

24. (AMC 8 2016) The digits 1, 2, 3, 4, and 5 are each used once to write a five-digit number  $PQRST$ . The three-digit number  $PQR$  is divisible by 4, the three-digit number  $QRS$  is divisible by 5, and the three-digit number  $RST$  is divisible by 3. What is  $P$ ?

- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** A

**Solution 1 (Modular Arithmetic)**

We see that since  $QRS$  is divisible by 5,  $S$  must equal either 0 or 5, but it cannot equal 0, so  $S = 5$ . We notice that since  $PQR$  must be even,  $R$  must be either 2 or 4. However, when  $R = 2$ , we see that  $T \equiv 2 \pmod{3}$ , which cannot happen because 2 and 5 are already used up; so  $R = 4$ . This gives  $T \equiv 3 \pmod{4}$ , meaning  $T = 3$ . Now, we see that  $Q$  could be either 1 or 2, but 14 is not divisible by 4, but 24 is. This means that  $Q = 2$  and  $P = \mathbf{(A) 1}$ .

Editor's note: If you are in a time constraint and can't get through the whole problem, you can, with this modular arithmetic, eliminate the ones' you know can't be in the first position.

**Solution 2**

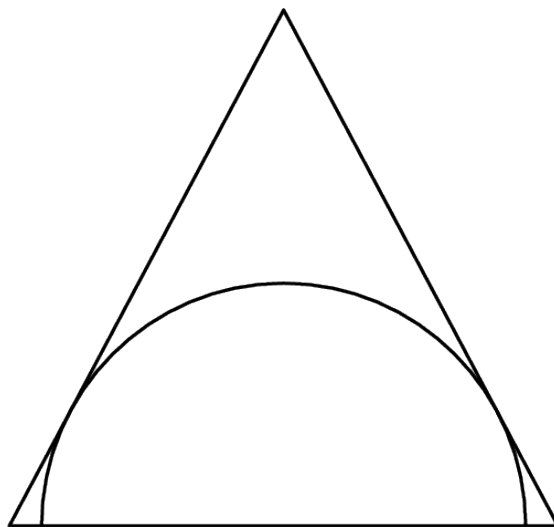
We know that out of  $PQRST$ ,  $QRS$  is divisible by 5. Therefore  $S$  is obviously 5 because  $QRS$  is divisible by 5. So we now have  $PQR5T$  as our number. Next, let's move on to the second piece of information that was given to us.  $RST$  is divisible by 3. So, according to the divisibility by 3 rule, the sum of  $RST$  has to be a multiple of 3. The only 2 big enough are 9 and 12 and since 5 is already given. The possible sums of  $RT$  are 4 and 7. So, the possible values for  $R$  are 1,3,4,3 and the possible values of  $T$  are 3,1,3,4. So, using this we can move on to the fact that  $PQR$  is divisible by 4. So, using that we know that  $R$  has to be even so 4 is the only possible value for  $R$ . Using that we also know that 3 is the only possible value for 3. So, we have  $PQRST = PQ453$  so the possible values are 1 and 2 for  $P$  and  $Q$ . Using the divisibility rule of 4 we know that  $QR$  has to be divisible by 4. So, either 14 or 24 are the possibilities, and 24 is divisible by 4. So the only value left for  $P$  is 1.  $P = \mathbf{(A) 1}$ .

**Solution 3**



We can simply try each of the answer choices, and we will see which one works. Trying  $P = \boxed{\text{(A)} 1}$ , if  $PQR$  is divisible by 4,  $QR$  must be divisible by four. Therefore,  $QR$  can only be 24, 52, or 32. However, since  $QRS$  is divisible by 5,  $S = 5$ , so  $QR$  cannot be 52. When  $QR = 32$ ,  $R = 2$ , the last requirement cannot be satisfied because  $R + S + T = 2 + 4 + 5 = 11$ , and 11 is not divisible by 3. However, when  $QR = 24$ ,  $R = 4$ , the last requirement can be satisfied. Hence, we can see that when  $P = \boxed{\text{(A)} 1}$ , there is one way to satisfy all three requirements, leading to a conclusion that  $P$  is  $\boxed{\text{(A)} 1}$ .

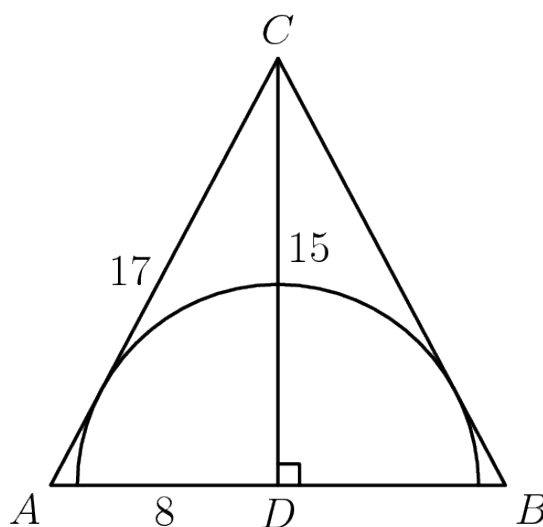
25. (AMC 8 2016) A semicircle is inscribed in an isosceles triangle with base 16 and height 15 so that the diameter of the semicircle is contained in the base of the triangle as shown. What is the radius of the semicircle?



- (A)  $4\sqrt{3}$     (B)  $\frac{120}{17}$     (C) 10    (D)  $\frac{17\sqrt{2}}{2}$     (E)  $\frac{17\sqrt{3}}{2}$

**Answer:** B

**Solution 1**

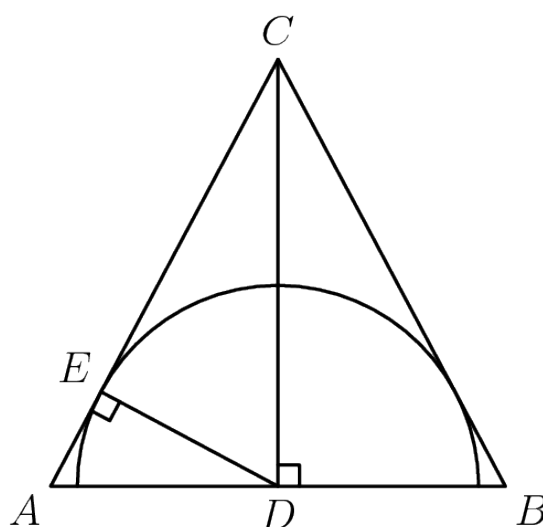


First, we drop a perpendicular, shown above, to the base of the triangle, cutting

the triangle into two congruent right triangles. This triangle is isosceles, which means perpendiculars are medians and vice versa. The base of the resulting right triangle is 8 for both sides, and the height is 15, as given. Using the Pythagorean theorem, we can find the length of the hypotenuse, or 17. Using the two legs of the right triangle, we find the area of the right triangle,  $60 \cdot \frac{60}{17}$  times 2 results in the radius, which is the height of the right triangle when using the hypotenuse as the base. Hence, the answer is  $\boxed{\text{(B)} \frac{120}{17}}$ .

- Note — There are several other following solutions below that use similar methods of finding the area two different ways shown in this solution. Try to eliminate any words (such as 'can' or 'would be') that make your solution less concise. Try to write 'sure of yourself.'

### Solution 2: Similar Triangles



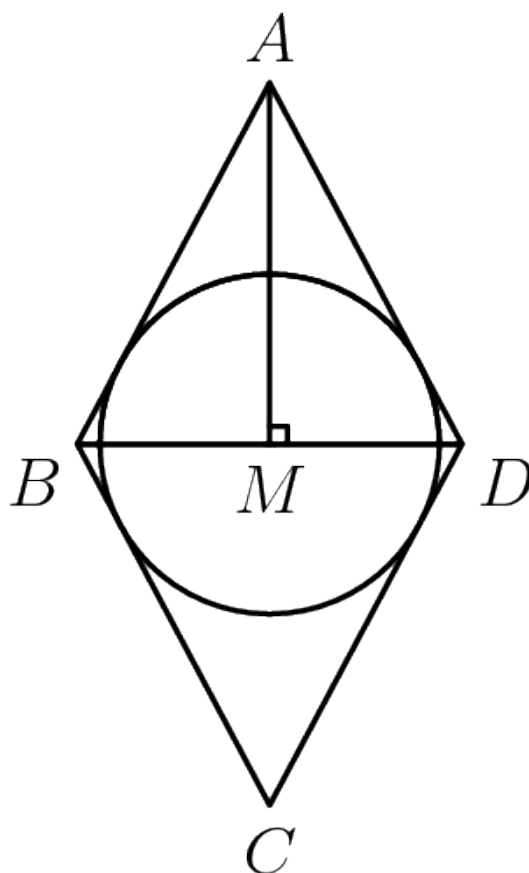
Let's call the triangle  $\triangle ABC$ , where  $AB = 16$  and  $AC = BC$ . Let's say that  $D$  is the midpoint of  $AB$  and  $E$  is the point where  $AC$  is tangent to the semicircle. We could also use  $BC$  instead of  $AC$  because of symmetry.

Notice that  $\triangle ACD \cong \triangle BCD$ , and are both 8-15-17 right triangles. We also know that we create a right angle with the intersection of the radius and a tangent line of a circle (or part of a circle). So, by AA similarity,  $\triangle AED \sim \triangle ADC$ , with  $\angle EAD \cong \angle DAC$  and  $\angle CDA \cong \angle DEA$ . This similarity means that we can create a proportion:  $\frac{AD}{AC} = \frac{DE}{CD}$ . We plug in  $AD = \frac{AB}{2} = 8$ ,  $AC = 17$ , and  $CD = 15$ . After

we multiply both sides by 15, we get  $DE = \frac{8}{17} \cdot 15 = \boxed{\text{(B)} \frac{120}{17}}$ .

(By the way, we could also use  $\triangle DEC \sim \triangle ADC$ .)

### Solution 3: Inscribed Circle



We'll call this triangle  $\triangle ABD$ . Let the midpoint of base  $BD$  be  $M$ . Divide the triangle in half by drawing a line from  $A$  to  $M$ . Half the base of  $\triangle ABD$  is  $\frac{16}{2} = 8$ . The height is 15, which is given in the question. Using the Pythagorean Triple 8-15-17, the length of each of the legs ( $AB$  and  $DA$ ) is 17.

Reflect the triangle over its base. This will create an inscribed circle in a rhombus  $ABCD$ . Because  $AB \cong DA$ ,  $BC \cong CD$ . Therefore  $AB = BC = CD = DA$ .

The semiperimeter  $s$  of the rhombus is  $\frac{AB+BC+CD+DA}{2} = \frac{(17)(4)}{2} = 34$ . Since the area of  $\triangle ABD$  is  $\frac{bh}{2}$ , the area  $[ABCD]$  of the rhombus is twice that, which is  $bh = (16)(15) = 240$ .

The **Formula for the Incircle of a Quadrilateral** is  $sr = [ABCD]$ . Substituting the semiperimeter and area into the equation,  $34r = 240$ . Solving this,  $r = \frac{240}{34} =$

$$\boxed{\text{(B)} \frac{120}{17}}.$$

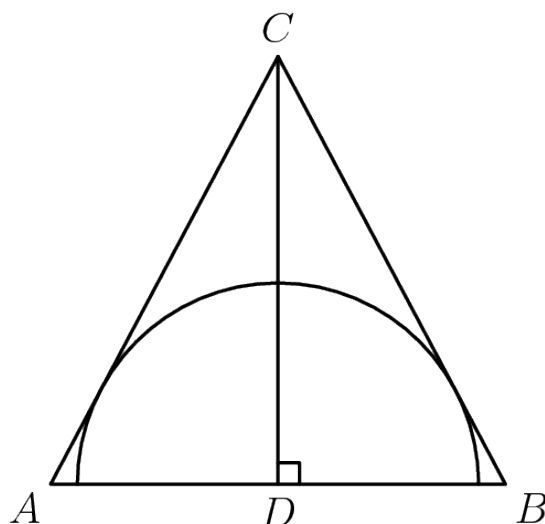
#### Solution 4: Inscribed Circle

Noting that we have a 8-15-17 triangle, we can find  $AE$  and  $CE$ . Let  $AE = x$ ,  $CE = 17 - x$ . Then by similar triangles (or "Altitude on Hypotenuse") we have  $15^2 = x * 17$ . Thus,  $AE = x = 225/17$ ,  $CE = 64/17$ . Now again by "Altitude on

Hypotenuse",  $r = \sqrt{AE * CE}$ . Therefore  $r = \boxed{\text{(B)} \frac{120}{17}}.$

#### Solution 5: Simple Trigonometry

Note: This solution uses **Trigonometric Concepts**



Denote the bottom left vertex of the isosceles triangle to be  $A$

Denote the bottom right vertex of the isosceles triangle to be  $B$

Denote the top vertex of the isosceles triangle to be  $C$

Drop an altitude from  $C$  to side  $AB$ . Denote the foot of the intersection to be  $D$ .

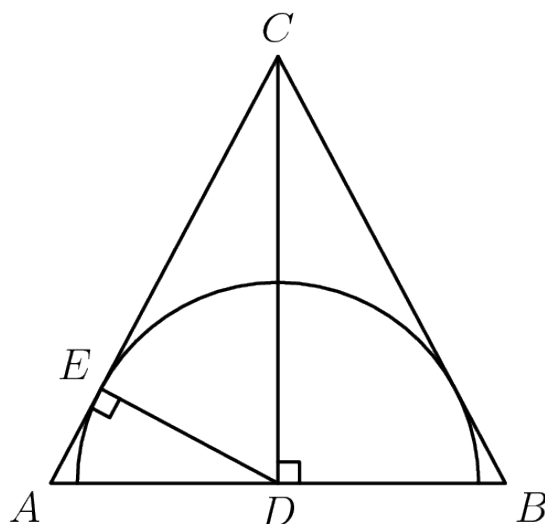
By the Pythagorean Theorem,  $AC = 17$ .

Now, we see that by  $\sin(x)$ ,  $\sin A = \frac{15}{17}$ .

This implies that  $\sin A = \frac{r}{8}$  ( $r$ =radius of semicircle).

Hence,  $r = \boxed{\text{(B)} \frac{120}{17}}$ .

**Solution 6: Area**



Credits for Asymptote go to whoever wrote this diagram up in Solution 2.

There are two ways to find the area of  $\triangle ABC$ . The first way is the most obvious, and that is to multiply the base times the height ( $16 \cdot 15$ ) and then divide it by two. The second way is, in a way, a little more complex. Note that  $\triangle ACD$  and  $\triangle BCD$  are congruent. This means that if we find the area of one triangle, we can

just multiply it's area by 2 and we've found the area of the larger triangle  $ABC$ . But since we always divide by two, as it is in the formula for finding the area of a triangle, the multiply by two and divide by two cancel out, giving us that if we just multiply base times height of  $\triangle ACD$ , we will get the area of  $\triangle ABC$ .

Now you might think: "But what is the other way of finding the area". Well that is  $AC$ , which would be the base, times  $DE$ , the radius, which would be the height.

The first way to find the area gives us the area of the  $\triangle ABC$ ,  $\frac{15 \cdot 16}{2} = 120$ . This gives us  $120 = AC \cdot r$  ( $r$  signifies radius). We can find  $AC$  using the Pythagorean Theorem on  $\triangle ACD$ . The two legs are 8 and 15, which gives us that the hypotenuse,  $AC$ , is equal to 17.

Now that we have an equation with only one variable for the radius,  $120 = 17r$ , we can just solve for the radius. We get  $r = \boxed{\text{(B)} \frac{120}{17}}$ .

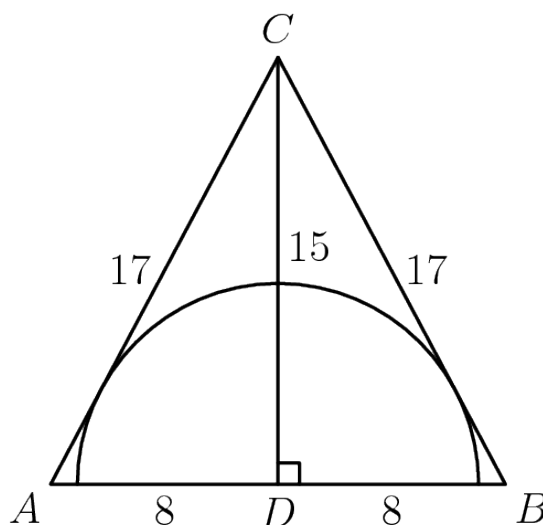
This may seem like a lengthy explanation, but doing it yourself when you know what to do, it actually takes very little time. Try it yourself!

### Solution 7

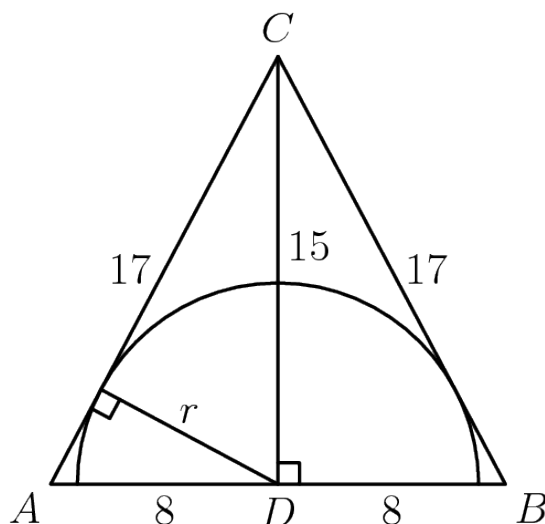
Let us draw altitude  $\overline{CD}$ . This cuts our base into line segments with length  $\frac{16}{2} = 8$ . Finding the area of the resulting triangles gives  $[\triangle ADC] = [\triangle BDC] = \frac{8 \cdot 15}{2} = 60$ . Since  $m\angle CDB = m\angle CDA = 90^\circ$ , we use the Pythagorean Theorem to find length  $\overline{AC}$ :

$$\begin{aligned} (\overline{AD})^2 + (\overline{CD})^2 &= (\overline{AC})^2 \\ 8^2 + 15^2 &= (\overline{AC})^2 \\ 64 + 225 &= (\overline{AC})^2 \\ 289 &= (\overline{AC})^2 \\ \sqrt{289} &= \sqrt{(\overline{AC})^2} \\ 17 &= \overline{AC}. \end{aligned}$$

Thus we have  $\overline{AC} = \overline{BC} = 17$ .



Letting 17 be the base of the triangle makes our height the radius of the semicircle:



Let  $r$  be the radius of the semicircle. Then we have  $\frac{17 \cdot r}{2} = 60 \implies 17 \cdot r = 120 \implies$

$\frac{17r}{17} = \frac{120}{17} \implies r = \boxed{\text{(B)} \frac{120}{17}}$ . **Remember: Area can help in places where you least expect**

solution by pog

### Solution 8: equations

Draw tangent lines to connect the centre of the circle to the equal sides. Since the angle is 90 degrees and the height is perpendicular to the side not equal with any of the other side lengths, the two triangles (Not the big one but the two smaller ones) are similar. We've got one common side that gives us the ratio is  $\frac{8}{15}$ . We've got a system of equations!

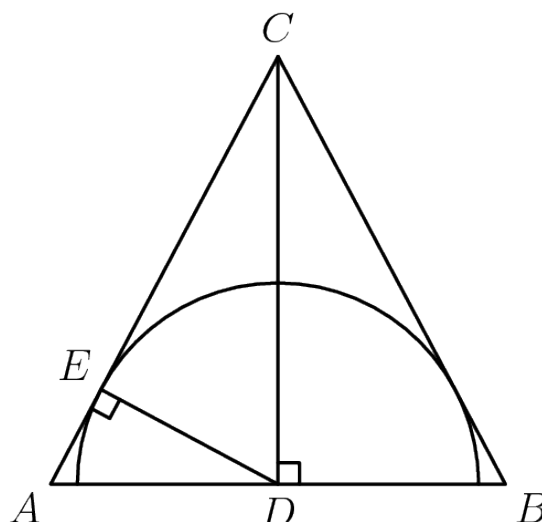
1.  $8/15 = x(\text{the radius})/y(\text{the side that is similar})$  2.  $x^2 + y^2 = 225$  (by the pythagorean theorem).

Solving the equations we get  $\boxed{\frac{120}{17}}$

created by justin6688

### Solution 9: Process of Elimination

Not a true solution, but in the interest of creativity and alternative methods, here is a way to cheese the solution in seconds.



First, notice that  $AC = 17$  (Pythagorean triple 8-15-17). Since all the side lengths of  $\triangle ADC$  are integers, and  $\triangle AED \sim \triangle ADC \sim \triangle DEC$ , all of the side length ratios are rational, and therefore,  $DE$  must be a rational value. We eliminate answer choices A, D, and E.

Finally, notice that answer choice C is impossible because that would mean that in  $\triangle AED$ , the leg  $DE = 10$  would be bigger than the hypotenuse  $AD = 8$ . The only answer choice remaining is **(B)**  $\frac{120}{17}$ .

### §33 AMC 8 2017

1. (AMC 8 2017) Which of the following values is the largest?

(A)  $2 + 0 + 1 + 7$       (B)  $2 \times 0 + 1 + 7$       (C)  $2 + 0 \times 1 + 7$       (D)  $2 + 0 + 1 \times 7$   
 (E)  $2 \times 0 \times 1 \times 7$

**Answer:** A

**Solution 1**

We will compute each expression.

A)  $2 + 0 + 1 + 7 = 10$

B)  $2 \times 0 + 1 + 7 = 8$

C)  $2 + 0 \times 1 + 7 = 9$

D)  $2 + 0 + 1 \times 7 = 9$

E)  $2 \times 0 \times 1 \times 7 = 0$

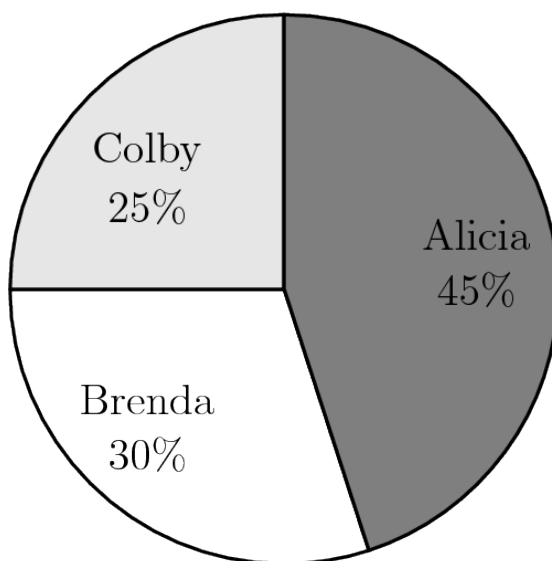
Ordering these, we get 10, 8, 9, 9, 0. Out of these, 10 is the largest number and option (A) adds up to 10. Therefore, the answer is **(A)**  $2 + 0 + 1 + 7$ .

- SBose

**Solution 2**

We immediately see that every one of the choices, except for A and D, has a number multiplied by 0. This will only make the expression's value smaller. We are left with A and D, but in D, 1 is multiplied by 7 to get 7, whereas in answer choice A, we get 8 out of 7 and 1, instead of 7. Therefore, **(A)**  $2 + 0 + 1 + 7$  is your answer.

2. (AMC 8 2017) Alicia, Brenda, and Colby were the candidates in a recent election for student president. The pie chart below shows how the votes were distributed among the three candidates. If Brenda received 36 votes, then how many votes were cast all together?



- (A) 70      (B) 84      (C) 100      (D) 106      (E) 120

**Answer:** E

**Solution 1**

Let  $x$  be the total amount of votes casted. From the chart, Brenda received 30% of the votes and had 36 votes. We can express this relationship as  $\frac{30}{100}x = 36$ . Solving for  $x$ , we get  $x = \boxed{\text{(E) } 120}$ .

**Solution 2**

We're being asked for the total number of votes cast – that represents 100% of the total number of votes. Brenda received 36 votes, which is  $\frac{30}{100} = \frac{3}{10}$  of the total number of votes. Multiplying 36 by  $\frac{10}{3}$ , we get the total number of votes, which is  $\boxed{\text{(E) } 120}$ .

**Solution 3**

If 36 votes is  $\frac{3}{10}$  of all the votes, we can divide that by 3 to get 12 as 10%, and then we can multiply the 12 by 10 to get to 120. So, the answer is  $\boxed{\text{(E) } 120}$ .

3. (AMC 8 2017) What is the value of the expression  $\sqrt{16\sqrt{8\sqrt{4}}}$ ?
- (A) 4      (B)  $4\sqrt{2}$       (C) 8      (D)  $8\sqrt{2}$       (E) 16

**Answer:** C

**Solution 1**

$$\sqrt{16\sqrt{8\sqrt{4}}} = \sqrt{16\sqrt{8 \cdot 2}} = \sqrt{16\sqrt{16}} = \sqrt{16 \cdot 4} = \sqrt{64} = \boxed{\text{(C) } 8}.$$

**Worse Solution** We solve the general form expression  $\sqrt{a\sqrt{b\sqrt{c}}}$ . Note,

$$\sqrt{a\sqrt{b\sqrt{c}}} = (a^4b^2c^1)^{\frac{1}{8}}$$



Thus our answer is,

$$(16^4 \cdot 8^2 4^1)^{\frac{1}{8}} = 16777216^{\frac{1}{8}} = 8$$

4. (AMC 8 2017) When 0.000315 is multiplied by 7,928,564 the product is closest to which of the following?

(A) 210      (B) 240      (C) 2100      (D) 2400      (E) 24000

**Answer:** D

**Solution**

We can approximate 7,928,564 to 8,000,000 and 0.000315 to 0.0003. Multiplying the two yields 2400. Thus, it shows our answer is **(D) 2400**.

5. (AMC 8 2017) What is the value of the expression  $\frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8}{1+2+3+4+5+6+7+8}$ ?
- (A) 1020      (B) 1120      (C) 1220      (D) 2240      (E) 3360

**Answer:** B

**Solution 1 (Brute Force)**

Directly calculating:

We evaluate both the top and bottom:  $\frac{40320}{36}$ . This simplifies to **(B) 1120**.

**Solution 2**

It is well known that the sum of all numbers from 1 to  $n$  is  $\frac{n(n+1)}{2}$ . Therefore, the denominator is equal to  $\frac{8 \cdot 9}{2} = 4 \cdot 9 = 2 \cdot 3 \cdot 6$ . Now, we can cancel the factors of 2, 3, and 6 from both the numerator and denominator, only leaving  $8 \cdot 7 \cdot 5 \cdot 4 \cdot 1$ . This evaluates to **(B) 1120**.

**Solution 3**

First, we evaluate  $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8$  to get 36. We notice that 36 is 6 squared, so we can factor the denominator like  $\frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8}{6 \cdot 6}$  then cancel the 6s out to get  $\frac{4 \cdot 5 \cdot 7 \cdot 8}{1}$ . Now that we have escaped fraction form, we multiply  $4 \cdot 5 \cdot 7 \cdot 8$ . Multiplying these, we get **(B) 1120**.

6. (AMC 8 2017) If the degree measures of the angles of a triangle are in the ratio 3 : 3 : 4, what is the degree measure of the largest angle of the triangle?

(A) 18      (B) 36      (C) 60      (D) 72      (E) 90

**Answer:** D

**Solution 1**

The sum of the ratios is 10. Since the sum of the angles of a triangle is  $180^\circ$ , the ratio can be scaled up to 54 : 54 : 72 ( $3 \cdot 18$  :  $3 \cdot 18$  :  $4 \cdot 18$ ). The numbers in the ratio 54 : 54 : 72 represent the angles of the triangle. The question asks for the largest, so the answer is **(D) 72**.

**Solution 2**

We can denote the angles of the triangle as  $3x$ ,  $3x$ ,  $4x$ . Due to the sum of the angles in a triangle,  $3x + 3x + 4x = 180^\circ \implies x = 18^\circ$ . The greatest angle is  $4x$  and after substitution we get **(D) 72**.

7. (AMC 8 2017) Let  $Z$  be a 6-digit positive integer, such as 247247, whose first three digits are the same as its last three digits taken in the same order. Which of the following numbers must also be a factor of  $Z$ ?

(A) 11      (B) 19      (C) 101      (D) 111      (E) 1111

**Answer:** A

**Solution 1 (detailed explanation)**

To check, if a number is divisible by 19, take its unit digit and multiply it by 2, then add the result to the rest of the number, and repeat this step until the number is reduced to two digits. If the result is divisible by 19, then the original number is also divisible by 19.

After we got 19 eliminated, we can see that the other options have a lot of 1's in them. The divisibility rule for 11 is add alternating digits up, then take the difference of them. The example number works like that. If we add variables, ABCDEF to make number ABCABC, we can see that  $(A+C+B) - (B+A+C) = 0$ . Which is divisible by 11, so our answer choice is **(A) 11**.

by: CHECKMATE2021 (edited by CHECKMATE2021)

**Solution 2**

We are given one of the numbers that can represent  $Z$ , so we can just try out the options to see which one is a factor of 247247. By the previous solution, 19 does not work even though it is a factor of the example number. We get **(A) 11**.

**Solution 3( similar to solution 1)**

Similar to solution 1, let  $Z = ABCABC$ . To prove it is divisible by 11, we can compute its alternating sum, which is  $A - B + C - A + B - C = 0$ , which is divisible by 11. Therefore, the answer is **(A) 11**.

**Solution 4**

We can find that all numbers like  $Z$  are divisible by 1001. 1001 is  $7 \times 11 \times 13$  and when we divide 11, we get a whole number. So, the answer is **(A) 11**.

8. (AMC 8 2017) Malcolm wants to visit Isabella after school today and knows the street where she lives but doesn't know her house number. She tells him, "My house number has two digits, and exactly three of the following four statements about it are true."

- (1) It is prime.
- (2) It is even
- (3) It is divisible by 7.
- (4) One of its digits is 9.

This information allows Malcolm to determine Isabella's house number. What is its units digit?

(A) 4      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** D

**Solution**

Notice that (1) cannot be true. Otherwise, the number would have to be prime and be either even or divisible by 7. This only happens if the number is 2 or 7,

neither of which are two-digit numbers, so we run into a contradiction. Thus, we must have (2), (3), and (4) be true. By (2), the 2-digit number is even, and thus, the digit in the tens place must be 9. The only even 2-digit number starting with 9 and divisible by 7 is 98, which has a units digit of **(D) 8**.

9. (AMC 8 2017) All of Marcy's marbles are blue, red, green, or yellow. One third of her marbles are blue, one fourth of them are red, and six of them are green. What is the smallest number of yellow marbles?

(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** D

**Solution 1**

The 6 green marbles and yellow marbles form  $1 - \frac{1}{3} - \frac{1}{4} = \frac{5}{12}$  of the total marbles. Now, suppose the total number of marbles is  $x$ . We know the number of yellow marbles is  $\frac{5}{12}x - 6$  and a positive integer. Therefore, 12 must divide  $x$ . Trying the smallest multiples of 12 for  $x$ , we see that when  $x = 12$ , we get there are  $-1$  yellow marbles, which is impossible. However when  $x = 24$ , there are  $\frac{5}{12} \cdot 24 - 6 = \mathbf{(D) 4}$  yellow marbles, which must be the smallest possible.

**Solution 2**

Since  $\frac{1}{3}$  of the marbles are blue and  $\frac{1}{4}$  are red, it is clear that the total number of marbles must be divisible by 12. If there are 12 marbles, then 4 are blue, 3 are red, and 6 are green, meaning that there are  $-1$  yellow marbles. This is impossible. Trying the next multiple of 12, 24, we find that 8 are blue, 6 are red, and 6 are green, meaning that the minimum number of yellow marbles is **(D) 4**.

**Solution 3**

Assume the total number of marbles is  $x$ . Since  $\frac{1}{3}$  of her marbles are blue the total number of blue marbles she has is  $\frac{x}{3}$ . Similarly she has  $\frac{x}{4}$  red marbles. Saying that the number of yellow marbles is  $y$  we get this equation  $\frac{x}{3} + \frac{x}{4} + 6 + y = x$ . Simplifying it we get  $\frac{7x}{12} + 6 + y = x$  which becomes  $6 + y = \frac{5x}{12}$  multiplying  $\frac{12}{5}$  on both sides we get  $\frac{12(6+y)}{5} = x$ . Because  $x$  is a natural number  $12(6+y)$  has to be divisible by 5. This means that  $6+y$  has to be divisible by 5. The value that satisfies that and is in the answer choices in **(D) 4**.

10. (AMC 8 2017) A box contains five cards, numbered 1, 2, 3, 4, and 5. Three cards are selected randomly without replacement from the box. What is the probability that 4 is the largest value selected?

(A)  $\frac{1}{10}$       (B)  $\frac{1}{5}$       (C)  $\frac{3}{10}$       (D)  $\frac{2}{5}$       (E)  $\frac{1}{2}$

**Answer:** C

**Solution 1 (combinational)**

There are  $\binom{5}{3}$  possible groups of cards that can be selected. If 4 is the largest card selected, then the other two cards must be either 1, 2, or 3, for a total  $\binom{3}{2}$  groups

of cards. Then, the probability is just  $\frac{\binom{3}{2}}{\binom{5}{3}} = \mathbf{(C) \frac{3}{10}}$ .

**Solution 2 (regular probability)**

$P(\text{no } 5) = \frac{4}{5} * \frac{3}{4} * \frac{2}{3} = \frac{2}{5}$ . This is the fraction of total cases with no fives.  $P(\text{no } 4 \text{ and no } 5) = \frac{3}{5} * \frac{2}{4} * \frac{1}{3} = \frac{6}{60} = \frac{1}{10}$ . This is the intersection of no fours and no fives. Subtract the fraction of no fours and no fives from that of no fives.  $\frac{2}{5} - \frac{1}{10} = \frac{3}{10} = \boxed{\text{(C)} \frac{3}{10}}$ .

### Solution 3 (Simple)

Pick any desirable situation. Let's say you first draw a 4, then a 3, then a 2. The probability of that is  $\frac{1}{5} * \frac{1}{4} * \frac{1}{3}$ , which is  $\frac{1}{60}$ . And for this case, it doesn't matter if you first draw a 3 or a 2. The probability of drawing a 4, a 3, and a 2 in any order is  $\frac{1}{60}$ .

Next, list other desirable situations. You can draw a 4, 3, 1, or a 4,2,1, regardless of order. So, our desirable situations are 4,3,2, then 4,3,1, and lastly 4,2,1, regardless of order.

Now, for each desirable situation, there are  $3! = 6$  ways to arrange the numbers. That means that there are 6 ways to arrange 4,3,1, then 6 ways to arrange 4,2,1, and 6 ways to arrange 4,3,2.  $6 * 3 = 18$ , which means, accounting for order, there are 18 desirable outcomes.

The probability of each desirable outcome is  $\frac{1}{60}$ . Since there are 18 of them, the probability that 4 is the highest value selected is  $\frac{1}{60} * 18$ , which equals  $\frac{18}{60}$ , which simplifies to the answer,  $\frac{3}{10}$ .

This might be a lengthy explanation, but the idea behind it is pretty simple, and it doesn't take long for it to find the answer. Wish you luck on the AMC8! Evžen D.

### Solution 4 (Complementary Counting)

First we have to find the probability that we choose a 5. One of the three numbers is 5 and the other two have to be chosen from the set  $\{1, 2, 3, 4\}$  so the number of sets with 5 as the largest number is  $\binom{4}{2} = 6$ . The number of ways to choose a set of three from  $\{1, 2, 3, 4, 5\}$  is  $\binom{5}{3} = 10$ , so that means that the probability that 4 is the highest number in the set is  $10/10 - 6/10 = 4/10$  right... nope! This is wrong because one of these sets that don't have 5 as the largest value also doesn't have 4 as its largest value! We have over counted! So, we find the number of ways where 4 isn't in the set either. There is only one possibility because  $\binom{3}{3} = 1$ . This means that there are only three ways to have a set with 4 as the largest value, so we find the probability to be  $\boxed{\text{(C)} \frac{3}{10}}$ .

### Solution 5 (List Count)

Step 1: List every possible group of 3 cards.

Since there are only 5 cards, list every possible set of 3 numbers:

1, 2, 3 1, 2, 4 1, 2, 5 1, 3, 4 1, 3, 5 1, 4, 5 2, 3, 4 2, 3, 5 2, 4, 5 3, 4, 5

Step 2: Find the groups where 4 is the biggest number.

1, 2, 4 (True) 1, 3, 4 (True) 2, 3, 4 (True)

Step 3: Turn it into a probability.

Probability just means: (number of groups that work) out of (total number of groups). So the probability is 3 out of 10, or  $3/10$ .

11. (*AMC 8 2017*) A square-shaped floor is covered with congruent square tiles. If the total number of tiles that lie on the two diagonals is 37, how many tiles cover the floor?

(A) 148      (B) 324      (C) 361      (D) 1296      (E) 1369

**Answer:** C

**Solution 1**

Since the number of tiles lying on both diagonals is 37, counting one tile twice, there are  $37 = 2x - 1 \implies x = 19$  tiles on each side, where  $x$  is the number of tiles on the side length of the square. This is because the number of tiles on the square's diagonal is equal to the number of tiles on the square's side length. Therefore, our answer is  $19^2 = \boxed{\text{(C) } 361}$ .

**Solution 2**

Visualize it as 4 separate diagonals connecting to one square in the middle. Each square on the diagonal corresponds to one square of horizontal/vertical distance (because it's a square). So, we figure out the length of each separate diagonal, multiply by two, and then add 1. (Realize that we can just join two of the separate diagonals on opposite sides together to save some time in calculations.) Therefore, the edge length is:

$$\frac{37 - 1}{4} \cdot 2 + 1 = 19$$

Thus, our solution is  $19^2 = \boxed{\text{(C) } 361}$ .

12. (*AMC 8 2017*) The smallest positive integer greater than 1 that leaves a remainder of 1 when divided by 4, 5, and 6 lies between which of the following pairs of numbers?

(A) 2 and 19      (B) 20 and 39      (C) 40 and 59      (D) 60 and 79      (E) 80 and 124

**Answer:** D

**Solution 1**

Since the remainder is the same for all numbers, then we will only need to find the lowest common multiple of the three given numbers, and add the given remainder. The  $\text{LCM}(4, 5, 6)$  is 60. Since  $60 + 1 = 61$ , that is in the range of  $\boxed{\text{(D) } 60 \text{ and } 79}$ .

**Solution 2**

Call the number we want to find  $n$ . We can say that

$$n \equiv 1 \pmod{4}$$

$$n \equiv 1 \pmod{5}$$

$$n \equiv 1 \pmod{6}$$

We can also say that  $n - 1$  is divisible by 4, 5, and 6. Therefore,  $n - 1 = \text{lcm}(4, 5, 6) = 60$ , so  $n = 60 + 1 = 61$  which is in the range of  $\boxed{\text{(D) } 60 \text{ and } 79}$ .

13. (*AMC 8 2017*) Peter, Emma, and Kyler played chess with each other. Peter won 4 games and lost 2 games. Emma won 3 games and lost 3 games. If Kyler lost 3 games, how many games did he win?

(A) 0    (B) 1    (C) 2    (D) 3    (E) 4

**Answer:** B**Solution**

Given  $n$  games, there must be a total of  $n$  wins and  $n$  losses. Hence,  $4+3+K = 2+3+3$  where  $K$  is Kyler's wins.  $K = 1$ , so our final answer is **(B) 1**. CHECKMATE2021

14. (AMC 8 2017) Chloe and Zoe are both students in Ms. Demeanor's math class. Last night, they each solved half of the problems in their homework assignment alone and then solved the other half together. Chloe had correct answers to only 80% of the problems she solved alone, but overall 88% of her answers were correct. Zoe had correct answers to 90% of the problems she solved alone. What was Zoe's overall percentage of correct answers?

(A) 89    (B) 92    (C) 93    (D) 96    (E) 98

**Answer:** C**Solution 1**

Let the number of questions that they solved alone be  $x$ . Let the percentage of problems they correctly solve together be  $a\%$ . As given,

$$\frac{80x}{100} + \frac{ax}{100} = \frac{2 \cdot 88x}{100}$$

.

Hence,  $a = 96$ .

Zoe got  $\frac{90x}{100} + \frac{ax}{100} = \frac{186x}{100}$  problems right out of  $2x$ . Therefore, Zoe got  $\frac{\frac{186x}{100}}{2x} = \frac{93}{100} =$  **(C) 93** percent of the problems correct.

**Solution 2**

Assume the total amount of problems is 100 per half homework assignment since we are dealing with percentages, not values. Then, we know that Chloe got 80 problems correct by herself and got 176 problems correct overall. We also know that Zoe had 90 problems she did correctly alone. We can see that the total amount of correct problems Chloe and Zoe did together was  $176 - 80 = 96$ . Therefore, Zoe did  $96 + 90 = 186$  problems out of 200 problems correctly. This is **(C) 93** percent.

**Solution 3**

In the problem, we can see that Chloe solved 80% of the problems she solved alone, but 88% of her answers are correct. If 80 and another number's average is 88, the other number must be 96. Then, Zoe solved 90% of the problems she did alone, but 96% of her answers were correct. Then, the average of 90 and 96 is **(C) 93**.

**Solution 4**

(Slightly different Solution) Suppose we said that there were 100 problems in their assignment. Then, Chloe had 40 correct and 10 incorrect on her portion, and 48 correct and 2 incorrect on the portion she and Zoe solved. Zoe has 45 correct and 5 incorrect on her portion, and 48 correct and 2 incorrect on the portion that she and Chloe solved. Then, Zoe has  $48 + 45 = 93$  correct answers out of 100, so the answer is **(C) 93**.

-HW73

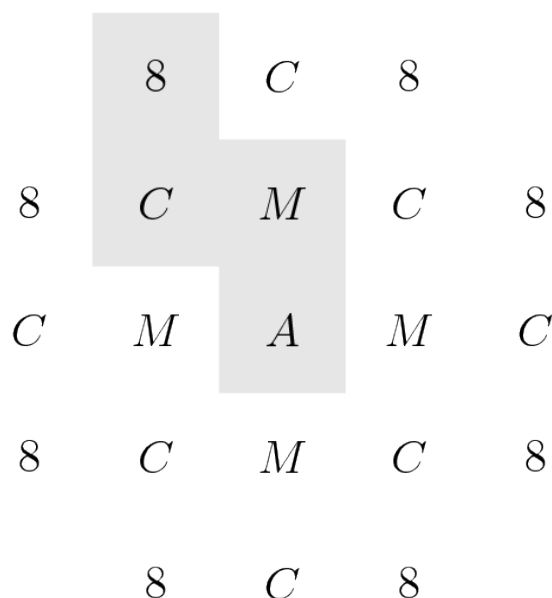
**Solution 5**

Let the total number of problems be  $t$ . Let the percentage of the number of problems that Chloe and Zoe did together and got right be  $x$ . As we can see, Chloe got 80% of  $\frac{1}{2}$  of the total problems right, hence,  $0.80 \cdot \frac{1}{2}t$ . We also know that Chloe got 88% of  $t$  problems right altogether, making it  $0.88 \cdot t$  total problems right. If we add  $x$  to the percentage of correct problems that Chloe solved alone, then that should be equal to the total number of problems that Chloe got right, making the equation:  $0.80 \cdot \frac{1}{2}t + x = 0.88 \cdot t$ . Solving that, we get  $x = 0.48t$ . We also know that Zoe got 90% of  $\frac{1}{2}$  of the total problems right, making it  $0.90 \cdot \frac{1}{2}t$ . We now add that amount to the percentage of problems that Chloe and Zoe got right together, making  $0.90 \cdot \frac{1}{2}t + 0.48t$ . Solving that, we get  $0.93t$ , which is equal to 93%; hence,

**(C) 93**.

-fn106068

15. (*AMC 8 2017*) In the arrangement of letters and numerals below, by how many different paths can one spell AMC8? Beginning at the A in the middle, a path allows only moves from one letter to an adjacent (above, below, left, or right, but not diagonal) letter. One example of such a path is traced in the picture.

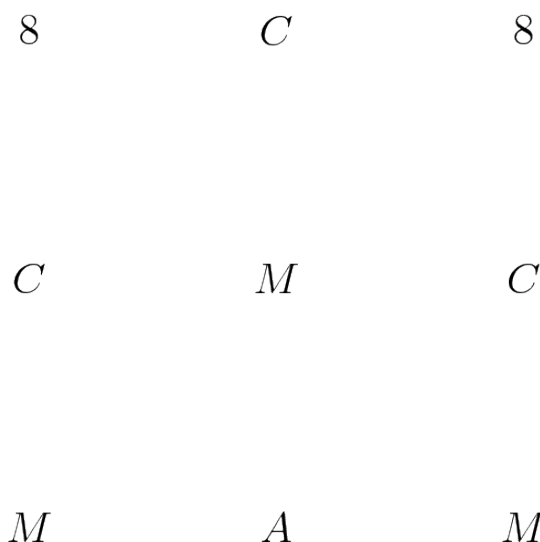


- (A) 8    (B) 9    (C) 12    (D) 24    (E) 36

**Answer:** D

**Solution 1 (Not Very Efficient)**

Notice that the upper-most section contains a 3 by 3 square that looks like:



It has 6 paths in which you can spell out AMC8. You will find four identical copies of this square in the figure, so multiply  $6 \cdot 4$  to get (D) 24 total paths.

**Solution 2**

There are three different kinds of paths that are on this diagram. The first kind is when you directly count  $A, M, C$  in a straight line. The second is when you count  $A$ , turn left or right to get  $M$ , then go up or down to count 8 and  $C$ . The third is the one where you start with  $A$ , move up or down to count  $M$ , turn left or right to count  $C$ , then move straight again to get 8.

There are 8 paths for each kind of path, making for  $8 \cdot 3 =$  (D) 24 paths.

**Solution 3 (easy and recommended)**

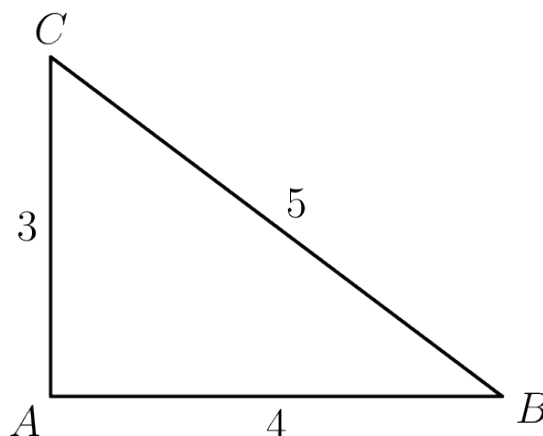
Notice that the  $A$  is adjacent to 4  $M$ s, each  $M$  is adjacent to 3  $C$ s, and each  $C$  is adjacent to 2 8's. So for each  $A$ , there are 4  $M$ s, and for each  $M$ , there are 3  $C$ s, and for each  $C$ , there are 2 8s. Thus, the answer is  $1 \cdot 4 \cdot 3 \cdot 2 =$  (D) 24.

**Solution 4**

We can do this problem by computing how many ways there are to get to each letter (in order). There is 1 way to get to the  $A$  in the center. We can only get to each of the other  $M$ s by going there from the  $A$ , so there is 1 way to get to each of the four  $M$ s. For the  $C$ s, we notice that four  $C$ s are surrounded by one  $M$ , and four  $C$ s are surrounded by two  $M$ s. Finally, each of the 8s is surrounded by one  $C$  with one way to get there and one  $C$  with two ways to get there. Therefore, there are three paths to any of the 8s. Since there are eight 8s, the answer is  $3 \cdot 8 =$  (D) 24.

16. (AMC 8 2017) In the figure below, choose point  $D$  on  $\overline{BC}$  so that  $\triangle ACD$  and  $\triangle ABD$  have equal perimeters. What is the area of  $\triangle ABD$ ?





- (A)  $\frac{3}{4}$     (B)  $\frac{3}{2}$     (C) 2    (D)  $\frac{12}{5}$     (E)  $\frac{5}{2}$

**Answer:** D

**Solution 1**

We know that the perimeters of the two small triangles are  $3 + CD + AD$  and  $4 + BD + AD$ . Setting both equal and using  $BD + CD = 5$ , we have  $BD = 2$  and  $CD = 3$ . Now, we simply have to find the area of  $\triangle ABD$ . Since  $\frac{BD}{CD} = \frac{2}{3}$ , we must have  $\frac{[ABD]}{[ACD]} = 2/3$ . Combining this with the fact that  $[ABC] = [ABD] + [ACD] =$

$$\frac{3 \cdot 4}{2} = 6, \text{ we get } [ABD] = \frac{2}{5}[ABC] = \frac{2}{5} \cdot 6 = \boxed{\text{(D)} \frac{12}{5}}.$$

**Solution 2**

Since  $\overline{AC}$  is 1 less than  $\overline{BC}$ ,  $\overline{CD}$  must be 1 more than  $\overline{BD}$  to equate the perimeter. Hence,  $\overline{BD} + \overline{BD} + 1 = 5$ , so  $\overline{BD} = 2$ . Therefore, the area of  $\triangle ABD$  is  $\frac{(2)(4)(\sin B)}{2} =$

$$4\left(\frac{3}{5}\right) = \boxed{\text{(D)} \frac{12}{5}}$$

**Solution 3 (Not Very Fast)**

Letting point  $A$  be  $(0,0)$  we get point  $B$  to be  $(4,0)$  and point  $C$  to be  $(0,3)$ . The line  $BC$  has a slope of  $-\frac{3}{4}$  and it has intercepts the  $y$ -axis at  $(0,3)$  so the equation for line  $BC$  is  $y = -\frac{3}{4}x + 3$ . Using the same methods from Solution 1 and Solution 2 you can find out that  $CD$  is 3. The equation of a circle centered at  $(0,3)$  with radius 3 is  $x^2 + (y - 3)^2 = 9$ . Plugging in  $y$  you get  $x^2 + (-\frac{3}{4}x)^2 = 9$ . Simplifying it you get  $x^2 + \frac{9}{16}x^2 = 9$ . Multiplying it by 16 and adding you get  $25x^2 = 144$  which becomes  $x = 12/5$  after solving for the positive answer. You can then plug in  $x = 12/5$  back into the slope intercept equation and you get  $y = -\frac{3}{4}(\frac{12}{5}) + 3$  which becomes  $y = 6/5$ . Letting that be the height you can see that the base is 4 so the

$$\text{area is } \frac{1}{2}\left(\frac{6}{5}\right)(4). \text{ Which becomes } \boxed{\text{(D)} \frac{12}{5}}.$$

Note: This uses coordinate geometry which could be useful when you don't know how to solve a geometric problem

17. (AMC 8 2017) Starting with some gold coins and some empty treasure chests, I tried to put 9 gold coins in each treasure chest, but that left 2 treasure chests empty. So instead I put 6 gold coins in each treasure chest, but then I had 3 gold coins left over. How many gold coins did I have?

- (A) 9    (B) 27    (C) 45    (D) 63    (E) 81

**Answer:** C

**Solution 1 (Using Equations)**

We can represent the amount of gold with  $g$  and the amount of chests with  $c$ . We can use the problem to make the following equations:

$$9c - 18 = g$$

$$6c + 3 = g$$

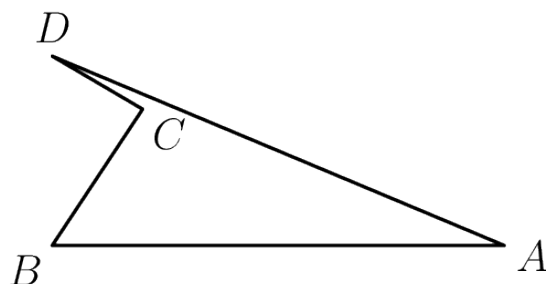
We do this because for 9 chests there are 2 empty and if 9 were in each 9 multiplied by 2 is 18 left.

Therefore,  $6c + 3 = 9c - 18$ . This implies that  $c = 7$ . We therefore have  $g = 45$ . So, our answer is **(C) 45**.

**Solution 2 (Using Answer Choices)**

With 9 coins, there are  $\frac{9}{9} + 2 = 1 + 2 = 3$  chests, by the first condition. These don't fit in with the second condition, so we move onto 27 coins. By the same first condition, there are 5 chests ( $\frac{27}{9} + 2$ ). This also doesn't fit with the second condition. So, onto 45 coins. The first condition implies that there are  $\frac{45}{9} + 2 = 7$  chests, which DOES fit with the second condition, since  $6 \cdot 7 + 3 = 42 + 3 = 45$ . Thus, the desired value is **(C) 45**.

18. (AMC 8 2017) In the non-convex quadrilateral  $ABCD$  shown below,  $\angle BCD$  is a right angle,  $AB = 12$ ,  $BC = 4$ ,  $CD = 3$ , and  $AD = 13$ .



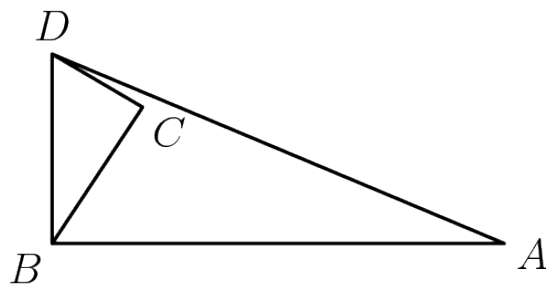
What is the area of quadrilateral  $ABCD$ ?

- (A) 12      (B) 24      (C) 26      (D) 30      (E) 36

**Answer:** B

**Solution 1**

We first connect point  $B$  with point  $D$ .



We can see that  $\triangle BCD$  is a 3-4-5 right triangle. We can also see that  $\triangle BDA$  is a right triangle, by the 5-12-13 Pythagorean triple. With these lengths, we can solve the problem. The area of  $\triangle BDA$  is  $\frac{5 \cdot 12}{2}$ , and the area of  $\triangle BCD$  is  $\frac{3 \cdot 4}{2}$ . Thus, the area of quadrilateral  $ABCD$  is  $30 - 6 = \boxed{\text{(B) } 24}$ .

### Solution 2

$\triangle BCD$  is a 3-4-5 right triangle. So the area of  $\triangle BCD$  is 6. Then we can use Heron's formula to compute the area of  $\triangle ABD$  whose sides have lengths 5, 12, and 13. The area of  $\triangle ABD = \sqrt{s(s-5)(s-12)(s-13)}$ , where  $s$  is the semi-perimeter of the triangle, that is  $s = (5 + 12 + 13)/2 = 15$ . Thus, the area of  $\triangle ABD$  is 30, so the area of  $ABCD$  is  $30 - 6 = \boxed{\text{(B) } 24}$ . —LarryFlora

19. (AMC 8 2017) For any positive integer  $M$ , the notation  $M!$  denotes the product of the integers 1 through  $M$ . What is the largest integer  $n$  for which  $5^n$  is a factor of the sum  $98! + 99! + 100!$ ?

(A) 23      (B) 24      (C) 25      (D) 26      (E) 27

**Answer:** D

### Solution

Factoring out  $98! + 99! + 100!$ , we have  $98!(1 + 99 + 99 \cdot 100)$ , which is  $98!(10000)$ . Next,  $98!$  has  $\left\lfloor \frac{98}{5} \right\rfloor + \left\lfloor \frac{98}{25} \right\rfloor = 19 + 3 = 22$  factors of 5. The 19 is because of all the multiples of 5. The 3 is because of all the multiples of 25. Now, 10,000 has 4 factors of 5, so there are a total of  $22 + 4 = \boxed{\text{(D) } 26}$  factors of 5.

Note: Do you know what formula this uses? Most AMC 8 test takers won't know it. It's **Legendre's Formula**.

20. (AMC 8 2017) An integer between 1000 and 9999, inclusive, is chosen at random. What is the probability that it is an odd integer whose digits are all distinct?

(A)  $\frac{14}{75}$       (B)  $\frac{56}{225}$       (C)  $\frac{107}{400}$       (D)  $\frac{7}{25}$       (E)  $\frac{9}{25}$

**Answer:** B

### Solution

There are 5 options for the last digit (1, 3, 5, 7, 9) as the integer must be odd. The first digit now has 8 options left (it can't be 0 or the same as the last digit). The second digit also has 8 options left (it can't be the same as the first or last digit). Finally, the third digit has 7 options (it can't be the same as the three digits that are already chosen).

Since there are 9,000 total integers, our answer is

$$\frac{8 \cdot 8 \cdot 7 \cdot 5}{9000} = \boxed{\text{(B) } \frac{56}{225}}.$$

21. (AMC 8 2017) Suppose  $a$ ,  $b$ , and  $c$  are nonzero real numbers, and  $a + b + c = 0$ . What are the possible value(s) for  $\frac{a}{|a|} + \frac{b}{|b|} + \frac{c}{|c|} + \frac{abc}{|abc|}$ ?

(A) 0      (B) 1 and  $-1$       (C) 2 and  $-2$       (D) 0, 2, and  $-2$       (E) 0, 1, and  $-1$

**Answer:** A

### Solution

There are 2 cases to consider:

Case 1: 2 of  $a$ ,  $b$ , and  $c$  are positive and the other is negative. Without loss of generality (WLOG), we can assume that  $a$  and  $b$  are positive and  $c$  is negative. In this case, we have that

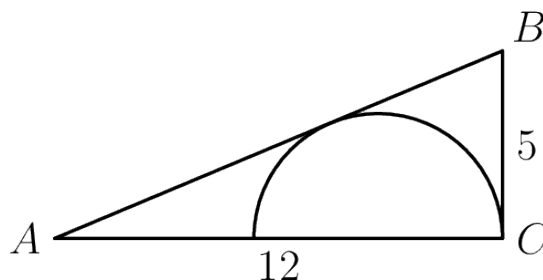
$$\frac{a}{|a|} + \frac{b}{|b|} + \frac{c}{|c|} + \frac{abc}{|abc|} = 1 + 1 - 1 - 1 = 0.$$

Case 2: 2 of  $a$ ,  $b$ , and  $c$  are negative and the other is positive. WLOG, we can assume that  $a$  and  $b$  are negative and  $c$  is positive. In this case, we have that

$$\frac{a}{|a|} + \frac{b}{|b|} + \frac{c}{|c|} + \frac{abc}{|abc|} = -1 - 1 + 1 + 1 = 0.$$

Note these are the only valid cases, for neither 3 negatives nor 3 positives would work as they cannot sum up to 0. In both cases, we get that the given expression equals **(A) 0**.

22. (AMC 8 2017) In the right triangle  $ABC$ ,  $AC = 12$ ,  $BC = 5$ , and angle  $C$  is a right angle. A semicircle is inscribed in the triangle as shown. What is the radius of the semicircle?

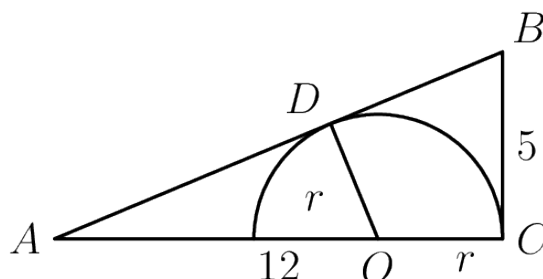


- (A)  $\frac{7}{6}$     (B)  $\frac{13}{5}$     (C)  $\frac{59}{18}$     (D)  $\frac{10}{3}$     (E)  $\frac{60}{13}$

**Answer: D**

**Solution 1 (Pythagorean Theorem)**

We can draw another radius from the center to the point of tangency. This angle,  $\angle ODB$ , is  $90^\circ$ . Label the center  $O$ , the point of tangency  $D$ , and the radius  $r$ .



Since  $ODBC$  is a kite, then  $DB = CB = 5$ . Also,  $AD = 13 - 5 = 8$ . By the **Pythagorean Theorem**,  $r^2 + 8^2 = (12 - r)^2$ . Solving,  $r^2 + 64 = 144 - 24r + r^2 \Rightarrow$

$$24r = 80 \Rightarrow \textbf{(D) } \frac{10}{3}.$$

### Solution 2 (Basic Trigonometry)

If we reflect triangle  $ABC$  over line  $AC$ , we will get isosceles triangle  $ABD$ . By the **Pythagorean Theorem**, we are capable of finding out that the  $AB = AD = 13$ . Hence,  $\tan \frac{\angle BAD}{2} = \tan \angle BAC = \frac{5}{12}$ . Therefore, as of triangle  $ABD$ , the radius of its inscribed circle  $r = \frac{\tan \frac{\angle BAD}{2} \cdot (AB + AD - BD)}{2} = \frac{\frac{5}{12} \cdot 16}{2} = \boxed{\text{(D)} \frac{10}{3}}$

### Solution 3

Similar to solution 2, we reflect  $\triangle ABC$  over line  $\overline{AC}$  and label the reflection of point  $B$  as  $D$ . As  $AB = AD = 13$  by the Pythagorean Theorem, we use the formula  $rs = A$ , where  $r$  is the inradius (what we're trying to find),  $s$  is the semiperimeter  $(\frac{AB + AD + BD}{2})$ , and  $A$  is the area of the triangle in which the incircle is inscribed in. Substitution gives:

$$\begin{aligned} r &= \frac{\frac{10 \cdot 12}{2}}{\frac{13 + 13 + 10}{2}} \\ r &= \frac{60}{18} \\ r &= \boxed{\text{(D)} \frac{10}{3}} \end{aligned}$$

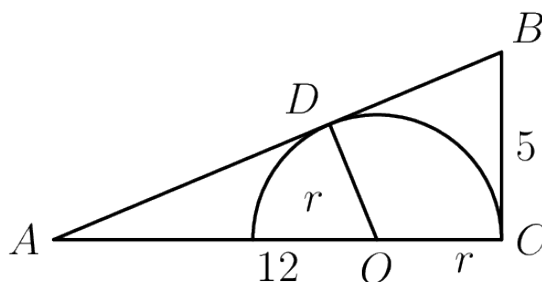
### Solution 4 (Area)

Using the labels from Solution 1, we can express the area of the entire triangle as  $[\triangle ABC] = [\triangle ADO] + [\triangle DBO] + [\triangle OBC]$ . Using  $Area_{\triangle} = \frac{1}{2}bh$  for each term yields  $\frac{1}{2}(AC)(CB) + \frac{1}{2}(AD)(DO) + \frac{1}{2}(BD)(DO) + \frac{1}{2}(OC)(CB)$ . By equal tangents,  $DB = CB = 5$ . By the **Pythagorean Theorem**,  $AB = \sqrt{12^2 + 5^2} = 13$ . It follows that  $AD = AB - DB = 13 - 5 = 8$ . Lastly, because both segments are radii of the semicircle,  $OD = OC = r$ . Plugging in, we get

$$\begin{aligned} \frac{1}{2}(12)(5) + \frac{1}{2}(8)(r) + \frac{1}{2}(5)(r) + \frac{1}{2}(r)(5) \\ 30 = 4r + \frac{5}{2}r + \frac{5}{2}r \\ 30 = 9r \\ r &= \boxed{\text{(D)} \frac{10}{3}} \end{aligned}$$

### Solution 5 (Similar Triangles)

As in Solution 1, we get the following diagram:



We notice that since both triangles  $ABC$  and  $AOD$  share angle  $A$ , they are similar. Then we can say that (since  $DB = CB = 5$  and  $AD = 13 - 5 = 8$ )

$$\frac{AD}{AC} = \frac{OD}{CB} \Rightarrow \frac{8}{12} = \frac{r}{5}. \text{ Solving, we get } r = \frac{40}{12} = \boxed{\text{(D)} \frac{10}{3}}. \text{ nidharshana29}$$

P.S: Full credit for the unedited asymptote goes to MrThinker who made it in Solution 1.

23. (AMC 8 2017) Each day for four days, Linda traveled for one hour at a speed that resulted in her traveling one mile in an integer number of minutes. Each day after the first, her speed decreased so that the number of minutes to travel one mile increased by 5 minutes over the preceding day. Each of the four days, her distance traveled was also an integer number of miles. What was the total number of miles for the four trips?

(A) 10      (B) 15      (C) 25      (D) 50      (E) 82

**Answer:** C

**Solution**

It is well known that Distance = Speed  $\cdot$  Time. In the question, we want distance. From the question, we have that the time is 60 minutes or 1 hour. By the equation derived from Distance = Speed  $\cdot$  Time, we have Speed =  $\frac{\text{Distance}}{\text{Time}}$ , so the speed is 1 mile per  $x$  minutes. Because we want the distance, we multiply the time and speed together yielding 60 mins  $\cdot \frac{1 \text{ mile}}{x \text{ mins}}$ . The minutes cancel out, so now we have  $\frac{60}{x}$  as our distance for the first day. The distance for the following days are:

$$\frac{60}{x}, \frac{60}{x+5}, \frac{60}{x+10}, \frac{60}{x+15}.$$

We know that  $x, x+5, x+10, x+15$  are all factors of 60, therefore,  $x=5$  because the factors have to be in an arithmetic sequence with the common difference being 5 and  $x=5$  is the only solution.

$$\frac{60}{5} + \frac{60}{10} + \frac{60}{15} + \frac{60}{20} = 12 + 6 + 4 + 3 = \boxed{\text{(C)} 25}.$$

24. (AMC 8 2017) Mrs. Sanders has three grandchildren, who call her regularly. One calls her every three days, one calls her every four days, and one calls her every five days. All three called her on December 31, 2016. On how many days during the next year did she not receive a phone call from any of her grandchildren?

(A) 78      (B) 80      (C) 144      (D) 146      (E) 152

**Answer:** D

**Solution 1 (Principle of Inclusion-Exclusion)**

We use Principle of Inclusion-Exclusion. There are 365 days in the year, and we subtract the days that she gets at least 1 phone call, which is

$$\left\lfloor \frac{365}{3} \right\rfloor + \left\lfloor \frac{365}{4} \right\rfloor + \left\lfloor \frac{365}{5} \right\rfloor.$$

To this result we add the number of days where she gets at least 2 phone calls in a day because we double subtracted these days, which is

$$\left\lfloor \frac{365}{12} \right\rfloor + \left\lfloor \frac{365}{15} \right\rfloor + \left\lfloor \frac{365}{20} \right\rfloor.$$

We now subtract the number of days where she gets three phone calls, which is  $\left\lfloor \frac{365}{60} \right\rfloor$ . Therefore, our answer is

$$365 - \left( \left\lfloor \frac{365}{3} \right\rfloor + \left\lfloor \frac{365}{4} \right\rfloor + \left\lfloor \frac{365}{5} \right\rfloor \right) + \left( \left\lfloor \frac{365}{12} \right\rfloor + \left\lfloor \frac{365}{15} \right\rfloor + \left\lfloor \frac{365}{20} \right\rfloor \right) - \left\lfloor \frac{365}{60} \right\rfloor = 365 - 285 + 72 - 6 = \boxed{\text{(D)} 146}$$

### Solution 2 (Least Common Multiple)

Note that  $\text{lcm}(3, 4, 5) = 60$ , so there is a cycle every 60 days.

As shown below, all days in a cycle that Mrs. Sanders receives a phone call from any of her grandchildren are colored in red, yellow, or green.

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |

The year 2017 has 365 days, or 6 cycles and 5 days.

- For each cycle, there are 24 days that Mrs. Sanders does not receive a phone call, as indicated by the white squares.
- For the last 5 days, there are 2 days that Mrs. Sanders does not receive a phone call, as indicated by the first 5 days in a cycle.

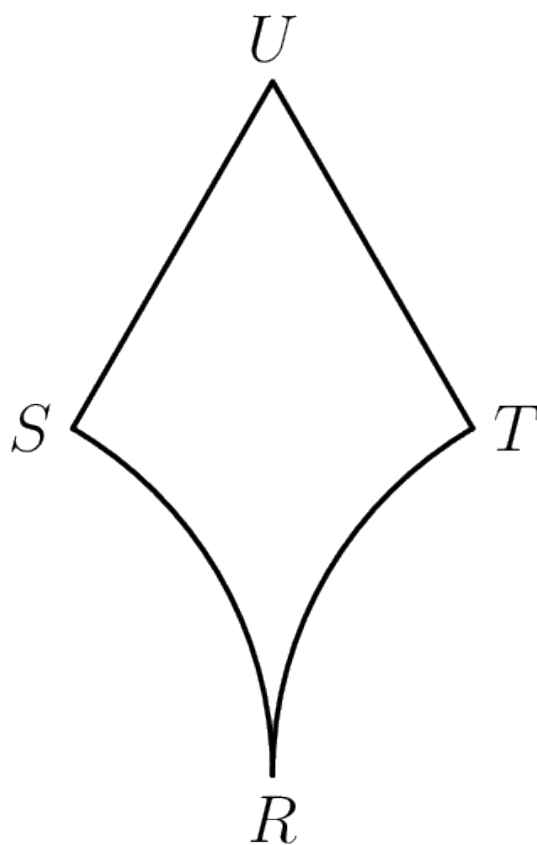
Together, the answer is  $24 \cdot 6 + 2 = \boxed{\text{(D)} 146}$ .

### Solution 3 (Linearity of Expectation)

For any randomly chosen day, there is a  $\frac{2}{3}$  chance the first child does not call her, a  $\frac{3}{4}$  chance the second child does not call her and a  $\frac{4}{5}$  chance the third child does not call her. So, in a randomly chosen day, there is a  $\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} = \frac{2}{5}$  chance no child calls her.

In particular, this "unrigorous" reasoning becomes rigorous, by linearity of expectation, when we take a sample of 360 days; over the first 360 days,  $360 \times \frac{2}{5} = 144$  days will go without a phone call. Now, we simply check the remaining days; on day 361 and 362, nobody calls her; on day 363, the child that calls every three days will call her; on day 364, the child that calls every four days will call her; on day 365, the child that calls every five days will call her. Thus, we add two more days to 144, to get our answer of  $\boxed{\text{(D)} 146}$ .

25. (AMC 8 2017) In the figure shown,  $\overline{US}$  and  $\overline{UT}$  are line segments each of length 2, and  $m\angle TUS = 60^\circ$ . Arcs  $\widehat{TR}$  and  $\widehat{SR}$  are each one-sixth of a circle with radius 2. What is the area of the region shown?

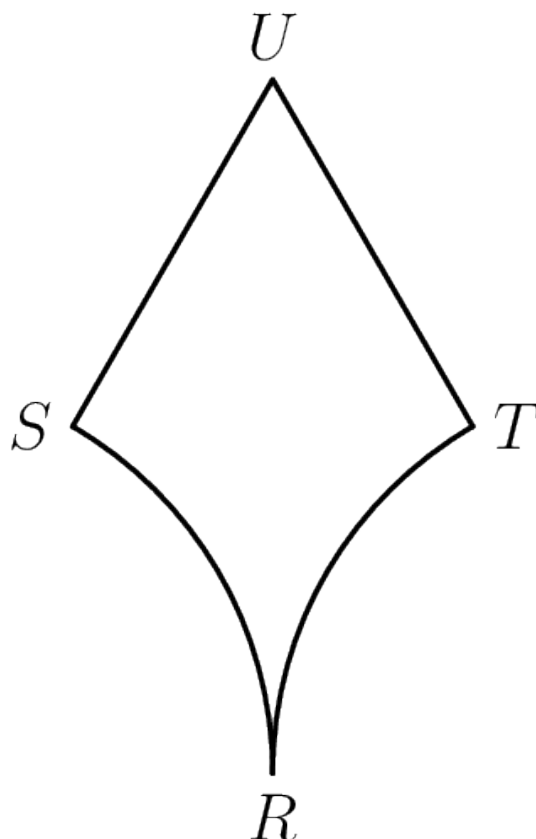


- (A)  $3\sqrt{3} - \pi$       (B)  $4\sqrt{3} - \frac{4\pi}{3}$       (C)  $2\sqrt{3}$       (D)  $4\sqrt{3} - \frac{2\pi}{3}$       (E)  $4 + \frac{4\pi}{3}$

**Answer:** B

**Solution 1**





In addition to the given diagram, we can draw lines  $\overline{SR}$  and  $\overline{RT}$ . The area of rhombus  $SRTU$  is half the product of its diagonals, which is  $\frac{2\sqrt{3} \cdot 2}{2} = 2\sqrt{3}$ . However, we have to subtract off the circular segments. The area of those can be found by computing the area of the circle with radius 2, multiplying it by  $\frac{1}{6}$ , then finally subtracting the area of an equilateral triangle with a side length 2 from the sector. The sum of the areas of the circular segments is  $2(\frac{4\pi}{6} - \sqrt{3})$ . The area of rhombus  $SRTU$  minus the circular segments is  $2\sqrt{3} - \frac{4\pi}{3} + 2\sqrt{3} = \boxed{\text{(B)} 4\sqrt{3} - \frac{4\pi}{3}}$ .

**Solution 2 (tiny bit intuitional)**

We can extend  $\overline{US}$ ,  $\overline{UT}$  to  $X$  and  $Y$ , respectively, such that  $X$  and  $Y$  are collinear to point  $R$ . Connect  $\overline{XY}$ . We can see points  $X$ ,  $Y$  are probably circle centers of arc  $SR$ ,  $TR$ , respectively. So,  $\overline{XS} = 2 = \overline{TY}$ . Thus,  $\triangle UXY$  is equilateral. The area of  $\triangle UXY$  is  $\frac{\sqrt{3}}{4} \cdot 4^2$ , or  $4\sqrt{3}$ , and both one sixth circles total up to  $\frac{4\pi}{3}$ . Finally, the answer is  $\boxed{\text{(B)} 4\sqrt{3} - \frac{4\pi}{3}}$ .

**§34 AMC 8 2018**

- (AMC 8 2018) An amusement park has a collection of scale models, with ratio 1 : 20, of buildings and other sights from around the country. The height of the United States Capitol is 289 feet. What is the height in feet of its replica to the nearest whole number?

(A) 14      (B) 15      (C) 16      (D) 18      (E) 20

**Answer:** A

**Solution 1(Good Ol' Ratios)**

You can see that since the ratio of real building's heights to the model building's height is  $1 : 20$ . We also know that the U.S Capitol is 289 feet in real life, so to find the height of the model, we divide by 20. That gives us 14.45 which rounds to 14. Therefore, to the nearest whole number, the duplicate is **(A) 14**.

**Solution 2**

We can compute  $\frac{289}{20}$  and round our answer to get **(A) 14**. It is basically Solution 1 without the ratio calculation. However, Solution 1 is referring further to the problem.

**Solution 3**

We know that  $20 \cdot 14 = 280$ , and that  $20 \cdot 15 = 300$ . These are the multiples of 20 around 289, and the closest one of those is 280. Therefore, the answer is  $\frac{280}{20} = \mathbf{(A) 14}$ .

2. (AMC 8 2018) What is the value of the product

$$\left(1 + \frac{1}{1}\right) \cdot \left(1 + \frac{1}{2}\right) \cdot \left(1 + \frac{1}{3}\right) \cdot \left(1 + \frac{1}{4}\right) \cdot \left(1 + \frac{1}{5}\right) \cdot \left(1 + \frac{1}{6}\right)?$$

- (A)  $\frac{7}{6}$     (B)  $\frac{4}{3}$     (C)  $\frac{7}{2}$     (D) 7    (E) 8

**Answer: D**

**Solution 1**

By adding up the numbers in each of the 6 parentheses, we get:

$$\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} \cdot \frac{7}{6}.$$

Using telescoping, most of the terms cancel out diagonally. We are left with  $\frac{7}{1}$  which is equivalent to 7. Thus, the answer would be **(D) 7**.

Also, you can also make the fractions  $\frac{7!}{6!}$ , which also yields **(D) 7**

**Solution 2 (use this when your brain no longer works)**

$$\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} \cdot \frac{7}{6} = \mathbf{(D) 7}. \text{ themathkidwoohooooo}$$

3. (AMC 8 2018) Students Arn, Bob, Cyd, Dan, Eve, and Fon are arranged in that order in a circle. They start counting: Arn first, then Bob, and so forth. When the number contains a 7 as a digit (such as 47) or is a multiple of 7 that person leaves the circle and the counting continues. Who is the last one present in the circle?

- (A) Arn    (B) Bob    (C) Cyd    (D) Dan    (E) Eve

**Answer: D**

**Solution 1(Simulation)**

The five numbers which cause people to leave the circle are 7, 14, 17, 21, and 27.

The most straightforward way to do this would be to draw out the circle with the people, and cross off people as you count.

Assuming the six people start with 1, Arn counts 7 so he leaves first. Then, Cyd counts 14 as there are 7 numbers to be counted from this point. Then, Fon, Bob, and Eve, count, 17, 21, and 27, respectively, so the last one standing is Dan. Hence, the answer would be **(D) Dan**.

**Solution 2**

Assign each person a position: Arn is in position 1, Bob in 2, etc. Note that if there are  $n$  people standing in a circle, a person will say a number that is congruent to their position modulo  $n$ . It follows that if there are  $k$  numbers to be said (with someone leaving on the  $k$ th), the person that leaves will stand in position  $k \pmod n$ .

The five numbers which cause people to leave the circle are 7, 14, 17, 21, and 27. Since there are initially 6 people in the circle, so the person who leaves stands in position  $7 \equiv 1 \pmod 6$ , or position 1.

Arn now leaves, and because Bob is the next person to say a number, we reassign the positions such that Bob is in position 1, Cyd in 2, and Fon in 5. There are 7 more numbers to be said, and someone leaves on the 7th. It follows that the person who leaves stands in position  $7 \equiv 2 \pmod 5$ .

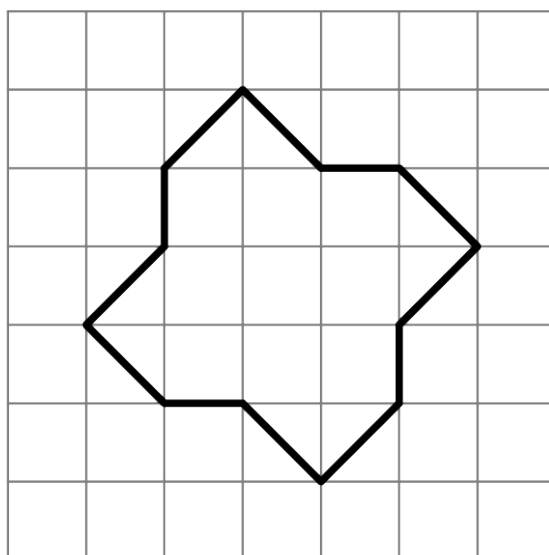
Cyd leaves the circle, and 4 people remain, namely Bob, Dan, Eve, and Fon. Assign the position numbers such that Dan has 1, Eve 2, etc. There are 3 more numbers to be said, and someone leaves on the 3rd. The person standing in position 3 (Fon) now leaves.

There are two people left, Eve and Dan, with Dan in position 1 now. There are 6 more numbers to be said, and Eve is standing in position  $6 \equiv 0 \pmod 2$ , so she leaves.

The last person standing is (D) Dan.

-Benedict T (countmath1)

4. (AMC 8 2018) The twelve-sided figure shown has been drawn on  $1\text{ cm} \times 1\text{ cm}$  graph paper. What is the area of the figure in  $\text{cm}^2$ ?



- (A) 12      (B) 12.5      (C) 13      (D) 13.5      (E) 14

**Answer:** C

**Solution 1**

We count  $3 \cdot 3 = 9$  unit squares in the middle, and 8 small triangles, which gives 4 rectangles each with an area of 1. Thus, the answer is  $9 + 4 =$  (C) 13.

**Solution 2**

We can see here that there are 9 total squares in the middle. We also see that the triangles that make the corners of the shape have an area half the squares' area. Then, we can easily find that each corner has an area of one square and there are 4 corners so we add that to the original 9 squares to get  $9 + 4 = \boxed{\text{(C) } 13}$ . That is how I did it.

**Solution 3**

We can apply Pick's Theorem here. There are 8 lattice points, and 12 lattice points on the boundary. Then,

$$8 + 12 \div 2 - 1 = \boxed{\text{(C) } 13}.$$

**Solution 4**

Use the [Shoelace Theorem](#) and get  $\boxed{\text{(C) } 13}$ .

5. (AMC 8 2018) What is the value of  $1 + 3 + 5 + \cdots + 2017 + 2019 - 2 - 4 - 6 - \cdots - 2016 - 2018$ ?

(A)  $-1010$       (B)  $-1009$       (C)  $1008$       (D)  $1009$       (E)  $1010$

**Answer:** E

**Solution 1**

Rearranging the terms, we get  $(1 - 2) + (3 - 4) + (5 - 6) + \cdots + (2017 - 2018) + 2019$ , and our answer is  $-1009 + 2019 = \boxed{\text{(E) } 1010}$ .

If you were stuck on this problem, refer to AoPS arithmetic lessons.

**Solution 2**

We can see that the last numbers of each of the sets (even numbers and odd numbers) have a difference of two. So, do the second last ones and so on. Now, all we need to find is the number of integers in any of the sets (I chose even) to get  $\boxed{\text{(E) } 1010}$ .

**Solution 3**

It is similar to the Solution 1: Rearranging the terms, we get  $1 + (3 - 2) + (5 - 4) + (6 - 5) \cdots + (2017 - 2016) + (2019 - 2018)$ , and our answer is  $1 + 1009 = \boxed{\text{(E) } 1010}$ .

**Solution 4**

Note that the sum of consecutive odd numbers can be expressed as a square, namely  $1 + 3 + 5 + 7 + \cdots + 2017 + 2019 = 1010^2$ . We can modify the negative numbers in the same way by adding 1 to each negative term, factoring a negative sign, and accounting for the extra 1's by subtracting 1009. We then have  $1010^2 - 1009^2 - 1009$ . Using difference of squares, we obtain  $(1010 + 1009)(1010 - 1009) - 1009 = 2019 - 1009 = \boxed{1010}$ .

**Solution 5**

The sum of odd numbers can be represented as the number of consecutive odds starting from one, squared. Since 2019 is  $(2019+1)/2$ , there are 1010 odd numbers. Therefore, the sum of the odd numbers is

$$1010^2 \quad \text{or} \quad 1010 \times 1010.$$

The sum of even numbers up to 2018 is calculated by noting that there are  $2018/2 = 1009$  even numbers. Using the formula for the sum of the first  $n$  even numbers,  $n(n+1)$ , we get

$$1009 \times (1009 + 1) \quad \text{or} \quad 1009 \times 1010.$$

The difference between these two sums is

$$1010 \times 1010 - 1009 \times 1010 = 1010.$$

Thus, the correct answer is  $-1009 + 2019 = \boxed{\text{(E)} 1010}$ .

6. (*AMC 8 2018*) On a trip to the beach, Anh traveled 50 miles on the highway and 10 miles on a coastal access road. He drove three times as fast on the highway as on the coastal road. If Anh spent 30 minutes driving on the coastal road, how many minutes did his entire trip take?
- (A) 50      (B) 70      (C) 80      (D) 90      (E) 100

**Answer:** C

**Solution 1**

Since Anh spends half an hour to drive 10 miles on the coastal road, his speed is

$$r = \frac{d}{t} = \frac{10}{0.5} = 20 \text{ mph. His speed on the highway then is } 60 \text{ mph.}$$

$50 \implies \boxed{\text{(C)} 80}$

**Solution 2**

Since Anh drives 3 times as fast on the highway, it takes him

$$\frac{1}{3} \text{ of the time to drive } 10 \text{ miles on the highway than on the coastal road.}$$

-UnstoppableGoddess (helped by qkddud)

7. (*AMC 8 2018*) The 5-digit number  $\underline{2} \underline{0} \underline{1} \underline{8} \underline{U}$  is divisible by 9. What is the remainder when this number is divided by 8?
- (A) 1      (B) 3      (C) 5      (D) 6      (E) 7

**Answer:** B

**Solution**

We use the property that the digits of a number must sum to a multiple of 9 if it are divisible by 9. This means  $2 + 0 + 1 + 8 + U$  must be divisible by 9. The only possible value for  $U$  then must be 7. Since we are looking for the remainder when divided by 8, we can ignore the thousands. The remainder when 187 is divided by 8 is  $\boxed{\text{(B)} 3}$ .

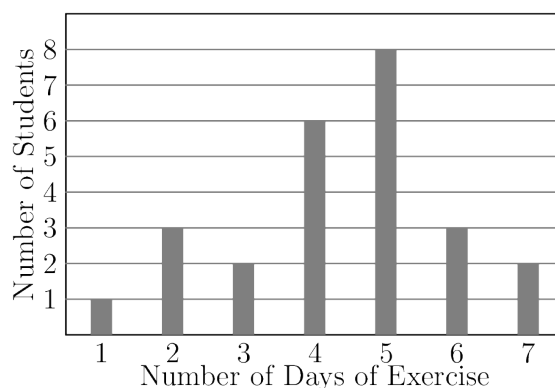
If you got stuck on this problem, refer to AOPS Number Theory. You're smart.

-InterstellarApex, Nivaar

Quick addon:

$$187 - 160 (8 \times 20) = 27. \quad 27 - 24 (8 \times 3) = 3. \quad \text{Therefore, the answer is } \boxed{\text{(B)} 3}.$$

8. (*AMC 8 2018*) Mr. Garcia asked the members of his health class how many days last week they exercised for at least 30 minutes. The results are summarized in the following bar graph, where the heights of the bars represent the number of students.



What was the mean number of days of exercise last week, rounded to the nearest hundredth, reported by the students in Mr. Garcia's class?

- (A) 3.50    (B) 3.57    (C) 4.36    (D) 4.50    (E) 5.00

**Answer:** C

**Solution**

The mean, or average number of days is the total number of days divided by the total number of students. The total number of days is  $1 \cdot 1 + 2 \cdot 3 + 3 \cdot 2 + 4 \cdot 6 + 5 \cdot 8 + 6 \cdot 3 + 7 \cdot 2 = 109$ . The total number of students is  $1 + 3 + 2 + 6 + 8 + 3 + 2 = 25$ . Hence,  $\frac{109}{25} = \boxed{\text{(C) } 4.36}$ .

This problem is a good example of times when the problem statement may give information that may seem important to solving the problem, but turns out it's just there to mess around with you.

In my opinion, I think that it's not a good example of that.

I don't think it is a good example either.

Agreed. Can someone explain another solution? Thanks.

-Flexo3168

9. (AMC 8 2018) Tyler is tiling the floor of his 12 foot by 16 foot living room. He plans to place one-foot by one-foot square tiles to form a border along the edges of the room and to fill in the rest of the floor with two-foot by two-foot square tiles. How many tiles will he use?

- (A) 48    (B) 87    (C) 91    (D) 96    (E) 120

**Answer:** B

**Solution 1**

He will place  $(12 \cdot 2) + ((16 - 2) \cdot 2) = 52$  tiles around the border. For the inner part of the room, we have  $10 \cdot 14 = 140$  square feet. Each tile takes up  $2^2$ , or 4 square feet, so he will use  $\frac{140}{4} = 35$  tiles for the inner part of the room. Thus, the answer is  $52 + 35 = \boxed{\text{(B) } 87}$ .

**Solution 2**

The area around the border:  $(12 \cdot 2) + (14 \cdot 2) = 52$ . The area of tiles around the border:  $1 \cdot 1 = 1$ . Therefore,  $\frac{52}{1} = 52$  is the number of tiles around the border.

The inner part will have  $(12 - 2)(16 - 2) = 140$ . The area of those tiles are  $2 \cdot 2 = 4$ .  $\frac{140}{4} = 35$  is the amount of tiles for the inner part. So,  $52 + 35 = \boxed{87}$ .

10. (AMC 8 2018) The *harmonic mean* of a set of non-zero numbers is the reciprocal of the average of the reciprocals of the numbers. What is the harmonic mean of 1, 2, and 4?

(A)  $\frac{3}{7}$       (B)  $\frac{7}{12}$       (C)  $\frac{12}{7}$       (D)  $\frac{7}{4}$       (E)  $\frac{7}{3}$

**Answer:** C

**Solution**

The sum of the reciprocals is  $\frac{1}{1} + \frac{1}{2} + \frac{1}{4} = \frac{7}{4}$ . Their average is  $\frac{7}{12}$ . Taking the reciprocal of this gives  $\boxed{\text{(C)} \frac{12}{7}}$ .

11. (AMC 8 2018) Abby, Bridget, and four of their classmates will be seated in two rows of three for a group picture, as shown.

|   |   |   |
|---|---|---|
| X | X | X |
| X | X | X |

If the seating positions are assigned randomly, what is the probability that Abby and Bridget are adjacent to each other in the same row or the same column?

(A)  $\frac{1}{3}$       (B)  $\frac{2}{5}$       (C)  $\frac{7}{15}$       (D)  $\frac{1}{2}$       (E)  $\frac{2}{3}$

**Answer:** C

**Solution 1**

There are a total of  $6!$  ways to arrange the kids.

Abby and Bridget can sit in 3 ways if they are adjacent in the same column:

|   |   |   |
|---|---|---|
| A | X | X |
| B | X | X |

|   |   |   |
|---|---|---|
| X | A | X |
| X | B | X |

|   |   |   |
|---|---|---|
| X | X | A |
| X | X | B |

For each of these seat positions, Abby and Bridget can switch seats, and the other 4 people can be arranged in  $4!$  ways which results in a total of  $3 \times 2 \times 4!$  ways to arrange them.

By the same logic, there are 4 ways for Abby and Bridget to be placed if they are adjacent in the same row: they can swap seats, and the other 4 people can be arranged in  $4!$  ways for a total of  $4 \times 2 \times 4!$  ways to arrange them.

We sum the 2 possibilities up to get  $\frac{(3 \cdot 2) \cdot 4! + (4 \cdot 2) \cdot 4!}{6!} = \frac{14 \cdot 4!}{6!} = \boxed{\frac{7}{15}}$  or (C).

If you got stuck on this problem, refer to AOPS Probability and Combinations.

### Solution 2

We can ignore other students, and treat Abby and Bridget as indistinguishable (since we only care about adjacency, not their order). Thus, the total number of ways is  $n(S) = {}_6C_2 = 15$ . In one row, they can be adjacent 2 ways:  $2 \cdot 2rows = 4$ . In one column, they can only be adjacent 1 way:  $1 \cdot 3cols = 3$ . Add these cases  $4 + 3 = 7$ , and therefore,  $P(\text{Abby and Bridget sitting adjacent})$  is  $\boxed{\text{(C)} \frac{7}{15}}$ .

### Solution 3

We can split the seating into two separate cases: if Abby is sitting on the corners, and if Abby isn't. If Abby is sitting in the corners, there is a  $\frac{2}{5}$  chance Bridget is sitting next to Abby, so there is a  $\frac{2}{5} \cdot \frac{4}{6} = \frac{4}{15}$  chance for the first case. Meanwhile, if Abby is sitting in the middle row, there is a  $\frac{3}{5}$  chance Bridget is sitting next to Abby, so there is a  $\frac{3}{5} \cdot \frac{2}{6} = \frac{1}{5}$  chance for the second case. Therefore,  $P(\text{Abby and Bridget are sitting adjacent to each other})$  is  $\frac{4}{15} + \frac{1}{5} = \frac{7}{15}$ , or  $\boxed{\text{C}}$ .

12. (AMC 8 2018) The clock in Sri's car, which is not accurate, gains time at a constant rate. One day as he begins shopping, he notes that his car clock and his watch (which is accurate) both say 12:00 noon. When he is done shopping, his watch says 12:30 and his car clock says 12:35. Later that day, Sri loses his watch. He looks at his car clock and it says 7:00. What is the actual time?  
 (A) 5 : 50      (B) 6 : 00      (C) 6 : 30      (D) 6 : 55      (E) 8 : 10

**Answer:** B

### Solution 1

We see that every 35 minutes the clock passes, the watch passes 30 minutes. That means that the clock is  $\frac{7}{6}$  as fast the watch, so we can set up proportions.  $\frac{\text{car clock}}{\text{watch}} = \frac{7}{6} = \frac{7 \text{ hours}}{x \text{ hours}}$ . Cross-multiplying we get  $x = 6$ . Now, this is obviously redundant, because we could just eyeball it to see that the watch would have passed 6 hours. But this method is better when the numbers are a bit more complex, which makes it both easier and reliable. Either way, our answer is  $\boxed{\text{(B)} 6 : 00}$ .

–BakedPotato66–Rishi09

### Solution 2

The ratio of the car clock to the watch clock is a ratio of 35:30 or 7:6 due to the clock being ahead by 5 minutes after 30 minutes has passed. This means that when the car clock passes 7 hours, the watch has passed 6 hours, meaning that the time would be  $\boxed{\text{(B)} 6 : 00}$ .

### Solution 3

From 12:00 pm (noon) to 7:00 pm, the car clock has passed 12 35-minute cycles ( $12 \times 35 = 420$ ) because 7 hours = 420 minutes. So, 12 30-minute cycles ( $12 \times 30 = 360$ ) for the watch time are 360 minutes, which is 6 hours. Therefore, 6 hours added to 12:00 pm (noon) makes the answer  $\boxed{\text{(B)} 6 : 00}$ .

### Solution 4 (a version of solution 3)

7 hours have passed since 12:00 pm and 7 hours = 420 minutes because there are 60 minutes in an hour. Because every 35 minutes, the clock is ahead by 5 minutes,



you need to divide 420 by 35 to find out how many times it happens. 420 divided by 35 is 12. Then you would multiply 12 by 5 because the clock is ahead by 5 minutes. 12 times 5 is 60, so that means that the clock is ahead by 60 minutes. In order to find the watch's time, you must find what was 60 minutes earlier than 7:00 which is **(B) 6 : 00**.

### Solution 5

If his car clock says it's 12:35 when it's actually 12:30, then that means that the car clock gains 10 minutes for every actual hour that has passed. Then, with this knowledge, you can make a chart:

Actual Time | Car Clock

12:00 | 12:00

1:00 | 1:10

2:00 | 2:20

3:00 | 3:30

2:00 | 2:20

3:00 | 3:30

4:00 | 4:40

5:00 | 5:50

6:00 | 7:00

As you can see, the answer is **(B) 6 : 00**. Evžen D.

13. (*AMC 8 2018*) Laila took five math tests, each worth a maximum of 100 points. Laila's score on each test was an integer between 0 and 100, inclusive. Laila received the same score on the first four tests, and she received a higher score on the last test. Her average score on the five tests was 82. How many values are possible for Laila's score on the last test?

(A) 4    (B) 5    (C) 9    (D) 10    (E) 18

**Answer:** A

### Solution 1

Say Laila gets a value of  $x$  on her first 4 tests, and a value of  $y$  on her last test. Thus,  $4x + y = 82 \cdot 5 = 410$ .

Because  $x$  and  $y$  are different,  $x$  must be less than 82 and  $y$  must be greater than 82. When  $x$  decreases by 1,  $y$  must increase by 4 to keep the total constant.

The greatest value for  $y$  is 98 (as  $y = 100$  would make  $x$  non-integer). In the range  $83 \leq y \leq 98$ , only 4 values for  $y$  result in integer values for  $x$ : 86, 90, 94 and 98. Thus, the answer is **(A) 4**.

### Solution 2

The average score is 82 which leads us to suppose that Laila got all 82 points for the tests. We know that Laila got the same points in the first four tests and they are all lower than the last test. Let the first four tests be 81 points, then the last test must be 86 points to keep the average fixed. Continue to decrement the first four tests to identify other possible combinations. The possible points for the fifth test are 86, 90, 94, 98. The answer is **(A) 4**.

14. (AMC 8 2018) Let  $N$  be the greatest five-digit number whose digits have a product of 120. What is the sum of the digits of  $N$ ?

(A) 15      (B) 16      (C) 17      (D) 18      (E) 20

**Answer:** D

**Solution 1**

If we start off with the first digit, we know that it can't be 9 since 9 is not a factor of 120. We go down to the digit 8, which does work since it is a factor of 120. Now, we have to know what digits will take up the remaining four spots. To find this result, just divide  $\frac{120}{8} = 15$ . The next place can be 5, as it is the largest factor, aside from 15. Consequently, our next three values will be 3, 1 and 1 if we use the same logic. Therefore, our five-digit number is 85311, so the sum is  $8 + 5 + 3 + 1 + 1 = 18 \Rightarrow \boxed{\text{(D) } 18}$ .

**Solution 2 (Factorial)**

120 is  $5!$ , so we have  $(5)(4)(3)(2)(1) = 120$ . (Alternatively, you could identify the prime factors  $(5)(3)(2)(2)(2) = 120$ .) Now look for the largest digit you can create by combining these factors.

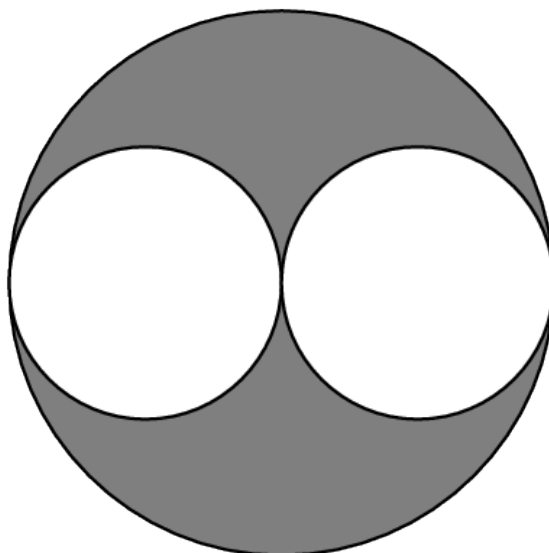
$$8 = 4 \cdot 2$$

Use this largest digit for the ten-thousands place: 8 \_ , \_ \_ \_

Next you use the 5 and the 3 for the next places: 85,3 \_ \_ (You can't use  $3 \cdot 2 = 6$  because the 2 was used to make 8.)

Fill the remaining places with 1: 85,311 =  $(5)(8)(3)(1)(1) = 120$   $8 + 5 + 3 + 1 + 1 = \boxed{\text{(D) } 18}$ .

15. (AMC 8 2018) In the diagram below, a diameter of each of the two smaller circles is a radius of the larger circle. If the two smaller circles have a combined area of 1 square unit, then what is the area of the shaded region, in square units?



(A)  $\frac{1}{4}$       (B)  $\frac{1}{3}$       (C)  $\frac{1}{2}$       (D) 1      (E)  $\frac{\pi}{2}$

**Answer:** D

**Solution 1**

Let the radius of the large circle be  $R$ . Then, the radius of the smaller circles are  $\frac{R}{2}$ . The areas of the circles are directly proportional to the square of the radii, so the ratio of the area of the small circle to the large one is  $\frac{1}{4}$  ( $\frac{R^2}{4}$  is  $\frac{1}{4}$  of  $R^2$ .) This means the combined area of the 2 smaller circles is half of the larger circle, and therefore the shaded region is equal to the combined area of the 2 smaller circles, which is **(D) 1**.

### Solution 2

Let the radius of the two smaller circles be  $r$ . It follows that the area of one of the smaller circles is  $\pi r^2$ . Thus, the area of the two inner circles combined would evaluate to  $2\pi r^2$  which is 1. Since the radius of the bigger circle is two times that of the smaller circles (the diameter), the radius of the larger circle in terms of  $r$  would be  $2r$ . The area of the larger circle would come to  $(2r)^2\pi = 4\pi r^2$ .

Subtracting the area of the smaller circles from that of the larger circle (since that would be the shaded region), we have

$$4\pi r^2 - 2\pi r^2 = 2\pi r^2 = 1.$$

Therefore, the area of the shaded region is **(D) 1**.

### Solution 3

The area of a small circle is  $\frac{1}{2} = \pi r^2$ . Solving, we get  $r = \sqrt{\frac{1}{2\pi}}$ .

The radius of the large circle is  $R = 2r$ . The area of the large circle is  $\pi R^2 = \pi(2r)^2 = 4\pi r^2 = 4\pi \frac{1}{2\pi} = 2$ .

Subtract the area of the small circles from the area of the large circle to get the area of the shaded region:  $2 - 1 = \mathbf{(D) 1}$ .

### Solution 4 (Similar to Solution 3)

To get the area of the small circles, we can get the equation  $\frac{1}{2} = \pi r^2$ . Solving for  $r$ , we get  $r = \sqrt{\frac{1}{2\pi}}$ . Then, we can get the radius of the big circle by doubling the small circle's radius, and that gives  $2\sqrt{\frac{1}{2\pi}}$ . Because the area of a circle is  $\pi r^2$ , you'll have to square the answer and multiple it by  $\pi$ . After you square it, you'll get  $4 \cdot \frac{1}{2\pi}$ , which equals to  $\frac{4}{2\pi}$ . Since the area of a circle is  $\pi r^2$ , we get  $\pi \cdot \frac{4}{2\pi}$ , and that equals  $\frac{4\pi}{2\pi}$ , which equals 2. Since each of the circles' area is  $\frac{1}{2}$ , the combined area of the small circles is 1. Since  $2 - 1 = 1$ , the area of the shaded region is **(D) 1**.

### Solution 5

Since the smaller circles are equivalent, the area of each of them is  $\frac{1}{2}$  square unit. Since the diameter of each of the two smaller circles is a radius of the larger circle, their areas must be in a ratio of 1 : 2. This tells us that the area of the larger circle is double the area of the smaller circle, giving us  $\frac{1}{2} \cdot 2 = \mathbf{(D) 1}$ .

16. (AMC 8 2018) Professor Chang has nine different language books lined up on a bookshelf: two Arabic, three German, and four Spanish. How many ways are there to arrange the nine books on the shelf keeping the Arabic books together and keeping the Spanish books together?

(A) 1440      (B) 2880      (C) 5760      (D) 182,440      (E) 362,880

**Answer:** C

**Solution 1**

To solve this, treat the two Arabic books as one unit and the four Spanish books as another unit. Along with the three German books, you now have five units to arrange.

You can arrange these five units in  $5!$  ways. Within the Arabic block, the two books can be arranged in  $2!$  ways, and within the Spanish block, the four books can be arranged in  $4!$  ways.

So now we try to find the german book possibilities, (Arabic block), (Spanish block), German book 1, German book 2, German book 3.  $5!$

Multiplying all these possibilities( $2!,4!,5!$ ) gives the total number of arrangements which is (C) 5760. ways to arrange the nine books while keeping the Arabic and Spanish books together.-TORPEDOROCKS

17. (*AMC 8 2018*) Bella begins to walk from her house toward her friend Ella's house. At the same time, Ella begins to ride her bicycle toward Bella's house. They each maintain a constant speed, and Ella rides 5 times as fast as Bella walks. The distance between their houses is 2 miles, which is 10,560 feet, and Bella covers  $2\frac{1}{2}$  feet with each step. How many steps will Bella take by the time she meets Ella?
- (A) 704      (B) 845      (C) 1056      (D) 1760      (E) 3520

**Answer:** A

**Solution 1 (Fast and Easy)**

Every 10 feet Bella goes, Ella goes 50 feet, which means a total of 60 feet. They need to travel that 60 feet  $10560 \div 60 = 176$  times to travel the entire 2 miles. Since Bella goes 10 feet 176 times, this means that she travels a total of 1760 feet. And since she walks 2.5 feet each step,  $1760 \div 2.5 =$  (A) 704

**Solution 2 (Use Answer Choices to our advantage)**

We know that Bella goes 2.5 feet per step and since Ella rides 5 times faster than Bella she must go 12.5 feet on her bike for every step of Bella's. For Bell

**Solution 3 (Fractions)**

We can turn  $2\frac{1}{2}$  into an improper fraction. It will then become  $\frac{5}{2}$ . Since Ella bikes 5 times faster, we multiply  $\frac{5}{2} \cdot 5 = \frac{25}{2}$ . Then we add  $\frac{5}{2} + \frac{25}{2}$  to find the distance they walk and bike together for each step of Bella's:  $\frac{30}{2} = 15$ . This means that they travel 15 ft. of distance for each step that Bella takes. Divide 10,560 by 15 to find that Bella takes 704 steps.

18. (*AMC 8 2018*) How many positive factors does 23,232 have?
- (A) 9      (B) 12      (C) 28      (D) 36      (E) 42

**Answer:** E

**Solution 1**

We can first find the prime factorization of 23,232, which is  $2^6 \cdot 3^1 \cdot 11^2$ . Now, we add one to our powers and multiply. Therefore, the answer is  $(6+1) \cdot (1+1) \cdot (2+1) = 7 \cdot 2 \cdot 3 =$  (E) 42

Note: 23232 is a large number, so we can look for shortcuts to factor it. One way to factor it quickly is to use 3 and 11 divisibility rules to observe that  $23232 = 3 \cdot 7744 = 3 \cdot 11 \cdot 704 = 3 \cdot 11^2 \cdot 64 = 3^1 \cdot 11^2 \cdot 2^6$ .

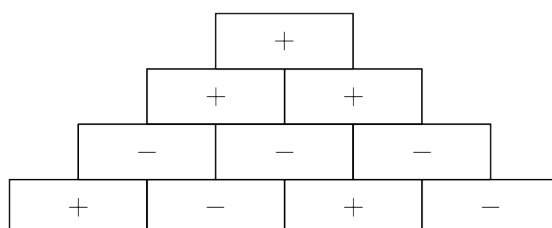
Another way is to spot the "32" and compute that  $23232 = 32 \cdot (101 + 10000/16) = 32 \cdot (101 + 5^4) = 32 \cdot 726 = 32 \cdot 11 \cdot 66$ .

A third way to factor it is to observe  $23232 = 24000 - 768$ . Factoring out the 3 gives us  $3(8000 - 256)$ . Since  $8000 = 2^6 \cdot 5^3$  and  $256 = 2^8$ , we have  $2^6 \cdot 3(5^3 - 2^2) = 2^6 \cdot 3(125 - 4) = 2^6 \cdot 3 \cdot 121 = 2^6 \cdot 3 \cdot 11^2$ .

**Solution 2 (Using  $264^2$ )**

Observe that  $69696 = 264^2$ , so this is  $\frac{1}{3}$  of  $264^2$  which is  $88 \cdot 264 = 11^2 \cdot 8^2 \cdot 3 = 11^2 \cdot 2^6 \cdot 3$ , which has  $3 \cdot 7 \cdot 2 = 42$  factors. The answer is **(E) 42**.

19. (AMC 8 2018) In a sign pyramid a cell gets a "+" if the two cells below it have the same sign, and it gets a "-" if the two cells below it have different signs. The diagram below illustrates a sign pyramid with four levels. How many possible ways are there to fill the four cells in the bottom row to produce a "+" at the top of the pyramid?



- (A) 2      (B) 4      (C) 8      (D) 12      (E) 16

**Answer:** C

**Solution 1**

You could just make out all of the patterns that make the top positive. In this case, you would have the following patterns:

$+++, -+-, ---, ++++, -+++, +-+-, ++-, -++$ . There are 8 patterns and so the answer is **(C) 8**.

-NinjaBoi2000

**Solution 2**

The top box is fixed by the problem.

Choose the left 3 bottom-row boxes freely. There are  $2^3 = 8$  ways.

Then the left 2 boxes on the row above are determined.

Then the left 1 box on the row above that is determined

Then the right 1 box on that row is determined.

Then the right 1 box on the row below is determined.

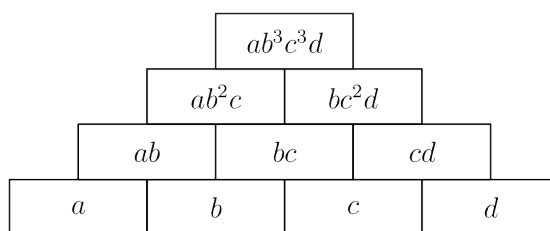
Then the right 1 box on the bottom row is determined, completing the diagram.

So the answer is **(C) 8**.

**Solution 3**

Let the plus sign represent 1 and the negative sign represent -1.

The four numbers on the bottom are  $a, b, c$ , and  $d$ , which are either 1 or -1.



Which means  $ab^3c^3d = 1$ . Since  $b$  and  $c$  are either 1 or -1,  $b^3 = b$  and  $c^3 = c$ . This shows that  $abcd = 1$ .

Therefore either  $a$ ,  $b$ ,  $c$ , and  $d$  are all positive or negative, or 2 are positive and 2 are negative.

There are 2 ways where  $a$ ,  $b$ ,  $c$ , and  $d$  are 1 (1, 1, 1, 1) and (-1, -1, -1, -1)

There are 6 ways where 2 variables are positive and 2 are negative: (1, 1, -1, -1), (1, -1, 1, -1), (-1, 1, 1, -1), (-1, -1, 1, 1), (-1, 1, -1, 1), and (-1, -1, 1, 1).

So the answer is (C) 8.

Note: This result can also be achieved by realizing that there are  $4!/2!2! = 6$  ways to arrange 2 negatives and 2 positives and 1 way each to arrange four of one sign.

#### Solution 4

The pyramid is built on the basic 3 blocks pattern: one above and two below. The basic pattern has four possible symbols and half of them have a + on the above, half of them have a - above. So, for the lowest layer with 4 blocks, there are  $2^4 = 16$  possible combinations and half of them will lead to a + (or -) on the top. The answer is  $16/2 =$ (C) 8.

If you notice this rule, you can give the answer whatever how many layers you have. The answer will be  $2^{n-1}$  for the layer with  $n$  blocks.

#### Solution 5, Casework

We can use casework to solve this problem. The only way for the top cell to have a + in it is if the third row of the pyramid (the one with 2 cells) is either -- or ++. First, let's pretend that the third row of the pyramid is ++. The only way for that to happen is if the second row (the one with 3 cells) is --- or +++. Now, let's pretend that the second is +++. That would have 2 possibilities for the first row (the one with 4 cells), ++++ and ----. Next, let's pretend that the second row is ---. That makes two more possibilities for the first row, -+-+ and +-+- . Now, let's pretend that the 3rd row is --, which means that the second row is either -+- or +-+. You will soon find that -+- find 2 possibilities for the first row, -++- or -+-+, and 2 possibilities for +-+, --++ and ++--. Together, we find that the answer is  $2 + 2 + 2 + 2 =$ (C) 8.

#### Solution 6 (Non-rigorous)

There are two choices for the uppermost: + or -. There are 16 cases in total ( $2^4$ ). Half of 16 is 8. The answer is (C) 8.

(Note: this is not recommended for solving, just an idea after solving.)

#### Solution 7 (Using probability)

If you pick the signs for two boxes at random, there is equal probability that the one above them will be either a + or a -. There are  $16(2^4)$  cases possible for the bottom four, so half of that, (C) 8, is the answer.

**Solution 8 (Very quick, uses formula)**

To get a plus on the top, you either need 4 negatives (----), 4 positives (++++), or 2 negative signs and 2 positive signs. So the number of ways you can get a plus on the top would be 2 plus the number of ways you can arrange 2 negative signs and 2 positive signs. The formula for distinct arrangements (permutations) in a set with repetition is :

$$P = \frac{n!}{n_1!n_2!n_3! \cdots n_k!}$$

So using this formula, you solve for  $4!/(2!*2!)$ , which equals 6. So the number of ways you get a plus on top would be  $2+6$ , which equals **(C) 8**. Evžen D.

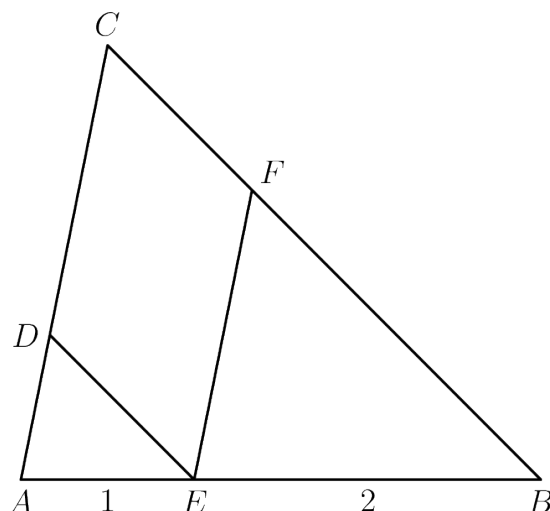
**Solution 9 (Modified version of Sol 1)**

The top block must be positive so below it two box must be of same sign. Now if we arrange the bottom row as following pattern such that according to that the row above bottom will be changed and so on , so that the 3rd two blocks are of same sign hence the top block will be positive.(Draw separate diagram and do yourself to understand )

$+-+, -+-, --, +++, -+++, +-+, ++-, -++$ .

There are 8 patterns .Hence **(C) 8**.

20. (AMC 8 2018) In  $\triangle ABC$ , point  $E$  is on  $\overline{AB}$  with  $AE = 1$  and  $EB = 2$ . Point  $D$  is on  $\overline{AC}$  so that  $\overline{DE} \parallel \overline{BC}$  and point  $F$  is on  $\overline{BC}$  so that  $\overline{EF} \parallel \overline{AC}$ . What is the ratio of the area of  $CDEF$  to the area of  $\triangle ABC$ ?



- (A)  $\frac{4}{9}$       (B)  $\frac{1}{2}$       (C)  $\frac{5}{9}$       (D)  $\frac{3}{5}$       (E)  $\frac{2}{3}$

**Answer:** A

**Solution 1**

By similar triangles, we have  $[ADE] = \frac{1}{9}[ABC]$ . Similarly, we see that  $[BEF] = \frac{4}{9}[ABC]$ . Using this information, we get  $[ACFE] = \frac{5}{9}[ABC]$ . Then, since

$[ADE] = \frac{1}{9}[ABC]$ , it follows that the  $[CDEF] = \frac{4}{9}[ABC]$ . Thus, the answer would be **(A)**

Sidenote:  $[ABC]$  denotes the area of triangle  $ABC$ . Similarly,  $[ABCD]$  denotes the area of figure  $ABCD$ .

**Solution 2**

Let  $a = DE$  and  $b =$  the height of  $\triangle ABC$ . We can extend  $\triangle ABC$  to form a parallelogram, which would equal  $3a \cdot 3b$ . The smaller parallelogram is  $a$  times  $2b$ . The smaller parallelogram is  $\frac{2}{9}$  of the larger parallelogram, so the answer would be  $\frac{2}{9} \cdot 2$ , since the triangle is  $\frac{1}{2}$  of the parallelogram, so the answer is **(A)**  $\frac{4}{9}$ .

By babyzombievillager with credits to many others who helped with the solution :D

**Solution 3**

$\triangle ADE \sim \triangle ABC \sim \triangle EFB$ . We can substitute  $\overline{DA}$  as  $\frac{1}{3}x$  and  $\overline{CD}$  as  $\frac{2}{3}x$ , where  $x$  is  $\overline{AC}$ . Side  $\overline{CB}$  having, distance  $y$ , has 2 parts also. And,  $\overline{CF}$  and  $\overline{FB}$  are  $\frac{1}{3}y$  and  $\frac{2}{3}y$  respectfully. You can consider the height of  $\triangle ADE$  and  $\triangle EFB$  as  $z$  and  $2z$  respectfully. The area of  $\triangle ADE$  is  $\frac{1 \cdot z}{2} = 0.5z$  because the area formula for a triangle is  $\frac{1}{2}bh$  or  $\frac{bh}{2}$ . The area of  $\triangle EFB$  will be  $\frac{2 \cdot 2z}{2} = 2z$ . So, the area of  $\triangle ABC$  will be  $\frac{3 \cdot (2z+z)}{2} = \frac{3 \cdot 3z}{2} = \frac{9z}{2} = 4.5z$ . The area of parallelogram  $CDEF$  will be  $4.5z - (0.5z + 2z) = 4.5z - 2.5z = 2z$ .

Parallelogram  $CDEF$  to  $\triangle ABC = \frac{2z}{4.5z} = \frac{2}{4.5} = \frac{4}{9}$ . The answer is **(A)**  $\frac{4}{9}$ .

(AMC 8 2018) How many positive three-digit integers have a remainder of 2 when divided by 6, a remainder of 5 when divided by 9, and a remainder of 7 when divided by 11?

- (A)** 1    **(B)** 2    **(C)** 3    **(D)** 4    **(E)** 5

**Answer:** E

**Solution 1**

Looking at the values, we notice that  $11 - 7 = 4$ ,  $9 - 5 = 4$  and  $6 - 2 = 4$ . This means we are looking for a value that is four less than a multiple of 11, 9, and 6. The least common multiple of these numbers is  $11 \cdot 3^2 \cdot 2 = 198$ , so the numbers that fulfill this can be written as  $198k - 4$ , where  $k$  is a positive integer. This value is only a three-digit integer when  $k$  is 1, 2, 3, 4 or 5, which gives 194, 392, 590, 788, and 986 respectively. Thus, we have 5 values, so our answer is **(E)** 5.

**Solution 2**

Let us create the equations:  $6x+2 = 9y+5 = 11z+7$ , and we know  $100 \leq 11z+7 < 1000$ , it gives us  $9 \leq z \leq 90$ , which is the range of the value of  $z$ . Because of  $6x+2 = 11z+7$ , then  $6x = 11z+5 = 6z+5(z+1)$ , so  $(z+1)$  must be a multiple of 6. Because of  $9y+5 = 11z+7$ , then  $9y = 11z+2 = 9z+2(z+1)$ , so  $(z+1)$  must also be a multiple of 9. Hence, the value of  $(z+1)$  must be a common multiple of 6 and 9, which means multiples of 18 (LCM of 6, 9). So, let's say  $z+1 = 18p$ ; then,  $9 \leq z = 18p-1 \leq 90$ , so  $1 \leq p \leq 91/18$  or  $1 \leq p \leq 5$ . Thus, the answer is **(E)** 5.

**Solution 3**

By the **Chinese Remainder Theorem**, we have that all solutions are in the form  $x = 198k + 194$  where  $k \in \mathbb{Z}$ . Counting the number of values, we get **(E)** 5.

**Solution 4**

We can use modular arithmetic. Set up the equations:  $x \equiv 2 \pmod{6}$ ,  $x \equiv 5 \pmod{9}$ , and  $x \equiv 7 \pmod{11}$ . These equations can also be written as  $x+4 \equiv 0 \pmod{6}$ ,  $x+4 \equiv 0 \pmod{9}$ , and  $x+4 \equiv 0 \pmod{11}$ . Since  $x+4$  is congruent to numbers 6, 9, and 11, then it must also be congruent to their LCM. Thus,  $x+4 \equiv 0 \pmod{198}$ , since 198 is the LCM of 6, 9, and 11. Since these numbers have to be three digits, they can only be 194, 392, 590, 788, and 986. This gives us the answer of **(E)** 5.

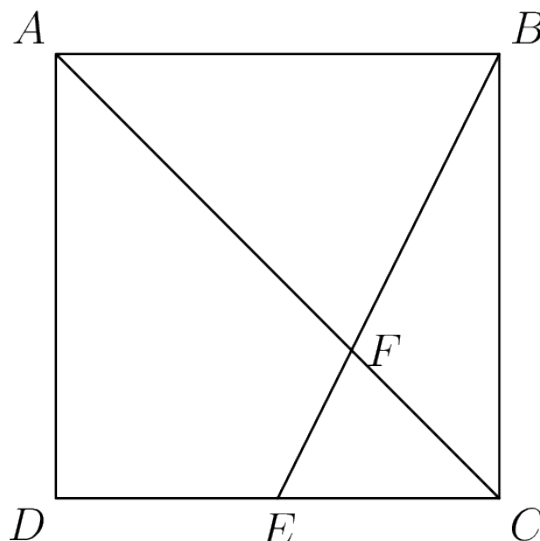
**Solution 5**

Let  $N$  be the three digit positive integer.  $N = 6a + 2 = 9b + 5 = 11c + 7$ . Then, we add four to all sides and write  $N + 4 = 6(a + 1) = 9(b + 1) = 11(c + 1)$ . Now, we know that  $N + 4$  is divisible by 6, 9, and 11. The LCM of 6, 9, and 11 is equal to 198, so



$N = 198k - 4$ . From this, we can figure out that  $N$  can be 5 different three digit numbers – 194, 392, 590, 788, and 986.  $\therefore$ , the answer is **(E) 5**.

(AMC 8 2018) Point  $E$  is the midpoint of side  $\overline{CD}$  in square  $ABCD$ , and  $\overline{BE}$  meets diagonal  $\overline{AC}$  at  $F$ . The area of quadrilateral  $AFED$  is 45. What is the area of  $ABCD$ ?



- (A) 100      (B) 108      (C) 120      (D) 135      (E) 144

**Answer:** B

**Solution 1**

We can use analytic geometry for this problem.

Let us start by giving  $D$  the coordinate  $(0,0)$ ,  $A$  the coordinate  $(0,1)$ , and so forth.  $\overline{AC}$  and  $\overline{EB}$  can be represented by the equations  $y = -x + 1$  and  $y = 2x - 1$ , respectively. Solving for their intersection gives point  $F$  coordinates  $(\frac{2}{3}, \frac{1}{3})$ .

Now, we can see that  $\triangle EFC$ 's area is simply  $\frac{\frac{1}{2} \cdot \frac{1}{3}}{2}$  or  $\frac{1}{12}$ . This means that pentagon  $ABCEF$ 's area is  $\frac{1}{2} + \frac{1}{12} = \frac{7}{12}$  of the entire square, and it follows that quadrilateral  $AFED$ 's area is  $\frac{5}{12}$  of the square.

The area of the square is then  $\frac{45}{\frac{5}{12}} = 9 \cdot 12 = \mathbf{(B) 108}$ .

**Solution 2**

Let  $x$  be the sides of the square.  $\triangle ABC$  has half the area of the square.  $\triangle FEC$  has base equal to half the square side length, and by AA Similarity with  $\triangle FBA$ , it has  $\frac{1}{1+2} = \frac{1}{3}$  the height, so has  $\frac{1}{12}$ th of the area of square ( $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{3}$  according to the triangle area formula). Thus, the area of the quadrilateral is  $1 - \frac{1}{3} - \frac{1}{12} = \frac{5}{12}$  of the area of the square. We get  $\frac{1}{3}$  because since the side length ratio of  $EC$  to  $AB$  is 2, the area ratio would be 4. Hence,  $\frac{1}{12} \cdot 4 = \frac{1}{3}$ .

The area of  $\triangle ADC$  is  $\frac{x \cdot x}{2}$ . Thus, the area of  $AFED$  equals to the area of  $\triangle ADC$  minus the area of  $\triangle FEC$ , which is  $\frac{1}{12}$  of  $x^2$  (the area of the square).

We can write the equation:  $\frac{x^2}{2} - \frac{x^2}{12} = 45$ .

Solving the equation, we get  $x = \sqrt{108}$ .

The area of the square is then  $\sqrt{108} \cdot \sqrt{108} = \mathbf{(B) 108}$ .

**Solution 3**

Extend  $\overline{AD}$  and  $\overline{BE}$  to meet at  $X$ . Drop an altitude from  $F$  to  $\overline{CE}$  and call it  $h$ . Also, call  $\overline{CE}$   $x$ . As stated before, we have  $\triangle ABF \sim \triangle CEF$ , so the ratio of their heights is

in a 1 : 2 ratio, making the altitude from  $F$  to  $\overline{AB}$   $2h$ . Note that this means that the side of the square is  $3h$ . In addition,  $\triangle XDE \sim \triangle XAB$  by AA Similarity in a 1 : 2 ratio. This means that the square's side length is  $2x$ , making  $3h = 2x$ .

Now, note that  $[ADEF] = [XAB] - [XDE] - [ABF]$ . We have  $[\triangle XAB] = (4x)(2x)/2 = 4x^2$ ,  $[\triangle XDE] = (x)(2x)/2 = x^2$ , and  $[\triangle ABF] = (2x)(2h)/2 = (2x)(4x/3)/2 = 4x^2/3$ . Subtracting makes  $[ADEF] = 4x^2 - x^2 - 4x^2/3 = 5x^2/3$ . We are given that  $[ADEF] = 45$ , so  $5x^2/3 = 45 \Rightarrow x^2 = 27$ . Therefore,  $x = 3\sqrt{3}$ , so our answer is  $(2x)^2 = 4x^2 = 4(27) = \boxed{\text{(B)} 108}$ .

#### Solution 4

Solution with Cartesian and Barycentric Coordinates:

We start with the following:

Claim: Given a square  $ABCD$ , let  $E$  be the midpoint of  $\overline{DC}$  and let  $BE \cap AC = F$ . Then  $\frac{AF}{FC} = 2$ .

Proof: We use Cartesian coordinates. Let  $D$  be the origin,  $A = (0, 1)$ ,  $C = (0, 1)$ ,  $B = (1, 1)$ . We have that  $\overline{AC}$  and  $\overline{EB}$  are governed by the equations  $y = -x + 1$  and  $y = 2x - 1$ , respectively. Solving,  $F = (\frac{2}{3}, \frac{1}{3})$ . The result follows.  $\square$

Now, we apply Barycentric Coordinates w.r.t.  $\triangle ACD$ . We let  $A = (1, 0, 0)$ ,  $D = (0, 1, 0)$ ,  $C = (0, 0, 1)$ . Then  $E = (0, \frac{1}{2}, \frac{1}{2})$ ,  $F = (\frac{1}{3}, 0, \frac{2}{3})$ .

In the barycentric coordinate system, the area formula is  $[XYZ] = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \cdot [ABC]$

where  $\triangle XYZ$  is a random triangle and  $\triangle ABC$  is the reference triangle. Using this, we find that

$$\frac{[FEC]}{[ACD]} = \frac{\begin{vmatrix} 0 & 0 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & 0 & \frac{2}{3} \end{vmatrix}}{\begin{vmatrix} 0 & 0 & 1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & 0 & \frac{2}{3} \end{vmatrix}} = \frac{1}{6}.$$

Let  $[FEC] = x$  so that  $[ACD] = 45 + x$ . Then, we have  $\frac{x}{x+45} = \frac{1}{6} \Rightarrow x = 9$ , so the answer is  $2(45 + 9) = \boxed{108}$ .

Note: Please do not learn Barycentric Coordinates for the AMC 8.

#### Solution 5 (easier)

$\triangle ABF \sim \triangle FEC$ , and the bases  $AB$  and  $EC$  are in the ratio 2 : 1, so  $\frac{[\triangle ABF]}{[\triangle FEC]} = (\frac{2}{1})^2 = \frac{4}{1}$ , and  $[\triangle ABF] = 4 \cdot [\triangle FEC]$

It is obvious that  $[AFED] + [\triangle FEC] = 45 + [\triangle FEC] = \frac{1}{2} \cdot [ABCD]$

and  $[AFED] + [\triangle ABF] = 45 + 4 \cdot [\triangle FEC] = \frac{3}{4} \cdot [ABCD]$

Solving the two equations we get  $[ABCD] = \boxed{\text{(B)} 108}$

- bbidev

#### Solution 6 (Similar to 5)

Since we know that point  $E$  is the midpoint of square with area  $4x^2$  (Side length of  $2x$ ) we can easily tell that  $\overline{EC}$  is equal to  $\overline{DE}$ . This especially helpful when we realize that  $\triangle ECF \sim \triangle ABF$ . Our setup is over, it is time to solve!

We can label the area of  $\triangle ECF$  as  $a$ , which means that  $\triangle ABF$  has an area of  $4a$  (This is because the width and height doubles which means the area is four times that of  $\triangle ECF$ ). Since we know that the area of quadrilateral  $ADEF$  is 45, this means that  $45 + a = 2x^2$  (half of the area due to diagonal).

Now we should find the area of trapezoid  $ABED$ . Since the bases are  $x$  and  $2x$  and the height is also  $2x$ , we can plug these values into the formula for the area of the trapezoid ( $\frac{b_1+b_2}{2} \cdot h$ ), and we will get the area as  $3x^2$  which is equal to  $45 + 4a$ . We now have our two equations:  $45 + a = 2x^2$  and  $45 + 4a = 3x^2$ .

We can now multiply the first equation by four so we can easily subtract the variable  $a$ . Doing this we acquire  $180 + 4a = 8x^2$ , and when we subtract both equations, we get  $135 = 5x^2$ . Remember, if we want to find the area, we have to look for  $4x^2$ , which is  $\frac{135}{5} \cdot 4$ . 135 divided by 5 is 27, and 27 times 4 is **(B)108**.

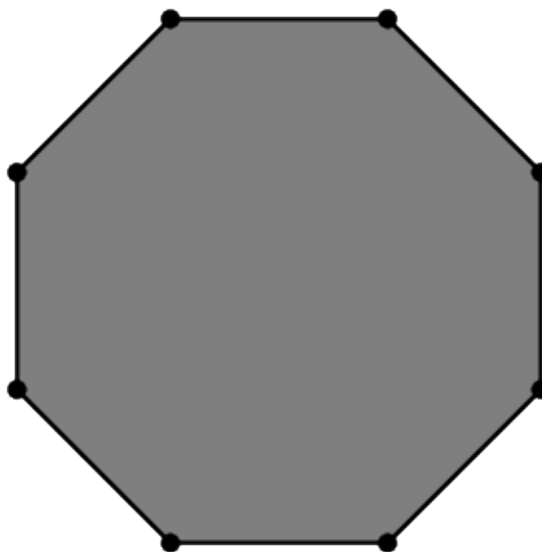
**Solution 7 (Basic Concepts)**

Set the length of  $EC$  to  $x$ . We now will proceed to find the altitude of triangle  $FEC$  if we consider line  $EC$  to be the base.

Notice that  $\triangle FEC$  and  $\triangle FAB$  are similar. Since the ratio of their bases is  $1 : 2$ , respectively, we can now set up a proportion:  $\frac{x}{y} = \frac{2x}{2x - y}$ . Cross-multiplying and simplifying gives the equation  $2x = 3y$ , or  $y = \frac{2}{3}x$ . From there, we can see that the area of  $\triangle FEC = \frac{1}{3}x^2$ .

Now, we subtract the area of  $\triangle FEC$  from  $\triangle ADC$  and set it equal to 45. Our equation then becomes  $2x^2 - \frac{1}{3}x^2 = 45$ , which upon solving for  $x$  yields  $x = 3\sqrt{3}$ , so the side length of the square is  $6\sqrt{3}$ , and squaring  $6\sqrt{3}$  gives **(B)108**.

(AMC 8 2018) From a regular octagon, a triangle is formed by connecting three randomly chosen vertices of the octagon. What is the probability that at least one of the sides of the triangle is also a side of the octagon?



- (A)  $\frac{2}{7}$     (B)  $\frac{5}{42}$     (C)  $\frac{11}{14}$     (D)  $\frac{5}{7}$     (E)  $\frac{6}{7}$

**Answer:** D

**Solution 1**

Firstly, we calculate the amount of all possible triangles. It is  $C_8^3 = 56$ . Then, we consider how many triangles have at least a edge which is also the edge of octagon.

If the triangle shares only one edge with the octagon, fix an edge, we can choose 4 different vertices for the triangle. Thus, there are  $8 \times 4 = 32$  possible triangles. If the triangle shares its two edges with the octagon, obviously, there are only 8 possible triangles.  $\frac{32+8}{56} = \textbf{(D)} \frac{5}{7}$ .

**Solution 2**

We split into two cases:

Case 1: Exactly one side is chosen. There are 4 lines that each contain 1 side of the octagon, and there are 8 points on the octagon. Thus, the total number of such lines is:

$$8 \cdot 4 = 32. \quad (1)$$

Case 2: Exactly two sides are chosen. The number of ways to choose 2 points on the octagon is:

$$\binom{8}{1} = 8. \quad (2)$$

Adding the results from both cases, the total is:

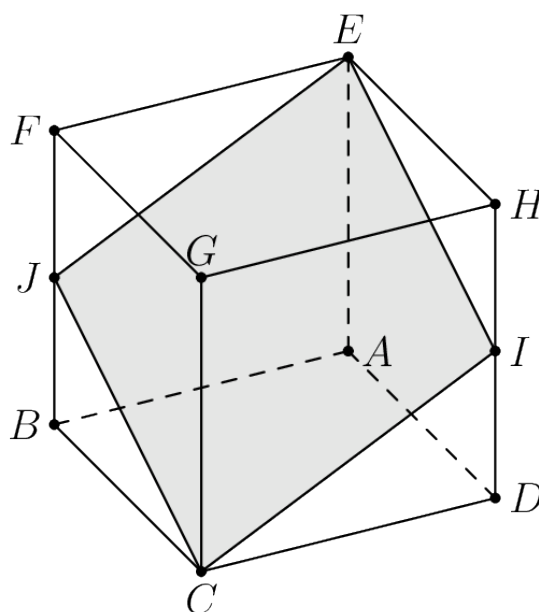
$$32 + 8 = \boxed{40}. \quad (3)$$

This is the numerator. To find the denominator we do  $C_8^3 = 56$ . Thus, our final answer is  $40/56 = \boxed{(D) \frac{5}{7}}$ .

-jb2015007

-FrankensteinQuixoteCabin

(AMC 8 2018) In the cube  $ABCDEFGH$  with opposite vertices  $C$  and  $E$ ,  $J$  and  $I$  are the midpoints of edges  $\overline{FB}$  and  $\overline{HD}$ , respectively. Let  $R$  be the ratio of the area of the cross-section  $EJCI$  to the area of one of the faces of the cube. What is  $R^2$ ?



- (A)  $\frac{5}{4}$     (B)  $\frac{4}{3}$     (C)  $\frac{3}{2}$     (D)  $\frac{25}{16}$     (E)  $\frac{9}{4}$

Answer: C

**Solution 1**

Note that  $EJCI$  is a rhombus by symmetry. Let the side length of the cube be  $s$ . By the Pythagorean theorem,  $EC = s\sqrt{3}$  and  $JI = s\sqrt{2}$ . Since the area of a rhombus is half the product of its diagonals, the area of the cross section is  $\frac{s^2\sqrt{6}}{2}$ . This gives  $R = \frac{\sqrt{6}}{2}$ .

Thus,  $R^2 = \boxed{(C) \frac{3}{2}}$ .

**Solution 2**

This time, instead of using a variable like we did in Solution 1,  $s$ , for the side length as in the above solution, choose an easy value for  $s$  such as 2. In the above solution,  $s^2$  cancels out in the end, so ultimately the answers are equivalent.

**Solution 3 (Coordinate Geometry)**

If the edges of the cube have same lengths  $a$ ,  $A$  is the origin,  $\vec{AD}$  is the positive  $x$  direction,  $\vec{AE}$  is the positive  $y$  direction, and  $\vec{AB}$  is the positive  $z$  direction. Therefore, we have  $E(0, a, 0)$ ,  $I(a, \frac{a}{2}, 0)$ ,  $C(a, 0, a)$ , and  $J(0, \frac{a}{2}, a)$ . Hence, we can figure out that:

$$|EC| = \sqrt{a^2 + a^2 + a^2} = \sqrt{3}a$$

$$|IJ| = \sqrt{a^2 + 0 + a^2} = \sqrt{2}a$$

Note that  $EJCI$  is a rhombus, so  $R = \frac{\sqrt{6}a^2}{2a^2} = \frac{\sqrt{6}}{2}$ . Finally, we can see that the answer is  $R^2 = \frac{6}{4} = \boxed{\text{(C)} \frac{3}{2}}$

**Solution 4 (AMC 10+ tactics)**

We can solve this with 3D Cartesian coordinates. Assume WLOG that the sides of the square are of length 2. Let  $C$  be the origin and let  $\vec{CD}$  be the positive  $x$  direction,  $\vec{CG}$  be the positive  $y$  direction, and  $\vec{CB}$  be the positive  $z$  direction. We find that  $I = (2, 1, 0)$ ,  $J = (0, 1, 2)$ ,  $E = (2, 2, 2)$ .

Notice that  $CI = CJ = EI = EJ = \sqrt{5}$  so  $EJCI$  is a rhombus. Furthermore, by the distance formula,  $IJ = \sqrt{8}$ .

By the Law of Cosines on  $\triangle JCI$  we have  $\cos \angle JCI = \frac{\sqrt{5}^2 + \sqrt{5}^2 - \sqrt{8}^2}{2 \cdot \sqrt{5} \cdot \sqrt{5}} = \frac{1}{5}$ . By the Law of Cosines on  $\triangle JEI$  we have  $\cos \angle JEI = \frac{\sqrt{5}^2 + \sqrt{5}^2 - \sqrt{8}^2}{2 \cdot \sqrt{5} \cdot \sqrt{5}} = \frac{1}{5}$ .

Brahmagupta's formula states given a quadrilateral  $ABCD$  with sides  $a, b, c, d$  then

$$[ABCD] = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd \cos^2 \left( \frac{B+D}{2} \right)}$$

where  $s = \frac{a+b+c+d}{2}$ . Using this formula, we find that

$$[EJCI] = \sqrt{(2\sqrt{5} - \sqrt{5})(2\sqrt{5} - \sqrt{5})(2\sqrt{5} - \sqrt{5})(2\sqrt{5} - \sqrt{5}) - \sqrt{5}^4 \left( \frac{1}{5} \right)^2} = 2\sqrt{6}.$$

Using Brahmagupta's formula again, we can find that  $[ABCD] = \sqrt{(4-2)(4-2)(4-2)(4-2) - 2^4 \cdot \cos^2 \left( \frac{B+D}{2} \right)} = \sqrt{16} = 4$ .

The answer is thus  $\left( \frac{2\sqrt{6}}{4} \right)^2 = \frac{3}{2}$  so we circle answer choice  $\boxed{\text{(C)} \frac{3}{2}}$

**Solution 5 (Not for use in a time-limited contest)**

We will use the following

Theorem: Suppose we have a quadrilateral with edges of length  $a, b, c, d$  (in that order) and diagonals of length  $p, q$ . Bretschneider's formula states that the area  $[ABCD] = \frac{1}{4} \cdot \sqrt{4p^2q^2 - (b^2 + d^2 - a^2 - c^2)^2}$ .

Proof: Here is one of my favorite proofs of this with vector geometry.

Suppose a quadrilateral has sides  $\vec{a}, \vec{b}, \vec{c}, \vec{d}$  such that  $\vec{a} + \vec{b} + \vec{c} + \vec{d} = \vec{0}$  and that the diagonals of the quadrilateral are  $\vec{p} = \vec{b} + \vec{c} = -\vec{a} - \vec{d}$  and  $\vec{q} = \vec{a} + \vec{b} = -\vec{c} - \vec{d}$ . The area of any such quadrilateral is  $\frac{1}{2}|\vec{p} \times \vec{q}|$ .

$$K = \frac{1}{2}|\vec{p} \times \vec{q}|$$

Lagrange's Identity states that  $|\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = |\vec{a} \times \vec{b}|^2 \implies \sqrt{|\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2} = |\vec{a} \times \vec{b}|$ . Therefore:

$$\begin{aligned} K &= \frac{1}{2} \sqrt{|\vec{p}|^2|\vec{q}|^2 - (\vec{p} \cdot \vec{q})^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - (2\vec{p} \cdot \vec{q})^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [2(\vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b})]^2} \\ &= \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [2\vec{b} \cdot (\vec{a} + \vec{b}) + 2\vec{c} \cdot (\vec{a} + \vec{b})]^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [-2\vec{b} \cdot (\vec{c} + \vec{d}) + 2\vec{c} \cdot (\vec{a} + \vec{b})]^2} \\ &= \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [-2\vec{b} \cdot \vec{c} - 2\vec{b} \cdot \vec{d} + 2\vec{a} \cdot \vec{c} + 2\vec{b} \cdot \vec{c}]^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [2\vec{a} \cdot \vec{c} - 2\vec{b} \cdot \vec{d}]^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - ([(\vec{a} + \vec{c}) \cdot (\vec{b} + \vec{d})])^2} \\ &= \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [|\vec{b}|^2 + |\vec{d}|^2 - |\vec{a}|^2 - |\vec{c}|^2 + (\vec{a} + \vec{c}) \cdot (\vec{a} + \vec{c}) - (\vec{b} + \vec{d}) \cdot (\vec{b} + \vec{d})]^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [|\vec{b}|^2 + |\vec{d}|^2 - |\vec{a}|^2 - |\vec{c}|^2]^2} \\ &= \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [|\vec{b}|^2 + |\vec{d}|^2 - |\vec{a}|^2 - |\vec{c}|^2 + |\vec{a} + \vec{c}|^2 - |(\vec{a} + \vec{c})|^2]^2} = \frac{1}{4} \sqrt{4|\vec{p}|^2|\vec{q}|^2 - [|\vec{b}|^2 + |\vec{d}|^2 - |\vec{a}|^2 - |\vec{c}|^2]^2} \end{aligned}$$

Then, if  $a, b, c, d$  represent  $|\vec{a}|, |\vec{b}|, |\vec{c}|, |\vec{d}|$  (and are thus the side lengths) while  $p, q$  represent  $|\vec{p}|, |\vec{q}|$  (and are thus the diagonal lengths), the area of a quadrilateral is:

$$K = \frac{1}{4} \sqrt{4p^2q^2 - (b^2 + d^2 - a^2 - c^2)^2}$$

□

Back to the problem. We use vectors. WLOG suppose the cube has sides of length 2. Let  $C$  be the origin and let  $\vec{CD}$  be the  $x$  direction,  $\vec{CG}$  be the  $y$  direction, and  $\vec{CB}$  be the  $z$  direction.

Then,  $\vec{CI} = \vec{JE} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$  and  $\vec{CJ} = \vec{IE} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ . Let  $\angle JCI = \theta$ . Dot product gives

$$\cos \theta = \frac{\left( \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right)}{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right| \left| \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right|} = \frac{1}{5}.$$

Hence,  $\vec{IJ} = \sqrt{|\vec{CI}|^2 + |\vec{CJ}|^2 - 2 \cdot \vec{CI} \cdot \vec{CJ} \cos \theta} = \sqrt{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right|^2 + \left| \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right|^2 - 2 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \cdot \frac{1}{5}} \iff$ 

$$|\vec{IJ}| = 2\sqrt{2}.$$

Now, notice that  $\vec{CE} = \vec{CI} + \vec{CJ} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$  so  $|\vec{CE}| = \left| \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \right| = 2\sqrt{3}$ .

Using Bretschneider's formula, we obtain  $[EJCI] = \frac{1}{4} \sqrt{4|\vec{IJ}|^2|\vec{CE}|^2 - (|\vec{CJ}|^2 + |\vec{IE}|^2 - |\vec{JE}|^2 - |\vec{CI}|^2)^2} = \frac{1}{4} \sqrt{4 \cdot (2\sqrt{2})^2 \cdot (2\sqrt{3})^2 - (5^2 + 5^2 - 5^2 - 5^2)^2} = 2\sqrt{6}$ .

Using Bretschneider's formula again, we can find that  $[ABCD] = \sqrt{(4-2)(4-2)(4-2)(4-2) - 2^4 \cdot \cos \theta} = \sqrt{16} = 4$ .

Hence, the answer is  $\left(\frac{2\sqrt{6}}{4}\right)^2 = \frac{3}{2}$  so we circle answer choice  $C$ .

Note: Of course this is a solution made by Franz Liszt, and also, holy shheeesshh this solution is long

### Solution 6

We can simply plug in values and find  $R$  to eliminate the variables and avoid the complicated algebra. Let's assume the side length of the cube is 4. The given figure is a rhombus because all its sides are equal, but the angles are not  $90^\circ$ . To find the area of the rhombus, we need to calculate the lengths of the two diagonals.

The formula for the space diagonal of a cube is  $\sqrt{3}s$ , so  $\overline{CE} = 4\sqrt{3}$ . By symmetry,  $\overline{IJ} = \overline{BD} = 4\sqrt{2}$ . Now, we can find the area of the rhombus using the formula for the area of a rhombus, which is the product of the diagonals divided by 2:

$$\begin{aligned}\text{Area} &= \frac{4\sqrt{3} \cdot 4\sqrt{2}}{2} \\ &= 8\sqrt{6}.\end{aligned}$$

Next, the area of one of the cube's faces is simply  $4^2 = 16$ . So,  $R$  is:

$$\left(\frac{8\sqrt{6}}{16}\right)^2 = \boxed{\text{(C)} \frac{3}{2}}.$$

-jb2015007

(Minor edits by FrankensteinQuixoteCabin)

**Note** Problem 21 of the 2008 AMC 10A was nearly identical to this question, except that in this question you have to look for the square of the area, not the actual area.

(AMC 8 2018) How many perfect cubes lie between  $2^8 + 1$  and  $2^{18} + 1$ , inclusive?

(A) 4      (B) 9      (C) 10      (D) 57      (E) 58

**Answer:** E

**Solution 1 (Brute force)**

First,  $2^8 + 1 = 257$ . Then,  $2^{18} + 1 = 262145$ . Now, we can see how many perfect cubes are between these two parameters. By guessing and checking, we find that it starts from 7 and ends with 64. Now, by counting how many numbers are between these, we find the answer to be  $\boxed{\text{(E)} 58}$ . (You can do this, but it will still take a few minutes. Try to do something else)

**Solution 2 (If you do not notice that )**

There is not so much guessing and checking after we find that it starts from 7 because  $7^3 = 343$ , which is over  $2^8 + 1 = 257$ . We can start guessing with the 10, answer C, as it is the middle value. Adding 10 to 7 gives us 17 and  $17^3 = 4,913$ , which is a bit low. So, we move "up" to 57, answer D. Adding 57 to 7 gives us 64 and  $64^3 = 262,144$ , which is perfect for the  $2^{18} + 1 = 262,145$ . But we are not done. Since it is inclusive, we must add 1 to this solution, which give us  $\boxed{\text{(E)} 58}$ .

**Solution 3 (Super fast!)**

We can immediately evaluate  $2^8 + 1$  which gives us 257 and the smallest perfect cube greater than that is  $343 = 7^3$ , and you also immediately notice that  $2^{18} = (2^6)^3 = 64^3$ . Now, all we have to do is count the cubes, which equals:

$$64 - 7 + 1 = \boxed{\text{(E)} 58} \tag{4}$$

-jb2015007

**Solution 4(Bascially the same as solution 3)**

We know that  $2^8 + 1 = 257$  which is only a little higher than  $6^3 = 216$ . This means that the lower bound of this is  $7^3$ .  $2^{18}$  is the same thing is  $64^3$ , which must be the upper bound. This results in  $64 - 7 + 1 = \boxed{\text{(E)} 58}$

## §35 AMC 8 2019

- (AMC 8 2019) Ike and Mike go into a sandwich shop with a total of \$30.00 to spend. Sandwiches cost \$4.50 each and soft drinks cost \$1.00 each. Ike and Mike plan to buy as many sandwiches as they can, and use any remaining money to buy

soft drinks. Counting both sandwiches and soft drinks, how many items will they buy?

- (A) 6      (B) 7      (C) 8      (D) 9      (E) 10

**Answer:** D

**Solution 1**

We know that the sandwiches cost 4.50 dollars. Guessing will bring us to multiplying 4.50 by 6, which gives us 27.00. Since they can spend 30.00 they have 3 dollars left. Since soft drinks cost 1.00 dollar each, they can buy 3 soft drinks, which makes them spend 30.00. Since they bought 6 sandwiches and 3 soft drinks, they bought a total of 9 items. Therefore, the answer is **(D) 9**.

- SBose

**Solution 2 (Using Algebra)**

Let  $s$  be the number of sandwiches and  $d$  be the number of sodas. We have to satisfy the equation of

$$4.50s + d = 30$$

In the question, it states that Ike and Mike buys as many sandwiches as possible. So, we drop the number of sodas for a while. We have:

$$\begin{aligned} 4.50s &= 30 \\ s &= \frac{30}{4.5} \\ s &= 6R3 \end{aligned}$$

We don't want a remainder so the maximum number of sandwiches is 6. The total money spent is  $6 \cdot 4.50 = 27$ . The number of dollar left to spent on sodas is  $30 - 27 = 3$  dollars. 3 dollars can buy 3 sodas leading us to a total of  $6 + 3 = 9$  items. Hence, the answer is **(D) 9**.

- by interactivemath

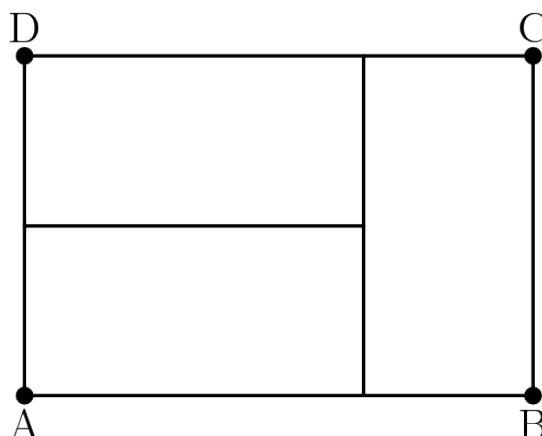
**Solution 3**

If the two could buy fractions of sandwiches, they'd be able to buy  $\frac{30.00 \text{ dollars}}{4.50 \frac{\text{dollars}}{\text{sandwich}}} = 6\frac{2}{3}$  sandwiches. But they can only buy a whole number of sandwiches, so they can buy 6, which costs  $6 \cdot \$4.50 = \$27$ . So they have \$3 left over, from which they can buy 3 drinks (since they each cost \$1, for a total item count of  $6 + 3 = 9$  items. The answer is **(D)**.

-Elihem

2. (AMC 8 2019) Three identical rectangles are put together to form rectangle  $ABCD$ , as shown in the figure below. Given that the length of the shorter side of each of the rectangles is 5 feet, what is the area in square feet of rectangle  $ABCD$ ?





- (A) 45      (B) 75      (C) 100      (D) 125      (E) 150

**Answer:** E

**Solution 1**

We can see that there are 2 rectangles lying on top of the other and that is the same as the length of one rectangle. Now we know that the shorter side is 5, so the bigger side is 10, if we do  $5 \cdot 2 = 10$ . Now we get the sides of the big rectangle being 15 and 10, so the area is **(E) 150**. avamarora

**Solution 2**

Using the diagram we find that the larger side of the small rectangle is 2 times the length of the smaller side. Therefore, the longer side is  $5 \cdot 2 = 10$ . So the area of the identical rectangles is  $5 \cdot 10 = 50$ . We have 3 identical rectangles that form the large rectangle. Therefore the area of the large rectangle is  $50 \cdot 3 =$  **(E) 150**. fath2012

**Solution 3**

We see that if the short sides are 5, the long side has to be  $5 \cdot 2 = 10$  because the long side is equal to the 2 short sides and because the rectangles are congruent. If that is to be, then the long side of the BIG rectangle(rectangle  $ABCD$ ) is  $10 + 5 = 15$  because long side + short side of the small rectangle is 15. The short side of rectangle  $ABCD$  is 10 because it is the long side of the short rectangle. Multiplying 15 and 10 together gets us  $15 \cdot 10$  which is **(E) 150**. mathboy282

**Solution 4**

We see that the 2 rectangles lying on top of each other give us the height of the rectangle. Using what we know, we can find out that the 2 rectangles put together is a square. So, we can infer that the length of the rectangles is 10. Adding that to the width of the third rectangle which is 5, we get that the length of the rectangle is 15. Multiplying 10 and 15 gives us  $15 \cdot 10$  which is **(E) 150**. awesomepony566

**Solution 5**

There are two rectangles lying on the side and one standing up. Given that one small side is 5, we can determine that two of the small sides make up a big side which means that the long side is equal to 10. The top side of the rectangle is made up of one small side and one long side, therefore the dimensions for the rectangle is  $10 \times 15$ . 10 multiplied by 15 is 150, hence the answer **(E) 150**. elenafan

**Remark** ALL THESE SOLUTIONS ARE VERY, VERY SIMILAR

3. (AMC 8 2019) Which of the following is the correct order of the fractions  $\frac{15}{11}$ ,  $\frac{19}{15}$ , and  $\frac{17}{13}$ , from least to greatest?

(A)  $\frac{15}{11} < \frac{17}{13} < \frac{19}{15}$  (B)  $\frac{15}{11} < \frac{19}{15} < \frac{17}{13}$  (C)  $\frac{17}{13} < \frac{19}{15} < \frac{15}{11}$  (D)  $\frac{19}{15} < \frac{15}{11} < \frac{17}{13}$  (E)  $\frac{19}{15} < \frac{17}{13} < \frac{15}{11}$

**Answer:** E

**Solution 1 (Bashing/Butterfly Method)**

We take a common denominator:

$$\frac{15}{11}, \frac{19}{15}, \frac{17}{13} = \frac{15 \cdot 15 \cdot 13}{11 \cdot 15 \cdot 13}, \frac{19 \cdot 11 \cdot 13}{15 \cdot 11 \cdot 13}, \frac{17 \cdot 11 \cdot 15}{13 \cdot 11 \cdot 15} = \frac{2925}{2145}, \frac{2717}{2145}, \frac{2805}{2145}.$$

Since  $2717 < 2805 < 2925$  it follows that the answer is (E)  $\frac{19}{15} < \frac{17}{13} < \frac{15}{11}$ .

Another approach to this problem is using the properties of one fraction being greater than another, also known as the **butterfly method**. That is, if  $\frac{a}{b} > \frac{c}{d}$ , then it must be true that  $a * d$  is greater than  $b * c$ . Using this approach, we can check for **at least** two distinct pairs of fractions and find out the greater one of

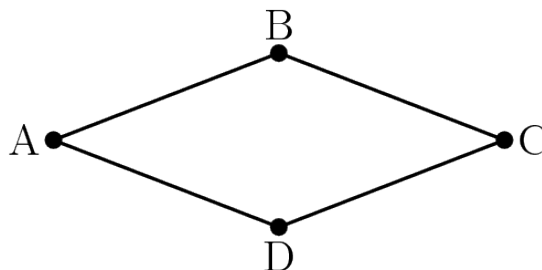
those two, logically giving us the expected answer of (E)  $\frac{19}{15} < \frac{17}{13} < \frac{15}{11}$ .

-xMidnightFirex

- Clearness by doulai1

- A

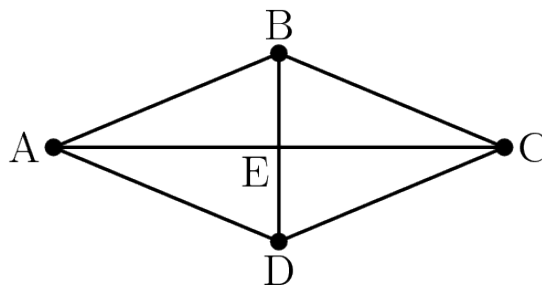
4. (AMC 8 2019) Quadrilateral  $ABCD$  is a rhombus with perimeter 52 meters. The length of diagonal  $\overline{AC}$  is 24 meters. What is the area in square meters of rhombus  $ABCD$ ?



(A) 60 (B) 90 (C) 105 (D) 120 (E) 144

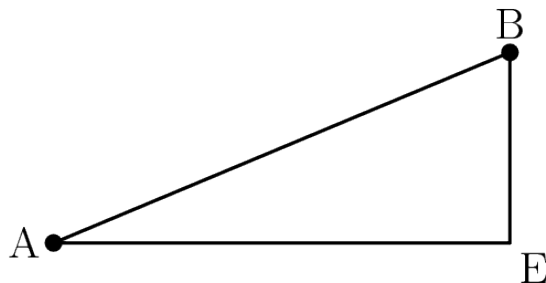
**Answer:** D

**Solution 1**



A rhombus has sides of equal length. Because the perimeter of the rhombus is 52, each side is  $\frac{52}{4} = 13$ . In a rhombus, diagonals are perpendicular and bisect each other, which means  $\overline{AE} = 12 = \overline{EC}$ .

Consider one of the



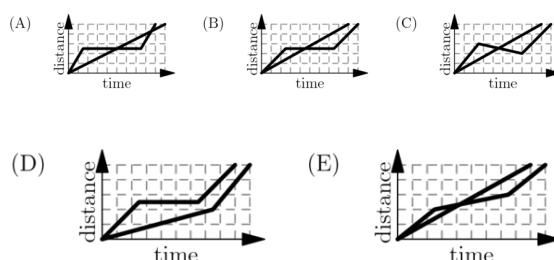
$\overline{AB} = 13$ , and  $\overline{AE} = 12$ . Using the Pythagorean theorem, we find that  $\overline{BE} = 5$ . You know the Pythagorean triple, (5, 12, 13).

Thus the values of the two diagonals are  $\overline{AC} = 24$  and  $\overline{BD} = 10$ . The area of a rhombus is  $= \frac{d_1 \cdot d_2}{2} = \frac{24 \cdot 10}{2} = 120$  **(D) 120**

### Solution 2

Right off the bat, we can see that the perimeter of the figure is 52. Dividing this by four, we can get that each side is equal to 13. By drawing a line perpendicular to the one given, we can split the figure into four right triangles. 12 ( $24/2$ ) is equal to the height of one small right triangle, and 13 is the slanted side. Using the Pythagorean theorem we can find that  $169$  ( $13$  squared)  $- 144$  ( $12$  squared)  $= 25$  (five squared). With this, we can determine that each small right triangle equals 30. Multiplying that by four we can get **(D) 120**.

5. (AMC 8 2019) A tortoise challenges a hare to a race. The hare eagerly agrees and quickly runs ahead, leaving the slow-moving tortoise behind, believing that he will win, the hare stops to take a nap. Meanwhile, the tortoise walks at a slow, steady pace for the entire race. The hare wakes up and runs to the finish line, only to find the tortoise already there. Which of the following graphs matches the description of the race, showing the distance  $d$  traveled by the two animals over time  $t$  from start to finish?

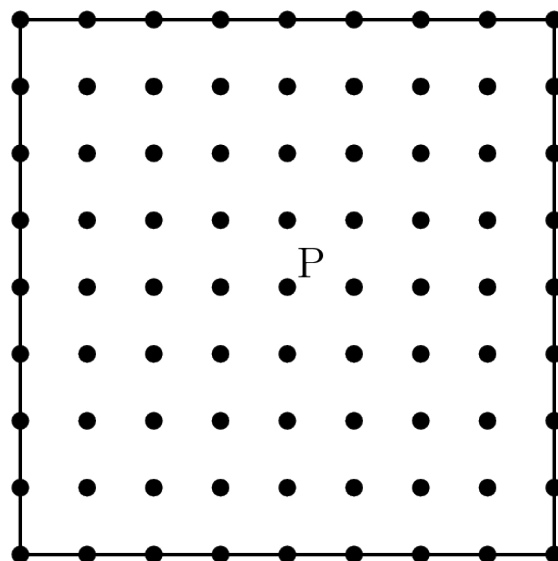


**Answer: B**

### Solution

First, the tortoise walks at a constant rate, ruling out (D). Second, when the hare is resting, the distance will stay the same, ruling out (E) and (C). Third, the tortoise wins the race, meaning that the non-constant one should go off the graph last, ruling out (A). The answer **(B)** is the only one left.

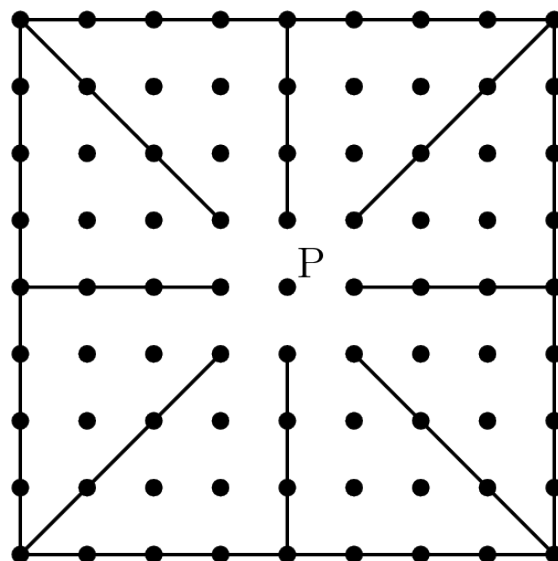
6. (AMC 8 2019) There are 81 grid points (uniformly spaced) in the square shown in the diagram below, including the points on the edges. Point  $P$  is in the center of the square. Given that point  $Q$  is randomly chosen among the other 80 points, what is the probability that the line  $PQ$  is a line of symmetry for the square?



- (A)  $\frac{1}{5}$     (B)  $\frac{1}{4}$     (C)  $\frac{2}{5}$     (D)  $\frac{9}{20}$     (E)  $\frac{1}{2}$

**Answer:** C

**Solution 1**



Lines of symmetry go through point  $P$ , and there are 8 directions the lines could go, and there are 4 dots at each direction.  $\frac{4 \times 8}{80} = \boxed{\text{(C)} \frac{2}{5}}$ .

**Solution 2**

Divide the grid into 4  $4 \times 5$  quadrants. Each row of 5 points has 1 point on a horizontal/vertical line of symmetry + 1 point on a diagonal line of symmetry:

$$\boxed{\text{(C)} \frac{2}{5}}.$$

7. (AMC 8 2019) Shauna takes five tests, each worth a maximum of 100 points. Her scores on the first three tests are 76, 94, and 87. In order to average 81 for all five tests, what is the lowest score she could earn on one of the other two tests?

(A) 48      (B) 52      (C) 66      (D) 70      (E) 74

**Answer:** A

**Solution 1**

We should notice that we can turn the information we are given into a linear equation and just solve for our set variables. I'll use the variables  $x$  and  $y$  for the scores on the last two tests.

$$\frac{76 + 94 + 87 + x + y}{5} = 81,$$

$$\frac{257 + x + y}{5} = 81.$$

We can now cross multiply to get rid of the denominator.

$$257 + x + y = 405,$$

$$x + y = 148.$$

Now that we have this equation, we will assign  $y$  as the lowest score of the two other tests, and so:

$$x = 100,$$

$$y = 48.$$

Now we know that the lowest score on the two other tests is (A) 48.

**Solution 2**

Right now, she scored 76, 94, and 87 points, for a total of 257 points. She wants her average to be 81 for her 5 tests, so she needs to score 405 points in total. This means she needs to score a total of  $405 - 257 = 148$  points in her next 2 tests. Since the maximum score she can get on one of her 2 tests is 100, the least possible score she can get is (A) 48.

Note: You can verify that 48 is the right answer because it is the lowest answer out of the 5. Since it is possible to get 48, we are guaranteed that that is the right answer.

**Solution 3**

We can compare each of the scores with the average of 81:  $76 \rightarrow -5, 94 \rightarrow +13, 87 \rightarrow +6, 100 \rightarrow +19$ ;

So the last one has to be  $-33$  (since all the differences have to sum to 0), which corresponds to  $81 - 33 = \span style="border: 1px solid black; padding: 0 2px;">48.$

**Solution 4**

We know that she scored 76, 94, and 87 points on her first 3 tests for a total of 257 points and that she wants her average to be 81 for her 5 total tests. Therefore, she needs to score a total of 405 points. In addition, one of the final 2 tests needs to be the maximum of 100 points, to make the final test score—the one that we are looking for—the lowest score possible for her to earn. We can now see here that the sum of  $76 + 94 + 100$  has a units digit of 0 and that the final test score must have a

units digit ending with a 5. Now, 87 needs to be added to a number that makes the sum divisible by 5. Among the answer choices of 48, 52, 66, 70, and 74, only **(A) 48** has a units digit that works. ( $48 + 87 = 135$ , giving us a units digit of 5.)

**Solution 5 (most intuitive)**

For Shauna to get an average of 81, she needs to score a total of  $81 \times 5 = 405$ . One of the final two tests must earn a score of 100 to ensure the final test score is the lowest possible.  $76 + 94 + 87 + 100 = 357$ , and  $405 - 357 = \mathbf{(A) 48}$ . themathkidwoohooooo

**Solution 6 (rocket speed)**

81 times 5 = 405. Adding the units digits of 76, 94, 87, and 100 gives 7. To end with five the fifth score ends with 8 so the answer is **(A) 48**.

8. (AMC 8 2019) Gilda has a bag of marbles. She gives 20% of them to her friend Pedro. Then Gilda gives 10% of what is left to another friend, Ebony. Finally, Gilda gives 25% of what is now left in the bag to her brother Jimmy. What percentage of her original bag of marbles does Gilda have left for herself?

(A) 20      (B)  $33\frac{1}{3}$       (C) 38      (D) 45      (E) 54

**Answer:** E

**Solution 1**

After Gilda gives 20% of the marbles to Pedro, she has 80% of the marbles left. If she then gives 10% of what's left to Ebony, she has  $(0.8 \times 0.9) = 72\%$  of what she had at the beginning. Finally, she gives 25% of what's left to her brother, so she has  $(0.75 \times 0.72) \mathbf{(E) 54}$  of what she had in the beginning left.

**Solution 2**

WOLG, suppose that Gilda has 100 marbles.

Then, she gives Pedro 20% of 100, which equals 20, leaving her with 80 marbles.

Out of 80 marbles, she gives 10% of 80, which is 8 to Ebony.

Thus, she remains with 72 marbles.

Then, she gives 25% of 72, which equals 18 to Jimmy, finally leaving her with 54.

And,  $\frac{54}{100} = 54\% = \mathbf{(E) 54}$ .

**Solution 3 (Cheese)**

Use this when you have about 10 minutes left and your brain does not function correctly.

Since she gave away 20% and 10% of what is left and then another 25% of what is actually left, we can do  $20 + 10 + 25$  or 55%. But it is actually going to be a bit more than 55% because 10% of what is left is not 10% of the total amount. So, the only option that is greater than  $100\% - 55\%$  is **(E) 54**.

9. (AMC 8 2019) Alex and Felicia each have cats as pets. Alex buys cat food in cylindrical cans that are 6 cm in diameter and 12 cm high. Felicia buys cat food in cylindrical cans that are 12 cm in diameter and 6 cm high. What is the ratio of the volume of one of Alex's cans to the volume one of Felicia's cans?

(A) 1 : 4      (B) 1 : 2      (C) 1 : 1      (D) 2 : 1      (E) 4 : 1

**Answer:** B

### Solution 1

Using the formula for the volume of a cylinder  $\pi r^2 h$ , we get Alex,  $108\pi$ , and Felicia,  $216\pi$ . We can quickly notice that  $\pi$  cancels out on both sides and that Alex's volume is  $1/2$  of Felicia's leaving  $1/2 = \boxed{\text{(B)} 1 : 2}$  as the answer.

### Solution 2

Using the formula for the volume of a cylinder, we get that the volume of Alex's can is  $3^2 \cdot 12 \cdot \pi$ , and that the volume of Felicia's can is  $6^2 \cdot 6 \cdot \pi$ . Now, we divide the volume of Alex's can by the volume of Felicia's can, so we get  $\frac{1}{2}$ , which is  $\boxed{\text{(B)} 1 : 2}$ .

-(Algebruh123)2020

### Solution 3

The ratio of the numbers is  $1/2$ . Looking closely at the formula  $r^2 * h * \pi$ , we see that the  $r * h * \pi$  will cancel, meaning that the ratio of them will be  $\frac{1(2)}{2(2)} = \boxed{\text{(B)} 1 : 2}$ .

-Lcz

### Solution 4

The second can is  $\cdot 2$  size in each of 2 dimensions, and  $\cdot 1/2$  size in 1 dimension.  $2^2/2 = \boxed{\text{(B)} 1 : 2}$ .

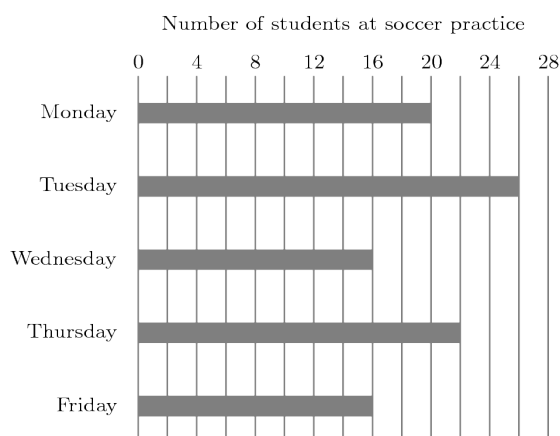
### Solution 5

Without calculating much, you can do  $(\pi r a^2) \cdot ha < -$  which is Alex's volume, with  $ra$  being Alex's radius ( $1/2 \cdot$  diameter), and  $ha$  being her cylinders height  $(\pi r f^2) \cdot hf < -$  which is Felicia's volume, with  $rf$  being Felicia's radius, and  $hf$  being her cylinders height. Since we need the ratio between Alexa's and Felicias, we can do  $(\pi r a^2) \cdot ha / (\pi r f^2) \cdot hf$  The  $\pi$  cancel out, then substitute back in the numbers, which gives you:

$$(3^2 \cdot 12) / (6^2 \cdot 6) = (9 \cdot 12) / (36 \cdot 6) = 18 / 36 = 1/2 = 1 : 2$$

-wahahaqueenie

10. (AMC 8 2019) The diagram shows the number of students at soccer practice each weekday during last week. After computing the mean and median values, Coach discovers that there were actually 21 participants on Wednesday. Which of the following statements describes the change in the mean and median after the correction is made?



- (A) The mean increases by 1 and the median does not change.  
 (B) The mean increases by 1 and the median increases by 1.  
 (C) The mean increases by 1 and the median increases by 5.  
 (D) The mean increases by 5 and the median increases by 1.  
 (E) The mean increases by 5 and the median increases by 5.

**Answer:** B

**Solution**

On Monday, 20 people come. On Tuesday, 26 people come. On Wednesday, 16 people come. On Thursday, 22 people come. Finally, on Friday, 16 people come.  $20 + 26 + 16 + 22 + 16 = 100$ , so the mean is 20. The median is (16, 16, 20, 22, 26) 20. The coach figures out that actually 21 people come on Wednesday. The new mean is 21, while the new median is (16, 20, 21, 22, 26) 21. Also, the median increases by 1 because now the median is 21 instead of 20. The median and mean both change, so the answer is **(B)**.

Another way to compute the change in mean is to notice that the sum increased by 5 with the correction. So, the average increased by  $5/5 = 1$ . Then, the median is computed the same way.

11. (AMC 8 2019) The eighth grade class at Lincoln Middle School has 93 students. Each student takes a math class, a foreign language class, or both. There are 70 eighth graders taking a math class, and there are 54 eighth graders taking a foreign language class. How many eighth graders take *only* a math class and *not* a foreign language class?
- (A) 16      (B) 23      (C) 31      (D) 39      (E) 70

**Answer:** D

**Solution 1**

Since each student must take a math class, a foreign language class, or both, all the students who do not take a foreign language class take a math class. So  $93 - 54 = \mathbf{(D) 39}$  students take a math class but not a foreign language class. (The fact that 70 students take a math class is irrelevant.)

-hailstone, edited by j314andrews

**Solution 2**

Let  $x$  be the number of students taking both a math and a foreign language class. By P-I-E, we get  $70 + 54 - x = 93$ .

Solving gives us  $x = 31$ .

But we want the number of students taking only a math class, which is  $70 - 31 = 39$ .

**(D) 39**

(i) ):):):):)

**Solution 3**

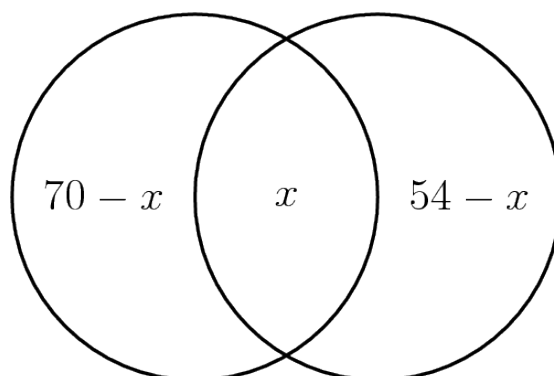
We have  $70 + 54 = 124$  people taking classes. However, we over-counted the number of people who take both classes. If we subtract the original amount of people who



take classes we get that 31 people took the two classes. To find the amount of people who took only math class, we subtract the people who didn't take only one math class, so we get  $70 - 31 = \boxed{\text{(D)} 39}$ .

-fath2012

**Solution 4**



We know that the sum of all three areas is 93. So, we have:

$$93 = 70 - x + x + 54 - x$$

$$93 = 70 + 54 - x$$

$$93 = 124 - x$$

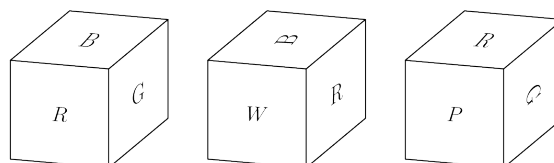
$$-31 = -x$$

$$x = 31$$

We are looking for the number of students in only math. This is  $70 - x$ . Substituting  $x$  with 31, our answer is  $\boxed{39}$ .

-mathnerdnair

12. (AMC 8 2019) The faces of a cube are painted in six different colors: red ( $R$ ), white ( $W$ ), green ( $G$ ), brown ( $B$ ), aqua ( $A$ ), and purple ( $P$ ). Three views of the cube are shown below. What is the color of the face opposite the aqua face?



- (A) red    (B) white    (C) green    (D) brown    (E) purple

**Answer:** A

**Solution 1**

$B$  is on the top, and  $R$  is on the side, and  $G$  is on the right side. That means that (image 2)  $W$  is on the left side. From the third image, you know that  $P$  must be on the bottom since  $G$  is sideways. That leaves us with the back, so the back must be  $A$ . The front is opposite of the back, so the answer is  $\boxed{\text{(A)} R}$ .

**Solution 2**

Looking closely, we can see that all faces are connected with  $R$  except for  $A$ . Thus, the answer is (A)  $R$ .

It is A, just draw it out! phoenixfire

**Solution 3**

From pic 1 and 2, we know that the  $G$ 's opposite is  $W$ . From pic 1 and 3, the  $B$ 's opposite is  $P$ . So the  $A$ 's opposite is (A)  $R$ .

13. (AMC 8 2019) A *palindrome* is a number that has the same value when read from left to right or from right to left. (For example, 12321 is a palindrome.) Let  $N$  be the least three-digit integer which is not a palindrome but which is the sum of three distinct two-digit palindromes. What is the sum of the digits of  $N$ ?

(A) 2      (B) 3      (C) 4      (D) 5      (E) 6

**Answer:** A

**Solution 1**

Note that the only positive 2-digit palindromes are multiples of 11, namely 11, 22, ..., 99. Since  $N$  is the sum of 2-digit palindromes,  $N$  is necessarily a multiple of 11. The smallest 3-digit multiple of 11 which is not a palindrome is 110, so  $N = 110$  is a candidate solution. We must check that 110 can be written as the sum of three distinct 2-digit palindromes; this suffices as  $110 = 77 + 22 + 11$ . Then,  $N = 110$ , and the sum of the digits of  $N$  is  $1 + 1 + 0 = \text{span style="border: 1px solid black; padding: 0 2px;">(A) 2.$

- There are other sets of 2-digit numbers that satisfy this rule. Some of them are  $110 = 11 + 33 + 66$  and  $110 = 22 + 33 + 55$

**Solution 2 (variant of Solution 1)**

We already know that two-digit palindromes can only be two-digit multiples of 11; which are: 11, 22, 33, 44, 55, 66, 77, 88, and 99. Since this is clear, we will need to find out the least multiple of 11 that is not a palindrome. Then, we start counting. 110... Aha! This multiple of 11, 110, not only isn't a palindrome, but it also is the sum of three distinct two-digit palindromes, for example:  $11 + 22 + 77$ ,  $22 + 33 + 55$ , and  $44 + 11 + 55$ ! The sum of  $N$ 's digits is  $1 + 1 + 0 = \text{span style="border: 1px solid black; padding: 0 2px;">(A) 2.$

Thank you to the writer of Solution 1 for inspiring me to create this!

EarthSaver 15:13, 11 June 2021 (EDT)

**Solution 3 (basically a version of the above solutions)**

As stated above, two-digit palindromes can only be two-digit multiples of 11. We can see that if we add anything that are multiples of 11 together, we will again get a multiple of 11. For instance,  $11 + 22 = 33$ . Since we know this fact and we are finding the smallest value possible, we can start with the first three-digit multiple of 11 which is 110. Since this is not a palindrome and can be the sum of 3 two-digit palindromes (see above solutions for more details), 110 fits the bill. We can see that the sum of 110's digits is  $1 + 1 + 0 = \text{span style="border: 1px solid black; padding: 0 2px;">(A) 2.$

**Solution 4 (FINALLY smth different lol)**

Two digit palindromes are two digit multiples of 11. When we take  $33 \times 3$ , we get 99, therefore 4 numbers that work for this would be 33, 33, 33 and 11. To convert this into three numbers, we can subtract 11 from the first 33, then add both 11s to the last 33 to get the set 22, 33 and 55.  $22 + 33 + 55 = 110$  and  $1 + 1 + 0 = \text{span style="border: 1px solid black; padding: 0 2px;">(A) 2.$

14. (*AMC 8 2019*) Isabella has 6 coupons that can be redeemed for free ice cream cones at Pete's Sweet Treats. In order to make the coupons last, she decides that she will redeem one every 10 days until she has used them all. She knows that Pete's is closed on Sundays, but as she circles the 6 dates on her calendar, she realizes that no circled date falls on a Sunday. On what day of the week does Isabella redeem her first coupon?

(A) Monday    (B) Tuesday    (C) Wednesday    (D) Thursday    (E) Friday

**Answer:** C

**Solution 1 ( Guess and Check! )**

Let Day 1 to Day2 denote a day where one coupon is redeemed and the day when the second coupon is redeemed.

If she starts on a Monday, she redeems her next coupon on Thursday.

Thursday to Sunday.

Thus, (A) Monday is incorrect.

If she starts on a Tuesday, she redeems her next coupon on Friday.

Friday to Monday.

Monday to Thursday.

Thursday to Sunday.

Thus (B) Tuesday is incorrect.

If she starts on a Wednesday, she redeems her next coupon on Saturday.

Saturday to Tuesday.

Tuesday to Friday.

Friday to Monday.

Monday to Thursday.

And on Thursday, she redeems her last coupon.

No Sunday occurred; thus, (C) Wednesday is correct.

Checking for the other options,

If she starts on a Thursday, she redeems her next coupon on Sunday.

Thus, (D) Thursday is incorrect.

If she starts on a Friday, she redeems her next coupon on Monday.

Monday to Thursday.

Thursday to Sunday.

Checking for the other options gave us negative results; thus, the answer is (C) Wednesday.

**Solution 2**

Let the days of the week be numbers (mod 7). 3 and 7 are coprime, so continuously adding 3 to a number (mod 7) will cycle through all numbers from 0 to 6. If a string of 6 numbers in this cycle does not contain 0, then if you minus 3 from the first number of this cycle, it will always be 0. So, the answer is (C) Wednesday.

**Solution 3**

Start counting on Sunday, to maximize the number of 10-day jumps before reaching Sunday again. Add the 10 ( $\equiv 3 \pmod{7}$ ) days to reach the first coupon day on **(C) Wednesday**.

Note: This only works when 7 and 3 are relatively prime. Otherwise, you'd prefer to start on a day whose distance from Sunday is relatively prime to the jump size.

#### Solution 4

Let Sunday be Day 0, Monday be Day 1, Tuesday be Day 2, and so forth. We see that Sundays fall on Day  $n$ , where  $n$  is a multiple of seven. If Isabella starts using her coupons on Monday (Day 1), she will fall on a Day that is a multiple of seven, a Sunday (her third coupon will be "used" on Day 21). Similarly, if she starts using her coupons on Tuesday (Day 2), Isabella will fall on a Day that is a multiple of seven (Day 42). Repeating this process, if she starts on Wednesday (Day 3), Isabella will first fall on a Day that is a multiple of seven, Day 63 (13, 23, 33, 43, 53 are not multiples of seven), but on her seventh coupon, of which she only has six. So, the answer is **(C) Wednesday**.

#### Solution 5

There is a ten day gap, so every time you are 3 days ahead. For Sunday to be the 7th day you must be 6 days ahead, so  $6 \times 3 = 18$ . 18 days prior is the same as 4 days prior to Sunday which is **(C) Wednesday**.

15. (AMC 8 2019) On a beach, 50 people are wearing sunglasses and 35 people are wearing caps. Some people are wearing both sunglasses and caps. If one of the people wearing a cap is selected at random, the probability that this person is also wearing sunglasses is  $\frac{2}{5}$ . If instead, someone wearing sunglasses is selected at random, what is the probability that this person is also wearing a cap?
- (A)  $\frac{14}{85}$     (B)  $\frac{7}{25}$     (C)  $\frac{2}{5}$     (D)  $\frac{4}{7}$     (E)  $\frac{7}{10}$

**Answer: B**

#### Solution 1

The number of people wearing caps and sunglasses is  $\frac{2}{5} \cdot 35 = 14$ . So then, 14 people out of the 50 people wearing sunglasses also have caps.

$$\frac{14}{50} = \text{(B)} \frac{7}{25}$$

#### Solution 2 (extremely freaking overcomplicated)

Let  $A$  be the event that a randomly selected person is wearing sunglasses, and let  $B$  be the event that a randomly selected person is wearing a cap. We can write  $P(A \cap B)$  in two ways:  $P(A)P(B|A)$  or  $P(B)P(A|B)$ . Suppose there are  $t$  people in total. Then

$$P(A) = \frac{50}{t}$$

and

$$P(B) = \frac{35}{t}.$$

Additionally, we know that the probability that someone is wearing sunglasses given that they wear a cap is  $\frac{2}{5}$ , so  $P(A|B) = \frac{2}{5}$ . We let  $P(B|A)$ , which is the quantity we want to find, be equal to  $x$ . Substituting in, we get

$$\frac{50}{t} \cdot x = \frac{35}{t} \cdot \frac{2}{5}$$

$$\begin{aligned}
&\Rightarrow 50x = 35 \cdot \frac{2}{5} \\
&\Rightarrow 50x = 14 \\
&\Rightarrow x = \frac{14}{50} \\
&= \boxed{\text{(B)} \frac{7}{25}}
\end{aligned}$$

Note: This solution makes use of the dependent events probability formula,  $P(A \cap B) = P(A)P(B|A)$ , where  $P(B|A)$  represents the probability that  $B$  occurs given that  $A$  has already occurred and  $P(A \cup B)$  represents the probability of both  $A$  and  $B$  happening.

16. (AMC 8 2019) Qiang drives 15 miles at an average speed of 30 miles per hour. How many additional miles will he have to drive at 55 miles per hour to average 50 miles per hour for the entire trip?

(A) 45      (B) 62      (C) 90      (D) 110      (E) 135

**Answer:** D

**Solution 1 (by looking at the answer choices)**

The only option that is easily divisible by 55 is 110, which gives 2 hours of travel. And, the formula is  $\frac{15}{30} + \frac{110}{55} = \frac{5}{2}$ .

And, Average Speed =  $\frac{\text{Total Distance}}{\text{Total Time}}$ .

Thus,  $\frac{125}{50} = \frac{5}{2}$ .

Both are equal and thus our answer is  $\boxed{\text{(D)} 110}$ .

**Solution 2**

To calculate the average speed, simply evaluate the total distance over the total time. Let the number of additional miles he has to drive be  $x$ . Therefore, the total distance is  $15 + x$  and the total time (in hours) is

$$\frac{15}{30} + \frac{x}{55} = \frac{1}{2} + \frac{x}{55}.$$

We can set up the following equation:

$$\frac{15 + x}{\frac{1}{2} + \frac{x}{55}} = 50.$$

Simplifying the equation, we get

$$15 + x = 25 + \frac{10x}{11}.$$

Solving the equation yields  $x = 110$ , so our answer is  $\boxed{\text{(D)} 110}$ .

**Solution 3**

If he travels 15 miles at a speed of 30 miles per hour, he travels for 30 min. Average rate is total distance over total time so  $(15 + d)/(0.5 + t) = 50$ , where  $d$  is the distance left to travel and  $t$  is the time to travel that distance. Solve for  $d$  to get  $d = 10 + 50t$ . You also know that he has to travel 55 miles per hour for some time,

so  $d = 55t$ . Plug that in for  $d$  to get  $55t = 10 + 50t$  and  $t = 2$  and since  $d = 55t$ ,  $d = 2 \cdot 55 = 110$ , the answer is **(D) 110**.

#### Solution 4

Let  $h$  be the amount of hours Qiang drives after his first 15 miles. Average speed, which we know is 50 mph, means total distance over total time. For 15 miles at 30 mph, the time taken is  $\frac{1}{2}$  hour, so the total time for this trip would be  $\frac{1}{2} + h$  hours. For the total distance, 15 miles are traveled in the first part and  $55h$  miles in the second. This gives the following equation:

$$\frac{15 + 55h}{\frac{1}{2} + h} = 50.$$

Cross multiplying, we get that  $15 + 55h = 50h + 25$ , and simple algebra gives  $h = 2$ . In 2 hours traveling at 55 mph, the distance traveled is  $\frac{2 \text{ hours}}{1} \cdot \frac{55 \text{ miles}}{1 \text{ hour}} = 2 \cdot 55 \text{ miles} = 110 \text{ miles}$ , which is choice **(D) 110**.

17. (AMC 8 2019) What is the value of the product

$$\left(\frac{1 \cdot 3}{2 \cdot 2}\right) \left(\frac{2 \cdot 4}{3 \cdot 3}\right) \left(\frac{3 \cdot 5}{4 \cdot 4}\right) \cdots \left(\frac{97 \cdot 99}{98 \cdot 98}\right) \left(\frac{98 \cdot 100}{99 \cdot 99}\right)?$$

- (A)  $\frac{1}{2}$     (B)  $\frac{50}{99}$     (C)  $\frac{9800}{9801}$     (D)  $\frac{100}{99}$     (E) 50

**Answer:** B

#### Solution 1 (telescoping)

We rewrite:

$$\frac{1}{2} \cdot \left(\frac{3 \cdot 2}{2 \cdot 3}\right) \left(\frac{4 \cdot 3}{3 \cdot 4}\right) \cdots \left(\frac{99 \cdot 98}{98 \cdot 99}\right) \cdot \frac{100}{99}$$

The middle terms cancel, leaving us with

$$\left(\frac{1 \cdot 100}{2 \cdot 99}\right) = \mathbf{(B) \frac{50}{99}}$$

#### Solution 2

If you calculate the first few values of the equation, all of the values tend to be around  $\frac{1}{2}$ , but are not equal to it. The answer closest to  $\frac{1}{2}$  but not equal to it is

$$\mathbf{(B) \frac{50}{99}}.$$

#### Solution 3

Rewriting the numerator and the denominator, we get  $\frac{100! \cdot 98!}{(99!)^2}$ . We can simplify by canceling  $99!$  on both sides, leaving us with:  $\frac{100 \cdot 98!}{2 \cdot 99!}$ . We rewrite  $99!$  as  $99 \cdot 98!$  and cancel  $98!$ , which gets  $\mathbf{(B) \frac{50}{99}}$ .

#### Solution 4

All of the terms have the form  $\frac{k^2-1}{k^2}$ , which is  $< 1$ , so the product is  $< 1$ , so we eliminate options (D) and (E). (C) is too close to 1 to be possible. The partial products seem to be approaching  $1/2$ , so we guess that  $1/2$  is the limit/asymptote,

and so any finite product would be slightly larger than  $1/2$ . Therefore, by process of elimination and a small guess, we get that the answer is **(B)**  $\frac{50}{99}$ .

### Solution 5

The product

$$\left(\frac{1 \cdot 3}{2 \cdot 2}\right) \left(\frac{2 \cdot 4}{3 \cdot 3}\right) \left(\frac{3 \cdot 5}{4 \cdot 4}\right) \cdots \left(\frac{98 \cdot 100}{99 \cdot 99}\right)$$

can be simplified by observing that in each individual fraction, the numerator and denominator contain factors that cancel out with adjacent terms. Specifically, the factor 3 in the numerator of the first fraction cancels with the 3 in the denominator of the second fraction, the 4 in the numerator of the second fraction cancels with the 4 in the denominator of the third fraction, and so on. This telescoping cancellation continues throughout the entire product.

After all the cancellations, only the first factor of the first fraction,  $\frac{1}{2}$ , and the last factor of the last fraction,  $\frac{100}{99}$ , remain. The value of the product is therefore:

$$\frac{1}{2} \cdot \frac{100}{99} = \frac{50}{99}.$$

Thus, the value of the product is: **(B)**  $\frac{50}{99}$ .

– Rayansh Mankad(SharpWhiz17)

18. (AMC 8 2019) The faces of each of two fair dice are numbered 1, 2, 3, 5, 7, and 8. When the two dice are tossed, what is the probability that their sum will be an even number?
- (A)**  $\frac{4}{9}$       **(B)**  $\frac{1}{2}$       **(C)**  $\frac{5}{9}$       **(D)**  $\frac{3}{5}$       **(E)**  $\frac{2}{3}$

**Answer:** C

### Solution 1 (Casework)

We have 2 dice with 2 evens and 4 odds on each die. For the sum to be even, the 2 rolls must be 2 odds or 2 evens.

Ways to roll 2 odds (Case 1): The total number of ways to obtain 2 odds on 2 rolls is  $4 * 4 = 16$ , as there are 4 possible odds on the first roll and 4 possible odds on the second roll.

Ways to roll 2 evens (Case 2): Similarly, we have  $2 * 2 = 4$  ways to obtain 2 evens. Probability is  $\frac{20}{36} = \frac{5}{9}$ , or **(C)**.

### Solution 2 (Complementary Counting)

We count the ways to get an odd. If the sum is odd, then we must have an even and an odd. The probability of an even is  $\frac{1}{3}$ , and the probability of an odd is  $\frac{2}{3}$ . We have to multiply by 2 because the even and odd can be in any order. This gets us  $\frac{4}{9}$ , so the answer is  $1 - \frac{4}{9} = \frac{5}{9} = \span style="border: 1px solid black; padding: 2px;">**(C)**.$

### Solution 3

To get an even, you must get either 2 odds or 2 evens. The probability of getting 2 odds is  $\frac{4}{6} \times \frac{4}{6}$ . The probability of getting 2 evens is  $\frac{2}{6} \times \frac{2}{6}$ . If you add them together, you get  $\frac{16}{36} + \frac{4}{36} = \span style="border: 1px solid black; padding: 2px;">**(C)**  $\frac{5}{9}$ .$

19. (AMC 8 2019) In a tournament there are six teams that play each other twice. A team earns 3 points for a win, 1 point for a draw, and 0 points for a loss. After all the games have been played it turns out that the top three teams earned the same number of total points. What is the greatest possible number of total points for each of the top three teams?

(A) 22      (B) 23      (C) 24      (D) 26      (E) 30

**Answer:** C

**Solution 1**

The teams play a total of  $2 \cdot \binom{6}{2} = 30$  games, and  $2 \cdot \binom{3}{2} = 6$  of these games only involve bottom-3 teams. So 24 games involve at least one top-3 team.

Since at most 3 points are awarded in each game, the top-3 teams could earn a total of at most  $24 \cdot 3 = 72$  points. Thus it is impossible for the top 3 teams to score more than  $\frac{72}{3} = 24$  points each.

It is possible for the top 3 teams to score (C) 24 points each. This occurs if each top-3 team wins all its games against bottom-3 teams, and each top-3 team wins 1 game and loses 1 game against each of the other 2 top-3 teams.

**Solution 2**

We can name the top three teams as  $A$ ,  $B$ , and  $C$ . We can see that (respective scores of)  $A = B = C$  because these teams have the same points. If we look at the matches that involve the top three teams, we see that there are some duplicates:  $AB$ ,  $BC$ , and  $AC$  come twice. In order to even out the scores and get the maximum score, we can say that in match  $AB$ ,  $A$  and  $B$  each win once out of the two games that they play. We can say the same thing for  $AC$  and  $BC$ . This tells us that each team  $A$ ,  $B$ , and  $C$  win and lose twice. This gives each team a total of  $3 + 3 + 0 + 0 = 6$  points. Now, we need to include the other three teams. We can label these teams as  $D$ ,  $E$ , and  $F$ . We can write down every match that  $A$ ,  $B$ , or  $C$  plays in that we haven't counted yet:  $AD$ ,  $AD$ ,  $AE$ ,  $AE$ ,  $AF$ ,  $AF$ ,  $BD$ ,  $BD$ ,  $BE$ ,  $BE$ ,  $BF$ ,  $BF$ ,  $CD$ ,  $CD$ ,  $CE$ ,  $CE$ ,  $CF$ , and  $CF$ . We can say  $A$ ,  $B$ , and  $C$  win each of these in order to obtain the maximum score that  $A$ ,  $B$ , and  $C$  can have. If  $A$ ,  $B$ , and  $C$  win all six of their matches,  $A$ ,  $B$ , and  $C$  will have a score of 18.  $18 + 6$  results in a maximum score of (C) 24

**Solution 3**

To start, we calculate how many games each team plays. Each team can play against 5 other teams twice, so there are 10 games that each team plays. So the answer is  $10 \cdot 3$  which is 30 But wait... if we want 3 teams to have the same amount of points, there can't possibly be a team who wins all their games. Let the top three teams be  $A$ ,  $B$ , and  $C$ .  $A$  plays  $B$  and  $C$  twice, so in order to maximize the games being played, we can split it 50 – 50 between the 4 games  $A$  plays against  $B$  or  $C$ . We find that we just subtract 2 games or 6 points. Therefore the answer is  $30 - 6$ , 24or (C) 24

**Solution 4**

Let the top 3 teams be  $A, B, C$  and the bottom teams to be  $D, E, F$ . There are 2 types of games the winners ( $A, B, C$ ) can play: Winners vs losers and Winner vs Winner.

Case 1: Winner vs Loser



For A, it would be  $AD, AE, AF$ . We want to maximize this or make A win in all 6 ( $3 \times 2$ ) games. Therefore, A would get  $6 \times 3$  points or 18.

Case 2: Winner vs Winner

There are  ${}_3P_2 = \frac{3!}{1!} = 6$  games they can play against each other. Since we want each 3 to have an equal amount. Each team should win 2 games and get 6 points. In total,  $18 + 6 = \boxed{\text{(C) } 24}$  RandomMathGuy500

### Solution 5

Name the teams A, B, C, D, E, and F. As we know that the top three teams, A, B, and C, tied, we can just focus on one, A. When A plays B or C, we want them to both win one and lose one, maximising the points for both teams. When A plays D, E, or F, we want A to win both times. This adds up to  $3 + 3(\text{top teams}) + 6 + 6 + 6$  (other teams), which equals 24 points. ParticlePhysics This solution is by far the fastest and probably most used

20. (AMC 8 2019) How many different real numbers  $x$  satisfy the equation

$$(x^2 - 5)^2 = 16?$$

- (A) 0      (B) 1      (C) 2      (D) 4      (E) 8

**Answer:** D

### Solution 1

We have that  $(x^2 - 5)^2 = 16$  if and only if  $x^2 - 5 = \pm 4$ . If  $x^2 - 5 = 4$ , then  $x^2 = 9 \implies x = \pm 3$ , giving 2 solutions. If  $x^2 - 5 = -4$ , then  $x^2 = 1 \implies x = \pm 1$ , giving 2 more solutions. All four of these solutions work, so the answer is  $\boxed{\text{(D) } 4}$ . Further, the equation is a **quartic** in  $x$ , so by the **Fundamental Theorem of Algebra**, there can be at most four real solutions.

### Solution 2

We can expand  $(x^2 - 5)^2$  to get  $x^4 - 10x^2 + 25$ , so now our equation is  $x^4 - 10x^2 + 25 = 16$ . Subtracting 16 from both sides gives us  $x^4 - 10x^2 + 9 = 0$ . Now, we can factor the left-hand side to get  $(x^2 - 9)(x^2 - 1) = 0$ . If  $x^2 - 9$  and/or  $x^2 - 1$  equals 0, then the entire left side will equal 0. Since the solutions can be both positive and negative, we have 4 solutions:  $-3, 3, -1, 1$  (we can find these solutions by setting  $x^2 - 9$  and  $x^2 - 1$  equal to 0 and solving for  $x$ ). So, the answer is  $\boxed{\text{(D) } 4}$ . Note: every positive root can also be shown as negative

### Solution 3

Subtract 16 from both sides and factor using difference of squares:

$$\begin{aligned}(x^2 - 5)^2 &= 16 \\(x^2 - 5)^2 - 16 &= 0 \\(x^2 - 5)^2 - 4^2 &= 0 \\[(x^2 - 5) - 4][(x^2 - 5) + 4] &= 0 \\(x^2 - 9)(x^2 - 1) &= 0 \\(x + 3)(x - 3)(x + 1)(x - 1) &= 0.\end{aligned}$$

Quite obviously, this equation has  $\boxed{\text{(D) } 4}$  solutions.

21. (AMC 8 2019) What is the area of the triangle formed by the lines  $y = 5$ ,  $y = 1 + x$ , and  $y = 1 - x$ ?

(A) 4      (B) 8      (C) 10      (D) 12      (E) 16

**Answer:** E

**Solution 1**

First, we need to find the coordinates where the graphs intersect.

We want the points  $x$  and  $y$  to be the same. Thus, we set  $5 = x + 1$ , and get  $x = 4$ . Plugging this into the equation,  $y = 1 - x$ ,  $y = 5$ , and  $y = 1 + x$  intersect at  $(4, 5)$ , we call this line  $x$ .

Doing the same thing, we get  $x = -4$ . Thus,  $y = 5$ . Also,  $y = 5$  and  $y = 1 - x$  intersect at  $(-4, 5)$ , and we call this line  $y$ .

It's apparent the only solution to  $1 - x = 1 + x$  is 0. Thus,  $y = 1$ .  $y = 1 - x$  and  $y = 1 + x$  intersect at  $(0, 1)$ , we call this line  $z$ .

Using the **Shoelace Theorem** we get:

$$\left( \frac{(20 - 4) - (-20 + 4)}{2} \right) = \frac{32}{2}$$

= So, our answer is **(E) 16**.

We might also see that the lines  $y$  and  $x$  are mirror images of each other. This is because, when rewritten, their slopes can be multiplied by  $-1$  to get the other. As the base is horizontal, this is an isosceles triangle with base 8, as the intersection points have a distance of 8. The height is  $5 - 1 = 4$ , so  $\frac{4 \cdot 8}{2} = \mathbf{(E)16}$ .

Warning: Do not use the distance formula for the base then use Heron's formula. It will take you half of the time you have left or maybe more!!

**Solution 2**

Graphing the lines, using the intersection points we found in Solution 1, we can see that the height of the triangle is 4, and the base is 8. Using the formula for the area of a triangle, we get  $\frac{4 \cdot 8}{2}$ , which is equal to **(E) 16**.

**Solution 3**

$y = x + 1$  and  $y = -x + 1$  have  $y$ -intercepts at  $(0, 1)$  and slopes of 1 and  $-1$ , respectively. Since the product of these slopes is  $-1$ , the two lines are perpendicular. From  $y = 5$ , we see that  $(-4, 5)$  and  $(4, 5)$  are the other two intersection points, and they are 8 units apart. By symmetry, this triangle is a  $45 - 45 - 90$  triangle, so the legs are  $4\sqrt{2}$  each and the area is  $\frac{(4\sqrt{2})^2}{2} = \mathbf{(E) 16}$ .

22. (AMC 8 2019) A store increased the original price of a shirt by a certain percent and then decreased the new price by the same percent. Given that the resulting price was 84% of the original price, by what percent was the price increased and decreased?

(A) 16      (B) 20      (C) 28      (D) 36      (E) 40

**Answer:** E

**Solution 1**

Suppose the fraction of discount is  $x$ . That means  $(1 - x)(1 + x) = 0.84$ ; so,  $1 - x^2 = 0.84$ , and  $(x^2) = 0.16$ , procuring  $x = 0.4$ . Therefore, the price was increased and decreased by 40%, or **(E) 40**.

### Solution 1.1

After the first increase by  $p$  percent, the shirt price became  $(1 + p)$  times greater than the original. Upon the decrease in  $p$  percent on this price, the shirt price became  $(1 - p)$  times less than  $(1 + p)$ , or  $(1 - p)(1 + p)$ . We know that this price is 84 percent of the original, so  $(1 - p)(1 + p) = 0.84$ .

From here, we can list the factors of 0.84 and see which are equidistant from 1. We see that 0.6 and 1.4 are both 0.4 from 1, so  $p = 0.4 = 40\%$ , or choice **(E) 40**.

### Solution 2 (Time-Consuming)

We can try out every option and see which one works. Wasting 30 mins of the 40 mins, by this method, we get **(E) 40**.

### Solution 3

Let  $x$  be the discount. We can also work in reverse such as  $(84)(\frac{100}{100-x})(\frac{100}{100+x}) = 100$ .

Thus,  $8400 = (100 + x)(100 - x)$ . Solving for  $x$  gives us  $x = 40, -40$ . But  $x$  has to be positive. Thus,  $x = 40$ .

### Solution 4 using the answer choices (similar to Solution 2)

Let our original cost be \$100. We are looking for a result of \$84, then. We try 16% and see it gets us higher than 84. We try 20% and see it gets us lower than 16 but still higher than 84. We know that the higher the percent, the less the value. We try 36, as we are not progressing much, and we are close! We try **40%**, and we have the answer; it worked.(OR: try (C) first to eliminate 2 answer choices)

### Solution 5 (A Variation of Solution 4)

Let our original cost be \$100, so we are looking for a whole number of \$84. Also, we can see that (A), (C), and (D) give us answers with decimals while we know that (B) and (E) give us whole numbers. Therefore, we only need to try these two: (B) \$100 increased by 20

Thus, 40% or **(E) 40** is the answer.

### Solution 6 (using factors)

First find the prime factors of 84. These are 2,2,3 and 7. Using this, try the answers. For example,  $100 \cdot 1.2 / 1.2$ . This is equal to  $120 \cdot 0.8$ . If you find the factors of 12 and 8, you find that there is no seven and too many 2's. So **(B) 20** is out. Using the same method, you can find that  $116 \cdot 0.84$  is out, as 84 already has all the factors of 84.  $128 \cdot 0.72$  and  $136 \cdot 0.64$  both don't have seven as a factor, so **(A) 16**, **(C) 28**, and **(D) 36** is out. This leaves  $140 \cdot 0.6$ , which has just the right amount of factors ( $2^2 \cdot 2, 3$ , and 7). Therefore, the answer is **(E) 40**

23. (AMC 8 2019) After Euclid High School's last basketball game, it was determined that  $\frac{1}{4}$  of the team's points were scored by Alexa and  $\frac{2}{7}$  were scored by Brittany. Chelsea scored 15 points. None of the other 7 team members scored more than 2 points. What was the total number of points scored by the other 7 team members?

(A) 10      (B) 11      (C) 12      (D) 13      (E) 14

**Answer:** B**Solution 1**

Given the information above, we start with the equation  $\frac{t}{4} + \frac{2t}{7} + 15 + x = t$ , where  $t$  is the total number of points scored and  $x \leq 14$  is the number of points scored by the remaining 7 team members, we can simplify to obtain the Diophantine equation  $x + 15 = \frac{13}{28}t$ , or  $28x + 28 \cdot 15 = 13t$ . Since  $t$  is necessarily divisible by 28, let  $t = 28u$  where  $u \geq 0$  and divide by 28 to obtain  $x + 15 = 13u$ . Then, it is easy to see  $u = 2$  ( $t = 56$ ) is the only candidate remaining, giving  $x = \boxed{\text{(B)}11}$ .

-scrabbler94

**Solution 2**

We first start by setting the total number of points as 28, since  $\text{LCM}(4, 7) = 28$ . However, we see that this does not work since we surpass the number of points just with the information given ( $28 \cdot \frac{1}{4} + 28 \cdot \frac{2}{7} + 15 = 30 (> 28)$ ). Next, we can see that the total number of points scored is 56 as, if it is more than or equal to 84, at least one of the others will score more than 2 points. With this, we have that Alexa, Brittany, and Chelsea score:  $56 \cdot \frac{1}{4} + 56 \cdot \frac{2}{7} + 15 = 45$ , and thus, the other seven players would have scored a total of  $56 - 45 = \boxed{\text{(B)}11}$ . (We see that this works since we could have 4 of them score 2 points, and the other 3 of them score 1 point.)

-aops5234 -Edited by [Penguin\\_Spellcaster](#)**Solution 3 — Modular Arithmetic**

Adding together Alexa's and Brittany's fractions, we get  $\frac{15}{28}$  as the fraction of the total number of points they scored together. However, this is just a ratio, so we can introduce a variable:  $\frac{15x}{28x}$  where  $x$  is the common ratio. Let  $y$  and  $z$  and  $w$  be the number of people who scored 1, 2, and 0 points, respectively. Writing an equation, we have  $\frac{13x}{28x} = 15 + y + 2z + 0w$ . We want all of our variables to be integers. Thus, we want  $15 + y + 2z = 0 \pmod{13}$ . Simplifying,  $y + 2z = 11 \pmod{13}$ . The only possible value, as this integer sum has to be less than  $7 \cdot 2 + 1 = 15$ , must be 11. Therefore,  $y + 2z = 11$ , and the answer is  $\boxed{\text{(B)} 11}$ .

- ab2024

**Solution 4: Answer choices**

We can rewrite the question as an algebraic equation:  $\frac{1}{4}T + \frac{2}{7}T + 15 + y = x$ , where  $x$  represents the total amount of points and  $y$  the amount of points the 7 other players scored. From there, we add the two fractions to get  $\frac{15}{28}x + 15 + y = x$ . Subtracting  $\frac{15}{28}x$  from both sides, we get  $\frac{13}{28}x = y + 15$ . We multiply each side by 28 to get rid of the denominator, in which we get  $13x = 420 + 28y$ . Now let's think of this logically. This equation is telling us that if you add 420 and 28 times the amount of points scored by the extra 7 players, you get 13 times the amount of points total. And since we have to have a whole number of points total, this means that  $420 + 28y$  must be divisible by 13. Plugging in all the answer choices for  $y$ , we find that the only answer that makes  $420 + 28y$  divisible by 13 is  $\boxed{\text{(B)} 11}$ .

**Solution 5 (modular congruences)**

As in the above, we can write the equation

$$\frac{1}{4}k + \frac{2}{7}k + 15 + y = k$$

where  $k$  is the total points, and  $y$  is the rest of the students' total. This simplifies to the equation

$$y + 15 = \frac{13}{28}k \implies 28y + 15 \cdot 28 = 13k$$

Since  $y, k$  are both integers, we have the modular congruence equation:

$$28y + 15 \equiv 0 \pmod{13} \implies 2y \equiv 11 \pmod{13}$$

Since the modular inverse of 2 (mod 13) is 7, this simplifies to:

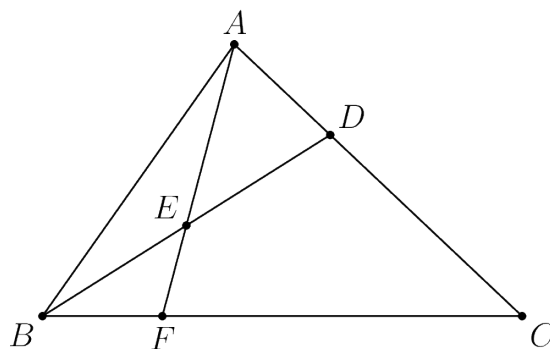
$$y \equiv 11 \cdot 7 \equiv 11 \pmod{13}$$

The question asks for the least value of  $y$ . This is 11, thus  $\boxed{(B)11}$  iretra

**Solution 6 (Because it doesn't have to be that complicated)**

We know from the problem that  $\frac{1}{4} + \frac{2}{7}$  of the points are already taken. Then we add them together yielding,  $\frac{15}{28}$ , which means there are  $\frac{13}{28}$  points left. Here is the magic part: Multiply it by 56, giving 26. And now we can subtract Chelsea's 15 points. That leaves us with 11. Therefore, The answer is  $\boxed{(B)11}$  P.S. Why do we multiply by 56? Since the answer is a whole number we have to multiply by a multiple of 28. We try different value and find 28 is too small and 84 is too big. Fluffy1234

24. (AMC 8 2019) In triangle  $ABC$ , point  $D$  divides side  $\overline{AC}$  so that  $AD : DC = 1 : 2$ . Let  $E$  be the midpoint of  $\overline{BD}$  and let  $F$  be the point of intersection of line  $BC$  and line  $AE$ . Given that the area of  $\triangle ABC$  is 360, what is the area of  $\triangle EBF$ ?



- (A) 24      (B) 30      (C) 32      (D) 36      (E) 40

**Answer: B**

25. (AMC 8 2019) Alice has 24 apples. In how many ways can she share them with Becky and Chris so that each of the three people has at least two apples?

- (A) 105      (B) 114      (C) 190      (D) 210      (E) 380

**Answer: C**

**Solution 1 (Stars and Bars/Sticks and Stones)**

Note: This solution uses the non-negative version for stars and bars. A solution using the positive version of stars is similar (first removing an apple from each person instead of 2).

This method uses the counting method of stars and bars (non-negative version). Since each person must have at least 2 apples, we can remove  $2 \cdot 3$  apples from the total that need to be sorted. With the remaining 18 apples, we can use stars and bars to determine the number of possibilities. Assume there are 18 stars in a row, and 2 bars, which will be placed to separate the stars into groups of 3. In total, there are 18 spaces for stars + 2 spaces for bars, for a total of 20 spaces. We can now do  $\binom{20}{2}$ . This is because if we choose distinct 2 spots for the bars to be placed, each combo of 3 groups will be different, and all apples will add up to 18. We can also do this because the apples are indistinguishable.  $\binom{20}{2}$  is 190, therefore the answer is (C) 190.

### Solution 2 (Answer Choices)

Consider an unordered triple  $(a, b, c)$  where  $a + b + c = 24$  and  $a, b, c$  are not necessarily distinct. Then, we will either have 1, 3, or 6 distinguishable ways to assign  $a, b$ , and  $c$  to Alice, Becky, and Chris. Thus, our answer will be  $x + 3y + 6z$  for some nonnegative integers  $x, y, z$ . Notice that we only have 1 way to assign the numbers  $a, b, c$  to Alice, Becky, and Chris when  $a = b = c$ . As this only happens 1 way ( $a = b = c = 8$ ), our answer is  $1 + 3y + 6z$  for some  $y, z$ . Finally, notice that this implies the answer is  $1 \pmod{3}$ . The only answer choice that satisfies this is (C) 190.

-BorealBear

### Solution 3

Since each person needs to have at least two apples, we can simply give each person two, leaving  $24 - 2 \times 3 = 18$  apples. For the remaining apples, if Alice is going to have  $a$  apples, Becky is going to have  $b$  apples, and Chris is going to have  $c$  apples, we have indeterminate equation  $a + b + c = 18$ . Currently, we can see that  $0 \leq a \leq 18$  where  $a$  is an integer, and when  $a$  equals any number in the range, there will be  $18 - a + 1 = 19 - a$  sets of values for  $b$  and  $c$ . Thus, there are  $19 + 18 + 17 + \cdots + 1 = \span style="border: 1px solid black; padding: 2px;">(C) 190 possible sets of values in total.$

## §36 AMC 8 2020

- (AMC 8 2020) Luka is making lemonade to sell at a school fundraiser. His recipe requires 4 times as much water as sugar and twice as much sugar as lemon juice. He uses 3 cups of lemon juice. How many cups of water does he need?  
(A) 6      (B) 8      (C) 12      (D) 18      (E) 24

**Answer:** E

### Solution 1 (Direct)

We have water : sugar : lemon juice =  $4 \cdot 2 : 2 : 1 = 8 : 2 : 1$ , so Luka needs  $3 \cdot 8 = \span style="border: 1px solid black; padding: 2px;">(E) 24 cups.$

### Solution 2 (Stepwise)

Since the amount of sugar is twice the amount of lemon juice, Luka uses  $3 \cdot 2 = 6$  cups of sugar.

Since the amount of water is 4 times the amount of sugar, he uses  $6 \cdot 4 = \span style="border: 1px solid black; padding: 2px;">(E) 24 cups of water.$

### Solution 3 (Combination of Solutions 1 and 2)

The ratio is Water : Sugar : Lemon Juice, or  $8 : 2 : 1$ . Since we know that Luka used 3 cups of lemon juice, he needs  $3 \cdot 2 = 6$  cups of sugar. Because the amount of water is 4 times the amount of sugar Luka needs, he will need  $6 \cdot 4 = \boxed{\text{(E)} 24}$  cups of water.

Thanks to MRENTHUSIASM for the inspiration!

EarthSaver 15:12, 11 June 2021 (EDT)

2. (*AMC 8 2020*) Four friends do yardwork for their neighbors over the weekend, earning \$15, \$20, \$25, and \$40, respectively. They decide to split their earnings equally among themselves. In total, how much will the friend who earned \$40 give to the others?

(A) \$5      (B) \$10      (C) \$15      (D) \$20      (E) \$25

**Answer:** C

**Solution**

The friends earn  $\$(15 + 20 + 25 + 40) = \$100$  in total. Since they decided to split their earnings equally, it follows that each person will get  $\$\left(\frac{100}{4}\right) = \$25$ . Since the friend who earned \$40 will need to leave with \$25, he will have to give  $\$(40 - 25) = \boxed{\text{(C)} \$15}$  to the others.

**Solution 2**

The friends earn  $\$(15 + 20 + 25 + 40) = \$100$  in total. Since they decided to split their earnings equally, it follows that each person will get  $\$\left(\frac{100}{4}\right) = \$25$ . The friend who earned 25 will not distribute anything, the friend that earned 15 needs 10, and the friend that earned 20 needs 5.  $10 + 5 = 15$ , therefore the answer is  $\$(40 - 25) = \boxed{\text{(C)} \$15}$ .

3. (*AMC 8 2020*) Carrie has a rectangular garden that measures 6 feet by 8 feet. She plants the entire garden with strawberry plants. Carrie is able to plant 4 strawberry plants per square foot, and she harvests an average of 10 strawberries per plant. How many strawberries can she expect to harvest? (A) 560      (B) 960      (C) 1120      (D) 1920      (E) 2560

**Answer:** D

**Solution 1**

Note that the unit of the answer is **strawberries**, which is the product of

- square feet
- plants per square foot
- strawberries per plant

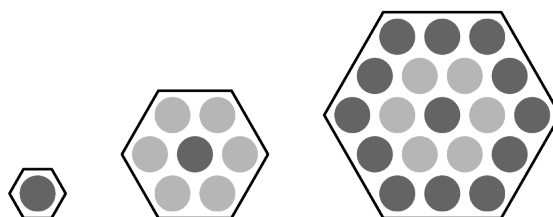
By conversion factors, we have

$$(6 \text{ ft} \cdot 8 \text{ ft}) \cdot \left(4 \frac{\text{plants}}{\text{ft}^2}\right) \cdot \left(10 \frac{\text{strawberries}}{\text{plant}}\right) = 6 \cdot 8 \cdot 4 \cdot 10 \text{ strawberries} = \boxed{\text{(D)} 1920} \text{ strawberries.}$$

**Solution 2**

The area of the garden is  $6 \cdot 8 = 48$  square feet. Since Carrie plants 4 strawberry plants per square foot, there are a total of  $48 \cdot 4 = 192$  strawberry plants, each of which produces 10 strawberries on average. Accordingly, she can expect to harvest  $192 \cdot 10 = \boxed{\text{(D)} 1920}$  strawberries.

4. (AMC 8 2020) Three hexagons of increasing size are shown below. Suppose the dot pattern continues so that each successive hexagon contains one more band of dots. How many dots are in the next hexagonal?



- (A) 35      (B) 37      (C) 39      (D) 43      (E) 49

**Answer:** B

**Solution 1 (Pattern of the Rows)**

Looking at the rows of each hexagon, we see that the first hexagon has 1 dot, the second has  $2 + 3 + 2$  dots, and the third has  $3 + 4 + 5 + 4 + 3$  dots. Given the way the hexagons are constructed, it is clear that this pattern continues. Hence, the fourth hexagon has  $4 + 5 + 6 + 7 + 6 + 5 + 4 = \boxed{\text{(B) } 37}$  dots.

**Solution 2 (Pattern of the Bands)**

The dots in the next hexagon have four bands. From innermost to outermost:

- The first band has 1 dot.
- The second band has 6 dots: 1 dot at each vertex of the hexagon.
- The third band has  $6 + 6 \cdot 1 = 12$  dots: 1 dot at each vertex of the hexagon and 1 other dot on each edge of the hexagon.
- The fourth band has  $6 + 6 \cdot 2 = 18$  dots: 1 dot at each vertex of the hexagon and 2 other dots on each edge of the hexagon.

Together, the answer is  $1 + 6 + 12 + 18 = \boxed{\text{(B) } 37}$ .

**Solution 3 (Pattern of the Bands)**

The first hexagon has 1 dot, the second hexagon has  $1 + 6$  dots, the third hexagon has  $1 + 6 + 12$  dots, and so on. The pattern continues since to go from hexagon  $n$  to hexagon  $(n + 1)$ , we add a new band of dots around the outside of the existing ones, with each side of the band having side length  $(n + 1)$ . Thus, the number of dots added is  $6(n + 1) - 6 = 6n$  (we subtract 6 as each of the corner hexagons in the band is counted as part of two sides.). We therefore predict that the fourth hexagon has  $1 + 6 + 12 + 18 = \boxed{\text{(B) } 37}$  dots.

**Remark**

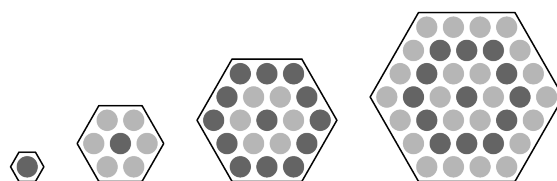
For positive integers  $n$ , let  $h_n$  denote the number of dots in the  $n$ th hexagon. We have  $h_1 = 1$  and  $h_{n+1} = h_n + 6n$ .

It follows that  $h_2 = 7$ ,  $h_3 = 19$ , and  $h_4 = 37$ .

**Solution 4 (Brute Force)**

From the full diagram below, the answer is  $\boxed{\text{(B) } 37}$ .





MRENTHUSIASM

**Solution 5 (Summation Notation)**

We solve this using the **Sigma Function** (summation notation).

The sequence starts at:

$$a_1 = 1$$

Each term increases by a growing difference:

$$a_2 = a_1 + 6 = 7$$

$$a_3 = a_2 + 12 = 19$$

$$a_4 = a_3 + 18 = 37$$

We observe that the difference added at each step forms an arithmetic sequence:

$$6, 12, 18, 24, \dots = 6k \text{ for } k = 1, 2, 3, \dots$$

To express this with the **Sigma Function**, we write the  $n$ -th term as:

$$a_n = 1 + \sum_{k=1}^{n-1} 6k = 1 + 3n(n-1)$$

This is a standard arithmetic sum, so we use the formula:

$$\sum_{k=1}^{n-1} k = \frac{(n-1)n}{2}$$

Substitute this in:

$$a_n = 1 + 6 \cdot \frac{(n-1)n}{2}$$

When we simplify:

$$a_n = 1 + 3n(n-1)$$

$$a_n = 1 + 3n(n-1)$$

Let's compute the 4th term using the sigma-based formula:

$$a_4 = 1 + 3 \cdot 4 \cdot (4-1) = 1 + 3 \cdot 12 = 1 + 36 = \boxed{37}$$

So the answer is 37.

**Solution 6 (Quick)**

Notice that the middle row, which is also the row with the most dots in it, has the same number of dots as the topmost or bottommost row multiplied by 2, then minus one. Since the new hexagon will have four dots at the topmost and bottommost rows, the middle row will have  $(4 \cdot 2) - 1 = 7$  dots. Also, notice that the rows increase in the number of dots from the topmost to the middle row, then decrease from the middle row to the bottommost row by 1. So, noticing these patterns, the amount of dots in the next hexagon is simply  $4 + 5 + 6 + 7 + 6 + 5 + 4 = \boxed{\text{(B) } 37}$  dots. Evžen D.

**Solution 7 (Fast Pattern Recognition)**

The first hexagon has 1 dot in it. The second hexagon has 7 dots in it. The third hexagon has 19 dots on it. From the first to the second, we add 6 hexagons. From the second to the third, we add 12 hexagons. Therefore, recognizing the pattern, from the third to the fourth case, we have to add 18 dots to the 19 we already have, resulting in our answer of **(B) 37** dots.

(AMC 8 2020) Three fourths of a pitcher is filled with pineapple juice. The pitcher is emptied by pouring an equal amount of juice into each of 5 cups. What percent of the total capacity of the pitcher did each cup receive?

- (A) 5      (B) 10      (C) 15      (D) 20      (E) 25

**Answer:** C

**Solution 1**

Each cup is filled with  $\frac{3}{4} \cdot \frac{1}{5} = \frac{3}{20}$  of the amount of juice in the pitcher, so the percentage is  $\frac{3}{20} \cdot 100 = \mathbf{(C) 15}$ .

**Solution 2**

The pitcher is  $\frac{3}{4}$  full, i.e. 75% full. Therefore each cup receives  $\frac{75}{5} = \mathbf{(C) 15}$  percent of the total capacity.

**Solution 3**

Assume that the pitcher has a total capacity of 100 ounces. Since it is filled three fourths with pineapple juice, it contains 75 ounces of pineapple juice, which means that each cup will contain  $\frac{75}{5} = 15$  ounces of pineapple juice. Since the total capacity of the pitcher was 100 ounces, it follows that each cup received 15% of the total capacity of the pitcher, yielding **(C) 15** as the answer.

(AMC 8 2020) Aaron, Darren, Karen, Maren, and Sharon rode on a small train that has five cars that seat one person each. Maren sat in the last car. Aaron sat directly behind Sharon. Darren sat in one of the cars in front of Aaron. At least one person sat between Karen and Darren. Who sat in the middle car?

- (A) Aaron      (B) Darren      (C) Karen      (D) Maren      (E) Sharon

**Answer:** A

**Solution 1**

Write the order of the cars as  $\square\square\square\square\square$ , where the left end of the row represents the back of the train and the right end represents the front. Call the people  $A, D, K, M$ , and  $S$  respectively. The first condition gives  $M\square\square\square\square$ , so we try  $MAS\square\square$ ,  $M\square AS\square$ , and  $M\square\square AS$ . In the first case, as  $D$  sat in front of  $A$ , we must have  $MASDK$  or  $MASKD$ , which do not comply with the last condition. In the second case, we obtain  $MKASD$ , which works, while the third case is obviously impossible since it results in there being no way for  $D$  to sit in front of  $A$ . It follows that, with the only possible arrangement being  $MKASD$ , the person sitting in the middle car is **(A) Aaron**.

**Solution 2**

Follow the first few steps of Solution 1. We must have  $M\square\square\square\square$ , and also have  $AS(\text{anything})D$  and  $K\dots D$ . There are only 4 spaces available for  $A, S, D, K$ , so the only possible arrangement of them is  $KASD$ , so the arrangement is  $MKASD$ , so the person in the middle car is **(A) Aaron**.

(AMC 8 2020) How many integers between 2020 and 2400 have four distinct digits arranged in increasing order? (For example, 2347 is one integer.)

- (A) 9      (B) 10      (C) 15      (D) 21      (E) 28

**Answer:** C

**Solution 1**

Firstly, we can observe that the second digit of such a number cannot be 1 or 2 because the digits must be distinct and in increasing order. The second digit also cannot be 4 as the number must be less than 2400, so the second digit must be 3. It remains to choose the latter two digits, which must be 2 distinct digits from  $\{4, 5, 6, 7, 8, 9\}$ . That can be done in  $\binom{6}{2} = \frac{6 \cdot 5}{2 \cdot 1} = 15$  ways; there is then only 1 way to order the digits, namely in increasing order. This means the answer is **(C) 15**.

**Solution 2 (without using the "choose" function)**

As in Solution 1, we find that the first two digits must be 23, and the third digit must be at least 4 because the digits can not repeat. If it is 4, then there are 5 choices for the last digit, namely 5, 6, 7, 8, or 9. Similarly, if the third digit is 5, there are 4 choices for the last digit, namely 6, 7, 8, and 9; if 6, there are 3 choices; if 7, there are 2 choices; and if 8, there is 1 choice. It follows that the total number of such integers is  $5 + 4 + 3 + 2 + 1 = \mathbf{(C) 15}$ .

(AMC 8 2020) Ricardo has 2020 coins, some of which are pennies (1-cent coins) and the rest of which are nickels (5-cent coins). He has at least one penny and at least one nickel. What is the difference in cents between the greatest possible and least possible amounts of money that Ricardo can have?

- (A) 8062      (B) 8068      (C) 8072      (D) 8076      (E) 8082

**Answer: C**

**Solution 1**

Clearly, the amount of money Ricardo has will be maximized when he has the maximum number of nickels. Since he must have at least one penny, the greatest number of nickels he can have is 2019, giving a total of  $(2019 \cdot 5 + 1)$  cents. Analogously, the amount of money he has will be least when he has the greatest number of pennies; as he must have at least one nickel, the greatest number of pennies he can have is also 2019, giving him a total of  $(2019 \cdot 1 + 5)$  cents. Hence the required difference is

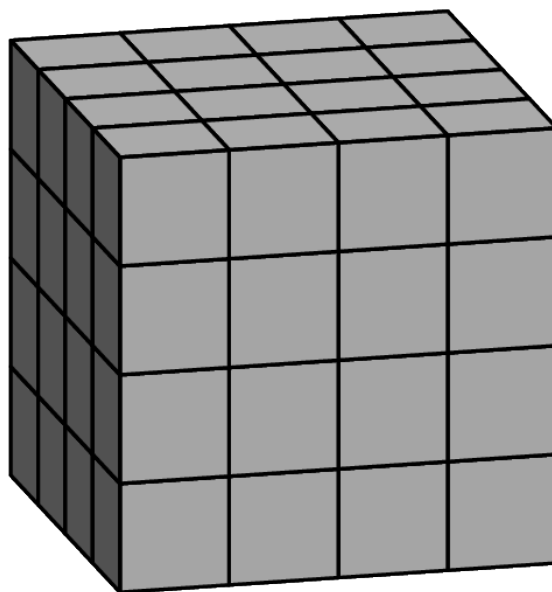
$$(2019 \cdot 5 + 1) - (2019 \cdot 1 + 5) = 2019 \cdot 4 - 4 = 4 \cdot 2018 = \mathbf{(C) 8072}$$

**Solution 2**

Suppose Ricardo has  $p$  pennies, so then he has  $(2020 - p)$  nickels. In order to have at least one penny and at least one nickel, we require  $p \geq 1$  and  $2020 - p \geq 1$ , i.e.  $1 \leq p \leq 2019$ . The number of cents he has is  $p + 5(2020 - p) = 10100 - 4p$ , so the maximum is  $10100 - 4 \cdot 1$  and the minimum is  $10100 - 4 \cdot 2019$ , and the difference is therefore

$$(10100 - 4 \cdot 1) - (10100 - 4 \cdot 2019) = 4 \cdot 2019 - 4 = 4 \cdot 2018 = \mathbf{(C) 8072}$$

(AMC 8 2020) Akash's birthday cake is in the form of a  $4 \times 4 \times 4$  inch cube. The cake has icing on the top and the four side faces, and no icing on the bottom. Suppose the cake is cut into 64 smaller cubes, each measuring  $1 \times 1 \times 1$  inch, as shown below. How many small pieces will have icing on exactly two sides?



- (A) 12    (B) 16    (C) 18    (D) 20    (E) 24

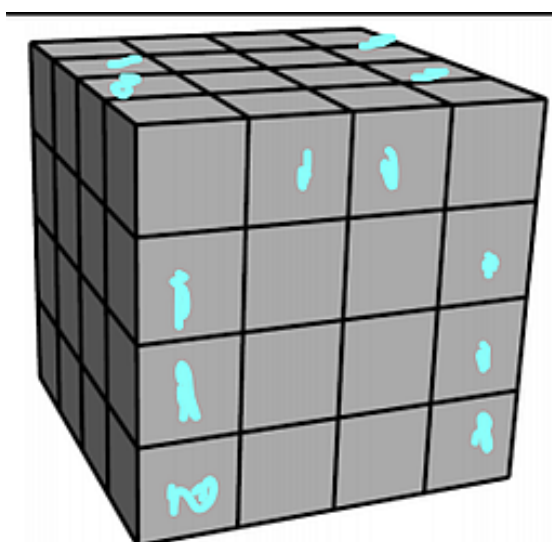
**Answer:** D

**Solution 1**

Notice that, for a small cube which does not form part of the bottom face, it will have exactly 2 faces with icing on them only if it is one of the 2 center cubes of an edge of the larger cube. There are  $12 - 4 = 8$  such edges (as we exclude the 4 edges of the bottom face), so this case yields  $2 \cdot 8 = 16$  small cubes. As for the bottom face, we can see that only the 4 corner cubes have exactly 2 faces with icing, so the total is  $16 + 4 = \boxed{\text{(D)} 20}$ .

**Solution 2**

The following diagram shows 12 of the small cubes having exactly 2 faces with icing on them; that is all of them except for those on the hidden face directly opposite the front face.



But the hidden face is an exact copy of the front face, so the answer is  $12 + 8 = \boxed{\text{(D)} 20}$ .

**Solution 3**

It is observable that only the middle-edged pieces of each face will have the icing exactly on two sides. There are 4 such pieces on each face. Considering 5 faces of the

cube (since the bottom is not iced), we can state that the number of such pieces of dimensions  $1 \times 1 \times 1$  will be  $5 \times 4 = 20$ . Therefore, our answer is 20.

**Solution 4 (For Rubik's Cubers)**

On a  $4 \times 4$  Rubik's cube, there are exactly 24 'edge' pieces, 8 'corners', and 24 'center' pieces. Edge pieces have 2 frosted faces (the ones on the bottom only have one, corners have 3 frosted faces, and centers have 1. So since we have 24 edges pieces, we subtract the 8 'edge' pieces on the bottom (they only have one frosted face), and then we add the 4 bottom 'corner' pieces (they also have frosted faces). we get  $24 - 8 + 4 = \boxed{\text{(D) } 20}$ .

-Solution by MismatchedCubing and Andrew\_Lu

(AMC 8 2020) Zara has a collection of 4 marbles: an Aggie, a Bumblebee, a Steelie, and a Tiger. She wants to display them in a row on a shelf, but does not want to put the Steelie and the Tiger next to one another. In how many ways can she do this?

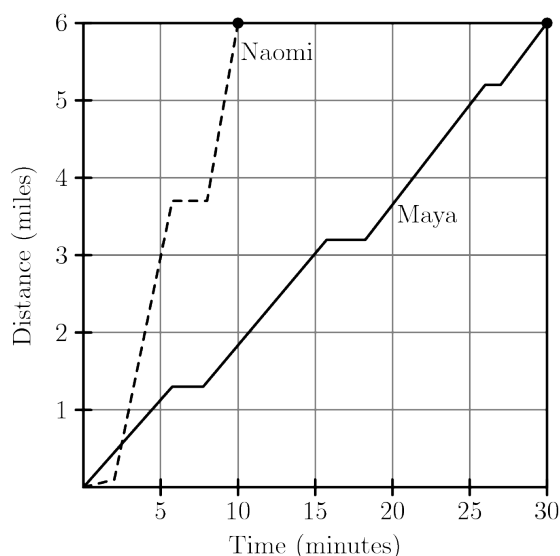
- (A) 6    (B) 8    (C) 12    (D) 18    (E) 24

**Answer: C**

**Solution 1**

Let the Aggie, Bumblebee, Steelie, and Tiger, be referred to by  $A, B, S$ , and  $T$ , respectively. If we ignore the constraint that  $S$  and  $T$  cannot be next to each other, we get a total of  $4! = 24$  ways to arrange the 4 marbles. We now simply have to subtract out the number of ways that  $S$  and  $T$  can be next to each other. If we place  $S$  and  $T$  next to each other in that order, then there are three places that we can place them, namely in the first two slots, in the second two slots, or in the last two slots (i.e.  $ST\square\square, \square ST\square, \square\square ST$ ). However, we could also have placed  $S$  and  $T$  in the opposite order (i.e.  $TS\square\square, \square TS\square, \square\square TS$ ). Thus there are 6 ways of placing  $S$  and  $T$  directly next to each other. Next, notice that for each of these placements, we have two open slots for placing  $A$  and  $B$ . Specifically, we can place  $A$  in the first open slot and  $B$  in the second open slot or switch their order and place  $B$  in the first open slot and  $A$  in the second open slot. This gives us a total of  $6 \times 2 = 12$  ways to place  $S$  and  $T$  next to each other. Subtracting this from the

(AMC 8 2020) After school, Maya and Naomi headed to the beach, 6 miles away. Maya decided to bike while Naomi took a bus. The graph below shows their journeys, indicating the time and distance traveled. What was the difference, in miles per hour, between Naomi's and Maya's average speeds?



- (A) 6      (B) 12      (C) 18      (D) 20      (E) 24

**Answer:** E

**Solution 1**

Naomi travels 6 miles in a time of 10 minutes, which is equivalent to  $\frac{1}{6}$  of an hour. Since speed =  $\frac{\text{distance}}{\text{time}}$ , her speed is  $\frac{6}{(\frac{1}{6})} = 36$  mph. By a similar calculation, Maya's speed is 12 mph, so the answer is  $36 - 12 = \boxed{\text{(E) } 24}$ .

**Solution 2 (Variant of Solution 1)**

Naomi's speed of 6 miles in 10 minutes is equivalent to  $6 \cdot 6 = 36$  miles per hour, while Maya's speed of 6 miles in 30 minutes (i.e. half an hour) is equivalent to  $6 \cdot 2 = 12$  miles per hour. The difference is consequently  $36 - 12 = \boxed{\text{(E) } 24}$ .

(AMC 8 2020) For a positive integer  $n$ , the factorial notation  $n!$  represents the product of the integers from  $n$  to 1. (For example,  $6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$ .) What value of  $N$  satisfies the following equation?

$$5! \cdot 9! = 12 \cdot N!$$

- (A) 10      (B) 11      (C) 12      (D) 13      (E) 14

**Answer:** A

**Solution 1**

We have  $5! = 2 \cdot 3 \cdot 4 \cdot 5$ , and  $2 \cdot 5 \cdot 9! = 10 \cdot 9! = 10!$ . Therefore, the equation becomes  $3 \cdot 4 \cdot 10! = 12 \cdot N!$ , and so  $12 \cdot 10! = 12 \cdot N!$ . Cancelling the 12s, it is clear that  $N = \boxed{\text{(A) } 10}$ .

**Solution 2 (variant of Solution 1)**

Since  $5! = 120$ , we obtain  $120 \cdot 9! = 12 \cdot N!$ , which becomes  $12 \cdot 10 \cdot 9! = 12 \cdot N!$  and thus  $12 \cdot 10! = 12 \cdot N!$ . We therefore deduce  $N = \boxed{\text{(A) } 10}$ .

**Solution 3 (using answer choices and elimination)**

We can see that the answers (B) to (E) contain a factor of 11, but there is no such factor of 11 in  $5! \cdot 9!$ . The factor 11 is in every answer choice except  $\boxed{\text{(A) } 10}$ , so four of the possible answers are eliminated. Therefore, the answer must be  $\boxed{\text{(A) } 10}$ . edited by HW73

**Solution 4**

We notice that  $5! \cdot 9! = (5!)^2 \cdot (9 \cdot 8 \cdot 7 \cdot 6)$ .

We know that  $5! = 120$ , so we have  $120(5! \cdot 9 \cdot 8 \cdot 7 \cdot 6) = 12 \cdot N!$

Isolating  $N!$  we have  $N! = 10 \cdot 5! \cdot 9 \cdot 8 \cdot 7 \cdot 6 \Rightarrow N! = 10! \Rightarrow N = \boxed{\text{(A) } 10}$ .

**Solution 5**

Notice that  $9!$  is equivalent to  $\frac{12!}{12 \cdot 11 \cdot 10}$ . In the expression  $\frac{12!}{12 \cdot 11 \cdot 10} \cdot 5 \cdot 4 \cdot 3 \cdot 2$ , cancel out 12, 4, 3, and 10, 5, 2. The resulting equation is  $\frac{12!}{11} = 12 \cdot N!$ . The equation gives  $12 \cdot 10! = 12 \cdot N!$ . Therefore,  $N = 10$ , so the answer is  $\boxed{\text{(A) } 10}$ .

(AMC 8 2020) Jamal has a drawer containing 6 green socks, 18 purple socks, and 12 orange socks. After adding more purple socks, Jamal noticed that there is now a 60% chance that a sock randomly selected from the drawer is purple. How many purple socks did Jamal add?

- (A) 6      (B) 9      (C) 12      (D) 18      (E) 24

**Answer:** B

**Solution 1**

After Jamal adds  $x$  purple socks, he has  $(18+x)$  purple socks and  $6+18+12+x = (36+x)$  total socks. This means the probability of drawing a purple sock is  $\frac{18+x}{36+x}$ , so we obtain

$$\frac{18+x}{36+x} = \frac{3}{5}$$

Since  $\frac{18+9}{36+9} = \frac{27}{45} = \frac{3}{5}$ , the answer is **(B) 9**.

**Solution 2 (variant of Solution 1)**

As in Solution 1, we have the equation  $\frac{18+x}{36+x} = \frac{3}{5}$ . Cross-multiplying yields  $90 + 5x = 108 + 3x \Rightarrow 2x = 18 \Rightarrow x = 9$ . Thus, Jamal added **(B) 9** purple socks.

**Solution 3 very fast**

The answer is 9

**Solution 4 (Testing Answer Choices)**

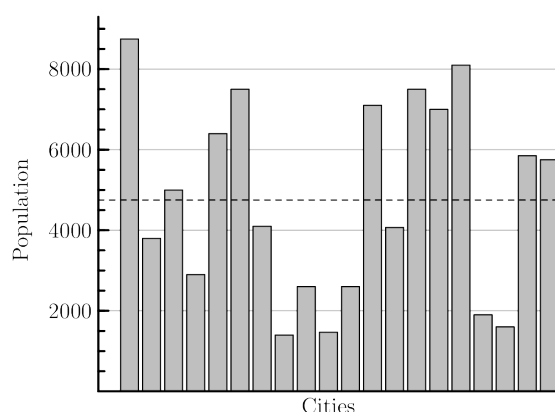
The probability that a randomly selected sock is purple is: The # of purple socks  $\div$  The total # of socks. We can test the answer choices until we find the correct answer.

[A] If Jamal adds 6 purple socks, he will have  $18 + 6 = 24$  purple socks. The probability of a randomly selected sock being purple after adding 6 socks is  $\frac{24}{6+24+12} = \frac{24}{42} = \frac{4}{7}$ , which isn't equal to  $\frac{3}{5}$ .

[B] If Jamal adds 9 purple socks, he will have  $18 + 9 = 27$  purple socks. The probability of a randomly selected sock being purple after adding 9 socks is  $\frac{27}{6+27+12} = \frac{27}{45} = \frac{3}{5}$ , which is indeed equal to  $\frac{3}{5}$ .

So our final answer is **(B) 9** purple socks.

(AMC 8 2020) There are 20 cities in the County of Newton. Their populations are shown in the bar chart below. The average population of all the cities is indicated by the horizontal dashed line. Which of the following is closest to the total population of all 20 cities?



- (A) 65,000      (B) 75,000      (C) 85,000      (D) 95,000      (E) 105,000

**Answer: D**

**Solution**

We can see that the dotted line is exactly halfway between 4,500 and 5,000, so it is at 4,750. As this is the average population of all 20 cities, the total population is simply  $4,750 \cdot 20 = \mathbf{(D) 95,000}$ .

**Solution 2**

The dotted line is slightly below 5000. Since  $5000 \cdot 20 = 100,000$ , the answer is slightly below this, so **(D) 95,000** would be the best choice to guess.

(AMC 8 2020) Suppose 15% of  $x$  equals 20% of  $y$ . What percentage of  $x$  is  $y$ ? (A) 5 (B) 35 (C) 75

**Answer:** C

**Solution 1**

Since  $20\% = \frac{1}{5}$ , multiplying the given condition by 5 shows that  $y$  is  $15 \cdot 5 = \boxed{\text{(C) } 75}$  percent of  $x$ .

**Solution 2**

Letting  $x = 100$  (without loss of generality), the condition becomes  $0.15 \cdot 100 = 0.2 \cdot y \Rightarrow 15 = \frac{y}{5} \Rightarrow y = 75$ . Clearly, it follows that  $y$  is 75% of  $x$ , so the answer is  $\boxed{\text{(C) } 75}$ .

**Solution 3**

We have  $15\% = \frac{3}{20}$  and  $20\% = \frac{1}{5}$ , so  $\frac{3}{20}x = \frac{1}{5}y$ . Solving for  $y$ , we multiply by 5 to give  $y = \frac{15}{20}x = \frac{3}{4}x$ , so the answer is  $\boxed{\text{(C) } 75}$ .

**Solution 4**

We are given  $0.15x = 0.20y$ , so we may assume without loss of generality that  $x = 20$  and  $y = 15$ . This means  $\frac{y}{x} = \frac{15}{20} = \frac{75}{100}$ , and thus the answer is  $\boxed{\text{(C) } 75}$ .

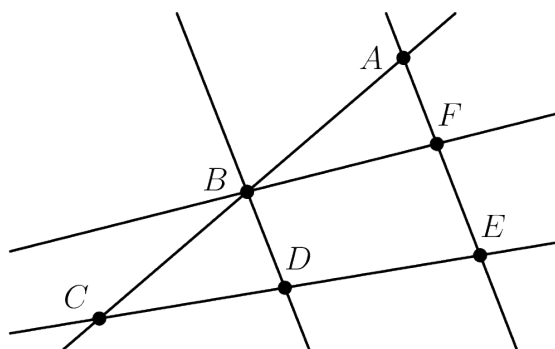
**Solution 5**

15% of  $x$  is  $0.15x$ , and 20% of  $y$  is  $0.20y$ . We put  $0.15x$  and  $0.20y$  into an equation, creating  $0.15x = 0.20y$  because 0.15 $x$  equals 0.20 $y$ . Solving for  $y$ , dividing 0.2 to both sides, we get  $y = \frac{15}{20}x = \frac{3}{4}x$ , so the answer is  $\boxed{\text{(C) } 75}$ .

**Solution 6**

15% of  $x$  can be written as  $\frac{15}{100}x$ , or  $\frac{15x}{100}$ . 20% of  $y$  can similarly be written as  $\frac{20}{100}y$ , or  $\frac{20y}{100}$ . So now,  $\frac{15x}{100} = \frac{20y}{100}$ . Using cross-multiplication, we can simplify the equation as:  $1500x = 2000y$ . Dividing both sides by 500, we get:  $3x = 4y$ .  $\frac{3}{4}$  is the same thing as 75%, so the answer is  $\boxed{\text{(C) } 75}$ .

(AMC 8 2020) Each of the points  $A, B, C, D, E$ , and  $F$  in the figure below represents a different digit from 1 to 6. Each of the five lines shown passes through some of these points. The digits along each line are added to produce five sums, one for each line. The total of the five sums is 47. What is the digit represented by B?



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

**Answer:** E

**Solution 1**

We can form the following expressions for the sum along each line:

$$\begin{cases} A + B + C \\ A + E + F \\ C + D + E \\ B + D \\ B + F \end{cases}$$



Adding these together, we must have  $2A + 3B + 2C + 2D + 2E + 2F = 47$ , i.e.  $2(A + B + C + D + E + F) + B = 47$ . Since  $A, B, C, D, E, F$  are unique integers between 1 and 6, we obtain  $A + B + C + D + E + F = 1 + 2 + 3 + 4 + 5 + 6 = 21$  (where the order doesn't matter as addition is commutative), so our equation simplifies to  $42 + B = 47$ . This means  $B = \boxed{\text{(E)} 5}$ . RJ5303707

**Solution 2**

Following the first few steps of Solution 1, we have  $2(A + C + D + E + F) + 3B = 47$ . Because an even number  $2(A + C + D + E + F)$  subtracted from an odd number (47) is always odd, we know that  $3B$  is odd, showing that  $B$  is odd. Now we know that  $B$  is 1, 3, or 5. If we try  $B = 1$ , we get  $43 = 47$ , which is not true. Also when we test  $B = 3$ , we get  $45 = 47$ , which is also false. Therefore, we end up getting  $B = \boxed{\text{(E)} 5}$ .

-minor edit by tonykuncheng

-minor edit by Lebron James

(AMC 8 2020) How many factors of 2020 have more than 3 factors? (As an example, 12 has 6 factors, namely 1, 2, 3, 4, 6, and 12.)

(A) 6      (B) 7      (C) 8      (D) 9      (E) 10

**Answer:** B

**Solution 1**

Since  $2020 = 2^2 \cdot 5 \cdot 101$ , we can simply list its factors:

1, 2, 4, 5, 10, 20, 101, 202, 404, 505, 1010, 2020.

There are 12 factors; only 1, 2, 4, 5, 101 don't have over 3 factors, so the remaining  $12 - 5 = \boxed{\text{(B)} 7}$  factors have more than 3 factors.

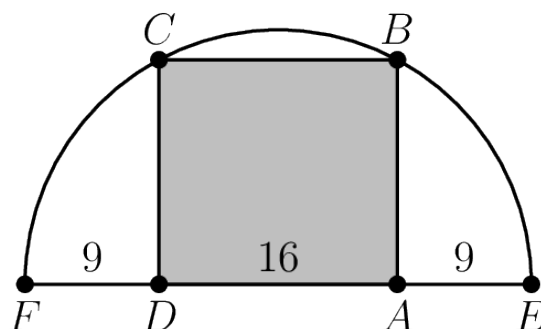
**Solution 2**

As in Solution 1, we prime factorize 2020 as  $2^2 \cdot 5 \cdot 101$ , and we recall the standard formula that the number of positive factors of an integer is found by adding 1 to each exponent in its prime factorization, and then multiplying these. Thus 2020 has  $(2 + 1)(1 + 1)(1 + 1) = 12$  factors. The only number which has one factor is 1. For a number to have exactly two factors, it must be prime, and the only prime factors of 2020 are 2, 5, and 101. For a number to have three factors, it must be a square of a prime (this follows from the standard formula mentioned above), and from the prime factorization, the only square of a prime that is a factor of 2020 is 4. Thus, there are 5 factors of 2020 which themselves have 1, 2, or 3 factors (namely 1, 2, 4, 5, and 101), so the number of factors of 2020 that have more than 3 factors is  $12 - 5 = \boxed{\text{(B)} 7}$ .

**Solution 3**

Let  $d(n)$  be the number of factors of  $n$ . We know by prime factorization that  $d(2020) = 12$ . These 12 numbers can be divided into unordered pairs  $a, b$  where  $ab = 2020$ . Since  $d(2020) = d(a)d(b)$ , one of  $d(a), d(b)$  has 3 or less factors and the other has 4 or more - in to total 6 factors of 2020 with more than 3 factors. However, this argument has exceptions where  $a$  and  $b$  share a nontrivial common factor, which in this case can only be two. There are two cases - One in which 5 and 101 divide the same factor, WLOG assumed to be  $a$ , so that  $d(a) = 2^3 > 3$  and  $d(b) = 2$ , as otherwise. In the other case,  $a = 5 \cdot 2$  and  $b = 101 \cdot 2$ , so that  $d(a) = d(b) = 4$ . This adds one factor with more than 3 factors, so the answer is  $\boxed{\text{(B)} 7}$ .

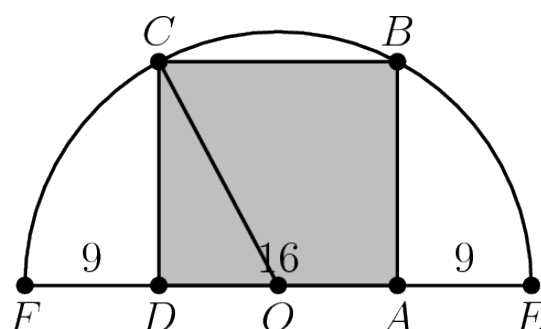
(AMC 8 2020) Rectangle  $ABCD$  is inscribed in a semicircle with diameter  $\overline{FE}$ , as shown in the figure. Let  $DA = 16$ , and let  $FD = AE = 9$ . What is the area of  $ABCD$ ?



- (A) 240    (B) 248    (C) 256    (D) 264    (E) 272

Answer: A

Solution 1 (Pythagorean Theorem)



Let  $O$  be the center of the semicircle. The diameter of the semicircle is  $9 + 16 + 9 = 34$ , so  $OC = 17$ . By symmetry,  $O$  is the midpoint of  $DA$ , so  $OD = OA = \frac{16}{2} = 8$ . By the Pythagorean theorem in right-angled triangle  $ODC$  (or  $OBA$ ), we have that  $CD$  (or  $AB$ ) is  $\sqrt{17^2 - 8^2} = 15$ . Accordingly, the area of  $ABCD$  is  $16 \cdot 15 = \boxed{\text{(A) } 240}$ .

Solution 2 (Coordinate Geometry)

Let the midpoint of segment  $FE$  be the origin. Evidently, point  $D = (-8, 0)$  and  $A = (8, 0)$ . Since points  $C$  and  $B$  share  $x$ -coordinates with  $D$  and  $A$  respectively, it suffices to find the  $y$ -coordinate of  $B$  (which will be the height of the rectangle) and multiply this by  $DA$  (which we know is 16). The radius of the semicircle is  $\frac{9+16+9}{2} = 17$ , so the whole circle has equation  $x^2 + y^2 = 289$ ; as already stated,  $B$  has the same  $x$ -coordinate as  $A$ , i.e. 8, so substituting this into the equation shows that  $y = \pm 15$ . Since  $y > 0$  at  $B$ , the  $y$ -coordinate of  $B$  is 15. Therefore, the answer is  $16 \cdot 15 = \boxed{\text{(A) } 240}$ .

(Note that the synthetic solution (Solution 1) is definitely faster and more elegant. However, this is the solution that you should use if you can't see any other easier strategy.)

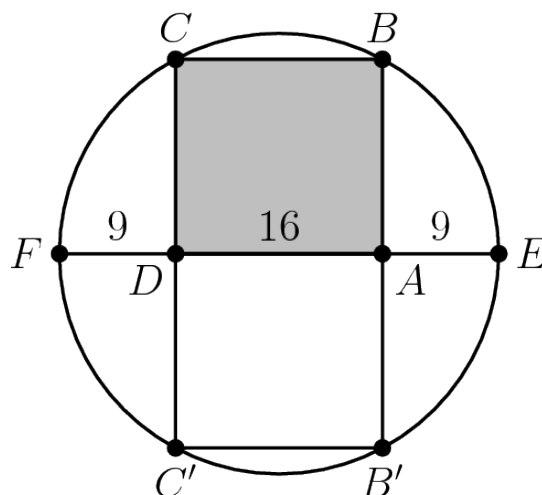
Solution 3

We can use a result from the Art of Problem Solving *Introduction to Algebra* book Sidenote: for a semicircle with diameter  $(1 + n)$ , such that the 1 part is on one side and the  $n$  part is on the other side, the height from the end of the 1 side (or the start of the  $n$  side) is  $\sqrt{n}$ . To use this formula, we scale the figure down by 9; this will give the height a length of  $\sqrt{\frac{16+9}{9}} = \sqrt{\frac{25}{9}} = \frac{5}{3}$ . Now, scaling back up by 9, the height  $DC$  is  $9 \cdot \frac{5}{3} = 15$ .

The answer is then  $15 \cdot 16 = \boxed{\text{(A) } 240}$ .-SweetMango77

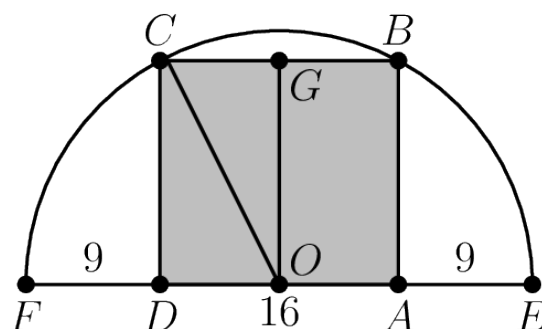
Solution 4 (Power of a Point Theorem)

Draw the other half of the circle as follows:



By the **Power of a Point Theorem**,  $FD \cdot DE = CD \cdot C'D$ . By symmetry,  $CD = C'D$ . We see that  $FD = 9$  and  $DE = 16 + 9 = 25$ . Substituting in these values,  $9 \cdot 25 = CD^2$ , giving  $CD^2 = 225$  and  $CD = 15$ . The area of the rectangle is therefore  $15 \cdot 16 = \boxed{\text{(A)} 240}$ .

**Solution 5 (Vertical Theorem)**



According to the Pythagorean Theorem and the Vertical Theorem, we can find out that  $OG = \sqrt{\left(\frac{2 \times 9 + 16}{2}\right)^2 - \left(\frac{16}{2}\right)^2} = 15$ . Therefore, the answer is  $15 \times 16 = \boxed{\text{(A)} 240}$ ;

(AMC 8 2020) A number is called flippy if its digits alternate between two distinct digits. For example, 2020 and 37373 are flippy, but 3883 and 123123 are not. How many five-digit flippy numbers are divisible by 15? (A) 3 (B) 4 (C) 5 (D) 6 (E) 8

**Answer:** B

**Solution 1**

A number is divisible by 15 precisely if it is divisible by 3 and 5. The latter means the last digit must be either 5 or 0, and the former means the sum of the digits must be divisible by 3. If the last digit is 0, the first digit would be 0 (because the digits alternate), which is not possible. Hence the last digit must be 5, and the number is of the form  $5 \square 5 \square 5$ . If the unknown digit is  $x$ , we deduce  $5 + x + 5 + x + 5 \equiv 0 \pmod{3} \Rightarrow 2x \equiv 0 \pmod{3}$ . We know  $2^{-1}$  exists modulo 3 because 2 is relatively prime to 3, so we conclude that  $x$  (i.e. the second and fourth digit of the number) must be a multiple of 3. It can be 0, 3, 6, or 9, so there are  $\boxed{\text{(B)} 4}$  options: 50505, 53535, 56565, and 59595.

**Solution 2 (variant of Solution 1, no mods)**

As in Solution 1, we find that such numbers must start with 5 and alternate with 5 (i.e. must be of the form  $5 \square 5 \square 5$ ), where the two digits between the 5s need to be the same. Call that digit  $x$ . For the number to be divisible by 3, the sum of the digits must

be divisible by 3; since the sum of the three 5s is 15, which is already a multiple of 3, it must also be the case that  $x + x = 2x$  is a multiple of 3. Thus, the problem reduces to finding the number of digits from 0 to 9 for which  $2x$  is a multiple of 3. This leads to  $x = 0, 3, 6$ , or  $9$ , so there are **(B) 4** possible numbers (namely 50505, 53535, 56565, and 59595).

**Solution 3**

After finding out that the last digit must be 5, the number is of the form  $5\square5\square5$ . If the unknown digit is  $x$ , we can find that one of the solutions to  $x$  is 0, since  $5 + 5 + 5$  is equal to 15, which is divisible by 3. After trying every one digit number, you'll notice that  $x$  must be a multiple of 3, meaning that  $x = 0, 3, 6$ , or  $9$ . 50505, 53535, 56565, and 59595 are the **(B) 4** solutions to this question.

**Solution 4 (mods)**

assume the number is  $ababa10101a + 1010b = 0(mod15)$

$$6a + 5b = 0(mod15)$$

$$a = 0(mod5)$$

$$5b = 0(mod15)$$

$$b = 0(mod3)$$

Solutions:  $(5, 0), (5, 3), (5, 6), (5, 9)$

**4**

(AMC 8 2020) A scientist walking through a forest recorded as integers the heights of 5 trees standing in a row. She observed that each tree was either 2 times as tall or half as tall as the one to its right. Unfortunately some of her data was lost when rain fell on her notebook. Her notes are shown below, with blanks indicating the missing numbers. Based on her observations, the scientist was able to reconstruct the lost data. What was the average height of the trees, in meters?

|                |             |
|----------------|-------------|
| Tree 1         | — meters    |
| Tree 2         | 11 meters   |
| Tree 3         | — meters    |
| Tree 4         | — meters    |
| Tree 5         | — meters    |
| Average height | — .2 meters |

(A) 22.2      (B) 24.2      (C) 33.2      (D) 35.2      (E) 37.2

**Answer:** B

**Solution 1**

We will show that 22, 11, 22, 44, and 22 meters are the heights of the trees from left to right. We are given that all tree heights are integers, so since Tree 2 has height 11 meters, we can deduce that Trees 1 and 3 both have a height of 22 meters. There are now three possible cases for the heights of Trees 4 and 5 (in order for them to be integers), namely heights of 11 and 22, 44 and 88, or 44 and 22. Checking each of these, in the first case, the average is 17.6 meters, which doesn't end in .2 as the problem requires. Therefore, we consider the other cases. With 44 and 88, the average is 37.4 meters, which again does not end in .2, but with 44 and 22, the average is 24.2 meters, which does. Consequently, the answer is **(B) 24.2**.

**Solution 2: EASY TO UNDERSTAND!!!**

Notice the average height of the trees ends with 0.2; therefore, the sum of all five heights of the trees must end with 1 or 6. ( $0.2 \cdot 5 = 1$ ) We already know Tree 2 is 11 meters tall. Both Tree 1 and Tree 3 must 22 meters tall - since neither can be 5.5 because according to the problem it says “integers”. Once again, apply our observation for solving for the Tree 4’s height. Tree 4 can’t be 11 meters for the sum of the five tree heights to still end with 1. Therefore, the Tree 4 is 44 meters tall. Now, Tree 5 can either be 22 or 88. Find the average height for both cases of Tree 5. Doing this, we realize the Tree 5 must be 22 for the average height to end with 0.2 and that the average height is **(B) 24.2**.

-by DummyAMC

### Solution 3

As in Solution 1, we shall show that the heights of the trees are 22, 11, 22, 44, and 22 meters. Let  $S$  be the sum of the heights, so that the average height will be  $\frac{S}{5}$  meters. We note that  $0.2 = \frac{1}{5}$ , so in order for  $\frac{S}{5}$  to end in .2,  $S$  must be one more than a multiple of 5. Moreover, as all the heights are integers, the heights of Tree 1 and Tree 3 are both 22 meters. At this point, our table looks as follows:

|                |               |
|----------------|---------------|
| Tree 1         | 22 meters     |
| Tree 2         | 11 meters     |
| Tree 3         | 22 meters     |
| Tree 4         | ___ meters    |
| Tree 5         | ___ meters    |
| Average height | ___ .2 meters |

If Tree 4 now has a height of 11, then Tree 5 would need to have height 22, but in that case  $S$  would equal 88, which is not 1 more than a multiple of 5. So we instead take Tree 4 to have height 44. Then the sum of the heights of the first 4 trees is  $22 + 11 + 22 + 44 = 99$ , so using a height of 22 for Tree 5 gives  $S = 121$ , which is 1 more than a multiple of 5 (whereas 88 gives  $S = 187$ , which is not). Thus the average height of the trees is  $\frac{121}{5} = \mathbf{(B) 24.2}$  meters.

### Solution 4

Since we know that the tree heights have to be integers, then it is immediate that Tree 1 and 3 have a height of 22. Now using the information given by the last column (that the average of the heights of the trees ends in .2), we can tell that the sum of all the heights of the trees ends in either 1 or 6, because those are the only numbers from 0 to 9 that are congruent to 1 after taking modulo 5. The two multiples of eleven (eleven because all of the tree heights have to be a multiple of eleven if they are integers) that come to mind are 66 and 121. Since the sum of the heights of Tree 1, 2, and 3 is already 55, we know that 66 is impossible to obtain. Then, we can decide with relative confidence that the answer should be  $\frac{121}{5} = \mathbf{(B) 24.2}$ .

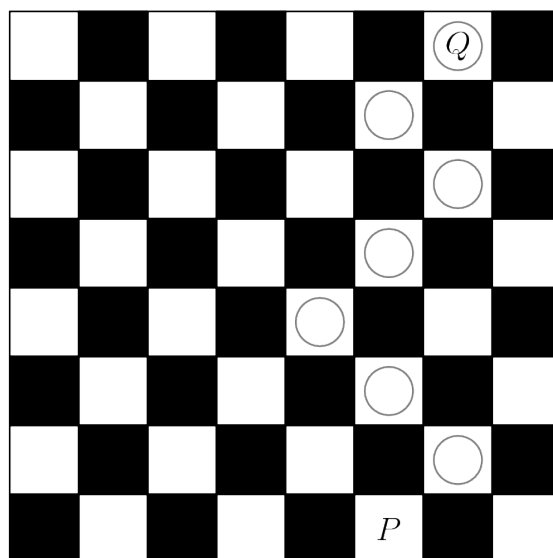
P.S. : We can check our solution by trying to obtain 121. Surely enough, we have Tree 1 = 22, Tree 2 = 11, Tree 3 = 22, Tree 4 = 44, and Tree 5 = 22. These all add up to get 121.

### Solution 5 (basic modular arithmetic)

Since her average ends with 0.2, the sum of her tree heights must be  $\equiv 1 \pmod{5}$ . If Tree 1’s height is  $a$ , Tree 3’s is  $b$ , Tree 4’s is  $c$ , and Tree 5’s is  $d$ , then we  $a$  and  $b$  are both 22, since all of the tree heights are integers. Now we have  $22 + 11 + 22 + c + d \equiv 1$

(mod 5). Simplifying, we get  $c + d \equiv 1 \pmod{5}$ . The only possible combination of  $c$  and  $d$  that abides the condition are 44 and 22, respectively. Adding these up, we get 121, and  $\frac{121}{5}$  is **(B) 24.2**.  
 -kn07

(AMC 8 2020) A game board consists of 64 squares that alternate in color between black and white. The figure below shows square  $P$  in the bottom row and square  $Q$  in the top row. A marker is placed at  $P$ . A step consists of moving the marker onto one of the adjoining white squares in the row above. How many 7-step paths are there from  $P$  to  $Q$ ? (The figure shows a sample path.)



- (A) 28      (B) 30      (C) 32      (D) 33      (E) 35

**Answer:** A

**Solution 1**

Notice that, from one of the 1 or 2 white squares immediately beneath it (since the marker can only move on white squares). This means that the number of ways to move from  $P$  to that square is the sum of the number of ways to move from  $P$  to each of the white squares immediately beneath it (also called the Waterfall Method). To solve the problem, we can accordingly construct the following diagram, where each number in a square is calculated as the sum of the numbers on the white squares immediately beneath that square (and thus will represent the number of ways to remove from  $P$  to that square, as already stated).

Solution/AMC\_8\_AMC\_2020\_21\_Solution\_Figure\_1.png

The answer is therefore **(A) 28**.

Note: This is a classic example of a problem involving **Pascal's triangle**.

### Solution 2

Suppose we "extend" the chessboard infinitely with 2 additional columns to the right, as shown below. The red line shows the right-hand edge of the original board.

Solution/AMC\_8\_AMC\_2020\_21\_Solution\_Figure\_2.png

The total number of paths from  $P$  to  $Q$ , including invalid paths which cross over the red line, is then the number of paths which make 4 steps up-and-right and 3 steps up-and-left, which is  $\binom{4+3}{3} = \binom{7}{3} = 35$ . We need to subtract the number of invalid paths, i.e. the number of paths that pass through  $X$  or  $Y$ . To get to  $X$ , the marker has to make 3 up-and-right steps, after which it can proceed to  $Q$  with 3 steps up-and-left and 1 step up-and-right. Thus, the number of paths from  $P$  to  $Q$  that pass through  $X$  is  $1 \cdot \binom{3+1}{3} = 4$ . Similarly, the number of paths that pass through  $Y$  is  $\binom{4+1}{1} \cdot 1 = 5$ . However, we have now double-counted the invalid paths which pass through both  $X$  and  $Y$ ; from the diagram, it is clear that there are only 2 of these (as the marker can get from  $X$  to  $Y$  by a step up-and-left and a step-up-and-right in either order). Hence the number of invalid paths is  $4 + 5 - 2 = 7$ , and the number of valid paths from  $P$  to  $Q$  is

$$35 - 7 = \boxed{\text{(A)} 28}.$$

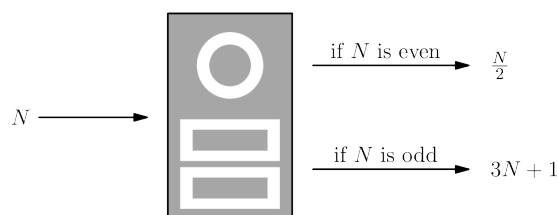
**Solution 3**

On any white square, we may choose to go left or right, as long as we do not cross over the border of the board. Call the moves  $L$  and  $R$  respectively. Every single legal path consists of 4  $R$ 's and 3  $L$ 's, so now all we have to find is the number of ways to order 4  $R$ 's and 3  $L$ 's in any way, which is  $\binom{7}{3} = 35$ . However, we originally promised that we will not go over the border, and now we have to subtract the paths that do go over the border. The paths that go over the border are any paths that start with  $RRR$  (1 path),  $RR$  (5 paths) and  $LRRR$  (1 path) so our final number of paths is  $35 - 7 = \boxed{\text{(A)} = 28}$ . PEKKA

**Solution 4: Intuitive Approach**

We label the rows starting from the bottom. At row 1, there is 1 way: at P. We draw all the possible ways to get to Q. There are two ways to choose for row 2, and another two ways to choose for row 3. However, you can go to the "edge" or the farthest possible square westward of Q, so you can't multiply by 2 again. Notice how, at the first step, we figured that the answer was even, so choice D and E are eliminated, and after the second row, we realized it must be a multiple of 4, so choice B is eliminated. When we get to the fourth row, we do not multiply by 2 again, since we have limited possibilities rather than multiplying by 2 again. Choice C implies that there are two possibilities per row; however, we know that if you go to the farthest possible, you only have one possibility, so it is not  $2^5 = 32$  so we know that the answer is choice  $\boxed{\text{(A)}}$ . hh99754539

(AMC 8 2020) When a positive integer  $N$  is fed into a machine, the output is a number calculated according to the rule shown below.



For example, starting with an input of  $N = 7$ , the machine will output  $3 \cdot 7 + 1 = 22$ . Then if the output is repeatedly inserted into the machine five more times, the final output is 26.

$$7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26$$

When the same 6-step process is applied to a different starting value of  $N$ , the final output is 1. What is the sum of all such integers  $N$ ?

$$N \rightarrow \underline{\hspace{1cm}} \rightarrow \underline{\hspace{1cm}} \rightarrow \underline{\hspace{1cm}} \rightarrow \underline{\hspace{1cm}} \rightarrow \underline{\hspace{1cm}} \rightarrow 1$$

- (A) 73      (B) 74      (C) 75      (D) 82      (E) 83

**Answer:** E

**Solution 1**

We start with final output of 1 and work backward, taking care to consider all possible inputs that could have resulted in any particular output. This produces following set of possibilities each stage:

$$\{1\} \rightarrow \{2\} \rightarrow \{4\} \rightarrow \{1, 8\} \rightarrow \{2, 16\} \rightarrow \{4, 5, 32\} \rightarrow \{1, 8, 10, 64\}$$

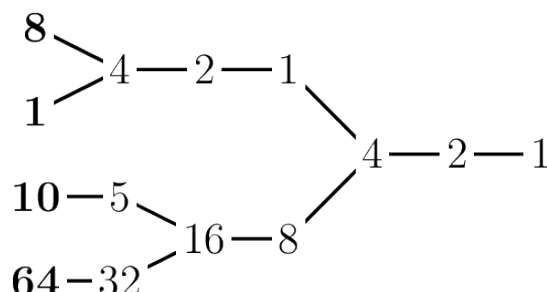
where, for example, 2 must come from 4 (as there is no integer  $n$  satisfying  $3n + 1 = 2$ ), but 16 could come from 32 or 5 (as  $\frac{32}{2} = 3 \cdot 5 + 1 = 16$ , and 32 is even while 5 is odd).



By construction, last set in this sequence contains all the numbers which will lead to number 1 to end of the 6-step process, and sum is  $1 + 8 + 10 + 64 = \boxed{\text{(E)} 83}$ .

**Solution 2 (variant of Solution 1)**

As in Solution 1, we work backwards from 1, this time showing the possible cases in a tree diagram:



The possible numbers are those at the "leaves" of the tree (the ends of the various branches), which are 1, 8, 64, and 10. Thus the answer is  $1 + 8 + 64 + 10 = \boxed{\text{(E)} 83}$ .

**Solution 3 (algebraic)**

We begin by finding the inverse of the function that the machine uses. Call the input  $I$  and the output  $O$ . If  $I$  is even,  $O = \frac{I}{2}$ , and if  $I$  is odd,  $O = 3I + 1$ . We can therefore see that  $I = 2O$  when  $I$  is even and  $I = \frac{O-1}{3}$  when  $I$  is odd. Therefore, starting with 1, if  $I$  is even,  $I = 2$ , and if  $I$  is odd,  $I = 0$ , but the latter is not valid since 0 is not actually odd. This means that the 2nd-to-last number in the sequence has to be 2. Now, substituting 2 into the inverse formulae, if  $I$  is even,  $I = 4$  (which is indeed even), and if  $I$  is odd,  $I = \frac{1}{3}$ , which is not an integer. This means the 3rd-to-last number in the sequence has to be 4. Substituting in 4, if  $I$  is even,  $I = 8$ , but if  $I$  is odd,  $I = 1$ . Both of these are valid solutions, so the 4th-to-last number can be either 1 or 8. If it is 1, then by the argument we have just made, the 5th-to-last number has to be 2, the 6th-to-last number has to be 4, and the 7th-to-last number, which is the first number, must be either 1 or 8. In this way, we have ultimately found two solutions:  $N = 1$  and  $N = 8$ .

On the other hand, if the 4th-to-last number is 8, substituting 8 into the inverse formulae shows that the 5th-to-last number is either 16 or  $\frac{7}{3}$ , but the latter is not an integer. Substituting 16 shows that if  $I$  is even,  $I = 32$ , but if  $I$  is odd,  $I = 5$ , and both of these are valid. If the 6th-to-last number is 32, then the first number must be 64, since  $\frac{31}{3}$  is not an integer; if the 6th-to-last number is 5, then the first number has to be 10, as  $\frac{4}{3}$  is not an integer. This means that, in total, there are 4 solutions for  $N$ , specifically, 1, 8, 10, and 64, which sum to  $\boxed{\text{(E)} 83}$ .

**Remark** This function is the basis for the [Collatz conjecture](#), which asserts that every positive integer  $n$  will eventually output the cycle 1, 4, 2, 1, 4, 2, ... when this function is repeatedly iterated starting with  $n$ . This conjecture remains unsolved but has been tested for all starting integers up to  $2^{71}$

(AMC 8 2020) Five different awards are to be given to three students. Each student will receive at least one award. In how many different ways can the awards be distributed?

- (A) 120      (B) 150      (C) 180      (D) 210      (E) 240

**Answer:** B

**Solution 1 (Constructive Counting)**

Firstly, observe that a single student can't receive 4 or 5 awards because this would mean that one of the other students receives no awards. Thus, each student must receive either 1, 2, or 3 awards. If a student receives 3 awards, then the other two students

must each receive 1 award; if a student receives 2 awards, then another student must also receive 2 awards and the remaining student must receive 1 award. We consider each of these two cases in turn.

If a student receives three awards, there are 3 ways to choose which student this is, and  $\binom{5}{3}$  ways to give that student 3 out of the 5 awards. Next, there are 2 students left and 2 awards to give out, with each student getting one award. There are clearly just 2 ways to distribute these two awards out, giving  $3 \cdot \binom{5}{3} \cdot 2 = 60$  ways to distribute the awards in this case.

In the other case, two students receive 2 awards and one student receives 1 award. We know there are 3 choices for which student gets 1 award. There are  $\binom{3}{1}$  ways to do this. Then, there are  $\binom{5}{2}$  ways to give the first student his two awards, leaving 3 awards yet to be distributed. There are then  $\binom{3}{2}$  ways to give the second student his 2 awards. Finally, there is only 1 student and 1 award left, so there is only 1 way to distribute this award. This results in  $\binom{5}{2} \cdot \binom{3}{2} \cdot 1 \cdot 3 = 90$  ways to distribute the awards in this case. Adding the results of these two cases, we get  $60 + 90 = \boxed{\text{(B) } 150}$ .

### Solution 2 (Version 2 of Solution 1)

Upon inspection (specified in the above solution), there are two cases of the distribution of awards to the students: one student gets 3 awards and the other each get 1 award or one student gets 1 award and the other two get 2 awards.

In the first case, there are  $\binom{3}{1} = 3$  ways to choose the person who gets 3 awards. From here, there are  $\binom{5}{3} = 10$  ways to choose the 3 awards from the 5 total awards. Now, one person has 2 choices for awards and the other has 1 choice for the award. Thus, the total number of ways to choose awards in this case is  $3 \cdot 10 \cdot 2 \cdot 1 = 60$ .

In the other case, there are  $\binom{3}{1} = 3$  ways to choose the person who gets 1 award, and 5 choices for his/her award. Then, one person has  $\binom{4}{2} = 6$  ways to have his/her awards and the other person has  $\binom{2}{2} = 1$  ways to have his/her awards. This gives  $3 \cdot 5 \cdot 6 \cdot 1 = 90$  ways for this case.

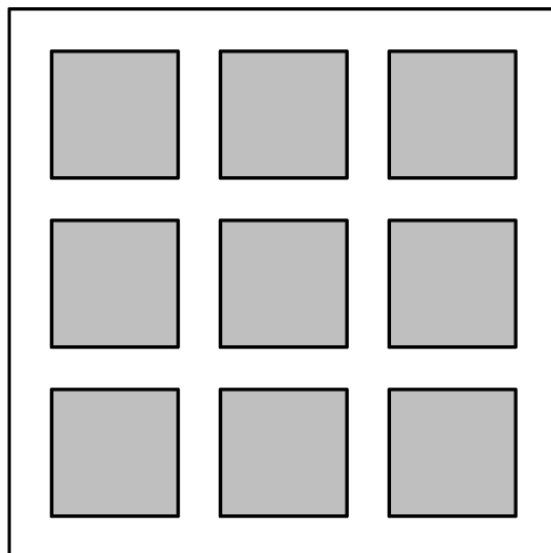
Adding these numbers together, we get  $60 + 90 = 150$  ways to distribute the awards, or choice  $\boxed{\text{(B) } 150}$ .

### Solution 3 (Complementary Counting)

Without the restriction that each student receives at least one award, we could take each of the awards and choose one of the 3 students to give it to. This would be  $3^5$  ways to distribute the awards in total. Now we need to subtract the cases where at least one student doesn't receive an award. If a student doesn't receive an award, there are 3 choices for which student that is, so  $2^5$  ways of choosing a student to receive each of the awards; in total,  $3 \cdot 32 = 96$ .

However, if 2 students both don't receive an award, then this case would be counted twice among the 96, so we need to add back in these cases. Said in other words, 2 students not receiving an award is equivalent to 1 student receiving 5 awards, and there are 3 choices for whom that student would be. To finish, the total number of ways to distribute the awards is  $243 - 96 + 3$ , or  $\boxed{\text{(B) } 150}$ .

(AMC 8 2020) A large square region is paved with  $n^2$  gray square tiles, each measuring  $s$  inches on a side. A border  $d$  inches wide surrounds each tile. The figure below shows the case for  $n = 3$ . When  $n = 24$ , the 576 gray tiles cover 64% of the area of the large square region. What is the ratio  $\frac{d}{s}$  for this larger value of  $n$ ?



- (A)  $\frac{6}{25}$     (B)  $\frac{1}{4}$     (C)  $\frac{9}{25}$     (D)  $\frac{7}{16}$     (E)  $\frac{9}{16}$

**Answer:** A

**Solution 1**

The area of the shaded region is  $(24s)^2$ . To find the area of the large square, we note that there is a  $d$ -inch border between each of the 23 pairs of consecutive squares, as well as from between the first/last squares and the large square, for a total of  $23 + 2 = 25$  times the length of the border, i.e.  $25d$ . Adding this to the total length of the consecutive squares, which is  $24s$ , the side length of the large square is  $(24s + 25d)$ , yielding the equation  $\frac{(24s)^2}{(24s+25d)^2} = \frac{64}{100}$ . Taking the square root of both sides (and using the fact that lengths are non-negative) gives  $\frac{24s}{24s+25d} = \frac{8}{10} = \frac{4}{5}$ , and cross-multiplying now gives  $120s = 96s + 100d \Rightarrow 24s = 100d \Rightarrow \frac{d}{s} = \frac{24}{100} = \boxed{\text{(A)} \frac{6}{25}}$ .

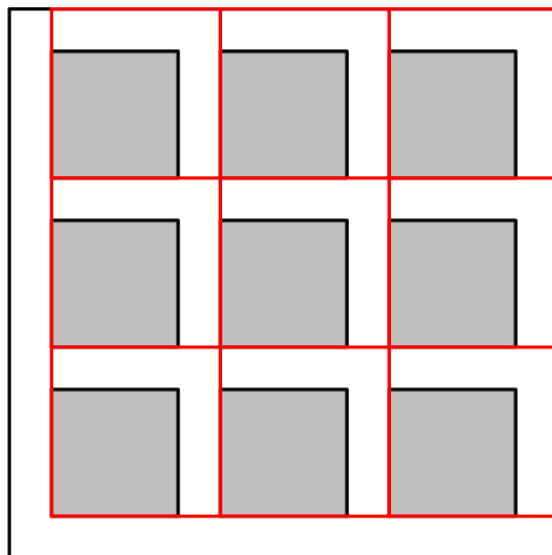
Note: Once we obtain  $\frac{24s}{24s+25d} = \frac{4}{5}$ , to ease computation, we may take the reciprocal of both sides to yield  $\frac{24s+25d}{24s} = 1 + \frac{25d}{24s} = \frac{5}{4}$ , so  $\frac{25d}{24s} = \frac{1}{4}$ . Multiplying both sides by  $\frac{24}{25}$  yields the same answer as before.

**Solution 2**

WLOG (Without Loss of Generality), we may let  $s = 1$  (since  $d$  will be determined by the scale of  $s$ , and we are only interested in the ratio  $\frac{d}{s}$ ). Then, as the total area of the 576 gray tiles is simply 576, the large square has area  $\frac{576}{0.64} = 900$ , making the side of the large square  $\sqrt{900} = 30$ . As in Solution 1, the side length of the large square consists of the total length of the gray tiles and 25 lots of the border, so the length of the border is  $d = \frac{30-24}{25} = \frac{6}{25}$ . Since  $\frac{d}{s} = d$  if  $s = 1$ , the answer is  $\boxed{\text{(A)} \frac{6}{25}}$ .

**Solution 3 (using answer choices)**

As in Solution 2, we let  $s = 1$  without loss of generality. For sufficiently large  $n$ , we can approximate the percentage of the area covered by the gray tiles by subdividing most of the region into congruent squares, as shown:



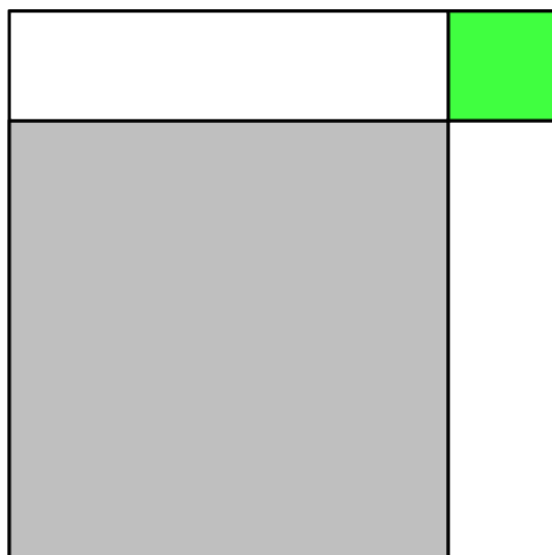
Each red square has side length  $(1 + d)$ , so by solving  $\frac{1^2}{(1+d)^2} = \frac{64}{100} \iff \frac{1}{1+d} = \frac{4}{5}$ , we obtain  $d = \frac{1}{4}$ . The actual fraction of the total area covered by the gray tiles will be slightly less than  $\frac{1}{(1+d)^2}$ , which implies  $\frac{1}{(1+d)^2} > \frac{64}{100} \iff \frac{1}{1+d} > \frac{4}{5} \iff d < \frac{1}{4}$ . Hence  $d$  (and thus  $\frac{d}{s}$ , since we are assuming  $s = 1$ ) is less than  $\frac{1}{4}$ , and the only choice that satisfies this is **(A)**  $\frac{6}{25}$ .

Note the answer cannot be  $\frac{1}{4}$  because this is only an approximation. The closest answer choice is A.

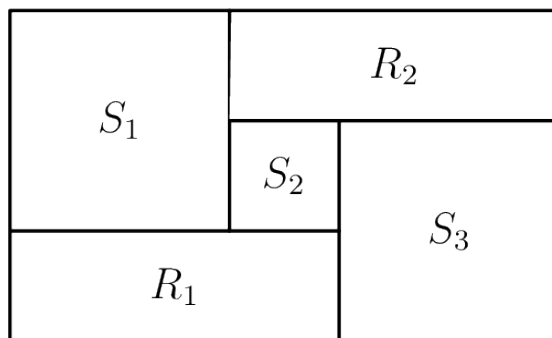
#### Solution 4

Pack all tiles together to form a square. As this square  $(24s \times 24s)$  covers 64% of the big square, that means its length is 80% long of the big square's  $(24s + 25d \times 24s + 25d)$ .

Thus,  $25d : 24s = 2 : 8$  implies  $\frac{d}{s} = \textbf{(A)} \frac{6}{25}$ .



(AMC 8 2020) Rectangles  $R_1$  and  $R_2$ , and squares  $S_1$ ,  $S_2$ , and  $S_3$ , shown below, combine to form a rectangle that is 3322 units wide and 2020 units high. What is the side length of  $S_2$  in units?



- (A) 651      (B) 655      (C) 656      (D) 662      (E) 666

**Answer:** A

**Solution 1**

Let the side length of each square  $S_k$  be  $s_k$ . Then, from the diagram, we can line up the top horizontal lengths of  $S_1$ ,  $S_2$ , and  $S_3$  to cover the top side of the large rectangle, so  $s_1 + s_2 + s_3 = 3322$ . Similarly, the short side of  $R_2$  will be  $s_1 - s_2$ , and lining this up with the left side of  $S_3$  to cover the vertical side of the large rectangle gives  $s_1 - s_2 + s_3 = 2020$ . We subtract the second equation from the first to obtain  $2s_2 = 1302$ , and thus  $s_2 = \boxed{\text{(A) } 651}$ .

**Solution 2**

Assuming that the problem is well-posed, it should be true in the particular case where  $S_1 \cong S_3$  and  $R_1 \cong R_2$ . Let the sum of the side lengths of  $S_1$  and  $S_3$  be  $x$ , and let the length of square  $S_2$  be  $y$ . We then have the system

$$\begin{cases} x + y = 3322 \\ x - y = 2020 \end{cases}$$

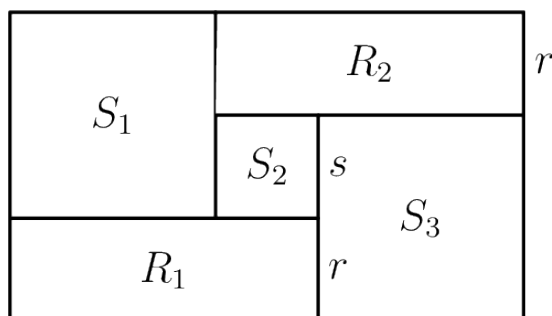
which we solve to determine  $y = \boxed{\text{(A) } 651}$ .

**Solution 3 (faster version of Solution 1)**

Since, for each pair of rectangles, the side lengths have a sum of 3322 or 2020 and a difference of  $S_2$ , the answer must be  $\frac{3322 - 2020}{2} = \frac{1302}{2} = \boxed{\text{(A) } 651}$ .

**Solution 4**

Let the side length of  $S_2$  be  $s$ , and the shorter side length of  $R_1$  and  $R_2$  be  $r$ . We have



From this diagram, it is evident that  $r + s + r = 2020$ . Also, the side length of  $S_1$  and  $S_3$  is  $r + s$ . Then,  $r + s + s + r + s = 3322$ . Now, we have 2 systems of equations.

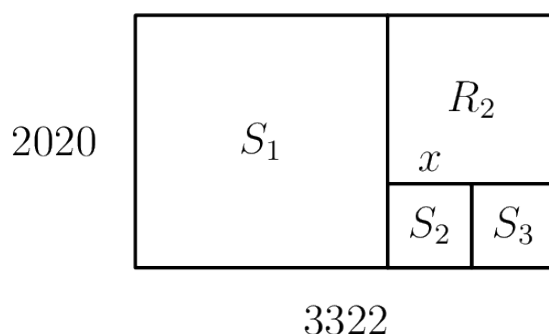
$$\begin{aligned} r + s + r &= 2020 \\ r + s + r + s + s &= 3322 \end{aligned}$$

We can see an  $r + s + r$  in the 2nd equation, so substituting that in gives us  $2020 + 2s = 3322 \Rightarrow 2s = 1302 \Rightarrow s = \boxed{\text{(A) } 651}$ .

**Solution 5 (Cheese)**

Note that the problem is underspecified. We can modify the diagram in any way as long as the relative positioning of the internal rectangles and squares is maintained. Here's an animation describing all such possible configurations:

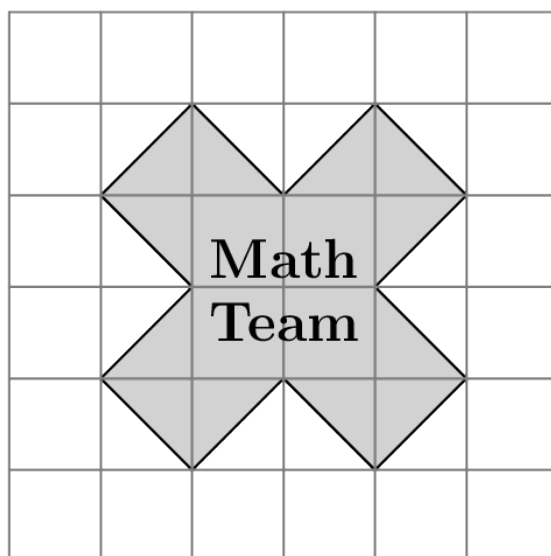
Let's pick the configuration where  $S_1$  is as large as possible and takes up the whole height of the rectangle. Then our diagram is just:



For  $S_2$  and  $S_3$  to share a top edge, they must actually be congruent! Therefore, the width of the entire rectangle, being given as 3322, is also  $2020 + 2x$ . Solving for  $x$ , we get  $x = \boxed{\text{(A) } 651}$ .

**§37 AMC 8 2022**

1. (*AMC 8 2022*) The Math Team designed a logo shaped like a multiplication symbol, shown below on a grid of 1-inch squares. What is the area of the logo in square inches?



(A) 10      (B) 12      (C) 13      (D) 14      (E) 15

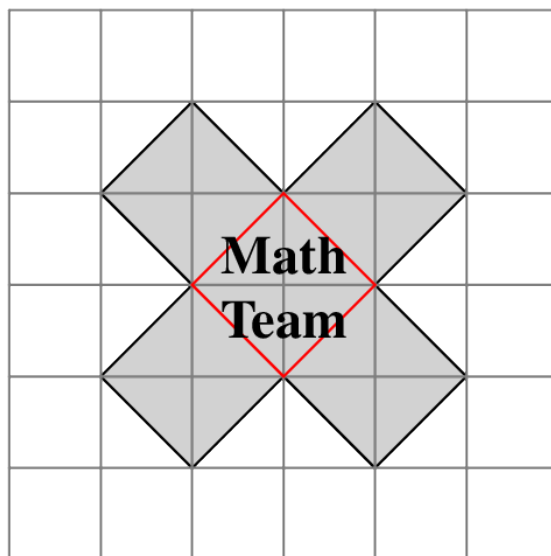
**Answer:** A

**Solution 1 (A quick solution.)**

We can see that there are 4 whole squares, since the area of each square will be 1,  $4 \cdot 1 = 4$ . Next, there are 12 half squares, and 2 half squares are 1 whole square, so  $12/2 = 6$  whole squares. The area of this will be  $6 \cdot 1 = 6$ . Finally, we can add the 2 numbers.  $4 + 6 = \boxed{\text{(A)} 10}$ .

**Solution 2**

Draw the following four lines as shown:



We see these lines split the figure into five squares with side length  $\sqrt{2}$ . Thus, the area is  $5 \cdot (\sqrt{2})^2 = 5 \cdot 2 = \boxed{\text{(A)} 10}$ .

**Solution 3**

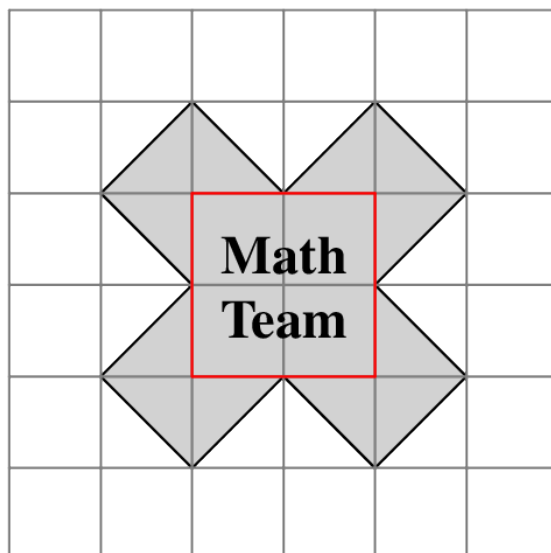
There are 5 lattice points in the interior of the logo and 12 lattice points on the boundary of the logo. Because of Pick's Theorem, the area of the logo is  $5 + \frac{12}{2} - 1 = \boxed{\text{(A)} 10}$ .

**Solution 4**

Notice that the area of the figure is equal to the area of the  $4 \times 4$  square subtracted by the 12 triangles that are half the area of each square, which is 1. The total area of the triangles not in the figure is  $12 \cdot \frac{1}{2} = 6$ , so the answer is  $16 - 6 = \boxed{\text{(A)} 10}$ .

**Solution 5**

Draw the following four lines as shown:



The area of the big square is 16, and the area of each triangle is 0.5. There are 12 of these triangles, so the total area of all the triangles is  $0.5 \cdot 12 = 6$ . Therefore, the area of the entire figure is  $16 + 6 = \boxed{\text{(A)} 22}$ .

**Solution 6 (Shoelace Theorem)**

The coordinates are  $(1, 2), (2, 1), (3, 2), (4, 1), (5, 2), (4, 3), (5, 4), (4, 5), (3, 4), (2, 5), (1, 4), (2, 3)$ . Use the **Shoelace Theorem** to get  $\boxed{\text{(A)} 10}$ .

**Solution 7 (Quick)**

If the triangles are rearranged such that the gaps are filled, there would be a 4 by 2 rectangle, and two 1 by 1 squares are present. Thus, the answer is  $\boxed{\text{(A)} 10}$ .

2. (AMC 8 2022) Consider these two operations:

$$a \blacklozenge b = a^2 - b^2$$

$$a \blackstar b = (a - b)^2$$

What is the value of  $(5 \blacklozenge 3) \blackstar 6$ ? (A)  $-20$  (B)  $4$  (C)  $16$  (D)  $100$  (E)  $220$

**Answer:** D

**Solution 1**

We can find a general solution to any  $((a \blacklozenge b) \blackstar c)$ .

$$\begin{aligned} & ((a \blacklozenge b) \blackstar c) \\ &= ((a^2 - b^2) \blackstar c) \\ &= (a^2 - b^2 - c)^2 \\ &= a^4 + b^4 - (a^2)(b^2) - 2(a^2)(c) - (b^2)(a^2) + 2(b^2)(c) + c^2 \\ &= 5^4 + 3^4 - (5^2)(3^2) - 2(5^2)(6) - (3^2)(5^2) + 2(3^2)(6) + 6^2 \\ &= 625 + 81 - 225 - 300 - 225 + 108 + 36 \\ &= \boxed{\text{(D)} 100} \end{aligned}$$



To time wasting

just go to solution 2

**Solution 2**

The problem is basically  $((5^2 - 3^2) - 6)^2$

$= (16 - 6)^2 = 10^2 = 100$  = option D

3. (AMC 8 2022) When three positive integers  $a$ ,  $b$ , and  $c$  are multiplied together, their product is 100. Suppose  $a < b < c$ . In how many ways can the numbers be chosen?

(A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** E

**Solution 1**

The positive divisors of 100 are

1, 2, 4, 5, 10, 20, 25, 50, 100.

It is clear that  $10 \leq c \leq 50$ , so we apply casework to  $c$ :

- If  $c = 10$ , then  $(a, b, c) = (2, 5, 10)$ .
- If  $c = 20$ , then  $(a, b, c) = (1, 5, 20)$ .
- If  $c = 25$ , then  $(a, b, c) = (1, 4, 25)$ .
- If  $c = 50$ , then  $(a, b, c) = (1, 2, 50)$ .

Together, the numbers  $a$ ,  $b$ , and  $c$  can be chosen in (E) 4 ways.

**Solution 2**

The positive divisors of 100 are

1, 2, 4, 5, 10, 20, 25, 50, 100.

We apply casework to  $a$ :

If  $a = 1$ , then there are 3 cases:

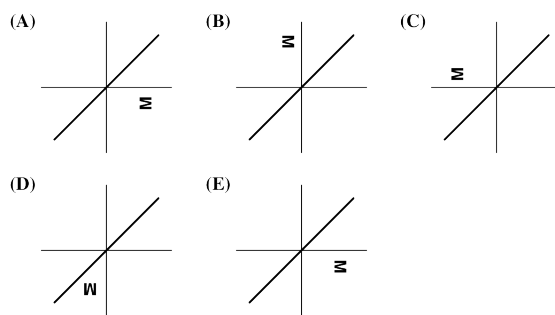
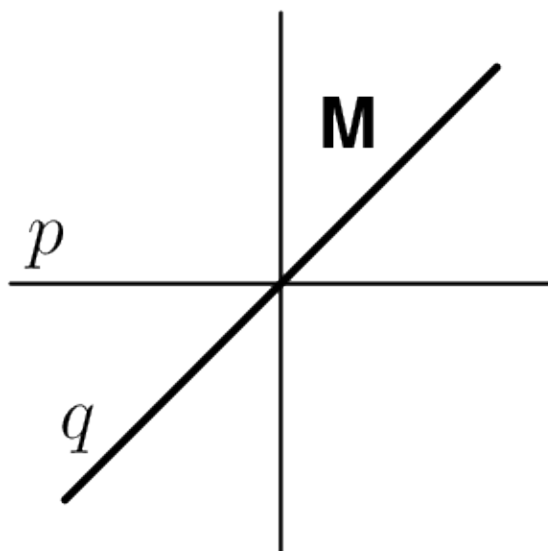
- $b = 2, c = 50$
- $b = 4, c = 25$
- $b = 5, c = 20$

If  $a = 2$ , then there is only 1 case:

- $b = 5, c = 10$

In total, there are  $3 + 1 =$ (E) 4 ways to choose distinct positive integer values of  $a, b, c$ .

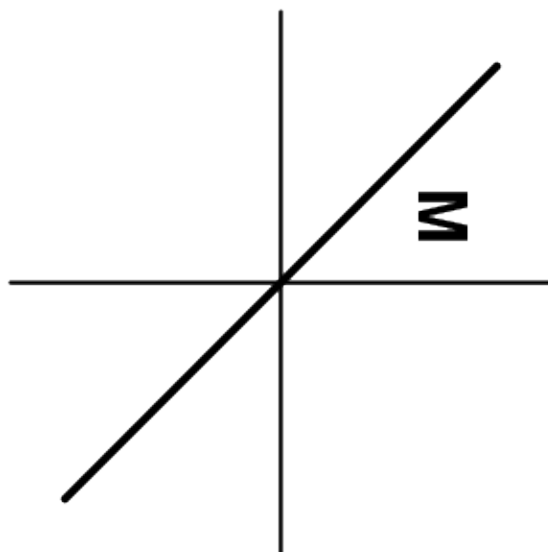
4. (AMC 8 2022) The letter **M** in the figure below is first reflected over the line  $q$  and then reflected over the line  $p$ . What is the resulting image?



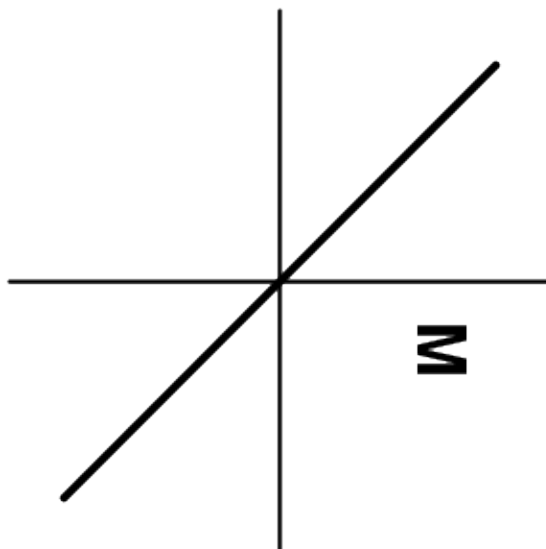
**Answer:** E

**Solution**

When  $M$  is first reflected over the line  $q$ , we obtain the following diagram:



When  $M$  is then reflected over the line  $p$ , we obtain the following diagram:



Therefore, the answer is **(E)**.

5. (AMC 8 2022) Anna and Bella are celebrating their birthdays together. Five years ago, when Bella turned 6 years old, she received a newborn kitten as a birthday present. Today the sum of the ages of the two children and the kitten is 30 years. How many years older than Bella is Anna?

(A) 1    (B) 2    (C) 3    (D) 4    (E) 5

**Answer:** C

**Solution**

Five years ago, Bella was 6 years old, and the kitten was 0 years old.

Today, Bella is 11 years old, and the kitten is 5 years old. It follows that Anna is  $30 - 11 - 5 = 14$  years old.

Therefore, Anna is  $14 - 11 =$  **(C) 3** years older than Bella.

**Solution 2**

We assume that

a) the current age of Anna = a

b) the current age of Bella = b

$$b - 5 = 6$$

$$\text{Thus, } b = 11$$

Kitten = 5 (as the it got it 5 years ago)

Since the total age of kitten and girls = 30,

$$\text{then } a = 30 - (11 + 5)$$

$$= 30 - 16$$

$$= 14 \text{ years.}$$

BUT THIS IS NOT THE FINAL ANSWER!!!

$$a - b = 14 - 11 = 3 = \text{option C}$$

6. (AMC 8 2022) Three positive integers are equally spaced on a number line. The middle number is 15, and the largest number is 4 times the smallest number. What is the smallest of these three numbers?

(A) 4      (B) 5      (C) 6      (D) 7      (E) 8

**Answer:** C

**Solution 1**

Let the smallest number be  $x$ . It follows that the largest number is  $4x$ .

Since  $x$ , 15, and  $4x$  are equally spaced on a number line, we have

$$4x - 15 = 15 - x$$

$$5x = 30$$

$$x = \boxed{\text{(C) } 6}.$$

MRENTHUSIASM

**Solution 2**

Let the common difference of the arithmetic sequence be  $d$ . Consequently, the smallest number is  $15 - d$  and the largest number is  $15 + d$ . As the largest number is 4 times the smallest number,  $15 + d = 60 - 4d \implies d = 9$ . Finally, we find that the smallest number is  $15 - 9 = \boxed{\text{(C) } 6}$ .

**Solution 3**

Let the smallest number be  $x$ . Because  $x$  and  $4x$  are equally spaced from 15, 15 must be the average. By adding  $x$  and  $4x$  and dividing by 2, we get that the mean is also  $2.5x$ . We get that  $2.5x = 15$ , and solving gets  $x = \boxed{\text{(C) } 6}$ .

**Solution 4**

Since the problem is giving us the options, we can try all of them. The differences MUST be equal because the numbers have to be evenly spaced on the number line. We see that 4 cannot work because  $15 - 4 = 11$ , and  $4 * 4 = 16$ .  $16 - 15$  is not 11, so this option does not work. Next, we can try 5. We see that this also does not work because  $15 - 5 = 10$ , and  $5 * 4 = 20$ .  $20 - 15 = 5$ , and this is not 10. We can then try 6 to see that it does work.  $15 - 6 = 9$ , and  $6 * 4 = 24$ .  $24 - 15 = 9$ . The differences are equivalent, so we can see that 6 indeed does work. Therefore, the answer is  $\boxed{\text{(C) } 6}$ .

7. (AMC 8 2022) When the World Wide Web first became popular in the 1990s, download speeds reached a maximum of about 56 kilobits per second. Approximately how many minutes would the download of a 4.2-megabyte song have taken at that speed? (Note that there are 8000 kilobits in a megabyte.)

(A) 0.6      (B) 10      (C) 1800      (D) 7200      (E) 36000

**Answer:** B

**Solution 1**

Notice that the number of kilobits in this song is  $4.2 \cdot 8000 = 8 \cdot 7 \cdot 6 \cdot 100$ .

We must divide this by 56 in order to find out how many seconds this song would take to download:  $\frac{8 \cdot 7 \cdot 6 \cdot 100}{56} = 600$ .

Finally, we divide this number by 60 because this is the number of seconds to get the answer  $\frac{600}{60} = \boxed{\text{(B) } 10}$ .

**Solution 2**

We seek a value of  $x$  that makes the following equation true, since every other quantity equals 1.

$$\frac{x \text{ min}}{4.2 \text{ mb}} \cdot \frac{56 \text{ kb}}{1 \text{ sec}} \cdot \frac{1 \text{ mb}}{8000 \text{ kb}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} = 1.$$

Solving yields  $x = \boxed{\text{(B)} 10}$ .

-Benedict T (countmath1)

8. (AMC 8 2022) What is the value of

$$\frac{1}{3} \cdot \frac{2}{4} \cdot \frac{3}{5} \cdots \frac{18}{20} \cdot \frac{19}{21} \cdot \frac{20}{22}?$$

- (A)  $\frac{1}{462}$     (B)  $\frac{1}{231}$     (C)  $\frac{1}{132}$     (D)  $\frac{2}{213}$     (E)  $\frac{1}{22}$

**Answer:** B

**Solution 1**

Note that common factors (from 3 to 20, inclusive) of the numerator and the denominator cancel. Therefore, the original expression becomes

$$\frac{1}{3} \cdot \frac{2}{4} \cdot \frac{3}{5} \cdots \frac{18}{20} \cdot \frac{19}{21} \cdot \frac{20}{22} = \frac{1 \cdot 2}{21 \cdot 22} = \frac{1}{21 \cdot 11} = \boxed{\text{(B)} \frac{1}{231}}.$$

**Solution 2**

The original expression becomes

$$\frac{20!}{22!/2!} = \frac{20! \cdot 2!}{22!} = \frac{20! \cdot 2}{20! \cdot 21 \cdot 22} = \frac{2}{21 \cdot 22} = \frac{1}{21 \cdot 11} = \boxed{\text{(B)} \frac{1}{231}}.$$

hh99754539

9. (AMC 8 2022) A cup of boiling water ( $212^{\circ}\text{F}$ ) is placed to cool in a room whose temperature remains constant at  $68^{\circ}\text{F}$ . Suppose the difference between the water temperature and the room temperature is halved every 5 minutes. What is the water temperature, in degrees Fahrenheit, after 15 minutes?

- (A) 77    (B) 86    (C) 92    (D) 98    (E) 104

**Answer:** B

**Solution**

Initially, the difference between the water temperature and the room temperature is  $212 - 68 = 144$  degrees Fahrenheit.

After 5 minutes, the difference between the temperatures is  $144 \div 2 = 72$  degrees Fahrenheit.

After 10 minutes, the difference between the temperatures is  $72 \div 2 = 36$  degrees Fahrenheit.

After 15 minutes, the difference between the temperatures is  $36 \div 2 = 18$  degrees Fahrenheit. At this point, the water temperature is  $68 + 18 = \boxed{\text{(B)} 86}$  degrees Fahrenheit.

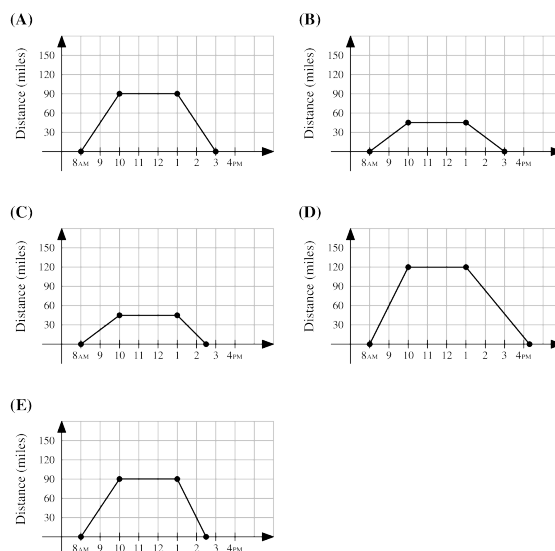
Remark

Alternatively, we can condense the solution above into the following equation:

$$68 + (212 - 68) \cdot \left(\frac{1}{2}\right)^{\frac{15}{5}} = 86.$$

MRENTHUSIASM MathFun1000

10. (AMC 8 2022) One sunny day, Ling decided to take a hike in the mountains. She left her house at 8 AM, drove at a constant speed of 45 miles per hour, and arrived at the hiking trail at 10 AM. After hiking for 3 hours, Ling drove home at a constant speed of 60 miles per hour. Which of the following graphs best illustrates the distance between Ling's car and her house over the course of her trip?



**Answer:** E

**Solution 1 (Analysis)**

Note that:

- At 8 AM, Ling's car was 0 miles from her house.
- From 8 AM to 10 AM, Ling drove to the hiking trail at a constant speed of 45 miles per hour. It follows that at 10 AM, Ling's car was  $45 \cdot 2 = 90$  miles from her house.
- From 10 AM to 1 PM, Ling did not move her car. It follows that at 1 PM, Ling's car was still 90 miles from her house.
- From 1 PM, Ling drove home at a constant speed of 60 miles per hour. So, she arrived home  $90 \div 60 = 1.5$  hour later. It follows that at 2 : 30 PM, Ling's car was 0 miles from her house.

Therefore, the answer is (E).

**Solution 2 (Elimination)**

Ling's trip took 2 hours, thus she traveled for  $2 \times 45 = 90$  miles. Choices (B), (C), and (D) are eliminated. Ling drove 45 miles per hour (mph) to the mountains,

and 60 mph back to her house, so the rightmost slope must be steeper than the leftmost one. Choice **(A)** is eliminated. This leaves us with **(E)**.

**Solution 3 (Elimination)**

Using the speed =  $\frac{\text{distance}}{\text{time}}$  formula, and plugging in the values 45 mph and 2 hrs, we get that the distance from Ling's house to the mountains is 90 miles. That means the first slope ends at 90, so choices **(B)**, **(C)**, and **(D)** are eliminated.

Furthermore, if the distance is 90 miles, and Ling is returning home at 60 mph, it must have taken her 1.5 hours. Adding 1.5 and 3 (how long she hiked for) to her arrival time, 10 AM, we see she must have come back home at 2 : 30 PM. Choice **(A)** is eliminated, so the only valid choice left is choice **(E)**.

11. (*AMC 8 2022*) Henry the donkey has a very long piece of pasta. He takes a number of bites of pasta, each time eating 3 inches of pasta from the middle of one piece. In the end, he has 10 pieces of pasta whose total length is 17 inches. How long, in inches, was the piece of pasta he started with?

**(A)** 34      **(B)** 38      **(C)** 41      **(D)** 44      **(E)** 47

**Answer:** D

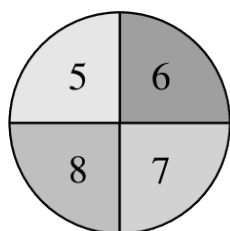
**Solution 1**

Let's say that the first strand of pasta he had is  $x$ . The first time he takes a bite of this strand will make it 2 pieces. This is  $x - 3$ . The second time he takes the bites of EACH strand will become  $(x - 3) * 3(2)$ . The 2 is for the two bites he took, 1 for each strand. This equation is showing that for every bite that he is taking, he is taking off 3 inches. This is being showed in the  $3(2)$ . He took 2 bites, each taking off 3 inches. Now, the problem is stating that there are 10 pieces of pasta and the TOTAL length is 17 inches. We have to plug in the total number of bites to the equation  $(x - 3) * 3( )$ . To find this out, we can see that for the number of strands there are, he took 1 less bite than that number. For example, if there are 2 pieces, he took 1 bite to get those. Since there are 10 pieces, he took 9 bites. The equation then becomes,  $(x - 3) * 3(9) = 17$ . To solve this,  $x = 44$ . Therefore, the answer is **(D) 44**.

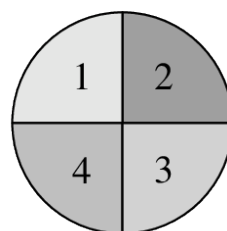
**Solution 2**

If there are 10 pieces of pasta, Henry took  $10 - 1 = 9$  bites. Each of these 9 bites took 3 inches of pasta out, so all his bites in total took away  $9 \cdot 3 = 27$  inches of pasta. Thus, the original piece of pasta was  $27 + 17 = \mathbf{(D) 44}$  inches long.

12. (*AMC 8 2022*) The arrows on the two spinners shown below are spun. Let the number  $N$  equal 10 times the number on Spinner A, added to the number on Spinner B. What is the probability that  $N$  is a perfect square number?



Spinner A



Spinner B

- (A)  $\frac{1}{16}$     (B)  $\frac{1}{8}$     (C)  $\frac{1}{4}$     (D)  $\frac{3}{8}$     (E)  $\frac{1}{2}$

**Answer:** B

**Solution 1**

First, we calculate that there are a total of  $4 \cdot 4 = 16$  possibilities. Now, we list all of two-digit perfect squares. 64 and 81 are the only ones that can be made using the spinner. Consequently, there is a  $\frac{2}{16} = \boxed{\text{(B)} \frac{1}{8}}$  probability that the number formed by the two spinners is a perfect square.

**Solution 2**

There are  $4 \cdot 4 = 16$  total possibilities of  $N$ . We know  $N = 10A + B$ , which  $A$  is a number from spinner  $A$ , and  $B$  is a number from spinner  $B$ . Also, notice that there are no perfect squares in the 50s or 70s, so only  $4 - 2 = 2$  values of  $N$  work, namely 64 and 81. Hence,  $\frac{2}{16} = \boxed{\text{(B)} \frac{1}{8}}$ .

**Solution 3**

We can see that the smallest number on the spinner 1 is 5. Possibilities for 5 are 5

Second one is 6 in which we get 61,62,63,64. We see that 64 is a square number. The next number is 7. Possibilities are 71,72,73, and 74. None of them are squares. Finally, we have 8. 81,82,83, and 84 are the numbers possible. 81 is a perfect square. As our trusty probability formula tells us, probability is the number of desired outcomes over possible outcomes. The possible outcomes are 16 and the desired outcomes are 2. The fraction is  $2/16$ . In simplest terms, it is  $1/8$ . Bringing to answer choice (B) I would like to stay anonymous

13. (AMC 8 2022) How many positive integers can fill the blank in the sentence below?  
 “One positive integer is \_\_\_\_\_ more than twice another, and the sum of the two numbers is 28.”  
 (A) 6    (B) 7    (C) 8    (D) 9    (E) 10

**Answer:** D

**Solution**

Let  $m$  and  $n$  be positive integers such that  $m > n$  and  $m + n = 28$ . It follows that  $m = 2n + d$  for some positive integer  $d$ . We wish to find the number of possible values for  $d$ .

By substitution, we have  $(2n + d) + n = 28$ , from which  $d = 28 - 3n$ . Note that  $n = 1, 2, 3, \dots, 9$  each generate a positive integer for  $d$ , so there are  $\boxed{\text{(D)} 9}$  possible values for  $d$ .

**Solution 2**

Let the positive integer be  $x$ , \_\_\_\_\_ be  $z$  and twice another  $2y$ . We can form the following equations:

$$x + y = 28 \quad x - 2y = z$$

We use elimination to get this equation:  $3y = 28 - z$ . The smallest that  $z$  can be is 1 ( $28 - 1 = 27$  (True)), and the largest 25 ( $28 - 25 = 3$  (True)).  $27 - 1 = 24$ , and since we only want multiples of 3 this will occur every 3 numbers.  $24 / 3 = 8$ . We are missing the 1 which we subtracted earlier, so our final answer will be 9 numbers, or  $\boxed{\text{(D)} 9}$ .



14. (AMC 8 2022) In how many ways can the letters in **BEEKEEPER** be rearranged so that two or more **E**s do not appear together?

(A) 1      (B) 4      (C) 12      (D) 24      (E) 120

**Answer:** D

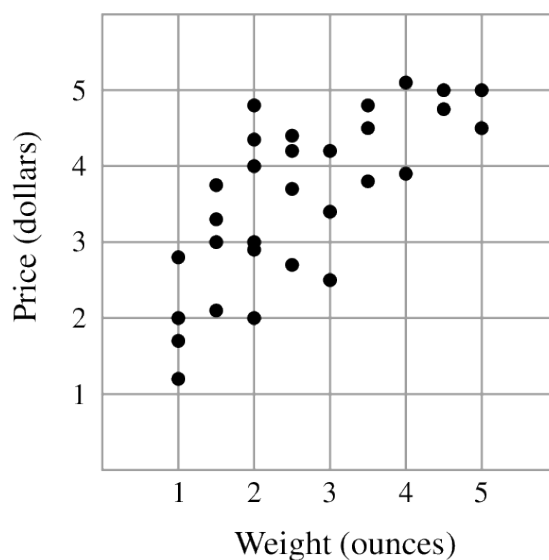
**Solution**

All valid arrangements of the letters must be of the form

**E \_ E \_ E \_ E \_ E.**

The problem is equivalent to counting the arrangements of **B**, **K**, **P**, and **R** into the four blanks, in which there are  $4! = \boxed{\text{(D) } 24}$  ways.

15. (AMC 8 2022) Laszlo went online to shop for black pepper and found thirty different black pepper options varying in weight and price, shown in the scatter plot below. In ounces, what is the weight of the pepper that offers the lowest price per ounce?

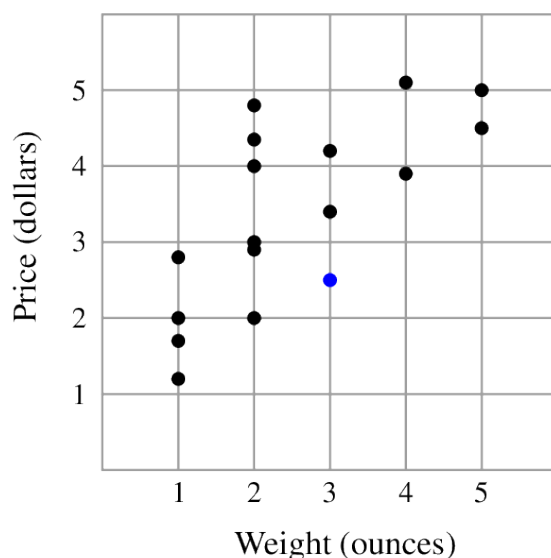


(A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** C

**Solution 1 (Elimination)**

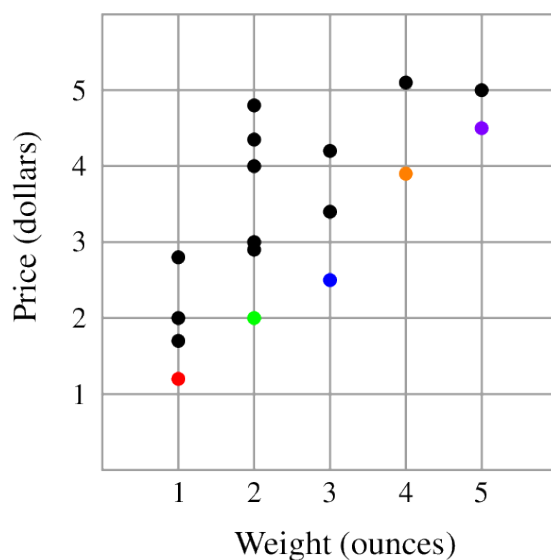
By the answer choices, we can disregard the points that do not have integer weights. As a result, we obtain the following diagram:



We then proceed in the same way that we had done in Solution 1. Following the steps, we figure out the blue dot that yields the lowest slope, along with passing the origin. We then can look at the x-axis (in this situation, the weight) and figure out it has **(C) 3** ounces.

### Solution 2 (Elimination)

We can find the lowest point in each line (1, 2, 3, 4, or 5) and find the price per pound. (Note that we don't need to find the points higher than the points below since we are finding the lowest price per pound.)



The red dot has a price per pound of something that is larger than 1. The green dot has a price per pound of 1. The blue dot has a price per pound of something like  $\frac{2.5}{3}$ . The orange dot has a price per pound that is less than 1, but is very close to it. The purple dot has a price per pound of something like  $\frac{4.5}{5}$ . We see that choices **(A)**, **(B)**, and **(D)** are eliminated. Also,  $\frac{4.5}{5} > \frac{2.5}{3}$  thus the answer is **(C) 3**.

16. (AMC 8 2022) Four numbers are written in a row. The average of the first two is

21, the average of the middle two is 26, and the average of the last two is 30. What is the average of the first and last of the numbers?

- (A) 24      (B) 25      (C) 26      (D) 27      (E) 28

**Answer:** B

**Solution 1 (Arithmetic)**

Note that the sum of the first two numbers is  $21 \cdot 2 = 42$ , the sum of the middle two numbers is  $26 \cdot 2 = 52$ , and the sum of the last two numbers is  $30 \cdot 2 = 60$ .

It follows that the sum of the four numbers is  $42 + 60 = 102$ , so the sum of the first and last numbers is  $102 - 52 = 50$ . Therefore, the average of the first and last numbers is  $50 \div 2 = \boxed{\text{(B) } 25}$ .

**Solution 2 (Algebra)**

Let  $a, b, c$ , and  $d$  be the four numbers in that order. We are given that

$$\frac{a+b}{2} = 21, \quad (1)$$

$$\frac{b+c}{2} = 26, \quad (2)$$

$$\frac{c+d}{2} = 30, \quad (3)$$

and we wish to find  $\frac{a+d}{2}$ .

We add (1) and (3), then subtract (2) from the result:

$$\frac{a+d}{2} = 21 + 30 - 26 = \boxed{\text{(B) } 25}.$$

MRENTHUSIASM

**Solution 3**

Let  $a, b, c$ , and  $d$  be the four numbers in that order. We are given that

$$\frac{a+b}{2} = 21, \quad (1)$$

$$\frac{b+c}{2} = 26, \quad (2)$$

$$\frac{c+d}{2} = 30, \quad (3)$$

and we wish to find  $\frac{a+d}{2}$ .

If we subtract (2) from (3), we get

$$\frac{c+d}{2} - \frac{b+c}{2} = \frac{d-b}{2} = 4.$$

So if we multiply by 2 then we get  $d = b + 8$ . Looking at our question:

$$\frac{a+d}{2} = \frac{a+b+8}{2} = \frac{a+b}{2} + \frac{8}{2} = 21 + 4 = \boxed{\text{(B) } 25}.$$

**Solution 4**

Let's label these 4 numbers  $A, B, C$ , and  $D$ .

$$A + B = 42 \quad B + C = 52 \quad C + D = 60$$

We can use elimination to solve this:  $A - C = -10$

and since

$$D + C = 60$$

then

$$A + D = 50$$

Finally, to find the average, divide by 2. The answer is 25, or  $B$

-OddPeppyBoots

17. (AMC 8 2022) If  $n$  is an even positive integer, the *double factorial* notation  $n!!$  represents the product of all the even integers from 2 to  $n$ . For example,  $8!! = 2 \cdot 4 \cdot 6 \cdot 8$ . What is the units digit of the following sum?

$$2!! + 4!! + 6!! + \cdots + 2018!! + 2020!! + 2022!!$$

- (A) 0      (B) 2      (C) 4      (D) 6      (E) 8

**Answer:** B

**Solution 1**

Notice that once  $n > 8$ , the units digit of  $n!!$  will be 0 because there will be a factor of 10. Thus, we only need to calculate the units digit of

$$2!! + 4!! + 6!! + 8!! = 2 + 8 + 48 + 48 \cdot 8.$$

We only care about units digits, so we have  $2 + 8 + 8 + 8 \cdot 8$ , which has the same units digit as  $2 + 8 + 8 + 4$ . The answer is **(B) 2**.

**Solution 2 (Solution 1 worded differently)**

We can see that after  $8!!$  in the sequence, the units digit is always 0 (every value after  $8!!$  is a multiple of 10). Therefore, our answer is the sum of the units digits of  $2!!$ ,  $4!!$ ,  $6!!$ , and  $8!!$  respectively. This sum is equal to  $2 + 8 + 8 + 4$ , or **(B) 2**.

18. (AMC 8 2022) The midpoints of the four sides of a rectangle are  $(-3, 0)$ ,  $(2, 0)$ ,  $(5, 4)$ , and  $(0, 4)$ . What is the area of the rectangle?
- (A) 20      (B) 25      (C) 40      (D) 50      (E) 80

**Answer:** C

**Solution 1**

The midpoints of the four sides of every rectangle are the vertices of a rhombus whose area is half the area of the rectangle: Note that the diagonals of the rhombus have the same lengths as the sides of the rectangle.

Let  $A = (-3, 0)$ ,  $B = (2, 0)$ ,  $C = (5, 4)$ , and  $D = (0, 4)$ . Note that  $A, B, C$ , and  $D$  are the vertices of a rhombus whose diagonals have lengths  $AC = 4\sqrt{5}$  and  $BD = 2\sqrt{5}$ . It follows that the dimensions of the rectangle are  $4\sqrt{5}$  and  $2\sqrt{5}$ , so the area of the rectangle is  $4\sqrt{5} \cdot 2\sqrt{5} = \mathbf{(C) 40}$ .

**Solution 2**

If a rectangle has area  $K$ , then the area of the quadrilateral formed by its midpoints is  $\frac{K}{2}$ .

Define points  $A, B, C$ , and  $D$  as Solution 1 does. Since  $A, B, C$ , and  $D$  are the midpoints of the rectangle, the rectangle's area is  $2[ABCD]$ . Now, note that  $ABCD$  is a parallelogram since  $AB = CD$  and  $\overline{AB} \parallel \overline{CD}$ . As the parallelogram's height from  $D$  to  $\overline{AB}$  is 4 and  $AB = 5$ , its area is  $4 \cdot 5 = 20$ . Therefore, the area of the rectangle is  $20 \cdot 2 = \boxed{\text{(C) } 40}$ .

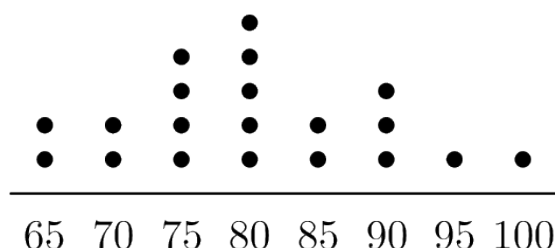
### Solution 3

Graph paper is not permitted on the AMC 8. Still, blank scratch paper is provided, and with that, you can draw a makeshift coordinate plane and "graph" some dots. It doesn't have to be very exact, but do make reasonable estimates on the locations of the points. Once you graph the dots and look around, it becomes apparent that the only rectangle that would make sense would be if the length were from  $(-3,0)$  to  $(5,4)$ , and the width would have to be from  $(0,4)$  to  $(2,0)$ . With that in mind, use the distance formula to solve for the length and width. The distance formula is as follows:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Simply put the coordinates into the formula to get the distance between the two points. Using the formula, the length is  $\sqrt{80}$ , and the width is  $\sqrt{20}$ . Now, multiply the length and width together to get  $\boxed{\text{(C) } 40}$ . Evžen D.

19. (AMC 8 2022) Mr. Ramos gave a test to his class of 20 students. The dot plot below shows the distribution of test scores.



Later Mr. Ramos discovered that there was a scoring error on one of the questions. He regraded the tests, awarding some of the students 5 extra points, which increased the median test score to 85. What is the minimum number of students who received extra points?

(Note that the *median* test score equals the average of the 2 scores in the middle if the 20 test scores are arranged in increasing order.)

- (A) 2    (B) 3    (C) 4    (D) 5    (E) 6

**Answer:** C

### Solution 1

We notice that 13 students have scores under 85 currently, and only 5 have scores over 85. We find the median of these two numbers, getting:

$$13 - 5 = 8$$

$$\frac{8}{2} = 4$$

$$13 - 4 = 9$$

Thus, we realize that 4 students must have their score increased by 5.

So, the correct answer is (C)4.

### Solution 2

We notice that if we number each of the dots from bottom to top in each column (65's, 70's, 75's, etc.) to find the median, the second and third dots in the "80's" column correspond to the tenth and eleventh values in the set (or the closest values to the median). To make the median now 85, we must add 5 to the tenth and eleventh values in the set (or the second and third dots in the "80's" column), as well as all the dots above it in the "80's" column. So, the correct answer is (C)4.

nidharshana29

20. (*AMC 8 2022*) The grid below is to be filled with integers in such a way that the sum of the numbers in each row and the sum of the numbers in each column are the same. Four numbers are missing. The number  $x$  in the lower left corner is larger than the other three missing numbers. What is the smallest possible value of  $x$ ?

|      |     |      |
|------|-----|------|
| $-2$ | $9$ | $5$  |
|      |     | $-1$ |
| $x$  |     | $8$  |

- (A)  $-1$     (B)  $5$     (C)  $6$     (D)  $8$     (E)  $9$

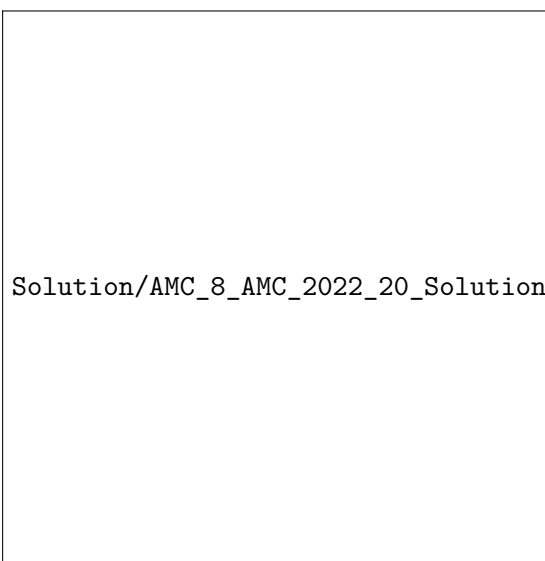
**Answer:** D

### Solution 1

The sum of the numbers in each row is 12. Consider the second row. In order for the sum of the numbers in this row to equal 12, the two shaded numbers must add up to 13:

|      |     |      |
|------|-----|------|
| $-2$ | $9$ | $5$  |
|      |     | $-1$ |
| $x$  |     | $8$  |

If two numbers add up to 13, one of them must be at least 7: If both shaded numbers are no more than 6, their sum can be at most 12. Therefore, for  $x$  to be larger than the three missing numbers,  $x$  must be at least 8. We can construct a working scenario where  $x = 8$ :



Solution/AMC\_8\_AMC\_2022\_20\_Solution\_Figure\_2.png

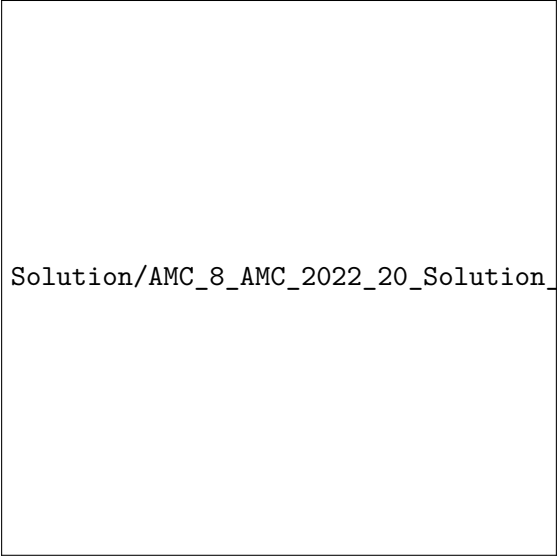
So, our answer is **(D) 8**.

### Solution 2

The sum of the numbers in each row is  $-2 + 9 + 5 = 12$ , and the sum of the numbers in each column is  $5 + (-1) + 8 = 12$ .

Let  $y$  be the number in the lower middle. It follows that  $x + y + 8 = 12$ , or  $x + y = 4$ .

We express the other two missing numbers in terms of  $x$  and  $y$ , as shown below:



Solution/AMC\_8\_AMC\_2022\_20\_Solution\_Figure\_3.png

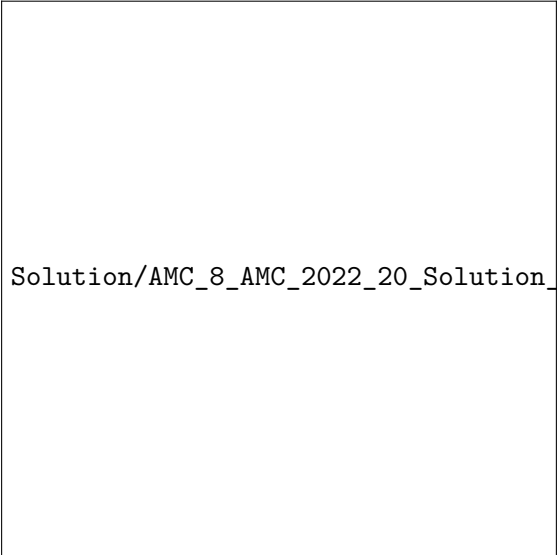
We have  $x > x - 1$ ,  $x > y + 10$ , and  $x > y$ . Note that the first inequality is true for all values of  $x$ . We only need to solve the second inequality so that the third inequality is true for all values of  $x$ . By substitution, we get  $x > (4 - x) + 10$ , from which  $x > 7$ .

Therefore, the smallest possible value of  $x$  is (D) 8.

### Solution 3

This is based on the Solution 2 above and it is perhaps a little simpler than that.

Let  $y$  be the number in the lower middle. Applying summation to first two columns yields the following.



Solution/AMC\_8\_AMC\_2022\_20\_Solution\_Figure\_4.png

Since  $x$  is greater than the other three, we have  $x > 14 - x$ , or  $x > 7$ .

Therefore, the smallest possible value of  $x$  is (D) 8.

### Solution 4 (Answer Choices)

Note that the sum of the rows and columns must be  $8 + 5 - 1 = 12$ . We proceed to test the answer choices.



Testing (A), when  $x = -1$ , the number above  $x$  must be 15, which contradicts the precondition that the numbers surrounding  $x$  is less than  $x$ .

Testing (B), the number above  $x$  is 9, which does not work.

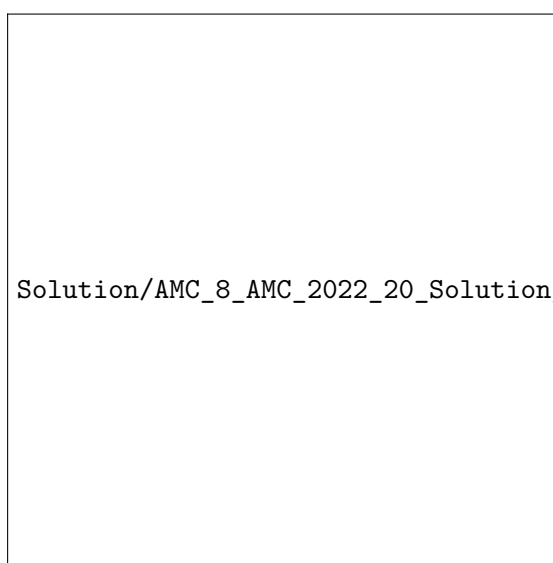
Testing (C), the number above  $x$  is 8, which does not work.

Testing (D), the number above  $x$  is 6, which *does* work. Hence, the answer is (D) 8.

We do not need to test (E), because the problem asks for the **smallest** value of  $x$ .

**Solution 5 (Super fast! No algebra and no testing any of the answer choices!)**

The sum of the numbers in each column and row should be  $5 + (-1) + 8 = 12$ . If we look at the 1<sup>st</sup> column, the gray squares (shown below) sum to  $12 - (-2) = 14$ .



If square  $x$  has to be **greater than or equal to** the three blank squares, then the least  $x$  can be is half the sum of the value of the gray squares, which is  $14 \div 2 = 7$ . But square  $x$  has to be **greater than** and **not greater than or equal to** the three blank squares, so the least  $x$  can be is  $7 + 1 = 8$ . Testing for the other rows and columns (it might be smaller than the other two squares!), we find that the smallest  $x$  can be is indeed 8; the other two squares are less than 8. Therefore, the answer is (D) 8.

21. (AMC 8 2022) Steph scored 15 baskets out of 20 attempts in the first half of a game, and 10 baskets out of 10 attempts in the second half. Candace took 12 attempts in the first half and 18 attempts in the second. In each half, Steph scored a higher percentage of baskets than Candace. Surprisingly they ended with the same overall percentage of baskets scored. How many more baskets did Candace score in the second half than in the first?

|         | First Half           | Second Half          |
|---------|----------------------|----------------------|
| Steph   | $\frac{15}{20}$      | $\frac{10}{10}$      |
| Candace | $\frac{\square}{12}$ | $\frac{\square}{18}$ |

- (A) 7    (B) 8    (C) 9    (D) 10    (E) 11

**Answer:** C

**Solution 1 (Inequalities)**

Let  $x$  be the number of shots that Candace made in the first half, and let  $y$  be the number of shots Candace made in the second half. Since Candace and Steph took the same number of attempts, with an equal percentage of baskets scored, we have  $x + y = 10 + 15 = 25$ . In addition, we have the following inequalities:

$$\frac{x}{12} < \frac{15}{20} \implies x < 9,$$

and

$$\frac{y}{18} < \frac{10}{10} \implies y < 18.$$

Pairing this up with  $x + y = 25$  we see the **only** possible solution is  $(x, y) = (8, 17)$ , for an answer of  $17 - 8 = \boxed{\text{(C) } 9}$ .

**Solution 2 (Answer Choices)**

Clearly, Steph made  $15 + 10 = 25$  shots in total. Also, due to parity reasons, the difference between the amount of shots Candace made in the first and second half must be odd. Thus, we can just test 7, 9, and 11, and after doing so we find that the answer is  $\boxed{\text{(C) } 9}$ .

**Solution 3**

Steph made 75 percent of his shots in the first half. He makes all of his shots in the second half. The most baskets Candace could have made in the first half is 8 baskets. The most she could have made in the second half is 17 baskets. Steph makes 25 and misses 5 baskets and the only way for Candace to make 25 shots is to make 8 in the first half and 17 in the second. Thus,  $17 - 8 = \boxed{\text{(C) } 9}$ .

22. (AMC 8 2022) A bus takes 2 minutes to drive from one stop to the next, and waits 1 minute at each stop to let passengers board. Zia takes 5 minutes to walk from one bus stop to the next. As Zia reaches a bus stop, if the bus is at the previous stop or has already left the previous stop, then she will wait for the bus. Otherwise she will start walking toward the next stop. Suppose the bus and Zia start at the same time toward the library, with the bus 3 stops behind. After how many minutes will Zia board the bus?



- (A) 17    (B) 19    (C) 20    (D) 21    (E) 23

**Answer:** A

**Solution 1**

Initially, suppose that the bus is at Stop 0 (starting point) and Zia is at Stop 3.

We construct the following table of 5-minute intervals:

| Time       | Bus's Location            | Zia's Location |
|------------|---------------------------|----------------|
| 5 Minutes  | Stop 2 (Waiting)          | Stop 4         |
| 10 Minutes | Midpoint of Stops 3 and 4 | Stop 5         |
| 15 Minutes | Stop 5 (Leaving)          | Stop 6         |

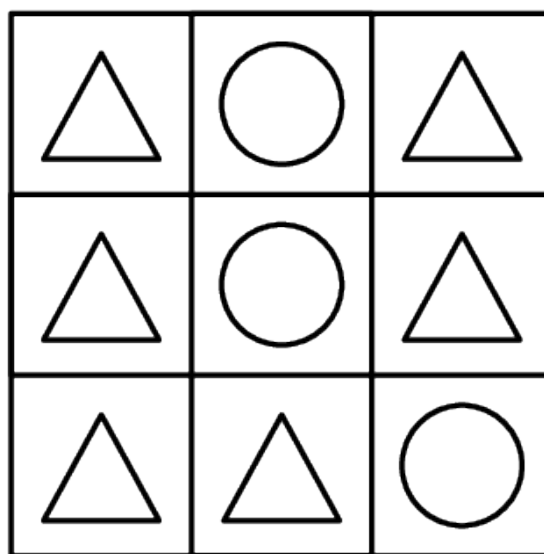
Note that Zia will wait for the bus after 15 minutes, and the bus will arrive 2 minutes later.

Therefore, the answer is  $15 + 2 = \boxed{\text{(A)} 17}$ .

**Solution 2**

Since Zia will wait for the bus if the bus is at the previous stop, we can create an equation to solve for when the bus is at the previous stop. The bus travels  $\frac{1}{3}$  of a stop per minute, and Zia travels  $\frac{1}{5}$  of a stop per minute. Now we create the equation,  $\frac{1}{3}m = \frac{1}{5}m + 3 - 1$  (the  $-1$  accounts for us wanting to find when the bus reaches the stop before Zia's). Solving, we find that  $m = 15$ . Now Zia has to wait 2 minutes for the bus to reach her, so our answer is  $15 + 2 = \boxed{\text{(A)} 17}$ .

23. (AMC 8 2022) A  $\triangle$  or  $\bigcirc$  is placed in each of the nine squares in a 3-by-3 grid. Shown below is a sample configuration with three  $\triangle$ s in a line.



How many configurations will have three  $\triangle$ s in a line and three  $\bigcirc$ s in a line?

- (A) 39      (B) 42      (C) 78      (D) 84      (E) 96

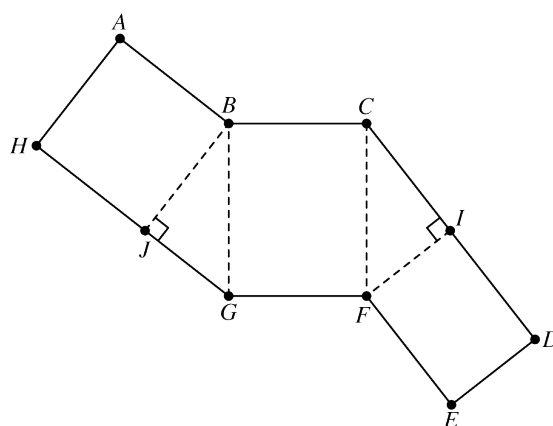
**Answer:** D

**Solution 1**

We will only consider cases where the three identical symbols are the same column, but at the end we shall double our answer as the same holds true for rows. There

are 3 ways to choose a column with all  $\bigcirc$ 's and 2 ways to choose a column with all  $\triangle$ 's. The third column can be filled in  $2^3 = 8$  ways. Therefore, we have a total of  $3 \cdot 2 \cdot 8 = 48$  cases. However, we overcounted the cases with 2 complete columns of with one symbol and 1 complete column with another symbol. This happens in  $2 \cdot 3 = 6$  cases.  $48 - 6 = 42$ . However, we have to remember to double our answer, giving us **(D) 84** ways to complete the grid.

24. (AMC 8 2022) The figure below shows a polygon  $ABCDEFGH$ , consisting of rectangles and right triangles. When cut out and folded on the dotted lines, the polygon forms a triangular prism. Suppose that  $AH = EF = 8$  and  $GH = 14$ . What is the volume of the prism?



- (A) 112      (B) 128      (C) 192      (D) 240      (E) 288

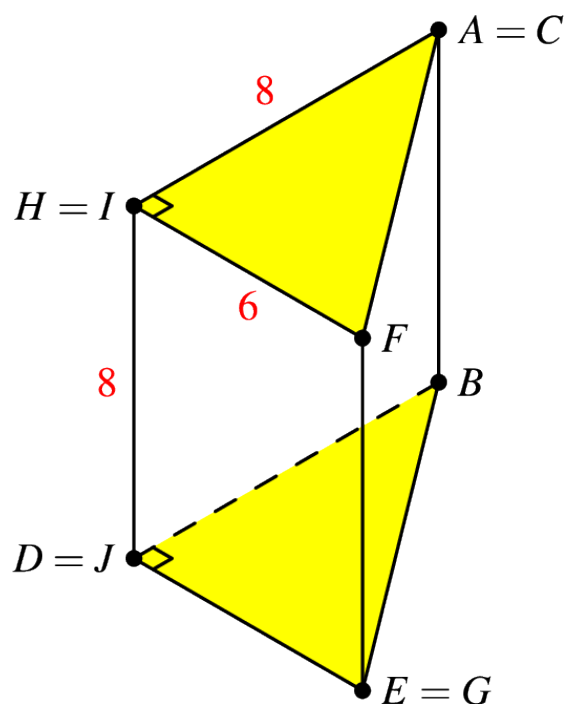
**Answer:** C

**Solution**

While imagining the folding,  $\overline{AB}$  goes on  $\overline{BC}$ ,  $\overline{AH}$  goes on  $\overline{CI}$ , and  $\overline{EF}$  goes on  $\overline{FG}$ . So,  $BJ = CI = 8$  and  $FG = BC = 8$ . Also,  $\overline{HJ}$  becomes an edge parallel to  $\overline{FG}$ , so that means  $HJ = 8$ .

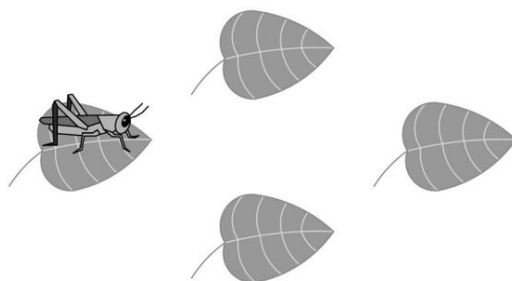
Since  $GH = 14$ , then  $JG = 14 - 8 = 6$ . So, the area of  $\triangle BJG$  is  $\frac{8 \cdot 6}{2} = 24$ . If we let  $\triangle BJG$  be the base, then the height is  $FG = 8$ . So, the volume is  $24 \cdot 8 = \mathbf{(C) 192}$ .

**Remark** After folding polygon  $ABCDEFGH$  on the dotted lines, we obtain the following triangular prism:



MRENTHUSIASM

25. (AMC 8 2022) A cricket randomly hops between 4 leaves, on each turn hopping to one of the other 3 leaves with equal probability. After 4 hops, what is the probability that the cricket has returned to the leaf where it started?



- (A)  $\frac{2}{9}$     (B)  $\frac{19}{80}$     (C)  $\frac{20}{81}$     (D)  $\frac{1}{4}$     (E)  $\frac{7}{27}$

**Answer:** E

**Solution 1 (Casework)**

Let  $A$  denote the leaf where the cricket starts and  $B$  denote a region containing the other 3 leaves. Note that:

- If the cricket is at  $A$ , then the probability that it hops to  $B$  next is 1.
- If the cricket is at  $B$ , then the probability that it hops to  $A$  next is  $\frac{1}{3}$ .
- If the cricket is at  $B$ , then the probability that it hops to  $B$  next is  $\frac{2}{3}$ .

We apply casework to the possible paths of the cricket:

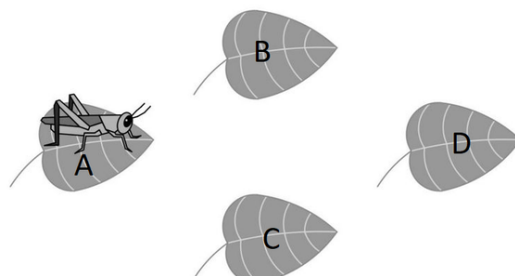
- a)  $A \rightarrow B \rightarrow A \rightarrow B \rightarrow A$  The probability for this case is  $1 \cdot \frac{1}{3} \cdot 1 \cdot \frac{1}{3} = \frac{1}{9}$ .
- b)  $A \rightarrow B \rightarrow B \rightarrow B \rightarrow A$  The probability for this case is  $1 \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{4}{27}$ .

Together, the probability that the cricket returns to  $A$  after 4 hops is  $\frac{1}{9} + \frac{4}{27} =$

(E)  $\frac{7}{27}$ .

### Solution 2 (Casework)

We can label the leaves as shown below:



Carefully counting cases, we see that there are 7 ways for the cricket to return to leaf  $A$  after four hops if its first hop was to leaf  $B$ :

- a)  $A \rightarrow B \rightarrow A \rightarrow B \rightarrow A$
- b)  $A \rightarrow B \rightarrow A \rightarrow C \rightarrow A$
- c)  $A \rightarrow B \rightarrow A \rightarrow D \rightarrow A$
- d)  $A \rightarrow B \rightarrow C \rightarrow B \rightarrow A$
- e)  $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$
- f)  $A \rightarrow B \rightarrow D \rightarrow B \rightarrow A$
- g)  $A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$

By symmetry, we know that there are 7 ways if the cricket's first hop was to leaf  $C$ , and there are 7 ways if the cricket's first hop was to leaf  $D$ . So, there are 21 ways in total for the cricket to return to leaf  $A$  after four hops.

Since there are  $3^4 = 81$  possible ways altogether for the cricket to hop to any other leaf four times, the answer is  $\frac{21}{81} =$  (E)  $\frac{7}{27}$ .

### Solution 3 (Complement)

There are always three possible leaves to jump to every time the cricket decides to jump, so there is a total number of  $3^4$  routes. Let  $A$  denote the leaf cricket starts at, and  $B, C, D$  be the other leaves. If we want the cricket to move to leaf  $A$  for its last jump, the cricket cannot jump to leaf  $A$  for its third jump. Also, considering that the cricket starts at leaf  $A$ , he cannot jump to leaf  $A$  for its first jump. Note that there are  $3 \cdot 2 = 6$  paths if the cricket moves to leaf  $A$  for its third jump. Therefore, we can conclude that the total number of possible paths for the cricket to return to leaf  $A$  after four jumps is  $3^3 - 6 = 21$ , so the answer is  $\frac{21}{3^4} = \frac{21}{81} =$  (E)  $\frac{7}{27}$ .

### Solution 4 (Recursion)

Denote  $P_n$  to be the probability that the cricket would return back to the first point after  $n$  hops. Then, we get the recursive formula

$$P_n = \frac{1}{3}(1 - P_{n-1})$$

because if the leaf is not on the target leaf, then there is a  $\frac{1}{3}$  probability that it will make it back.

With this formula and the fact that  $P_1 = 0$  (After one hop, the cricket can never be back to the target leaf.), we have

$$P_2 = \frac{1}{3}, P_3 = \frac{2}{9}, P_4 = \frac{7}{27},$$

so our answer is  $\boxed{\text{(E)} \frac{7}{27}}$ .

### Solution 5 (Dynamic Programming)

Let  $A$  denote the leaf cricket starts at, and  $B, C, D$  be the other leaves, similar to Solution 2.

Let  $A(n)$  be the probability the cricket lands on  $A$  after  $n$  hops,  $B(n)$  be the probability the cricket lands on  $B$  after crawling  $n$  hops, etc.

Note that  $A(1) = 0$  and  $B(1) = C(1) = D(1) = \frac{1}{3}$ . For  $n \geq 2$ , the probability that the cricket land on each leaf after  $n$  hops is  $\frac{1}{3}$  the sum of the probability the cricket land on other leaves after  $n - 1$  hops. So, we have

$$\begin{aligned} A(n) &= \frac{1}{3} \cdot [B(n-1) + C(n-1) + D(n-1)], \\ B(n) &= \frac{1}{3} \cdot [A(n-1) + C(n-1) + D(n-1)], \\ C(n) &= \frac{1}{3} \cdot [A(n-1) + B(n-1) + D(n-1)], \\ D(n) &= \frac{1}{3} \cdot [A(n-1) + B(n-1) + C(n-1)]. \end{aligned}$$

It follows that  $A(n) = B(n-1) = C(n-1) = D(n-1)$ .

We construct the following table:

| $n$ | $A(n)$         | $B(n)$          | $C(n)$          | $D(n)$          |
|-----|----------------|-----------------|-----------------|-----------------|
| 1   | 0              | $\frac{1}{3}$   | $\frac{1}{3}$   | $\frac{1}{3}$   |
| 2   | $\frac{1}{3}$  | $\frac{2}{9}$   | $\frac{2}{9}$   | $\frac{2}{9}$   |
| 3   | $\frac{2}{9}$  | $\frac{7}{27}$  | $\frac{7}{27}$  | $\frac{7}{27}$  |
| 4   | $\frac{7}{27}$ | $\frac{20}{81}$ | $\frac{20}{81}$ | $\frac{20}{81}$ |

Therefore, the answer is  $A(4) = \boxed{\text{(E)} \frac{7}{27}}$ .

### Solution 6 (Generating Function)

Assign the leaves to 0, 1, 2, and 3 modulo 4, and let 0 be the starting leaf. We then use generating functions with relation to the change of leaves. For example, from 3 to 1 would be a change of 2, and from 1 to 2 would be a change of 1. This generating function is equal to  $(x + x^2 + x^3)^4$ . It is clear that we want the coefficients in the form of  $x^{4n}$ , where  $n$  is a positive integer. One application of roots of unity filter gives us a successful case count of  $\frac{81+1+1+1}{4} = 21$ .

Therefore, the answer is  $\frac{21}{3^4} = \boxed{\text{(E)} \frac{7}{27}}.$

**Solution 7 (Also Generating Functions)**

Let the leaves be  $(0,0)$ ,  $(0,1)$ ,  $(1,0)$ , and  $(1,1)$  on the coordinate plane, with the cricket starting at  $(0,0)$ . Then we write a generating function. We denote  $x$  a change in the  $x$ -value of the cricket, and similarly for  $y$ . Then our generating function is  $(x + y + xy)^4$ , and we wish to compute the number of terms in which the exponents of both  $x$  and  $y$  are even. To do this, we first square to get  $(x^2 + y^2 + x^2y^2 + 2xy + 2x^2y + 2xy^2)^2$ . Note that every term squared will give even powers for  $x$  and  $y$ , so that gives us  $3 + 4 \cdot 3 = 15$ . Then every combination of  $x^2, y^2$ , and  $x^2y^2$  will also give us even powers for  $x$  and  $y$ , so that yields 6 more terms, for a total of 21. Now in total there  $3^4 = 81$  possible sequences, so  $21/81$  gives us the answer of  $\boxed{\text{(E)} \frac{7}{27}}.$

**Remark** This problem is a reduced version of [1985 AIME Problem 12](#), changing 7 steps into 4 steps.

This problem is also similar to [2003 AIME II Problem 13](#).

We count the number of *favorable* sequences in which the cricket is on the starting leaf after the fourth hop.

**Case 1: The cricket returns to the starting leaf exactly once.**

Since the cricket cannot remain on the same leaf, the only way for it to return exactly once and still end at the starting leaf is for the return to occur on the *fourth hop*.

- Hop 1: From the starting leaf, there are 3 choices.
- Hop 2: The cricket is on a non-starting leaf; it must avoid returning, so there are 2 choices.
- Hop 3: Again, the cricket is on a non-starting leaf and must avoid returning, giving 2 choices.
- Hop 4: The cricket must return to the starting leaf, which is 1 favorable choice.

Thus the number of favorable sequences in Case 1 is

$$3 \cdot 2 \cdot 2 \cdot 1 = 12.$$

**Case 2: The cricket returns to the starting leaf exactly twice.**

With four hops and no ability to stay in place, the only possible return pattern is

$$\text{start} \rightarrow \text{other} \rightarrow \text{start} \rightarrow \text{other} \rightarrow \text{start},$$

which corresponds to hops 2 and 4 being returns.

- Hop 1: From the starting leaf, there are 3 choices.
- Hop 2: The cricket must return to the starting leaf (1 choice).
- Hop 3: From the starting leaf, there are again 3 choices.



- Hop 4: The cricket must return to the starting leaf (1 choice).

Thus the number of favorable sequences in Case 2 is

$$3 \cdot 1 \cdot 3 \cdot 1 = 9.$$

**Conclusion.**

The total number of favorable sequences is

$$12 + 9 = 21.$$

Therefore, the desired probability is

$$\frac{21}{81} = \boxed{\frac{7}{27}}.$$

### §38 AMC 8 2023

1. (*AMC 8 2023*) What is the value of  $(8 \times 4 + 2) - (8 + 4 \times 2)$ ?

(A) 0    (B) 6    (C) 10    (D) 18    (E) 24

**Answer:** D

**Solution**

By the **order of operations**, we have

$$(8 \times 4 + 2) - (8 + 4 \times 2) = (32 + 2) - (8 + 8) = 34 - 16 = \boxed{\text{(D)} 18}.$$

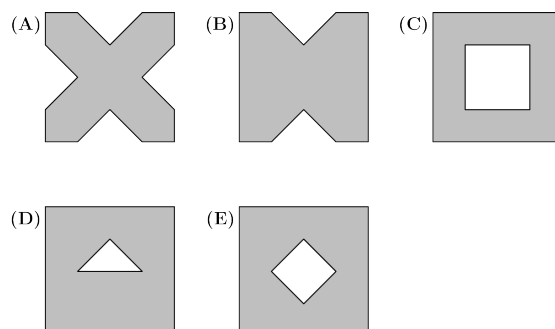
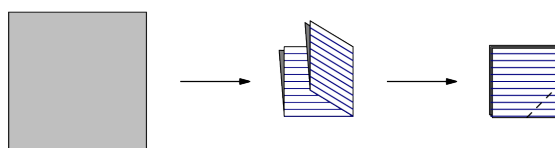
apex304, TaeKim, MRENTHUSIASM, sandville\_kris

**Solution 2**

We can simplify the expression above in another way:

$$(8 \times 4 + 2) - (8 + 4 \times 2) = 8 \times 4 + 2 - 8 - 4 \times 2 = 32 + 2 - 8 - 8 = 34 - 16 = \boxed{\text{(D)} 18}.$$

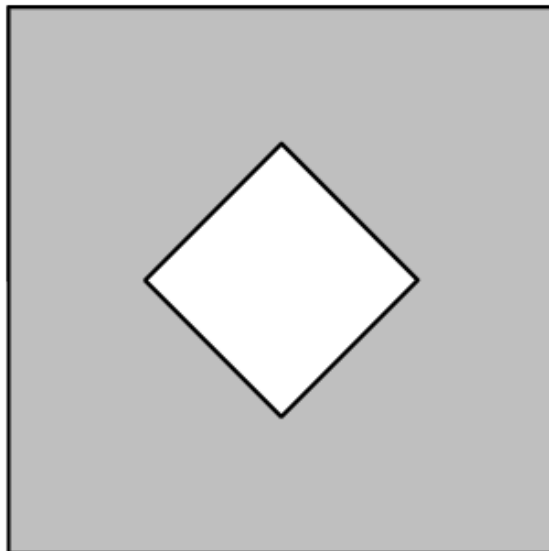
2. (*AMC 8 2023*) A square piece of paper is folded twice into four equal quarters, as shown below, then cut along the dashed line. When unfolded, the paper will match which of the following figures?



**Answer:** E

**Solution**

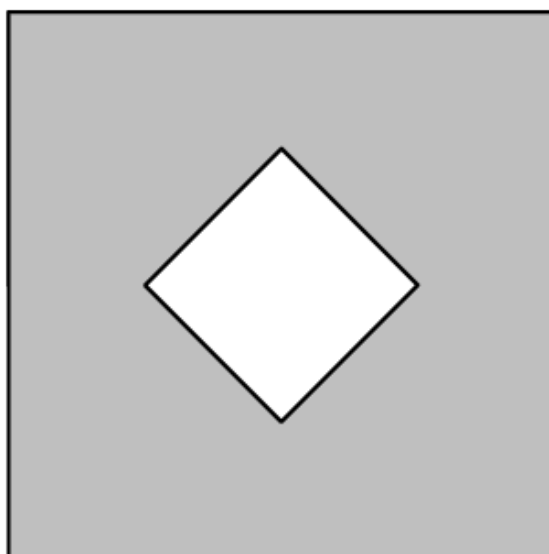
Notice that when we unfold the paper along the vertical fold line, we get the following shape:



Therefore the answer is **(E)**.

**Solution 2**

When you fold the paper in that specific way, it shows the top left square. That means that any cut you make on that folded part will look like that on that top left square. Since it is folded into four parts, the cut will reflect on all of the other 3 parts. Since there is a cut diagonally on the bottom right of that square, that will reflect on the other squares, making the shape of **(E)**



3. (AMC 8 2023) *Wind chill* is a measure of how cold people feel when exposed to wind outside. A good estimate for wind chill can be found using this calculation

$$(\text{wind chill}) = (\text{air temperature}) - 0.7 \times (\text{wind speed}),$$

where temperature is measured in degrees Fahrenheit ( $^{\circ}\text{F}$ ) and the wind speed is measured in miles per hour (mph). Suppose the air temperature is  $36^{\circ}\text{F}$  and the wind speed is 18 mph. Which of the following is closest to the approximate wind chill?

- (A) 18      (B) 23      (C) 28      (D) 32      (E) 35

**Answer:** B

**Solution 1**

By substitution, we have

$$\begin{aligned} (\text{wind chill}) &= 36 - 0.7 \times 18 \\ &= 36 - 12.6 \\ &= 23.4 \\ &\approx \boxed{\text{(B) } 23}. \end{aligned}$$

apex304, SohumUttamchandani, wuwang2002, TaeKim, Cxruptedpat, MREN-THUSIASM

**Solution 2**

0.7 is very close to  $\frac{2}{3}$  - therefore, we can substitute  $\frac{2}{3}$  into the equation to get

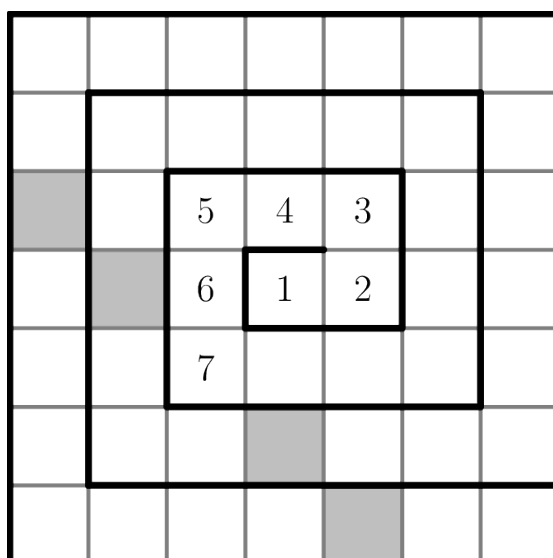
$$36 - \frac{2}{3} \cdot 18 = 36 - 12 = 24.$$

As  $\frac{2}{3}$  is slightly less than 0.7, the correct answer is slightly less than 24. Therefore, the answer is  $\boxed{\text{(B) } 23}$ .

**Solution 3**

0.7 is equivalent to  $\frac{7}{10}$ .  $\frac{7}{10}$  times 18 is  $\frac{126}{10}$ , which is 12.6. Since the air temperature is 36 degrees Fahrenheit,  $36 - 12.6$  is 23.4, which is the closest to  $\boxed{\text{(B) } 23}$ .

4. (*AMC 8 2023*) The numbers from 1 to 49 are arranged in a spiral pattern on a square grid, beginning at the center. The first few numbers have been entered into the grid below. Consider the four numbers that will appear in the shaded squares, on the same diagonal as the number 7. How many of these four numbers are prime?



- (A) 0    (B) 1    (C) 2    (D) 3    (E) 4

**Answer:** D

**Solution 1**

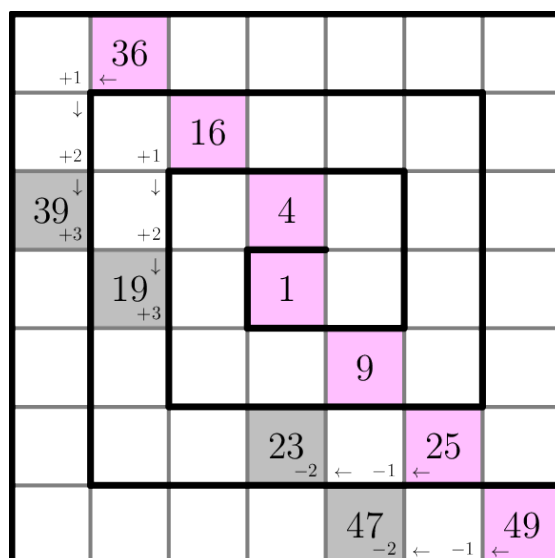
We fill out the grid, as shown below:

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| 37 | 36 | 35 | 34 | 33 | 32 | 31 |
| 38 | 17 | 16 | 15 | 14 | 13 | 30 |
| 39 | 18 | 5  | 4  | 3  | 12 | 29 |
| 40 | 19 | 6  | 1  | 2  | 11 | 28 |
| 41 | 20 | 7  | 8  | 9  | 10 | 27 |
| 42 | 21 | 22 | 23 | 24 | 25 | 26 |
| 43 | 44 | 45 | 46 | 47 | 48 | 49 |

From the four numbers that appear in the shaded squares, **(D) 3** of them are prime: 19, 23, and 47.

**Solution 2**

Note that given time constraint, it's better to only count from perfect squares (in pink), as shown below:



From the four numbers that appear in the shaded squares, **(D) 3** of them are prime: 19, 23, and 47.

### Solution 3

Since the last shaded square is very close to the final number, which is 49, we can easily subtract two instead of having to count all the way there. Then, we can count (:P).

5. (AMC 8 2023) A lake contains 250 trout, along with a variety of other fish. When a marine biologist catches and releases a sample of 180 fish from the lake, 30 are identified as trout. Assume that the ratio of trout to the total number of fish is the same in both the sample and the lake. How many fish are there in the lake?
- (A) 1250    (B) 1500    (C) 1750    (D) 1800    (E) 2000

**Answer:** B

### Solution

Note that

$$\frac{\text{number of trout}}{\text{total number of fish}} = \frac{30}{180} = \frac{1}{6}.$$

So, the total number of fish is 6 times the number of trout. Since the lake contains 250 trout, there are  $250 \cdot 6 = \text{(B) 1500}$  fish in the lake.

6. (AMC 8 2023) The digits 2, 0, 2, and 3 are placed in the expression below, one digit per box. What is the maximum possible value of the expression?

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} \begin{array}{|c|} \hline \square \\ \hline \end{array} \times \begin{array}{|c|} \hline \square \\ \hline \end{array} \begin{array}{|c|} \hline \square \\ \hline \end{array}$$

- (A) 0    (B) 8    (C) 9    (D) 16    (E) 18

**Answer:** C

**Solution 1**

First, let us consider the case where 0 is a base: This would result in the entire expression being 0. Contrastingly, if 0 is an exponent, we will get a value greater than 0.  $3^2 \times 2^0 = 9$  is greater than  $2^3 \times 2^0 = 8$  and  $2^2 \times 3^0 = 4$ . Therefore, the answer is **(C) 9**.

**Solution 2**

The maximum possible value of using the digits 2, 0, 2, and 3: We can maximize our value by keeping the 3 and 2 together in one power (the biggest with the biggest and the smallest with the smallest). This shows  $3^2 \times 2^0 = 9 \times 1 = 9$ . (We don't want  $0^2$  because that is 0.) It is going to be **(C) 9**.

**Solution 3**

Trying all 12 distinct orderings, we see that the only possible values are 0, 4, 8, and 9, the greatest of which is **(C) 9**.

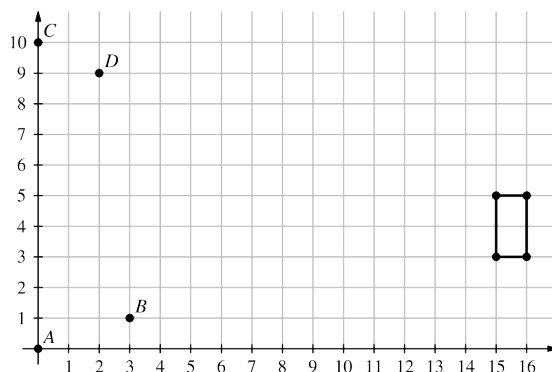
**Solution 4**

There are two 2's and one 3. To make use of them all, use 2 and 3 as the bases and 2 and 0 as the exponents.

**Solution 5**

If 0 is a base, then the whole expression becomes 0. Therefore, 0 must be an exponent. But since  $2^0 = 3^0 = 1$ , we do not want to waste our 3 on the 0. We will have a 3 and a 2 left, and since  $3^2$  is greater than  $2^3$ , we get  $3^2 \cdot 2^0 = 9$ .

7. (AMC 8 2023) A rectangle, with sides parallel to the  $x$ -axis and  $y$ -axis, has opposite vertices located at  $(15, 3)$  and  $(16, 5)$ . A line is drawn through points  $A(0, 0)$  and  $B(3, 1)$ . Another line is drawn through points  $C(0, 10)$  and  $D(2, 9)$ . How many points on the rectangle lie on at least one of the two lines?

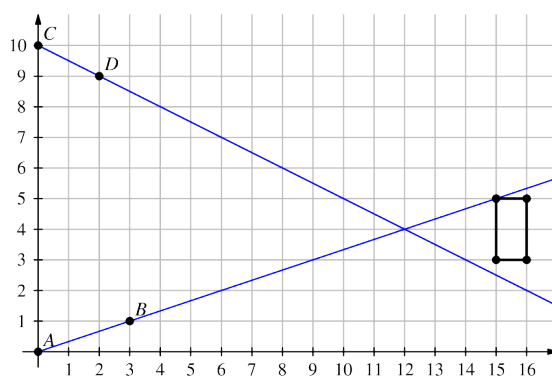


- (A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer: B**

**Solution 1**

If we extend the lines, we have the following diagram:



Therefore, we see that the answer is **(B) 1**.

### Solution 2

Note that the  $y$ -intercepts of line  $AB$  and line  $CD$  are 0 and 10. If the analytic expression for line  $AB$  is  $y = k_1x$ , and the analytic expression for line  $CD$  is  $y = k_2x + 10$ , we have equations:  $3k_1 = 1$  and  $2k_2 + 10 = 9$ . Solving these equations, we can find out that  $k_1 = \frac{1}{3}$  and  $k_2 = -\frac{1}{2}$ . Therefore, we can determine that the expression for line  $AB$  is  $y = \frac{1}{3}x$  and the expression for line  $CD$  is  $y = -\frac{1}{2}x + 10$ . When  $x = 15$ , the coordinates that line  $AB$  and line  $CD$  pass through are  $(15, 5)$  and  $(15, \frac{5}{2})$ , and  $(15, 5)$  lies perfectly on one vertex of the rectangle while the  $y$  coordinate of  $(15, \frac{5}{2})$  is out of the range  $3 \leq y \leq 5$  (lower than the bottom left corner of the rectangle  $(15, 3)$ ). Considering that the  $y$  value of the line  $CD$  will only decrease, and the  $y$  value of the line  $AB$  will only increase, there will not be another point on the rectangle that lies on either of the two lines. Thus, we can conclude that the answer is **(B) 1**.

8. (AMC 8 2023) Lola, Lolo, Tiya, and Tiyo participated in a ping pong tournament. Each player competed against each of the other three players exactly twice. Shown below are the win-loss records for the players. The numbers 1 and 0 represent a win or loss, respectively. For example, Lola won five matches and lost the fourth match. What was Tiyo's win-loss record?

| Player | Result |
|--------|--------|
| Lola   | 111011 |
| Lolo   | 101010 |
| Tiya   | 010100 |
| Tiyo   | ?????? |

(A) 000101      (B) 001001      (C) 010000      (D) 010101      (E) 011000

**Answer:** A

### Solution 1

We can calculate the total number of wins (1's) by seeing how many matches were played, which is 12 matches played. Then, we can calculate the # of wins already on the table, which is  $5 + 3 + 2 = 10$ , so there are  $12 - 10 = 2$  wins left in the mystery player. Now, we will make the key observation that there is only 2 wins (1's) per column as there are 2 winners and 2 losers in each round. Strategically looking through the columns counting the 1's and putting our own 2 1's when the column isn't already full yields **(A) 000101**.

**Solution 2 (Similar to 1 but More Detailed)**

In total, there will be  $\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$  games because there are  $\binom{4}{2}$  ways to choose a pair of people from the four players. And, each player will play each other player exactly twice. Each of these 12 games will have 1 winner and 1 loser, so there will be a total of 12 1's and 12 0's in the win-loss table. Therefore, Tiyo will have  $12 - 10 = 2$  1's and  $12 - 8 = 4$  0's in his record.

Now, all we have to do is figure out the order of these 1's and 0's. In every round, there are two games; the players are split into two pairs and the people in those pairs play each other. Thus, every round should have 2 winners and 2 losers which means that every column of the win-loss table should have 2 1's and 2 0's. Looking at the filled-in table so far, we see that columns 4 and 6 need one more 1, so Tiyo must have 1's in those columns and 0's gone in the others. Therefore, our answer is **(A) 000101**.

Remark: Note that we can skip straight to the reasoning used in the second paragraph (which is Solution 3 below).

TRICK!:(We can just figure out the missing one by looking at the other 3 numbers provided)

Minor edit - Runningmouse

**Solution 3 (Faster)**

We can look one by one. We see that Lola and Lolo won the first game and Tiya lost. This shows that Tiyo must have lost as well because the results must be 2 wins and 2 losses. We use the same logic for games 2 and 3, giving us 0's again. We look at the choices, and we see A is the only that starts with 3 0's

This shows our answer is **(A) 000101**.

**Solution 4 (Patterns)**

We look at the scores. Notice how every column of numbers represents one round. We see that each column either has 2 wins and 1 loss, or 2 losses and 1 win. So, we come to the conclusion that Tiyo's score must make it 2:2. So, now we solve. The first column has two wins already, and only one loss, so we decide that Tiyo lost (0). The second column also has two wins, so we give Tiyo 0. The third column follows this as well. In the fourth column, we see that Lola and Lolo both lost, and Tiya is the only one who won (1), so Tiyo also won (1). The fifth column follows the same as the first three, and the last column shows that Tiyo also won. So, Tiyo got: 000101. The only option that fits this is A.

**Solution 5 (Easiest Solution)**

There should be two wins and two losses in each column of the graph of the win-loss records. Thus, for the first column, there are already two wins and a loss, which means that, for the column to have the same amount of wins and losses, Tiyo would have to lose. Use the same logic for the other columns, and find that the answer is A, or 000101. SuperVince1

**Solution 6 (Fastest Solution)**

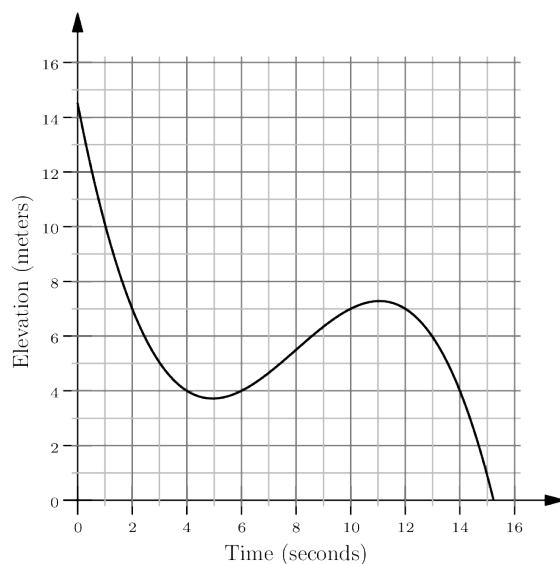
Each player has six games and there are four players so there will be  $6 \times 4 = 24$  games. Total wins and losses of all players should be equal so there should be 12 total wins and losses among all players. By counting, it is known that there are already 10 total wins and 8 total losses. Tiyo's games must make each of these 12 so he will have 2 wins and 4 losses.



However, this leaves three different answers. To tell which is correct, we look at the 4th game of all players. There is 1 win and 2 losses, forcing Tiyo's 4th game to be a win.

Therefore, the answer is **(A)** 000101.

9. (AMC 8 2023) Malaika is skiing on a mountain. The graph below shows her elevation, in meters, above the base of the mountain as she skis along a trail. In total, how many seconds does she spend at an elevation between 4 and 7 meters?

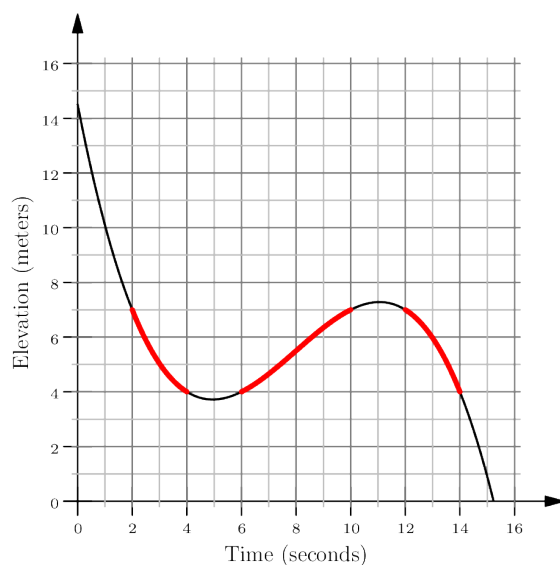


- (A) 6      (B) 8      (C) 10      (D) 12      (E) 14

**Answer:** B

**Solution 1**

We mark the time intervals in which Malaika's elevation is between 4 and 7 meters in red, as shown below:



The requested time intervals are:

- from the 2nd to the 4th seconds
- from the 6th to the 10th seconds
- from the 12th to the 14th seconds

In total, Malaika spends  $(4 - 2) + (10 - 6) + (14 - 12) = \boxed{\text{(B)} 8}$  seconds at such elevation.

### Solution 2

Notice that the entire section between the 2 second mark and the 14 second mark is between the 4 and 7 feet elevation level except the 2 seconds where she skis just under the 4 feet mark and when she skis just above the 7 feet mark, making the answer  $14 - 2 - 2 - 2 = \boxed{\text{(B)} 8}$ .

10. (*AMC 8 2023*) Harold made a plum pie to take on a picnic. He was able to eat only  $\frac{1}{4}$  of the pie, and he left the rest for his friends. A moose came by and ate  $\frac{1}{3}$  of what Harold left behind. After that, a porcupine ate  $\frac{1}{3}$  of what the moose left behind. How much of the original pie still remained after the porcupine left?

(A)  $\frac{1}{12}$       (B)  $\frac{1}{6}$       (C)  $\frac{1}{4}$       (D)  $\frac{1}{3}$       (E)  $\frac{5}{12}$

**Answer:** D

### Solution 1

Note that:

- Harold ate  $\frac{1}{4}$  of the pie. After that,  $1 - \frac{1}{4} = \frac{3}{4}$  of the pie was left behind.
- The moose ate  $\frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4}$  of the pie. After that,  $\frac{3}{4} - \frac{1}{4} = \frac{1}{2}$  of the pie was left behind.
- The porcupine ate  $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$  of the pie. After that,  $\frac{1}{2} - \frac{1}{6} = \boxed{\text{(D)} \frac{1}{3}}$  of the pie was left behind.

More simply, we can condense the solution above into the following equation:

$$\left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{3}\right) = \frac{3}{4} \cdot \frac{2}{3} \cdot \frac{2}{3} = \frac{1}{3}.$$

### Solution 2

Because Harold ate  $\frac{1}{4}$  of the pie, there must be  $\frac{3}{4}$  of the pie he had at the start left. Because the moose ate  $\frac{1}{3}$  of the pie, there must be  $\frac{2}{3}$  of the pie the moose had at the start left. Because the porcupine ate  $\frac{1}{3}$  of the pie, there must be  $\frac{2}{3}$  of the pie the porcupine had at the start left. Therefore, the answer must be  $\frac{3}{4} \times \frac{2}{3} \times \frac{2}{3} = \boxed{\text{(D)} \frac{1}{3}}$  of the pie left.

**Remark** This problem is a great example of the implementation of **remaining portions**. This concept, in short, refers to the method of finding out what remains of a whole when part of it is taken. If you look closely at our solution, we can see that the denominator is the same when we said Harold ate  $\frac{1}{4}$  of the pie and we got  $\frac{3}{4}$  of the pie left. We can also see the numerator is just the denominator minus the old denominator.

11. (*AMC 8 2023*) NASA's Perseverance Rover was launched on July 30, 2020. After traveling 292,526,838 miles, it landed on Mars in Jezero Crater about 6.5 months later. Which of the following is closest to the Rover's average interplanetary speed in miles per hour?

(A) 6,000      (B) 12,000      (C) 60,000      (D) 120,000      (E) 600,000

**Answer:** C

**Solution 1**

Note that 6.5 months is approximately  $6.5 \cdot 30 \cdot 24$  hours. Therefore, the speed (in miles per hour) is

$$\frac{292,526,838}{6.5 \cdot 30 \cdot 24} \approx \frac{300,000,000}{6.5 \cdot 30 \cdot 24} = \frac{10,000,000}{6.5 \cdot 24} \approx \frac{10,000,000}{6.4 \cdot 25} = \frac{10,000,000}{160} = 62,500 \approx \boxed{\text{(C) } 60,000}.$$

As the answer choices are far apart from each other, we can ensure that the approximation is correct.

**Solution 2.1**

Note that  $292,526,838 \approx 300,000,000$  miles. We also know that 6.5 months is approximately  $6.5 \cdot 30 \cdot 24$  hours. Now, we can calculate the speed in miles per hour, which we find is about

$$\frac{300,000,000}{6.5 \cdot 30 \cdot 24} = \frac{10,000,000}{6.5 \cdot 24} = \frac{10,000,000}{13 \cdot 12} = \frac{10,000,000}{156} \approx \frac{10,000,000}{150} \approx \frac{200,000}{3} \approx \boxed{\text{(C) } 60,000}.$$

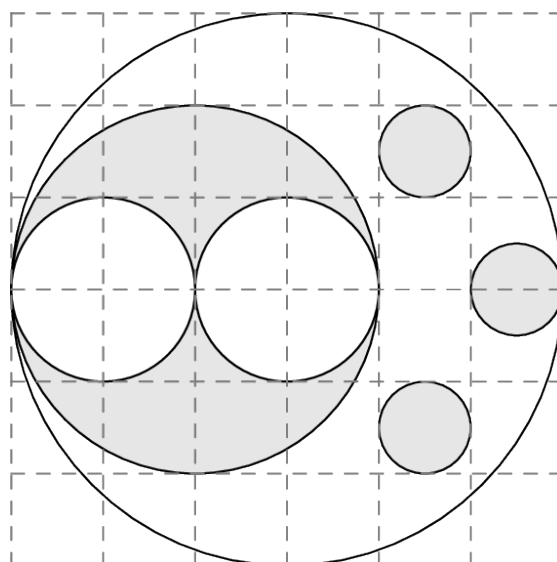
MathFun1000

**Solution 3**

The rover traveled about  $292,526,838 \approx 300,000,000$  miles. 6.5 months is about  $6 \cdot 30 + 15 = 180 + 15 = 195 \approx 200$  days. 200 days is about  $200 \cdot 24 \approx 200 \cdot 25 = 5,000$  hours.  $\frac{300,000,000}{5,000} = \boxed{\text{(C) } 60,000}$

**Remark** This problem is a great example of a situation where rounding all the numbers to ones we can easily work with is an excellent first step. In any competitive timed test, especially the AMC, if you see problems that require you to choose answers that are the closest, this is a sign to you that you can speed through this problem and move on quickly. The rounding being:  $\frac{300000000}{200 \cdot 20}$  Nivaar

12. (*AMC 8 2023*) The figure below shows a large white circle with a number of smaller white and shaded circles in its interior. What fraction of the interior of the large white circle is shaded?



- (A)  $\frac{1}{4}$     (B)  $\frac{11}{36}$     (C)  $\frac{1}{3}$     (D)  $\frac{19}{36}$     (E)  $\frac{5}{9}$

**Answer:** B

**Solution 1**

First, the total area of the radius 3 circle is simply just  $9 \cdot \pi$  when using our area of a circle formula.

Now from here, we have to find our shaded area. This can be done by adding the areas of the  $\frac{1}{2}$ -radius circles and add; then, take the area of the 1 radius circles and subtract that from the area of the 2 radius circle to get our resulting complex shape area. Adding these up, we will get  $3 \cdot \frac{1}{4}\pi + 4\pi - \pi - \pi = \frac{3}{4}\pi + 2\pi = \frac{11\pi}{4}$ .

So, our answer is  $\frac{\frac{11\pi}{4}}{9\pi} = \boxed{\text{(B)} \frac{11}{36}}$ .

Minor edits by Shriyans Chowdhury

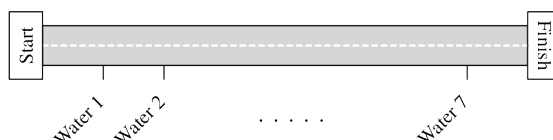
**Solution 2**

Pretend each circle is a square. The large shaded circle is a square with area 16 units<sup>2</sup>, and the two white circles inside it each have areas of 4 units<sup>2</sup>, which adds up to 8 units<sup>2</sup>. The three small shaded circles become three squares with area 1 units<sup>2</sup>, and add up to 3 units<sup>2</sup>. Adding the areas of the shaded circles (19) and subtracting the areas of the white circles (8), we get 11 units<sup>2</sup>. Since the largest white circle in which all these other circles are becomes a square that has area

36 units<sup>2</sup>, our answer is  $\boxed{\text{(B)} \frac{11}{36}}$ .

-yume\_mitaLaTeX (edits -apex304, CoOlPoTaToEs, KGINSPLECTORBOI, SlimeKnight, rickastleylover99)

13. (*AMC 8 2023*) Along the route of a bicycle race, 7 water stations are evenly spaced between the start and finish lines, as shown in the figure below. There are also 2 repair stations evenly spaced between the start and finish lines. The 3rd water station is located 2 miles after the 1st repair station. How long is the race in miles?



- (A) 8      (B) 16      (C) 24      (D) 48      (E) 96

**Answer:** D

**Solution**

Suppose that the race is  $d$  miles long. The water stations are located at

$$\frac{d}{8}, \frac{2d}{8}, \dots, \frac{7d}{8}$$

miles from the start, and the repair stations are located at

$$\frac{d}{3}, \frac{2d}{3}$$

miles from the start.

We are given that  $\frac{3d}{8} = \frac{d}{3} + 2$ , from which

$$\begin{aligned} \frac{9d}{24} &= \frac{8d}{24} + 2 \\ \frac{d}{24} &= 2 \\ d &= \boxed{\text{(D) } 48}. \end{aligned}$$

apex304, SohUmUttamchandani, wuwang2002, TaeKim, Cxrupptedpat, MRENTHUSI-ASM

14. (AMC 8 2023) Nicolas is planning to send a package to his friend Anton, who is a stamp collector. To pay for the postage, Nicolas would like to cover the package with a large number of stamps. Suppose he has a collection of 5-cent, 10-cent, and 25-cent stamps, with exactly 20 of each type. What is the greatest number of stamps Nicolas can use to make exactly \$7.10 in postage? (Note: The amount \$7.10 corresponds to 7 dollars and 10 cents. One dollar is worth 100 cents.)

- (A) 45      (B) 46      (C) 51      (D) 54      (E) 55

**Answer:** E

**Solution 1**

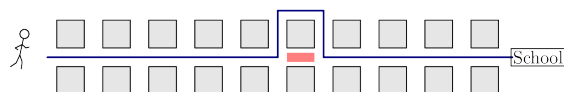
Let's use the most stamps to make 7.10. We have 20 of each stamp, 5-cent (nickels), 10-cent (dimes), and 25-cent (quarters).

If we want the highest number of stamps, we must have the highest number of the smaller value stamps (like the coins above). We can use 20 nickels and 20 dimes to bring our total cost to  $7.10 - 3.00 = 4.10$ . However, when we try to use quarters, the 25 cents don't fit evenly, so we have to give back 15 cents to make the quarter amount 4.25. The most efficient way to do this is to give back a 10-cent (dime) stamp and a 5-cent (nickel) stamp to have 38 stamps used so far. Now, we just use  $\frac{425}{25} = 17$  quarters to get a grand total of  $38 + 17 = \boxed{\text{(E) } 55}$ .

**Solution 2**

The value of his entire stamp collection is 8 dollars. To make \$7.10 with stamps, he should remove 90 cents worth of stamps with as few stamps as possible. To do this, he should start by removing as many 25 cent stamps as possible as they have the greatest denomination. He can remove at most 3 of these stamps. He still has to remove  $90 - 25 \cdot 3 = 15$  cents worth of stamps. This can be done with one 5 and 10 cent stamp. In total, he has  $20 \cdot 3 = 60$  stamps in his entire collection. As a result, the maximum number of stamps he can use is  $20 \cdot 3 - 5 = \boxed{\text{(E)} 55}$ .

15. (AMC 8 2023) Viswam walks half a mile to get to school each day. His route consists of 10 city blocks of equal length and he takes 1 minute to walk each block. Today, after walking 5 blocks, Viswam discovers he has to make a detour, walking 3 blocks of equal length instead of 1 block to reach the next corner. From the time he starts his detour, at what speed, in mph, must he walk, in order to get to school at his usual time?



- (A) 4      (B) 4.2      (C) 4.5      (D) 4.8      (E) 5

**Answer:** B

### Solution 1

Note that Viswam walks at a constant speed of 60 blocks per hour as he takes 1 minute to walk each block. After walking 5 blocks, he has taken 5 minutes and has 5 minutes remaining to walk 7 blocks. Therefore, he must walk at a speed of  $7 \cdot 60 \div 5 = 84$  blocks per hour to get to school on time, from the time he starts his detour. Since he normally walks  $\frac{1}{2}$  mile, which is equal to 10 blocks, 1 mile is equal to 20 blocks. Therefore, he must walk at  $84 \div 20 = 4.2$  mph from the time he starts his detour to get to school on time, so the answer is  $\boxed{\text{(B)} 4.2}$ .

### Solution 2

Viswam walks 10 blocks, or half a mile, in 10 minutes. Therefore, he walks at a rate of 3 mph. When he takes his detour, he must travel 7 blocks instead of 5. Our final equation is  $\frac{7}{5} \times 3 = \frac{21}{5} = \boxed{\text{(B)} 4.2}$ .

Note: In the second half of his journey, he is travelling  $7/5$  as far as he originally expected to, so he needs to travel at  $7/5$  the speed to get there on time.

### Solution 3

When calculating the speed of his normal route, we get:

$$\frac{0.5 \text{ miles}}{10 \text{ minutes}} = \frac{3 \text{ miles}}{60 \text{ minutes}}$$

Keep in mind that his route is 10 blocks long. Since we count 7 blocks when the detour starts, than this would mean that he has  $\frac{7}{10}$  miles left to walk, considering the normal amount of blocks that he usually walks. Multiplying by 6 to get the rate, we get,  $\frac{42}{10} = 4.2$

Therefore, the answer is  $\boxed{\text{(B)} 4.2}$ .

16. (AMC 8 2023) The letters P, Q, and R are entered into a  $20 \times 20$  table according to the pattern shown below. How many Ps, Qs, and Rs will appear in the completed table?

|          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ |
| Q        | R        | P        | Q        | R        | $\dots$  |
| P        | Q        | R        | P        | Q        | $\dots$  |
| R        | P        | Q        | R        | P        | $\dots$  |
| Q        | R        | P        | Q        | R        | $\dots$  |
| P        | Q        | R        | P        | Q        | $\dots$  |

- (A) 132 Ps, 134 Qs, 134 Rs (B) 133 Ps, 133 Qs, 134 Rs (C) 133 Ps, 134 Qs, 133 Rs  
 (D) 134 Ps, 132 Qs, 134 Rs (E) 134 Ps, 133 Qs, 133 Rs

**Answer:** C

### Solution 1 (Finding a Pattern)

In our  $5 \times 5$  grid, there are 8, 9 and 8 of the letters P, Q, and R, respectively, and in a  $2 \times 2$  grid, there are 1, 2 and 1 of the letters P, Q, and R, respectively. We see that in both grids, there are  $x, x + 1$ , and  $x$  of the P, Q, and R, respectively. This is because in any  $n \times n$  grid with  $n \equiv 2 \pmod{3}$ , there are  $x, x + 1$ , and  $x$  of the P, Q, and R, respectively. We can see that the only answer choice which satisfies this condition is **(C) 133 Ps, 134 Qs, 133 Rs**.

### Solution 2

Since  $20 \equiv 2 \pmod{3}$  and Q is in the 2nd diagonal, it is also in the 20th diagonal, and so we find that there are  $2(2 + 5 + 8 + 11 + 14 + 17) + 20 = 134$  Qs. Since all the P's and R's are symmetric, the answer is **(C) 133 Ps, 134 Qs, 133 Rs**.

### Solution 3

Notice that rows  $x$  and  $x + 3$  are the same, for any  $1 \leq x \leq 17$ . Additionally, rows 1, 2, and 3 collectively contain the same number of Ps, Qs, and Rs, because the letters are just substituted for one another. Therefore, the number of Ps, Qs, and Rs in the first 18 rows is 120. The first row has 7 Ps, 7 Qs, and 6 Rs, and the second row has 6 Ps, 7 Qs, and 7 Rs. Adding these up, we obtain **(C) 133 Ps, 134 Qs, 133 Rs**.

### Solution 5

This solution refers to the full diagram in Solution 4.

Note the Q diagonals are symmetric. The R and P diagonals are not symmetric, but are reflections of each other about the Q diagonals:

- The upper Q diagonal of length 2 is surrounded by a P diagonal of length 3 and an R diagonal of length 1.
- The lower Q diagonal of length 2 is surrounded by a P diagonal of length 1 and an R diagonal of length 3.

When looking at a pair of  $Q$  diagonals of the same length  $x$ , there is a total of  $2x$  Rs and Ps next to these 2 diagonals.

The main diagonal of 20 Qs has 19 Ps and 19 Rs next to it. Thus, the total is  $x + 1$  Qs,  $x$  Ps,  $x$  Rs. Therefore, the answer is **(C) 133 Ps, 134 Qs, 133 Rs**.

### Solution 6 (Modular Arithmetic)

0-Benedict T (countmath1)

### Solution 7 (Answer Choices, Fast, Same as Solution 1)

We can first observe that  $P$  shows on diagonals increasing or decreasing by 3.

It starts at 1, 4, 7, 9... and increases in the form  $3x - 2$ . Using our answer choices, (B) and (C) are the only fits.

$Q$  is like this as well, increasing 2, 5, 8, 11.... This means  $Q$  has to be in the form of  $3x - 1$ . Testing this out leads us with **(C) 133 Ps, 134 Qs, 133 Rs**.

### Solution 8 (Intuitive, Fastest)

When we find the letters at the corners, we see that there are 2 other corners with Q's and 1 with R. There are 2 corners with Q and 1 with P and R. Therefore, it makes sense that there should be more Q's than other letters. Only **(C) 133 Ps, 134 Qs, 133 Rs** has more Q's than other letters.

**Only do it this way if there are 30 seconds left on the clock, as it may not always work!**

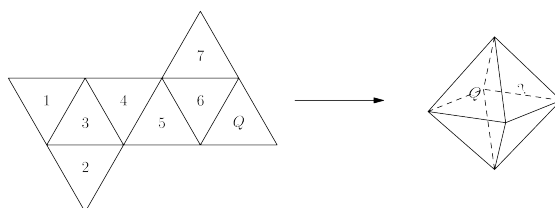
### Solution 9 (Think and Label)

Since there are 3 letters, there must be a multiple of 3 for the height of the array for the amount of letters to be the same. However, there is one less ( $21 - 1 = 20$ ) so there will be one less letter in each column compared to the other two. Looking at the diagram, the first column starts with a  $P$  then a  $Q$ . Since it will miss the third letter,  $R$ , we write this as  $-R$ . We do this process for the remaining 20 columns, and applying the same logic if the first three are  $-R$ ,  $-P$  and  $-Q$ , there will be one less  $-Q$  meaning that there will be an extra  $Q$ . The only answer that has in extra  $Q$  is **(C) 133 Ps, 134 Qs, 133 Rs**

### Solution 10 (Fast and easy to understand)

Since it is a 20 by 20 grid, there will be 400 letters in total. Since there are a total of 3 letters, in the order  $Q, R, P$  we have to divide 400 by 3, which gives us 133 with a remainder of 1. Since  $Q$  is the first letter, there will be 134  $Q$ 's, 133  $R$ 's and 133  $P$ 's, so the answer is **(C) 133 Ps, 134 Qs, 133 Rs**

17. (AMC 8 2023) A regular octahedron has eight equilateral triangle faces with four faces meeting at each vertex. Jun will make the regular octahedron shown on the right by folding the piece of paper shown on the left. Which numbered face will end up to the right of  $Q$ ?



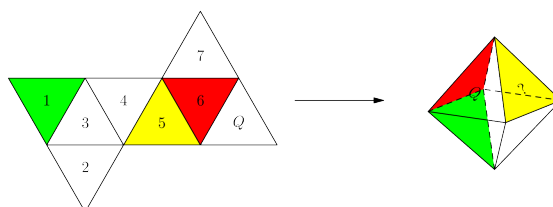
- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5



**Answer:** A

**Solution 1**

We color face 6 red and face 5 yellow. Note that from the octahedron, face 5 and face ? do not share anything in common. From the net, face 5 shares at least one vertex with all other faces except face 1, which is shown in green:



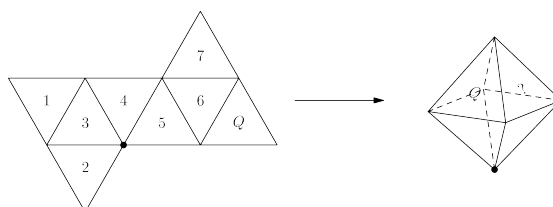
Therefore, the answer is **(A) 1**.

**Solution 2**

We label the octohedron going triangle by triangle until we reach the ? triangle. The triangle to the left of the  $Q$  should be labeled with a 6. Underneath triangle 6 is triangle 5. The triangle to the right of triangle 5 is triangle 4 and further to the right is triangle 3. Finally, the side of triangle 3 under triangle  $Q$  is 2, so the triangle to the right of  $Q$  is **(A) 1**.

**Solution 3**

Notice that the triangles labeled 2, 3, 4, and 5 make the bottom half of the octahedron, as shown below:



Therefore, **(B)**, **(C)**, **(D)**, and **(E)** are clearly not the correct answer. Thus, the only choice left is **(A) 1**.

**Solution 4 (Really Simple Reasoning)**

The first half of the octahedron will need 4 triangles connected to one another to form it. We can choose the triangles 4, 5, 6, and 7 and form the half around the vertex they all share. That leaves triangles 1, 3, 2, and  $Q$  to form the second half. Triangle 3 will definitely share its sides with triangles 1 and 2, leaving them to share their second side with triangle  $Q$ . Since triangle  $Q$  will certainly share its left side with triangle 2, the only triangle left to share its right side is triangle **(A) 1**

**Solution 5 (brute force)**

Trace the net on your scratch paper, rip it out, and fold it. You'll see the answer is **(A) 1** This is not recommended.

18. (AMC 8 2023) Greta Grasshopper sits on a long line of lily pads in a pond. From any lily pad, Greta can jump 5 pads to the right or 3 pads to the left. What is the fewest

number of jumps Greta must make to reach the lily pad located 2023 pads to the right of her starting point?

- (A) 405      (B) 407      (C) 409      (D) 411      (E) 413

**Answer:** D

**Solution 1**

We have 2 directions going 5 right or 3 left. We can assign a variable to each of these directions. We can call going right 1 direction  $X$  and we can call going 1 left  $Y$ . We can build an equation of  $5X - 3Y = 2023$ , where we have to limit the number of moves we do. We can do this by making more of our moves the 5 move turn than the 3 move turn. The first obvious step is to go some amount of moves in the right direction then subtract off in the left direction to land on 2023. The least amount of 3's added to 2023 to make a multiple of 5 is 4 as  $2023 + 4(3) = 2035$ . So now, we have solved the problem as we just go  $\frac{2035}{5} = 407$  hops right, and just do 4 more hops left. Yielding  $407 + 4 = \boxed{\text{(D) 411}}$  as our answer.

**Solution 2**

Notice that  $2023 \equiv 3 \pmod{5}$ , and jumping to the left increases the value of Greta's position  $\pmod{5}$  by 2. Therefore, the number of jumps to the left must be 4  $\pmod{5}$ . As the number of jumps to the left increases, so does the number of jumps to the right, we must minimize both, which occurs when we jump 4 to the left and 407 to the right. The answer is  $\boxed{\text{(D) 411}}$ .

**Solution 3**

$5y - 2023$  must be divisible by 3. The smallest value of  $y$  that will achieve this is 407, which lands it at 2035. After that, it takes 4 jumps back, making a total of  $\boxed{\text{(D) 411}}$ .

**Solution 3.1**

Here is Solution 3 but worded differently: We can to go back  $3x$  steps from  $5y$  steps we took past 2023. The other way to say this is we go  $3x$  steps from 2023 to get to  $5y$ . What is the least value of  $3x$ ? We need  $2023 + 3x$  to end in a 5 or 0. The least value for  $3x$ , which makes  $2023 + 3x = 2035$ , is 4.  $\frac{2035}{5} = 407$ . Therefore,  $407 + 4 = \boxed{\text{(D) 411}}$ .

**Solution 4 (Easy to understand and simple)**

Greta needs to jump 2023 steps to the right. Since we want (a multiple of 5)- (a multiple of 3) that ends in a 3. The smallest this can be is  $15 - 12 = 3$ .

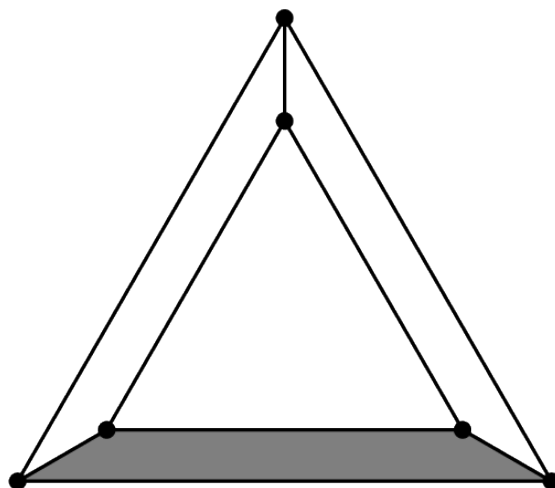
We can apply this to the problem; the closest number that ends in 15 from 2023 is 2015. (Note that 2015 is 403 jumps (every 5 jumps to the right is one lily pad, we will use this later.)

From this, we know that to get to 2003 steps, Greta would have to jump 12 pads to the left, which would take 4 jumps from the lily pad (every 3 jumps to the left is one lily pad).

Lastly, to get to 2023, it would take 20 pads to the right, which would be 4 jumps from the lily pads as well.

In total, this is  $403 + 4 + 4 = 411$  , or  $\boxed{\text{(D) 411}}$ .

19. (AMC 8 2023) An equilateral triangle is placed inside a larger equilateral triangle so that the region between them can be divided into three congruent trapezoids, as shown below. The side length of the inner triangle is  $\frac{2}{3}$  the side length of the larger triangle. What is the ratio of the area of one trapezoid to the area of the inner triangle?



- (A) 1 : 3      (B) 3 : 8      (C) 5 : 12      (D) 7 : 16      (E) 4 : 9

**Answer:** C

**Solution 1**

All equilateral triangles are similar. For the outer equilateral triangle to the inner equilateral triangle, since their side-length ratio is  $\frac{3}{2}$ , their area ratio is  $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$ . It follows that the area ratio of three trapezoids to the inner equilateral triangle is  $\frac{9}{4} - 1 = \frac{5}{4}$ , so the area ratio of one trapezoid to the inner equilateral triangle is

$$\frac{5}{4} \cdot \frac{1}{3} = \frac{5}{12} = \boxed{\text{(C) } 5 : 12}.$$

apex304, SohumUttamchandani, wuwang2002, TaeKim, Cxrupptedpat, MRENTHUSI-ASM, JubilantJelly

**Solution 2**

Subtracting the larger equilateral triangle from the smaller one yields the sum of the three trapezoids. Since the ratio of the side lengths of the larger to the smaller one is 3 : 2, we can set the side lengths as 3 and 2, respectively. So, the sum of the trapezoids is  $\frac{9\sqrt{3}}{4} - \frac{4\sqrt{3}}{4} = \frac{5\sqrt{3}}{4}$ . We are also told that the three trapezoids are congruent, thus the area of each of them is  $\frac{1}{3} \cdot \frac{5\sqrt{3}}{4} = \frac{5\sqrt{3}}{12}$ . Hence, the ratio is  $\frac{\frac{5\sqrt{3}}{12}}{\frac{\sqrt{3}}{4}} = \frac{5}{12} = \boxed{\text{(C) } 5 : 12}$ .

20. (AMC 8 2023) Two integers are inserted into the list 3, 3, 8, 11, 28 to double its range. The mode and median remain unchanged. What is the maximum possible sum of the two additional numbers?

- (A) 56      (B) 57      (C) 58      (D) 60      (E) 61

**Answer:** D

**Solution 1**

To double the range, we must find the current range, which is  $28 - 3 = 25$ , to then double to:  $2(25) = 50$ . Since we do not want to change the median, we need to get a value less than 8 (as 8 would change the mode) for the smaller, which will be 7. This fixes 53 for the larger value. Anything less than 3 is not beneficial to the optimization because you want to get the largest range without changing the mode. So, taking our optimal values of 7 and 53, we have an answer of  $7 + 53 = \boxed{\text{(D) } 60}$ .

21. (AMC 8 2023) Alina writes the numbers  $1, 2, \dots, 9$  on separate cards, one number per card. She wishes to divide the cards into 3 groups of 3 cards so that the sum of the numbers in each group will be the same. In how many ways can this be done?

(A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** C

### Solution 1

The group with 5 must have the two other numbers adding up to 10, since the sum of all the numbers is  $(1 + 2 + \dots + 9) = \frac{9(10)}{2} = 45$ . The sum of the numbers in each group must therefore be  $\frac{45}{3} = 15$ . We can have  $(1, 5, 9)$ ,  $(2, 5, 8)$ ,  $(3, 5, 7)$ , or  $(4, 5, 6)$ . With the first group, we have  $(2, 3, 4, 6, 7, 8)$  left over. The only way to form a group of 3 numbers that add up to 15 is with  $(3, 4, 8)$  or  $(2, 6, 7)$ . One of the possible arrangements is therefore  $(1, 5, 9)(3, 4, 8)(2, 6, 7)$ . Then, with the second group, we have  $(1, 3, 4, 6, 7, 9)$  left over. With these numbers, there is no way to form a group of 3 numbers adding to 15. Similarly, with the third group there is  $(1, 2, 4, 6, 8, 9)$  left over and we can make a group of 3 numbers adding to 15 with  $(1, 6, 8)$  or  $(2, 4, 9)$ . Another arrangement is  $(3, 5, 7)(1, 6, 8)(2, 4, 9)$ . Finally, the last group has  $(1, 2, 3, 7, 8, 9)$  left over. There is no way to make a group of 3 numbers adding to 15 with this, so the arrangements are  $(1, 5, 9)(3, 4, 8)(2, 6, 7)$  and  $(3, 5, 7)(1, 6, 8)(2, 4, 9)$ . So there are **(C) 2** sets that can be formed.

### Solution 2

The sum of the numbers across all equally valued sets is  $(1 + 2 + \dots + 9) = \frac{9(10)}{2} = 45$ . The value of the numbers in each set would be  $\frac{45}{3} = 15$ . We know that the numbers 9, 8, and 7 must belong in different sets, as putting any 2 numbers in 1 set will either pass or match the limit of 15 per set, and we would then still need to add 1 more number after that. Note that these numbers must be distinct, as Alina only has 1 of each number, and order does not matter in the sets. Starting with the set that includes the number 9, the next two numbers must add up to 6, and there are **2** ways of doing this  $(2, 4)(1, 5)$ . Note we cannot use any number past 6, as those numbers must be used in the other sets. The next set, which includes the number 8, must have two numbers that add up to 7, and there are **3** ways to do this  $(2, 5)(1, 6)(3, 4)$ . The final set, which includes the number 7, must have 2 numbers that sum up to 8, and there are **2** ways to do this  $(2, 6)(3, 5)$ . Now we have found the number of ways in which each set sums up to 15. To find the number of ways in which all three sets sum up to 15 concurrently, we must take the minimum of 2, 3, and 2, which gives us an answer of **(C) 2** triplets of sets with 3 values, in which each set sum to the same amount

### Solution 3

Note that each group of numbers should sum to  $\frac{1+2+3+4+5+6+7+8+9}{3} = 15$ . Thus, this is equivalent to asking, “How many ways can you fill in a three by three magic square with the integers 1 through 9?” since we can take the three rows of the magic square as our three groups. If you have closely studied magic squares, you might know that in a three by three magic square that is to be filled in with the integers 1 through 9, the center of the square would be 5 (the average of the numbers), and the numbers in the corners should be even(\*). The such pairs (disregarding order) are  $(2, 8)$  and  $(4, 6)$ . Let’s fix the position of 2 to be the top left corner. This would make 8 in the bottom right corner. We can have either 4 or 6 to be in the top right corner, for a total of **(C) 2** such groups of three. (The groups are  $(8, 3, 4)(1, 5, 7)(6, 9, 2)$  and  $(8, 1, 6)(3, 5, 7)(4, 9, 2)$ .) Note that if we had instead fixed the position of 4, 6, or 8, they would correspond to one of the two

cases, just in a different configuration.

(\*)We can prove this using proof by contradiction. Label the nine small squares within the magic square from  $a$  to  $i$  from left to right, top to bottom. Firstly, we know that  $a + i$  and  $c + g$  sum to 10 since the center square is 5. Thus,  $a$  and  $i$  must have the same parity, and so must  $c$  and  $g$ . Suppose that  $a$  and  $c$  have different parity. Since  $a + b + c = 15$ ,  $b$  must be even. By a similar argument,  $h$  must also be even, and so must  $d$  and  $f$ . Our initial assumption is that one of  $a$  and  $c$  is odd and the other is even; however, we end up with six even numbers needed to fill in the square, but there are only four even integers from 1 to 9. Now suppose that all of  $a, c, g$ , and  $i$  are odd. This would make each of  $b, d, f$ , and  $h$  odd, but clearly we do not have enough odd numbers to make all nine numbers odd. Thus, each corner square must be even.

#### Solution 4(Trial and Error)

To start, we must find the value that all three groups must add to, which is  $\frac{1+2+3+4+5+6+7+8+9}{3} = 15$ . Then, we can notice that the numbers 1, 2, and 3 must be in different groups because two of them and any other digit 1-9 added together cannot produce 15 (For example,  $2 + 3 = 5$  which means that we need a 10 to sum to 15. In the same way, 7, 8, and 9 cannot be placed in the same group because putting them in the same group would make the numbers greater than 15 (For example,  $7 + 8 = 15$ , meaning that it would be impossible to add another number with making the sum greater than 15. Now, we can just use trial and error to solve the problem (this is feasible because there are only  $3 * 2 = 6$  ways to pair a number 1 – 3 and 7 – 9 and then add another number 4 – 6 in order to sum to 15. By trial and error, we see that 1 and 7 must not be paired together because to have a sum of 15, there must be another 7 in the group (The problem only lets us use 1 of each number). This eliminates two of the options we are considering. Now, we only have to check the 4 options where 7 is not in a group with 1.

By trial and error, we can see that only two cases work:

1. (1, 8, 6); (2, 9, 4); (3, 7, 5) and 2. (1, 9, 5); (2, 7, 6); (3, 8, 4)

Because we only have two options, the answer is C (Too bashy, not recommended in the real test if you encounter these problems)

#### Solution 5

I don't know for sure, but this problem could be the same as picking 3 pairs of lines in a magic square. The only 3 sets that work are either all columns or all rows, as diagonals have only 2 pairs, and they overlap. Giving us 2, or C (If there are any errors with this, please let me know)

CubiksRube's solution is indeed correct to some degree. We know that a magic square has 8 lines where the numbers add up to the same: 15. As the question has stated, we need 3 groups of 3 cards for 1 possibility. There are 2 groups of three groups of three numbers - the horizontal three and the vertical 3 - and they both add to the same number. We don't care about the diagonals because there are only 2 of them and we need 3 groups for 1 possibility.

We also know that this is the only 2 possibilities because otherwise there would be multiple magic squares.

Hence the answer (C) 2

#### Solution 6, short, quick and easy

We know that the groups will have to be  $\frac{1+2+3+4+5+6+7+8+9}{3} = 15$ . For each of the greatest 3 numbers (7, 8, 9) there are 2 possible ways to get a total of 15 for each group.

Example (9):  $15 = 9 + 5 + 1$ , or  $9 + 4 + 2$ . This will be same for 7 and 8.

Therefore, there are 2 ways this can be done, or **(C) 2**.

22. (AMC 8 2023) In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term is 4000. What is the first term?

(A) 1      (B) 2      (C) 4      (D) 5      (E) 10

**Answer:** D

**Solution 1**

In this solution, we will use trial and error to solve. 4000 can be expressed as  $200 \times 20$ . We divide 200 by 20 and get 10, divide 20 by 10 and get 2, and divide 10 by 2 to get **(D) 5**. No one said that they have to be in ascending order!

Solution by [ILoveMath31415926535](#) and clarification edits by apex304 Anonymous question here, if 5 is first term, how do you get 10?

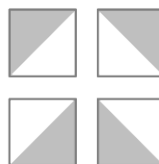
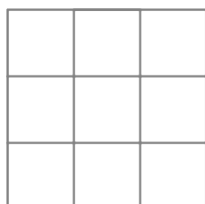
(answer to anonymous question) "No one said they had to be in ascending order!" means after 5, the term doesn't need to increase each time.

**Solution 2**

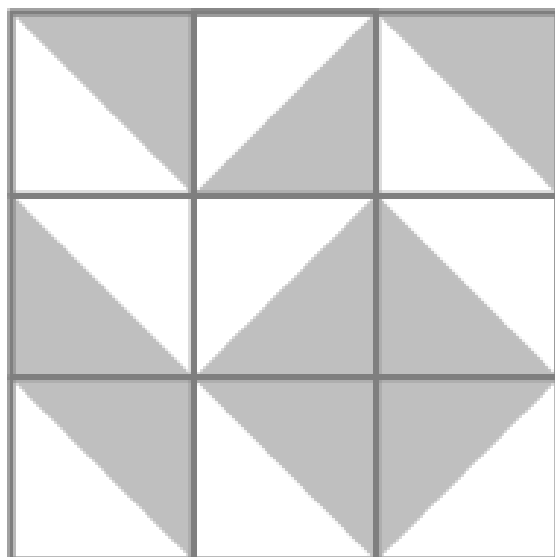
Consider the first term is  $A$  and the second term is  $B$ . Then, the following terms will be  $AB$ ,  $AB^2$ ,  $A^2B^3$  and  $A^3B^5$ . Notice that  $4000 = 2^5 \times 5^3$ , then we obtain  $A = \mathbf{(D) 5}$  and  $B = 2$ .

Solution by Slimeknight, edited by XingZheZhong

23. (AMC 8 2023) Each square in a  $3 \times 3$  grid is randomly filled with one of the 4 gray and white tiles shown below on the right.



What is the probability that the tiling will contain a large gray diamond in one of the smaller  $2 \times 2$  grids? Below is an example of such tiling.



- (A)  $\frac{1}{1024}$     (B)  $\frac{1}{256}$     (C)  $\frac{1}{64}$     (D)  $\frac{1}{16}$     (E)  $\frac{1}{4}$

**Answer:** C

**Solution 1**

There are 4 cases that the tiling will contain a large gray diamond in one of the smaller  $2 \times 2$  grids, as shown below:



There are  $4^5$  ways to decide the 5 white squares for each case, and the cases do not have any overlap.

So, the requested probability is

$$\frac{4 \cdot 4^5}{4^9} = \frac{4^6}{4^9} = \frac{1}{4^3} = \boxed{\text{(C)} \frac{1}{64}}.$$

apex304, TaeKim, MRENTHUSIASM

**Solution 2**

Note that the middle tile can be any of the four tiles. The gray part of the middle tile points towards one of the corners, and for the gray diamond to appear the three adjacent tiles must all be perfect. Thus, the solution is  $\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \boxed{\text{(C)} \frac{1}{64}}.$

**Solution 3**

Note that each tile must be in its precise place. Because of that, each diamond has a  $\left(\frac{1}{4}\right)^4$  chance of appearing. And since there are 4 placements, our solution is  $\frac{1}{4^4} \cdot 4 = \boxed{\text{(C)} \frac{1}{64}}.$

**Solution 4 (Linearity of Expectation)**

Let  $S_1, S_2, S_3$ , and  $S_4$  denote the 4 smaller  $2 \times 2$  squares within the  $3 \times 3$  square in some order. For each  $S_i$ , let  $X_i = 1$  if it contains a large gray diamond tiling and  $X_i = 0$  otherwise. This means that  $\mathbb{E}[X_i]$  is the probability that square  $S_i$  has a large gray

diamond, so  $\mathbb{E}[X_1 + X_2 + X_3 + X_4]$  is our desired probability. However, since there is only one possible way to arrange the squares within every  $2 \times 2$  square to form such a tiling, we have  $\mathbb{E}[X_i] = \left(\frac{1}{4}\right)^4 = \frac{1}{256}$  for all  $i$  (as each of the smallest tiles has 4 possible arrangements), and from the linearity of expectation we get

$$\mathbb{E}[X_1 + X_2 + X_3 + X_4] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \mathbb{E}[X_3] + \mathbb{E}[X_4] = \frac{1}{256} + \frac{1}{256} + \frac{1}{256} + \frac{1}{256} = \boxed{\text{(C)} \frac{1}{64}}.$$

eibc

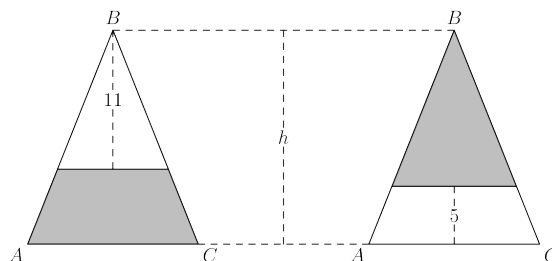
Remark 1: This method might be too advanced for the AMC 8, and is probably unnecessary (refer to the other solutions for simpler techniques).

Remark 2: Note that Probability and Expected Value are equivalent in this problem since there will never be two diamonds on one tiling. i.e.  $X_1 + X_2 + X_3 + X_4 \leq 1$ .

### Solution 5 (Simple and Quick)

We can start by finding the probability of making a diamond with the squares in ONLY the top-left corner of the square. There are  $4 \times 4 \times 4 \times 4 = 256$  total cases of putting four random tiles in the top-left corner. Since there is only one favorable case (the case of making a diamond) out of 256 total, the probability of making a diamond is  $\frac{1}{256}$ . Now, we can simply multiply this by four to account for all corners of the square to get  $4 \times \frac{1}{256}$  or  $\boxed{\text{(C)} \frac{1}{64}}$ .

24. (AMC 8 2023) Isosceles triangle  $ABC$  has equal side lengths  $AB$  and  $BC$ . In the figures below, segments are drawn parallel to  $\overline{AC}$  so that the shaded portions of  $\triangle ABC$  have the same area. The heights of the two unshaded portions are 11 and 5 units, respectively. What is the height  $h$  of  $\triangle ABC$ ?



- (A) 14.6      (B) 14.8      (C) 15      (D) 15.2      (E) 15.4

**Answer:** A

### Solution 1

First, we notice that the smaller isosceles triangles are similar to the larger isosceles triangles. We can find that the area of the gray area in the first triangle is  $[ABC] \cdot \left(1 - \left(\frac{11}{h}\right)^2\right)$ . Similarly, we can find that the area of the gray part in the second triangle is  $[ABC] \cdot \left(\frac{h-5}{h}\right)^2$ . These areas are equal, so  $1 - \left(\frac{11}{h}\right)^2 = \left(\frac{h-5}{h}\right)^2$ . Simplifying yields  $10h = 146$  so  $h = \boxed{\text{(A)} 14.6}$ .

### Solution 2

We can call the length of  $AC$  as  $X$ . Therefore, the length of the base of the triangle with height 11 is  $\frac{11}{H} = \frac{A}{X}$ . Therefore, the base of the smaller triangle is  $\frac{11X}{H}$ . We find that the area of the trapezoid is  $\frac{HX}{2} - \frac{11^2X}{2H}$ .



Using similar triangles once again, we find that the base of the shaded triangle is  $\frac{H-5}{H} = \frac{B}{X}$ . Therefore, the area is  $(H-5)(HX-5X)/2H$ .

Since the areas are the same, we find that  $(HX)/2 - 121X/2H = (H-5)(HX-5X)/2H$ . Multiplying each side by  $2H$ , we get  $H^2x/X - 121X = h^2X - 5HX - 5HX + 25X$ . Therefore, we can subtract  $25X + H^2x/X$  from both sides, and get  $-146X = -10hx$ . Finally, we divide both sides by  $-x$  and get  $10h = 146$ .  $h$  is **(A) 14.6**.

Solution by CHECKMATE2021 (edits by someone that is unknown)

### Solution 3 (Faster)

Since the length of AC does not matter, we can assume the height of triangle ABC is  $h$ . We assume that the bases of the first triangle are  $h$  and 11 and the bases of the second triangle are  $h-5$  and  $h$ . Therefore, the area of the trapezoid in the first diagram is  $h^2/2 - \frac{11^2}{2}$ .

The area of the triangle in the second diagram is now  $\frac{(h-5)^2}{2}$ .

Therefore,  $\frac{h^2-11^2}{2} = \frac{(h-5)^2}{2}$ . Multiplying both sides by 2, we get  $h^2 - 121 = h^2 - 10h + 25$ . Subtracting  $h^2 + 25$  from both sides, we get  $-146 = -10h$  and  $h$  is **(A) 14.6**.

Solution by ILoveMath31415926535, and CHECKMATE2021 and London\_System

25. (AMC 8 2023) Fifteen integers  $a_1, a_2, a_3, \dots, a_{15}$  are arranged in order on a number line. The integers are equally spaced and have the property that

$$1 \leq a_1 \leq 10, \quad 13 \leq a_2 \leq 20, \quad \text{and} \quad 241 \leq a_{15} \leq 250.$$

What is the sum of digits of  $a_{14}$ ? **(A) 8**      **(B) 9**      **(C) 10**      **(D) 11**      **(E) 12**

**Answer:** A

### Solution 1

We can find the possible values of the common difference by finding the numbers which satisfy the conditions. To do this, we find the minimum of the last two:  $241 - 20 = 221$ , and the maximum:  $250 - 13 = 237$ . There is a difference of 13 between them, so only 17 and 18 work, as  $17 \cdot 13 = 221$ , so 17 satisfies  $221 \leq 13x \leq 237$ . The number 18 is similarly found. 19, however, is too much.

Now, we check with the first and last equations using the same method. We know  $241 - 10 \leq 14x \leq 250 - 1$ . Therefore,  $231 \leq 14x \leq 249$ . We test both values we just got, and we can realize that 18 is too large to satisfy this inequality. On the other hand, we can now find that the difference will be 17, which satisfies this inequality.

The last step is to find the first term. We know that the first term can only be from 1 to 3 since any larger value would render the second inequality invalid. Testing these three, we find that only  $a_1 = 3$  will satisfy all the inequalities. Therefore,  $a_{14} = 13 \cdot 17 + 3 = 224$ . The sum of the digits is therefore **(A) 8**.

### Solution 2 (complicated but fast)

Step 1: Express everything in terms of  $a_1$  and the common difference  $d$

Since the 15 integers are equally spaced, they form an arithmetic sequence:  $a_1, a_1 + d, a_1 + 2d, \dots, a_1 + 14d$ .

So:  $a_2 = a_1 + d, \quad a_{15} = a_1 + 14d$ .

Step 2: Translate the given inequalities

We are told:  $1 \leq a_1 \leq 10$ ,  $13 \leq a_2 \leq 20$ ,  $241 \leq a_{15} \leq 250$ .

Step 3: Find the possible range for  $a_{15} - a_2$

We have:  $a_{15} - a_2 = (a_1 + 14d) - (a_1 + d) = 13d$ .

Now, based on the ranges given:  $a_{15}$  can be between 241 and 250,  $a_2$  can be between 13 and 20.

So the smallest possible value of  $a_{15} - a_2$  happens when  $a_{15}$  is smallest and  $a_2$  is largest:  $241 - 20 = 221$ . The largest possible value of  $a_{15} - a_2$  happens when  $a_{15}$  is largest and  $a_2$  is smallest:  $250 - 13 = 237$ .

Hence,  $221 \leq a_{15} - a_2 \leq 237$ .

Step 4: Solve for  $d$

Since  $a_{15} - a_2 = 13d$ ,  $221 \leq 13d \leq 237$ . Divide through by 13:  $17 \leq d \leq 18.23$ . Because  $d$  is an integer,  $d$  can only be 17 or 18.

Step 5: Check each possible  $d$

Case 1:  $d = 17$  Then  $a_2 = a_1 + 17$  implies  $13 \leq a_1 + 17 \leq 20 \Rightarrow -4 \leq a_1 \leq 3$ . But since  $1 \leq a_1 \leq 10$ , the overlap gives  $1 \leq a_1 \leq 3$ .

Now check  $a_{15} = a_1 + 14(17) = a_1 + 238$ . We need  $241 \leq a_{15} \leq 250$ :  $241 \leq a_1 + 238 \leq 250 \Rightarrow 3 \leq a_1 \leq 12$ . The overlap of  $1 \leq a_1 \leq 3$  and  $3 \leq a_1 \leq 12$  is  $a_1 = 3$ .

So  $a_1 = 3, d = 17$ .

Case 2:  $d = 18$  Then  $a_2 = a_1 + 18$  implies  $13 \leq a_1 + 18 \leq 20 \Rightarrow -5 \leq a_1 \leq 2$ . Within  $1 \leq a_1 \leq 10$ , this gives  $a_1 = 1$  or  $2$ .

But  $a_{15} = a_1 + 14(18) = a_1 + 252$ , so  $a_{15} \geq 1 + 252 = 253 > 250$ , which doesn't satisfy the given range. Hence  $d = 18$  is impossible.

Step 6: Compute  $a_{14}$

With  $a_1 = 3$  and  $d = 17$ ,  $a_{14} = a_1 + 13d = 3 + 13(17) = 3 + 221 = 224$ .

Step 7: Find the sum of digits

$2 + 2 + 4 = 8$ .

(True) Final Answer:

8

### Solution 3(simplified)

We are given 15 equally spaced integers, with  $a_1$  between 1 and 10,  $a_2$  between 13 and 20, and  $a_{15}$  between 241 and 250. The integers form an arithmetic sequence, so we use the formula  $a_n = a_1 + (n - 1) \cdot d$ , where  $d$  is the common difference. From the given information, we calculate the range for  $d$  and find that  $d = 17$  works. Substituting  $a_1 = 3$  and  $d = 17$  into the formula, we find  $a_{14} = 3 + 13 \cdot 17 = 224$ . The problem asks for the sum of the digits of  $a_{14}$ , so we sum the digits of 224, which are 2, 2, and 4, giving  $2 + 2 + 4 = 8$ , and the final answer is 8.

### Solution 4 (Simple, intuitive, and fast)

This solution is similar to Solution 1, but useful if you are not familiar with inequalities. Since the integers form an arithmetic sequence, the distance from  $a_1$  to  $a_{15}$  is  $14d$ , so  $a_{15} = a_1 + 14d$ . We want  $14d$  to be the multiple of 14 closest to but not exceeding 241 (since  $a_1$  cannot be negative), and  $14 \cdot 17 = 238$  is that value, so  $d = 17$ . Because  $a_2 = a_1 + d$  must be at most 20, we get  $a_1 + 17 \leq 20$ , so  $a_1 \leq 3$ ; if  $a_1$  were 4 or larger, then  $a_2$  would exceed 20. Returning to  $a_{15} = 14d + a_1$ , we have  $a_{15} = 238 + a_1$ , and since

$a_{15} \geq 241$ , this forces  $a_1 = 3$ , giving  $a_{15} = 241$ . Subtracting 17 then gives  $a_{14} = 224$ , whose digits sum to  $2 + 2 + 4 = 8$ , so the answer is **(A) 8**.

-ChamzC19

### §39 AMC 8 2024

1. (AMC 8 2024) What is the ones digit of

$$222,222 - 22,222 - 2,222 - 222 - 22 - 2 = ?$$

- (A) 0    (B) 2    (C) 4    (D) 6    (E) 8

**Answer:** B

**Solution 1**

We can rewrite the expression as  $222,222 - (22,222 + 2,222 + 222 + 22 + 2)$ . We note that the units digit of  $22,222 + 2,222 + 222 + 22 + 2$  is 0 because all the units digits of the five numbers are 2 and  $5 \cdot 2 = 10$ , which has a units digit of 0. Now, we have something with a units digit of 0 subtracted from 222,222, and so the units digit of this expression is **(B) 2**.

**Solution 2**

$$222,222 - 22,222 = 200,000$$

$$200,000 - 2,222 = 197778$$

$$197778 - 222 = 197556$$

$$197556 - 22 = 197534$$

$$197534 - 2 = 197532$$

So our answer is **(B) 2**. (Note that brute forcing takes lots of time and the AMC8 is a timed test)

**Solution 3**

We only care about the units digits. Thus,  $2 - 2$  ends in 0,  $0 - 2$  after regrouping (10-2) ends in 8,  $8 - 2$  ends in 6,  $6 - 2$  ends in 4, and  $4 - 2$  ends in **(B) 2**.

**Solution 4**

We just take the units digit of each and subtract, adding an extra ten to the first number so we don't get a negative number:

$$(12 - 2) - (2 + 2 + 2 + 2) = 10 - 8 = \mathbf{(B) 2}$$

**Solution 5**

$$222,222 - 22,222 - 2,222 - 222 - 22 - 2 \equiv 2 - 2 - 2 - 2 - 2 \equiv -8 \equiv \mathbf{(B) 2} \pmod{10}$$

**Solution 6**

We can ignore the other digits and just do  $22 - 2 - 2 - 2 - 2 - 2$ . Because you are subtracting five 2s and  $2 \cdot 5 = 10$ , you subtract 10 from 22. This gives us 12, so the last digit is **(B) 2**.

2. (AMC 8 2024) What is the value of this expression in decimal form?

$$\frac{44}{11} + \frac{110}{44} + \frac{44}{1100}$$

- (A) 6.4      (B) 6.504      (C) 6.54      (D) 6.9      (E) 6.94

**Answer:** C

**Solution 1**

We see that  $\frac{44}{11}$  is 4;  $\frac{110}{44}$  simplifies to  $\frac{5}{2}$ , which is 2.5;

and  $\frac{44}{1100}$  simplifies to  $\frac{1}{25}$ , which is 0.04;

$4 + 2.5 + 0.04$  reveals

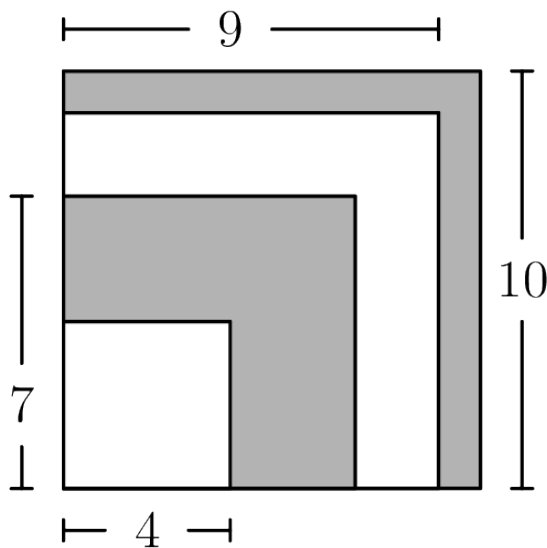
$$\frac{44}{11} + \frac{110}{44} + \frac{44}{1100}$$

is (C) 6.54.

**Solution 2 (not recommended)**

Using brute force we can make  $\frac{44}{11}$  into  $\frac{4400}{1100}$ ,  $\frac{110}{44}$  into  $\frac{2750}{1100}$ , and  $\frac{44}{1100}$ , and then we add and get  $\frac{7194}{1100}$  and this simplifies to (C) 6.54

3. (AMC 8 2024) Four squares of side length 4, 7, 9, and 10 are arranged in increasing size order so that their left edges and bottom edges align. The squares alternate in color white-gray-white-gray, respectively, as shown in the figure. What is the area of the visible gray region in square units?



- (A) 42      (B) 45      (C) 49      (D) 50      (E) 52

**Answer:** E

**Solution 1**

We work inwards. The area of the outer shaded square is the area of the whole square minus the area of the second largest square. The area of the inner shaded region is the area of the third largest square minus the area of the smallest square. The sum of these areas is

$$10^2 - 9^2 + 7^2 - 4^2 = 100 - 81 + 49 - 16 = 19 + 33 = \text{ (E) 52 }$$

This problem appears multiple times in various math competitions including the AMC and MATHCOUNTS.

-Benedict (countmath1) Nivaar and anabel.disher

### Solution 2

In solution 1, we can use **Difference of squares** to get the answer, rather than calculating the squares of the numbers:

$$10^2 - 9^2 + 7^2 - 4^2 = (10-9)(10+9) + (7-4)(7+4) = 1 \cdot 19 + 3 \cdot 11 = 19 + 33 = \boxed{\text{(E)} 52}$$

minor edits by Sowthistle2011 :)

### Solution 3

Solve for the areas for each square inside the larger one, and then subtract the areas of the white regions from the areas of each square:

$$(10^2 + 7^2) - (9^2 + 4^2) = 149 - 97 = \boxed{\text{(E)} 52}$$

-MrTechguyPCMT and anabel.disher

### Solution 4

We can calculate it as the sum of the areas of 2 smaller trapezoids and 2 larger trapezoids.

$$2 \left( \frac{(7+4)(7-4)}{2} \right) + 2 \left( \frac{(10+9)(10-9)}{2} \right) = 10^2 - 9^2 + 7^2 - 4^2 = 19 + 33 = \boxed{\text{(E)} 52}$$

-SahanWijetunga

4. (*AMC 8 2024*) When Yunji added all the integers from 1 to 9, she mistakenly left out a number. Her incorrect sum turned out to be a square number. What number did Yunji leave out?

(A) 5      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** E

### Solution

basically the one below i guess just explained a lil different. so yes we add 1 - 9 to get 45. then we see what's the closet square, which is obviously 36 (i hope we have our squares memorized by now) and then we do 45 minus 36 to get 9  
 $\boxed{\text{(E)} 9}$ . pereena

Adding numbers 1 through 9 gives us 45. This was her expected sum, but what she got was a perfect square. Since she got that perfect square sum by forgetting a number, that sum is less than 45. The square number right under 45 is 36. So  $45 - 36 = 9$ . So the solution is  $\boxed{\text{(E)} 9}$ .

5. (*AMC 8 2024*) Aaliyah rolls 2 standard 6-sided dice. She notices that the product of the two numbers rolled is a multiple of 6. Which of the following integers cannot be the sum of the two numbers rolled?

(A) 5      (B) 6      (C) 7      (D) 8      (E) 9

**Answer:** B

**Solution 1**

First, figure out all pairs of numbers whose product is 6. Then, using the process of elimination, we can find the following:

(A) is possible:  $2 \times 3$  (C) is possible:  $1 \times 6$  (D) is possible:  $2 \times 6$

The only integer that cannot be the sum is **(B) 6**.

We can prove that B is correct because the only ordered pairs that add to 6 is (1, 5), (5, 1), (2, 4), (4, 2), and (3, 3), and these multiply to 5, 8, and 9.

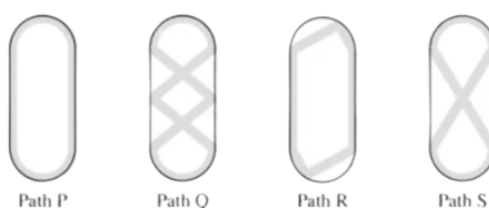
Slight changes VersatileGopher38

**Solution 2**

First, see that in order for the two numbers to have a product of 6, it must either have 6 as a factor or have factors that have a product of 6 (Notably  $2 \times 3$ ).

When we look at the answer choices, we see that answers C, D, E all are more than 6, so they can all be added as a sum of 6. Answer A works because  $2 + 3 = 5$ , so 6 is the only answer choice that doesn't work. Thus, the answer is **(B) 6**.

6. (AMC 8 2024) Sergei skated around an ice rink, gliding along different paths. The gray lines in the figures below show four of the paths labeled P, Q, R, and S. What is the sorted order of the four paths from shortest to longest?



- (A) P, Q, R, S    (B) P, R, S, Q    (C) Q, S, P, R    (D) R, P, S, Q    (E) R, S, P, Q

**Answer: D**

**Solution 1**

You can measure the lengths of the paths until you find a couple of guaranteed true inferred statements as such: Q is greater than S, P is greater than R, and R and P are the smallest two, therefore the order is R, P, S, Q. Thus we get the answer **(D) R, P, S, Q**.

- U-King

**Solution 2 (Intuitive)**

Obviously Path Q is the longest path, followed by Path S.

So, it is down to Paths P and R.

Notice that curved lines are always longer than the straight ones that meet their endpoints, therefore Path P is longer than Path R.

Thus, the order from shortest to longest is **(D) R, P, S, Q**.

**Solution 3 (Indepth version of solution 1)**

We can compare paths that look similar, P with R and Q with S.

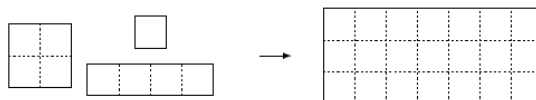
$R$  is shorter than  $P$  because it replaces curved lines in  $P$  with straight lines, and straight lines are always the shortest distance between two points.

The two straight lines in  $S$  can be seen as paths going diagonally straight across the rink, and the straight lines in  $Q$  can be seen as initially going diagonally across the rink, before taking a short detour and then finishing its path. Since a straight line between two points is shorter than zig-zagging lines, path  $S$  is shorter than path  $Q$ .

The only difference between path  $P$  and path  $S$  is the straight lines, with  $P$  going up and down while  $S$  goes diagonally. A right triangle can be drawn from a line from  $P$  and a line from  $S$ . We can see that the straight lines in  $P$  act as the leg of the right triangle while the hypotenuse is the straight lines from  $S$ . The legs of a right triangle are always less than the hypotenuse, so  $P$  is shorter than  $S$ .

Putting together these inequality statements, the answer is **(D)**  $R, P, S, Q$ .

7. (AMC 8 2024) A  $3 \times 7$  rectangle is covered without overlap by 3 shapes of tiles:  $2 \times 2$ ,  $1 \times 4$ , and  $1 \times 1$ , shown below. What is the minimum possible number of  $1 \times 1$  tiles used?



- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** E

### Solution 1

We can eliminate B, C, and D, because they are not 21 subtracted by any multiple of 4. Finally, we see that there is no way to have A, so the solution is **(E) 5**.

### Solution 2

Let  $x$  be the number of  $1 \times 1$  tiles. There are 21 squares and each  $2 \times 2$  or  $1 \times 4$  tile takes up 4 squares, so  $x \equiv 1 \pmod{4}$ , so it is either 1 or 5. Color the columns, starting with red, then blue, and alternating colors, ending with a red column. There are 12 red squares and 9 blue squares, but each  $2 \times 2$  and  $1 \times 4$  shape takes up an equal number of blue and red squares, so there must be 3 more  $1 \times 1$  tiles on red squares than on blue squares, which is impossible if there is just one, so the answer is **(E) 5**, which can easily be confirmed to work.

### Solution 3

Suppose there are  $a$  different  $2 \times 2$  tiles,  $b$  different  $4 \times 1$  tiles and  $c$  different  $1 \times 1$  tiles. Since the areas of these tiles must total up to 21 (area of the whole grid), we have

$$4a + 4b + c = 21.$$

Reducing modulo 4 gives  $c \equiv 1 \pmod{4}$ , or  $c = 1$  or  $c = 5$ .

If  $c = 1$ , then  $a + b = 5$ . After some testing, there is no valid pair  $(a, b)$  that works, so the answer must be **(E) 5**, which can be constructed in many ways.

-Benedict T (countmath1)

### Solution 4

Since the rectangle is a  $3 \times 7$  grid, its total area (or the number of  $1 \times 1$  squares) is 21. To minimize the number of  $1 \times 1$  squares used, we begin by filling the area with larger shapes. Specifically, we can use two  $2 \times 2$  squares, each covering 4 squares, and two  $1 \times 4$  rectangles, each also covering 4 squares. This gives us a total of  $4 + 4 + 4 + 4 = 16$  squares filled by larger shapes. After filling 16 squares, we have 5 squares left. We can then fill in 5 additional  $1 \times 1$  squares, leaving us with a total of 5  $1 \times 1$  squares.

Thus, the least number of  $1 \times 1$  squares required is **(E) 5**. -AADHYA2012

8. (AMC 8 2024) On Monday, Taye has \$2. Every day, he either gains \$3 or doubles the amount of money he had on the previous day. How many different dollar amounts could Taye have on Thursday, 3 days later?

(A) 3      (B) 4      (C) 5      (D) 6      (E) 7

**Answer:** D

**Solution 1**

How many dollar values could be on the first day? Only 2 dollars. The second day, you can either add 3 dollars, or double, so you can have 5 dollars, or 4. For each of these values, you have 2 values for each. For 5 dollars, you have 10 dollars or 8, and for 4 dollars, you have 8 dollars or 7 dollars. Now, you have 2 values for each of these. For 10 dollars, you have 13 dollars or 20, for 8 dollars, you have 16 dollars or 11, for 8 dollars, you have 16 dollars or 11, and for 7 dollars, you have 14 dollars or 10.

On the final day, there are 11, 11, 16, and 16 repeating, leaving you with  $8 - 2 =$  **(D) 6** different values.

**Solution 2**

Continue as in Solution 1 to get 7, 8, or 10 dollars by the 2nd day. The only way to get the same dollar amount occurring twice by branching (multiply by 2 or adding 3) from here is if  $7 + 3 = 10 \cdot 2$  or  $7 + 3 = 8 \cdot 2$  which both aren't true. Hence our answer is  $3 \cdot 2 =$  **(D) 6**.

**Solution 3 (Brute Force)**

Let  $a$  represent Taye adding 3 dollars and  $d$  represent Taye doubling his money. We can list out all the possibilities:  $a \Rightarrow a \Rightarrow a = 11$ ,  $a \Rightarrow a \Rightarrow d = 16$ ,  $a \Rightarrow d \Rightarrow a = 12$ ,  $a \Rightarrow d \Rightarrow d = 20$ ,  $d \Rightarrow a \Rightarrow a = 10$ ,  $d \Rightarrow a \Rightarrow d = 14$ ,  $d \Rightarrow d \Rightarrow a = 11$ ,  $d \Rightarrow d \Rightarrow d = 16$ . The unique values are 10, 11, 12, 14, 16, 20. Thus, our desired answer is **(D) 6**.

9. (AMC 8 2024) All the marbles in Maria's collection are red, green, or blue. Maria has half as many red marbles as green marbles and twice as many blue marbles as green marbles. Which of the following could be the total number of marbles in Maria's collection?

(A) 24      (B) 25      (C) 26      (D) 27      (E) 28

**Answer:** E

**Solution 1**

Since Maria has half as many red marbles as green, we can call the number of red marbles  $x$ , and the number of green marbles  $2x$ . Since she has half as many green marbles as blue, we can call the number of blue marbles  $4x$ . Adding them up, we



have:  $7x$  marbles. The number of marbles therefore must be a multiple of 7, as  $x$  represents an integer, so the only possible answer is **(E) 28**.

Minor Edit - Runningmouse

### Solution 2

Suppose Maria has  $g$  green marbles and  $t$  total marbles. She then has  $\frac{g}{2}$  red marbles and  $2g$  blue marbles. Altogether, Maria has

$$g + \frac{g}{2} + 2g = \frac{7g}{2} = t$$

marbles, so  $g = \frac{2t}{7}$ , so  $t$  must be a multiple of 7. The only multiple of 7 in the answer choices is **(E) 28**.

-Benedict T (countmath1) and anabel.disher

### Solution 3

This solution is similar to Solution 1. The most simplest and effective way to solve this problem is equations. We have the following equations:

$2r = g$  and  $2g = b$  We essentially want to find out the total number of mables in Maria's collection or in equation terms  $(r + g + b)$ . Converting that equation and substituting the values we get  $r + 2r + 4r = 7r$  Now, we know the total is a multiple of 7 leaving our answer as **(E) 28**. -AADHYA2012

10. (AMC 8 2024) In January 1980 the Moana Loa Observation recorded carbon dioxide ( $CO_2$ ) levels of 338 ppm (parts per million). Over the years the average  $CO_2$  reading has increased by about 1.515 ppm each year. What is the expected  $CO_2$  level in ppm in January 2030? Round your answer to the nearest integer.
- (A) 399      (B) 414      (C) 420      (D) 444      (E) 459

**Answer:** B

### Solution 1

This is a time period of  $2030 - 1980 = 50$  years, so we can expect the ppm to increase by  $50 * 1.515 = 75.75 \approx 76$  ppm. Since we started with 338 ppm, we have  $76 + 338 =$  **(B) 414**.

-ILoveMath31415926535

### Solution 2

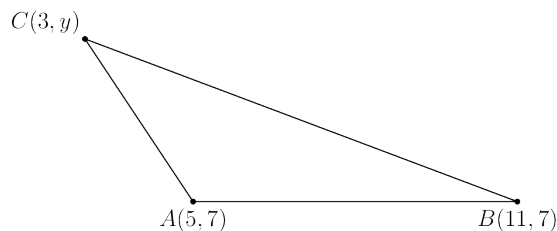
$2030 - 1980 = 50$  years. The ppm level in 2030 is  $338 + 50 * 1.515 = 338 + 75.75 = 413.75 \approx$  **(B) 414**.

### Solution 3

From Jan 1980 = 338 ppm increase  
 $= 1.515$  ppm increased / year  
 $= 50$  yrs (in this case)  $= 1.515 \times 50$   
 $= 75.75 \approx 76$

Thus, 2030 prediction  $= 338 + 76 = 414 =$  option B

11. (AMC 8 2024) The coordinates of  $\triangle ABC$  are  $A(5, 7)$ ,  $B(11, 7)$ , and  $C(3, y)$ , with  $y > 7$ . The area of  $\triangle ABC$  is 12. What is the value of  $y$ ?



- (A) 8      (B) 9      (C) 10      (D) 11      (E) 12

**Answer:** D

**Solution 1**

Since the triangle has a base of 6, we can plug in that value as the base. Then, we can solve the equation for the height. Doing so gives us,

$$\frac{6h}{2} = 3h = 12.$$

This means that  $h = 4$ , so that means that we have to add 4 to the  $y$ -coordinate. So the answer is  $7 + 4 = \boxed{(D)11}$ .

**Solution 2**

By the Shoelace Theorem,  $\triangle ABC$  has area

$$\frac{1}{2} |(y \cdot 11 + 7 \cdot 5 + 7 \cdot 3) - (3 \cdot 7 + 11 \cdot 7 + 5 \cdot y)| = \frac{1}{2} |(11y + 56) - (98 + 5y)| = \frac{1}{2} |6y - 42|.$$

From the problem, this is equal to 12. We now solve for  $y$ .

$$\frac{1}{2} |6y - 42| = 12 \quad |6y - 42| = 24 \quad 6y - 42 = 24 \text{ OR } 6y - 42 = -24 \quad 6y = 66 \text{ OR } 6y = 18 \\ y = 11 \text{ OR } y = 3$$

However, since, as stated in the problem,  $y > 7$ , our only valid solution is  $\boxed{(D)11}$ .

**Solution 3**

As in the figure, the triangle is determined by the vectors  $\begin{bmatrix} -2 \\ y-7 \end{bmatrix}$  and  $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ . Recall that the absolute value of the determinant of these vectors is the area of the parallelogram determined by those vectors; the triangle has half the area of that parallelogram. Then we must have that  $\frac{1}{2} \begin{vmatrix} -2 & y-7 \\ 6 & 0 \end{vmatrix} = 12 \implies \begin{vmatrix} -2 & y-7 \\ 6 & 0 \end{vmatrix} = \pm 24$ . Expanding the determinants, we find that  $-6(y-7) = 24$  or  $-6(y-7) = -24$ . Solving each equation individually, we find that  $y = 3$  or  $y = 11$ . However, the problem states that  $y > 7$ , so the only valid solution is  $\boxed{(D)11}$ .

**Solution 4**

Draw a rectangle so that the hypotenuse is the diagonal of the rectangle. The base is 8 and the height is  $y - 7$ , so the area is  $8(y - 7) = 8y - 56$ .

Inside the rectangle, there are 2 extra right triangles along with the original triangle. The larger right triangle has an area of half the rectangle, so it has an area of

$$\frac{8y-56}{2} = 4y - 28$$

The smaller right triangle has a base of 2 and a height of  $y - 7$ , so its area is

$$\frac{2(y-7)}{2} = y - 7$$

Subtracting the extra right triangles from the area of the rectangle, you get

$$(8y - 56) - (4y - 28) - (y - 7) = 3y - 21$$

Since the problem told us that the original triangle had an area of 12, you get the equation

$$3y - 21 = 12 \quad 3y = 33 \quad y = 11$$

So the answer to the problem is **D**.

### Solution 5

If the area of  $\triangle ABC$  is 12, then the base times height must be equal to 24. We can see that the base is equal to 6, subtracting the smaller  $x$  value from the larger  $x$  value of the two coordinates. So the change of the  $y$  value from point  $A$  to point  $C$  must be 4. Adding 4 to point  $A$ 's  $y$  value gives us 11, so the answer is  $C$ .

- mathdragonmochi

12. (AMC 8 2024) Rohan keeps a total of 90 guppies in 4 fish tanks.

- There is 1 more guppy in the 2nd tank than the 1st tank.
- There are 2 more guppies in the 3rd tank than the 2nd tank.
- There are 3 more guppies in the 4th tank than the 3rd tank.

How many guppies are in the 4th tank?

(A) 20      (B) 21      (C) 23      (D) 24      (E) 26

**Answer:** E

### Solution 1

Let  $x$  denote the number of guppies in the first tank.

Then, we have the following for the number of guppies in the rest of the tanks:

- The number of guppies in the second tank is  $x + 1$
- The number of guppies in the third tank is  $x + 1 + 2$
- The number of guppies in the fourth tank is  $x + 1 + 2 + 3$

The number of guppies in all of the tanks combined is 90, so we can write the equation

$$(x) + (x + 1) + (x + 1 + 2) + (x + 1 + 2 + 3) = 90.$$

Simplifying the equation gives

$$4x + 10 = 90.$$

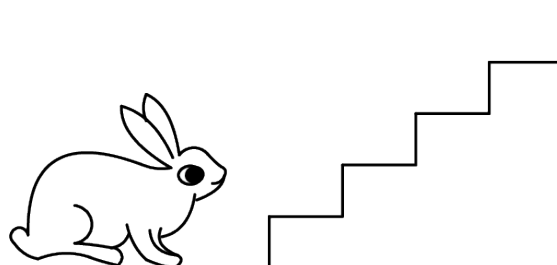
Solving the resulting equation gives  $x = 20$ , so the number of guppies in the fourth tank is  $20 + 1 + 2 + 3 = \boxed{\text{(E)} 26}$ .

### Solution 2

Suppose there are no guppies in the first tank. Then, the number of guppies in the other tanks are 1, 3, and 6, or 10 guppies in total. We need to add  $90 - 10 = 80$

guppies into 4 tanks or 20 guppies in each tank, so the number of guppies in the fourth tank is  $20 + 6 = \boxed{\text{(E)} 26}$ .

13. (AMC 8 2024) Buzz Bunny is hopping up and down a set of stairs, one step at a time. In how many ways can Buzz Bunny start on the ground, make a sequence of 6 hops, and end up back on the ground? (For example, one sequence of hops is up-up-down-down-up-down.)



- (A) 4      (B) 5      (C) 6      (D) 8      (E) 12

**Answer:** B

**Solution 1**

Looking at the answer choices, you see that you can list them out. Doing this gets you:

*UUDDUD UDUDUD UUUDDD UDUUDD UUDUDD*

Counting all the paths listed above gets you  $\boxed{\text{(B)} 5}$ .

**Solution 2**

Any combination can be written as some re-arrangement of *UUUDDD*. Clearly we must end going down, and start going up, so we need the number of ways to insert 2 *U*'s and 2 *D*'s into *U* \_\_\_\_ *D*. There are  $\binom{4}{2} = 6$  ways, but we have to remove the case *UDDUUD*, giving us  $\boxed{\text{(B)} 5}$ .

- We know there are no more cases since there will be at least one *U* before we have a *D* (from the first *U*), at least two *U*'s before two *D*'s (since we removed the one case), and at least three *U*'s before three *D*'s, as we end with the third *D*.

with minor edits from shaopl2022

**Solution 3 (pretty good, but needs more explanation on the catalan numbers)**

These numbers are clearly the Catalan numbers. Since we have 6 steps, we need the third Catalan number, which is  $\boxed{\text{(B)} 5}$ . andliu766

with minor edits from shaopl2022

**Solution 4 (slightly confusing, but it was how i did it too)**

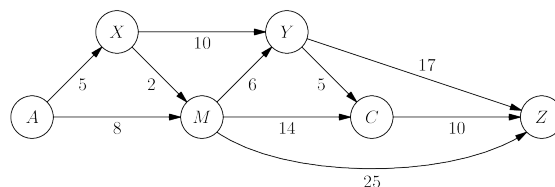
The first step must be a *U*, last step must be a *D*. After the third step we can get only position 3 or position 1. In the first case there is only one way: *UUUDDD*. In the second case we have two way to get this position *UDU* and *UUD*. Similarly, we have two way return to position 0 (*UDD* and *DUD*). Therefore, we have  $1 + 2 \cdot 2 = \boxed{\text{(B)} 5}$ .

with minor edits from shaopl2022

**Solution 5**

Realize that 2 of the solutions are  $UDUDUD$ , and  $UUUDDDD$  which both consist of 3  $U$ 's and 3  $D$ 's. Then, there are five gaps where the  $U$ 's can go in the  $D$ 's so the answer is **(B) 5**.

14. (AMC 8 2024) What is the distance of the shortest route from A to Z?



- (A) 28      (B) 29      (C) 30      (D) 31      (E) 32

**Answer:** A

### Solution 1

We can simply see that path  $A \rightarrow X \rightarrow M \rightarrow Y \rightarrow C \rightarrow Z$  will give us the smallest value. Adding,  $5 + 2 + 6 + 5 + 10 = \boxed{28}$ . This is nice as it's also the smallest value, solidifying our answer.

You can also simply brute-force it or sort of think ahead - for example, getting from A to M can be done 2 ways;  $A \rightarrow X \rightarrow M$  ( $5 + 2$ ) or  $A \rightarrow M$  (8), so you should take the shorter route ( $5 + 2$ ). Another example is M to C, two ways - one is  $6 + 5$  and the other is 14. Take the shorter route. After this, you need to consider a few more times - consider if  $5 + 10$  ( $Y \rightarrow C \rightarrow Z$ ) is greater than 17 ( $Y \rightarrow Z$ ), which it is not, and consider if 25 ( $M \rightarrow Z$ ) is greater than  $14 + 10$  ( $M \rightarrow C \rightarrow Z$ ) or  $6 + 5 + 10$  ( $M \rightarrow Y \rightarrow C \rightarrow Z$ ) which it is not. TLDR:  $5 + 2 + 6 + 5 + 10 = \boxed{28}$ . [Note: This is probably just the thinking behind the solution.] Double-note: As MaxyMoosy said, since this answer is the smallest one, it has to be the right answer.

### Solution 2 (Advanced)

We can execute **Dijkstra's algorithm** by hand to find the shortest path from A to every other town, including Z. This effectively proves that, assuming we execute the algorithm correctly, that we will have found the shortest distance. The distance estimates for each step of the algorithm (from A to each node) are shown below:

| Current node | A | M | C        | X | Y        | Z         |
|--------------|---|---|----------|---|----------|-----------|
| A            | 0 | 8 | $\infty$ | 5 | $\infty$ | $\infty$  |
| X            | 0 | 7 | $\infty$ | 5 | 15       | $\infty$  |
| M            | 0 | 7 | 21       | 5 | 13       | 32        |
| Y            | 0 | 7 | 18       | 5 | 13       | 30        |
| C            | 0 | 7 | 18       | 5 | 13       | 28        |
| Z            | 0 | 7 | 18       | 5 | 13       | <b>28</b> |

The steps are as follows: starting with the initial node A, set  $d(A) = 0$  and  $d(v) = \infty$  for all  $v \in \{M, C, X, Y, Z\}$  where  $d(v)$  indicates the distance from A to  $v$ . Consider the outgoing edges  $(A, X)$  and  $(A, M)$  and update the distance estimates  $d(X) = 5$  and  $d(M) = 8$ , completing the first row of the table.

The node X is the unvisited node with the lowest distance estimate, so we will consider X and its outgoing edges  $(X, Y)$  and  $(X, M)$ . The distance estimate  $d(Y)$  equals  $d(X) + 10 = 15$ , and the distance estimate  $d(M)$  updates to  $d(X) + 2 = 7$ ,

because  $7 < 8$ . This completes the second row of the table. Repeating this process for each unvisited node (in order of its distance estimate) yields the correct distance  $d(Z) = \boxed{\text{(A)} 28}$  once the algorithm is complete.

**Solution 3 (Variant of Solution 1)**

From  $A$ , we want to find the shortest route to  $Z$ , so we want to try to find the shortest path through each node (not necessarily all of them). We should follow the arrows, since all of them lead to  $Z$ . From  $A$ , there are 2 paths we can take, to  $M$  (8), or to  $X$  (5). We travel to  $X$ , since  $5 < 8$ . From  $X$ , we go to  $M$  ( $2 < 10$ ), we go to  $Y$  ( $6 < 14 < 25$ ), we go to  $C$ , ( $5 < 17$ ) and finally go to  $Z$ . Adding up gives  $5 + 2 + 6 + 5 + 10 = 28 = \boxed{\text{(A)}}$ .

**Solution 4 (On paper)**

We can cross off a few routes:

- $AM$  can be crossed off because  $AXM$  is shorter
- $XY$  can be crossed off because  $XMY$  is shorter
- $YZ$  can be crossed off because  $YCZ$  is shorter
- $MC$  can be crossed off because  $MYC$  is shorter
- $MZ$  can be crossed off because  $MCZ$  is shorter

Finally, we are left with a single path  $AXMYCZ$  from  $A$  to  $Z$  which adding it up gives  $28 = \boxed{\text{(A)}}$ .

15. (AMC 8 2024) Let the letters  $F, L, Y, B, U, G$  represent distinct digits. Suppose  $\overline{F L Y F L Y}$  is the GREATEST number that satisfies the equation

$$8 \cdot \overline{F L Y F L Y} = \overline{B U G B U G}.$$

What is the value of  $\overline{F L Y} + \overline{B U G}$ ?

- (A) 1089      (B) 1098      (C) 1107      (D) 1116      (E) 1125

**Answer:** C

**Solution 1**

Let  $\overline{FLY}$  be the 3-digit number  $100F + 10L + Y$ .

Repeating a 3-digit block twice gives

$$\overline{FLYFLY} = 1000 \cdot \overline{FLY} + \overline{FLY} = 1001 \cdot \overline{FLY}.$$

Similarly,

$$\overline{BUGBUG} = 1001 \cdot \overline{BUG}.$$

Given  $8 \cdot \overline{FLYFLY} = \overline{BUGBUG}$ , substitute:

$$8(1001\overline{FLY}) = 1001\overline{BUG}.$$

Divide by 1001:

$$8\overline{FLY} = \overline{BUG}.$$

Since  $\overline{BUG}$  is a 3-digit number,

$$100 \leq 8\overline{FLY} \leq 999 \quad \Rightarrow \quad 13 \leq \overline{FLY} \leq 124.$$

To maximize  $\overline{FLYFLY}$ , maximize  $\overline{FLY}$ .

Try the largest possible value:

$$\overline{FLY} = 123 \Rightarrow \overline{BUG} = 8 \cdot 123 = 984.$$

The digits  $\{1, 2, 3, 9, 8, 4\}$  are all distinct, so this works and is maximal.

Therefore,

$$\overline{FLY} + \overline{BUG} = 123 + 984 = 1107.$$

1107

### Solution 2

The divisibility rule for 8 is that the last 3 digits need to be divisible by 8. That means that  $BUG$  needs to be a multiple of 8 with distinct digits. And since multiplying 8 by  $FLY$  gives  $BUG$ , we know that the number needs to be a product of 8 and a 3 digit number. From the more foundational multiplication facts, we know that  $125 \cdot 8 = 1000$ . Moving down from there, we can try  $124 \cdot 8 = 992$ . This won't work because 992 does not have distinct digits. Next, we can try  $123 \cdot 8 = 984$ . 1, 2, 3, 9, 8, and 4 are all distinct digits. That means that  $FLY = 123$  and  $BUG = 984$ .  $123 + 984 = 1107$ , so the answer is C. Note that while guessing and checking multiples of 8 sounds like an unreliable strategy, we know that the number needs to be  $8 \cdot FLYFLY$ , which would for certain be a large number.

-mathdragonmochi

16. (AMC 8 2024) Minh enters the numbers 1 through 81 into the cells of a  $9 \times 9$  grid in some order. She calculates the product of the numbers in each row and column. What is the least number of rows and columns that could have a product divisible by 3?

(A) 8      (B) 9      (C) 10      (D) 11      (E) 12

**Answer:** D

### Solution 1

Note you can swap/rotate any configuration of rows, such that all the rows and columns that have a product of 3 are in the top left. Hence the points are bounded by a  $a \times b$  rectangle. This has  $ab$  area and  $a + b$  rows and columns divisible by 3. We want  $ab \geq 27$  and  $a + b$  minimized.

If  $ab = 27$ , we achieve minimum with  $a + b = 9 + 3 = 12$ .

If  $ab = 28$ , our best is  $a + b = 7 + 4 = 11$ . Note if  $a + b = 10$ ,  $ab = 25$ . Because  $25 < 27$ , there is no smaller answer, and we get (D)11.

- SahanWijetunga vockey(minor edits) phy6(minor edits)

### Solution 2

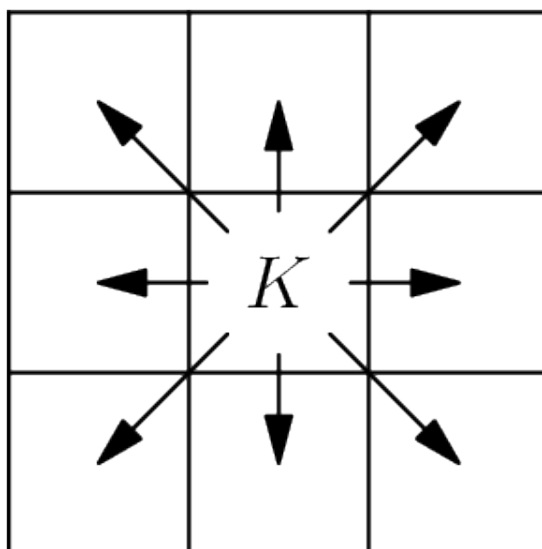
For a row or column to have a product divisible by 3, there must be a multiple of 3 in the row or column. To create the least amount of rows and columns with multiples of 3, we must find a way to keep them all together, to minimize the total number of rows and columns with multiples of threes in it. From 1 to 81, there are 27 multiples of 3 ( $81/3 = 27$ ). So we have to fill 27 cells with numbers that are multiples of 3. If we put 25 of these numbers in a  $5 \times 5$  grid, there would be 5 rows

and 5 columns (10 in total), with products divisible by 3. However, we have 27 numbers, so 2 numbers still need to be put in the  $9 \times 9$  grid. If we put both numbers in the 6th column, but one in the first row, and one in the second row, (next to the 5 by 5 grid already filled), we would have a total of 6 columns now, and still 5 rows with products that are multiples of 3. Since  $6 + 5 = 11$ , the answer is **(D)11**

### Solution 3

In the numbers 1 to 81, there are 27 multiples of three. In order to minimize the rows and columns, the best way is to make a square. However, the closest square is 25, meaning there are two multiples of three remaining. However, you can place these multiples right above the  $5 \times 5$  square, meaning the answer is **(D)11** eJust be better Shriyans Chowdhury (minor configuration error)

17. (AMC 8 2024) A chess king is said to *attack* all squares one step away from it (basically any square right next to it in any direction), horizontally, vertically, or diagonally. For instance, a king on the center square of a  $3 \times 3$  grid attacks all 8 other squares, as shown below. Suppose a white king and a black king are placed on different squares of  $3 \times 3$  grid so that they do not attack each other. In how many ways can this be done?



- (A) 20      (B) 24      (C) 27      (D) 28      (E) 32

**Answer:** E

### Solution 1

If you place a king in any of the 4 corners, the other king will have 5 spots to go and there are 4 corners, so  $5 \times 4 = 20$ . If you place a king in any of the 4 edges, the other king will have 3 spots to go and there are 4 edges so  $3 \times 4 = 12$ . That gives us  $20 + 12 = 32$  spots for the other king to go into in total. So **(E)32** is the answer. andliu766, captaindiamond868 (minor edits) AliceDubbleYou

### Solution 2

We see that the center is not a viable spot for either of the kings to be in, as it would attack all nearby squares.

This gives three combinations:



Corner-corner: There are 4 corners, and none of them are touching orthogonally or diagonally, so it's  $\binom{4}{2} = 6$

Corner-edge: For each corner, there are two edges that don't border it,  $4 \cdot 2 = 8$

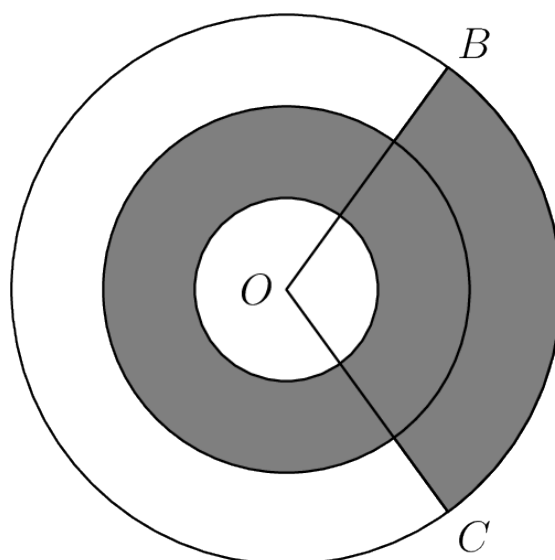
Edge-edge: The only possible combinations of this that work are top-bottom and left-right edges, so 2 for this type

$$6 + 8 + 2 = 16$$

### Solution 3 (Complementary Counting)

We see that the center is off-limits for the kings. Additionally, the kings cannot be next to each other. There are 24 ways for the kings to be placed adjacently: 16 corner-edge permutations and 8 edge-edge permutations. In total, there are  $8P2 = 56$  ways to arrange the kings conditions aside. Therefore, our answer is  $56 - 24 = \boxed{\text{(E)}32}$

18. (AMC 8 2024) Three concentric circles centered at  $O$  have radii of 1, 2, and 3. Points  $B$  and  $C$  lie on the largest circle. The region between the two smaller circles is shaded, as is the portion of the region between the two larger circles bounded by central angles  $BOC$ , as shown in the figure below. Suppose the shaded and unshaded regions are equal in area. What is the measure of  $\angle BOC$  in degrees?



- (A) 108    (B) 120    (C) 135    (D) 144    (E) 150

**Answer:** A

### Solution 1

Let  $x = \angle BOC$ .

We see that the shaded region is the inner ring plus a sector  $x^\circ$  of the outer ring. Using the formula for the area of a circle ( $A = \pi r^2$ ), we find that the area of  $x$  is  $(4\pi - \pi) + \frac{x}{360}(9\pi - 4\pi)$ . This simplifies to  $3\pi + \frac{x}{360}(5\pi)$ .

The unshaded portion is comprised of the smallest circle plus the sector  $(360 - x)^\circ$  of the outer ring, which evaluates to  $\pi + \frac{360-x}{360}(5\pi)$ .

We are told these are equal. Therefore,  $3\pi + \frac{x}{360}(5\pi) = \pi + \frac{360-x}{360}(5\pi)$ . Solving for  $x$  reveals  $x = \boxed{\text{(A)} 108}$ .

### Solution 2

Notice that for the 3rd most outer ring of the circle, the ratio of the shaded region to non-shaded region is the ratio of  $\angle BOC$  to  $360 - \angle BOC$ . With that, all we need to do is solve for the shaded region.

The inner most circle has radius 1, and the second circle has radius 2. Therefore, the first shaded area has  $4\pi - \pi = 3\pi$  area. The circle has total area  $9\pi$ , so the other shaded region must have  $1.5\pi$  area, as the non-shaded and shaded area is equivalent. So for the 3rd outer ring, the total area is  $9\pi - 4\pi = 5\pi$ , so the non-shaded part of the outer ring is  $5\pi - 1.5\pi = 3.5\pi$ .

Now as said before, the ratio of these two areas is the ratio of  $\angle BOC$  and  $360 - \angle BOC$ . So,  $\frac{3.5}{1.5} = \frac{7}{3}$ . We have  $7x : 3x$  where  $7x + 3x = 360$ ,  $x = 36$ , so our answer is  $3x = 108$ , (A)108.

### Solution 3 (Ruler Tool)

The AMC 8 test allows a ruler tool that you can rotate and drag. You can use the tool to make a straight segment (which we know is 180 degrees), and we let the angle of desire be  $x$ . We can estimate that  $180 - x$  is just about 30 degrees short of  $x$  itself, so  $x - 30 = 180 - x$ , solving gives  $x = 105$ , therefore the closest answer is (A) 108.

- Note: This isn't the most accurate method to use on AMC 8s, this is just a quick method if time is short or you do not know how to do the problem and want to guess at it with your best shot.

### Solution 4

Suppose the desired angle is some fraction  $x$  of the total degree measure of the circle. We now compile a list of the shaded and unshaded areas. The inner circle of radius 1 is completely unshaded, so it contributes 1 to the unshaded area. (Everything will be a multiple of  $\pi$ , so we omit it.) The inner annulus has area  $2^2 - 1^2 = 3$ , which it contributes to the shaded area. The outer annulus has a total area of  $3^2 - 2^2 = 5$ ; the fraction  $x$  is shaded, so the shaded portion of the outer annulus contributes  $5x$  to the shaded area, while the other  $1 - x$  fraction is unshaded, so the unshaded portion contributes  $5(1 - x)$  to the unshaded area. We now equate and solve.

$$1 + 5(1 - x) = 3 + 5x$$

Upon solving, we find that  $x = \frac{3}{10}$ , so the degree measure is  $360 \cdot \frac{3}{10} = \span style="border: 1px solid black; padding: 0 2px;">(A)108.$

### Solution 5

Let  $\theta = \angle BOC$ ,  $A_1$  be the area of the inner annulus,  $A_2$  be the area of the outer annulus sector  $BOC$  and  $A_s$  the total area of the shaded regions. Then,

$$A_1 + A_2 = A_s$$

$A_s$  is half the total area so  $A_s = \frac{\pi \cdot 3^2}{2} = \frac{9\pi}{2}$  The area of an annulus with outer radius  $R$  and inner radius  $r$  is given by:

$$A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$$

So the area of the inner annulus is  $A_1 = \pi(2^2 - 1^2) = 3\pi$  The area of the outer annulus sector is

$$A_2 = \frac{\theta}{360^\circ} \times \pi(3^2 - 2^2) = \frac{\theta}{360^\circ} \times 5\pi$$

Then we have:

$$3\pi + \frac{\theta}{360^\circ} \times 5\pi = \frac{9\pi}{2}$$

Solving for  $\theta$ :

$$\frac{\theta}{360^\circ} \times 5\pi = \frac{3\pi}{2}$$

$$\theta = \frac{3\pi}{2} \times \frac{360^\circ}{5\pi}$$

$$\theta = \frac{3 \times 360^\circ}{10}$$

So our answer is  $\theta = 3 \times 36^\circ$ , (A)108

19. (AMC 8 2024) Jordan owns 15 pairs of sneakers. Three fifths of the pairs are red and the rest are white. Two thirds of the pairs are high-top and the rest are low-top. The red high-top sneakers make up a fraction of the collection. What is the least possible value of this fraction?



- (A) 0      (B)  $\frac{1}{5}$       (C)  $\frac{4}{15}$       (D)  $\frac{1}{3}$       (E)  $\frac{2}{5}$

**Answer:** C

**Solution 1**

Jordan has 10 high top sneakers, and 6 white sneakers. We would want as many white high-top sneakers as possible, so we set 6 high-top sneakers to be white. Then, we have  $10 - 6 = 4$  red high-top sneakers, so the answer is  $\frac{4}{15}$ .

**Solution 2**

We first start by finding the number of red and white sneakers.  $\frac{3}{5} \times 15 = 9$  red sneakers, so 6 are white. Then  $\frac{2}{3} \times 15 = 10$  are high top sneakers, so 5 are low top sneakers. Now think about 15 slots, and the first 10 are labeled high-top sneakers. If we insert the last 5 sneakers as red sneakers, there are 4 leftover red sneakers.

Putting those 4 sneakers as high top sneakers, we have our answer as C, or  $\frac{4}{15}$ .

-ermwhatthesigma

-slight change by the-guy-with-the"W"hairline-and-a-not-goofy-looking-gyat-with-so-much-recoil-it-bounces-to-the-nearest-blackhole...

**Solution 3**

There are  $\frac{3}{5} \cdot 15 = 9$  red pairs of sneakers and 6 white pairs. There are also  $\frac{2}{3} \cdot 15 = 10$  high-top pairs of sneakers and 5 low-top pairs. Let  $r$  be the number of red high-top sneakers and let  $w$  be the number of white high-top sneakers. It follows that there are  $9 - r$  red pairs of low-top sneakers and  $6 - r$  white pairs. We must have  $9 - r \leq 5$ , in order to have a valid amount of white sneakers. Solving

this inequality gives  $r \geq 4$ , so the smallest possible value for  $r$  is 4. This means that there would be  $9 - 4 = 5$  pairs of low-top red sneakers, so there are 0 pairs of low-top white sneakers and 6 pairs of high top white sneakers. This checks out perfectly, so the smallest fraction is  $(C) \frac{4}{15}$ .

-hola

#### Solution 4

$15 \times \frac{3}{5} = 9$ , which is the number of red pairs of sneakers. Then,  $\frac{2}{3} \times 15 = 10$ , so there are ten pairs of high-top sneakers. If there are ten high-top sneakers, there are five low top sneakers. To have the lowest amount of high-top red sneakers, all five of the low-top sneakers need to be red. There are four red sneakers remaining so they have to be high-top. So the answer is  $(C) \frac{4}{15}$ .

Answer by AliceDubbleYou

-Minor edit by angieeverfree

-Minor edit by SmartyDigits

#### Solution 5

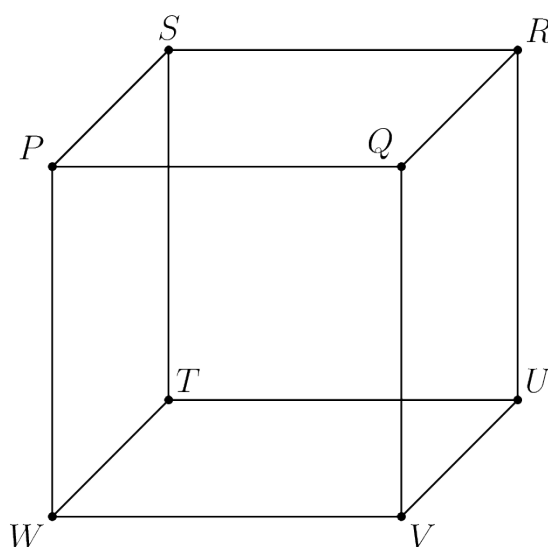
Let the number of red high-top sneakers be  $x$ . Making a two way table we get:

|          | Red     | White    | Total |
|----------|---------|----------|-------|
| High-top | $x$     | $10 - x$ | 10    |
| Low-top  | $9 - x$ | $x - 4$  | 5     |
| Total    | 9       | 5        | 14    |

the table we can see that  $4 \leq x \leq 9$ . Therefore the minimum value is when  $x = 4$ .

So the answer is  $(C) \frac{4}{15}$ .

20. (AMC 8 2024) Any three vertices of the cube  $PQRSTU VW$ , shown in the figure below, can be connected to form a triangle. (For example, vertices  $P, Q$ , and  $R$  can be connected to form  $\triangle PQR$ .) How many of these triangles are equilateral and contain  $P$  as a vertex?



- (A) 0      (B) 1      (C) 2      (D) 3      (E) 6

**Answer:** D

**Solution 1**

The only equilateral triangles that can be formed are through the diagonals of the faces of the square. From P you have 3 possible vertices that are possible to form a diagonal through one of the faces. Therefore, there are 3 possible triangles. So the answer is (D) 3 Math645, andliu766, e\_\_\_\_\_

**Solution 2**

Each other compatible point must be an even number of edges away from P, so the compatible points are R, V, and T. Therefore, we must choose two of the three points, because P must be a point in the triangle. So, the answer is  $\binom{3}{2} =$ (D) 3

-ILoveMath31415926535

**Solution 3 (arduous and not recommended)**

List them out- you get  $PRV$ ,  $PRT$ , and  $PVT$ . Therefore, the answer is (D) 3

-Anonymussrvusdmathstudent234234

**Solution 4 (Easy)**

After looking at the cube, we realize that an equilateral triangle can only be formed by three lines that form a diagonal along a face of the cube (such as  $PV$ ). Because the problem has a condition that one of the triangle's vertex must be on  $P$ , the three diagonals that can be formed are  $PT$ ,  $PR$ , and  $PV$ .

Now, we can choose any of 2 out of the 3 lines we have listed, and connect any of them with another line (for example, if we choose  $PT$  and  $PR$ , the third diagonal is  $RT$ ). Thus, there are 3 ways to choose the 3 diagonals, so the answer is 3, or

(D) 3

**Video Solutions**

21. (*AMC 8 2024*) A group of frogs (called an *army*) is living in a tree. A frog turns green when in the shade and turns yellow when in the sun. Initially, the ratio of green to yellow frogs was 3 : 1. Then 3 green frogs moved to the sunny side and 5 yellow frogs moved to the shady side. Now the ratio is 4 : 1. What is the difference between the number of green frogs and the number of yellow frogs now?

(A) 10      (B) 12      (C) 16      (D) 20      (E) 24

**Answer:** E

**Solution 1**

Let the initial number of green frogs be  $g$  and the initial number of yellow frogs be  $y$ . Since the ratio of the number of green frogs to yellow frogs is initially 3 : 1,  $g = 3y$ . Now, 3 green frogs move to the sunny side and 5 yellow frogs move to the shade side, thus the new number of green frogs is  $g + 2$  and the new number of yellow frogs is  $y - 2$ . We are given that  $\frac{g+2}{y-2} = \frac{4}{1}$ , so  $g + 2 = 4y - 8$ , since  $g = 3y$ , we have  $3y + 2 = 4y - 8$ , so  $y = 10$  and  $g = 30$ . Thus the answer is  $(g + 2) - (y - 2) = 32 - 8 =$ (E) 24.

-anonchalantdreadhead

**Solution 2**

Since the original ratio is 3 : 1 and the new ratio is 4 : 1, the number of frogs must be a multiple of 12, the only solutions left are (B) and (E).

Let's start with 12 frogs:

We must have 9 frogs in the shade and 3 frogs in the sun. After the change, there would be 11 frogs in the shade and 1 frog in the sun, which is not a 4 : 1 ratio.

Therefore the answer is: (E) 24.

-ILoveMath31415926535

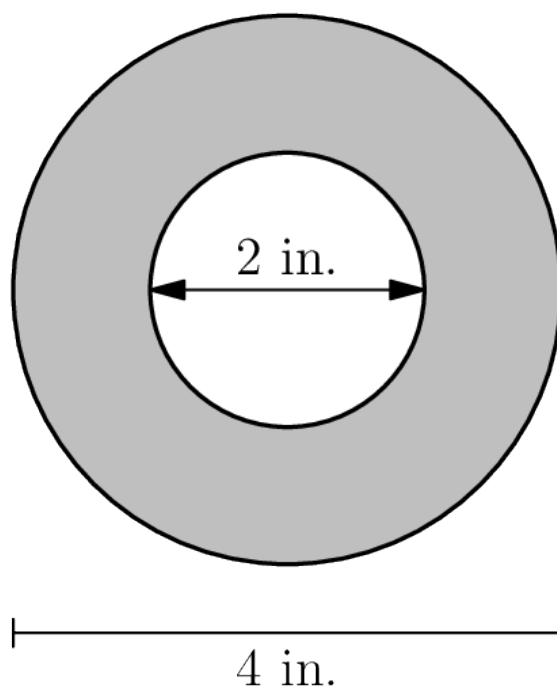
**Solution 3 (Simple and easy to make sense of)**

The ratio of  $g$  (green) to  $y$  (yellow) frogs is 3 : 1. When 3 green frogs move to the sunny side and 5 yellow frogs move to the shady side, the ratio becomes  $g + 2 : y - 2$  which is 4 : 1.

So earlier,  $\frac{3}{4}$ , or  $\frac{15}{20}$ , of the total frogs were green. Now,  $\frac{4}{5}$ , or  $\frac{16}{20}$ , of the total frogs are green. When the 2 frogs transferred from the yellow side to the green, the green side gained  $\frac{1}{20}$  of the total amount of frogs. So,  $2 = \frac{1}{20}a$ , where  $a$  is the total number of frogs. Solving for  $a$  we get  $a = 40$ .

If the ratio 4 : 1 has a total of 40, then we can multiply each of them by  $\frac{40}{(4+1)}$ , or 8, and find that there are 32 green frogs and 8 yellow frogs. Therefore, the difference between the green and yellow frogs is (E) 24

22. (AMC 8 2024) A roll of tape is 4 inches in diameter and is wrapped around a ring that is 2 inches in diameter. A cross section of the tape is shown in the figure below. The tape is 0.015 inches thick. If the tape is completely unrolled, approximately how long would it be? Round your answer to the nearest 100 inches.



- (A) 300    (B) 600    (C) 1200    (D) 1500    (E) 1800

**Answer:** B

**Solution 1**

The roll of tape is  $1/0.015 \approx 66$  layersthick. Inordertofindthetotallength, wehavetofindtheaverageof Multiplied by  $3\pi \cdot 66$  gives approximately (B) 600.

**Solution 2**

There are about  $\frac{1}{0.015} = \frac{200}{3}$  "full circles" of tape, and with average circumference of  $\frac{4+2}{2}\pi = 3\pi$ ,  $\frac{200}{3} \cdot 3\pi = 200\pi$ , which means the answer is  $\boxed{600}$ .

**Solution 3**

We can figure out the length of the tape by considering the side of the tape as a really thin rectangle that has a width of 0.015 inches. The side of the tape is wrapped into an annulus (The shaded region between 2 circles with the same center), meaning the area of the shaded region is equal to the area of the really thin rectangle.

The area of the shaded region is  $\pi(\frac{4}{2})^2 - \pi(\frac{2}{2})^2 = 3\pi$ , and we divide that by 0.015 to get  $200\pi$ . Approximating  $\pi$  to be 3, we get the final answer to be  $200 \cdot 3 = \text{(B)} 600$ .

**Solution 4**

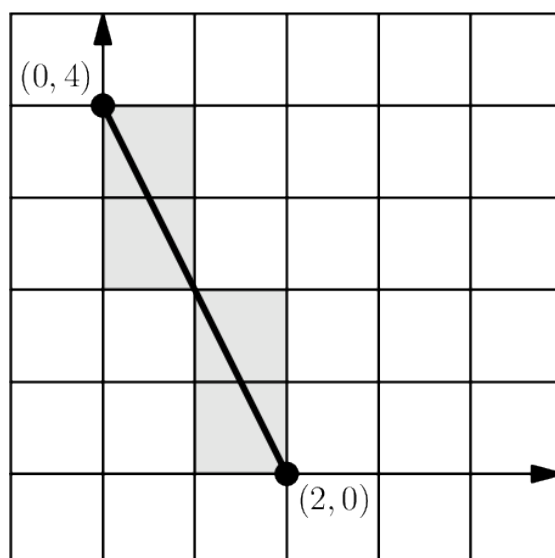
The volume of the tape is always the same, but we can either calculate it when the tape is unrolled as a really long, thin rectangular prism, or we can calculate it as a cylinder with a hole cut out of it. When we calculate it as a long rectangular prism, we can say that the length is  $X$  (this is what the problem wants!) and the width is  $Y$ . Then, the volume is, of course,  $0.015 \cdot X \cdot Y$ . Now, notice that the "width" of our rectangular prism is also the "height" of our cylinder with a hole cut out of it. Then, we can calculate the volume as base times height, or in this case,  $3\pi \cdot Y$ . Now, since the volume always stays the same, we know that  $3\pi \cdot Y = 0.015 \cdot X \cdot Y$ . Cancelling the  $Y$ 's give us an equation for  $X$ , and if we approximate  $\pi$  as 3, then  $X = \boxed{600}$ .

**Solution 5**

If you cannot notice that the average diameter is 3, you can still solve this problem by the following method.

The same with solution 1, we have  $\frac{1000}{0.015}$  layers of tape. If we consider every layers with the diameter 2, the length should be  $\frac{1000}{0.015} 2\pi \approx 400$ . If the diameter is seen as 4, the length should be 800. So, the length is between 400 and 800, the only possible answer is  $\boxed{600}$ .

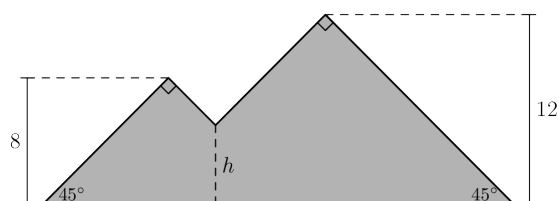
23. (AMC 8 2024) Rodrigo has a very large sheet of graph paper. First he draws a line segment connecting point  $(0, 4)$  to point  $(2, 0)$  and colors the 4 cells whose interiors intersect the segment, as shown below. Next Rodrigo draws a line segment connecting point  $(2000, 3000)$  to point  $(5000, 8000)$ . How many cells will he color this time?



- (A) 6000    (B) 6500    (C) 7000    (D) 7500    (E) 8000

**Answer:** C

24. (AMC 8 2024) Jean has made a piece of stained glass art in the shape of two mountains, as shown in the figure below. One mountain peak is 8 feet high while the other peak is 12 feet high. Each peak forms a  $90^\circ$  angle, and the straight sides form a  $45^\circ$  angle with the ground. The artwork has an area of 183 square feet. The sides of the mountain meet at an intersection point near the center of the artwork,  $h$  feet above the ground. What is the value of  $h$ ?

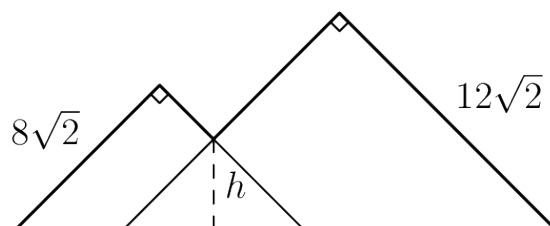


- (A) 4    (B) 5    (C)  $4\sqrt{2}$     (D) 6    (E)  $5\sqrt{2}$

**Answer:** B

**Solution 1**

Extend the "inner part" of the mountain so that the image is two right triangles that overlap in a third right triangle as shown.



The side length of the largest right triangle is  $12\sqrt{2}$ , which means its area is 144. Similarly, the area of the second largest right triangle is 64 (the side length is

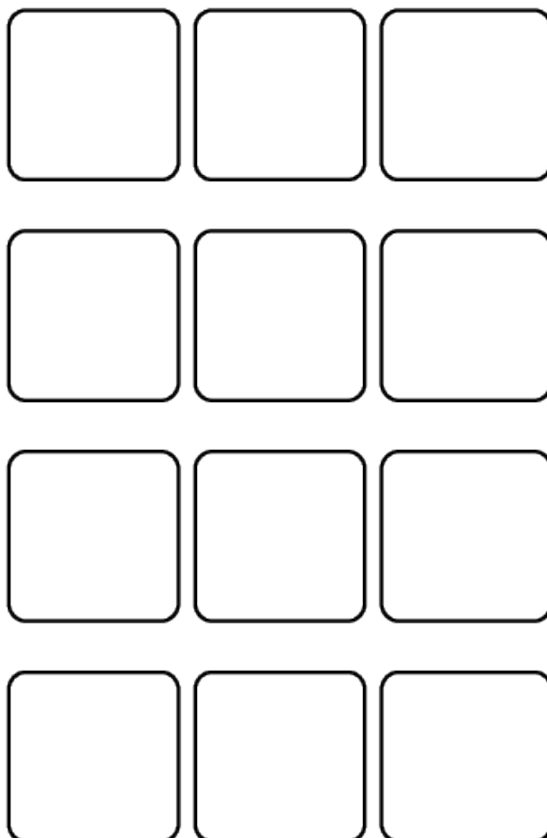


$8\sqrt{2}$ ), and the area of the overlap is  $h^2$  (the side length is  $h\sqrt{2}$ ). Because the right triangles have a side ratio of  $1:1:\sqrt{2}$ . Thus,

$$144 + 64 - h^2 = 183,$$

which means that the answer is **(B) 5**.

25. (AMC 8 2024) An airplane has 4 rows of seats with 3 seats in each row. Eight passengers have boarded the plane and are distributed randomly among the seats. A married couple is next to board. What is the probability there will be 2 adjacent seats in the same row for the couple?



- (A)  $\frac{8}{15}$     (B)  $\frac{32}{55}$     (C)  $\frac{20}{33}$     (D)  $\frac{34}{55}$     (E)  $\frac{8}{11}$

**Answer: C**

**Solution 1 (Complementary Counting Casework)**

Suppose the passengers are indistinguishable. There are  $\binom{12}{8} = 495$  total ways to distribute the passengers. We proceed with complementary counting, and instead, will count the number of passenger arrangements such that the couple cannot sit anywhere. Consider the partitions of 8 among the rows of 3 seats, to make our lives easier, assuming they are non-increasing. We have  $(3, 3, 2, 0)$ ,  $(3, 3, 1, 1)$ ,  $(3, 2, 2, 1)$ ,  $(2, 2, 2, 2)$ .

For the first partition, clearly, the couple will always be able to sit in the row with 0 occupied seats, so we have 0 cases here.

For the second partition, there are  $\frac{4!}{2!2!} = 6$  ways to permute the partition. Now the rows with exactly 1 passenger must be in the middle, so this case generates 6 cases.

For the third partition, there are  $\frac{4!}{2!} = 12$  ways to permute the partition. For rows with 2 passengers, there are  $\binom{3}{2} = 3$  ways to arrange them in the row so that the couple cannot sit there. The row with 1 passenger must be in the middle. We obtain  $12 \cdot 3^2 = 108$  cases.

For the fourth partition, there is 1 way to permute the partition. As said before, rows with 2 passengers can be arranged in 3 ways, so we obtain  $3^4 = 81$  cases.

Collectively, we obtain a total of  $6 + 108 + 81 = 195$  cases. The final probability is

$$1 - \frac{195}{495} = \boxed{(C) \frac{20}{33}}.$$

### Solution 2 (Straightforward Casework)

Suppose the passengers are indistinguishable.

You could think the question as: if 4 empty seats are placed, what is the probability that there are 2 adjacent seats open?

We proceed by casework.

Case 1: There is exactly one pair of open seats. Then the other seat in that row must be occupied. The other two empty seats are distributed across the remaining 3 rows without being adjacent, which is  $\binom{9}{2} - 6 = 30$  cases per pair of open seats for  $30 \cdot 8 = 240$  total cases.

Case 2: There is one row of open seats. 4 ways to choose the row and 9 to choose the final empty seat for  $4 \cdot 9 = 36$  cases.

Case 3: There are 2 independent pairs of open seats. Choose the 2 rows, then the placement of each pair within each row for  $\binom{4}{2} \cdot 2^2 = 24$  cases.

In total, we get  $240 + 36 + 24 = 300$  cases total for a probability of

$$\frac{300}{\binom{12}{4}} = \frac{300}{495} = \boxed{(C) \frac{20}{33}}$$

### Solution 3 (Complementary Casework on Middle Seats)

We notice that if we have a middle seat in a row, then the couple cannot sit in that row. So, we perform complementary casework.

Case 1: Four people sitting in middle seats.

In this case, there are 4 people left to order, and 8 seats. This gives  $\binom{8}{4}$  total combinations for this case.

Case 2: Three people sitting in middle seats.

In this case, there are  $\binom{4}{3}$  ways to permute the rows in which the middle seat is occupied. For the row in which the people do not occupy the middle row, we must have two people sitting at the ends of the rows to guarantee the couple cannot sit there. So, for the rest of the 3 people, there are 6 possible seats. So, there are  $\binom{4}{3} \cdot \binom{6}{3}$  total combinations.

Case 3: Two people sitting in middle seats.

In this case, there are  $\binom{4}{2}$  ways to permute the rows in which the middle seat is occupied. For the rows in which the people do not occupy the middle row, we must have two people sitting at the ends of the rows to guarantee the couple cannot sit there. So, for the rest of the 2 people, there are 4 possible seats. So, there are  $\binom{4}{2} \cdot \binom{4}{2}$  total combinations.

Case 4: One person sitting in a middle seat

In this case, there are  $\binom{4}{1}$  ways to permute the rows in which the middle seat is occupied. For the rows in which the people do not occupy the middle row, we must have two people sitting at the ends of the rows to guarantee the couple cannot sit there. So, for the rest of the last person, there are 2 possible seats. So, there are  $\binom{4}{1} \cdot \binom{2}{1}$  total combinations.

Case 5: Zero people sitting in a middle seat

In this case, we must have every person sitting at the ends of the seats. So, there is only 1 combination.

In total, we have

$$\binom{8}{4} + \binom{4}{3} \cdot \binom{6}{3} + \binom{4}{2} \cdot \binom{4}{2} + \binom{4}{1} \cdot \binom{2}{1} + 1$$

combinations, or 195 combinations. The final step is to find the total amount of combinations without restrictions. This is simply  $\binom{12}{4} = 495$ . So, finally employing complementary counting, we have that the probability that there will be 2 adjacent seats for the couple is

$$1 - \frac{195}{495} = \frac{20}{33}.$$

#### Solution 4 (Get lucky)

There are  $12 \cdot 11 = 132$  for two people (the married couple) to be seated. This will be our denominator.

There are 8 pairs of seats that are next to each other in the diagram (2 per row; left-middle and middle-right). This will be our numerator.

Since there are  $8 + 2 = 10$  total people on the plane, we should multiply our numerator by that to account for all ways the 10 people could be seated (e.x. the husband and the wife could be switched around and it would still work, same applies to the other passengers)

Therefore, our numerator is  $8 \cdot 10 = 80$ .

This creates the fraction  $\frac{80}{132}$ , which simplifies to

$$\boxed{(C) \frac{20}{33}}.$$

Remark: This solution is flawed because if there are  $9 + 2 = 11$  people on the plane, the probability could not be calculated this way, because intuitively, the probability should be decreased (as the probability for 9 people and 1 couple is  $\frac{19}{55}$ ), not increased (to be  $\frac{2}{3}$ ). -ericz

### Solution 5

Consider the couple seated together and there should be 8 seated ways (2 ways in each row). And the other 8 people can be seated in other 10 seats randomly.

There will be total  $8 \cdot P(10, 2)$

Consider two double counting cases.

Case I: the other 8 people are seated in (1, 3) (2, 3), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (4, 3)

It was double counted for couple's seats (1, 1) (1, 2) and (2, 1), (2, 2)

There will be  $\frac{8 \cdot 6}{2} \times P(8, 8)$

Case II: one whole row is empty and the other 8 people are randomly seated in other rows

It was double counted for couple's seats such as (1, 1) (1, 2) and (1, 2) (1, 3)

There will be  $4 \cdot P(9, 8)$

So the probability is  $\frac{8 \cdot P(10, 2) - \frac{8 \cdot 6}{2} \cdot P(8, 8) - 4 \cdot P(9, 8)}{P(12, 8)}$

which simplifies to

$$\boxed{(C) \frac{20}{33}}$$

- Orlando Liu Cupertino Middle School

### Solution 6 (A Directed Graph)

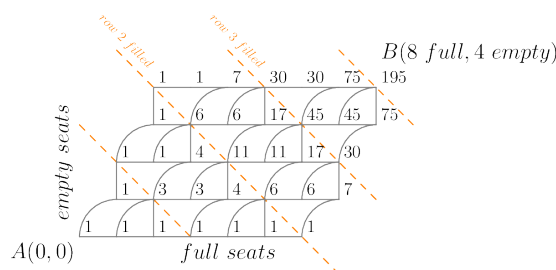
Notice there are  $\binom{12}{4} = 495$  ways to distribute 4 empty seats and 8 occupied seats without constraints, and that this may be represented as the number of paths from  $A$  to  $B$  in the directed graph below. This graph is not a diagram of the plane; rather, a node at coordinates  $(x, y)$  refers to the number of (*occupied*, *empty*) seats that have been placed since leaving  $A$ .

Beginning from  $A$  we move up one node when adding an empty seat or right one node when adding an occupied seat as we fill the airplane by row. After three such moves, for example, we arrive at the leftmost dashed line, indicating row one is complete. The number adjacent to a node indicates the number of unique paths to a node and is simply the sum of its incoming paths. We arrive at  $B(8 \text{ full}, 4 \text{ empty})$  with a total of 495 unconstrained combinations as expected.



Next, let's modify our graph to accommodate the complementary case where no adjacent empty seats are allowed. Each empty seat must be followed by a full seat —

excepting the final end seat in a row — and is indicated by a path rising and curving right. Once again, the number of paths to a node is the sum of incoming paths, and we thus arrive at 195 ways to distribute 4 empty seats with none adjacent.



Our lucky couple's final probability of sitting together is  $1 - \frac{195}{495} = \boxed{(C) \frac{20}{33}}$ .

### Solution 7 (Perfect for a Generating Function)

Let's count the complementary case of distributing empty seats with none adjacent. Beginning with the first row, there is 1 way to fill the row with no empty seats. There are 3 ways to complete the row with 1 empty seat. There is 1 way to complete the row with 2 empty seats. There is no way to fit 3 non-adjacent empty seats.

Let's represent these values as a polynomial where the exponents represent the number of empty seats and the coefficients represent the associated number of possible combinations. For the first row, we have  $1x^2 + 3x^1 + 1x^0$ . We have no intention of assigning or determining any value for  $x$ . Our polynomial simply keeps our calculations organized.

Let's consider the case of one empty seat in the second row. There are 3 positions it could be in, so we'd represent that as  $3x^1$ , of course. If we ask how many combinations exist for a single empty seat in the first row and a single empty seat in the second row, we multiply the  $3x^1$  from the first row by the  $3x^1$  from the second row and arrive at  $9x^2$ , i.e., 9 combinations of 2 empty seats with one in each row.

Are there other ways to arrive at 2 empty seats anywhere in the first two rows? Yes, we could have  $1x^2$  in the first row and  $1x^0$  in the second row, or vice versa. So the total number of ways to populate 2 empty seats in the first two rows is  $1x^2 \cdot 1x^0 + 3x^1 \cdot 3x^1 + 1x^0 \cdot 1x^2 = 11x^2$ .

Notice that multiplying our original polynomial  $1x^2 + 3x^1 + 1x^0$  by the same polynomial for the second row produces the  $11x^2$  term above, and similarly produces a count of the combinations for every other possible number of empty seats in the first two rows, i.e.,  $(1x^2 + 3x^1 + 1x^0)^2 = 1x^4 + 6x^3 + 11x^2 + 6x^1 + 1x^0$ ; how wonderful.

|          |        |        |         |        |        |
|----------|--------|--------|---------|--------|--------|
|          |        |        | $1x^2$  | $3x^1$ | $1x^0$ |
| $\times$ |        |        | $1x^2$  | $3x^1$ | $1x^0$ |
|          |        |        | <hr/>   |        |        |
|          |        |        | 1       | 3      | 1      |
|          |        | 3      | 9       | 3      |        |
| $+$      | 1      | 3      | 1       |        |        |
|          |        | <hr/>  |         |        |        |
|          | $1x^4$ | $6x^3$ | $11x^2$ | $6x^1$ | $1x^0$ |

We conclude that  $(1x^2 + 3x^1 + 1x^0)^4$  will contain our answer when considering all four rows of our airplane. Or, using our prior product,  $(1x^4 + 6x^3 + 11x^2 + 6x^1 + 1x^0)^2$  would be even easier to compute, especially since we only need to make calculations resulting in  $x^4$  terms, i.e.,  $x^4 \cdot x^0$  and  $x^3 \cdot x^1$ , etc. We proceed with  $2(1x^4 \cdot 1x^0 + 6x^3 \cdot 6x^1) + 11x^2 \cdot 11x^2 = 195x^4$ . There are 195 ways to fit four empty seats in our airplane with no two adjacent.

|          |        |         |         |          |          |          |         |         |        |
|----------|--------|---------|---------|----------|----------|----------|---------|---------|--------|
|          |        |         |         |          | $1x^4$   | $6x^3$   | $11x^2$ | $6x^1$  | $1x^0$ |
| $\times$ |        |         |         |          | $1x^4$   | $6x^3$   | $11x^2$ | $6x^1$  | $1x^0$ |
|          |        |         |         |          | 1        | 6        | 11      | 6       | 1      |
|          |        |         |         | 6        | 36       | 66       | 36      | 6       |        |
|          |        |         | 11      | 66       | 121      | 66       | 11      |         |        |
|          |        | 6       | 36      | 66       | 36       | 6        |         |         |        |
| +        | 1      | 6       | 11      | 6        | 1        |          |         |         |        |
|          | $1x^8$ | $12x^7$ | $58x^6$ | $144x^5$ | $195x^4$ | $144x^3$ | $58x^2$ | $12x^1$ | $1x^0$ |

Dividing this by the number of unconstrained ways to fit four empty seats in our plane gives the probability that our couple cannot sit together.

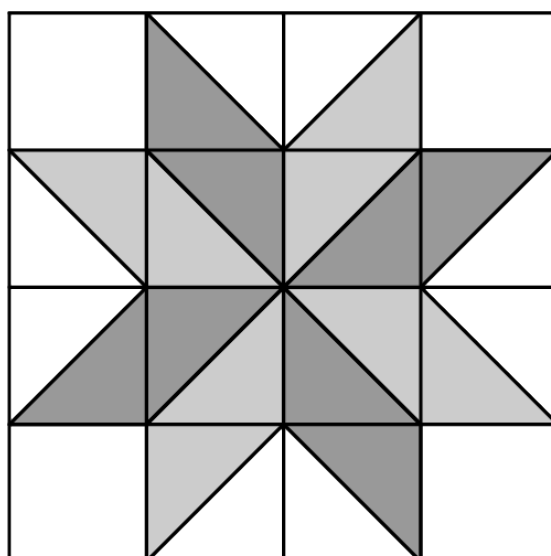
To compute the number of unconstrained ways to fit four empty seats anywhere in our airplane, we have a choice of calculating  $(1x^3 + 3x^2 + 3x^1 + 1x^0)^4$  or  $(1x^1 + 1x^0)^{12}$  or perhaps just  $\binom{12}{4} = 495$  this time.

Our married couple's final probability of sitting together is  $1 - \frac{195}{495} = \boxed{(C) \frac{20}{33}}$ .

**Other Solution** You see that the denom should be 495 because 12 choose 8, and then you find the answer choices with a denom of a divisor of 495, A and C. Then, you see that  $8/15$  would be too obvious, so you choose C.-This strat is luck based. Do not try in real life.

## §40 AMC 8 2025

- (AMC 8 2025) The eight-pointed star, shown in the figure below, is a popular quilting pattern. What percent of the entire  $4 \times 4$  grid is covered by the star?



(A) 40      (B) 50      (C) 60      (D) 75      (E) 80

**Answer:** B

**Solution 1**

we count the amount of shaded half triangles, 8. we half this to get the amount of full squares it equals. then we add 4 for the four full squares in the middle. this gets us a total of eight full squares, out of a total of 16. so  $8 / 16$  is simplified to  $1 / 2$  or **(B) 50** percent of the entire grid. pereena

Each of the unshaded triangles has base length 2 and height 1, so they all have area  $\frac{2 \cdot 1}{2} = 1$ . Each of the unshaded unit squares has area 1. The area of the shaded region is equal to the area of the entire grid minus the area of the unshaded region, or  $4^2 - 4 \cdot 1 - 4 \cdot 1 = 8$ . The star is then  $\frac{8}{16} = \frac{1}{2} = \frac{50}{100}$ , or **(B) 50** percent of the entire grid.—DummyAMC

**Solution 2**

There are 16 total squares in the diagram and each square has 2 triangles whose areas are half the area of a unit square. Thus, the total number of triangles in the diagram is  $4^2 \cdot 2 = 32$  triangles. There are 16 shaded triangles in the diagram, so the area of the star is  $\frac{16}{32} = \frac{1}{2} = \frac{50}{100}$ , or **(B) 50** percent. Pi\_in\_da\_box

**Solution 3**

There are 4 squares that are entirely shaded and 4 squares that have no shading. This cancels them out. The rest of the squares are half-half. Therefore the shaded region is **(B) 50** percent of the grid.

**Solution 4**

Note that we can move one triangle from each of the four cells in the middle to each of the four corners. This will leave every cell in the grid with one triangle each, and each triangle has an area of half the area of each cell. Thus, our answer must equal to  $\frac{1}{2}$ , and so our answer is **(B) 50**. derekwang2048

**Solution 5**

The shaded area is a  $2 \times 2$  square in the middle of the figure combined with 8 small triangles. Since each small triangle is  $\frac{1}{2}$  of a unit square, the star's area is equal to the area of  $4 + 8 \cdot \frac{1}{2} = 8$  unit squares, which **(B) 50** percent of the grid.

-vockey

**Solution 6**

You can move the extra half a squares and make 4 squares so there are  $8/16$ . The answer is **(B) 50** percent of 16.

William

2. (AMC 8 2025) The table below shows the Ancient Egyptian hieroglyphs that were used to represent different numbers.

|                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |  |  |
| 100,000                                                                             | 10,000                                                                              | 1,000                                                                               | 100                                                                                 | 10                                                                                   | 1                                                                                     |

For example, the number 32 was represented by the hieroglyphs  $\cap \cap \cap ||$ . What number is represented by the following combination of hieroglyphs?



- (A) 1, 423      (B) 10, 423      (C) 14, 023      (D) 14, 203      (E) 14, 230

**Answer:** B

**Solution 1**

The first hieroglyph is worth 10,000, the next 4 are worth  $100 \cdot 4 = 400$ , the next 2 are worth  $10 \cdot 2 = 20$ , and the last 3 are worth  $1 \cdot 3 = 3$ . Therefore, the answer is  $10,000 + 400 + 20 + 3 = \boxed{\text{(B)} 10,423}$

**Solution 2**

$$1 \cdot 10^4 + 0 \cdot 10^3 + 4 \cdot 10^2 + 2 \cdot 10^1 + 3 \cdot 10^0 = 10423$$

**Solution 3 (fast version of Solution x)**

From a first inspection, you may notice that there is no thousands hieroglyph. The only option with zero in the thousands place is  $\boxed{\text{(B)} 10,423}$

3. (AMC 8 2025) *Buffalo Shuffle-o* is a card game in which all the cards are distributed evenly among all players at the start of the game. When Annika and 3 of her friends play *Buffalo Shuffle-o*, each player is dealt 15 cards. Suppose 2 more friends join the next game. How many cards will be dealt to each player?
- (A) 8      (B) 9      (C) 10      (D) 11      (E) 12

**Answer:** C

**Solution 1**

We start with Annika and 3 of her friends playing, meaning that there are 4 players. This must mean that there is a total of  $4 \cdot 15 = 60$  cards. If 2 more players joined, there would be 6 players, and since the cards need to be split evenly, this would mean that each player gets  $\frac{60}{6} = \boxed{\text{(C)} 10}$  *Buffalo Shuffle-O* cards each meaning that the final answer is 10.

i agree pereena

4. (AMC 8 2025) Lucius is counting backward by 7s. His first three numbers are 100, 93, and 86. What is his 10th number?
- (A) 30      (B) 37      (C) 42      (D) 44      (E) 47

**Answer:** B

**Solution 1**

We plug  $a = 100$ ,  $d = -7$  and  $n = 10$  into the formula  $a + d(n - 1)$  for the  $n$ th term of an arithmetic sequence whose first term is  $a$  and common difference is  $d$  to get  $100 - 7(10 - 1) = \boxed{\text{(B)} 37}$ .

**Solution 2**



Since we want to find the 9th number Lucius says after he says 100, 7 is subtracted from his number 9 times, so our answer is  $100 - (9 \cdot 7) = \boxed{\text{(B)} 37}$

**Solution 3**

Using **brute force** and counting backward by 7s, we have 100, 93, 86, 79, 72, 65, 58, 51, 44,  $\boxed{\text{(B)} 37}$ .

Note that this solution is not practical and very time-consuming.

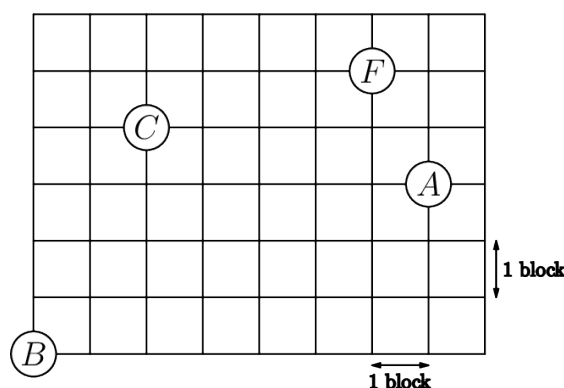
**Solution 4**

This can be thought of as an arithmetic sequence. Knowing that our first term is 100, we have to add 7 to get to our 0th term, 107. Our answer is then  $107 - 10 \cdot 7 = \boxed{\text{(B)} 37}$ .

**Solution 5**

Setting this as a function  $f(x) = 100 - 7(x - 1)$  where  $x$  is the  $x$ th number, we can plug in 10 for  $x$ . Quick arithmetic gives us the answer of  $\boxed{\text{(B)} 37}$ .

5. (AMC 8 2025) Betty drives a truck to deliver packages in a neighborhood whose street map is shown below. Betty starts at the factory (labeled  $F$ ) and drives to location  $A$ , then  $B$ , then  $C$ , before returning to  $F$ . What is the shortest distance, in blocks, she can drive to complete the route?

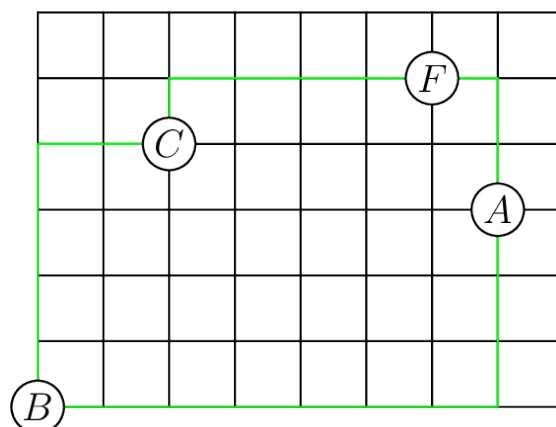


- (A) 20      (B) 22      (C) 24      (D) 26      (E) 28

**Answer: C**

**Solution 1**

Each shortest possible path from  $A$  to  $B$  follows the edges of the rectangle. The following path outlines a path of  $\boxed{\text{(C)} 24}$  units:



**Solution 2**

We can find the shortest distance using a line diagonally from one point to the other, creating a sort of slope, then find the sum of rise and run of the slope, which happens to be the shortest distance, repeat this process until you get back to Point  $F$ , and you should get  $2 + 1 + 3 + 7 + 4 + 2 + 1 + 4$ , which is equal to

**(C) 24**. Imhappy62789

**Solution 3**

To find the shortest distance Betty drives, we apply Taxicab Geometry (also known as the  $L_1$  metric). In a grid system, the distance between any two points  $(x_1, y_1)$  and  $(x_2, y_2)$  is defined as  $d = |x_1 - x_2| + |y_1 - y_2|$ .

Let the bottom-left corner of the grid be the origin  $(0, 0)$ . Based on the provided map, we assign coordinates to the given locations:

$$B = (0, 0) \quad C = (2, 4) \quad F = (6, 5) \quad A = (7, 3)$$

Betty's route follows the sequence  $F \rightarrow A \rightarrow B \rightarrow C \rightarrow F$ . We calculate the Manhattan distance for each leg of the journey:

$$F(6, 5) \text{ to } A(7, 3): |6 - 7| + |5 - 3| = 1 + 2 = 3$$

$$A(7, 3) \text{ to } B(0, 0): |7 - 0| + |3 - 0| = 7 + 3 = 10$$

$$B(0, 0) \text{ to } C(2, 4): |0 - 2| + |0 - 4| = 2 + 4 = 6$$

$$C(2, 4) \text{ to } F(6, 5): |2 - 6| + |4 - 5| = 4 + 1 = 5$$

The total shortest distance is the sum of these segments:

$$3 + 10 + 6 + 5 = 24$$

The shortest distance Betty can drive to complete the route is **24**.

6. (AMC 8 2025) Sekou writes down the numbers 15, 16, 17, 18, 19. After he erases one of his numbers, the sum of the remaining four numbers is a multiple of 4. Which number did he erase?

(A) 15      (B) 16      (C) 17      (D) 18      (E) 19

**Answer:** C

**Solution 1**

The sum of all five numbers is 85. Since 85 is 1 more than a multiple of 4, the number being subtracted must be 1 more than a multiple of 4. Thus, the answer is **(C) 17**. Gavin\_Deng

**Solution 2 - Creativity**

Notice the 10's place as you erase one number; you get 40, which is a multiple of 4, so you can ignore the 10's place in the numbers provided altogether because it will always be a multiple of 40. This means that the only relevant numbers are the 1's digit ones. If you add the numbers now, you get  $5 + 6 + 7 + 8 + 9 = 35$ . The closest multiple of 4 underneath 35 which can be achieved as a result of subtracting either 5, 6, 7, 8, or 9 is 28, which is achieved by subtracting 7, therefore the answer is **(C) 17**.

**Solution 3**

The sum of the residues of these numbers modulo 4 is  $-1 + 0 + 1 + 2 + 3 = 5 \equiv 1 \pmod{4}$ . Hence, the number being subtracted must be congruent to 1 modulo 4. The only such answer is **(C) 17**.

**Note:** To those who are confused about why, for 15, it said the residue modulo 4 is  $-1$  instead of 3 (because of 12), look here.  $-1 \equiv 3 \pmod{4}$ . The same principle above applies even if you replace  $-1$  with 3; two numbers are in the same residue class if their difference is a multiple of the modulus ( $3 - (-1) = 4$ , which is  $1 \times 4$ ).

#### Solution 4

The sum of all the numbers here is equivalent to  $17 \cdot 5$ . Subtracting 17 gives us  $17 \cdot 4$ , which is clearly a multiple of 4. Therefore, the answer is **(C) 17**. themathkidwoohooooo

#### Solution 5

Since the sum of  $15 + \cdots + 19$  is odd, we immediately exclude B and D. We further note that if A is true, then E is true. Hence, the answer is C.

#### Solution 6

We can use the fact that  $(a + b) \pmod{n} = [(a \pmod{n}) + (b \pmod{n})] \pmod{n}$ . Notice that 15, 16, 17, 18, 19 dividing by 4 have remainders 3, 0, 1, 2, 3. Their sum is 9. It is easy to see that  $9 - 1 = 8$  is divisible by 4 and so **(C) 17** is the correct answer.

#### Solution 7

If we use brute force and add  $15 + 16 + 17 + 18 + 19$ , we get 85. When we subtract every number from 85, we get 70, 69, 68, 67, 66, we see that 68 is divisible by 4, so **(C) 17** is the correct answer.

7. (AMC 8 2025) On the most recent exam on Prof. Xochi's class,

- 5 students earned a score of at least 95%
- 13 students earned a score of at least 90%
- 27 students earned a score of at least 85%
- 50 students earned a score of at least 80%

How many students earned a score of at least 80% and less than 90%?

(A) 8      (B) 14      (C) 22      (D) 37      (E) 45

**Answer:** D

#### Solution 1

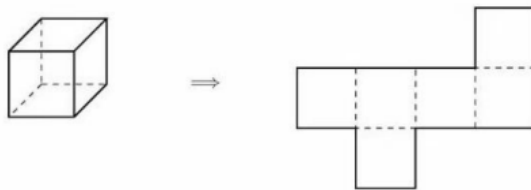
50 people scored at least 80%, and out of these 50 people, 13 of them earned a score that was not less than 90%, so the number of people that scored in between at least 80% and less than 90% is  $50 - 13 = \textbf{(D) 37}$ .

#### Solution 2

Let  $a$  denote the number of people who had a score of at least 85, but less than 90, and let  $b$  denote the number of people who had a score of at least 80 but less than 85. Our answer is equal to  $a + b$ . We find  $a = 27 - 13 = 14$ , while  $b = 50 - 27 = 23$ . Thus, the answer is  $23 + 14 = \textbf{(D) 37}$ .

-vockey

8. (AMC 8 2025) Isaiah cuts open a cardboard cube along some of its edges to form the flat shape shown on the right, which has an area of 18 square centimeters. What is the volume of the cube in cubic centimeters?



- (A)  $3\sqrt{3}$     (B) 6    (C) 9    (D)  $6\sqrt{3}$     (E)  $9\sqrt{3}$

**Answer:** A

**Solution 1**

Each of the 6 faces of the cube have equal area, so the area of each face is equal to  $\frac{18}{6} = 3$ , making the side length  $\sqrt{3}$ . From this, we can see that the volume of the cube is  $\sqrt{3}^3 = \boxed{\text{(A) } 3\sqrt{3}}$

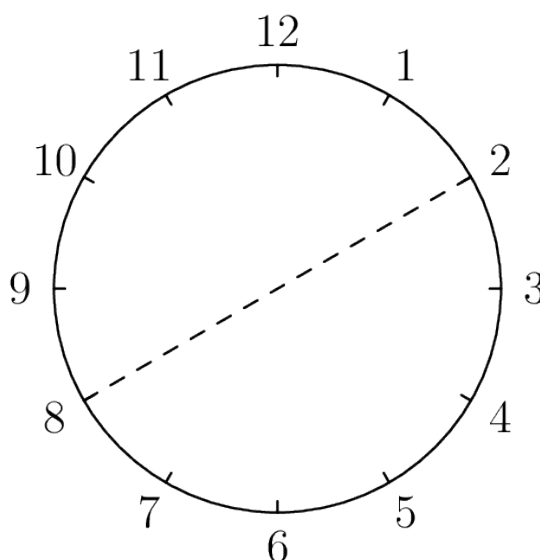
**Solution 2**

Let the side length of the cube shown be equal to  $s$  centimeters. The surface area of this cube is equal to the area of the net of the cube, which is equal to 18 square centimeters. The surface area of this cube is also  $6s^2$  square centimeters, so we have

$$6s^2 = 18 \implies s^2 = \frac{18}{6} = 3 \implies s = \sqrt{3}.$$

However, we aren't done. We have found that the side length of the cube is  $\sqrt{3}cm$ , but the question asks for the volume of the cube, which is equal to  $s^3 = \sqrt{3}^3 = \boxed{\text{(A) } 3\sqrt{3}}$  cubic centimeters.

9. (AMC 8 2025) Ningli looks at the 6 pairs of numbers directly across from each other on a clock. She takes the average of each pair of numbers. What is the average of the resulting 6 numbers?



- (A) 5    (B) 6.5    (C) 8    (D) 9.5    (E) 12

**Answer:** B

**Solution 1**

This solution uses **Gauss Summation**.

Our answer is

$$\frac{\frac{1+7}{2} + \frac{2+8}{2} + \cdots + \frac{6+12}{2}}{6} = \frac{\frac{1}{2}((1+7) + (2+8) + \cdots + (6+12))}{6} \quad (5)$$

$$(6)$$

$$= \frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12}{2 \cdot 6} \quad (7)$$

$$(8)$$

$$= \frac{\frac{12 \cdot 13}{2}}{2 \cdot 6} \quad (9)$$

$$(10)$$

$$= \frac{78}{12} \quad (11)$$

$$(12)$$

$$= \boxed{\text{(B)} 6.5} \quad (13)$$

**Solution 2**

We proceed with **brute force**. All of the pairs of opposite numbers on the clock are (12, 6), (1, 7), (2, 8), (3, 9), (4, 10), (5, 11), where order doesn't matter. The averages of each of these pairs are 9, 4, 5, 6, 7, and 8 respectively, and the average these numbers is  $\frac{9+4+5+6+7+8}{6} = \boxed{\text{(B)} 6.5}$

**Solution 3 (most efficient)**

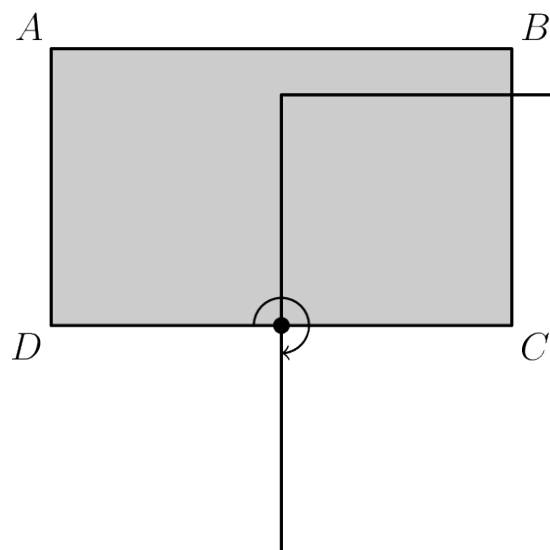
The problem is asking for the average of all 12 numbers. To find the average of all 12 numbers, you find the sum of all the integers from 1 to 12 which is 78, and divide it by 12 because there are 12 terms. Therefore, the answer is  $\frac{78}{12} = \boxed{\text{(B)} 6.5}$ .

You could also see that the average of 2 and 8 is 5, only one more that its consecutive pair, 1 and 7 which is 4. Assuming that this is the pattern, we can find the average 4 to 9 (6 consecutive integers starting with 4), and we that find our answer is  $\boxed{\text{(B)} 6.5}$ .

**Solution 4 (easiest to do in your head)**

You can see that the average of each pair increases by 1 as you go around the circle;  $\frac{1+7}{2} = 4$ ,  $\frac{2+8}{2} = 5$ , and etcetera up to  $\frac{6+12}{2} = 9$ . Therefore, the average of all the pairs is simply the average of the lowest and highest pairs since the average of each pair increases by the same amount as you go around the circle. Hence, the answer is the average of 4 and 9:  $\frac{4+9}{2} = \boxed{\text{(B)} 6.5}$ .

10. (AMC 8 2025) In the figure below,  $ABCD$  is a rectangle with sides of length  $AB = 5$  inches and  $AD = 3$  inches. Rectangle  $ABCD$  is rotated  $90^\circ$  clockwise around the midpoint of side  $DC$  to give a second rectangle. What is the total area, in square inches, covered by the two overlapping rectangles?



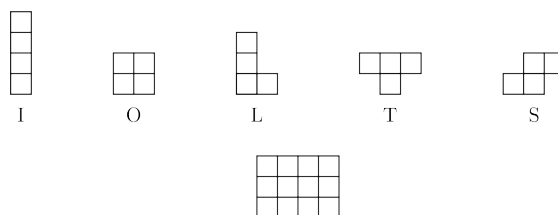
- (A) 21      (B) 22.25      (C) 23      (D) 23.75      (E) 25

**Answer:** D

**Solution 1**

The area of each rectangle is  $5 \cdot 3 = 15$ . Then the sum of the areas of the two regions is the sum of the areas of the two rectangles, minus the area of their overlap. To find the area of the overlap, we note that the region of overlap is a square, each of whose sides have length 2.5 (as they are formed by the midpoint of one of the long sides, the vertex, and also, since it is rotated 90 degrees). Then the answer is  $15 + 15 - 2.5^2 = \boxed{\text{(D) } 23.75}$ .

11. (AMC 8 2025) A *tetromino* consists of four squares connected along their edges. There are five possible tetromino shapes, *I*, *O*, *L*, *T*, and *S*, shown below, which can be rotated or flipped over. Three tetrominoes are used to completely cover a  $3 \times 4$  rectangle. At least one of the tiles is an *S* tile. What are the other two tiles?

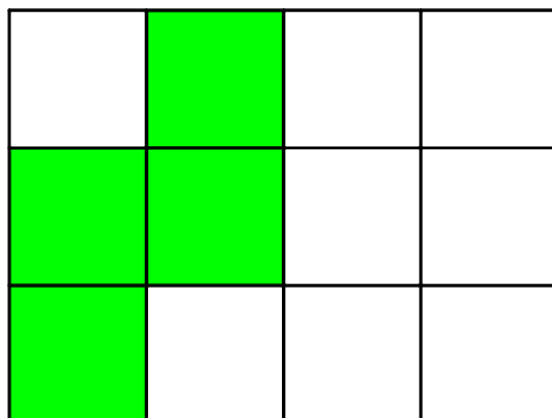


- (A) *I* and *L*      (B) *I* and *T*      (C) *L* and *L*      (D) *L* and *S*      (E) *O* and *T*

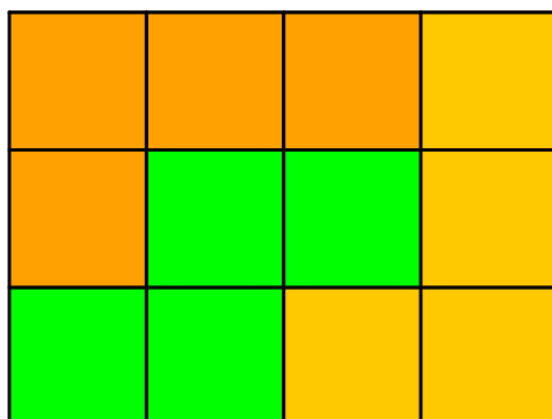
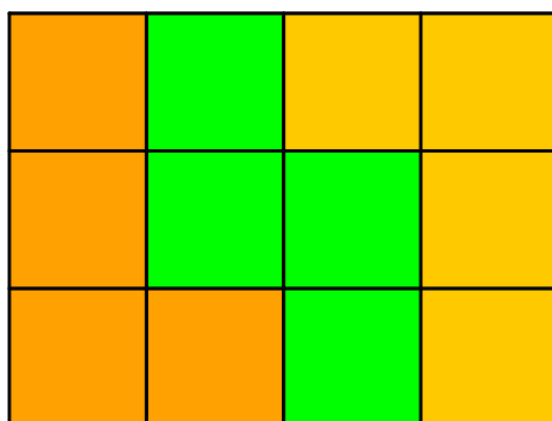
**Answer:** C

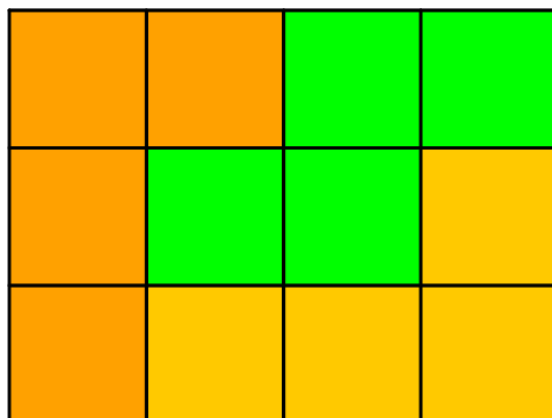
**Solution 1**

The  $3 \times 4$  rectangle allows for 7 possible places to put the *S* piece, with each possible placement having an inverted version. One of the cases looks like this:



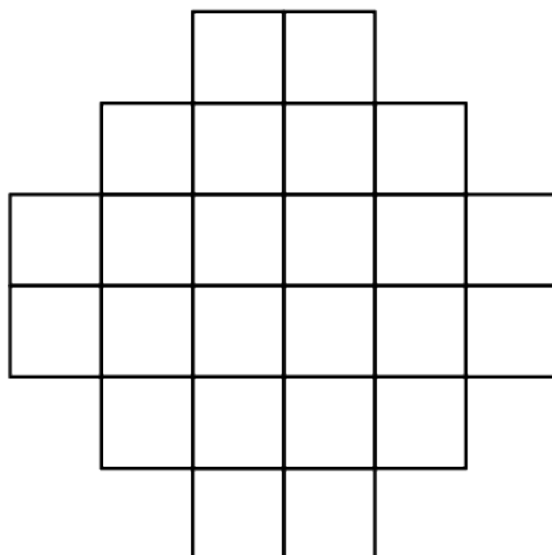
As you can see, there is a hole in the top left corner of the board, which would be impossible to fill using the tetrominos. There are three cases in which a hole isn't created; the S lies flat in the bottom left corner, it lies flat in the top right corner, or it stands upright in the center. All three tilings are shown below.





For each of the inverted cases, the L pieces can be inverted along with the S piece. Because the only cases that fill the rectangle after the S is placed are the ones that use two L pieces, the answer must be **(C) L and L**. bubbly617

12. (AMC 8 2025) The region shown below consists of 24 squares, each with side length 1 centimeter. What is the area, in square centimeters, of the largest circle that can fit inside the region, possibly touching the boundaries?



- (A)  $3\pi$     (B)  $4\pi$     (C)  $5\pi$     (D)  $6\pi$     (E)  $8\pi$

**Answer:** C

**Solution 1**

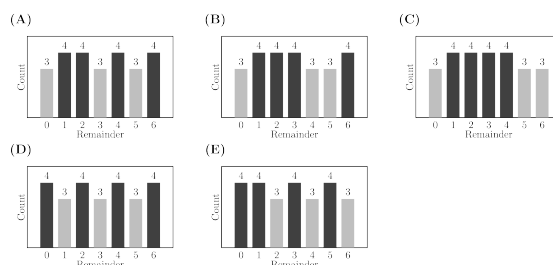
The largest circle that can fit inside the figure has its center in the middle of the figure and will be tangent to the figure in 8 points. By the Pythagorean Theorem, the distance from the center to one of these 8 points is  $\sqrt{2^2 + 1^2} = \sqrt{5}$ , so the area of this circle is  $\pi\sqrt{5}^2 = \mathbf{(C) \ 5\pi}$ .

**Solution 2**

Draw the circle in the grid and analyze the radius. Its radius is a little more than 2 but a lot less than 2.5, so the area is a little more than  $4\pi$ . So, the area of the circle is **(C)  $5\pi$**  with a radius of approximately 2.23.- leafy



13. (AMC 8 2025) Each of the even numbers  $2, 4, 6, \dots, 50$  is divided by 7. The remainders are recorded. Which histogram displays the number of times each remainder occurs?



**Answer:** A

### Solution 1

Let's take the numbers 2 through 14 (evens). The remainders will be 2, 4, 6, 1, 3, 5, and 0. This sequence keeps repeating itself over and over. We can take  $\text{floor}(50/14) = 3$ , so after the number 42, every remainder has been achieved 3 times. However, since 44, 46, 48, and 50 are left, the remainders of those will be 2, 4, 6, and 1 respectively. The only histogram in which those 4 numbers are set at 4 is histogram **(A)**.

### Solution 2

Writing down all the remainders gives us

$$2, 4, 6, 1, 3, 5, 0, 2, 4, 6, 1, 3, 5, 0, 2, 4, 6, 1, 3, 5, 0, 2, 4, 6, 1.$$

In this list, there are 3 numbers with remainder 0, 4 numbers with remainder 1, 4 numbers with remainder 2, 3 numbers with remainder 3, 4 numbers with remainder 4, 3 numbers with remainder 5, and 4 numbers with remainder 6. Manually computation of every single term can be avoided by recognizing the pattern alternates from 0, 2, 4, 6 to 1, 3, 5 and there are 25 terms. The only histogram that matches this is **(A)**.

### Solution 3

First, we find all of the multiples of 7 that are even, and are therefore, in this list. Knowing that 7 is odd, and that  $\text{odd} + \text{odd} = \text{even}$ , we can find all of the even multiples of 7 by simply finding all of the multiples of 14 that fit on this list. Doing this, we end up with

$$14, 28, 42.$$

Now, we can clearly see that there are only 3 numbers in this list with 0 as a remainder. This disproves 2 of our 5 answers immediately. Since our remaining answers are identical until we reach 3 as a remainder, we can skip right to there. Now, we need to find all even numbers that leave a remainder of 3 when divided by 7. To do this, we add 3 to all ODD multiples of 7 because  $\text{odd} + \text{odd} = \text{even}$ . This gives us

$$10, 24, 38.$$

Note that we don't add 3 to 49 because that exceeds 50. This shows us that there are only 3 numbers on this list that have a remainder of 3 when divided by 7, disproving histograms B and C, which say that there are 4 of these numbers. This reveals the answer to be **(A)**.

14. (AMC 8 2025) A number  $N$  is inserted into the list 2, 6, 7, 7, 28. The mean is now twice as great as the median. What is  $N$ ?

(A) 7      (B) 14      (C) 20      (D) 28      (E) 34

**Answer:** E

**Solution 1**

We see the median is 7. Since the mean is two times as great as the median, the mean must be  $7 \times 2 = 14$ . The sum of the numbers without  $N$  is  $2 + 6 + 7 + 7 + 28 = 8 + 14 + 28 = 50$ . Using the sum, we then have:

$$\frac{N+50}{6} = 14.$$

Multiplying both sides by 6, we get:

$$N + 50 = 14 \times 6 = 84 \quad N = 84 - 50 = \boxed{\text{(E) } 34}$$

The median of the list is 7, so the mean of the new list will be  $7 \cdot 2 = 14$ . Since there are 6 numbers in the new list, the sum of the 6 numbers will be  $14 \cdot 6 = 84$ . Therefore,  $2 + 6 + 7 + 7 + 28 + N = 84 \implies N = \boxed{\text{(E) } 34}$

**Solution 2**

Since the average right now is 10, and the median is 7, we see that  $N$  must be larger than 10, which means that the median of the 6 resulting numbers should be 7, making the mean of these 14. We can do  $2 + 6 + 7 + 7 + 28 + N = 14 \cdot 6 = 84$ .  $50 + N = 84$ , so  $N = \boxed{\text{(E) } 34}$

i feel this explanation doesn't make much sense anonymous

This explanation could be better explained anonymous

**Solution 3**

We try out every option by inserting each number into the list. After trying out each number, we get  $\boxed{\text{(E) } 34}$

Note that this is very time-consuming and it is not the most practical solution.

**Solution 4**

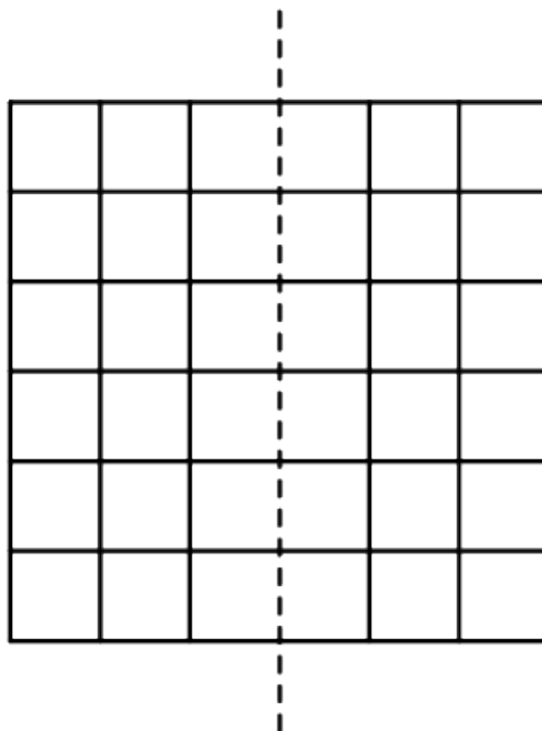
We could use answer choices to solve this problem. The sum of the 5 numbers is 50. If you add 7 to the list, 57 is not divisible by 6, therefore it will not work. Same thing applies to 14 and 20. The only possible choices left are 28 and 34. Now you check 28. You see that 28 doesn't work because  $(28 + 50) \div 6 = 13$  and 13 is not twice of the median, which is still 7. Therefore, only choice left is  $\boxed{\text{(E) } 34}$

**Solution 5**

We see that in the list, both 7s are 3rd and 4th. So, no matter what number we plug in, we will have two 7s in 3rd and 4th. So, the median will always be 7. That means we have to find the option that makes the mean 14. If the mean is 14, the sum of all the six numbers is 84. The sum of the five numbers we are given is 50, so the answer must be

$$\boxed{\text{(E) } 34}$$

15. (AMC 8 2025) Kei draws a 6-by-6 grid. He colors 13 of the unit squares silver and the remaining squares gold. Kei then folds the grid in half vertically, forming pairs of overlapping unit squares. Let  $m$  and  $M$  equal the least and greatest possible number of gold-on-gold pairs, respectively. What is the value of  $m + M$ ?



- (A) 12      (B) 14      (C) 16      (D) 18      (E) 20

**Answer:** C

**Solution 1**

We can see that the least number of gold-on-gold pairs will be obtained when the 13 silver squares are placed on the two sides so that they don't overlap when folded over (because then it will minimize the number of gold-on-golds). We can see that if we split them up 6 and 7 on both sides, and then fold it, the number of gold-on-golds will be  $18 - 13 = 5$ .

The maximum number of gold-on-golds will be achieved when the silver squares overlap when folded over, which will increase the number of gold-on-golds. If we align 6 silver squares with each other on each side, and put the last one somewhere else, we get the maximum is  $18 - 7 = 11$ . Therefore, the answer is  $11 + 5 = \boxed{\text{(C) } 16}$ .

16. (AMC 8 2025) Five distinct integers from 1 to 10 are chosen, and five distinct integers from 11 to 20 are chosen. No two numbers differ by exactly 10. What is the sum of the ten chosen numbers?

- (A) 95      (B) 100      (C) 105      (D) 110      (E) 115

**Answer:** C

**Solution**

Note that **for no two numbers to differ by 10, every number chosen must have a different units digit**. To make computations easier, we can choose  $(1, 2, 3, 4, 5)$  from the first group and  $(16, 17, 18, 19, 20)$  from the second group. Then the sum evaluates to  $1 + 2 + 3 + 4 + 5 + 16 + 17 + 18 + 19 + 20$  (rearrange  $1 + 19 + 2 + 18 + \dots$ ) =  $\boxed{\text{(C) } 105}$ .

**Another Way To Compute** For  $1 + 2 + 3 + 4 + 5 + 16 + 17 + 18 + 19 + 20$ , we can add the first term and the last term, which is 21. If we add the second term and the

second-to-last term it is also 21. There are 5 pairs that sum to 21, so the answer is  $21 \times 5$  which equals  $\boxed{(C) 105}$ .

- leafy

**Similar solution** One efficient method is to quickly add  $(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$ , which is 55. Then because you took 50 in total away from  $(16, 17, 18, 19, 20)$ , you add 50.  $55 + 50 = \boxed{(C) 105}$ .

**Solution 4 (efficient)**

To solve this problem, I started with the easiest/smallest case possible. In my opinion, that was

$$1 + 2 + 3 + 4 + 5 + 16 + 17 + 18 + 19 + 20.$$

I then solved this equation quickly using Little Gauss's method, rearranging that into

$$1 + 19 + 2 + 18 + 3 + 17 + 4 + 16 + 5 + 20.$$

I simplified this into

$$20 + 20 + 20 + 20 + 25.$$

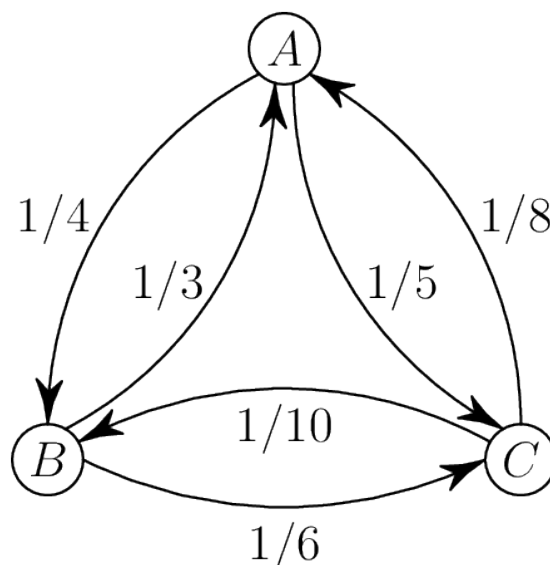
Solving this simple equation gives us the answer, which is  $\boxed{(C) 105}$ .

**Solution 5 (logic: for me it took about 1.5 mins)**

Start with 1. If you choose it, you cannot choose 11, so you must now pick 12. If you choose 12, then you can't choose 2, so you must now pick 13. If you choose 13, you cannot choose 3, so you must pick 4. If you choose 4, then you cannot choose 14 so you have to go to 15. Then you can't choose 5 so you have to go to 6. Then you can't choose 16, so you have to go to 17. Then you can't choose 7, so you have to choose 18. Then you can't choose 8, so you have to go to 9. Finally, you cannot choose 19, so you must go to 20. This makes the equation  $1 + 12 + 3 + 14 + 5 + 16 + 7 + 18 + 9 + 20 = 105$ , hence  $\boxed{(C) 105}$ .

Sorry if it is a bit confusing its just how I solved it so wanted to share!-N.

17. (AMC 8 2025) In the land of Markovia, there are three cities:  $A$ ,  $B$ , and  $C$ . There are 100 people who live in  $A$ , 120 who live in  $B$ , and 160 who live in  $C$ . Everyone works in one of the three cities, and a person may work in the same city where they live. In the figure below, an arrow pointing from one city to another is labeled with the fraction of people living in the first city who work in the second city. (For example,  $\frac{1}{4}$  of the people who live in  $A$  work in  $B$ .) How many people work in  $A$ ?



- (A) 55    (B) 60    (C) 85    (D) 115    (E) 160

**Answer:** D

**Solution 1**

There are  $100 \cdot (\frac{1}{4} + \frac{1}{5}) = 100 \cdot \frac{9}{20} = 45$  people who do not work in city  $A$  that live in city  $A$ , meaning that  $100 - 45 = 55$  people who live in city  $A$  work in city  $A$ . There are  $\frac{1}{3} \cdot 120 = 40$  people who live in city  $B$  and work in  $A$ , as well as  $\frac{1}{8} \cdot 160 = 20$  people who live in city  $C$  that work in city  $A$ . Therefore, the answer is  $55 + 40 + 20 = \boxed{\text{(D)} 115}$ .

**Solution 2**

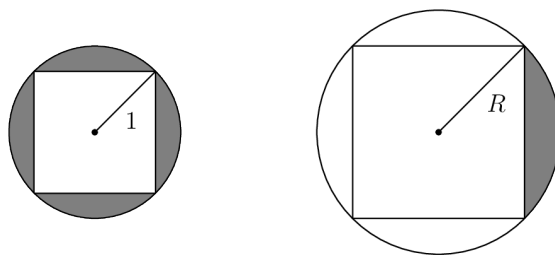
We could also make an equation. Let's denote the number of people that live in city  $A$  as  $a$ ,  $B$  as  $b$ , and  $C$  as  $c$ . If  $x$  denotes the number of people working in city  $A$ , then  $x = \frac{1}{8}c + \frac{1}{3}b + a - \frac{1}{4}a - \frac{1}{5}a$ . This is because  $\frac{1}{8}$  of the people from City  $C$  and  $\frac{1}{3}$  of the people from City  $B$  work in city  $A$  as shown in the image. We also know that  $\frac{9}{20}$  of the people living in  $A$  work in other cities. We are already given the values of variables  $a$ ,  $b$ , and  $c$  as 100, 120, and 160 respectively. Plug the values it into the main equation like this:  $x = 160 \cdot \frac{1}{8} + 120 \cdot \frac{1}{3} + 100 - 100 \cdot \frac{1}{4} - 100 \cdot \frac{1}{5}$ . We solve for it and get our answer  $\boxed{\text{(D)} 115}$ .

**Remark**

- (i) <i>Main article: [Markov Chains](#)</i>

This model is known as the [Markov Chain](#), a type of stochastic process that models systems where the next state depends only on the current state, not on the sequence of events that preceded it. This is known as the Markov property (memoryless property).

18. (AMC 8 2025) The region between a circle of radius 1 and an inscribed square is shaded. Another circle has radius  $R$ . A quarter of the region between this second circle and its inscribed square is shaded. If the two shaded regions have the same area, what is the value of  $R$ ?



- (A)  $\sqrt{2}$     (B) 2    (C)  $2\sqrt{2}$     (D) 4    (E)  $4\sqrt{2}$

**Answer:** B

**Solution 1**

The area of the shaded region in the circle on the left is the area of the circle minus the area of the square, or  $(\pi - 2)$ . The shaded area in the circle on the right is  $\frac{1}{4}$  of the area of the circle minus the area of the square, or  $\frac{\pi R^2 - 2R^2}{4}$ , which can be factored as  $\frac{R^2(\pi - 2)}{4}$ . Since the shaded areas are equal to each other, we have  $\pi - 2 = \frac{R^2(\pi - 2)}{4}$ , which simplifies to  $R^2 = 4$ . Taking the square root, we have  $R = \boxed{\text{(B) } 2}$ .

**Solution 2 (Trig)**

The side length of the smaller square is  $\sqrt{2}$ , which follows from the Pythagorean Theorem since the square is inscribed in a circle of radius 1, making the diagonal equal to 2. Thus, the area of the square is  $(\sqrt{2})^2 = 2$ . The circle in which the square is inscribed has radius 1, so its area is  $\pi \cdot 1^2 = \pi$ . Therefore, the area of the shaded region (the part of the circle not covered by the square) is  $\pi - 2$ .

Now, consider the larger square, which is inscribed in a circle of radius  $R$ . This square can be divided into four congruent right triangles by drawing lines from the center of the circle to the vertices of the square. Each triangle has legs of length  $R$  and includes a  $90^\circ$  angle at the center. Using the triangle area formula  $\frac{1}{2}ab\sin(C)$  with  $a = b = R$  and  $C = 90^\circ$ , each triangle has area  $\frac{1}{2}R^2$ . Since there are four such triangles, the area of the square is  $4 \cdot \frac{1}{2}R^2 = 2R^2$ . The area of the circle is  $\pi R^2$ , so the area of the shaded region (the part of the circle outside the square) is  $\frac{1}{4}(\pi R^2 - 2R^2) = \frac{1}{4}R^2(\pi - 2)$ .

We are told that this shaded region also has area equal to  $\pi - 2$ , so we set up the equation  $\frac{1}{4}R^2(\pi - 2) = \pi - 2$ . Assuming  $\pi - 2 \neq 0$ , we divide both sides by  $\pi - 2$ , giving  $\frac{1}{4}R^2 = 1$ , which leads to  $R^2 = 4$ , and finally  $R = 2$ . Thus, the answer is  $\boxed{\text{(B) } 2}$ .

**Solution 3**

We start with the first area. Since the square is inscribed, its diagonal is 2  $\implies$  its side length is  $\sqrt{2} \implies$  its area is 2, therefore the first area is  $\pi - 2$ . The second area is  $\frac{R^2\pi - 2R^2}{4}$ , found in a similar manner. Writing and solving the equation, we have

$$\pi - 2 = \frac{R^2\pi - 2R^2}{4} \implies 4(\pi - 2) = R^2(\pi - 2) \implies R = 2.$$

The answer is  $\boxed{\text{(B) } 2}$  Tacos\_are\_yummy\_1

**Solution 4 (Using similarity)**

Since the two figures are similar, and the 4 regions in the first circle are equal in area to one of the four regions in the second circle, the second figure's area is 4 times the first figure's area. Therefore, the side lengths are multiplied by a factor of  $\sqrt{4} = 2$ , so the answer is  $1 \cdot 2 = \boxed{\text{(B)}2}$ .

**Solution 5**

(Really similar to Solution 1, but has a more larger explanation for the answer) First find the area of the smaller circle with the known radius. Since the radius is 1, the area of the circle is  $\pi$ , and finding the side length of the square would equal to  $\sqrt{2}$ . Squaring that gives us the area of the square which is 2, and the shaded area of the first one is equal to  $\pi - 2$ . The second area can be shortened to  $\frac{R^2\pi}{4} - \frac{(\sqrt{2R^2})^2}{4}$ , since we only need  $\frac{1}{4}$ th of the sections. We then make them equal, which makes this equation:

$$\pi - 2 = \frac{R^2\pi}{4} - \frac{(\sqrt{2R^2})^2}{4}$$

Simplifying this equation would be  $4\pi - 8 = R^2\pi - 2R^2$ . We can factor out the  $R^2$  which makes  $4\pi - 8 = R^2(\pi - 2)$ . Dividing out the  $\pi - 2$  would give us 4 on the left side.  $4 = R^2$  which leaves us to choice  $\boxed{\text{(B)} 2}$ . Imhappy62789

**Solution 6**

We are given two circles, each with an inscribed square. In the first circle, the radius is 1 unit, and the region between the circle and the inscribed square is shaded. In the second circle, one quarter of the region between the circle and the inscribed square is shaded. The shaded areas in both circles are the same. Our goal is to determine the radius  $R$  of the second circle.

**Step 1: Set up a ratio for the areas of the shaded regions**

We know that the shaded region in the first circle (with radius 1) and the shaded region in the second circle (with radius  $R$ ) are proportional in terms of the total area between the circle and the inscribed square. The shaded area in the second circle is one quarter of the total area between the circle and the square, while the shaded area in the first circle is the entire region.

**Step 2: Relate the areas to the radius**

Let's recall that the area between the circle and the inscribed square is proportional to the square of the radius of the circle because the square's side length scales with the radius of the circle.

So, in the first circle with radius 1:

$$\text{Area between circle and square} \propto 1^2 = 1$$

and the shaded area is proportional to the entire area between the circle and the square, i.e., 1.

In the second circle with radius  $R$ :

$$\text{Area between circle and square} \propto R^2$$

and the shaded area is one-quarter of this total area, so it is proportional to  $\frac{1}{4}R^2$ .

**Step 3: Set up the ratio of shaded areas**

We are told that the shaded areas in both circles are the same, so we set up the following equation:

$$1 = \frac{1}{4}R^2$$

**Step 4: Solve for  $R$**

Multiply both sides of the equation by 4:

$$4 = R^2$$

Taking the square root of both sides:

$$R = 2$$

Thus, the radius of the second circle is (B) 2.

**Solution 7**

Notice that if you shaded the rest of the region between the square and the circle, then the area would be 4 times the area of the shaded region in the smaller circle. Since the radius changes by a factor of  $k^2$ ,  $k^2 = 4$ , solving for gets you 2. Hence, the answer is

(B) 2

**Solution 8 (Very similar to the ones above)**

We want to find a quick and easy formula that can be applied to this (not necessarily a formula, but a rule). We take the area of the circle and subtract it from the area of the square and you will find the area of the 4 outer shaded regions. This rule is basically  $\pi r^2 - 2r^2 = r^2(\pi - 2)$ . We plug in the radius from the first figure and compute the answer to be  $\pi - 2$ . Since this represents the area of ONE region in the second figure, we have to multiply our result by four which gets us  $4\pi - 8$  or to make it easier, you can factor the four and make it  $4(\pi - 2)$ . Using our rule, we can set both equations equal to each other.

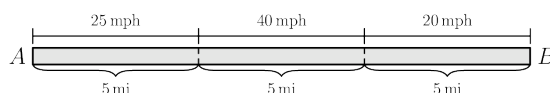
$$R^2(\pi - 2) = 4(\pi - 2) \quad R^2 = 4 \quad R = \text{(B)2}$$

**Solution 9 (Do Not Ever Use This)**

Both area formulas,  $\pi r^2$  and  $s^2$  are quadratic, therefore multiplying radius/side length by 2 gives 4x the area. The shaded region of the larger figure is 4 times the smaller, therefore the answer is (B)2.

(For anyone wondering the LaTeX for the answer box is putting this thing: "(AnswerLetter)AnswerNumber" between two *signs.*) *thethmathkidwoohooooo(plsdon'tusetthis)*

19. (AMC 8 2025) Two towns,  $A$  and  $B$ , are connected by a straight road, 15 miles long. Traveling from town  $A$  to town  $B$ , the speed limit changes every 5 miles: from 25 to 40 to 20 miles per hour (mph). Two cars, one at town  $A$  and one at town  $B$ , start moving toward each other at the same time. They drive at exactly the speed limit in each portion of the road. How far from town  $A$ , in miles, will the two cars meet?



- (A) 7.75      (B) 8      (C) 8.25      (D) 8.5      (E) 8.75

**Answer:** D

**Solution 1**

The first car, moving from town  $A$  at 25 miles per hour, takes  $\frac{5}{25} = \frac{1}{5}$  hours = 12 minutes. The second car, traveling a



20. (*AMC 8 2025*) Sarika, Dev, and Rajiv are sharing a large block of cheese. They take turns cutting off half of what remains and eating it: first Sarika eats half of the cheese, then Dev eats half of the remaining half, then Rajiv eats half of what remains, then back to Sarika, and so on. They stop when the cheese is too small to see. About what fraction of the original block of cheese does Sarika eat in total?

(A)  $\frac{4}{7}$       (B)  $\frac{3}{5}$       (C)  $\frac{2}{3}$       (D)  $\frac{3}{4}$       (E)  $\frac{7}{8}$

**Answer:** A

**Solution 1**

Let the total amount of cheese be 1. We will track the amount of cheese Sarika eats throughout the process.

**First Round:** Sarika eats half of the total cheese, so she eats:

$$\frac{1}{2}.$$

**Second Round:** Dev eats half of what remains after Sarika's turn, which is:

$$\frac{1}{4}.$$

**Third Round:** Rajiv eats half of the remaining cheese after Dev's turn, which is:

$$\frac{1}{8}.$$

At the end of the first round, the total cheese eaten is:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}.$$

We observe that Sarika's consumption follows a **geometric sequence**. In the first round, she eats  $\frac{1}{2}$ , and in subsequent rounds, she eats half of what remains from the previous round. This gives the following series for Sarika's total consumption:

$$\frac{1}{2} + \frac{1}{16} + \frac{1}{128} + \cdots$$

This is a geometric series with first term  $\frac{1}{2}$  and common ratio  $\frac{1}{8}$ . The sum  $S$  of this infinite geometric series is given by the formula:

$$S = \frac{a}{1 - r},$$

where  $a$  is the first term and  $r$  is the common ratio. Substituting  $a = \frac{1}{2}$  and  $r = \frac{1}{8}$ :

$$S = \frac{\frac{1}{2}}{1 - \frac{1}{8}} = \frac{\frac{1}{2}}{\frac{7}{8}} = \frac{1}{2} \times \frac{8}{7} = \frac{4}{7}.$$

Thus, Sarika eats  $\frac{4}{7}$  of the original block of cheese. The correct answer is:

$$\boxed{\text{(A)} \frac{4}{7}}.$$

**Solution 2 (Estimation)**

Sarika eats  $1/2$  of the original cheese, and then because the others eat  $1/4$  and  $1/8$ , she eats  $1/16$  next, and then  $1/128$ , and then so on. Since the values later are going to be too small to make a huge difference, we can use these 3 values. She ate  $(64 + 8 + 1)/128 = 73/128$ . We can replace the 73 with a 72 for now, so  $72/128 = 9/16$ , which simplifies to around 56.25. Since there is a little bit more of the cheese to be accounted for, the amount that she eats will be around  $\boxed{\text{(A)} \frac{4}{7}}$ .

### Solution 3 Speed

Her first 2 bites will be  $\frac{1}{2}$  and  $\frac{1}{16}$ . Everything else won't matter that much, so the answer will be closest to  $\frac{9}{16}, \frac{4}{7}$ .

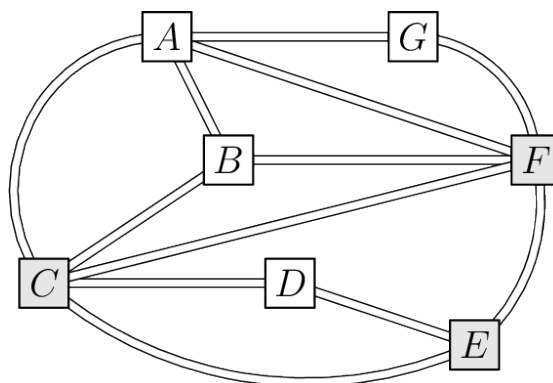
### Solution 4 (Basically Video Solution)

Let Sarika eat a total of  $x$  units of cheese. We can then see that Dev will eat a total of  $\frac{x}{2}$  units and Rajiv will eat  $\frac{x}{4}$  units. Our desired answer is  $\frac{x}{x + \frac{x}{2} + \frac{x}{4}}$ , or  $\boxed{\text{(A)} \frac{4}{7}}$ .

Key Idea: Let  $x$  be the fraction eaten by Sarika. Then Dev eats  $\frac{x}{2}$  and Rajiv eats  $\frac{x}{4}$ . Hence  $x + \frac{x}{2} + \frac{x}{4} = 1$  so solving for  $x$  we get

$$x = \boxed{\text{(A)} \frac{4}{7}}$$

21. (AMC 8 2025) The Konigsberg School has assigned grades 1 through 7 to pods A through G, one grade per pod. Some of the pods are connected by walkways, as shown in the figure below. The school noticed that each pair of connected pods has been assigned grades differing by 2 or more grade levels. (For example, grades 1 and 2 will not be in pods directly connected by a walkway.) What is the sum of the grade levels assigned to pods C, D, E, and F?



- (A) 12    (B) 13    (C) 14    (D) 15    (E) 16

Answer: A

### Solution 1

The key observation for this solution is to observe that pods C and F both have degree 5; that is, they are connected to five other pods. This implies that C and F must contain grades 1 and 7 in either order (if C or F contained any of grades 2, 3, 4, 5, or 6, then there would only be four possible grades for five pods, a contradiction). It does not

matter which grade  $C$  and  $F$  gets (see Solution 2 below), so we will assume that pod  $C$  is assigned grade 1, and pod  $F$  is assigned grade 7.

Next, pod  $D$  is the only pod which is not adjacent to pod  $F$ , so pod  $D$  must be assigned grade 6. Similarly, pod  $G$  must be assigned grade 2.

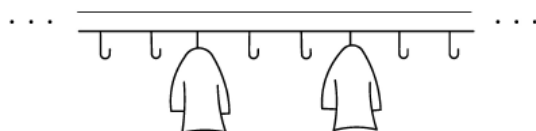
Lastly, we need to assign grades 3, 4, and 5 to pods  $A$ ,  $B$ , and  $E$ . Note that  $A$  and  $B$  are adjacent; therefore, pods  $A$  and  $B$  must contain grades 3 and 5. We conclude that  $E$  must be assigned grade 4. The requested sum is  $1 + 4 + 7 = \text{(A)} 12$ .

### Solution 2

Suppose that we replaced each label  $x$  with  $8 - x$ . That is, we replaced the grade  $a$  assigned to  $A$  with the grade  $8 - a$ , the grade  $b$  assigned to  $B$  with  $8 - b$ , and so on. We claim that if the initial labeling is valid (i.e., each pair of connected pods has grades differing by 2 or more), then so will the new labeling. This follows as if  $|x - y| \geq 2$ , then  $|(8 - x) - (8 - y)| = |-x + y| = |x - y| \geq 2$ . This shows that if there is an assignment  $(a, b, c, d, e, f, g)$  of grades to the pods, where  $\{a, b, \dots, g\} = \{1, 2, \dots, 7\}$ , then there is a second, valid assignment.

But because this is an AMC 8 problem and there is only one correct answer, we can conclude that  $c + e + f = (8 - c) + (8 - e) + (8 - f)$ . Simplifying and solving for  $c + e + f$  gives  $c + e + f = \text{(A)} 12$ .

22. (AMC 8 2025) A classroom has a row of 35 coat hooks. Paulina likes coats to be equally spaced, so that there is the same number of empty hooks before the first coat, after the last coat, and between every coat and the next one. Suppose there is at least 1 coat and at least 1 empty hook. How many different numbers of coats can satisfy Paulina's pattern?



- (A) 2      (B) 4      (C) 5      (D) 7      (E) 9

**Answer:** D

### Solution 1

Suppose there are  $c$  coats on the rack. Notice that there are  $c + 1$  "gaps" formed by these coats, each of which must have the same number of empty spaces (say,  $k$ ). Then the values  $k$  and  $c$  must satisfy  $c + k(c + 1) = 35 \implies kc + k + c = 35$ . We now use **Simon's Favorite Factoring Trick** as follows:

$$kc + k + c = 35$$

$$\implies kc + k + c + 1 = 36$$

$$\implies (k + 1)(c + 1) = 36$$

Our only restrictions now are that  $k > 0 \implies k + 1 > 1$  and  $c > 0 \implies c + 1 > 1$ . Other than that, each factor pair of 36 produces a valid solution  $(k, c)$ , which in turn uniquely determines an arrangement. Since 36 has 9 factors, our answer is  $9 - 2 = \boxed{\text{(D)} 7}$ . cxsmi

### Solution 2

Say Paulina placed  $n$  coats. That will divide the 35 hooks into  $n + 1$  spaces and  $35 - n$  empty hooks. Therefore,

$$n + 1 \mid 35 - n$$

The values of  $n$  that satisfy this are

$$n \in 1, 2, 3, 5, 8, 11, 17$$

The answer is (D) 7.

### Solution 3

Let the number of empty hooks before the first coat be denoted as  $n$ .

This means that there are  $n$  empty hooks between all the coats. In every situation, there are  $n + 1$  gaps, since  $n$  cannot be zero.

To solve the problem, subtract  $2n$  from 35 (accounting for the  $n$  empty hooks at the beginning and  $n$  empty hooks at the end). Then subtract 1 (for the first occupied hook). After that, divide the remaining number of hooks by  $n + 1$  (the number of hooks between coats). We do this because the first occupied hook is always after  $n$  hooks; we just need to check if the remaining hooks can be evenly divided by the number of gaps.

To clarify:

- If  $n = 1$ :

$$35 - 2n = 33$$

$$33 - 1 = 32$$

$$\frac{32}{n + 1} = 16$$

Since 16 is an integer, this situation satisfies the conditions.

- If  $n = 2$ :

We can skip the the third step, as each time you add 1 to  $n$ , it decreases by 2 from the previous situation.

$$\frac{30}{n + 1} = 16$$

This situation also satisfies the conditions.

Continue this process until the difference exceeds  $35 - (2n + 1)$ . We find that 7 situations satisfy the conditions. Therefore, the answer is 7.

*Note: This method is slow but easy to check and understand.*

### Solution 4 (Similar to Solution 1 but without Simon's Favorite Factoring Trick)

Suppose there are  $n$  occupied hooks on the rack. Observe that there are  $n + 1$  intervals, each of which must have the same number of empty hooks. Let's suppose  $m$  is the number of empty hooks plus the occupied hook to its right. Then, we have the equation

$$(n + 1)m - 1 = 35$$

$$\implies (n + 1)m = 36$$

Our restrictions are  $n > 0$  and  $m > 1$ . 36 has 9 factors, and two pairs do not meet the restriction. Our answer is then  $9 - 2 = \boxed{\text{(D)} 7}$ .

**Solution 5 (neat trick)**

Note that, for any possible valid arrangement of coats, there is always a set number of empty coat hangers between the coats. This is never 0, since then there would be no empty hangers on either side. So suppose there is  $n$  hangers between every coat. Then, there is one coat every  $n+1$  hangers. For any value of  $n+1$ , divide it into 35, the amount of hangers. If

$$35/(n+1)$$

has a remainder of  $n$ , then that would be a valid arrangement for that value of  $n+1$ . This works for values of 2, 3, 4, 6, 9, 12, and 18. This means there are  $\boxed{\text{(D)} 7}$  total possible arrangements.

-Arm123

23. (AMC 8 2025) How many four-digit numbers have all three of the following properties?

- (I) The tens and ones digit are both 9.
- (II) The number is 1 less than a perfect square.
- (III) The number is the product of exactly two prime numbers.

(A) 0      (B) 1      (C) 2      (D) 3      (E) 4

**Answer:** B

**Solution 1**

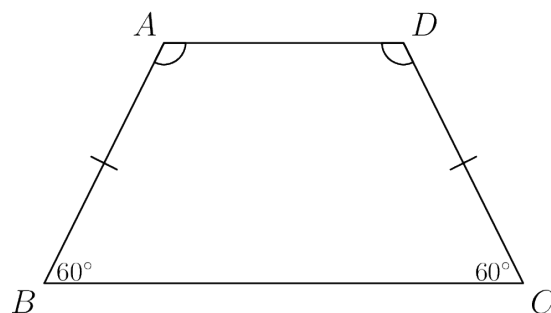
The "Condition (II)" perfect square must end in "00" because  $\dots 99 + 1 = \dots 00$ . *Condition (I)*. Four-digit perfect squares ending in "00" are 40, 50, 60, 70, 80, 90, which when squaring these numbers, gets you (1600, 2500, 3600, 4900, 6400, 8100) respectively.

*Condition (II)* also says the number is in the form  $n^2 - 1$ . By the **<b>Difference of Squares</b>**,  $n^2 - 1 = (n+1)(n-1)$ . Hence:

- $40^2 - 1 = (39)(41)$
- $50^2 - 1 = (49)(51)$
- $60^2 - 1 = (59)(61)$
- $70^2 - 1 = (69)(71)$
- $80^2 - 1 = (79)(81)$
- $90^2 - 1 = (89)(91)$

On this list, the only number that is the product of 2 prime numbers (condition 3) is  $60^2 - 1 = (59)(61)$ , so the answer is  $\boxed{\text{(B)} 1}$ .

24. (AMC 8 2025) In trapezoid  $ABCD$ , angles  $B$  and  $C$  measure  $60^\circ$  and  $AB = DC$ . The side lengths are all positive integers, and the perimeter of  $ABCD$  is 30 units. How many non-congruent trapezoids satisfy all of these conditions?



- (A) 0    (B) 1    (C) 2    (D) 3    (E) 4

**Answer:** E

**Solution 1**

Let  $a$  be the length of the shorter base, and let  $b$  be the length of the longer one. Note that these two parameters, along with the angle measures and the fact that the trapezoid is isosceles, uniquely determine a trapezoid. We drop perpendiculars down from the endpoints of the top base. Then the length from the foot of this perpendicular to either vertex will be half the difference between the lengths of the two bases, or  $\frac{b-a}{2}$ . Now, since we have a 30-60-90 triangle and this side length corresponds to the "30" part, the length of the hypotenuse (one of the legs) is  $2 \cdot \frac{b-a}{2} = b-a$ . Then the perimeter of the trapezoid is  $2(b-a) + a + b = 3b - a = 30$ . The only other stipulation for this trapezoid to be valid is that  $b > a$  (which was our assumption). We can now easily count the valid pairs  $(a, b)$ , yielding  $(3, 11), (6, 12), (9, 13), (12, 14)$ . It is clear that proceeding further would cause  $a \geq b$ , so we have **(E) 4** valid trapezoids. cxsmi

**Solution 2**

Let  $x$  be the length of  $AB$  and  $DC$ , and let  $b$  be the length of the shorter base. Because  $\angle B$  and  $\angle C = 60^\circ$ , the length of the longer base is  $b + \frac{x}{2} + \frac{x}{2} = b + x$ . Therefore, the perimeter is  $3x + 2b = 30$ . The number of positive integer pairs  $(x, b)$  is  $(2, 12), (4, 9), (6, 6), (8, 3)$ , meaning the answer is **(E) 4**.

way where you don't have to test pairs:

We can rearrange  $3x + 2b = 30$  to  $\frac{30-3x}{2} = b$ . We want  $\frac{30-3x}{2}$  to be a positive integer so  $b$  is a positive integer. 30 is a multiple of 3 and  $-3x$  is just subtracting multiples of 3's from 30 so every number on the numerator of  $\frac{30-3x}{2}$  is a multiple of 3 under 30 ( $x$  is also a positive integer so it can't be larger than 30 or not be a multiple of 3). For  $30 - 3x$  to be divisible by 2 it must be even, so we are finding even multiples of 3 under 30 which are 24, 18, 12, and 6. Each number will give us a valid  $x$  and a valid  $b$  pair giving us 4 solutions.

**Solution 3**

Let  $x$  be the length of the legs of the trapezoid. Draw the angle bisectors of the 120 degree angles. Now you have 2 equilateral triangles, and another figure in between them. Let  $x + a$  be the length of the shorter base of the trapezoid, and let  $2x + a$  be the length of the longer base. Since the perimeter of the trapezoid is 30,  $5x + 2a = 30$ . Since  $x > 0$ , and  $a$  can be negative as long as  $x + a > 0$ , you get 4 solutions for  $(x, a)$ , namely  $(2, 10), (4, 5), (6, 0), (8, -5)$ . Any solution with  $x \geq 10$  would lead to  $x + a \leq 0$ . Hence, the answer is **(E) 4**

**Solution 4**

Drop altitudes from angle  $A$  and angle  $D$ . Then, two  $30-60-90$  triangles and a rectangle are created. Let the hypotenuse of both of the triangles equal  $x$  and let side  $AD$  equal  $y$ . Then, the distance from  $B$  and  $C$  to the feet of the altitudes that are closest to them is  $\frac{x}{2}$ , and the distance between the feet of the two altitudes would be  $y$ . The rest is similar to Solution 2 Soupboy0

### Solution 5 (SIMILAR TO SOLUTION 4)

We give lengths to the base of the  $30-60-90$  triangles on both sides as the trapezoid as variable  $x$ . We give lengths of Line  $AB$  and the other side as variable  $y$ . When we add them, we have  $6x + 2y = 30$  from the sides. If we let  $x = 5$ , the other sides won't work so it has to be integers less than 5 which is 4 which is the answer.- Spacepandamath13

### Solution 6

We're given a **trapezoid  $ABCD$**  with these facts:

- Angles  $\angle B = \angle C = 60^\circ$
- $AB = DC$  (the legs are equal)
- All side lengths are **positive integers**
- **Perimeter** = 30 units
- We're to find how many **non-congruent** trapezoids satisfy all this.

—

#### Step 1: Understand the trapezoid

Because  $\angle B = \angle C = 60^\circ$ , and those angles are **adjacent to the bases**, the trapezoid is **isosceles**.

Let:

- $AB = DC = x$  (legs—equal)
- $AD = a$ ,  $BC = b$  (bases—not necessarily equal)
- So: **Perimeter** =  $a + b + 2x = 30$

—

#### Step 2: Use geometry

From the angle info, drop perpendiculars from points  $A$  and  $D$  to base  $BC$ . This creates two right triangles with angles of  $60^\circ$ , so the **height** of the trapezoid is:

$$\text{height} = x \cdot \sin(60^\circ) = x \cdot \frac{\sqrt{3}}{2}$$

And the **projection** of the leg on the base is:

$$\text{horizontal part} = x \cdot \cos(60^\circ) = x \cdot \frac{1}{2}$$

So the total difference between the longer base and shorter base is:

$$b - a = 2 \cdot \left( x \cdot \frac{1}{2} \right) = x \quad \Rightarrow \quad b = a + x$$

—





4

– ChatGPT 4

– Edited by JubilantJelly

### Solution 7

Explanations and Reasoning to Similar Solutions

Note that trapezoid  $ABCD$  is isosceles with  $B$  and  $C$  being the bottom 2 vertices. The shape can be dissected into 3 shapes with 2 congruent right triangles separated by a rectangle. This is done by dropping an altitude, a perpendicular point on the base from each of the 2 top vertices, named  $E$  and  $F$ , respectively below  $A$  and  $D$ . The property that angles  $\angle B$  and  $\angle C$  measure  $60^\circ$  is significant here, forming  $30-60-90$  triangles ( $\triangle ABE$  and  $\triangle DCF$ ). Now the assumed larger <sup>a</sup> side  $BC$  is separated into 3 sections by the 3 shapes. Or  $BC = BE + EF + CF$ . <sup>b</sup> Recall that  $\triangle ABE$  and  $\triangle DCF$  are  $30-60-90$  triangles, the side length ratios are  $1 : \sqrt{3} : 2$ . Naming the length of the shorter side of the right triangles as  $a$ , and the length of side  $AD$  as  $b$ , then the trapezoid's side lengths can be expressed in only 2 variables. Now knowing  $AD = b$ ,  $DC = 2a$  <sup>c</sup>,  $BC = a + b + a = 2a + b$ , and  $AC = 2a$ , the perimeter is  $b + 2a + 2a + b + 2a = 2b + 6a = 30$ . Simplifying,  $3a + b = 15$ , only 4 pairs of values satisfy this condition, namely  $(1, 12)$ ,  $(2, 9)$ ,  $(3, 6)$ ,  $(4, 3)$ .

Thus there (E) 4 possible non-congruent trapezoids.

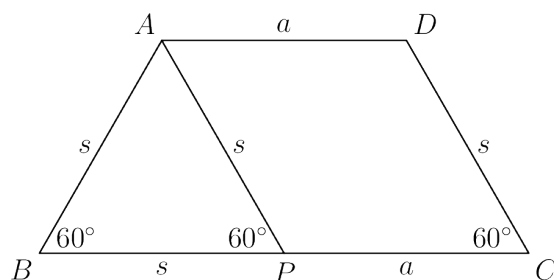
<sup>a</sup> The question asks for non-congruent trapezoids, and rotation preserves congruence so the assumption is always true WLOG.

<sup>b</sup> The shortest side of  $\triangle ABE$ , the length of  $AD$  (shape  $ADFE$  is a rectangle so  $FE = AD$ ), and the shortest side of  $\triangle DCF$ .

<sup>c</sup>  $AB = DC = 2a$ ,  $AB$  and  $DC$  are the hypotenuse of triangles  $\triangle ABE$  and  $\triangle DCF$ . moracle

### Solution 8

Let  $a = AD$  and  $s = AB = DC$ . Draw a segment from  $A$  to  $\overline{BC}$  parallel to  $\overline{DC}$ , meeting  $\overline{BC}$  at point  $P$ , forming parallelogram  $ADCP$ . It follows that  $AP = DC = s$  and  $PC = AD = a$ . Isosceles  $\triangle ABP$  has base angles measuring  $60^\circ$ , so it is equilateral and  $BP = s$ .



Then the perimeter of  $ABCD$  is  $L = 2a + 3s = 30$ . To satisfy this equation with positive integer values, the number  $a$  must be divisible by 3 because  $2a = 3(10 - s)$ . The only possible values for  $a$  are 3, 6, 9, and 12, each value corresponding to a different trapezoid. Greater multiples of 3 will not lead to a positive integer value for  $s$ . Therefore there are 4 different trapezoids that satisfy the conditions.

The table below lists the side lengths  $a$ ,  $s$ , and  $a + s$  for these 4 trapezoids, along with their perimeters.

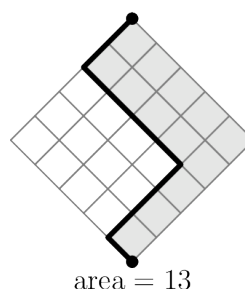
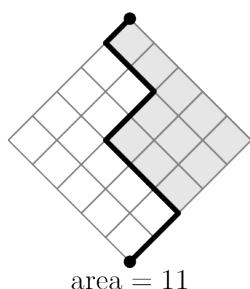
| $a$ | $s$ | $a + s$ | $L = 2a + 3s$                 |
|-----|-----|---------|-------------------------------|
| 3   | 8   | 11      | $2 \cdot 3 + 3 \cdot 8 = 30$  |
| 6   | 6   | 12      | $2 \cdot 6 + 3 \cdot 6 = 30$  |
| 9   | 4   | 13      | $2 \cdot 9 + 3 \cdot 4 = 30$  |
| 12  | 2   | 14      | $2 \cdot 12 + 3 \cdot 2 = 30$ |

**Solution 9 (Similar to Solutions 4 and 5)**

We can draw altitudes from A and B and create a rectangle and 2  $30 - 60 - 90$  triangles. Two sides of each triangle are on the trapezoid. We can label each of the hypotenuses of the small  $30 - 60 - 90$  triangles as  $2a$  and each of the short sides of the triangles as  $a$ . Then, the perimeter of the trapezoid becomes  $2(2a + a) + a$  positive integer (we'll call this integer  $n$ )  $= 30$ . Simplifying, we get  $6a + n = 30$ , and since  $n$  is not 0,  $a$  must be a positive integer less than 5, which yields  $\boxed{4}$  options: 1, 2, 3, and 4.

**solution 10** Extend lines AB and CD until they intersect. Calling that point E, you can find out that triangle EBC is equilateral because the angles at B and C are 60 degrees. Assume that side  $BC = a$ , then BE and CE are also equal to  $a$ . Assuming that  $AE = x$ , then DE must also be  $x$  because triangle AED is equilateral. This also means that  $AD = x$ . Since  $CD = CE - DE$ , we have  $CD = a - x$ . Similarly,  $AB = a - x$  because  $AB = CD$ . This means the perimeter of the trapezoid ABCD is  $a + (a - x) + (a - x) + x$ . The problem tells us that the perimeter is 30 so we get the equation  $3a - x = 30$ . Adding  $x$  and dividing by 3 you get  $a = 10 - x/3$ . Since  $x$  is a side length, it must be an integer. That means  $x$  can any integer between 0 and 30 exclusive. Side length  $a$  must also be an integer, that means  $x$  must be a multiple of 3. So  $x$  can be 3, 6, 9, 12, 15, 18, 21, 24, 27. From the image given to us, we see that  $BC = a$  is greater than  $AD = x$ . That means  $x$  can't be greater than or equal to 15 because that would cause  $x$  to be greater than  $a$ . This leaves us with 4 values of  $x$  which are 3, 6, 9, 12. That means there are 4 non-congruent trapezoids that satisfy those conditions. This means the answer is  $\boxed{\text{(E)} 4}$ .

25. (AMC 8 2025) Makayla finds all the possible ways to draw a path in a  $5 \times 5$  diamond-shaped grid. Each path starts at the bottom of the grid and ends at the top, always moving one unit northeast or northwest. She computes the area of the region between each path and the right side of the grid. Two examples are shown in the figures below. What is the sum of the areas determined by all possible paths?



- (A) 2520      (B) 3150      (C) 3840      (D) 4730      (E) 5050

**Answer:** B

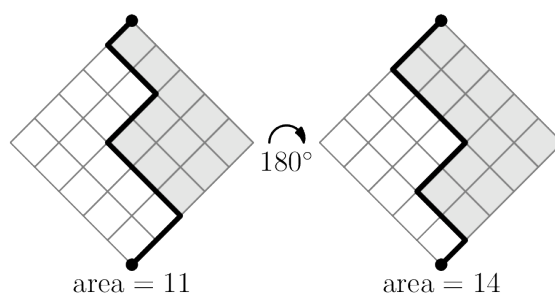
**Solution 1**

**Step 1:** To find the total number of paths, observe that all paths will have 10 total steps. We have to choose which 5 of these steps will be NE (the rest will be NW). So

the total number of paths is  $\binom{10}{5}$ . The formula for **combinations** is:  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$  and  $\binom{10}{5} = \frac{10!}{5! \times 5!} = 252$ .

**Step 2:** Each path splits the total area of 25 in two parts. So, for any path that gives area =  $A$ , you can find a unique "*sister*" path that has an area =  $25 - A$  (in other words, the pair of paths have a combined area of 25). Possible ways to define the "*sister*" path are:

- Rotate the entire grid  $180^\circ$



- Swap each step of the original paths (for example, each NW becomes NE) (this is a reflection over the diagonal)

**Step 3:** There are a few ways to get from this observation to the total area:

- There are  $252/2 = 126$  pairs of such paths, and the total area of each pair is 25. So the total area given by all paths is  $126 \times 25$ .
- Each of the 252 paths gives an area of 25 if you also count the "*sister*" paths. Since each "*sister*" path is also one of the 252, you have to divide by 2 to avoid double counting. So the total area given by all paths is  $\frac{252 \times 25}{2}$ .
- Note that the average area of two "*sister*" paths is  $\frac{25}{2}$ , so you can think about every path having this area *on average*. So the total area given by all paths is  $252 \times \frac{25}{2}$ .

The final answer is **(B) 3150**.

**Note:** This problem has a bijection (or 1-1 correspondence), check out **Intermediate Counting Probability, Chapter 4**, and **Introduction to Counting Probability, Chapter 5**

### Solution 2

If we test this problem on a smaller  $2 \times 2$  diamond, we have 6 ways to go from  $A$  to  $B$ , and the total area is  $0 + 1 + 2 + 2 + 3 + 4 = 12$ , so the average area is  $\frac{12}{6} = 2$ , which is also the area of the diamond  $2 \times 2 = 4$  divided by 2. If we assume this is true for a  $5 \times 5$  diamond, then the average area is  $\frac{25}{2}$ . The number of paths from  $A$  to  $B$  is  $\binom{10}{5} = 252$ , and  $252 \cdot \frac{25}{2} = \mathbf{(B) 3150}$ .

This is technically flawed, since the area of a diamond is length times width.

### Solution 3 Easier to Motivate

Most other solutions don't explain how they got all the cases, as well as require an insight that's somewhat hard to think of, so I'll explain another in detail.

If we rotate the grid 45 degrees clockwise we can have a  $5 \times 5$  grid where we can move up and right. There can only be one horizontal line segment in each column at a certain  $y$  coordinate. We'll denote the  $y$  coordinates as 0, 1, 2, 3, 4, 5. Once we choose the  $y$

coordinates for each column, we have a unique path. However, we can't move down which means that columns cannot have a higher y coordinate than the ones to the right. This is the same as distributing 5 balls in six boxes labeled 0, 1, 2, 3, 4, 5. For example, if we get 2 in 0, 2 in 1, and 1 in 4, then the order would be 0, 0, 1, 1, 4. This is one unique path, and the total number of paths is represented by  $6+5-1$  choose  $6-1 = 252$ . Also, the sum of the y-coordinates represents the area, which means we want the average sum of the y-coordinates. This is  $5 * (0 + 5)/2 = 25/2$ , and  $25/2 * 252 = \boxed{3150}$ .

#### Solution 4

As found, there are 252 total paths. However, we can take advantage of symmetry here. We consider the total area of the left side of the grid in each of the 252 paths. Since the total area of the left side of the grid in each of the 252 paths is equal to the total area of the right side of the grid in each of the 252 paths, and the two total areas sum to  $(252)(25) = 6300$ , then the total area of the right side of the grid is  $6300/2 = \boxed{3150}$ .

#### Solution 5 (maybe cheese)

We notice that there are  $\binom{10}{5} = 252$  different paths on the grid. We also notice that the average area enclosed by a path is  $\frac{25}{2}$ , the area of the square divided by 2. This means that the answer is  $252 \cdot \frac{25}{2} = \boxed{3150}$ .

#### Solution 6 (q-analog)

q-analog basic definitions are for  $k \in \mathbb{N}$ , then

$$[k]_q = 1 + q + \dots + q^{k-1}$$

with

$$[0]_q = 1$$

The q-factorial of  $n \in \mathbb{N}$  is defined as

$$[n]!_q = [n]_q \cdot [n-1]_q \cdots [0]_q$$

The q-binomial theorem states

$$\binom{n}{k}_q = \frac{[n]!_q}{[k]!_q [n-k]!_q}$$

and finally for q-analog of lattice paths aka q-lattice path, we have the q-lattice path theorem which states for any  $a, b \in \mathbb{N}$ ,

$$\sum_{P \in L(a,b)} q^{\text{area}(P)} = \binom{a+b}{a}_q$$

To get the sum of the areas we have to differentiate the q-Lattice Path and evaluate at  $q = 1$ . For this problem our lattice path is  $L(5, 5)$  and we calculate the q-binomial as  $q^{25} + q^{24} + 2q^{23} + 3q^{22} + 5q^{21} + 7q^{20} + 9q^{19} + \dots$  and take the derivative  $\frac{d}{dq} \binom{10}{5}_q \Big|_{q=1}$  to get  $25q^{24} + 24q^{23} + 46q^{22} + 66q^{21} + 105q^{20} + 140q^{19} + 171q^{18} + \dots$  and then plug in the 1 to finally get  $\boxed{3150}$ . Lopkiloim

shoutout to <https://enumeration.ca/>

#### Solution 7 (Rigorous Representation Theoretical)

Let  $\mathcal{P}_{a,b}$  be the set of lattice paths from  $(0,0)$  to  $(a,b)$  using north steps  $N = (0,1)$  and east steps  $E = (1,0)$ . There are

$$|\mathcal{P}_{a,b}| = \binom{a+b}{a}$$

such paths.

For a path  $P \in \mathcal{P}_{a,b}$  list its steps as the word  $\mathbf{w} = w_1 \cdots w_{a+b}$  with each  $w_i \in \{E, N\}$ . Define the lattice area

$$\text{area}(P) = \sum_{i=1}^a y_i(P),$$

where  $y_i(P)$  is the  $y$ -coordinate just before the  $i$ -th  $E$ . Equivalently,

$$\text{area}(\mathbf{w}) = \iota(\mathbf{w}) := |\{(i, j) : i < j, w_i = E, w_j = N\}|.$$

The symmetric group  $S_{a+b}$  acts by permuting positions of the word, and the action is transitive; in fact

$$\mathcal{P}_{a,b} \cong S_{a+b}/(S_a \times S_b), \quad M := \mathbb{C}[\mathcal{P}_{a,b}] \cong \text{Ind}_{S_a \times S_b}^{S_{a+b}} \mathbf{1}.$$

Fix an ordered pair of positions  $(i, j)$  with  $i < j$ . Among all words, exactly half have  $(w_i, w_j) = (E, N)$  and half have  $(N, E)$ . Hence

$$\sum_{\mathbf{w} \in \mathcal{P}_{a,b}} \iota(\mathbf{w}) = \binom{a+b}{2} \binom{a+b-2}{a-1} = \frac{ab}{2} \binom{a+b}{a},$$

since for each ordered pair  $(i, j)$  we set  $w_i = E, w_j = N$  and distribute the remaining  $a-1$   $E$ 's and  $b-1$   $N$ 's among the other  $a+b-2$  positions in  $\binom{a+b-2}{a-1}$  ways.

Therefore

$$\sum_{P \in \mathcal{P}_{a,b}} \text{area}(P) = \binom{a+b}{a} \frac{ab}{2}$$

and the average lattice area under such paths is  $ab/2$ .

### Solution 8 (bash)

Find every single case and add them up, getting 3150

It's an annoying, but it's better than doing nothing right? (Yes I did this in-contest) (No you didn't)

## §41 AMC 8 2026

1. (AMC 8 2026) What is the value of the following expression?

$$1+2-3+4+5-6+7+8-9+10+11-12 \quad \text{(A) } 18 \quad \text{(B) } 21 \quad \text{(C) } 24 \quad \text{(D) } 27 \quad \text{(E) } 30$$

**Answer:** A

### Solution 1

We can use mental calculations to get  $1+2-3+4+5-6+7+8-9+10+11-12$   
 $= \boxed{\text{(A)} 18}.$

**Simple trick** To solve the expression  $1 + 2 - 3 + 4 + 5 - 6 + 7 + 8 - 9 + 10 + 11 - 12$ , we can define a function  $f(x)$  whose derivative  $f'(x)$  evaluated at  $x = 1$  yields the original sum. By grouping the terms into blocks of three, we have:

$$f(x) = (x + x^2 - x^3) + (x^4 + x^5 - x^6) + (x^7 + x^8 - x^9) + (x^{10} + x^{11} - x^{12})$$

$$f(x) = (x + x^2 - x^3)(1 + x^3 + x^6 + x^9)$$

Let  $u(x) = x + x^2 - x^3$  and  $v(x) = 1 + x^3 + x^6 + x^9$ . Using the Product Rule,  $f'(x) = u'(x)v(x) + u(x)v'(x)$ . Now we evaluating these components at  $x = 1$ :

$$u(1) = 1 + 1 - 1 = 1 \quad u'(1) = 1 + 2(1) - 3(1)^2 = 0 \quad v(1) = 1 + 1 + 1 + 1 = 4 \\ v'(1) = 3(1)^2 + 6(1)^5 + 9(1)^8 = 18$$

Substituting these values into the product rule gives:

$$f'(1) = (0 \cdot 4) + (1 \cdot 18) = \boxed{18}$$

### Solution 2

$$1 + 2 - 3 + 4 + 5 - 6 + 7 + 8 - 9 + 10 + 11 - 12 = \boxed{18}$$

### Solution 3

Note that  $2 - 3 = 5 - 6 = 8 - 9 = 11 - 12 = -1$ , so we have a commutative sum of  $-4$ . Then, the sum is just  $(1 + 4 + 7 + 10) - 4 = 1 + 7 + 10 = \boxed{18}$ .

### Solution 4 (Break the problem into chunks!)

Notice that we can break this up into chunks of 3 numbers.  $1 + 2 - 3 = 0$ ,  $4 + 5 - 6 = 3$ ,  $7 + 8 - 9 = 6$ , and  $10 + 11 - 12 = 9$ , these are all multiples of 3 so we get  $3 + 6 + 9 = 3 \times 1 + 3 \times 2 + 3 \times 3 = 3 \times (1 + 2 + 3) = 3 \times 6 = \boxed{18}$

### Solution 5

$$1 + 2 - 3 + 4 + 5 - 6 + 7 + 8 - 9 + 10 + 11 - 12 = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + \\ 9 + 10 + 11 + 12 - 3 \times 2 - 6 \times 2 - 9 \times 2 - 12 \times 2 = \frac{12 \times 13}{2} - 3 + 6 + 9 + 12 \times 2 = \\ 78 - 60 = 18, \text{ so our answer is } \boxed{\text{(A)}18}.$$

### Solution 6 (CRITICAL THINKING!!)

Notice that  $1 + 2 - 3$  and  $4 + 5 - 6$  have one similarity; the latter is the former plus  $3 + 3 - 3 = 3$ , because  $(1 + 3) + (2 + 3) - (3 + 3)$ . This means that expression 2 is  $3 + 3 - 3 = 3 + 0 = 3$  greater than the first, which is 3. By trivial,  $1 + 2 - 3 = 0$ . This means that the sum can be expressed as  $1 + 2 - 3 + (1 + 3) + (2 + 3) - (3 + 3)$ . This can even be extended further to the 7+8-9 part and the 10+11-12 part, so the expression can simplify down to  $0 + 3 + 3 - 3 + 6 + 6 - 6 + 9 + 9 - 9$ , or  $0 + 3 + 6 + 9 = \boxed{18}$ . You can also use the pattern for the negatives. For example  $1 - 2 = -1$  the negative one can really help.

### Solution 7 (Cancellation)

We see that many of the numbers in the equation can be cancelled, so we take:

$$1 + 2 - 3 = 0 \quad 4 + 5 - 9 = 0 \quad -6 - 12 + 8 + 10 = 0$$

Which leaves...

$$7 + 11 = \boxed{18}$$

This is a very simple method for quick, concise mental math, especially on pencil and paper tests.

**Solution 8 (Grouping by -1)**

You can see that from the question, you can group like:

$$1 + (2 - 3) + 4 + (5 - 6) + 7 + (8 - 9) + 10 + (11 - 12)$$

Which simplifies to:

$$1 - 1 + 4 - 1 + 7 - 1 + 10 - 1$$

And quickly, we can group it into:

$$(1 - 1) + (4 - 1) + (7 - 1) + (10 - 1)$$

and is equal to...

$$3 + 6 + 9$$

And then the answer is 18

2. (AMC 8 2026) In the array shown below, 3s are surrounded by 2s, which are then surrounded by a border of 1s. What is the sum of the numbers in the array?

$$\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 & 2 & 1 \\ 1 & 2 & 3 & 3 & 3 & 2 & 1 \\ 1 & 2 & 2 & 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{array}$$

- (A) 49      (B) 51      (C) 53      (D) 55      (E) 57

**Answer:** C

**Solution 1**

Notice that if we label the rows  $A_1, A_2, A_3, \dots, A_5$  top down, then, if we denote  $S(A_n)$  to be the sum of the  $A_n$  row,  $A_1 = A_5$ ,  $A_2 = A_4$ , and  $A_3 = A_3$  through symmetry. Thus the total sum is  $2A_1 + 2A_2 + A_3$ .

Then, it is obvious that the sum of  $A_1 = 7$ , the sum of  $A_2 = 2 + 2(5) = 12$ , and the sum of  $A_3 = 2(3) + 3(3) = 5(3) = 15$ . The answer is then  $2 \cdot 7 + 2 \cdot 12 + 15 = 14 + 15 + 24 = 29 + 24 = \text{(C) } 53$ .

**Solution 2 (brute force)**

You can just add up all the numbers in the array, giving (C) 53. Baihly2024

**Solution 3**

We have 20 ones, 12 twos, and 3 threes, so we have  $20 \times 1 + 12 \times 2 + 3 \times 3 = \text{(C) } 53$  as our answer.

**Solution 4**

Looking at the answer choices, we observe that each has a different units digit. Therefore, it is sufficient to determine the sum modulo 10.

There are two rows consisting of five 2's each (the second and fourth rows), whose total contribution is  $(2 + 2 + 2 + 2 + 2) + (2 + 2 + 2 + 2 + 2) = 20 \equiv 0 \pmod{10}$ .

In the middle row, four of the entries sum to  $2 + 2 + 3 + 3 = 10 \equiv 0 \pmod{10}$ , leaving one remaining 3.

The border of the array consists of 20 entries equal to 1, whose sum is  $20 \equiv 0 \pmod{10}$ .

Thus, the entire sum is congruent to 3 (mod 10). Among the answer choices, only (C) 53 ends in 3.

Therefore, the sum of the numbers in the array is 53.

**Solution 5**

The sum of the entries in this array can be found by counting each entry of the whole  $5 \times 7$  array once, each entry of the inner  $3 \times 5$  array a second time, and finally each entry of the innermost  $1 \times 3$  array a third time. This gives a total sum of  $5 \times 7 + 3 \times 5 + 1 \times 3 = 35 + 15 + 3 =$  (C) 53

-j314andrews

**Solution 6 (smart application of brute force)**

We can see there are 20 '1's, so we get  $20 \cdot 1 = 20$ . Then, we notice there are 12 '2's, yielding a total of  $12 \cdot 2 = 24$ . Finally, we notice there are 3 '3's, yielding a total of  $3 \cdot 3 = 9$ . Adding all sums up, we arrive at a grand total of  $20 + 24 + 9 = 53$ . So our answer is (C) 53.

**Solution 7 (Formula)**

In this solution we use a formula to find the sum of all entries in the array:

R = Number of rows (5)

C = Number of Columns (7)

S = Difference between each inlaid ring of numbers ( $3 - 2 = 1, 2 - 1 = 1$ , so  $s = 1$ )

T = starting number: the outer ring value (1)

$(RC - (R - 2)(C - 2))T + ((R - 2)(C - 2) - (R - 4)(C - 4))(T + 1) + ((R - 4)(C - 4) - (R - 6)(C - 6))(T + 2)$

This of course equals (C) 53.

You could undoubtedly simplify this using more variable, but I find this a straightforward sequence for these purposes. I understand the correlation between T, S, and the number that you subtract from the variables R and C, so this could be an opportunity for another simplification.

I suggest you try this with more arrays to possibly come up with an easier formula.

3. (AMC 8 2026) Haruki has a piece of wire that is 24 centimeters long. He wants to bend it to form each of the following shapes, one at a time.
- A regular hexagon with side length 5 cm.
  - A square of area  $36 \text{ cm}^2$ .
  - A right triangle whose legs are 6 and 8 cm long.

Which of the shapes can Haruki make?

- (A) Triangle only      (B) Hexagon and square only  
 (C) Hexagon and triangle only      (D) Square and triangle only  
 (E) Hexagon, triangle, and square

**Answer:** D

**Solution 1**

To solve this problem we can find the perimeter of each of the given spaces and check if it is less than 24.



The first shape is a regular hexagon with side length 5. The hexagon has a perimeter of  $6 \cdot 5 = 30$  centimeters. The second shape is a square with area 36 or a side length of 6. Then the perimeter of all 4 sides is 24 centimeters. Finally, we need to find the perimeter of a right triangle with legs 6 and 8 centimeters. The hypotenuse of the right triangle is 10 so the total perimeter is  $6 + 8 + 10 = 24$  centimeters.

Thus, only the square and the triangle can be made with 24 centimeters of wire so the answer is **(D) Square and triangle only**.

**Solution 2 (Easy and simple, no need to know the sides of the triangle!)**

First, consider the regular hexagon. A regular hexagon with side length 5 cm has perimeter  $6 \cdot 5 = 30$  cm, which exceeds the available 24 cm of wire. Therefore, the hexagon cannot be formed, eliminating answer choices **(B)**, **(C)**, and **(E)**.

Next, consider the square. A square with area  $36 \text{ cm}^2$  has side length  $\sqrt{36} = 6$  cm, so its perimeter is  $4 \cdot 6 = 24$  cm. Thus, the square can be formed.

At this point, the only remaining answer choices are **(A)** and **(D)**. But since **(A) Triangle only** does not include the square and **(D)** does, the correct answer must be **(D)**.

Therefore, the shapes Haruki can make are the square and the triangle, and the answer is **(D) Square and triangle only**. Conveniently, the triangle is listed last, so the shapes can be evaluated in the given order.

4. (AMC 8 2026) Brynn's savings decreased by 20% in July, then increased by 50% in August. Brynn's savings are now what percent of the original amount?
- (A)** 80      **(B)** 90      **(C)** 100      **(D)** 110      **(E)** 120

**Answer:** E

**Solution 1**

Let  $x$  be Byron's original amount. We see that Byron had  $0.8x$  in July. Then, he increased his savings by HALF of how much he had as of July which is  $0.4x$ . Adding  $0.4x + 0.8x$ , we get  $1.2x$ , so his August savings are **(E) 120**% of  $x$ .

We see that the change is  $(1 - 0.2)(1 + 0.5) = 0.8 \cdot 1.5 = 1.2$ . That is an increase of 20%, so his August savings are **(E) 120**% of his original savings.

**Solution 2**

Let's assume Brynn has \$100 at the start. In July, he lost 20%, so he lost \$20, so he has  $100 - 20 = 80$  dollars left. 50 percent of 80 dollars is 40 dollars, so he gained 40 dollars in August, so now he has  $80 + 40 = 120$  dollars, which is 120% of 100 dollars, so our answer is **(E) 120**%.

We can also do it just with percentages. He has 100% then loses 20%, meaning he now has 80%. Then he gains 50 percent of our NEW amount (which is 80 percent) and 50 percent of 80 is 40 percent, so we add 40 and 80 to get our answer, **(E) 120**%.

You can also do it with fractions. You can say you have  $\frac{100}{100}$  at the beginning, then losing 20% would result in  $\frac{80}{100}$ . 50% of  $\frac{80}{100}$  is  $\frac{40}{100}$ . Adding  $\frac{40}{100}$  to the original  $\frac{80}{100}$  gives  $\frac{120}{100}$ , which is **(E) 120**% of  $\frac{100}{100}$ .

**Solution 3**

Let Brynn's savings be  $x$ . Then, in July from  $x$  his savings went down  $x - \frac{1}{5}x = \frac{4}{5}x$ . From that amount, it increased 50% in August, meaning we have  $\frac{8}{10}x + \frac{4}{10}x = \frac{12}{10}x = \frac{6}{5}x$  remaining. Of the total  $x$ , which is  $\frac{6}{5}x$ , we see that the  $x$  cancel out and the increase is  $\frac{6}{5}$ , or  $\boxed{120}\%$

5. (AMC 8 2026) Casey went on a road trip that covered 100 miles, stopping for a lunch break along the way. The trip took 3 hours in total and her average speed while driving was 40 miles per hour (mph). In minutes, how long was the lunch break?
- (A) 15      (B) 30      (C) 40      (D) 45      (E) 60

**Answer:** B

**Solution**

The average speed is  $\frac{\text{Total distance}}{\text{Total time}}$ , or  $\frac{100}{3-t}$ . Thus  $\frac{100}{3-t} = 40 \Rightarrow 3 - t = \frac{5}{2} \Rightarrow t = \frac{1}{2}$ , which is  $\boxed{\text{(B) } 30}$  minutes.

nah we can just do it like this. 100 miles at 40 miles per hour means we can do 1 mile in 1.5 minutes. so we multiply 100 by 1.5 to get 150 minutes or 2 hours and 30 minutes. (this is how long it took her to drive, and if the total time is three hours we do 3 minus 2.5 to get our answer  $\boxed{\text{(B) } 30}$  minutes.

thanks pookies pereena

**Solution 2**

Average speed is total distance over total time. Thus, the total time spent driving is  $t$  and the total distance is 100, the average speed is  $100/t = 40$ . Solving, we have that  $t = 100/40 = 10/4 = 5/2$  hours, or 2 hours and 30 minutes. She drove for 3 hours, and thus her lunch break was  $\boxed{30}$  minutes.

**Solution 3**

If the trip took 3 hours in total and her average speed was 40 miles per hour, we know she would have traveled 120 miles. Since she only traveled 100 miles with a lunch break, she would have traveled 20 miles if she didn't have a lunch break. Since her average speed was 40 miles per hour, the lunch break took  $60 * \frac{20}{40}$  minutes. So the answer is  $\boxed{30}$  minutes.

**Solution 4**

Since Casey drove 100 miles at a constant rate of 40 mph, we can say it took her  $100/40 = 2.5$  hours to drive to his/her destination WITHOUT a break. The problem mentioned the trip took 3 hours, so  $3 - 2.5$  hours yields a total of 0.5 hours or 30 minutes, so the answer is  $\boxed{\text{(B) } 30}$ .

**Solution 5**

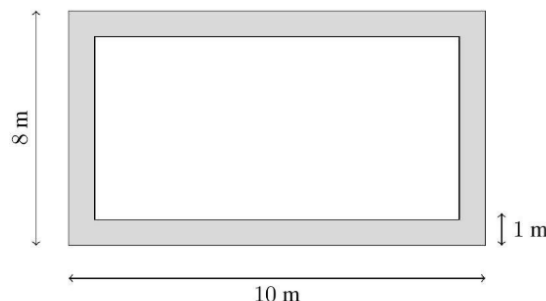
Using dimensional analysis, her time spent driving is

$$100 \text{ miles} \cdot \frac{1 \text{ hour}}{40 \text{ miles}} = 2.5 \text{ hours.}$$

Therefore, she spent  $3 - 2.5 = 0.5$  hours during her lunch break or

$$0.5 \text{ hours} \cdot \frac{60 \text{ minutes}}{1 \text{ hour}} = \boxed{\text{(B) } 30} \text{ minutes.}$$

6. (AMC 8 2026) Peter lives near a rectangular field that is filled with blackberry bushes. The field is 10 meters long and 8 meters wide, and Peter can reach any blackberries that are within 1 meter of an edge of the field. The portion of the field he can reach is shaded in the figure below. What fraction of the area of the field can Peter reach?



- (A)  $\frac{1}{6}$     (B)  $\frac{1}{4}$     (C)  $\frac{1}{3}$     (D)  $\frac{3}{8}$     (E)  $\frac{2}{5}$

**Answer:** E

**Solution**

We can calculate this reachable area to be Total area – Unreachable area. The total area is  $8 \cdot 10 = 80$ , and the unreachable area is  $(8 - 2)(10 - 2) = 6 \cdot 8 = 48$ .

Thus, our answer is  $\frac{80-48}{80} = \frac{32}{80} = \boxed{\text{(E)} \frac{2}{5}}$ .

**Solution 2**

Each "region" is one unit from an edge. Thus, let  $l$  be the length and  $w$  be the width of the unshaded region. Then, it is obvious that  $l = 10 - 2 = 8$  and  $w = 8 - 2 = 6$ , and the unshaded regions dimensions are  $8 \times 6$ .

Then, the area of the shaded region is the area of the total region subtracted by the area of the unshaded region, or  $80 - 48 = 32 \text{ un}^2$ . Then, the ratio is simply  $32/80$ , or  $\boxed{\frac{2}{5}}$ .

**Solution 3**

By rearranging the diagram, the unreachable (white) region can be moved to the lower-left corner without changing the shaded area. The reachable region then forms an L-shaped figure composed of two rectangles.

One rectangle has dimensions  $8 \times 2$  and the other has dimensions  $10 \times 2$ . These rectangles overlap in a  $2 \times 2$  square, which must be subtracted once.

Thus, the area of the shaded region is  $(8 \cdot 2) + (10 \cdot 2) - (2 \cdot 2) = 2(10 + 8 - 2)$ .

The total area of the field is  $10 \cdot 8 = 80$ . Dividing, the fraction of the field Peter can reach is  $\frac{2(10 + 8 - 2)}{80} = \frac{10 + 8 - 2}{40} = \frac{16}{40} = \boxed{\text{(E)} \frac{2}{5}}$ .

**Solution 4**

We can first start off by finding the unreachable area. The problem tells us that the distance between the reachable and unreachable area is 1, and since there are 2, we get the dimensions of the unreachable to be  $8 \cdot 6$ , because  $8 = 10 - 2$ , and  $6 = 8 - 2$ . Therefore, the area of the unreachable is 48. However, the problem asks

us to find the fraction of area he can reach, so we need to get area of the entire field and subtract by 48. Since the entire field is  $8 \cdot 10 = 80$ , we get  $80 - 48 = 32$ . So the area of the part Peter can reach is 32. Finally, to convert this to a fraction of the entire field, we can do  $\frac{32}{80}$ , so our answer is  $\frac{32}{80} = \boxed{\text{(E)} \frac{2}{5}}$ .

To add on, its just like calculating the total area- the unwanted rectangular area.

7. (AMC 8 2026) Mika would like to estimate how far she can ride a new model of electric bike on a fully charged battery. She completed two trips totaling 40 miles. The first trip used  $\frac{1}{2}$  of the total battery power, while the second trip used  $\frac{3}{10}$  of the total battery power. How many miles can this electric bike go on a fully charged battery?

(A) 45      (B) 48      (C) 50      (D) 52      (E) 55

**Answer:** C

**Solution**

We can see that the total battery spent is  $\frac{1}{2} + \frac{3}{10} = \frac{5+3}{10} = \frac{4}{5}$ . We can thus write the proportion  $\frac{40}{\frac{4}{5}} = \frac{x}{1}$ . Since  $\frac{40}{\frac{4}{5}} = 40 \cdot \frac{5}{4} = \boxed{50}$

**Solution 2**

The fully charged battery has a total battery power  $P$ . There is a bijective analysis that if one uses a set speed for both ways in a directional plane, the amount  $P$  is the farthest they can travel. Then, after the first trip,  $\frac{1}{2}P$  is used. Then, after the second,  $\frac{3}{10}P$  is used. Thus, the total distance traveled (40 miles) corresponds to the total fraction of battery used:

$$\frac{1}{2}P + \frac{3}{10}P = 40$$

To clear the denominators, we multiply by 10:

$$5P + 3P = 400$$

$$8P = 400$$

$$P = 50$$

The answer is clearly  $\boxed{50}$ .

**Solution 3**

Start with combining the battery fractions listed in the problem, and we compare that to the number of miles she rode.

$$\frac{1}{2} + \frac{3}{10} = \frac{8}{10}$$

So 40 miles is  $\frac{8}{10}$  of her battery power.

$$\frac{8}{10}x = 40$$

Multiply both sides by  $\frac{10}{8}$ :

$$\frac{10}{8} \cdot \frac{8}{10}x = 40 \cdot \frac{10}{8}$$

$$x = \frac{400}{8}$$

The answer is 50.

8. (AMC 8 2026) A poll asked a number of people if they liked solving mathematics problems. Exactly 74% answered “yes.” What is the fewest possible number of people who could have been asked the question?

(A) 10      (B) 20      (C) 25      (D) 50      (E) 100

**Answer:** D

**Solution 1**

Well, we use our critical thinking skills to know that you can’t have .5 or .7 people (or any fraction for that matter). So, we will be like me while taking this test and do each answer individually (starting from smallest) until one gives us a whole number. which in this case that is (D) 50. (although this is quite slow so I’d use the other methods posted) pereena come on let’s think positively, ok?

**Solution 2**

Let  $n$  be the number of people. Then,  $\frac{74}{100}n$  like to do math (I think I am one of them).  $\frac{74}{100}$  simplifies down to  $\frac{37}{50}$ , so the remaining people who don’t like math (haters just hate I guess) would also fall under the 50 denominator category. The answer is 50

**Solution 3**

We start by trying every answer from smallest to greatest. We have that 10 is the smallest choice, so we try this. If we polled 10 people, then 7.6 must have said yes, which is clearly impossible, since fractional people can’t exist so we discard this answer. If we continue in this matter, we can find that the answer is (D) 50.

**Solution 4**

If 10 people are polled, then 74 must be divisible by 10, but it’s not, so it is not 10 people. If 20 people are polled, then 74 must be divisible by 5, but it’s not, so it is not 20 people. If 25 people are polled, then 74 must be divisible by 4, but it’s not, so it is not 25 people. If 50 people are polled, then 76 must be divisible by 2, which it is, so our answer is (D) 50.

**Solution 5**

Our percent is 74 Assuming the people are in whole numbers...

If it is A) 10, then the percent will have to be a multiple of  $\frac{100}{10} = 10$

If it is B) 20, then the percent will have to be a multiple of  $\frac{100}{20} = 5$

If it is C) 25, then the percent will have to be a multiple of  $\frac{100}{25} = 4$

It it is D) 50, then the percent will have to be a multiple of  $\frac{100}{50} = 2$

If it is E) 100, then the percent will have to be a multiple of  $\frac{100}{100} = 1$

In this case, 74 only fits D and E, and because D is less, (D) 50 is the correct answer.

**Solution 6**

We can see that  $74\% = \frac{74}{100}$  by definition. This fraction can be simplified to  $\frac{37}{50}$ , meaning 37 out of 50 people said "Yes". Since  $\gcd(37, 50) = 1$ , if this fraction was divided any further, we would have fractional numerator and denominator, which is clearly impossible. Therefore, the minimum number of people surveyed was (D) 50. Ingeniouspeacock36

9. (AMC 8 2026) What is the value of this expression?

$$\frac{\sqrt{16\sqrt{81}}}{\sqrt{81\sqrt{16}}}$$

- (A)  $\frac{4}{9}$       (B)  $\frac{2}{3}$       (C) 1      (D)  $\frac{3}{2}$       (E)  $\frac{9}{4}$

**Answer:** B

**Solution 1**

The inner square root cannot be negative. If the outer square root is negative, the denominator is also negative, resulting in a positive fraction, an identity. Thus, we only consider positive squares, in which the numerator simplifies to  $\sqrt{16 \cdot 9} = \sqrt{16}\sqrt{9} = 4 \cdot 3 = 12$ , and the denominator simplifies to  $\sqrt{81 \cdot 4} = \sqrt{81}\sqrt{4} = 9 \cdot 2 = 18$ . The answer is  $\frac{12}{18} = \frac{4}{6} = \frac{2}{3}$ .

(Other solutions weren't explicit with logic).

**Solution 2**

Since  $\sqrt{81} = 9$  and  $\sqrt{16} = 4$ , we can simplify this expression to  $\frac{\sqrt{16 \cdot 9}}{\sqrt{81 \cdot 4}} = \frac{\sqrt{144}}{\sqrt{324}}$ , which can be calculated to be  $\frac{12}{18} = \frac{2}{3}$ . (B)  $\frac{2}{3}$ .

**Solution 3**

$\frac{\sqrt{16\sqrt{81}}}{\sqrt{81\sqrt{16}}} = \frac{4\sqrt{\sqrt{81}}}{9\sqrt{\sqrt{16}}} = \frac{4 \times 3}{9 \times 2} = \frac{12}{18} = \frac{2}{3}$ , so our answer is (B)  $\frac{2}{3}$ .

**Solution 4**

$\frac{\sqrt{16\sqrt{81}}}{\sqrt{81\sqrt{16}}} = \frac{\sqrt{16 \cdot 9}}{\sqrt{81 \cdot 4}}$  Notice that  $16 = 4^2$ ,  $9 = 3^2$ ,  $81 = 9^2$ , and  $4 = 2^2$ . We also know that  $x^2y^2 = (xy)^2$ . So this equals  $\frac{\sqrt{4^2 \cdot 3^2}}{\sqrt{9^2 \cdot 2^2}} = \frac{\sqrt{(4 \cdot 3)^2}}{\sqrt{(9 \cdot 2)^2}} = \frac{\sqrt{12^2}}{\sqrt{18^2}} = \frac{12}{18} = \frac{2}{3}$ , so our answer is (B)  $\frac{2}{3}$ .

**Solution 5**

Since  $\sqrt[3]{81} = 3$ , and  $\sqrt{16} = 4$ , the numerator of  $\frac{\sqrt{16\sqrt{81}}}{\sqrt{81\sqrt{16}}}$  becomes  $4 \cdot 3 = 12$ . To find the denominator's value, we know that  $\sqrt[3]{16} = 2$ , and that  $\sqrt{81} = 9$ , the denominator of  $\frac{\sqrt{16\sqrt{81}}}{\sqrt{81\sqrt{16}}}$  yields  $9 \cdot 2 = 18$ . We are left with  $\frac{12}{18}$ . After further simplification, our answer is (B)  $\frac{2}{3}$ . Ingeniouspeacock36

10. (AMC 8 2026) Five runners completed the grueling Xmarathon: Luke, Melina, Nico, Olympia, and Pedro.

- Nico finished 11 minutes behind Pedro.

- Olympia finished 2 minutes ahead of Melina, but 3 minutes behind Pedro.
- Olympia finished 6 minutes ahead of Luke.

Which runner finished fourth?

- (A) Luke      (B) Melina      (C) Nico      (D) Olympia      (E) Pedro

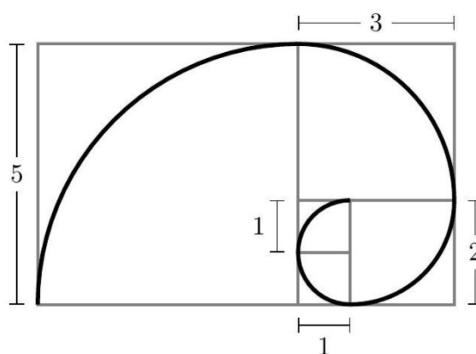
**Answer:** A

### Solution 1

Let the respective places be  $L$ ,  $M$ ,  $N$ ,  $O$ , and  $P$  respectively. Then,  $P = 11 + N$ ,  $O - 2 = M$ ,  $O + 3 = P$ , and  $O = 6 + L$ . Then, substitution for  $P$  gives us  $O + 3 = N + 11$ , in which  $O = N + 8$ . Then,  $N + 6 = M$ ,  $N + 8 = 6 + L \Rightarrow N + 2 = L$ . So  $N$  finished  $m$  minutes behind everyone (that was ok), and thus was last.

Now,  $O = 6 + L$ , in which

11. (AMC 8 2026) Squares of side length 1, 1, 2, 3, and 5 are arranged to form the rectangle shown below. A curve is drawn by inscribing a quarter circle in each square and joining the quarter circles in order, from shortest to longest. What is the length of the curve?



- (A)  $4\pi$       (B)  $6\pi$       (C)  $\frac{13}{2}\pi$       (D)  $8\pi$       (E)  $13\pi$

**Answer:** B

### Solution 2 (No Fibonacci)

For those who don't know Fibonacci's Spiral, One could see that between the two unit squares, the curvature is  $\frac{1}{2} \cdot 2\pi(1) = \pi$ . For the unit 2 square, the curve is  $\frac{1}{4} \cdot 2\pi(2) = \pi$ . For the unit three square, is is  $\frac{1}{4} \cdot 2\pi(3) = \frac{3}{2}\pi$ . For the unit 5 square, we have  $\frac{1}{4} \cdot 2\pi \cdot 5 = \frac{10}{4}\pi = \frac{5}{2}\pi$ . The length of the curve is  $2\pi + \frac{8}{2}\pi = 2\pi + 4\pi = \boxed{6\pi}$ .

### Solution 3 (Fibonacci)

For those who know Fibonacci's Spiral, the length of the curve  $L_n = \frac{\pi}{2} \sum_{k=1}^n F_k$  where  $F_k$  is the  $k$ th Fibonacci Number. Here,  $n$  is the number of squares the curve traverses. Then,  $L_5 = \frac{\pi}{2} \cdot (12) = \boxed{6\pi}$ .

### Solution 4 (Straightforward)

Note that the circumference of a quarter arc is  $\frac{1}{4} \cdot 2 \cdot \pi \cdot r$  Which is the same thing as  $\frac{1}{2} \cdot \pi \cdot r$

The first part of the spiral has a quarter circle with the radius of 5 and so the formula would be

$$\frac{1}{2} \cdot \pi \cdot 5 = \frac{5}{2}\pi$$

The second part of the spiral has a quarter circle with the radius of 3 and so the formula is

$$\frac{1}{2} \cdot \pi \cdot 3 = \frac{3}{2}\pi$$

The third part of the spiral has a quarter circle with the radius of 2 and so the formula is

$$\frac{1}{2} \cdot \pi \cdot 2 = \pi$$

The fourth and fifth part can be viewed as 2 quarter circles with radius 1, so they are essentially one semicircle with radius 1. The formula then would be  $\pi \cdot \frac{1}{2} \cdot 2 \cdot r = \pi$ . Therefore, it is just  $\pi$ .

Adding them all up, we get

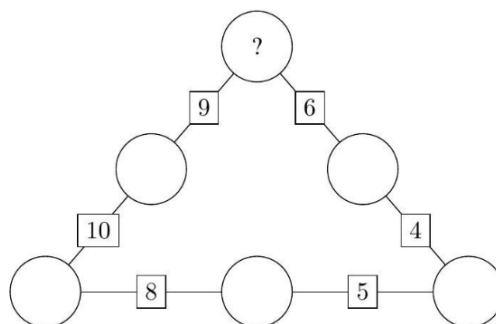
$$\frac{5}{2}\pi + \frac{3}{2}\pi + 1\pi + 1\pi = \frac{8}{2}\pi + 2\pi = 6\pi$$

Therefore our final answer is

$$\boxed{6\pi}.$$

-Razorpro10

12. (AMC 8 2026) In the figure below, each circle will be filled with a digit from 1 to 6. Each digit must appear exactly once. The sum of the digits in neighboring circles is shown in the box between them. What digit must be placed in the top circle?



- (A) 2    (B) 3    (C) 4    (D) 5    (E) it is impossible to fill the circles

**Answer:** D

**Solution**

Let  $O$  denote an odd number and  $E$  denote an even. Then, we know that  $O + E = O$ , and thus the two neighboring circles that form sums 9 and 5 must be one of each odd and even. To obtain 5, we have  $1 + 4$  or  $2 + 3$ . To obtain 9, we have  $3 + 6$ ,  $4 + 5$ . Consider each case separately.

Case 1: If  $1 + 4$  makes the sum of 5, then the two numbers that make 9 must be  $3 + 6$ . Then, 3 is forced on top, and we require another 3. However, the problem doesn't allow this, so this is impossible.



Case 2:  $2 + 3$  is on 5, meaning that  $4 + 5$  is on 9. Then, we have two subcases.

Case A: 4 is on top. Then, we require 2 to complete the sum of 6, which is not possible as  $2 + 3$  is on 5.

Case B: 5 is on top, meaning that we require 1 right below, 3 on the bottom right, 2 to complete the sum of 5, and the rest are uniquely determined.

The number is 5.

### Solution 2

In the bottom-left corner, we have 2 numbers that add up to 10 filled in from the digits from 1 to 6. There are two ways you can do this, 5 and 5 or 6 and 4. Since we can only use each integer once, the first case doesn't work, so we must have the second case. Now we need to figure out the ordering. If we have 4 by the 8, the other circle must be 4, but again, we can only use each positive integer once, so the 6 must be by the 8, and the 4 must be by the 10 and the 9. Notice that we can already find the number that needs to go into the circle we want, which is (D) 5 since  $9 - 4 = 5$ . Baihly2024

### Solution 3

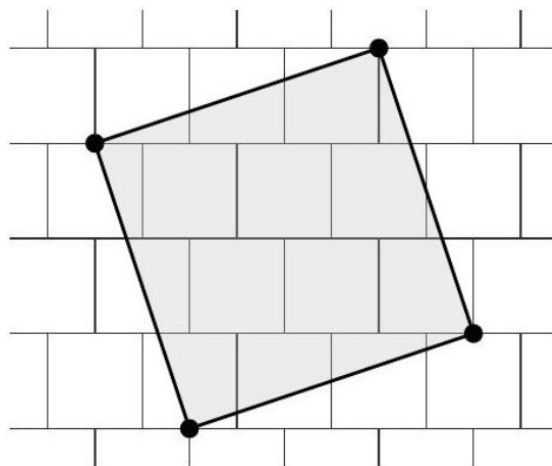
We do trial and error, and because it is a shorter problem it doesn't take long. we can find that the only possibility is (D) 5

### Solution 4 (Prob the fastest)

Let's set space 1 to the top, space 2 to the middle left, space 3 to the middle right, space 4 to the bottom left, space 5 to the bottom middle, and space 6 to the bottom right. Note that I am only doing this to better explain this to you, the numbering of spaces is just a waste of time on the AMC 8.

Look at space 3 and space 6. The numbers must sum to 4. The two numbers cannot be 2 and 2, as each digit must repeat only once, therefore the only number pair that satisfies this is 1 and 3. Now we must figure out where to place these numbers. We can now look at space 1 and space 3. If we say that space 3 is equal to 1, then space 1 must be equal to 5. That is one possible solution. Now, if we made space 3 equal to 3 instead, space 1 must be equal to 3 as well, which contradicts the problem. Therefore, the only possible solution is 5. Now, we must check and solve the rest of the triangle to make sure that there is a solution, and the answer is not E. We can now find that the final answer is (D) 5.

13. (AMC 8 2026) The figure below shows a tiling of  $1 \times 1$  unit squares. Each row of unit squares is shifted horizontally by half a unit relative to the row above it. A shaded square is drawn on top of the tiling. Each vertex of the shaded square is a vertex of one of the unit squares. In square units, what is the area of the shaded square?



- (A) 10      (B)  $\frac{21}{2}$       (C)  $\frac{32}{3}$       (D) 11      (E)  $\frac{34}{3}$

**Answer:** A

**Solution**

Label the vertices  $A$ ,  $B$ ,  $C$ ,  $D$ , in clockwise order. Then, we use coordinate geometry to easily solve this. Let point  $A$  be on  $(0,0)$ . Then, point  $B$  is 1 unit up and 3 right of  $(0,0)$ , or  $(3,1)$ . The figure is said to be a square. and so we just calculate the distance from  $(0,0)$  to  $(3,1)$ . Using the distance formula, we have  $\sqrt{(1-0)^2 + (3-0)^2} = \sqrt{1+9} = \sqrt{10}$ . The area is just this value squared, or  $\boxed{10}$ .

**Solution 2**

Since the side length of a square is 1 unit, we can use Pythagoras on a side length (note that we only need to find one side since the shape is a square). If we do it on the left side, we get  $\sqrt{3^2 + (0.5 + 0.5)^2} = \sqrt{10}$ . Therefore, the final area is the side length squared, which is  $(\sqrt{10})^2 = \boxed{\text{(A) } 10}$ . Baihly2024

**Solution 3**

Notice that the top-left side of our square is the hypotenuse of a right triangle with legs of length 1cm and 3cm. Using the Pythagorean Theorem, the side length of our square is  $\sqrt{1^2 + 3^2} = \sqrt{10}\text{cm}$ , so the area is  $\sqrt{10}^2 = 10\text{cm}^2$ , so we obtain  $\boxed{\text{(A) } 10}$ .

**Solution 4(Quick And Easy)**

Notice that we can construct a 4x4 square that passes tangent to each corner. The area of this square is 16. This square forms 4 congruent right triangles, with a base of 3 and height of 1. The area of one of these triangles is 1.5, and you can multiply that by 4 to get the total area of these triangles, 6. Subtract that from the area of the constructed square to get  $\boxed{\text{(A) } 10}$ . as your answer.

It would look something like

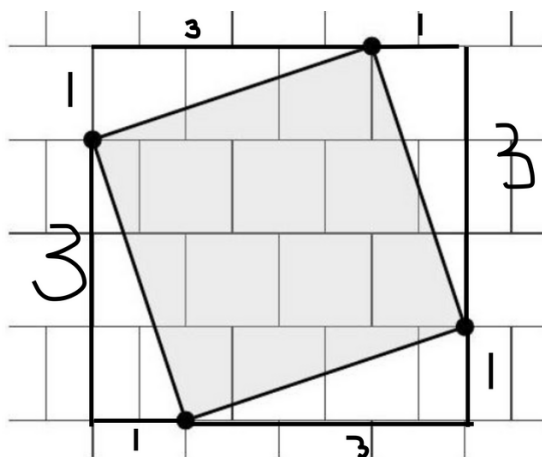


image by -N.

### Solution 5

Similar to Solution 4, we construct one right triangle outside the square. We can see that it has side lengths 3 and 1. Using the Pythagorean Theorem we find that the hypotenuse, which is also a side of the original square, is  $\sqrt{10}$ . Now, we can find the area of the square by squaring that, which yields **(A) 10**.

### Solution 6 (Without Pythagoras or Distance Formula)

- Note: this solution is unreliable and only use it if you have no other methods to solve.

We can count the number of small squares inside the large square and add 1 to our running count if the square is more than half shaded, add  $\frac{1}{2}$  if it is exactly (or almost exactly)  $\frac{1}{2}$  shaded, and add 0 if it is less than half shaded. Counting the squares, we have 5 fully shaded, 4 more than half, and 2 exactly half.  $5 + 4 + 2 \times \frac{1}{2} = \mathbf{(A)10}$

14. (AMC 8 2026) Jami picked three equally spaced integer numbers on the number line. The sum of the first and the second numbers is 40, while the sum of the second and third numbers is 60. What is the sum of all three numbers?
- (A) 70    (B) 75    (C) 80    (D) 85    (E) 90

**Answer:** B

### Solution

Since the integers are equally spaced, they form an arithmetic sequence, so we can label the middle term  $a$ , the smallest one  $a - d$  and the largest one  $a + d$ . We are given that  $(a - d) + a = 2a - d = 40$  and  $a + (a + d) = 2a + d = 60$ . Solving,  $(2a - d) + (2a + d) = 4a = 100$  so  $a = 25$ . The problem asks for the sum of all 3 numbers, which is  $(a - d) + a + (a + d) = 3a = 3 \cdot 25 = \mathbf{(B)75}$ .

### Solution 2

The numbers  $a$ ,  $b$ , and  $c$  are equally spaced by a distance  $k$  on the number line. Then, we have  $b = a + k$  and  $c = a + 2k$ . Then, the sum of the first and second numbers is  $a + a + k = 40 = 2a + k = 40$ , and the sum of the second and third numbers is  $a + k + a + 2k = 2a + 3k = 60$ . Then, solving we obtain  $-2k = -20$ ,  $k = 10$ , and thus  $2a + 10 = 40$ ,  $a = 15$ . Then,  $b = 25$ , and  $c = 35$ . The sum of all three is  $15 + 25 + 35 = 70 + 5 = \mathbf{(B)75}$ .

**Solution 3**

The difference of the sum of the first two integers and the sum of the last two integers differs by 20. Hence, the common difference of the arithmetic sequence must be 10. The only way for the sum of two integers in an arithmetic sequence with common difference 10 to have a sum ending in 0 is for all the integers to end in either 0 or 5.

Testing 5 and 15 and 10 and 20 does not meet the required sum for the first two integers in the sequence. However, the sum of 15 and 25 is equivalent to 40, and adding 35(25 + 10) to 25 matches up with the sum of the last two integers, 60. Consequently, the arithmetic sequence is 15, 25, 35, with a common difference of 10. Adding up all the terms in the sequence gives the answer,  $15 + 25 + 35 = \boxed{\text{(B)}75}$

**Solution 4, (the most complicated, just math)**

$$x + y = 40$$

(equation 1)

$$y + z = 60$$

(equation 2)

$$x = a, y = a + d, z = a + 2d$$

$$x + x + d = 40$$

(equation 1)

$$x + d + x + 2d = 40$$

(equation 2)

$$(2x + 3d) - (2x + d) = 20$$

(eqs. 1 and 2 subtracted)

$$2d = 20, d = 10$$

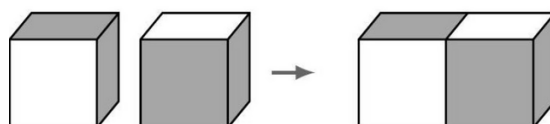
$$2a + 10 = 40, a = 15$$

(substitute)

$$a + a + d + a + 2d = 15 + 15 + 10 + 15 + 20 = \boxed{75}$$

(plug in and substitute)

15. (AMC 8 2026) Elijah has a large collection of identical wooden cubes which are white on 4 faces and gray on 2 faces that share an edge. He glues some cubes together face-to-face. The figure below shows 2 cubes being glued together, leaving 3 gray faces visible. What is the fewest number of cubes that he could glue together to ensure that no gray faces are visible, no matter how he rotates the figure?



- (A) 4    (B) 6    (C) 8    (D) 9    (E) 27

**Answer:** A

**Solution 1 (2D Plane)**

Create a square in which two adjacent sides are colored red. Now, connect each of  $n$  squares such that each red side is tangent to another. What is the minimum number required?

By solving this bijection, we solve the problem above. Through experimentation, each square, assuming the colored sides are the rightmost and bottom of a square, must have another square on the side and below. Thus we obtain 3 squares.

However, the top right square must have one more square below it, giving us a final count of  $3 + 1 = \boxed{4}$  minimum cubes.

**Solution 2 (omg so easy)**

We can have 4 cubes with all their gray faces facing each other, and that will make no gray faces visible. Therefore, our answer is **(A) 4**. Note that this is also the minimum answer choice.

**Solution 3**

Every time we add a new cube, there are 2 colored faces that are added, but only 1 colored face on 2 cubes that are covered. If we start with 1 cube, we see adding another cube colored-on-colored can cause both remaining colored faces to face in the same direction. So if we have 2  $2 \times 1$  cube formations, by aligning all of the colored faces onto one plane we get the minimum of  $\boxed{4}$ . That is pretty nice because we won't have any doubts about whether it could be lower.

16. (AMC 8 2026) Consider all positive four-digit integers consisting of only even digits. What fraction of these integers are divisible by 4?

(A)  $\frac{1}{4}$     (B)  $\frac{2}{5}$     (C)  $\frac{1}{2}$     (D)  $\frac{3}{5}$     (E)  $\frac{3}{4}$

**Answer: D**

**Solution 1**

The divisibility rule for 4 is that the last 2 digits must be a multiple of 4. Because of this, we can "discard" the first two digits, as the last two digits are the only ones that matter (this doesn't actually change the answer, since for every possible case that the first two digits can be, there are the same number of cases the last two digits can be, so the ratio of cases that work to the cases that don't work is still the same). We can also notice that the tens digit also doesn't matter, since every single digit in the number must be even. Therefore, we only need to care about the ones digit. We have the even digits 0, 2, 4, 6, and 8, and there are 3 digits that are divisible by 4, namely 0, 4, and 8, out of 5 total digits, so we have the final fraction

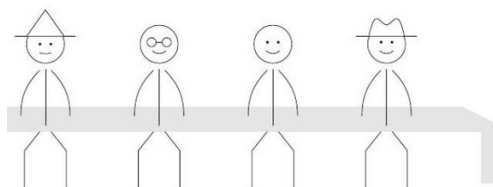
**(D)  $\frac{3}{5}$** . Baihly2024

**Solution 2**

A number is divisible by 4 if and only if the last two digits are divisible by 4. Then, the third and fourth digits must be divisible by 4. So, those possibilities are  $\{(00), (04), (08), (12), (16), (20), \dots, (96)\}$ . There are 25 cases in this, however, some of these are odd! Luckily, each odd tens digit would have an even units digit of either 2 or 6 on the third or fourth multiples, meaning each sequence of 5 between 4 and 20 actually contributes 3 to the count, giving us  $5(3) - 1 + 1 = 15$  actual cases. Then, the remaining two numbers are to be consisting of the digits  $\{0, 2, 4, 6, 8\}$ . The first digit has 4 choices, and the next has 5, giving a total of 20 unique

combinations. These 20 combinations can go with any of the other 24 combinations, giving 300 total combinations. There are a total of  $4 \times 5 \times 5 \times 5 = 125 \times 4 = 500$  four digit integers whose digits are all even. The fraction is  $300/500 = \boxed{3/5}$

17. (AMC 8 2026) Four students are seated in a row. They chat with the people sitting next to them, then rearrange themselves so that they are no longer seated next to any of the same people. How many rearrangements are possible?



- (A) 2      (B) 4      (C) 9      (D) 12      (E) 24

**Answer:** A

**Solution 1 (AMC 10 Bijection)**

Label the human beings A, B, C, and D in that order from left to right. It doesn't matter the original configuration. Now, the problem is:

How many four letter strings ABCD exist such that A is not next to B, B not next to C, and C is not next to D.

We can count the number of strings manually (or just do casework), to obtain  $\boxed{2}$ .

This is incredibly similar to one AMC 10 problem, that asked how many four letter strings abcd have no two consecutive letters in the alphabet next to one another.

**Solution 2**

Let's say the people are A, B, C, and D, and they seated in the order ABCD at first. A cannot be next to B, B cannot be next to C, and C cannot be next to D, so our only possibilities are CADB and BDAC as orders, so our answer is  $\boxed{(A)2}$ .

**Solution 3**

Since B sits next to both A and C, B can only sit next to D. The only way this is possible is if B sits at either the front or the back. We can apply this same logic to C. There are now two cases: BDAC or CADB, which is the former with its order reversed. Therefore, our answer is  $\boxed{(A)2}$ .

18. (AMC 8 2026) In how many ways can 60 be written as the sum of two or more consecutive odd positive integers that are arranged in increasing order?

- (A) 1      (B) 2      (C) 3      (D) 4      (E) 5

**Answer:** B

**Solution 1 (Casework)**

Let  $n$  be the first number in such a series, and let  $k$  be the number of terms in the series. Then the sum of the series is  $n + (n + 2) + \dots + (n + 2(k - 1)) = kn + 2\left(\frac{(k-1)k}{2}\right) = kn + k(k - 1)$ . The goal is thus to find all solutions to the equation  $kn + k(k - 1) = 60$ , where  $n$  is an odd positive integer and  $k > 2$  is an integer.

Consider the following cases:

Case 1:  $k = 2$ . Then  $2n + 2 = 60$ , which yields a valid solution in  $n = 29$  and the series  $29 + 31$ .

Case 2:  $k = 3$ . Then  $3n + 6 = 60$ , which yields an invalid solution in  $n = 18$ .

Case 3:  $k = 4$ . Then  $4n + 12 = 60$ , which yields an invalid solution in  $n = 12$ .

Case 4:  $k = 5$ . Then  $5n + 20 = 60$ , which yields an invalid solution in  $n = 8$ .

Case 5:  $k = 6$ . Then  $6n + 30 = 60$ , which yields a valid solution in  $n = 5$  and the series  $5 + 7 + 9 + 11 + 13 + 15$ .

Case 6:  $k = 7$ . Then  $7n + 42 = 60$ , which yields an invalid solution in  $n = \frac{18}{7}$ .

Case 7:  $k = 8$ . Then  $8n + 56 = 60$ , which yields an invalid solution in  $n = \frac{1}{2}$ .

Case 8:  $k \geq 9$ . Then  $nk + k(k - 1) > k(k - 1) > 60$ , so there will not be any more valid solutions.

Therefore, there are (B) 2 such series.

### Solution 2

We can write the sum of  $n$  integers as

$$a + (a + 2) + (a + 4) + \cdots + (a + 2(n - 1)).$$

This sum can be simplified algebraically to  $an + 2 \cdot \frac{n(n-1)}{2} = an + n(n - 1) = n(a + n - 1)$ . Thus,

$$n(a + n - 1) = 60.$$

We are given that  $a$  is odd, along with all the other terms. If  $n$  was odd as well, we would have the sum be odd as well, and 60 is clearly not odd. Thus, we know  $n$  is even.

Trying different values of  $n$  from 2 onwards, we find the following valid ordered pairs for  $(a, n)$ :  $(29, 2), (5, 6)$ . Therefore, there are only (B) 2 solutions that meet the conditions of the problem.

Note: The solutions we found here can be written as  $29 + 31$  and  $5 + 7 + 9 + 11 + 13 + 15$ .

### Solution 3 (Manual Ordered Pairing [MOP])

A sum of consecutive odd positive integers can also be represented as an arithmetic sequence, in this case, with common difference 2 and the first term being odd. The sum of an arithmetic sequence can also be represented as the product of the average term in the sequence and the number of terms. Therefore, we have  $60 = (\text{average term of the sequence}) \cdot (\text{number of terms})$ . We know that the number of terms must be even, since if it was odd, then the sum of the sequence would be an odd number. We also know that the average term of the sequence must be even, because since there are an even number of terms, then the average term is the median of the two "middle" terms, which must be even. Therefore, we must have it such that (average term of the sequence) and (number of terms) multiply to 60, while both factors have to be even. This gives 4 ordered pairs  $(x, y)$  with  $x$  being the average term of the sequence and  $y$  being the number of terms:  $(2, 30), (6, 10), (10, 6), (30, 2)$ . However,  $(2, 30)$  gives an arithmetic sequence with some numbers in the sequence being negative, and since the problem specifies we need positive integers, we discard this case. Similarly,  $(6, 10)$  doesn't work either. The other cases work though, so we have a total of (B) 2 cases. Baihly2024

### Solution 4 (Nice language, a little long)

Let's write the possible pairs down, in the form (# of numbers, average of the sequence). This will work, because the number of terms times the average is the sum of the numbers (proof provided at bottom, but it's intuitive). Now we have:

(1, 60), (2, 30), (3, 20), (4, 15), (5, 12), (6, 10), (10, 6), (12, 5), (15, 4), (20, 3), (30, 2), (60, 1)

That's a lot! Let's eliminate the obvious losers: We know (1, 60) is out, because we need at least two numbers. (60, 1) is also out because that's 60 odd integers that average to 1, so some will have to be negative (and that violates the rules!). Checking out (2, 30), that works with 29 and 31. We can see (10, 6) and anything to the right of that won't work, because they will have to contain more negative numbers to the left of them; (10, 6) has to have numbers  $-3, -1, 1, 3, 5$  on the negative side. We want to make sure (6, 10) hasn't turned to the negative side, so we try write it down: 5, 7, 9, 11, 13, 15. That works!

So far, we have:

eliminated: (1, 60), (10, 6), (12, 5), (15, 4), (20, 3), (30, 2), (60, 1)

confirmed : (2, 30), (6, 10)

checking : (3, 20), (4, 15), (5, 12)

Wait a minute! Three odd consecutive integers can't average to an even! If we try:  $((2x - 1) + (2x + 1) + (2x + 3))/3 = (6x + 3)/3 = (2x + 1)$ , which is an odd number! That eliminates (3, 20) and (5, 12). Finally, we see that 4 odd numbers can't average to an odd, because  $(2x - 1) + (2x - 3) + (2x + 1) + (2x + 3) = 8x$ , an even number. So that eliminates the last one, and we are left with 2 confirmed cases.

**Proof used in solution 4 (trivial)** Say we have a list on integers  $[x_1, x_2, x_3, \dots, x_n]$ . Then there are  $n$  terms. The average of this list is  $(x_1 + x_2 + \dots + x_n)/n$ , and when we multiply it by  $n$ , we get  $x_1 + x_2 + \dots + x_n$ , the sum of the list. Therefore, the number of terms times the average of the terms, is the sum of the terms.

### Solution 5 (Difference of Squares)

Recall that for integers  $n \geq 0$ , the sum of the first  $n$  odd positive integers is  $n^2$ . Therefore, the sum of the  $a^{\text{th}}$  through  $b^{\text{th}}$  odd numbers, where  $a \leq b$  are positive integers, is  $b^2 - (a - 1)^2$ .

Let  $c = a - 1$ , so  $c$  is an integer such that  $0 < c < b$ . Then each series of consecutive odd positive integers summing to 60 corresponds to a solution to the equation  $b^2 - c^2 = 60$ , where  $b > c$  are positive integers. Factoring yields  $(b + c)(b - c) = 60$ . Since  $b + c$  and  $b - c$  must either be both even or both odd, they are both even. That is,  $b + c = 2m$  and  $b - c = 2n$  for some positive integers  $m > n$ . Since  $(2m)(2n) = 4mn = 60$ ,  $mn = 15$ . The only values of  $(m, n)$  satisfying this equation are (15, 1) and (5, 3). The corresponding values of  $(a, b)$  are (15, 16) and (3, 8), which yield  $5 + 7 + 9 + 11 + 13 + 15$  and  $29 + 31$ , respectively. Therefore, there are

**(B) 2** such solutions.

-j314andrews

### Solution 6 (Consecutive Number Addition)

You know that adding consecutive numbers can be represented by  $(x) + (x + 1) + (x + 2) \dots$ . So, adding consecutive odd numbers would be represented by



$(x) + (x + 2) + (x + 4) \dots$ , since every other number is odd. However, every other number is also even. So you must be careful as to what number  $x$  is so you know if this series will be even or odd. Split this problem up into 5 cases (the answer choices really help out since the max is 5).

Case 1 (two consecutive numbers):  $(x) + (x + 2) = 60$ ,  $2x + 2 = 60$ ,  $x = 29$  which satisfies the odd and consecutive constraint.

Case 2 (three consecutive numbers):  $(x) + (x + 2) + (x + 4) = 60$ ,  $3x + 6 = 60$ ,  $x = 18$  which doesn't satisfy the odd and consecutive constraint.

Case 3 (four consecutive numbers):  $(x) + (x + 2) + (x + 4) + (x + 6) = 60$ ,  $4x + 12 = 60$ ,  $x = 12$  which doesn't satisfy the odd and consecutive constraint.

Case 4 (five consecutive numbers):  $(x) + (x + 2) + (x + 4) + (x + 6) + (x + 8) = 60$ ,  $5x + 20 = 60$ ,  $x = 8$  which doesn't satisfy the odd and consecutive constraint.

Case 5 (six consecutive numbers):  $(x) + (x + 2) + (x + 4) + (x + 6) + (x + 8) + (x + 10) = 60$ ,  $6x + 30 = 60$ ,  $x = 5$  which satisfies the odd and consecutive constraint.

Therefore, there are 2 cases out of the 5 which satisfy the constraints, so there are **(B) 2** such solutions.

-N.

### Solution 7 (straightforward)

First, note that the sum of an odd number of odd integers is odd. Since 60 is even, the number of consecutive odd integers,  $n$ , must be even.

For  $n = 2$ , the two numbers are centered around  $\frac{60}{2} = 30$ , so they are  $30 \pm 1$ , giving 29 and 31. Their sum is 60, so this works.

For  $n = 4$ , the numbers are centered around  $\frac{60}{4} = 15$ , so they are  $15 \pm 1$  and  $15 \pm 3$ , giving 12, 14, 16, 18. These are not odd, so this case is invalid.

For  $n = 6$ , the numbers are centered around  $\frac{60}{6} = 10$ , so they are  $10 \pm 1$ ,  $10 \pm 3$ ,  $10 \pm 5$ , giving 5, 7, 9, 11, 13, 15. These are consecutive odd positive integers whose sum is 60, so this works.

For  $n = 8$ , we have  $\frac{60}{8} = 7.5$ , which is not an integer, so this case is impossible.

For  $n = 10$ , the numbers would be centered around  $\frac{60}{10} = 6$ , giving  $6 \pm 1$ ,  $6 \pm 3$ ,  $6 \pm 5$ ,  $6 \pm 7$ ,  $6 \pm 9$ . This includes negative numbers, which are not allowed since the problem asks for positive integers.

Thus, there are exactly 2 valid representations, so the answer is **(B) 2**.

### Solution 8 (Factors and Mean)

Let  $a$  be the first odd positive integer, then

$$a + (a + 2) + (a + 4) + \dots + (a + 2(n - 1)) = 60$$

$$na + (n - 1)n = 60$$

$$n^2 + (a - 1)n = 60$$

The mean of the consecutive odd numbers

$$n + (a - 1) = \frac{60}{n}$$

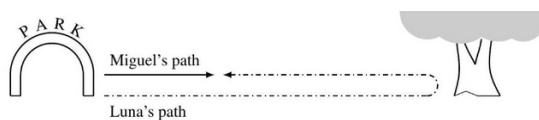
Note:  $60 = 2^2 \times 3 \times 5$ ;  $n$  is even as sum of odd number of odd numbers is odd;  $(a - 1)$  is even

The mean  $\frac{60}{n} = n + (a - 1)$  is an even integer and hence  $n$  could not be a multiple of 4.

The possible values of  $n = \{2, 2 \times 3, 2 \times 5, 2 \times 3 \times 5\}$ , But when  $n = 10, 30$  the mean  $\frac{60}{n}$  will be smaller than  $n$  such that  $a$  could not be positive integer.

Therefore,  $n = 2, 6$  and the answer is **(B) 2**.

19. (AMC 8 2026) Miguel is walking with his dog, Luna. When they reach the entrance to a park, Miguel throws a ball straight ahead and continues to walk at a steady pace. Luna sprints toward the ball, which stops by a tree. As soon as the dog reaches the ball, she brings it back to Miguel. Luna runs 5 times faster than Miguel walks. What fraction of the distance between the entrance and the tree does Miguel cover by the time Luna brings him the ball?



- (A)  $\frac{1}{6}$     (B)  $\frac{1}{5}$     (C)  $\frac{1}{4}$     (D)  $\frac{1}{3}$     (E)  $\frac{2}{5}$

**Answer:** D

**Solution**

Let Miguel's speed equal  $s$ . That makes Luna's speed  $5s$ . Let the distance between the park entrance and the tree be  $d$ . The total distance covered by Miguel and Luna is  $s + 5s = 6s$ , which is also equal to  $2d$ . Thus  $6s = 2d \Rightarrow s = \boxed{\text{(D)} \frac{1}{3}} d$ .

**Solution 2**

The distance between the park entrance and the tree is  $x$ . Then, the rate that Miguel walks is  $v$ , and Luna runs at  $5v$ . As Luna runs the distance  $x$  and back, she runs a total of  $2x - m$  (where  $m$  is the amount that Miguel walks). Then, because  $d/s = t$  (where  $d$  is distance,  $t$  is time, and  $s$  is speed), It takes Luna  $\frac{2x-m}{5v}$  time to reach back to Miguel. Also,  $d = st$ , and by that time, Miguel traveled  $m$ , so  $m = v \left( \frac{2x-m}{5v} \right)$ , or  $5m = 2x - m$ ,  $6m = 2x$ ,  $3m = x$ . The total distance from the park to the tree is  $3m$ , and Miguel traversed  $m$ , giving us a total of  $\frac{m}{3m} = \boxed{\frac{1}{3}}$  the distance traveled.

**Solution 3 (Set the Distance and Speed!)**

The distance doesn't seem to matter as if it did, then the problem would have told the distance, so let's say the distance is 30 units. Also the speed of Luna doesn't matter as the ratio of Luna's speed to Miguel's speed is 5:1, always the same. We can say that Luna goes 5 units a second, then that means that Miguel goes  $\frac{5}{5} = 1$  units a second. Luna will reach the tree after  $\frac{30}{5} = 6$  seconds. When Luna reaches the tree Miguel will be  $6 \times 1 = 6$  units from the entrance of the park. We can set up the equations:  $x + 6$  and  $-5x + 30$ , where  $x$  represents seconds as Miguel is 6 units from the entrance is walking away at a rate of 1 unit a second and Luna is 30 units from the entrance and is walking towards the entrance at a rate of 5 units a

second. We can now set both equations equal together to find out how far away from the entrance they meet.  $x + 6 = -5x + 30 \implies 6x = 24 \implies x = 4$ . This means that they meet 4 seconds after Luna reaches the tree, this means they meet after  $4 + 6 = 10$  seconds as Luna takes 6 seconds to reach the tree. In 10 seconds Miguel travels  $10 \times 1 = 10$  units, and  $\frac{10}{30} = \frac{1}{3}$ , giving us  $\boxed{\frac{1}{3}}$  Shaarav14 for Carnage MS

#### Solution 4

Since Luna is 5 times faster than Miguel, Miguel gets  $\frac{1}{5}$  of the way to the tree by the time Luna reaches the ball. The ratio of Miguel's speed to Luna's is 1:5, and the distance between Miguel and the tree is  $\frac{4}{5}$  the distance between the entrance and the tree. So Miguel walks another  $\frac{4}{5} \times \frac{1}{6}$  of the way from the entrance to the tree by the time Luna comes back to him. Therefore, Miguel walks  $\frac{1}{5} + \frac{4}{30} = \boxed{\frac{1}{3}}$  of the way to the tree from the entrance.-Pillowwillowtree01

#### Solution 5 (Quick)

Since Luna runs 5 times faster than Miguel walks, by the time Luna reaches the tree, Miguel has walked  $\frac{1}{5}$  of the distance from the entrance to the tree. This eliminates answer choices (A) and (B).

This is a multiple-choice problem with the remaining answers listed in increasing order, so we test the middle option, (D), which corresponds to Miguel walking  $\frac{1}{3}$  of the distance.

If Miguel walks a total distance of  $\frac{1}{3}$ , then the additional distance he walks while Luna is running back toward him is  $\frac{1}{3} - \frac{1}{5} = \frac{2}{15}$ . Since Luna runs 5 times faster than Miguel walks, she covers  $5 \cdot \frac{2}{15} = \frac{2}{3}$  of the distance on her return.

Thus, Luna runs the entire distance to the tree and  $\frac{2}{3}$  of the distance back, while Miguel walks  $\frac{1}{3}$  of the distance forward, so they meet at the same point. Therefore, the fraction of the distance Miguel covers is  $\boxed{\frac{1}{3}}$ .

#### Solution 6

Because Luna is running five times as fast as Miguel, when the two meet, Luna will have run five times as far. We can call the distance from the place where Miguel threw the ball to the tree  $X$ , and the distance that Miguel runs  $Y$ . When the two meet they will have run a combined total of  $2X$ . Miguel runs  $Y$ , and because she runs five times as fast, Luna runs  $5Y$ . Their combined total is therefore  $Y + 5Y = 6Y$ . So  $6Y = 2x$ . The value we are trying to find is  $\frac{Y}{X}$

We divide both sides of our equation by  $6x$  to get  $\frac{6Y}{6X} = \frac{2X}{6X}$ . This can be simplified to get  $\frac{Y}{X} = \frac{1}{3}$ . So the fraction of the distance to the tree that Miguel runs is  $\frac{Y}{X} = \boxed{\frac{1}{3}}$

#### Solution 7 - Cheese

If you do not know how to use normal math to solve this question, you can also use a ruler and utilize the provided diagram. This gives that the meeting point is  $\frac{1}{3}$  in regard to distance to the tree, which means that the answer is  $\boxed{(D) \frac{1}{3}}$  **Note:** Even though test proctors, and essentially everybody, says that the diagrams and

images in competition math competitions like AMC 8 are not necessarily drawn to scale, they almost always are. This question isn't an exception. There are some exceptions, though (like the 2022 AMC 8 Problem 24.) So do not rely on this method, unless you have no other choice.

20. (AMC 8 2026) The land of Catania uses gold coins and silver coins. Gold coins are 1 mm thick and silver coins are 3 mm thick. In how many ways can Taylor make a stack of coins that is 8 mm tall using any arrangement of gold and silver coins, assuming order matters?

(A) 3      (B) 7      (C) 10      (D) 13      (E) 16

**Answer:** D

**Solution 1**

A stack containing 3 or more silver coins is at least 9 mm tall, so a stack of height 8 mm may contain at most 2 silver coins.

An 8 mm stack with 0 silver coins contains 8 gold coins. There is  $\binom{8}{0} = 1$  such stack.

An 8 mm stack with 1 silver coin contains 5 gold coins. There are  $\binom{6}{1} = 6$  such stacks.

An 8 mm stack with 2 silver coins contains 2 gold coins. There are  $\binom{4}{2} = 6$  such stacks.

Thus there are  $1 + 6 + 6 = \boxed{\text{(D)} 13}$  different 8 mm stacks.

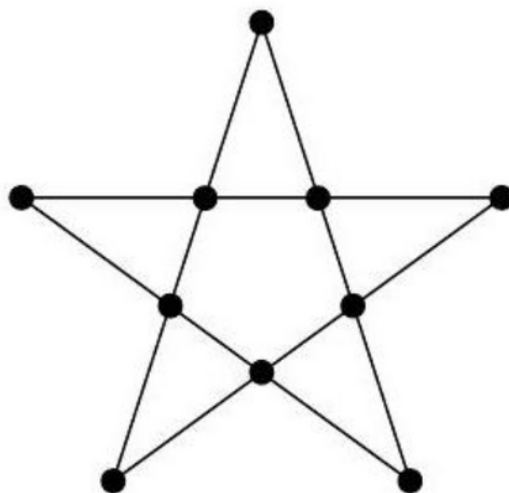
**Solution 2 (Recursion)**

Let  $a_k$  be the number of different stacks of height  $k$  mm. Then  $a_1 = 1$ , as the only possible 1 mm stack is a single gold coin. Also  $a_2 = 1$ , as the only possible 2 mm stack is two gold coins, while  $a_3 = 2$ , as the two possible 3 mm stacks are either three gold coins or one silver coin.

If  $k \geq 4$ , a stack of height  $k$  mm can be formed by placing a stack of height  $(k - 1)$  mm on a gold coin or placing a stack of height  $(k - 3)$  mm on a silver coin. Therefore,  $a_k = a_{k-1} + a_{k-3}$ . So  $a_4 = 3$ ,  $a_5 = 4$ ,  $a_6 = 6$ ,  $a_7 = 9$ , and  $a_8 = \boxed{\text{(D)} 13}$ .

-j314andrews

21. (AMC 8 2026) Charlotte the spider is walking along a web shaped like a 5-pointed star, shown in the figure below. The web has 5 outer points and 5 inner points. Each time Charlotte reaches a point, she randomly chooses a neighboring point and moves to that point. Charlotte starts at one of the outer points and makes 3 moves (re-visiting points is allowed). What is the probability she is now at one of the outer points of the star?



- (A)  $\frac{1}{5}$     (B)  $\frac{1}{4}$     (C)  $\frac{2}{5}$     (D)  $\frac{1}{2}$     (E)  $\frac{3}{5}$

**Answer:** B

### Solution 1

The graph undergoes **rotational symmetry** meaning when scaled on a factor  $k$ , here  $k = 5$  of the points, it is the same as considering a fixed starting position of Charlotte. WLOG let this starting point be the topmost one. On the second turn, she cannot be on an outer point. Then she is forced to an inner point in a 1 chance. From any outer point, she must move to an outer point with a  $\frac{1}{2}$  chance. Finally, from that outer point, she must move to an inner point with  $\frac{1}{2}$  chance. Therefore, the probability is  $\boxed{\frac{1}{4}}$ .

In general, the probabilistic depends on if the pentagram is on some  $n$ th moveset. For instance,  $p_0 = 1$ , and all  $p_n = \frac{1}{2}(1 - p_{n-1})$ . The term  $(1 - p_{n-1})$  is of the point that to get to an outer point one must come from an inner point.

Note to SquishBJ from Pinotation: You are absolutely correct, thank you for taking the time to review my solution. Constructive counting is a swing and miss here due to the fact the question doesn't ask for strict tournaments rather cases based on dependency. Good catch :>

### Solution 2

We begin by noticing that from an outer point, the next move must be to an inner point as both adjacent points are inner. Next, there are four adjacent points - two are inner, two are outer. Thus, there is a  $\frac{1}{2}$  probability for both. The final move depends on which type of point was the last one - if it was outer, the next one would have to be inner, so that gives us a probability of 0 for an outer point. However, if it was inner, there is, again, a  $\frac{1}{2}$  probability that the next one is outer.

Therefore our final probability is  $1 \cdot (\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 0) = \frac{1}{2} \cdot \frac{1}{2} = \boxed{\text{(B)} \frac{1}{4}}$ .

### Solution 3 (Markov Chains)

Note that there are only two states due to five-fold symmetry. Hence, we can create a Markov Chain with two states: outer and inner points. Transitioning from an outer point to an inner point happens with probability 1, and transitioning from

an inner point to an outer point happens with probability  $\frac{1}{2}$ . The spider always changes from states, and it never stays on the same state in two consecutive moves. The spider starts on an outer point, so the probability at Move 0 for outer points is 1.

Enumerating the values for 3 moves using our recursive rules above:

| Move | Outer         | Inner         |
|------|---------------|---------------|
| 0    | 1             | 0             |
| 1    | 0             | 1             |
| 2    | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 3    | $\frac{1}{4}$ | $\frac{3}{4}$ |

We aim to find the probability of the spider being at an outer point on Move 3.

Thus, the answer is  $\boxed{\text{(B)}\frac{1}{4}}$ .

#### Solution 4 (Recursive)

Let  $p_n$  be the probability that Charlotte is at an outer point after  $n$  moves.

From the symmetry of the star graph we derive that the probabilities are:

- Outer  $\rightarrow$  Inner with probability 1
- Inner  $\rightarrow$  Outer with probability  $\frac{1}{2}$
- Inner  $\rightarrow$  Inner with probability  $\frac{1}{2}$

If Charlotte is at an outer point after  $n + 1$  moves, she must have been at an inner point after  $n$  moves and then moved outward. Therefore

$$p_{n+1} = \frac{1}{2}(1 - p_n)$$

The initial condition is

$$p_0 = 1$$

since she starts at an outer point.

Rewrite the recurrence as

$$p_{n+1} = \frac{1}{2} - \frac{1}{2}p_n.$$

First, we find the constant equilibrium solution  $c$ :

$$c = \frac{1}{2} - \frac{1}{2}c$$

$$\begin{aligned}\frac{3}{2}c &= \frac{1}{2} \\ c &= \frac{1}{3}.\end{aligned}$$

$$q_n = p_n - \frac{1}{3}.$$

Substituting into the recurrence,

$$p_{n+1} = \frac{1}{2} - \frac{1}{2} \left( q_n + \frac{1}{3} \right) = \frac{1}{3} - \frac{1}{2} q_n.$$

Thus

$$q_{n+1} = -\frac{1}{2} q_n.$$

We now have a clean equation.

$$q_n = q_0 \left( -\frac{1}{2} \right)^n.$$

Now we do some simple algebra.

$$\begin{aligned}q_0 &= p_0 - \frac{1}{3} = 1 - \frac{1}{3} = \frac{2}{3}, \\ q_n &= \frac{2}{3} \left( -\frac{1}{2} \right)^n. \\ p_n &= q_n + \frac{1}{3} = \frac{1}{3} + \frac{2}{3} \left( -\frac{1}{2} \right)^n.\end{aligned}$$

Now we plug in  $n = 3$ :

$$p_3 = \frac{1}{3} + \frac{2}{3} \left( -\frac{1}{2} \right)^3 = \frac{1}{3} - \frac{1}{12} = \frac{1}{4}.$$

Thus the probability Charlotte is at an outer point after 3 moves is  $\boxed{\frac{1}{4}}$ .

**Remark** This solution turns a recursive sequence into a geometric sequence. It's overkill for this question, but it could be used if we are finding the probability after a large number of moves, e.g., 2026 moves.

### Solution 5 (Quick)

Like solution 1 and 2 we know that we can choose one edge point to start on. From any edge point there are two possibilities to go to, we can just use one of these possibilities due to symmetry, making the problem easier to solve. Once we go to this point we can see that there is a  $\frac{1}{2}$  probability that you go to an edge point and a  $\frac{1}{2}$  probability that you stay in a center point. If you go to an edge point there is a 0 probability that you can end on an edge point. While if you stay in the middle, there is a  $\frac{1}{2}$  probability that you go to an edge point on your 3rd move. We can calculate this using symmetry without loss of generality making this problem much easier to solve. Finally our answer is  $\frac{1}{2} * \frac{1}{2} = \frac{1}{4}$  which is  $\boxed{B : \frac{1}{4}}$

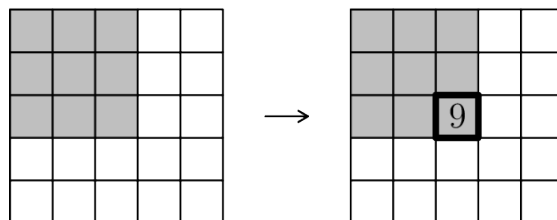
(PS. Make sure to check whether you can use symmetry for a counting problem before you actually use it.) SShreyas

22. (AMC 8 2026) The integers from 1 to 25 are arbitrarily separated into five groups of 5 numbers each. The median of each group is identified. Let  $M$  equal the median of the five medians. What is the least possible value of  $M$ ?

(A) 9    (B) 10    (C) 12    (D) 13    (E) 14

**Answer:** A

**Solution 1**



Consider using a five by five square table to solve this problem. The rows represent the five groups, the center column is the median of each group and the center square cell is the least possible value of  $M$ . The only numbers that really matter is the ones in the shaded area, the rest can be ignored. In order to find the least possible value of  $M$  the smallest numbers, 1...9, need to be in these nine square cells. 9 is the smallest number that is able to satisfy both median of a group and median of five medians. Final answer, the least possible value of  $M$  is **(A) 9**. TutorJack

**Solution 2 (Quick and easy)**

If a group has median  $m$ , then we must have that 3 of the numbers in that group are  $\leq m$ . Since there are 5 different groups, 3 groups must have a median  $\leq M$ , so there are at least  $3 \cdot 3 = 9$  numbers that are  $\leq M$ . Since there are at least 9 numbers that are  $\leq M$ , we have  $M$  at minimum **(A) 9**. Baihly2024

**Solution 3 (Solution 1 Rigorously)**

Let the groupings be  $G_1, G_2, \dots, G_5$ . Let the medians of each of the groupings be  $m_1, m_2, m_3, m_4$ , and  $m_5$  in that order. Let  $M$  be the median of those medians. Then, we wish to minimize  $M$ . This can be possible if we have three elements in  $G_1$  less than  $M$ , three elements in  $G_2$  less than  $M$ , and three elements in  $G_3$  less than an equal to  $M$  are there (as the last number itself is  $M$ ). Thus, the smallest way to choose 9 such numbers is  $\{1, 2, 3, 4, \dots, 9\}$ , in which  $M$  is the largest of these, and thus the answer is **9**.

**Solution 4**

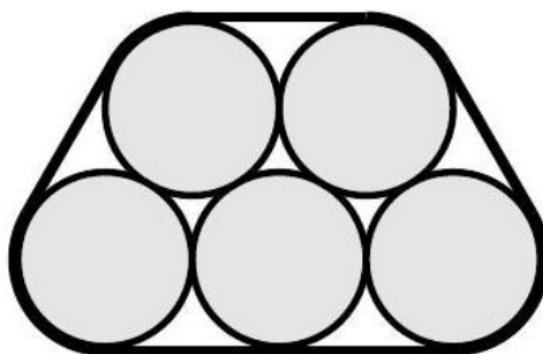
If you want to prove a solution one of the easiest ways is just to list an example of it. A median has to be greater than the first two numbers of the group. Then we just list the groups out. This includes:

(1, 2, 3, 16, 17), (4, 5, 6, 18, 19), (7, 8, 9, 20, 21), (10, 11, 12, 22, 23), (13, 14, 15, 24, 25)

Then according to this triple we are able to find the  $M$  is **(A) 9**.

23. (AMC 8 2026) Lakshmi has 5 round coins of diameter 4 centimeters. She arranges the coins in 2 rows on a table top, as shown below, and wraps an elastic band tightly around them. In centimeters, what will be the length of the band?





- (A)  $2\pi + 20$       (B)  $\frac{5}{2}\pi + 20$       (C)  $4\pi + 20$       (D)  $\frac{9}{2}\pi + 20$       (E)  $5\pi + 20$

**Answer:** C

**Solution 1**

```
unitsize(5mm); draw(circle((2,2),2)); o)); draw(circle((10,2),2)); draw
```

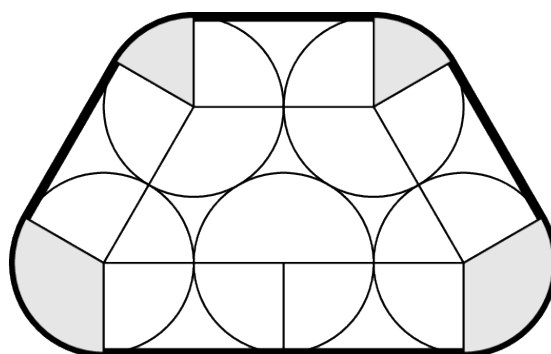
In order to find the length of the elastic band, perpendicular lines are added to the figure to divide the band into sections for easier calculation. After the additions, the length of band consist of five identical lines and a full circle, the four shaded areas. The total length is five lines, the diameter of the coins, plus the circumference of a coin, so  $5d + 4\pi$ . Finally, the length of the elastic band is (C)  $4\pi + 20$ . TutorJack

**Solution 2 (Pure Geometry)**

As shown in the figure above, we just connect some lines and do relations to find the straight parts of the bands (That's easy). Now, we need to find the four curves. The three central circles form a triangle with angle measures  $60^\circ$ . Extend the lines of that triangle and draw two perpendicular lines going through the top left circle's center. Then, it becomes obvious the angle measure for the two topmost arcs of the band is  $60^\circ$ . Thus, the length of the curve is  $\frac{1}{6} \cdot \pi \cdot 2 \cdot 2 = \frac{2}{3}\pi$ . There are two of these, giving us a total of  $\frac{4}{3}\pi$ . Now, the bottom two. We do the same process and this time find out the measure of the arcs is  $120^\circ$ , meaning the length of the curve is  $\frac{1}{3} \cdot 2 \cdot \pi \cdot 2 = \frac{4}{3}\pi$ . There are two such curves, and thus the total length is  $\frac{8}{3}\pi$ . The total curvature length is  $\frac{12}{3}\pi = 4\pi$ . The straight parts is just additive to  $4 + 8 + 4 + 4 = 20$ . Thus, the length of the band is  $20 + 4\pi$ .

**Solution 3 (Using the answer choices, quickest and best for competitions)**

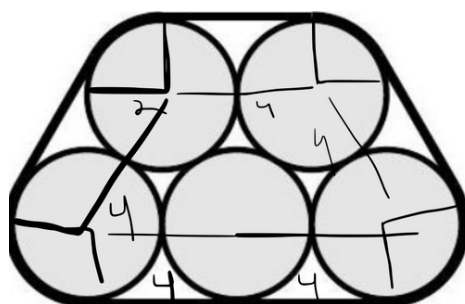
If you look at the answer choices you will notice that all the constants are the same, meaning that we just have to find the coefficient of pi. To do this we can form an isosceles trapezoid with the centers of the four corner circles.



Once we do this we can split the trapezoid into three equilateral triangles, after doing this we can solve for the curved lengths. The bottom 2 curved lengths are each  $120^\circ$  as  $360 - 90 - 90 - 60 = 120$ , because this is an isosceles trapezoid the top two angles must be the same, thus they are both  $60^\circ$  as  $\frac{360 - (120 + 120)}{2} = 60$  (We can also solve for this like we did for the bottom two by doing  $360 - 90 - 90 - 60 - 60 = 60$ )  $60 + 60 + 120 + 120 = 360$  so the total curved part is the same as the circumference of the circle which is diameter times  $\pi$  which is  $4\pi$ , so the answer is  $(C) 4\pi + 20$ .

PS. For problems like these where a band is wrapped around congruent circles, the curved part is always equal to the circumference of a circle.

### Solution 5 (better visualization)



There are 4 fourths of a circle with radius 2, so there is one circle with radius 2. Finding the circumference gives  $4\pi$ , and adding the 4's created by matching the centers of one circle to another by using the radius property gives (it's always equal if it comes from the center)  $(C) 4\pi + 20$ .

-N.

### Solution 6 (Cheese)

We know that the coefficient of  $\pi$  must be  $2 \times 2$  because the angles in the trapezoid add to  $360^\circ$ , the circles are the same size, and the formula for the circumference of a circle is  $\pi d$ . The straight bits add to the rational bit and the curved bits add to the irrationals (with  $\pi$ ). Therefore, the coefficient is  $4\pi$  so the correct answer is  $(C) 4\pi + 20$ .

24. (AMC 8 2026) The factorial notation  $n!$  is defined as the product of the first  $n$  positive integers. (For example,  $3! = 1 \cdot 2 \cdot 3 = 6$ .) Define the superfactorial of a positive number, denoted by  $n^!$ , to be the product of the factorials of the first  $n$  integers. (For example,  $3^! = 1! \cdot 2! \cdot 3! = 12$ .) How many factors of 7 appear in the prime factorization of  $51^!$ , the superfactorial of 51?

(A) 147      (B) 150      (C) 156      (D) 168      (E) 171

Answer: E

### Solution 1 (Solution 2 simplified) - Come and Go Solution

From  $7! - 13!$ , there is one power of 7. From  $14! - 20!$ , there are 2 powers.  $21! - 27!$ , 3 powers.  $28! - 34!$ , 4 powers. From  $35! - 41!$ , 5 powers. From  $42! - 48!$ , 6 powers. Finally, from  $49!$  to  $51!$ , there are 8 powers (as each one contributes  $7^2$ ). Thus, in a superfactorial, the number of powers is just  $7 + 14 + 21 + 28 + 35 + 42 + 24$  powers total, or  $171$ .

In another way, we can think of it like 7 multiples. For example, from seven to thirteen there happens to exist 7 7's. Later from 14 to 20 there happens to exist 7\*2 sevens. When we continue to move on to 49 and 51, there exists an extra seven because of the squares. Adding them up we should get 171 as our answer.

Note for Jellycat88: That logic is flawed. Definition of multiples always implies one factor of seven. your idea works as long as the value is not a perfect square. for instance, 14! through 21!, each corresponds two multiples of seven, those being 7 and 14. when you count that extra square, you are counting a power not a multiple, because from 49 to 51, we obtain only seven multiples, 7,14,21,28,35,42,49. However, from 49 through 51, we obtain eight **powers**. it is a minor yet huge difference, which is why your logic is slightly flawed.

### Solution 2 - Thought Process

We begin by noticing that for the first few factorials (1! to 6!) there are no factors of 7, because the first multiple of 7 is 7 itself. So, these numbers contribute 0 sevens.

The next 7 factorials (7! to 13!) all have one factor of 7 in them, since every number less than or equal to  $n$  is multiplied in  $n!$ . These all contribute 7 sevens.

For the next 7 factorials (14! to 20!), there are two factors of 7 (namely 7 and 14). These numbers contribute 14 sevens.

If we continue with this method, we see that once we hit  $7k!$  we have  $7k$  sevens contributed, since there are 7 numbers in between. This will work from 1! to 48!, because when we get to  $49! = (7^2)!$ , we have one extra 7 for every number, since we have an extra factor of 7.

This is only for 3 numbers, 49!, 50!, and 51! (our final factorial). They have  $7 + 1 = 8$  factors of 7, because they have 7 multiples of 7 and one extra from 49. These three contribute a total of  $3 \cdot 8 = 24$  sevens.

Finally, sum these numbers. Most are just the first 6 multiples of 7, and we can calculate this sum to be  $0 + 7 + 14 + 21 + \dots + 42 = 7 \cdot \frac{6(6+1)}{2} = 7 \cdot \frac{42}{2} = 7 \cdot 21 = 147$ . The final step is to add the final  $3 \cdot 8 = 24$  sevens to get our answer,  $147 + 24 = \boxed{\text{(E)} 171}$ .

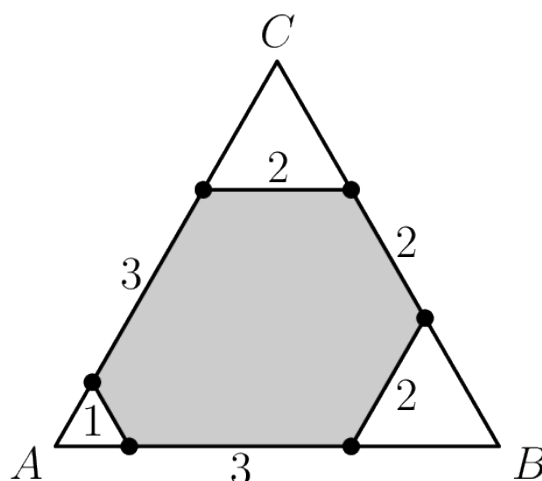
### Solution 3

If  $51! = 1! \cdot 2! \cdot \dots \cdot 51! = (1)(1 \cdot 2)(1 \cdot 2 \cdot 3)(\dots)(1 \cdot 2 \cdot 3 \cdot \dots \cdot 51)$  is expanded as such, each integer  $k \leq 51$  will appear once in each of the expansions of  $\{k!, \dots, 51!\}$ , for a total of  $52 - k$  appearances.

Therefore, in this expansion, 7 appears 45 times, 14 appears 38 times, 21 appears 31 times, 28 appears 24 times, 35 appears 17 times, 42 appears 10 times, and 49 appears 3 times. Except for 49, which contributes 2 copies of the factor 7, each of these terms contributes 1 copy of the factor 7, for a total of  $45 + 38 + 31 + 24 + 17 + 10 + 3 \cdot 2 = \boxed{\text{(E)} 171}$  copies of the factor 7.

-j314andrews

25. (AMC 8 2026) In an equiangular hexagon, all interior angles measure  $120^\circ$ . An example of such a hexagon with side lengths 2, 3, 1, 3, 2, and 2 is shown below, inscribed in equilateral triangle  $ABC$ . Consider all equiangular hexagons with positive integer side lengths that can be inscribed in  $\triangle ABC$ , with all six vertices on the sides of the triangle. What is the total number of such hexagons? Hexagons that differ only by a rotation or a reflection are considered the same.



- (A) 4    (B) 5    (C) 6    (D) 7    (E) 8

**Answer:** E

**Solution 1 (Recursion)**

Let the side lengths of the hexagon be  $a, b, c, d, e, f$ . Obviously, each one of these is to be in between 1, 2, or 3 (see solution 2 for proof). Every 2 can be followed by a 2 or a 3. Every 3 must be followed by a 1 or 2, and every 1 must be followed by a 3. The question is now how many strings of length 6 satisfy these conditions, in which the first and last numbers also satisfy these conditions.

To count that, we use recursion. Let  $a_n$  be a string of length  $n$  ending in 1,  $b_n$  is a string of length  $n$  ending in 2, and  $c_n$  is a string that ends in 3. For  $a_{n+1}$ , we have that it must come from  $c_n$ , so  $a_{n+1} = c_n$ . Additionally,  $b_{n+1} = b_n + c_n$ , and  $c_{n+1} = a_n$ . Then,  $a_1 = 1$ ,  $b_1 = 1$ , and  $c_1 = 1$ . Thus,  $a_2 = 1$ ,  $b_2 = 2$ , and  $c_2 = 1$ ;  $a_3 = 1$ ,  $b_3 = 3$ , and  $c_3 = 1$ ;  $a_4 = 1$ ,  $b_4 = 4$ ,  $c_4 = 1$ ;  $a_5 = 1$ ,  $b_5 = 5$ ,  $c_5 = 1$ ; and finally  $a_6 = 1$ ,  $b_6 = 6$ , and  $c_6 = 1$ .

Our answer is  $a_6 + b_6 + c_6$ , or  $1 + 6 + 1 = \boxed{8}$ .

**Solution 2**

An equiangular hexagon has all interior angles  $120^\circ$ , so consecutive sides turn by  $60^\circ$  each time. Place an equilateral triangle  $ABC$  of side length 6 (this is the side length in the given example since  $1 + 3 + 2 = 6$  along each side).

Since all six vertices of the hexagon lie on the sides of  $\triangle ABC$ , each side of  $\triangle ABC$  contains exactly two consecutive vertices of the hexagon. Hence three alternating sides of the hexagon lie along the sides of  $\triangle ABC$ .

Look near vertex  $A$ . Between the two sides  $AB, AC$  and the hexagon there is a triangular gap whose vertex is  $A$ . Because  $\angle A = 60^\circ$  and the hexagon makes  $120^\circ$  interior angles, each of the two angles of the gap adjacent to the hexagon is  $60^\circ$ . Therefore the gap at  $A$  is an equilateral triangle. Do the same at  $B$  and  $C$ : the hexagon is obtained by “cutting off three equilateral corner triangles.”

Let the side lengths of the three cut-off equilateral triangles at  $A, B, C$  be

$$x, y, z \quad (\text{positive integers}).$$

Along side  $AB$  of length 6, the two cuts remove lengths  $x$  (near  $A$ ) and  $y$  (near  $B$ ), so the remaining middle segment on  $AB$  (which is a side of the hexagon) has

length  $6 - x - y$ . Similarly the other two middle segments on  $BC$  and  $CA$  have lengths  $6 - y - z$  and  $6 - z - x$ .

Thus, going around the hexagon, its six side lengths are

$$(6 - x - y, y, 6 - y - z, z, 6 - z - x, x),$$

so all are positive if

$$x + y \leq 5, \quad y + z \leq 5, \quad z + x \leq 5,$$

with  $x, y, z \in \mathbb{Z}_{>0}$ .

Two such hexagons that differ by a rotation or reflection of the equilateral triangle correspond exactly to permuting  $(x, y, z)$ , so we count solutions up to permutation. We may assume

$$1 \leq x \leq y \leq z.$$

Now enumerate all triples  $(x, y, z)$  with  $x \leq y \leq z$  satisfying the pair-sum constraints.

If  $x = 1$ , then  $1 + y \leq 5 \Rightarrow y \leq 4$  and  $1 + z \leq 5 \Rightarrow z \leq 4$ . Also  $y + z \leq 5$ . The valid triples are:

$$(1, 1, 1), (1, 1, 2), (1, 1, 3), (1, 1, 4), (1, 2, 2), (1, 2, 3).$$

(For example,  $(1, 2, 4)$  fails since  $2 + 4 > 5$ .)

If  $x = 2$ , then  $2 + y \leq 5 \Rightarrow y \leq 3$  and  $2 + z \leq 5 \Rightarrow z \leq 3$ . Also  $y \leq z$  and  $y + z \leq 5$ . The valid triples are:

$$(2, 2, 2), (2, 2, 3).$$

If  $x \geq 3$ , then  $x + y \geq 3 + 3 = 6 > 5$ , impossible.

So there are

$$6 + 2 = 8$$

unordered triples, hence 8 distinct hexagons up to rotation/reflection.

8

### Solution 3 (Casework)

We can already get  $(1, 1, 1)$  and  $(2, 2, 2)$  if we want equal sides (the case  $(3, 3, 3)$  doesn't work as it makes a triangle)

Then, we can compute  $(1, 1, 2)$ ,  $(2, 2, 3)$ ,  $(2, 2, 1)$ ,  $(1, 1, 4)$ ,  $(1, 1, 3)$  for cases of the same side length (and also two of the same)

Finally, we find the cause  $(1, 2, 3)$  for all unique numbers.

This gives us a total of (E) 8 total cases.

### Solution 4 (Burnside's Lemma, unnecessarily long)

Let  $X$  be the set of all possible orientations of the hexagon (where reflections and rotations are different), let  $\mathcal{O}$  denote the set of all orbits of  $X$  under  $D_3$  where  $D_3$  is the dihedral group of order 3. Burnside's tells us that

$$|\mathcal{O}| = \frac{1}{|D_3|} \sum_{g \in D_3} |\text{Fix}(g)|$$

where  $\text{Fix}(g) = \{x \mid g \cdot x = x\}$ , the fixed points under the action  $g$ .  $|\mathcal{O}|$  is the solution to our problem in this case, as every rotation or reflection of the hexagon is represented in  $D_3$ . More accurately,  $D_3$  represents the symmetries of the triangle, and the hexagon is the triangle with the vertices cut out.

First, let's find  $|X|$ . Letting the sides of the hexagon be  $a, b, c, d, e, f$ , we note that  $b = 6 - a - c, d = 6 - c - e, f = 6 - e - a$ . Since  $0 < a, b, c, d, e, f < 6$ ,  $1 \leq a + c, c + e, e + a \leq 5$ . We can now do casework:

$a = 1$ : We have  $1 \leq c \leq 4, 1 \leq e \leq 4$ , and  $1 \leq c + e \leq 5$ . Letting  $x = c$  and  $y = e$ , we manually check the lattice points in a square in the  $xy$ -plane with coordinates  $(1, 1)$  to  $(4, 4)$  for  $x + y > 5$ . 6 elements satisfy this inequality, so they are the solutions  $(c, e)$  which don't work - therefore, there are  $16 - 6 = 10$  working solutions  $(c, e)$  satisfying our inequalities.

$a = 2$ :  $1 \leq c \leq 3, 1 \leq e \leq 3, 1 \leq c + e \leq 5$ . In a similar fashion, in the  $3 \times 3$  square of lattice points only  $(3, 3)$  fails to meet  $c + e \leq 5$  so 8 solutions work from here.

$a = 3$ :  $1 \leq c \leq 2, 1 \leq e \leq 2, 1 \leq c + e \leq 5$ . Obviously every solution works here, so 4 solutions add to the count.

$a = 4$ : Here,  $(c, e) = (1, 1)$  is the only working solution.

The final count is 23. Therefore,  $|X| = 23$ .

There are three conjugacy classes of  $D_3$ :  $\{e\}, \{R, R^{-1}\}$ , and  $\{T, TR, TR^2\}$ . We picture  $D_3$  as a triangle with each of the vertices labeled  $a, c, e$  in some order. (The hexagon is the equilateral triangle of side 6 with smaller equilateral triangles of length  $a, c, e$  cut out of the corners.  $\text{Fix}(g) = \text{Fix}(h)$  if  $g$  and  $h$  are in the same conjugacy class, so it suffices to find  $|\text{Fix}(e)|, |\text{Fix}(R)|$ , and  $|\text{Fix}(T)|$ .

Every "vertex" is in its own orbit after the  $e$ -action, meaning that every element is completely fixed, so  $|\text{Fix}(e)| = |X|$ .

The orbit of  $R$  contains all the vertices, meaning for an element to be fixed, every vertex must be the same, resulting in two triangles:  $(1, 1, 1)$  and  $(2, 2, 2)$ .

Finally, the  $T$ -action fixes one of the vertices and swaps the other two. WLOG let  $a$  be the fixed vertex and  $c, e$  be swapped - meaning that they must be equal for the triangle to be fixed. Our equations become  $1 \leq a + c \leq 5, 1 \leq 2c \leq 5$ , or that  $c \in 1, 2$ .  $c = 1$  generates 4 solutions, while  $c = 2$  generates 3. Therefore, 7 triangles are fixed by the  $T$ -action, and  $|\text{Fix}(T)| = 7$ .

Plugging all this into Burnside's,

$$\begin{aligned} |\mathcal{O}| &= \frac{1}{|D_3|} \sum_{g \in D_3} |\text{Fix}(g)| \\ |\mathcal{O}| &= \frac{1}{6} (1 \cdot 23 + 2 \cdot 2 + 3 \cdot 7) \\ |\mathcal{O}| &= \frac{1}{6} (48) \\ |\mathcal{O}| &= \boxed{(E) \ 8}. \end{aligned}$$

### Solution 5 (Intuitive, Simple)

Observe that the side length of  $\triangle ABC$  is 6. A hexagon is equiangular if and only if all "outer triangles, *formed by two adjacent vertices of the hexagon and one vertex of the equilateral triangle, are equilateral*. We perform casework on each equilateral triangle and obtain 8 different configurations. Hence  $\boxed{8}$ .